
Foundational Mathematics

A Textbook

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Foundational Mathematics

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This edition was prepared with LaTeX. Corrections identified after publication may be incorporated into dated revisions.

Preface

When I began this book, I did not expect to develop it into a complete volume. I had previously started several projects that remained unfinished—not because I considered them poor, but because I had not found the passion needed to carry them through.

During my foundation studies, I came to realise how important it is for students to master the mathematical ideas introduced at this stage. Foundation mathematics provides the gateway to a wide range of mathematical topics encountered in later study.

This book brings together material taught by Centres for Foundation Studies, matriculation colleges, and other institutions offering foundation programmes. Their syllabuses vary considerably. At PASUM, for example, Physical Science students study Bernoulli differential equations and inverse hyperbolic functions, topics that are not generally included in matriculation courses. These differences encouraged me to combine the content offered by the various institutions into a single volume.

I also decided to retain several topics associated with earlier curricula, including linear law, and to introduce subjects such as graph theory that are not always taught at foundation level. Although some of this material extends beyond a typical pre-university syllabus, it provides useful preparation for degree-level study and allows interested readers to explore further.

The book is intended primarily for students undertaking foundation-level study, including matriculation, Cambridge A Level, and equivalent programmes. It may also be useful for undergraduates who wish to refresh their current knowledge, as well as to general readers who enjoy mathematics.

I have no intention of making a profit from this work or of seeking publicity through it. My aim is simply to encourage readers to recognise the importance of mathematics, not only in their future studies but also in the way it shapes everyday life. Whether or not we notice it, mathematics is everywhere, and I hope this book helps readers appreciate both its usefulness and its beauty.

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How to Use This Book

The chapters are arranged in four parts that move from algebraic foundations to modelling, calculus, and discrete or computational topics. Most readers should study the chapters within a part in order, although instructors may select a shorter route suited to their programme.

Each chapter opens with prerequisites and learning outcomes. Worked examples develop the principal methods, while the chapter review collects key results and warns against common errors. Practice exercises generally progress from routine skills to applications and modelling. Concise answers are provided in Appendix F; they are intended for checking work after a complete solution has been attempted.

Appendix A explains recurring symbols and unit conventions, and Appendix D collects frequently used formulae. The subject index provides a second route to definitions and examples when a topic appears in more than one chapter.

The symbol conventions and calculator settings appropriate to a problem should always be checked before calculation. Unless a question requests a decimal approximation, exact forms should be retained for as long as possible. Numerical answers should include suitable units and a stated degree of accuracy where rounding is required.

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Introduction

Mathematics carries an unfortunate reputation among many students. Its very mention may recall frustration at school, long nights spent preparing for examinations, or the persistent belief that the subject is suited only to a small group of naturally gifted people. Watching someone evaluate a definite integral such as $\int_a^b f(x) dx$ without a second thought, or working through a proof line by line without hesitation, can reinforce the impression that mathematical ability is fixed at the outset. This impression, although widespread and understandable, doesn't really hold up.

Mathematics is not, by any means, an arbitrary collection of rules to be fully memorised, nor is it primarily a technique for rapid mental computation. It is, more precisely, a formal language developed over centuries to describe structure, quantity, and change with precision. Each definition, theorem, and notational convention you will encounter in the chapters that follow has a history: it was introduced to resolve a specific problem, to generalise an existing result, or to remove an ambiguity in earlier reasoning. Mathematics is best regarded not as an obstacle reserved for specialists, but as a cumulative intellectual tradition, one that is built incrementally by many contributors over a long period of time, and accessible to any student willing to engage with it systematically and in the proper order.

The applications of mathematics can be found in nearly every field of inquiry, often more extensively than students initially thought. In physics, it provides the formal apparatus by which the laws of motion are expressed, as in the relation $\mathbf{F} = m\mathbf{a}$, which relates force, mass, and acceleration in a way that holds from a dropped ball to the orbit of planets. In chemistry, mathematical models are used to describe the rates at which molecular species react and the conditions under which chemical systems reach equilibrium. In computer science, mathematics shapes the logical foundations underlying algorithm design, software correctness, and the architectures of modern artificial intelligence systems, from search algorithms to neural networks. In economics and psychology, quantitative methods are used to model decision-making, to analyse the behaviour of markets, and to characterise human behaviour under conditions of uncertainty and incomplete information. Mathematics, in this sense, functions as an underlying infrastructure of the sciences, whether or not it is explicitly acknowledged within a given field of study, and noticing this is often the first real reason to want to study it.

Despite this ubiquity, moving from secondary school to university-level quantitative work is genuinely hard for a lot of students (like myself), and that's worth saying plainly rather than glossing over. School curricula tends to focus on procedure: you are trained to apply a given formula correctly under examination conditions, with comparatively little attention paid to the underlying justification for that formula, or to the circumstances under which it may fail to apply. Then all of a sudden, in the university level, you are expected to analyse functions in the abstract, to derive results from first principles, and to construct multi-step arguments independently, often with far less scaffolding than before. This shift feels less like a gradual step up and it is a common source of difficulty even

among capable and well-prepared students (and top-scorers too). Such difficulty should not be interpreted as evidence of insufficient aptitude on your part, but rather, it reflects a structural gap in how mathematical reasoning is typically introduced prior to university study which is, more or so, a gap in instruction, not in you.

It is worth noting that this level of difficulty is by no means unique to any one student, and recognising this can itself be a useful corrective to the discouragement it often produces. The experience is a near-universal feature of the transition into rigorous quantitative coursework, encountered in some form by the majority of students who go on to develop genuine proficiency in the subject. Apparent fluency among one's peers is frequently a matter of prior exposure, practised presentation, or simple confidence under pressure, rather than an absence of underlying difficulty. Feeling confused at this stage isn't unusual, but it is characteristic of the learning process itself, and almost everyone who has worked through this content has felt it to some degree.

Back to this book, all chapters have been explained as clearly and thoroughly as possible (and as I can) without extending unnecessarily beyond the boundaries of pre-university (i.e. STPM, foundation, matriculation and all programmes that's equivalent to A-level) mathematics. The chapters are arranged in a way that the fundamental ideas introduced near the beginning support the more advanced material that appears later. Here's what all the chapters have to offer:

Chapter 1 introduces readers to the idea of the real number system, including rational and irrational numbers, set and interval notation, indices, surds, and logarithms. These ideas supply much of the notation and algebraic language required throughout the rest of the book.

Chapter 2 develops the concept of a function, in which readers will examine domains and ranges, common families of functions, transformations, composite functions, piecewise functions and inverse functions. It establishes the connection between algebraic rules and their graphical representations.

Chapter 3 studies polynomials, their arithmetic operations, methods of dividing polynomials, factorisation, the remainder and factor theorems, and partial fractions. Readers are encouraged to understand deeply the ideas of partial fractions as it will be used again in calculus especially.

Chapter 4 extends the algebraic techniques of the preceding chapters to inequalities involving quadratic, cubic, rational, and absolute-value expressions.

Chapter 5 introduces matrices and their operations before proceeding to determinants, inverses, and the solution of systems of simultaneous linear equations. This, along with vector is the gateway to a subject called Linear Algebra, which is essential in the world of computer science.

Chapter 6 develops the idea of trigonometry from the measurement of angles and the basic trigonometric functions to graphs, identities, equations, and inverse trigonometric functions.

Chapter 7 explains how non-linear relationships may be transformed into straight-line form, a technique that allows curved data to be analysed using the familiar tools of linear regression. It introduces scatter plots, lines of best fit, and linear law, together with the interpretation of gradients and intercepts once a model has been fitted to experimental or

observed data.

Chapter 8 examines the coordinate geometry of circles and conic sections, including parabolas, ellipses, and hyperbolas. Each curve is studied through its standard algebraic form, its principal geometric features, and its parametric equations, which describe the curve in terms of an auxiliary variable rather than a direct relation between x and y .

Chapter 9 introduces the principal ideas of probability and statistics, beginning with the rules of counting through permutations and combinations, which underpin the calculation of probabilities in situations involving a finite number of outcomes. From this foundation, the chapter proceeds to the binomial, normal, Poisson, and exponential distributions, examining the circumstances under which each arises, their defining parameters, and their use in modelling discrete and continuous random phenomena. This chapter, then ends with the idea of hypothesis testing, which is important in the making of a census.

Chapter 10 extends the real number system to complex numbers and develops their algebraic, polar, exponential, and geometric forms, together with De Moivre's theorem and Euler's formula. This chapter marks a shift from regarding numbers purely as quantities towards regarding them as points possessing their own geometric structure.

Chapter 11 introduces hyperbolic functions, their identities, graphs, and inverse functions, while examining their similarities to and differences from the circular trigonometric functions

Chapter 12 begins the study of differential calculus through the idea of limits and first principles, then develops the standard rules of differentiation of different functions and forms together with their applications to rates of change and optimisation.

Chapter 13 introduces integral calculus as the inverse process of differentiation. It covers the principal methods of integration, Fundamental Theorem of Calculus, indefinite and definite integrals, and applications involving accumulated quantities and areas (including Disc and Washer methods), in which a quantity is summed continuously rather than in discrete steps.

Chapter 14 brings differentiation and integration together through differential equations. It introduces several important classes of first-order equations and demonstrates their use in modelling growth, decay, mixing, and other systems subject to continuous change.

Chapter 15 studies sequences and series, including convergence, binomial and power series, and the Taylor and Maclaurin expansions. This material draws directly on earlier calculus: convergence rests on the idea of a limit, and the coefficients of a Taylor or Maclaurin expansion come from repeated differentiation, allowing a function to be approximated, and in favourable cases reconstructed exactly, by an infinite polynomial built from its derivatives.

In **Chapter 16**, readers study vectors in two and three dimensions, including vector operations, products, relative motion, and vector-valued differentiation and integration. This chapter unites algebraic, geometric, and calculus-based methods of describing magnitude, direction, and motion.

Chapter 17 (taken from Malaysia's DSKP curriculum) provides an introduction to graph theory, examining vertices and edges, paths, cycles, trees, planar graphs, colouring,

shortest paths, and spanning trees. It offers a first encounter with the discrete mathematics underlying computer science, optimisation, and network analysis, and differs markedly in character from the calculus-oriented chapters that precede it.

Chapter 18 introduces logical reasoning through propositions, truth tables, implications, quantifiers, valid arguments, common fallacies, and elementary methods of mathematical proof. This chapter is concerned less with computation than with the disciplined construction of argument, a skill that underlies the reasoning employed throughout the rest of the book, whether or not it is made explicit.

The bonus **Chapter 19** introduces computing with R. It develops the basic syntax required to work with vectors and data frames, to import and visualise data, to evaluate probability distributions, to carry out one-sample hypothesis tests, and to perform matrix calculations, thereby providing an opportunity to apply several ideas from earlier chapters, particularly Chapters 5 and 9, in a computational setting. This chapter is very useful for students who will continue their studies in analytical mathematics, statistical mathematics and equivalent.

The syllabus of the Centre for Foundation Studies in Science, Universiti Malaya (PASUM), forms the main basis for the book. Material from other Malaysian and international pre-university programmes has been folded in too, so readers can focus on whatever their own programme actually requires while still having the rest of the chapters available as extension material, for anyone who wants to go deeper into a topic or get ahead for later study.

Each chapter's exercises were arranged in the order from basic skills up to application-based problems, so you get a chance to nail down a technique before it's thrown at you in a less familiar setting. The appendices collect the notation, trigonometric and hyperbolic identities, statistical tables, exercise answers, and references used through the book.

The material draws on textbooks, lecture notes, curricula, and open educational resources from institutions including Universiti Malaya, MIT OpenCourseWare, Harvard University, and the London School of Economics and Political Science. Full details are in the reference list, and if a topic here interests you beyond what this book covers, it's worth going to those original sources directly.

Altogether, the chapters are meant to build on each other rather than sit as separate topics. It helps to move between algebraic, graphical, numerical, and geometric ways of looking at a problem, since a single question often makes more sense, and is easier to check, once you've seen it from more than one angle.

Part I

Algebra and Functions

Chapter 1

Number System

Chapter guide

Prerequisites. Arithmetic with integers, fractions, decimals, and basic algebraic notation.

Learning outcomes. By the end of the chapter, readers should be able to classify real numbers, use set and interval notation, manipulate indices, surds, and logarithms, and solve equations involving these forms.

The real number system is normally studied near the beginning of a foundation mathematics course because it supplies the language and algebraic structure used throughout the subject.

A number system represents quantities using specified symbols and rules. This chapter concentrates on the real numbers, which divide into two disjoint classes: rational numbers and irrational numbers. Real numbers provide the principal setting for elementary algebra, calculus, geometry, and measurement.

The selection and sequencing of topics follow commonly used Malaysian pre-university treatments of number, algebra, indices, surds, and logarithms (Chai et al. 2024; Ministry of Education Malaysia 2018; Wong Mee Kiong et al. 2019; Zaini bin Musa et al. 2020).

Number notation did not arise in a single step. Different cultures developed ways to record quantity, place value, zero, and symbolic calculation. The timeline in Figure 1.1 is deliberately conceptual rather than a complete history: its purpose is to show how several ideas contribute to the notation used today.

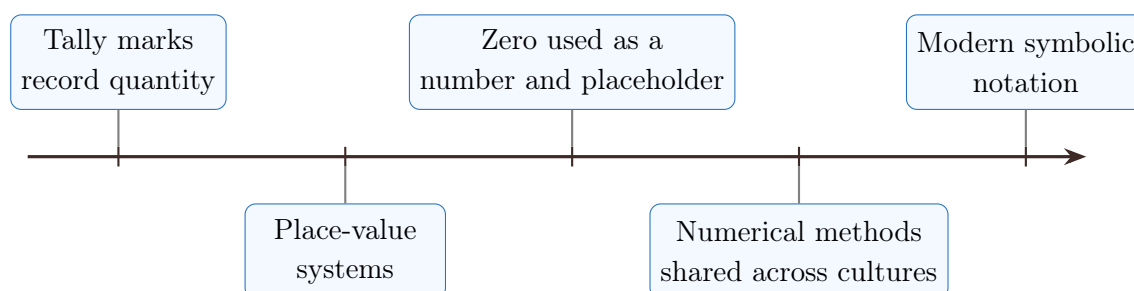


Figure 1.1: A simplified conceptual timeline of ideas that contributed to modern number notation. These developments occurred across multiple cultures and often overlapped.

1.1 Rational numbers

Informally, a rational number is a number that can be expressed as a fraction of two integers.

Definition 1.1: Rational number.

A real number is rational if it can be expressed as a quotient of two integers with a non-zero denominator. The set of rational numbers is

$$\mathbb{Q} = \left\{ \frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0 \right\}. \quad (1.1)$$

Examples include 2, $\frac{4}{5}$, and -7 . The number $\sqrt{2}$ is not rational.

1.1.1 Natural numbers

The notation for the natural numbers is not completely universal: some texts include zero, while others begin with one. Throughout this book, the natural numbers begin with one:

$$\mathbb{N} = \{1, 2, 3, \dots\}. \quad (1.2)$$

These are the positive counting numbers. They are contained in the integers, rational numbers, real numbers, and complex numbers.

1.1.2 Whole numbers

The *whole numbers* are obtained by adjoining zero to the natural numbers:

$$\mathbb{W} = \{0, 1, 2, 3, \dots\}. \quad (1.3)$$

The symbol $\mathbb{W}$ is common in school mathematics but is not universal; the same set is also written $\mathbb{N}_0$. Under the convention adopted in this book, $0 \notin \mathbb{N}$ but $0 \in \mathbb{W}$.

1.1.3 Integers

The integers consist of zero, the positive natural numbers, and their negatives. Define

$$\mathbb{Z}^+ = \{1, 2, 3, \dots\}, \quad (1.4)$$

$$\mathbb{Z}^- = \{-1, -2, -3, \dots\}. \quad (1.5)$$

Consequently,

$$\mathbb{Z} = \mathbb{Z}^- \cup \{0\} \cup \mathbb{Z}^+. \quad (1.6)$$

1.1.4 Prime numbers

A *prime number* is a natural number greater than 1 whose only positive divisors are 1 and itself. A natural number greater than 1 that is not prime is called a *composite number*. The number 1 is neither prime nor composite.

There is no universally adopted symbol for the set of prime numbers. In this chapter, it may be denoted by $\mathcal{P}$ and written in set-builder form as

$$\mathcal{P} = \{n \in \mathbb{N} \mid n > 1 \text{ and } (d \mid n \implies d \in \{1, n\}) \text{ for every } d \in \mathbb{N}\}. \quad (1.7)$$

Prime factorisation will be used later when simplifying integers, indices, and surds.

1.2 Irrational numbers

An irrational number is a real number that cannot be written as a quotient of two integers.

Definition 1.2: Irrational number.

A real number is irrational if it cannot be expressed as $\frac{a}{b}$, where $a, b \in \mathbb{Z}$ and $b \neq 0$. The set of irrational numbers is

$$\mathbb{R} \setminus \mathbb{Q}. \quad (1.8)$$

1.2.1 Proving that a number is irrational

The following elementary lemma makes the parity step in the irrationality argument explicit.

Lemma 1.1 (Parity of a square). *For every integer n , the integer n^2 is even if and only if n is even.*

Proof. If n is even, write $n = 2k$ for some $k \in \mathbb{Z}$. Then $n^2 = 2(2k^2)$, so n^2 is even. Conversely, suppose that n is not even. Then $n = 2k + 1$ for some $k \in \mathbb{Z}$, and

$$n^2 = (2k + 1)^2 = 2(2k^2 + 2k) + 1,$$

which is odd. Therefore an even square cannot come from an odd integer, so n^2 even implies n even. $\square$

Theorem 1.2. *The number $\sqrt{2}$ is irrational.*

Proof. Suppose, for contradiction, that $\sqrt{2}$ is rational. It can then be written in lowest terms as

$$\sqrt{2} = \frac{a}{b},$$

where $a, b \in \mathbb{N}$ and $\gcd(a, b) = 1$. Squaring gives

$$2 = \frac{a^2}{b^2}, \quad \text{so} \quad a^2 = 2b^2.$$

Thus a^2 is even, so the lemma implies that a is even. Write $a = 2c$ for some $c \in \mathbb{N}$. Substitution yields

$$4c^2 = 2b^2, \quad \text{so} \quad b^2 = 2c^2.$$

The lemma now implies that b is even. Hence 2 divides both a and b , contradicting the choice $\gcd(a, b) = 1$. Therefore $\sqrt{2}$ is irrational. $\square$

1.3 Visualising the real number sets

The principal inclusions among the common real number sets are shown in Figure 1.2. The irrational numbers occupy the part of $\mathbb{R}$ outside $\mathbb{Q}$.

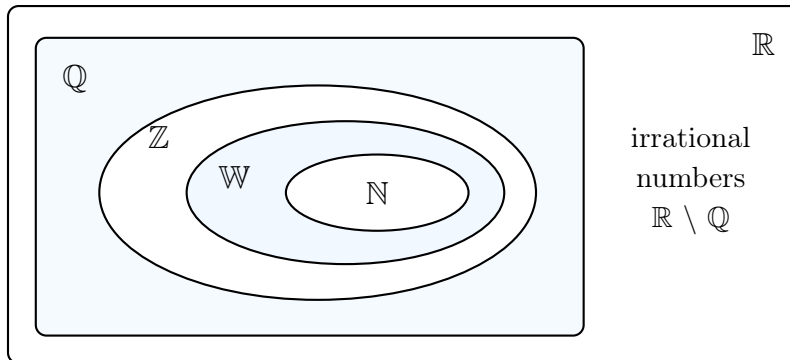


Figure 1.2: The nesting of the natural, whole, integer, rational, and real number sets under the convention used in this book.

Thus,

$$\mathbb{N} \subsetneq \mathbb{W} \subsetneq \mathbb{Z} \subsetneq \mathbb{Q} \subsetneq \mathbb{R}, \quad (1.9)$$

$$\mathbb{R} = \mathbb{Q} \cup (\mathbb{R} \setminus \mathbb{Q}), \quad \mathbb{Q} \cap (\mathbb{R} \setminus \mathbb{Q}) = \emptyset. \quad (1.10)$$

1.4 Interval notation

An interval is a set containing every real number between specified endpoints. For example, $[0, 1]$ contains 0, 1, and every real number between them; it is called the *unit interval*. An interval may exclude both endpoints, include both endpoints, or include exactly one endpoint.

Intervals can be represented using inequalities, interval notation, and number-line diagrams.

1.4.1 Inequalities

An **inequality** compares quantities whose values need not be equal. Inequalities are particularly useful for describing intervals because they state directly which real numbers lie to the left or right of an endpoint. The four principal inequality symbols are

$$<, \quad >, \quad \leq, \quad \geq.$$

The symbols $<$ and $>$ give **strict inequalities**. For example, $x < 4$ excludes the endpoint 4. The symbols $\leq$ and $\geq$ give **non-strict inequalities**; thus $x \leq 4$ includes the endpoint.

Two inequalities may be combined to place both a lower and an upper bound on a variable. If $a < b$, then

$$a < x < b$$

means that both $a < x$ and $x < b$ must hold. It therefore describes all real numbers strictly between a and b . Endpoints may be included separately, as in $a \leq x < b$ or $a < x \leq b$.

Inequality	Description	Interval notation
$a < x < b$	Between a and b ; neither endpoint is included	(a, b)
$a \leq x \leq b$	Between a and b ; both endpoints are included	$[a, b]$
$a < x \leq b$	a is excluded and b is included	$(a, b]$
$a \leq x < b$	a is included and b is excluded	$[a, b)$
$x < a$	All real numbers less than a	$(-\infty, a)$
$x \geq a$	All real numbers greater than or equal to a	$[a, \infty)$

The word *and* corresponds to the intersection of two conditions. For example, $x > -2$ and $x \leq 5$ combine to give $-2 < x \leq 5$. The word *or* corresponds to a union. Thus

$$x < -1 \quad \text{or} \quad x \geq 3$$

describes two separate intervals. When infinity occurs, its endpoint is always excluded because ∞ and $-\infty$ are not real numbers.

Example 1.1: Representing a bounded interval with inequalities

The interval

$$[-3, 4)$$

contains -3 but does not contain 4 . Its inequality representation is

$$\boxed{-3 \leq x < 4}.$$

Equivalently, this says that $x \geq -3$ *and* $x < 4$.

1.4.2 Types of intervals

For real numbers $a < b$, the principal bounded intervals are as follows.

1. An **open interval** (a, b) contains all real numbers between a and b but excludes both endpoints. For example, $(0, 2)$ is the set $\{x \in \mathbb{R} \mid 0 < x < 2\}$.
2. A **closed interval** $[a, b]$ contains all real numbers between a and b , including both endpoints. Thus $[0, 3] = \{x \in \mathbb{R} \mid 0 \leq x \leq 3\}$.

3. A **half-open interval** includes exactly one endpoint. For example, $(2, 9]$ excludes 2 and includes 9; equivalently, $(2, 9] = \{x \in \mathbb{R} \mid 2 < x \leq 9\}$.

1.4.3 Number line

A **number line** is a geometric representation of the real numbers. A point called the **origin** is assigned the value zero. Numbers increase as we move to the right of the origin and decrease as we move to the left. Consequently, positive numbers appear on the right-hand side, while negative numbers appear on the left-hand side.

The distance between consecutive integers is chosen to represent one unit. Once this scale has been established, every real number can be placed at a unique point on the line. These include integers such as -4 and 3 , rational numbers such as $-\frac{15}{4}$ and $\frac{20}{9}$, and irrational numbers such as $\sqrt{2}$, e , and π .

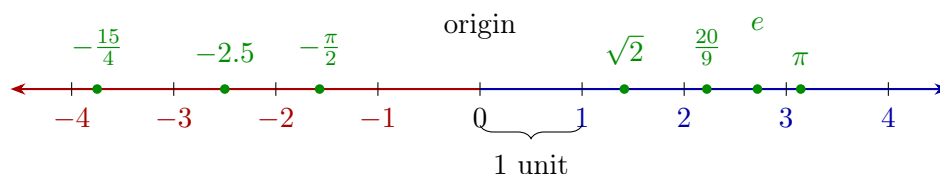


Figure 1.3: Rational and irrational numbers represented on the real number line.

The real number line extends indefinitely in both directions, from negative infinity to positive infinity. In interval notation, the set of all real numbers is written as

$$\mathbb{R} = (-\infty, \infty).$$

Parentheses must be used because $-\infty$ and ∞ are not real numbers and therefore cannot be included as endpoints.

1.5 Set notation and relationships among sets

A *set* is a well-defined collection of distinct objects. The objects contained in a set are called its *elements* or *members*. Sets are usually denoted by capital letters such as A , B , and S , while their elements are written inside braces.

For example,

$$A = \{1, 2, 3, 4\}$$

is the set whose elements are 1, 2, 3, and 4. The order in which elements are listed does not matter, and repeating an element does not create a new element. Thus,

$$\{1, 2, 3\} = \{3, 2, 1\} = \{1, 1, 2, 3\}.$$

1.5.1 Common set notation

The most frequently used symbols are summarised below.

Table 1.1: Basic notation for sets.

Symbol	Name	Meaning
$\{ \}$	Set braces	Enclose the elements or defining condition of a set.
$x \in A$	Element of	x is an element of A .
$x \notin A$	Not an element of	x is not an element of A .
$\emptyset$	Empty set	The unique set containing no elements.
U	Universal set	The set of all objects under consideration.
$n(A)$	Number of elements	The number of elements in a finite set A . The notation $ A $ is also sometimes used.
$A = B$	Equal sets	A and B contain exactly the same elements.
$A \subseteq B$	Subset	Every element of A is also an element of B .
$A \subsetneq B$	Proper subset	$A \subseteq B$, but $A \neq B$.
$A \not\subseteq B$	Not a subset	At least one element of A is not in B .
$A \cup B$	Union	Elements that belong to A , to B , or to both.
$A \cap B$	Intersection	Elements that belong to both A and B .
$A \setminus B$	Set difference	Elements in A that are not in B .
A^c	Complement	Elements of U that do not belong to A .

Throughout this book, $A \subseteq B$ means that every element of A belongs to B . The notation $A \subsetneq B$ is used when A is a proper subset of B , meaning that $A \subseteq B$ but $A \neq B$.

1.5.2 Ways to describe a set

Roster form

In roster or listing form, the elements are written explicitly:

$$P = \{2, 3, 5, 7\}.$$

This says that P is the set of prime numbers below 10.

Set-builder form

In set-builder notation, a set is described by a property that its elements must satisfy:

$$A = \{x \in S \mid x \text{ satisfies a stated condition}\}. \quad (1.11)$$

The vertical bar $|$ is read as “such that.” A colon may also be used. For example,

$$\begin{aligned}
E &= \{x \in \mathbb{Z} \mid x \text{ is even}\}, \\
I &= \{x \in \mathbb{R} \mid -1 < x \leq 3\} = (-1, 3], \\
Q &= \left\{ \frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0 \right\}.
\end{aligned}$$

Set-builder notation must specify a domain whenever the condition alone could be ambiguous. For instance, the possible values represented by $\{x \mid x^2 < 4\}$ depend on whether x is chosen from the integers or the real numbers.

1.5.3 Membership, equality, and subsets

Membership and subset notation describe different relationships. If

$$A = \{1, 2, 3\},$$

then

$$1 \in A, \quad \{1\} \subseteq A,$$

but $\{1\} \in A$ is false because the set $\{1\}$ itself is not listed as an element of A .

Two sets are equal when they contain exactly the same elements. For example,

$$\{1, 2, 3\} = \{3, 1, 2\},$$

because the order of the elements is irrelevant. Every set is a subset of itself, and the empty set is a subset of every set.

1.5.4 Operations on sets

Let

$$A = \{1, 2, 3, 4\} \quad \text{and} \quad B = \{3, 4, 5\}.$$

Their union contains every element occurring in either set:

$$A \cup B = \{1, 2, 3, 4, 5\}.$$

Their intersection contains only their common elements:

$$A \cap B = \{3, 4\}.$$

Their differences depend on the order:

$$A \setminus B = \{1, 2\}, \quad B \setminus A = \{5\}.$$

Two sets are *disjoint*, or *mutually exclusive*, if they have no common elements:

$$\boxed{A \cap B = \emptyset.} \quad (1.12)$$

The complement of A is always defined relative to a universal set U :

$$\boxed{A^c = U \setminus A.} \quad (1.13)$$

For example, if $U = \{1, 2, 3, 4, 5, 6\}$ and $A = \{1, 2, 3, 4\}$, then $A^c = \{5, 6\}$.

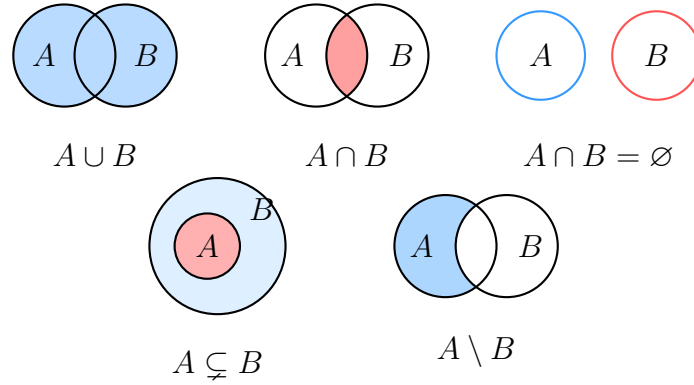


Figure 1.4: Common relationships and operations between two sets.

1.5.5 Useful set identities

The following identities are particularly useful when interpreting Venn diagrams and probability statements:

$$\begin{aligned} A \cup \emptyset &= A, & A \cap U &= A, \\ A \cup U &= U, & A \cap \emptyset &= \emptyset, \\ A \cup A^c &= U, & A \cap A^c &= \emptyset. \end{aligned}$$

De Morgan's laws describe the complements of unions and intersections:

$$\boxed{(A \cup B)^c = A^c \cap B^c, \quad (A \cap B)^c = A^c \cup B^c.} \quad (1.14)$$

In words, the complement of “ A or B ” is “not A and not B ”, while the complement of “ A and B ” is “not A or not B ”. These statements are often used when translating between set notation and probability notation.

1.5.6 Counting elements in sets

The notation $n(A)$ represents the number of elements in a finite set A . When two sets overlap, adding $n(A)$ and $n(B)$ counts the elements in their intersection twice. Therefore,

$$\boxed{n(A \cup B) = n(A) + n(B) - n(A \cap B).} \quad (1.15)$$

If A and B are disjoint, then $n(A \cap B) = 0$, so

$$n(A \cup B) = n(A) + n(B).$$

For example, suppose a class of 30 students contains 18 students studying physics, 15 studying chemistry, and 7 studying both. The number studying at least one of the two subjects is

$$n(P \cup C) = 18 + 15 - 7 = 26.$$

Hence, the number studying neither subject is

$$30 - 26 = 4.$$

In a two-set Venn diagram, it is usually safest to enter the intersection first. The numbers belonging only to A and only to B can then be found by subtraction, followed by the number outside both sets.

1.5.7 Relationships among common number sets

Using the convention $\mathbb{N} = \{1, 2, 3, \dots\}$ and $\mathbb{W} = \{0, 1, 2, 3, \dots\}$, the familiar number sets satisfy

$$\boxed{\mathbb{N} \subsetneq \mathbb{W} \subsetneq \mathbb{Z} \subsetneq \mathbb{Q} \subsetneq \mathbb{R}.} \quad (1.16)$$

The irrational numbers form the complement of $\mathbb{Q}$ within $\mathbb{R}$:

$$\mathbb{R} \setminus \mathbb{Q}.$$

Consequently,

$$\boxed{\mathbb{R} = \mathbb{Q} \cup (\mathbb{R} \setminus \mathbb{Q}), \quad \mathbb{Q} \cap (\mathbb{R} \setminus \mathbb{Q}) = \emptyset.} \quad (1.17)$$

Thus, every real number is either rational or irrational, but no real number can be both.

1.6 Properties of real numbers

The real numbers form an *ordered field*. This means that addition and multiplication obey the familiar field laws and that the ordering of real numbers is compatible with these operations. For all $a, b, c \in \mathbb{R}$:

1. **Closure:** $a + b \in \mathbb{R}$ and $ab \in \mathbb{R}$.
2. **Commutativity:** $a + b = b + a$ and $ab = ba$.
3. **Associativity:** $(a + b) + c = a + (b + c)$ and $(ab)c = a(bc)$.
4. **Distributivity:** $a(b + c) = ab + ac$.

5. Identity elements:

- (a) 0 is the additive identity: $a + 0 = 0 + a = a$.
- (b) 1 is the multiplicative identity: $a \cdot 1 = 1 \cdot a = a$.

6. Inverse elements:

- (a) Every $a \in \mathbb{R}$ has an additive inverse $-a$, for which $a + (-a) = 0$.
- (b) Every non-zero $a \in \mathbb{R}$ has a multiplicative inverse $a^{-1} = 1/a$, for which $aa^{-1} = 1$.

7. Compatibility with order: if $a < b$, then $a + c < b + c$; if, in addition, $c > 0$, then $ac < bc$.**1.7 Indices, surds and logarithms**

Indices, surds, and logarithms extend familiar arithmetic notation and provide essential techniques for algebra, functions, and calculus. Some readers will have encountered these topics previously; the treatment below reviews their fundamental laws while developing the precision required at pre-university level.

1.7.1 Indices

An **index** (plural: **indices**), also called an **exponent** or a **power**, provides a compact way of writing repeated multiplication. For a positive integer n ,

$$a^n = \underbrace{a \times a \times \cdots \times a}_{n \text{ factors}}. \quad (1.18)$$

In the expression a^n , the number or expression a is the **base**, while n is the **index** or **exponent**. For example,

$$3^4 = 3 \times 3 \times 3 \times 3 = 81.$$

Here, 3 is the base and 4 is the index. The expression is read as “three to the power of four” or “three raised to the fourth power.” In particular, an exponent of 1 leaves the base unchanged:

$$a^1 = a.$$

Some examples of positive integer indices are

$$\begin{aligned}
5^2 &= 5 \times 5 = 25, \\
2^3 &= 2 \times 2 \times 2 = 8, \\
3^4 &= 3 \times 3 \times 3 \times 3 = 81, \\
2^5 &= 2 \times 2 \times 2 \times 2 \times 2 = 32, \\
13^1 &= 13.
\end{aligned}$$

The notation is not restricted to numerical bases. For example, $x^4 = x \times x \times x \times x$. The base may also be a product, a fraction, or an algebraic expression, provided that brackets are used to show the intended base clearly.

Laws of indices

Let a and b be real numbers, and let m and n be integers. Whenever all the expressions involved are defined, the principal laws of indices are as follows.

Law	Rule	Conditions
Product of powers	$a^m a^n = a^{m+n}$	The bases must be the same.
Quotient of powers	$\frac{a^m}{a^n} = a^{m-n}$	$a \neq 0$.
Power of a power	$(a^m)^n = a^{mn}$	Multiply the indices.
Power of a product	$(ab)^n = a^n b^n$	The index applies to every factor.
Power of a quotient	$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$	$b \neq 0$.
Zero index	$a^0 = 1$	$a \neq 0$.
Negative index	$a^{-n} = \frac{1}{a^n}$	$a \neq 0$.
Fractional index	$a^{m/n} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$	The required roots must be real.

Table 1.2: The principal laws of indices.

The product law follows directly from repeated multiplication. For positive integers m and n ,

$$a^m a^n = \underbrace{a \times \cdots \times a}_{m \text{ factors}} \underbrace{a \times \cdots \times a}_{n \text{ factors}} = a^{m+n}.$$

Similarly, common factors cancel when powers of the same non-zero base are divided. For example,

$$\frac{a^7}{a^3} = \frac{a \times a \times a \times a \times a \times a \times a}{a \times a \times a} = a^4 = a^{7-3}.$$

These laws apply only in the form stated. Powers with different bases cannot usually be combined merely by adding or subtracting their indices.

Zero and negative indices

For $a \neq 0$, the quotient law gives

$$\frac{a^n}{a^n} = a^{n-n} = a^0.$$

Since every non-zero number divided by itself is 1, it follows that

$$a^0 = 1, \quad a \neq 0. \quad (1.19)$$

A negative index indicates a reciprocal. For a positive integer n ,

$$a^{-n} = \frac{1}{a^n}, \quad a \neq 0. \quad (1.20)$$

For example,

$$2^{-3} = \frac{1}{2^3} = \frac{1}{8}, \quad \frac{1}{x^{-4}} = x^4.$$

A negative exponent does *not* make the value negative; it takes the reciprocal of the corresponding positive power. Thus $2^{-3} = 1/8$, whereas $-2^3 = -8$.

The base 0 requires special care. Although $0^n = 0$ for every positive integer n , expressions such as 0^{-n} involve division by zero and are undefined. The expression 0^0 is also left undefined in elementary algebra.

Fractional indices

Fractional indices provide another notation for roots. If n is a positive integer, then

$$a^{1/n} = \sqrt[n]{a}. \quad (1.21)$$

More generally,

$$a^{m/n} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m. \quad (1.22)$$

The denominator n gives the root, while the numerator m gives the power. For example,

$$\begin{aligned}
 27^{2/3} &= \left(\sqrt[3]{27}\right)^2 = 3^2 = 9, \\
 16^{3/4} &= \left(\sqrt[4]{16}\right)^3 = 2^3 = 8, \\
 16^{-3/4} &= \frac{1}{16^{3/4}} = \frac{1}{8}.
 \end{aligned}$$

When working only with real numbers, an even root requires a non-negative radicand. Thus $16^{1/2} = 4$ is real, but $(-16)^{1/2}$ is not real. Odd roots of negative numbers are real; for instance,

$$(-8)^{2/3} = \left(\sqrt[3]{-8}\right)^2 = (-2)^2 = 4.$$

Worked examples

Example 1.2

Simplify $\frac{x^7 x^{-2}}{x^3}$, where $x \neq 0$.

$$\begin{aligned}
 \frac{x^7 x^{-2}}{x^3} &= x^{7+(-2)-3} \\
 &= x^2.
 \end{aligned}$$

Example 1.3

Express $\frac{(3a^2 b^{-1})^2}{9a^{-1}}$ using positive indices only.

$$\begin{aligned}
 \frac{(3a^2 b^{-1})^2}{9a^{-1}} &= \frac{9a^4 b^{-2}}{9a^{-1}} \\
 &= a^{4-(-1)} b^{-2} \\
 &= \frac{a^5}{b^2}.
 \end{aligned}$$

Example 1.4

Simplify $\sqrt[3]{x^8}$, assuming $x \geq 0$.

$$\begin{aligned}
 \sqrt[3]{x^8} &= x^{8/3} \\
 &= x^{2+2/3} \\
 &= x^2 \sqrt[3]{x^2}.
 \end{aligned}$$

Example 1.5

Solve $2^{x+1} = 16$.

Since $16 = 2^4$, both sides can be written with the same base:

$$\begin{aligned}2^{x+1} &= 2^4, \\x + 1 &= 4, \\x &= 3.\end{aligned}$$

The laws of indices govern multiplication and division, not addition and subtraction. In general,

$$a^m + a^n \neq a^{m+n} \quad \text{and} \quad (a + b)^n \neq a^n + b^n.$$

For example, $2^2 + 2^3 = 4 + 8 = 12$, not $2^5 = 32$, and $(2 + 3)^2 = 25$, not $2^2 + 3^2 = 13$. Brackets are also important:

$$(-2)^4 = 16, \quad -2^4 = -(2^4) = -16.$$

Therefore, before applying any index law, first identify the complete base and the operation joining the terms.

1.7.2 Surds

A **surd** is an irrational number expressed using a root. In elementary algebra, it commonly has the form

$$\sqrt[n]{a}, \tag{1.23}$$

where $n \geq 2$ is a positive integer and a is a rational number that is not a perfect n th power. The symbol $\sqrt{}$ is called the **radical symbol**, n is the **index** of the root, and a is the **radicand**. When the index is 2, it is normally omitted, so $\sqrt[n]{a}$ is written simply as $\sqrt{a}$.

For example,

$$\sqrt{2}, \quad \sqrt{5}, \quad \sqrt[3]{7}$$

are surds. However,

$$\sqrt{9} = 3 \quad \text{and} \quad \sqrt[3]{8} = 2$$

are not surds because their values are rational. Although π is irrational, it is also not a surd because it is not expressed as an irrational root of a rational number.

Surds allow exact values to be retained. For instance, the diagonal of a unit square has exact length $\sqrt{2}$, whereas 1.414 is only a decimal approximation. Unless an approximation

is requested, answers involving surds should therefore be left in exact form.

Surds are closely related to fractional indices:

$$\sqrt[n]{a} = a^{1/n}, \quad \sqrt[n]{a^m} = a^{m/n}, \quad (1.24)$$

whenever the expressions are defined in the real number system.

Real roots and their domains

If n is even, $\sqrt[n]{a}$ is real only when $a \geq 0$, and the radical symbol represents the non-negative principal root. Thus

$$\sqrt{25} = 5,$$

not ± 5 . The equation $x^2 = 25$ has the two solutions $x = \pm 5$, but the symbol $\sqrt{25}$ itself denotes only 5.

If n is odd, negative radicands are permitted. For example,

$$\sqrt[3]{-125} = -5.$$

Consequently, $\sqrt{-6}$ is not a real number and is not considered a real surd in this chapter.

Properties of roots

The following rules connect roots with products, quotients, powers, and nested radicals. All quantities are assumed to satisfy the stated real-domain conditions.

Property	Rule	Conditions
Product rule	$\sqrt[n]{ab} = \sqrt[n]{a} \sqrt[n]{b}$	If n is even, $a, b \geq 0$.
Quotient rule	$\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$	$b \neq 0$; even roots must be real.
Nested roots	$\sqrt[m]{\sqrt[n]{a}} = \sqrt[mn]{a}$	All roots involved must be real.
Odd-index power	$\sqrt[n]{a^n} = a$	n is odd.
Even-index power	$\sqrt[n]{a^n} = a $	n is even.

Table 1.3: Properties of real n th roots.

The absolute value in the final rule is essential because an even root is non-negative. For example,

$$\sqrt{(-4)^2} = \sqrt{16} = 4 = |-4|,$$

not -4 . By contrast,

$$\sqrt[3]{(-4)^3} = -4.$$

Some applications of the product and quotient rules are

$$\begin{aligned}\sqrt[3]{(-8)(27)} &= \sqrt[3]{-8}\sqrt[3]{27} = (-2)(3) = -6, \\ \sqrt[4]{\frac{16}{81}} &= \frac{\sqrt[4]{16}}{\sqrt[4]{81}} = \frac{2}{3}, \\ \sqrt{\sqrt[3]{729}} &= \sqrt[6]{729} = 3.\end{aligned}$$

Simplifying surds

A surd is in **simplest form** when its radicand contains no perfect power factor corresponding to the index, there is no fraction under the radical sign, and there is no surd in the denominator.

To simplify a surd, factor the radicand into a perfect power and a remaining factor, then apply the product rule. For square roots,

$$\sqrt{k^2a} = |k|\sqrt{a}.$$

When $k > 0$, this is usually written as $k\sqrt{a}$. For example,

$$\begin{aligned}\sqrt{72} &= \sqrt{36 \times 2} = 6\sqrt{2}, \\ \sqrt{180} &= \sqrt{36 \times 5} = 6\sqrt{5}, \\ \sqrt[3]{54} &= \sqrt[3]{27 \times 2} = 3\sqrt[3]{2}, \\ \sqrt[4]{48} &= \sqrt[4]{16 \times 3} = 2\sqrt[4]{3}.\end{aligned}$$

When simplifying a fraction under a square root, the quotient rule may first be used:

$$\sqrt{\frac{18}{25}} = \frac{\sqrt{18}}{\sqrt{25}} = \frac{3\sqrt{2}}{5}.$$

Like surds

Surds can be added or subtracted only after they have been simplified to the same radical part. Such terms are called **like surds**. They combine in the same way as like algebraic terms:

$$p\sqrt[n]{a} + q\sqrt[n]{a} = (p + q)\sqrt[n]{a}. \quad (1.25)$$

For example,

$$\begin{aligned}
3\sqrt{8} - \sqrt{18} + 2\sqrt{50} &= 3(2\sqrt{2}) - 3\sqrt{2} + 2(5\sqrt{2}) \\
&= 6\sqrt{2} - 3\sqrt{2} + 10\sqrt{2} \\
&= 13\sqrt{2}.
\end{aligned}$$

The terms $\sqrt{2}$ and $\sqrt{3}$ are unlike surds, so an expression such as $4\sqrt{2} + 3\sqrt{3}$ cannot be simplified by addition.

Multiplying surds

Surds are multiplied by multiplying their coefficients and radicands, then simplifying the result. For example,

$$\begin{aligned}
(3\sqrt{2})(4\sqrt{6}) &= 12\sqrt{12} \\
&= 24\sqrt{3}.
\end{aligned}$$

Products involving brackets are expanded using the ordinary distributive law. In particular,

$$\begin{aligned}
(\sqrt{a} + \sqrt{b})^2 &= a + 2\sqrt{ab} + b, \\
(\sqrt{a} - \sqrt{b})^2 &= a - 2\sqrt{ab} + b.
\end{aligned}$$

The expressions $\sqrt{a} + \sqrt{b}$ and $\sqrt{a} - \sqrt{b}$ are called **conjugates**. Their product is rational whenever a and b are rational:

$$(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b}) = a - b. \quad (1.26)$$

For example,

$$(2 + \sqrt{3})(2 - \sqrt{3}) = 4 - 3 = 1.$$

Rationalising denominators

To **rationalise a denominator** means to rewrite a fraction so that no surd remains in its denominator. The value of the fraction is unchanged because both its numerator and denominator are multiplied by the same non-zero expression.

For a single square-root denominator, multiply by the required root:

$$\begin{aligned}\frac{5}{2\sqrt{3}} &= \frac{5}{2\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}} \\ &= \frac{5\sqrt{3}}{6}.\end{aligned}$$

More generally, if $a > 0$ and $0 < m < n$, then

$$\frac{1}{\sqrt[n]{a^m}} \cdot \frac{\sqrt[n]{a^{n-m}}}{\sqrt[n]{a^{n-m}}} = \frac{\sqrt[n]{a^{n-m}}}{a}.$$

For a denominator containing two terms, multiply by its conjugate. For example,

$$\begin{aligned}\frac{3}{\sqrt{5} - \sqrt{2}} &= \frac{3}{\sqrt{5} - \sqrt{2}} \cdot \frac{\sqrt{5} + \sqrt{2}}{\sqrt{5} + \sqrt{2}} \\ &= \frac{3(\sqrt{5} + \sqrt{2})}{5 - 2} \\ &= \sqrt{5} + \sqrt{2}.\end{aligned}$$

Likewise,

$$\begin{aligned}\frac{4}{3 + \sqrt{5}} &= \frac{4(3 - \sqrt{5})}{(3 + \sqrt{5})(3 - \sqrt{5})} \\ &= \frac{4(3 - \sqrt{5})}{9 - 5} \\ &= 3 - \sqrt{5}.\end{aligned}$$

Worked examples

Example 1.6

Simplify $\sqrt{12} + \sqrt{27} - \sqrt{75}$.

$$\begin{aligned}\sqrt{12} + \sqrt{27} - \sqrt{75} &= 2\sqrt{3} + 3\sqrt{3} - 5\sqrt{3} \\ &= 0.\end{aligned}$$

Example 1.7

Expand and simplify $(3 + 2\sqrt{2})(1 - \sqrt{2})$.

$$\begin{aligned}(3 + 2\sqrt{2})(1 - \sqrt{2}) &= 3 - 3\sqrt{2} + 2\sqrt{2} - 4 \\ &= -1 - \sqrt{2}.\end{aligned}$$

Example 1.8

Rationalise and simplify $\frac{2 + \sqrt{3}}{2 - \sqrt{3}}$.

$$\begin{aligned}\frac{2 + \sqrt{3}}{2 - \sqrt{3}} &= \frac{(2 + \sqrt{3})^2}{(2 - \sqrt{3})(2 + \sqrt{3})} \\ &= \frac{4 + 4\sqrt{3} + 3}{4 - 3} \\ &= 7 + 4\sqrt{3}.\end{aligned}$$

Example 1.9

Solve $x + \sqrt{x} = 6$ for $x \geq 0$.

Let $u = \sqrt{x}$, so that $x = u^2$ and $u \geq 0$. Then

$$\begin{aligned}u^2 + u &= 6, \\ u^2 + u - 6 &= 0, \\ (u + 3)(u - 2) &= 0.\end{aligned}$$

Since $u \geq 0$, $u = 2$. Therefore $x = u^2 = 4$. Substitution confirms that $4 + \sqrt{4} = 6$. Roots do not distribute over addition or subtraction. In general,

$$\sqrt{a + b} \neq \sqrt{a} + \sqrt{b} \quad \text{and} \quad \sqrt{a - b} \neq \sqrt{a} - \sqrt{b}.$$

For example, $\sqrt{9 + 16} = 5$, whereas $\sqrt{9} + \sqrt{16} = 7$. It is also important to remember that $\sqrt{a^2} = |a|$, not necessarily a , and that unlike surds cannot be combined by addition. Each radical should therefore be simplified before any attempt is made to collect terms.

1.7.3 Logarithms

A **logarithm** is the inverse operation of exponentiation. For a positive base $a \neq 1$ and a positive number x ,

$$\log_a x = y \iff a^y = x. \quad (1.27)$$

The expression on the left is called the **logarithmic form**, while the expression on the right is called the **index form** or **exponential form**. In $\log_a x = y$, a is the **base**, x

is the **argument**, and y is the logarithm. The statement is read as “the logarithm of x to the base a is y .”

For example,

$$\begin{aligned}\log_2 8 = 3 &\iff 2^3 = 8, \\ \log_5 25 = 2 &\iff 5^2 = 25, \\ \log_2 \left(\frac{1}{8}\right) = -3 &\iff 2^{-3} = \frac{1}{8}, \\ \log_9 3 = \frac{1}{2} &\iff 9^{1/2} = 3.\end{aligned}$$

Thus a logarithm is an exponent. It answers the question: “To what power must the base be raised to obtain the argument?”

Restrictions on logarithms

For a real logarithm $\log_a x$ to be defined, the following restrictions must hold:

$$a > 0, \quad a \neq 1, \quad \text{and} \quad x > 0. \quad (1.28)$$

The base cannot be negative because a negative base does not produce a real-valued exponential function for every real exponent. The base also cannot be 1, since $1^y = 1$ for every real y and therefore cannot be inverted to obtain logarithms of other positive numbers.

The argument must be positive. Since $a^y > 0$ for every real y whenever $a > 0$, no real exponent can make a^y equal to zero or to a negative number. Consequently,

$$\log_a 0 \quad \text{and} \quad \log_a(-x), \quad x > 0,$$

are undefined in the real number system. The value of a logarithm itself may nevertheless be positive, zero, or negative.

The logarithmic function $y = \log_a x$ has domain $(0, \infty)$ and range $\mathbb{R}$. Its graph passes through $(1, 0)$ and $(a, 1)$ and has the y -axis, $x = 0$, as a vertical asymptote. It is the inverse of $y = a^x$, so the two graphs are reflections of one another in the line $y = x$. If $a > 1$, $\log_a x$ is increasing; if $0 < a < 1$, it is decreasing.

Laws of logarithms

Let $a > 0$ with $a \neq 1$, and let x and y be positive real numbers. The principal laws of logarithms are as follows.

Law	Rule	Conditions
Product law	$\log_a(xy) = \log_a x + \log_a y$	$x, y > 0$.
Quotient law	$\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$	$x, y > 0$.
Power law	$\log_a(x^r) = r \log_a x$	$x > 0, r \in \mathbb{R}$.
Logarithm of 1	$\log_a 1 = 0$	Since $a^0 = 1$.
Logarithm of the base	$\log_a a = 1$	Since $a^1 = a$.
Inverse law I	$\log_a(a^r) = r$	$r \in \mathbb{R}$.
Inverse law II	$a^{\log_a x} = x$	$x > 0$.
Change of base	$\log_a x = \frac{\log_b x}{\log_b a}$	$b > 0, b \neq 1$.

Table 1.4: The principal laws of logarithms.

The product and quotient laws follow from the corresponding laws of indices. Suppose

$$u = \log_a x \quad \text{and} \quad v = \log_a y.$$

Then $x = a^u$ and $y = a^v$. Hence

$$xy = a^u a^v = a^{u+v},$$

so $\log_a(xy) = u + v = \log_a x + \log_a y$. Similarly,

$$\frac{x}{y} = \frac{a^u}{a^v} = a^{u-v},$$

which gives the quotient law.

Several useful consequences follow immediately:

$$\begin{aligned} \log_a\left(\frac{1}{x}\right) &= -\log_a x, \\ \log_a \sqrt[n]{x} &= \frac{1}{n} \log_a x, \\ \log_a\left(\frac{x^p}{y^q}\right) &= p \log_a x - q \log_a y. \end{aligned}$$

Common logarithms and change of base

In most pre-university textbooks and on scientific calculators, a logarithm written without an explicit base denotes the **common logarithm**, whose base is 10:

$$\log x = \log_{10} x.$$

For example,

$$\log 1000 = 3 \quad \text{because} \quad 10^3 = 1000.$$

The change-of-base formula allows a logarithm with any valid base to be evaluated using a calculator. For example,

$$\log_2 7 = \frac{\log 7}{\log 2}.$$

To derive the formula, let $c = \log_a x$, so $a^c = x$. Taking logarithms to any valid base b gives

$$\begin{aligned} \log_b(a^c) &= \log_b x, \\ c \log_b a &= \log_b x, \\ c &= \frac{\log_b x}{\log_b a}. \end{aligned}$$

Since $c = \log_a x$, the required result follows.

Expanding and condensing logarithmic expressions

The logarithm laws can be used to expand one logarithm into several terms. For $x, y, z > 0$,

$$\begin{aligned} \log_a \left(\frac{x^3 \sqrt{y}}{z^2} \right) &= \log_a(x^3) + \log_a(y^{1/2}) - \log_a(z^2) \\ &= 3 \log_a x + \frac{1}{2} \log_a y - 2 \log_a z. \end{aligned}$$

The same laws can be applied in reverse to **condense** several logarithms into one:

$$\begin{aligned} 2 \log_a x - \frac{1}{2} \log_a y + \log_a z &= \log_a(x^2) - \log_a(\sqrt{y}) + \log_a z \\ &= \log_a \left(\frac{x^2 z}{\sqrt{y}} \right). \end{aligned}$$

All arguments must remain positive throughout these manipulations.

Logarithmic equations and inequalities

A simple logarithmic equation may be converted directly to index form. For example,

$$\begin{aligned}\log_3(x-1) &= 2, \\ x-1 &= 3^2, \\ x &= 10.\end{aligned}$$

When logarithms with the same base are equal, their positive arguments are equal:

$$\log_a f(x) = \log_a g(x) \implies f(x) = g(x), \quad (1.29)$$

provided $f(x) > 0$ and $g(x) > 0$.

When several logarithms occur, they may first be combined. For example,

$$\begin{aligned}\log_2 x + \log_2(x-2) &= 3, \\ \log_2(x(x-2)) &= 3, \\ x(x-2) &= 2^3, \\ x^2 - 2x - 8 &= 0, \\ (x-4)(x+2) &= 0.\end{aligned}$$

The original logarithms require $x > 2$, so $x = -2$ must be rejected. The only solution is $x = 4$. Checking the domain is essential because algebraic manipulation may produce inadmissible solutions.

For logarithmic inequalities, the direction depends on the base. If $a > 1$, then

$$\log_a f(x) > \log_a g(x) \iff f(x) > g(x).$$

If $0 < a < 1$, the logarithmic function is decreasing, so the direction of the inequality is reversed. In either case, both arguments must be positive.

Worked examples

Example 1.10

Evaluate $\log_2 64 - \log_2 4 + \log_2 \left(\frac{1}{2}\right)$.

$$\begin{aligned}\log_2 64 - \log_2 4 + \log_2 \left(\frac{1}{2}\right) &= 6 - 2 + (-1) \\ &= 3.\end{aligned}$$

Example 1.11

Simplify $\log_3 54 - \log_3 2$ without using a calculator.

$$\begin{aligned}\log_3 54 - \log_3 2 &= \log_3 \left(\frac{54}{2} \right) \\ &= \log_3 27 \\ &= 3.\end{aligned}$$

Example 1.12

Given that $\log_a 2 = p$ and $\log_a 3 = q$, express $\log_a 72$ in terms of p and q .
Since $72 = 2^3 \times 3^2$,

$$\begin{aligned}\log_a 72 &= \log_a (2^3 \times 3^2) \\ &= 3 \log_a 2 + 2 \log_a 3 \\ &= 3p + 2q.\end{aligned}$$

Example 1.13

Solve $\log_5(2x - 1) = 2$.

$$\begin{aligned}2x - 1 &= 5^2, \\ 2x &= 26, \\ x &= 13.\end{aligned}$$

The argument is positive at $x = 13$, so the solution is valid.
The product law does not apply to addition. In general,

$$\log_a(x + y) \neq \log_a x + \log_a y.$$

Similarly, the power law must not be confused with raising the logarithm itself to a power:

$$\log_a(x^r) = r \log_a x, \quad \text{but generally} \quad (\log_a x)^r \neq r \log_a x.$$

Arguments must always be checked for positivity, and the base must be positive and unequal to 1. These restrictions apply before and after any logarithmic expression is expanded, condensed, or solved.

Euler's number

Euler's number, denoted by e , is the irrational constant

$$e = 2.718281828459 \dots$$

It is the natural base for exponential growth and decay. One way of defining it is through the limit

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n. \quad (1.30)$$

This limit arises from continuously compounded interest: if one unit of money earns interest at an annual rate of 100%, compounded n times in a year, its value after one year is

$$\left(1 + \frac{1}{n}\right)^n,$$

which approaches e as the number of compounding periods increases without bound. The exponential function e^x obeys the ordinary laws of indices, including

$$e^0 = 1, \quad e^{x+y} = e^x e^y, \quad e^{-x} = \frac{1}{e^x}.$$

It is always positive, has domain $\mathbb{R}$ and range $(0, \infty)$, and passes through $(0, 1)$. Its special importance in calculus comes from the property

$$\frac{d}{dx} e^x = e^x;$$

that is, its rate of change is equal to its own value. Consequently, models of continuous growth and decay commonly take the form

$$P(t) = P_0 e^{kt},$$

where P_0 is the initial value and k is a constant. Positive k represents growth, while negative k represents decay.

Natural logarithms

The **natural logarithm** is the logarithm to the base e . It is written as $\ln x$ rather than $\log_e x$:

$$\ln x = \log_e x. \quad (1.31)$$

Therefore,

$$\ln x = y \iff e^y = x. \quad (1.32)$$

For example,

$$\begin{aligned}\ln e &= 1, \\ \ln(e^3) &= 3, \\ \ln\left(\frac{1}{e^2}\right) &= -2, \\ \ln \sqrt{e} &= \frac{1}{2}.\end{aligned}$$

Like every real logarithm, $\ln x$ is defined only when $x > 0$. Its domain is $(0, \infty)$ and its range is $\mathbb{R}$. The graph passes through $(1, 0)$ and $(e, 1)$, is increasing for all $x > 0$, and has the y -axis as a vertical asymptote. Since $y = \ln x$ and $y = e^x$ are inverse functions, their graphs are reflections of one another in the line $y = x$.

Laws of natural logarithms

The ordinary logarithm laws apply with base e . For $x, y > 0$ and $r \in \mathbb{R}$,

$$\ln(xy) = \ln x + \ln y, \quad (1.33)$$

$$\ln\left(\frac{x}{y}\right) = \ln x - \ln y, \quad (1.34)$$

$$\ln(x^r) = r \ln x. \quad (1.35)$$

The inverse relationship between e^x and $\ln x$ gives

$$\ln(e^x) = x, \quad e^{\ln x} = x \quad (x > 0). \quad (1.36)$$

Other useful special values include

$$\ln 1 = 0, \quad \ln e = 1, \quad \ln\left(\frac{1}{x}\right) = -\ln x.$$

Natural logarithms may also be used in the change-of-base formula:

$$\log_a x = \frac{\ln x}{\ln a}, \quad a > 0, \quad a \neq 1, \quad x > 0. \quad (1.37)$$

This is the form most commonly used by scientific calculators when a logarithm with an arbitrary base is required.

Solving equations involving e and $\ln$

Natural logarithms are particularly useful for solving equations in which the unknown appears in an exponent. Taking $\ln$ of both sides allows the power law to bring the exponent down as a coefficient.

For example, to solve $e^{2x} = 7$,

$$\begin{aligned}\ln(e^{2x}) &= \ln 7, \\ 2x &= \ln 7, \\ x &= \frac{1}{2} \ln 7.\end{aligned}$$

Conversely, a natural-logarithmic equation can be rewritten in exponential form. For example,

$$\begin{aligned}\ln(2x - 1) &= 3, \\ 2x - 1 &= e^3, \\ x &= \frac{e^3 + 1}{2}.\end{aligned}$$

The argument $2x - 1$ is positive at this value, so the solution is valid.

When several natural logarithms occur, they may first be combined. Consider

$$\ln x + \ln(x - 3) = \ln 4.$$

The original equation requires $x > 3$. Using the product law,

$$\begin{aligned}\ln(x(x - 3)) &= \ln 4, \\ x(x - 3) &= 4, \\ x^2 - 3x - 4 &= 0, \\ (x - 4)(x + 1) &= 0.\end{aligned}$$

The algebraic candidates are $x = 4$ and $x = -1$, but only $x = 4$ satisfies the domain restriction.

Continuous growth and decay

If a quantity is modelled by

$$P(t) = P_0 e^{kt},$$

then natural logarithms can be used to isolate time. Assuming $P(t) > 0$ and $P_0 > 0$,

$$\begin{aligned}\frac{P(t)}{P_0} &= e^{kt}, \\ \ln\left(\frac{P(t)}{P_0}\right) &= kt, \\ t &= \frac{1}{k} \ln\left(\frac{P(t)}{P_0}\right), \quad k \neq 0.\end{aligned}$$

For exponential growth, the time required for a quantity to double is found by setting $P(t) = 2P_0$:

$$t_{\text{double}} = \frac{\ln 2}{k}, \quad k > 0.$$

Similarly, if $k < 0$, the half-life is

$$t_{1/2} = \frac{\ln 2}{|k|}.$$

Calculus properties

The natural logarithm has a particularly simple derivative:

$$\frac{d}{dx} \ln x = \frac{1}{x}, \quad x > 0. \quad (1.38)$$

More generally, by the chain rule,

$$\frac{d}{dx} \ln f(x) = \frac{f'(x)}{f(x)},$$

where $f(x) > 0$. The corresponding antiderivative is

$$\int \frac{1}{x} dx = \ln |x| + C, \quad x \neq 0. \quad (1.39)$$

The absolute value is required because the integrand $1/x$ is defined on both the positive and negative parts of the real number line. These calculus properties will be developed further in the chapters on differentiation and integration.

Natural logarithms do not distribute over addition or subtraction:

$$\ln(x + y) \neq \ln x + \ln y.$$

The identity $\ln(x^r) = r \ln x$ assumes $x > 0$. If x may be negative, then for $x \neq 0$ the correct real identity is

$$\ln(x^2) = 2 \ln |x|.$$

Every proposed solution to an equation involving $\ln$ must be checked to ensure that each logarithmic argument is positive.

1.8 Solving problems involving indices, surds and logarithms

Problems involving indices, surds, and logarithms often appear different on the surface, but they are governed by the same central idea: rewrite the expressions into a compatible form before attempting to solve the equation. The most important part of the process is to preserve the domain of the original problem.

1.8.1 A general problem-solving strategy

A reliable approach is as follows.

1. **State the restrictions.** Check denominators, even roots, logarithmic arguments, and logarithmic bases before manipulating the equation.
2. **Simplify the structure.** Rewrite powers using a common base, simplify surds, or combine logarithms using the appropriate laws.
3. **Choose a suitable transformation.** This may involve a substitution, isolating a radical, converting between logarithmic and exponential forms, or taking logarithms of both sides.
4. **Solve the resulting equation.** After the transformation, the problem will often reduce to a linear, quadratic, or polynomial equation.
5. **Check every candidate in the original equation.** This is essential because squaring, clearing denominators, or combining logarithms may introduce values that are not permitted.

Problem type	Useful first step	Final check
Exponential equation	Rewrite with a common base, substitute for a repeated power, or take logarithms.	Exponential expressions must remain positive.
Surd equation	State the real domain, isolate one radical, and square carefully.	Substitute because squaring may create extraneous roots.
Logarithmic equation	Require every argument to be positive, then combine or convert to exponential form.	Check all arguments and bases in the original equation.
Exponential or logarithmic inequality	Determine whether the relevant function is increasing or decreasing.	Intersect the result with the original domain.

Table 1.5: A summary of common problem-solving approaches.

1.8.2 Equations involving indices

Rewriting with a common base

If both sides can be expressed using the same positive base $a \neq 1$, then the indices may be equated:

$$a^{f(x)} = a^{g(x)} \implies f(x) = g(x).$$

For example, solve $3^{2x-1} = 27$.

Since $27 = 3^3$,

$$\begin{aligned} 3^{2x-1} &= 3^3, \\ 2x - 1 &= 3, \\ x &= 2. \end{aligned}$$

Substitution for repeated powers

When an equation contains powers such as a^{2x} and a^x , use $a^{2x} = (a^x)^2$ and introduce a substitution.

Consider

$$4^x - 5(2^x) + 4 = 0.$$

Since $4^x = (2^x)^2$, let $u = 2^x$. Notice that $u > 0$. Then

$$\begin{aligned} u^2 - 5u + 4 &= 0, \\ (u - 1)(u - 4) &= 0. \end{aligned}$$

Thus $u = 1$ or $u = 4$. Returning to 2^x ,

$$\begin{aligned} 2^x = 1 &\implies x = 0, \\ 2^x = 4 &\implies x = 2. \end{aligned}$$

Hence the solutions are $x = 0$ and $x = 2$.

Taking logarithms

If the bases cannot be made equal, take logarithms of both sides. For example, solve $5^{x+1} = 7$.

$$\begin{aligned}\ln(5^{x+1}) &= \ln 7, \\ (x+1)\ln 5 &= \ln 7, \\ x &= \frac{\ln 7}{\ln 5} - 1.\end{aligned}$$

An equation of the form $a^{f(x)} = c$ has no real solution when $a > 0$ and $c \leq 0$, because $a^{f(x)}$ is always positive.

1.8.3 Equations involving surds

The usual method for a surd equation is to isolate one radical and then raise both sides to the required power. Every solution must be checked in the original equation.

One radical

Solve

$$\sqrt{x+5} = x-1.$$

The left-hand side is non-negative, so the equation requires $x-1 \geq 0$; hence $x \geq 1$. Squaring both sides gives

$$\begin{aligned}x+5 &= (x-1)^2, \\ x+5 &= x^2-2x+1, \\ x^2-3x-4 &= 0, \\ (x-4)(x+1) &= 0.\end{aligned}$$

The algebraic candidates are $x = 4$ and $x = -1$. The second value violates $x \geq 1$. Substitution verifies that

$$\sqrt{4+5} = 4-1 = 3,$$

so the only solution is $x = 4$.

Two radicals

Solve

$$\sqrt{x+1} + \sqrt{x-2} = 3.$$

The domain is $x \geq 2$. Isolate one radical:

$$\sqrt{x+1} = 3 - \sqrt{x-2}.$$

Squaring gives

$$\begin{aligned}x + 1 &= 9 - 6\sqrt{x - 2} + x - 2, \\6\sqrt{x - 2} &= 6, \\\sqrt{x - 2} &= 1, \\x &= 3.\end{aligned}$$

Checking in the original equation,

$$\sqrt{3 + 1} + \sqrt{3 - 2} = 2 + 1 = 3.$$

Therefore $x = 3$ is valid.

1.8.4 Equations involving logarithms

The domain should be determined before the logarithm laws are applied. In particular, every logarithmic argument must be positive.

Combining logarithms

Solve

$$\log_2(x - 1) + \log_2(x - 3) = 3.$$

The domain conditions $x - 1 > 0$ and $x - 3 > 0$ give $x > 3$. Applying the product law,

$$\begin{aligned}\log_2((x - 1)(x - 3)) &= 3, \\(x - 1)(x - 3) &= 2^3, \\x^2 - 4x + 3 &= 8, \\x^2 - 4x - 5 &= 0, \\(x - 5)(x + 1) &= 0.\end{aligned}$$

The candidates are $x = 5$ and $x = -1$. Since $x > 3$, the only solution is $x = 5$.

Changing logarithmic bases

Solve

$$\log_2 x + \log_4 x = 6.$$

The domain is $x > 0$. By the change-of-base formula,

$$\log_4 x = \frac{\log_2 x}{\log_2 4} = \frac{1}{2} \log_2 x.$$

Hence

$$\begin{aligned}
 \log_2 x + \frac{1}{2} \log_2 x &= 6, \\
 \frac{3}{2} \log_2 x &= 6, \\
 \log_2 x &= 4, \\
 x &= 16.
 \end{aligned}$$

A variable logarithmic base

Solve $\log_x 16 = 4$. The base restrictions are $x > 0$ and $x \neq 1$. Converting to exponential form gives

$$x^4 = 16.$$

The algebraic equation has real solutions $x = \pm 2$, but a logarithmic base must be positive. Therefore the only valid solution is $x = 2$.

1.8.5 Mixed transformations

Some problems require more than one idea. The expression should first be rewritten using the connection between powers, roots, and logarithms.

Example 1.14

Solve $\sqrt{2^x} = 8$.

Using $\sqrt{a} = a^{1/2}$,

$$\begin{aligned}
 2^{x/2} &= 8, \\
 2^{x/2} &= 2^3, \\
 \frac{x}{2} &= 3, \\
 x &= 6.
 \end{aligned}$$

Example 1.15

Solve $3^{\log_3(2x-1)} = 9$.

The logarithmic argument requires $2x - 1 > 0$. Using the inverse law $a^{\log_a u} = u$,

$$\begin{aligned}
 2x - 1 &= 9, \\
 x &= 5.
 \end{aligned}$$

The value $x = 5$ satisfies the domain restriction.

Example 1.16

Solve $\ln \sqrt{x} = 2$.

The domain is $x > 0$. Converting to exponential form,

$$\begin{aligned}\sqrt{x} &= e^2, \\ x &= e^4.\end{aligned}$$

1.8.6 Exponential and logarithmic inequalities

The direction of an inequality depends on whether the relevant exponential or logarithmic function is increasing or decreasing.

If $a > 1$, both a^x and $\log_a x$ are increasing, so the direction is preserved. If $0 < a < 1$, both functions are decreasing, so the direction is reversed when their arguments or indices are compared.

Example 1.17

Solve $2^{x-1} > 8$.

$$\begin{aligned}2^{x-1} &> 2^3, \\ x - 1 &> 3, \\ x &> 4.\end{aligned}$$

Example 1.18

Solve $\left(\frac{1}{3}\right)^{2x} < 9$.

Rewrite both sides using the increasing base 3:

$$\begin{aligned}3^{-2x} &< 3^2, \\ -2x &< 2, \\ x &> -1.\end{aligned}$$

The inequality reverses when the final line is divided by -2 .

Example 1.19

Solve $\log_2(x - 1) \leq 3$.

The domain is $x > 1$. Since the base 2 is greater than 1,

$$\begin{aligned}x - 1 &\leq 2^3, \\x &\leq 9.\end{aligned}$$

Combining this with the domain gives

$$1 < x \leq 9.$$

Example 1.20

Solve $\log_{1/2}(x + 1) > -2$.

The domain is $x > -1$. Since $0 < 1/2 < 1$, the logarithmic function is decreasing, so the inequality reverses:

$$\begin{aligned}x + 1 &< \left(\frac{1}{2}\right)^{-2}, \\x + 1 &< 4, \\x &< 3.\end{aligned}$$

Therefore,

$$-1 < x < 3.$$

1.8.7 An application to compound interest

If a principal P is invested at an annual interest rate r , compounded n times per year, its value after t years is

$$A = P \left(1 + \frac{r}{n}\right)^{nt}. \quad (1.40)$$

To solve for time, divide by P and take natural logarithms:

$$\begin{aligned}\frac{A}{P} &= \left(1 + \frac{r}{n}\right)^{nt}, \\ \ln\left(\frac{A}{P}\right) &= nt \ln\left(1 + \frac{r}{n}\right), \\ t &= \frac{\ln(A/P)}{n \ln(1 + r/n)}.\end{aligned}$$

Suppose RM 5,000 is invested at 4% per year, compounded monthly. The time required for the investment to reach RM 6,500 is

$$\begin{aligned}t &= \frac{\ln(6500/5000)}{12 \ln(1 + 0.04/12)} \\&\approx 6.57 \text{ years.}\end{aligned}$$

Chapter summary and common pitfalls

Key ideas. Real numbers are organised through nested sets; interval notation records endpoint inclusion; and the laws of indices, surds, and logarithms provide equivalent ways to rewrite numerical expressions.

Common pitfalls. Do not treat infinity as an endpoint, cancel terms across addition, ignore domain restrictions on roots and logarithms, or accept solutions introduced by squaring without checking them.

1.9 Practice and exercises

1. Is zero a rational number? Write it in the form $\frac{p}{q}$, where p and q are integers and $q \neq 0$.
2. Let r be rational and let s be irrational.
 - a) Must $r + s$ be irrational? Justify your answer using the definition of a rational number.
 - b) Must rs be irrational? Give a counterexample if the answer is no, and state an additional condition on r that guarantees rs is irrational.
3. State whether each statement is true or false. Justify every false statement.
 - a) Every natural number is a whole number.
 - b) Every integer is a whole number.
 - c) Every rational number is a whole number.
 - d) Every irrational number is a real number.
 - e) Every real number is an irrational number.
 - f) $\frac{\sqrt{5}}{4}$ is a rational number.
 - g) There is at least one integer strictly between any two distinct integers.
 - h) There are only finitely many rational numbers between 7 and 8.
 - i) The square of every irrational number is irrational.
 - j) $\frac{\sqrt{18}}{\sqrt{2}}$ is irrational because neither $\sqrt{18}$ nor $\sqrt{2}$ is an integer.
4. Classify each number as rational or irrational. Simplify the expression and justify your classification.
 - a) $\sqrt{225}$
 - b) $2\sqrt{20}$
 - c) $\sqrt{\frac{16}{50}}$

d) $\frac{\sqrt{45}}{\sqrt{125}}$

5. For each equation, determine whether every possible real value of the variable is rational or irrational. State the possible values where appropriate.

a) $a^2 = 7$

b) $b^2 = 16$

c) $c^2 = 0.09$

d) $d^2 = \frac{13}{9}$

6. Write three different rational numbers lying strictly between each pair.

a) -3 and -2

b) 0.20 and 0.21

c) $\frac{3}{8}$ and $\frac{1}{2}$

7. Give one rational number and one irrational number lying strictly between each pair. Briefly justify that both numbers lie in the stated interval.

a) 1 and 2

b) 0 and 0.2

c) $\frac{2}{5}$ and $\frac{3}{5}$

d) $\sqrt{3}$ and $\sqrt{5}$

8. Express each decimal in the form $\frac{p}{q}$, where p and q are integers and $q \neq 0$.

a) 0.45

b) $0.\overline{36}$

c) $1.2\overline{7}$

d) 0.0032

Hence, using an algebraic argument, show that $0.\overline{18} = \frac{2}{11}$.

9. Draw a suitable number line and mark the following numbers as accurately as possible:

$$5, \quad 5.4, \quad -\frac{7}{3}, \quad -\frac{9}{4}.$$

On a second number line, locate $\sqrt{6}$, $\sqrt{11}$, and $\sqrt{19}$ by first identifying the two consecutive integers between which each surd lies.

10. Show on the real line:

- a) $[-3, 0) \cap (-3, \pi)$
- b) $(-1, e]$
- c) $(-\infty, 5] \cup [-5, \infty)$

11. Simplify each of the following:

- a. $\{x : -2 < x < 4\} \cap \{x : 3 < x < 6\}$
- b. $\{x : x < 3\} \cup \{x : -2 < x < 5\}$
- c. $[-3, 1) \cap [0, 4)$
- d. $(-2, 2) \cup [1, 3]$
- e. $\{x : -5 < x < 1\} \cap \{x : -8 \leq x \leq 3\}$
- f. $\{x : 1 \leq x \leq 4\} \cup \{x : 0 < x < 2\}$
- g. $[2, 9) \cap (-4, 7]$
- h. $(4, \infty) \cup [-2, 6]$

12. State the property that justifies each statement.

- a) $5(3 + 7) = 5(3) + 5(7)$
- b) $(6 + 2) + 8 = (2 + 6) + 8$
- c) $(5a)(b) = 5(ab)$
- d) $(ab) \left(\frac{1}{ab}\right) = 1$
- e) $(x + y) + (-x) = (-x) + (x + y)$
- f) $(-x) + (x + y) = (-x + x) + y$
- g) $(-x + x) + y = 0 + y$
- h) $0 + y = y$

13. Johan wanted to work out $1273 - 497$ but did not have a calculator. He remembered learning about the associative law in school, so he worked out the answer as follows:

$1273 - 497$	Line 1
$= 1273 - (500 - 3)$	Line 2
$= (1273 - 500) - 3$	Line 3
$= 773 - 3$	Line 4
$= 770$	Line 5

Johan's friend, Bienda, told him that she thought Johan's answer was wrong, and that the correct answer should be 776. Which line of Johan's working was wrong?

14. Without using a calculator, evaluate:

(a) $(0.01)^{-1/2}$

(b) $(0.008)^{2/3}$

(c) $\left(-\frac{1}{5}\right)^{-3}$

15. Simplify:

(a) $\frac{(2x^3y^{-4})^2}{8xy^{-6}}$

(b) $\frac{9x^2y^5}{6y^{-3}} \left(\frac{1}{xy^{-4}}\right)^2$

(c) $\frac{1}{\sqrt{y}}$

16. Simplify without using a calculator:

(a) $\sqrt{\frac{3}{5}} \cdot \sqrt{\frac{75}{27}}$

(b) $8\sqrt{2} - \sqrt{32} + 5\sqrt{98}$

(c) $\frac{3\sqrt{12}}{4\sqrt[3]{54}}$

(d) $\sqrt{80} - 2\sqrt{45} + 3\sqrt{5}$

(e) $\frac{\sqrt{32}}{4} + \frac{\sqrt{50}}{5}$

(f) $(\sqrt{5} - \sqrt{2})^2$

(g) $2\sqrt{6} + \sqrt{24} + \frac{5}{\sqrt{6}}$

17. Simplify without using a calculator:

(a) $\log_3(27\sqrt{3})$

(b) $\log_{0.5}(0.0625)$

(c) $\log_3 64 + \log_3 4$

(d) $\log_4 8 + \log_4 2$

(e) $\log_{1/3} 27$

(f) $\log_5 \left(\frac{125}{\sqrt{5}} \right)$

18. Solve:

(a) $5^{x-2} = 625$

(b) $3^{2x+2} + 3^{x+3} - 10 = 0$

(c) $\sqrt{7+x} = x - 5$

(d) $2^{2x} - 5(2^x) + 4 = 0$

(e) $4^{x+1} = 8^{2x-1}$

(f) $\sqrt{2x+3} = x$

(g) $3^x + 3^{-x} = \frac{10}{3}$

19. Solve the following logarithmic equations:

(a) $\log_5 y = 2$

(b) $\ln(x - 5) = 1$

(c) $\log x = 2 \log 3 + \log(5x - 2)$

(d) $\log_3(3x) - \frac{1}{2} \log_3 4 = \log_3(x - 1)$

(e) $\log_2(x - 1) + \log_2(x - 3) = 3$

(f) $\ln x - \ln(x - 2) = \ln 3$

(g) $2 \log x = \log(9x - 8)$

(h) $\log_3(x + 5) = 1 + \log_3(x - 1)$

20. What is the nature of the roots of the quadratic equations below?

(a) $x^2 - 6x + 4 = 0$

(b) $2x^2 + 4x + 2 = 0$

(c) $x^2 - 3x + 5 = 0$

21. Find the range of values of m for which the line $y = mx + 2$ is able to cut the curve $y = 3x^2 - mx + 5$.

22. Find the real solutions of the following quadratic-type equations:

(a) $4x^{-4} - 7x^{-2} - 2 = 0$

(b) $\frac{1}{t-1} + \frac{1}{t-4} = \frac{5}{4}$

(c) $\frac{2x}{x-3} + \frac{5}{x+3} = \frac{36}{x^2-9}$

23. A lumber company has 1,200,000 trees and harvests 7% of the remaining trees each year. How many years will it take for half of the original trees to remain if the relationship between the number of trees, n , and time, t , in years is

$$n = 1,200,000(1 - 0.07)^t?$$

24. Simplify the following expressions. Give each final answer using positive indices only.

(a) $\frac{3^{4x} \times 3^{2x}}{3^{-x}}$

(b) $\frac{5^{m-1} - 5^{m+1}}{5^{m+2}}$

(c) $\frac{4^{n-2} + 4^{n+1}}{16}$

(d) $a^5b^{-2} \times a^{-3}b^7$

(e) $(pq^3)^2 \times p^4q^{-1}$

(f) $(2x^{-2})^3 \times (4xy)^2$

(g) $\frac{(2r^3s)^4(r^2)^3}{r^{17}s^2}$

(h) $(u^2v^{-3})^3 \times v^5$

(i) $(3^{-2}xy^2)^2 \div (3xy)^{-1}$

(j) $18c^{-5}d^4 \div 6c^2d^{-1}$

(k) $b^{1/4} \times 3b^{-1/2}$

(l) $\sqrt[3]{z^5} \times \sqrt[4]{z^{-2}}$

25. Use the laws of indices to show that:

(a) $8^{2p-1} = \frac{64^p}{8}$

(b) $25^{q+1} = 25(5^{2q})$

(c) $3^{k+2} + 3^{k+1} + 3^k$ is divisible by 13 for every positive integer k .

26. Solve the following exponential equations:

(a) $9^{x-2} = 27^{x+1}$

(b) $5^{x+2} - 5^{x+1} = 100$

(c) $16^{x-1} = \frac{8^{2x}}{4}$

27. Answer the following questions involving exponential models. Give non-integer numerical answers to a suitable degree of accuracy.

- (a) A tennis ball is dropped from a height of 2.5 m. After each bounce, its maximum height is 82% of the preceding maximum height. The height after n bounces is modelled by

$$h_n = 2.5(0.82)^n.$$

Find the height before the first bounce and after the seventh bounce. Hence determine the first value of n for which $h_n < 0.5$ m.

- (b) A yeast culture initially contains 240 cells and triples every hour. Its population after t hours is

$$N(t) = 240(3^t).$$

Find the population after five hours and determine when the population reaches 58,320 cells.

- (c) A city's population was approximately 1.8 million in 2024 and is expected to increase by 2.2% each year. Using

$$P = 1.8(1.022)^t,$$

where t is the number of years after 2024 and P is measured in millions, estimate the population in 2042.

- (d) Nurul invests RM 15,000 in an account paying 6% annual compound interest. Determine the value of the investment after nine years using

$$A = 15,000(1.06)^t.$$

28. Complete the following questions involving surds.

- (a) In triangle PQR , $\angle PQR = 60^\circ$, $PQ = 2\sqrt{3}$ cm, and $QR = 4\sqrt{3}$ cm. Find the exact length of PR .
- (b) The perpendicular sides of a right-angled triangle have lengths $(4 - \sqrt{3})$ cm and $(4 + \sqrt{3})$ cm. Find its exact area and the exact length of its hypotenuse.

- (c) Solve

$$1 + \sqrt{y} = 4 + 2\sqrt{5}.$$

Express the answer in the form $a + b\sqrt{5}$, where a and b are rational numbers.

- (d) Solve each equation and check every answer in the original equation.

(i) $\sqrt{5 - 4x} + 3x = 0$

(ii) $\sqrt{x + 3} + \sqrt{x - 1} = 4$

(iii) $\sqrt{3x + 4} - \sqrt{3x - 2} = 2$

Additional grouped practice

The following questions collect related calculations under one number. This allows each exercise to test a complete skill while avoiding repeated instructions.

29. Classify each number into the smallest applicable set among $\mathbb{N}$, $\mathbb{W}$, $\mathbb{Z}$, $\mathbb{Q}$, the irrational numbers, and $\mathbb{R}$:

- (a) 17;
(b) 0;
(c) -8 ;
(d) $0.\overline{27}$;
(e) $\sqrt{49}$;
(f) $\sqrt{12}$;
(g) $\pi - 3$.

30. Arrange in ascending order:

- (a) $-\sqrt{5}$, -2.3 , $-\frac{9}{4}$, -2 ;
(b) $\frac{2}{3}$, 0.67 , $\sqrt{0.45}$, $\frac{7}{10}$.

31. Convert between inequality, interval, and set-builder notation:

- (a) $-4 < x \leq 3$;
(b) $[2, \infty)$;
(c) $\{x \in \mathbb{R} : |x - 1| < 5\}$;
(d) $(-\infty, -2) \cup [4, \infty)$.

32. Let $A = [-3, 4)$, $B = (1, 7]$, and $C = (-\infty, 0]$. Find $A \cap B$, $A \cup B$, $A \cap C$, $(A \cup B) \cap C$, and $A \setminus B$.
33. In a group of 50 students, 31 study Mathematics, 24 study Physics, and 12 study both. Find the numbers studying only Mathematics, only Physics, at least one subject, and neither subject.
34. For $S = \{1, 2, 3, 4\}$:
- (a) list all two-element subsets;
 - (b) state the number of subsets and proper subsets;
 - (c) determine whether $\emptyset \subseteq S$ and $\emptyset \in S$.
35. Verify De Morgan's laws using $U = \{1, 2, 3, 4, 5, 6\}$, $A = \{1, 2, 4\}$, and $B = \{2, 3, 5\}$.
36. Name the real-number property used in each step:
- (a) $7 + x = x + 7$;
 - (b) $(ab)c = a(bc)$;
 - (c) $4(y + 2) = 4y + 8$;
 - (d) $z + 0 = z$;
 - (e) $q(1/q) = 1$, $q \neq 0$.
37. Solve and illustrate each solution on a number line:
- (a) $|x - 3| = 5$;
 - (b) $|2x + 1| < 7$;
 - (c) $|x + 4| \geq 2$.
38. Write in scientific notation and evaluate:
- (a) 0.0000725;
 - (b) 8,430,000;
 - (c) $(3.2 \times 10^5)(4 \times 10^{-3})$;
 - (d) $(6 \times 10^{-8})/(1.5 \times 10^2)$.
39. Convert to fractions in lowest terms:
- (a) $0.\overline{7}$;
 - (b) $0.1\overline{6}$;
 - (c) $2.\overline{45}$;

(d) $0.12\bar{3}$.

40. Prove that $\sqrt{3}$ is irrational. Hence prove that $5 + 2\sqrt{3}$ is irrational.

41. Evaluate without a calculator:

(a) $16^{3/4}$;

(b) $27^{-2/3}$;

(c) $81^{1/2}27^{1/3}$;

(d) $32^{2/5}$.

42. Simplify using positive indices:

(a) $(x^3y^{-2})^2(x^{-1}y^5)$;

(b) $\frac{a^{-2}b^4}{a^3b^{-1}}$;

(c) $\frac{(2p^2q^{-1})^3}{4p^{-1}q^2}$.

43. Express using radical notation and simplify where possible:

(a) $x^{3/2}$;

(b) $a^{5/3}$;

(c) $64^{2/3}$;

(d) $y^{-3/4}$.

44. Simplify, assuming $a > 0$ where required:

(a) $\sqrt{72}$;

(b) $3\sqrt{50} - 2\sqrt{8}$;

(c) $\sqrt{12}\sqrt{15}$;

(d) $\sqrt[3]{54}$.

45. Expand and simplify:

(a) $(\sqrt{7} + \sqrt{3})^2$;

(b) $(2\sqrt{5} - \sqrt{2})(2\sqrt{5} + \sqrt{2})$;

(c) $(3 - \sqrt{6})^2$.

46. Rationalise the denominator:

(a) $5/\sqrt{3}$;

- (b) $4/(3 + \sqrt{5})$;
- (c) $1/(\sqrt{7} - \sqrt{2})$;
- (d) $(2 + \sqrt{3})/(2 - \sqrt{3})$.

47. Solve and check:

- (a) $\sqrt{x+4} = x - 2$;
- (b) $\sqrt{2x+3} = x$;
- (c) $\sqrt{x+8} - \sqrt{x} = 2$.

48. Solve the index equations:

- (a) $2^{x+3} = 64$;
- (b) $9^x = 3^{x+4}$;
- (c) $4^x - 5(2^x) + 4 = 0$.

49. Solve the exponential inequalities:

- (a) $2^{x-1} > 8$;
- (b) $(1/3)^{2x} \leq 27$;
- (c) $5^{x+1} < 1$.

50. Simplify:

- (a) $\frac{2^{n+2} + 2^n}{2^{n-1}}$;
- (b) $\frac{3^{2m} - 9^{m-1}}{3^{2m-2}}$;
- (c) $(\sqrt{a})^3/a^{-1/2}$.

51. Evaluate exactly:

- (a) $\log_2(1/8)$;
- (b) $\log_{10}(0.001)$;
- (c) $\ln e^5$;
- (d) $\log_9 3$.

52. Expand completely:

- (a) $\log_a(x^3y/\sqrt{z})$;
- (b) $\ln[(x-1)^2/(x+2)^4]$.

53. Condense into one logarithm:

(a) $2 \ln x + \ln y - 3 \ln z$;

(b) $\frac{1}{2} \log_a p + 2 \log_a q$.

54. Use change of base to express $\log_7 20$ using natural logarithms, and estimate it to four decimal places.

55. Solve, respecting the logarithm domains:

(a) $\log_2(x - 1) = 3$;

(b) $\ln x + \ln(x - 2) = \ln 3$;

(c) $\log_3(x + 5) - \log_3 x = 1$.

56. Solve:

(a) $\log_2 x \geq 4$;

(b) $\ln(x - 1) < 0$;

(c) $\log_{1/2}(x + 2) > -1$.

57. Solve exactly using logarithms:

(a) $3^x = 10$;

(b) $5e^{2x} = 17$;

(c) $2^{x+1} = 7^{x-1}$.

58. Prove from the laws of logarithms that $\log_a b \log_b c = \log_a c$, where all quantities are defined.

59. Evaluate exactly:

(a) $\log_2 \sqrt{8}$;

(b) $\log_3(1/\sqrt[3]{9})$;

(c) $\ln \sqrt{e}$.

60. An amount grows according to $A = P(1 + r)^t$. Rearrange the model to make t the subject, then find t when $A = 2P$ and $r = 0.08$.

61. Solve and express the answers in interval notation:

(a) $-2 < 3x + 1 \leq 10$;

(b) $x^2 < 9$;

(c) $(x - 1)(x + 4) \geq 0$.

62. Write in set-builder notation:

- (a) the positive even integers;
- (b) rational numbers between -1 and 1 ;
- (c) real numbers at least 4 units from 2.

63. Solve:

- (a) $|3x - 2| \leq 10$;
- (b) $2 < |x + 1| < 5$;
- (c) $|x| + |x - 2| = 4$.

64. Determine the values of k for which $x^2 + kx + 9 = 0$ has

- (a) equal real roots;
- (b) two distinct real roots;
- (c) no real roots.

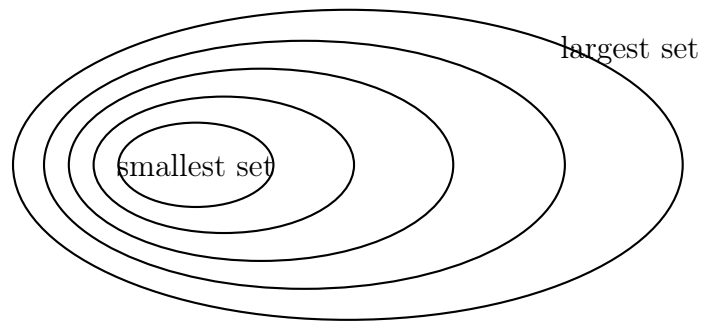
65. State the domain of each expression:

- (a) $\sqrt{7 - 3x}$;
- (b) $1/\sqrt{x + 5}$;
- (c) $\ln(4 - x^2)$;
- (d) $\sqrt{x - 1}/(x - 4)$.

66. State whether each assertion is always, sometimes, or never true:

- (a) rational + irrational is rational;
- (b) irrational $\times$ irrational is rational;
- (c) the square root of an integer is irrational;
- (d) a terminating decimal is rational.

67. Copy and complete the nesting diagram with $\mathbb{N}$, $\mathbb{W}$, $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$, then place -4 , 0 , $3/5$, $\sqrt{2}$, and 9 in the smallest appropriate region.



68. Use prime factorisation to find the HCF and LCM of
- (a) 84 and 126;
 - (b) 72, 90, and 120.
69. Test divisibility and justify your answers:
- (a) Is 43857 divisible by 3 or 9?
 - (b) Is 7260 divisible by 4, 5, and 11?
 - (c) Find the possible digit(s) a if $57a24$ is divisible by 9.
70. Round 38.7462 to 3 significant figures and 2 decimal places, and state the corresponding lower and upper bounds in each case.
71. A measured length is 12.4 cm correct to the nearest 0.1 cm. Write its error interval and find the greatest possible percentage error.
72. Simplify completely:
- (a) $\frac{\sqrt{48} + \sqrt{75}}{\sqrt{3}}$;
 - (b) $\frac{2^{x+2} - 2^x}{2^{x-1}}$;
 - (c) $\log_2 12 + \log_2 6 - \log_2 9$.
73. Without decimal approximations, determine which is larger:
- (a) $3\sqrt{5}$ or $2\sqrt{13}$;
 - (b) $\log_2 7$ or $\log_3 10$;
 - (c) $2^{1/3}$ or $3^{1/4}$.
74. Solve the mixed equation

$$\log_2(x - 1) + \log_2(x + 1) = 3$$

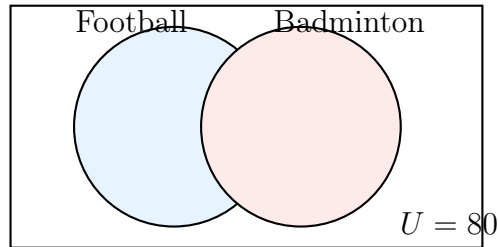
and then simplify $\sqrt{x^2 - 1}$ at the valid solution.

75. Prove that $\sqrt{2} + \sqrt{3}$ is irrational, then rationalise and simplify $1/(\sqrt{3} - \sqrt{2})$.

Application-based problems

76. A freezer must remain between -22°C and -18°C , inclusive. Express the permitted temperatures using an inequality, interval notation, and a number-line sketch.
77. A recipe uses flour and sugar in the ratio 5 : 2. If 875 g of flour is used, find the sugar required and the fraction of the mixture that is sugar.
78. A map has scale 1 : 50,000. Two towns are 7.6 cm apart on the map. Find their actual separation in kilometres.
79. A memory device contains 2^{36} bytes. Express this as a multiple of 2^{30} bytes and state the corresponding number of gibibytes.
80. Earthquake magnitude is modelled by $M = \log_{10}(I/I_0)$. Compare the intensities of earthquakes with magnitudes 6.8 and 4.3.
81. The pH of a solution is $-\log_{10}[H^+]$. Find the pH when $[H^+] = 2.5 \times 10^{-4} \text{ mol L}^{-1}$ and the concentration when pH = 8.2.
82. Sound level is $L = 10 \log_{10}(I/I_0)$. Find the intensity ratio corresponding to an increase of 35 dB.
83. A town has 48,000 residents and grows by 1.8% annually. Estimate its population after 12 years and the first whole year in which it exceeds 60,000.
84. A machine worth RM 90,000 loses 14% of its value each year. Find its value after five years and the time taken to fall below half its original value.
85. RM 12,000 is invested at 4.8% annual compound interest. Find the value after eight years and the doubling time.
86. A price of RM 250 rises by 3.5% each year. Find the price after six years and the total percentage increase.
87. A medicine has half-life 6 hours. If the initial amount is 160 mg, write an exponential model and find the amount after 15 hours.
88. A ball is dropped from 3 m and rebounds to 70% of its preceding height. Find the height after the sixth bounce and the first bounce number for which the height is below 0.2 m.

89. A bacterial culture begins with 500 cells and doubles every 45 minutes. Find the population after 6 hours and the time required to exceed 100,000 cells.
90. The diagram represents a survey of 80 students. Forty-five play football, 38 play badminton, and 17 play both.



Fill the four regions and find the probability that a randomly selected student plays neither sport.

91. In a class, 28 students study Biology, 22 study Chemistry, and 15 study both. If every student studies at least one of the two subjects, find the class size and the probability that a randomly chosen student studies exactly one subject.
92. A component is acceptable when its diameter d satisfies $|d - 12.00| \leq 0.03$ mm. Give the acceptable interval and decide whether diameters 11.96, 11.98, and 12.031 mm pass inspection.
93. A square tile has area 180 cm^2 . Find its exact side length and exact diagonal length in simplest surd form.
94. A rectangular screen measures $(8 + \sqrt{3})$ cm by $(8 - \sqrt{3})$ cm. Find its exact area and diagonal length.
95. An electrical resistance is $R = 12/(\sqrt{5} - 1)$ ohms. Rationalise the expression and give an exact simplified value.
96. Light travels at $3.00 \times 10^8 \text{ m s}^{-1}$. Find the distance travelled in 2.5 milliseconds, in scientific notation and ordinary notation.
97. The mean Earth–Sun distance is 1.496×10^{11} m. Express the distance for 8 such journeys in scientific notation and kilometres.
98. A loan balance is modelled by $B = 25,000(1.006)^n$. After how many months will the balance first exceed RM 30,000?
99. A radioactive sample follows $M = 75(0.92)^t$ grams. Find the mass after 10 years and the time at which only 20 g remains.

100. Plan A grows RM 5,000 at 5% annually; Plan B grows RM 6,500 at 3% annually. Write both models, determine when their values are equal, and state which plan is larger before and after that time.

Chapter 2

Functions

Chapter guide

Prerequisites. Algebraic manipulation, Cartesian coordinates, equations, inequalities, and set notation.

Learning outcomes. Readers should be able to distinguish domain, codomain, and range; recognise and transform common function families; combine and compose functions; and determine inverses with suitable domain restrictions.

The ideas developed in Chapter 1 underpin the study of functions. A function provides a precise rule for associating permitted inputs with outputs, allowing mathematical relationships to be represented algebraically, numerically, graphically, and verbally. Functions are used throughout mathematics, science, engineering, economics, and computing.

This chapter introduces relations, functions, domains, codomains, and ranges before developing common function families, graph transformations, algebraic operations, composition, and inverse functions. The practice section then brings these ideas together in both routine and application-based problems. The terminology follows standard set-theoretic and pre-university treatments of functions (Encyclopaedia Britannica 2026; Schwarz 2021).

2.1 Introduction to functions

A function is commonly described as a rule or correspondence that associates each permitted input with exactly one output. The input variable is often called the *independent variable*, while the output variable is called the *dependent variable* (Encyclopaedia Britannica 2026). In 1837, the German mathematician Johann Peter Gustav Lejeune Dirichlet gave an influential formulation of this idea (O'Connor and Robertson 2000):

“If a variable y is so related to a variable x that whenever a numerical value is assigned to x , there is a rule according to which a unique value of y is determined, then y is said to be a function of the independent variable x .”

—Peter Dirichlet, 1837



Figure 2.1: Johann Peter Gustav Lejeune Dirichlet. Source: Wikimedia Commons, *File:Dirichlet.jpg*; public-domain work, marked with the Creative Commons Public Domain Mark 1.0.

Modern notation separates the sets from their elements. If X is the input set and Y is the target set, then

$$f: X \rightarrow Y, \quad x \mapsto f(x),$$

means that the function f assigns the element $x \in X$ to the element $f(x) \in Y$. The equation $y = f(x)$ names the corresponding output by y .

2.1.1 Relations and functions

Definition 2.1: Relation.

Let X and Y be sets. A *relation* from X to Y is any set of ordered pairs (x, y) with $x \in X$ and $y \in Y$; equivalently, it is a subset of the Cartesian product $X \times Y$.

Definition 2.2: Function.

A *function* $f: X \rightarrow Y$ is a relation that assigns every $x \in X$ to exactly one $y \in Y$. In logical notation,

$$\forall x \in X, \quad \exists! y \in Y \text{ such that } (x, y) \in f. \quad (2.1)$$

Different inputs are allowed to produce the same output; such behaviour is called *many-to-one*. However, one input cannot produce two different outputs. A relation with a missing output for some element of its stated domain is also not a function on that domain.

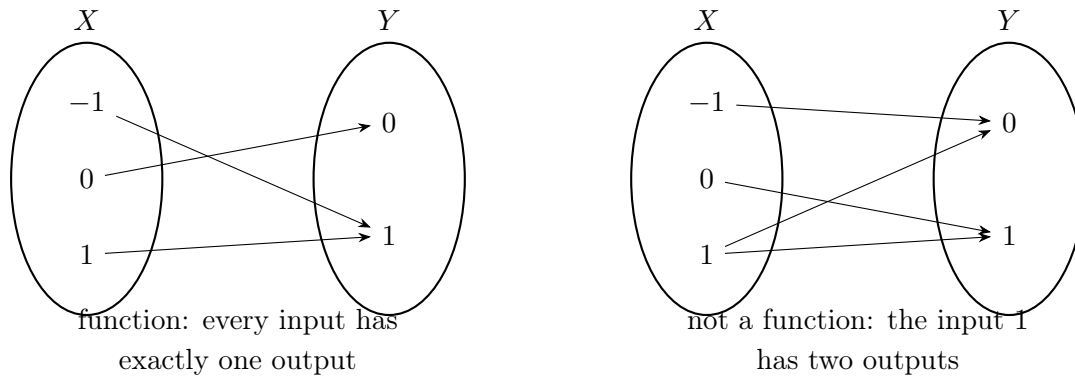


Figure 2.2: A function may be many-to-one, but each input must have exactly one output.

2.2 Domain and range

Each function f has its own domain and range. When we say that each function has its own domain and range, we are pointing to the fundamental identity of that function.

2.2.1 Domain

Definition 2.3: Domain of a function.

Let $f: X \rightarrow Y$. The *domain* of f is the stated input set X . Thus,

$$D(f) = X. \quad (2.2)$$

If only a real-valued formula is given and no domain is stated, its *natural real domain* is normally taken to be

$$D(f) = \{x \in \mathbb{R} \mid f(x) \text{ exists as a real number}\}. \quad (2.3)$$

Geometrically, the domain consists of the real values represented along the x -axis for which the graph contains at least one point. It is therefore the projection of the graph of the function onto the x -axis.

2.2.2 Codomain

Definition 2.4: Codomain of a function.

Let $f: X \rightarrow Y$. The *codomain* of f is the target set Y specified as part of the definition of the function. In symbols,

$$\text{codom}(f) = Y, \quad f(x) \in Y \quad \text{for every } x \in X. \quad (2.4)$$

Every actual output must lie in the codomain, although some elements of the codomain need not be attained. Consequently,

$$R(f) \subseteq \text{codom}(f).$$

The codomain and range are equal precisely when f is surjective.

2.2.3 Range

Definition 2.5: Range of a function.

Let $f: X \rightarrow Y$. The *range*, also called the *image*, of f is the set of all values in Y that are produced by at least one input from X . It is defined by

$$R(f) = f(X) = \{f(x) \mid x \in X\} = \{y \in Y \mid \text{there exists } x \in X \text{ such that } y = f(x)\}. \quad (2.5)$$

Therefore, the range is always a subset of the codomain:

$$R(f) \subseteq Y.$$

If every element of Y is attained, then $R(f) = Y$ and the function is called *surjective*, or *onto*.

The domain and range can also be read geometrically. The domain is the horizontal projection of the graph onto the x -axis, while the range is its vertical projection onto the y -axis. In Figure 2.3, the graph exists only for $-1 \leq x \leq 3$ and takes values from 0 to 4.

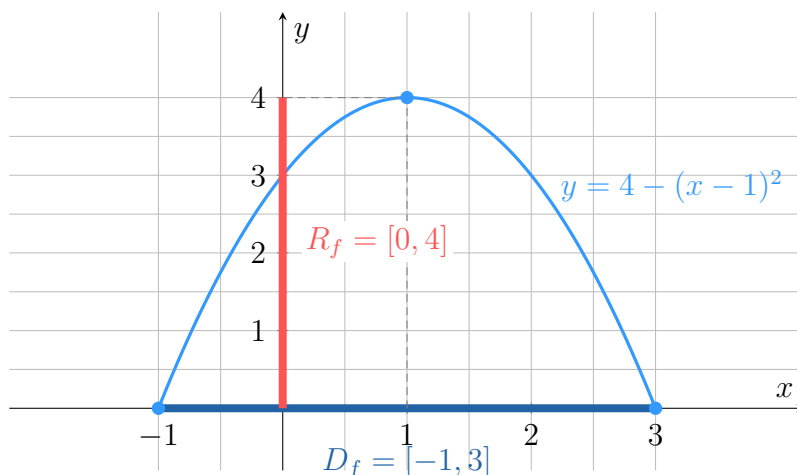


Figure 2.3: The domain and range are the horizontal and vertical projections of a graph onto the coordinate axes.

2.2.4 One-to-one, onto, and bijective functions

Definition 2.6: One-to-one function.

A function $f: X \rightarrow Y$ is *one-to-one*, or *injective*, if different inputs always produce different outputs. Equivalently,

$$f(x_1) = f(x_2) \implies x_1 = x_2. \quad (2.6)$$

A function is *onto*, or *surjective*, when every element of its codomain is attained, so that $R(f) = Y$. A function that is both injective and surjective is called *bijective*.

Bijjective functions pair the elements of the domain and codomain without omissions or repetitions and therefore possess inverse functions.

2.2.5 Vertical and horizontal line tests

The *vertical line test* determines whether a plane curve represents y as a function of x : every vertical line must meet the graph at most once. The *horizontal line test* determines whether a function is one-to-one: every horizontal line must meet its graph at most once.

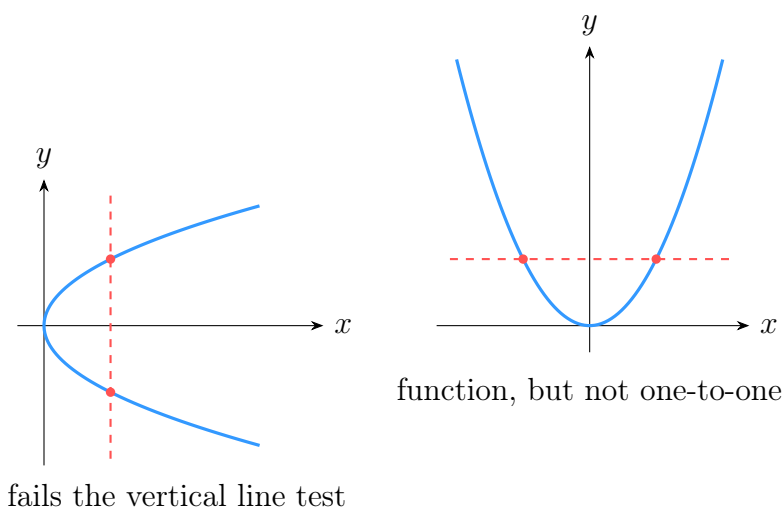


Figure 2.4: The sideways parabola fails the vertical line test, while $y = x^2$ passes the vertical line test but fails the horizontal line test.

2.3 Transformations of functions

Translations, dilations, and reflections can be combined in the standard form

$$g(x) = af(b(x - h)) + k, \quad a, b \neq 0. \quad (2.7)$$

If (x, y) lies on the graph of $y = f(x)$, then its corresponding point on $y = g(x)$ is

$$(x, y) \mapsto \left(h + \frac{x}{b}, k + ay\right). \quad (2.8)$$

This point mapping prevents the common error of treating horizontal and vertical factors in the same way.

Parameter	Effect on the graph of $y = f(x)$
h	Translate right by h units if $h > 0$, or left by $ h $ units if $h < 0$.
k	Translate upward by k units if $k > 0$, or downward by $ k $ units if $k < 0$.
a	Multiply vertical coordinates by a : stretch if $ a > 1$, compress if $0 < a < 1$, and reflect about the x -axis if $a < 0$.
b	Divide horizontal coordinates by b : compress if $ b > 1$, stretch if $0 < b < 1$, and include a horizontal reflection if $b < 0$.

Table 2.1: The parameters in $g(x) = af(b(x - h)) + k$.

2.3.1 Translations

For a vertical translation, $g(x) = f(x) + k$ and

$$(x, y) \mapsto (x, y + k).$$

Thus $k > 0$ shifts the graph upward, while $k < 0$ shifts it downward.

For a horizontal translation, $g(x) = f(x - h)$ and

$$(x, y) \mapsto (x + h, y).$$

Thus $h > 0$ shifts the graph to the right, while $h < 0$ shifts it to the left. The sign appears reversed inside the function because the original output $f(x)$ occurs on the transformed graph at the input $x + h$.

2.3.2 Dilations of functions

A *dilation* changes the scale of a graph without translating it. It may stretch or compress the graph vertically, horizontally, or in both directions. The effect is easiest to understand by following how a point on the original graph moves.

Vertical dilation

Suppose that (x, y) lies on $y = f(x)$, so that $y = f(x)$. For the transformed function

$$g(x) = af(x),$$

the input remains unchanged while every output is multiplied by a . Therefore,

$$(x, y) \mapsto (x, ay). \tag{2.9}$$

For $a > 0$:

- if $a > 1$, the graph is stretched vertically by scale factor a ;
- if $0 < a < 1$, the graph is compressed vertically by scale factor a .

The dilation is measured from the x -axis. If $a < 0$, the scale factor is $|a|$ and the graph is also reflected about the x -axis. When $a \neq 0$, the x -intercepts are unchanged because $af(x) = 0$ precisely when $f(x) = 0$.

Horizontal dilation

Now consider

$$g(x) = f(bx), \quad b \neq 0.$$

If (x, y) lies on $y = f(x)$, then the same output y occurs in the new graph when the input is x/b . Hence,

$$(x, y) \mapsto \left(\frac{x}{b}, y\right). \quad (2.10)$$

For $b > 0$:

- if $b > 1$, the graph is compressed horizontally by scale factor $1/b$;
- if $0 < b < 1$, the graph is stretched horizontally by scale factor $1/b$.

The reciprocal is important: multiplying the input by 2 halves every horizontal coordinate. If $b < 0$, the graph also undergoes a reflection about the y -axis. The y -intercept is unchanged because $g(0) = f(0)$.

Transformation	Point mapping	Effect for a positive factor
$y = af(x)$	$(x, y) \mapsto (x, ay)$	Vertical stretch if $a > 1$; vertical compression if $0 < a < 1$.
$y = f(bx)$	$(x, y) \mapsto (x/b, y)$	Horizontal compression if $b > 1$; horizontal stretch if $0 < b < 1$.

Table 2.2: Vertical and horizontal dilations of $y = f(x)$.

For the combined transformation

$$g(x) = af(bx),$$

the coordinate mapping is

$$(x, y) \mapsto \left(\frac{x}{b}, ay\right). \quad (2.11)$$

If the original domain and range are D_f and R_f , respectively, then

$$\begin{aligned} D_g &= \{x \in \mathbb{R} : bx \in D_f\}, \\ R_g &= \{ay : y \in R_f\}. \end{aligned}$$

Example 2.1: Combining two dilations

Let

$$f(x) = (x - 1)^2 + 1$$

and define $g(x) = 3f(2x)$. Describe the transformation and determine the image of the vertex of f .

The factor 2 inside the function produces a horizontal compression by scale factor $1/2$. The factor 3 outside the function produces a vertical stretch by scale factor 3. Thus,

$$(x, y) \mapsto \left(\frac{x}{2}, 3y\right).$$

The vertex of f is $(1, 1)$, so its image is

$$(1, 1) \mapsto \left(\frac{1}{2}, 3\right).$$

Algebraically,

$$g(x) = 3[(2x - 1)^2 + 1],$$

which confirms that the transformed graph has vertex $(1/2, 3)$.

Figure 2.5 illustrates the difference between multiplying the output and multiplying the input. A vertical stretch changes the heights of points, whereas a horizontal compression changes their horizontal positions.

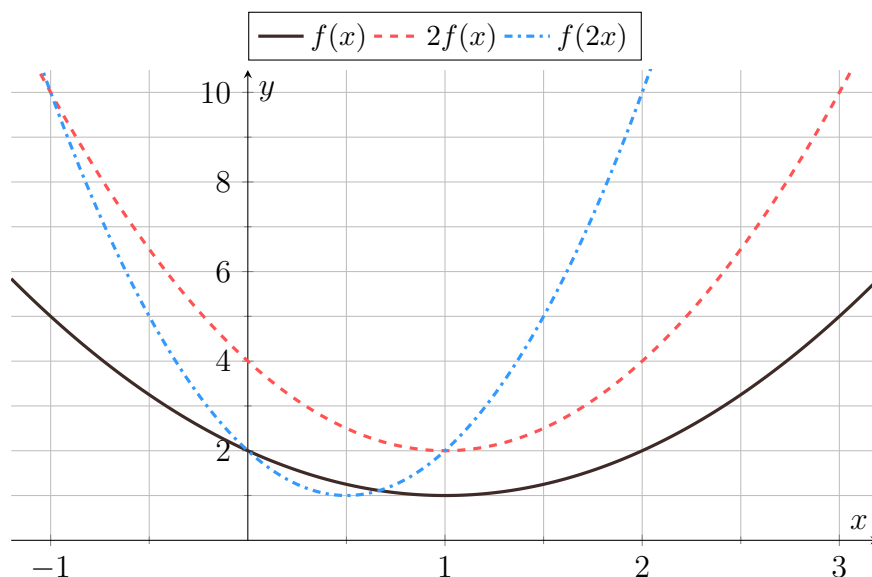


Figure 2.5: For $f(x) = (x - 1)^2 + 1$, multiplying the output gives a vertical stretch, while multiplying the input gives a horizontal compression.

2.3.3 Reflections of functions

A function can be reflected about either the x -axis or the y -axis.

1. The graph $y = -f(x)$ is the reflection of $y = f(x)$ about the x -axis. Its point mapping is

$$(x, y) \mapsto (x, -y).$$

2. The graph $y = f(-x)$ is the reflection of $y = f(x)$ about the y -axis. Its point mapping is

$$(x, y) \mapsto (-x, y).$$

Figure 2.6 compares these operations on one non-symmetric graph. Using a non-symmetric function makes the two reflections visibly different: $-f(x)$ changes every output sign, whereas $f(-x)$ changes every input sign.

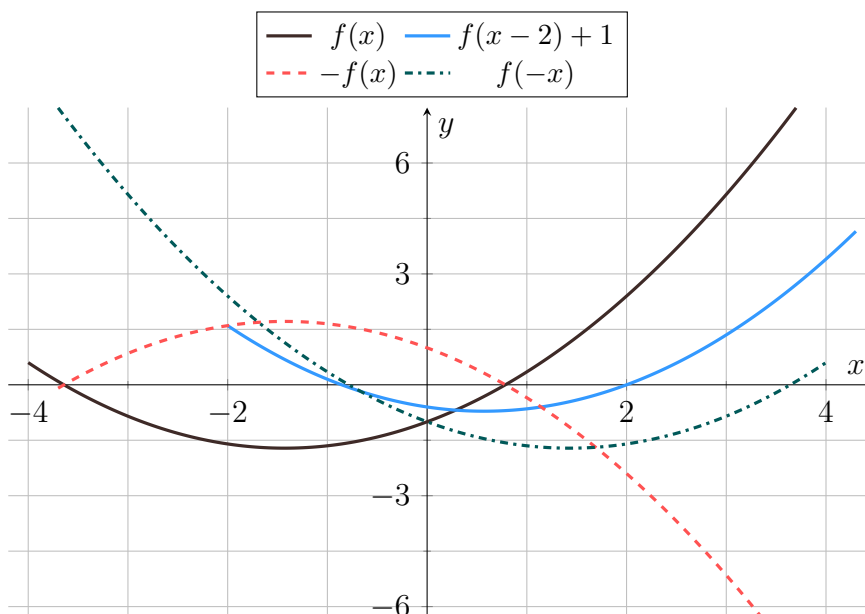


Figure 2.6: A translation and the two coordinate-axis reflections of $f(x) = 0.35x^2 + x - 1$.

For the general transformation $g(x) = af(b(x - h)) + k$, applying the horizontal and vertical scalings or reflections before the translations reproduces the point mapping in Equation (2.8).

2.4 Linear functions

A linear function has a graph that is a straight line. In pre-university mathematics, its general form is

$$f(x) = mx + c, \tag{2.12}$$

where $m, c \in \mathbb{R}$. The same form is sometimes written as $f(x) = ax + b$; the coefficients a and b play the same roles as m and c , respectively.

Strictly speaking, when $m \neq 0$, $f(x) = mx + c$ is a polynomial of degree one. When $m = 0$, the function becomes the constant function

$$f(x) = c,$$

whose graph is a horizontal line. In elementary graphing, both types are usually discussed together as straight-line functions.

Symbol	Name	Interpretation
x	Independent variable	The input plotted along the horizontal axis.
$f(x)$ or y	Dependent variable	The output plotted along the vertical axis.
m	Gradient or slope	The change in y produced by a one-unit change in x .
c	y -intercept	The value of the function when $x = 0$.

Table 2.3: Terminology for the linear function $f(x) = mx + c$.

The **basic linear function** is

$$f(x) = mx,$$

which passes through the origin. Adding c translates this graph vertically by c units. If $c > 0$, the graph shifts upward; if $c < 0$, it shifts downward.

For $m \neq 0$, the domain and range are both the set of real numbers:

$$D_f = \mathbb{R}, \quad R_f = \mathbb{R}.$$

For the constant function $f(x) = c$, the domain remains $\mathbb{R}$ but its range is the single-element set $\{c\}$.

Intercepts and graphing

The y -intercept is obtained by setting $x = 0$:

$$f(0) = c.$$

Hence the graph crosses the y -axis at $(0, c)$. If $m \neq 0$, the x -intercept is found by setting $f(x) = 0$:

$$\begin{aligned} mx + c &= 0, \\ x &= -\frac{c}{m}. \end{aligned}$$

Thus the graph crosses the x -axis at $(-\frac{c}{m}, 0)$. Since exactly one straight line passes through two distinct points, a linear graph can be drawn by plotting any two points and joining them with a straight line.

Consider

$$f(x) = -3x + 2.$$

Its gradient is -3 , its y -intercept is $(0, 2)$, and its x -intercept is $(\frac{2}{3}, 0)$. The negative gradient means that the graph falls from left to right.

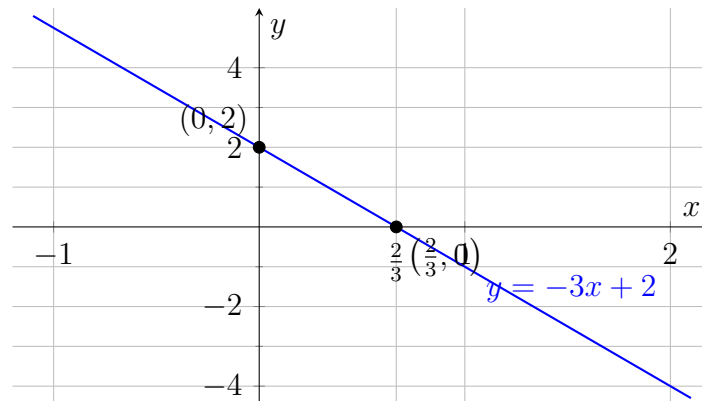


Figure 2.7: The graph of $f(x) = -3x + 2$.

Linear functions and transformations

Let the basic function be $f(x) = 3x$. The function

$$g(x) = -3x + 2$$

can be written as

$$g(x) = -f(x) + 2 = f(-x) + 2.$$

Therefore, the graph of g may be obtained by reflecting the graph of f about either coordinate axis and then translating it upward by 2 units. For a function of the form $f(x) = mx$, the two reflections give the same result because

$$-f(x) = f(-x).$$

There are also infinitely many equivalent sequences of translations. For example,

$$\begin{aligned} g(x) &= -f(x) + 2, \\ g(x) &= -f(x + 1) + 5, \\ g(x) &= -f(x - 1) - 1. \end{aligned}$$

These correspond respectively to:

- (a) reflecting and shifting upward by 2 units;
- (b) reflecting, shifting left by 1 unit, and shifting upward by 5 units; and
- (c) reflecting, shifting right by 1 unit, and shifting downward by 1 unit.

All three procedures produce the same final graph.

2.4.1 Gradient, m

The gradient measures the rate of change of y with respect to x . For two distinct points (x_1, y_1) and (x_2, y_2) on a non-vertical line,

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}, \quad x_2 \neq x_1. \quad (2.13)$$

The numerator represents the vertical change, or *rise*, while the denominator represents the horizontal change, or *run*. Reversing the order of both points does not change the gradient:

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{y_1 - y_2}{x_1 - x_2}.$$

Gradient	Behaviour	Description
$m > 0$	Increasing	The line rises from left to right.
$m < 0$	Decreasing	The line falls from left to right.
$m = 0$	Constant	The line is horizontal.
Undefined	Vertical	$\Delta x = 0$; a vertical line is not the graph of a function of x .

Table 2.4: Geometrical interpretation of the gradient.

The larger $|m|$ is, the steeper the line. For example, $y = 5x$ is steeper than $y = 2x$, while $y = -5x$ and $y = 5x$ have the same steepness but opposite directions.

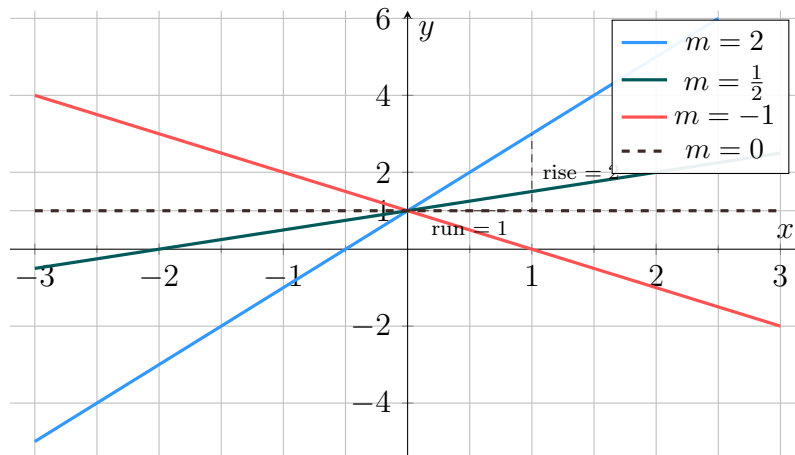


Figure 2.8: Lines through the same y -intercept illustrate how the sign and magnitude of the gradient control direction and steepness.

Forms of the equation of a line

The **slope–intercept form** is

$$y = mx + c.$$

If the gradient m and one point (x_1, y_1) are known, the **point–slope form** is

$$y - y_1 = m(x - x_1). \quad (2.14)$$

If two points are known, first calculate

$$m = \frac{y_2 - y_1}{x_2 - x_1},$$

and then substitute either point into the point–slope form. A line may also be written in the standard form

$$Ax + By = C.$$

When $B \neq 0$, this can be rearranged into

$$y = -\frac{A}{B}x + \frac{C}{B},$$

so its gradient is $-A/B$.

Parallel and perpendicular lines

Two non-vertical lines are parallel if and only if their gradients are equal:

$$m_1 = m_2.$$

Two non-vertical lines are perpendicular if their gradients are negative reciprocals:

$$m_1 m_2 = -1. \quad (2.15)$$

Thus a line with gradient $m \neq 0$ has a perpendicular gradient $-1/m$. Horizontal and vertical lines are also perpendicular to one another.

Worked examples

Example 2.2

Find the equation of the line passing through $(2, 5)$ and $(6, -3)$.

The gradient is

$$m = \frac{-3 - 5}{6 - 2} = \frac{-8}{4} = -2.$$

Using $(2, 5)$ in the point-slope form,

$$\begin{aligned} y - 5 &= -2(x - 2), \\ y &= -2x + 9. \end{aligned}$$

Example 2.3

Find the equation of the line passing through $(2, -1)$ and parallel to $3x - 2y = 6$.

Rearranging the given line,

$$y = \frac{3}{2}x - 3,$$

so the required gradient is $3/2$. Therefore,

$$\begin{aligned} y + 1 &= \frac{3}{2}(x - 2), \\ y &= \frac{3}{2}x - 4. \end{aligned}$$

Example 2.4

Find the equation of the line through $(2, -1)$ that is perpendicular to $3x - 2y = 6$.

The perpendicular gradient is $-2/3$. Hence

$$\begin{aligned} y + 1 &= -\frac{2}{3}(x - 2), \\ y &= -\frac{2}{3}x + \frac{1}{3}. \end{aligned}$$

Gradient as a rate of change

In an applied problem, the gradient carries units. Suppose the cost C , in ringgit, of travelling d kilometres is

$$C(d) = 3d + 5.$$

The gradient 3 represents a charge of RM3 per kilometre, while the y -intercept 5 represents a fixed initial charge of RM5. This interpretation of gradient as a constant rate of change will later connect linear functions with differentiation.

2.5 Quadratic functions

A **quadratic function** is a polynomial function of degree two. Its general form is

$$f(x) = ax^2 + bx + c, \quad a, b, c \in \mathbb{R}, \quad a \neq 0. \quad (2.16)$$

The condition $a \neq 0$ is essential. If $a = 0$, the quadratic term disappears and the function becomes linear or constant. The graph of a quadratic function is called a **parabola**.

2.5.1 The basic quadratic function

The basic form is

$$f(x) = ax^2, \quad a \neq 0.$$

Its vertex is the origin $(0, 0)$ and its axis of symmetry is the y -axis, whose equation is $x = 0$. Its domain is always

$$D_f = \mathbb{R}.$$

The sign of a determines the direction in which the parabola opens:

$$R_f = \begin{cases} [0, \infty), & a > 0, \\ (-\infty, 0], & a < 0. \end{cases}$$

If $a > 0$, the vertex is a minimum point; if $a < 0$, it is a maximum point. The magnitude $|a|$ controls the width of the parabola:

- if $|a| > 1$, the graph is narrower than $y = x^2$;
- if $0 < |a| < 1$, the graph is wider than $y = x^2$; and
- replacing a by $-a$ reflects the graph about the x -axis.

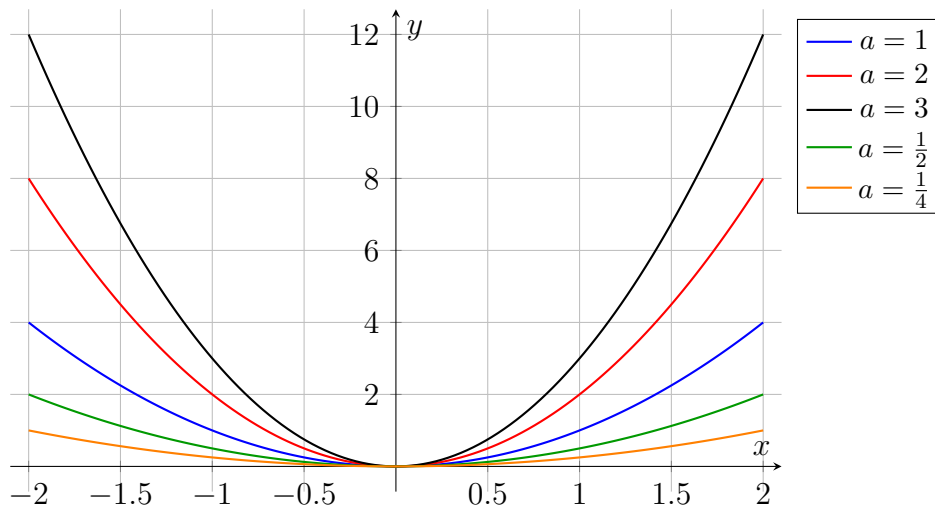


Figure 2.9: The effect of positive values of a on $f(x) = ax^2$.

All the graphs in the figure have domain $\mathbb{R}$, range $[0, \infty)$, vertex $(0, 0)$, and axis of symmetry $x = 0$. Only their widths differ.

2.5.2 Forms of a quadratic function

A quadratic function can be written in several equivalent forms. Each form makes different features of the graph immediately visible.

Form	Equation	Information shown directly
Basic form	$f(x) = ax^2$	Vertex $(0, 0)$ and axis $x = 0$.
Standard or general form	$f(x) = ax^2 + bx + c$	Leading coefficient a and y -intercept $(0, c)$.
Vertex or completed-square form	$f(x) = a(x - h)^2 + k$	Vertex (h, k) and axis of symmetry $x = h$.
Factored, root, or intercept form	$f(x) = a(x - r_1)(x - r_2)$	Real zeros r_1, r_2 and x -intercepts $(r_1, 0), (r_2, 0)$.

Table 2.5: The principal forms of a quadratic function.

In the factored form, r_1 and r_2 are real roots. If the roots are equal, the function is written as

$$f(x) = a(x - r)^2,$$

and the graph touches the x -axis at the repeated root r . A quadratic with no real roots cannot be expressed in factored form using real linear factors.

The corresponding quadratic equation is obtained by setting the function equal to zero. Hence its common forms are

$$\begin{aligned} ax^2 + bx + c &= 0, \\ a(x - h)^2 + k &= 0, \\ a(x - r_1)(x - r_2) &= 0. \end{aligned}$$

2.5.3 Vertex, axis of symmetry, and intercepts

For the standard form $f(x) = ax^2 + bx + c$, the y -intercept is $(0, c)$. The axis of symmetry is

$$x = -\frac{b}{2a}. \quad (2.17)$$

The vertex lies on this axis. Its coordinates are therefore

$$\left(-\frac{b}{2a}, f\left(-\frac{b}{2a}\right)\right). \quad (2.18)$$

Equivalently, the vertex is

$$\left(-\frac{b}{2a}, \frac{4ac - b^2}{4a}\right). \quad (2.19)$$

In vertex form $f(x) = a(x - h)^2 + k$, the vertex is (h, k) . Notice that the sign inside the bracket is opposite to the x -coordinate of the vertex. For example, the vertex of $2(x - 3)^2 + 5$ is $(3, 5)$, whereas the vertex of $2(x + 3)^2 + 5$ is $(-3, 5)$.

The range follows directly from the vertex form:

$$R_f = \begin{cases} [k, \infty), & a > 0, \\ (-\infty, k], & a < 0. \end{cases}$$

2.5.4 Completing the square

Completing the square converts the standard form into vertex form. The central identity is

$$x^2 + px = \left(x + \frac{p}{2}\right)^2 - \left(\frac{p}{2}\right)^2. \quad (2.20)$$

This works because half the coefficient of x becomes the constant inside the squared bracket.

When the coefficient of x^2 is 1

Write $x^2 - 6x + 11$ in completed-square form.

Half the coefficient of x is -3 . Therefore,

$$\begin{aligned}x^2 - 6x + 11 &= (x - 3)^2 - 9 + 11 \\&= (x - 3)^2 + 2.\end{aligned}$$

Thus the corresponding graph has vertex $(3, 2)$, axis of symmetry $x = 3$, and minimum value 2.

When the coefficient of x^2 is not 1

First factor the coefficient of x^2 from the terms involving x . For example,

$$\begin{aligned}2x^2 + 8x - 3 &= 2(x^2 + 4x) - 3 \\&= 2((x + 2)^2 - 4) - 3 \\&= 2(x + 2)^2 - 11.\end{aligned}$$

The vertex is $(-2, -11)$, the axis of symmetry is $x = -2$, and the graph has a minimum value of -11 .

If the leading coefficient is negative, the same procedure applies:

$$\begin{aligned}-3x^2 + 12x + 1 &= -3(x^2 - 4x) + 1 \\&= -3((x - 2)^2 - 4) + 1 \\&= -3(x - 2)^2 + 13.\end{aligned}$$

This graph has vertex $(2, 13)$ and a maximum value of 13.

Completing the square in the general case

Starting with $ax^2 + bx + c$, where $a \neq 0$,

$$\begin{aligned}ax^2 + bx + c &= a\left(x^2 + \frac{b}{a}x\right) + c \\&= a\left[\left(x + \frac{b}{2a}\right)^2 - \left(\frac{b}{2a}\right)^2\right] + c \\&= a\left(x + \frac{b}{2a}\right)^2 + c - \frac{b^2}{4a} \\&= a\left(x + \frac{b}{2a}\right)^2 + \frac{4ac - b^2}{4a}.\end{aligned}$$

This derivation confirms that

$$h = -\frac{b}{2a}, \quad k = \frac{4ac - b^2}{4a}.$$

2.5.5 Solving quadratic equations

A quadratic equation may be solved by factorisation, completing the square, or the quadratic formula. Graphically, its real solutions are the x -coordinates at which the parabola meets the x -axis.

Factorisation

Solve $x^2 - x - 6 = 0$.

$$\begin{aligned}x^2 - x - 6 &= 0, \\(x - 3)(x + 2) &= 0.\end{aligned}$$

By the zero-product property,

$$x = 3 \quad \text{or} \quad x = -2.$$

Completing the square

Solve $x^2 + 4x - 1 = 0$.

$$\begin{aligned}x^2 + 4x &= 1, \\x^2 + 4x + 4 &= 5, \\(x + 2)^2 &= 5, \\x + 2 &= \pm\sqrt{5}, \\x &= -2 \pm \sqrt{5}.\end{aligned}$$

The $\pm$ sign is required because both positive and negative square roots must be considered when solving an equation of the form $u^2 = k$.

Deriving the quadratic formula

Theorem 2.1 (Quadratic formula). *Let $a, b, c \in \mathbb{R}$ with $a \neq 0$, and set $\Delta = b^2 - 4ac$. If $\Delta \geq 0$, the real solutions of $ax^2 + bx + c = 0$ are exactly*

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}. \quad (2.21)$$

If $\Delta < 0$, the equation has no real solution.

Proof. Multiplying $ax^2 + bx + c = 0$ by the non-zero number $4a$ and completing the square gives a chain of equivalent equations:

$$\begin{aligned}
ax^2 + bx + c &= 0, \\
4a^2x^2 + 4abx + 4ac &= 0, \\
(2ax + b)^2 &= b^2 - 4ac = \Delta.
\end{aligned}$$

If $\Delta < 0$, this last equation is impossible over $\mathbb{R}$ because a real square is non-negative. If $\Delta \geq 0$, it is equivalent to

$$2ax + b = \pm\sqrt{\Delta},$$

which yields the two displayed values of x . Because every preceding step was reversible, these values are solutions and no real solutions have been lost. The same formula describes the complex roots after square roots of negative numbers are introduced in Chapter 10. $\square$

2.5.6 The discriminant and the nature of the roots

The expression

$$\Delta = b^2 - 4ac \tag{2.22}$$

is called the **discriminant**. It determines the number and type of real roots.

Discriminant	Roots	Graphical meaning
$\Delta > 0$	Two distinct real roots	The parabola crosses the x -axis twice.
$\Delta = 0$	One repeated real root	The parabola touches the x -axis at its vertex.
$\Delta < 0$	No real roots	The parabola does not meet the x -axis.

Table 2.6: The discriminant and the nature of quadratic roots.

The three cases in the table are shown together in Figure 2.10. The coefficient of x^2 is positive in every panel, so each parabola opens upward; only its vertical position relative to the x -axis changes the number of real roots.

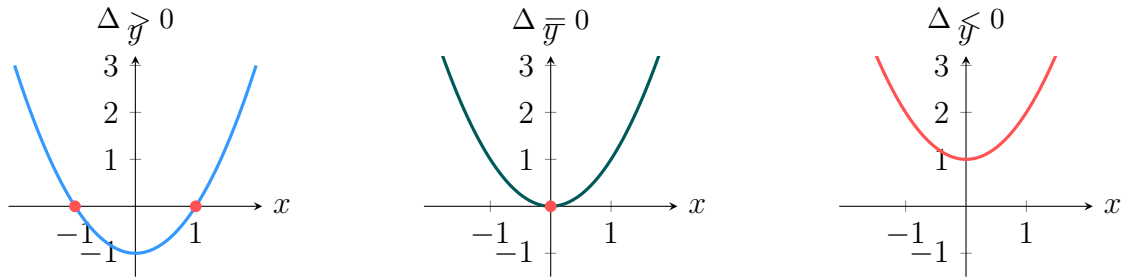


Figure 2.10: The discriminant counts the intersections of a quadratic graph with the x -axis.

For example, consider $2x^2 + 4x + 2 = 0$. Its discriminant is

$$\Delta = 4^2 - 4(2)(2) = 16 - 16 = 0.$$

Therefore it has one repeated real root. Indeed,

$$2x^2 + 4x + 2 = 2(x + 1)^2,$$

so the repeated root is $x = -1$.

If the roots are α and β , then comparing

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

gives the useful relationships

$$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}. \quad (2.23)$$

2.5.7 Connecting the different forms

Consider

$$f(x) = x^2 - 4x - 5.$$

Its three principal forms are

$$\begin{aligned} f(x) &= x^2 - 4x - 5 && \text{standard form,} \\ &= (x - 2)^2 - 9 && \text{vertex form,} \\ &= (x - 5)(x + 1) && \text{factored form.} \end{aligned}$$

Each form reveals different information:

- the standard form gives the y -intercept $(0, -5)$;
- the vertex form gives the vertex $(2, -9)$ and axis $x = 2$; and
- the factored form gives the x -intercepts $(-1, 0)$ and $(5, 0)$.

Since the coefficient of x^2 is positive, the parabola opens upward. Its domain is $\mathbb{R}$ and its range is $[-9, \infty)$.

2.5.8 Sketching a quadratic graph

To sketch $f(x) = ax^2 + bx + c$ accurately:

1. determine whether the graph opens upward or downward from the sign of a ;
2. find the axis of symmetry and vertex;
3. find the y -intercept $(0, c)$;
4. find the real x -intercepts, if any;
5. plot one or two additional symmetric points if necessary; and
6. draw a smooth parabola through the points.

The graph is symmetric about its axis, so points equally far from the axis have the same y -coordinate:

$$f(h - d) = f(h + d)$$

for a parabola whose axis is $x = h$.

2.5.9 Maximum and minimum applications

The vertex gives the maximum or minimum value of a quadratic model. Suppose a rectangular enclosure uses 20 metres of fencing for two widths and one length, with the fourth side along a wall. If each width is x metres, then the length is $20 - 2x$ and the area is

$$\begin{aligned} A(x) &= x(20 - 2x) \\ &= -2x^2 + 20x \\ &= -2(x - 5)^2 + 50. \end{aligned}$$

The coefficient of the squared term is negative, so the vertex gives a maximum. The greatest possible area is

$$50 \text{ m}^2,$$

obtained when the width is 5 metres and the length is 10 metres.

2.6 Cubic functions

A *cubic function* is a polynomial function whose highest power of the variable is three. Its general form is

$$f(x) = ax^3 + bx^2 + cx + d, \quad a, b, c, d \in \mathbb{R}, \quad a \neq 0. \quad (2.24)$$

The condition $a \neq 0$ is essential. If $a = 0$, the cubic term disappears and the function becomes quadratic, linear, or constant. The coefficient a is called the *leading coefficient*, while $d = f(0)$ is the constant term and gives the y -intercept $(0, d)$.

Every real cubic function has

$$D_f = \mathbb{R} \quad \text{and} \quad R_f = \mathbb{R}.$$

There are no restrictions on the input because a polynomial is defined for every real number. The range is also all real numbers because the two ends of a cubic graph extend without bound in opposite vertical directions.

2.6.1 End behaviour

For very large positive or negative values of x , the leading term ax^3 dominates the remaining terms. Consequently, the sign of a determines the end behaviour.

Leading coefficient	As $x \rightarrow -\infty$	As $x \rightarrow \infty$
$a > 0$	$f(x) \rightarrow -\infty$	$f(x) \rightarrow \infty$
$a < 0$	$f(x) \rightarrow \infty$	$f(x) \rightarrow -\infty$

Table 2.7: End behaviour of a cubic function.

Thus, a cubic with a positive leading coefficient rises overall from the bottom left to the top right, whereas a cubic with a negative leading coefficient falls overall from the top left to the bottom right. Local bends may occur between the ends, so “rises overall” does not necessarily mean that the function is increasing at every point.

2.6.2 The basic cubic function

The basic family of cubic functions is

$$f(x) = ax^3, \quad a \in \mathbb{R}, \quad a \neq 0. \quad (2.25)$$

Its graph passes through the origin, which is its only intercept and its point of inflection. For this particular family,

$$f(-x) = a(-x)^3 = -ax^3 = -f(x).$$

Therefore, $f(x) = ax^3$ is an *odd function*; its graph has rotational symmetry of 180° about the origin. Equivalently, $-f(x) = f(-x)$. This identity is not true for every cubic function, since the terms bx^2 and d may destroy symmetry about the origin.

For $a > 0$, the graph is increasing for all real x . Increasing the value of a produces a vertical stretch, making the graph appear steeper away from the origin. A value $0 < a < 1$ produces a vertical compression. If $a < 0$, the graph is reflected in the x -axis and is decreasing for all real x .

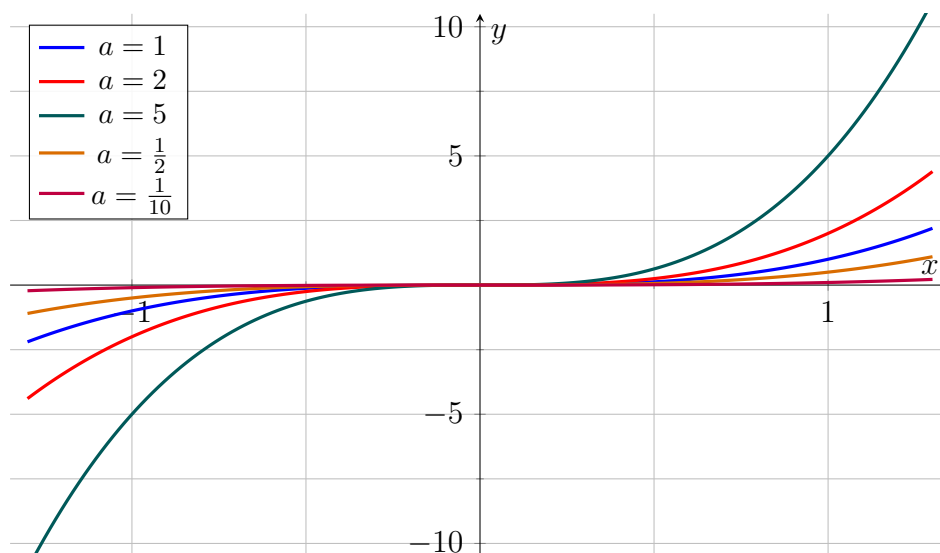


Figure 2.11: The family $f(x) = ax^3$ for several positive values of a .

All five graphs in the figure have domain $\mathbb{R}$ and range $\mathbb{R}$. They also pass through $(0, 0)$ and have the same end behaviour; only their vertical scale changes.

2.6.3 Useful forms of a cubic function

As with quadratics, different algebraic forms reveal different features of a cubic function.

Form	Expression	Information shown
General or expanded	$f(x) = ax^3 + bx^2 + cx + d$	Leading coefficient and y -intercept $(0, d)$
Factored	$f(x) = a(x - r_1)(x - r_2)(x - r_3)$	Real roots and their multiplicities, when the cubic splits into real linear factors
Transformed basic cubic	$f(x) = a(x - h)^3 + k$	Translation, reflection, vertical scale, and inflection point (h, k)

Table 2.8: Common forms of a cubic function.

Not every cubic can be written as $a(x - h)^3 + k$. That form describes a translation and vertical scaling of the basic graph $y = x^3$. In A-Level questions, the expanded form is useful for identifying the leading coefficient and y -intercept, the factored form is useful for finding roots, and the transformed form is useful for describing graph transformations.

2.6.4 Transformations of the basic cubic

Consider

$$g(x) = a(x - h)^3 + k, \quad a \neq 0. \quad (2.26)$$

Starting from $y = x^3$:

- h translates the graph horizontally: to the right if $h > 0$ and to the left if $h < 0$;
- k translates the graph vertically: upward if $k > 0$ and downward if $k < 0$;
- $|a| > 1$ gives a vertical stretch and $0 < |a| < 1$ gives a vertical compression; and
- $a < 0$ reflects the graph in the x -axis.

The centre of rotational symmetry and point of inflection move from $(0, 0)$ to (h, k) . The domain and range remain $\mathbb{R}$.

Example 2.5: Transforming a cubic graph

Describe the graph of

$$g(x) = -2(x - 1)^3 + 3$$

in relation to $y = x^3$.

Solution. The graph is stretched vertically by a factor of 2, reflected in the x -axis, translated one unit to the right, and translated three units upward. Its point of inflection is $(1, 3)$. Since the leading coefficient is negative, the graph falls overall from left to right.

2.6.5 Roots and the factor theorem

The roots or zeros of a cubic are the solutions of

$$f(x) = 0.$$

They are the x -coordinates at which the graph meets the x -axis. Every cubic polynomial with real coefficients has at least one real root because its two ends extend in opposite vertical directions. It may have one, two, or three distinct real roots; the case of two distinct real roots occurs when one of them is repeated. If repeated roots are counted according to their multiplicities, a cubic has three roots over the complex numbers.

The factor theorem states that

$$f(r) = 0 \quad \Longleftrightarrow \quad (x - r) \text{ is a factor of } f(x). \quad (2.27)$$

Once one root is known, polynomial division can reduce the cubic to a quadratic factor, which can then be solved by factorisation, completing the square, or the quadratic formula. In many A-Level questions, a root or linear factor is supplied. Otherwise, simple integer values may be tested using the factor theorem; for a monic polynomial with integer coefficients, factors of the constant term are sensible values to test first.

Example 2.6: Solving a cubic equation

Solve

$$x^3 - 4x^2 + x + 6 = 0.$$

Solution. Testing the possible integer roots shows that $x = 2$ is a root because

$$2^3 - 4(2)^2 + 2 + 6 = 0.$$

Hence, $(x - 2)$ is a factor. Division or comparison of coefficients gives

$$\begin{aligned} x^3 - 4x^2 + x + 6 &= (x - 2)(x^2 - 2x - 3) \\ &= (x - 2)(x - 3)(x + 1). \end{aligned}$$

Therefore,

$$\boxed{x = -1, 2, 3}.$$

If no simple exact factor is available, a root may be estimated graphically or by a numerical method when this is required by the syllabus or question.

2.6.6 Multiplicity and the behaviour at a root

A root may occur more than once in the factorisation. The number of times its factor is repeated is called its *multiplicity*.

Multiplicity	Typical factor	Graphical behaviour at $x = r$
1	$(x - r)$	The graph crosses the x -axis.
2	$(x - r)^2$	The graph touches the x -axis and turns around.
3	$(x - r)^3$	The graph crosses the x -axis while flattening at the root.

Table 2.9: Root multiplicity and the corresponding graph behaviour.

More generally, a graph crosses the axis at a root of odd multiplicity and touches without crossing at a root of even multiplicity.

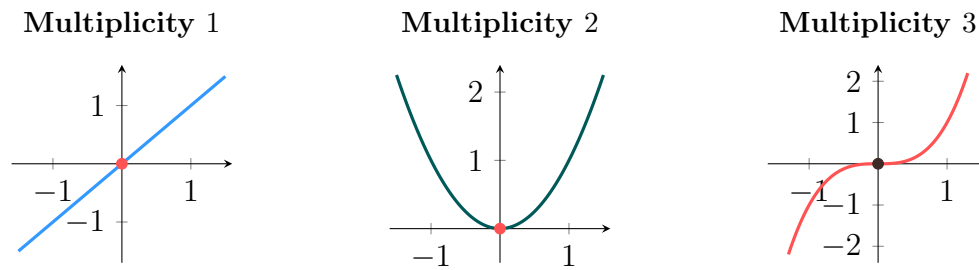


Figure 2.12: Local graph behaviour at roots of multiplicity 1, 2, and 3: crossing, touching, and flattened crossing.

Example 2.7: Repeated roots

Consider

$$f(x) = (x - 1)^2(x + 2).$$

The root $x = 1$ has multiplicity two, so the graph touches the x -axis and turns there. The root $x = -2$ has multiplicity one, so the graph crosses the x -axis there. The leading coefficient is positive, so the graph begins at the bottom left and ends at the top right.

2.6.7 Stationary points and turning points

A cubic graph may have two turning points or no turning points. It can never have more than two because the derivative of a cubic is quadratic. For

$$f(x) = ax^3 + bx^2 + cx + d,$$

the derivative is

$$f'(x) = 3ax^2 + 2bx + c. \quad (2.28)$$

Stationary points occur where $f'(x) = 0$. The discriminant of this quadratic derivative is

$$(2b)^2 - 4(3a)c = 4(b^2 - 3ac).$$

Therefore:

- if $b^2 - 3ac > 0$, there are two distinct stationary points: one local maximum and one local minimum;
- if $b^2 - 3ac = 0$, there is one repeated stationary point, which is a stationary point of inflection; and
- if $b^2 - 3ac < 0$, there are no stationary points and the cubic is monotonic.

This derivative method will be developed more fully in the chapters on calculus. When calculus is not yet available, turning points can still be estimated from a graph or a table of values.

2.6.8 The point of inflection

Every cubic graph has a point of inflection, where its concavity changes. The second derivative is

$$f''(x) = 6ax + 2b.$$

Setting $f''(x) = 0$ gives the x -coordinate of the point of inflection:

$$x = -\frac{b}{3a}. \quad (2.29)$$

Hence, the point of inflection is

$$\left(-\frac{b}{3a}, f\left(-\frac{b}{3a}\right)\right).$$

This point is also the centre of the cubic graph's 180° rotational symmetry. An inflection point need not be stationary; it is stationary only when its gradient is zero.

Example 2.8: Stationary points and inflection point

Find the stationary points and point of inflection of

$$f(x) = x^3 - 3x^2 - 9x + 5.$$

Solution. Differentiate:

$$\begin{aligned} f'(x) &= 3x^2 - 6x - 9 \\ &= 3(x - 3)(x + 1). \end{aligned}$$

Thus, the stationary points occur at $x = -1$ and $x = 3$. Their coordinates are

$$f(-1) = 10, \quad f(3) = -22.$$

Since $f''(x) = 6x - 6$, we have $f''(-1) < 0$ and $f''(3) > 0$. Therefore, $(-1, 10)$ is a local maximum and $(3, -22)$ is a local minimum.

For the point of inflection,

$$6x - 6 = 0 \implies x = 1,$$

and $f(1) = -6$. Hence, the point of inflection is $(1, -6)$.

2.6.9 Sketching a cubic graph

A reliable sketch can be constructed using the following information:

1. Identify the sign of the leading coefficient and draw the correct end behaviour.

2. Find the y -intercept by calculating $f(0)$.
3. Solve $f(x) = 0$ to locate the x -intercepts, noting the multiplicity of each root.
4. Find or estimate the stationary points, if any.
5. Locate the point of inflection when the required methods are available.
6. Join the features with a smooth continuous curve. A polynomial graph has no breaks, holes, or vertical asymptotes.

Example 2.9: Sketching from a factorisation

Sketch the main features of

$$f(x) = x^3 - 4x = x(x - 2)(x + 2).$$

Solution. The roots are -2 , 0 , and 2 , each with multiplicity one, so the graph crosses the x -axis at all three points. Its y -intercept is also $(0, 0)$. The leading coefficient is positive, so the graph extends from the bottom left to the top right. Furthermore,

$$f(-x) = -f(x),$$

so the graph is symmetric under a 180° rotation about the origin. These features determine the familiar S-shaped cubic with one local maximum between -2 and 0 and one local minimum between 0 and 2 .

2.7 Rational functions

A rational function is formed by dividing one polynomial by another. Its graph may contain excluded inputs, vertical asymptotes, horizontal or oblique asymptotes, and removable discontinuities called holes. These features make domain analysis especially important.

Definition 2.7: Rational function.

A rational function has the form

$$f(x) = \frac{P(x)}{Q(x)}, \quad (2.30)$$

where P and Q are polynomials and $Q(x) \neq 0$. Its domain consists of all real numbers for which the denominator is non-zero.

For example,

$$f(x) = \frac{x + 2}{x^2 - 9}$$

has denominator

$$x^2 - 9 = (x - 3)(x + 3),$$

so its domain is

$$D_f = \mathbb{R} \setminus \{-3, 3\}.$$

2.7.1 The basic reciprocal function

The simplest rational function is

$$f(x) = \frac{a}{x}, \quad a \neq 0. \quad (2.31)$$

Its domain and range are

$$D_f = \mathbb{R} \setminus \{0\}, \quad R_f = \mathbb{R} \setminus \{0\}.$$

The axes are asymptotes:

- $x = 0$, the y -axis, is the vertical asymptote;
- $y = 0$, the x -axis, is the horizontal asymptote.

The graph approaches an asymptote indefinitely but never intersects either axis. Since

$$f(-x) = -f(x),$$

the function is odd and its graph has 180° rotational symmetry about the origin.

If $a > 0$, the two branches lie in Quadrants I and III. If $a < 0$, they lie in Quadrants II and IV. Increasing $|a|$ moves corresponding points farther from the coordinate axes.

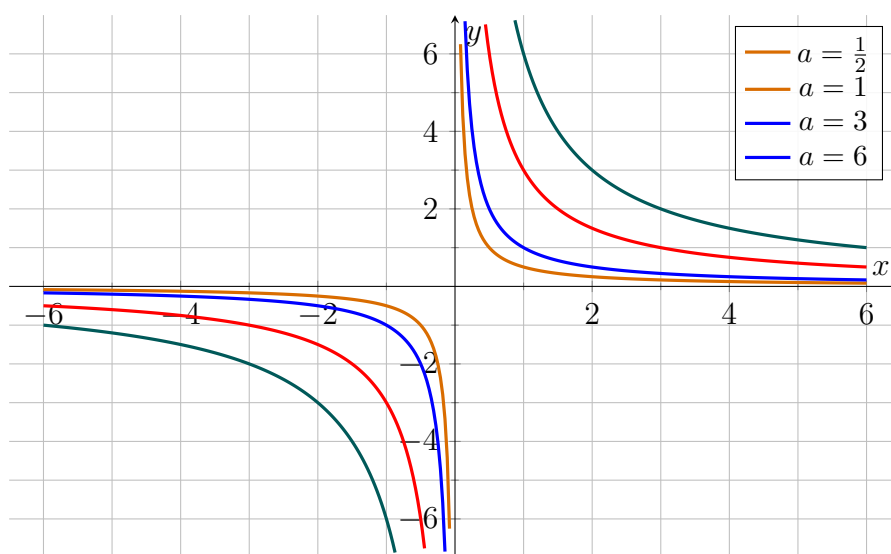


Figure 2.13: The reciprocal family $f(x) = a/x$ for several positive values of a .

All four graphs in Figure 2.13 have the same domain, range, and asymptotes. For a fixed positive x , a larger value of a produces a larger output.

2.7.2 Translations of the reciprocal graph

A translated reciprocal function has the form

$$g(x) = \frac{a}{x - h} + k, \quad a \neq 0. \quad (2.32)$$

Starting from $y = a/x$:

- h translates the graph horizontally;
- k translates the graph vertically;
- $|a|$ controls the scale; and
- changing the sign of a reflects the graph in either asymptote.

The vertical and horizontal asymptotes are

$$x = h \quad \text{and} \quad y = k,$$

respectively. Therefore,

$$D_g = \mathbb{R} \setminus \{h\}, \quad R_g = \mathbb{R} \setminus \{k\}. \quad (2.33)$$

Their point of intersection, (h, k) , is the centre of the graph's 180° rotational symmetry. Indeed, for any non-zero t ,

$$g(h+t) + g(h-t) = 2k.$$

Example 2.10: Translation left and downward

Consider

$$g(x) = \frac{2}{x+1} - 2.$$

Since $x+1 = x - (-1)$, the graph of $y = 2/x$ is translated one unit to the left and two units downward. Thus,

$$\begin{aligned} D_g &= \mathbb{R} \setminus \{-1\}, \\ R_g &= \mathbb{R} \setminus \{-2\}. \end{aligned}$$

The asymptotes are $x = -1$ and $y = -2$, and the centre is $(-1, -2)$.

To find the x -intercept,

$$\begin{aligned} \frac{2}{x+1} - 2 &= 0, \\ \frac{2}{x+1} &= 2, \\ x &= 0. \end{aligned}$$

Also, $g(0) = 0$, so both intercepts occur at the origin.

Example 2.11: Rewriting a linear-over-linear function

Consider

$$h(x) = \frac{3x-4}{x-2}.$$

Rewrite the numerator as

$$3x - 4 = 3(x - 2) + 2.$$

Hence,

$$h(x) = 3 + \frac{2}{x - 2}.$$

The graph of $y = 2/x$ is translated two units to the right and three units upward. Therefore,

$$D_h = \mathbb{R} \setminus \{2\},$$

$$R_h = \mathbb{R} \setminus \{3\}.$$

Its asymptotes are $x = 2$ and $y = 3$, and its centre is $(2, 3)$.

The intercepts are

$$3x - 4 = 0 \implies x = \frac{4}{3},$$

$$h(0) = 2.$$

Thus, the x -intercept is $(\frac{4}{3}, 0)$ and the y -intercept is $(0, 2)$.

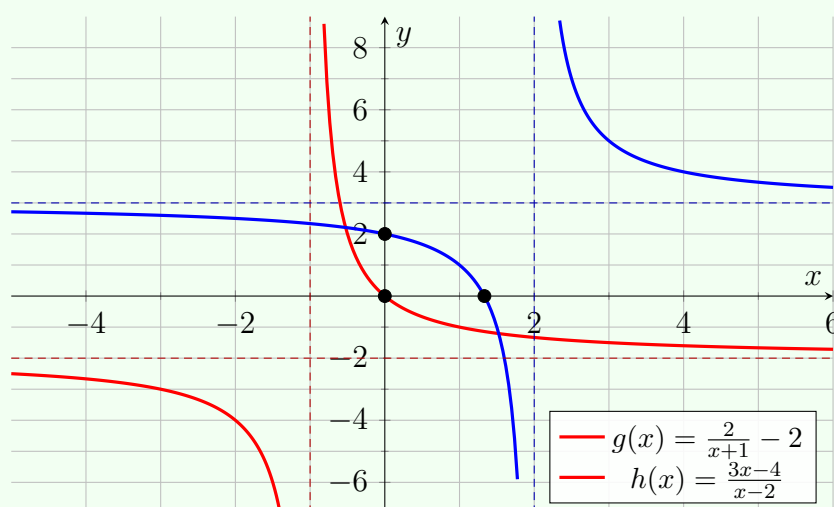


Figure 2.14: Two transformations of the reciprocal graph.

2.7.3 Linear-over-linear rational functions

Consider

$$f(x) = \frac{ax + b}{cx + d}, \quad c \neq 0. \quad (2.34)$$

If $ad - bc \neq 0$, the function can be written in translated reciprocal form:

$$\frac{ax + b}{cx + d} = \frac{a}{c} + \frac{bc - ad}{c(cx + d)}. \quad (2.35)$$

Therefore, its asymptotes are

$$x = -\frac{d}{c} \quad \text{and} \quad y = \frac{a}{c}.$$

Its domain and range are

$$\mathbb{R} \setminus \left\{ -\frac{d}{c} \right\} \quad \text{and} \quad \mathbb{R} \setminus \left\{ \frac{a}{c} \right\},$$

respectively.

Example 2.12: Reading the main features

For

$$f(x) = \frac{2x + 1}{x - 3},$$

division gives

$$f(x) = 2 + \frac{7}{x - 3}.$$

Thus, the vertical asymptote is $x = 3$, the horizontal asymptote is $y = 2$, and the centre is $(3, 2)$. The intercepts are

$$\left(-\frac{1}{2}, 0 \right) \quad \text{and} \quad \left(0, -\frac{1}{3} \right).$$

2.7.4 Intercepts of a rational function

For

$$f(x) = \frac{P(x)}{Q(x)},$$

an x -intercept occurs where

$$P(x) = 0,$$

provided that the denominator is also non-zero. The y -intercept is $f(0)$, provided that 0 belongs to the domain.

For example,

$$r(x) = \frac{(x - 1)(x + 4)}{(x - 2)(x + 3)}$$

has x -intercepts at $x = 1$ and $x = -4$, vertical asymptotes at $x = 2$ and $x = -3$, and y -intercept

$$r(0) = \frac{(-1)(4)}{(-2)(3)} = \frac{2}{3}.$$

2.7.5 Holes and cancelled factors

A denominator zero does not always produce a vertical asymptote. First factor the numerator and denominator and cancel any common factor, while retaining the corresponding excluded input.

Example 2.13: A removable discontinuity

Consider

$$f(x) = \frac{x^2 - 1}{x - 1}.$$

Then

$$\begin{aligned} f(x) &= \frac{(x - 1)(x + 1)}{x - 1} \\ &= x + 1, \quad x \neq 1. \end{aligned}$$

The graph is the straight line $y = x + 1$ with a hole at $(1, 2)$. Since the factor $x - 1$ cancels completely, $x = 1$ is not a vertical asymptote.

In contrast,

$$q(x) = \frac{x + 1}{(x - 1)(x + 2)}$$

has no common factor, so $x = 1$ and $x = -2$ are vertical asymptotes.

2.7.6 Horizontal asymptotes

Let n be the degree of the numerator and m the degree of the denominator. For a rational function in simplified form:

- if $n < m$, the horizontal asymptote is $y = 0$;
- if $n = m$, the horizontal asymptote is the ratio of the leading coefficients;
- if $n > m$, there is no horizontal asymptote.

For example,

$$\begin{aligned} \frac{2x + 1}{x^2 + 4} &\text{ has horizontal asymptote } y = 0, \\ \frac{3x^2 - x + 1}{2x^2 + 5} &\text{ has horizontal asymptote } y = \frac{3}{2}. \end{aligned}$$

A graph may cross a horizontal asymptote. The asymptote describes the end behaviour as $x \rightarrow \pm\infty$; it is not generally an excluded output. For the special translated reciprocal function $a/(x - h) + k$, however, the value $y = k$ is excluded from the range.

2.7.7 Oblique asymptotes

If the degree of the numerator is exactly one greater than the degree of the denominator, polynomial division produces an oblique or slant asymptote.

Example 2.14: Finding an oblique asymptote

For

$$f(x) = \frac{x^2 + 1}{x - 1},$$

division gives

$$f(x) = x + 1 + \frac{2}{x - 1}.$$

Therefore, $x = 1$ is the vertical asymptote and

$$y = x + 1$$

is the oblique asymptote.

More generally, polynomial division may produce a polynomial asymptote when the numerator has degree greater than the denominator.

2.7.8 Behaviour near a vertical asymptote

The sign of a rational expression determines whether a branch approaches $+\infty$ or $-\infty$. For example,

$$f(x) = \frac{1}{x - 2}$$

satisfies

$$\begin{aligned} f(x) &\longrightarrow -\infty && \text{as } x \longrightarrow 2^-, \\ f(x) &\longrightarrow +\infty && \text{as } x \longrightarrow 2^+. \end{aligned}$$

Testing one value on each side of an asymptote is often sufficient to place the branches correctly in a sketch.

2.7.9 Solving rational equations

To solve a rational equation:

1. state every excluded value;
2. multiply by the lowest common denominator;
3. solve the resulting equation; and
4. reject any value excluded from the original equation.

Example 2.15: Clearing denominators

Solve

$$\frac{2}{x - 1} + 1 = \frac{3}{x - 1}.$$

The original equation requires $x \neq 1$. Multiplying by $x - 1$ gives

$$\begin{aligned} 2 + (x - 1) &= 3, \\ x &= 2. \end{aligned}$$

Since 2 is not excluded, the solution is

$$\boxed{x = 2}.$$

2.7.10 Sketching a rational function

A reliable sketch can be constructed using the following procedure:

1. Factor and simplify, recording all excluded inputs before cancellation.
2. State the domain and identify any holes.
3. Find the vertical asymptotes from the remaining denominator factors.
4. Determine the horizontal, oblique, or polynomial asymptote.
5. Find the x - and y -intercepts.
6. Test points in the intervals separated by the vertical asymptotes.
7. Draw smooth branches that approach the required asymptotes.

When working with rational functions, remember that:

- denominator zeros must be excluded from the domain;
- a cancelled denominator factor produces a hole, not a vertical asymptote;
- uncanceled denominator zeros usually produce vertical asymptotes;
- horizontal asymptotes describe end behaviour and may sometimes be crossed; and
- solutions obtained after clearing denominators must be checked in the original equation.

2.8 Exponential functions

An exponential function is a function in which the independent variable appears in the exponent. Exponential functions describe quantities that grow or decrease by a constant multiplicative factor over equal intervals, such as compound interest, population growth, radioactive decay, and cooling.

Definition 2.8: Exponential function.

An exponential function has the form

$$f(x) = Ab^x, \tag{2.36}$$

where $A \neq 0$, $b > 0$, and $b \neq 1$. The constant b is called the base.

The restriction $b > 0$ ensures that b^x is real for every real value of x . The base $b = 1$ is excluded because $1^x = 1$ is constant rather than exponential growth or decay.

The usual laws of indices remain valid:

$$\begin{aligned} b^m b^n &= b^{m+n}, & \frac{b^m}{b^n} &= b^{m-n}, \\ (b^m)^n &= b^{mn}, & b^{-n} &= \frac{1}{b^n}. \end{aligned}$$

2.8.1 The basic exponential graph

The natural exponential function is

$$f(x) = e^x, \quad (2.37)$$

where

$$e \approx 2.718281828.$$

Its domain and range are

$$D_f = \mathbb{R}, \quad R_f = (0, \infty).$$

The graph passes through $(0, 1)$ because $e^0 = 1$. It has no x -intercept, since e^x is positive for every real x . The x -axis, $y = 0$, is a horizontal asymptote:

$$e^x \longrightarrow 0 \quad \text{as} \quad x \longrightarrow -\infty.$$

As $x \rightarrow \infty$, the value of e^x increases without bound.

More generally, for $f(x) = b^x$:

- if $b > 1$, the function is increasing and represents exponential growth;
- if $0 < b < 1$, the function is decreasing and represents exponential decay.

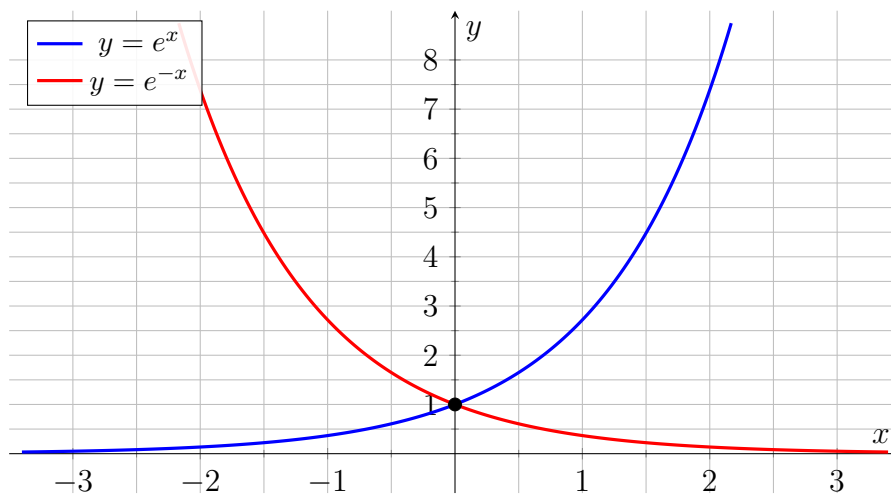


Figure 2.15: The exponential growth function e^x and decay function e^{-x} .

The graph of $y = e^{-x}$ is the reflection of $y = e^x$ in the y -axis. Both graphs have domain $\mathbb{R}$, range $(0, \infty)$, y -intercept $(0, 1)$, and horizontal asymptote $y = 0$.

2.8.2 Transformations of exponential graphs

A transformed exponential function is commonly written as

$$g(x) = Ab^{x-h} + k, \quad A \neq 0, \quad b > 0, \quad b \neq 1. \quad (2.38)$$

Starting from $y = b^x$:

- h translates the graph horizontally;
- k translates the graph vertically;
- $|A|$ produces a vertical stretch or compression; and
- $A < 0$ reflects the graph in the x -axis.

The domain remains $\mathbb{R}$. Since $b^{x-h} > 0$, the range is

$$R_g = \begin{cases} (k, \infty), & A > 0, \\ (-\infty, k), & A < 0, \end{cases} \quad (2.39)$$

and the horizontal asymptote is

$$y = k.$$

Example 2.16: Translation to the right and downward

Consider

$$g(x) = e^{x-1} - 2.$$

The graph of $y = e^x$ is translated one unit to the right and two units downward. Hence,

$$D_g = \mathbb{R}, \quad R_g = (-2, \infty),$$

and the horizontal asymptote is $y = -2$.

The y -intercept is

$$g(0) = e^{-1} - 2 = \frac{1}{e} - 2.$$

To find the x -intercept,

$$\begin{aligned} e^{x-1} - 2 &= 0, \\ e^{x-1} &= 2, \\ x - 1 &= \ln 2, \\ x &= 1 + \ln 2. \end{aligned}$$

Therefore, the x -intercept is $(1 + \ln 2, 0)$.

Example 2.17: Translation to the left and upward

Consider

$$h(x) = e^{x+2} + 3.$$

The graph is translated two units to the left and three units upward. Thus,

$$D_h = \mathbb{R}, \quad R_h = (3, \infty),$$

and the horizontal asymptote is $y = 3$. Its y -intercept is

$$h(0) = e^2 + 3.$$

There is no x -intercept because $h(x) > 3$ for every real x .

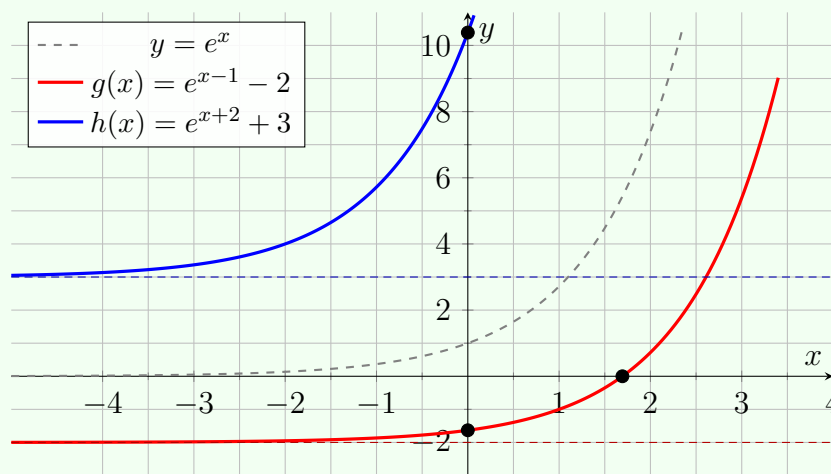


Figure 2.16: Translations of the basic natural exponential graph.

2.8.3 The effect of a coefficient

For

$$f(x) = Ab^x,$$

the coefficient A determines the y -intercept because

$$f(0) = A.$$

For example, the function

$$p(x) = -2e^x + 4$$

is reflected in the x -axis, stretched vertically by a factor of 2, and translated four units upward. Its domain is $\mathbb{R}$, its range is $(-\infty, 4)$, and its horizontal asymptote is $y = 4$. Its intercepts are

$$\begin{aligned} p(0) &= 2, \\ -2e^x + 4 &= 0 \implies x = \ln 2. \end{aligned}$$

2.8.4 Solving exponential equations

The method used depends on the form of the equation.

Writing both sides with the same base

Example 2.18: Equal bases

Solve

$$3^{2x-1} = 27.$$

Since $27 = 3^3$,

$$3^{2x-1} = 3^3,$$

$$2x - 1 = 3,$$

$$x = 2.$$

Taking logarithms

If the two sides cannot conveniently be written with the same base, take a logarithm of both sides.

Example 2.19: Using logarithms

Solve

$$5^x = 12.$$

Taking natural logarithms gives

$$\ln(5^x) = \ln 12,$$

$$x \ln 5 = \ln 12,$$

$$x = \frac{\ln 12}{\ln 5} \approx 1.544.$$

Using a substitution

Equations containing related exponential terms may be converted into quadratic equations.

Example 2.20: A quadratic in disguise

Solve

$$4^x - 5(2^x) + 4 = 0.$$

Since $4^x = (2^x)^2$, let $u = 2^x$, where $u > 0$. Then

$$u^2 - 5u + 4 = 0,$$

$$(u - 1)(u - 4) = 0.$$

Thus, $u = 1$ or $u = 4$. Therefore,

$$\begin{aligned} 2^x = 1 &\implies x = 0, \\ 2^x = 4 &\implies x = 2. \end{aligned}$$

2.8.5 Exponential inequalities

The direction of an inequality depends on whether the base represents growth or decay. For positive b with $b \neq 1$,

$$\begin{aligned} b^u < b^v &\iff u < v, & b > 1, \\ b^u < b^v &\iff u > v, & 0 < b < 1. \end{aligned}$$

For example,

$$2^{x+1} > 16$$

becomes

$$\begin{aligned} 2^{x+1} &> 2^4, \\ x+1 &> 4, \\ x &> 3. \end{aligned}$$

On the other hand,

$$\left(\frac{1}{2}\right)^{x-1} \leq \left(\frac{1}{2}\right)^3$$

gives $x - 1 \geq 3$, because the base lies between 0 and 1. Hence, $x \geq 4$.

2.8.6 Exponential growth and decay models

If a quantity changes by the same percentage during each equal time interval, it can be modelled by

$$Q(t) = Q_0 b^t, \tag{2.40}$$

where Q_0 is the initial quantity and b is the factor per unit time.

For a growth rate of r per unit time,

$$b = 1 + r,$$

while for a decay rate of r ,

$$b = 1 - r.$$

The percentage rate must be written as a decimal; for example, $6\% = 0.06$.

Example 2.21: Percentage growth

A population initially contains 2400 organisms and grows by 5% each year. After t years,

$$P(t) = 2400(1.05)^t.$$

After 8 years,

$$P(8) = 2400(1.05)^8 \approx 3546.$$

Example 2.22: Percentage decay

A machine worth RM 50,000 loses 12% of its value each year. Its value after t years is

$$V(t) = 50\,000(0.88)^t.$$

After 5 years,

$$V(5) = 50\,000(0.88)^5 \approx \text{RM } 26,387.$$

2.8.7 Continuous exponential models

Continuous growth or decay is commonly written as

$$Q(t) = Q_0 e^{kt}, \quad (2.41)$$

where k is the continuous growth constant. The model represents growth if $k > 0$ and decay if $k < 0$.

If T_d is the doubling time, then

$$T_d = \frac{\ln 2}{k}, \quad k > 0.$$

If $T_{1/2}$ is the half-life, then

$$T_{1/2} = \frac{\ln 2}{|k|}, \quad k < 0.$$

Example 2.23: Half-life

A radioactive sample has mass 80 g and a half-life of 6 years. A suitable model is

$$M(t) = 80 \left(\frac{1}{2} \right)^{t/6}.$$

After 18 years, three half-lives have passed, so

$$M(18) = 80 \left(\frac{1}{2} \right)^3 = 10 \text{ g}.$$

2.8.8 Compound interest

If a principal P is invested at an annual nominal interest rate r , compounded n times per year for t years, the accumulated amount is

$$A = P \left(1 + \frac{r}{n} \right)^{nt}. \quad (2.42)$$

For continuous compounding,

$$A = Pe^{rt}. \quad (2.43)$$

Example 2.24: Compound interest

RM 3,000 is invested at 4.8% per annum, compounded monthly, for 5 years. Then

$$\begin{aligned} A &= 3000 \left(1 + \frac{0.048}{12} \right)^{12(5)} \\ &\approx \text{RM } 3,811.92. \end{aligned}$$

2.8.9 Exponential and logarithmic functions

The exponential function $y = b^x$ and the logarithmic function $y = \log_b x$ are inverses. Therefore, their graphs are reflections in the line $y = x$, and

$$\log_b(b^x) = x, \quad b^{\log_b x} = x, \quad x > 0. \quad (2.44)$$

This relationship is developed further in Section 2.8.

2.9 Logarithmic functions

A logarithmic function reverses an exponential function. Logarithms are used when the unknown quantity appears as an exponent and are therefore important in algebra, scientific modelling, and later work in calculus.

Definition 2.9: Logarithm.

Let $b > 0$ with $b \neq 1$. For $x > 0$,

$$y = \log_b x \iff b^y = x. \quad (2.45)$$

Thus, $\log_b x$ is the exponent to which the base b must be raised to produce x .

For example,

$$\log_2 8 = 3 \quad \text{because} \quad 2^3 = 8,$$

and

$$\log_{10} 0.01 = -2 \quad \text{because} \quad 10^{-2} = 0.01.$$

The restriction $x > 0$ is essential: the logarithm of zero or a negative real number is not defined in the real number system.

2.9.1 Common and natural logarithms

Two bases occur especially often:

$$\begin{array}{ll} \log x = \log_{10} x, & \text{the common logarithm,} \\ \ln x = \log_e x, & \text{the natural logarithm,} \end{array}$$

where

$$e \approx 2.71828.$$

Unless a base is shown, the notation $\log x$ commonly means base 10, while $\ln x$ always means base e . Some useful exact values are

$$\begin{array}{lll} \log_b 1 = 0, & \log_b b = 1, & \log_b(b^r) = r, \\ \ln 1 = 0, & \ln e = 1, & \ln(e^r) = r. \end{array}$$

2.9.2 The basic logarithmic graph

The basic natural-logarithm function is

$$f(x) = \ln x. \tag{2.46}$$

Its principal features are:

- the domain is $(0, \infty)$;
- the range is $\mathbb{R}$;
- the y -axis, $x = 0$, is a vertical asymptote;
- the graph crosses the x -axis at $(1, 0)$;
- there is no y -intercept; and
- the function is increasing, but its rate of increase gradually decreases.

Hence,

$$D_f = (0, \infty), \quad R_f = \mathbb{R}.$$

As x approaches zero from the right, $\ln x$ decreases without bound:

$$\ln x \longrightarrow -\infty \quad \text{as} \quad x \longrightarrow 0^+.$$

More generally, $y = \log_b x$ passes through $(1, 0)$ and has vertical asymptote $x = 0$. Its direction depends on the base:

- if $b > 1$, the function is increasing;
- if $0 < b < 1$, the function is decreasing.

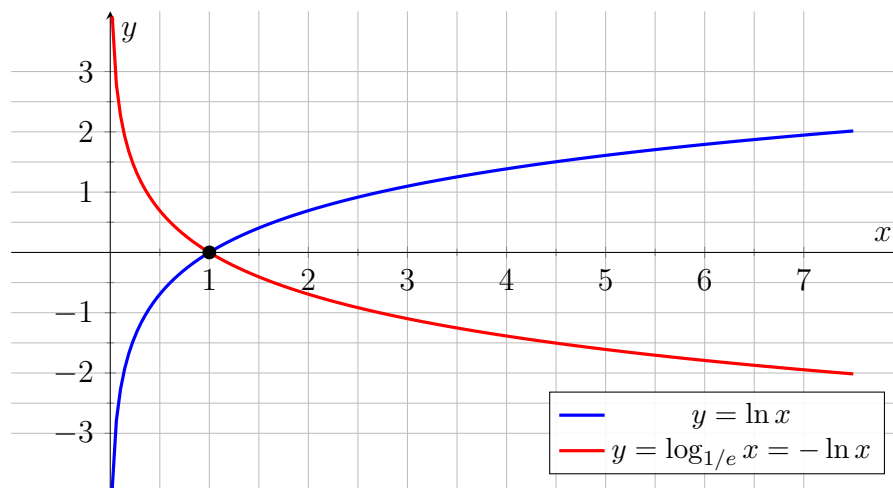


Figure 2.17: Increasing and decreasing logarithmic functions.

2.9.3 The inverse relationship with exponential functions

Since $y = \log_b x$ is equivalent to $x = b^y$, the functions

$$f(x) = b^x \quad \text{and} \quad f^{-1}(x) = \log_b x$$

are inverses. Their graphs are reflections of one another in the line $y = x$. Their domains and ranges are interchanged:

$$\begin{aligned} D_{b^x} &= \mathbb{R}, & R_{b^x} &= (0, \infty), \\ D_{\log_b x} &= (0, \infty), & R_{\log_b x} &= \mathbb{R}. \end{aligned}$$

This inverse relationship gives the identities

$$\log_b(b^x) = x \quad \text{and} \quad b^{\log_b x} = x, \quad x > 0. \quad (2.47)$$

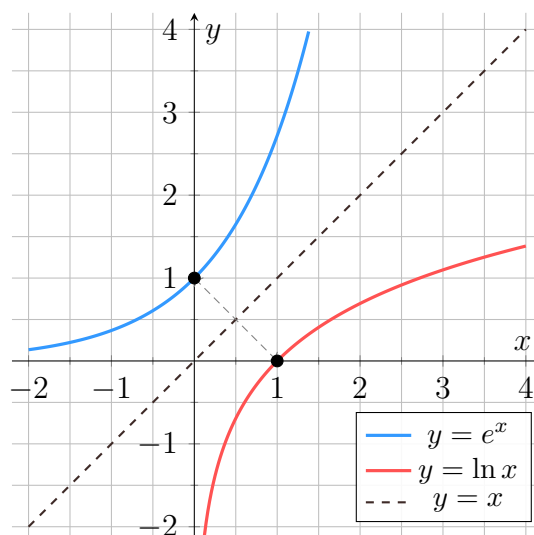


Figure 2.18: The graphs of $y = e^x$ and $y = \ln x$ are mirror images in the line $y = x$; corresponding points have their coordinates interchanged.

2.9.4 Transformations of the logarithmic graph

A transformed natural-logarithm function may be written as

$$g(x) = a \ln(x - h) + k, \quad a \neq 0. \quad (2.48)$$

Starting from $y = \ln x$:

- h translates the graph horizontally;
- k translates the graph vertically;
- $|a|$ produces a vertical stretch or compression; and
- $a < 0$ reflects the graph in the x -axis.

The logarithm requires $x - h > 0$, so

$$D_g = (h, \infty), \quad R_g = \mathbb{R},$$

and the vertical asymptote is $x = h$. A vertical translation changes neither the domain nor the vertical asymptote.

Example 2.25: Translation to the right and downward

Consider

$$g(x) = \ln(x - 1) - 2.$$

The graph of $y = \ln x$ is translated one unit to the right and two units downward. Therefore,

$$D_g = (1, \infty), \quad R_g = \mathbb{R},$$

and its vertical asymptote is $x = 1$. To find the x -intercept,

$$\begin{aligned}\ln(x - 1) - 2 &= 0, \\ \ln(x - 1) &= 2, \\ x - 1 &= e^2, \\ x &= 1 + e^2.\end{aligned}$$

Thus, the x -intercept is $(1 + e^2, 0)$. There is no y -intercept because $x = 0$ is outside the domain.

Example 2.26: Translation to the left and upward

Consider

$$h(x) = \ln(x + 2) + 3.$$

The graph is translated two units to the left and three units upward. Hence,

$$D_h = (-2, \infty), \quad R_h = \mathbb{R},$$

and its vertical asymptote is $x = -2$. Its intercepts are found from

$$\begin{aligned}h(0) &= \ln 2 + 3, \\ \ln(x + 2) + 3 &= 0 \implies x = e^{-3} - 2.\end{aligned}$$

Therefore, the y -intercept is $(0, \ln 2 + 3)$ and the x -intercept is $(e^{-3} - 2, 0)$.

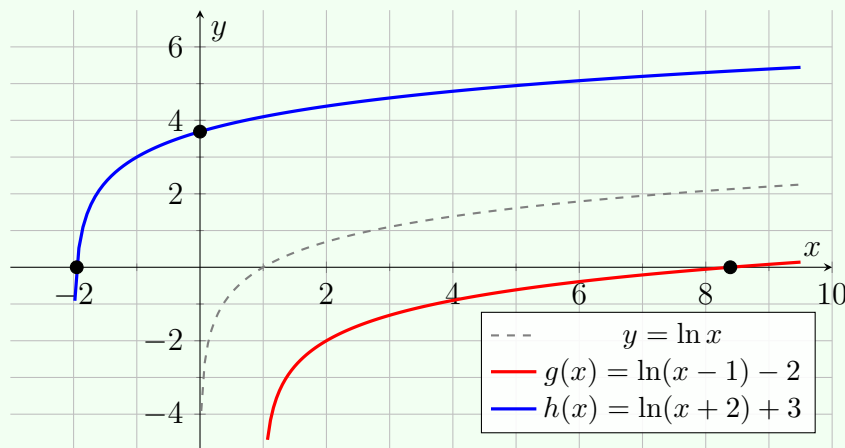


Figure 2.19: Translations of the basic natural-logarithm graph.

If the argument contains a coefficient, such as

$$q(x) = \ln(c(x - h)) + k,$$

the complete argument must be positive:

$$c(x - h) > 0.$$

When $c < 0$, the allowed inputs lie to the left of $x = h$, and the graph is reflected horizontally.

2.9.5 Laws of logarithms

For $M > 0$, $N > 0$, and any real number r ,

$$\log_b(MN) = \log_b M + \log_b N, \quad (2.49)$$

$$\log_b\left(\frac{M}{N}\right) = \log_b M - \log_b N, \quad (2.50)$$

$$\log_b(M^r) = r \log_b M. \quad (2.51)$$

These laws convert multiplication into addition, division into subtraction, and powers into coefficients. They do not apply to a sum or difference; in general,

$$\log_b(M + N) \neq \log_b M + \log_b N.$$

Example 2.27: Expanding a logarithm

For $x > 0$ and $y > 0$,

$$\begin{aligned} \ln\left(\frac{x^3\sqrt{y}}{5}\right) &= \ln(x^3) + \ln(y^{1/2}) - \ln 5 \\ &= 3 \ln x + \frac{1}{2} \ln y - \ln 5. \end{aligned}$$

Example 2.28: Condensing logarithms

Provided all logarithmic arguments are positive,

$$\begin{aligned} 2 \ln x - \ln(x - 1) &= \ln(x^2) - \ln(x - 1) \\ &= \ln\left(\frac{x^2}{x - 1}\right). \end{aligned}$$

The original expression requires both $x > 0$ and $x - 1 > 0$, so its domain is $x > 1$.

2.9.6 Change of base

Calculators usually provide keys for $\log$ and $\ln$. A logarithm with any valid base may be evaluated using

$$\log_b x = \frac{\log x}{\log b} = \frac{\ln x}{\ln b}. \quad (2.52)$$

For example,

$$\log_2 7 = \frac{\ln 7}{\ln 2} \approx 2.807.$$

2.9.7 Determining the domain

Every logarithmic argument must be strictly positive. For example, the function

$$f(x) = \ln(3x - 6)$$

requires

$$3x - 6 > 0,$$

so $D_f = (2, \infty)$.

When several logarithms occur, every condition must hold simultaneously. For

$$p(x) = \ln(x + 1) + \ln(4 - x),$$

the restrictions are

$$x + 1 > 0 \quad \text{and} \quad 4 - x > 0.$$

Therefore,

$$D_p = (-1, 4).$$

2.9.8 Solving logarithmic equations

The main methods are to rewrite the equation in exponential form or to use the logarithm laws to combine terms. Every proposed solution must satisfy the original positive-argument restrictions.

Example 2.29: Converting to exponential form

Solve

$$\log_3(x - 1) = 2.$$

Using the definition of a logarithm,

$$x - 1 = 3^2 = 9,$$

so

$$\boxed{x = 10}.$$

This value is valid because $x - 1 > 0$.

Example 2.30: Combining two logarithms

Solve

$$\ln x + \ln(x - 3) = \ln 4.$$

The original equation requires $x > 3$. Combining the logarithms gives

$$\begin{aligned}\ln(x(x-3)) &= \ln 4, \\ x(x-3) &= 4, \\ x^2 - 3x - 4 &= 0, \\ (x-4)(x+1) &= 0.\end{aligned}$$

The algebraic candidates are $x = 4$ and $x = -1$, but only $x = 4$ satisfies $x > 3$. Hence,

$$\boxed{x = 4}.$$

2.9.9 Logarithmic inequalities

The direction of an inequality depends on the base. If $u > 0$ and $v > 0$, then

$$\begin{aligned}\log_b u < \log_b v &\iff u < v, & b > 1, \\ \log_b u < \log_b v &\iff u > v, & 0 < b < 1.\end{aligned}$$

For example,

$$\log_2(x-1) > 3$$

requires $x > 1$. Since the base 2 is greater than 1,

$$\begin{aligned}x-1 &> 2^3, \\ x &> 9.\end{aligned}$$

2.10 Radical functions

A radical function contains the independent variable inside a root. The square-root function is the most common example at SPM and pre-university level, although cube-root and higher-root functions follow the same general principles.

Definition 2.10: Radical function.

A radical function is a function that can be written in a form involving

$$f(x) = a\sqrt[n]{x-h} + k, \quad n \geq 2, \quad (2.53)$$

where n is an integer and the expression under the root determines which real inputs are permitted.

The parity of the index n is important:

- if n is even, the radicand must be non-negative for the output to be real;
- if n is odd, every real radicand is allowed.

2.10.1 The basic square-root function

The basic square-root function is

$$f(x) = \sqrt{x}. \quad (2.54)$$

Because the square root of a negative real number is not real, x must satisfy $x \geq 0$. Furthermore, the principal square root is non-negative. Hence,

$$D_f = [0, \infty), \quad R_f = [0, \infty).$$

The graph begins at the endpoint $(0, 0)$, increases as x increases, and gradually becomes less steep. It passes through the convenient points

$$(0, 0), \quad (1, 1), \quad (4, 2), \quad (9, 3), \quad (16, 4).$$

These points are obtained by choosing perfect-square inputs.

2.10.2 The family $f(x) = a\sqrt{x}$

Consider

$$f(x) = a\sqrt{x}, \quad a \neq 0. \quad (2.55)$$

The coefficient a changes only the output values, so the domain remains $[0, \infty)$. Its effect is as follows:

- $|a| > 1$ produces a vertical stretch;
- $0 < |a| < 1$ produces a vertical compression;
- $a < 0$ reflects the graph in the x -axis.

The range is

$$R_f = \begin{cases} [0, \infty), & a > 0, \\ (-\infty, 0], & a < 0. \end{cases} \quad (2.56)$$

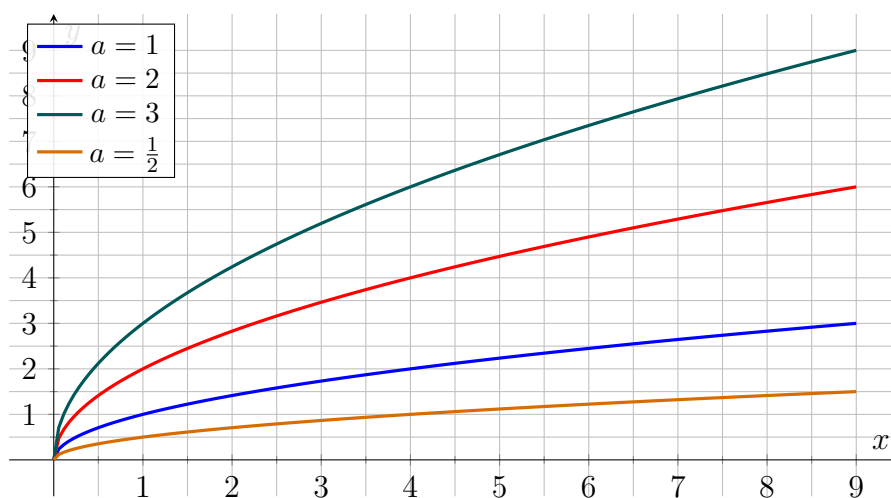


Figure 2.20: The family $f(x) = a\sqrt{x}$ for several positive values of a .

Example 2.31: Comparing vertical scales

The functions

$$\sqrt{x}, \quad 2\sqrt{x}, \quad 3\sqrt{x}, \quad \text{and} \quad \frac{1}{2}\sqrt{x}$$

all have domain $[0, \infty)$ and range $[0, \infty)$. They share the endpoint $(0, 0)$. The graph of $3\sqrt{x}$ is the greatest for every $x > 0$, while $\frac{1}{2}\sqrt{x}$ is the most vertically compressed.

2.10.3 Transformations of the square-root graph

A transformed square-root function is commonly written as

$$g(x) = a\sqrt{x - h} + k, \quad a \neq 0. \quad (2.57)$$

Starting from $y = \sqrt{x}$:

- h translates the graph horizontally, right when $h > 0$ and left when $h < 0$;
- k translates the graph vertically;
- $|a|$ produces a vertical stretch or compression; and
- $a < 0$ reflects the graph in the x -axis.

The endpoint is (h, k) . Since $x - h \geq 0$, the domain is

$$D_g = [h, \infty).$$

The range depends on the sign of a :

$$R_g = \begin{cases} [k, \infty), & a > 0, \\ (-\infty, k], & a < 0. \end{cases} \quad (2.58)$$

Thus, the endpoint gives the minimum value when $a > 0$ and the maximum value when $a < 0$.

Example 2.32: Translation of $\sqrt{x}$

Consider

$$g(x) = \sqrt{x + 2} + 3.$$

Since $x + 2 = x - (-2)$, the graph of $y = \sqrt{x}$ is translated two units to the left and three units upward. Its endpoint is $(-2, 3)$, and

$$D_g = [-2, \infty), \quad R_g = [3, \infty).$$

The graph has no x -intercept because its minimum value is 3. Its y -intercept is

$$g(0) = \sqrt{2} + 3.$$

Example 2.33: Stretching and translating

Consider

$$h(x) = 2\sqrt{x-2} - 2.$$

The basic graph is translated two units to the right, stretched vertically by a factor of 2, and translated two units downward. Its endpoint is $(2, -2)$, and

$$D_h = [2, \infty), \quad R_h = [-2, \infty).$$

To find the x -intercept,

$$\begin{aligned} 2\sqrt{x-2} - 2 &= 0, \\ \sqrt{x-2} &= 1, \\ x &= 3. \end{aligned}$$

Therefore, the x -intercept is $(3, 0)$. There is no y -intercept because 0 is not in the domain.

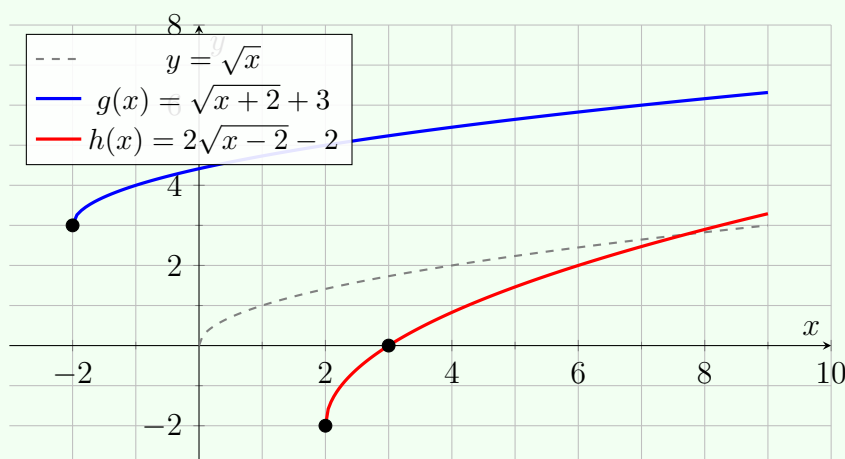


Figure 2.21: Two transformations of the basic square-root graph.

2.10.4 Determining the domain

For a square-root function, require the radicand to be non-negative. For example,

$$f(x) = \sqrt{5-2x}$$

is real when

$$\begin{aligned} 5 - 2x &\geq 0, \\ x &\leq \frac{5}{2}. \end{aligned}$$

Hence,

$$D_f = \left(-\infty, \frac{5}{2}\right], \quad R_f = [0, \infty).$$

The endpoint is $(\frac{5}{2}, 0)$ and the graph extends to the left.

When the radicand is nonlinear, solve the corresponding inequality.

Example 2.34: A quadratic radicand

Find the domain of

$$q(x) = \sqrt{x^2 - 5x + 6}.$$

The radicand must satisfy

$$\begin{aligned} x^2 - 5x + 6 &\geq 0, \\ (x - 2)(x - 3) &\geq 0. \end{aligned}$$

The product is non-negative outside the roots, so

$$D_q = (-\infty, 2] \cup [3, \infty).$$

If an even root occurs in a denominator, the radicand must be strictly positive. For instance,

$$r(x) = \frac{1}{\sqrt{x-1}}$$

has domain $(1, \infty)$, because $x = 1$ would make the denominator zero.

2.10.5 Square-root functions and inverse functions

The function $y = \sqrt{x}$ is the inverse of $y = x^2$ only when the quadratic is restricted to $x \geq 0$. Without this restriction, $y = x^2$ is not one-to-one and therefore has no inverse function on all of $\mathbb{R}$.

For

$$f(x) = a\sqrt{x-h} + k, \quad a \neq 0,$$

write

$$\begin{aligned} y - k &= a\sqrt{x-h}, \\ \left(\frac{y-k}{a}\right)^2 &= x-h. \end{aligned}$$

After interchanging x and y ,

$$f^{-1}(x) = \left(\frac{x-k}{a}\right)^2 + h, \tag{2.59}$$

with the domain of f^{-1} restricted to the range of f .

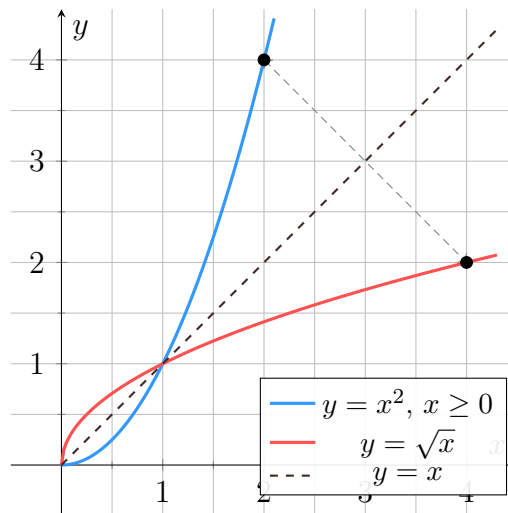


Figure 2.22: Restricting $y = x^2$ to $x \geq 0$ makes it one-to-one, and its inverse $y = \sqrt{x}$ is obtained by reflection in $y = x$.

2.10.6 Solving radical equations

To solve an equation containing a square root:

1. isolate the radical expression;
2. square both sides;
3. solve the resulting equation; and
4. check every candidate in the original equation.

The final check is essential because squaring may introduce extraneous solutions.

Example 2.35: Checking for an extraneous root

Solve

$$\sqrt{2x+3} = x.$$

The right-hand side must be non-negative, so $x \geq 0$. Squaring gives

$$\begin{aligned} 2x + 3 &= x^2, \\ x^2 - 2x - 3 &= 0, \\ (x - 3)(x + 1) &= 0. \end{aligned}$$

The candidates are $x = 3$ and $x = -1$. Only $x = 3$ satisfies both $x \geq 0$ and the original equation. Therefore,

$$\boxed{x = 3}.$$

2.10.7 Odd-root functions

The basic cube-root function

$$f(x) = \sqrt[3]{x}$$

accepts every real input and produces every real output. Thus,

$$D_f = \mathbb{R}, \quad R_f = \mathbb{R}.$$

It is an odd function and has rotational symmetry of 180° about the origin. More generally,

$$g(x) = a\sqrt[3]{x-h} + k$$

has centre of rotational symmetry (h, k) and retains domain and range $\mathbb{R}$. The same domain principle applies to every odd-index root.

2.10.8 Sketching a radical function

To sketch a square-root function $y = a\sqrt{x-h} + k$:

1. locate the endpoint (h, k) ;
2. determine the domain from the radicand;
3. use the sign and magnitude of a to determine orientation and vertical scale;
4. transform several convenient points from $y = \sqrt{x}$; and
5. state the domain, range, and intercepts.

2.11 Absolute value (modulus) functions

The absolute value, or modulus, of a real number measures its distance from zero on the real number line. Since distance is never negative, $|x|$ is always non-negative. Absolute-value functions are useful for representing distance, reflection, and rules that change at a particular input.

Definition 2.11: Absolute value.

For every real number x ,

$$|x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0. \end{cases} \quad (2.60)$$

Thus, $|x|$ is the distance between x and 0, while $|x - a|$ is the distance between x and a .

Some elementary properties are

$$\begin{aligned} |x| &\geq 0, & |-x| &= |x|, \\ |xy| &= |x||y|, & \left|\frac{x}{y}\right| &= \frac{|x|}{|y|}, \quad y \neq 0, \\ |x|^2 &= x^2. \end{aligned}$$

In general, $|x + y|$ is not equal to $|x| + |y|$. Instead, the triangle inequality states that

$$|x + y| \leq |x| + |y|.$$

2.11.1 The basic absolute-value function

The basic absolute-value function is

$$f(x) = |x|. \quad (2.61)$$

Its graph consists of two straight-line rays meeting at the origin. It is V-shaped, with vertex $(0, 0)$ and axis of symmetry $x = 0$. Its domain and range are

$$D_f = \mathbb{R}, \quad R_f = [0, \infty).$$

The function decreases on $(-\infty, 0]$ and increases on $[0, \infty)$. It is even because $f(-x) = |-x| = |x| = f(x)$, so its graph is symmetric about the y -axis.

2.11.2 The family $f(x) = a|x|$

For

$$f(x) = a|x|, \quad a \neq 0, \quad (2.62)$$

the coefficient a controls the scale and orientation of the graph:

- $|a| > 1$ gives a vertical stretch, so the graph appears narrower;
- $0 < |a| < 1$ gives a vertical compression, so the graph appears wider;
- $a > 0$ gives an upward-opening graph with minimum value 0; and
- $a < 0$ reflects the graph in the x -axis and gives maximum value 0.

Every graph in this family has domain $\mathbb{R}$, vertex $(0, 0)$, and axis of symmetry $x = 0$. Its range is

$$R_f = \begin{cases} [0, \infty), & a > 0, \\ (-\infty, 0], & a < 0. \end{cases} \quad (2.63)$$

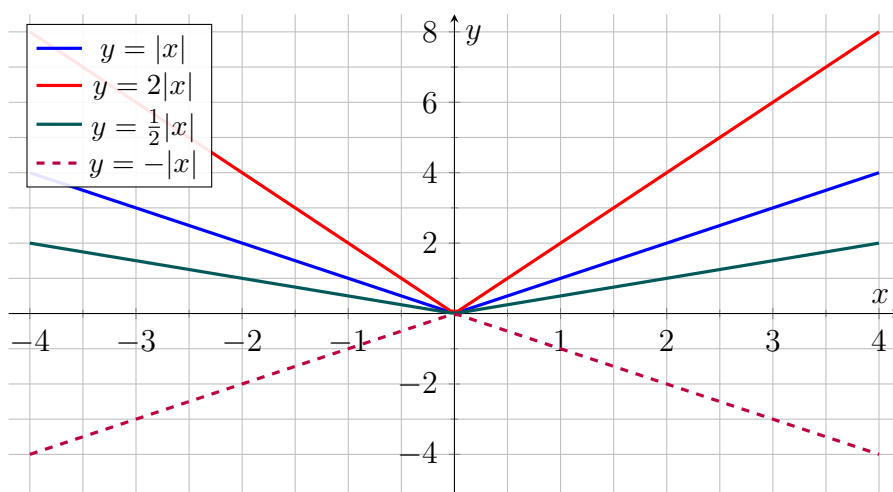


Figure 2.23: The graph of $y = a|x|$ for several values of a .

Example 2.36: Reflection and vertical scaling

For

$$f(x) = -\frac{3}{2}|x|,$$

the factor $\frac{3}{2}$ produces a vertical stretch and the negative sign reflects the graph in the x -axis. The graph opens downward, has vertex $(0, 0)$, domain $\mathbb{R}$, and range $(-\infty, 0]$. Its maximum value is 0.

2.11.3 Translations and the general form

A transformed absolute-value function is commonly written as

$$g(x) = a|x - h| + k, \quad a \neq 0. \quad (2.64)$$

Starting from $y = |x|$:

- h translates the graph horizontally, right when $h > 0$ and left when $h < 0$;
- k translates it vertically, upward when $k > 0$ and downward when $k < 0$;
- $|a|$ controls the vertical stretch or compression; and
- $a < 0$ reflects the graph in the x -axis.

The vertex is (h, k) and the axis of symmetry is $x = h$. The domain is $\mathbb{R}$, while

$$R_g = \begin{cases} [k, \infty), & a > 0, \\ (-\infty, k], & a < 0. \end{cases} \quad (2.65)$$

Thus, k is the minimum value when $a > 0$ and the maximum value when $a < 0$. Care is needed with horizontal translations: $|x + 3| = |x - (-3)|$ has its vertex at $x = -3$.

Example 2.37: Transforming $2|x|$

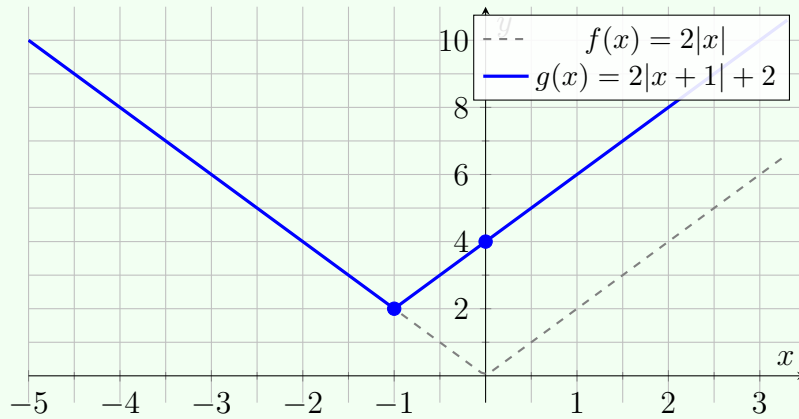
Consider

$$g(x) = 2|x + 1| + 2.$$

The graph of $y = 2|x|$ is translated one unit left and two units upward. Its vertex is $(-1, 2)$, its axis of symmetry is $x = -1$, and

$$D_g = \mathbb{R}, \quad R_g = [2, \infty).$$

Since $g(0) = 4$, the y -intercept is $(0, 4)$. The graph has no x -intercepts because its minimum value is 2.

Figure 2.24: Translation of $y = 2|x|$ to obtain $y = 2|x + 1| + 2$.

2.11.4 Piecewise form and gradients

The general form may be written without modulus notation as

$$a|x - h| + k = \begin{cases} -a(x - h) + k, & x < h, \\ a(x - h) + k, & x \geq h. \end{cases} \quad (2.66)$$

The gradient to the left of the vertex is $-a$, while the gradient to the right is a . The graph is continuous at $x = h$, but has a sharp corner there when $a \neq 0$.

For example,

$$2|x + 1| + 2 = \begin{cases} -2x, & x < -1, \\ 2x + 4, & x \geq -1. \end{cases}$$

2.11.5 Intercepts

For $g(x) = a|x - h| + k$, the y -intercept is

$$g(0) = a|h| + k.$$

The x -intercepts satisfy

$$|x - h| = -\frac{k}{a}.$$

Therefore, there are two x -intercepts when $-k/a > 0$, one when $-k/a = 0$, and none when $-k/a < 0$. When two exist, they are symmetric about $x = h$.

2.11.6 Finding an equation from the vertex

If the vertex (h, k) is known, begin with

$$f(x) = a|x - h| + k.$$

A second point determines a .

Example 2.38: Finding the equation

Suppose the vertex is $(2, -1)$ and the graph passes through $(5, 8)$. Then

$$f(x) = a|x - 2| - 1.$$

Substituting $(5, 8)$ gives

$$8 = a|5 - 2| - 1,$$

$$9 = 3a,$$

$$a = 3.$$

Hence,

$$f(x) = 3|x - 2| - 1.$$

2.11.7 The graphs of $y = |f(x)|$ and $y = f(|x|)$

These transformations have different effects:

- For $y = |f(x)|$, retain the portions where $f(x) \geq 0$ and reflect every part below the x -axis upward.
- For $y = f(|x|)$, retain the part of $y = f(x)$ for $x \geq 0$ and reflect it in the y -axis. The result is an even function.

For instance, if $f(x) = x^2 - 4$, then

$$|f(x)| = |x^2 - 4|$$

reflects the portion of the parabola below the x -axis upward. However,

$$f(|x|) = |x|^2 - 4 = x^2 - 4,$$

so the second transformation does not change this already-even function.

2.11.8 Equations and distance

For a real expression $u(x)$ and $c \geq 0$,

$$|u(x)| = c \iff u(x) = c \text{ or } u(x) = -c. \quad (2.67)$$

If $c < 0$, there is no real solution. The equation $|x - a| = d$, where $d \geq 0$, means that x is exactly d units from a , so

$$x = a - d \quad \text{or} \quad x = a + d.$$

Absolute-value inequalities are developed further in the chapter on inequalities.

2.11.9 Sketching an absolute-value graph

To sketch $y = a|x - h| + k$:

1. identify the vertex (h, k) and axis of symmetry $x = h$;
2. use the sign of a to determine whether the graph opens upward or downward;
3. use $|a|$ to determine the steepness of the two rays;
4. find the intercepts when required; and
5. state the domain and range.

2.12 Piecewise functions

A piecewise function uses different formulae on different parts of its domain. Each formula is accompanied by a condition that specifies the inputs for which that formula is used. Piecewise definitions are useful when a single algebraic expression cannot conveniently describe the entire function.

Definition 2.12: Piecewise function.

A function f is called a *piecewise function* if its rule is given by two or more expressions, each applying to a stated subset of the domain. A typical definition has the form

$$f(x) = \begin{cases} f_1(x), & x \in D_1, \\ f_2(x), & x \in D_2, \\ \vdots & \vdots \\ f_n(x), & x \in D_n. \end{cases}$$

The sets $D_1, D_2, \dots, D_n$ specify which rule is used. Together, they form the domain of f . If two conditions include the same input, their formulae must give the same output there; otherwise the definition does not describe a function.

For example, the modulus function may be written as

$$|x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0. \end{cases} \quad (2.68)$$

The two formulae meet at $x = 0$ and together cover every real input.

2.12.1 Selecting the correct rule

To evaluate a piecewise function, first determine which condition contains the input. Only then should the input be substituted into the corresponding formula. Particular care is required at boundary values such as $x = 1$ or $x = 2$.

Example 2.39: Evaluating at boundary values

Let

$$f(x) = \begin{cases} x + 4, & x < -1, \\ x^2, & -1 \leq x < 2, \\ 7 - x, & x \geq 2. \end{cases}$$

Find $f(-3)$, $f(-1)$, $f(2)$, and $f(4)$.

Solution. Since $-3 < -1$, the first rule applies:

$$f(-3) = -3 + 4 = 1.$$

The value -1 is included in the middle interval, so

$$f(-1) = (-1)^2 = 1.$$

The middle rule excludes $x = 2$, while the third rule includes it. Hence,

$$f(2) = 7 - 2 = 5, \quad f(4) = 7 - 4 = 3.$$

An inequality containing $<$ or $>$ produces an open endpoint, while an inequality containing $\leq$ or $\geq$ produces a closed endpoint. On a graph, open endpoints are marked with hollow circles and closed endpoints with solid circles.

2.12.2 Sketching a piecewise graph

A piecewise graph is drawn one part at a time. For each part:

1. identify the basic graph represented by the formula;
2. restrict it to the interval stated beside the formula;
3. mark each boundary point as open or closed; and
4. check whether neighbouring pieces meet, cross, or leave a jump.

Example 2.40: A constant–quadratic–constant function

Sketch

$$g(x) = \begin{cases} 4, & x < 0, \\ 4 + 3x - x^2, & 0 \leq x \leq 4, \\ 0, & x > 4. \end{cases}$$

The middle piece is a downward-opening parabola. Completing the square gives

$$4 + 3x - x^2 = \frac{25}{4} - \left(x - \frac{3}{2}\right)^2,$$

so its maximum occurs at $(\frac{3}{2}, \frac{25}{4})$. The three pieces agree at their junctions because

the quadratic gives $g(0) = 4$ and $g(4) = 0$.

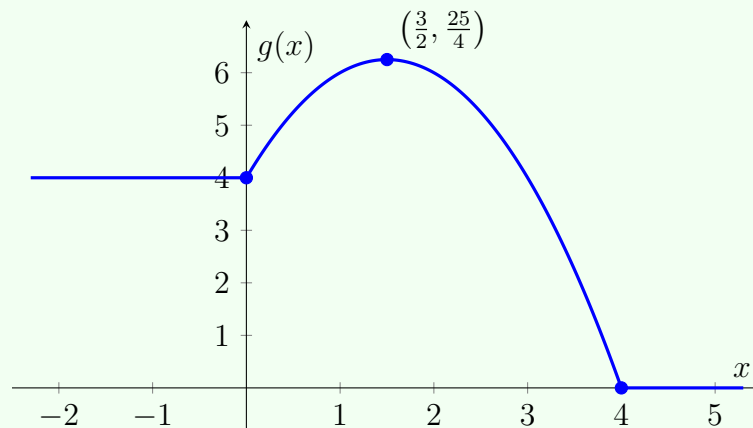


Figure 2.25: The graph of a function defined by three pieces.

The domain is $\mathbb{R}$ and the range is

$$\left[0, \frac{25}{4}\right].$$

2.12.3 Domain, range, and endpoint behaviour

The domain of a piecewise function is the union of the input intervals stated in its definition. Its range is the union of the output sets produced by the individual pieces. An endpoint excluded from one piece may still belong to the function if it is included by another piece.

Example 2.41: Domain and range from three pieces

Consider

$$h(x) = \begin{cases} x^2, & 0 \leq x < 1, \\ 2 - x, & 1 \leq x < 2, \\ x - 1, & 2 \leq x \leq 3. \end{cases}$$

The intervals combine to give

$$D_h = [0, 3].$$

For the first piece, $0 \leq h(x) < 1$. For the second piece, $0 < h(x) \leq 1$. For the third piece, $1 \leq h(x) \leq 2$. Taking the union,

$$R_h = [0, 2].$$

At $x = 1$, the left-hand values approach 1 and the second piece gives $h(1) = 1$, so the graph joins there. At $x = 2$, however, the second piece approaches 0 while the third piece gives $h(2) = 1$, producing a jump.

2.12.4 Continuity at a junction

Suppose the rule changes at $x = a$. The function is continuous at the junction if

$$\lim_{x \rightarrow a^-} f(x) = f(a) = \lim_{x \rightarrow a^+} f(x). \quad (2.69)$$

At pre-university level, this is often tested by substituting a into the two neighbouring formulae. A failure of the two sides to agree creates a jump discontinuity or a removable discontinuity, depending on the graph.

Example 2.42: Determining constants for continuity

The function

$$p(x) = \begin{cases} ax + 1, & x < 2, \\ x^2 + b, & x \geq 2 \end{cases}$$

is continuous at $x = 2$ and satisfies $p(2) = 7$. Find a and b .

Solution. Since the second rule includes $x = 2$,

$$\begin{aligned} p(2) &= 2^2 + b = 7, \\ b &= 3. \end{aligned}$$

For continuity, the left-hand expression must also approach 7:

$$\begin{aligned} 2a + 1 &= 7, \\ a &= 3. \end{aligned}$$

Thus $a = 3$ and $b = 3$.

2.12.5 Modulus functions as piecewise functions

For any expression $u(x)$,

$$|u(x)| = \begin{cases} u(x), & u(x) \geq 0, \\ -u(x), & u(x) < 0. \end{cases} \quad (2.70)$$

The boundary occurs where $u(x) = 0$. Rewriting a modulus expression in piecewise form is useful for sketching graphs and solving equations.

Example 2.43: Solving a modulus equation piecewise

Solve

$$|2x - 3| = x + 1.$$

The expression $2x - 3$ changes sign at $x = \frac{3}{2}$.

If $x < \frac{3}{2}$, then $|2x - 3| = 3 - 2x$, so

$$\begin{aligned} 3 - 2x &= x + 1, \\ x &= \frac{2}{3}, \end{aligned}$$

which satisfies $x < \frac{3}{2}$.

If $x \geq \frac{3}{2}$, then $|2x - 3| = 2x - 3$, so

$$\begin{aligned} 2x - 3 &= x + 1, \\ x &= 4, \end{aligned}$$

which satisfies $x \geq \frac{3}{2}$. Therefore,

$$x = \frac{2}{3} \text{ or } x = 4.$$

2.12.6 Constructing a piecewise model

Real-world charges and rates often change after a threshold. The condition beside each rule must reflect exactly when that rate applies.

Example 2.44: A parking-charge model

A car park charges RM 4 for up to two hours, RM 1.50 for each additional hour from two to six hours, and RM 3 for each hour beyond six hours. For a parking time of t hours, the charge may be modelled by

$$C(t) = \begin{cases} 4, & 0 < t \leq 2, \\ 4 + 1.5(t - 2), & 2 < t \leq 6, \\ 10 + 3(t - 6), & t > 6. \end{cases}$$

The formulae agree at the thresholds: the second expression gives RM 10 at $t = 6$, which is the starting value of the third expression.

For $t = 5$,

$$C(5) = 4 + 1.5(5 - 2) = \text{RM } 8.50.$$

Furthermore, $C(t) \leq 16$ throughout the first two intervals. In the final interval,

$$\begin{aligned} 10 + 3(t - 6) &\leq 16, \\ t &\leq 8. \end{aligned}$$

Hence a charge not exceeding RM 16 corresponds to $0 < t \leq 8$.

2.13 Algebraic operations on functions

Functions may be combined by applying ordinary arithmetic to their outputs. If the same input x is admitted by both f and g , then the values $f(x)$ and $g(x)$ may be added, subtracted, multiplied, or divided to form a new function. The domain of the new function must be determined together with its formula.

Definition 2.13: Algebraic operations on functions.

Let f and g be real-valued functions. Their sum, difference, product, and quotient are defined by

$$\begin{aligned}(f + g)(x) &= f(x) + g(x), \\(f - g)(x) &= f(x) - g(x), \\(fg)(x) &= f(x)g(x), \\ \left(\frac{f}{g}\right)(x) &= \frac{f(x)}{g(x)}, \quad g(x) \neq 0.\end{aligned}$$

If D_f and D_g denote the domains of f and g , respectively, then

$$\begin{aligned}D_{f+g} &= D_{f-g} = D_{fg} = D_f \cap D_g, \\ D_{f/g} &= \{x \in D_f \cap D_g \mid g(x) \neq 0\}.\end{aligned}$$

The notation fg means the product of the two functions; it does not mean composition. Thus,

$$(fg)(x) = f(x)g(x), \quad (f \circ g)(x) = f(g(x)).$$

These expressions are generally different.

2.13.1 Finding the formula of the new function

To combine two functions, substitute their formulae into the required operation and simplify carefully. Brackets are particularly important in a difference or quotient because the entire expression for $g(x)$ is being subtracted or used as the denominator.

Example 2.45: Operations on two polynomial functions

Let

$$f(x) = 2x - 3, \quad g(x) = x^2 + 1.$$

Find the four algebraic combinations of f and g .

Solution.

$$\begin{aligned}(f + g)(x) &= (2x - 3) + (x^2 + 1) = x^2 + 2x - 2, \\(f - g)(x) &= (2x - 3) - (x^2 + 1) = -x^2 + 2x - 4, \\(fg)(x) &= (2x - 3)(x^2 + 1) = 2x^3 - 3x^2 + 2x - 3, \\ \left(\frac{f}{g}\right)(x) &= \frac{2x - 3}{x^2 + 1}.\end{aligned}$$

Both original functions have domain $\mathbb{R}$. Moreover, $x^2 + 1 > 0$ for every real x , so every one of the four new functions also has domain $\mathbb{R}$.

The order matters for subtraction and division:

$$f - g = -(g - f), \quad \frac{f}{g} \neq \frac{g}{f}$$

in general. Addition and multiplication, however, are commutative whenever both sides are defined:

$$f + g = g + f, \quad fg = gf.$$

2.13.2 Determining the domain

At SPM and pre-university level, most domain restrictions arise from the following rules:

- a polynomial is defined for every real number;
- an even root requires its radicand to be non-negative;
- a denominator must be non-zero; and
- the argument of a logarithm must be positive.

For a sum, difference, or product, retain every restriction imposed by either original function. For a quotient, retain those restrictions and also remove every value that makes the divisor $g(x)$ equal to zero.

Example 2.46: A radical and a polynomial

Let

$$f(x) = \sqrt{x + 3} - 2, \quad g(x) = x^2 - 4.$$

Find $(f + g)(x)$, $(f - g)(x)$, $(fg)(x)$, and $\left(\frac{f}{g}\right)(x)$. State the domain of each function.

Solution. The square root requires $x + 3 \geq 0$, so

$$D_f = [-3, \infty), \quad D_g = \mathbb{R}.$$

Therefore,

$$\begin{aligned}(f + g)(x) &= \sqrt{x + 3} - 2 + x^2 - 4 = x^2 + \sqrt{x + 3} - 6, \\(f - g)(x) &= \sqrt{x + 3} - 2 - (x^2 - 4) = \sqrt{x + 3} - x^2 + 2, \\(fg)(x) &= (\sqrt{x + 3} - 2)(x^2 - 4), \\ \left(\frac{f}{g}\right)(x) &= \frac{\sqrt{x + 3} - 2}{x^2 - 4}.\end{aligned}$$

The first three functions have the common domain

$$[-3, \infty).$$

For the quotient, $x^2 - 4 \neq 0$, so $x \neq -2$ and $x \neq 2$. Consequently,

$$D_{f/g} = [-3, \infty) \setminus \{-2, 2\}.$$

2.13.3 Restrictions that remain after simplification

A simplified formula does not restore values excluded from the original definition. This point frequently appears in examination questions involving rational functions.

Example 2.47: A cancelled factor

Let

$$f(x) = x^2 - 9, \quad g(x) = x - 3.$$

Then

$$\begin{aligned}\left(\frac{f}{g}\right)(x) &= \frac{x^2 - 9}{x - 3} \\ &= \frac{(x - 3)(x + 3)}{x - 3} \\ &= x + 3, \quad x \neq 3.\end{aligned}$$

Although the simplified expression $x + 3$ is meaningful at $x = 3$, the original quotient is not. Hence,

$$D_{f/g} = \mathbb{R} \setminus \{3\}.$$

Graphically, the quotient has the same straight-line graph as $y = x + 3$ except for a hole at $(3, 6)$.

2.13.4 Evaluating algebraic combinations

If a belongs to the required domain, an operation may be evaluated either by first obtaining the new formula or by evaluating the original functions separately:

$$\begin{aligned}(f + g)(a) &= f(a) + g(a), \\ (fg)(a) &= f(a)g(a), \\ \left(\frac{f}{g}\right)(a) &= \frac{f(a)}{g(a)}, \quad g(a) \neq 0.\end{aligned}$$

Example 2.48: Evaluation and an excluded input

Suppose

$$f(x) = x + 1, \quad g(x) = x - 2.$$

At $x = 4$,

$$\begin{aligned}(f + g)(4) &= f(4) + g(4) = 5 + 2 = 7, \\ (fg)(4) &= f(4)g(4) = 5(2) = 10, \\ \left(\frac{f}{g}\right)(4) &= \frac{5}{2}.\end{aligned}$$

However, $\left(\frac{f}{g}\right)(2)$ is undefined because $g(2) = 0$.

2.13.5 Scalar multiples and powers

For a real constant k , the scalar multiple kf is defined by

$$(kf)(x) = kf(x).$$

For a positive integer n , the pointwise power of a function is

$$[f(x)]^n = f(x)f(x) \cdots f(x).$$

To avoid ambiguity, this book writes $[f(x)]^2$ for the pointwise square and $f^{\circ 2}(x) = f(f(x))$ for the second iterate. Some sources use $f^2(x)$ for iteration, so their intended meaning must be read from context.

2.13.6 Useful algebraic laws

Where all expressions are defined, functions obey the familiar algebraic laws

$$\begin{aligned} f + (g + h) &= (f + g) + h, \\ f \cdot (g \cdot h) &= (f \cdot g) \cdot h, \\ f \cdot (g + h) &= f \cdot g + f \cdot h, \\ (f + g) - (f + h) &= g - h. \end{aligned}$$

These identities may help simplify expressions, but they do not remove the need to state the domain.

2.14 Composite functions

A composite function is formed when the output of one function is used as the input of another. If a value x is first processed by g and the result $g(x)$ is then processed by f , the final output is $f(g(x))$.

Definition 2.14: Composite function.

Let $g: X \rightarrow Y$ and $f: Y \rightarrow Z$. The composite function $f \circ g: X \rightarrow Z$ is defined by

$$(f \circ g)(x) = f(g(x)). \quad (2.71)$$

It may be read as f composed with g , f after g , or f of g .

The notation is read from right to left: g acts first and f acts second. The composition is defined only when x belongs to the domain of g and $g(x)$ belongs to the domain of f .

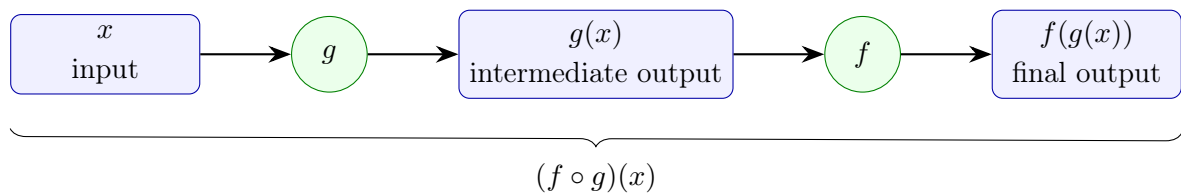


Figure 2.26: The order of operations in the composite function $f \circ g$.

2.14.1 Order of composition

Composition is generally *not commutative*. In other words,

$$f \circ g \neq g \circ f$$

in most cases, because changing the order changes which function acts on the input first.

Example 2.49: Why order matters

Let

$$f(x) = x + 2, \quad g(x) = x^2.$$

Then

$$\begin{aligned}(f \circ g)(x) &= f(x^2) = x^2 + 2, \\ (g \circ f)(x) &= g(x + 2) = (x + 2)^2.\end{aligned}$$

Therefore, $(f \circ g)(x) \neq (g \circ f)(x)$. The first composition squares the input before adding 2, while the second adds 2 before squaring.

2.14.2 Domain and range of a composite function

The domain of $f \circ g$ is not always simply the domain of g . The output from g must also be an acceptable input for f . Formally,

$$D_{f \circ g} = \{x \in D_g \mid g(x) \in D_f\}. \quad (2.72)$$

The range satisfies

$$R_{f \circ g} = \{f(g(x)) \mid x \in D_{f \circ g}\} \subseteq R_f.$$

To determine a composite domain:

1. begin with the restrictions required by the inner function;
2. require the output of the inner function to lie in the domain of the outer function; and
3. retain these restrictions even if the composite expression later simplifies.

Example 2.50: A radical restriction

Let

$$f(x) = \sqrt{x}, \quad g(x) = 4 - x^2.$$

For $f \circ g$,

$$(f \circ g)(x) = \sqrt{4 - x^2}.$$

The radicand must be non-negative:

$$\begin{aligned}4 - x^2 &\geq 0, \\ x^2 &\leq 4.\end{aligned}$$

Hence,

$$D_{f \circ g} = [-2, 2], \quad R_{f \circ g} = [0, 2].$$

In the reverse order,

$$\begin{aligned}(g \circ f)(x) &= g(\sqrt{x}) \\ &= 4 - (\sqrt{x})^2 \\ &= 4 - x.\end{aligned}$$

Although the final expression is linear, the original requirement $x \geq 0$ remains. Therefore,

$$D_{g \circ f} = [0, \infty), \quad R_{g \circ f} = (-\infty, 4].$$

Example 2.51: A denominator restriction

Let

$$f(x) = \frac{1}{x}, \quad g(x) = x - 2.$$

Then

$$\begin{aligned}(f \circ g)(x) &= \frac{1}{x-2}, & D_{f \circ g} &= \mathbb{R} \setminus \{2\}, \\ (g \circ f)(x) &= \frac{1}{x} - 2, & D_{g \circ f} &= \mathbb{R} \setminus \{0\}.\end{aligned}$$

The domains differ because the inner function is different in each order.

2.14.3 A complete polynomial example

Consider the following example.

Example 2.52: Comparing two compositions

Given

$$f(x) = 1 - x^2, \quad g(x) = 2x - 1,$$

find the domains and ranges of f and g , both compositions, their domains and ranges, and the values $(f \circ g)(3)$ and $(g \circ f)(-2)$.

Solution. Both functions are polynomials, so

$$D_f = D_g = \mathbb{R}.$$

The parabola $f(x) = 1 - x^2$ has maximum value 1, while the non-constant linear function g takes every real value. Thus,

$$R_f = (-\infty, 1], \quad R_g = \mathbb{R}.$$

First,

$$\begin{aligned}
 (f \circ g)(x) &= f(2x - 1) \\
 &= 1 - (2x - 1)^2 \\
 &= -4x^2 + 4x \\
 &= 1 - 4 \left(x - \frac{1}{2} \right)^2.
 \end{aligned}$$

Next,

$$\begin{aligned}
 (g \circ f)(x) &= g(1 - x^2) \\
 &= 2(1 - x^2) - 1 \\
 &= 1 - 2x^2.
 \end{aligned}$$

Therefore,

$$(f \circ g)(x) \neq (g \circ f)(x).$$

Both composites are quadratic functions defined for every real x , and both have maximum value 1. Hence,

$$\begin{aligned}
 D_{f \circ g} &= D_{g \circ f} = \mathbb{R}, \\
 R_{f \circ g} &= R_{g \circ f} = (-\infty, 1].
 \end{aligned}$$

Finally,

$$\begin{aligned}
 (f \circ g)(3) &= 1 - (2(3) - 1)^2 = -24, \\
 (g \circ f)(-2) &= 1 - 2(-2)^2 = -7.
 \end{aligned}$$

2.14.4 Evaluating a composition at a given input

A numerical value may be found by working from the inside outward or by first obtaining the composite formula.

Example 2.53: Evaluating from the inside outward

Let

$$f(x) = 3x - 1, \quad g(x) = x^2 + 2.$$

To find $(f \circ g)(-2)$, calculate

$$\begin{aligned}
 g(-2) &= (-2)^2 + 2 = 6, \\
 f(g(-2)) &= f(6) = 3(6) - 1 = 17.
 \end{aligned}$$

Thus,

$$(f \circ g)(-2) = 17.$$

2.14.5 Composition of three functions

For three functions f , g , and h ,

$$(f \circ g \circ h)(x) = f(g(h(x))). \quad (2.73)$$

The innermost function acts first. Function composition is associative:

$$f \circ (g \circ h) = (f \circ g) \circ h,$$

provided that all required domains are satisfied.

For example, if

$$f(x) = 2x, \quad g(x) = x + 1, \quad h(x) = x^2,$$

then

$$\begin{aligned} (f \circ g \circ h)(x) &= f(g(h(x))) \\ &= f(g(x^2)) \\ &= f(x^2 + 1) \\ &= 2x^2 + 2. \end{aligned}$$

2.14.6 Repeated or iterated functions

When a function is composed with itself, the notation

$$f^2 = f \circ f$$

is sometimes used. In this context, $f^2(x)$ means $f(f(x))$; it does *not* mean $[f(x)]^2$. Similarly,

$$f^3(x) = f(f(f(x))).$$

Example 2.54: Iterating a linear function

Let $f(x) = 2x - 1$. Then

$$\begin{aligned} f^2(x) &= f(2x - 1) = 2(2x - 1) - 1 = 4x - 3, \\ f^3(x) &= f(4x - 3) = 2(4x - 3) - 1 = 8x - 7. \end{aligned}$$

2.14.7 Finding an unknown component function

Sometimes a composite function and one of its components are known. The other component can then be found by substitution and algebra.

Example 2.55: Determining g from $f \circ g$

Suppose

$$f(x) = 2x + 3 \quad \text{and} \quad (f \circ g)(x) = 6x - 1.$$

Since $f(g(x)) = 2g(x) + 3$,

$$2g(x) + 3 = 6x - 1,$$

$$2g(x) = 6x - 4,$$

$$g(x) = 3x - 2.$$

Therefore,

$$\boxed{g(x) = 3x - 2}.$$

If f has an inverse and

$$f(g(x)) = h(x),$$

then applying f^{-1} gives

$$g(x) = f^{-1}(h(x)),$$

subject to the appropriate domain restrictions.

2.15 Inverse functions

An inverse function reverses the action of the original function. This is possible only when each output of the original function comes from at most one input; hence the original function must be one-to-one. If necessary, its codomain may be restricted to its range.

Definition 2.15: Inverse function.

Let $f: D_f \rightarrow R_f$ be one-to-one. The *inverse function* $f^{-1}: R_f \rightarrow D_f$ is defined by

$$f^{-1}(y) = x \iff f(x) = y. \quad (2.74)$$

Consequently,

$$D_{f^{-1}} = R_f, \quad R_{f^{-1}} = D_f. \quad (2.75)$$

The notation f^{-1} denotes an inverse function, not a reciprocal:

$$f^{-1}(x) \neq \frac{1}{f(x)}$$

in general. The reciprocal $1/f$ performs division, whereas f^{-1} reverses the input–output correspondence.

The graph of $y = f^{-1}(x)$ is the reflection of the graph of $y = f(x)$ in the line $y = x$, because each ordered pair (x, y) is replaced by (y, x) .

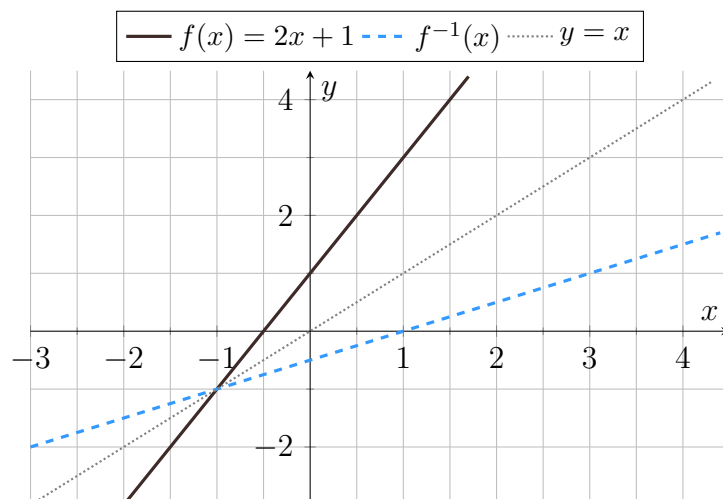


Figure 2.27: The graphs of a one-to-one function and its inverse are reflections in the line $y = x$.

To find an inverse algebraically:

1. write $y = f(x)$;
2. solve the equation for x in terms of y ;
3. interchange the variable names x and y ; and
4. state the domain and range of both functions.

The horizontal line test should be applied before this procedure. If the function is not one-to-one, its domain must be restricted appropriately.

Example 2.56: Finding and verifying a linear inverse

Let $f(x) = 3x - 5$. Starting with $y = 3x - 5$ gives

$$\begin{aligned} y + 5 &= 3x, \\ x &= \frac{y + 5}{3}. \end{aligned}$$

Therefore,

$$f^{-1}(x) = \frac{x + 5}{3}.$$

Both functions have domain and range $\mathbb{R}$. Moreover,

$$\begin{aligned} (f^{-1} \circ f)(x) &= \frac{(3x - 5) + 5}{3} = x, \\ (f \circ f^{-1})(x) &= 3 \left(\frac{x + 5}{3} \right) - 5 = x. \end{aligned}$$

Example 2.57: Restricting a quadratic before inversion

Consider

$$f(x) = (x - 2)^2 + 1, \quad x \geq 2.$$

The restriction $x \geq 2$ selects the increasing half of the parabola, so f is one-to-one.

From $y = (x - 2)^2 + 1$,

$$(x - 2)^2 = y - 1,$$

$$x - 2 = \sqrt{y - 1},$$

$$x = 2 + \sqrt{y - 1}.$$

The positive square root is selected because $x \geq 2$. Hence,

$$f^{-1}(x) = 2 + \sqrt{x - 1},$$

with

$$D_f = [2, \infty),$$

$$R_f = [1, \infty),$$

$$D_{f^{-1}} = [1, \infty),$$

$$R_{f^{-1}} = [2, \infty).$$

This extends the discussion in Section 2.10.5.

Example 2.58: Inverse of a linear-over-linear function

Let

$$f(x) = \frac{2x + 3}{x - 1}, \quad x \neq 1.$$

Solving $y = (2x + 3)/(x - 1)$ for x gives

$$y(x - 1) = 2x + 3,$$

$$x(y - 2) = y + 3,$$

$$x = \frac{y + 3}{y - 2}.$$

Therefore,

$$f^{-1}(x) = \frac{x + 3}{x - 2}, \quad x \neq 2.$$

The original function has domain $\mathbb{R} \setminus \{1\}$ and range $\mathbb{R} \setminus \{2\}$; these sets are interchanged for the inverse.

Finally, the inverse identities are

$$(f^{-1} \circ f)(x) = x, \quad x \in D_f, \quad (2.76)$$

$$(f \circ f^{-1})(x) = x, \quad x \in R_f. \quad (2.77)$$

The stated domains matter because the input of f^{-1} must belong to the range of f .

Chapter summary and common pitfalls

Key ideas. A function assigns exactly one output to every input in its domain. The domain is the input set, the codomain is the stated target set, and the range is the subset actually attained. Transformations move or rescale graphs, composition applies functions from the inside outward, and an inverse reverses a one-to-one function.

Common pitfalls. Do not confuse the codomain with the range. Horizontal factors act reciprocally, the order of composition matters, and f^{-1} denotes an inverse rather than $1/f$. Domains must be checked after algebraic operations and compositions, and a non-injective function requires a suitable restriction before it has an inverse function.

2.16 Practice and exercises

Convention. For a transformation written as $g(x) = af(b(x - h)) + k$, use the point mapping $(x, y) \mapsto (h + x/b, k + ay)$. Equivalently, apply horizontal and vertical dilations or reflections before the translations unless otherwise stated.

Linear, quadratic, cubic, and rational functions

1. State whether each statement is **TRUE** or **FALSE**.

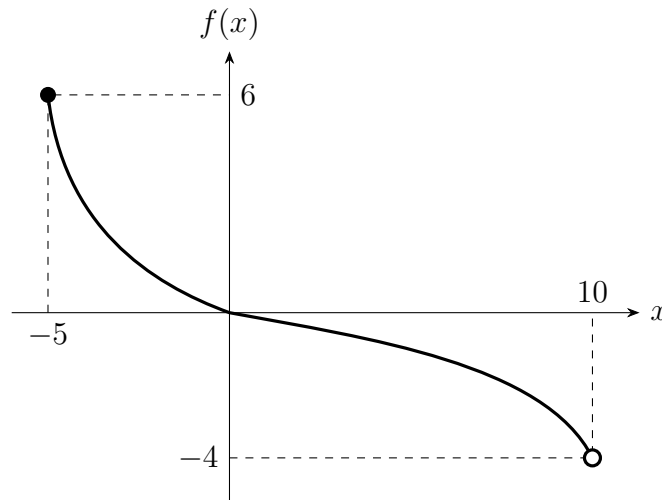
- i. The domain of a linear function in general is always all real numbers.
- ii. The graph of $f(x) = (x - 2)^2 + 3$ is obtained from the basic quadratic graph $y = x^2$ by shifting two units to the left and three units upward.
- iii. The range of the basic cubic function $f(x) = x^3$ is all real numbers.
- iv. The rational function $f(x) = \frac{1}{x}$ has domain $\mathbb{R}$.
- v. Translating the graph of $f(x) = x$ vertically by four units does not change its domain or range.

2. Fill in each blank.

- i. The domain of $f(x) = x^2$ is the set of all _____ numbers.
- ii. The graph of $f(x) = (x + 3)^2 - 1$ is obtained from $y = x^2$ by shifting _____ units to the left and _____ unit(s) downward.
- iii. The range of the basic quadratic function $f(x) = x^2$ is $f(x) \geq$ _____.
- iv. The function $f(x) = \frac{1}{x}$ has a vertical asymptote at _____.
- v. Translating $f(x) = x$ two units downward gives the new function

$$g(x) = \underline{\hspace{2cm}}.$$

3. The graph of $y = f(x)$ is shown below.



- a) State the domain and range of $y = f(x)$.
- b) Sketch each graph and state its domain and range.
 - i) $g(x) = -f(x)$
 - ii) $h(x) = -f(-x)$
 - iii) $k(x) = f(3 - x)$
 - iv) $p(x) = -f(-x) - 2$
 - v) $q(x) = -f(x - 2) + 3$
 - vi) $r(x) = -f(1 - x) - 2$

4. The graph of

$$g(x) = -\frac{1}{3}x + 2$$

can be obtained from the graph of $f(x) = \frac{1}{3}x$ using infinitely many combinations of reflections and translations.

- a) List five different combinations of transformations that produce the graph of g from the graph of f .
 - b) Plot the five transformed graphs and $g(x)$ using suitable graphing software. State your observation and conclusion.
5. Find $g(x)$ in terms of x for each transformation.
- a) $f(x) = 3x$ is reflected in one coordinate axis, shifted two units to the right, and then three units downward.
 - b) $f(x) = \frac{1}{2}x^2$ is reflected in the x -axis, shifted one unit to the left, and then two units upward.

- c) $f(x) = 3x^2$ is shifted two units to the right and then one unit downward.
- d) $f(x) = \frac{1}{2}x^3$ is shifted three units to the left and then two units upward.
- e) $f(x) = 2x^3$ is shifted one unit to the right and then three units downward.
- f) $f(x) = \frac{2}{x}$ is shifted two units to the right and then one unit downward.
- g) $f(x) = \frac{3}{x}$ is reflected in one coordinate axis, shifted two units to the left, and then two units downward.

Plot $f(x)$ and $g(x)$ for parts (a), (b), and (g). Then summarise how each of the following is represented algebraically:

- a) reflection in one axis, the x -axis, the y -axis, and both axes;
- b) translation to the left, right, upward, and downward.

6. Explain how the graph of g is obtained from the graph of f in each case.

- a) $f(x) = \frac{1}{2}x^2$, $g(x) = \frac{1}{2}x^2 - 2x - 1$
- b) $f(x) = x^2$, $g(x) = 4x - x^2 - 2$
- c) $f(x) = \frac{3}{x}$, $g(x) = \frac{3}{x+3} - 2$
- d) $f(x) = \frac{1}{2x}$, $g(x) = \frac{4x-13}{2x-6}$

7. Sketch each graph using transformations. Indicate all x -intercepts and y -intercepts that exist, and state the domain and range.

- a) $g(x) = -3x + 1$
- b) $h(x) = 3 + 2x - x^2$
- c) $p(x) = 2x^2 - 4x + 3$
- d) $p(x) = (x - 1)^3 + 2$
- e) $k(x) = \frac{2x - 4}{x - 3}$
- f) $p(x) = \frac{3x + 5}{x + 2}$
- g) For $g(x) = \frac{1 - x}{1 + 2x}$, suppose the horizontal asymptote is the constant function $f(x) = A$. Determine A and verify it graphically.

Extension. Using suitable graphing software, plot $f(x) = e^x$, $g(x) = \ln x$, and $y = x$ on the same axes.

- a) Describe the relationship between the graphs of e^x and $\ln x$.

- b) State the domain and range of each function.
- c) Plot $h(x) = \sqrt{x}$ and $k(x) = |x|$. Explain why the domain and range of h are non-negative and why the range of k is non-negative.

Exponential, logarithmic, radical, and modulus functions

8. State whether each statement is **TRUE** or **FALSE**.

- i. The domain of $f(x) = e^x$ is all real numbers.
- ii. The range of $f(x) = \ln x$ is $x > 0$.
- iii. The graph of $f(x) = |x|$ is symmetric about the y -axis.
- iv. The domain of $h(x) = \sqrt{x-3}$ is $x > 3$.
- v. Multiplying $f(x) = \sqrt{x}$ by -1 reflects its graph in the x -axis.
- vi. The graph of $g(x) = e^{x+2}$ is the graph of $f(x) = e^x$ shifted two units upward.
- vii. The range of $p(x) = |x-5|$ is $[0, \infty)$.
- viii. The domain of $k(x) = \ln(x+4)$ is $x > 4$.

9. Fill in each blank.

- i. The domain of $r(x) = \ln(x-7)$ is x _____.
- ii. The range of $t(x) = e^{-x}$ is _____.
- iii. The graph of $c(x) = |x|+4$ is obtained from $f(x) = |x|$ by the following translation:
_____.

iv. For $j(x) = \sqrt{9-x}$, complete

$$D_j = \text{_____}, \quad R_j = \text{_____}.$$

v. The graph of $p(x) = e^x - 5$ is obtained from $f(x) = e^x$ by the following translation:
_____.

vi. The function $g(x) = -\sqrt{x+2}$ is obtained from $f(x) = \sqrt{x}$ by reflecting in _____ and shifting _____.

vii. The graphs of $p(x) = e^x$ and $q(x) = \ln x$ are reflections of one another in the line _____.

10. Each statement below is false. Explain why and give the correct statement.

- i. The domain of $f(x) = \ln x$ is $x \in \mathbb{R}$.

- ii. The graph of $g(x) = e^x$ touches the x -axis.
- iii. The graph of $k(x) = |x - 2|$ has vertex $(0, 2)$.
- iv. The function $r(x) = \sqrt{x - 5}$ is defined for $x > 5$.
- v. The functions $p(x) = \ln x$ and $q(x) = e^x$ have the same domain and range.

11. Given

$$f(x) = 2\sqrt{2}, \quad g(x) = x\sqrt{2}, \quad h(x) = 2\sqrt{x},$$

state the domain and range of each function. Sketch all three graphs on the same coordinate system, indicate their intercepts, and determine the common point of intersection.

12. Find $g(x)$ in terms of x after performing the stated transformations.

- a) $f(x) = \ln x$ is shifted three units left and four units downward.
- b) $f(x) = \ln x$ is reflected in both axes, shifted two units right, and three units upward.
- c) $f(x) = e^x$ is shifted three units left and four units upward.
- d) $f(x) = e^{2x}$ is reflected in both axes, shifted one unit right, and two units downward.
- e) $f(x) = \sqrt{x}$ is reflected in both axes, shifted two units left, and three units upward.
- f) $f(x) = \sqrt{5}\sqrt{x}$ is reflected in the y -axis, shifted three units right, and one unit downward.
- g) $f(x) = 2|x|$ is shifted four units right and two units downward.
- h) $f(x) = 3|x|$ is reflected in the x -axis, shifted two units left, and four units upward.

13. Explain how the graph of g is obtained from the graph of f .

- a) $f(x) = \ln x, \quad g(x) = -\ln(1 - x) + 2$
- b) $f(x) = e^x, \quad g(x) = -e^{x+2} - 3$
- c) $f(x) = 3\sqrt{x}, \quad g(x) = -3 + \sqrt{9 - 9x}$
- d) $f(x) = 2|x|, \quad g(x) = |3 - 2x| + 3$

14. Sketch each graph using transformations. Indicate every x -intercept and y -intercept that exists, and state the domain and range.

- a) $s(x) = -2 - \ln(3 - x)$
- b) $t(x) = \ln(-2 - x) + 4$

c) $r(x) = 2 - e^{-x+2}$

d) $p(x) = -\sqrt{3-x} - 2$

e) $h(x) = -|2-4x| + 2$

f) $w(x) = -\ln(-2x-5) + 3$

g) $q(x) = 2 - e^{-3x+2}$

h) $r(x) = \sqrt{2x+4} - 2$

i) $g(x) = -5|x+3| - 5$

Piecewise functions, operations, and compositions

15. Sketch each function and determine its range.

a) $f(x) = \frac{|9x|}{5x}$

b) $g(x) = |2x+3| - 3x$

c)

$$k(x) = \begin{cases} \frac{2}{x-2} + 2, & x < 2, \\ 2(x-2)^2, & x \geq 2 \end{cases}$$

d)

$$r(x) = \begin{cases} 6-x, & x < 3, \\ \sqrt{x-2}, & 3 \leq x < 6, \\ 5-2x, & x > 6 \end{cases}$$

e)

$$s(x) = \begin{cases} -5, & x \geq 2, \\ 2|x-1| - 3, & -2 < x < 2, \\ \sqrt{2-x}, & x \leq -2 \end{cases}$$

f)

$$q(x) = \begin{cases} \sqrt{1-2x}, & x < -4, \\ 2, & -4 \leq x < 3, \\ \frac{3}{x} - 2, & x \geq 3 \end{cases}$$

16. Given

$$f(x) = \begin{cases} -\sqrt{2-2x}, & x < -1, \\ 4x+3, & x \geq -1 \end{cases} \quad g(x) = \begin{cases} x^2+1, & x \geq 2, \\ -4, & x < 2, \end{cases}$$

obtain a piecewise expression for $(f+g)(x)$. Sketch its graph and determine its range.

17. Given

$$p(x) = \begin{cases} 2x - 1, & x \leq -3, \\ x^2 + 3, & x > -3 \end{cases} \quad q(x) = \begin{cases} x^2 + 5, & x > 1, \\ 3, & x \leq 1, \end{cases}$$

obtain a piecewise expression for $(q - p)(x)$. Sketch its graph and determine its range.

18. Find $(f + g)(x)$, $(f - g)(x)$, $(fg)(x)$, and $\left(\frac{f}{g}\right)(x)$. State the domain of each resulting function.

a) $f(x) = \sqrt{2x + 3}$, $g(x) = x^2 - 2$

b) $f(x) = \ln(x + 5) + 2$, $g(x) = 2|x - 3| - 4$

19. State the domain of each function.

a) $f(x) = 2x - \sqrt{4x - 3}$

b) $h(x) = \frac{x^2 - 4}{|x - 2| - 3}$

c) $g(x) = \frac{2x + 3}{\sqrt{3x + 2}}$

d) $k(x) = \frac{\sqrt{3x + 10}}{2x + 3} + \frac{2x - 1}{(x - 3)^2 - 1}$

20. Find $(f \circ g)(x)$, $(g \circ f)(x)$, $(f \circ g)(-3)$, and $(g \circ f)(2)$ for each pair.

a) $f(x) = 2x - 3$, $g(x) = x^2 - 2x$

b) $f(x) = \frac{2}{1 - 2x}$, $g(x) = \sqrt{1 - 2x}$

c) $f(x) = |3 - x| + 2$, $g(x) = 3$

d) $f(x) = \ln(x + 3) - 2$, $g(x) = 3 - e^{x+2}$

Additional mixed practice

The questions in this section group closely related skills so that each number tests a complete idea rather than repeating the same instruction many times.

21. Decide whether each relation defines y as a function of x :

(a) $\{(-2, 1), (0, 3), (2, 1)\}$;

(b) $\{(1, 4), (1, -4), (3, 2)\}$;

(c) $x^2 + y^2 = 9$;

(d) $y = \sqrt{9 - x^2}$.

22. Find the domain of each function:

(a) $\frac{3x+1}{x^2-9}$;

(b) $\sqrt{5-2x}$;

(c) $\ln(x^2-4)$;

(d) $\frac{\sqrt{x+1}}{x-2}$.

23. State the range of each function:

(a) $(x-3)^2+2$;

(b) $-2(x+1)^2+5$;

(c) $\sqrt{x+4}-3$;

(d) $4-e^x$.

24. If $f(x) = 2x^2 - 3x + 1$, evaluate $f(-2)$, $f(a)$, $f(a+1)$, and $f(a+1) - f(a)$.

25. Describe fully the transformation from $y = f(x)$ to each graph:

(a) $y = f(x-4) + 3$;

(b) $y = -f(x+2)$;

(c) $y = f(-x) - 5$;

(d) $y = -f(3-x) + 1$.

26. Write the transformed equation when $f(x) = x^2$ is

(a) shifted right 3 and down 4;

(b) reflected in the x -axis and shifted left 2;

(c) stretched vertically by factor 3 and shifted up 1.

27. For each line, state its gradient and intercepts:

(a) $2x + 3y = 12$;

(b) $y = -4x + 7$;

(c) $x = 5$;

(d) $y = -3$.

28. Find the equation of the line

(a) through $(2, -1)$ with gradient 3;

(b) through $(-2, 5)$ and $(4, -1)$;

- (c) parallel to $2x - y = 6$ through $(1, 4)$;
- (d) perpendicular to $y = -\frac{1}{2}x + 3$ through the origin.
29. For each quadratic, state its vertex, axis of symmetry, and y -intercept:
- (a) $(x - 2)^2 - 5$;
- (b) $-3(x + 1)^2 + 4$;
- (c) $x^2 - 6x + 7$;
- (d) $2(x - 4)^2 + 1$;
- (e) $-\frac{1}{2}(x + 3)^2 + 6$;
- (f) $2x^2 + 8x - 3$.
30. Complete the square:
- (a) $x^2 + 8x + 3$;
- (b) $2x^2 - 12x + 5$;
- (c) $-x^2 + 4x + 1$;
- (d) $3x^2 + 18x - 7$;
- (e) $-2x^2 - 8x + 5$;
- (f) $\frac{1}{2}x^2 - 3x + 4$.
31. Solve over the real numbers and state the number of distinct roots:
- (a) $x^2 - 5x + 6 = 0$;
- (b) $4x^2 + 4x + 1 = 0$;
- (c) $x^2 + 2x + 5 = 0$;
- (d) $2x^2 - 7x + 3 = 0$;
- (e) $9x^2 - 12x + 4 = 0$;
- (f) $x^2 - 2x - 1 = 0$.
32. Find the quadratic with roots -2 and 5 and leading coefficient 2 . Hence find its y -intercept and vertex.
33. For $p(x) = x^3 - 4x$, find all intercepts, determine whether p is even, odd, or neither, and describe its end behaviour.
34. For each factored polynomial, state every zero and its multiplicity, and say whether the graph crosses or touches the axis:
- (a) $(x + 2)^2(x - 1)$;

(b) $-(x-3)^3(x+1)^2$.

35. Find the vertical and horizontal asymptotes and all intercepts:

(a) $\frac{2x+1}{x-3}$;

(b) $\frac{4}{x+2} - 1$;

(c) $\frac{x^2-1}{x-1}$, noting any removable discontinuity.

(d) $\frac{3x-6}{x+1}$;

(e) $\frac{6}{x-4} + 2$;

(f) $\frac{x}{x^2-4}$.

36. Solve:

(a) $\frac{1}{x-1} = 2$;

(b) $\frac{x+3}{x-2} = 4$;

(c) $\frac{2}{x} + \frac{1}{x-1} = 0$.

(d) $\frac{3}{x+2} = 1$;

(e) $\frac{2x-1}{x+1} = 3$;

(f) $\frac{1}{x-2} + \frac{1}{x+2} = 1$.

37. Evaluate 2^x at $x = -2, -1, 0, 1, 2$, and use the values to state its intercept, asymptote, domain, and range.

38. Expand or condense as directed:

(a) expand $\ln\left(\frac{x^2\sqrt{x+1}}{x-3}\right)$;

(b) condense $2\ln x - \frac{1}{2}\ln(x+4)$;

(c) express $\log_2 32 = 5$ in exponential form.

(d) expand $\log_3\left(\frac{9x^4}{\sqrt{y}}\right)$;

(e) condense $3\log_5 a + \log_5 b - 2\log_5 c$;

(f) express $10^{-3} = 0.001$ in logarithmic form.

39. Solve and check for extraneous roots:

- (a) $\sqrt{x+5} = x-1$;
- (b) $\sqrt{2x+3} = 5$;
- (c) $\sqrt{x} + \sqrt{x-5} = 5$.
- (d) $\sqrt{3x+1} = 4$;
- (e) $\sqrt{x+6} = x$;
- (f) $\sqrt{x+2} + \sqrt{x-1} = 3$.

40. Solve:

- (a) $|2x-3| = 7$;
- (b) $|x+1| < 4$;
- (c) $|3-x| \geq 5$.
- (d) $|3x+2| = 8$;
- (e) $|2x-1| \leq 5$;
- (f) $|x-4| > 3$.

Exponential, logarithmic, radical, and modulus review

41. For $g(x) = -2e^{x-1} + 3$, state the transformations from e^x , domain, range, horizontal asymptote, and y -intercept.

42. Solve:

- (a) $3^{x+1} = 81$;
- (b) $4^x = \frac{1}{8}$;
- (c) $5^{2x} = 7$ (give an exact logarithmic answer).
- (d) $2^{x-2} = 8$;
- (e) $9^{x+1} = 27$;
- (f) $e^{2x-1} = 6$ (give an exact answer).

43. Expand completely:

- (a) $\ln\left(\frac{x^3\sqrt{y}}{z^2}\right)$;
- (b) $\log_a\left(\frac{(u+1)^2}{\sqrt{v}}\right)$.
- (c) $\log_2\left(\frac{8x^2}{y}\right)$;

(d) $\ln\left(\frac{\sqrt[3]{p^2q}}{r^4}\right);$

(e) $\log_b\left(\frac{m}{n^2\sqrt{k}}\right).$

44. Condense to a single logarithm:

(a) $3\ln x + \ln y - 2\ln z;$

(b) $\frac{1}{2}\log_a p - \log_a q.$

(c) $2\ln u + \frac{1}{2}\ln v;$

(d) $\log_3 r + 2\log_3 s - \log_3 t;$

(e) $4\log_b x - \frac{1}{3}\log_b y.$

45. Solve, observing all domain restrictions:

(a) $\ln(x-1) = \ln 5;$

(b) $\log_2 x + \log_2(x-2) = 3;$

(c) $\ln x + \ln(x+3) = \ln 4.$

(d) $\log_3(x+1) = 2;$

(e) $\ln(x-2) + \ln(x+2) = \ln 5;$

(f) $\log_5 x - \log_5(x-4) = 1.$

46. Find the domain of

(a) $\ln(7-2x);$

(b) $\ln[(x-1)(x+4)];$

(c) $\frac{1}{\ln(x+2)}.$

(d) $\log_3(5-x);$

(e) $\ln(x^2-4);$

(f) $\frac{1}{\ln(2x-1)}.$

47. Describe the transformations, domain, range, and endpoint of $y = -3\sqrt{2-x} + 4$.

48. Solve and reject any extraneous solutions:

(a) $\sqrt{x+1} + 1 = x;$

(b) $\sqrt{3x-2} = x;$

(c) $\sqrt{x+6} = \sqrt{2x-1}.$

49. Write each function without modulus signs by using cases:

(a) $|x - 4|$;

(b) $2|3 - x| - 1$;

(c) $|x + 2| + |x - 1|$.

50. For $y = -2|x + 3| + 5$, state the vertex, axis of symmetry, intercepts, domain, and range.

51. Find all intersections of

(a) $y = x + 2$ and $y = x^2$;

(b) $y = 2^x$ and $y = 4$;

(c) $y = |x|$ and $y = x + 2$.

52. Show algebraically that e^x and $\ln x$ are inverse functions, and state how their graphs are related.

53. Complete a feature table giving the domain, range, intercepts, and asymptotes of e^x , $\ln x$, $\sqrt{x}$, $|x|$, and $1/x$.

54. Find the constants:

(a) $f(x) = ab^x$, $f(0) = 3$, $f(2) = 12$;

(b) $g(x) = \log_2(x - h) + k$ has vertical asymptote $x = 3$ and passes through $(4, -2)$.

(c) $p(x) = cd^x$, $p(1) = 6$, $p(3) = 54$;

(d) $q(x) = ae^{kx}$, $q(0) = 4$, $q(2) = 4e^6$;

(e) $r(x) = \log_3(x - h) + k$ has vertical asymptote $x = -2$ and passes through $(1, 2)$.

55. Solve the mixed equations:

(a) $e^{2x} = 7$;

(b) $\ln(x^2) = \ln 9$;

(c) $|e^x - 2| = 1$.

(d) $e^{2x} - 5e^x + 4 = 0$;

(e) $\ln(x - 1) + \ln(x + 1) = \ln 8$;

(f) $|\ln x| = \ln 4$.

Piecewise functions, operations, compositions, and inverses

56. For

$$f(x) = \begin{cases} x + 3, & x < 0, \\ x^2, & 0 \leq x < 2, \\ 5 - x, & x \geq 2, \end{cases}$$

evaluate $f(-4)$, $f(0)$, $f(3/2)$, $f(2)$, and $f(6)$.

57. Sketch the function in Question 56 and state its domain, range, intercepts, and any discontinuities.

58. Find a and b so that

$$f(x) = \begin{cases} ax + 1, & x < 2, \\ x^2 + b, & x \geq 2 \end{cases}$$

is continuous at $x = 2$ and satisfies $f(2) = 7$.

59. Write a piecewise function for a parking charge of RM 4 for the first hour and RM 2.50 for each additional hour or part thereof, for stays of at most four hours. Clearly state the time intervals used.

60. Let $f(x) = \sqrt{x-1}$ and $g(x) = x + 2$. Find $f + g$, $f - g$, fg , and f/g , together with their domains.

61. Simplify each quotient and retain all restrictions inherited from the original expression:

(a) $\frac{x^2 - 9}{x - 3}$;

(b) $\frac{x^2 + x - 6}{x^2 - 4}$;

(c) $\frac{\sqrt{x} - 2}{x - 4}$.

62. If $f(x) = 3x - 1$ and $g(x) = x^2 + 2$, find $f \circ g$, $g \circ f$, and the values $(f \circ g)(-2)$ and $(g \circ f)(1)$.

63. Find the domain of each composite:

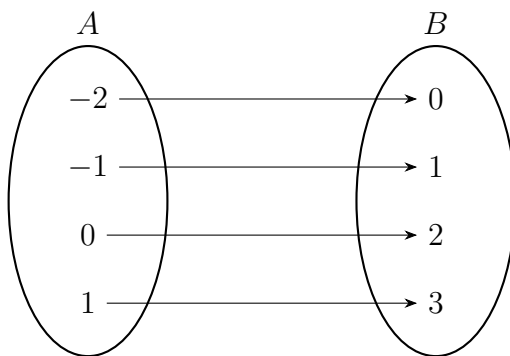
(a) $f(g(x))$ for $f(x) = \sqrt{x}$ and $g(x) = 5 - 2x$;

(b) $g(f(x))$ for $f(x) = \ln x$ and $g(x) = 1/(x - 2)$;

(c) $f(g(x))$ for $f(x) = \ln x$ and $g(x) = x^2 - 9$.

64. Given $f(x) = x + 1$, $g(x) = 2x$, and $h(x) = x^2$, find $f \circ g \circ h$, $h \circ g \circ f$, and both values at $x = 2$.

65. For $f(x) = 2x - 3$, find $f^{\circ 2}(x)$, $f^{\circ 3}(x)$, and a formula for $f^{\circ n}(x)$.
66. Given $f(x) = 2x + 1$ and $(f \circ g)(x) = 6x - 7$, determine $g(x)$.
67. Find the inverse of each linear function and state its domain:
- (a) $f(x) = 5x - 4$;
 - (b) $g(x) = (3 - x)/2$.
68. Restrict $f(x) = (x - 2)^2 + 1$ to $x \geq 2$, then find f^{-1} and state the domain and range of both functions.
69. Find the inverse of $f(x) = \frac{2x + 3}{x - 1}$ and state all restrictions.
70. Verify by composition that $f(x) = 3x + 2$ and $g(x) = (x - 2)/3$ are inverses.
71. Decide whether each function is one-to-one on its stated domain:
- (a) x^3 on $\mathbb{R}$;
 - (b) x^2 on $\mathbb{R}$;
 - (c) x^2 on $[0, \infty)$;
 - (d) $|x - 1|$ on $[1, \infty)$.
72. If $f(x) = x^2 - 1$ and $g(x) = 2x + 3$, solve $(f \circ g)(x) = 15$ and $(g \circ f)(x) = 5$.
73. Let $f(x) = ax + b$ and $g(x) = 2x - 1$. If $(f \circ g)(x) = 6x + 5$, find a and b , then calculate $(g \circ f)(x)$.
74. The mapping diagram shows a relation from A to B .

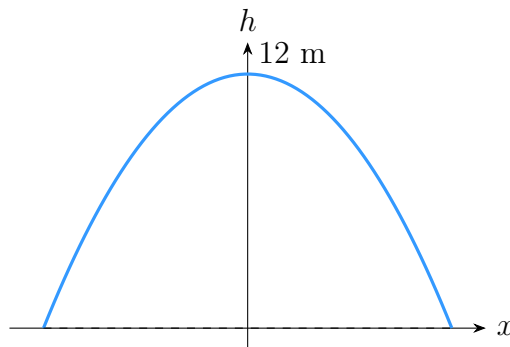


State the rule, determine whether it is one-to-one, and draw the inverse mapping. Is the inverse also a function?

75. For $f(x) = \sqrt{x + 1}$ and $g(x) = x^2 - 1$ restricted to $x \geq 0$, find both composites, simplify them carefully, and explain why the restrictions matter.

Application-based problems

76. A taxi charges RM 4 when a journey begins and RM 1.80 per kilometre. Write the fare $F(d)$, find $F(12)$, and determine the greatest whole number of kilometres affordable with RM 30.
77. A mobile-data plan costs RM 25 for up to 10 GB and RM 3 for each additional GB. Write a piecewise cost function and find the costs of using 7 GB and 16 GB.
78. Temperature conversions satisfy $F(C) = \frac{9}{5}C + 32$. Find $F(25)$, obtain the inverse function, and convert 95°F to Celsius.
79. A small business has cost $C(x) = 1200 + 18x$ and revenue $R(x) = 30x$. Form the profit function, find the break-even output, and state when the business makes a profit.
80. The height of a ball is $h(t) = -5t^2 + 20t + 1$ metres. Find its initial height, maximum height and the time at which the maximum occurs.
81. A farmer uses 80 m of fencing for three sides of a rectangular field beside a river. If x is the width perpendicular to the river, form the area function and find its maximum.
82. A bridge arch is modelled by $h(x) = 12 - \frac{1}{3}x^2$, where x is the horizontal distance in metres from the centre.



Find the ground-level span and the height 3 m from the centre.

83. A medicine concentration is $M(t) = 80(0.75)^t$ mg L^{-1} . Find $M(4)$ and determine the first whole hour at which the concentration is below 20 mg L^{-1} .
84. A population follows $P(t) = 5000(1.04)^t$. Find the population after 10 years and the time required to exceed 7000.
85. An investment of RM 8,000 earns 5% annually. Write its value function, find its value after 6 years, and find the doubling time.

86. The acidity of a solution is $\text{pH} = -\log_{10}[H^+]$. Find the pH when $[H^+] = 3.2 \times 10^{-5}$ and find $[H^+]$ when $\text{pH} = 6$.
87. Sound level is $L = 10 \log_{10}(I/I_0)$. By how many decibels does the level increase when the intensity is multiplied by 100?
88. A cooling model is $T(t) = 22 + 68e^{-0.12t}$. Find $T(10)$ and the time when the temperature first reaches 30°C .
89. Light intensity is modelled by $I(d) = 900/d^2$. Find $I(3)$, determine the distance for which $I = 36$, and interpret the restriction on the domain.
90. The average cost per item is $A(x) = 12 + 600/x$. Find $A(50)$ and $A(200)$, and interpret the horizontal and vertical asymptotes.
91. A response curve is $R(c) = \frac{100c}{c+20}$ for concentration $c \geq 0$. Find $R(20)$, solve $R(c) = 80$, and state the limiting response.
92. A car's stopping distance is $d(v) = 0.006v^2 + 0.25v$, with v in km h^{-1} . Find $d(80)$ and solve for the positive speed when $d = 60$ m.
93. The radius of a circular garden is $r(A) = \sqrt{A/\pi}$. Explain why this is the inverse of $A(r) = \pi r^2$ only when $r \geq 0$, and find the radius of a $100\pi \text{ m}^2$ garden.
94. A manufactured rod is acceptable when its length x satisfies $|x - 50| \leq 0.08$ cm. Express the acceptable interval and decide whether rods of lengths 49.91 cm and 50.06 cm pass inspection.
95. The distance of a point x on a number line from a sensor at $x = 7$ is $D(x) = |x - 7|$. Find the points whose distance is 12 and the region whose distance is at most 3.
96. Electricity costs RM 18 for the first 100 kW h and RM 0.28 per kW h thereafter. Write a continuous piecewise cost function and calculate the bill for 250 kW h.
97. A courier charges according to

$$C(m) = \begin{cases} 6, & 0 < m \leq 1, \\ 6 + 2.5(m - 1), & 1 < m \leq 5, \\ 16 + 1.5(m - 5), & m > 5. \end{cases}$$

Find $C(0.8)$, $C(4)$, and $C(8)$, and sketch the graph, marking open and closed endpoints.

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98. A sensor outputs voltage $V(T) = 0.04T + 0.5$, and a converter reports $D(V) = 250V - 125$. Find $(D \circ V)(T)$ and the displayed value at $T = 30^\circ\text{C}$.
99. A calibration curve is $S(c) = \frac{5c + 2}{c + 1}$ for $c \geq 0$. Find its inverse and use it to recover c when the sensor reading is $S = 4.4$.
100. A rental company compares $A(d) = 45 + 0.35d$ and $B(d) = 70 + 0.20d$, where d is the distance driven in kilometres. Find the break-even distance and state which plan is cheaper before and after that distance.

Chapter 3

Polynomials

Chapter guide

Prerequisites. Laws of indices, algebraic expansion, factorisation, and elementary functions.

Learning outcomes. Readers should be able to classify and operate on polynomials, perform polynomial division, apply the remainder and factor theorems, solve polynomial equations, and decompose rational expressions into partial fractions.

Expressions such as $3x^3 - 4x$ and $4x^2 - 5x + 1$ occur throughout algebra, calculus, numerical analysis, and mathematical modelling. They are examples of **polynomials**. This chapter develops their terminology and algebra before studying division, the remainder and factor theorems, factorisation, polynomial equations, and partial-fraction decomposition. Its scope is aligned with Malaysian secondary and foundation mathematics references (Wong Mee Kiong et al. 2019; Hashim and Mohamad Sukri 2025–2026).

3.1 Definition and terminology

Definition 3.1: Polynomial.

A polynomial in the variable x is an expression of the form

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0, \quad (3.1)$$

where n is a non-negative integer, the coefficients $a_0, a_1, \dots, a_n$ are constants, and $a_n \neq 0$.

The integer n is the **degree** of the polynomial, a_n is its **leading coefficient**, $a_n x^n$ is its **leading term**, and a_0 is its **constant term**. A non-zero constant polynomial has degree 0. The zero polynomial has every coefficient equal to zero, and its degree is normally left undefined.

Missing powers are understood to have coefficient zero. For example,

$$5x^4 - 2x + 7 = 5x^4 + 0x^3 + 0x^2 - 2x + 7.$$

This observation is particularly important when polynomial division is performed.

3.1.1 Recognising polynomials

The exponents of the variable must be non-negative integers. Coefficients may be any real numbers unless a different coefficient set is specified.

Expression	Polynomial?	Reason
$2x^5 - 7x + 1$	Yes	Degree 5, leading coefficient 2
$-3x^2 + \sqrt{2}x - 7$	Yes	Real coefficients are allowed
$2x^{-1} - 7x + 1$	No	Contains a negative exponent
$\sqrt{x} + x$	No	Contains the exponent 1/2
$3^x + 1$	No	The variable occurs in an exponent
$\frac{x^2 + 1}{x - 1}$	No	The expression is a rational function

Table 3.1: Recognising polynomial expressions.

A one-term polynomial is a **monomial**, a two-term polynomial is a **binomial**, and a three-term polynomial is a **trinomial**. These names describe the number of non-zero terms, not the degree.

3.1.2 Polynomial functions, values, zeros, and roots

Substituting $x = c$ into $p(x)$ gives the value $p(c)$. If

$$p(c) = 0,$$

then c is called a **zero** or **root** of p , and $(c, 0)$ is an x -intercept of the graph $y = p(x)$. The terms “root” and “zero” are therefore algebraic descriptions of the same number.

For example, if

$$p(x) = (x + 1)(x - 2)(x + 3),$$

then $p(-1) = p(2) = p(-3) = 0$. Its roots are -1 , 2 , and -3 .

3.1.3 Leading-term behaviour

For very large $|x|$, the leading term $a_n x^n$ determines the end behaviour of the graph. In particular,

- an even-degree polynomial has ends pointing in the same direction;
- an odd-degree polynomial has ends pointing in opposite directions;
- a positive leading coefficient makes the right-hand end rise; and
- a negative leading coefficient makes the right-hand end fall.

Figure 3.1 collects the four possible end-behaviour patterns. Lower-degree terms can create turning points, but they do not change these directions as $|x|$ becomes large.

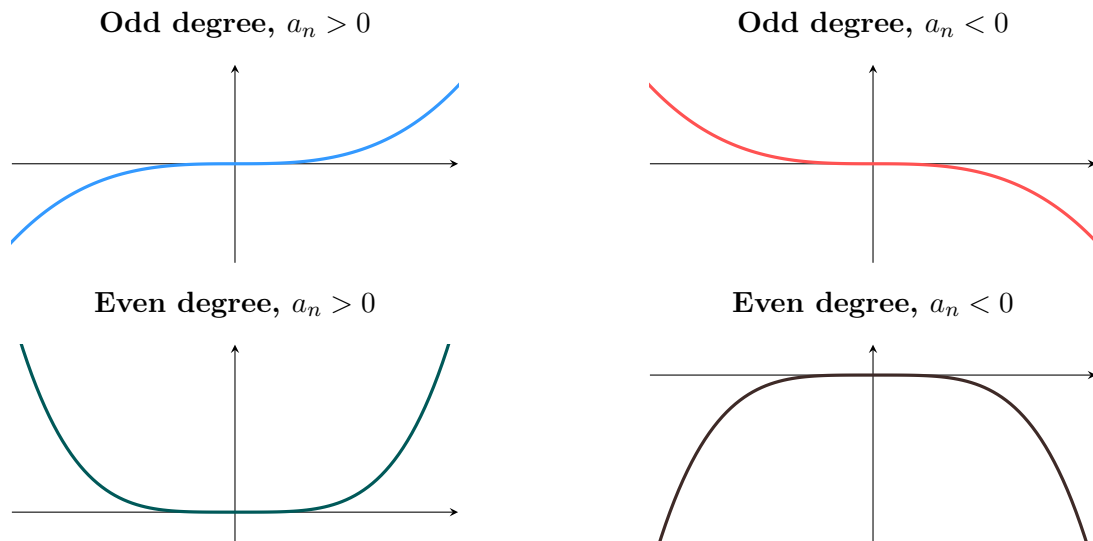


Figure 3.1: The four end-behaviour patterns determined by the parity of the degree and the sign of the leading coefficient.

This provides a useful check when sketching a factorised polynomial.

3.2 Algebraic operations on polynomials

Polynomials are closed under addition, subtraction, and multiplication: the result of each operation is another polynomial.

3.2.1 Addition and subtraction

Only like powers may be combined. Let

$$f(x) = 2x^5 - 7x - 9, \quad g(x) = 3x + 5.$$

Then

$$\begin{aligned} f(x) + g(x) &= 2x^5 - 7x - 9 + 3x + 5 \\ &= \boxed{2x^5 - 4x - 4}. \end{aligned}$$

Similarly, if

$$a(x) = 3x^3 + 2x^2 + 8, \quad b(x) = 3x^3 + 5x + 1,$$

then

$$\begin{aligned} a(x) - b(x) &= 3x^3 + 2x^2 + 8 - (3x^3 + 5x + 1) \\ &= \boxed{2x^2 - 5x + 7}. \end{aligned}$$

The brackets are essential because subtraction changes the sign of every term in the second polynomial.

3.2.2 Multiplication

Use the distributive law and the laws of indices. For example,

$$\begin{aligned}(2x^2 - 3x + 1)(x + 4) &= 2x^3 + 8x^2 - 3x^2 - 12x + x + 4 \\ &= \boxed{2x^3 + 5x^2 - 11x + 4}.\end{aligned}$$

The degree of a product of two non-zero polynomials is the sum of their degrees:

$$\deg(fg) = \deg f + \deg g.$$

3.3 The polynomial division algorithm

Division of polynomials extends ordinary integer division.

Definition 3.2: Polynomial division algorithm.

Let $p(x)$ and $g(x)$ be polynomials with $g(x) \neq 0$. There exist unique polynomials $q(x)$ and $r(x)$ such that

$$\boxed{p(x) = g(x)q(x) + r(x)}, \quad (3.2)$$

where either $r(x) = 0$ or $\deg r < \deg g$.

Here p is the **dividend**, g the **divisor**, q the **quotient**, and r the **remainder**. Equivalently,

$$\frac{p(x)}{g(x)} = q(x) + \frac{r(x)}{g(x)}.$$

$p(x)$
dividend

=

 $g(x)$
divisor

×

 $q(x)$
quotient

+

 $r(x)$
remainder

the part divided exactly

Figure 3.2: The structure of polynomial division.

Long division works for every non-zero polynomial divisor. Synthetic division is a shortened method for a linear divisor.

3.4 Polynomial long division

The procedure mirrors numerical long division:

1. arrange the dividend and divisor in descending powers;

2. insert zero coefficients for missing powers;
3. divide the leading term of the current dividend by the leading term of the divisor;
4. multiply the divisor by this quotient term and subtract;
5. bring down the remaining terms and repeat; and
6. stop when the remainder has lower degree than the divisor.

Example 3.1: linear divisor

$$x^2 + x - 6 = (x - 2)(x + 3),$$

so division by $x - 2$ gives quotient $x + 3$ and remainder 0.

Example 3.2: quadratic divisor

Divide

$$p(x) = x^4 + 2x^2 - 8x + 3$$

by $g(x) = x^2 - 3x + 1$. Write the missing cubic term explicitly:

$$x^4 + 0x^3 + 2x^2 - 8x + 3.$$

The successive leading terms of the quotient are x^2 , $3x$, and 10:

$$\begin{aligned} x^4 + 0x^3 + 2x^2 - 8x + 3 - x^2(x^2 - 3x + 1) &= 3x^3 + x^2 - 8x + 3, \\ 3x^3 + x^2 - 8x + 3 - 3x(x^2 - 3x + 1) &= 10x^2 - 11x + 3, \\ 10x^2 - 11x + 3 - 10(x^2 - 3x + 1) &= 19x - 7. \end{aligned}$$

Therefore,

$$x^4 + 2x^2 - 8x + 3 = (x^2 - 3x + 1)(x^2 + 3x + 10) + (19x - 7).$$

The remainder is valid because its degree, 1, is less than the degree of the quadratic divisor.

3.5 Synthetic division

Synthetic division records only the coefficients. For division by $x - c$, write c at the left, bring down the first coefficient, and repeatedly multiply by c and add.

Example 3.3

Divide $x^2 + x - 6$ by $x - 2$:

$$\begin{array}{r|rrr} 2 & 1 & 1 & -6 \\ & & 2 & 6 \\ \hline & 1 & 3 & 0 \end{array}$$

The bottom row gives quotient coefficients 1, 3 followed by the remainder. Thus

$$x^2 + x - 6 = (x - 2)(x + 3).$$

3.5.1 Divisors of the form $ax - b$

Since

$$ax - b = a \left(x - \frac{b}{a} \right),$$

synthetic division uses $c = b/a$. The synthetic quotient is the quotient for $x - b/a$, so it must then be divided by a to obtain the quotient for $ax - b$. The remainder is unchanged.

Example 3.4

Divide $8x^3 + 4x + 3$ by $2x - 1$. Use $c = 1/2$ and include the missing x^2 coefficient:

$$\begin{array}{r|rrrr} \frac{1}{2} & 8 & 0 & 4 & 3 \\ & & 4 & 2 & 3 \\ \hline & 8 & 4 & 6 & 6 \end{array}$$

The quotient for $x - 1/2$ is $8x^2 + 4x + 6$. Dividing it by 2 gives the quotient for $2x - 1$:

$$\boxed{8x^3 + 4x + 3 = (2x - 1)(4x^2 + 2x + 3) + 6.}$$

Synthetic division also works with complex values of c , although complex numbers and their algebra are developed formally in Chapter 10.

3.6 The remainder theorem

Theorem 3.1 (Remainder theorem). *Let $p(x)$ be a polynomial and let c be a number. When $p(x)$ is divided by $x - c$, the remainder is $p(c)$.*

Proof. The polynomial division algorithm gives unique polynomials $q(x)$ and $r(x)$ such that

$$p(x) = (x - c)q(x) + r(x),$$

where either $r(x) = 0$ or $\deg r < \deg(x - c) = 1$. Thus $r(x)$ is a constant, say r . Evaluating the identity at $x = c$ gives

$$p(c) = (c - c)q(c) + r = r.$$

Hence the remainder is $p(c)$. □

For example, if $p(x) = 2x^2 - 5x - 1$, the remainder after division by $x - 3$ is

$$p(3) = 2(3)^2 - 5(3) - 1 = \boxed{2}.$$

No division is required.

For a divisor $ax - b$, use its zero $x = b/a$. Hence the remainder is

$$\boxed{p\left(\frac{b}{a}\right)}.$$

3.6.1 Remainder after division by a quadratic

If the divisor is $(x - a)(x - b)$ with $a \neq b$, then the remainder has the form $r(x) = mx + n$. If $p(a)$ and $p(b)$ are known, substitute the two values to determine m and n .

For example, suppose division of $p(x)$ by $x - 2$ gives remainder 5, while division by $x + 1$ gives remainder -4 . The remainder after division by $(x - 2)(x + 1)$ is $mx + n$, where

$$\begin{aligned} 2m + n &= 5, \\ -m + n &= -4. \end{aligned}$$

Thus $m = 3$ and $n = -1$, so the required remainder is

$$\boxed{3x - 1}.$$

3.7 The factor theorem

The factor theorem is the zero-remainder case of the remainder theorem.

Theorem 3.2 (Factor theorem). *For a polynomial $p(x)$ and a number c ,*

$$\boxed{p(c) = 0 \iff x - c \text{ is a factor of } p(x).}$$

Proof. By the remainder theorem, division by $x - c$ gives

$$p(x) = (x - c)q(x) + p(c)$$

for some polynomial $q(x)$. If $p(c) = 0$, then $p(x) = (x - c)q(x)$, so $x - c$ is a factor. Conversely, if $p(x) = (x - c)q(x)$, evaluating at $x = c$ gives $p(c) = 0$. This proves both directions of the equivalence. □

For a factor $ax - b$, test the value $x = b/a$.

Example 3.5

Show that $2x - 1$ is a factor of

$$p(x) = 2x^3 + 7x^2 + 2x - 3.$$

Since the zero of $2x - 1$ is $1/2$,

$$\begin{aligned} p\left(\frac{1}{2}\right) &= 2\left(\frac{1}{2}\right)^3 + 7\left(\frac{1}{2}\right)^2 + 2\left(\frac{1}{2}\right) - 3 \\ &= 0. \end{aligned}$$

Therefore $2x - 1$ is a factor.

Example 3.6: finding an unknown coefficient

Given that $x - 5$ is a factor of

$$p(x) = x^3 - x^2 - 32x + a,$$

the factor theorem gives

$$\begin{aligned} p(5) &= 125 - 25 - 160 + a = 0, \\ a &= \boxed{60}. \end{aligned}$$

3.8 Factorising polynomials

The following statements are equivalent:

$$p(c) = 0, \quad x - c \text{ is a factor}, \quad c \text{ is a root.}$$

This equivalence provides a systematic way to factorise higher-degree polynomials:

1. find or test a possible root c ;
2. use the factor theorem to confirm that $p(c) = 0$;
3. divide by $x - c$ or the corresponding linear factor; and
4. factorise the lower-degree quotient.

Example 3.7

Factorise

$$p(x) = x^3 - 3x^2 - 13x + 15,$$

given that $x - 1$ is a factor. Synthetic division gives

$$p(x) = (x - 1)(x^2 - 2x - 15).$$

Since $x^2 - 2x - 15 = (x - 5)(x + 3)$,

$$p(x) = (x - 1)(x - 5)(x + 3).$$

The roots are 1, 5, and -3 .

3.8.1 The rational-root theorem

Theorem 3.3 (Rational-root theorem). *Let*

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$$

have integer coefficients with $a_n \neq 0$. If s/t is a rational root, where $s, t \in \mathbb{Z}$, $t > 0$, and $\gcd(s, t) = 1$, then

$$s \mid a_0, \quad t \mid a_n.$$

Proof. Substituting $x = s/t$ and multiplying by t^n gives

$$a_n s^n + a_{n-1} s^{n-1} t + \cdots + a_1 s t^{n-1} + a_0 t^n = 0.$$

Rearranging shows that s divides $a_0 t^n$. Since $\gcd(s, t) = 1$, we also have $\gcd(s, t^n) = 1$, and therefore $s \mid a_0$. Similarly,

$$a_n s^n = -t(a_{n-1} s^{n-1} + \cdots + a_1 s t^{n-2} + a_0 t^{n-1}),$$

so t divides $a_n s^n$. Because $\gcd(t, s^n) = 1$, it follows that $t \mid a_n$. □

This theorem produces a finite list of candidates. Each candidate must still be tested; appearing on the list does not guarantee that it is a root.

For example, possible rational roots of

$$2x^3 - 3x^2 - 8x + 12$$

include $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12$ and their halves. Testing $x = 2$ gives zero, and division yields

$$2x^3 - 3x^2 - 8x + 12 = (x - 2)(2x^2 + x - 6) = \boxed{(x - 2)(2x - 3)(x + 2)}.$$

3.8.2 Repeated roots and multiplicity

If $(x - c)^m$ is a factor but $(x - c)^{m+1}$ is not, then c is a root of **multiplicity** m .

- At a root of odd multiplicity, the graph crosses the x -axis.
- At a root of even multiplicity, the graph touches the axis and turns back.
- A multiplicity greater than 1 makes the graph flatter near the root.

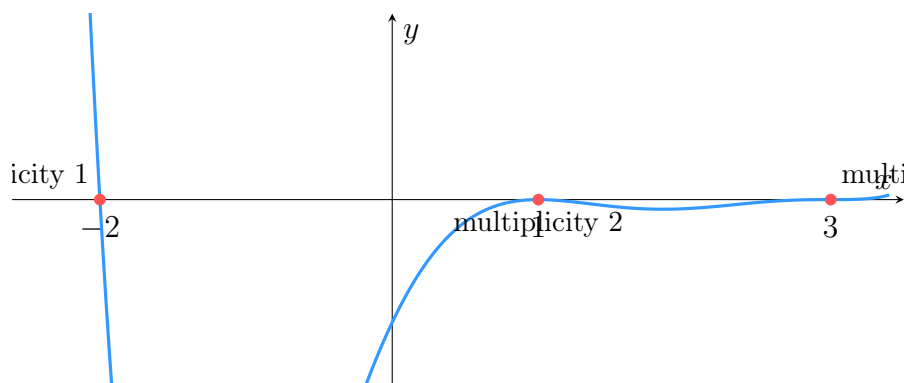


Figure 3.3: Typical behaviour at roots of different multiplicities for a scaled version of $(x + 2)(x - 1)^2(x - 3)^3$.

3.8.3 Constructing a polynomial from its roots

If the roots are $r_1, r_2, \dots, r_n$, with multiplicity included, then

$$p(x) = A(x - r_1)(x - r_2) \cdots (x - r_n), \quad A \neq 0.$$

The value of A is the leading coefficient. If the polynomial is monic, $A = 1$. For example, the monic cubic with roots -2 , 1 , and 3 is

$$\begin{aligned} p(x) &= (x + 2)(x - 1)(x - 3) \\ &= \boxed{x^3 - 2x^2 - 5x + 6}. \end{aligned}$$

3.9 Solving polynomial equations

To solve $p(x) = 0$, first express p as a product of factors and then use the zero-product property. For example,

$$\begin{aligned} x^3 - 4x^2 - x + 4 &= x^2(x - 4) - 1(x - 4) \\ &= (x^2 - 1)(x - 4) \\ &= (x - 1)(x + 1)(x - 4). \end{aligned}$$

Hence

$$x^3 - 4x^2 - x + 4 = 0 \implies \boxed{x = -1, 1, 4}.$$

A polynomial of degree n has at most n distinct real roots and exactly n complex roots when multiplicity is counted. Some of those roots may be non-real.

3.10 Rational functions and partial fractions

Definition 3.3: Rational function.

A rational function is a quotient

$$R(x) = \frac{P(x)}{Q(x)}, \quad Q(x) \neq 0,$$

where P and Q are polynomials.

The fraction is **proper** if $\deg P < \deg Q$ and **improper** if $\deg P \geq \deg Q$. An improper fraction must first be divided to obtain

$$\text{polynomial} + \text{proper rational fraction}.$$

Partial-fraction decomposition reverses the process of combining algebraic fractions: it writes a proper rational function as a sum of simpler fractions. This is especially useful in integration, differential equations, and inverse transforms.

3.10.1 Forms of a partial-fraction decomposition

First factorise the denominator completely over the real numbers. Include one term for every power of every repeated factor.

Denominator factor	Terms required
Distinct linear factors $(x - a)(x - b)$	$\frac{A}{x - a} + \frac{B}{x - b}$
Repeated linear factor $(x - a)^m$	$\frac{A_1}{x - a} + \frac{A_2}{(x - a)^2} + \cdots + \frac{A_m}{(x - a)^m}$
Irreducible quadratic $x^2 + bx + c$	$\frac{Ax + B}{x^2 + bx + c}$
Repeated irreducible quadratic $[x^2 + bx + c]^m$	one linear numerator over each power from 1 to m

Table 3.2: The principal real partial-fraction forms.

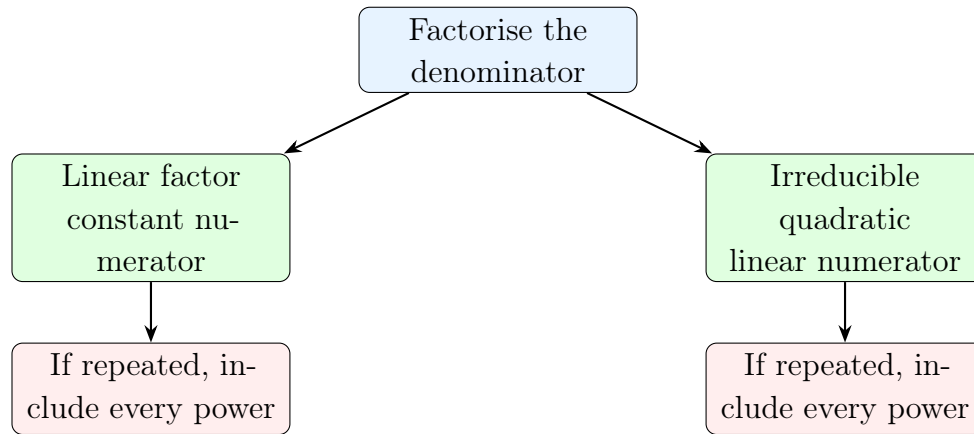


Figure 3.4: Choosing the numerator and denominator terms in a decomposition.

After writing the correct form, multiply by the common denominator. The unknown constants may then be found by substituting convenient values or by comparing coefficients.

3.10.2 Distinct linear factors

Decompose

$$\frac{x}{(x+3)(x-2)}.$$

Write

$$\frac{x}{(x+3)(x-2)} = \frac{A}{x+3} + \frac{B}{x-2}.$$

Multiplication by $(x+3)(x-2)$ gives

$$x = A(x-2) + B(x+3).$$

Substituting $x = -3$ gives $A = 3/5$, while $x = 2$ gives $B = 2/5$. Therefore,

$$\boxed{\frac{x}{(x+3)(x-2)} = \frac{3}{5(x+3)} + \frac{2}{5(x-2)}}.$$

3.10.3 Repeated linear factors

For

$$\frac{3x+5}{(x-1)^2},$$

both powers of the repeated factor are required:

$$\frac{3x+5}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2}.$$

Thus

$$3x + 5 = A(x - 1) + B.$$

Comparing coefficients gives $A = 3$ and $B = 8$, so

$$\frac{3x + 5}{(x - 1)^2} = \frac{3}{x - 1} + \frac{8}{(x - 1)^2}.$$

3.10.4 An irreducible quadratic factor

Decompose

$$\frac{2x^2 + 3x + 4}{x(x^2 + 1)}.$$

Because $x^2 + 1$ has no real linear factors, its numerator must be linear:

$$\frac{2x^2 + 3x + 4}{x(x^2 + 1)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 1}.$$

Multiplying through gives

$$\begin{aligned} 2x^2 + 3x + 4 &= A(x^2 + 1) + x(Bx + C) \\ &= (A + B)x^2 + Cx + A. \end{aligned}$$

Hence $A = 4$, $C = 3$, and $B = -2$. Therefore,

$$\frac{2x^2 + 3x + 4}{x(x^2 + 1)} = \frac{4}{x} + \frac{-2x + 3}{x^2 + 1}.$$

3.10.5 Improper rational fractions

Decompose

$$\frac{2x^3 + 3x^2 + 4x + 5}{x^2 + x}.$$

Polynomial division gives

$$\frac{2x^3 + 3x^2 + 4x + 5}{x^2 + x} = 2x + 1 + \frac{3x + 5}{x(x + 1)}.$$

Now

$$\frac{3x + 5}{x(x + 1)} = \frac{5}{x} - \frac{2}{x + 1}.$$

Thus

$$\boxed{\frac{2x^3 + 3x^2 + 4x + 5}{x^2 + x} = 2x + 1 + \frac{5}{x} - \frac{2}{x + 1}}.$$

The restrictions from the original denominator, here $x \neq 0, -1$, remain in force after decomposition.

3.10.6 Checking a decomposition

Combining the proposed simpler fractions over a common denominator should recover the original rational function. For example,

$$\frac{4x^3 + 2x^2 + 3x + 1}{x^4 + x^3 + x^2 + x} = \boxed{\frac{1}{x} + \frac{2}{x + 1} + \frac{x}{x^2 + 1}}.$$

Since $x^4 + x^3 + x^2 + x = x(x + 1)(x^2 + 1)$, combining the right-hand side verifies the identity.

Chapter summary and common pitfalls

Key ideas. Degree and leading coefficient control end behaviour; division links a polynomial to its quotient and remainder; and roots, factors, and partial fractions provide complementary representations.

Common pitfalls. Preserve descending powers and zero coefficients during division, distinguish multiplicity from the number of distinct roots, and match the correct numerator form to each partial-fraction denominator.

3.11 Practice and exercises

Terminology and operations

- Complete the following classification and terminology tasks.
 - State whether $4x^5 - 3x + 1$, $2x^{-1} + x$, $\sqrt{2}x^3 - x$, and $3^x + 1$ are polynomials. For every polynomial, state its degree and leading coefficient.
 - For $p(x) = -3x^6 + 2x^4 - 7$, state the degree, leading coefficient, constant term, and end behaviour.
- Let $f(x) = 2x^3 - x + 4$ and $g(x) = x^2 + 3x - 2$.
 - Find $f + 2g$, $f - g$, and fg .
 - Evaluate $p(-2)$ for $p(x) = 3x^4 - 5x^2 + 2x - 1$.
- Construct and expand each requested polynomial.
 - the monic cubic whose roots are -2 , 1 , and 3 ;
 - the monic polynomial of least degree whose roots are 2 , with multiplicity 2 , and -3 ;
 - the monic quartic whose roots are -1 , 1 , and 0 , with 0 having multiplicity 2 .

Division, remainders, and factors

- Perform the indicated division and state both quotient and remainder.
 - Use long division for $(x^4 + 2x^2 - 8x + 3) \div (x^2 - 3x + 1)$.
 - Use synthetic division for $(x^2 + x - 6) \div (x - 2)$.
 - Verify the dividend identity for part (a).
- Divide by the non-monic linear divisor.
 - $(8x^3 + 4x + 3) \div (2x - 1)$;
 - $(5x^3 + 2x^2 - 3x + 1) \div (3x + 1)$.

6. Find each remainder without carrying out full polynomial division.

(a) $(2x^2 - 5x - 1) \div (x - 3)$;

(b) $(2x^2 - 5x - 1) \div (2x + 3)$;

(c) $(x^3 - 4x + 7) \div (3x - 2)$.

7. Determine the unknown coefficient using the factor theorem.

(a) $x - 4$ is a factor of $x^3 + kx^2 - 13x + 12$;

(b) $x - 5$ is a factor of $x^3 - x^2 - 32x + a$;

(c) $x + 2$ is a factor of $x^3 + mx^2 - 4x + 12$.

8. Complete the factor and quotient calculations.

(a) Show that $2x - 1$ is a factor of $2x^3 + 7x^2 + 2x - 3$, and find the quotient.

(b) Factorise $x^3 - 3x^2 - 13x + 15$, given that $x - 1$ is a factor.

9. A polynomial gives remainder 5 on division by $x - 2$ and remainder -4 on division by $x + 1$.

(a) Find its remainder on division by $(x - 2)(x + 1)$.

(b) State the value of this remainder polynomial at $x = 0$.

Factorisation and polynomial equations

10. Solve each cubic equation over the real numbers.

(a) $2x^3 + 7x^2 + 2x - 3 = 0$;

(b) $x^3 - 4x^2 - x + 4 = 0$;

(c) $x^3 - 3x^2 - 13x + 15 = 0$.

11. Complete the higher-degree and cubic factorisations.

(a) Find all zeros of $x^4 - 5x^2 + 4$.

(b) Factorise completely $2x^3 - 3x^2 - 8x + 12$ and state its zeros.

12. For $p(x) = (x + 2)(x - 1)^2(x - 3)^3$, state every root and multiplicity, describe whether the graph crosses or touches the axis, give its end behaviour, and find the y -intercept.

Partial fractions

13. Decompose each fraction with distinct linear factors.

(a) $\frac{x}{(x + 3)(x - 2)}$;

(b) $\frac{5x + 1}{(x - 1)(x + 2)}$;

(c) $\frac{2x + 7}{(x - 2)(x + 3)}$.

14. Decompose each fraction, first performing division where necessary.

(a) $\frac{3x + 5}{(x - 1)^2}$;

(b) $\frac{2x^2 + 3x + 4}{x(x^2 + 1)}$;

(c) $\frac{2x^3 + 3x^2 + 4x + 5}{x^2 + x}$.

15. Complete the following structural partial-fraction tasks.

(a) Decompose $\frac{x^2 + 1}{(x - 1)(x + 1)(x + 2)}$.

(b) Verify that

$$\frac{4x^3 + 2x^2 + 3x + 1}{x^4 + x^3 + x^2 + x} = \frac{1}{x} + \frac{2}{x + 1} + \frac{x}{x^2 + 1}.$$

(c) Write the correct decomposition form for $\frac{P(x)}{(x - 1)^3(x^2 + 4)^2}$, assuming the fraction is proper. Do not find the constants.

Additional mixed practice

The following questions consolidate several closely related skills within the same numbered problem. This avoids repeating near-identical questions while still providing substantial practice.

16. State whether each expression is a polynomial. If it is, give its degree and leading coefficient.

(a) $5x^6 - \pi x + 2$

(b) $\frac{x^2 + 1}{x}$

(c) $\sqrt{3}x^2 + 1$

(d) $x^{2/3} + 1$

(e) -8

(f) 0

17. State the degree, leading coefficient, and constant term of each polynomial.

(a) $-4x^8 + 3x^3 - x + 5$

- (b) $7 - 2x^5$
 - (c) $x^2(x - 3)^4$
 - (d) $(2x + 1)^3(x^2 + 4)$
18. Write $5 - 2x^3 + 7x^5 - x$ in descending powers and list all of its coefficients, including zeros for missing powers.
19. For $p(x) = 2x^4 - x^3 + 3x - 6$, find $p(0)$, $p(1)$, $p(-1)$, and $p(2)$.
20. Let $p(x) = ax^2 + bx + c$. Find $p(x)$ if $p(0) = 1$, $p(1) = 4$, and $p(-1) = 6$.
21. Let $f(x) = x^3 - 2x + 1$ and $g(x) = 2x^2 + x - 3$. Find
- (a) $2f - g$;
 - (b) $f + 3g$;
 - (c) $g - f$.
22. Expand and simplify.
- (a) $(x - 4)(x + 2)$
 - (b) $(3x^2 + x - 2)(x - 1)$
 - (c) $(x^2 - 2x + 3)(x^2 + 2x + 3)$
23. If $f(x) = x^2 - 1$ and $g(x) = x + 3$, find $f(g(x))$ and $g(f(x))$.
24. Given
- $$f(x) + g(x) = 4x^2 - 2x + 6, \quad f(x) - g(x) = 2x^2 + 6x - 4,$$
- find $f(x)$ and $g(x)$.
25. Suppose $\deg p = 4$ and $\deg q = 2$. State the degree of pq , $p + q$, and p^2q . What can be said about the degree of $p - q^2$?
26. Classify each polynomial as even, odd, or neither.
- (a) $x^6 - 3x^2 + 5$
 - (b) $2x^5 - x$
 - (c) $x^4 + x$
 - (d) the zero polynomial
27. For $p(x) = 3(x + 2)^3(x - 1)^2$, state its zeros and their multiplicities, describe the graph at each zero, and find the y -intercept.

28. Construct the requested polynomial.

- (a) the monic cubic with roots -3 , 0 , and 2 ;
- (b) a cubic with leading coefficient -2 , a double root at 1 , and another root at -4 ;
- (c) the monic polynomial of least degree with roots $\pm\sqrt{5}$.

29. Describe the end behaviour of a polynomial whose leading term is

- (a) $2x^7$;
- (b) $-x^4$;
- (c) $-3x^5$;
- (d) $6x^8$.

30. Find a and b if

$$(ax + b)(x + 3) = 2x^2 + 7x + 3.$$

Division, remainders, and factorisation review

31. Use long division to find the quotient and remainder in each case.

- (a) $(2x^4 - 3x^3 + 5x - 7) \div (x^2 + 1)$
- (b) $(x^4 - 1) \div (x^2 - 1)$
- (c) $(3x^3 + 2x^2 - x + 5) \div (x^2 + x - 2)$

32. Use synthetic division.

- (a) $(2x^5 - 3x^3 + x - 4) \div (x - 2)$
- (b) $(x^4 + 4x^2 + 1) \div (x + 1)$
- (c) $(x^5 - 32) \div (x - 2)$

33. Divide by a non-monic linear divisor.

- (a) $(4x^4 - x^2 + 3) \div (2x - 1)$
- (b) $(3x^3 + 5x^2 - 2x + 7) \div (3x + 2)$

34. A polynomial is divided by $x^2 - 2x + 3$. Its quotient is $2x + 5$ and its remainder is $4x - 1$. Find the dividend.

35. The polynomial $p(x) = x^3 + ax^2 + bx - 5$ gives quotient $x^2 + 2x - 3$ when divided by $x + 1$. Find a , b , and the remainder.

36. Use the remainder theorem to find the remainder in each case.

- (a) $(x^5 - 2x + 3) \div (x - 2)$

- (b) $(3x^4 + x - 1) \div (x + 1)$
 - (c) $(2x^3 - 5x^2 + 4) \div (x - 3)$
 - (d) $(x^6 + 1) \div (x + 2)$
37. Find the remainder without performing division.
- (a) $(x^3 + 2x - 5) \div (2x - 1)$
 - (b) $(2x^4 - x + 7) \div (3x + 1)$
38. The polynomial $x^3 + ax^2 + bx - 6$ has factors $x - 1$ and $x - 2$. Find a and b .
39. A polynomial gives remainder 4 on division by $x + 1$ and remainder -5 on division by $x - 2$. Find the remainder after division by $(x + 1)(x - 2)$.
40. When $p(x)$ is divided by $x(x - 1)(x - 2)$, its remainder is $ax^2 + bx + c$. Given $p(0) = 1$, $p(1) = 3$, and $p(2) = 9$, find the remainder.
41. Find the remainder when $x^{101} + x^2 + 1$ is divided by $x^2 - 1$.
42. Find k so that the stated divisibility condition holds.
- (a) $x^2 + kx + 6$ is divisible by $x - 2$;
 - (b) $2x^3 + kx^2 - 8$ is divisible by $x + 2$;
 - (c) $x^3 - 4x + k$ is divisible by $x - 1$.
43. The polynomial $p(x) = x^3 + kx - 2$ gives the same remainder when divided by $x - 1$ and by $x + 1$. Find k .
44. Find the remainder when $5x^8 - 2x^5 + 3x^2 - x + 4$ is divided by $x^2 - 1$.
45. Use Horner's method or synthetic evaluation to calculate $p(10)$ for $p(x) = 3x^5 - 2x^4 + x^2 - 7$.
46. Factorise completely by grouping.
- (a) $x^3 + 2x^2 - 9x - 18$
 - (b) $2x^3 - 5x^2 - 8x + 20$
 - (c) $x^3 - 4x^2 - x + 4$
47. Use the supplied factor to factorise completely.
- (a) $x^3 + 2x^2 - 5x - 6$, given $x - 2$
 - (b) $x^4 - 5x^3 + 5x^2 + 5x - 6$, given $x - 1$

- (c) $3x^3 + x^2 - 12x - 4$, given $3x + 1$
48. List all possible rational roots, without testing them.
- (a) $2x^3 - 3x^2 - 8x + 12$
(b) $3x^3 + x^2 - 12x - 4$
49. Factorise completely.
- (a) $x^3 - 7x + 6$
(b) $2x^3 + x^2 - 8x - 4$
(c) $x^3 - 4x^2 - 7x + 10$
50. Factorise each polynomial using a substitution such as $u = x^2$ or $u = x^4$.
- (a) $x^4 - 10x^2 + 9$
(b) $2x^4 - 5x^2 - 3$
(c) $x^8 - 17x^4 + 16$
51. Express each polynomial using repeated factors.
- (a) $x^4 - 2x^3 + x^2$
(b) $x^4 + 2x^3 - 3x^2 - 4x + 4$
(c) $x^5 - 3x^4 + 3x^3 - x^2$
52. Solve over the real numbers.
- (a) $x^3 + 2x^2 - 5x - 6 = 0$
(b) $x^4 - 10x^2 + 9 = 0$
(c) $2x^3 + x^2 - 8x - 4 = 0$
53. Construct the polynomial described.
- (a) the monic cubic with a double root at 1 and a root at -2 ;
(b) a cubic with leading coefficient 4 and roots $0, -1, 3$;
(c) the monic polynomial of least degree that touches the x -axis at both $x = -2$ and $x = 2$.
54. Describe the roots, multiplicities, and local axis behaviour of
- (a) $(x + 3)^2(x - 1)^3$;
(b) $-2(x + 1)(x - 2)^4$.
55. Determine the intervals on which $p(x) = (x + 2)(x - 1)^2(x - 3)$ is positive and negative.

Further partial-fraction practice

56. Decompose each fraction with distinct linear factors.

(a) $\frac{7x - 1}{(x - 2)(x + 1)}$

(b) $\frac{4x + 5}{(x + 2)(x - 3)}$

(c) $\frac{2x - 9}{(x - 4)(x + 3)}$

57. Decompose each fraction with a repeated linear factor.

(a) $\frac{2x + 1}{(x + 2)^2}$

(b) $\frac{x^2 + 4x + 7}{(x - 1)^3}$

58. Decompose.

(a) $\frac{3x^2 + 5x + 7}{(x - 1)(x^2 + 1)}$

(b) $\frac{2x^3 + x^2 + 5x + 6}{(x - 1)^2(x^2 + 4)}$

59. First perform polynomial division, then decompose the proper fraction.

(a) $\frac{x^3 + 2x^2 + 5x + 7}{x^2 - 1}$

(b) $\frac{3x^4 + x^3 + 2x + 1}{x(x + 1)}$

60. Complete the following mixed tasks.

(a) Write the correct decomposition form for $\frac{P(x)}{(x - 2)^2(x^2 + 9)}$.

(b) Combine $\frac{2}{x - 1} - \frac{3}{x + 2}$ into one fraction.

(c) Show that $\frac{1}{n(n + 1)} = \frac{1}{n} - \frac{1}{n + 1}$.

Application-based problems

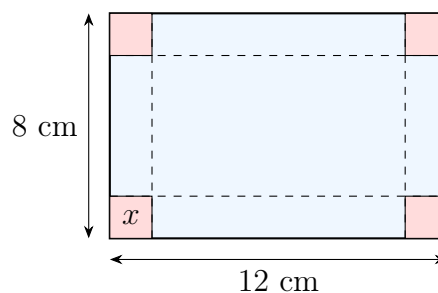
61. A company sells x units at a total revenue $R(x) = x(50 - x)$ ringgit. Its cost is $C(x) = 10x + 300$ ringgit. Find the profit polynomial and the break-even production levels. State when the model predicts a positive profit.

62. The height of a projectile is

$$h(t) = -5t^2 + 20t + 25$$

metres after t seconds. Find its height after 2 seconds and the physically meaningful time at which it reaches the ground.

63. A rectangular garden has width x metres and length $x + 3$ metres. Its area is 70 m^2 . Form and solve a polynomial equation to find its dimensions.
64. Squares of side x centimetres are cut from the four corners of a 12 cm by 8 cm sheet and the sides are folded to form an open box.



Derive and expand the volume polynomial. Find the volume when $x = 1$, and explain which zeros of the polynomial mark the physical endpoints of the construction.

65. A rectangular photograph has dimensions x centimetres by $x - 1$ centimetres. Its surrounding rectangular frame has outside dimensions $x + 4$ by $x + 3$. Find the polynomial for the visible frame area and evaluate it when $x = 10$.
66. A 10 m by 6 m garden is surrounded by a path of uniform width x metres. The path has area 80 m^2 . Form a polynomial equation and find the width of the path.
67. Three consecutive positive integers have product 120. Represent the situation with a cubic polynomial and find the integers.
68. A simplified bacterial-count model is

$$N(t) = 100 + t(t + 1)(t + 2),$$

where t is measured in hours. Find the first non-negative time at which the model predicts $N(t) = 220$.

69. The cross-section of an arch is modelled by

$$h(x) = -(x - 2)(x - 8).$$

Find the points where the arch meets the ground, its horizontal span, and the height at $x = 5$.

70. A quadratic calibration curve passes through $(0, 1)$, $(1, 4)$, and $(2, 9)$. Find the polynomial.

71. A sequence of measured values is shown below.

n	0	1	2	3	4
$p(n)$	3	8	15	24	?

Use finite differences to identify a quadratic polynomial and predict $p(4)$.

72. A square picture of side x centimetres is surrounded by a square border, making the outside side length $x + 4$. If the border area is 96 cm^2 , find x .

73. A cylindrical vessel has radius x centimetres and height $x + 3$ centimetres. Its volume is $54\pi \text{ cm}^3$. Form the polynomial equation and find the positive radius.

74. A profit index is modelled by

$$P(x) = x^3 - 6x^2 + 11x - 6.$$

Factorise the model, identify its break-even levels, and determine where $P(x) > 0$ for $x \geq 0$.

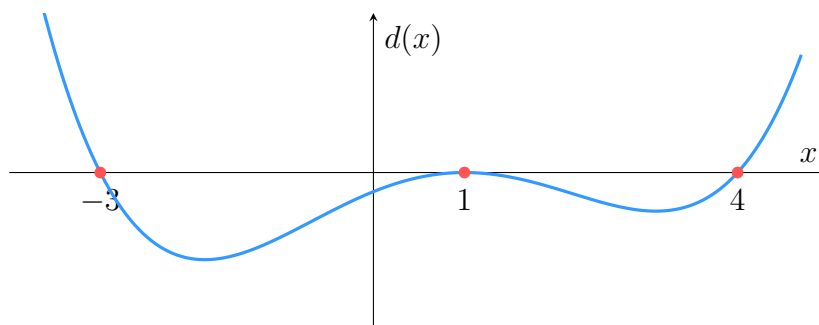
75. The volume of a package is

$$V(x) = x^3 + 5x^2 + 2x - 8.$$

One edge has length $x + 4$. Use polynomial division and factorisation to find the other two edge expressions. State the physical restriction on x and find the volume when $x = 3$.

76. The vertical displacement of a flexible bridge deck is modelled by

$$d(x) = 0.04(x + 3)(x - 1)^2(x - 4).$$



Identify every zero-displacement position, its multiplicity, and whether the profile crosses or touches the reference line there.

77. A two-compartment concentration model is

$$C(t) = \frac{7t + 23}{(t + 2)(t + 5)}, \quad t \geq 0.$$

Decompose $C(t)$ into partial fractions and find $C(0)$ and $C(5)$.

78. A crate has volume polynomial

$$V(x) = x^3 + 7x^2 + 14x + 8,$$

and one side has length $x + 2$. Find the other two linear dimensions and the volume when $x = 2$.

79. The polynomial $D(x) = 4x^3 + 2x^2 + 3x + 1$ stores the decimal digits 4, 2, 3, 1 as coefficients. Evaluate $D(10)$ and explain the result.

80. A manufacturing tolerance is modelled by

$$T(x) = 2x^3 - 9x^2 + 4x + 15.$$

Verify that $x = 3$ is a zero, factorise $T(x)$ completely, and state all non-negative zero-tolerance settings.

81. A particle's displacement is

$$s(t) = t^3 - 6t^2 + 9t.$$

Factorise $s(t)$ and find all non-negative times when the particle is at the origin. Distinguish the repeated time.

82. A circular garden has outer radius $x + 2$ metres and inner radius x metres. Find and simplify the polynomial expression for the annular area, and evaluate it at $x = 3$.

83. A sensor output is $S(x) = x^3 - 4x + 10$. Without division, find the remainder when $S(x)$ is divided by $x - 2$, and interpret it as the sensor output at the calibration setting $x = 2$.

84. A staged correction system loses a fraction $1/[k(k + 1)]$ at stage k . Use partial fractions to find

$$L_N = \sum_{k=1}^N \frac{1}{k(k + 1)},$$

and calculate L_9 .

85. An average-cost model is

$$A(x) = \frac{2x^2 + 7x + 3}{(x+1)(x+3)}, \quad x \geq 0.$$

Determine whether the fraction is proper, divide and decompose it, find $A(1)$, and state the value approached by $A(x)$ as x becomes very large.

Extended applications and modelling

86. A theatre charges RM p per ticket. At that price, the predicted attendance is $800 - 20p$ people.

- Form and expand the revenue polynomial $R(p)$.
- Find the ticket prices for which the revenue is RM 6,000.
- State the meaningful price interval and interpret the zeros of the revenue model.

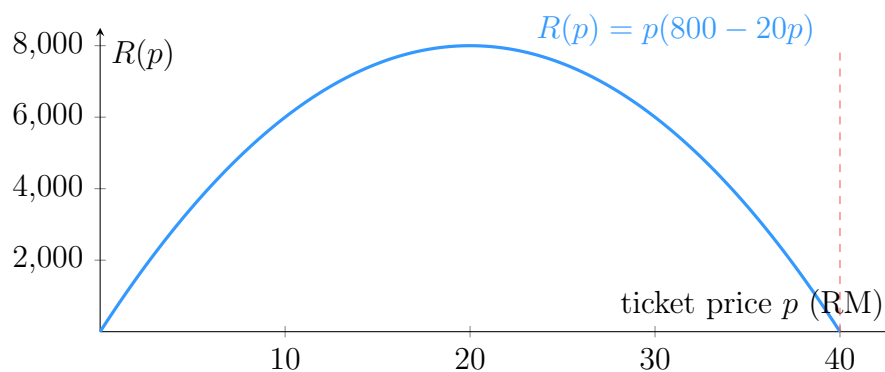


Figure 3.5: The revenue model is meaningful only while the predicted attendance is non-negative.

87. A test vehicle's fuel-efficiency score at speed v km h⁻¹ is modelled by a quadratic. The score is 14 at both 40 km h⁻¹ and 80 km h⁻¹, and its maximum value is 16 at 60 km h⁻¹.

- Find the quadratic in vertex form and expanded form.
- Find the speeds at which the score is 15.
- State the maximum score and the speed at which it occurs.

88. A rectangular crate has dimensions x , $x + 1$, and $x + 4$ metres and volume 160 m³. Form the cubic equation, factorise it, and find the physical dimensions.

89. The volume $V(h)$ of liquid in a calibrated reservoir is modelled by $V(h) = ah^3 + bh^2 + ch$, where h is the measured depth. The readings are $V(1) = 6$, $V(2) = 20$, and $V(3) = 48$. Find the calibration polynomial and predict $V(4)$.

90. A medicine concentration is modelled by

$$C(t) = \frac{5t + 14}{(t + 1)(t + 4)}, \quad t \geq 0.$$

Decompose $C(t)$ into partial fractions, find $C(0)$ and $C(2)$, and state the limiting concentration predicted as $t \rightarrow \infty$.

91. A digital sensor records the following calibration values.

n	0	1	2	3	4
$p(n)$	2	5	16	41	86

Use finite differences to show that a cubic model is appropriate, find $p(n)$, and predict $p(5)$.

92. A camera corrects horizontal lens distortion using

$$y = x + 0.002x^3.$$

Show that the correction function is odd. Find the corrected positions and displacements $y - x$ for objects detected at $x = 10$ and $x = -10$.

93. An ecological index is modelled by

$$P(t) = 500 + 2t(t - 3)(t - 8), \quad t \geq 0.$$

Find when the index equals its baseline value 500, determine when it lies above or below the baseline, and calculate $P(5)$.

94. A simplified filter response is

$$H(f) = \frac{(f + 1)(f + 2)}{(f + 2)(f + 3)}.$$

Simplify the expression while retaining every restriction from the original model. Find $H(1)$, identify the removable discontinuity, and state the limiting value as $f \rightarrow \infty$.

95. A rectangular solar farm has perimeter 60 m and area 176 m². If its width is x and does not exceed its length, form and solve a polynomial equation to find its dimensions.

96. The height of a test rocket is modelled by

$$h(t) = t(8 - t)(t + 2),$$

where t is measured in seconds. Expand the polynomial, find every zero, identify the physically relevant flight interval, and calculate $h(4)$.

97. A gear-test index is $T(n) = n^3 - 7n + 6$. Use the rational-root theorem to factorise $T(n)$, find the non-negative zero-index settings, and determine where $T(n)$ is positive or negative for $n \geq 0$.

98. A message checksum is represented by

$$M(x) = x^7 + 2x^4 - 3x + 5.$$

Find the remainder when $M(x)$ is divided by $x^2 - 1$, using the relations $x^{2k} \equiv 1$ and $x^{2k+1} \equiv x$ modulo $x^2 - 1$.

99. A delivery-cost polynomial is

$$C(x) = 2x^3 + 9x^2 + 13x + 6.$$

Given that $x + 2$ is a factor, use division and further factorisation to express $C(x)$ as a product of three linear factors. Find $C(2)$.

100. A 12 cm-wide metal strip is folded upward along two lines to form a rectangular channel of depth x cm and base width $12 - 2x$ cm.

- (a) Form the cross-sectional area polynomial $A(x)$ and state its physical domain.
- (b) Complete the square to find the greatest possible area.
- (c) Find the two fold depths that produce area 16 cm^2 .

Chapter 4

Inequalities

Chapter guide

Prerequisites. Algebraic manipulation, factorisation, number-line notation, and graphs of elementary functions.

Learning outcomes. Readers should be able to solve linear, polynomial, rational, and absolute-value inequalities; express solution sets using intervals; and verify solutions using sign diagrams or graphs.

An equation asks when two expressions have the same value. An inequality asks when one expression is larger or smaller than another. Consequently, an equation often has isolated solutions, whereas the solution of an inequality is commonly an interval or a union of intervals. Inequalities are used to describe limits, tolerances, feasible regions, safe operating conditions, and the ranges in which one model exceeds another.

This chapter develops a single idea that works across linear, quadratic, cubic, rational, and absolute-value inequalities: identify the values at which the sign of an expression can change, divide the number line at those values, and determine which intervals satisfy the required comparison. The chapter's methods and notation are consistent with standard Malaysian pre-university algebra treatments (Ministry of Education Malaysia 2018; Hashim and Mohamad Sukri 2025–2026).

4.1 Definition and notation of inequalities

Definition 4.1: Inequality.

An inequality is a mathematical statement comparing the order or relative size of two quantities. Its solution set consists of all values that make the comparison true.

The four principal inequality symbols are summarised below.

Symbol	Meaning	Endpoint convention
$a < b$	a is less than b	endpoint excluded
$a > b$	a is greater than b	endpoint excluded
$a \leq b$	a is less than or equal to b	endpoint included
$a \geq b$	a is greater than or equal to b	endpoint included

A strict inequality uses $<$ or $>$; equality is not permitted at its boundary. A non-strict inequality uses $\leq$ or $\geq$ and includes the boundary whenever the original expression is

defined there. On a number line, an open circle represents an excluded endpoint and a filled circle represents an included endpoint.

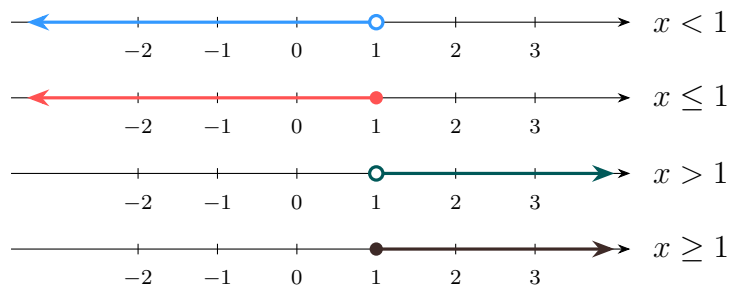


Figure 4.1: Open and closed endpoints on a number line.

The word *and* means that both conditions must hold, so their solution sets are intersected. The word *or* means that at least one condition must hold, so their solution sets are united. For example,

$$x > -2 \text{ and } x \leq 5 \iff -2 < x \leq 5 \iff (-2, 5].$$

4.2 Properties of inequalities

Let $a, b, c \in \mathbb{R}$ and suppose that $a < b$. The fundamental order properties are

$$\begin{aligned} a + c &< b + c, \\ c > 0 &\implies ac < bc, \\ c < 0 &\implies ac > bc. \end{aligned}$$

Thus adding or subtracting the same real number preserves the direction of an inequality. Multiplication or division by a positive number also preserves the direction, but multiplication or division by a negative number reverses it. For example,

$$-3x < 12 \implies x > -4.$$

The reversal is necessary: multiplying the true statement $2 < 5$ by -1 produces $-2 > -5$.

Other useful properties include

$$\begin{aligned} a < b \text{ and } b < c &\implies a < c && \text{(transitivity),} \\ a < b \text{ and } c < d &\implies a + c < b + d, \\ 0 < a < b &\implies \frac{1}{a} > \frac{1}{b}. \end{aligned}$$

Operations such as taking reciprocals or squaring require sign information. For instance, $-3 < 2$ but $(-3)^2 > 2^2$. It is therefore unsafe to square both sides of an arbitrary inequality without first checking the signs.

Example 4.1: a compound linear inequality

Solve $-6 < 3x - 2 \leq 10$.

Adding 2 throughout and then dividing by the positive number 3 gives

$$-4 < 3x \leq 12 \implies -\frac{4}{3} < x \leq 4.$$

Hence the solution set is $(-4/3, 4]$.

4.3 A common sign-analysis principle

After moving all terms to one side, many inequalities take the form

$$F(x) > 0, \quad F(x) < 0, \quad F(x) \geq 0, \quad \text{or} \quad F(x) \leq 0.$$

If F is factored, its sign can change only at a zero of a factor or at a point where F is undefined. These values are called *critical values*. They divide the number line into intervals on each of which the sign is constant.

A reliable sign-chart procedure is as follows.

1. Move every term to one side and write 0 on the other.
2. Factor the expression as fully as possible.
3. List the zeros and, for rational expressions, all excluded values.
4. Arrange the critical values on a number line.
5. Determine the sign on each interval, either from the factors or by testing one convenient value.
6. Select the intervals having the required sign and decide which endpoints may be included.

At a zero of odd multiplicity, the sign changes. At a zero of even multiplicity, the graph touches the axis and the sign does not change. A point at which a denominator is zero is always excluded, even when the inequality is non-strict.

4.4 Quadratic inequalities

A quadratic inequality can be written in the form

$$ax^2 + bx + c > 0, \quad ax^2 + bx + c < 0, \quad ax^2 + bx + c \geq 0, \quad \text{or} \quad ax^2 + bx + c \leq 0, \quad a \neq 0.$$

It may be solved graphically, with a sign chart, or by marking the roots on a number line. All three methods express the same fact: the quadratic is positive where its parabola lies above the x -axis and negative where it lies below the axis.

Example 4.2: solving a quadratic inequality

Solve $x^2 + 2x > 15$.

First write the inequality in standard form and factorise:

$$x^2 + 2x - 15 > 0 \iff (x + 5)(x - 3) > 0.$$

The critical values are -5 and 3 . The product is positive when both factors have the same sign, which occurs outside the roots. Therefore

$$x < -5 \quad \text{or} \quad x > 3, \quad x \in (-\infty, -5) \cup (3, \infty).$$

If the original sign were $\leq$, the solution would instead be $[-5, 3]$.

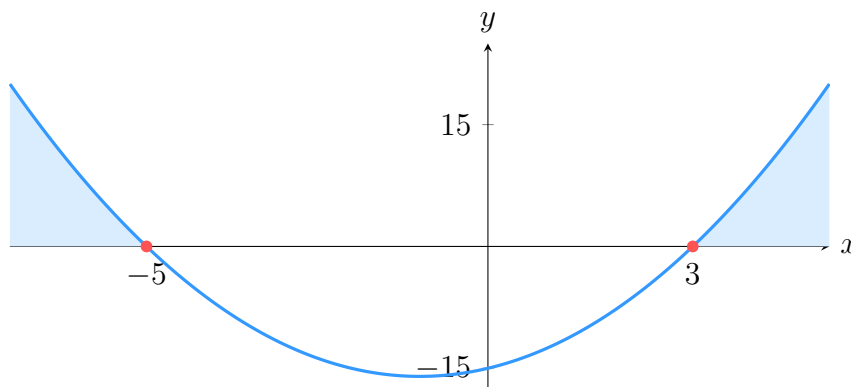


Figure 4.2: The shaded parts show where $x^2 + 2x - 15 > 0$.

The discriminant $\Delta = b^2 - 4ac$ also provides useful information. If $\Delta < 0$, the quadratic has no real zeros and keeps the sign of a for every real x . If $\Delta = 0$, it has a repeated root and does not change sign there. If $\Delta > 0$, it has two distinct roots; an upward-opening parabola is positive outside the roots and negative between them, with the pattern reversed for a downward-opening parabola.

For example, $x^2 + 2x + 5 = (x + 1)^2 + 4$ is positive for every real x , whereas $-(x - 2)^2 \leq 0$ for every real x and is equal to zero only at $x = 2$.

4.5 Cubic and higher-degree inequalities

For a polynomial inequality, factorisation reveals the real zeros and their multiplicities. Arrange the zeros in increasing order and determine the sign on the intervening intervals. A simple zero, or any zero of odd multiplicity, causes a sign change; a zero of even multiplicity does not.

Example 4.3: a cubic inequality

Solve

$$x^3 - x^2 - 4x + 4 < 0.$$

Factor by grouping:

$$x^2(x - 1) - 4(x - 1) = (x - 1)(x^2 - 4) = (x + 2)(x - 1)(x - 2).$$

The sign chart is

x	$(-\infty, -2)$	-2	$(-2, 1)$	1	$(1, 2)$	2	$(2, \infty)$
$(x + 2)(x - 1)(x - 2)$	$-$	0	$+$	0	$-$	0	$+$

Hence

$$x \in (-\infty, -2) \cup (1, 2).$$

The endpoints are omitted because the inequality is strict.

Repeated factors must be handled carefully. For instance, $(x - 2)^2(x + 1)$ has a double zero at 2, so the sign does not change there. Because $(x - 2)^2$ is positive except at 2, the inequality

$$(x - 2)^2(x + 1) < 0$$

has solution $x < -1$.

For expressions such as $x^4 - 5x^2 + 4$, a substitution can reduce the degree. Setting $u = x^2$ gives

$$u^2 - 5u + 4 = (u - 1)(u - 4),$$

after which the restrictions $u = x^2 \geq 0$ and the corresponding values of x must be restored.

4.6 Rational inequalities

A rational inequality compares a quotient of polynomials with zero, for example

$$\frac{P(x)}{Q(x)} \geq 0, \quad Q(x) \neq 0.$$

Both the zeros of P and the zeros of Q are critical values. Numerator zeros may be included in a non-strict inequality, but denominator zeros are never included because the original expression is undefined there.

It is generally unsafe to clear a variable denominator by multiplication: its sign may be unknown, and a negative denominator would reverse the inequality. Instead, combine the terms into one rational expression, factor the numerator and denominator, and use a sign chart.

Example 4.4: a rational sign chart

Solve

$$\frac{x+1}{x-3} < 0.$$

The numerator is zero at $x = -1$, while $x = 3$ is excluded. The signs are

x	$(-\infty, -1)$	-1	$(-1, 3)$	3	$(3, \infty)$
$x+1$	—	0	+	+	+
$x-3$	—	—	—	0	+
$\frac{x+1}{x-3}$	+	0	—	undefined	+

Therefore the quotient is negative precisely when

$$x \in (-1, 3).$$

Cancellation does not remove a restriction from the original expression. For example,

$$\frac{x^2 - 1}{x - 1} \geq 0, \quad x \neq 1,$$

simplifies to $x + 1 \geq 0$ only for $x \neq 1$. Its solution is consequently

$$[-1, 1) \cup (1, \infty),$$

not the uninterrupted interval $[-1, \infty)$.**4.7 Inequalities involving absolute values****4.7.1 Absolute-value notation and distance****Definition 4.2: Absolute value.**For a real number x ,

$$|x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0. \end{cases}$$

Thus $|x|$ is the distance of x from 0 on the real number line.More generally, $|x - a|$ is the distance between x and a . Therefore

$$|x - a| < r$$

describes points within distance r of a , whereas $|x - a| > r$ describes points more than distance r from a .

4.7.2 Properties of absolute values

For real a, b and nonzero b ,

$$\begin{aligned} |a| &= |-a| \geq 0, \\ |a - b| &= |b - a|, \\ |ab| &= |a| |b|, \\ \left| \frac{a}{b} \right| &= \frac{|a|}{|b|}, \\ |a| &= \sqrt{a^2}, \\ |a + b| &\leq |a| + |b|. \end{aligned}$$

The last result is the triangle inequality. A related result, the reverse triangle inequality, is

$$||a| - |b|| \leq |a - b|.$$

4.7.3 Equations involving absolute values

If $c \geq 0$, then

$$|u| = c \iff u = c \text{ or } u = -c.$$

If $c < 0$, the equation has no solution because an absolute value cannot be negative.

Example 4.5: solving by cases

Solve $|3x - 4| = 5$.

The two possible cases are

$$3x - 4 = 5 \quad \text{or} \quad 3x - 4 = -5.$$

They give $x = 3$ or $x = -1/3$. Both values satisfy the original equation.

When both sides are absolute values, squaring is valid because both sides are non-negative. For example,

$$|8x - 3| = |7 - 6x|$$

is equivalent to $(8x - 3)^2 = (7 - 6x)^2$, or to the two cases $8x - 3 = 7 - 6x$ and $8x - 3 = -(7 - 6x)$. Thus $x = 5/7$ or $x = -2$.

4.7.4 Inequalities involving absolute values

For $c > 0$, the basic equivalences are

$$\begin{aligned} |u| < c &\iff -c < u < c, \\ |u| \leq c &\iff -c \leq u \leq c, \\ |u| > c &\iff u < -c \text{ or } u > c, \\ |u| \geq c &\iff u \leq -c \text{ or } u \geq c. \end{aligned}$$

Before applying these forms, inspect the right-hand side. For example, $|u| < c$ has no solution when $c \leq 0$, while $|u| > c$ is true for every real u when $c < 0$.

Example 4.6: an absolute value of a quadratic

Solve

$$|x^2 - 4x + 2| \leq 2.$$

Using $|u| \leq 2 \iff -2 \leq u \leq 2$ gives

$$-2 \leq x^2 - 4x + 2 \leq 2.$$

The left inequality becomes $(x - 2)^2 \geq 0$, which is always true. The right inequality becomes $x(x - 4) \leq 0$, giving

$$x \in [0, 4].$$

For the related inequality $|x^2 - 4x + 2| \geq 2$, the solution is $(-\infty, 0] \cup \{2\} \cup [4, \infty)$.

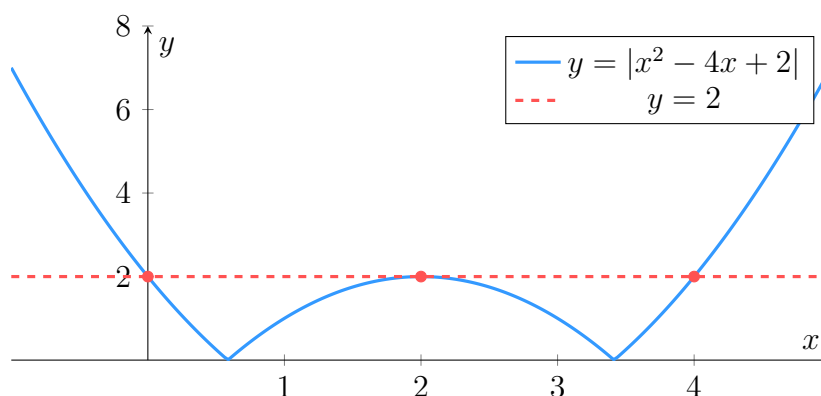


Figure 4.3: The graph lies at or below $y = 2$ for $0 \leq x \leq 4$.

Because absolute values are non-negative, comparisons such as $|f(x)| < |g(x)|$ may be squared without changing their meaning. For example,

$$\begin{aligned} |x - 3| > |2x + 5| &\iff (x - 3)^2 > (2x + 5)^2 \\ &\iff (x + 8)(3x + 2) < 0, \end{aligned}$$

so $-8 < x < -2/3$.

For rational expressions, domain restrictions must still be retained. To solve

$$\frac{1}{|x - 4|} \leq \frac{3}{|x + 1|},$$

first note that $x \neq 4, -1$. Multiplying by the positive quantity $|x - 4||x + 1|$ gives $|x + 1| \leq 3|x - 4|$. Squaring and simplifying yields

$$(x + 1)^2 \leq 9(x - 4)^2,$$

whose boundary values are $11/4$ and $13/2$. Restoring the excluded values, the solution is

$$(-\infty, -1) \cup (-1, 11/4] \cup [13/2, \infty).$$

Finally, when the right side contains a variable expression, its sign is part of the problem. In

$$|5x - 6| < 2x + 1,$$

the right side must be positive. Solving

$$-(2x + 1) < 5x - 6 < 2x + 1$$

gives $5/7 < x < 7/3$, which already satisfies $2x + 1 > 0$.

4.8 Choosing a method and checking solutions

Type	Recommended first approach
Linear or compound	Isolate the variable; reverse the sign after multiplying or dividing by a negative quantity.
Quadratic	Move to zero, factor or use the discriminant, then read the sign from the parabola or a sign chart.
Cubic or polynomial	Factor fully and track sign changes at roots, including their multiplicities.
Rational	Combine into one quotient, record denominator restrictions, and construct a sign chart.
Absolute value	Use distance equivalences, cases, a graph, or squaring when both sides are known to be non-negative.

Before finalising a solution, check that every endpoint satisfies the original inequality, that strict endpoints have been omitted, and that no value making a denominator zero has been included. A quick substitution from each proposed interval is often enough to detect a reversed sign or a missing restriction.

Chapter summary and common pitfalls

Key ideas. Critical values divide the number line into intervals on which an expression has a constant sign. A sign diagram or graph then identifies the required solution set.

Common pitfalls. Reverse an inequality when multiplying or dividing by a negative quantity, exclude values that make a denominator zero, and use open endpoints for strict inequalities.

4.9 Practice and exercises

Give solution sets in interval notation unless another form is requested. When a rational expression is involved, state any excluded values explicitly.

Linear inequalities, intervals, and fundamental properties

1. Convert between inequality and interval notation.

- (a) $x \leq 3$;
- (b) $-2 < x < 5$;
- (c) $[1, \infty)$;
- (d) $(-\infty, -4) \cup [2, \infty)$;
- (e) $x \neq 6$;
- (f) $-7 \leq x < 0$.

2. Find the values satisfying each pair or alternative of inequalities.

- (a) $x > 1$ and $x \leq 4$;
- (b) $1 - x \leq -2$ and $2x + 3 > 4x - 5$;
- (c) $2x > x - 3$ and $1 - x > 2x + 4$;
- (d) $x > 3$ or $x \geq -5$;
- (e) $2x > x - 3$ or $1 - x > 2x - 5$;
- (f) $4 \leq 2x \leq 10$ or $3 < x < 11$.

3. Solve the linear inequalities.

- (a) $-2x \geq 8$;
- (b) $3(2x + 1/5) \geq 3x - 2/5$;
- (c) $\frac{y}{3} - \frac{y-2}{2} \leq \frac{y}{4} - 4$;
- (d) $\frac{2}{3}(y+4) - \frac{y}{4} > \frac{1}{2}(3-y) + \frac{y}{6}$.

4. Solve the compound inequalities.

(a) $-6 < 3y - 2 \leq 4$;

(b) $24 \leq \frac{2}{3}(y - 5) < 36$;

(c) $y - 1 < 3y + 1 < y + 5$.

5. Solve each simultaneous nonlinear inequality.

(a) $3y + 4 \leq y^2 - 6 \leq 9 - 2y$;

(b) $3y < 2y - y^2 \leq 1$.

6. Correct each statement if necessary.

(a) Dividing $-3x < 12$ by -3 gives $x < -4$.

(b) If $a < b$ and $c < 0$, then $ac < bc$.

(c) If $0 < a < b$, then $1/a < 1/b$.

(d) If $a < b$, then $a^2 < b^2$.

7. Solve each system or alternative.

(a) $2x - 1 \leq 7$ and $5 - x < 8$;

(b) $x + 4 > 0$ and $3x - 2 \geq x + 6$;

(c) $4 - 2x \leq 0$ or $3x + 1 < -5$.

8. Given $2 \leq x < 5$, find the sharp range of

(a) $3x - 1$;

(b) $-2x + 4$;

(c) x^2 ;

(d) $1/x$.

9. List the integer solutions and state their number.

(a) $-3 < x \leq 4$;

(b) $2 \leq (3n + 1)/2 < 8$;

(c) $n^2 < 20$.

10. Write the solution represented by each number line in both inequality and interval notation.

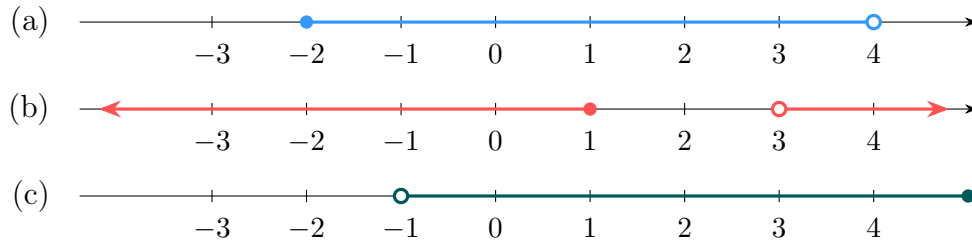


Figure 4.4: Number-line intervals for Question 10.

11. Solve and state whether the solution is all real numbers or the empty set.

(a) $3(x - 2) < 3x + 1$;

(b) $2(x + 4) > 2x + 9$;

(c) $5 - 2x \leq 5 - 2x$.

12. Count the integer solutions.

(a) $x^2 - 5x + 4 \leq 0$;

(b) $(x + 3)(x - 2) > 0$ for $-6 \leq x \leq 6$;

(c) $-10 < 3x + 2 \leq 11$.

13. Solve.

(a) $\frac{2x - 1}{3} \leq \frac{x + 5}{2}$;

(b) $\frac{3 - 2x}{5} > \frac{x + 1}{3}$;

(c) $1 - \frac{x - 2}{4} \geq \frac{2x + 3}{6}$.

14. Solve.

(a) $4(2x - 3) - 5 \leq 3(x + 2) + x$;

(b) $2 - 3(1 - 2x) > 5x + 7$;

(c) $5(x - 1) + 2 < 5x - 3$.

15. Explain why adding the same real number preserves an inequality, multiplying by a positive number preserves its direction, and multiplying by a negative number reverses its direction. Give one numerical example of each rule.

Quadratic, cubic, and higher-degree inequalities

16. Solve.

(a) $x^2 > 4$;

- (b) $x^2 \leq 8x$;
(c) $(1 - x)(4 - x) > x + 11$;
(d) $x^2 - 4x > -3$.
17. Solve $x^2 + 2x - 8 < 0$ graphically, and verify the result by factorisation and a sign chart.
18. Solve $x^2 - 10 > 3x$ by the graphical, sign-chart, and number-line methods.
19. Find the values of k for which the stated inequality holds for every real x .
- (a) $x^2 + kx + 9 > 0$;
(b) $x^2 - 4x + k \geq 0$;
(c) $-x^2 + kx - 4 < 0$.
20. Solve without factorising, using the discriminant and the sign of the leading coefficient.
- (a) $x^2 + 2x + 5 > 0$;
(b) $x^2 + 2x + 5 < 0$;
(c) $-2x^2 + 4x - 5 \leq 0$.
21. Solve, paying attention to repeated roots.
- (a) $(x - 3)^2 \leq 0$;
(b) $(2x + 1)^2 > 0$;
(c) $-(x + 4)^2 \geq 0$.
22. Determine where the first expression is no greater than the second.
- (a) $x^2 - 2x - 3$ and $2x + 5$;
(b) $2x^2 + x$ and $x^2 + 5x - 5$;
(c) $x + 1$ and $3 - x^2$.
23. Solve.
- (a) $(x + 3)(2 - x)(1 - 2x) > 0$;
(b) $x^3 + 5 \geq 5x^2 + x$;
(c) $(x^2 - 4x + 3)(x^2 - 3) < 0$.
24. Solve $(2 - x)(x + 3)(x + 4) > 0$ and sketch a cubic graph consistent with the sign changes.
25. Solve $x^3 - 2x^2 \geq 5x - 6$.

26. Solve $x^3 + x^2 - 25x - 25 \leq 0$.
27. Solve, noting the effect of even and odd multiplicities.
- (a) $(x - 2)^2(x + 1) < 0$;
 - (b) $(x + 3)^3(x - 1)^2 \geq 0$;
 - (c) $x^2(x - 4)(x + 2) \leq 0$.
28. Solve.
- (a) $(x^2 - 1)(x^2 - 9) > 0$;
 - (b) $(x^2 - 4)^2 \leq 0$;
 - (c) $(x - 1)(x + 2)(x^2 + 4) < 0$.
29. Construct a sign chart and solve.
- (a) $(x + 2)(x - 1)(x - 4) \leq 0$;
 - (b) $-(x + 3)^2(x - 2) > 0$;
 - (c) $(2x - 1)^3(x + 5) \geq 0$.
30. Use the substitution $u = x^2$.
- (a) $x^4 - 5x^2 + 4 \leq 0$;
 - (b) $x^4 - 13x^2 + 36 > 0$;
 - (c) $x^4 + 3x^2 - 4 \geq 0$.
31. Determine the unknown parameter or roots.
- (a) The solution of $x^2 - (k + 3)x + 3k \leq 0$ is $[3, 7]$. Find k .
 - (b) The solution of $(x - a)(x - b) > 0$ is $(-\infty, -2) \cup (6, \infty)$. Find a and b .
32. Solve $x^2 - 4x \leq 5 - x$. Interpret the result as the values for which the graph of $y = x^2 - 4x$ lies below the line $y = 5 - x$.
33. Find the values of k for which the inequality has at least one real solution.
- (a) $x^2 - 6x + k \leq 0$;
 - (b) $x^2 + kx + 4 < 0$;
 - (c) $(x - k)^2 + 1 \leq 0$.
34. Factorise first, then solve.
- (a) $x^3 - 4x^2 - x + 4 \geq 0$;

(b) $x^3 + 2x^2 - 5x - 6 < 0$.

35. Construct the least-degree monic polynomial having simple zeros at -2 and 4 and a double zero at 1 . Hence solve $p(x) \leq 0$.

Rational inequalities

36. Solve.

(a) $x + 2 \geq \frac{30}{x + 1}$;

(b) $\frac{2}{x + 1} > \frac{1}{x - 2}$;

(c) $\frac{3x + 1}{x - 4} \leq 1$;

(d) $\frac{x}{(x + 1)^2} > 0$.

37. Solve.

(a) $\frac{y^2 + 5y}{y - 3} \geq 0$;

(b) $\frac{2x}{x + 3} \leq 1$;

(c) $\frac{3}{x - 3} \leq \frac{2}{x + 2}$.

38. Solve $\frac{x}{(4 - x)^2} \geq 0$.

39. Solve $2x \geq \frac{1}{x}$.

40. Solve $\frac{x^2 + x - 6}{x - 4} \geq 0$.

41. Solve

$$\frac{x - 5}{2x + 1} > \frac{4x}{2x + 1} - 3.$$

42. Solve $\frac{x - 4}{x + 5} < 4$.

43. Solve $\frac{x^2 - 2x - 15}{x - 2} \geq 0$.

44. Simplify cautiously and retain every restriction from the original expression.

(a) $\frac{x^2 - 1}{x - 1} > 0$;

(b) $\frac{x^2 - 4}{x + 2} \leq 0;$

(c) $\frac{x^2 - 9}{x - 3} \geq 0.$

45. Solve, noting the repeated factors in the denominator or numerator.

(a) $\frac{x - 2}{(x + 1)^2} \leq 0;$

(b) $\frac{(x + 4)^2}{x - 1} > 0;$

(c) $\frac{(x - 3)^2}{(x + 2)^3} \geq 0.$

46. Construct a complete sign chart and solve.

(a) $\frac{(2x - 1)(x + 3)}{(x - 4)(x + 1)} < 0;$

(b) $\frac{(x - 1)^2(x + 5)}{x - 2} \geq 0;$

(c) $\frac{x^2 + 1}{x^2 - 4} \leq 0.$

47. Find the parameter.

(a) $\frac{x - k}{x - 2} > 0$ has solution $(-\infty, 2) \cup (5, \infty).$

(b) $\frac{x + 1}{x - k} \leq 0$ has solution $[-1, 3).$

48. Solve the reciprocal inequalities.

(a) $1/x > 1/3;$

(b) $2/x \leq -1;$

(c) $1/(x - 1) \geq 1/(x + 2).$

49. Solve each system or alternative.

(a) $\frac{x - 1}{x + 2} \geq 0$ and $x < 4;$

(b) $\frac{x}{x - 3} < 0$ or $x \geq 5;$

(c) $\frac{1}{x + 1} > 0$ and $\frac{x - 2}{x + 1} \leq 0.$

50. The graph shows $f(x) = 1/(x - 1)$ and $g(x) = 2/(x + 2)$. Use it to estimate, and then algebraically determine, the values for which $f(x) \leq g(x)$.

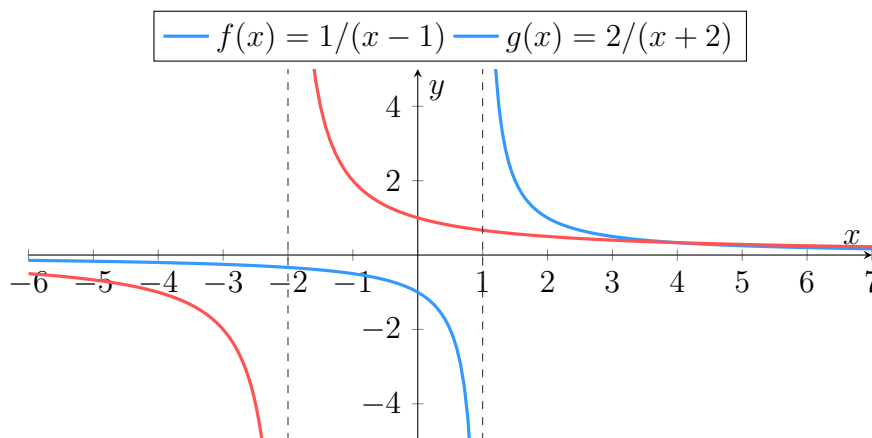


Figure 4.5: Two rational functions for Question 50.

Absolute-value equations and inequalities

51. Decide whether each statement is always true. If it is false, give a counterexample or the necessary condition.

- (a) $|ab| = |a||b|$;
- (b) $|a + b| = |a| + |b|$;
- (c) $|a|^2 = a^2$;
- (d) $|a| < b$ if and only if $-b < a < b$;
- (e) $|a - b|$ is the distance between a and b .

52. Solve.

- (a) $|y + 8| = 3$;
- (b) $|3x - 1| = -5$;
- (c) $|5y - 7| = 11$.

53. Solve.

- (a) $|6y| - 2y = 8$;
- (b) $|x^2 - 2x + 1| = x - 1$;
- (c) $|2x + 5| = |15 - 6x|$.

54. Solve the grouped linear absolute-value equations.

- (a) $|4x - 7| = 9$;
- (b) $|2 - 5x| = 0$;
- (c) $|3x + 1| = 4$.

55. Solve.

(a) $|x^2 - 5x + 6| = 0$;

(b) $|x^2 - 4| = 5$;

(c) $|x^2 - 2x| = 3$.

56. Solve $|3x - 4| = 5$ by splitting into cases, by graphing, and by squaring both sides. Explain why all three methods agree.

57. Solve $|8x - 3| = |7 - 6x|$.

58. Solve.

(a) $|x^2 - 3| < 2$;

(b) $|x^2 - 2x| \geq 24$;

(c) $\left| \frac{x+1}{x-1} \right| \leq 1$.

59. Use sketches of the relevant absolute-value graphs to solve.

(a) $|x/2 + 2| > 5$;

(b) $|x + 3| \geq 2|x - 3|$.

60. Solve $|3x + 4| < |x + 4|$ by cases, by graphing, and by squaring.

61. Solve by squaring both sides.

(a) $|3x - 9| \geq |2x + 1|$;

(b) $|x| > |3x - 4|$;

(c) $|x - 1| \leq |3x - 6|$.

62. Solve $|x^2 - 4x + 2| \leq 2$ using both the compound-inequality and graphical methods.

63. Solve $|x^2 - 4x + 2| \geq 2$ and compare its solution with that of Question 62.

64. Solve

$$\frac{1}{|x-4|} \leq \frac{3}{|x+1|}.$$

65. Solve $|x - 3| > |2x + 5|$ by at least two methods.

66. Solve $|5x - 6| < 2x + 1$, first identifying the values for which the right-hand side is positive.

67. Solve.

- (a) $|x - 2| + |x + 1| \leq 5$;
- (b) $|x| + |x - 4| > 6$.

68. Solve the nested absolute-value problems.

- (a) $||x| - 3| \leq 1$;
- (b) $||2x - 1| - 4| > 2$;
- (c) $||x + 2| - 1| = 2$.

69. Interpret the parameter geometrically.

- (a) Solve $|x - a| < r$ for $r > 0$.
- (b) If $|x - a| \leq r$ has solution $[-2, 8]$, find a and r .
- (c) Find the solution of $|2x - 5| < k$ for $k > 0$.

70. Solve.

- (a) $||x - 1| - 2| = 0$;
- (b) $||x| - 2| = 1$;
- (c) $|x + 1| = 2 - |x + 1|$.

71. Solve.

- (a) $|2x - 1| < |x + 5|$;
- (b) $|x + 2| \geq |3x - 1|$;
- (c) $|x - 4| \leq 2|x + 1|$.

72. Solve.

- (a) $\left| \frac{x - 2}{x + 1} \right| < 1$;
- (b) $\left| \frac{1}{x - 3} \right| \geq 2$;
- (c) $\left| \frac{2x + 1}{x - 4} \right| > 3$.

73. Write each function without absolute-value signs. Hence answer the additional request.

- (a) $|2x - 4|$;
- (b) $|x + 1| - |x - 2|$, and solve when it equals 1;
- (c) $|x^2 - 4|$.

74. Find each union or intersection.

- (a) $|x - 2| \leq 3$ and $|x + 1| < 2$;
(b) $|2x - 1| > 3$ and $|x| \leq 4$;
(c) $|x - 5| \leq 2$ or $|x + 2| < 1$.
75. Prove the triangle inequality $|a + b| \leq |a| + |b|$. Hence derive the reverse triangle inequality $||a| - |b|| \leq |a - b|$.

Applications and modelling

76. A chair is designed to be 100 cm high with an error of no more than 2 cm. Write an absolute-value inequality for an acceptable height h and state the allowable range.
77. A shaft should have diameter 12.00 mm with tolerance 0.03 mm. Write the acceptance inequality and interval.
78. A cold-storage room should remain within 2°C of -18°C . Find the safe temperature interval and an inequality that triggers an alarm outside it.
79. An elevator carries an operator of mass 75 kg and passengers whose average mass is 68 kg. If its load limit is 800 kg, find the greatest whole number of passengers.
80. A function venue costs RM 1,200 plus RM 45 per attendee. If the budget is RM 3,000, find the greatest possible number of attendees.
81. Phone plan A costs $\text{RM } 25 + 0.08m$ and plan B costs $\text{RM } 40 + 0.03m$, where m is the monthly number of minutes. For what usage is plan A no more expensive than plan B?
82. A small firm's profit in thousands of ringgit is

$$P(x) = -2x^2 + 120x - 1000,$$

where x is the number of hundreds of units sold. Find the sales range for which the firm makes a positive profit.

83. A projectile has height $h(t) = -5t^2 + 30t + 2$ metres. During what time interval is it at least 32 m above the ground?
84. A rectangular garden has width x metres and length $x + 4$ metres. Find the possible widths if its area must be at least 96 m^2 .
85. A simplified braking-distance model is $d(v) = 0.05v^2 + 0.5v$ metres. Find the allowable non-negative speeds if the available stopping distance is at most 60 m.

86. A parabolic arch has height

$$y = 12 - \frac{x^2}{9}$$

metres. A vehicle is 4 m wide and centred below the arch. Find its greatest possible height, and the permitted height if a 0.5 m vertical safety clearance is required.

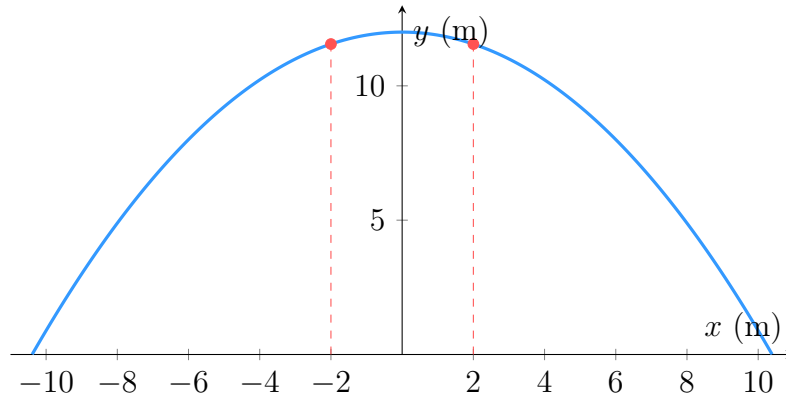


Figure 4.6: Parabolic clearance for Question 86.

87. A medicine concentration is modelled by $C(t) = 40/(t + 2)$ milligrams per litre for $t \geq 0$. Determine when the concentration lies between 5 and 10 milligrams per litre, inclusive.
88. The average production cost of x components is $A(x) = 500/x + 8$ ringgit per component. How many components must be made for the average cost to be at most RM 13?
89. In a simple circuit, $I = 12/(R + 2)$ amperes, where $R \geq 0$ is a resistance in ohms. Find the resistance range for which $1 \leq I \leq 3$.
90. A tank contains 20 litres of solution with 4 kilograms of dissolved material. If x litres of pure concentrate are added, find the values of $x \geq 0$ for which the final concentration is between 25% and 40%.
91. Points on a straight road are represented by their coordinate x in kilometres. Find the locations within 7 km of the marker at $x = 12$.
92. A component passes both inspections only if

$$|L - 50| \leq 0.4 \quad \text{and} \quad |L - 50.2| \leq 0.3.$$

Find the common acceptable length range in millimetres.

93. Two service centres lie at coordinates 2 and 10 on a straight road. Find the points that are strictly closer to the first centre.
94. A signal deviation is $e(t) = t^2 - 6t + 5$. Find the times for which $|e(t)| \leq 4$.

95. A standardised reading is accepted when $|z| \leq 2.5$. Classify -3.1 , -2.5 , 0.4 , and 2.6 as accepted or flagged.
96. A qualifying score must be within 8 marks of 80 and must also be at least 75. Find the qualifying score interval.
97. Squares of side x centimetres are removed from the corners of a 12 cm by 8 cm card. Find the values of x for which the open box has volume at least 60 cm^3 .
98. A parking permit costs RM 900 plus RM 50 for each reserved space. A department may spend at most RM 1,200. Find the possible whole-number values for the number of reserved spaces.
99. A 2 m metal bar has length

$$L(T) = 2[1 + 12 \times 10^{-6}(T - 20)]$$

metres at temperature $T^\circ\text{C}$. Find the temperatures for which its length differs from 2 m by at most 0.001 m.

100. A workshop's daily profit is

$$P(n) = -n^2 + 40n - 300, \quad 5 \leq n \leq 50,$$

where n is the number of completed jobs.

- (a) Find the whole-number values of n for which the workshop does not make a loss.
- (b) Find the number of jobs giving maximum profit and state that profit.

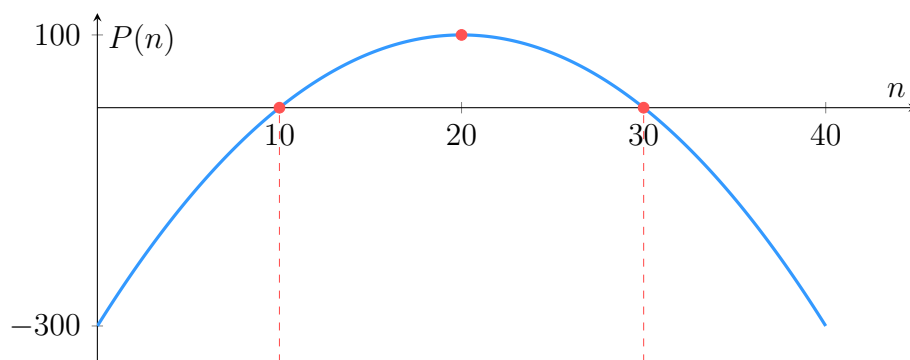


Figure 4.7: Profit and the break-even points in Question 100.

Chapter 5

Matrices

Chapter guide

Prerequisites. Simultaneous linear equations and basic algebra.

Learning outcomes. Readers should be able to identify matrix types, perform valid matrix operations, evaluate determinants, find inverses, and use matrices to solve systems of linear equations and represent transformations.

Matrices arrange data in rows and columns and provide a compact language for many mathematical and computational processes. They are fundamental to linear algebra and are used to solve systems of equations, represent transformations, organise data, and perform calculations in computer programs.

This chapter introduces matrix notation, arithmetic, determinants, inverses, and the solution of simultaneous linear equations.

The treatment is informed by matriculation and foundation-level presentations of matrices and systems of equations (Chai et al. 2024; Jalaluddin, Mazlin et al. 2025–2026).

5.1 Introduction to matrices

A **matrix** is a rectangular array of numbers arranged in rows and columns and enclosed by brackets or parentheses. For example,

$$\begin{bmatrix} 2 & 3 & 7 \\ 1 & 4 & 5 \end{bmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 4 & 8 \\ 2 & 3 & 7 \\ 6 & 5 & 1 \end{pmatrix}$$

are matrices. The first has two rows and three columns, while the second has three rows and three columns.

5.1.1 Order of a matrix

The size, or **order**, of a matrix is determined by its number of rows and columns. If a matrix A has m rows and n columns, then A is called an $m \times n$ matrix. The notation $m \times n$ is read as “ m by n ,” where the first number gives the number of rows and the second gives the number of columns.

Thus, the matrices above have orders 2×3 and 3×3 , respectively. A matrix with the same number of rows and columns is called a **square matrix**.

5.1.2 Entries of a matrix

A general $m \times n$ matrix may be written as

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = [a_{ij}]_{m \times n}. \quad (5.1)$$

Each number in a matrix is called an **entry**, or **element**. The notation a_{ij} denotes the entry in the i th row and j th column. For instance, a_{23} is the entry in the second row and third column.

The **leading entry** of a nonzero row is its first nonzero entry when read from left to right. Leading entries are particularly important when matrices are reduced to echelon form.

The **main diagonal**, also called the **leading diagonal**, consists of the entries

$$a_{11}, a_{22}, a_{33}, \dots, a_{rr}, \quad r = \min(m, n).$$

For a square matrix of order n , the main diagonal is $a_{11}, a_{22}, \dots, a_{nn}$.

Figure 5.1 shows how the two subscripts of an entry identify its position. The row number is always written first and the column number second.

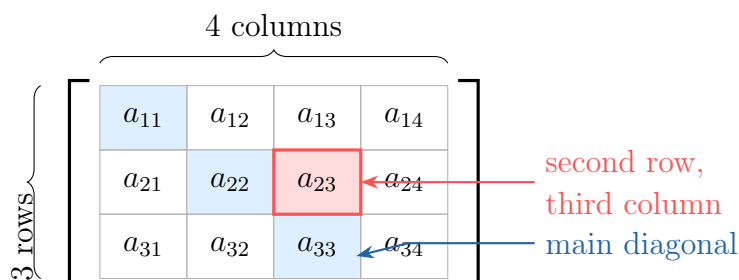
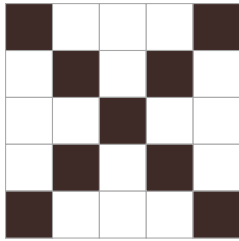


Figure 5.1: Anatomy of a 3×4 matrix. The entry a_{23} and the main diagonal are highlighted.

Matrices can also encode images. In a binary image, an entry 1 may denote a dark pixel and an entry 0 a light pixel. The position of each matrix entry then determines the position of the corresponding pixel.



binary pixel image

$$P = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Each row and column of P corresponds to a row and column of pixels in the image.

Figure 5.2: A 5×5 binary image represented by a matrix. Digital images use the same principle with many more pixels and intensity values.

5.2 Types of matrices

Matrices may be classified according to their order and the arrangement of their entries. Some common types are described below.

1. Row and column matrices

A **row matrix**, also called a row vector, contains only one row and therefore has order $1 \times n$. For example,

$$R = [2 \quad 5 \quad 1].$$

A **column matrix**, also called a column vector, contains only one column and therefore has order $m \times 1$. For example,

$$C = \begin{bmatrix} 1 \\ 0 \\ 6 \end{bmatrix}.$$

2. Square matrix

A square matrix has the same number of rows and columns. A square matrix with n rows and n columns is said to be of order n . For example,

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

is a square matrix of order 2.

3. Zero matrix

A zero matrix, or null matrix, is a matrix in which every entry is zero. It is commonly denoted by O or $O_{m \times n}$. For example,

$$O_{2 \times 2} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

4. Diagonal matrix

A diagonal matrix is a square matrix in which every entry outside the main diagonal is zero. The entries on the main diagonal may be zero or nonzero. For example,

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

5. Identity matrix

An identity matrix is a diagonal matrix whose entries on the main diagonal are all 1. The identity matrix of order n is denoted by I_n . For example,

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

If A is a square matrix of order n , then multiplication by the identity matrix leaves A unchanged:

$$AI_n = I_nA = A.$$

6. Upper triangular matrix

An upper triangular matrix is a square matrix in which every entry below the main diagonal is zero. Thus, $a_{ij} = 0$ whenever $i > j$. For example,

$$U = \begin{bmatrix} 1 & 4 & 5 \\ 0 & 2 & 6 \\ 0 & 0 & 3 \end{bmatrix}.$$

7. Lower triangular matrix

A lower triangular matrix is a square matrix in which every entry above the main diagonal is zero. Thus, $a_{ij} = 0$ whenever $i < j$. For example,

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 2 & 0 \\ 5 & 6 & 3 \end{bmatrix}.$$

8. Symmetric matrix

A symmetric matrix is a square matrix whose entries are unchanged when reflected across the main diagonal. Equivalently,

$$a_{ij} = a_{ji} \quad \text{for all } i, j.$$

For example,

$$S = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}.$$

A matrix is symmetric precisely when $S^T = S$, where S^T denotes its transpose.

5.3 Arithmetic operations of matrices

Matrices support arithmetic operations, but the dimensions of the operands determine whether each operation is defined.

5.3.1 Addition and subtraction of matrices

Two matrices can be added or subtracted only when they have the same order. If

$$A = [a_{ij}]_{m \times n} \quad \text{and} \quad B = [b_{ij}]_{m \times n},$$

then their sum and difference are found by adding or subtracting corresponding entries:

$$A + B = [a_{ij} + b_{ij}]_{m \times n}, \quad (5.2)$$

$$A - B = [a_{ij} - b_{ij}]_{m \times n}. \quad (5.3)$$

Example 5.1

Let

$$A = \begin{bmatrix} 3 & 5 \\ 2 & -3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 4 \\ 3 & 0 \end{bmatrix}.$$

Find $A + B$ and $A - B$.

Solution. Adding and subtracting corresponding entries gives

$$A + B = \begin{bmatrix} 3+1 & 5+4 \\ 2+3 & -3+0 \end{bmatrix} = \begin{bmatrix} 4 & 9 \\ 5 & -3 \end{bmatrix},$$

$$A - B = \begin{bmatrix} 3-1 & 5-4 \\ 2-3 & -3-0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ -1 & -3 \end{bmatrix}.$$

If A , B , and C have the same order and O is the zero matrix of that order, matrix addition satisfies

$$A + B = B + A \quad (\text{commutative law}), \quad (5.4)$$

$$A + (B + C) = (A + B) + C \quad (\text{associative law}), \quad (5.5)$$

$$A + O = O + A = A \quad (\text{additive identity}), \quad (5.6)$$

$$A + (-A) = (-A) + A = O \quad (\text{additive inverse}). \quad (5.7)$$

5.3.2 Scalar multiplication of matrices

A **scalar** is an ordinary number. If k is a scalar and $A = [a_{ij}]$, then the scalar product kA is obtained by multiplying every entry of A by k :

$$kA = [ka_{ij}]. \quad (5.8)$$

For example, if

$$A = \begin{bmatrix} 1 & -2 \\ 4 & 3 \end{bmatrix},$$

then

$$2A = 2 \begin{bmatrix} 1 & -2 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 2 & -4 \\ 8 & 6 \end{bmatrix}.$$

For matrices A and B of the same order and scalars k, k_1 , and k_2 ,

$$k(A + B) = kA + kB, \quad (5.9)$$

$$(k_1 + k_2)A = k_1A + k_2A, \quad (5.10)$$

$$k_1(k_2A) = (k_1k_2)A. \quad (5.11)$$

5.3.3 Matrix multiplication

The product of a row matrix and a column matrix with the same number of entries is obtained by multiplying corresponding entries and then adding the results. For example,

$$\begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = ac + bd. \quad (5.12)$$

Hence,

$$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} = 1(2) + 2(-1) + 3(4) = 12.$$

More generally, the product AB is defined only when the number of columns of A equals the number of rows of B . If

A has order $m \times p$ and B has order $p \times n$,

then AB exists and has order $m \times n$:

$$\underset{m \times p}{A} \underset{p \times n}{B} = \underset{m \times n}{AB}. \quad (5.13)$$

Each entry of AB is obtained by multiplying a row of A by a column of B . If $A = [a_{ij}]$ and $B = [b_{ij}]$, then

$$(AB)_{ij} = \sum_{r=1}^p a_{ir} b_{rj}. \quad (5.14)$$

Example 5.2

Let

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & -1 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -3 \\ 5 & 1 \\ 4 & 2 \end{bmatrix}.$$

Since A has order 2×3 and B has order 3×2 , the product AB exists and has order 2×2 . Applying the row-by-column rule,

$$\begin{aligned} AB &= \begin{bmatrix} 1(1) + 0(5) + 2(4) & 1(-3) + 0(1) + 2(2) \\ 3(1) + (-1)(5) + 4(4) & 3(-3) + (-1)(1) + 4(2) \end{bmatrix} \\ &= \begin{bmatrix} 9 & 1 \\ 14 & -2 \end{bmatrix}. \end{aligned}$$

Figure 5.3 isolates one calculation from the product. The highlighted second row of A is paired with the highlighted first column of B , producing the entry in row 2, column 1 of AB .

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & -1 & 4 \end{bmatrix} \times B = \begin{bmatrix} 1 & -3 \\ 5 & 1 \\ 4 & 2 \end{bmatrix} = AB = \begin{bmatrix} 9 & 1 \\ 14 & -2 \end{bmatrix}$$

$$3(1) + (-1)(5) + 4(4) = 14$$

row 2 times column 1

Figure 5.3: The row-by-column calculation of the $(2, 1)$ entry of AB .

The order of the factors is important. Consider

$$P = \begin{bmatrix} 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}, \quad Q = \begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix}.$$

The product PQ is not defined because the three columns of P do not match the two rows of Q . However, QP is defined and has order 2×3 :

$$\begin{aligned} QP &= \begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 14 & 8 & 2 \\ 9 & 5 & 1 \end{bmatrix}. \end{aligned}$$

5.3.4 Properties of matrix multiplication

Whenever the orders of the matrices make the stated products and sums meaningful, matrix multiplication satisfies the following properties:

$$A(B + C) = AB + AC \quad (\text{left distributive law}), \quad (5.15)$$

$$(A + B)C = AC + BC \quad (\text{right distributive law}), \quad (5.16)$$

$$A(BC) = (AB)C \quad (\text{associative law}), \quad (5.17)$$

$$AI = IA = A \quad (\text{multiplicative identity}). \quad (5.18)$$

If O is a zero matrix of the required order, then

$$AO = O \quad \text{and} \quad OA = O. \quad (5.19)$$

In general, matrix multiplication is *not commutative*; even when both products exist,

$$AB \neq BA \quad (5.20)$$

may occur. The two products can also have different orders, or one may exist while the other does not.

If A is a square matrix, its positive integer powers are defined by repeated multiplication:

$$A^n = \underbrace{A \cdot A \cdots A}_{n \text{ factors}}. \quad (5.21)$$

For positive integers p and q ,

$$A^p A^q = A^{p+q}, \quad (5.22)$$

$$(A^p)^q = A^{pq}. \quad (5.23)$$

Furthermore, $I^n = I$ for every positive integer n .

Example 5.3

Let

$$D = \begin{bmatrix} -2 & 0 \\ 0 & 3 \end{bmatrix}.$$

Then

$$D^2 = \begin{bmatrix} 4 & 0 \\ 0 & 9 \end{bmatrix}, \quad D^3 = \begin{bmatrix} -8 & 0 \\ 0 & 27 \end{bmatrix}.$$

The pattern gives

$$D^n = \begin{bmatrix} (-2)^n & 0 \\ 0 & 3^n \end{bmatrix} \quad (5.24)$$

for every positive integer n .

5.4 Transpose of matrices

The **transpose** of a matrix is obtained by interchanging its rows and columns. If A is an $m \times n$ matrix, then its transpose, denoted by A^T , is an $n \times m$ matrix. In entry notation,

$$(A^T)_{ij} = a_{ji}. \quad (5.25)$$

For example, if

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix},$$

then

$$A^T = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}.$$

The transpose may be viewed as a reflection of the entries across the main diagonal. In particular, the entry in position (i, j) moves to position (j, i) .

$$A = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} \xrightarrow{\text{transpose}} A^T = \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}$$

$$f = a_{23} \qquad f = (A^T)_{32}$$

Figure 5.4: Transposition interchanges rows and columns, moving each a_{ij} to position (j, i) .

5.4.1 Properties of transposes

Let A and B be matrices of suitable orders and let k be a scalar. Then

$$(kA)^T = kA^T, \quad (5.26)$$

$$(A^T)^T = A, \quad (5.27)$$

$$(A + B)^T = A^T + B^T, \quad (5.28)$$

$$(A - B)^T = A^T - B^T, \quad (5.29)$$

$$(AB)^T = B^T A^T. \quad (5.30)$$

The last property reverses the order of the factors. This reversal is essential because matrix multiplication is generally not commutative.

Example 5.4

Let

$$A = \begin{bmatrix} 1 & 3 & 0 \\ 0 & 2 & 4 \\ 5 & 0 & 6 \end{bmatrix}.$$

Verify that $(kA)^T = kA^T$.

Solution. Multiplying first and then transposing gives

$$\begin{aligned} (kA)^T &= \begin{bmatrix} k & 3k & 0 \\ 0 & 2k & 4k \\ 5k & 0 & 6k \end{bmatrix}^T \\ &= \begin{bmatrix} k & 0 & 5k \\ 3k & 2k & 0 \\ 0 & 4k & 6k \end{bmatrix}. \end{aligned}$$

On the other hand,

$$\begin{aligned}
 kA^T &= k \begin{bmatrix} 1 & 0 & 5 \\ 3 & 2 & 0 \\ 0 & 4 & 6 \end{bmatrix} \\
 &= \begin{bmatrix} k & 0 & 5k \\ 3k & 2k & 0 \\ 0 & 4k & 6k \end{bmatrix}.
 \end{aligned}$$

Therefore, $(kA)^T = kA^T$.

Example 5.5

Suppose

$$A^T = \begin{bmatrix} 1 & -2 & 5 \\ 4 & 1 & -1 \\ -3 & 2 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 4 \\ 3 & -1 \\ 0 & -3 \end{bmatrix}.$$

Find $(B^T A)^T$ as a single matrix.

Solution. Using the transpose-of-a-product property,

$$(B^T A)^T = A^T (B^T)^T = A^T B.$$

Hence,

$$\begin{aligned}
 (B^T A)^T &= \begin{bmatrix} 1 & -2 & 5 \\ 4 & 1 & -1 \\ -3 & 2 & 0 \end{bmatrix} \begin{bmatrix} 2 & 4 \\ 3 & -1 \\ 0 & -3 \end{bmatrix} \\
 &= \begin{bmatrix} -4 & -9 \\ 11 & 18 \\ 0 & -14 \end{bmatrix}.
 \end{aligned}$$

5.4.2 Symmetric and skew-symmetric matrices

A matrix B is **symmetric** if it is square and

$$B^T = B. \quad (5.31)$$

Equivalently, its entries satisfy $b_{ij} = b_{ji}$ for every i and j . Thus, the entries on opposite sides of the main diagonal are equal. For example,

$$B = \begin{bmatrix} 1 & -1 & 3 \\ -1 & 4 & 2 \\ 3 & 2 & 3 \end{bmatrix}$$

is symmetric because

$$B^T = \begin{bmatrix} 1 & -1 & 3 \\ -1 & 4 & 2 \\ 3 & 2 & 3 \end{bmatrix} = B.$$

A square matrix B is **skew-symmetric** if

$$B^T = -B. \quad (5.32)$$

Equivalently, $b_{ij} = -b_{ji}$ for every i and j . Every diagonal entry of a skew-symmetric matrix must be zero because

$$b_{ii} = -b_{ii} \implies b_{ii} = 0.$$

For example, let

$$B = \begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & -3 \\ -2 & 3 & 0 \end{bmatrix}.$$

Then

$$B^T = \begin{bmatrix} 0 & -1 & -2 \\ 1 & 0 & 3 \\ 2 & -3 & 0 \end{bmatrix} = -B,$$

so B is skew-symmetric.

5.5 Determinant of matrices

A **determinant** is a scalar value associated with a square matrix. The determinant of a matrix A is denoted by

$$\det(A) \quad \text{or} \quad |A|.$$

Determinants are defined only for square matrices.

5.5.1 Determinant of a 2×2 matrix

If

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix},$$

then

$$\boxed{|A| = a_{11}a_{22} - a_{12}a_{21}}. \quad (5.33)$$

Thus, the product along the downward diagonal from left to right is reduced by the product along the other diagonal.

For a 2×2 matrix, the determinant also has a geometric meaning. If the columns of A transform the unit coordinate vectors, then $|\det(A)|$ is the factor by which areas are scaled. A negative determinant additionally indicates that orientation has been reversed.

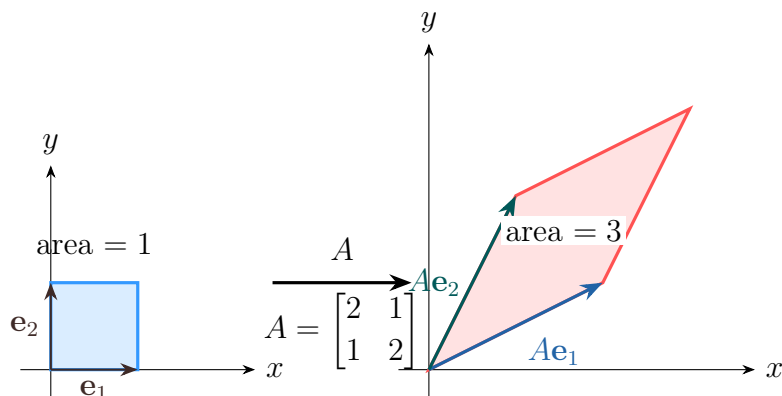


Figure 5.5: The transformation A maps the unit square to a parallelogram whose area is $|\det(A)| = |2(2) - 1(1)| = 3$.

Example 5.6

Given

$$A = \begin{bmatrix} 2 & 4 \\ -2 & -3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 \\ 3 & 6 \end{bmatrix},$$

find $|A|$ and $|B|$.

Solution.

$$|A| = 2(-3) - 4(-2) = -6 + 8 = 2,$$

$$|B| = 1(6) - 1(3) = 3.$$

5.5.2 Minors and cofactors

Let A be a square matrix. The **minor** of the entry a_{ij} , denoted by M_{ij} , is the determinant of the matrix obtained by deleting the i th row and the j th column of A .

The **cofactor** of a_{ij} is denoted by C_{ij} and is defined by

$$C_{ij} = (-1)^{i+j} M_{ij}. \quad (5.34)$$

For a 3×3 matrix, the cofactor signs follow the alternating pattern

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}.$$

Example 5.7

Let

$$A = \begin{bmatrix} 3 & -1 & 4 \\ -2 & 5 & 2 \\ 1 & 4 & -3 \end{bmatrix}.$$

Find M_{12} and M_{33} .

Solution. To obtain M_{12} , delete the first row and second column:

$$M_{12} = \begin{vmatrix} -2 & 2 \\ 1 & -3 \end{vmatrix} = (-2)(-3) - 2(1) = 4.$$

To obtain M_{33} , delete the third row and third column:

$$M_{33} = \begin{vmatrix} 3 & -1 \\ -2 & 5 \end{vmatrix} = 3(5) - (-1)(-2) = 13.$$

The corresponding cofactors are $C_{12} = -4$ and $C_{33} = 13$.

5.5.3 Cofactor expansion

The determinant of a square matrix may be evaluated by expanding along any row or any column. For a 3×3 matrix, expansion along the i th row gives

$$|A| = a_{i1}C_{i1} + a_{i2}C_{i2} + a_{i3}C_{i3} = \sum_{j=1}^3 a_{ij}C_{ij}. \quad (5.35)$$

Expansion along the j th column gives

$$|A| = a_{1j}C_{1j} + a_{2j}C_{2j} + a_{3j}C_{3j} = \sum_{i=1}^3 a_{ij}C_{ij}. \quad (5.36)$$

A row or column containing several zeros is usually the most convenient choice.

Example 5.8

Evaluate

$$\begin{vmatrix} 2 & 1 & -3 \\ 4 & 0 & 1 \\ 1 & 2 & 3 \end{vmatrix}.$$

Solution. Expanding along the first row,

$$\begin{aligned}
|A| &= 2 \begin{vmatrix} 0 & 1 \\ 2 & 3 \end{vmatrix} - 1 \begin{vmatrix} 4 & 1 \\ 1 & 3 \end{vmatrix} + (-3) \begin{vmatrix} 4 & 0 \\ 1 & 2 \end{vmatrix} \\
&= 2(0 - 2) - (12 - 1) - 3(8 - 0) \\
&= -4 - 11 - 24 \\
&= -39.
\end{aligned}$$

Example 5.9

Evaluate

$$\begin{vmatrix} 3 & 0 & 1 \\ 2 & 4 & 0 \\ -3 & 1 & 5 \end{vmatrix}.$$

Solution. Expanding along the first row,

$$\begin{aligned}
|B| &= 3 \begin{vmatrix} 4 & 0 \\ 1 & 5 \end{vmatrix} - 0 \begin{vmatrix} 2 & 0 \\ -3 & 5 \end{vmatrix} + 1 \begin{vmatrix} 2 & 4 \\ -3 & 1 \end{vmatrix} \\
&= 3(20) + (2 + 12) \\
&= 74.
\end{aligned}$$

5.5.4 Diagonal expansion for a 3×3 matrix

For a 3×3 matrix, the determinant may also be calculated using the diagonal method, commonly called **Sarrus' rule**. If

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix},$$

then

$$\begin{aligned}
|A| &= a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} \\
&\quad - a_{13}a_{22}a_{31} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33}.
\end{aligned} \tag{5.37}$$

Figure 5.6 displays the bookkeeping behind this formula. The first two columns are copied to the right: products along the three downward diagonals are added, and products along the three upward diagonals are subtracted.

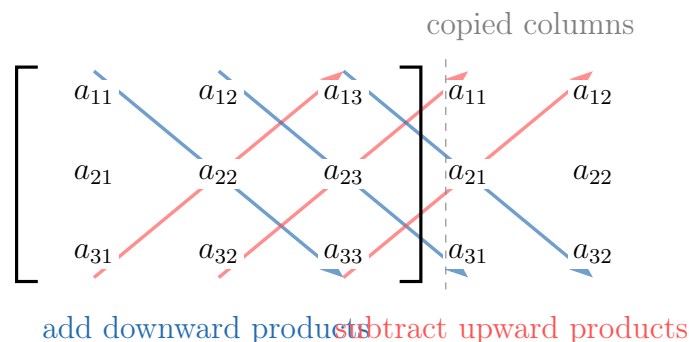


Figure 5.6: Sarrus' rule for a 3×3 determinant. It is not valid for determinants of higher order.

For example,

$$\begin{aligned}
 \begin{vmatrix} 2 & 1 & -3 \\ 4 & 0 & 1 \\ 1 & 2 & 3 \end{vmatrix} &= 2(0)(3) + 1(1)(1) + (-3)(4)(2) \\
 &\quad - (-3)(0)(1) - 2(1)(2) - 1(4)(3) \\
 &= 0 + 1 - 24 - 0 - 4 - 12 \\
 &= -39,
 \end{aligned}$$

which agrees with the cofactor expansion. Sarrus' rule applies only to determinants of order 3; it must not be extended directly to larger matrices.

5.5.5 Properties of determinants

Let A and B be square matrices of order n and let k be a scalar. The following properties are useful:

1. Multiplying the entire matrix by k multiplies each of its n rows by k . Therefore,

$$|kA| = k^n |A|. \quad (5.38)$$

2. The determinant of a product is the product of the determinants:

$$|AB| = |A||B|. \quad (5.39)$$

3. A matrix and its transpose have the same determinant:

$$|A^T| = |A|. \quad (5.40)$$

4. Interchanging two rows, or interchanging two columns, reverses the sign of the determinant.

5. Multiplying one row or one column by k multiplies the determinant by k .
6. Adding a multiple of one row to another row does not change the determinant. The analogous result holds for columns.
7. If two rows or two columns are identical, or proportional, then the determinant is zero.
8. If A is upper or lower triangular, then its determinant is the product of its diagonal entries:

$$|A| = a_{11}a_{22} \cdots a_{nn}. \quad (5.41)$$

Example 5.10

Let

$$A = \begin{bmatrix} 1 & -1 & 0 \\ -3 & x-2 & 2 \\ 0 & -2 & x+1 \end{bmatrix}$$

and suppose that $|A| = -5$. Find:

- (a) the value of x ;
- (b) $|A^T|$;
- (c) $|A^3|$.

Solution. Expanding along the first row,

$$\begin{aligned} |A| &= \begin{vmatrix} x-2 & 2 \\ -2 & x+1 \end{vmatrix} + \begin{vmatrix} -3 & 2 \\ 0 & x+1 \end{vmatrix} \\ &= (x-2)(x+1) + 4 - 3(x+1) \\ &= x^2 - 4x - 1. \end{aligned}$$

Since $|A| = -5$,

$$\begin{aligned} x^2 - 4x - 1 &= -5, \\ x^2 - 4x + 4 &= 0, \\ (x-2)^2 &= 0. \end{aligned}$$

Therefore,

$$\boxed{x = 2}.$$

Using the determinant properties,

and

$$|A^T| = |A| = -5$$

$$|A^3| = |A|^3 = (-5)^3 = -125.$$

5.6 Inverse of matrices

Let A be a square matrix of order n . If there exists a matrix B of the same order such that

$$AB = BA = I_n, \quad (5.42)$$

then B is called the **inverse** of A . The inverse of A is denoted by A^{-1} , so

$$AA^{-1} = A^{-1}A = I_n. \quad (5.43)$$

An inverse therefore reverses the action of the original matrix. Applying A and then A^{-1} has the same net effect as applying the identity matrix.

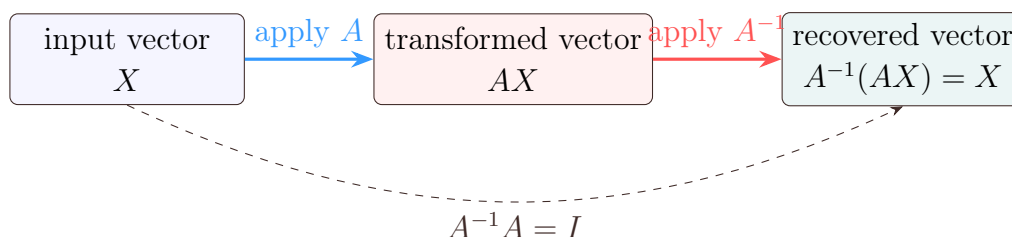


Figure 5.7: The inverse matrix undoes the transformation produced by A .

Example 5.11

Let

$$A = \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & -5 \\ -1 & 2 \end{bmatrix}.$$

Show that A is the inverse of B .

Solution. We calculate both products:

$$\begin{aligned} AB &= \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 3 & -5 \\ -1 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 6 - 5 & -10 + 10 \\ 3 - 3 & -5 + 6 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \end{aligned}$$

and

$$\begin{aligned}
 BA &= \begin{bmatrix} 3 & -5 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix} \\
 &= \begin{bmatrix} 6-5 & 15-15 \\ -2+2 & -5+6 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.
 \end{aligned}$$

Therefore, $AB = BA = I_2$, and hence

$$A = B^{-1} \quad \text{and} \quad B = A^{-1}.$$

5.6.1 Inverse of a 2×2 matrix

Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

Consider the matrix

$$\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Multiplying gives

$$\begin{aligned}
 \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} &= \begin{bmatrix} ad-bc & 0 \\ 0 & ad-bc \end{bmatrix} \\
 &= (ad-bc)I_2.
 \end{aligned}$$

Since $|A| = ad - bc$, division by the determinant gives the identity matrix whenever $|A| \neq 0$. Therefore,

$$\boxed{A^{-1} = \frac{1}{|A|} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}} \quad \text{where} \quad |A| = ad - bc \neq 0. \quad (5.44)$$

To find the inverse of a 2×2 matrix:

1. interchange the two entries on the main diagonal;
2. reverse the signs of the two off-diagonal entries;
3. divide every entry by the determinant.

A square matrix with a nonzero determinant is called **nonsingular**, or **invertible**. If $|A| = 0$, then A^{-1} does not exist and A is called **singular**.

Example 5.12

Find the inverse of

$$A = \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix}.$$

Solution. First calculate the determinant:

$$|A| = 3(2) - 1(4) = 2.$$

Since $|A| \neq 0$, the inverse exists. Therefore,

$$\begin{aligned} A^{-1} &= \frac{1}{2} \begin{bmatrix} 2 & -1 \\ -4 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 1 & -\frac{1}{2} \\ -2 & \frac{3}{2} \end{bmatrix}. \end{aligned}$$

Indeed,

$$AA^{-1} = A^{-1}A = I_2.$$

5.6.2 The adjugate matrix

Let A be a square matrix and let C_{ij} be the cofactor of its entry a_{ij} . The **cofactor matrix** of A is

$$C = \begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{bmatrix}.$$

The **adjugate** of A , sometimes called the classical adjoint, is the transpose of its cofactor matrix:

$$\text{adj}(A) = C^T. \quad (5.45)$$

Example 5.13

Find the adjugate of

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 0 & 3 & 1 \\ 1 & 0 & 2 \end{bmatrix}.$$

Solution. The cofactors are

$$\begin{aligned}
C_{11} &= \begin{vmatrix} 3 & 1 \\ 0 & 2 \end{vmatrix} = 6, & C_{12} &= -\begin{vmatrix} 0 & 1 \\ 1 & 2 \end{vmatrix} = 1, & C_{13} &= \begin{vmatrix} 0 & 3 \\ 1 & 0 \end{vmatrix} = -3, \\
C_{21} &= -\begin{vmatrix} -1 & 3 \\ 0 & 2 \end{vmatrix} = 2, & C_{22} &= \begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix} = 1, & C_{23} &= -\begin{vmatrix} 2 & -1 \\ 1 & 0 \end{vmatrix} = -1, \\
C_{31} &= \begin{vmatrix} -1 & 3 \\ 3 & 1 \end{vmatrix} = -10, & C_{32} &= -\begin{vmatrix} 2 & 3 \\ 0 & 1 \end{vmatrix} = -2, & C_{33} &= \begin{vmatrix} 2 & -1 \\ 0 & 3 \end{vmatrix} = 6.
\end{aligned}$$

Therefore, the cofactor matrix is

$$C = \begin{bmatrix} 6 & 1 & -3 \\ 2 & 1 & -1 \\ -10 & -2 & 6 \end{bmatrix},$$

and hence

$$\operatorname{adj}(A) = C^T = \begin{bmatrix} 6 & 2 & -10 \\ 1 & 1 & -2 \\ -3 & -1 & 6 \end{bmatrix}.$$

5.6.3 Finding an inverse using the adjugate

For every square matrix A ,

$$A \operatorname{adj}(A) = \operatorname{adj}(A)A = |A|I. \quad (5.46)$$

Consequently, if $|A| \neq 0$, then

$$A^{-1} = \frac{1}{|A|} \operatorname{adj}(A). \quad (5.47)$$

Example 5.14

Find the inverse of

$$A = \begin{bmatrix} 2 & 3 & 0 \\ -5 & 0 & 4 \\ 0 & 2 & 1 \end{bmatrix}$$

using the adjugate method.

Solution. First,

$$\begin{aligned}
 |A| &= 2 \begin{vmatrix} 0 & 4 \\ 2 & 1 \end{vmatrix} - 3 \begin{vmatrix} -5 & 4 \\ 0 & 1 \end{vmatrix} \\
 &= 2(-8) - 3(-5) \\
 &= -1.
 \end{aligned}$$

Thus, the inverse exists. The cofactor matrix is

$$C = \begin{bmatrix} -8 & 5 & -10 \\ -3 & 2 & -4 \\ 12 & -8 & 15 \end{bmatrix},$$

so

$$\text{adj}(A) = C^T = \begin{bmatrix} -8 & -3 & 12 \\ 5 & 2 & -8 \\ -10 & -4 & 15 \end{bmatrix}.$$

Therefore,

$$\begin{aligned}
 A^{-1} &= \frac{1}{-1} \begin{bmatrix} -8 & -3 & 12 \\ 5 & 2 & -8 \\ -10 & -4 & 15 \end{bmatrix} \\
 &= \boxed{\begin{bmatrix} 8 & 3 & -12 \\ -5 & -2 & 8 \\ 10 & 4 & -15 \end{bmatrix}}.
 \end{aligned}$$

5.6.4 Elementary row operations

An **elementary row operation** is one of the following reversible operations:

1. interchange two rows:

$$R_i \leftrightarrow R_j;$$

2. multiply a row by a nonzero scalar k :

$$R_i \longrightarrow kR_i;$$

3. add a scalar multiple of one row to another:

$$R_i \longrightarrow R_i + kR_j.$$

Matrices connected by a sequence of elementary row operations are called **row-equivalent**.

5.6.5 Finding an inverse using elementary row operations

To find A^{-1} by row reduction, write A beside the identity matrix of the same order to form the augmented matrix

$$[A \mid I].$$

Apply elementary row operations until the left-hand side becomes the identity:

$$[A \mid I] \longrightarrow [I \mid A^{-1}]. \quad (5.48)$$

If the left-hand side cannot be reduced to the identity, then A is singular and has no inverse.

Example 5.15

Use elementary row operations to find the inverse of

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}.$$

Solution. Begin with

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right].$$

Eliminate the entry above the leading 1 in the third column by applying

$$R_2 \longrightarrow R_2 - 2R_3.$$

This gives

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & -2 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] = [I_3 \mid A^{-1}].$$

Hence,

$$A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}.$$

5.7 System of linear equations with three variables

A system of three linear equations in the variables x , y , and z may be written as

$$\begin{aligned}a_{11}x + a_{12}y + a_{13}z &= b_1, \\a_{21}x + a_{22}y + a_{23}z &= b_2, \\a_{31}x + a_{32}y + a_{33}z &= b_3.\end{aligned}$$

This system has the matrix form

$$AX = B, \tag{5.49}$$

where

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

Here, A is the **coefficient matrix**, X is the **variable matrix**, and B is the **constant matrix**.

Example 5.16

The system

$$\begin{aligned}2x - y + 3z &= 4, \\x + 2y - z &= 1, \\3x + y - 2z &= 2\end{aligned}$$

can be written as

$$\begin{bmatrix} 2 & -1 & 3 \\ 1 & 2 & -1 \\ 3 & 1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}.$$

5.7.1 Solution using the inverse matrix

If A is nonsingular, then A^{-1} exists. Premultiplying $AX = B$ by A^{-1} gives

$$\begin{aligned}A^{-1}AX &= A^{-1}B, \\IX &= A^{-1}B.\end{aligned}$$

Therefore,

$$\boxed{X = A^{-1}B}. \tag{5.50}$$

This method cannot be used when $|A| = 0$.

Example 5.17

Use the inverse matrix to solve

$$x + 2y + 7z = 1,$$

$$x + 3y = 2,$$

$$-y + 8z = 3.$$

Solution. The matrix equation is

$$\begin{bmatrix} 1 & 2 & 7 \\ 1 & 3 & 0 \\ 0 & -1 & 8 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

The determinant of the coefficient matrix is 1, and its inverse is

$$A^{-1} = \begin{bmatrix} 24 & -23 & -21 \\ -8 & 8 & 7 \\ -1 & 1 & 1 \end{bmatrix}.$$

Hence,

$$\begin{aligned} X = A^{-1}B &= \begin{bmatrix} 24 & -23 & -21 \\ -8 & 8 & 7 \\ -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \\ &= \begin{bmatrix} -85 \\ 29 \\ 4 \end{bmatrix}. \end{aligned}$$

Therefore,

$$\boxed{x = -85, \quad y = 29, \quad z = 4}.$$

5.7.2 Gauss–Jordan elimination

The Gauss–Jordan method solves a system by reducing its augmented matrix. Starting from

$$[A \mid B],$$

apply elementary row operations until the coefficient portion is in reduced row-echelon form. For a system with a unique solution, the reduction has the form

$$[A \mid B] \longrightarrow [I \mid X]. \quad (5.51)$$

Example 5.18

Solve by Gauss–Jordan elimination:

$$\begin{aligned}x + z &= 5, \\3x + y + 2z &= 11, \\3x + 2y + 2z &= 10.\end{aligned}$$

Solution. Form the augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 5 \\ 3 & 1 & 2 & 11 \\ 3 & 2 & 2 & 10 \end{array} \right].$$

Apply $R_2 \rightarrow R_2 - 3R_1$ and $R_3 \rightarrow R_3 - 3R_1$:

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 5 \\ 0 & 1 & -1 & -4 \\ 0 & 2 & -1 & -5 \end{array} \right].$$

Next, apply $R_3 \rightarrow R_3 - 2R_2$:

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 5 \\ 0 & 1 & -1 & -4 \\ 0 & 0 & 1 & 3 \end{array} \right].$$

Finally, apply $R_1 \rightarrow R_1 - R_3$ and $R_2 \rightarrow R_2 + R_3$:

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{array} \right].$$

Thus,

$$\boxed{x = 2, \quad y = -1, \quad z = 3}.$$

5.7.3 Possible solution sets

Let n be the number of variables and let $[A \mid B]$ be the augmented matrix.

1. The system has a **unique solution** if

$$\text{rank}(A) = \text{rank}([A \mid B]) = n.$$

For a square coefficient matrix, $|A| \neq 0$ guarantees this case.

2. The system has **infinitely many solutions** if

$$\text{rank}(A) = \text{rank}([A \mid B]) < n.$$

At least one variable is then free.

3. The system has **no solution** if

$$\text{rank}(A) < \text{rank}([A \mid B]).$$

Row reduction reveals a contradictory row such as

$$[0 \ 0 \ 0 \mid c], \quad c \neq 0.$$

The same three possibilities can be seen most simply with two equations in two variables. In three variables, the corresponding objects are planes rather than lines, but the algebraic classification is unchanged.

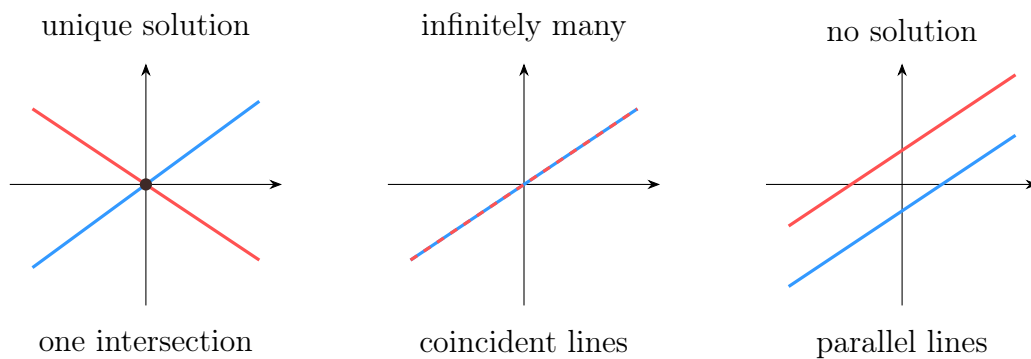


Figure 5.8: Geometric analogues of the three possible solution sets for a linear system.

5.7.4 Cramer's rule

Cramer's rule uses determinants to solve a square system $AX = B$ when $|A| \neq 0$.

For the two-variable system

$$a_1x + b_1y = c_1,$$

$$a_2x + b_2y = c_2,$$

let

$$\Delta = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}.$$

Then

$$x = \frac{\begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix}}{\Delta}, \quad y = \frac{\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}}{\Delta}. \quad (5.52)$$

For a three-variable system, define $\Delta = |A|$. Let Δ_x , Δ_y , and Δ_z be the determinants obtained by replacing, respectively, the first, second, and third columns of A by B . Then

$$x = \frac{\Delta_x}{\Delta}, \quad y = \frac{\Delta_y}{\Delta}, \quad z = \frac{\Delta_z}{\Delta}. \quad (5.53)$$

Example 5.19

Use Cramer's rule to solve

$$\begin{aligned} x + y - z &= 0, \\ 2x - 3y + z &= 1, \\ 2x + y + 2z &= 7. \end{aligned}$$

Solution. The coefficient determinant is

$$\Delta = \begin{vmatrix} 1 & 1 & -1 \\ 2 & -3 & 1 \\ 2 & 1 & 2 \end{vmatrix} = -17.$$

Replacing each column in turn gives

$$\Delta_x = \begin{vmatrix} 0 & 1 & -1 \\ 1 & -3 & 1 \\ 7 & 1 & 2 \end{vmatrix} = -17,$$

$$\Delta_y = \begin{vmatrix} 1 & 0 & -1 \\ 2 & 1 & 1 \\ 2 & 7 & 2 \end{vmatrix} = -17,$$

$$\Delta_z = \begin{vmatrix} 1 & 1 & 0 \\ 2 & -3 & 1 \\ 2 & 1 & 7 \end{vmatrix} = -34.$$

Therefore,

$$x = \frac{-17}{-17} = 1, \quad y = \frac{-17}{-17} = 1, \quad z = \frac{-34}{-17} = 2.$$

Hence,

$$x = 1, \quad y = 1, \quad z = 2.$$

Chapter summary and common pitfalls

Key ideas. Matrix dimensions determine which operations are valid. Determinants test invertibility, inverses solve suitable square systems, and matrix products encode successive linear operations.

Common pitfalls. Matrix multiplication is generally not commutative; entrywise multiplication is not the standard matrix product; and a zero determinant means that an inverse does not exist.

5.8 Practice and exercises

1. State the type of each of the following matrices:

(a) (15)

(b) $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

(c) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

(d) $\begin{pmatrix} 22 & 0 & 0 \\ 44 & 11 & 0 \\ 11 & 77 & 33 \end{pmatrix}$

(e) $(2 \ 0 \ 8)$

(f) $\begin{pmatrix} 3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 3 \end{pmatrix}$

(g) $\begin{pmatrix} 1 & 5 & 8 \\ 5 & 1 & 7 \\ 8 & 7 & 1 \end{pmatrix}$

(h) $\begin{pmatrix} 6 & 5 & h \\ G & 23 & 3 \\ K & s & 19 \end{pmatrix}$

(i) $\begin{pmatrix} 0 & 4 & -3 \\ -4 & 0 & 2 \\ 3 & -2 & 0 \end{pmatrix}$

(j) $\begin{pmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{pmatrix}$

(k) $\begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix}$

$$(l) \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$$

$$(m) \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

2. Find a general 3×3 matrix that satisfies each stated condition:

- (a) $a_{ij} = 0$ if $i \neq j$;
- (b) $a_{ij} = 0$ if $i < j$;
- (c) $a_{ij} = 0$ if $i > j$;
- (d) $a_{ij} = 0$ if $|i - j| > 1$.

3. Given

$$A = \begin{pmatrix} 2 & 3 \\ -1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & -1 \\ 4 & 2 \end{pmatrix}, \quad C = \begin{pmatrix} 2 & -3 \\ 1 & 2 \\ 4 & -1 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 3 \\ -1 & 2 \\ 3 & 4 \end{pmatrix},$$

determine whether each expression is defined. Evaluate every expression that is defined, where M is a scalar and I is an identity matrix of suitable order.

- (a) $A + B$
- (b) $2C + 3B$
- (c) $4C - 2D$
- (d) $6D$
- (e) $M(A + B)$
- (f) BD
- (g) $D^T C^T$
- (h) $(AB^T)^T$
- (i) IA
- (j) For

$$E = \begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix}, \quad F = \begin{pmatrix} 4 & -2 \\ 0 & 5 \end{pmatrix},$$

calculate $2E - F$ and $3F + E$.

- (k) Suppose P , Q , and R have orders 2×3 , 3×4 , and 4×2 , respectively. Determine which of PQ , QR , RP , PR , QP , and RQ are defined, and state the order of every defined product.

4. State whether each of the following statements is **true** or **false**, giving a reason. Suppose that A and B have the same order, C is an $m \times n$ matrix, D is an $n \times p$ matrix, $\mathbf{0}$ is a zero matrix, and k and m are scalars whenever the stated operations are defined.

- (a) $A + \mathbf{0} = \mathbf{0} + A$ follows from the associative property.
- (b) $A + (-A) = (-A) + A$.
- (c) $mA + kA = A(m + k)$.
- (d) $CD = DC$.
- (e) $CD = \mathbf{0}$ implies that $C = \mathbf{0}$ and $D = \mathbf{0}$.
- (f) $A(B + C) = AB + AC$ whenever the products are defined.
- (g) $(AB)C = A(BC)$ whenever the products are defined.
- (h) $A\mathbf{0} = \mathbf{0}$ whenever the product is defined.
- (i) $AB = AC$ implies $B = C$.

5. Arrange the matrices

$$\begin{pmatrix} 5 & 2 \end{pmatrix}, \quad \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix}, \quad \begin{pmatrix} 1 & -3 & 2 \\ 4 & 1 & 5 \end{pmatrix}$$

so that their product is a 2×2 matrix. Hence, find the product.

6. Suppose

$$G = \begin{pmatrix} 3 & -1 \\ 2 & 0 \end{pmatrix}$$

and $G^2 + pG + qI = \mathbf{0}$, where I is the identity matrix and $\mathbf{0}$ is the zero matrix. Find p and q .

7. If

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad B = \begin{pmatrix} u & v & w \\ x & y & z \end{pmatrix},$$

show directly that $(AB)^T = B^T A^T$.

8. Evaluate each determinant by cofactor expansion along the stated row.

- (a) Along the second column:

$$\begin{vmatrix} 3 & -2 & -8 \\ -2 & 0 & -3 \\ 1 & 0 & -4 \end{vmatrix}.$$

(b) Along the first row:

$$\begin{vmatrix} 1 & 2 & 0 \\ 3 & -1 & 2 \\ 2 & 1 & 1 \end{vmatrix}.$$

9. Use the properties of determinants. For parts (a)–(e), state whether the statement is **true** or **false** and give a reason. Then answer part (f).

(a)

$$\begin{vmatrix} 1 & 2 & 3 \\ 5 & 4 & 2 \\ 9 & 7 & 5 \end{vmatrix} = \begin{vmatrix} 1 & 5 & 9 \\ 2 & 4 & 7 \\ 3 & 2 & 5 \end{vmatrix}.$$

(b)

$$\begin{vmatrix} 1 & 2 & 7 \\ 8 & 6 & 4 \\ 9 & 9 & 1 \end{vmatrix} = \begin{vmatrix} 8 & 6 & 4 \\ 1 & 2 & 7 \\ 9 & 9 & 1 \end{vmatrix}.$$

(c)

$$\begin{vmatrix} 2 & 2 & 9 \\ 0 & 7 & 5 \\ 4 & 4 & 5 \end{vmatrix} = - \begin{vmatrix} 9 & 2 & 2 \\ 5 & 7 & 0 \\ 5 & 4 & 4 \end{vmatrix}.$$

(d)

$$\begin{vmatrix} 2 & 7 & 2 \\ 5 & 1 & 5 \\ 7 & 5 & 7 \end{vmatrix} = 0.$$

(e)

$$\begin{vmatrix} 3 & 7 & 5 \\ 0 & 7 & 5 \\ 0 & 0 & 1 \end{vmatrix} = 3 \times 7 \times 1.$$

(f) If $\det A = 6$, a matrix B is obtained by interchanging two rows, multiplying one row by -2 , and then adding four times one row to another. Find $\det B$.

10. Given

$$A = \begin{pmatrix} 3 & 1 & -1 \\ 2 & 4 & 1 \\ 7 & -5 & 5 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 & 0 \\ 1 & -1 & 2 \\ 3 & 1 & 1 \end{pmatrix},$$

find A^{-1} and B^{-1} using each of the following methods:

- (a) the adjoint method;
- (b) elementary row operations.

11. Solve each system using the inverse-matrix method, calculating the adjoint of the

coefficient matrix.

(a)

$$\begin{aligned}2x &= 3 + 4y, \\ y + 3x &= 5.\end{aligned}$$

(b)

$$\begin{aligned}-2x + 2y + 3z &= 1, \\ x - y &= 3, \\ y + 4z + 2 &= 0.\end{aligned}$$

(c) $2x + y = 7, \quad x - y = 2.$

(d)

$$x + y = 3, \quad 2x - y + z = 3, \quad x + 2z = 7.$$

12. Solve each system using Gauss–Jordan elimination.

(a)

$$\begin{aligned}x + 2y &= 3, \\ x + 3 &= y.\end{aligned}$$

(b)

$$\begin{aligned}2x - y + z &= 8, \\ x + y + 2z - 4 &= 0, \\ 3x - 3y - z &= 10.\end{aligned}$$

(c) $3x + 2y = 8, \quad x - y = 1.$

(d)

$$x + y + z = 6, \quad 2x - y + z = 3, \quad x + 2y - z = 2.$$

13. Solve each system of linear equations using Cramer's rule.

(a)

$$\begin{aligned}7x &= 2y + 3, \\ y + 3x &= 5.\end{aligned}$$

(b)

$$\begin{aligned}4x + 5y &= 2, \\11x + y + 2z &= 3, \\x + 5y + 2z &= 1.\end{aligned}$$

(c) $3x - 2y = 4, \quad 5x + y = 7.$

(d)

$$x + y + z = 6, \quad 2x - y + z = 3, \quad x + 2y - z = 2.$$

14. Aishah bought a marker pen, two files, and an examination pad for RM 36. Balqis bought three marker pens, two examination pads, and a file for RM 43. Camelia bought three marker pens, an examination pad, and four files for RM 74. Suppose that x , y , and z represent the respective prices of a marker pen, an examination pad, and a file.

- (a) Write a system of linear equations in x , y , and z .
- (b) Write the system in matrix form.
- (c) Use Cramer's rule to find the price of each item.

15. Two companies, P and Q , award cash prizes for leadership, innovation, and punctuality. Company P awards a total of RM 8,500 to four employees for leadership, five for innovation, and two for punctuality. Company Q awards a total of RM 4,550 to three employees for leadership, two for innovation, and one for punctuality. The sum of the three prize amounts awarded to an employee is RM 2,150.

Let x , y , and z be the respective prize amounts for leadership, innovation, and punctuality.

- (a) Write the corresponding system of linear equations.
- (b) Find x , y , and z using Gauss–Jordan elimination.

Additional grouped practice

The following questions group closely related matrix skills under one number. Show sufficient working, especially when using determinants, inverses, or row operations.

16. Let

$$A = \begin{pmatrix} 2 & -1 & 4 \\ 0 & 3 & 5 \end{pmatrix}.$$

State the order of A , and find a_{12} , a_{23} , the second column, and the first row.

17. Find x , y , and z if

$$\begin{pmatrix} x+1 & 2y \\ z-3 & 4 \end{pmatrix} = \begin{pmatrix} 5 & 6 \\ 1 & 4 \end{pmatrix}.$$

18. Construct the 3×4 matrix $A = (a_{ij})$ for which $a_{ij} = 2i - j$.

19. Solve the matrix equation $3X - 2A = B$, where

$$A = \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 4 & 3 \\ -2 & 5 \end{pmatrix}.$$

20. Calculate AB for

$$A = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 \\ 0 & -1 \\ 4 & 2 \end{pmatrix}.$$

21. For

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix},$$

calculate AB and BA . What do the results demonstrate?

22. If

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

find A^2 , conjecture a formula for A^n , and hence find A^5 .

23. Let

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}.$$

Evaluate $A^2 - 4A + 4I$.

24. Given

$$A = \begin{pmatrix} 1 & -2 & 0 \\ 3 & 4 & 5 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 & 2 \\ -1 & 3 & 4 \end{pmatrix},$$

find A^T and $(A+B)^T$, and verify that $(A+B)^T = A^T + B^T$.

25. Decompose

$$A = \begin{pmatrix} 2 & 3 \\ -1 & 4 \end{pmatrix}$$

as $A = S + K$, where S is symmetric and K is skew-symmetric.

26. Consider

$$M = \begin{pmatrix} a & 2b \\ -6 & c \end{pmatrix}.$$

Find the restrictions on a , b , and c for M to be

- (a) symmetric;
- (b) skew-symmetric.

27. Calculate

$$\begin{vmatrix} 3 & -2 \\ 5 & 4 \end{vmatrix},$$

and find k if $\begin{vmatrix} k & 1 \\ 4 & 2 \end{vmatrix} = 0$.

28. Let

$$A = \begin{pmatrix} 2 & -1 & 4 \\ 0 & 3 & 5 \\ 0 & 0 & -2 \end{pmatrix}.$$

Find $\det A$, $\det(A^2)$, and $\det(3A)$.

29. Find all real values of k for which

$$\begin{pmatrix} k & 1 & 0 \\ 2 & k & 1 \\ 0 & 3 & k \end{pmatrix}$$

is singular.

30. Suppose A and B are invertible 3×3 matrices with $\det A = -2$ and $\det B = 5$. Find

- (a) $\det(A^T B)$;
- (b) $\det(A^{-1})$;
- (c) $\det(2A)$;
- (d) $\det(B^3)$.

31. For

$$A = \begin{pmatrix} 2 & -1 & 3 \\ 0 & 4 & 5 \\ 1 & 2 & -2 \end{pmatrix},$$

find the minor M_{12} and the cofactors C_{12} , C_{23} , and C_{31} .

32. Find the adjugate and inverse of

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{pmatrix}.$$

33. Find the inverse of

$$A = \begin{pmatrix} 4 & 1 \\ 2 & 1 \end{pmatrix},$$

and use it to solve $A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 9 \\ 5 \end{pmatrix}$.

34. Verify by multiplication that

$$\begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix}$$

is the inverse of $\begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}$.

35. Determine the values of k for which

$$A = \begin{pmatrix} k & 2 \\ 3 & k \end{pmatrix}$$

is invertible, and state A^{-1} for those values.

36. Solve $AX = B$, where

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix}, \quad B = \begin{pmatrix} 4 & 1 \\ 7 & 2 \end{pmatrix}.$$

37. Solve $XA = B$, where

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 5 & 2 \\ 3 & 1 \end{pmatrix}.$$

38. State whether each identity is true or false for invertible matrices and non-zero scalars. Correct every false identity.

- (a) $(AB)^{-1} = A^{-1}B^{-1}$;
- (b) $(A^T)^{-1} = (A^{-1})^T$;
- (c) $(kA)^{-1} = kA^{-1}$;
- (d) A is invertible if and only if $\det A \neq 0$.

39. Reduce

$$\left(\begin{array}{cc|c} 1 & 2 & 5 \\ 3 & 7 & 16 \end{array} \right)$$

to reduced row-echelon form and state the corresponding solution.

40. Find the reduced row-echelon form and rank of

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{pmatrix}.$$

41. State the ranks of

$$\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}, \quad I_3, \quad \mathbf{0}_{2 \times 4}.$$

42. Classify each augmented matrix as representing a system with a unique solution, no solution, or infinitely many solutions. State the solution set where it is immediate.

(a) $\left(\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{array} \right);$

(b) $\left(\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 0 & 1 \end{array} \right);$

(c) $\left(\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 0 & 0 \end{array} \right).$

43. Determine whether each system is inconsistent or dependent, and describe its solution set.

(a) $x + y = 2, 2x + 2y = 5;$

(b) $x + y = 2, 2x + 2y = 4.$

44. For the system $x + y = 2, 2x + 2y = k$, determine the values of k that give no solution, infinitely many solutions, or a unique solution.

45. Explain when the inverse-matrix method and Cramer's rule may be used. State one advantage of Gauss-Jordan elimination when the coefficient matrix is singular or non-square.

46. Write the system

$$2x - y + 3z = 4, \quad x + 4y - z = 2, \quad 3x + 2z = 5$$

in the form $A\mathbf{x} = \mathbf{b}$.

47. Write the system represented by

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 0 & 3 & 5 & -2 \\ 2 & -1 & 0 & 7 \end{array} \right).$$

48. Find a , b , c , and d if

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 5 \\ 3 & 7 \end{pmatrix}.$$

49. For $D = \text{diag}(2, -1, 3)$, write down D^n and find D^4 .

50. Verify and name the special power property of each matrix:

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

51. Evaluate

$$\begin{vmatrix} 0 & 2 & 0 \\ 3 & 1 & 4 \\ 0 & 5 & 6 \end{vmatrix}$$

by choosing the most efficient row or column for cofactor expansion.

52. Use the adjugate method to find the inverse of

$$\begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}.$$

53. Use elementary row operations to find the inverse of

$$\begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}.$$

54. Let

$$E = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

Calculate EA and AE , and describe the row or column operation caused by each multiplication.

55. Prove, using the rule for the transpose of a product, that $(ABC)^T = C^T B^T A^T$ whenever the products are defined.

- 56.** A 4×4 matrix has determinant 3. It is transformed by one row interchange, multiplication of one row by -2 , and a row replacement operation. Find the determinant of the resulting matrix.

- 57.** The matrix

$$A = \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix}$$

satisfies $A^2 - 3A + 2I = \mathbf{0}$. Use this relation to express A^5 as a linear expression in A and I , and then calculate A^5 .

- 58.** Find a and b from

$$\begin{pmatrix} a+b & a-b \\ 2a & b \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 4 & 2 \end{pmatrix}.$$

- 59.** The reduced row-echelon form of an augmented matrix is

$$\left(\begin{array}{ccc|c} 1 & 0 & 2 & -1 \\ 0 & 1 & -3 & 4 \\ 0 & 0 & 0 & 0 \end{array} \right).$$

Identify the pivot and free variables, state the rank, and write the solution parametrically.

- 60.** Show that

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix}$$

is row-equivalent to I_2 by stating a sequence of elementary row operations.

- 61.** Solve the homogeneous system

$$x + 2y - z = 0, \quad 2x + 4y - 2z = 0, \quad y + z = 0,$$

and give a basis vector for its non-zero solutions.

- 62.** The equations

$$x + y + z = 3, \quad 2x + 2y + 2z = 6, \quad x - y = 1$$

are dependent. Find their complete solution set.

- 63.** For

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix},$$

calculate AB , $\det A$, and A^{-1} , then solve $A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ 11 \end{pmatrix}$.

Application-based questions

The next twelve questions replace repeated computational drills with applications that use matrix arithmetic, products, systems, inverses, transition matrices, and determinants.

- 64.** Three students receive marks in coursework, a test, and a final examination:

$$S = \begin{pmatrix} 80 & 70 & 90 \\ 65 & 88 & 76 \\ 92 & 84 & 81 \end{pmatrix}.$$

If the weighting vector is $w = (0.3, 0.3, 0.4)^T$, find the weighted-mark vector Sw and identify the highest mark.

- 65.** Two travellers record spending in US dollars, euros, and Singapore dollars:

$$E = \begin{pmatrix} 200 & 150 & 300 \\ 120 & 220 & 180 \end{pmatrix}.$$

Using the conversion vector $c = (4.70, 5.10, 3.50)^T$ ringgit per unit, find each traveller's total spending in ringgit and their combined spending.

- 66.** An investment portfolio holds RM 5,000, RM 3,000, and RM 2,000 in three assets. The columns of

$$R = \begin{pmatrix} 0.04 & -0.02 \\ 0.06 & 0.01 \\ -0.01 & 0.05 \end{pmatrix}$$

give the returns in two economic scenarios. Use a matrix product to find the portfolio profit in each scenario.

- 67.** A factory first converts two raw materials into three components, then assembles two products. The matrices

$$R = \begin{pmatrix} 2 & 1 & 3 \\ 1 & 2 & 1 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 2 \\ 2 & 1 \\ 1 & 1 \end{pmatrix}$$

give raw-material units per component and component units per product.

(a) Find RC and interpret its order.

(b) Find the raw materials required for the production vector $p = (10, 8)^T$.

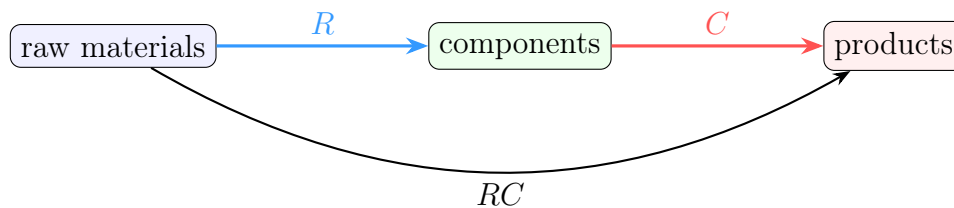


Figure 5.9: Two-stage production matrices for Question 67.

68. A simple colour-correction matrix is

$$T = \begin{pmatrix} 1 & 0.1 & 0 \\ 0 & 0.9 & 0.1 \\ 0 & 0 & 1.1 \end{pmatrix}.$$

Apply it to the RGB pixel vector $(120, 80, 40)^T$ and interpret the three entries of the result as corrected red, green, and blue intensities.

69. Three vertical supports carry a total load of 1200 N. Their forces x , y , and z satisfy

$$x + y + z = 1200, \quad 2x - y = 0, \quad y - z = 100.$$

Write the system in matrix form and solve for the three support forces.

70. A pharmacy mixes 2%, 5%, and 8% solutions. It needs 50 L containing 2.2 L of active ingredient, and the volume of the 5% solution must be twice that of the 8% solution. Form and solve a 3×3 system for the three volumes.

71. A weather model has column transition matrix

$$P = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix}$$

for sunny and rainy days. Find the stationary distribution of 1000 long-run observations by solving $Pv = v$ together with the total condition.

72. A hospital schedules x , y , and z nurses on morning, afternoon, and night shifts. There are 29 nurses in total, the morning shift has four more nurses than the afternoon shift, and the afternoon shift has twice as many as the night shift. Form and solve a matrix system.

73. A service has x , y , and z basic, standard, and premium subscriptions. The totals satisfy

$$x + y + z = 1000, \quad 20x + 35y + 50z = 30500, \quad x + 2y + 4z = 1900.$$

Solve for the number of subscriptions of each type.

74. Map coordinates are obtained from ground coordinates by

$$\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Find the inverse transformation and recover (x, y) when $(u, v) = (13, 14)$.

75. A three-dimensional packaging transformation has matrix

$$T = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

Find $\det T$ and hence the volume of the transformed package when the original volume is 20 m^3 .

76. Two shops record the quantities sold of three products in

$$Q = \begin{pmatrix} 10 & 8 & 5 \\ 6 & 12 & 4 \end{pmatrix}.$$

If the price vector is $p = (4, 3, 5)^T$ ringgit, use a matrix product to find the revenue of each shop and the combined revenue.

77. A warehouse stores three materials. Its initial stock, deliveries, and dispatches are

$$s = \begin{pmatrix} 120 \\ 80 \\ 100 \end{pmatrix}, \quad d = \begin{pmatrix} 30 \\ 20 \\ 10 \end{pmatrix}, \quad q = \begin{pmatrix} 25 \\ 35 \\ 15 \end{pmatrix}.$$

Find the closing stock vector and identify the material with the lowest closing stock.

78. Three delivery routes use two vehicle classes. Their daily load matrix is

$$L = \begin{pmatrix} 3 & 4 \\ 5 & 2 \\ 1 & 6 \end{pmatrix},$$

and the cost per load is $c = (120, 150)^T$ ringgit. Find the route-cost vector and the total daily cost.

79. A meal contains two servings of food A , one of B , and three of C . The nutrient matrix,

whose columns represent protein and fibre, is

$$N = \begin{pmatrix} 3 & 1 \\ 2 & 4 \\ 1 & 2 \end{pmatrix}.$$

Use a matrix product to find the meal's total protein and fibre.

- 80.** A theatre sold 350 tickets. Adult tickets cost RM 12 and student tickets cost RM 7, producing RM 3,200. Form and solve a matrix equation for the numbers of adult and student tickets.
- 81.** A laboratory needs 30 litres of a 32% solution by mixing 20% and 50% solutions. Form a 2×2 system and determine the required volume of each solution.
- 82.** The currents I_1 and I_2 in a simple circuit satisfy

$$2I_1 - I_2 = 5, \quad -I_1 + 3I_2 = 5.$$

Solve the system using a matrix method.

- 83.** A workshop makes tables and chairs. A table requires two labour units and four wood units; a chair requires three labour units and two wood units. There are 120 labour units and 160 wood units available. Use matrices to find the production quantities that use all resources.
- 84.** The prices of a sandwich, drink, and fruit cup are x , y , and z ringgit. Three orders give

$$2x + y + z = 13, \quad x + 3y + 2z = 17, \quad 3x + 2y + z = 20.$$

Find the three prices.

- 85.** Traffic can travel in both directions between junctions A and B . The hourly flows are shown below; a sensor also reports $y = 2x$.

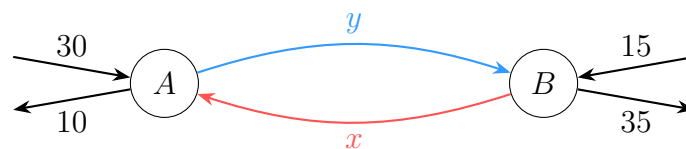


Figure 5.10: Hourly traffic flow at two connected junctions.

Write the flow-conservation equations and use the sensor relation to find x and y .

86. A message vector $v = (3, 5)^T$ is encoded using

$$K = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

Find the encoded vector, calculate K^{-1} , and demonstrate how the original message is recovered.

87. A population is divided between two locations. With column state vectors, its transition matrix is

$$P = \begin{pmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{pmatrix}.$$

Starting from $(100, 0)^T$, find the populations after one and two transitions.

88. A simple age-structured population model uses

$$M = \begin{pmatrix} 0 & 2 \\ 0.5 & 0.8 \end{pmatrix}, \quad v_0 = \begin{pmatrix} 200 \\ 100 \end{pmatrix},$$

where the entries count juveniles and adults. Find v_1 and v_2 .

89. In a two-sector production model, total output x satisfies $x = Ax + d$, where

$$A = \begin{pmatrix} 0.2 & 0.1 \\ 0.3 & 0.1 \end{pmatrix}, \quad d = \begin{pmatrix} 72 \\ 42 \end{pmatrix}.$$

Find the required total-output vector.

90. A 2×2 greyscale image has pixel matrix

$$P = \begin{pmatrix} 50 & 100 \\ 150 & 200 \end{pmatrix}.$$

Increasing every pixel value by 20 is represented by adding $20\mathbf{1}$, where $\mathbf{1}$ is the all-ones matrix. Find the adjusted image and its transpose.

91. The rows of

$$R = \begin{pmatrix} 6 & 2 & 2 \\ 5 & 4 & 1 \\ 4 & 3 & 3 \end{pmatrix}$$

give wins, draws, and losses for three teams. With three points for a win and one for a draw, use a matrix product to rank the teams by points.

92. A network has adjacency matrix

$$A = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}.$$

Find the degree vector $A\mathbf{1}$ and the number of two-step walks from vertex 1 to vertex 4.

93. The linear transformation

$$T = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$$

acts on the triangle with vertices $(0, 0)$, $(1, 0)$, and $(0, 1)$. Find the image vertices and compare the original and transformed areas.

94. A vector $v = (3, 1)^T$ is first rotated 90° anticlockwise and then reflected in the x -axis.

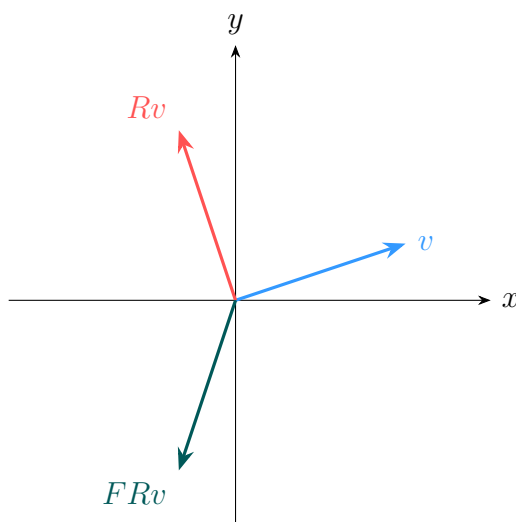
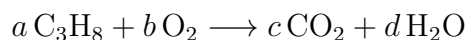


Figure 5.11: A rotation followed by a reflection.

Write the rotation and reflection matrices and calculate both transformed vectors.

95. Use a determinant to find the area of the triangle with vertices $(1, 1)$, $(5, 2)$, and $(3, 6)$.

96. Balance the combustion equation



by forming a homogeneous linear system. Give the least positive integer values of a , b , c , and d .

97. Find the quadratic $p(x) = ax^2 + bx + c$ passing through $(0, 1)$, $(1, 4)$, and $(2, 9)$ by

solving a matrix system.

- 98.** A sensor is calibrated by $y = ax^2 + bx + c$. Measurements at $x = -1, 0, 2$ give $y = 2, 1, 5$, respectively. Determine the calibration equation.

- 99.** A factory allocates x , y , and z batches to three products. The requirements are

$$x + y + z = 100, \quad 2x + y + 3z = 200, \quad x + 2y + z = 130.$$

Find the allocation and verify it in the original equations.

- 100.** A supply plan uses x , y , and z batches and must satisfy

$$x + y + z = 90, \quad 2x + y = 100, \quad x + 2z = 70.$$

Solve the system. If the respective costs are RM 5,000, RM 4,000, and RM 6,000 per batch, calculate the total cost using a row-by-column product.

Part II

Geometry, Modelling, and Data

Chapter 6

Trigonometry

Chapter guide

Prerequisites. Right-angled triangles, coordinate geometry, algebra, and function graphs.

Learning outcomes. Readers should be able to work in degrees and radians, use unit-circle definitions and exact values, transform and solve trigonometric equations, prove identities, and interpret inverse trigonometric functions.

Trigonometry studies the relationship between angles, coordinates, and ratios. Its elementary definitions arise from right-angled triangles, but the unit-circle interpretation extends the subject to angles of any size and provides the natural foundation for trigonometric graphs, identities, equations, and inverse functions.

Throughout this chapter, angles are measured in radians unless a degree symbol is shown explicitly.

Trigonometry is especially useful when a distance cannot be measured directly. In triangulation, a known baseline and two measured angles determine the position of a remote point, as illustrated in Figure 6.1.

The notation, identities, and equation-solving conventions used here follow standard Malaysian secondary and pre-university references (Ministry of Education Malaysia 2018; Wong Mee Kiong et al. 2019; Zaini bin Musa et al. 2020).

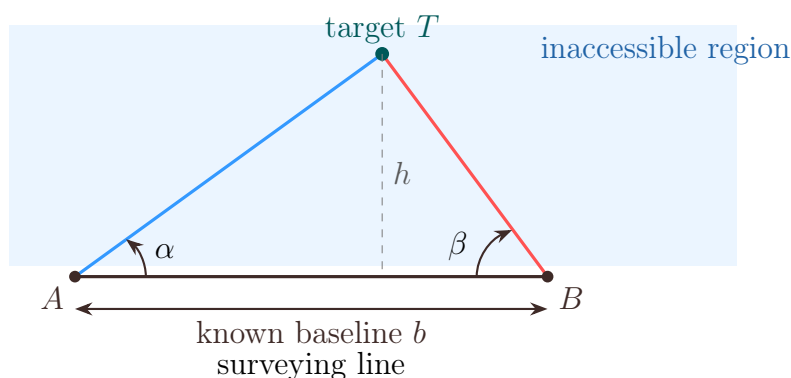


Figure 6.1: Triangulation locates an inaccessible target using a measured baseline and two angles. The sine rule can then determine the remaining lengths of the triangle.

6.1 Angles

An angle is formed by two rays with a common endpoint, called the *vertex*. When an angle is placed in **standard position**, its vertex is at the origin and its initial side lies along the positive x -axis. An anticlockwise rotation produces a positive angle, whereas a clockwise rotation produces a negative angle.

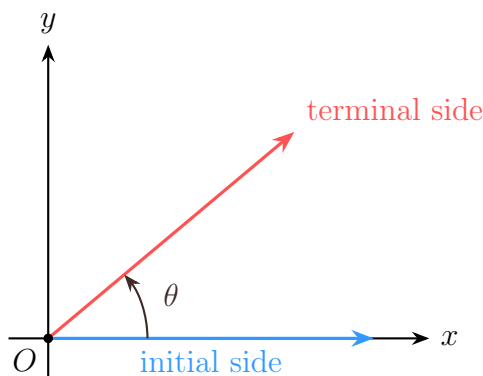


Figure 6.2: An angle θ in standard position.

Angles are commonly classified as follows:

Type	Size
acute	$0^\circ < \theta < 90^\circ$
right	$\theta = 90^\circ$
obtuse	$90^\circ < \theta < 180^\circ$
straight	$\theta = 180^\circ$
reflex	$180^\circ < \theta < 360^\circ$
complete	$\theta = 360^\circ$

6.1.1 Degree and radian measure

One complete revolution measures 360° or 2π radians. Hence,

$$\boxed{180^\circ = \pi \text{ radians}}, \quad \boxed{\theta_{\text{rad}} = \theta_{\text{deg}} \frac{\pi}{180}}, \quad \boxed{\theta_{\text{deg}} = \theta_{\text{rad}} \frac{180}{\pi}}. \quad (6.1)$$

For example,

$$\begin{aligned} 135^\circ &= 135 \frac{\pi}{180} = \frac{3\pi}{4}, \\ -660^\circ &= -660 \frac{\pi}{180} = -\frac{11\pi}{3}, \\ \frac{4\pi}{3} &= \frac{4\pi}{3} \frac{180}{\pi} = 240^\circ. \end{aligned}$$

Radians are dimensionless and are the standard angular unit in calculus. Whenever derivatives, integrals, or power series involve trigonometric functions, their familiar formulae assume radian measure.

6.1.2 Quadrantal, coterminal, and reference angles

A **quadrantal angle** has its terminal side on a coordinate axis. In radians, these angles are integer multiples of $\pi/2$.

Angles that share a terminal side are **coterminal**. If $k \in \mathbb{Z}$, then

$$\boxed{\theta + 360^\circ k} \quad \text{or} \quad \boxed{\theta + 2\pi k} \quad (6.2)$$

represents every angle coterminal with θ .

The **reference angle** α is the acute angle between the terminal side and the x -axis. For $0 \leq \theta < 2\pi$,

$$\alpha = \begin{cases} \theta, & 0 \leq \theta \leq \pi/2, \\ \pi - \theta, & \pi/2 \leq \theta \leq \pi, \\ \theta - \pi, & \pi \leq \theta \leq 3\pi/2, \\ 2\pi - \theta, & 3\pi/2 \leq \theta < 2\pi. \end{cases}$$

For instance, 520° is coterminal with 160° , so its reference angle is 20° . Similarly, -210° is coterminal with 150° , whose reference angle is 30° .

6.2 Trigonometric functions

Let the terminal side of θ pass through a point $P(x, y)$, and let

$$r = \sqrt{x^2 + y^2} > 0.$$

The six trigonometric functions are defined by

$$\begin{aligned} \sin \theta &= \frac{y}{r}, & \cos \theta &= \frac{x}{r}, & \tan \theta &= \frac{y}{x}, & (6.3) \\ \csc \theta &= \frac{r}{y}, & \sec \theta &= \frac{r}{x}, & \cot \theta &= \frac{x}{y}, & (6.4) \end{aligned}$$

where a denominator must not be zero.

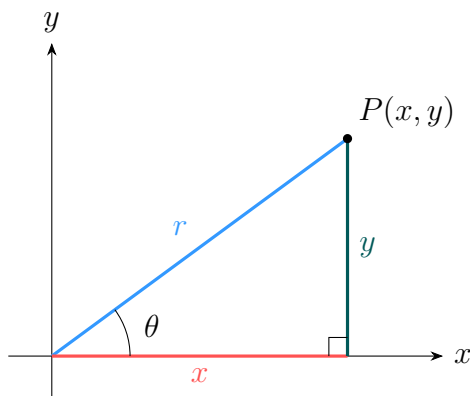


Figure 6.3: Coordinate definitions of the trigonometric ratios.

On the **unit circle**, $r = 1$, so the point at angle θ is

$$\boxed{P = (\cos \theta, \sin \theta)} \quad (6.5)$$

This identity is the bridge between circle geometry and trigonometric graphs.

6.2.1 Reciprocal, quotient, and cofunction identities

The definitions immediately give

$$\csc \theta = \frac{1}{\sin \theta}, \quad \sec \theta = \frac{1}{\cos \theta}, \quad \cot \theta = \frac{1}{\tan \theta}, \quad (6.6)$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}, \quad \cot \theta = \frac{\cos \theta}{\sin \theta}. \quad (6.7)$$

Complementary angles interchange sine and cosine, tangent and cotangent, and secant and cosecant:

$$\sin \left(\frac{\pi}{2} - \theta \right) = \cos \theta, \quad \cos \left(\frac{\pi}{2} - \theta \right) = \sin \theta, \quad (6.8)$$

$$\tan \left(\frac{\pi}{2} - \theta \right) = \cot \theta, \quad \sec \left(\frac{\pi}{2} - \theta \right) = \csc \theta. \quad (6.9)$$

6.2.2 Signs in the four quadrants

The signs of x and y determine the signs of the trigonometric functions. The mnemonic *All Students Take Calculus* records which functions are positive in Quadrants I, II, III, and IV:

Quadrant	Positive functions	Coordinate signs
I	all six	$x > 0, y > 0$
II	sin, csc	$x < 0, y > 0$
III	tan, cot	$x < 0, y < 0$
IV	cos, sec	$x > 0, y < 0$

For example,

$$\cos 225^\circ = -\cos 45^\circ = -\frac{\sqrt{2}}{2}, \quad \csc 405^\circ = \csc 45^\circ = \sqrt{2}.$$

6.2.3 Even and odd functions

Reflection in the x -axis on the unit circle gives

$$\sin(-\theta) = -\sin \theta, \quad \cos(-\theta) = \cos \theta, \quad \tan(-\theta) = -\tan \theta, \quad (6.10)$$

$$\csc(-\theta) = -\csc \theta, \quad \sec(-\theta) = \sec \theta, \quad \cot(-\theta) = -\cot \theta. \quad (6.11)$$

Thus sine, tangent, cosecant, and cotangent are odd functions, while cosine and secant are even.

6.2.4 Special angles

The exact values at 0 , $\pi/6$, $\pi/4$, $\pi/3$, and $\pi/2$ follow from the 30° – 60° – 90° and 45° – 45° – 90° triangles.

θ	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$
$\sin \theta$	0	$1/2$	$\sqrt{2}/2$	$\sqrt{3}/2$	1
$\cos \theta$	1	$\sqrt{3}/2$	$\sqrt{2}/2$	$1/2$	0
$\tan \theta$	0	$1/\sqrt{3}$	1	$\sqrt{3}$	undefined

Table 6.1: Exact trigonometric values at the principal special angles.

Example 6.1: Finding all six functions from one ratio

Suppose $\sin \theta = -3/5$ and θ lies in Quadrant IV. A reference triangle has opposite side -3 , hypotenuse 5 , and adjacent side $\sqrt{5^2 - 3^2} = 4$. Therefore,

$$\begin{aligned} \cos \theta &= \frac{4}{5}, & \tan \theta &= -\frac{3}{4}, & \csc \theta &= -\frac{5}{3}, \\ \sec \theta &= \frac{5}{4}, & \cot \theta &= -\frac{4}{3}. \end{aligned}$$

6.3 Trigonometric graphs

The sine and cosine functions have domain $\mathbb{R}$, range $[-1, 1]$, and period 2π . Sine is odd and cosine is even. The tangent function has period π , range $\mathbb{R}$, zeros at $x = k\pi$, and vertical asymptotes at $x = \pi/2 + k\pi$.

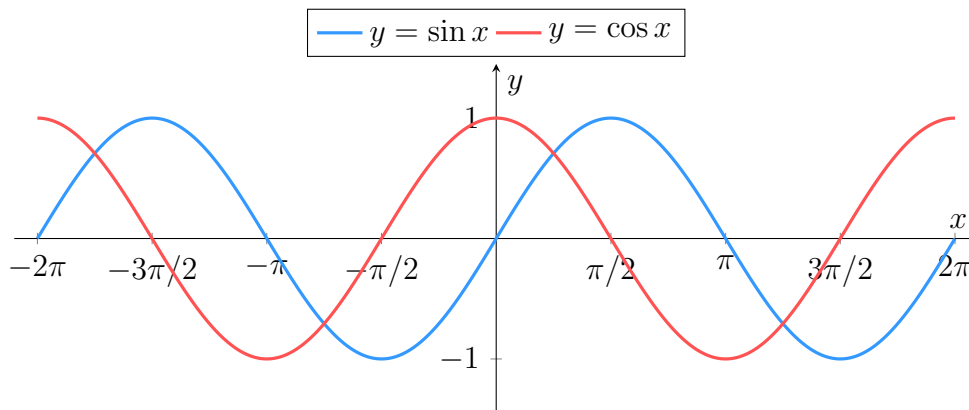


Figure 6.4: The basic sine and cosine graphs.

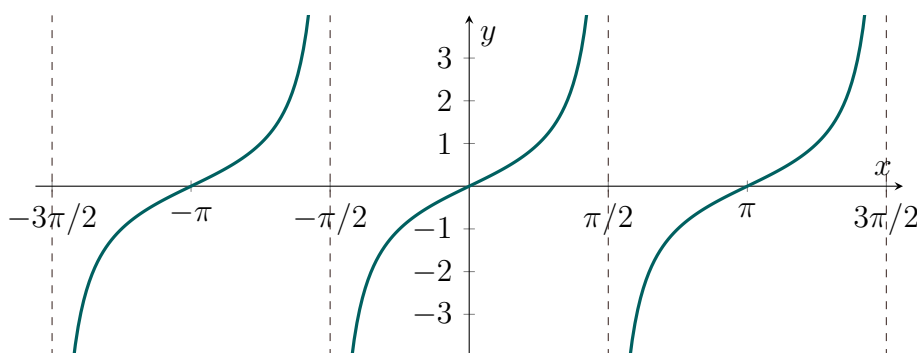


Figure 6.5: The tangent graph and its vertical asymptotes.

The reciprocal graphs $y = \sec x$, $y = \csc x$, and $y = \cot x$ inherit their undefined points from the zeros of cosine, sine, and tangent, respectively. In particular,

$$\begin{aligned}\sec x &\in (-\infty, -1] \cup [1, \infty), \\ \csc x &\in (-\infty, -1] \cup [1, \infty).\end{aligned}$$

6.3.1 Graph transformations

For

$$y = a \sin(b(x - h)) + d \quad \text{or} \quad y = a \cos(b(x - h)) + d, \quad (6.12)$$

the principal graph features are

$$\begin{aligned}
 \text{amplitude} &= |a|, \\
 \text{period} &= \frac{2\pi}{|b|}, \\
 \text{horizontal shift} &= h, \\
 \text{midline} &= y = d, \\
 \text{range} &= [d - |a|, d + |a|].
 \end{aligned}$$

For tangent and cotangent, the period is $\pi/|b|$, and there is no amplitude because their ranges are unbounded.

Example 6.2: Reading a transformed graph

For

$$y = -3 \sin(2(x - \pi/2)) + 1,$$

the amplitude is 3, the period is π , the graph is shifted right by $\pi/2$, reflected in its midline, and shifted upward by 1. Its midline is $y = 1$ and its range is $[-2, 4]$.

6.4 Angles of elevation and depression

An **angle of elevation** is measured upward from an observer's horizontal line of sight. An **angle of depression** is measured downward. Since horizontal lines are parallel, an angle of depression often equals the corresponding angle of elevation by alternate interior angles.

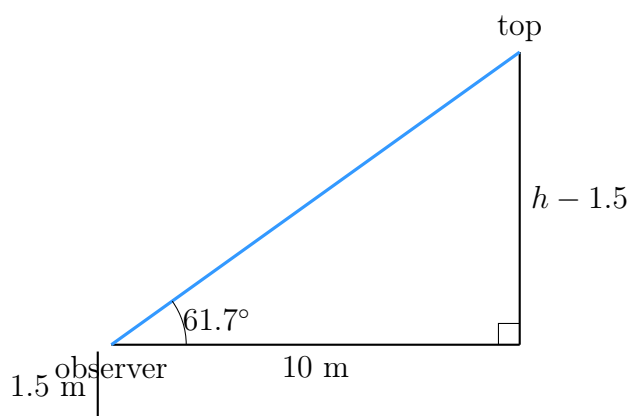


Figure 6.6: A flagpole observed from eye level 1.5 m above the ground.

In Figure 6.6,

$$\tan 61.7^\circ = \frac{h - 1.5}{10},$$

so

$$h = 10 \tan 61.7^\circ + 1.5 \approx \boxed{20.072 \text{ m}}.$$

When two observations are made from points separated by a known distance, draw one right triangle for each observation. The unknown distance shared by the triangles can then be eliminated algebraically.

6.5 Trigonometric identities

An identity is an equation that is true for every value in the common domain of its two sides. Proving an identity is different from solving an equation: one usually begins with the more complicated side and transforms it until it matches the other side.

6.5.1 Pythagorean identities

On the unit circle, $(x, y) = (\cos \theta, \sin \theta)$ and $x^2 + y^2 = 1$. Therefore,

$$\boxed{\sin^2 \theta + \cos^2 \theta = 1.} \quad (6.13)$$

Dividing by $\cos^2 \theta$ and $\sin^2 \theta$, respectively, gives

$$\boxed{1 + \tan^2 \theta = \sec^2 \theta}, \quad (6.14)$$

$$\boxed{1 + \cot^2 \theta = \csc^2 \theta}. \quad (6.15)$$

For example,

$$\begin{aligned} \csc \theta - \cos \theta \cot \theta &= \frac{1}{\sin \theta} - \cos \theta \frac{\cos \theta}{\sin \theta} \\ &= \frac{1 - \cos^2 \theta}{\sin \theta} \\ &= \frac{\sin^2 \theta}{\sin \theta} \\ &= \sin \theta. \end{aligned}$$

6.5.2 Compound-angle identities

The sine, cosine, and tangent of a sum or difference are

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B, \quad (6.16)$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B, \quad (6.17)$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}, \quad (6.18)$$

where all expressions involved are defined.

One geometric derivation begins with two unit-circle points $P = (\cos A, \sin A)$ and $Q = (\cos B, \sin B)$. Comparing the squared chord length PQ^2 with the chord subtending $A - B$ gives

$$\cos(A - B) = \cos A \cos B + \sin A \sin B.$$

The remaining formulae follow by replacing an angle with its negative and using the even-odd identities.

Example 6.3: Exact values from compound angles

Since $15^\circ = 45^\circ - 30^\circ$,

$$\begin{aligned} \sin 15^\circ &= \sin 45^\circ \cos 30^\circ - \cos 45^\circ \sin 30^\circ \\ &= \frac{\sqrt{2}}{2} \frac{\sqrt{3}}{2} - \frac{\sqrt{2}}{2} \frac{1}{2} \\ &= \frac{\sqrt{6} - \sqrt{2}}{4}. \end{aligned}$$

6.5.3 Double- and triple-angle identities

Setting $B = A$ in the compound-angle identities gives

$$\sin 2A = 2 \sin A \cos A, \quad (6.19)$$

$$\cos 2A = \cos^2 A - \sin^2 A = 1 - 2 \sin^2 A = 2 \cos^2 A - 1, \quad (6.20)$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}. \quad (6.21)$$

Applying the compound identities once more yields

$$\cos 3A = 4 \cos^3 A - 3 \cos A, \quad (6.22)$$

$$\sin 3A = 3 \sin A - 4 \sin^3 A. \quad (6.23)$$

If $\cos A = -3/5$ and $\pi/2 < A < \pi$, then $\sin A = 4/5$, so

$$\sin 2A = -\frac{24}{25}, \quad \cos 2A = -\frac{7}{25}, \quad \tan 2A = \frac{24}{7}.$$

6.5.4 Half-angle identities

Rearranging the double-angle formulae gives

$$\sin \frac{A}{2} = \pm \sqrt{\frac{1 - \cos A}{2}}, \quad (6.24)$$

$$\cos \frac{A}{2} = \pm \sqrt{\frac{1 + \cos A}{2}}, \quad (6.25)$$

$$\tan \frac{A}{2} = \pm \sqrt{\frac{1 - \cos A}{1 + \cos A}} = \frac{\sin A}{1 + \cos A} = \frac{1 - \cos A}{\sin A}. \quad (6.26)$$

The sign is determined by the quadrant containing $A/2$; it is not chosen arbitrarily. For example, 112.5° lies in Quadrant II, so

$$\begin{aligned} \cos 112.5^\circ &= -\sqrt{\frac{1 + \cos 225^\circ}{2}} \\ &= -\frac{1}{2}\sqrt{2 - \sqrt{2}}. \end{aligned}$$

6.5.5 Product-to-sum identities

Adding and subtracting the compound-angle formulae gives

$$\sin A \cos B = \frac{1}{2} [\sin(A + B) + \sin(A - B)], \quad (6.27)$$

$$\cos A \sin B = \frac{1}{2} [\sin(A + B) - \sin(A - B)], \quad (6.28)$$

$$\cos A \cos B = \frac{1}{2} [\cos(A + B) + \cos(A - B)], \quad (6.29)$$

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]. \quad (6.30)$$

Thus,

$$\begin{aligned} \sin 3\theta \cos 2\theta &= \frac{1}{2}(\sin 5\theta + \sin \theta), \\ 2 \cos 6\theta \cos 3\theta &= \cos 9\theta + \cos 3\theta. \end{aligned}$$

6.5.6 Sum-to-product identities

Writing $A = (K + M)/2$ and $B = (K - M)/2$ in the preceding identities gives

$$\sin K + \sin M = 2 \sin \frac{K+M}{2} \cos \frac{K-M}{2}, \quad (6.31)$$

$$\sin K - \sin M = 2 \cos \frac{K+M}{2} \sin \frac{K-M}{2}, \quad (6.32)$$

$$\cos K + \cos M = 2 \cos \frac{K+M}{2} \cos \frac{K-M}{2}, \quad (6.33)$$

$$\cos K - \cos M = -2 \sin \frac{K+M}{2} \sin \frac{K-M}{2}. \quad (6.34)$$

For example,

$$\cos 3\theta - \cos 5\theta = 2 \sin 4\theta \sin \theta.$$

6.6 Trigonometric equations

Solving a trigonometric equation requires both algebra and periodicity. A calculator may produce a principal value, but it does not automatically produce every solution.

6.6.1 Linear form

For $-1 \leq a \leq 1$, let $\alpha = \sin^{-1} a$ or $\alpha = \cos^{-1} a$ as appropriate. General solutions may be expressed as

$$\sin x = a \implies x = \alpha + 2k\pi \quad \text{or} \quad x = \pi - \alpha + 2k\pi, \quad (6.35)$$

$$\cos x = a \implies x = \pm\alpha + 2k\pi, \quad (6.36)$$

$$\tan x = a \implies x = \tan^{-1} a + k\pi, \quad (6.37)$$

where $k \in \mathbb{Z}$. When an interval is prescribed, substitute enough integer values of k to capture every solution in that interval.

Example 6.4: Solving a linear trigonometric equation

Solve $\cos 2x = -\sqrt{2}/2$ for $0 \leq x \leq 2\pi$. Since cosine is negative in Quadrants II and III,

$$2x = \frac{3\pi}{4} + 2k\pi \quad \text{or} \quad 2x = \frac{5\pi}{4} + 2k\pi.$$

Therefore,

$$x = \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{11\pi}{8}, \frac{13\pi}{8}.$$

Equations with a common factor must be treated carefully. For example,

$$\begin{aligned}\sin x \cos x &= \cos x \\ \cos x(\sin x - 1) &= 0.\end{aligned}$$

Dividing by $\cos x$ would incorrectly discard every solution for which $\cos x = 0$.

6.6.2 Quadratic form

Equations such as

$$a \sin^2 x + b \sin x + c = 0$$

are solved by substituting $u = \sin x$, solving the resulting quadratic, and then rejecting roots outside $[-1, 1]$. The same method applies to cosine and tangent.

For example,

$$\begin{aligned}4 \sin^2 x - 5 \sin x + 1 &= 0 \\ (4 \sin x - 1)(\sin x - 1) &= 0.\end{aligned}$$

Thus $\sin x = 1/4$ or $\sin x = 1$. On $0^\circ \leq x \leq 180^\circ$,

$$x \approx 14.478^\circ, \quad 90^\circ, \quad 165.522^\circ.$$

6.6.3 Addition form

An expression

$$a \cos x + b \sin x$$

can be reduced to a single sinusoid. If

$$a \cos x + b \sin x = R \cos(x - \alpha),$$

then comparison with $R \cos x \cos \alpha + R \sin x \sin \alpha$ gives

$$\boxed{R = \sqrt{a^2 + b^2}}, \quad \boxed{R \cos \alpha = a, \quad R \sin \alpha = b}. \quad (6.38)$$

The quadrant of α must agree with the signs of a and b . Equivalently, one may use $R \sin(x + \beta)$.

For

$$f(x) = \cos x - 3 \sin x,$$

we have $R = \sqrt{10}$ and may choose $\alpha = \tan^{-1}(-3)$, so

$$f(x) = \sqrt{10} \cos(x - \alpha).$$

Hence the maximum and minimum values are $\sqrt{10}$ and $-\sqrt{10}$. This reduction also converts equations involving $a \cos x + b \sin x$ into ordinary cosine or sine equations.

6.7 Inverse trigonometric functions

The periodic trigonometric functions are not one-to-one on their full domains. An inverse function exists only after restricting each original function to an interval on which it is one-to-one.

The notation $\sin^{-1} x$ denotes inverse sine, not $1/\sin x$. The reciprocal of sine is $\csc x$.

Definition 6.1: Principal value

The principal value of an inverse trigonometric function is the unique angle in its chosen principal range whose trigonometric value equals the given number.

The standard principal branches used in this book are shown below.

Function	Domain	Principal range
$y = \sin^{-1} x$	$[-1, 1]$	$[-\pi/2, \pi/2]$
$y = \cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
$y = \tan^{-1} x$	$\mathbb{R}$	$(-\pi/2, \pi/2)$
$y = \cot^{-1} x$	$\mathbb{R}$	$(0, \pi)$
$y = \sec^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$[0, \pi] \setminus \{\pi/2\}$
$y = \csc^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$[-\pi/2, \pi/2] \setminus \{0\}$

Table 6.2: Domains and principal ranges of inverse trigonometric functions.

For example,

$$\begin{aligned}\sin^{-1} \left(\frac{1}{\sqrt{2}} \right) &= \frac{\pi}{4}, \\ \cos^{-1} \left(-\frac{1}{2} \right) &= \frac{2\pi}{3}, \\ \tan^{-1} \left(-\frac{1}{\sqrt{3}} \right) &= -\frac{\pi}{6}.\end{aligned}$$

The last answer is $-\pi/6$, rather than $5\pi/6$, because inverse tangent must return a value in $(-\pi/2, \pi/2)$.

6.7.1 Graphs of inverse trigonometric functions

The graph of an inverse function is the reflection of the original restricted graph in the line $y = x$. Consequently, the domain and range are interchanged.

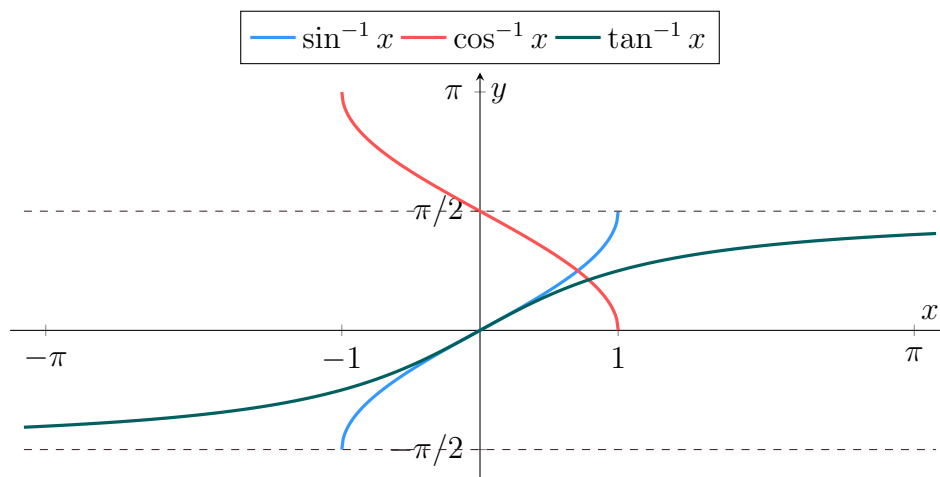


Figure 6.7: The principal inverse-sine, inverse-cosine, and inverse-tangent graphs.

Inverse tangent has horizontal asymptotes $y = \pm\pi/2$. Inverse secant and inverse cosecant have two branches because their domains contain two disjoint intervals.

6.7.2 Composition properties

Applying a trigonometric function after its inverse always recovers the input, provided the input lies in the inverse function's domain:

$$\sin(\sin^{-1} x) = x, \quad -1 \leq x \leq 1, \quad (6.39)$$

$$\cos(\cos^{-1} x) = x, \quad -1 \leq x \leq 1, \quad (6.40)$$

$$\tan(\tan^{-1} x) = x, \quad x \in \mathbb{R}. \quad (6.41)$$

The reverse composition requires the angle to lie in the principal range:

$$\sin^{-1}(\sin \theta) = \theta, \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2},$$

$$\cos^{-1}(\cos \theta) = \theta, \quad 0 \leq \theta \leq \pi,$$

$$\tan^{-1}(\tan \theta) = \theta, \quad -\frac{\pi}{2} < \theta < \frac{\pi}{2}.$$

Outside these ranges, the angle must first be replaced by the coterminal or symmetry-related angle in the correct principal range. For example,

$$\sin^{-1}\left(\sin \frac{5\pi}{6}\right) = \sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6},$$

not $5\pi/6$.

6.7.3 Reciprocal and symmetry properties

With the principal branches in Table 6.2,

$$\csc^{-1} x = \sin^{-1} \left(\frac{1}{x} \right), \quad |x| \geq 1, \quad (6.42)$$

$$\sec^{-1} x = \cos^{-1} \left(\frac{1}{x} \right), \quad |x| \geq 1. \quad (6.43)$$

For inverse cotangent, care is needed because of its range $(0, \pi)$:

$$\cot^{-1} x = \begin{cases} \tan^{-1}(1/x), & x > 0, \\ \pi/2, & x = 0, \\ \pi + \tan^{-1}(1/x), & x < 0. \end{cases}$$

The odd and complementary properties include

$$\sin^{-1}(-x) = -\sin^{-1} x, \quad (6.44)$$

$$\tan^{-1}(-x) = -\tan^{-1} x, \quad (6.45)$$

$$\cos^{-1}(-x) = \pi - \cos^{-1} x, \quad (6.46)$$

$$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}, \quad (6.47)$$

$$\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}. \quad (6.48)$$

For $|x| \geq 1$, the chosen branches also give

$$\sec^{-1} x + \csc^{-1} x = \frac{\pi}{2}.$$

6.7.4 Algebraic evaluation of inverse compositions

Let $\theta = \sin^{-1} x$. Since $-\pi/2 \leq \theta \leq \pi/2$, we know $\cos \theta \geq 0$, so

$$\boxed{\cos(\sin^{-1} x) = \sqrt{1 - x^2}}. \quad (6.49)$$

Similarly,

$$\tan(\sin^{-1} x) = \frac{x}{\sqrt{1 - x^2}}, \quad (6.50)$$

$$\sin(\cos^{-1} x) = \sqrt{1 - x^2}, \quad (6.51)$$

$$\tan(\cos^{-1} x) = \frac{\sqrt{1 - x^2}}{x}, \quad (6.52)$$

$$\sin(\tan^{-1} x) = \frac{x}{\sqrt{1 + x^2}}, \quad (6.53)$$

$$\cos(\tan^{-1} x) = \frac{1}{\sqrt{1 + x^2}}, \quad (6.54)$$

with the natural domain restrictions and signs inherited from the principal ranges.

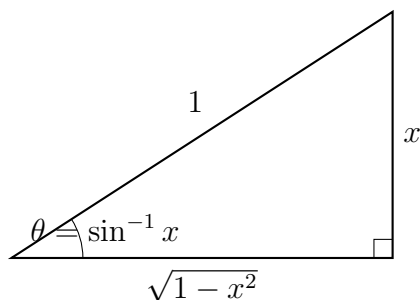


Figure 6.8: A reference triangle for expressions involving $\sin^{-1} x$.

Example 6.5: Simplifying a nested inverse expression

Let $\theta = \sin^{-1} x$. Then

$$\cos(\sin^{-1} x) = \cos \theta = \sqrt{1 - x^2}.$$

Also,

$$\cot(\csc^{-1} x) = \frac{\sqrt{x^2 - 1}}{\operatorname{sgn}(x)}, \quad |x| \geq 1,$$

where the sign is determined by the principal range of $\csc^{-1} x$.

6.7.5 Addition identities involving inverse tangent

If $\alpha = \tan^{-1} x$ and $\beta = \tan^{-1} y$, then

$$\tan(\alpha + \beta) = \frac{x + y}{1 - xy}. \quad (6.55)$$

Therefore,

$$\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x + y}{1 - xy} \right)$$

whenever the sum on the left lies in the principal range of inverse tangent. Otherwise, an appropriate multiple of π must be added. This branch check is essential.

For example, since both angles are positive and their sum is acute,

$$\begin{aligned} \tan^{-1} \frac{1}{7} + \tan^{-1} \frac{1}{13} &= \tan^{-1} \left(\frac{\frac{1}{7} + \frac{1}{13}}{1 - \frac{1}{91}} \right) \\ &= \boxed{\tan^{-1} \frac{2}{9}}. \end{aligned}$$

As another example,

$$\begin{aligned}\cos\left(\sin^{-1}\frac{3}{5} + \frac{\pi}{2}\right) &= -\sin\left(\sin^{-1}\frac{3}{5}\right) \\ &= \boxed{-\frac{3}{5}}.\end{aligned}$$

Chapter summary and common pitfalls

Key ideas. The unit circle extends trigonometric ratios to all real angles. Identities transform expressions, periodicity generates complete solution sets, and inverse functions use restricted domains.

Common pitfalls. Check whether an angle is in degrees or radians, include all periodic solutions, retain the correct quadrant sign, and do not cancel trigonometric expressions when the cancelled factor may be zero.

6.8 Practice and exercises

Angles, exact values, and graphs

1. Convert to radians:

- (a) 225° ;
- (b) -150° ;
- (c) 810° .

2. Convert to degrees:

- (a) $7\pi/6$;
- (b) $-5\pi/4$;
- (c) $11\pi/3$.

3. Complete the angle and coterminal-angle review.

- (a) For -750° , find a coterminal angle in $[0^\circ, 360^\circ)$ and its reference angle.
- (b) For $13\pi/4$, find a coterminal angle in $[0, 2\pi)$ and its reference angle.
- (c) For 520° , find a coterminal angle in $[0^\circ, 360^\circ)$ and its reference angle.
- (d) Identify the quadrantal angles in

$$60^\circ, 185^\circ, \frac{5\pi}{2}, 270^\circ, -660^\circ, -270^\circ, 3\pi, -\pi, 905^\circ.$$

- (e) Give two positive and two negative angles coterminal with 120° .

4. Find the exact values of

- (a) $\sin 150^\circ$;
- (b) $\cos 225^\circ$;
- (c) $\tan 330^\circ$;
- (d) $\csc(7\pi/6)$;
- (e) $\sec(-\pi/3)$;

- (f) $\cot(3\pi/4)$;
- (g) $\sin 135^\circ$;
- (h) $\tan 300^\circ$;
- (i) $\sin(-330^\circ)$;
- (j) $\cos(5\pi/4)$;
- (k) $\sin(-7\pi/6)$;
- (l) $\tan(-5\pi/3)$.

5. Determine all six trigonometric ratios in each case.

- (a) The terminal side of θ passes through $(-5, 12)$.
- (b) $\cos \theta = 2/\sqrt{13}$ and θ is acute. Hence also find $\cot(\pi/2 - \theta)$ and $\csc(\theta - \pi/2)$.

6. Determine the quadrant and the remaining trigonometric ratios.

- (a) $\sin \theta = -3/5$ and θ lies in Quadrant IV.
- (b) $\tan(\pi/2 - \theta) = 2/3$ and $\sec \theta < 0$.

7. Simplify:

- (a) $\sin(\pi/2 - x)$;
- (b) $\cos(-x)$;
- (c) $\tan(x - \pi/2)$;
- (d) $\csc(-x)$.

Use the unit circle to justify the cofunction and negative-angle formulae, then state which of the six trigonometric functions are even and which are odd.

8. Find the exact values of $\sin 15^\circ$ and $\cos 75^\circ$.

9. A flagpole is observed from eye level 1.5 m above the ground. The observer is 10 m from its base and measures an elevation angle of 61.7° . Find the pole's height to three decimal places.

10. A tree casts an 8.8 m shadow when the Sun's rays make an angle of 49° with the horizontal. Find the tree's height.

11. For $y = -3\sin(2(x - \pi/2)) + 1$, state the amplitude, period, horizontal shift, midline, and range.

12. State the domain, range, period, parity, zeros, and asymptotes of $y = \tan x$.

Identities and exact manipulation

13. Verify $\csc x - \cos x \cot x = \sin x$.

14. Verify

$$\frac{\tan x + \cot x}{\sec x \csc x} = 1.$$

15. Prove that

$$\frac{\sin x}{1 + \cos x} + \frac{1 + \cos x}{\sin x} = 2 \csc x.$$

16. Find the exact values of $\sin 75^\circ$, $\cos 15^\circ$, and $\tan 105^\circ$.

17. Given $\cos A = 5/13$ with A acute and $\sin B = -7/25$ with B in Quadrant III, find $\sin(A + B)$.

18. Given $\cos A = -3/5$ and $\pi/2 < A < \pi$, find $\sin 2A$, $\cos 2A$, and $\tan 2A$.

19. Prove $\cos 3x = 4 \cos^3 x - 3 \cos x$.

20. Use half-angle identities to find the exact values of

(a) $\cos 112.5^\circ$;

(b) $\tan 67.5^\circ$;

(c) $\sin 22.5^\circ$;

(d) $\cos 22.5^\circ$;

(e) $\tan 22.5^\circ$.

21. Convert to sums or differences:

(a) $\sin 3x \cos 2x$;

(b) $\sin 5x \sin 2x$;

(c) $2 \cos 6x \cos 3x$;

(d) $\sin 2x \cos 3x$;

(e) $\cos 3x \sin 6x$;

(f) $\sin 4x \sin x$.

22. Convert to products:

(a) $\cos 3x - \cos 5x$;

(b) $\cos 6x + \cos 3x$;

(c) $\sin 4x + \sin 2x$;

- (d) $\cos 5x - \cos 3x$;
- (e) $\cos 3x + \cos 6x$;
- (f) $\sin 7x - \sin 3x$.

23. Verify

$$\frac{\sin 4x + \sin 2x}{\cos 4x - \cos 2x} = -\cot x.$$

Trigonometric equations

24. Solve $\sin x = \sqrt{3}/2$ for $0 \leq x \leq 2\pi$.

25. Solve the double-angle equations.

- (a) $\cos 2x = -\sqrt{2}/2$ for $0 \leq x \leq 2\pi$;
- (b) $\cos 2x = -1/2$ for $0 \leq x \leq 2\pi$.

26. Solve $\tan 3x = 1/\sqrt{3}$ for $0 \leq x \leq \pi$.

27. Solve $\sqrt{3} + 2 \cos 2x = 0$ for all real x .

28. Solve $\sin x \cos x = \cos x$ for $0 \leq x \leq 2\pi$.

29. Solve the equations involving $\sin x$.

- (a) $4 \sin^2 x - 5 \sin x + 1 = 0$ for $0^\circ \leq x \leq 180^\circ$;
- (b) $4 \sin^2 x - 3 = 0$ for $-\pi \leq x \leq 3\pi$.

30. Solve the equations involving $\cos x$ and $\sec x$.

- (a) $4 \cos^2 x - 3 = 0$ for all real x ;
- (b) $\sec^2 x = 2$ for $-\pi \leq x \leq \pi$.

31. Express $3 \cos x + 4 \sin x$ in the form $R \cos(x - \alpha)$, where $R > 0$ and $0 < \alpha < \pi/2$. Hence state its maximum and minimum values.

32. Express $\cos x - 3 \sin x$ as one cosine function and solve $\cos x - 3 \sin x = 1$ for $0 \leq x < 2\pi$.

Inverse trigonometric functions

33. State the domain and principal range of $\sin^{-1} x$, $\cos^{-1} x$, $\tan^{-1} x$, $\sec^{-1} x$, and $\csc^{-1} x$.

34. Find the principal values:

- (a) $\sin^{-1}(1/\sqrt{2})$;
- (b) $\cos^{-1}(-1/2)$;
- (c) $\tan^{-1}(-1/\sqrt{3})$;
- (d) $\sec^{-1}(-2)$;
- (e) $\csc^{-1}(2)$.

35. Evaluate:

- (a) $\sin(\sin^{-1}(3/5))$;
- (b) $\cos(\cos^{-1}(-4/7))$;
- (c) $\tan(\tan^{-1} 5)$.

36. Evaluate:

- (a) $\sin^{-1}(\sin(5\pi/6))$;
- (b) $\cos^{-1}(\cos(7\pi/4))$;
- (c) $\tan^{-1}(\tan(3\pi/4))$.

37. Prove $\sin^{-1} x + \cos^{-1} x = \pi/2$ for $-1 \leq x \leq 1$.

38. Simplify $\cos(\sin^{-1} x)$ and $\tan(\sin^{-1} x)$.

39. Simplify $\sin(\cos^{-1} x)$ and $\cos(\tan^{-1} x)$.

40. Find the exact value of $\cos(\sin^{-1}(3/5) + \pi/2)$.

41. Prove

$$\tan^{-1} \frac{1}{7} + \tan^{-1} \frac{1}{13} = \tan^{-1} \frac{2}{9}.$$

42. Determine all $x \in [-1, 1]$ satisfying $\sin^{-1} x = \cos^{-1} x$.

43. Solve $\tan^{-1} x = \pi/6$ and $\cos^{-1} x = 2\pi/3$.

44. Explain why $\sin^{-1}(\sin \theta) = \theta$ is not true for every real θ , and state the interval on which it is true.

Consolidated additional practice

The following questions group closely related skills into single exercises. They draw on the tutorial material while avoiding repeated one-step questions.

45. Verify each identity on every value for which both sides are defined.

- (a) $(\sin x + \cos x)^2 + (\sin x - \cos x)^2 = 2$;
(b) $\frac{1 - \sin x}{\cos x} = \frac{\cos x}{1 + \sin x}$;
(c) $\frac{\sec x - \cos x}{\tan x} = \sin x$.
46. On one set of axes, sketch $y = \sin x$, $y = \cos x$, $y = \cos(\pi/2 - x)$, and $y = \tan x$ for $-2\pi \leq x \leq 2\pi$. Mark intercepts and vertical asymptotes, then use the graphs to explain the cofunction identity in Question 7.
47. Let $f(x) = \sin^2 x + \cos x$.
- (a) Rewrite f entirely in terms of $\cos x$ and complete the square.
(b) Find its period and range.
(c) Find its zeros in exact inverse-trigonometric form.
48. Suppose
- $$\tan \alpha = -\frac{5}{12}, \quad -\frac{\pi}{2} < \alpha < \frac{\pi}{2}, \quad \cos \beta = -\frac{12}{13}, \quad 0 \leq \beta \leq \pi.$$
- Determine the quadrants of α and β , then find the exact value of $\sec(\alpha - \beta)$.
49. Starting with
- $$7 \tan \theta + 2 \cos \theta = 5 \sec \theta,$$
- derive a quadratic equation in $\sin \theta$ and solve for $0 \leq \theta \leq \pi$.
50. Solve $3 \cos \theta = \cot \theta$ for $0^\circ \leq \theta < 360^\circ$. Give non-quadrantal answers to the nearest 0.001° .
51. Solve
- $$2 \tan^2 \theta + 5 \tan \theta - 3 = 0$$
- for $0^\circ \leq \theta < 360^\circ$, to the nearest 0.001° .
52. Express $\sin 6x + \sin 2x$ as a product. Hence prove
- $$\sin 6x + \sin 2x - \sin 4x = 4 \cos 2x \cos 3x \sin x.$$
53. Write $5 \sin x - 12 \cos x$ in the form $R \sin(x - \alpha)$, where $R > 0$ and $0 < \alpha < \pi/2$. State its maximum and minimum values and the corresponding general values of x . Find the acute solution of $5 \sin x - 12 \cos x = 2$.
54. Write $3 \cos x + 4 \sin x$ in the form $R \cos(x - \alpha)$. State its extrema and determine whether $3 \cos x + 4 \sin x = 2$ has an acute solution.

55. Write $8 \cos x - 15 \sin x$ in the form $R \cos(x + \alpha)$. State its extrema and find the acute solution of $8 \cos x - 15 \sin x = 2$.

56. Find the domains of

$$f(x) = \tan^{-1} x, \quad g(x) = \cos^{-1}(3x), \quad h(x) = f(x) - g(x).$$

57. If $\theta = \tan^{-1}(4/3)$, find the exact values of the other five trigonometric ratios of θ .

58. If $\theta = \sec^{-1}(13/5)$, find the exact values of the other five trigonometric ratios of θ .

59. Evaluate exactly.

- (a) $\sin\left(2 \tan^{-1} \frac{2}{3}\right)$;
- (b) $\cos\left(\sin^{-1} \frac{3}{5} + \sec^{-1} 2\right)$;
- (c) $\tan\left(\frac{\pi}{4} + \cot^{-1} \frac{4}{5}\right)$;
- (d) $\sin\left(\tan^{-1} \sqrt{3} + \cos^{-1} \frac{1}{\sqrt{2}}\right)$.

60. Evaluate exactly.

- (a) $\sin\left(\sec^{-1} \frac{2}{\sqrt{3}}\right)$;
- (b) $\cos\left(2 \sin^{-1} \frac{3}{5} + \frac{\pi}{2}\right)$;
- (c) $\cos\left(\tan^{-1} \frac{4}{3} - \csc^{-1} \frac{3}{\sqrt{5}}\right)$;
- (d) $\csc(\sec^{-1} 3) + \tan(\cos^{-1}(1/2))$.

61. Simplify each expression algebraically and state the indicated interval on which the simplified form is valid.

- (a) $\cos\left(\sin^{-1} \frac{x-1}{x}\right)$ for $x \geq 1/2$;
- (b) $\sin\left(\cos^{-1} \frac{x+1}{\sqrt{x^2-1}}\right)$ for $x < -1$.

62. Prove the inverse-function symmetry formulae

$$\sin^{-1}(-x) = -\sin^{-1} x, \quad \cos^{-1}(-x) = \pi - \cos^{-1} x,$$

and show that

$$\sec^{-1} x + \csc^{-1} x = \frac{\pi}{2} \quad (|x| \geq 1).$$

63. Evaluate exactly, paying careful attention to principal ranges:

- (a) $\tan^{-1} 1 + \tan^{-1} 2$;
- (b) $\tan^{-1} 2 - \tan^{-1} 1$.

Explain why the ordinary tangent addition formula alone does not decide which branch of $\tan^{-1}$ is required.

64. Find the principal values:

- (a) $\sin^{-1}(\sin(7\pi/6))$;
- (b) $\cos^{-1}(\cos(-2\pi/3))$;
- (c) $\tan^{-1}(\tan(5\pi/6))$;
- (d) $\sec^{-1}(\sec(-2\pi/3))$.

65. Evaluate exactly:

- (a) $\cos(\tan^{-1}(3/4))$;
- (b) $\tan(\cos^{-1}(-5/13))$;
- (c) $\sin(\sec^{-1}(-5/4))$;
- (d) $\csc(\tan^{-1}(12/5))$.

Applications and modelling

The first ten questions below replace repeated symbolic drills with applications from robotics, navigation, mechanics, physiology, optics, and transportation.

66. A planar robotic arm has two links. The first link is 4 units long and makes an angle of 30° with the positive x -axis. The second link is 3 units long and is turned a further 60° from the first.

- (a) Find the coordinates of the joint and the end effector.
- (b) Find the distance and direction of the end effector from the origin.

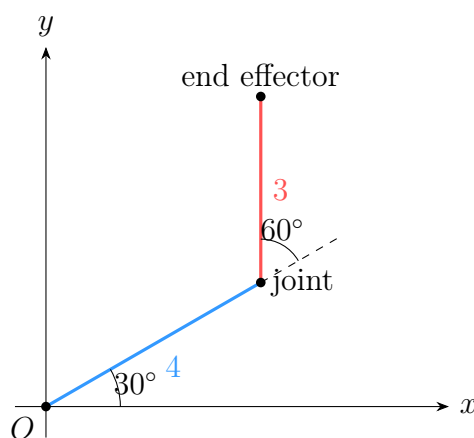


Figure 6.9: Two-link robotic arm for Question 66.

67. A survey boat travels 12 km due east and then 9 km at 40° north of east.

- (a) Resolve the total displacement into east and north components.
 - (b) Find the straight-line distance and direction from the starting point.
68. An aircraft has an airspeed of 180 km h^{-1} due north while a wind blows at 40 km h^{-1} due east. Find the aircraft's ground speed and its direction east of north.
69. The displacement of a piston is modelled by

$$s(t) = 12 + 5 \sin(4\pi t - \pi/6)$$

centimetres, where t is in seconds.

- (a) State the midline, amplitude, frequency, and period.
 - (b) Find the first time $t \geq 0$ at which the piston is at maximum displacement.
 - (c) During the first period, find the times when $s(t) = 12$.
70. A simplified breathing model gives lung volume

$$V(t) = 2.8 + 0.6 \sin\left(\frac{\pi t}{2}\right)$$

litres, where t is in seconds.

- (a) State the range and breathing-cycle period.
 - (b) Find the first time when $V = 3.1 \text{ L}$.
 - (c) During the first cycle, determine when $V \geq 3.1 \text{ L}$.
71. The ideal power received by a solar panel is $P = 2.4 \cos \theta \text{ kW}$, where $0^\circ \leq \theta \leq 90^\circ$ is the angle between the Sun's rays and the panel's normal. Find the incidence angles for which the received power is at least 1.8 kW .
72. Surveyors observe a triangular site ABC . They measure $AB = 180 \text{ m}$, $\angle A = 52^\circ$, and $\angle C = 73^\circ$. Find $\angle B$, BC , and AC .
73. Snell's law is $n_1 \sin \theta_1 = n_2 \sin \theta_2$.
- (a) A ray passes from air ($n_1 = 1.00$) into glass ($n_2 = 1.50$) with incidence angle 45° . Find the refracted angle.
 - (b) Find the critical angle for a ray travelling from the same glass into air.
74. The magnitude of the torque produced by a force is $\tau = rF \sin \theta$. A 120 N force is applied 0.35 m from a pivot.
- (a) Find the maximum possible torque.
 - (b) For $0^\circ \leq \theta \leq 180^\circ$, find the angles for which $\tau \geq 30 \text{ N m}$.

75. A frictionless banked road satisfies

$$\tan \theta = \frac{v^2}{rg}.$$

Find the required banking angle for $v = 20 \text{ m s}^{-1}$, $r = 80 \text{ m}$, and $g = 9.8 \text{ m s}^{-2}$. Then find the design speed when $\theta = 30^\circ$ and the other quantities are unchanged.

76. Buildings A and B stand on level ground 35 m apart. From the roof of A , the angle of depression to the foot of B is 34° and the angle of elevation to the roof of B is 24° . Find the heights of both buildings.

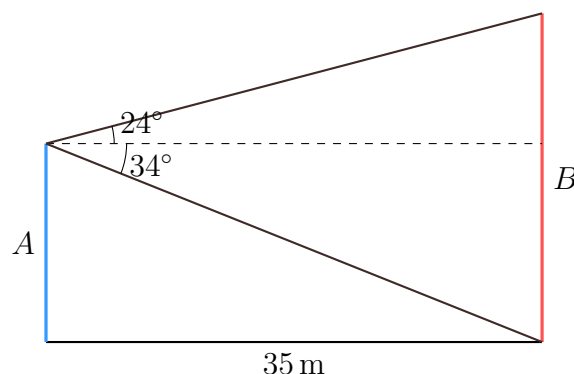


Figure 6.10: Angles observed from the roof of building A .

77. Two observations of the top of a building are made from points on the same straight level road. The points are 35 ft apart, and the nearer point gives an angle of elevation of 74.2° while the farther point gives 51.8° . Find the building height and the distance from the nearer point to its foot.
78. Two vertical poles, 10 m and 15 m high, stand 25 m apart. A wire joins the top of each pole to the foot of the other. At what height above the ground do the wires cross?
79. A stationary ground-level camera is a horizontal distance x from a missile's vertical flight path. During a tracked interval, the missile rises from altitude b to altitude $a + b$. Show that the angle swept out by the camera's line of sight is

$$\theta = \cot^{-1}\left(\frac{x}{a+b}\right) - \cot^{-1}\left(\frac{x}{b}\right).$$

State a physically meaningful domain for x , a , and b .

80. A triangle has side lengths 4, 2, and 3 units. Find, to the nearest degree, the angle opposite the side of length 4.
81. A lighthouse lamp is 60 m above sea level. The angle of depression to a boat is 12° . Assuming level water, find the horizontal distance from the boat to the lighthouse.

82. An access ramp must rise 0.9 m and may make an angle of at most 5° with the horizontal. Find the minimum horizontal length of ground required.
83. A Ferris wheel has radius 18 m, its centre is 20 m above the ground, and it completes one revolution every 40 s. A passenger starts at the lowest point.
- Construct a cosine model for the passenger's height $h(t)$.
 - Find the height after 10 s.
 - During the first revolution, when is the passenger 29 m above the ground?

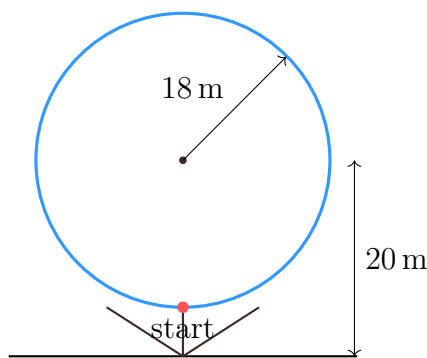


Figure 6.11: Ferris-wheel model for Question 83.

84. At a harbour, the water depth varies between 1.4 m and 5.2 m with a period of 12 hours. High tide occurs at $t = 0$.
- Construct a cosine model for the depth.
 - Find the depth after three hours.
 - Find the first two times after high tide when the depth is 2.0 m.
85. The number of daylight hours is modelled by

$$D(t) = 12 + 1.5 \cos\left(\frac{2\pi(t - 172)}{365}\right),$$

where t is the day number. Interpret the amplitude, midline, period, and phase shift, and state the predicted longest and shortest days.

86. A sound wave is represented by

$$y(t) = 0.004 \sin(440\pi t + \pi/3).$$

Find its amplitude, angular frequency, ordinary frequency, period, and horizontal phase shift.

87. An alternating voltage is $V(t) = 240 \sin(100\pi t)$ volts. Find its maximum magnitude and frequency, and determine the first positive time at which $V = 120$ volts.

88. Neglecting air resistance, the horizontal range of a projectile is

$$R = \frac{u^2 \sin 2\theta}{g}.$$

For $u = 20 \text{ m s}^{-1}$ and $g = 9.8 \text{ m s}^{-2}$, find the range at $\theta = 30^\circ$ and the angle and value of the greatest possible range.

89. Two walking routes of lengths 8 km and 11 km leave the same point with an included angle of 65° . How far apart are their endpoints?
90. A symmetrical roof has half-span 5.5 m and vertical rise 3.2 m. Find its pitch angle and the sloping rafter length.
91. A drone is 120 m above level ground and 160 m horizontally from an observer. Find the angle of elevation and the line-of-sight distance.
92. A point moves anticlockwise on a circle of radius 5 according to $\theta(t) = \pi t/6$, where t is in seconds and it starts on the positive x -axis. Write its coordinate functions and find its position when $t = 4$.
93. Monthly temperature is modelled by

$$T(t) = 27 + 4 \cos\left(\frac{2\pi(t-2)}{12}\right),$$

where $t = 1$ represents January. Find the predicted maximum and minimum, the months in which they occur, and the temperature in May.

94. Use a product-to-sum identity to resolve the signal $\sin 8t \cos 3t$ into two sine waves. State their angular frequencies.
95. Use a sum-to-product identity to rewrite

$$\cos(102\pi t) + \cos(98\pi t).$$

Identify the carrier frequency and the beat frequency.

96. Combine the signal $6 \cos t + 8 \sin t$ into one cosine function. State its amplitude and phase shift.
97. A camera is 20 m horizontally from a vertical object. Relative to the camera lens, the bottom and top of the object are 5 m and 17 m higher, respectively. Find the vertical angle subtended by the object.
98. A 10 m ladder has its foot 6 m from a vertical wall. Find its angle with the ground and the height reached.

99. From an aircraft 2000 m above level ground, the angles of depression to the near and far banks of a straight river are 39° and 27° . Find the river width, assuming both banks lie on the same side of the point directly beneath the aircraft.
100. A solar installation has idealised output

$$P(t) = 50 + 30 \sin\left(\frac{\pi(t-6)}{12}\right), \quad 0 \leq t \leq 24,$$

where P is in kilowatts and t is hours after midnight.

- State the maximum and minimum predicted outputs and when they occur.
- Determine the time interval during which $P(t) \geq 65$.
- Explain one limitation of this sinusoidal model in a real solar installation.

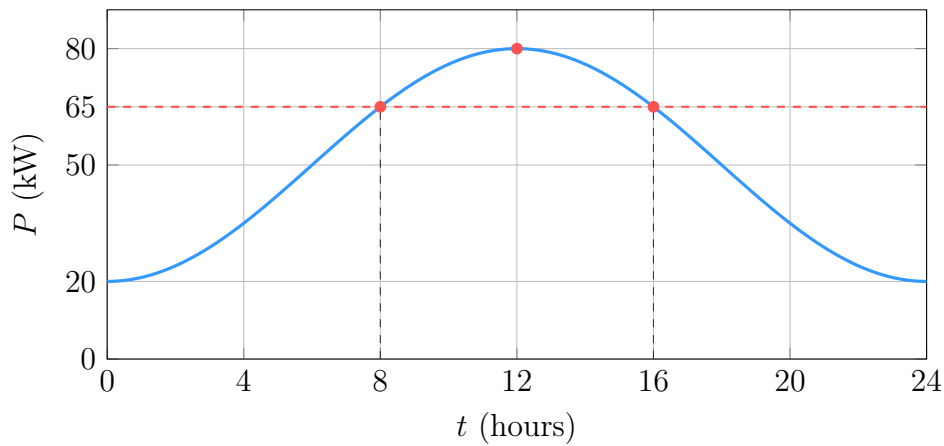


Figure 6.12: The idealised daily-output model and the 65 kW threshold.

Chapter 7

Linearisation and Linear Law

Chapter guide

Prerequisites. Functions, logarithms, coordinate graphs, gradients, and straight-line equations.

Learning outcomes. Readers should be able to choose transformations that linearise common models, plot transformed variables, determine constants from gradients and intercepts, and interpret the resulting model in its original variables.

Many relationships encountered in mathematics and the sciences are not linear when they are expressed in their original variables. Their graphs may be parabolas, reciprocal curves, exponential curves, or power curves. Even so, a suitable change of variables can often reveal an underlying straight line. This process is useful because the gradient and vertical intercept of a straight line are easy to estimate and interpret.

This chapter follows the learning progression of the Malaysian KSSM Additional Mathematics curriculum for linear law (Ministry of Education Malaysia 2018). It distinguishes linear and non-linear relationships, constructs and interprets lines of best fit, transforms common non-linear relationships into linear form, and uses the resulting straight lines to determine unknown constants and solve problems.

It is important to understand what a transformation does. The original relationship does not suddenly become linear in its original variables. Instead, new variables are defined so that the transformed data satisfy an equation of the form

$$Y = mX + c. \quad (7.1)$$

The straight line is therefore drawn on transformed axes, labelled X and Y , rather than necessarily on the original x - and y -axes.

7.1 Linear and non-linear relationships

Definition 7.1: Linear relationship.

A relationship between two variables is linear if its graph is a straight line. In suitable variables x and y , it may be written as

$$y = mx + c,$$

where m and c are constants. The constant m is the gradient and c is the vertical intercept.

Definition 7.2: Non-linear relationship.

A relationship is non-linear if its graph is not a straight line. Its rate of change is not constant over the interval being considered.

For exact data whose x -values are equally spaced, the first differences of the y -values provide a quick test. Constant first differences indicate a linear relationship. If the first differences vary, the relationship is non-linear, although a different transformation may make it linear.

Feature	Linear relationship	Non-linear relationship
Graph	Straight line	Curve, or a graph whose gradient changes
Rate of change	Constant	Variable
Equally spaced data	Constant first differences	First differences are generally not constant
Typical equation	$y = mx + c$	$y = ax^2 + b$, $y = ab^x$, $y = ax^n$, or $y = \frac{a}{x} + b$

Table 7.1: Comparison of linear and non-linear relationships.

7.1.1 Recognising a relationship from a table

Consider the following two sets of exact data.

x	0	1	2	3	4
y_1	2	5	8	11	14
y_2	1	3	9	19	33

For y_1 , every increase of one unit in x produces an increase of three units in y_1 . Hence $y_1 = 3x + 2$ is linear. For y_2 , the first differences are 2, 6, 10, 14, so the relationship is not linear in x and y_2 . In fact, $y_2 = 2x^2 + 1$, which becomes linear when x^2 is used as the horizontal variable.

Experimental data rarely lie exactly on a mathematical curve. Small departures may arise from measurement error, rounding, or natural variation. Consequently, linearity should be judged from the overall pattern rather than from exact equality of every difference.

7.2 Scatter plots and the line of best fit

A scatter plot represents paired observations (x, y) as separate points. If the points lie close to a straight line, a linear model may be suitable. The line used to describe the overall trend is called a *line of best fit*.

Definition 7.3: Line of best fit.

A line of best fit is a straight line that represents the general trend of a set of data. When drawn by inspection, the points should be distributed as evenly as possible above and below the line.

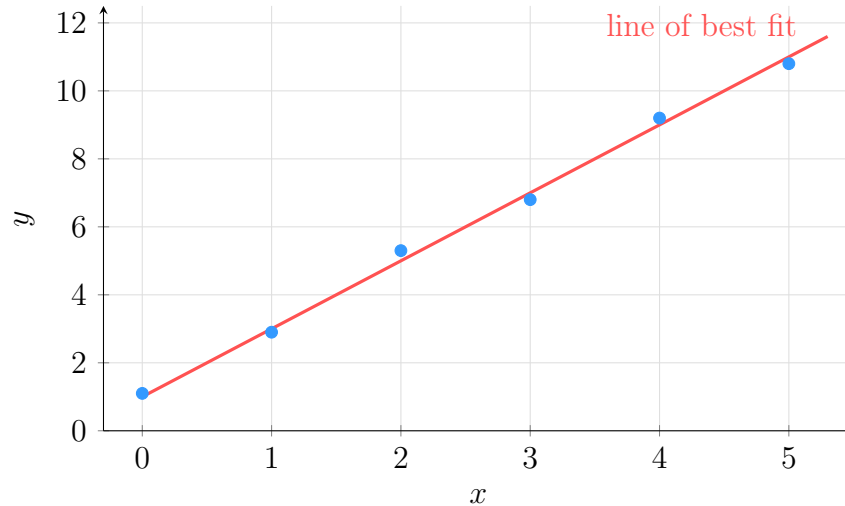


Figure 7.1: A scatter plot with a line of best fit drawn by inspection.

A sensible line of best fit should follow these principles:

- It should pass through the centre of the cluster rather than join the first and last observations.
- The points should be reasonably balanced above and below it.
- It need not pass through any particular data point.
- It should not be forced through the origin unless the context or the model requires a zero intercept.
- An isolated outlier should be investigated before it is allowed to influence the line strongly.

Technology may produce a regression line, whereas inspection produces a visually fitted line. Their equations need not be identical, but both should describe the same general trend when the linear model is appropriate.

7.2.1 Equation of a line of best fit

After drawing the line, choose two convenient points *on the line* that are far apart. These points do not have to be original observations. If they are (X_1, Y_1) and (X_2, Y_2) , then

$$m = \frac{Y_2 - Y_1}{X_2 - X_1}, \quad c = Y_1 - mX_1. \quad (7.2)$$

Using widely separated points reduces the effect of reading error. The equation is then written as $Y = mX + c$, with the transformed axis labels used exactly as they appear on the graph.

7.2.2 Interpreting the gradient and intercept

The gradient represents the change in Y for each unit increase in X . Its units are therefore

$$\frac{\text{units of } Y}{\text{units of } X}.$$

The intercept is the estimated value of Y when $X = 0$. This value is meaningful only if $X = 0$ is permitted by the model and relevant to the context. Estimates made within the observed range are called *interpolations*. Predictions beyond the observed range are *extrapolations* and should be treated more cautiously.

7.3 The principle of linearisation

Definition 7.4: Linearisation.

Linearisation is the process of defining transformed variables X and Y so that a non-linear relationship between x and y takes the straight-line form

$$Y = mX + c.$$

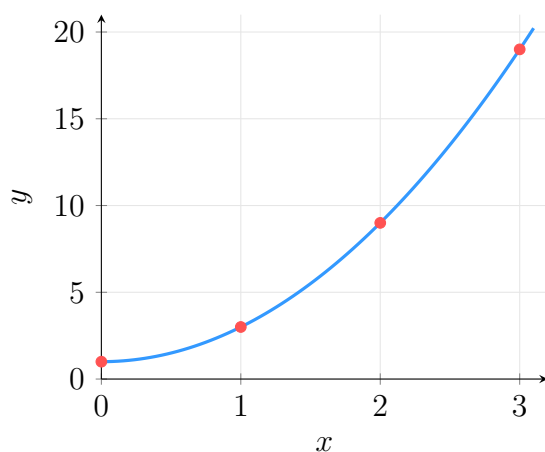
For example, suppose

$$y = 2x^2 + 1.$$

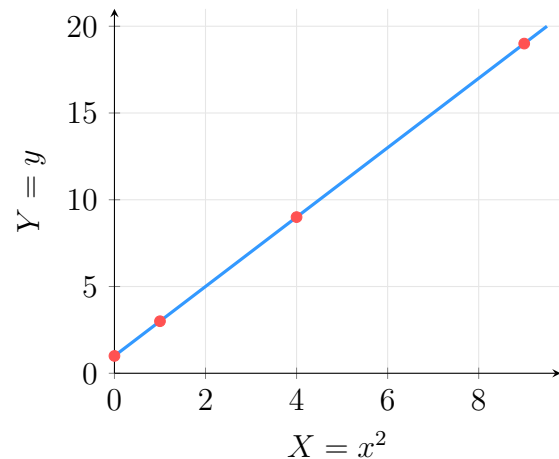
The graph of y against x is a parabola. If $X = x^2$ and $Y = y$, however, then

$$Y = 2X + 1,$$

which is a straight line with gradient 2 and intercept 1.



Original graph: y against x



Transformed graph: Y against X

Figure 7.2: Linearising the relationship $y = 2x^2 + 1$.

7.3.1 A systematic method

To linearise a relationship and interpret its graph:

1. Rearrange the given equation until it resembles $Y = mX + c$.
2. Identify the complete expressions represented by X and Y .
3. Transform every pair of data values consistently.
4. Label the axes with the transformed quantities and their units.
5. Plot the transformed points and draw a line of best fit.
6. Determine the gradient and intercept from the line.
7. Compare them with m and c to recover the original constants.
8. Check that the result respects all domain restrictions and is sensible in the original context.

7.4 Common transformations

The correct transformation is obtained from algebra, not from memorising the shape of a graph alone. Nevertheless, the following forms occur frequently.

Original relation	Linear form	X	Y	m	c
$y = ax^2 + b$	$Y = aX + b$	x^2	y	a	b
$y = a\sqrt{x} + b$	$Y = aX + b$	$\sqrt{x}$	y	a	b
$y = \frac{a}{x} + b$	$Y = aX + b$	$\frac{1}{x}$	y	a	b
$y = ax + \frac{b}{x}$	$xy = ax^2 + b$	x^2	xy	a	b
$y = \frac{a}{x + b}$	$\frac{1}{y} = \frac{1}{a}x + \frac{b}{a}$	x	$\frac{1}{y}$	$\frac{1}{a}$	$\frac{b}{a}$
$y = ax^n$	$\log y = n \log x + \log a$	$\log x$	$\log y$	n	$\log a$
$y = Ab^x$	$\log y = x \log b + \log A$	x	$\log y$	$\log b$	$\log A$
$y = Ae^{kx}$	$\ln y = kx + \ln A$	x	$\ln y$	k	$\ln A$

Table 7.2: Frequently used linearising transformations.

For logarithmic transformations, all quantities inside logarithms must be positive. The same logarithm base must be used throughout a calculation. Either common logarithms or natural logarithms may be used for a power law, but the intercept must be converted back using the same base.

7.5 Algebraic techniques for obtaining linear form

7.5.1 Direct substitution

If the non-linear expression already appears as a single term, it can often be treated as a new variable immediately.

Example 7.1: A quadratic transformation

The variables x and y satisfy

$$y = ax^2 + b.$$

A graph of y against x^2 has gradient 3.4 and vertical intercept -1.2 . Determine a and b .

Let $X = x^2$ and $Y = y$. Then

$$Y = aX + b.$$

Comparison with $Y = mX + c$ gives

$$a = m = 3.4, \quad b = c = -1.2.$$

Thus the original relationship is

$$y = 3.4x^2 - 1.2.$$

7.5.2 Multiplication or division

Sometimes the required straight-line variables become visible only after the equation has been multiplied or divided by a variable.

Example 7.2: Removing a reciprocal term

Suppose

$$y = ax + \frac{b}{x}, \quad x \neq 0.$$

Show how a straight-line graph may be obtained. If its gradient is 4 and its vertical intercept is -3 , determine the original equation.

Multiplying by x gives

$$xy = ax^2 + b.$$

Hence choose

$$Y = xy, \quad X = x^2.$$

The straight-line form is $Y = aX + b$, so $a = 4$ and $b = -3$. Therefore

$$y = 4x - \frac{3}{x}.$$

As another example, dividing $y = ax^2 + bx$ by x , where $x \neq 0$, gives

$$\frac{y}{x} = ax + b.$$

Thus a graph of y/x against x is linear, with gradient a and intercept b .

7.5.3 Taking reciprocals

For a relationship containing a variable in the denominator, taking the reciprocal of both sides can expose a linear form.

Example 7.3: A reciprocal model

The relationship between x and y is

$$y = \frac{a}{x + b}.$$

A graph of $1/y$ against x has gradient 0.2 and intercept 0.6. Find a and b .

Taking reciprocals gives

$$\frac{1}{y} = \frac{x + b}{a} = \frac{1}{a}x + \frac{b}{a}.$$

Therefore

$$\frac{1}{a} = 0.2 \implies a = 5,$$

and

$$\frac{b}{a} = 0.6 \implies b = 3.$$

Hence

$$y = \frac{5}{x + 3}.$$

7.5.4 Taking logarithms

Logarithms turn multiplication into addition and powers into coefficients. They are therefore particularly useful for power and exponential models.

For the power law

$$y = ax^n, \quad a > 0, x > 0, y > 0,$$

taking logarithms gives

$$\log y = n \log x + \log a. \quad (7.3)$$

Consequently, a graph of $\log y$ against $\log x$ has gradient n and intercept $\log a$.

Example 7.4: Recovering a power law

A graph of $\log_{10} y$ against $\log_{10} x$ is a straight line with gradient 1.5 and vertical intercept 0.3010. Determine the relationship between x and y .

From

$$\log_{10} y = n \log_{10} x + \log_{10} a,$$

we obtain $n = 1.5$ and

$$\log_{10} a = 0.3010.$$

Thus $a = 10^{0.3010} \approx 2$, and the relationship is

$$y = 2x^{3/2}.$$

For the exponential model

$$y = Ae^{kx}, \quad A > 0,$$

taking natural logarithms gives

$$\ln y = kx + \ln A. \quad (7.4)$$

A graph of $\ln y$ against x therefore has gradient k and intercept $\ln A$.

Example 7.5: Recovering an exponential model

A graph of $\ln y$ against x has equation

$$\ln y = -0.4x + \ln 12.$$

Exponentiating both sides gives

$$\begin{aligned} y &= e^{-0.4x + \ln 12} \\ &= e^{\ln 12} e^{-0.4x} \\ &= 12e^{-0.4x}. \end{aligned}$$

The negative gradient indicates exponential decay.

7.6 Constructing a model from data

Suppose experimental observations suggest a relationship of the form

$$y = ax^2 + b.$$

The measured values are shown below.

x	1	2	3	4	5
y	5.1	10.8	21.2	34.7	53.3

Define $X = x^2$ and $Y = y$. The transformed table is

$X = x^2$	1	4	9	16	25
$Y = y$	5.1	10.8	21.2	34.7	53.3

The transformed points lie close to a straight line. A fitted equation might be

$$Y \approx 2.01X + 2.94.$$

Comparing this with $Y = aX + b$ gives $a \approx 2.01$ and $b \approx 2.94$. Rounding to values consistent with the precision of the data suggests the model

$$y \approx 2x^2 + 3.$$

The fitted line does not pass through every transformed point because the data contain measurement variation. This is expected in an experimental setting.

Figure 7.3 displays the same observations before and after the substitution $X = x^2$. The curvature in the original plot becomes an approximately straight-line pattern in the transformed plot.

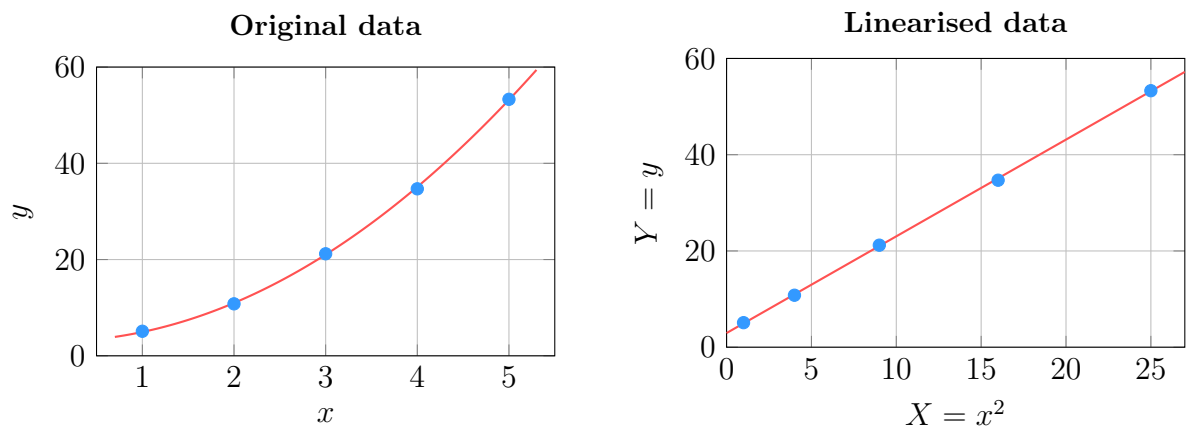


Figure 7.3: The substitution $X = x^2$ converts a quadratic relationship into an approximately linear one while preserving the observations.

7.7 Applications of linear law

Linearisation is useful whenever the constants in a non-linear model have a practical meaning. Examples include exponential growth and decay, power-law scaling, resistance models, cooling data, and relationships between physical quantities obtained experimentally.

Example 7.6: An exponential decay application

The amount M of a substance remaining after time t is modelled by

$$M = M_0 e^{-kt},$$

where $M_0 > 0$ and $k > 0$. A graph of $\ln M$ against t has gradient -0.18 and vertical intercept 4.20 .

Comparison with

$$\ln M = -kt + \ln M_0$$

gives

$$k = 0.18 \quad \text{and} \quad M_0 = e^{4.20} \approx 66.7.$$

Hence

$$M \approx 66.7e^{-0.18t}.$$

For example, at $t = 5$,

$$M \approx 66.7e^{-0.9} \approx 27.1.$$

When solving an application, the final answer should return to the original variables and include suitable units. Any prediction should also be checked against the observed range and the assumptions of the model.

Chapter summary and common pitfalls

Key ideas. Linearisation defines new variables so that a non-linear model becomes $Y = mX + c$. The gradient and intercept must then be translated back into the constants of the original model.

Common pitfalls. Label transformed axes rather than the original variables, transform every data value consistently, and do not interpret a gradient until it has been matched to the derived straight-line equation.

7.8 Practice and exercises

Unless otherwise stated, use an appropriate line of best fit when working with measured data and give constants to three significant figures. Related algebraic drills are grouped as subparts; later questions emphasise choosing, interpreting, and evaluating models in context.

A. Foundations and graphical reasoning

1. Determine whether each exact data set represents a linear relationship. Give a reason and, where possible, find its equation.

(a)	x	0	1	2	3	4
	y	7	11	15	19	23

(b)	x	0	1	2	3	4
	y	2	5	10	17	26

(c)	x	1	2	4	7
	y	3	5	9	15

2. The points $(1, 4)$, $(3, 10)$, $(6, 19)$, and $(9, 28)$ are exact. Show that they are collinear and find the equation of the line.
3. Plot the observations below, draw a line of best fit, and estimate its equation. Compare your estimate with the least-squares line $y = 1.983x + 2.043$.

x	0	1	2	3	4	5
y	2.1	3.8	6.2	7.9	10.2	11.8

4. A fitted line for temperature is $T = -1.6t + 84$, where T is in degrees Celsius and t is in minutes. Interpret the gradient and intercept, estimate T when $t = 25$, and classify the estimate if the observations covered $0 \leq t \leq 20$.
5. A calibration line passes through $(2, 7)$ and $(10, 31)$, where the horizontal variable is pressure in kPa and the vertical variable is sensor output in mV. Find its equation and state the units of its gradient.

6. For the model $\hat{y} = 2x + 1$, calculate the residual $y - \hat{y}$ at every observation and comment on their balance.

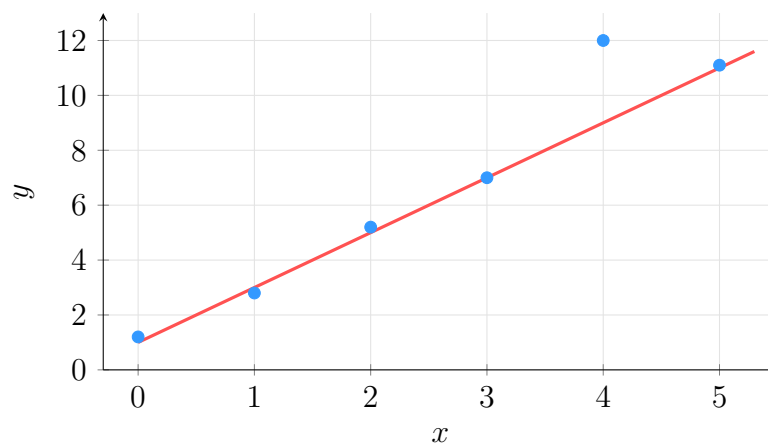
x	0	1	2	3
y	1.2	2.8	5.3	6.7

7. Two proposed models give residuals, in increasing order of x ,

$$A : 0.2, -0.1, 0.0, 0.1, -0.2, \quad B : -1.0, 0.5, 1.0, 0.5, -1.0.$$

Which model is more consistent with a straight-line relationship? Explain.

8. Data were collected for $4 \leq x \leq 18$. Classify predictions at $x = 6$, $x = 18$, $x = 2$, and $x = 25$ as interpolation or extrapolation.
9. A line fitted to battery capacity C against battery age t has intercept 102%. Explain why the intercept can be useful even though a physical capacity greater than 100% may appear questionable.
10. Explain when a line of best fit should be forced through the origin. Give one example where doing so is justified and one where it is not.
11. The graph below includes one observation that departs markedly from the general trend.



Identify the possible outlier, estimate its residual from the drawn line, and explain what should be checked before excluding it.

12. A scatter plot has a strong curved pattern with very little scatter. Explain why strong association alone does not justify fitting a straight line in the original variables.
13. State the correct horizontal and vertical axis labels for each plot:
- (a) y against x^2 ;

- (b) $\ln y$ against x ;
 - (c) $1/y$ against $1/x$;
 - (d) $\log_{10} y$ against $\log_{10} x$.
14. Distinguish exact mathematical data from measured experimental data. How does this distinction affect the way a straight line is fitted?
15. Match each observed shape to a useful first transformation: an upward parabola, exponential decay, a reciprocal curve, and a power law on positive axes. Choose from x^2 , $\ln y$, $1/x$, and the pair $(\log x, \log y)$.

B. Grouped linearisation techniques

16. Rewrite each relationship as $Y = mX + c$ by changing only the horizontal variable:
- (a) $y = ax^3 + b$;
 - (b) $y = a\sqrt{x} + b$;
 - (c) $y = a/x + b$.
17. Obtain a straight-line form by multiplying through:
- (a) $y = ax + b/x$;
 - (b) $y = a + b/x^2$;
 - (c) $y = (ax + b)/x$.
18. Obtain a straight-line form by dividing through:
- (a) $y = ax^2 + bx$;
 - (b) $y = ax^3 + bx^2$;
 - (c) $xy = ax^2 + bx$.
19. Obtain a straight-line form by taking reciprocals:
- (a) $y = a/(x + b)$;
 - (b) $y = 1/(ax + b)$;
 - (c) $y = a/(x^2 + b)$.
20. Obtain a straight-line form by taking logarithms:
- (a) $y = ax^n$;
 - (b) $y = Ab^x$;

(c) $y = Ae^{kx}$.

21. If $y = 1/(ax + b)$, state the graph whose gradient is a and intercept is b .
22. If $y = x/(ax + b)$, show that plotting x/y against x gives a straight line. Identify its gradient and intercept.
23. If $y = a/(x^2 + b)$, show that a graph of $1/y$ against x^2 is linear. Express its gradient and intercept in terms of a and b .
24. If $y = ax/(x + b)$, show that a graph of $1/y$ against $1/x$ is linear. Express its gradient and intercept in terms of a and b .
25. For $y = ax^n$, explain why both x and y must be positive when the model is linearised with real logarithms.
26. For $y = Ab^x$, write the straight-line equations obtained by using (a) common logarithms and (b) natural logarithms. State how A and b are recovered in each case.
27. For $y = Ae^{-kx}$, where $A, k > 0$, identify the gradient and intercept of a graph of $\ln y$ against x . What feature of the line indicates decay?
28. A known background value c is present in $y - c = ax^n$. State a logarithmic transformation that can be used and the restrictions it requires.
29. Find suitable transformed variables for:
 - (a) $\sqrt{y} = ax + b$;
 - (b) $1/y = a/x + b$;
 - (c) $x/y = ax + b$.
30. The equation $y = a + b/x$ can be linearised in two equivalent ways. Give one method using $1/x$ and another using multiplication by x .
31. A line with axes $Y = xy$ and $X = x^2$ has equation $Y = mX + c$. Write the corresponding relationship between x and y .
32. State the domain restrictions needed before applying:
 - (a) $\ln y$;
 - (b) $\log x$ and $\log y$ together;
 - (c) $1/x$;

- (d) $1/y$.
33. Explain why changing from $\log_{10}$ to $\ln$ changes the numerical intercept of a power-law plot but does not change the exponent n .
34. Complete the interpretation for each transformed line:
- (a) $y = ax^2 + b$, plotted as y against x^2 ;
 - (b) $y = a/(x + b)$, plotted as $1/y$ against x ;
 - (c) $y = ax^n$, plotted as $\ln y$ against $\ln x$;
 - (d) $y = Ae^{kx}$, plotted as $\ln y$ against x .
35. Select one appropriate transformed plot for each proposed model: $y = ax^4 + b$, $y = a + b/x$, $y = a/(x + b)$, $y = ax^n$, and $y = Ab^x$.

C. Recovering constants and equations

36. The graph of y against x^2 for $y = ax^2 + b$ passes through $(2, 7)$ and $(8, 25)$. Find a and b .
37. For $y = ax + b/x$, the graph of xy against x^2 is $Y = 2.5X - 4$. Find a , b , and y when $x = 4$.
38. For $y = a/(x + b)$, a graph of $1/y$ against x has gradient 0.125 and intercept 0.5. Determine a and b .
39. A power law $y = ax^n$ gives $\log_{10} y = 2.4 \log_{10} x - 0.6990$. Determine a and n .
40. For $y = Ab^x$, a graph of $\log_{10} y$ against x has gradient 0.1761 and intercept 0.4771. Find A and b .
41. A graph of $\ln y$ against x passes through $(1, 2.6)$ and $(5, 1.0)$. Assuming $y = Ae^{kx}$, find A and k and state whether the model represents growth or decay.
42. For $y = ax^2 + bx$, the graph of y/x against x has equation $Y = 3X - 2$. Find a and b .
43. For $y = a/(x^2 + b)$, the graph of $1/y$ against x^2 is $Y = 0.2X + 0.4$. Find a and b .
44. For $y = ax/(x + b)$, the graph of $1/y$ against $1/x$ is $Y = 0.4X + 0.2$. Find a and b .
45. For $y = x/(ax + b)$, the graph of x/y against x is $Y = 0.3X + 0.6$. Find a and b .

46. The exact data below satisfy $y = ax^2 + b$. Transform the table and determine a and b .

x	1	2	3	4
y	5	14	29	50

47. The exact data below satisfy $y = ax + b/x$. Form a table of x^2 and xy , then determine a and b .

x	1	2	3	4
y	-1	2.5	5	7.25

48. Positive variables obey a power law. Use the data to determine its exact equation.

x	1	2	4	8
y	3	12	48	192

49. Determine the exponential equation represented by

x	0	1	2	3
y	5	10	20	40.

State the gradient and intercept of the corresponding graph of $\log_{10} y$ against x .

50. Determine the exponential-decay equation represented by $y = 80, 40, 20, 10$ at $x = 0, 1, 2, 3$. Give both the forms Ab^x and Ae^{kx} .

51. Recover the original model from each transformed line:

- (a) axes $Y = y$, $X = x^3$, with $Y = 2X - 1$;
- (b) axes $Y = 1/y$, $X = x$, with $Y = 0.25X + 0.5$;
- (c) axes $Y = \ln y$, $X = \ln x$, with $Y = 1.5X + \ln 3$.

52. A transformed straight line passes through $(1, 4)$ and $(9, 20)$. Find its gradient and intercept. If the axes are $Y = y$ and $X = x^2$, write the original equation.

53. On a graph of $\ln M$ against time t in hours, what are the units of the gradient? Explain why $\ln M$ itself is treated as dimensionless in the model.

54. Two students estimate a gradient using points read from the same line. One uses $(1.0, 2.2)$ and $(9.0, 18.1)$; the other uses $(3.0, 6.0)$ and $(7.0, 14.2)$. Calculate both estimates and explain why widely separated points are normally preferred.

55. A transformed model is $\ln C = -0.25t + \ln 80$. Write C in terms of t and estimate C when $t = 6$.

D. Fitting, checking, and comparing models

56. The measured data are believed to satisfy $y = ax^2 + b$:

x	1	2	3	4	5
y	5.1	10.8	21.2	34.7	53.3.

Fit a straight line to y against x^2 , estimate a and b , and suggest a simple model consistent with the precision of the data.

57. The concentration data below are believed to follow $C = C_0e^{-kt}$:

t	0	1	2	3	4
C	50.0	37.2	27.5	20.4	15.1.

Fit $\ln C$ against t , estimate C_0 and k , and predict C at $t = 5$.

58. A liquid cools in a room at 22°C . Test the model $T - 22 = Ae^{-kt}$ using

t (min)	0	2	4	6	8	10
T ($^\circ\text{C}$)	70.0	54.1	43.3	35.9	30.8	27.4.

State the transformed plot, estimate A and k , and predict T at $t = 15$ minutes.

59. The positive data $(x, y) = (1, 5), (2, 11), (3, 21), (4, 35)$ may follow either $y = ax^2 + b$ or $y = Ax^n$. Describe both tests and decide which model is better supported by the exact data.
60. Model A gives residuals $0.1, -0.1, 0, 0.1, -0.1$; model B gives $-1, 0.5, 1, 0.5, -1$. Sketch or describe the two residual patterns and explain why curvature in a residual plot signals an inadequate transformation.
61. Fit a line to
 $(0, 1.1), (1, 3.0), (2, 5.1), (3, 13.0), (4, 9.2), (5, 11.1)$.
 Then refit after temporarily omitting $(3, 13.0)$. Compare the two lines and state why omission must be justified by context rather than by convenience.
62. A fitted model is $\hat{y} = 1.5x + 0.8$. For observations $(0, 1.0), (1, 2.1), (2, 4.2), (3, 5.0)$, calculate all residuals and identify the observation with the largest absolute residual.
63. A power-law fit gives $\log_{10} y = 1.30 \log_{10} x + 0.602$. Write the original model and estimate y when $x = 5$.
64. Explain why a model fitted for $2 \leq x \leq 8$ may give an unreliable prediction at $x = 50$, even when its transformed points are nearly collinear.

65. Design a brief investigation to test whether the period T of an oscillator follows $T = aL^n$. Specify the measurements, transformed columns, graph, constants obtained, and one source of uncertainty.

E. Application-based modelling

66. A pendulum experiment gives the line $\ln T = 0.5 \ln L + \ln 2$, with T in seconds and L in metres. Find the model, identify the exponent, and estimate T when $L = 2$ m.

67. A force sensor records

x (cm)	0	1	2	3
F (N)	2	5	8	11.

Determine the calibration equation, interpret its gradient and intercept, and explain why no non-linear transformation is needed.

68. The terminal speed v of a small sphere is modelled by $v = ar^n$. A fitted line is $\ln v = 1.8 \ln r + \ln 0.6$. Determine a and n , then estimate v when $r = 2$ in the stated units.
69. The mass M of geometrically similar specimens is modelled by $M = aL^n$. A common-log plot gives $\log_{10} M = 3 \log_{10} L - 1.3010$. Find the model and estimate M when $L = 20$ cm.
70. A bacterial culture follows $N = N_0 e^{kt}$. Its transformed line is $\ln N = 0.35t + \ln 500$. Find N_0 , the growth constant, and the doubling time.
71. Radioactive mass follows $M = M_0 e^{-kt}$. A graph of $\ln M$ against time has intercept $\ln 60$ and gradient -0.18 per year. Find the half-life and the mass after eight years.
72. A medicine concentration follows $C = C_0 e^{-kt}$. A patient has $C_0 = 60 \text{ mg L}^{-1}$ and $k = 0.12 \text{ h}^{-1}$. State the transformed straight line and calculate C after eight hours.
73. A drink at 80°C cools in a 22°C room according to $T - 22 = 58e^{-0.15t}$, with t in minutes. State the line obtained by plotting $\ln(T - 22)$ against t and estimate T after 12 minutes.
74. A capacitor discharges according to $V = V_0 e^{-t/(RC)}$. Its $\ln V$ against t graph has intercept $\ln 12$ and gradient -0.25 s^{-1} . Find V_0 , the time constant RC , and V after six seconds.
75. Light intensity in a material satisfies $I = I_0 e^{-\mu x}$. A straight-line plot gives intercept $\ln 200$ and gradient -0.35 cm^{-1} . Find the thickness that reduces the intensity to one tenth of its initial value.

76. A point source is expected to obey $I = ad^n$. Explain how a log-log plot tests the inverse-square law. If the fitted line is $\ln I = -2 \ln d + \ln 180$, find I at $d = 5$ m.
77. For a simple pendulum, $T = 2\pi\sqrt{L/g}$. Show that T^2 plotted against L has gradient $4\pi^2/g$. If the measured gradient is $4.024 \text{ s}^2 \text{ m}^{-1}$, estimate g .
78. An orbital model is $T = ar^{3/2}$. State a direct transformed plot that avoids logarithms. If its gradient is 0.5 in consistent units, find T when $r = 4$.
79. The resistance of a wire of fixed length is modelled by $R = a/d^2$, where d is diameter. State the linear plot. If $a = 0.060 \text{ } \Omega \text{ mm}^2$, find R when $d = 0.5$ mm.
80. A flow rate is modelled by $Q = ar^n$. A log-log graph has gradient 4 and intercept $\ln 1.2$. Find the model and estimate Q when $r = 1.5$ cm.
81. The thin-lens equation is $1/f = 1/u + 1/v$. Show that plotting $1/v$ against $1/u$ gives a straight line. State its gradient and explain how f is found from the intercept.
82. At constant temperature, measured gas pressure is modelled by $P = a/V + b$. A plot of P against $1/V$ has gradient 240 and intercept 5. Find P when $V = 8$ L and interpret b .
83. The Arrhenius equation is $k = Ae^{-E/(RT)}$. Show that $\ln k$ plotted against $1/T$ has gradient $-E/R$. If the gradient is -5200 K , estimate E using $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$.
84. A reaction rate satisfies $r = aC^n$. A log-log fit gives $n = 1.5$ and $a = 0.08$ in consistent units. Estimate the rate at $C = 3$ and explain what the exponent means.
85. Enzyme data follow
- $$\frac{1}{v} = \frac{K_m}{V_{\max}} \frac{1}{[S]} + \frac{1}{V_{\max}}.$$
- A plot of $1/v$ against $1/[S]$ has gradient 0.75 and intercept 0.25. Find $V_{\max}$ and K_m .
86. A town population follows $P = P_0 b^t$. A transformed graph gives $\ln P = t \ln 1.08 + \ln 2000$. Find the annual percentage growth and predict the population after six years.
87. A machine depreciates according to $V = V_0 b^t$. Its value is RM 12,000 initially and is multiplied by 0.82 each year. State the common-log line and estimate the value after six years.
88. An investment obeys $A = P(1+r)^t$. Explain how a graph of $\ln A$ against t can estimate P and r . Find the value after ten years when $P = \text{RM } 5,000$ and $r = 0.06$.

89. A price index follows $P = P_0 b^t$. If $P_0 = 240$ and the annual multiplier is 1.045, state the transformed line and estimate P after eight years.
90. The time per manufactured unit follows a learning curve $t = an^b$. A log-log fit gives $a = 0.8$ h and $b = 0.75$. Estimate the time for $n = 40$ and interpret the positive exponent.
91. Production cost follows $C = am^n$. A log-log line has gradient 0.6 and intercept $\ln 4.5$. Find the cost when $m = 15$ and state the percentage interpretation of the exponent.
92. Aerodynamic drag is modelled by $F = av^n$. A log-log fit gives $a = 2.0 \times 10^{-5}$ and $n = 2$. Find F at $v = 30 \text{ m s}^{-1}$ and identify the physical meaning of doubling v under this model.
93. A vehicle's stopping distance is modelled by $d = av^2 + b$. State the transformed plot. If its gradient is 0.04 and intercept is 3, find d at $v = 25 \text{ m s}^{-1}$ and interpret b .
94. A temperature sensor obeys $R = aT + b$ over its calibrated range. A fitted line is $R = 0.38T + 100$. Interpret both constants and explain why using the model far outside the calibrated range is risky.
95. Illumination from a lamp follows $I = a/d^2$. Determine the linear plot and, if $a = 720$, find I at $d = 6$ m.
96. Solar flux is modelled by $F = ar^{-2}$, where r is measured in astronomical units. If $F = 1361 \text{ W m}^{-2}$ at $r = 1$, estimate the flux at $r = 1.52$ and state the gradient expected on a log-log plot.
97. Crater diameter follows $D = aE^n$. A fit gives $a = 0.012$ and $n = 0.28$ in consistent units. Estimate D for $E = 10^6$ and explain why a given percentage increase in E does not produce the same percentage increase in D .
98. Wind-turbine power is modelled by $P = av^n$. A log-log fit has gradient 3.1 and intercept $\ln 0.42$. Find the model, estimate P at $v = 8 \text{ m s}^{-1}$, and compare the fitted exponent with the ideal cubic law.
99. Positive data may follow either $y = ax^2 + b$ or $y = Ax^n$. Describe a fair graphical comparison using y against x^2 , $\ln y$ against $\ln x$, residual plots, and the same observations.
100. An environmental pollutant is measured every two days:

t (d)	0	2	4	6	8	10	12
M (mg)	120	89	66	50	37	28	21.

Test $M = M_0 e^{-kt}$ by fitting $\ln M$ against t . Estimate M_0 , k , the half-life, and M at day 15. Finally, give one reason the day-15 prediction should be reported cautiously.

Chapter 8

Coordinate Geometry and Conic Sections

Chapter guide

Prerequisites. Cartesian coordinates, distance and midpoint formulae, quadratic equations, and graph transformations.

Learning outcomes. Readers should be able to identify conic sections from their equations, derive and use standard forms, determine key geometric features, and work with parametric representations and tangents.

In this chapter, we examine the four principal conic sections: the circle, parabola, ellipse, and hyperbola. Geometrically, these curves can be produced by intersecting a plane with a double cone. Analytically, they arise from second-degree equations in x and y . Their equations may look similar, but each curve is characterised by a different distance condition.

Conic	Defining locus property
Circle	Constant distance from one fixed point, the centre.
Parabola	Equal distance from a fixed point and a fixed line.
Ellipse	Constant sum of distances from two fixed points.
Hyperbola	Constant absolute difference of distances from two fixed points.

The central skill is not merely recognising a memorised formula. This chapter translates between geometric information and equations, completes squares to reveal hidden centres and vertices, uses intersections to solve geometric problems, and sketches each curve from its principal features. Parametric equations then describe a moving point on a conic and make it possible to control direction, starting position, and the portion of a curve that is traced.

The coordinate and vector approaches used in this chapter draw on standard treatments of circles, conic sections, and Euclidean geometry (Dawkins 2022; Auroux 2010; Marsden and Tromba 2026).

8.1 Circles

A **circle** is the locus of all points P in a plane that are a fixed distance from a fixed point. The fixed point is the **centre**, and the fixed distance is the **radius**. Unlike an ellipse or

hyperbola, a circle has the same width in every direction and therefore has no preferred axis.

Let the centre of a circle be $C(h, k)$, and let $P(x, y)$ be any point on the circle. Since the distance from the centre to every point on the circle is the constant radius r , we have

$$CP = r.$$

Using the distance formula,

$$\sqrt{(x - h)^2 + (y - k)^2} = r.$$

Squaring both sides gives the **standard equation of a circle** with centre (h, k) and radius r :

$$(x - h)^2 + (y - k)^2 = r^2. \quad (8.1)$$

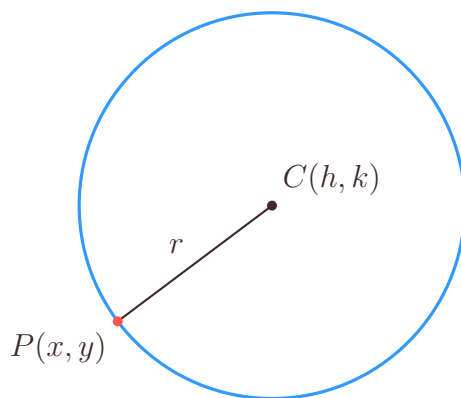


Figure 8.1: A circle with centre $C(h, k)$ and radius r .

When the centre is at the origin, $(h, k) = (0, 0)$, the standard equation becomes

$$x^2 + y^2 = r^2. \quad (8.2)$$

Expanding the standard equation gives

$$\begin{aligned} (x - h)^2 + (y - k)^2 &= r^2, \\ x^2 - 2hx + h^2 + y^2 - 2ky + k^2 &= r^2, \\ x^2 + y^2 - 2hx - 2ky + (h^2 + k^2 - r^2) &= 0. \end{aligned}$$

If $F = h^2 + k^2 - r^2$, this may be written as

$$x^2 + y^2 - 2hx - 2ky + F = 0. \quad (8.3)$$

Therefore, a circle written in this form has centre (h, k) and radius

$$r = \sqrt{h^2 + k^2 - F}. \quad (8.4)$$

More generally, if the equation is written as

$$x^2 + y^2 + Dx + Ey + F = 0,$$

then its centre and radius are

$$\left(-\frac{D}{2}, -\frac{E}{2}\right) \quad \text{and} \quad r = \sqrt{\frac{D^2 + E^2}{4} - F}, \quad (8.5)$$

provided that $D^2 + E^2 - 4F > 0$.

The sign of the squared radius describes three possibilities:

- i. If $D^2 + E^2 - 4F > 0$, the equation represents a real circle.
- ii. If $D^2 + E^2 - 4F = 0$, the radius is zero and the circle has collapsed to a single point.
- iii. If $D^2 + E^2 - 4F < 0$, there are no real points satisfying the equation.

8.1.1 Completing the square and sketching a circle

When a circle is given in expanded form, completing the square exposes its centre and radius. The identities

$$x^2 + Dx = \left(x + \frac{D}{2}\right)^2 - \frac{D^2}{4}, \quad y^2 + Ey = \left(y + \frac{E}{2}\right)^2 - \frac{E^2}{4}$$

show why the centre coordinates are $(-D/2, -E/2)$.

Example 8.1: recovering the centre and radius

Write $x^2 + y^2 - 6x + 8y - 11 = 0$ in standard form.

Group the x - and y -terms and complete both squares:

$$\begin{aligned} (x^2 - 6x) + (y^2 + 8y) &= 11, \\ (x - 3)^2 - 9 + (y + 4)^2 - 16 &= 11, \\ (x - 3)^2 + (y + 4)^2 &= 36. \end{aligned}$$

Thus the circle has centre $(3, -4)$ and radius 6. A sketch can be made by marking the centre and the four extreme points $(9, -4)$, $(-3, -4)$, $(3, 2)$, and $(3, -10)$.

To test a point $Q(u, v)$ against the circle $(x - h)^2 + (y - k)^2 = r^2$, compare

$$(u - h)^2 + (v - k)^2 \quad \text{with} \quad r^2.$$

A smaller value places Q inside, equality places it on the circle, and a larger value places it outside.

8.1.2 Intersection between circle and a straight line

The points of intersection between a circle and a straight line can be found by solving their equations simultaneously. One variable is first eliminated, usually by substituting the equation of the line into the equation of the circle. This produces a quadratic equation, and every real solution corresponds to a point at which the line meets the circle.

Suppose the resulting quadratic equation is

$$Ax^2 + Bx + C = 0,$$

with discriminant

$$\Delta = B^2 - 4AC.$$

The geometric relationship between the line and the circle is determined by the value of Δ :

- i. If $\Delta > 0$, the quadratic equation has two distinct real roots. Therefore, the line intersects the circle at two different points, P and Q .
- ii. If $\Delta = 0$, the quadratic equation has one repeated real root. Therefore, the line touches the circle once at P and is a tangent to the circle at that point.
- iii. If $\Delta < 0$, the quadratic equation has no real roots. Therefore, the line does not intersect the circle.

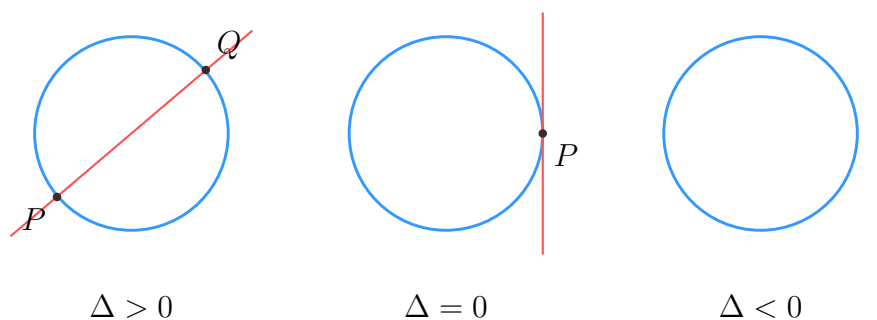


Figure 8.2: The three possible relationships between a circle and a straight line.

There is also a direct geometric test. For a line

$$Ax + By + C = 0$$

and a circle with centre (h, k) and radius r , the perpendicular distance from the centre to the line is

$$d = \frac{|Ah + Bk + C|}{\sqrt{A^2 + B^2}}.$$

The line is a secant when $d < r$, a tangent when $d = r$, and an external line when $d > r$. This method is often faster than substitution when only the type of intersection is required.

8.1.3 Chords and radical axes

A **chord** joins two points of a circle. The perpendicular from the centre to a chord bisects the chord; conversely, the perpendicular bisector of any non-diameter chord passes through the centre. These facts frequently turn coordinate problems into right-triangle calculations.

For a circle C , define the power of a point $P(u, v)$ by

$$\text{Pow}_C(P) = (u - h)^2 + (v - k)^2 - r^2.$$

The power is negative inside the circle, zero on it, and positive outside it. If two circles are given, the locus of points having equal powers with respect to them is called their **radical axis**. Subtracting the two circle equations cancels the x^2 and y^2 terms, so the radical axis is a straight line. When the circles intersect, it is precisely the line containing their common chord.

Example 8.2: finding a radical axis

For

$$C_1 : x^2 + y^2 = 25, \quad C_2 : (x - 4)^2 + y^2 = 9,$$

expand C_2 and subtract the equations. This gives $x = 4$. Substitution into C_1 gives $y = \pm 3$, so the common chord has endpoints $(4, 3)$ and $(4, -3)$.

8.1.4 Equation of tangent and normal to a circle

At any point on a circle, the tangent is perpendicular to the radius drawn to the point of contact. The normal is perpendicular to the tangent and therefore lies along the same straight line as the radius through that point.

Consider the circle

$$(x - h)^2 + (y - k)^2 = r^2,$$

and let $P(x_1, y_1)$ be a point on the circle. Since the radius CP is normal to the tangent, the radius vector

$$\langle x_1 - h, y_1 - k \rangle$$

is perpendicular to every direction vector of the tangent. Hence, the equation of the tangent at P is

$$(x_1 - h)(x - x_1) + (y_1 - k)(y - y_1) = 0. \quad (8.6)$$

Because P lies on the circle, $(x_1 - h)^2 + (y_1 - k)^2 = r^2$. The tangent equation may therefore also be written as

$$(x_1 - h)(x - h) + (y_1 - k)(y - k) = r^2. \quad (8.7)$$

The normal passes through both $C(h, k)$ and $P(x_1, y_1)$. Its equation is

$$(x_1 - h)(y - k) = (y_1 - k)(x - h). \quad (8.8)$$

These vector forms remain valid when the tangent or normal is vertical, so they are generally more convenient than using gradients. If the circle is centred at the origin, the equations simplify to

$$\begin{aligned} xx_1 + yy_1 &= r^2 && \text{(tangent),} \\ x_1y - y_1x &= 0 && \text{(normal).} \end{aligned}$$

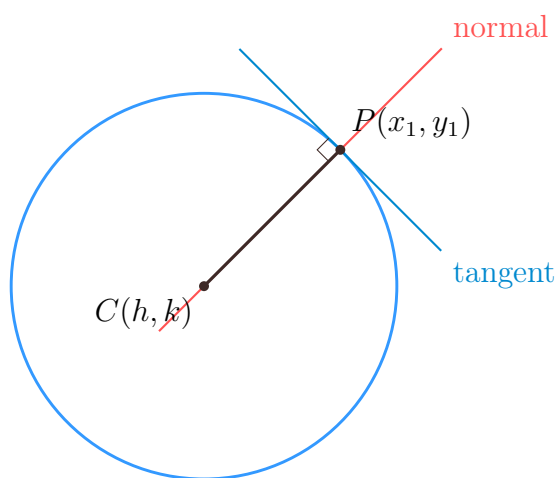


Figure 8.3: The tangent at P is perpendicular to the radius CP , while the normal passes through C and P .

Example 8.3: tangent and normal at a point

For $x^2 + y^2 = 10$ at $P(1, 3)$, the tangent formula gives

$$x + 3y = 10.$$

The normal is the line through the origin and $(1, 3)$, namely $y = 3x$. Notice that the tangent gradient is $-1/3$ and the normal gradient is 3 , so their product is -1 .

8.1.5 Length of tangent to a circle

Let $T(m, n)$ be a point outside a circle with centre $C(h, k)$ and radius r . Suppose the tangent from T touches the circle at S . Since a tangent is perpendicular to the radius at the point of contact, triangle CST is right-angled at S .

By the Pythagorean theorem,

$$\begin{aligned}CT^2 &= CS^2 + ST^2, \\ST^2 &= CT^2 - CS^2.\end{aligned}$$

Now,

$$CT = \sqrt{(m-h)^2 + (n-k)^2} \quad \text{and} \quad CS = r.$$

Therefore, the length of the tangent is

$$ST = \sqrt{(m-h)^2 + (n-k)^2 - r^2}. \quad (8.9)$$

This formula gives a real, positive length only when

$$(m-h)^2 + (n-k)^2 > r^2,$$

which means that T lies outside the circle. If equality holds, T lies on the circle and the tangent length is zero. If the left-hand side is smaller than r^2 , then T lies inside the circle and no real tangent can be drawn from it.

For a circle written in the general form

$$x^2 + y^2 + Dx + Ey + F = 0,$$

the tangent length from $T(m, n)$ can be found directly from

$$ST = \sqrt{m^2 + n^2 + Dm + En + F}. \quad (8.10)$$

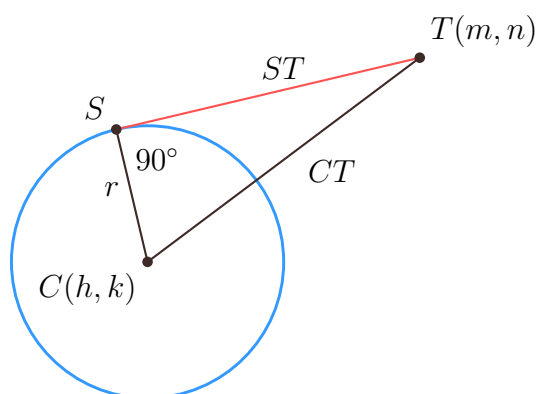


Figure 8.4: The tangent length follows from the right triangle CST .

Two tangents drawn from the same external point have equal lengths. If they touch the circle at S_1 and S_2 , then the right triangles CTS_1 and CTS_2 have the same hypotenuse CT and the same radius, so $TS_1 = TS_2$. This symmetry is useful when constructing tangent routes around a circular boundary or proving that two line segments are equal.

8.2 Parabola

A **parabola** is the locus of all points P in a plane that are equidistant from a fixed point, called the **focus**, and a fixed straight line, called the **directrix**. The point at which the parabola changes direction is called the **vertex**, and the line passing through the vertex and focus is called the **axis of symmetry**.

Consider a parabola with vertex at the origin, focus $F(0, a)$, and directrix $y = -a$. If $P(x, y)$ is any point on the parabola, its distance from the focus equals its perpendicular distance from the directrix. Therefore,

$$\sqrt{x^2 + (y - a)^2} = |y + a|.$$

Squaring and simplifying gives

$$\begin{aligned} x^2 + (y - a)^2 &= (y + a)^2, \\ x^2 + y^2 - 2ay + a^2 &= y^2 + 2ay + a^2, \\ x^2 &= 4ay. \end{aligned}$$

Hence, the standard equation of a vertical parabola with vertex at the origin is

$$x^2 = 4ay. \quad (8.11)$$

When $a > 0$, the parabola opens upwards; when $a < 0$, it opens downwards. If the vertex is translated to (h, k) , then the focus is $(h, k + a)$ and the directrix is $y = k - a$. Repeating the same distance argument gives

$$(x - h)^2 = 4a(y - k). \quad (8.12)$$

The **latus rectum** is the chord that passes through the focus and is parallel to the directrix. For the parabola $x^2 = 4ay$, the latus rectum lies on $y = a$. Substituting $y = a$ into the equation gives

$$x^2 = 4a^2,$$

so its endpoints are $(2a, a)$ and $(-2a, a)$. Therefore, its length is

$$4|a|. \quad (8.13)$$

For the translated parabola $(x - h)^2 = 4a(y - k)$, the latus rectum has endpoints

$$(h + 2a, k + a) \quad \text{and} \quad (h - 2a, k + a).$$

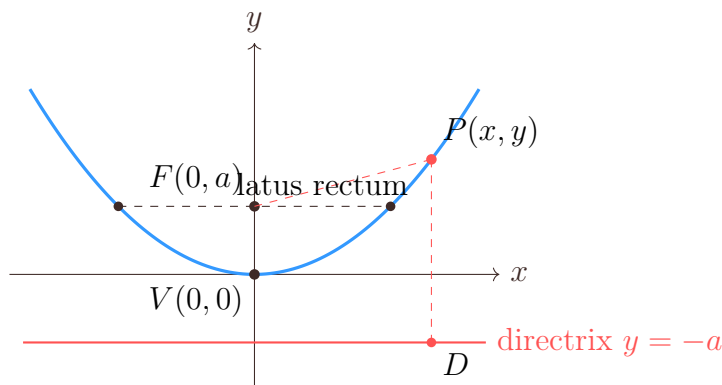


Figure 8.5: The principal features of the parabola $x^2 = 4ay$. For every point P on the curve, $PF = PD$.

8.2.1 Important terms in a parabola

For either standard form, the parameter $|a|$ is the distance from the vertex to the focus and also the perpendicular distance from the vertex to the directrix. The sign of a determines the direction in which the parabola opens.

The **focus** is the fixed point used in the locus definition of the parabola. The **directrix** is the fixed line from which every point on the parabola has the same distance as it has from the focus. The **axis of symmetry** is the line through the vertex and focus; it is perpendicular to the directrix and divides the parabola into two mirror-image halves. The **vertex** is the point where the parabola meets its axis and is located halfway between the focus and the directrix, measured along the axis.

Feature	Vertical parabola	Horizontal parabola
Equation	$(x - h)^2 = 4a(y - k)$	$(y - k)^2 = 4a(x - h)$
Vertex	(h, k)	(h, k)
Focus	$(h, k + a)$	$(h + a, k)$
Directrix	$y = k - a$	$x = h - a$
Axis of symmetry	$x = h$	$y = k$
Direction	Up if $a > 0$; down if $a < 0$	Right if $a > 0$; left if $a < 0$
Latus rectum	Length $4 a $	Length $4 a $

Table 8.1: Important features of vertical and horizontal parabolas.

8.2.2 Standard equations of parabola

The four common standard equations are most clearly stated by taking $a > 0$, so that a always represents the positive distance from the vertex to the focus. The sign placed before $4a$ then determines the direction in which the parabola opens.

Opens	Equation	Vertex	Focus	Directrix
Upward	$(x - h)^2 = 4a(y - k)$	(h, k)	$(h, k + a)$	$y = k - a$
Downward	$(x - h)^2 = -4a(y - k)$	(h, k)	$(h, k - a)$	$y = k + a$
Right	$(y - k)^2 = 4a(x - h)$	(h, k)	$(h + a, k)$	$x = h - a$
Left	$(y - k)^2 = -4a(x - h)$	(h, k)	$(h - a, k)$	$x = h + a$

Table 8.2: Standard equations of parabolas, where $a > 0$.

The squared variable indicates the orientation of the axis. If the x -expression is squared, the axis is vertical; if the y -expression is squared, the axis is horizontal. A positive coefficient on the right produces an upward- or right-opening parabola, whereas a negative coefficient produces a downward- or left-opening parabola.

When an equation is not already in standard form, the following workflow is reliable:

1. Identify which variable is squared. This determines whether the axis is vertical or horizontal.
2. Move the linear expression in the unsquared variable to the other side.
3. Complete the square in the squared variable.
4. Factor the right-hand side into the form $4a(y - k)$ or $4a(x - h)$.
5. Read the vertex, focus, directrix, and direction of opening before sketching.

Example 8.4: converting to standard form

Consider

$$x^2 - 6x - 8y + 25 = 0.$$

Completing the square in x gives

$$\begin{aligned} x^2 - 6x + 25 &= 8y, \\ (x - 3)^2 + 16 &= 8y, \\ (x - 3)^2 &= 8(y - 2). \end{aligned}$$

Thus $4a = 8$, so $a = 2$. The vertex is $(3, 2)$, the focus is $(3, 4)$, the directrix is $y = 0$, and the parabola opens upwards.

The coefficient $4a$ controls the width. For two parabolas having the same vertex and orientation, a larger $|a|$ produces a wider curve because the squared coordinate grows more slowly for a given displacement along the axis. At a vertical distance s from the vertex of $(x - h)^2 = 4a(y - k)$ with $as > 0$, the half-width is

$$|x - h| = 2\sqrt{|as|}.$$

This formula is especially useful for arches, reflectors, and clearance problems.

Example 8.5

Find the equation of the parabola with vertex $(0, 0)$ and focus $(2, 0)$. Then, sketch the parabola.

Solution. Since the focus lies two units to the right of the vertex, the parabola opens to the right and has

$$a = 2.$$

The standard equation of a right-opening parabola with vertex at the origin is

$$y^2 = 4ax.$$

Substituting $a = 2$ gives

$$\boxed{y^2 = 8x}. \quad (8.14)$$

Its axis of symmetry is the x -axis, its focus is $(2, 0)$, and its directrix is $x = -2$. The latus rectum has endpoints $(2, 4)$ and $(2, -4)$.

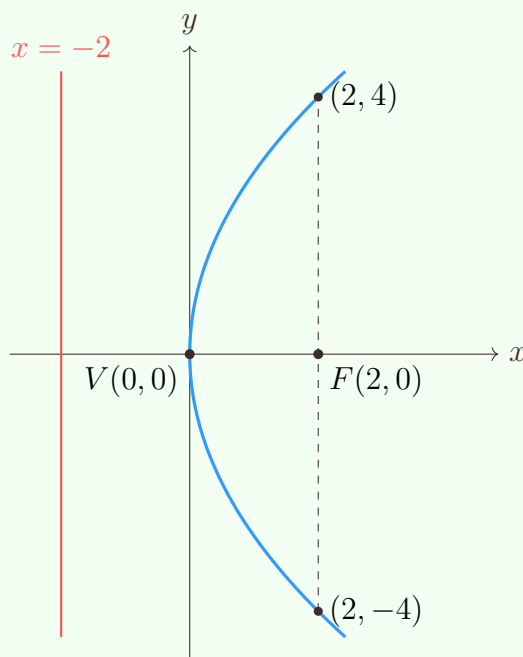


Figure 8.6: The parabola $y^2 = 8x$.

8.2.3 Intersections and chords of a parabola

To intersect a parabola with a line, substitute the line equation into the parabola. The resulting quadratic has the same geometric interpretation as in the circle case: two real roots give two intersection points, a repeated root indicates tangency, and no real roots means the line misses the curve.

For example, intersect $x^2 = 4y$ with $y = x + 2$:

$$x^2 = 4x + 8 \implies x^2 - 4x - 8 = 0.$$

Hence $x = 2 \pm 2\sqrt{3}$, and the corresponding points are

$$(2 \pm 2\sqrt{3}, 4 \pm 2\sqrt{3}),$$

with matching signs.

A line perpendicular to the axis cuts a chord whose midpoint lies on the axis. For $x^2 = 4ay$, the horizontal line $y = k$ has endpoints $(\pm 2\sqrt{ak}, k)$ when $ak \geq 0$, so the chord length is

$$4\sqrt{ak}.$$

The case $k = 0$ gives the vertex as a repeated intersection.

8.2.4 A sketching checklist for parabolas

Before drawing a parabola, record the following information:

1. the vertex and axis of symmetry;
2. the sign and magnitude of a ;
3. the focus and directrix, which must lie equal distances from the vertex on opposite sides;
4. the latus-rectum endpoints, which provide two accurately placed points on the curve;
5. any relevant intercepts or intersections.

The curve should be smooth, symmetric about its axis, and open only in the direction indicated by the focus.

8.3 Ellipse

An **ellipse** is the locus of all points $P(x, y)$ in a plane for which the sum of the distances from two fixed points is constant. The two fixed points are called the **foci**, denoted by F_1 and F_2 . Thus,

$$PF_1 + PF_2 = 2a, \tag{8.15}$$

where $2a$ is the constant length and is also the length of the major axis.

The **major axis** is the longest chord of the ellipse. It passes through both foci and the centre. The **centre** is the midpoint of the line segment joining F_1 and F_2 . The **minor axis** passes through the centre and is perpendicular to the major axis. Its endpoints, together with the endpoints of the major axis, are the vertices of the ellipse.

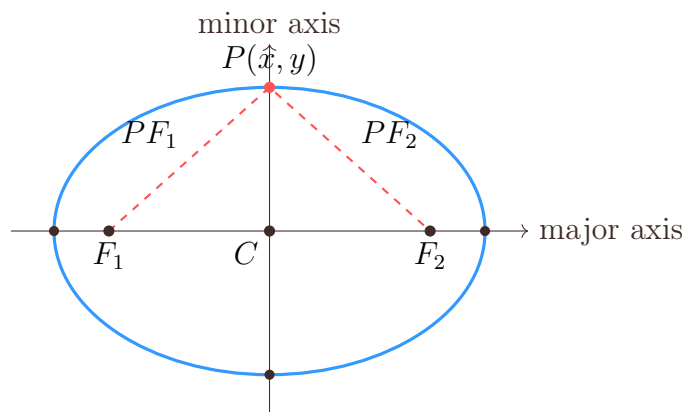


Figure 8.7: An ellipse is defined by the constant sum $PF_1 + PF_2$.

8.3.1 Equation of ellipse

Consider an ellipse centred at the origin with its major axis along the x -axis. Let its foci be

$$F_1(-c, 0) \quad \text{and} \quad F_2(c, 0),$$

and let $P(x, y)$ be any point on the ellipse. By the locus definition,

$$PF_1 + PF_2 = 2a,$$

where $a > c > 0$. Using the distance formula,

$$\sqrt{(x+c)^2 + y^2} + \sqrt{(x-c)^2 + y^2} = 2a.$$

Rearranging and squaring once gives

$$\begin{aligned} \sqrt{(x+c)^2 + y^2} &= 2a - \sqrt{(x-c)^2 + y^2}, \\ 4a\sqrt{(x-c)^2 + y^2} &= 4a^2 - 4cx, \\ \sqrt{(x-c)^2 + y^2} &= a - \frac{c}{a}x. \end{aligned}$$

Squaring again and simplifying,

$$\begin{aligned} (x-c)^2 + y^2 &= \left(a - \frac{c}{a}x\right)^2, \\ \left(1 - \frac{c^2}{a^2}\right)x^2 + y^2 &= a^2 - c^2. \end{aligned}$$

Define

$$b^2 = a^2 - c^2, \tag{8.16}$$

or equivalently,

$$c^2 = a^2 - b^2. \quad (8.17)$$

Since $1 - c^2/a^2 = b^2/a^2$, the equation becomes

$$\boxed{\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1}, \quad a > b > 0. \quad (8.18)$$

This is the standard equation of a horizontal ellipse centred at the origin. Setting $y = 0$ gives $x = \pm a$, while setting $x = 0$ gives $y = \pm b$. Therefore, its major vertices are $(\pm a, 0)$ and its minor vertices are $(0, \pm b)$.

The quantities a , b , and c have distinct geometric meanings:

- a is the semi-major axis and is the greatest of the three;
- b is the semi-minor axis;
- c is the distance from the centre to either focus.

The relation $a^2 = b^2 + c^2$ may be visualised using the right triangle from a focus to a co-vertex. If $a = b$, then $c = 0$ and the two foci coincide at the centre; the ellipse becomes a circle. As c approaches a , the ellipse becomes increasingly elongated.

The dimensionless quantity

$$e = \frac{c}{a}, \quad 0 \leq e < 1,$$

is called the **eccentricity**. A circle has $e = 0$, while ellipses with eccentricity closer to 1 are more stretched along their major axes.

For an ellipse centred at (h, k) , the standard equations are

$$\frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} = 1 \quad (\text{horizontal major axis}), \quad (8.19)$$

$$\frac{(x - h)^2}{b^2} + \frac{(y - k)^2}{a^2} = 1 \quad (\text{vertical major axis}), \quad (8.20)$$

where $a > b > 0$ and $c = \sqrt{a^2 - b^2}$ in both cases.

Feature	Horizontal major axis	Vertical major axis
Centre	(h, k)	(h, k)
Major vertices	$(h \pm a, k)$	$(h, k \pm a)$
Minor vertices	$(h, k \pm b)$	$(h \pm b, k)$
Foci	$(h \pm c, k)$	$(h, k \pm c)$
Major-axis length	$2a$	$2a$
Minor-axis length	$2b$	$2b$

Table 8.3: Principal features of the two ellipse orientations.

The **latus rectum** of an ellipse is a chord through a focus and perpendicular to the major axis. For the horizontal ellipse centred at the origin, set $x = c$ in

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

Using $c^2 = a^2 - b^2$ gives

$$y = \pm \frac{b^2}{a}.$$

Hence, the endpoints of the latus rectum through $(c, 0)$ are

$$\left(c, \frac{b^2}{a}\right) \quad \text{and} \quad \left(c, -\frac{b^2}{a}\right),$$

and its length is

$$\frac{2b^2}{a}. \quad (8.21)$$

The same length applies to a vertical ellipse when a continues to denote the semi-major axis.

8.3.2 Converting a general ellipse to standard form

The coefficients of x^2 and y^2 in an axis-aligned ellipse have the same sign. Completing the square separately in x and y , then dividing so that the right-hand side is 1, reveals the centre and axes.

Example 8.6: completing two squares

Write

$$9x^2 + 4y^2 - 54x + 45 = 0$$

in standard form.

$$\begin{aligned} 9(x^2 - 6x) + 4y^2 &= -45, \\ 9((x - 3)^2 - 9) + 4y^2 &= -45, \\ 9(x - 3)^2 + 4y^2 &= 36. \end{aligned}$$

Therefore

$$\frac{(x - 3)^2}{4} + \frac{y^2}{9} = 1.$$

The centre is $(3, 0)$, the major axis is vertical, $a = 3$, $b = 2$, and $c = \sqrt{5}$. The vertices are $(3, \pm 3)$ and the foci are $(3, \pm \sqrt{5})$.

8.3.3 Point position and sketching an ellipse

For an ellipse in standard form, substitute a point into the normalised left-hand side. For example, for

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1,$$

define

$$S = \frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2}.$$

The point is inside when $S < 1$, on the ellipse when $S = 1$, and outside when $S > 1$.

A labelled sketch should be built in this order:

1. mark the centre;
2. identify the larger denominator and draw the major axis in that direction;
3. mark the vertices at distance a and the co-vertices at distance b ;
4. calculate $c = \sqrt{a^2 - b^2}$ and mark the foci;
5. draw a smooth closed curve through the four axis endpoints.

The foci must always lie inside the ellipse on the major axis. Confusing the larger denominator with the x -denominator is a common error: the larger denominator determines the major-axis direction, regardless of which variable it accompanies.

8.4 Hyperbola

A **hyperbola** is the locus of all points P in a plane for which the absolute difference between the distances from two fixed points is constant. The fixed points are called the **foci**, denoted by F_1 and F_2 . Thus,

$$|PF_1 - PF_2| = 2a. \quad (8.22)$$

A hyperbola consists of two separate, mirror-image curves called **branches**. The **centre** is the midpoint of the foci. The **vertices** are the points on the branches closest to the centre, each at a distance a from it. The **transverse axis** passes through the centre, vertices, and foci, while the **conjugate axis** passes through the centre and is perpendicular to the transverse axis. The **asymptotes** are straight lines that the branches approach as they extend towards infinity.

8.4.1 Derivation of the standard equation

Consider a horizontal hyperbola centred at the origin with foci

$$F_1(-c, 0) \quad \text{and} \quad F_2(c, 0),$$

where $c > a > 0$. For a point $P(x, y)$ on the right branch, the distance condition may be written as

$$PF_1 - PF_2 = 2a.$$

Using the distance formula,

$$\sqrt{(x+c)^2 + y^2} - \sqrt{(x-c)^2 + y^2} = 2a.$$

Rearranging and squaring once gives

$$\begin{aligned}\sqrt{(x+c)^2 + y^2} &= 2a + \sqrt{(x-c)^2 + y^2}, \\ 4cx - 4a^2 &= 4a\sqrt{(x-c)^2 + y^2}, \\ \sqrt{(x-c)^2 + y^2} &= \frac{c}{a}x - a.\end{aligned}$$

Squaring again and simplifying,

$$\begin{aligned}(x-c)^2 + y^2 &= \left(\frac{c}{a}x - a\right)^2, \\ \left(\frac{c^2}{a^2} - 1\right)x^2 - y^2 &= c^2 - a^2.\end{aligned}$$

Define

$$b^2 = c^2 - a^2, \tag{8.23}$$

so that

$$c^2 = a^2 + b^2. \tag{8.24}$$

The equation then becomes

$$\boxed{\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1}. \tag{8.25}$$

The minus sign is the quickest algebraic distinction between a hyperbola and an ellipse. For a hyperbola, $c^2 = a^2 + b^2$, so the foci lie farther from the centre than the vertices. Its eccentricity is

$$e = \frac{c}{a} > 1.$$

This contrasts with the ellipse, for which $c < a$ and $e < 1$. The denominator a^2 always belongs to the positive squared term and therefore identifies the transverse-axis direction; a need not be the larger numerical semi-axis.

8.4.2 Standard equations and principal features

For a hyperbola centred at (h, k) , the two standard orientations are

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1 \quad (\text{horizontal transverse axis}), \quad (8.26)$$

$$\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1 \quad (\text{vertical transverse axis}), \quad (8.27)$$

where $a, b > 0$ and $c = \sqrt{a^2 + b^2}$.

Feature	Horizontal hyperbola	Vertical hyperbola
Centre	(h, k)	(h, k)
Vertices	$(h \pm a, k)$	$(h, k \pm a)$
Foci	$(h \pm c, k)$	$(h, k \pm c)$
Transverse-axis length	$2a$	$2a$
Conjugate-axis length	$2b$	$2b$
Asymptotes	$y - k = \pm \frac{b}{a}(x - h)$	$y - k = \pm \frac{a}{b}(x - h)$
Latus rectum length	$\frac{2b^2}{a}$	$\frac{2b^2}{a}$

Table 8.4: Principal features of horizontal and vertical hyperbolas.

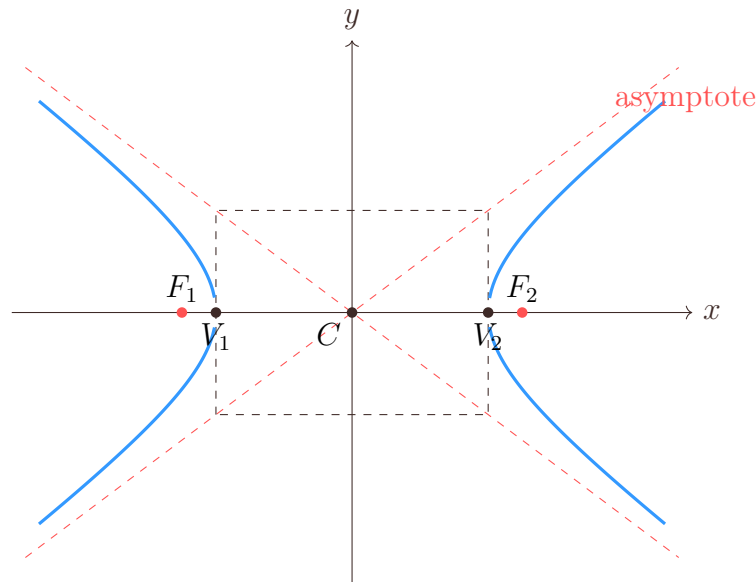


Figure 8.8: A horizontal hyperbola with its vertices, foci, auxiliary rectangle, and asymptotes.

The dashed rectangle in Figure 8.8 is the **auxiliary rectangle**. For a horizontal hyperbola, its corners are $(h \pm a, k \pm b)$. Its diagonals are the asymptotes. The branches begin at the vertices, remain outside this rectangle in the transverse direction, and approach the diagonals without meeting them.

8.4.3 Converting a general hyperbola to standard form

An axis-aligned hyperbola has squared terms with opposite signs. Complete the squares, move the constant to the other side, and divide by a positive number so that the right-hand side becomes 1.

Example 8.7: revealing a vertical hyperbola

Convert

$$4x^2 - 9y^2 - 8x - 36y - 68 = 0$$

to standard form.

$$\begin{aligned} 4(x^2 - 2x) - 9(y^2 + 4y) &= 68, \\ 4((x - 1)^2 - 1) - 9((y + 2)^2 - 4) &= 68, \\ 9(y + 2)^2 - 4(x - 1)^2 &= 36. \end{aligned}$$

Hence

$$\frac{(y + 2)^2}{4} - \frac{(x - 1)^2}{9} = 1.$$

The centre is $(1, -2)$, the transverse axis is vertical, $a = 2$, $b = 3$, and $c = \sqrt{13}$. The vertices are $(1, 0)$ and $(1, -4)$, and the asymptotes are

$$y + 2 = \pm \frac{2}{3}(x - 1).$$

8.4.4 Equations of the asymptotes

For the horizontal hyperbola

$$\frac{(x - h)^2}{a^2} - \frac{(y - k)^2}{b^2} = 1,$$

rearranging gives

$$(y - k)^2 = \frac{b^2}{a^2} ((x - h)^2 - a^2).$$

Hence,

$$y - k = \pm \frac{b(x - h)}{a} \sqrt{1 - \frac{a^2}{(x - h)^2}}.$$

As $|x| \rightarrow \infty$, the square-root factor approaches 1. Therefore, the asymptotes are

$$\boxed{y - k = \pm \frac{b}{a}(x - h)}. \quad (8.28)$$

By the same reasoning, the asymptotes of the vertical hyperbola are

$$\boxed{y - k = \pm \frac{a}{b}(x - h)}. \quad (8.29)$$

8.4.5 Latus rectum of a hyperbola

The **latus rectum** is a chord through a focus and perpendicular to the transverse axis. For the horizontal hyperbola centred at the origin, set $x = c$ in

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

Then

$$\begin{aligned} \frac{c^2}{a^2} - \frac{y^2}{b^2} &= 1, \\ y^2 &= b^2 \left(\frac{c^2 - a^2}{a^2} \right). \end{aligned}$$

Since $c^2 - a^2 = b^2$,

$$y = \pm \frac{b^2}{a}.$$

Thus, the latus rectum through $(c, 0)$ has endpoints

$$\left(c, \frac{b^2}{a} \right) \quad \text{and} \quad \left(c, -\frac{b^2}{a} \right),$$

and its length is

$$\boxed{\frac{2b^2}{a}}. \quad (8.30)$$

8.4.6 Point position and sketching a hyperbola

For a horizontal hyperbola, the normalised quantity

$$H = \frac{(x - h)^2}{a^2} - \frac{(y - k)^2}{b^2}$$

equals 1 on the curve. Unlike the ellipse test, $H < 1$ does not by itself mean “inside”, because a hyperbola does not enclose a bounded region. It is usually clearer to determine whether a point lies on the curve by direct substitution and to describe its position relative to the branches and asymptotes.

A reliable sketching sequence is:

1. complete the square and identify the centre;
2. locate the positive squared term to determine the transverse-axis direction;

3. mark the vertices at distance a from the centre;
4. draw the auxiliary rectangle using half-widths a and b ;
5. extend its diagonals to form the asymptotes;
6. calculate $c = \sqrt{a^2 + b^2}$ and mark the foci;
7. draw the two branches through the vertices, approaching the asymptotes.

The curve never crosses the conjugate axis. A horizontal hyperbola has no y -axis-direction vertices, and a vertical hyperbola has no x -axis-direction vertices.

8.5 Parametric equation

A parametric equation expresses related quantities, such as the coordinates of a point, as functions of an independent variable called a **parameter**. Instead of writing y directly as a function of x , we write

$$x = f(t), \quad y = g(t).$$

The ordered pair $(f(t), g(t))$ determines the position of a moving point. Eliminating t reveals the Cartesian curve, but the parametric description contains additional information:

- the permitted range of t determines which portion is traced;
- increasing t determines the direction of travel;
- the form of f and g determines the starting position and how the motion progresses along the curve.

Different parametrisations can describe the same Cartesian curve. For example, $x = \cos t$, $y = \sin t$ traces the unit circle once on $0 \leq t \leq 2\pi$, whereas $x = \cos 2t$, $y = \sin 2t$ traces it twice over the same parameter interval.

8.5.1 Parametric equation of a circle

Consider a circle of radius r centred at the origin. A point $P(x, y)$ on the circle may be described using the parameter t by

$$x = r \cos t, \quad y = r \sin t. \quad (8.31)$$

Eliminating the parameter verifies the Cartesian equation. Since

$$\cos t = \frac{x}{r} \quad \text{and} \quad \sin t = \frac{y}{r},$$

the identity $\cos^2 t + \sin^2 t = 1$ gives

$$\left(\frac{x}{r}\right)^2 + \left(\frac{y}{r}\right)^2 = 1, \\ x^2 + y^2 = r^2.$$

Thus, $x = r \cos t$ and $y = r \sin t$ are parametric equations of the circle $x^2 + y^2 = r^2$.

More generally, for a circle with centre (h, k) and radius r , the parametric equations are

$$\boxed{x = h + r \cos t, \quad y = k + r \sin t}. \quad (8.32)$$

Indeed,

$$\cos t = \frac{x - h}{r} \quad \text{and} \quad \sin t = \frac{y - k}{r},$$

so

$$\left(\frac{x - h}{r}\right)^2 + \left(\frac{y - k}{r}\right)^2 = 1,$$

which is equivalent to

$$(x - h)^2 + (y - k)^2 = r^2.$$

As t varies from 0 to 2π , the point P travels once anticlockwise around the circle, beginning at $(h + r, k)$.

8.5.2 Parametric equation of a parabola

For the horizontal parabola

$$(y - k)^2 = 4a(x - h),$$

a convenient parametrisation is

$$\boxed{x = h + at^2, \quad y = k + 2at}, \quad t \in \mathbb{R}. \quad (8.33)$$

To eliminate the parameter, use

$$t = \frac{y - k}{2a}.$$

Then

$$\begin{aligned} x - h &= at^2, \\ &= a \left(\frac{y - k}{2a} \right)^2, \\ &= \frac{(y - k)^2}{4a}, \end{aligned}$$

which gives

$$(y - k)^2 = 4a(x - h).$$

Similarly, for the vertical parabola

$$(x - h)^2 = 4a(y - k),$$

the parametric equations are

$$\boxed{x = h + 2at, \quad y = k + at^2}, \quad t \in \mathbb{R}. \quad (8.34)$$

Indeed, $t = (x - h)/(2a)$, and hence

$$\begin{aligned} y - k &= at^2, \\ &= a \left(\frac{x - h}{2a} \right)^2, \\ &= \frac{(x - h)^2}{4a}, \end{aligned}$$

so

$$(x - h)^2 = 4a(y - k).$$

In these parametrisations, $a > 0$ produces a right- or upward-opening parabola, while $a < 0$ produces a left- or downward-opening parabola.

8.5.3 Parametric equation of an ellipse

For the ellipse centred at (h, k) with horizontal major axis,

$$\frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} = 1, \quad a > b > 0,$$

the parametric equations are

$$\boxed{x = h + a \cos t, \quad y = k + b \sin t}. \quad (8.35)$$

Indeed,

$$\cos t = \frac{x - h}{a} \quad \text{and} \quad \sin t = \frac{y - k}{b}.$$

Using $\cos^2 t + \sin^2 t = 1$ gives

$$\left(\frac{x - h}{a} \right)^2 + \left(\frac{y - k}{b} \right)^2 = 1,$$

which is precisely

$$\frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} = 1.$$

For an ellipse with a vertical major axis,

$$\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1,$$

the corresponding parametrisation is

$$\boxed{x = h + b \cos t, \quad y = k + a \sin t}. \quad (8.36)$$

As t varies from 0 to 2π , the point travels once anticlockwise around the ellipse.

8.5.4 Parametric equation of a hyperbola

For the horizontal hyperbola

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1,$$

a trigonometric parametrisation is

$$\boxed{x = h + a \sec t, \quad y = k + b \tan t}. \quad (8.37)$$

Since

$$\sec t = \frac{x-h}{a} \quad \text{and} \quad \tan t = \frac{y-k}{b},$$

the identity $\sec^2 t - \tan^2 t = 1$ gives

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1.$$

For the vertical hyperbola

$$\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1,$$

the corresponding parametrisation is

$$\boxed{x = h + b \tan t, \quad y = k + a \sec t}. \quad (8.38)$$

Indeed,

$$\tan t = \frac{x-h}{b} \quad \text{and} \quad \sec t = \frac{y-k}{a},$$

so $\sec^2 t - \tan^2 t = 1$ gives

$$\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1.$$

These parametrisations require $\cos t \neq 0$; equivalently,

$$t \neq \frac{\pi}{2} + n\pi, \quad n \in \mathbb{Z}.$$

Different intervals of t trace the two separate branches of the hyperbola.

For the horizontal form, intervals with $\sec t > 0$ trace the right branch, while intervals with $\sec t < 0$ trace the left branch. A single interval crossing $\pi/2 + n\pi$ is not permitted because the coordinates become undefined there.

8.5.5 Eliminating a parameter and interpreting the result

The method used to eliminate a parameter depends on its form:

1. If one equation is linear in t , solve it for t and substitute.
2. If sine and cosine appear, use $\sin^2 t + \cos^2 t = 1$.
3. If tangent and secant appear, use $\sec^2 t - \tan^2 t = 1$.
4. After obtaining the Cartesian equation, return to the original parameter range to determine whether the entire curve is traced.

Example 8.8: a parametrised parabola

Let

$$x = 1 + 3t, \quad y = 2t^2 - 4, \quad t \in \mathbb{R}.$$

Since $t = (x - 1)/3$,

$$y = \frac{2}{9}(x - 1)^2 - 4.$$

Equivalently,

$$(x - 1)^2 = \frac{9}{2}(y + 4).$$

Because every real value of t is allowed, the whole parabola is traced. The vertex occurs at $t = 0$, giving $(1, -4)$. Negative t traces the left side towards the vertex, and positive t traces the right side away from it.

Example 8.9: a restricted arc

The equations

$$x = 2 + 5 \cos t, \quad y = -1 + 5 \sin t, \quad 0 \leq t \leq \pi$$

satisfy $(x - 2)^2 + (y + 1)^2 = 25$. However, $\sin t \geq 0$ on the given interval, so only the upper semicircle is traced, from $(7, -1)$ through $(2, 4)$ to $(-3, -1)$.

8.6 Applications of conic sections

Conic sections occur whenever distance, reflection, orbital motion, or a quadratic profile controls a design. The equation is rarely the starting point in an application. More often, physical information supplies a centre, focus, axis, width, depth, or constant distance, and these data must first be translated into a suitable conic model.

8.6.1 Circular boundaries, coverage, and safe zones

Circles model regions determined by a fixed maximum distance: communication coverage around a transmitter, a rotating arm, a circular track, or an exclusion zone around hazardous equipment. Tangency is important when a road, pipe, or line of sight must just avoid the circular region.

Example 8.10: classifying a road near a protected zone

A protected zone is modelled by

$$(x - 4)^2 + (y + 2)^2 = 49,$$

and a proposed straight road is $3x + 4y = 30$. The centre is $(4, -2)$ and the radius is 7. The centre-to-road distance is

$$d = \frac{|3(4) + 4(-2) - 30|}{5} = \frac{26}{5}.$$

Since $26/5 < 7$, the road is a secant and passes through the protected zone. A tangent redesign would require the centre-to-road distance to equal 7.

8.6.2 Parabolic reflectors and arches

A parabola has a reflection property: rays parallel to its axis are reflected through the focus, and rays emitted from the focus are reflected parallel to the axis. This is why parabolic profiles are used in satellite dishes, headlamps, radio telescopes, solar cookers, and directional microphones.

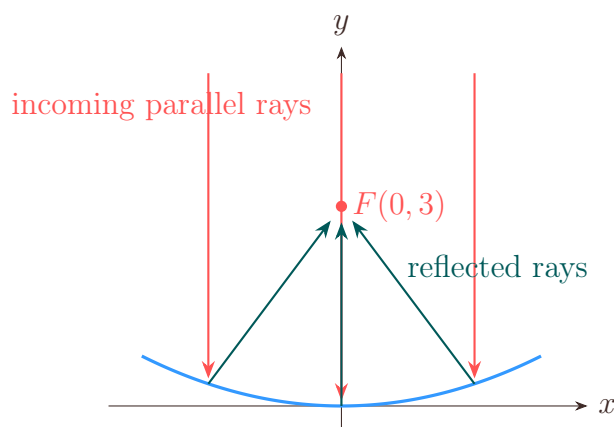


Figure 8.9: Parallel rays reflected by a parabolic profile pass through its focus.

Example 8.11: locating a dish receiver

A dish is 6 m wide and 0.75 m deep. Place the vertex at the origin and model the cross-section by $x^2 = 4ay$. At the rim, $(x, y) = (3, 0.75)$, so

$$9 = 4a(0.75) = 3a.$$

Hence $a = 3$ and the focus is $(0, 3)$. The receiver should be placed three metres from the vertex along the axis.

Parabolic arches are usually modelled with a downward-opening equation. If an arch has vertex $(0, H)$ and meets the ground at $x = \pm W/2$, these three points determine a model of the form $y = H - kx^2$. Vehicle clearance must be checked at the vehicle edges, not only at its centre.

8.6.3 Elliptical acoustics and orbital paths

The reflection property of an ellipse sends a ray leaving one focus towards the other focus after reflection from the boundary. Whispering galleries and some medical reflectors use this focusing effect. The same geometry appears in orbital models: one focus of a planetary ellipse is occupied by the central star rather than by the centre of the ellipse.

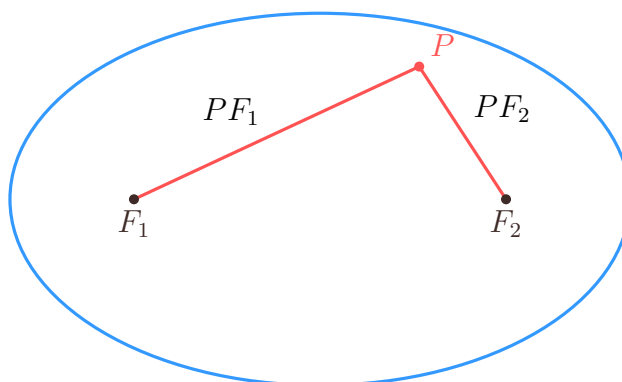


Figure 8.10: The focal path F_1PF_2 has constant total length for every point P on an ellipse.

Example 8.12: acoustic foci of an elliptical room

An elliptical room is 26 m long and 10 m wide. Taking its centre as the origin gives $a = 13$ and $b = 5$. Therefore

$$c = \sqrt{13^2 - 5^2} = 12,$$

and a suitable model is

$$\frac{x^2}{169} + \frac{y^2}{25} = 1.$$

The acoustic foci are $(\pm 12, 0)$, which are 24 m apart. Any boundary path from one focus to the other has total length $2a = 26$ m.

For an orbit with nearest and farthest focal distances $r_{\min}$ and $r_{\max}$,

$$a - c = r_{\min}, \quad a + c = r_{\max}.$$

Adding and subtracting these equations gives a and c , after which $b^2 = a^2 - c^2$ determines the complete orbital ellipse.

8.6.4 Hyperbolic localisation and structural profiles

A difference in distances naturally creates a hyperbola. If two stations receive a signal at different times, the measured arrival-time difference can be converted into a distance difference. Every possible source location then lies on one branch of a hyperbola with the stations as foci. Combining measurements from additional station pairs narrows the source to an intersection point.

Example 8.13: a two-station location curve

Stations are placed at $(-10, 0)$ and $(10, 0)$, and a source is known to be 12 units farther from one station than from the other. Then $c = 10$ and $2a = 12$, so $a = 6$. Since

$$b^2 = c^2 - a^2 = 100 - 36 = 64,$$

the possible locations satisfy

$$\frac{x^2}{36} - \frac{y^2}{64} = 1.$$

The right branch corresponds to sources nearer the station at $(10, 0)$; the left branch corresponds to sources nearer the station at $(-10, 0)$.

Hyperbolic profiles also appear in cooling towers and some optical systems. The narrowest cross-section occurs at the vertices. Farther from the centre, the sides widen while approaching the asymptotes, making the asymptotes useful for understanding the large-scale shape of the structure.

8.6.5 Parametric models of motion

Parametric equations are especially natural when time is the parameter. A point moving anticlockwise around a circle of radius r with centre (h, k) and period T can be modelled by

$$x = h + r \cos\left(\frac{2\pi t}{T}\right), \quad y = k + r \sin\left(\frac{2\pi t}{T}\right),$$

provided it begins at the rightmost point when $t = 0$.

Example 8.14: a Ferris-wheel model

A Ferris wheel has radius 12 m, centre $(0, 14)$, and period 30 s. A rider beginning at the rightmost point is described by

$$x = 12 \cos\left(\frac{\pi t}{15}\right), \quad y = 14 + 12 \sin\left(\frac{\pi t}{15}\right).$$

At $t = 7.5$ s the rider is at the top, $(0, 26)$; at $t = 15$ s the rider is at $(-12, 14)$; and the minimum height is 2 m at $t = 22.5$ s.

The same principle applies to elliptical motion by replacing the common radius with two semi-axis lengths. Changing the sign of the sine term reverses the direction, while

adding a phase inside the trigonometric functions changes the starting point.

8.6.6 A practical modelling workflow

An application rarely announces which conic equation should be used. A reliable modelling process is therefore as important as the algebra:

1. Choose an origin and coordinate axes that exploit symmetry. Placing a centre or vertex at the origin usually removes unnecessary linear terms.
2. Translate the defining condition into geometry. A fixed distance suggests a circle; equal distances from a point and a line suggest a parabola; a constant sum or difference of focal distances suggests an ellipse or hyperbola.
3. Use the given dimensions to determine the parameters. Widths, heights, depths, vertices, foci, and boundary points should be substituted into the model rather than estimated from a sketch.
4. Retain the physical restrictions. A bridge arch may use only the upper part of a parabola, a receiver may occupy one branch of a hyperbola, and a parameter interval may trace only part of a curve.
5. Check the model against every supplied measurement and interpret the final result with units. Algebraically valid points outside the physical domain must be discarded.

The equation is consequently only one part of a complete answer. A sound solution also states what its parameters mean, identifies the relevant portion of the conic, and explains how the requested distance, location, clearance, or time follows from the model. As a dimensional check, quantities added or compared must have compatible units; for example, the two sides of a distance equation must both represent lengths.

Chapter summary and common pitfalls

Key ideas. Each conic is characterised by a locus condition and a standard equation. Completing the square exposes translations and scales, while parametrisations describe points and motion along the curve.

Common pitfalls. Keep the signs distinguishing ellipses from hyperbolas, identify the correct major or transverse axis, and apply translations to both the equation and the coordinates of geometric features.

8.7 Practice and exercises

Give exact values unless an approximation is requested. Sketches should show all principal features and use equal scales on the coordinate axes whenever this is geometrically appropriate.

Circles

1. State the centre and radius of each circle.

(a) $(x - 3)^2 + (y + 2)^2 = 25$;

(b) $x^2 + (y - 4)^2 = 9$;

(c) $(x + 1)^2 + (y - 5)^2 = 12$;

(d) $x^2 + y^2 = 18$.

2. Complete the square. Hence state the centre and radius, or explain why no real circle is represented.

(a) $x^2 + y^2 - 6x + 8y - 11 = 0$;

(b) $x^2 + y^2 + 4x - 10y + 13 = 0$;

(c) $2x^2 + 2y^2 - 8x + 12y - 24 = 0$;

(d) $x^2 + y^2 + 2x - 4y + 10 = 0$.

3. Find the equation of the circle described.

(a) Centre $(-2, 3)$ and radius 4;

(b) centre at the origin and passing through $(3, -4)$;

(c) centre $(1, -1)$ and tangent to the line $x = 5$;

(d) centre $(2, 3)$ and tangent to the line $y = -1$.

4. Find the equation of the circle whose diameter has the stated endpoints.

(a) $(0, 2)$ and $(4, -4)$;

(b) $(-3, 1)$ and $(5, 1)$;

- (c) $(-2, -2)$ and $(4, 6)$.
5. Find the equation of the circle through each set of three points.
- (a) $(1, 0)$, $(-1, 0)$, and $(0, 1)$;
(b) $(0, 0)$, $(4, 0)$, and $(0, 6)$;
(c) $(2, 1)$, $(6, 1)$, and $(4, 5)$.
6. Relative to $(x - 2)^2 + (y + 1)^2 = 25$, classify each point as inside, on, or outside the circle:
- $(2, 4)$, $(5, 3)$, $(0, 0)$, $(8, -1)$, $(-2, -4)$.
7. Find all intercepts with the coordinate axes.
- (a) $x^2 + y^2 = 16$;
(b) $(x - 2)^2 + (y + 1)^2 = 10$;
(c) $x^2 + y^2 - 6x - 4y = 0$.
8. Find the points of intersection of each circle and line.
- (a) $x^2 + y^2 = 25$ and $y = 3$;
(b) $(x - 2)^2 + (y + 1)^2 = 10$ and $y = x - 1$;
(c) $x^2 + y^2 - 4x + 2y - 4 = 0$ and $x = 2$.
9. Find all values of the parameter for which the line is tangent to the circle.
- (a) $y = k$ and $x^2 + y^2 = 25$;
(b) $y = x + c$ and $x^2 + y^2 = 8$;
(c) $x + 2y = k$ and $(x - 1)^2 + (y + 2)^2 = 9$.
10. Find the equation of the tangent at the given point.
- (a) $x^2 + y^2 = 10$ at $(1, 3)$;
(b) $x^2 + y^2 - 6x - 2y + 5 = 0$ at $(2, 3)$;
(c) $(x + 2)^2 + (y - 1)^2 = 25$ at $(1, 5)$.
11. Find the equation of the normal at each point in Question 10.
12. The endpoints of a diameter are $P(0, 2)$ and $Q(4, -4)$. Determine the circle and find the tangent and normal at P .
13. Find the length of a tangent from the stated external point.
- (a) $Q(3, 6)$ to $(x - 8)^2 + (y - 1)^2 = 9$;

- (b) $Q(2, -3)$ to $x^2 + y^2 - 10x - 4y + 12 = 0$;
(c) $Q(10, 2)$ to $(x - 2)^2 + (y - 2)^2 = 16$.
14. From $P(13, 0)$, draw the two tangents to $x^2 + y^2 = 25$. Find their equations, their points of contact, and their common length.
15. Find the radical axis of each pair of circles. For part (a), also find the endpoints of the common chord.
- (a) $x^2 + y^2 = 25$ and $(x - 4)^2 + y^2 = 9$;
(b) $x^2 + y^2 - 4x = 0$ and $x^2 + y^2 + 2y - 8 = 0$.
16. The circle $(x - 2)^2 + (y + 1)^2 = 25$ is compared with the lines

$$3x + 4y = 20, \quad 3x + 4y = 27, \quad 3x + 4y = 32.$$

Classify each line as a secant, tangent, or external line without solving simultaneous equations.

17. Find the equation of each circle.
- (a) Radius 3, tangent to both axes, centre in the first quadrant;
(b) radius 4, centre on $y = 2x$ in the first quadrant, tangent to the y -axis;
(c) tangent to the x -axis at $(2, 0)$, lying above the axis, and passing through $(5, 4)$.
18. Determine and identify each locus.
- (a) Points at distance 5 from $(2, -1)$;
(b) points equidistant from $(-2, 0)$ and $(4, 0)$;
(c) points whose distance from $(-2, 0)$ is twice their distance from $(1, 0)$.
19. For $x^2 + y^2 = r^2$, use the radius through (x_1, y_1) to derive the tangent equation $xx_1 + yy_1 = r^2$. Explain geometrically why the tangent and radius are perpendicular.
20. A circular garden has boundary

$$x^2 + y^2 - 20x - 12y + 100 = 0.$$

Find its centre and diameter, the tangent at the northernmost point, and the area of a two-metre-wide path constructed outside the garden.

Parabolas

21. State the vertex, focus, directrix, axis, and direction of opening.

- (a) $x^2 = 4y$;
 - (b) $(x - 3)^2 = -(y + 1)$;
 - (c) $(y - 2)^2 = 8(x + 1)$;
 - (d) $y^2 + 2y - x = 0$.
22. Write each equation in standard form. Hence state its vertex and focal parameter a .
- (a) $x^2 + 8x = 4y - 8$;
 - (b) $x^2 - 6x - 8y + 25 = 0$;
 - (c) $y^2 - 4y + 8x + 12 = 0$;
 - (d) $2x^2 + 12x + 8y + 10 = 0$.
23. Find the equation of the parabola from the given data.
- (a) Vertex $(0, 0)$, focus $(0, 3)$;
 - (b) vertex $(2, -1)$, focus $(5, -1)$;
 - (c) vertex $(-2, 4)$, focus $(-2, 1)$;
 - (d) vertex $(1, 2)$, directrix $x = -1$.
24. Find the equation and focus of each parabola.
- (a) Horizontal axis, vertex $(-1, 2)$, through $(-5, -2)$;
 - (b) vertical axis, vertex $(2, 2)$, through $(4, -1)$;
 - (c) vertical axis, vertex $(0, -2)$, through $(4, 2)$.
25. Find the focus, endpoints of the latus rectum, and latus-rectum length.
- (a) $(x - 1)^2 = 12(y + 2)$;
 - (b) $(y + 3)^2 = -16(x - 2)$;
 - (c) $y^2 = 20x$.
26. Find all points of intersection.
- (a) $x^2 = 4y$ and $y = x + 2$;
 - (b) $(y - 1)^2 = 8(x + 1)$ and $x = 2$;
 - (c) $(x - 1)^2 = 4(y + 1)$ and $y = 3$.
27. A circle of radius r is centred at the focus of $(x - h)^2 = 4a(y - k)$, where $a > 0$. Show that it touches the parabola only when $r = a$, and identify the point of contact.
28. Show that the circle whose diameter is the latus rectum of $(y - 2)^2 = 4a(x - 3)$ is tangent to the directrix. State the point of tangency.

29. Use the focus–directrix definition to derive the equation of each parabola.

- (a) Focus $(0, 2)$, directrix $y = -2$;
- (b) focus $(3, 0)$, directrix $x = -1$;
- (c) focus $(-2, 1)$, directrix $x = 4$.

30. Determine the value of p .

- (a) $x^2 = 4py$ passes through $(6, 3)$;
- (b) $(y + 1)^2 = 4p(x - 2)$ passes through $(6, 7)$;
- (c) $(x - 3)^2 = 4p(y + 2)$ passes through $(-1, -1)$.

31. Match each equation to its description, and justify the match:

$$x^2 = 8y, \quad y^2 = -12x, \quad (x - 2)^2 = -4(y + 1), \quad (y + 3)^2 = 16(x - 1).$$

The descriptions are: opens right, opens left, opens upwards, and opens downwards.

- 32. For the parabola $x^2 = 8y$, determine the number of intersections with the horizontal line $y = k$ when $k > 0$, $k = 0$, and $k < 0$.
- 33. Compare $x^2 = 4y$ and $(x - 2)^2 = 4(y + 1)$. Describe the translation taking the first graph to the second, and compare their foci and directrices.
- 34. Find the intersections of $x^2 = 4y$ and $x^2 + y^2 = 5$. Explain why the other algebraic root obtained after substitution must be rejected.
- 35. A horizontal chord of $x^2 = 4ay$, $a > 0$, lies on $y = k$. Derive its length. Hence find k when the chord has length 8 in $x^2 = 4y$.
- 36. A parabola has axis $x = 3$, vertex $(3, -2)$, and latus-rectum length 12. Find the two possible equations and their foci.

Ellipses

37. State the centre, orientation, vertices, co-vertices, and foci.

- (a) $\frac{x^2}{25} + \frac{y^2}{9} = 1$;
- (b) $\frac{(x - 2)^2}{16} + \frac{(y + 1)^2}{25} = 1$;
- (c) $\frac{(x + 3)^2}{36} + \frac{(y - 2)^2}{20} = 1$;

(d) $\frac{(x-1)^2}{9} + \frac{(y-4)^2}{16} = 1.$

38. Complete the square and write in standard form. State the centre, semi-axis lengths, and foci.

(a) $9x^2 + 4y^2 - 54x + 45 = 0;$

(b) $16x^2 + 9y^2 - 96x + 36y + 36 = 0;$

(c) $4x^2 + 9y^2 + 16x - 54y + 61 = 0.$

39. Find the equation of the ellipse described.

(a) Centre $(2, -2)$, vertex $(7, -2)$, focus $(4, -2)$;

(b) centre $(1, 2)$, focus $(4, 2)$, through $(1, 3)$;

(c) centre $(1, 2)$, vertex $(4, 2)$, through $(1, 3)$;

(d) vertices $(4, 3)$ and $(4, 9)$, focus $(4, 8)$.

40. Find the equation from the foci and major-axis length.

(a) Foci $(5, 1)$ and $(-1, 1)$, major-axis length 8;

(b) foci $(0, 4)$ and $(0, -4)$, major-axis length 10.

41. Find the equation of the ellipse with the stated axis endpoints.

(a) Vertices $(-6, 2)$ and $(4, 2)$; co-vertices $(-1, -1)$ and $(-1, 5)$;

(b) vertices $(3, -5)$ and $(3, 7)$; co-vertices $(-1, 1)$ and $(7, 1)$.

42. An ellipse centred at the origin passes through the stated points. Find its equation and foci.

(a) $(2, 0)$ and $(1, 3\sqrt{3}/2)$;

(b) $(0, 4)$ and $(3, 0)$.

43. Find the endpoints and length of each latus rectum.

(a) $\frac{x^2}{25} + \frac{y^2}{9} = 1;$

(b) $\frac{(x-2)^2}{16} + \frac{(y+1)^2}{25} = 1.$

44. For $x^2/16 + y^2/9 = 1$, classify $(4, 0)$, $(0, 3)$, $(2, 2)$, $(5, 0)$, and $(0, 0)$ as inside, on, or outside the ellipse.

45. Find the intersections with $x^2/25 + y^2/9 = 1$.

(a) $y = 0;$

- (b) $x = 3$;
(c) $y = 3x/5$.
46. Starting from the focal definition, explain why the constant sum of distances is $2a$ and why $c^2 = a^2 - b^2$.
47. Find the major- and minor-axis lengths and the enclosed area.
(a) $x^2/36 + y^2/16 = 1$;
(b) $(x - 2)^2/9 + (y + 1)^2/25 = 1$;
(c) $4x^2 + 9y^2 = 36$.
48. Verify the focal-sum property for the points $(0, 3)$ and $(3, 12/5)$ on $x^2/25 + y^2/9 = 1$.
49. Determine $k > 0$ so that the point lies on the ellipse.
(a) $(3, 2)$ on $x^2/25 + y^2/k = 1$;
(b) $(2, 3)$ on $x^2/k + y^2/25 = 1$.
50. Find the area of the smallest axis-parallel rectangle containing each ellipse in Question 47.
51. Show that the circle $x^2 + y^2 = 25$ touches the ellipse $x^2/25 + y^2/9 = 1$ only at its two horizontal vertices.
52. An ellipse has centre $(2, -1)$, horizontal major axis of length 12, and minor axis of length 8. Find its equation, vertices, co-vertices, and foci.

Hyperbolas

53. State the orientation, centre, vertices, foci, and asymptotes.
(a) $\frac{x^2}{16} - \frac{y^2}{25} = 1$;
(b) $\frac{(x + 2)^2}{9} - \frac{(y - 1)^2}{25} = 1$;
(c) $\frac{y^2}{36} - \frac{(x + 2)^2}{9} = 1$;
(d) $\frac{(y - 3)^2}{16} - \frac{(x - 1)^2}{20} = 1$.
54. Complete the square and write in standard form. Hence state the centre and orientation.
(a) $4x^2 - 25y^2 - 32x + 164 = 0$;

- (b) $16x^2 - y^2 + 64x - 2y + 67 = 0$;
(c) $9x^2 - 4y^2 + 18x + 16y - 43 = 0$.
55. Find the equation of the hyperbola described.
- (a) Centre $(1, 2)$, vertex $(4, 2)$, focus $(6, 2)$;
(b) centre $(-2, 1)$, vertex $(-2, 5)$, focus $(-2, 6)$;
(c) centre at the origin, vertex $(3, 0)$, asymptotes $y = \pm 2x$.
56. Construct the hyperbola from its asymptotes and a vertex.
- (a) Centre $(0, 0)$, asymptotes $y = \pm 3x/4$, vertex $(4, 0)$;
(b) centre $(2, -1)$, vertical transverse axis, asymptotes $y + 1 = \pm 2(x - 2)$, vertex $(2, 2)$.
57. Find the endpoints and length of the latus recta.
- (a) $x^2/16 - y^2/9 = 1$;
(b) $y^2/25 - x^2/11 = 1$.
58. Show that the conjugate-axis length of $16x^2 - y^2 + 64x - 2y + 67 = 0$ is one unit. Find its asymptotes.
59. Find all intersections with $x^2/9 - y^2/4 = 1$.
- (a) $y = 0$;
(b) $x = 5$;
(c) $y = x$.
60. A hyperbola has centre $(1, 2)$ and a vertical transverse axis. The ratio of the distance between the foci to the distance between the vertices is $5 : 3$. Find the relationship between a and b , then give one possible equation.
61. Determine the missing denominator.
- (a) $x^2/16 - y^2/b^2 = 1$ has foci $(\pm 5, 0)$;
(b) $(y - 2)^2/9 - (x + 1)^2/b^2 = 1$ has foci $(-1, 7)$ and $(-1, -3)$.
62. Verify the constant-difference property for the vertex $(3, 0)$ and the point $(5, 16/3)$ on $x^2/9 - y^2/16 = 1$.
63. Determine the unknown semi-axis.
- (a) $(x - 1)^2/a^2 - (y + 2)^2/9 = 1$ passes through $(6, -2)$;

- (b) $(y - 3)^2/16 - (x + 1)^2/b^2 = 1$ passes through $(1, 8)$.
64. Explain how the auxiliary rectangle produces the asymptotes of a hyperbola. Apply the construction to $(x - 2)^2/25 - (y + 1)^2/9 = 1$.
65. Prove that the branches of $x^2/a^2 - y^2/b^2 = 1$ approach $y = \pm bx/a$ as $|x|$ becomes large.
66. The foci of a hyperbola are $(-5, 0)$ and $(5, 0)$, and the absolute difference of focal distances is 6. Find its equation.
67. A horizontal hyperbola has centre at the origin, vertices $(\pm 4, 0)$, and passes through $(5, 3)$. Find its equation, foci, and asymptotes.
68. Determine whether each equation represents a horizontal hyperbola, vertical hyperbola, pair of intersecting lines, or no real curve.
- (a) $x^2/9 - y^2/4 = 1$;
 - (b) $y^2/16 - x^2/25 = 1$;
 - (c) $x^2/9 - y^2/4 = 0$;
 - (d) $x^2/9 + y^2/4 = -1$.

Parametric equations

69. Eliminate t and identify the curve.
- (a) $x = 4 - 2t, y = 3 + 6t - 4t^2$;
 - (b) $x = 1 + t, y = 2t^2 - 3$;
 - (c) $x = 2t - 1, y = t^2 + 4$.
70. Eliminate the parameter and state the centre and radius.
- (a) $x = 3 \sin \theta, y = 3 \cos \theta$;
 - (b) $x = 2 + 4 \cos t, y = -1 + 4 \sin t$;
 - (c) $x = -3 + 5 \sin u, y = 2 - 5 \cos u$.
71. Eliminate the parameter and identify the ellipse.
- (a) $x = 3 \sin(2\theta), y = -4 \cos(2\theta)$;
 - (b) $x = 2 + \sin(\theta/3), y = -5 + 4 \cos(\theta/3)$;
 - (c) $x = -1 + 5 \cos t, y = 3 + 2 \sin t$.
72. Eliminate the parameter and identify the hyperbola. State excluded parameter values.

- (a) $x = 1 + 2 \tan \alpha$, $y = 3 \sec \alpha - 4$;
 - (b) $x = 4 \sec t$, $y = 3 \tan t$;
 - (c) $x = -2 + 2 \tan u$, $y = 1 + 5 \sec u$.
73. Eliminate t and describe which part of the Cartesian curve is traced.
- (a) $x = t + 1/t$, $y = t^2 + 1/t^2$, $t \neq 0$;
 - (b) $x = t - 1/t$, $y = t^2 + 1/t^2$, $t \neq 0$.
74. Find the points corresponding to the listed parameter values.
- (a) $x = 2 + 3 \cos t$, $y = -1 + 3 \sin t$ at $t = 0, \pi/2, \pi, 3\pi/2$;
 - (b) $x = 4 \cos t$, $y = 2 \sin t$ at $t = \pi/6, \pi/2, \pi$.
75. Find all parameter values giving the stated point.
- (a) $x = 5 \cos t$, $y = 5 \sin t$ at $(3, 4)$;
 - (b) $x = 2 \tan t$, $y = 3 \sec t$ at $(2, 3\sqrt{2})$;
 - (c) $x = 1 + 2s$, $y = 3 - s$ at $(5, 1)$.
76. State the direction and portion traced as the parameter increases.
- (a) $x = \cos t$, $y = \sin t$, $0 \leq t \leq 2\pi$;
 - (b) $x = \cos t$, $y = -\sin t$, $0 \leq t \leq 2\pi$;
 - (c) $x = 3 \cos t$, $y = 2 \sin t$, $0 \leq t \leq \pi$.
77. Find all intersections between the parametric curve and the line.
- (a) $x = 4 \cos t$, $y = 4 \sin t$ and $y = x$;
 - (b) $x = t$, $y = t^2$ and $y = 2x$;
 - (c) $x = 5 \cos t$, $y = 3 \sin t$ and $x = 0$.
78. Give a convenient parametrisation.
- (a) $(x - 2)^2 + (y + 1)^2 = 25$;
 - (b) $(x + 1)^2/16 + (y - 3)^2/4 = 1$;
 - (c) $(x - 2)^2 = 8(y + 1)$;
 - (d) $x^2/9 - y^2/16 = 1$.
79. The parametrisation $x = \cos(2t)$, $y = \sin(2t)$ is used for $0 \leq t \leq 2\pi$. Identify the curve, state its direction, and determine how many times it is traced.
80. A point moves on $x = 50 \cos(\pi t/30)$, $y = 30 \sin(\pi t/30)$, where t is measured in seconds. Find its positions at $t = 0, 15, 30, 45$ and the time of its first return to the initial point.

Applications and modelling

81. A Ferris wheel has radius 18 m and its centre is 20 m above the ground. It completes one revolution in 40 s, beginning at the rightmost point and moving anticlockwise. Write parametric equations and find the rider's height at $t = 10, 20, 30$ s.
82. A circular running track has outer radius 50 m and inner radius 45 m. Find the area of the track and the length of one lap measured along its midline.
83. A horizontal cylindrical tank has circular cross-section of radius 5 m. Find the width of the water surface when the water depth is (a) 2 m and (b) 4 m.
84. A protected circular zone has centre $(4, 3)$ and radius 2 km. Two straight access routes from $P(10, 3)$ are tangent to the boundary. Find the route length and both boundary contact points.
85. A straight road $y = x - 3$ crosses a circular coverage region $(x - 2)^2 + (y + 1)^2 = 100$. Find the entry and exit points and the length of road lying inside the region.
86. A satellite dish is 4.8 m wide and 0.6 m deep. Model its cross-section by a parabola with vertex at the origin and vertical axis. Find the focus and state where the receiver should be placed.

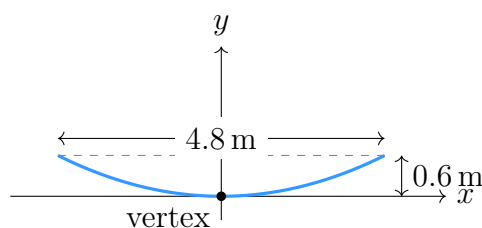


Figure 8.11: Parabolic dish cross-section for Question 86.

87. A parabolic bridge arch is modelled by $y = 9 - x^2/16$, where distances are in metres. Find its ground-level width and the maximum height of a centred vehicle of width 8 m if a vertical safety clearance of 0.5 m is required.
88. A solar cooker has cross-section $y = x^2/12$. Find the focus and the opening width at a depth of 4 m.
89. An elliptical running track has major-axis length 120 m and minor-axis length 80 m. Place its centre at the origin. Find its equation, foci, and enclosed area.
90. A planet's nearest and farthest distances from its star are 40 and 160 million kilometres. Assuming the star lies at one focus, find the semi-major axis, focal distance, semi-minor axis, and a suitable equation of the orbit.

91. An elliptical whispering gallery is 30 m long and 18 m wide. Find the positions of its two acoustic foci and the distance between them.
92. A lithotripter reflector is modelled by $x^2/169 + y^2/25 = 1$. Find its foci and the constant total path length from one focus to the reflector and then to the other focus.
93. A cooling-tower cross-section is modelled by

$$\frac{x^2}{36} - \frac{(y-50)^2}{900} = 1,$$

where distances are in metres. Find the minimum tower width, its width at height 80 m, and the equations of the asymptotes.

94. Two navigation stations are at $(-6, 0)$ and $(6, 0)$. A vessel's absolute difference in distance from the stations is 8 km. Find the locus equation and identify which branch contains points closer to the station at $(6, 0)$.
95. A hyperbolic telescope mirror has cross-section $y^2/16 - x^2/9 = 1$. Find its vertices, foci, asymptotes, and the distance from the upper vertex to the nearer focus.
96. A drone circles a point $(100, 50)$ at radius 30 m once every 12 minutes, beginning at the easternmost point and moving anticlockwise. Write parametric equations and find its position after 3, 6, and 9 minutes.
97. A satellite follows

$$x = 7000 \cos\left(\frac{2\pi t}{90}\right), \quad y = 5000 \sin\left(\frac{2\pi t}{90}\right),$$

where distances are in kilometres and t is in minutes. Find the orbital period, the foci, and the positions at $t = 0, 22.5, 45$.

98. A water jet follows $x = 6t$, $y = 12t - 3t^2$, with distances in metres. Eliminate t , find the greatest height, and determine where the stream returns to ground level.
99. Two tunnel designs have the same central height of 8 m:

$$y = 8 - \frac{x^2}{8} \quad \text{and} \quad x^2 + y^2 = 64, \quad y \geq 0.$$

Compare their clearances at the edges of a centred 6 m-wide vehicle and state which design gives more clearance.

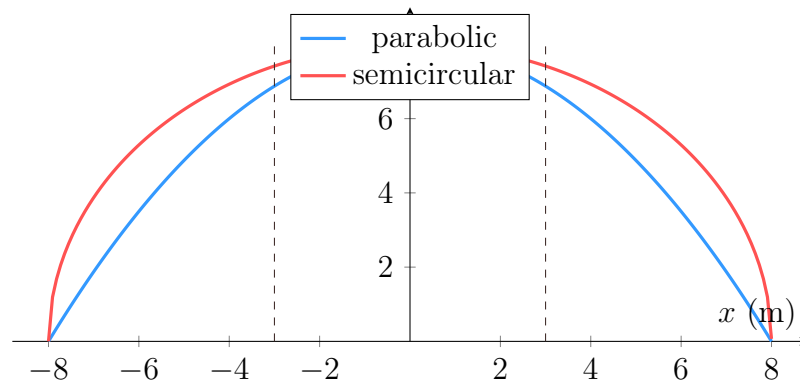


Figure 8.12: Tunnel profiles for Question 99.

100. A landscaped site has elliptical boundary $x^2/100 + y^2/64 = 1$, a circular fountain $x^2 + y^2 = 16$, and a service road $y = 6$.
- Classify the two boundaries and state their principal dimensions.
 - Find where the road enters and leaves the site.
 - Does the road meet the fountain? Justify your answer.
 - Give a parametrisation of the site boundary.

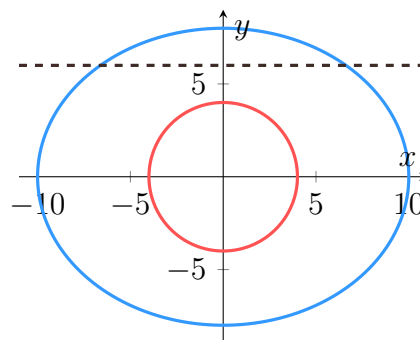


Figure 8.13: Site boundary, fountain, and service road for Question 100.

Chapter 9

Probability and Statistics

Chapter guide

Prerequisites. Fractions, algebra, indices, summation notation, and elementary data summaries.

Learning outcomes. Readers should be able to count outcomes, calculate probabilities, use discrete and continuous distributions, describe sampling behaviour, construct confidence intervals, and carry out elementary hypothesis tests.

Probability provides a mathematical language for uncertainty, while statistics uses data to describe populations and to make decisions in the presence of uncertainty. The two subjects are closely connected: probability models tell us how random quantities behave, and statistical inference uses those models to learn from samples.

This chapter begins with counting methods and the laws of probability. It then develops discrete and continuous random variables, introduces several standard probability distributions, and concludes with sampling distributions, confidence intervals, and hypothesis tests for a population mean. Definitions, notation, and tabular conventions are based on established introductory statistics references (Moore et al. 2021; Murdoch and Barnes 1998; Jalaluddin, Hashim et al. 2025–2026).

9.1 Counting techniques

Counting is often the first step in a probability calculation. If all outcomes are equally likely, then a probability can be found by dividing the number of favourable outcomes by the total number of possible outcomes.

9.1.1 The multiplication principle

Suppose a process consists of several successive choices. If the first choice can be made in n_1 ways, the second in n_2 ways, and so on, then the entire process can be completed in

$$n_1 n_2 \cdots n_k \tag{9.1}$$

ways, provided that the number of choices available at each stage is as stated.

Example 9.1

A cafe offers two types of rice, five vegetables, eight main dishes, and three desserts. The number of meals containing one item from each category is

$$2 \times 5 \times 8 \times 3 = 240.$$

If a code has four positions and each position may contain any digit from 0 to 9, repetition is allowed and there are 10^4 possible codes. If digits may not be repeated, there are $10 \times 9 \times 8 \times 7$ codes.

9.1.2 Factorials and permutations

For a non-negative integer n ,

$$n! = n(n-1)(n-2) \cdots 2 \cdot 1, \quad 0! = 1.$$

A **permutation** is an arrangement in which order matters. The number of ways to arrange all n distinct objects is $n!$. The number of ordered arrangements of r objects selected from n distinct objects is

$${}_nP_r = \frac{n!}{(n-r)!}. \quad (9.2)$$

If repetition is allowed in an ordered selection of length r from n available objects, there are n^r arrangements.

When some objects are identical, division removes the overcounting. If there are n objects in total, with n_1 identical objects of one kind, n_2 of a second kind, and so on, then the number of distinct arrangements is

$$\frac{n!}{n_1!n_2! \cdots n_k!}. \quad (9.3)$$

Example 9.2

The word STATISTICS contains ten letters: three S's, three T's, two I's, one A, and one C. Hence the number of distinct linear arrangements is

$$\frac{10!}{3!3!2!} = 50\,400.$$

Restrictions are commonly handled by treating a group as one block or by subtracting unwanted arrangements from the total. For example, if two particular people must stand together among eight people, regard the pair as one object. The pair can be internally arranged in two ways, so the number of arrangements is $2(7!)$. If the pair must be separated, the answer is $8! - 2(7!)$.

9.1.3 Circular permutations

Rotating everyone around a circular table does not produce a new relative arrangement. Fixing one person therefore gives

$$(n - 1)! \quad (9.4)$$

arrangements of n distinct objects around a circle. Clockwise and anticlockwise arrangements are different when people sit around a table. If reflections are also considered identical, as for a necklace that may be turned over, the count is usually $(n - 1)!/2$ for $n > 2$.

9.1.4 Combinations

A **combination** is a selection in which order does not matter. The number of ways to select r objects from n distinct objects is

$$\binom{n}{r} = {}_nC_r = \frac{n!}{r!(n - r)!}. \quad (9.5)$$

The identities

$$\binom{n}{r} = \binom{n}{n - r}, \quad \binom{n}{r} + \binom{n}{r + 1} = \binom{n + 1}{r + 1}$$

are often useful.

Example 9.3

A team of three is chosen from six male and eight female students. A team with more female than male students contains either two females and one male or three females. Therefore,

$$\binom{8}{2} \binom{6}{1} + \binom{8}{3} = 224.$$

Since there are $\binom{14}{3} = 364$ unrestricted teams, the probability that a randomly selected team has more female than male students is

$$\frac{224}{364} = \frac{8}{13}.$$

9.2 Experiments, events, and probability

A **random experiment** is a process whose result is uncertain. An individual result is an **outcome**; the set of all possible outcomes is the **sample space**, denoted by S ; and an **event** is a subset of S .

For example, when two coins are tossed,

$$S = \{HH, HT, TH, TT\}.$$

The event of obtaining exactly one head is $\{HT, TH\}$.

For every event A ,

$$0 \leq P(A) \leq 1, \quad P(S) = 1, \quad P(A') = 1 - P(A), \quad (9.6)$$

where A' denotes the complement of A . If all outcomes are equally likely,

$$P(A) = \frac{n(A)}{n(S)}. \quad (9.7)$$

An impossible event has probability 0, while a certain event has probability 1.

9.2.1 Union, intersection, and the addition rule

The event $A \cap B$ occurs when both A and B occur. The event $A \cup B$ occurs when at least one of A and B occurs. Since the overlap must not be counted twice,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B). \quad (9.8)$$

Events are **mutually exclusive** when they cannot occur together. In that case, $A \cap B = \emptyset$ and

$$P(A \cup B) = P(A) + P(B). \quad (9.9)$$

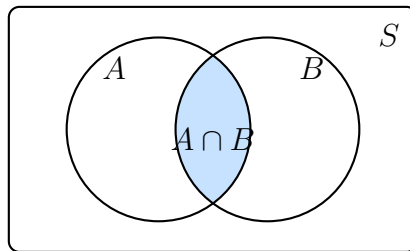


Figure 9.1: The intersection is the overlap counted in both $P(A)$ and $P(B)$.

Example 9.4

If $P(A) = 0.44$, $P(B) = 0.37$, and $P(A \cup B) = 0.67$, then

$$P(A \cap B) = 0.44 + 0.37 - 0.67 = 0.14.$$

9.2.2 Cards, tables, and counting probabilities

A standard deck contains 52 cards in four suits of 13. Hearts and diamonds are red; clubs and spades are black. Each suit has one ace, the numbers 2 to 10, and the three picture cards: jack, queen, and king.

Two-way tables are useful when a population is classified by two variables. For example,

	Male	Female	Total
Pass	57	62	119
Fail	12	19	31
Total	69	81	150

Thus $P(\text{fail}) = 31/150$, while $P(\text{fail} \mid \text{male}) = 12/69$. The denominator changes in the conditional probability because only male candidates remain in the reduced sample space.

9.2.3 Conditional probability

The probability of B given that A has occurred is

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)}, \quad P(A) > 0. \quad (9.10)$$

Rearranging gives the multiplication rule

$$P(A \cap B) = P(A)P(B \mid A) = P(B)P(A \mid B). \quad (9.11)$$

When sampling without replacement, the probabilities generally change after each selection. A tree diagram keeps track of these conditional probabilities. Suppose a bag contains five red and seven blue marbles, and two marbles are drawn without replacement.

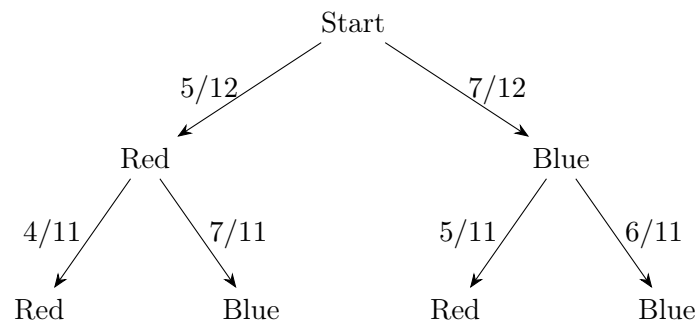


Figure 9.2: Probabilities along a path are multiplied; probabilities of disjoint paths are added.

For example,

$$\begin{aligned}
 P(\text{same colour}) &= P(RR) + P(BB) \\
 &= \frac{5}{12} \frac{4}{11} + \frac{7}{12} \frac{6}{11} = \frac{31}{66}.
 \end{aligned}$$

9.2.4 Independent events

Events A and B are **independent** when the occurrence of either does not change the probability of the other. Equivalent conditions are

$$P(B | A) = P(B), \quad P(A \cap B) = P(A)P(B). \quad (9.12)$$

For independent events $A_1, \dots, A_n$,

$$P(A_1 \cap \dots \cap A_n) = \prod_{i=1}^n P(A_i).$$

Mutual exclusivity and independence are different ideas. Two mutually exclusive events with positive probabilities cannot be independent, because one event makes the other impossible.

9.2.5 The total probability rule and Bayes' theorem

Suppose $A_1, A_2, \dots, A_n$ form a partition of the sample space: they are mutually exclusive and their union is S . For any event E ,

$$P(E) = \sum_{i=1}^n P(A_i)P(E | A_i). \quad (9.13)$$

Bayes' theorem reverses a conditional probability:

$$P(A_j | E) = \frac{P(A_j)P(E | A_j)}{\sum_{i=1}^n P(A_i)P(E | A_i)}. \quad (9.14)$$

Example 9.5

Factories A , B , and C supply 50%, 30%, and 20% of a company's chargers. Their defective rates are 10%, 5%, and 3%, respectively. If D is the event that a charger is defective,

$$\begin{aligned} P(D) &= (0.50)(0.10) + (0.30)(0.05) + (0.20)(0.03) \\ &= 0.071. \end{aligned}$$

If a selected charger is defective, then

$$P(A | D) = \frac{(0.50)(0.10)}{0.071} = \frac{50}{71} \approx 0.7042.$$

The posterior probability is high because factory A supplies half of all chargers and has the highest defective rate.

9.3 Random variables and probability distributions

A **random variable** assigns a numerical value to each outcome of a random experiment. Upper-case letters such as X denote random variables; lower-case letters such as x denote

particular values.

A discrete random variable takes separate, countable values, such as the number of defective items in a sample. A continuous random variable may take any value in an interval, such as a lifetime, mass, or distance.

9.3.1 Discrete probability distributions

The **probability mass function** (pmf) of a discrete random variable is

$$f(x) = P(X = x).$$

It must satisfy

$$f(x) \geq 0, \quad \sum_x f(x) = 1. \quad (9.15)$$

Some texts call f a probability distribution function. The term pmf helps distinguish it from a continuous probability density function.

Example 9.6

A biased spinner has the distribution

x	1	2	3	4	5
$P(X = x)$	0.15	0.24	k	0.25	0.19

Since the probabilities sum to 1, $k = 0.17$. Therefore,

$$\begin{aligned} P(X \geq 4) &= 0.25 + 0.19 = 0.44, \\ P(2 < X \leq 4) &= 0.17 + 0.25 = 0.42. \end{aligned}$$

9.3.2 The cumulative distribution function

For any random variable X , the **cumulative distribution function** (cdf) is

$$F(t) = P(X \leq t). \quad (9.16)$$

For a discrete random variable,

$$F(t) = \sum_{x \leq t} P(X = x).$$

It follows that

$$\begin{aligned} P(a < X \leq b) &= F(b) - F(a), \\ P(X = x) &= F(x) - F(x^-), \end{aligned}$$

where $F(x^-)$ is the value immediately to the left of x . A discrete cdf is a non-decreasing step function.

For the spinner above,

$$F(t) = \begin{cases} 0, & t < 1, \\ 0.15, & 1 \leq t < 2, \\ 0.39, & 2 \leq t < 3, \\ 0.56, & 3 \leq t < 4, \\ 0.81, & 4 \leq t < 5, \\ 1, & t \geq 5. \end{cases}$$

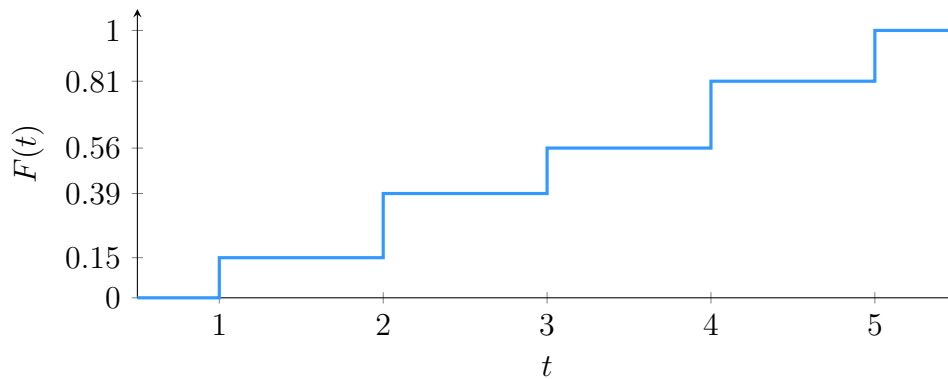


Figure 9.3: The cdf of a discrete random variable increases by $P(X = x)$ at each possible value x .

9.3.3 Expectation, variance, mode, and median

The **expectation**, or mean, of a discrete random variable is

$$\mu = E(X) = \sum_x xP(X = x). \quad (9.17)$$

More generally, if g is a function of X ,

$$E[g(X)] = \sum_x g(x)P(X = x). \quad (9.18)$$

The variance and standard deviation are

$$\text{Var}(X) = E[(X - \mu)^2] \quad (9.19)$$

$$= E(X^2) - [E(X)]^2, \quad (9.20)$$

$$\sigma = \sqrt{\text{Var}(X)}. \quad (9.21)$$

For constants a and b ,

$$E(aX + b) = aE(X) + b, \quad (9.22)$$

$$\text{Var}(aX + b) = a^2 \text{Var}(X). \quad (9.23)$$

Adding a constant shifts every value but does not change the spread; multiplying by a multiplies the standard deviation by $|a|$ and the variance by a^2 .

For the spinner distribution,

$$\begin{aligned} E(X) &= 1(0.15) + 2(0.24) + 3(0.17) + 4(0.25) + 5(0.19) \\ &= 3.09, \\ E(X^2) &= 11.39, \\ \text{Var}(X) &= 11.39 - (3.09)^2 = 1.8419. \end{aligned}$$

The **mode** is the value with greatest probability, so the mode is 4. A median is a value that divides the probability into two halves. Since $F(2) = 0.39$ and $F(3) = 0.56$, a median is 3.

Expectation is a long-run average and need not be a possible value of the random variable. For a fair die, for example, $E(X) = 3.5$, although a single throw can never score 3.5.

9.4 Continuous random variables

A continuous random variable is described by a **probability density function** (pdf) f . A valid density satisfies

$$f(x) \geq 0, \quad \int_{-\infty}^{\infty} f(x) dx = 1. \quad (9.24)$$

Probabilities are areas under the density curve:

$$P(a \leq X \leq b) = \int_a^b f(x) dx. \quad (9.25)$$

For a continuous random variable, a single point has zero probability. Hence

$$P(a < X < b) = P(a \leq X < b) = P(a < X \leq b) = P(a \leq X \leq b).$$

The value $f(x)$ is a density, not a probability. It may exceed 1 on a short interval, provided that the total area is 1.

9.4.1 Continuous cumulative distributions

The cdf of a continuous random variable is

$$F(t) = P(X \leq t) = \int_{-\infty}^t f(x) dx. \quad (9.26)$$

Where F is differentiable,

$$F'(x) = f(x).$$

Furthermore,

$$P(a < X \leq b) = F(b) - F(a).$$

Example 9.7

Let

$$f(x) = \begin{cases} \frac{6-2x}{9}, & 0 \leq x \leq 3, \\ 0, & \text{otherwise.} \end{cases}$$

The area under the density is

$$\int_0^3 \frac{6-2x}{9} dx = \left[\frac{6x-x^2}{9} \right]_0^3 = 1.$$

Therefore,

$$F(x) = \begin{cases} 0, & x < 0, \\ \frac{6x-x^2}{9}, & 0 \leq x \leq 3, \\ 1, & x > 3. \end{cases}$$

For example,

$$P(1 \leq X \leq 2) = F(2) - F(1) = \frac{3}{9} = \frac{1}{3}.$$

The median m satisfies $F(m) = 1/2$, giving

$$m = 3 - \frac{3}{\sqrt{2}} \approx 0.879.$$

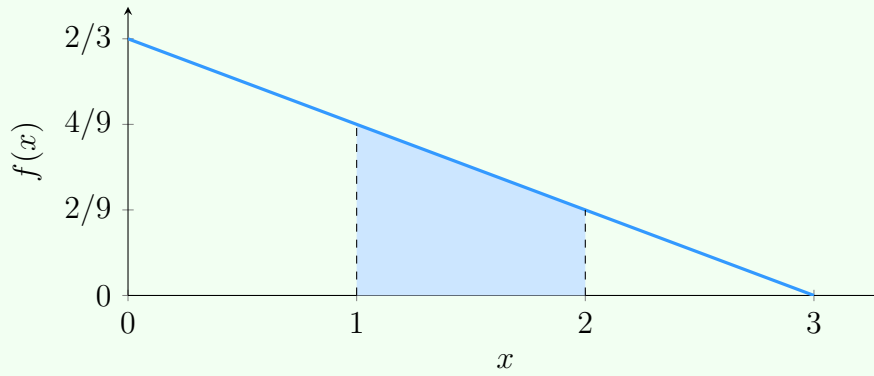


Figure 9.4: For a continuous variable, probability is represented by area under the density curve.

9.4.2 Continuous expectation and variance

For a continuous random variable,

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx, \quad (9.27)$$

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f(x) dx, \quad (9.28)$$

$$\text{Var}(X) = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx \quad (9.29)$$

$$= E(X^2) - [E(X)]^2. \quad (9.30)$$

The transformation rules for $E(aX + b)$ and $\text{Var}(aX + b)$ are the same as in the discrete case. For the triangular density above,

$$E(X) = 1, \quad E(X^2) = \frac{3}{2}, \quad \text{Var}(X) = \frac{1}{2}.$$

9.5 The binomial distribution

A binomial model applies when the following conditions hold.

1. There is a fixed number n of trials.
2. Each trial has two outcomes, labelled success and failure.
3. The trials are independent.
4. The probability of success is the same on every trial.

If X counts the number of successes, p is the probability of success, and $q = 1 - p$, then

$$X \sim B(n, p)$$

and

$$P(X = x) = \binom{n}{x} p^x q^{n-x}, \quad x = 0, 1, \dots, n. \quad (9.31)$$

The combination chooses which x trials are successes, while $p^x q^{n-x}$ is the probability of any particular arrangement of those successes and failures. The mean, variance, and standard deviation are

$$E(X) = np, \quad \text{Var}(X) = npq, \quad \sigma = \sqrt{npq}. \quad (9.32)$$

Example 9.8

Ten per cent of a manufacturer's phones are defective. Five phones are selected independently from a very large production line. If X is the number of defective phones, then $X \sim B(5, 0.1)$. Hence

$$\begin{aligned} P(X = 1) &= \binom{5}{1} (0.1)(0.9)^4 = 0.32805, \\ P(X \geq 1) &= 1 - P(X = 0) = 1 - (0.9)^5 = 0.40951, \\ P(X > 1) &= 1 - P(X \leq 1) = 0.08146. \end{aligned}$$

Probability statements should be translated carefully:

Phrase	Symbol	Phrase	Symbol
exactly r	$X = r$	at most r	$X \leq r$
at least r	$X \geq r$	fewer than r	$X < r$
more than r	$X > r$	from a to b inclusive	$a \leq X \leq b$

The cumulative binomial tables at the end of this book give $P(X \leq r)$. Other probabilities can be obtained by subtraction or complementation. When $p > 0.5$, it is sometimes easier to count failures. If $Y = n - X$, then

$$Y \sim B(n, 1 - p).$$

For example, $P(X \leq 3) = P(Y \geq n - 3)$.

The binomial model describes many two-outcome settings, such as the number of patients experiencing a side effect, fraudulent transactions in a batch, or items returned to a retailer. It is inappropriate if the trials are not independent or if the success probability changes substantially from trial to trial.

9.6 The Poisson distribution

The Poisson distribution models the number of random occurrences in a fixed interval of time, distance, area, or volume. The occurrences should be independent, and the average rate should remain constant over the interval. Examples include arrivals at a service counter, defects along a roll of material, or accidents at a junction.

If X has mean number of occurrences $\lambda > 0$, then

$$X \sim \text{Po}(\lambda)$$

and

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots \quad (9.33)$$

Unlike the binomial distribution, there is no finite upper limit for X . Its mean and variance are equal:

$$E(X) = \lambda, \quad \text{Var}(X) = \lambda, \quad \sigma = \sqrt{\lambda}. \quad (9.34)$$

The parameter must be scaled to match the interval. If the mean is 3 calls per minute, the mean is 15 calls per five minutes and 180 calls per hour.

Example 9.9

A car breaks down an average of three times per month. With $X \sim \text{Po}(3)$,

$$\begin{aligned} P(X = 2) &= e^{-3} \frac{3^2}{2!} \approx 0.2240, \\ P(X \leq 1) &= P(X = 0) + P(X = 1) \\ &= e^{-3}(1 + 3) \approx 0.1991. \end{aligned}$$

9.6.1 Poisson approximation to the binomial distribution

If n is large, p is small, and $\lambda = np$ is of moderate size, then

$$B(n, p) \approx \text{Po}(np).$$

A common practical guideline is that the approximation is reasonable when $n > 20$ and $np < 5$. If p is close to 1, apply the approximation to the number of failures, whose probability is $q = 1 - p$.

Example 9.10

Each of 100 letters has probability 0.03 of being delivered to the wrong house. The exact model is $X \sim B(100, 0.03)$, while the Poisson approximation has $\lambda = 3$. Thus

$$\begin{aligned}
 P(X \geq 4) &\approx 1 - P(\text{Po}(3) \leq 3) \\
 &= 1 - e^{-3} \left(1 + 3 + \frac{3^2}{2!} + \frac{3^3}{3!} \right) \\
 &\approx 0.3528.
 \end{aligned}$$

9.7 The normal distribution

A continuous random variable X follows a normal distribution with mean μ and variance σ^2 if

$$X \sim N(\mu, \sigma^2)$$

and its density is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{x - \mu}{\sigma} \right)^2 \right], \quad -\infty < x < \infty. \quad (9.35)$$

The normal curve is unimodal and symmetric about μ . Changing μ translates the curve, while increasing σ makes it wider and lower. The total area is 1, and symmetry gives $P(X < \mu) = P(X > \mu) = 1/2$.

9.7.1 The empirical rule

For a normal distribution, approximately 68% of values lie within one standard deviation of the mean, 95% within two, and 99.7% within three.

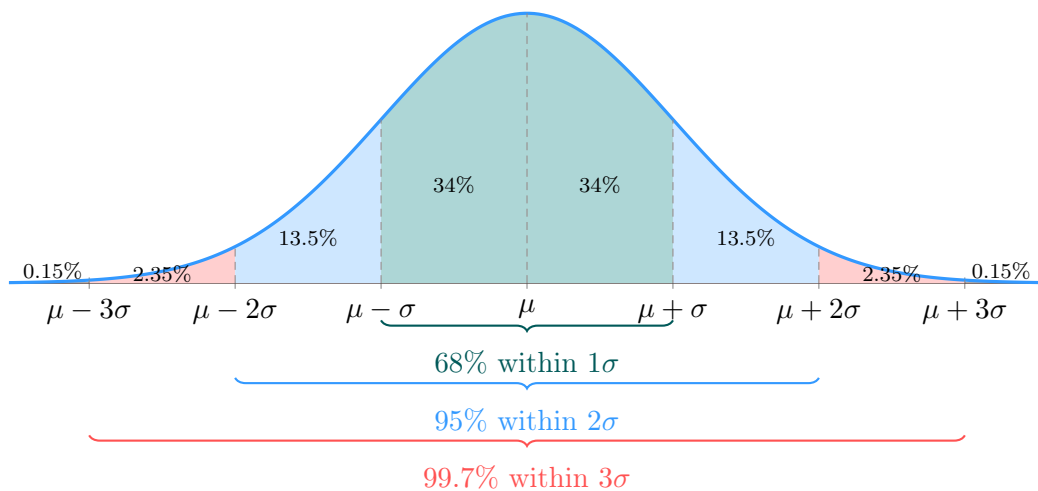


Figure 9.5: The empirical 68–95–99.7 rule, showing both the individual standard-deviation bands and the cumulative central areas.

9.7.2 Standardisation and normal tables

The **standard normal variable** has distribution $Z \sim N(0, 1)$. Any normal random variable can be standardised using

$$Z = \frac{X - \mu}{\sigma}. \quad (9.36)$$

If $\Phi(z) = P(Z \leq z)$, symmetry gives

$$\Phi(-z) = 1 - \Phi(z).$$

Therefore,

$$\begin{aligned} P(X < a) &= \Phi\left(\frac{a - \mu}{\sigma}\right), \\ P(X > a) &= 1 - \Phi\left(\frac{a - \mu}{\sigma}\right), \\ P(a < X < b) &= \Phi\left(\frac{b - \mu}{\sigma}\right) - \Phi\left(\frac{a - \mu}{\sigma}\right). \end{aligned}$$

Example 9.11

Examination scores are normally distributed with mean 60 and standard deviation 10. Then

$$\begin{aligned} P(X < 49) &= P\left(Z < \frac{49 - 60}{10}\right) \\ &= P(Z < -1.10) \approx 0.1357. \end{aligned}$$

Also,

$$\begin{aligned} P(68.2 < X < 88.7) &= P(0.82 < Z < 2.87) \\ &= \Phi(2.87) - \Phi(0.82) \\ &\approx 0.2040. \end{aligned}$$

9.7.3 Finding an unknown value or parameter

Normal tables can also be used in reverse. If $P(X < a) = p$, first find the standard normal quantile z satisfying $\Phi(z) = p$, and then use

$$a = \mu + z\sigma.$$

Example 9.12

Tyre lifetimes are normal with mean 7500 km, and 5% last less than 6000 km. Since the lower 5% standard normal quantile is approximately -1.645 ,

$$\frac{6000 - 7500}{\sigma} = -1.645,$$

so $\sigma \approx 912$ km.

9.7.4 Normal approximation to the binomial distribution

If $X \sim B(n, p)$ and both np and nq exceed 5, then

$$X \approx Y, \quad Y \sim N(np, npq). \quad (9.37)$$

The binomial variable is discrete but the normal variable is continuous. A **continuity correction** moves each integer boundary by 0.5 so that a whole binomial bar is represented by the corresponding area under the normal curve.

Binomial probability	Corrected normal probability
$P(X = r)$	$P(r - 0.5 < Y < r + 0.5)$
$P(X \leq r)$	$P(Y < r + 0.5)$
$P(X < r)$	$P(Y < r - 0.5)$
$P(X \geq r)$	$P(Y > r - 0.5)$
$P(X > r)$	$P(Y > r + 0.5)$
$P(a \leq X \leq b)$	$P(a - 0.5 < Y < b + 0.5)$

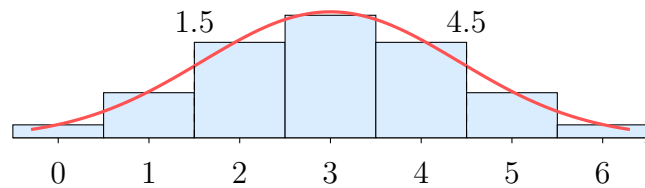


Figure 9.6: To approximate $P(2 \leq X \leq 4)$, use the continuous interval $1.5 < Y < 4.5$.

Example 9.13

If $X \sim B(40, 0.6)$, then $\mu = 24$ and $\sigma = \sqrt{9.6} \approx 3.098$. Hence

$$\begin{aligned} P(X \geq 20) &\approx P(Y > 19.5) \\ &= P\left(Z > \frac{19.5 - 24}{3.098}\right) \\ &= P(Z > -1.453) \approx 0.9268. \end{aligned}$$

9.8 The continuous uniform distribution

If every interval of equal length within $[a, b]$ is equally likely, then X has a continuous uniform distribution, written $X \sim U(a, b)$, with

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b, \\ 0, & \text{otherwise.} \end{cases} \quad (9.38)$$

Consequently,

$$P(c < X < d) = \frac{d-c}{b-a}, \quad E(X) = \frac{a+b}{2}, \quad \text{Var}(X) = \frac{(b-a)^2}{12}. \quad (9.39)$$

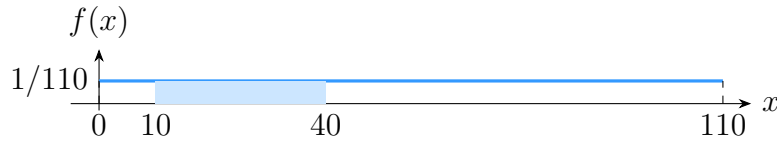


Figure 9.7: For a uniform density, probability is proportional to interval length.

Example 9.14

The time between customer arrivals is uniform from 0 to 110 seconds. Then

$$\begin{aligned} P(X < 25) &= \frac{25}{110} \approx 0.2273, \\ P(10 < X < 40) &= \frac{30}{110} \approx 0.2727, \\ P(X > 55) &= \frac{55}{110} = 0.5. \end{aligned}$$

The mean is 55 seconds and the standard deviation is $110/\sqrt{12} \approx 31.75$ seconds.

9.9 The exponential distribution

The exponential distribution models the waiting time until the next event in a Poisson process. If events occur at rate λ per unit time, then the waiting time T has density

$$f(t) = \begin{cases} \lambda e^{-\lambda t}, & t \geq 0, \\ 0, & t < 0, \end{cases} \quad (9.40)$$

and cdf

$$F(t) = P(T \leq t) = 1 - e^{-\lambda t}, \quad t \geq 0. \quad (9.41)$$

Thus

$$P(T > t) = e^{-\lambda t}, \quad E(T) = \frac{1}{\lambda}, \quad \text{Var}(T) = \frac{1}{\lambda^2}. \quad (9.42)$$

The exponential distribution is **memoryless**:

$$P(T > s + t \mid T > s) = P(T > t).$$

The units must be consistent. If the rate is per hour and the waiting time is given in minutes, convert one of them before substituting.

Example 9.15

Customers arrive at an ATM at a rate of 20 per hour. The probability that the next customer arrives within three minutes, or 0.05 hour, is

$$P(T \leq 0.05) = 1 - e^{-20(0.05)} = 1 - e^{-1} \approx 0.6321.$$

Within twelve minutes, or 0.2 hour, the probability is $1 - e^{-4} \approx 0.9817$.

If an exponentially distributed lifetime has mean 16 months, then $\lambda = 1/16$. The 80th percentile t satisfies

$$1 - e^{-t/16} = 0.8,$$

so $t = -16 \ln(0.2) \approx 25.75$ months.

9.10 Using R for probability and statistics

R is a programming language for numerical and statistical work. It may be used directly, while RStudio provides an optional graphical interface. An object may be created with either `<-` or `=`; R is case sensitive, and text following `#` is a comment.

```
n <- 15
a <- 12 + 14
(20 - 4) / 2
7^2
exp(2)
log(10)      # natural logarithm
sqrt(100)
```

Vectors are created with `c`, a colon, `seq`, or `rep`. Arithmetic and mathematical functions act element by element.

```
x <- c(1, 7, 32, 16)
y <- 1:10
z <- seq(1, 5, by = 0.5)
w <- rep(1:3, 4)
sqrt(c(100, 225, 400))
```

```
z[3]
z[z > 2.8]
length(z)
```

Useful summaries include `sum`, `mean`, `median`, `var`, `sd`, `sort`, and `summary`. Setting a random-number seed makes a simulation reproducible.

```
set.seed(13)
x <- rnorm(10, mean = 10, sd = 3)
mean(x)
sd(x)
summary(x)
```

For standard distributions, R uses four related prefixes:

Prefix	Purpose	Example
d	pmf or density	<code>dbinom(3, 5, 0.1)</code>
p	cumulative probability $P(X \leq x)$	<code>pnorm(1.96)</code>
q	quantile corresponding to a probability	<code>qnorm(0.975)</code>
r	simulated random values	<code>rpois(8, 3)</code>

The distribution names include `binom`, `pois`, `norm`, `unif`, and `exp`. For example, the phone problem from the binomial section can be evaluated by

```
dbinom(3, size = 5, prob = 0.1)      # P(X = 3)
1 - pbinom(1, size = 5, prob = 0.1)  # P(X > 1)
pbinom(1, size = 5, prob = 0.1)      # P(X < 2)
1 - pbinom(0, size = 5, prob = 0.1)  # P(X >= 1)
pbinom(2, size = 5, prob = 0.1)      # P(X <= 2)
```

The function `choose(n,r)` evaluates $\binom{n}{r}$, while `factorial(n)` evaluates $n!$. R should support the mathematical reasoning rather than replace the identification of an appropriate model or the interpretation of the result.

9.11 Populations, samples, and sampling methods

A **population** is the entire collection of individuals or items of interest. A **sample** is a subset selected from that population. A numerical summary of a population is a **parameter**; a numerical summary calculated from a sample is a **statistic**.

Measure	Population parameter	Sample statistic
Mean	μ	$\bar{x}$
Variance	σ^2	s^2
Standard deviation	σ	s
Proportion	p	$\hat{p}$

Sampling is usually less costly, less time-consuming, and more practical than measuring every member of a population. Its usefulness depends on whether the sample represents the target population.

9.11.1 Probability sampling

In a probability sample, selection probabilities are known. This permits the use of probability theory to quantify sampling uncertainty.

1. In a **simple random sample**, every sample of a given size has the same probability of selection.
2. In a **systematic sample**, a random starting point is chosen and then every k th item is selected. A hidden periodic pattern in the population list may cause bias.
3. In a **stratified sample**, the population is divided into relatively homogeneous strata and a random sample is taken from each stratum. This ensures representation of important groups.
4. In a **cluster sample**, naturally occurring groups are randomly selected, and all or some members of the selected clusters are studied. This can reduce travel or administrative cost.

9.11.2 Non-probability sampling

In a non-probability sample, the probability of selection is unknown, so the usual inferential guarantees do not automatically apply.

1. A **convenience sample** uses individuals who are easy to reach.
2. A **purposive** or **judgement sample** deliberately selects individuals with specified expertise or characteristics.
3. A **quota sample** fills predetermined numbers from selected categories without random selection within the categories.
4. A **snowball sample** asks participants to recruit other participants and is useful for difficult-to-reach populations.

Large sample size does not repair a systematically biased selection method. Coverage errors, non-response, leading questions, and voluntary-response bias must be considered separately from random sampling error.

9.12 Sampling distributions

Suppose a random sample $X_1, X_2, \dots, X_n$ is taken from a population with mean μ and variance σ^2 . The sample mean is

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}.$$

If every possible sample of size n were selected and its mean calculated, the distribution of those means would be the **sampling distribution of the sample mean**. It satisfies

$$E(\bar{X}) = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}, \quad \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}. \quad (9.43)$$

The quantity $\sigma/\sqrt{n}$ is the **standard error** of the mean. It decreases as n increases, so larger random samples tend to give more precise estimates of μ .

If the population is normal, then

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

for every sample size n .

9.12.1 The central limit theorem

For independent observations from a population with finite mean and variance, the distribution of the standardised sample mean

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \quad (9.44)$$

approaches $N(0, 1)$ as n increases, even when the population itself is not normal. The guideline $n \geq 30$ is often used, but the required size depends on the amount of skewness and the presence of outliers. A nearly symmetric population may require fewer observations; a strongly skewed or heavy-tailed population may require more.

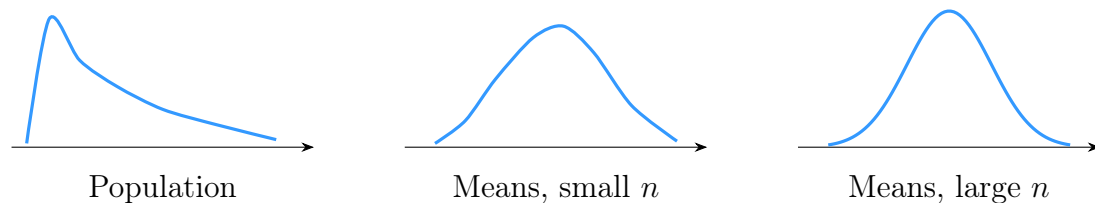


Figure 9.8: Sample means become more nearly normal and less variable as the sample size increases.

Example 9.16

A population has mean 50 and standard deviation 12. For random samples of size 36,

$$E(\bar{X}) = 50, \quad \sigma_{\bar{X}} = \frac{12}{\sqrt{36}} = 2.$$

Assuming the sampling distribution is normal or the central limit theorem is applicable,

$$\begin{aligned} P(48 < \bar{X} < 53) &= P(-1 < Z < 1.5) \\ &\approx 0.7745. \end{aligned}$$

9.13 Estimation of a population mean

A **point estimate** is a single statistic used to estimate a parameter; for example, $\bar{x}$ estimates μ . An **interval estimate** gives a range of plausible parameter values. A confidence interval has the general form

$$\text{point estimate} \pm \text{margin of error.} \quad (9.45)$$

A $(1 - \alpha)100\%$ confidence level corresponds to central area $1 - \alpha$ and tail area $\alpha/2$ on each side. Common standard normal critical values are

Confidence level	α	$z_{\alpha/2}$
90%	0.10	1.645
95%	0.05	1.960
99%	0.01	2.576

9.13.1 Known population standard deviation

If σ is known and the population is normal, or the sample is large enough for the central limit theorem, then a confidence interval for μ is

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}. \quad (9.46)$$

Example 9.17

A random sample of 11 circuits from a normal population has mean resistance 2.20 ohms. Past testing gives $\sigma = 0.35$ ohm. A 95% confidence interval is

$$\begin{aligned} 2.20 \pm 1.96 \left(\frac{0.35}{\sqrt{11}} \right) &= 2.20 \pm 0.207 \\ &= (1.993, 2.407). \end{aligned}$$

For 99% confidence, the larger critical value produces the wider interval

$$2.20 \pm 2.576 \left(\frac{0.35}{\sqrt{11}} \right) = (1.928, 2.472).$$

For a desired margin of error E , the sample size must satisfy

$$n \geq \left(\frac{z_{\alpha/2} \sigma}{E} \right)^2, \quad (9.47)$$

and the result is rounded up to the next integer.

9.13.2 Unknown population standard deviation

When σ is unknown, replace it with the sample standard deviation

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}$$

and use Student's t distribution:

$$\bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}. \quad (9.48)$$

The t distribution is symmetric and bell-shaped but has heavier tails than the standard normal distribution. Its shape depends on the degrees of freedom $n - 1$ and approaches the standard normal curve as n increases.

For a small sample, the population should be approximately normal with no severe outliers. For a large sample, the procedure is more robust because of the central limit theorem.

Example 9.18

A random sample has $n = 25$, $\bar{x} = 50$, and $s = 8$. For 95% confidence and 24 degrees of freedom, $t_{0.025, 24} \approx 2.064$. Therefore,

$$\begin{aligned} 50 \pm 2.064 \left(\frac{8}{\sqrt{25}} \right) &= 50 \pm 3.302 \\ &= (46.698, 53.302). \end{aligned}$$

9.13.3 Interpreting a confidence interval

A 95% confidence procedure captures the true parameter in about 95% of repeated random samples. After a particular interval has been calculated, the unknown fixed parameter either lies inside it or it does not. It is therefore better to say, “We are 95% confident that the population mean lies between the two limits,” rather than assigning a 95% probability to the fixed parameter.

Increasing the confidence level widens an interval. Increasing the sample size narrows it, because the standard error is proportional to $1/\sqrt{n}$.

9.14 Hypothesis testing about a population mean

A statistical hypothesis is a claim about a population parameter. The **null hypothesis** H_0 is the claim tested, while the **alternative hypothesis** H_1 or H_a describes the departure of interest. Equality belongs in H_0 .

Question	Null hypothesis	Alternative hypothesis
Has the mean changed?	$H_0 : \mu = \mu_0$	$H_1 : \mu \neq \mu_0$
Has the mean increased?	$H_0 : \mu = \mu_0$	$H_1 : \mu > \mu_0$
Has the mean decreased?	$H_0 : \mu = \mu_0$	$H_1 : \mu < \mu_0$

The first is a two-tailed test; the others are right- and left-tailed tests. The **significance level** α is the probability of rejecting a true null hypothesis. Common values are 0.10, 0.05, and 0.01.

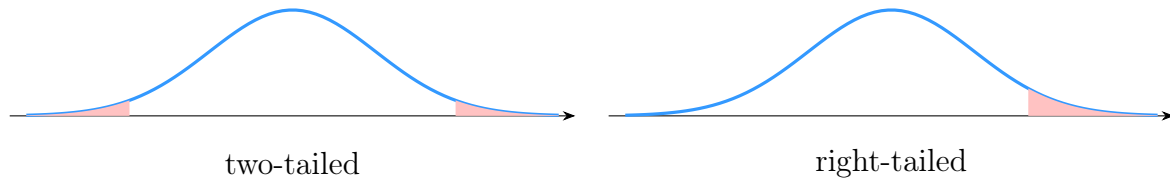


Figure 9.9: At the 5% level, a two-tailed z test uses critical values ± 1.96 , while a right-tailed test uses 1.645.

9.14.1 Test statistics

If the population standard deviation is known, use

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}. \quad (9.49)$$

If σ is unknown, use

$$T = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}, \quad T \sim t_{n-1} \quad \text{under } H_0. \quad (9.50)$$

For large samples, a standard normal calculation with s in place of σ is a close approximation, but the t procedure remains valid and is the more systematic choice.

9.14.2 A hypothesis-testing procedure

1. State H_0 and H_1 in terms of the population parameter.
2. Choose the significance level α before examining the result.
3. Check the sampling assumptions and calculate the test statistic.
4. Use a critical region or calculate the p -value.
5. Reject H_0 if the statistic lies in the critical region, or if $p \leq \alpha$. Otherwise, do not reject H_0 .
6. State the conclusion in the context of the question.

The p -value is the probability, assuming H_0 is true, of obtaining a test statistic at least as extreme as the observed one in the direction specified by H_1 . A small p -value indicates that the observed data are difficult to reconcile with H_0 .

Do not write that H_0 has been proved true. A non-significant result means only that the sample does not provide sufficient evidence against H_0 at the chosen significance level.

9.14.3 Worked one-sample tests

Example 9.19: known standard deviation

A sample of 100 light bulbs has mean lifetime 470 hours. The population standard deviation is 25 hours. Test

$$H_0 : \mu = 480 \quad \text{against} \quad H_1 : \mu < 480$$

at $\alpha = 0.05$. The test statistic is

$$z = \frac{470 - 480}{25/\sqrt{100}} = -4.00.$$

The lower-tail critical value is -1.645 . Since $-4.00 < -1.645$, reject H_0 . There is strong evidence at the 5% level that the population mean lifetime is below 480 hours.

Example 9.20: unknown standard deviation

The mean hotel-room price is claimed to be RM 168 per night. A random sample of 25 hotels has $\bar{x} = 172.50$ and $s = 15.40$. Test for a difference at $\alpha = 0.05$:

$$H_0 : \mu = 168, \quad H_1 : \mu \neq 168.$$

The statistic is

$$t = \frac{172.50 - 168}{15.40/\sqrt{25}} \approx 1.461, \quad \text{d.f.} = 24.$$

The two-tailed critical values are approximately ± 2.064 . Since 1.461 is not in the rejection region, do not reject H_0 . The sample does not give sufficient evidence at the 5% level that the mean differs from RM 168.

9.14.4 Critical-value, confidence-interval, and p-value approaches

The three approaches lead to the same decision when applied consistently.

- (i) The **critical-value approach** compares the test statistic with the boundary of the rejection region.
- (ii) The **p-value approach** rejects H_0 when $p \leq \alpha$.
- (iii) For a two-tailed test at significance level α , the **confidence-interval approach** rejects $H_0 : \mu = \mu_0$ precisely when the $(1 - \alpha)100\%$ confidence interval excludes μ_0 .

The confidence-interval equivalence does not use the same central interval for a one-tailed test without adjustment. For one-tailed claims, a one-sided confidence bound or the direct p-value approach is clearer.

9.14.5 Type I and Type II errors

A **Type I error** occurs when a true H_0 is rejected; its probability is α . A **Type II error** occurs when a false H_0 is not rejected; its probability is denoted by β . The probability $1 - \beta$ of rejecting a false null hypothesis is the **power** of the test. For a fixed sample size, reducing α generally increases the risk of a Type II error. Increasing the sample size can reduce both risks for a specified meaningful departure from H_0 .

Chapter summary and common pitfalls

Key ideas. Probability models quantify random variation, while statistical inference uses sample information to estimate or test population claims. Every distribution and procedure has assumptions and parameters.

Common pitfalls. Distinguish independent from mutually exclusive events, state distribution parameters, use the correct tail in a test, and interpret a confidence interval or p -value in context rather than as a probability that a fixed hypothesis is true.

9.15 Practice and exercises

Unless otherwise stated, give exact answers where appropriate and numerical probabilities to four decimal places.

Counting and elementary probability

1. A meal consists of one drink, one main course, and one dessert. A restaurant offers 6 drinks, 9 main courses, and 5 desserts. How many different meals are possible?
2. A four-digit code is formed from the digits $0, 1, \dots, 9$.
 - (a) How many codes are possible if repetition is allowed?
 - (b) How many are possible if no digit may be repeated?
 - (c) How many have no repeated digit and begin with a non-zero digit?
3. Find the number of distinct arrangements of each word:
 - (a) PROBABILITY;
 - (b) MATRICULATION.
4. Eleven different compact discs consist of six pop, three jazz, and two classical discs. In how many arrangements on a shelf are the three jazz discs together?
5. Find the number of ways in which
 - (a) six people can sit around a round table;
 - (b) ten differently coloured beads can form a necklace if reflected arrangements are considered identical.
6. A committee of five is selected from six boys and four girls. Find the probability that it contains
 - (a) three boys and two girls;

- (b) more boys than girls.
7. A number is selected uniformly from the first 250 positive integers. Find the probability that it is divisible by 2 or by 7.
8. Two fair dice are rolled. Find the probability that
- (a) their sum is 10;
 - (b) their sum is divisible by 6;
 - (c) either of these events occurs.
9. In a class of 30, every student studies Geography or Science or both. Twenty study Geography and 21 study Science. Find the probability that a randomly selected student
- (a) studies both subjects;
 - (b) studies only Geography;
 - (c) studies Geography given that the student studies Science.
10. A bag contains five red and seven blue marbles. Two are drawn without replacement. Find the probability of
- (a) red followed by blue;
 - (b) two marbles of the same colour;
 - (c) at least one red marble.
11. A person chooses route A with probability $1/4$ and route B otherwise. The probabilities of being late on the two routes are $2/3$ and $1/3$, respectively. Find
- (a) the probability of being late;
 - (b) the probability that route B was used, given that the person was late.
12. Seventy per cent of seeds are treated. A treated seed germinates with probability 0.8, while an untreated seed germinates with probability 0.3. Find the probability that
- (a) a randomly selected seed germinates;
 - (b) a germinated seed was treated.

Random variables and distributions

13. A random variable has distribution

x	0	1	2	3	4
$P(X = x)$	0.26	k	$3k$	0.05	0.09

Find k , $E(X)$, and $\text{Var}(X)$. If $Y = 2X - 7$, find $E(Y)$ and $\text{Var}(Y)$.

14. A fair die is rolled. If a 6 appears, Azie wins RM 6; otherwise she loses RM 1. Find her expected gain per game. Is the game favourable to her in the long run?

15. A continuous random variable has density

$$f(x) = \begin{cases} \frac{x(6-x)}{36}, & 0 \leq x \leq 6, \\ 0, & \text{otherwise.} \end{cases}$$

Verify that this is a density and find $P(2 \leq X \leq 3.5)$, $P(X < 3)$, and $P(X > 5)$.

16. A continuous random variable has density

$$f(t) = \begin{cases} k/t^4, & t \geq 1, \\ 0, & t < 1. \end{cases}$$

Find k , the cdf, the mean, and $P(2.5 \leq T \leq 3.5)$.

17. If $X \sim B(20, 0.2)$, find

- (a) $P(X = 4)$;
- (b) $P(X \leq 3)$;
- (c) $P(X \geq 1)$;
- (d) the mean and standard deviation.

18. Four per cent of eggs in a shipment arrive damaged. Let X be the number damaged among 500 eggs. State a suitable model and find its mean and standard deviation.

19. Calls arrive at a switchboard at an average rate of 180 per hour. Find the probability that a randomly selected minute has

- (a) fewer than six calls;
- (b) more than two calls;
- (c) more than two calls, given that fewer than six arrive.

20. Accidents at a junction occur at an average rate of four per week. Find the probability of

- (a) at most one accident in one week;
- (b) exactly three in a fortnight;
- (c) no accident in three weeks.

21. Use a Poisson approximation to estimate the probability that at least four of 100 independently delivered letters go to the wrong house when the error probability is 0.03.
22. If $Z \sim N(0, 1)$, find
- (a) $P(Z > 2.43)$;
 - (b) $P(Z < -2.35)$;
 - (c) $P(-1 < Z < 2)$;
 - (d) the value a such that $P(Z < a) = 0.8485$.
23. If $X \sim N(80, 100)$, find
- (a) $P(X \leq 100)$;
 - (b) $P(65 < X \leq 80)$;
 - (c) $P(|X - 80| \leq 10)$.
24. Examination marks are normally distributed with mean μ and standard deviation σ . Thirty per cent are below 45, while fifteen per cent exceed 65. Find μ and σ .
25. If $X \sim B(80, 0.3)$, use a normal approximation with a continuity correction to estimate
- (a) $P(X = 14)$;
 - (b) $P(X \geq 15)$;
 - (c) $P(11 \leq X \leq 16)$.
26. A waiting time is uniformly distributed between 0 and 110 seconds. Find the probability that it is
- (a) less than 25 seconds;
 - (b) between 10 and 40 seconds;
 - (c) more than 55 seconds.
- Also find its mean and standard deviation.
27. A checkout completes service at an exponential rate of 10 customers per hour. Find the probability that service takes
- (a) fewer than four minutes;
 - (b) more than eleven minutes;
 - (c) between four and nine minutes.

Sampling, estimation, and testing

28. For each situation, identify the population and the sample.
- (a) Thirty points on one car are selected to study its paint thickness.
 - (b) Seven classrooms are randomly selected for air-quality tests at an elementary school.
 - (c) One hundred seniors are surveyed about a school's cafeteria.
29. Explain one advantage and one possible weakness of simple random, systematic, stratified, cluster, convenience, and snowball sampling.
30. A population has mean 72 and standard deviation 18. For samples of size 81, find the mean and standard error of $\bar{X}$, and calculate $P(69 < \bar{X} < 75)$.
31. A sample of 200 tea boxes has mean mass 101.0 g and standard deviation 2.78 g. Construct a 99% confidence interval for the population mean mass.
32. A random sample of 12 students has mean GPA 2.71 and sample standard deviation 0.51. Assuming a normal population, construct a 90% confidence interval for the population mean GPA.
33. A modified steel bar has known population standard deviation 100 lb. A sample of 25 bars has mean yield point 8439 lb. Construct a 90% confidence interval for the true mean yield point.
34. A credit manager claims that the mean unpaid balance exceeds RM 400. A random sample of 172 balances has mean RM 407 and standard deviation RM 38. Test the claim at the 5% significance level.
35. The mean height of female college basketball players is claimed to be 69.5 inches. A random sample of 25 players has mean 70.2 inches and sample variance 4.41 square inches. Assuming normality, test at the 1% level whether the population mean differs from 69.5 inches.
36. A new machine is claimed to raise production above 250 units per hour. Ten randomly selected hours have mean production 256 and sample standard deviation 6. Assuming normality, test the claim at the 5% level.
37. A simple random sample of 106 body temperatures has mean 98.20°F. The population standard deviation is known to be 0.62°F. At the 5% level, test the claim that the population mean is 98.6°F using
- (a) the critical-value method;

- (b) a confidence interval;
- (c) the p-value method.

Further counting and probability

38. Find the numbers of distinct arrangements of
- (a) MISSISSIPPI;
 - (b) STATISTICS.
39. Twelve students consist of five science and seven arts students. Find the number of ways to
- (a) select a president, vice-president, and secretary;
 - (b) select a committee of four;
 - (c) select a committee of four containing at least two science students.
40. Four different mathematics books, three different physics books, and two different chemistry books are placed on a shelf. Find the number of arrangements when
- (a) books of each subject remain together;
 - (b) only the mathematics books remain together;
 - (c) two specified mathematics books are not adjacent.
41. Eight friends sit around a circular table. Find the number of arrangements when
- (a) there is no restriction;
 - (b) Aina and Ben sit together;
 - (c) Aina and Ben do not sit together.
42. From a standard 52-card deck, find the number of five-card hands containing
- (a) exactly two hearts;
 - (b) at least one ace;
 - (c) a full house.
43. Two fair dice are rolled. Given that the sum is at least 9, find the probability that
- (a) the sum is 10;
 - (b) at least one die shows 6;
 - (c) the dice show the same number.

44. A box contains six red, five blue, and four green counters. Three are selected without replacement. Find the probability of
- (a) one counter of each colour;
 - (b) no green counter;
 - (c) at least two red counters.
45. Events A and B satisfy $P(A) = 0.55$, $P(B) = 0.40$, and $P(A \cap B) = 0.25$. Find $P(A \cup B)$, $P(A | B)$, and $P(B | A)$, and determine whether the events are independent.
46. The records of 120 commuters are shown below.

	Late	On time	Total
Car	18	42	60
Bus	12	48	60
Total	30	90	120

Find the probabilities of bus and late, late given car, and car given on time. Decide whether travel method and punctuality are independent.

47. Components come from factories A, B, C in proportions 0.60, 0.30, 0.10, with defect rates 0.02, 0.04, 0.07, respectively. Find the probability of a defect and, given a defect, the probability that it came from each factory.

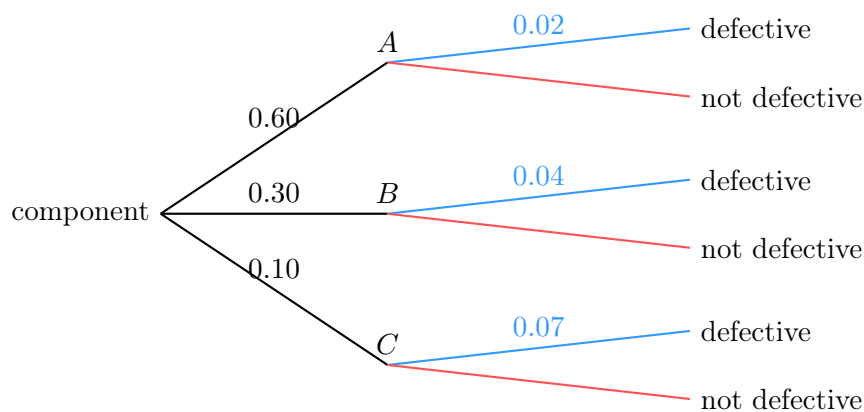


Figure 9.10: Supplier and defect branches for Question 47.

48. In a group of 100 students, 52 study Biology, 47 Chemistry, and 40 Physics. The pairwise intersections contain 20, 18, and 15 students, and 8 study all three. Find the numbers studying at least one, exactly two, and none of the subjects.
49. A point is selected uniformly from the square $0 \leq x, y \leq 10$. Let R be the region $x + y \leq 6$. Find $P((X, Y) \in R)$ and $P((X, Y) \in R | X \leq 4)$.

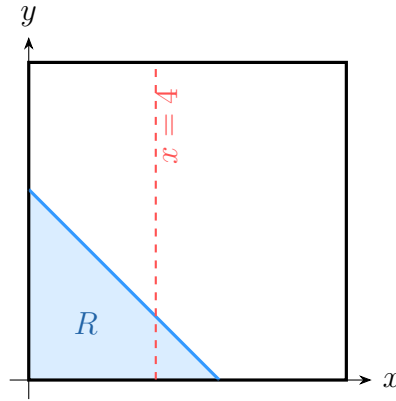


Figure 9.11: The geometric-probability region for Question 49.

50. An eight-character code uses 26 uppercase letters and ten digits, with repetition allowed. Find the number of codes that
- (a) begin with a letter;
 - (b) begin with a letter and contain exactly two digits;
 - (c) begin with a letter and contain at least one digit.

Further random variables and distributions

51. A discrete variable has

x	-1	0	1	2
$P(X = x)$	0.10	0.30	0.40	0.20

Construct its cdf and find $P(-1 < X \leq 1)$, a median, and the mode.

52. Three fair coins are tossed and X is the number of heads. List the distribution of X and find its mean and variance.
53. A game pays RM 12 with probability 0.10, RM 4 with probability 0.25, and nothing otherwise. Find the expected payout, fair entry fee, and expected organiser profit for a RM 3 fee.
54. A continuous variable has density $f(x) = k(1 - x^2)$ for $0 \leq x \leq 1$. Find k , the cdf, $P(0.2 < X < 0.8)$, and $E(X)$.
55. A continuous variable has density $f(x) = kx$ for $0 \leq x \leq 2$. Find k , the cdf, median, mean, and variance.
56. A binomial variable has mean 12 and variance 8.4. Find n, p .
57. If $X \sim B(12, 0.3)$, find $P(X = 4)$, $P(X \leq 2)$, $P(3 \leq X \leq 6)$, and $P(X \geq 1)$.

58. Devices fail independently with probability 0.08. Find the least number tested so that observing at least one failure has probability at least 0.95.
59. Messages arrive at 2.4 per ten minutes. Find the probabilities of no message in five minutes, exactly two in five minutes, and at least five in thirty minutes.
60. If $X \sim \text{Po}(3)$, find $P(X = 2 \mid X \leq 3)$ and $P(X \geq 4 \mid X \geq 2)$.
61. Use a Poisson approximation for $X \sim \text{B}(200, 0.015)$ to estimate $P(X = 0)$, $P(X \leq 2)$, and $P(X \geq 5)$.
62. For $Z \sim N(0, 1)$, find z with $P(Z < z) = 0.975$, $a > 0$ with $P(|Z| < a) = 0.90$, and b with $P(Z > b) = 0.01$.
63. A normal variable satisfies $P(X < 40) = 0.1587$ and $P(X < 70) = 0.9772$. Find its mean and standard deviation.
64. Packet masses are $N(500, 4^2)$ grams. Find the probability of a mass from 492 to 508 g, the expected number outside this interval in 1000 packets, and the fifth percentile.
65. For $X \sim \text{B}(100, 0.55)$, use a continuity-corrected normal approximation to estimate $P(50 \leq X \leq 60)$.

Further sampling and inference

66. Identify the sampling method: every fiftieth register entry after a random start; samples from every programme; all students in four randomly selected tutorial groups; volunteers responding online.
67. Identify likely bias and suggest an improvement when cafeteria satisfaction is sampled only inside the cafeteria, internet access only through an online form, and sleep only among students at an 8 a.m. lecture.
68. A population has mean 120 and standard deviation 30. For $n = 36$, find the mean and standard error of $\bar{X}$ and $P(115 < \bar{X} < 128)$.
69. For population standard deviation 20, compare samples of sizes 25 and 100 by finding their standard errors and the probabilities that $\bar{X}$ is within 3 units of the population mean.
70. A skewed population has mean 50 and standard deviation 12. Use the central limit theorem with $n = 64$ to estimate $P(\bar{X} > 53)$.

71. A sample of 64 has mean 42.5 and known population standard deviation 8. Construct a 95% confidence interval for the mean.
72. The measurements 12.1, 11.8, 12.4, 12.0, 11.9, 12.3, 12.2, 11.7 form a normal random sample. Construct a 95% confidence interval for the mean.
73. Find the least sample size for margin of error at most 3 at 95% confidence when $\sigma = 15$.
74. Given $n = 100$, $\bar{x} = 75$, and $\sigma = 20$, construct 90%, 95%, and 99% confidence intervals and compare widths.
75. State H_0, H_1 for claims that a mean is below 50, differs from 120, and exceeds 8.
76. For $H_0 : \mu = 500$ mL against $H_1 : \mu < 500$ mL, describe Type I and Type II errors and the power of the test.
77. With known $\sigma = 7$, $n = 49$, and $\bar{x} = 102.4$, test $H_0 : \mu = 100$ against $H_1 : \mu > 100$ at 1%.
78. For a normal population, $n = 16$, $\bar{x} = 74.2$, and $s = 5.6$. Test $H_0 : \mu = 72$ against $H_1 : \mu \neq 72$ at 5%.
79. A test gives p-value 0.032. State the decisions at levels 0.05 and 0.01 and explain the difference.
80. A sample has $n = 36$, $\bar{x} = 19.2$, and known $\sigma = 3$. Test $H_0 : \mu = 20$ against $H_1 : \mu \neq 20$ at 5% using a statistic, p-value, and confidence interval.

Application-based problems

81. A genetic trait occurs independently with probability 0.20 in each of five offspring. Find the probabilities of exactly three, at least one, and at most two offspring having the trait.
82. An emergency team is formed from eight doctors, six nurses, and four technicians.
 - (a) Count the six-person teams containing exactly two members from each profession.
 - (b) For such a team, count the selections when one of the two selected doctors is also appointed leader.
83. An airport loses 1.8 bags per day on average. Find the probabilities of no loss in one day, at least three losses over two days, and exactly five losses in seven days.

84. Emergency calls arrive at 12 per hour. In ten minutes, find the probabilities of at most two calls, more than three calls, and exactly two given at most three.
85. Web-server interarrival time is exponential with mean 30 seconds. Find the probabilities of a request within 20 seconds, after 60 seconds, and after 90 seconds given that 60 seconds have passed.
86. A safety system contains components A and B in parallel, followed by component C in series. Their reliabilities are 0.90, 0.85, and 0.95, respectively.
- Find the probability that the system operates successfully.
 - Find the probability that the system fails.
 - Recalculate the system reliability if component C is upgraded to reliability 0.98.

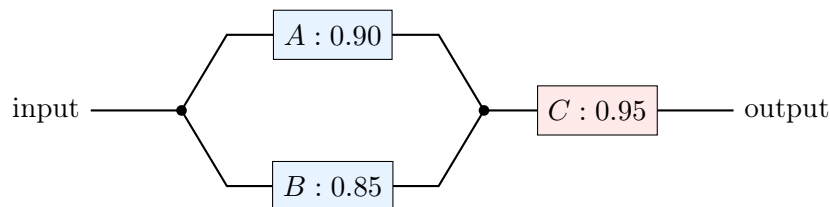


Figure 9.12: Parallel components followed by a series component for Question 86.

87. Battery lifetime is exponential with median 18 months. Find its rate, the probability of lasting beyond 24 months, and a warranty period under which only 10% fail.
88. Bus waiting time is uniform from 0 to 15 minutes. Find the probabilities of waiting below five minutes and from four to ten minutes, together with the mean and standard deviation.
89. Bottle fill is $N(1.00, 0.015^2)$ litres. Find the probability of a fill outside $[0.97, 1.03]$, the expected number outside in 200 bottles, and the 99th percentile.
90. Examination scores are $N(68, 12^2)$. Find the top-10% cutoff, the proportion below 40, and the central 95% interval.
91. A disease prevalence is 0.02. A test has sensitivity 0.95 and specificity 0.90. Find $P(+)$, $P(\text{disease} \mid +)$, and $P(\text{no disease} \mid -)$.

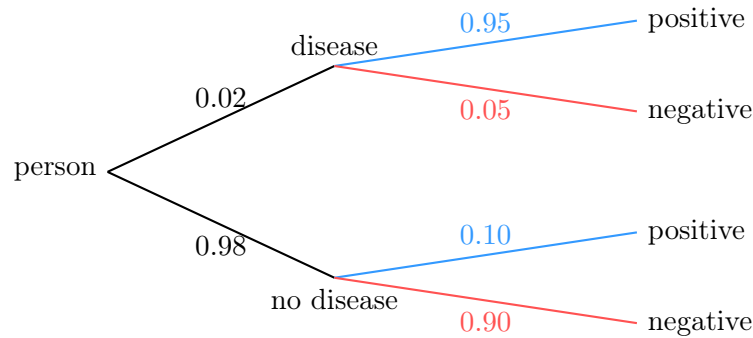


Figure 9.13: Screening-test branches for Question 91.

92. An insurer pays RM 0, RM 1,000, or RM 5,000 with probabilities 0.85, 0.10, and 0.05. Administration costs RM 120. Find the expected claim, break-even premium, and expected profit for a RM 600 premium.
93. An airline sells 105 tickets for 100 seats. Each passenger appears independently with probability 0.92. State a model, find the expected attendance, and find the probability somebody is denied boarding.
94. A water-quality instrument has known $\sigma = 1.8 \text{ mg L}^{-1}$. For $n = 36$ and $\bar{x} = 7.4$, construct a 95% interval and test $H_0 : \mu = 8$ against $H_1 : \mu < 8$ at 5%.
95. Twelve normal tensile strengths have $\bar{x} = 502 \text{ MPa}$ and $s = 18 \text{ MPa}$. Construct a 95% interval and test whether μ differs from 500 MPa at 5%.
96. Nine normal recovery times have $\bar{x} = 6.8$ days and $s = 1.2$ days. Test at 5% whether the mean is below 7.5 days, stating the assumptions.
97. Select and justify a binomial, Poisson, normal, uniform, or exponential model for: defects among 40 chips; calls in five minutes; memoryless component lifetime; measurement error from many small effects; and a wait equally likely between two departures.
98. A school has 2000 students, 500 in each of four year groups. Design a stratified sample of 200 and explain why surveying only library users could bias estimated study time.
99. Parcels come from depots A, B, C with source probabilities 0.50, 0.30, and 0.20, respectively. Their late-delivery rates are 0.04, 0.07, and 0.12. Find the late probability, $P(C \mid \text{late})$, and the probability that none of ten independent parcels is late.
100. Sixteen normal pollutant readings have $\bar{x} = 48.7$ and $s = 4.0$. Construct a 95% interval, test $H_0 : \mu = 50$ against $H_1 : \mu < 50$ at 5%, and describe a Type II error.

Chapter 10

Complex Numbers

Chapter guide

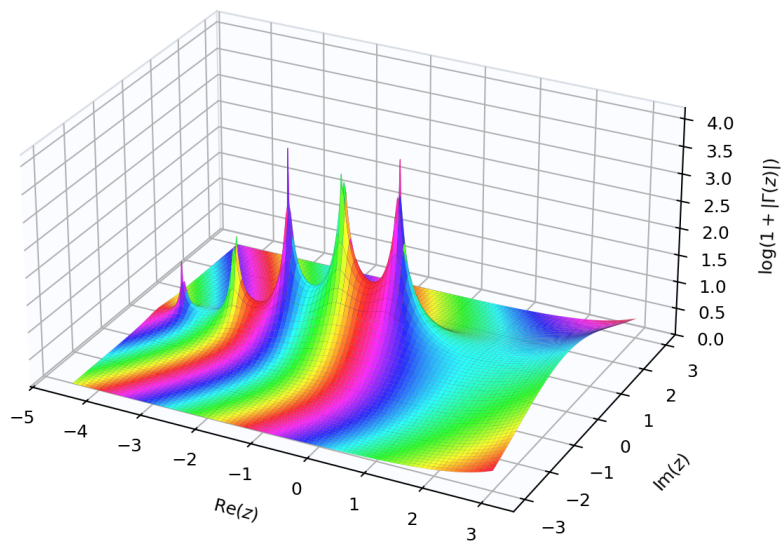
Prerequisites. Real-number algebra, polynomial equations, trigonometry, radians, and coordinate geometry.

Learning outcomes. Readers should be able to perform complex arithmetic, use conjugates and moduli, move between Cartesian, polar, and exponential forms, apply De Moivre's theorem, and interpret loci in the Argand plane.

This chapter extends the real number system to the complex number system. The real numbers studied in Chapter 1 remain present as a special subset of this wider system.

A complex number contains a real part and an imaginary part. This chapter examines complex arithmetic, geometric representation, identities, and fundamental theorems, following the scope of the relevant foundation syllabus (Abu Bakar et al. 2025–2026).

Magnitude and phase of the complex Gamma function



Height shows magnitude; hue shows complex argument. Capped peaks occur at poles on the non-positive real axis.

Figure 10.1: A visualisation of the complex Gamma function. Height represents a capped logarithmic magnitude and colour represents complex argument; the peaks above the non-positive integers correspond to poles. Original figure generated for this book using SciPy and Matplotlib.

Complex numbers are indispensable in many branches of pure mathematics and also have substantial applications in physics and engineering, particularly in quantum mechanics, oscillations, signal analysis, and electrical circuit theory.

10.1 Introduction to complex numbers

A complex number has a real part and an imaginary part. It may therefore be represented by a point in a two-dimensional plane whose horizontal and vertical axes correspond to those two parts. A complex number z can be written as

$$z = a + bi$$

where $\text{Re}(z) = a$, $\text{Im}(z) = b$, and $i^2 = -1$.

Complex numbers provide solutions to polynomial equations that have no real solutions. More generally, the fundamental theorem of algebra states that every non-constant polynomial with complex coefficients has at least one complex zero.

For example, the equation $(x + 1)^2 = -9$ has no real solution because the square of a real number cannot be negative. Over the complex numbers, however, it has the solutions $x = -1 + 3i$ and $x = -1 - 3i$.

10.2 Identities of complex numbers

The following operations are defined componentwise for complex numbers.

10.2.1 Equality of complex numbers

For two complex numbers $z_1 = a + bi$ and $z_2 = c + di$,

$$a + bi = c + di$$

if and only if $a = c$ and $b = d$.

10.2.2 Sum of complex numbers

Given two complex numbers $z_1 = a + bi$ and $z_2 = c + di$,

$$(a + bi) + (c + di) = (a + c) + (b + d)i.$$

10.2.3 Difference of complex numbers

Given two complex numbers $z_1 = a + bi$ and $z_2 = c + di$,

$$(a + bi) - (c + di) = (a - c) + (b - d)i.$$

10.2.4 Product of complex numbers

Given two complex numbers $z_1 = a + bi$ and $z_2 = c + di$,

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i.$$

10.2.5 Division of complex numbers

Given two complex numbers $z_1 = a + bi$ and $z_2 = c + di$, where $z_2 \neq 0$,

$$\begin{aligned} \frac{a + bi}{c + di} &= \frac{a + bi}{c + di} \cdot \frac{c - di}{c - di} \\ &= \frac{(a + bi)(c - di)}{c^2 + d^2} \\ &= \frac{(ac + bd) + (bc - ad)i}{c^2 + d^2}. \end{aligned}$$

10.3 Conjugates of complex numbers

Complex conjugation reverses the sign of the imaginary part. It is central to division, modulus calculations, polynomial roots, and symmetry in the Argand plane.

Definition 10.1: Complex conjugate.

If $z = a + bi$ is a complex number, its conjugate $\bar{z}$ is defined by

$$\bar{z} = \overline{a + bi} = a - bi.$$

10.3.1 Properties of complex conjugates

The principal algebraic properties of conjugation are listed below.

1. **Product with its conjugate.** If $z = a + bi$, then

$$z\bar{z} = a^2 + b^2.$$

2. **Equality of real parts.** A complex number and its conjugate have the same real part:

$$\operatorname{Re}(z) = \operatorname{Re}(\bar{z}).$$

3. **Double conjugation.** Taking the conjugate twice returns the original number:

$$\overline{\bar{z}} = z.$$

4. **Conjugate of a sum.**

$$\overline{z + w} = \bar{z} + \bar{w}.$$

5. **Conjugate of a product.**

$$\overline{zw} = \bar{z}\bar{w}.$$

10.4 Powers of i

Recall that every complex number has the form

$$z = a + bi$$

where $\operatorname{Re}(z) = a$, $\operatorname{Im}(z) = b$, and $i^2 = -1$.

Since $i^2 = -1$, it follows that $i^4 = (i^2)^2 = 1$. The powers of i therefore repeat with period four:

$$i, \quad -1, \quad -i, \quad 1, \quad i, \dots$$

Example 10.1

Evaluate i^{27} .

$$\begin{aligned} i^{27} &= i^{4(6)+3} \\ &= (i^4)^6 i^3 \\ &= 1^6(-i) \\ &= -i. \end{aligned}$$

10.5 Quadratic equations with a negative discriminant

Consider a quadratic equation with real coefficients,

$$ax^2 + bx + c = 0, \quad a \neq 0.$$

Its solutions are given by the quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}. \quad (10.1)$$

The expression

$$\Delta = b^2 - 4ac \quad (10.2)$$

is called the *discriminant*. It determines the nature of the roots before the equation is solved.

Value of the discriminant	Nature of the roots
$\Delta > 0$	Two distinct real roots
$\Delta = 0$	One repeated real root
$\Delta < 0$	Two non-real complex conjugate roots

Table 10.1: The discriminant and the nature of the roots of a quadratic equation.

When $\Delta < 0$, write

$$\Delta = -d, \quad d > 0.$$

Since $i^2 = -1$, we have $\sqrt{-d} = i\sqrt{d}$. The quadratic formula then becomes

$$x = \frac{-b \pm i\sqrt{4ac - b^2}}{2a}. \quad (10.3)$$

The two roots have the same real part and imaginary parts of equal magnitude but opposite signs. They therefore form a *complex conjugate pair*.

Example 10.2: solving a monic quadratic

Solve

$$x^2 - 4x + 13 = 0.$$

Here, $a = 1$, $b = -4$, and $c = 13$. The discriminant is

$$\Delta = (-4)^2 - 4(1)(13) = 16 - 52 = -36.$$

Therefore,

$$\begin{aligned} x &= \frac{-(-4) \pm \sqrt{-36}}{2(1)} \\ &= \frac{4 \pm 6i}{2} \\ &= 2 \pm 3i. \end{aligned}$$

Hence, the roots are

$$\boxed{x = 2 + 3i \quad \text{or} \quad x = 2 - 3i.}$$

On an Argand diagram, these roots are reflections of each other in the real axis.

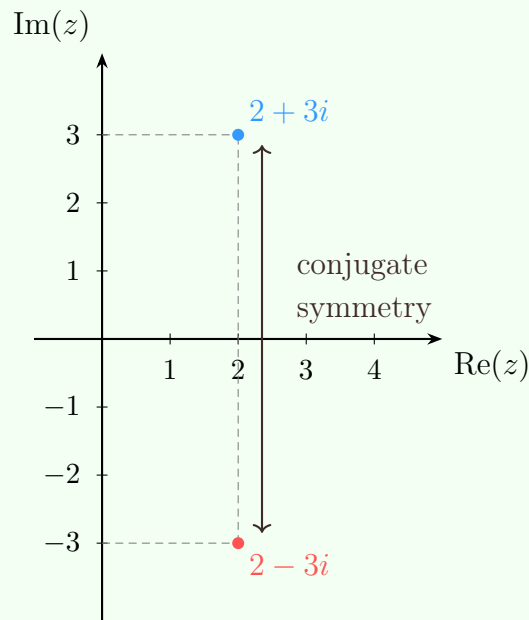


Figure 10.2: The conjugate roots $2 + 3i$ and $2 - 3i$ are symmetric about the real axis.

Example 10.3: simplifying an irrational imaginary part

Solve

$$2x^2 + 4x + 5 = 0.$$

The discriminant is

$$\Delta = 4^2 - 4(2)(5) = 16 - 40 = -24.$$

Using $\sqrt{-24} = i\sqrt{24} = 2i\sqrt{6}$,

$$\begin{aligned}
 x &= \frac{-4 \pm \sqrt{-24}}{4} \\
 &= \frac{-4 \pm 2i\sqrt{6}}{4} \\
 &= -1 \pm \frac{\sqrt{6}}{2}i.
 \end{aligned}$$

Thus,

$$x = -1 + \frac{\sqrt{6}}{2}i \quad \text{or} \quad x = -1 - \frac{\sqrt{6}}{2}i.$$

Why complex roots occur in conjugate pairs

Suppose a polynomial $p(x)$ has real coefficients and $p(z) = 0$. Taking the complex conjugate of both sides gives

$$\overline{p(z)} = \bar{0} = 0.$$

Because all the coefficients are real, conjugation passes through the polynomial operations, so

$$\overline{p(z)} = p(\bar{z}).$$

Therefore, $p(\bar{z}) = 0$. Hence, whenever $z = p + qi$ is a non-real root of a polynomial with real coefficients, its conjugate $\bar{z} = p - qi$ is also a root.

Forming a quadratic equation from complex roots

If a quadratic equation with real coefficients has one root $p + qi$, where $q \neq 0$, then its other root must be $p - qi$. The corresponding quadratic factorisation is

$$\begin{aligned}
 [x - (p + qi)][x - (p - qi)] &= [(x - p) - qi][(x - p) + qi] \\
 &= (x - p)^2 - (qi)^2 \\
 &= (x - p)^2 + q^2.
 \end{aligned}$$

For example, the quadratic whose roots are $3 + 2i$ and $3 - 2i$ is

$$\begin{aligned}
 [x - (3 + 2i)][x - (3 - 2i)] &= 0, \\
 (x - 3)^2 + 4 &= 0,
 \end{aligned}$$

$$x^2 - 6x + 13 = 0.$$

The conjugate terms cancel during expansion, leaving a polynomial with real coefficients.

Checking the roots using their sum and product

For $ax^2 + bx + c = 0$ with roots α and β ,

$$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}.$$

If $\alpha = p + qi$ and $\beta = p - qi$, then

$$\alpha + \beta = 2p, \quad \alpha\beta = p^2 + q^2,$$

which are both real, as expected for a quadratic with real coefficients.

10.6 Polar form of a complex number

A complex number extends the one-dimensional real number line to a two-dimensional plane. Its real part is plotted horizontally and its imaginary part vertically. This representation is called an *Argand diagram*. Thus $z = a + bi$ corresponds to the ordered pair (a, b) .

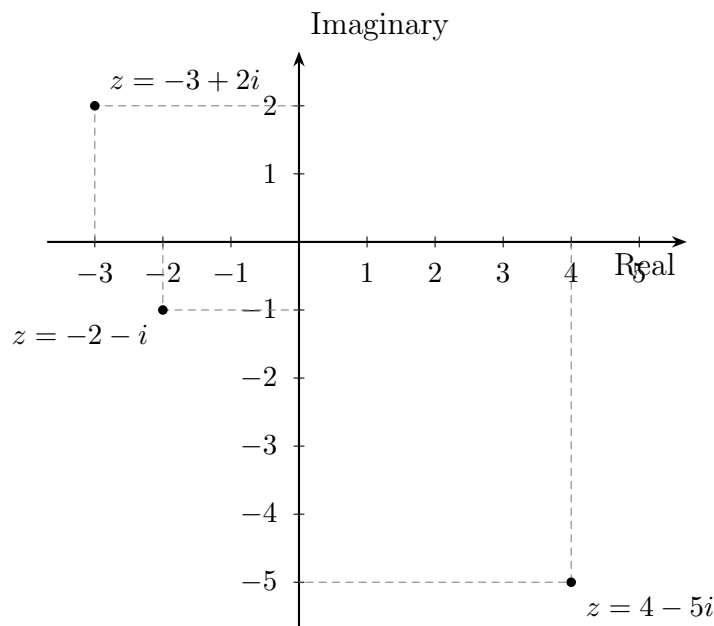


Figure 10.3: Complex numbers represented on an Argand diagram.

A complex number $z = a + bi$ is said to be in **standard form**, also known as Cartesian form. On an Argand diagram, it is represented by the point (a, b) , where a gives its horizontal position along the real axis and b gives its vertical position along the imaginary axis. Instead of describing the point using these two Cartesian coordinates, we may describe it using its distance and direction from the origin. This representation is called

the **polar form**, the **modulus–argument form**, or the **trigonometric form** of a complex number.

The distance from the origin to the point (a, b) is called the **modulus** or **absolute value** of z , denoted by $|z|$. By Pythagoras' theorem,

$$r = |z| = \sqrt{a^2 + b^2}.$$

The angle θ measured anticlockwise from the positive real axis to the line joining the origin to (a, b) is called an **argument** of z . It is denoted by $\arg(z)$ and satisfies

$$\tan \theta = \frac{b}{a},$$

with the quadrant of (a, b) taken into account when choosing the value of θ .

Since $a = r \cos \theta$ and $b = r \sin \theta$, the Cartesian form can be rewritten as

$$\begin{aligned} z &= a + bi \\ &= r \cos \theta + i(r \sin \theta) \\ &= r(\cos \theta + i \sin \theta). \end{aligned}$$

Therefore, the polar form of a complex number is

$$\boxed{z = r(\cos \theta + i \sin \theta)},$$

where $r = |z|$ and $\theta = \arg(z)$.

For example, consider $z = 3 + 4i$. This complex number corresponds to the point $(3, 4)$ on the Argand diagram. Its modulus is

$$|z| = \sqrt{3^2 + 4^2} = 5,$$

Because the point lies in the first quadrant, its argument is

$$\theta = \tan^{-1} \left(\frac{4}{3} \right) \approx 53.13^\circ \quad \text{or} \quad 0.927 \text{ radians.}$$

Hence, the polar form of $z = 3 + 4i$ is

$$z = 5(\cos 53.13^\circ + i \sin 53.13^\circ).$$

10.6.1 Algebra of polar form

Polar form makes the multiplication and division of complex numbers particularly simple. Suppose

$$\begin{aligned} z_1 &= r_1(\cos \theta_1 + i \sin \theta_1), \\ z_2 &= r_2(\cos \theta_2 + i \sin \theta_2). \end{aligned}$$

When two complex numbers are multiplied, their moduli are multiplied and their arguments are added. Direct expansion and the angle-sum identities give

$$\begin{aligned} z_1 z_2 &= r_1 r_2 (\cos \theta_1 + i \sin \theta_1)(\cos \theta_2 + i \sin \theta_2) \\ &= r_1 r_2 [\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2 \\ &\quad + i(\sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2)] \\ &= r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]. \end{aligned}$$

Thus,

$$\boxed{z_1 z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]}.$$

Similarly, when one nonzero complex number is divided by another, their moduli are divided and their arguments are subtracted. Thus, provided that $z_2 \neq 0$,

$$\boxed{\frac{z_1}{z_2} = \frac{r_1}{r_2} [\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)]}.$$

The reciprocal of a nonzero complex number follows directly from the division rule. If

$$z = r(\cos \theta + i \sin \theta),$$

then its reciprocal has modulus $1/r$ and argument $-\theta$. Hence,

$$\boxed{\frac{1}{z} = \frac{1}{r}(\cos \theta - i \sin \theta)}, \quad z \neq 0.$$

10.7 De Moivre's theorem

The expression $\cos \theta + i \sin \theta$ is sometimes abbreviated to $\text{cis } \theta$. De Moivre's theorem describes all of its integer powers.

Theorem 10.1 (De Moivre's theorem). *For every real number θ and every integer n ,*

$$(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta).$$

Proof. For $n = 0$, both sides equal 1. For positive integers, use induction. The case $n = 1$ is immediate. Assume the result holds for $n = k$. Then the angle-sum identities give

$$\begin{aligned} (\cos \theta + i \sin \theta)^{k+1} &= [\cos(k\theta) + i \sin(k\theta)](\cos \theta + i \sin \theta) \\ &= \cos((k+1)\theta) + i \sin((k+1)\theta). \end{aligned}$$

Thus the result holds for every positive integer. Finally, if $n = -m$ with $m > 0$, the reciprocal rule gives

$$\begin{aligned} (\cos \theta + i \sin \theta)^{-m} &= \frac{1}{\cos(m\theta) + i \sin(m\theta)} \\ &= \cos(-m\theta) + i \sin(-m\theta). \end{aligned}$$

Hence the formula also holds for negative integers. $\square$

Corollary 10.2. *If $z = r(\cos \theta + i \sin \theta)$ and n is a positive integer, then*

$$z^n = r^n [\cos(n\theta) + i \sin(n\theta)].$$

If $z \neq 0$, the same formula holds for every integer n .

Proof. Raise $r(\cos \theta + i \sin \theta)$ to the power n and apply de Moivre's theorem. Negative powers additionally require $r \neq 0$. $\square$

10.8 Euler's formula

Euler's formula connects the complex exponential with the trigonometric functions. Angles in the formula are measured in radians.

Theorem 10.3 (Euler's formula). *For every real number θ ,*

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

Proof. Using the absolutely convergent power-series definitions permits the even and odd terms to be regrouped:

$$\begin{aligned} e^{i\theta} &= \sum_{n=0}^{\infty} \frac{(i\theta)^n}{n!} \\ &= \sum_{k=0}^{\infty} \frac{(-1)^k \theta^{2k}}{(2k)!} + i \sum_{k=0}^{\infty} \frac{(-1)^k \theta^{2k+1}}{(2k+1)!} \\ &= \cos \theta + i \sin \theta. \end{aligned}$$

$\square$

Consequently, for $r \geq 0$,

$$re^{i\theta} = r(\cos \theta + i \sin \theta).$$

10.9 Functions of complex numbers

This section extends the exponential, trigonometric, and logarithmic functions to complex inputs and examines their algebraic and geometric behaviour.

10.9.1 Applications of Euler's formula and De Moivre's theorem

This subsection provides a preliminary look at how Euler's formula and De Moivre's theorem can be used to derive trigonometric identities and simplify powers of trigonometric functions. Recall Euler's formula,

$$e^{ix} = \cos x + i \sin x.$$

Consider the exponential $e^{i(x+y)}$. By Euler's formula,

$$e^{i(x+y)} = \cos(x+y) + i \sin(x+y). \quad (10.4)$$

However, the laws of indices also give

$$\begin{aligned} e^{i(x+y)} &= e^{ix} e^{iy} \\ &= (\cos x + i \sin x)(\cos y + i \sin y) \\ &= \cos x \cos y - \sin x \sin y \\ &\quad + i(\sin x \cos y + \cos x \sin y). \end{aligned}$$

Comparing the real and imaginary parts with Equation (10.4) produces the familiar angle-sum identities

$$\begin{aligned} \cos(x+y) &= \cos x \cos y - \sin x \sin y, \\ \sin(x+y) &= \sin x \cos y + \cos x \sin y. \end{aligned}$$

The corresponding difference identities follow by replacing y with $-y$ and using $\cos(-y) = \cos y$ and $\sin(-y) = -\sin y$.

Trigonometric addition and subtraction identities

$$\begin{aligned} \sin(x+y) &= \sin x \cos y + \cos x \sin y, \\ \sin(x-y) &= \sin x \cos y - \cos x \sin y, \\ \cos(x+y) &= \cos x \cos y - \sin x \sin y, \\ \cos(x-y) &= \cos x \cos y + \sin x \sin y. \end{aligned}$$

Euler's formula can also be used to express the sine and cosine functions in exponential form. Replacing x with $-x$ gives

$$e^{-ix} = \cos x - i \sin x.$$

Adding this equation to $e^{ix} = \cos x + i \sin x$ eliminates the sine terms, whereas subtracting it eliminates the cosine terms. Hence,

$$\boxed{\cos x = \frac{e^{ix} + e^{-ix}}{2}} \quad \text{and} \quad \boxed{\sin x = \frac{e^{ix} - e^{-ix}}{2i}}.$$

These formulae are especially useful when powers of sine and cosine must be rewritten in terms of multiple angles.

Example 10.4

Express $\cos^3 x$ in terms of $\cos x$ and $\cos 3x$.

Using the exponential form of cosine,

$$\begin{aligned}\cos^3 x &= \left(\frac{e^{ix} + e^{-ix}}{2} \right)^3 \\ &= \frac{1}{8} (e^{3ix} + 3e^{ix} + 3e^{-ix} + e^{-3ix}) \\ &= \frac{1}{8} [(e^{3ix} + e^{-3ix}) + 3(e^{ix} + e^{-ix})].\end{aligned}$$

Since $e^{it} + e^{-it} = 2 \cos t$, it follows that

$$\begin{aligned}\cos^3 x &= \frac{1}{8} (2 \cos 3x + 6 \cos x) \\ &= \boxed{\frac{3 \cos x + \cos 3x}{4}}.\end{aligned}$$

10.9.2 The n -th roots of a complex number

Recall that a nonzero complex number may be written in Cartesian form as $z = a + bi$, in polar form as $z = r(\cos \theta + i \sin \theta)$, or in Euler form as $z = re^{i\theta}$, where $r = |z| > 0$ and $\theta = \arg(z)$. By De Moivre's theorem,

$$[r(\cos \theta + i \sin \theta)]^n = r^n (\cos n\theta + i \sin n\theta).$$

We now consider the reverse problem: given a complex number w , find every complex number z satisfying

$$z^n = w,$$

where n is a positive integer. These solutions are called the **n -th roots** of w .

Let the nonzero complex number w have modulus R and principal argument ϕ . Since coterminal angles differ by integer multiples of 2π , all possible arguments of w are $\phi + 2\pi k$, where $k \in \mathbb{Z}$. Thus,

$$w = Re^{i(\phi+2\pi k)}.$$

Taking the n -th root gives

$$\begin{aligned}z_k &= R^{1/n} e^{i(\phi+2\pi k)/n} \\ &= R^{1/n} \left[\cos \left(\frac{\phi + 2\pi k}{n} \right) + i \sin \left(\frac{\phi + 2\pi k}{n} \right) \right].\end{aligned}$$

Only n distinct roots are produced, so it is sufficient to take $k = 0, 1, 2, \dots, n-1$.

The n -th roots of a complex number

If $w = R(\cos \phi + i \sin \phi)$, where $R > 0$, then its n distinct n -th roots are

$$z_k = R^{1/n} \left[\cos \left(\frac{\phi + 2\pi k}{n} \right) + i \sin \left(\frac{\phi + 2\pi k}{n} \right) \right],$$

where $k = 0, 1, 2, \dots, n-1$.

All the roots have the same modulus, $R^{1/n}$, and consecutive roots differ in argument by $2\pi/n$. Geometrically, they are equally spaced around a circle centred at the origin. If $w = 0$, its only n -th root is $z = 0$.

Example 10.5

Find the cube roots of 8.

We may write $8 = 8(\cos 0 + i \sin 0)$. Here, $R = 8$, $\phi = 0$, and $n = 3$. Therefore,

$$z_k = 2 \left[\cos \left(\frac{2\pi k}{3} \right) + i \sin \left(\frac{2\pi k}{3} \right) \right], \quad k = 0, 1, 2.$$

The three roots are

$$\begin{aligned} z_0 &= 2, \\ z_1 &= -1 + i\sqrt{3}, \\ z_2 &= -1 - i\sqrt{3}. \end{aligned}$$

10.9.3 Complex exponential function

For $z = a + bi$, the complex exponential is defined by

$$e^z = e^{a+bi} = e^a e^{ib}.$$

Euler's formula gives $e^{ib} = \cos b + i \sin b$. Consequently,

$$|e^z| = e^a, \quad \arg(e^z) = b + 2\pi k, \quad k \in \mathbb{Z}.$$

$$\begin{aligned} e^{a+bi} &= e^a (\cos b + i \sin b) \\ &= e^a \cos b + i e^a \sin b. \end{aligned}$$

In particular, $e^{z+2\pi i} = e^z$, so the complex exponential has period $2\pi i$.

Example 10.6

Find the exact Cartesian form of

$$e^{-1+\frac{\pi}{3}i}.$$

Using $e^{a+bi} = e^a(\cos b + i \sin b)$,

$$\begin{aligned} e^{-1+\frac{\pi}{3}i} &= e^{-1} \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) \\ &= \frac{1}{e} \left(\frac{1}{2} + i \frac{\sqrt{3}}{2} \right) \\ &= \boxed{\frac{1}{2e} + \frac{\sqrt{3}}{2e}i}. \end{aligned}$$

10.9.4 Complex cosine and sine functions

The exponential identities for real sine and cosine extend naturally to a complex argument $z = a + bi$.

Evaluating cosine of a complex number

From Euler's formula,

$$\begin{aligned} e^{ix} &= \cos x + i \sin x, \\ e^{-ix} &= \cos x - i \sin x. \end{aligned}$$

Adding these equations eliminates the sine terms:

$$e^{ix} + e^{-ix} = 2 \cos x.$$

Therefore, cosine can be expressed in exponential form as

$$\cos x = \frac{e^{ix} + e^{-ix}}{2}.$$

This definition also applies when the input is complex. Let $z = a + bi$. Then

$$\begin{aligned} \cos z &= \frac{e^{iz} + e^{-iz}}{2} \\ &= \frac{e^{i(a+bi)} + e^{-i(a+bi)}}{2} \\ &= \frac{e^{-b+ia} + e^{b-ia}}{2} \\ &= \frac{e^{-b}(\cos a + i \sin a) + e^b(\cos a - i \sin a)}{2}. \end{aligned}$$

Grouping the real and imaginary terms gives

$$\begin{aligned}\cos(a + bi) &= \frac{(e^b + e^{-b}) \cos a}{2} + i \frac{(e^{-b} - e^b) \sin a}{2} \\ &= \cos a \cosh b - i \sin a \sinh b.\end{aligned}$$

Hence,

$$\boxed{\cos(a + bi) = \cos a \cosh b - i \sin a \sinh b}.$$

Its real and imaginary parts are therefore

$$\begin{aligned}\operatorname{Re}(\cos z) &= \cos a \cosh b, \\ \operatorname{Im}(\cos z) &= -\sin a \sinh b.\end{aligned}$$

Example 10.7

Evaluate $\cos(2 - 2i)$.

Using

$$\cos(a + bi) = \cos a \cosh b - i \sin a \sinh b,$$

with $a = 2$ and $b = -2$, we obtain

$$\begin{aligned}\cos(2 - 2i) &= \cos 2 \cosh(-2) - i \sin 2 \sinh(-2) \\ &= \cos 2 \cosh 2 + i \sin 2 \sinh 2.\end{aligned}$$

Therefore, the exact Cartesian form is

$$\boxed{\cos(2 - 2i) = \cos 2 \cosh 2 + i \sin 2 \sinh 2}.$$

Evaluating sine of a complex number

From Euler's formula,

$$\begin{aligned}e^{ix} &= \cos x + i \sin x, \\ e^{-ix} &= \cos x - i \sin x.\end{aligned}$$

Subtracting the second equation from the first gives

$$e^{ix} - e^{-ix} = 2i \sin x.$$

Therefore, sine can be expressed in exponential form as

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i}.$$

Let $z = a + bi$. Applying this definition to a complex input gives

$$\begin{aligned}\sin z &= \frac{e^{iz} - e^{-iz}}{2i} \\ &= \frac{e^{i(a+bi)} - e^{-i(a+bi)}}{2i} \\ &= \frac{e^{-b+ia} - e^{b-ia}}{2i} \\ &= \frac{e^{-b}(\cos a + i \sin a) - e^b(\cos a - i \sin a)}{2i}.\end{aligned}$$

Grouping the real and imaginary terms gives

$$\begin{aligned}\sin(a + bi) &= \frac{(e^{-b} - e^b) \cos a + i(e^{-b} + e^b) \sin a}{2i} \\ &= \frac{(e^b + e^{-b}) \sin a}{2} + i \frac{(e^b - e^{-b}) \cos a}{2} \\ &= \sin a \cosh b + i \cos a \sinh b.\end{aligned}$$

Hence,

$$\boxed{\sin(a + bi) = \sin a \cosh b + i \cos a \sinh b}.$$

Its real and imaginary parts are therefore

$$\operatorname{Re}(\sin z) = \sin a \cosh b,$$

$$\operatorname{Im}(\sin z) = \cos a \sinh b.$$

Example 10.8

Evaluate $\sin(e^{\frac{\pi}{2}i})$.

By Euler's formula,

$$e^{\frac{\pi}{2}i} = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2} = i.$$

Therefore,

$$\begin{aligned}\sin(e^{\frac{\pi}{2}i}) &= \sin(i) \\ &= \sin(0 + i) \\ &= \sin 0 \cosh 1 + i \cos 0 \sinh 1 \\ &= i \sinh 1 \\ &= \boxed{\frac{i}{2} (e - e^{-1})}.\end{aligned}$$

10.9.5 Complex logarithmic function

For real numbers, the natural logarithm is the inverse of the exponential function. In the complex plane, however, the exponential function is periodic because

$$e^{z+2\pi ki} = e^z, \quad k \in \mathbb{Z}.$$

Consequently, a nonzero complex number has infinitely many logarithmic values.

Let $z \neq 0$ have modulus $|z|$ and argument $\arg(z)$. Its Euler form, including every coterminal argument, is

$$z = |z|e^{i(\arg(z)+2\pi k)}, \quad k \in \mathbb{Z}.$$

Taking the logarithm gives

$$\begin{aligned} \log z &= \log(|z|e^{i(\arg(z)+2\pi k)}) \\ &= \ln |z| + i(\arg(z) + 2\pi k), \quad k \in \mathbb{Z}. \end{aligned}$$

Thus, the general complex logarithm is

$$\boxed{\log z = \ln |z| + i(\arg(z) + 2\pi k)}, \quad k \in \mathbb{Z}.$$

To obtain a single value, we choose the principal argument $\text{Arg}(z) \in (-\pi, \pi]$. The resulting **principal complex logarithm** is

$$\boxed{\text{Log } z = \ln |z| + i \text{Arg}(z)}.$$

The complex logarithm is not defined at $z = 0$ because $|0| = 0$ and $\ln 0$ is undefined.

Example 10.9

Find the general complex logarithm and the principal complex logarithm of

$$z = 2e^{-\frac{\pi}{3}i}.$$

Here, $|z| = 2$ and the principal argument is $\text{Arg}(z) = -\pi/3$. Therefore, the general complex logarithm is

$$\begin{aligned} \log z &= \ln 2 + i\left(-\frac{\pi}{3} + 2\pi k\right) \\ &= \boxed{\ln 2 + i\frac{(6k-1)\pi}{3}}, \quad k \in \mathbb{Z}. \end{aligned}$$

The principal complex logarithm is obtained by taking $k = 0$:

$$\boxed{\text{Log } z = \ln 2 - \frac{\pi}{3}i}.$$

Example 10.10

Find the complex numbers z satisfying

$$\operatorname{Log}(z^2 - 4) = \ln 2 + \pi i.$$

Exponentiating both sides gives

$$\begin{aligned} z^2 - 4 &= e^{\ln 2 + \pi i} \\ &= 2(\cos \pi + i \sin \pi) \\ &= -2. \end{aligned}$$

Hence,

$$\begin{aligned} z^2 &= 2, \\ z &= \boxed{\pm\sqrt{2}}. \end{aligned}$$

10.10 Geometric interpretation of complex equations

A complex equation $f(z)$ can also be represented geometrically.

10.10.1 Circle

Let z be a variable complex number and let z_1 be a fixed complex number. The equation

$$\boxed{|z - z_1| = r}, \quad r > 0,$$

represents a circle in the Argand plane with centre z_1 and radius r .

To understand this geometrically, regard z and z_1 as position vectors from the origin. Their difference, $z - z_1$, is the displacement vector from the fixed point z_1 to the variable point z . Therefore, $|z - z_1|$ is the distance between these two points. Requiring this distance to remain equal to the constant r produces the locus of a circle.

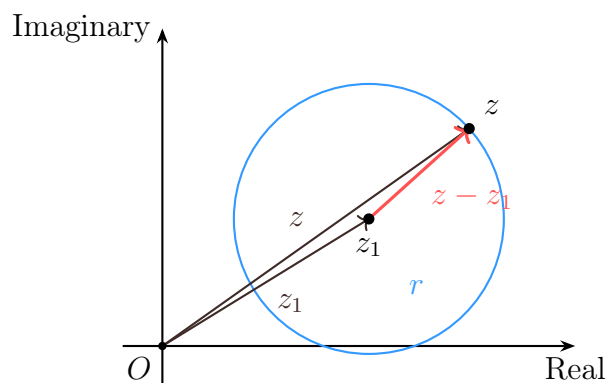


Figure 10.4: The locus $|z - z_1| = r$ is a circle centred at z_1 .

The same equation can be expressed in Cartesian form. Let

$$z = x + iy \quad \text{and} \quad z_1 = a + ib.$$

Then

$$\begin{aligned} |z - z_1| &= |(x - a) + i(y - b)| \\ &= \sqrt{(x - a)^2 + (y - b)^2}. \end{aligned}$$

Thus,

$$|z - z_1| = r \iff \boxed{(x - a)^2 + (y - b)^2 = r^2},$$

which is the Cartesian equation of a circle with centre (a, b) and radius r .

10.10.2 Perpendicular bisector

Let z_1 and z_2 be two distinct fixed complex numbers, and let z be a variable complex number. The equation

$$\boxed{|z - z_1| = |z - z_2|}$$

states that the point z is equally distant from the fixed points z_1 and z_2 . The locus of every point satisfying this condition is the perpendicular bisector of the line segment joining z_1 and z_2 .

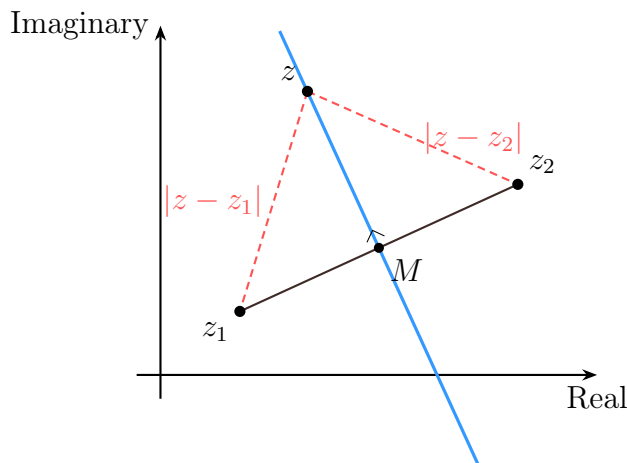


Figure 10.5: The locus $|z - z_1| = |z - z_2|$ is the perpendicular bisector of $z_1 z_2$.

To obtain the Cartesian equation, let

$$z = x + iy, \quad z_1 = a + ib, \quad z_2 = c + id.$$

Then

$$\begin{aligned} |z - z_1| &= |z - z_2|, \\ \sqrt{(x - a)^2 + (y - b)^2} &= \sqrt{(x - c)^2 + (y - d)^2}. \end{aligned}$$

Squaring both sides and simplifying gives

$$2(c - a)x + 2(d - b)y = c^2 + d^2 - a^2 - b^2.$$

This is a linear equation, confirming that the locus is a straight line.

10.10.3 Half-line

Let z_1 be a fixed complex number and let z be variable. The equation

$$\arg(z - z_1) = \theta, \quad -\pi < \theta \leq \pi,$$

represents a half-line, or ray, beginning at z_1 and extending in the direction that makes an angle θ with the positive real axis.

The difference $z - z_1$ is the displacement vector from z_1 to z . Requiring its argument to equal the fixed angle θ fixes the direction of this vector, while its modulus may take any positive value. The initial point z_1 is excluded because $z - z_1 = 0$ there, and $\arg(0)$ is undefined.

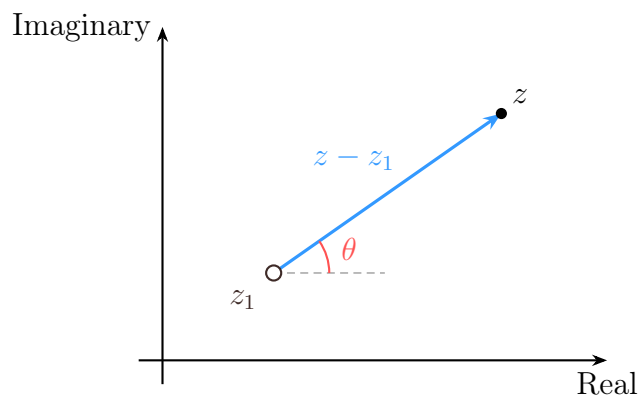


Figure 10.6: The locus $\arg(z - z_1) = \theta$ is a half-line from z_1 , excluding z_1 .

For a Cartesian description, let

$$z = x + iy \quad \text{and} \quad z_1 = a + ib.$$

Then

$$\arg((x - a) + i(y - b)) = \theta.$$

When $\cos \theta \neq 0$, the supporting straight line is

$$y - b = (x - a) \tan \theta.$$

To select only the required half-line, the direction must also satisfy

$$(x - a) \cos \theta + (y - b) \sin \theta > 0.$$

Equivalently, the locus may be written parametrically as

$$(x, y) = (a, b) + t(\cos \theta, \sin \theta), \quad t > 0.$$

10.11 Complex inequalities

Complex numbers do not possess a natural order compatible with their field operations, so expressions such as $z_1 < z_2$ are generally undefined. Inequalities are instead applied to real-valued quantities associated with complex numbers, particularly their moduli, real parts, imaginary parts, and arguments.

The principal types of loci considered here correspond to the geometric equations introduced earlier:

1. Circle or disc:

$$|z - (a + bi)| < r.$$

This represents all points whose distance from the fixed point $a + bi$ is less than r ; hence, it describes the interior of a circle.

2. Region determined by a perpendicular bisector:

$$|z - (a + bi)| \geq |z - (c + di)|.$$

This represents the half-plane containing points that are at least as close to $c + di$ as they are to $a + bi$.

3. Angular region determined by a half-line:

$$\arg(z - (a + bi)) \leq \theta.$$

This selects points whose displacement from $a + bi$ has an argument no greater than θ , subject to the chosen principal-argument interval.

When sketching these loci, the style of the boundary is important:

- A **solid boundary** is used for $\leq$ or $\geq$ because equality is included.
- A **dashed boundary** is used for $<$ or $>$ because equality is excluded.

After drawing the appropriate boundary, the required side or region is shaded according to the inequality.

10.11.1 Inequalities in circle

The complex equation of a circle is

$$|z - (a + bi)| = r$$

This equation states that the distance from the variable point z to the fixed centre $a + bi$ is exactly r .

Replacing the equality with an inequality produces four possibilities:

1. **Interior excluding the boundary:**

$$|z - (a + bi)| < r.$$

The distance from z to the centre is less than r . Hence, z lies inside the circle, while the circular boundary is excluded.

2. **Exterior excluding the boundary:**

$$|z - (a + bi)| > r.$$

The distance from z to the centre is greater than r . Hence, z lies outside the circle, while the circular boundary is excluded.

3. **Interior including the boundary:**

$$|z - (a + bi)| \leq r.$$

The point z lies inside or on the circle. This region is called a **closed disc**.

4. **Exterior including the boundary:**

$$|z - (a + bi)| \geq r.$$

The point z lies outside or on the circle.

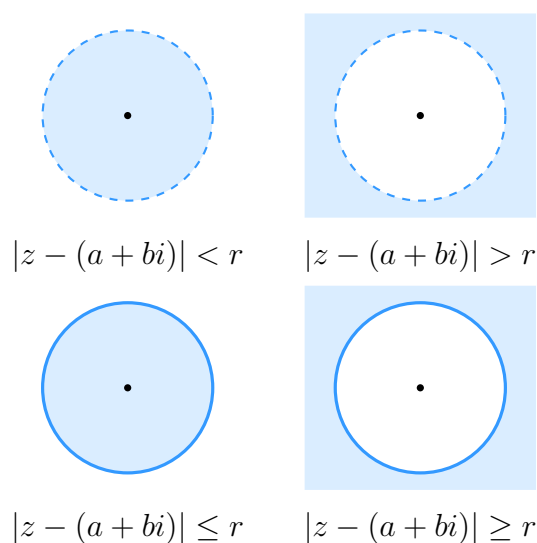


Figure 10.7: The four regions determined by a circle inequality. Dashed boundaries are excluded; solid boundaries are included.

10.11.2 Perpendicular bisector

Recall that the equation

$$|z - (a + bi)| = |z - (c + di)|$$

represents the perpendicular bisector of the segment joining the fixed points $a + bi$ and $c + di$. Replacing the equality with an inequality selects one of the two half-planes determined by this line.

There are four possibilities:

1. **Closer to $a + bi$, excluding the boundary:**

$$|z - (a + bi)| < |z - (c + di)|.$$

The locus is the half-plane containing $a + bi$, with the perpendicular bisector excluded.

2. **Closer to or equidistant from $a + bi$:**

$$|z - (a + bi)| \leq |z - (c + di)|.$$

The locus is the half-plane containing $a + bi$, including the perpendicular bisector.

3. **Closer to $c + di$, excluding the boundary:**

$$|z - (a + bi)| > |z - (c + di)|.$$

The locus is the half-plane containing $c + di$, with the perpendicular bisector excluded.

4. **Closer to or equidistant from $c + di$:**

$$|z - (a + bi)| \geq |z - (c + di)|.$$

The locus is the half-plane containing $c + di$, including the perpendicular bisector.

For the Cartesian form, let $z = x + iy$ and define

$$L(x, y) = 2(c - a)x + 2(d - b)y - (c^2 + d^2 - a^2 - b^2).$$

Then

$$\begin{aligned} |z - (a + bi)| < |z - (c + di)| &\iff L(x, y) < 0, \\ |z - (a + bi)| \leq |z - (c + di)| &\iff L(x, y) \leq 0, \\ |z - (a + bi)| > |z - (c + di)| &\iff L(x, y) > 0, \\ |z - (a + bi)| \geq |z - (c + di)| &\iff L(x, y) \geq 0. \end{aligned}$$

The boundary $L(x, y) = 0$ is drawn dashed for strict inequalities and solid for inclusive inequalities.

10.11.3 Half-line

An inequality involving an argument is meaningful only after a branch has been chosen. Here, we use the principal argument

$$-\pi < \text{Arg}(z - (a + bi)) \leq \pi.$$

The fixed point $a + bi$ acts as the vertex, and the ray

$$\text{Arg}(z - (a + bi)) = \theta$$

forms the main boundary of the angular region, where $-\pi < \theta \leq \pi$. The point $z = a + bi$ is excluded because the argument of zero is undefined.

There are four possibilities:

1. Arguments less than θ :

$$\text{Arg}(z - (a + bi)) < \theta.$$

Equivalently,

$$-\pi < \text{Arg}(z - (a + bi)) < \theta.$$

The boundary ray at angle θ is excluded and is therefore drawn dashed.

2. Arguments less than or equal to θ :

$$\text{Arg}(z - (a + bi)) \leq \theta.$$

Equivalently,

$$-\pi < \text{Arg}(z - (a + bi)) \leq \theta.$$

The boundary ray at angle θ is included and is therefore drawn solid.

3. Arguments greater than θ :

$$\text{Arg}(z - (a + bi)) > \theta.$$

Equivalently,

$$\theta < \text{Arg}(z - (a + bi)) \leq \pi.$$

The boundary ray at angle θ is excluded and is therefore drawn dashed.

4. Arguments greater than or equal to θ :

$$\text{Arg}(z - (a + bi)) \geq \theta.$$

Equivalently,

$$\theta \leq \text{Arg}(z - (a + bi)) \leq \pi.$$

The boundary ray at angle θ is included and is therefore drawn solid.

In each case, the required angular sector is shaded. Positive arguments are measured anticlockwise from the positive real direction, while negative arguments are measured clockwise.

Chapter summary and common pitfalls

Key ideas. Complex numbers may be represented algebraically or geometrically. Multiplication combines moduli and arguments, conjugation supports division, and De Moivre's theorem generates powers and roots.

Common pitfalls. Arguments differ by multiples of 2π , an n th-root problem normally has n distinct roots, and inequalities do not order complex numbers in the same way as real numbers.

10.12 Practice and exercises

Use $\operatorname{cis} \theta = \cos \theta + i \sin \theta$ where this notation is convenient. Give principal arguments in $(-\pi, \pi]$ unless another range is specified, and show the significant features of every requested Argand diagram.

Foundations, polar form, powers, and roots

1. Explain why the set of complex numbers can be regarded as the universal set for the number systems discussed in this chapter. Draw a Venn diagram that includes the sets of complex numbers, real numbers, irrational numbers, and whole numbers.

2. Solve the following equations:

(a) $(2x - 1)^2 + 9 = 0$

(b) $x^2 - 2x + 5 = 0$

3. Find the real and imaginary parts of each expression:

(a) $(1 + 2i)(3 - i)$

(b) $\frac{5 - i}{1 - i}$

(c) i^{-5}

4. Evaluate i^{4n} for $n \in \mathbb{Z}^+$. Hence, explain step by step how to evaluate i^n for any $n \in \mathbb{Z}^+$.

5. If

$$x + iy = (-1 + \sqrt{2}i)(-1 - \sqrt{2}i),$$

find x and y .

6. If $\operatorname{Im}(z) < 0$ and $\operatorname{Re}(z) = 0$, determine the principal argument of z .
7. For each of the following complex numbers, calculate its modulus and principal argument in radians. Then sketch its Argand diagram and write the number in polar and exponential forms.

(a) $z = 3 - 3\sqrt{3}i$

(b) $z = i\sqrt{3}$

(c) $z = -2\sqrt{3} + 2i$

8. Express each complex number in exponential form:

(a) $3 \left(\cos \frac{\pi}{4} - i \sin \frac{\pi}{4} \right)$

(b) $-3\sqrt{3} - 3i$

(c) $-4 + 4\sqrt{3}i$

9. Calculate the real and imaginary parts of:

(a) $3e^{-\frac{\pi}{3}i}$

(b) $2e^{-\frac{3\pi}{4}i}$

(c) $4e^{\frac{5\pi}{6}i}$

(d) $3e^{-\frac{\pi}{2}i}$

10. For each pair, find the product and quotient using exponential form.

(a) $z_1 = \sqrt{3} + i$ and $z_2 = -1 + i$

(b) $z_1 = -1 - i$ and $z_2 = 1 - \sqrt{3}i$

11. Find:

(a) $\left| \frac{(-2 - 2i)(1 - i\sqrt{3})}{\sqrt{3} - i} \right|$

(b) $\arg \left(\frac{3 - 3i}{(-2 + 2i)(-1 - i)} \right)$

12. Express each expression in both exponential and Cartesian form.

(a) $\frac{2}{\left(\cos \frac{\pi}{6} - i \sin \frac{\pi}{6} \right)^3}$

(b) $\left[2 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) \right]^5$

13. Calculate:

(a) $\arg \left[\left(\cos \frac{\pi}{6} - i \sin \frac{\pi}{6} \right)^5 \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right)^7 \right]$

(b) the modulus and argument of

$$\frac{\left[3 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right) \right]^5}{\left[2 \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right) \right]^3}.$$

14. Simplify to terms in $\cos x$ and $\sin x$.

- (a) $(\cos 7x + i \sin 7x)(\cos 5x - i \sin 5x)$
 (b) $\frac{\cos 2x + i \sin 2x}{\cos 3x - i \sin 3x}$
 (c) $\frac{\cos x - i \sin x}{(\cos 3x + i \sin 3x)(\cos 2x - i \sin 2x)}$

15. Let $z_1 = -4 + 4i$ and $z_2 = -2\sqrt{3} + 2i$. Write the Cartesian form for $(z_1)^9$ and $(z_2)^8$.
 16. Consider the complex numbers

$$z_1 = 2e^{-\frac{5\pi}{6}i} \quad \text{and} \quad z_2 = 4e^{-\frac{3\pi}{4}i}.$$

Find $|z_1 + z_2|$. Hence, show that

$$\cos \frac{11\pi}{12} = -\frac{\sqrt{2}(\sqrt{3} + 1)}{4}.$$

17. Show that $e^{ix} + e^{-ix} = 2 \cos x$ and $e^{ix} - e^{-ix} = 2i \sin x$. Deduce that, for a positive integer n ,

- (a) $e^{inx} + e^{-inx} = 2 \cos(nx)$,
 (b) $e^{inx} - e^{-inx} = 2i \sin(nx)$.

Hence, express $\cos(4x)$ as a sum of powers of $\cos x$ only and $\sin(4x)$ as a sum of products of powers of $\sin x$ and $\cos x$.

18. Prove:

- (a) $\cos(5\theta) = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta$,
 (b) $\sin(5\theta) = 16 \sin^5 \theta - 20 \sin^3 \theta + 5 \sin \theta$.

19. Find the square roots of

$$z = 2 + 2\sqrt{3}i.$$

20. Find the cube roots of the following numbers.

- (a) $z = -3 + 3i$
 (b) $z = 8 \left(\cos \frac{\pi}{3} - i \sin \frac{\pi}{3} \right)$
 (c) $z = 27e^{-\frac{3\pi}{4}i}$

21. What are the tenth roots of 1? How many of these are on the real axis? On the imaginary axis? In the first quadrant? In the other quadrants?

22. Find all z for which $z^5 = -32i$ and $\text{Im}(z) > 0$.

Quadratics, conjugates, and algebraic structure

23. Solve each equation.

(a) $x^2 + 4x + 13 = 0$;

(b) $2x^2 - 4x + 10 = 0$;

(c) $3x^2 + 12x + 39 = 0$;

(d) $x^2 + 9 = 0$.

24. Form a quadratic equation with real integer coefficients having the stated conjugate roots.

(a) $2 \pm 5i$;

(b) $-1 \pm \sqrt{3}i$;

(c) $\frac{3}{2} \pm \frac{1}{2}i$.

25. Construct a monic polynomial of least degree with real coefficients having the stated roots.

(a) $2, 1 + 3i, 1 - 3i$;

(b) $i, -i, 2 + i, 2 - i$.

26. Determine the real values of k for which

(a) $x^2 + 2kx + k^2 + 4 = 0$ has non-real roots;

(b) $x^2 + kx + 9 = 0$ has a repeated root;

(c) $x^2 + (k - 1)x + k = 0$ has two non-real roots.

27. Solve the quadratic equations with complex coefficients.

(a) $z^2 - (1 + i)z + i = 0$;

(b) $z^2 + 2iz - 1 = 0$;

(c) $z^2 - (2 + i)z + (1 + i) = 0$.

28. Use the sum and product of roots.

(a) State the sum and product of the roots of $2z^2 - 6z + 13 = 0$.

(b) A monic quadratic with real coefficients has root $-2 + 5i$. Find its other root and its equation.

(c) Find two numbers whose sum is -4 and whose product is 13 .

29. Factorise $z^4 + 6z^2 + 25$ completely over the complex numbers and into irreducible quadratic factors over the real numbers.

30. Solve $z^2 - 4z + 13 = 0$, plot both roots, and describe their symmetry, common real part, separation, and distance from the origin.
31. The equation $z^2 + (a + 1)z + (b - 2) = 0$ has roots $-2 \pm 3i$. Find the real numbers a and b .
32. A damped oscillator has characteristic equation $r^2 + 6r + 25 = 0$. Find the roots and identify the decay rate and angular frequency represented by them.
33. Solve for z .
- (a) $z + 2\bar{z} = 9 - i$;
 - (b) $3z - \bar{z} = 4 + 8i$;
 - (c) $2z + 3\bar{z} = 10 + 2i$.
34. Find z under each set of conditions.
- (a) $\operatorname{Re}(z) = 4$, $|z| = 5$, and $\operatorname{Im}(z) > 0$;
 - (b) $\operatorname{Im}(z) = -5$, $|z| = 13$, and $\operatorname{Re}(z) > 0$;
 - (c) $z + \bar{z} = -6$, $z\bar{z} = 25$, and $\operatorname{Im}(z) > 0$.
35. Let $z = 2 - 3i$ and $w = -1 + 4i$. Verify numerically that

$$\overline{z + w} = \bar{z} + \bar{w}, \quad \overline{zw} = \bar{z}\bar{w},$$

and explain why these identities hold in general.

Argand geometry and polar operations

36. Plot $1 + i$, $-2 + 2i$, $-3 - i$, and $2 - \sqrt{3}i$ on one Argand diagram. State the quadrant or axis containing each point.
37. For each number, find its modulus, principal argument, polar form, and exponential form.
- (a) $1 + i$;
 - (b) $-\sqrt{3} + i$;
 - (c) $-2 - 2i$;
 - (d) $-4i$.
38. Convert to Cartesian form.
- (a) $2 \operatorname{cis}(2\pi/3)$;
 - (b) $3 \operatorname{cis}(-\pi/6)$;

(c) $4e^{3\pi i/4}$;

(d) $5e^{-\pi i/2}$.

39. Rewrite each expression using a positive modulus and a principal argument.

(a) $-2 \operatorname{cis}(\pi/3)$;

(b) $3 \operatorname{cis}(5\pi/4)$;

(c) $2e^{7\pi i/3}$.

40. Give the principal argument and all arguments of $-1 + i$, $-i$, and $-2 - 2\sqrt{3}i$.

41. If $z = 5e^{2\pi i/3}$, write $\bar{z}$, $1/z$, and $-z$ in exponential form using principal arguments.

42. Find the product of $2 \operatorname{cis}(\pi/6)$ and $3 \operatorname{cis}(-\pi/4)$ in polar and Cartesian forms.

43. Find the quotient

$$\frac{6 \operatorname{cis}(5\pi/6)}{2 \operatorname{cis}(\pi/3)}$$

in polar and Cartesian forms.

44. Let $z_1 = 2 \operatorname{cis}(\pi/6)$, $z_2 = 3 \operatorname{cis}(-\pi/4)$, and $z_3 = 4 \operatorname{cis}(2\pi/3)$. Find $z_1 z_2 / z_3$.

45. Points A and B represent $1 + 2i$ and $-3 + 5i$. Find AB and the complex number representing the midpoint of AB .

46. Point P divides the segment joining $A = 1 + i$ to $B = 7 - 2i$ internally in the ratio $AP : PB = 2 : 1$. Find the complex number representing P .

47. Find the area of the triangle whose vertices represent 0 , $3 + i$, and $1 + 4i$.

48. Let $u = 1 + 2i$ and $v = 4 - 2i$. Use $\operatorname{Re}(u\bar{v})$ to determine whether the corresponding vectors are perpendicular.

49. Write the vertices of the regular hexagon centred at the origin, having circumradius 2 and one vertex at 2, in exponential form.

50. Let $u = e^{\pi i/6}$. Determine all positive integers n for which u^n is real and all positive integers n for which it is purely imaginary. Also find the least positive n for which $(e^{2\pi i/5})^n = 1$.

Euler's formula, complex functions, and logarithms

51. Starting from Euler's formula, prove $e^{ix} + e^{-ix} = 2 \cos x$ and $e^{ix} - e^{-ix} = 2i \sin x$.

52. Express $\cos x$ and $\sin x$ in terms of complex exponentials, and use the formulae to evaluate them at $x = \pi/3$.
53. Simplify.
- (a) $e^{i\pi} + 1$;
 - (b) $e^{i\pi/3}e^{-i\pi/6}$;
 - (c) $e^{2\pi i/3}/e^{-\pi i/3}$.
54. Find all real x satisfying
- (a) $e^{ix} = 1$;
 - (b) $e^{ix} = -1$;
 - (c) $e^{ix} = i$;
 - (d) $e^{ix} = -i$.
55. Evaluate in Cartesian form.
- (a) $e^{2+i\pi/3}$;
 - (b) $e^{-1-i\pi}$;
 - (c) $e^{\ln 3 + 5\pi i/6}$.
56. Use the complex definitions of sine and cosine to find $\cos(2i)$ and $\sin(-i)$.
57. Prove that
- $$\cos(iy) = \cosh y, \quad \sin(iy) = i \sinh y$$
- for real y .
58. Find the principal logarithm of $1 + i$, -4 , and $-i$.
59. Find all complex solutions of
- (a) $e^z = 1$;
 - (b) $e^z = -1$;
 - (c) $e^z = 2i$.
60. Show by taking $z = w = -1$ that the principal logarithm need not satisfy $\operatorname{Log}(zw) = \operatorname{Log} z + \operatorname{Log} w$.
61. Use De Moivre's theorem to derive the standard formulae for $\cos 3x$, $\sin 3x$, $\cos 4x$, and $\sin 4x$.
62. Use complex multiplication to obtain exact values of $\cos 15^\circ$, $\sin 15^\circ$, $\cos 75^\circ$, and $\sin 75^\circ$.

Complex loci and inequalities

63. Describe and sketch each locus.

(a) $|z - (2 - i)| = 3$;

(b) $|2z + 4i| = 6$;

(c) $|z + 1 - 3i| = \sqrt{5}$.

64. Describe and sketch the regions

(a) $|z - (1 + i)| \leq 2$;

(b) $|z + 2| > 3$;

(c) $1 < |z - i| \leq 4$.

65. Find the Cartesian equation of

$$|z - (1 + i)| = |z - (5 - 3i)|$$

and explain its geometric meaning.

66. Show that $|z| = 2|z - 3|$ is a circle. Find its centre and radius.

67. Describe the half-line $\arg(z - (1 - i)) = \pi/3$, including its initial point and direction.

68. Sketch the sector

$$-\frac{\pi}{4} \leq \text{Arg}(z + 2) \leq \frac{\pi}{4}.$$

State whether the vertex is included.

69. Sketch the set satisfying $-1 \leq \text{Re}(z) \leq 3$ and $\text{Im}(z) > 2$.

70. Represent and sketch the rectangle $-2 \leq \text{Re}(z) \leq 3$, $1 < \text{Im}(z) < 4$, carefully indicating its included and excluded edges.

71. Find the half-plane represented by $|z - 1| < |z + 3|$.

72. Find the points satisfying $|z| = 5$ and $\text{Im}(z) = 3$.

73. Find the points satisfying $|z - (2 + i)| = \sqrt{10}$ and $\text{Re}(z) = 5$.

74. Find the intersections of the circles $|z| = 5$ and $|z - 6| = 5$.

75. Under $w = (1 + i)z - 2$, find the images of 0, 1, i , and $1 + i$. Describe the geometric transformation.

76. Find the image of the circle $|z - 1| = 2$ under $w = 2iz + 3$.
77. Describe and sketch $|z - (1 + i)| \leq 2$ together with $\operatorname{Re}(z) \geq 1$.
78. Identify the locus represented by each equation.
- (a) $z\bar{z} - 2\operatorname{Re}(z) = 3$;
 - (b) $\operatorname{Im}((1 - i)z) = 2$;
 - (c) $\operatorname{Re}\left(\frac{z}{1 + i}\right) = 1$.

Applications and modelling

79. Three balanced voltage phasors are 120 , $120e^{2\pi i/3}$, and $120e^{4\pi i/3}$ volts. Find their resultant and explain the significance of the answer.
80. A series circuit has resistance 8Ω , inductive reactance 10Ω , and capacitive reactance 4Ω . Its impedance is $Z = R + i(X_L - X_C)$. Find Z in Cartesian and polar forms. If the applied voltage phasor is 100 volts, find the current phasor $I = V/Z$.
81. Two sinusoidal signals of the same frequency have phasors 3 and $4i$. Find the amplitude and phase of their resultant.
82. A simple digital filter has frequency response $H(\omega) = 1 - e^{-i\omega}$. Find its gain $|H|$ and phase at $\omega = \pi/3$.
83. Multiplication by $\sqrt{3} + i$ models a planar enlargement followed by a rotation. State the scale factor and rotation angle, then find the image of the point $z = 2 - i$.
84. A graphics program maps every point by $w = e^{i\pi/4}z + (2 - i)$. Find the images of the triangle with vertices 0 , 2 , and $2i$, and describe the transformation.
85. A vessel travels 6 km east and then 8 km north. Represent the displacements by complex numbers and find the magnitude and principal argument of the resultant displacement.
86. A robot follows $z_{n+1} = iz_n + 1$ from $z_0 = 0$. Find z_1, z_2, z_3, z_4 and describe the resulting path.
87. A square in a computer graphic has one vertex at $2 + i$ relative to its centre at the origin. Successive vertices are generated by multiplication by i . Find all four vertices.
88. Five equal antenna signals have phases separated by $2\pi/5$. Write their phasors using fifth roots of unity and find their sum.

89. A load has impedance $Z = 12 + 5i\Omega$ and is supplied by a voltage phasor of 130 volts. Find the current, its phase angle, and the power factor $\cos(\arg Z)$.

90. In a series RLC circuit,

$$Z = R + i \left(\omega L - \frac{1}{\omega C} \right).$$

Find the positive resonant angular frequency for $L = 0.2\text{ H}$ and $C = 0.005\text{ F}$, when the imaginary part of Z is zero.

91. The displacement of a damped system satisfies the characteristic equation $r^2 + 4r + 13 = 0$. Find the roots and interpret their real and imaginary parts as decay rate and angular frequency.

92. The point A is represented by $1 + 2i$. Rotate A through 60° anticlockwise about the origin to obtain B . Find B and verify that $OA = OB$.

93. A point $z = 3 - 2i$ is reflected successively in the real axis, the imaginary axis, and the origin. Give the complex number representing each reflected image.

94. A radar target is 20 km from the origin at a bearing represented by an angle of 150° from the positive real axis. Give its Cartesian complex coordinate.

95. Two coherent waves have phasors $4e^{i\pi/6}$ and $4e^{-i\pi/6}$. Find the resultant amplitude and phase, and describe the interference.

96. In a reciprocal mapping $w = 1/z$, find the image of $z = 3 + 4i$. Compare the moduli and arguments of z and w .

97. Starting with $z_0 = 0$, compute four iterations of $z_{n+1} = z_n^2 + i$. Decide whether the values observed so far remain bounded.

98. Two unit clock-hand vectors point at angles 0 and $\pi/2$. Represent their tips by complex numbers and find the distance between the tips.

99. Two beacons are represented by -3 and 3 . A receiver is equidistant from them and exactly 5 units from the first beacon. Find all possible receiver positions.

100. Radio stations A and B are represented by -4 and 4 . A service point z must satisfy the following conditions:

- (a) it is equidistant from A and B ;
- (b) it lies on the boundary $|z + 4| = 5$ of station A 's coverage;
- (c) its direction from A satisfies $-\pi/4 \leq \text{Arg}(z + 4) \leq \pi/4$.

Find all service points satisfying all three conditions and explain the geometry using the diagram.

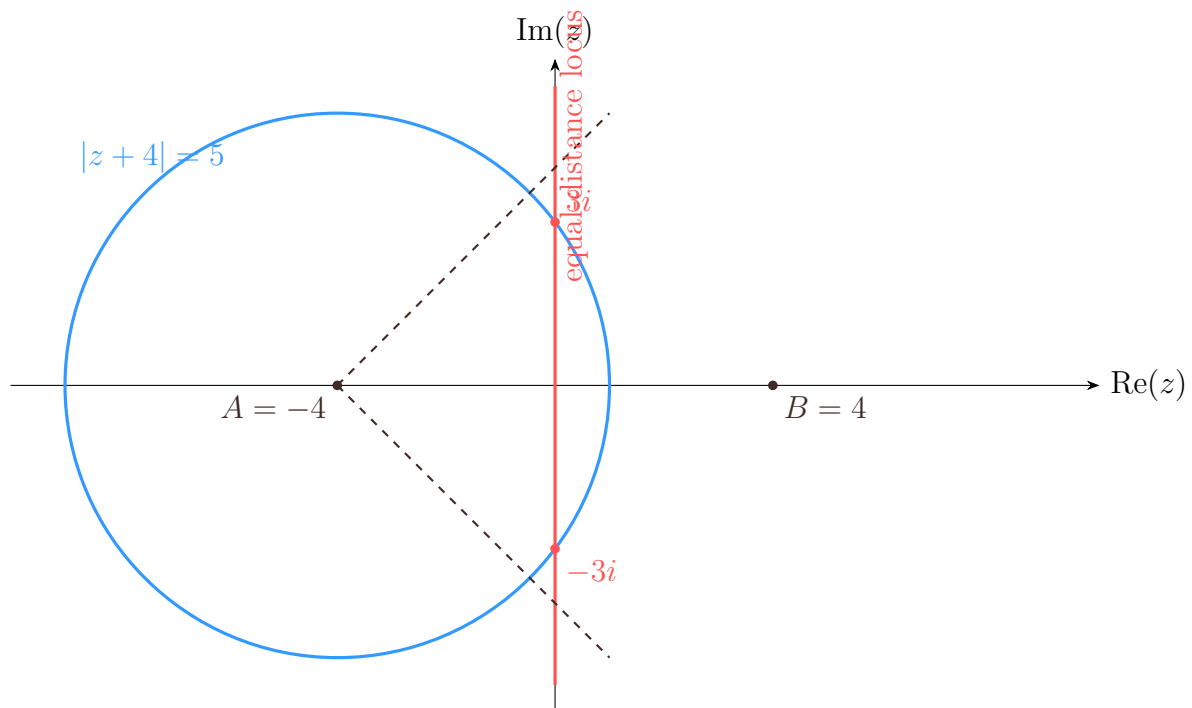


Figure 10.8: Coverage circle, equal-distance line, and permitted angular sector for Question 100.

Part III

Calculus and Series

Chapter 11

Hyperbolic Functions

Chapter guide

Prerequisites. Exponential and logarithmic functions, trigonometric identities, graph transformations, and elementary calculus.

Learning outcomes. Readers should be able to use exponential definitions of hyperbolic functions, derive and apply their identities, interpret their graphs, differentiate and integrate them, and work with inverse hyperbolic functions.

The functions studied so far, from constant and polynomial functions to trigonometric functions, will already be familiar from secondary-school mathematics. Hyperbolic functions form another important family, with definitions and identities closely related to exponential functions. The presentation follows standard foundation and introductory-calculus treatments (Abu Bakar et al. 2025–2026; Neill 2018).

11.1 Definition of hyperbolic functions

Definition 11.1: Hyperbolic functions.

Hyperbolic functions are analogues of the ordinary trigonometric functions, but their geometry is associated with a hyperbola rather than a circle. The basic hyperbolic functions are combinations of e^x and e^{-x} . In particular, e^x decomposes into an even part and an odd part:

$$e^x = \frac{e^x + e^{-x}}{2} + \frac{e^x - e^{-x}}{2}$$

where $(e^x - e^{-x})/2 = \sinh x$ is odd and $(e^x + e^{-x})/2 = \cosh x$ is even.

Many hyperbolic identities resemble trigonometric identities, but differences in sign arise from the contrasting equations of the unit hyperbola and unit circle.

11.2 Defining $y = \sinh x$

$\sinh x$ is called the *hyperbolic sine* of x and is commonly pronounced “shine” or “sinch.” It is defined by

$$\sinh x = \frac{e^x - e^{-x}}{2}. \quad (11.1)$$

The graph can be understood by writing the definition as

$$\sinh x = \frac{e^x}{2} + \left(-\frac{e^{-x}}{2}\right).$$

Thus, $y = \sinh x$ is obtained by adding the ordinates of the two exponential curves $y = \frac{1}{2}e^x$ and $y = -\frac{1}{2}e^{-x}$. At $x = 0$, these two contributions cancel:

$$\sinh 0 = \frac{e^0 - e^0}{2} = 0.$$

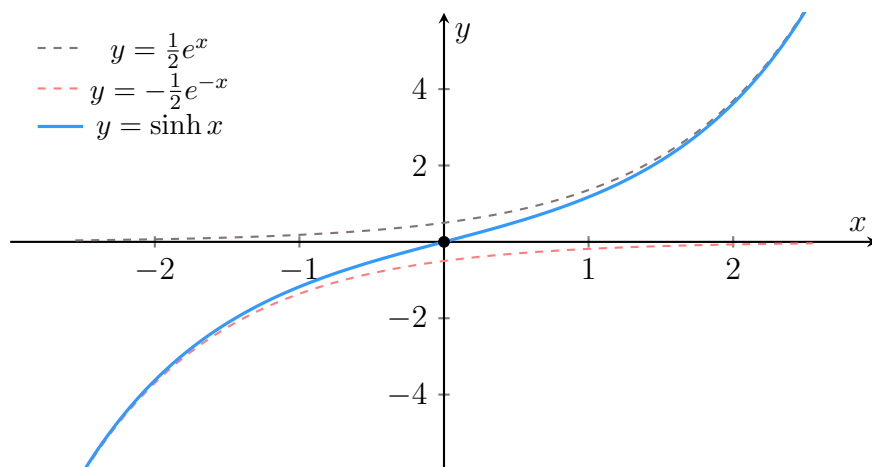


Figure 11.1: The graph of $y = \sinh x$ formed from its two exponential components.

For large positive x , the term e^{-x} approaches zero, so $\sinh x$ behaves approximately like $\frac{1}{2}e^x$ and increases rapidly. For large negative x , the term e^x approaches zero, so $\sinh x$ behaves approximately like $-\frac{1}{2}e^{-x}$ and becomes increasingly negative.

The function is odd. Indeed,

$$\begin{aligned} \sinh(-x) &= \frac{e^{-x} - e^x}{2} \\ &= -\frac{e^x - e^{-x}}{2} \\ &= -\sinh x. \end{aligned}$$

Consequently, the graph has rotational symmetry of order two about the origin. Some useful values are

x	-2	-1	0	1	2
$\sinh x$	-3.627	-1.175	0	1.175	3.627

Table 11.1: Selected values of $\sinh x$, rounded to three decimal places.

The main features of $y = \sinh x$ are therefore:

- its domain and range are both $\mathbb{R}$;

- it crosses both axes at $(0, 0)$;
- it is an odd function, since $\sinh(-x) = -\sinh x$;
- it is increasing for every real value of x ;
- unlike $\sin x$, it is not periodic and is not restricted to the interval $[-1, 1]$.

11.3 Defining $y = \cosh x$

$\cosh x$ is called the *hyperbolic cosine* of x and is commonly pronounced “cosh.” It is defined by

$$\cosh x = \frac{e^x + e^{-x}}{2}. \quad (11.2)$$

This definition may be written as

$$\cosh x = \frac{e^x}{2} + \frac{e^{-x}}{2},$$

so the graph is formed by adding the ordinates of two positive exponential curves. At $x = 0$,

$$\cosh 0 = \frac{e^0 + e^0}{2} = 1.$$

Since both terms in the definition are positive, $\cosh x$ is positive for every real x .

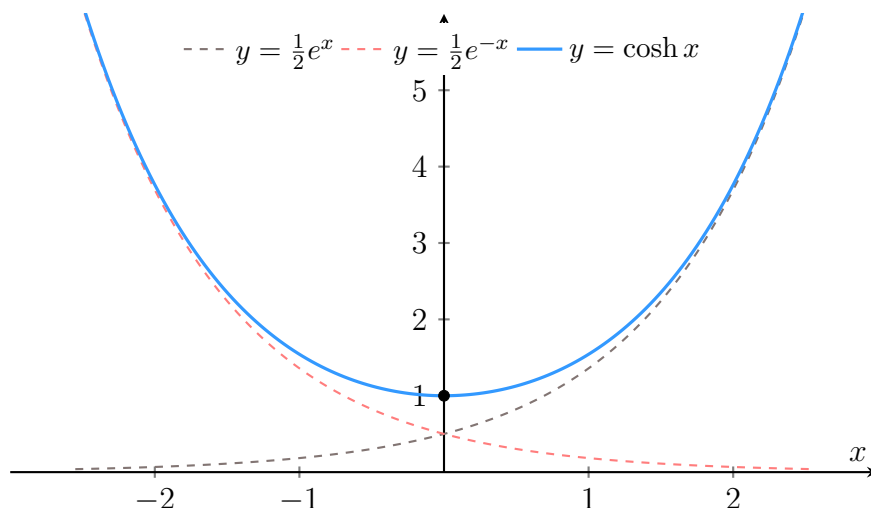


Figure 11.2: The graph of $y = \cosh x$ formed from its two exponential components.

One important physical occurrence of the hyperbolic cosine is the *catenary*: the curve formed by a uniform flexible chain hanging under its own weight. After suitable translation and scaling, its equation is $y = a \cosh(x/a)$. Although a shallow catenary may resemble a parabola, the two curves are not identical.

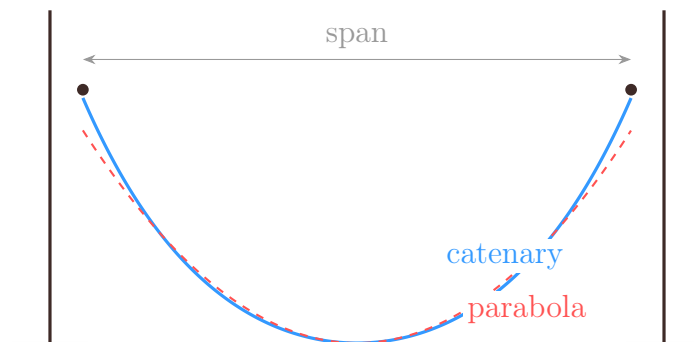


Figure 11.3: A freely hanging uniform chain forms a catenary, modelled by a translated and scaled hyperbolic cosine. The dashed parabola is included to show the small but meaningful difference between the curves.

For large positive x , e^{-x} approaches zero, so $\cosh x$ behaves approximately like $\frac{1}{2}e^x$. For large negative x , e^x approaches zero, so $\cosh x$ behaves approximately like $\frac{1}{2}e^{-x}$. Hence, the graph rises on both sides of its minimum point.

The function is even because

$$\begin{aligned}\cosh(-x) &= \frac{e^{-x} + e^x}{2} \\ &= \frac{e^x + e^{-x}}{2} \\ &= \cosh x.\end{aligned}$$

Consequently, the graph is symmetric about the y -axis. Some useful values are

x	-2	-1	0	1	2
$\cosh x$	3.762	1.543	1	1.543	3.762

Table 11.2: Selected values of $\cosh x$, rounded to three decimal places.

The main features of $y = \cosh x$ are:

- its domain is $\mathbb{R}$ and its range is $[1, \infty)$;
- it crosses the y -axis at $(0, 1)$ and has no x -intercepts;
- it is an even function, since $\cosh(-x) = \cosh x$;
- it decreases as x approaches 0 from the left and increases for $x > 0$;
- its minimum value is 1, attained when $x = 0$;
- unlike $\cos x$, it is not periodic and is never negative.

11.4 Defining $y = \tanh x$

$\tanh x$ is called the *hyperbolic tangent* of x and is commonly pronounced “tanch.” It is defined as the quotient of the hyperbolic sine and hyperbolic cosine:

$$\boxed{\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}.} \quad (11.3)$$

Since $\cosh x \geq 1$, the denominator is never zero. Therefore, $\tanh x$ is defined for every real value of x . At the origin,

$$\tanh 0 = \frac{1 - 1}{1 + 1} = 0.$$

To study the behaviour for large positive x , divide the numerator and denominator by e^x :

$$\tanh x = \frac{1 - e^{-2x}}{1 + e^{-2x}}.$$

As x increases, e^{-2x} approaches zero and hence $\tanh x$ approaches 1. Similarly, as x becomes increasingly negative, $\tanh x$ approaches -1 .

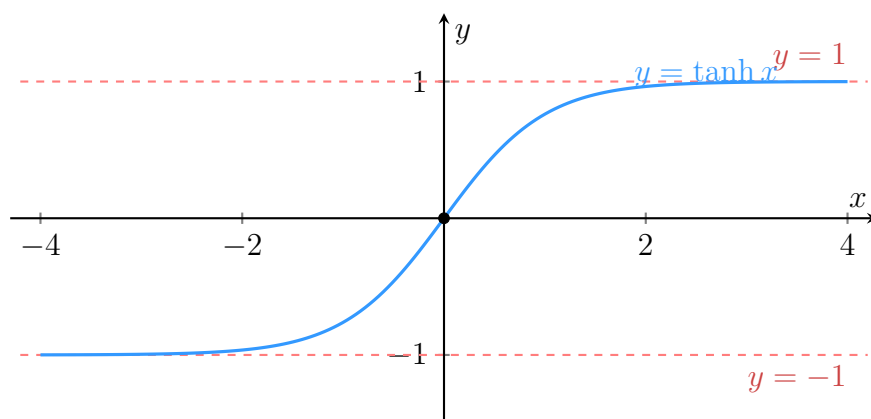


Figure 11.4: The graph of $y = \tanh x$ and its horizontal asymptotes.

The function is odd because $\sinh x$ is odd while $\cosh x$ is even:

$$\tanh(-x) = \frac{\sinh(-x)}{\cosh(-x)} = \frac{-\sinh x}{\cosh x} = -\tanh x.$$

Some useful values are

x	-2	-1	0	1	2
$\tanh x$	-0.964	-0.762	0	0.762	0.964

Table 11.3: Selected values of $\tanh x$, rounded to three decimal places.

The main features of $y = \tanh x$ are:

- its domain is $\mathbb{R}$ and its range is $(-1, 1)$;
- it crosses both axes at $(0, 0)$;
- it is an odd function, since $\tanh(-x) = -\tanh x$;
- it is increasing for every real value of x ;
- it has horizontal asymptotes $y = 1$ and $y = -1$;
- unlike $\tan x$, it is not periodic and has no vertical asymptotes.

11.5 Hyperbolic function identities

The hyperbolic functions satisfy identities that resemble the familiar trigonometric identities. However, an important sign difference appears in the fundamental identity: hyperbolic functions are associated with the hyperbola $x^2 - y^2 = 1$, rather than the circle $x^2 + y^2 = 1$.

11.5.1 Reciprocal hyperbolic functions

By analogy with the reciprocal trigonometric functions, the hyperbolic secant, hyperbolic cosecant, and hyperbolic cotangent are defined by

$$\operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}}, \quad (11.4)$$

$$\operatorname{cosech} x = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}}, \quad (11.5)$$

$$\operatorname{coth} x = \frac{1}{\tanh x} = \frac{\cosh x}{\sinh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}}. \quad (11.6)$$

The abbreviation $\operatorname{csch} x$ is also commonly used in place of $\operatorname{cosech} x$. Since $\cosh x$ is never zero, $\operatorname{sech} x$ is defined for every real x . By contrast, $\sinh 0 = \tanh 0 = 0$, so $\operatorname{cosech} x$ and $\operatorname{coth} x$ are undefined when $x = 0$.

Function	Domain	Range	Parity
$\operatorname{sech} x$	$\mathbb{R}$	$(0, 1]$	Even
$\operatorname{cosech} x$	$\mathbb{R} \setminus \{0\}$	$\mathbb{R} \setminus \{0\}$	Odd
$\operatorname{coth} x$	$\mathbb{R} \setminus \{0\}$	$(-\infty, -1) \cup (1, \infty)$	Odd

Table 11.4: Domains, ranges, and symmetries of the reciprocal hyperbolic functions.

For example, $\operatorname{sech} x$ is even because

$$\operatorname{sech}(-x) = \frac{1}{\cosh(-x)} = \frac{1}{\cosh x} = \operatorname{sech} x.$$

Similarly, the reciprocals of the odd functions $\sinh x$ and $\tanh x$ are odd wherever they are defined.

11.5.2 The fundamental identity

The most important hyperbolic identity is

$$\boxed{\cosh^2 x - \sinh^2 x = 1.} \quad (11.7)$$

It follows directly from the exponential definitions:

$$\begin{aligned} \cosh^2 x - \sinh^2 x &= \left(\frac{e^x + e^{-x}}{2} \right)^2 - \left(\frac{e^x - e^{-x}}{2} \right)^2 \\ &= \frac{e^{2x} + 2 + e^{-2x} - (e^{2x} - 2 + e^{-2x})}{4} \\ &= 1. \end{aligned}$$

Dividing the fundamental identity by $\cosh^2 x$ gives

$$1 - \frac{\sinh^2 x}{\cosh^2 x} = \frac{1}{\cosh^2 x}.$$

Therefore,

$$\boxed{1 - \tanh^2 x = \operatorname{sech}^2 x.} \quad (11.8)$$

For $x \neq 0$, dividing instead by $\sinh^2 x$ gives

$$\frac{\cosh^2 x}{\sinh^2 x} - 1 = \frac{1}{\sinh^2 x}.$$

Hence,

$$\boxed{\coth^2 x - 1 = \operatorname{cosech}^2 x.} \quad (11.9)$$

These three forms should be remembered together:

$$\boxed{\begin{aligned} \cosh^2 x - \sinh^2 x &= 1, \\ 1 - \tanh^2 x &= \operatorname{sech}^2 x, \\ \coth^2 x - 1 &= \operatorname{cosech}^2 x. \end{aligned}}$$

Example 11.1: Simplifying an expression

Simplify $(\cosh x - \sinh x)(\cosh x + \sinh x)$.

Using the difference of two squares,

$$\begin{aligned}(\cosh x - \sinh x)(\cosh x + \sinh x) &= \cosh^2 x - \sinh^2 x \\ &= \boxed{1}.\end{aligned}$$

In fact, the exponential definitions also give

$$\cosh x + \sinh x = e^x, \quad \cosh x - \sinh x = e^{-x},$$

whose product is again 1.

Example 11.2: Using an identity

Simplify

$$\frac{1 - \tanh^2 x}{\operatorname{sech} x}.$$

Since $1 - \tanh^2 x = \operatorname{sech}^2 x$,

$$\frac{1 - \tanh^2 x}{\operatorname{sech} x} = \frac{\operatorname{sech}^2 x}{\operatorname{sech} x} = \boxed{\operatorname{sech} x}.$$

11.5.3 Addition and subtraction formulae

For all real x and y ,

$$\boxed{\sinh(x \pm y) = \sinh x \cosh y \pm \cosh x \sinh y.} \quad (11.10)$$

$$\boxed{\cosh(x \pm y) = \cosh x \cosh y \pm \sinh x \sinh y.} \quad (11.11)$$

The signs in the hyperbolic sine and cosine formulae match the sign on the left. In particular,

$$\begin{aligned}\sinh(x - y) &= \sinh x \cosh y - \cosh x \sinh y, \\ \cosh(x - y) &= \cosh x \cosh y - \sinh x \sinh y.\end{aligned}$$

To verify the addition formula for hyperbolic sine, use the exponential definitions:

$$\begin{aligned}
\sinh x \cosh y + \cosh x \sinh y &= \frac{(e^x - e^{-x})(e^y + e^{-y})}{4} \\
&\quad + \frac{(e^x + e^{-x})(e^y - e^{-y})}{4} \\
&= \frac{e^{x+y} - e^{-(x+y)}}{2} \\
&= \sinh(x+y).
\end{aligned}$$

Dividing the formula for $\sinh(x \pm y)$ by the corresponding formula for $\cosh(x \pm y)$ gives

$$\tanh(x \pm y) = \frac{\tanh x \pm \tanh y}{1 \pm \tanh x \tanh y} \quad (11.12)$$

whenever the denominator is nonzero.

Example 11.3: Applying an addition formula

Suppose

$$\sinh a = \frac{3}{4}, \quad \cosh a = \frac{5}{4},$$

and

$$\sinh b = \frac{5}{12}, \quad \cosh b = \frac{13}{12}.$$

Then

$$\begin{aligned}
\sinh(a+b) &= \sinh a \cosh b + \cosh a \sinh b \\
&= \left(\frac{3}{4}\right) \left(\frac{13}{12}\right) + \left(\frac{5}{4}\right) \left(\frac{5}{12}\right) \\
&= \frac{39}{48} + \frac{25}{48} \\
&= \frac{4}{3}.
\end{aligned}$$

The supplied values are consistent with the fundamental identity because

$$\left(\frac{5}{4}\right)^2 - \left(\frac{3}{4}\right)^2 = 1 \quad \text{and} \quad \left(\frac{13}{12}\right)^2 - \left(\frac{5}{12}\right)^2 = 1.$$

11.5.4 Double-angle and power identities

Setting $y = x$ in the addition formulae gives

$$\begin{aligned}
 \sinh 2x &= 2 \sinh x \cosh x, \\
 \cosh 2x &= \cosh^2 x + \sinh^2 x, \\
 \tanh 2x &= \frac{2 \tanh x}{1 + \tanh^2 x}.
 \end{aligned}
 \tag{11.13}$$

Using $\cosh^2 x - \sinh^2 x = 1$, the formula for $\cosh 2x$ can be written in two other useful forms:

$$\cosh 2x = 2 \cosh^2 x - 1 = 1 + 2 \sinh^2 x. \tag{11.14}$$

Rearranging these equations gives the power identities

$$\cosh^2 x = \frac{\cosh 2x + 1}{2}, \quad \sinh^2 x = \frac{\cosh 2x - 1}{2}. \tag{11.15}$$

These formulae are useful when simplifying expressions and, later, when integrating powers of hyperbolic functions.

Example 11.4: Expressing a power using a double angle

Write $3 \cosh^2 x - 2 \sinh^2 x$ in terms of $\cosh 2x$.

Substituting the power identities,

$$\begin{aligned}
 3 \cosh^2 x - 2 \sinh^2 x &= 3 \left(\frac{\cosh 2x + 1}{2} \right) - 2 \left(\frac{\cosh 2x - 1}{2} \right) \\
 &= \frac{3 \cosh 2x + 3 - 2 \cosh 2x + 2}{2} \\
 &= \frac{\cosh 2x + 5}{2}.
 \end{aligned}$$

11.5.5 Summary of identities

Type	Identity
Fundamental	$\cosh^2 x - \sinh^2 x = 1$
Reciprocal forms	$1 - \tanh^2 x = \operatorname{sech}^2 x$, $\coth^2 x - 1 = \operatorname{cosech}^2 x$
Addition	$\sinh(x \pm y) = \sinh x \cosh y \pm \cosh x \sinh y$
Addition	$\cosh(x \pm y) = \cosh x \cosh y \pm \sinh x \sinh y$
Addition	$\tanh(x \pm y) = \frac{\tanh x \pm \tanh y}{1 \pm \tanh x \tanh y}$
Double angle	$\sinh 2x = 2 \sinh x \cosh x$, $\tanh 2x = \frac{2 \tanh x}{1 + \tanh^2 x}$
Double angle	$\cosh 2x = \cosh^2 x + \sinh^2 x = 2 \cosh^2 x - 1 = 1 + 2 \sinh^2 x$
Power	$\cosh^2 x = \frac{\cosh 2x + 1}{2}$, $\sinh^2 x = \frac{\cosh 2x - 1}{2}$

Table 11.5: A-level hyperbolic identities.

11.6 Inverse hyperbolic functions

In general, by reflecting the graphs of the hyperbolic functions about the line $y = x$ with some appropriate restrictions, we can establish the corresponding inverse hyperbolic functions.

11.6.1 Principal inverse functions

An inverse function is obtained by interchanging the input and output. Its graph is therefore the reflection of the original graph in the line $y = x$. The functions $\sinh x$, $\tanh x$, $\operatorname{cosech} x$, and $\coth x$ are one-to-one on their real domains. However, $\cosh x$ and $\operatorname{sech} x$ are even functions, so their domains are restricted to $x \geq 0$ when defining their principal inverses.

The superscript -1 denotes a functional inverse in this section. It does not denote a reciprocal. For example,

$$\sinh^{-1} x \neq \frac{1}{\sinh x}, \quad \frac{1}{\sinh x} = \operatorname{cosech} x.$$

Function	Domain	Range	Logarithmic form
$\sinh^{-1} x$	$\mathbb{R}$	$\mathbb{R}$	$\ln \left(x + \sqrt{x^2 + 1} \right)$
$\cosh^{-1} x$	$[1, \infty)$	$[0, \infty)$	$\ln \left(x + \sqrt{x^2 - 1} \right)$
$\tanh^{-1} x$	$(-1, 1)$	$\mathbb{R}$	$\frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$
$\operatorname{cosech}^{-1} x$	$\mathbb{R} \setminus \{0\}$	$\mathbb{R} \setminus \{0\}$	$\ln \left(\frac{1}{x} + \sqrt{1 + \frac{1}{x^2}} \right)$
$\operatorname{sech}^{-1} x$	$(0, 1]$	$[0, \infty)$	$\ln \left(\frac{1 + \sqrt{1 - x^2}}{x} \right)$
$\coth^{-1} x$	$(-\infty, -1) \cup (1, \infty)$	$\mathbb{R} \setminus \{0\}$	$\frac{1}{2} \ln \left(\frac{x+1}{x-1} \right)$

Table 11.6: Principal real inverse hyperbolic functions.

Logarithmic forms of the inverse hyperbolic functions

$$\begin{aligned}
 \sinh^{-1} x &= \ln \left(x + \sqrt{x^2 + 1} \right), & x &\in \mathbb{R}, \\
 \cosh^{-1} x &= \ln \left(x + \sqrt{x^2 - 1} \right), & x &\geq 1, \\
 \tanh^{-1} x &= \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right), & |x| &< 1, \\
 \operatorname{cosech}^{-1} x &= \ln \left(\frac{1}{x} + \frac{\sqrt{1+x^2}}{|x|} \right), & x &\neq 0, \\
 \operatorname{sech}^{-1} x &= \ln \left(\frac{1 + \sqrt{1-x^2}}{x} \right), & 0 &< x \leq 1, \\
 \coth^{-1} x &= \frac{1}{2} \ln \left(\frac{x+1}{x-1} \right), & |x| &> 1.
 \end{aligned}$$

The principal value of $\cosh^{-1} x$ is nonnegative. When solving the equation $\cosh y = x$ without imposing the principal-range restriction, both values

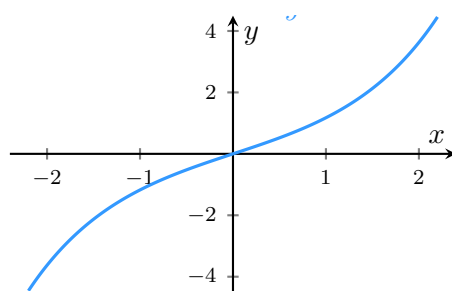
$$y = \pm \cosh^{-1} x = \pm \ln \left(x + \sqrt{x^2 - 1} \right)$$

must be considered.

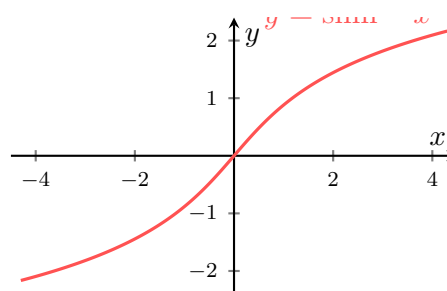
11.6.2 Inverse hyperbolic sine

Since $\sinh x$ is strictly increasing on $\mathbb{R}$, no restriction is needed. Thus

$$y = \sinh^{-1} x \iff x = \sinh y.$$



Hyperbolic sine



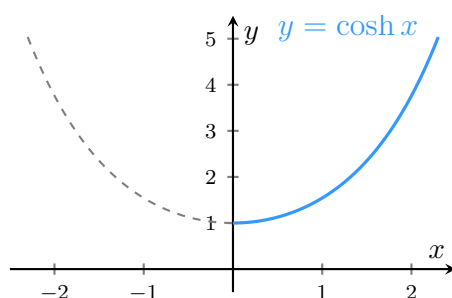
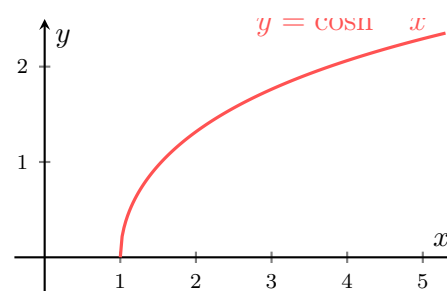
Inverse hyperbolic sine

Figure 11.5: The graphs of $y = \sinh x$ and $y = \sinh^{-1} x$.

11.6.3 Inverse hyperbolic cosine

The graph of $\cosh x$ is symmetric about the y -axis. To obtain a function with an inverse, its domain is restricted to $x \geq 0$. Therefore,

$$y = \cosh^{-1} x \iff x = \cosh y, \quad y \geq 0.$$

Restricted branch $x \geq 0$ 

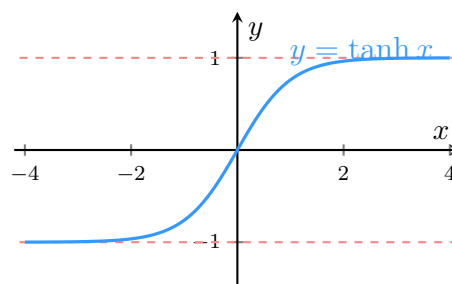
Inverse hyperbolic cosine

Figure 11.6: The principal inverse of $\cosh x$ uses the branch $x \geq 0$.

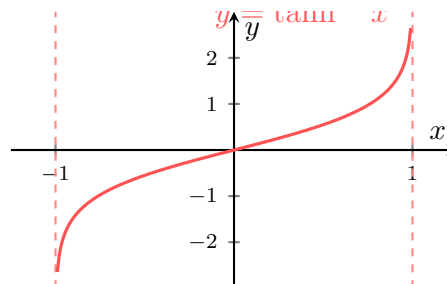
11.6.4 Inverse hyperbolic tangent

The function $\tanh x$ is strictly increasing with range $(-1, 1)$. Hence,

$$y = \tanh^{-1} x \iff x = \tanh y, \quad -1 < x < 1.$$



Hyperbolic tangent



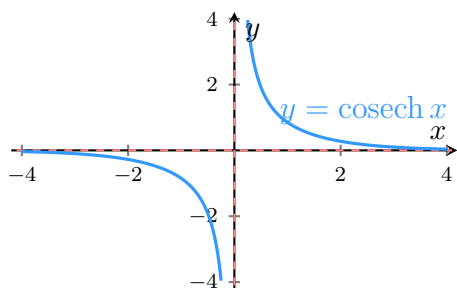
Inverse hyperbolic tangent

Figure 11.7: The horizontal asymptotes of $\tanh x$ become the domain boundaries of $\tanh^{-1} x$.

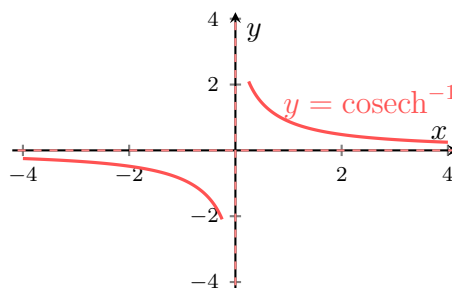
11.6.5 Inverse hyperbolic cosecant

The hyperbolic cosecant is one-to-one on $\mathbb{R} \setminus \{0\}$. Its inverse satisfies

$$y = \operatorname{cosech}^{-1} x \iff x = \operatorname{cosech} y.$$



Hyperbolic cosecant



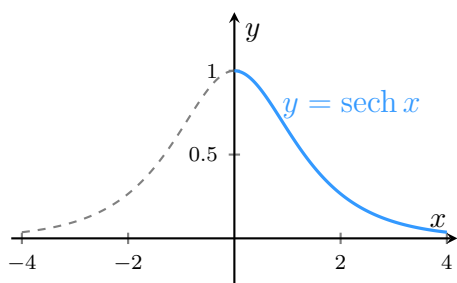
Inverse hyperbolic cosecant

Figure 11.8: The graphs of hyperbolic cosecant and its inverse.

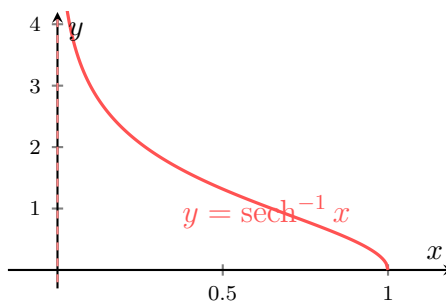
11.6.6 Inverse hyperbolic secant

Since $\operatorname{sech} x$ is even, its principal inverse is defined using the restriction $x \geq 0$:

$$y = \operatorname{sech}^{-1} x \iff x = \operatorname{sech} y, \quad y \geq 0.$$



Restricted branch $x \geq 0$



Inverse hyperbolic secant

Figure 11.9: The principal inverse of $\operatorname{sech} x$ uses the branch $x \geq 0$.

11.6.7 Inverse hyperbolic cotangent

The hyperbolic cotangent maps its two real branches onto $(-\infty, -1) \cup (1, \infty)$. Its inverse is defined by

$$y = \operatorname{coth}^{-1} x \iff x = \operatorname{coth} y.$$

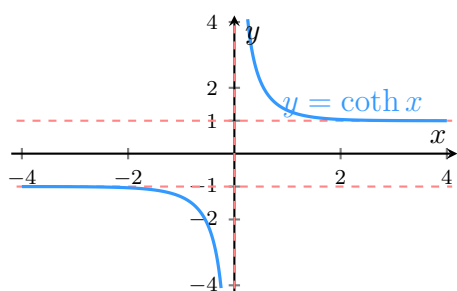
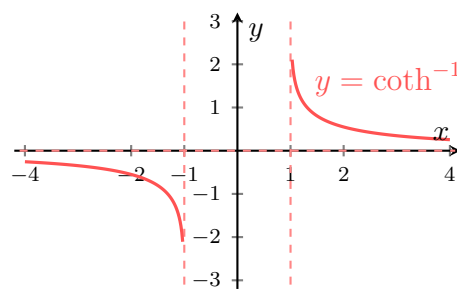
**Hyperbolic cotangent****Inverse hyperbolic cotangent**

Figure 11.10: The graphs of hyperbolic cotangent and its inverse.

Chapter summary and common pitfalls

Key ideas. Hyperbolic functions are built from exponentials and satisfy identities resembling, but not identical to, trigonometric identities. Their inverses can be expressed logarithmically.

Common pitfalls. Remember the sign in $\cosh^2 x - \sinh^2 x = 1$, distinguish inverse functions from reciprocals, and check the domains of logarithmic inverse-hyperbolic formulae.

11.7 Practice and exercises

Unless otherwise stated, give exact answers and work over the real numbers. The notation $\sinh^{-1}$, $\cosh^{-1}$, and $\tanh^{-1}$ denotes inverse hyperbolic functions, not reciprocals.

Definitions and fundamental properties

1. Write each function in terms of e^x and e^{-x} :

- (a) $\sinh x$;
- (b) $\cosh x$;
- (c) $\tanh x$;
- (d) $\operatorname{sech} x$;
- (e) $\operatorname{cosech} x$;
- (f) $\coth x$.

2. Evaluate, where defined,

$$\sinh 0, \quad \cosh 0, \quad \tanh 0, \quad \operatorname{sech} 0, \quad \operatorname{cosech} 0, \quad \coth 0.$$

3. Given $e^x = 3$, find the exact values of all six hyperbolic functions of x .

4. Given $\sinh x = 5/12$ and $x > 0$, find

- (a) $\cosh x$;
- (b) $\tanh x$;
- (c) $\operatorname{sech} x$;
- (d) $\operatorname{cosech} x$;
- (e) $\coth x$.

5. Given $\cosh x = 13/5$ and $x < 0$, find the other five hyperbolic functions of x .

6. Given $\tanh x = -3/5$, determine $\sinh x$, $\cosh x$, and the three reciprocal hyperbolic functions.

7. Find exact values at the stated arguments:

- (a) $\sinh(\ln 2)$ and $\cosh(\ln 2)$;
- (b) $\tanh(\ln 3)$ and $\operatorname{sech}(\ln 3)$;
- (c) $\operatorname{cosech}(\ln 4)$ and $\coth(\ln 4)$.

8. Show from the definitions that

$$e^x = \cosh x + \sinh x, \quad e^{-x} = \cosh x - \sinh x.$$

Hence express $e^{2x} + e^{-2x}$ and $e^{2x} - e^{-2x}$ using hyperbolic functions.

9. Simplify:

- (a) $(\cosh x + \sinh x)(\cosh x - \sinh x)$;
- (b) $\cosh^2 x - 1$;
- (c) $1 - \tanh^2 x$;
- (d) $\coth^2 x - 1$.

10. Prove the parity statements

- (a) $\sinh(-x) = -\sinh x$;
- (b) $\cosh(-x) = \cosh x$;
- (c) $\tanh(-x) = -\tanh x$;
- (d) $\operatorname{sech}(-x) = \operatorname{sech} x$.

11. State the domain, range, parity, intercepts, and horizontal asymptotes, where applicable, of

- (a) $\sinh x$;
- (b) $\cosh x$;
- (c) $\tanh x$;
- (d) $\operatorname{sech} x$.

12. Solve over the real numbers:

- (a) $\sinh x = 0$;
- (b) $\sinh x = 3/4$;
- (c) $\sinh x = -5/12$.

13. Solve over the real numbers:

- (a) $\cosh x = 1$;

(b) $\cosh x = 5/4$;

(c) $\cosh x = 1/2$.

14. Solve over the real numbers:

(a) $\tanh x = 1/2$;

(b) $\tanh x = -3/5$;

(c) $\tanh x = 2$.

15. Solve, stating any excluded values:

(a) $\operatorname{sech} x = 4/5$;

(b) $\operatorname{cosech} x = 12/5$;

(c) $\coth x = -5/3$.

Identities and equations

16. Prove $\cosh^2 x - \sinh^2 x = 1$ directly from the exponential definitions.

17. Derive

$$1 - \tanh^2 x = \operatorname{sech}^2 x, \quad \coth^2 x - 1 = \operatorname{cosech}^2 x.$$

18. Expand using the addition and subtraction formulae:

(a) $\sinh(x + y)$;

(b) $\sinh(x - y)$;

(c) $\cosh(x + y)$;

(d) $\cosh(x - y)$;

(e) $\tanh(x + y)$.

19. Suppose $\sinh a = 3/4$, $\cosh a = 5/4$, $\sinh b = 5/12$, and $\cosh b = 13/12$. Find

(a) $\sinh(a + b)$;

(b) $\cosh(a + b)$;

(c) $\sinh(a - b)$;

(d) $\cosh(a - b)$;

(e) $\tanh(a + b)$.

20. Given $\tanh a = 1/2$ and $\tanh b = 1/3$, find $\tanh(a + b)$, $\tanh(a - b)$, and $\tanh 2a$.

21. Given $\sinh x = 3/4$ and $x > 0$, find $\sinh 2x$, $\cosh 2x$, and $\tanh 2x$.

22. Given $\cosh 2x = 7$ and $x > 0$, find $\sinh x$, $\cosh x$, and $\tanh x$.
23. Express in terms of $\cosh 2x$:
- (a) $\cosh^2 x$;
 - (b) $\sinh^2 x$;
 - (c) $5 \cosh^2 x - 3 \sinh^2 x$;
 - (d) $2 \sinh^2 x + 7$.
24. Prove:
- (a) $\cosh(x + y) + \cosh(x - y) = 2 \cosh x \cosh y$;
 - (b) $\cosh(x + y) - \cosh(x - y) = 2 \sinh x \sinh y$;
 - (c) $\sinh(x + y) + \sinh(x - y) = 2 \sinh x \cosh y$.
25. Simplify:
- (a) $\frac{1 - \tanh^2 x}{\operatorname{sech} x}$;
 - (b) $\frac{\cosh 2x - 1}{\sinh 2x}$;
 - (c) $\frac{\cosh x + \sinh x}{\cosh x - \sinh x}$;
 - (d) $(\coth x - \operatorname{cosech} x)(\coth x + \operatorname{cosech} x)$.
26. Find all possible values of $\sinh x$ satisfying $12 \cosh^2 x + 7 \sinh x = 24$.
27. Solve over $\mathbb{R}$:
- (a) $2 \sinh^2 x - \sinh x - 1 = 0$;
 - (b) $2 \cosh^2 x - 5 \cosh x + 2 = 0$;
 - (c) $3 \tanh^2 x + 2 \tanh x - 1 = 0$.
28. Solve using exponential definitions:
- (a) $\sinh x = 4$;
 - (b) $\cosh x = 2$;
 - (c) $\tanh x = 3/4$.
29. Solve:
- (a) $\cosh 2x = 5$;
 - (b) $\sinh 2x = 3$;

(c) $\tanh 2x = 1/2$.

30. Solve:

(a) $\sinh(x + \ln 2) = 0$;

(b) $\cosh(x - 1) = \cosh 3$;

(c) $\tanh(2x + 1) = \tanh 5$.

31. Derive

$$\sinh 3x = 3 \sinh x + 4 \sinh^3 x, \quad \cosh 3x = 4 \cosh^3 x - 3 \cosh x.$$

32. Given $\sinh x = 5/12$ and $x > 0$, find exact values of $\sinh 3x$, $\cosh 3x$, and $\cosh 5x$.

33. Express each as a polynomial in the indicated function:

(a) $\cosh 4x$ in $\cosh x$;

(b) $\sinh 4x$ in $\sinh x$ and $\cosh x$;

(c) $\tanh 3x$ in $\tanh x$.

34. Let $C_n = \cosh(nx)$. Show that

$$C_{n+1} = 2 \cosh x C_n - C_{n-1}.$$

Use the recurrence to obtain $\cosh 4x$ from C_0 and C_1 .

35. Prove that for every real x ,

(a) $\cosh x \geq 1$;

(b) $|\sinh x| < \cosh x$;

(c) $|\tanh x| < 1$.

Graphs and transformations

36. Sketch $y = \sinh x$, marking its intercept, symmetry, domain, and range.

37. Sketch $y = \cosh x$, marking its minimum point, symmetry, domain, and range.

38. Sketch $y = \tanh x$, marking its intercept and both horizontal asymptotes.

39. Sketch $y = \operatorname{sech} x$, marking its maximum point, symmetry, and horizontal asymptote.

40. On separate axes, sketch $y = \operatorname{cosech} x$ and $y = \coth x$. State all asymptotes and excluded values.

41. Describe the transformations from the appropriate basic graph:
- (a) $y = 2 \sinh(x - 1) + 3$;
 - (b) $y = -\cosh(2x) + 1$;
 - (c) $y = \tanh(x + 2) - 1$;
 - (d) $y = 3 \operatorname{sech}(x/2)$.
42. State the intercepts, centre or turning point, range, and asymptotes where applicable:
- (a) $y = \sinh(x - 2)$;
 - (b) $y = 4 - 2 \cosh x$;
 - (c) $y = 3 \tanh(2x) + 1$.
43. Find the intersections of
- (a) $y = \sinh x$ and $y = 0$;
 - (b) $y = \cosh x$ and $y = 2$;
 - (c) $y = \tanh x$ and $y = 1/2$.
44. State the number of real roots of each equation and justify your answer from a graph:
- (a) $\cosh x = -2$;
 - (b) $\operatorname{sech} x = 1/2$;
 - (c) $\tanh x = 1$;
 - (d) $\sinh x = 7$.
45. The point $P(t) = (\cosh t, \sinh t)$ moves on the unit hyperbola.
- (a) Verify that $P(t)$ lies on $X^2 - Y^2 = 1$.
 - (b) Find $P(0)$ and $P(\ln 2)$.
 - (c) Explain geometrically why $P(-t)$ is the reflection of $P(t)$ in the X -axis.
46. Complete a table of $\sinh x$, $\cosh x$, and $\tanh x$ at $x = -2, -1, 0, 1, 2$, correct to three decimal places. Use it to compare their growth and symmetry.
47. Determine the limiting behaviour as $x \rightarrow \infty$ and $x \rightarrow -\infty$ of
- (a) $\sinh x$;
 - (b) $\cosh x$;
 - (c) $\tanh x$;
 - (d) $\operatorname{sech} x$.

48. Match each description to a function:

- (a) even, positive, maximum 1, tending to 0 at both ends;
- (b) odd, vertical asymptote $x = 0$, horizontal asymptotes $y = \pm 1$;
- (c) odd, range $\mathbb{R}$, one intercept;
- (d) even, minimum 1, no horizontal asymptote.

49. Explain how the graphs of a one-to-one hyperbolic function and its inverse are related. Illustrate the statement for $y = \sinh x$ and $y = \sinh^{-1} x$.

50. Explain why $\cosh x$ and $\operatorname{sech} x$ require domain restrictions before inverses are defined. State the principal branch used for each.

Inverse hyperbolic functions

51. Explain the difference between $\sinh^{-1} x$ and $1/\sinh x$. Write the reciprocal using unambiguous notation.

52. State the domain and range of each principal inverse: $\sinh^{-1}$, $\cosh^{-1}$, $\tanh^{-1}$, $\operatorname{cosech}^{-1}$, sech^{-1} , and coth^{-1} .

53. Evaluate exactly:

- (a) $\sinh^{-1} 0$;
- (b) $\cosh^{-1} 1$;
- (c) $\tanh^{-1} 0$;
- (d) $\operatorname{sech}^{-1} 1$.

54. Use $\sinh^{-1} x = \ln(x + \sqrt{x^2 + 1})$ to evaluate

- (a) $\sinh^{-1}(3/4)$;
- (b) $\sinh^{-1}(5/12)$;
- (c) $\sinh^{-1}(-3/4)$.

55. Use $\cosh^{-1} x = \ln(x + \sqrt{x^2 - 1})$ to evaluate

- (a) $\cosh^{-1}(5/4)$;
- (b) $\cosh^{-1}(13/12)$;
- (c) $\cosh^{-1} 2$.

56. Use $\tanh^{-1} x = \frac{1}{2} \ln((1+x)/(1-x))$ to evaluate

- (a) $\tanh^{-1}(1/2)$;

- (b) $\tanh^{-1}(-3/5)$;
- (c) $\tanh^{-1}(5/13)$.

57. Evaluate exactly:

- (a) $\operatorname{cosech}^{-1}(4/3)$;
- (b) $\operatorname{cosech}^{-1}(-12/5)$;
- (c) $\operatorname{cosech}^{-1} 1$.

58. Evaluate exactly:

- (a) $\operatorname{sech}^{-1}(4/5)$;
- (b) $\operatorname{sech}^{-1}(12/13)$;
- (c) $\operatorname{sech}^{-1}(1/2)$.

59. Evaluate exactly:

- (a) $\operatorname{coth}^{-1} 2$;
- (b) $\operatorname{coth}^{-1}(-5/3)$;
- (c) $\operatorname{coth}^{-1}(13/5)$.

60. Prove that

$$\sinh^{-1} x = \ln \left(x + \sqrt{1 + x^2} \right).$$

61. Derive the logarithmic form of $\cosh^{-1} x$, stating why $x \geq 1$ and why the positive square root is selected.

62. Derive the logarithmic form of $\tanh^{-1} x$, stating its domain.

63. Derive logarithmic forms for $\operatorname{cosech}^{-1} x$, $\operatorname{sech}^{-1} x$, and $\operatorname{coth}^{-1} x$ from the corresponding reciprocal equations.

64. Simplify, respecting domains:

- (a) $\sinh(\sinh^{-1} x)$;
- (b) $\cosh(\cosh^{-1} x)$;
- (c) $\tanh(\tanh^{-1} x)$;
- (d) $\operatorname{sech}(\operatorname{sech}^{-1} x)$.

65. Simplify, respecting principal branches:

- (a) $\sinh^{-1}(\sinh x)$;

- (b) $\cosh^{-1}(\cosh x)$;
- (c) $\tanh^{-1}(\tanh x)$;
- (d) $\operatorname{sech}^{-1}(\operatorname{sech} x)$.

66. If $u = \sinh^{-1} x$, express $\cosh u$, $\tanh u$, and $\operatorname{sech} u$ algebraically in terms of x .
67. If $u = \cosh^{-1} x$, express $\sinh u$, $\tanh u$, and $\operatorname{cosech} u$ algebraically in terms of x .
68. If $u = \tanh^{-1} x$, express $\cosh u$, $\sinh u$, and $\operatorname{sech} u$ algebraically in terms of x .
69. Solve:

- (a) $\sinh^{-1} x = \ln 3$;
- (b) $\cosh^{-1} x = \ln 2$;
- (c) $\tanh^{-1} x = \frac{1}{2} \ln 5$.

70. Solve, or state that no real solution exists:

- (a) $\cosh^{-1}(2x - 1) = 0$;
- (b) $\tanh^{-1}(3x) = \ln 2$;
- (c) $\operatorname{sech}^{-1}(x/2) = \ln 3$.

71. Prove that $\sinh^{-1}(-x) = -\sinh^{-1} x$ and $\tanh^{-1}(-x) = -\tanh^{-1} x$.

72. Show that

$$\cosh^{-1}(\cosh x) = |x| \quad \text{and} \quad \operatorname{sech}^{-1}(\operatorname{sech} x) = |x|.$$

73. Prove, for real x , that

$$\sinh^{-1} x = \tanh^{-1} \left(\frac{x}{\sqrt{1+x^2}} \right).$$

74. Prove, for $x \geq 1$, that

$$\cosh^{-1} x = \sinh^{-1} \sqrt{x^2 - 1}.$$

75. Use an addition identity to show that, whenever the expressions are defined,

$$\tanh^{-1} u + \tanh^{-1} v = \tanh^{-1} \left(\frac{u+v}{1+uv} \right),$$

subject to the correct principal-value condition.

76. Write each logarithm as an inverse hyperbolic value:

- (a) $\ln 2$;

- (b) $\ln(2 + \sqrt{3})$;
- (c) $\frac{1}{2} \ln 3$;
- (d) $\ln((3 + \sqrt{5})/2)$.

77. Find the real values of x for which each expression is defined:

- (a) $\cosh^{-1}(2x - 3)$;
- (b) $\tanh^{-1}(x^2 - 2)$;
- (c) $\operatorname{sech}^{-1}(1 - x)$;
- (d) $\coth^{-1}(2x)$.

78. Simplify:

- (a) $\sinh(2 \sinh^{-1} x)$;
- (b) $\cosh(2 \sinh^{-1} x)$;
- (c) $\tanh(2 \tanh^{-1} x)$;
- (d) $\cosh(3 \cosh^{-1} x)$.

79. Solve

$$\sinh^{-1} x + \sinh^{-1}(3/4) = \ln 2.$$

80. Find x , or state that no real solution exists:

- (a) $\sinh^{-1}(3/4) + \sinh^{-1} x = \sinh^{-1}(4/3)$;
- (b) $\cosh^{-1} x + \cosh^{-1}(3/4) = -\cosh^{-1}(4/3)$;
- (c) $\tanh^{-1}(2/5) - \tanh^{-1} x = \sinh^{-1}(3/5)$.

Applications and modelling

81. A cable is modelled by $y = 12 \cosh(x/12) - 12$, where distances are in metres. Find its height at $x = 0$ and $x = 12$, and find the two values of x at which the cable is 6 m above its lowest point.
82. A symmetric bridge cable is modelled by $y = 8 \cosh(x/8)$ between towers at $x = \pm 10$.

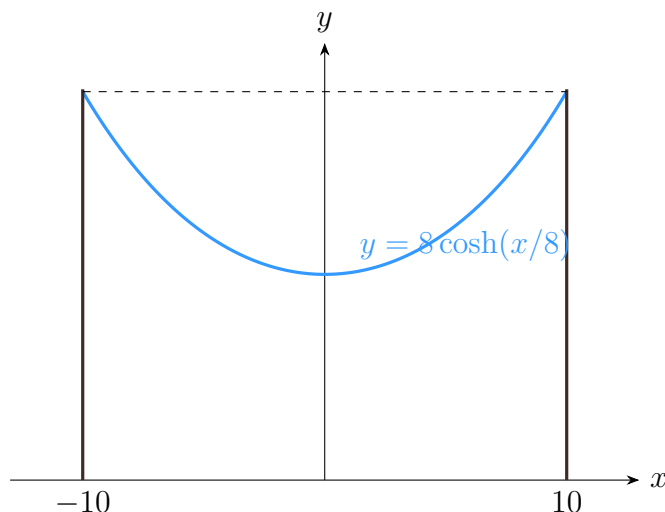


Figure 11.11: The cable and towers in Question 82.

Find the cable height at the centre and at each tower, then calculate the vertical sag from tower level to the centre.

83. The length of a symmetric hanging chain between $x = -L/2$ and $x = L/2$ is given by $S = 2a \sinh(L/(2a))$. For $a = 10$ m and $L = 12$ m, find S . If instead $S = 15$ m, express L exactly using an inverse hyperbolic function.
84. An inverted catenary arch has height $h(x) = 20 - 5 \cosh(x/5)$ metres. Find its central height, its ground-level span, and the height at $x = 4$ m.
85. A point on the right branch of $X^2 - Y^2 = 1$ is represented by $(X, Y) = (\cosh t, \sinh t)$.
- Find the point when $t = \ln 2$.
 - Find t when the point is $(5/4, 3/4)$.
 - Explain why increasing t moves the point away from the vertex $(1, 0)$.
86. In special relativity, a dimensionless speed is written $\beta = \tanh \theta$, where θ is the rapidity. The Lorentz factor is $\gamma = \cosh \theta$ and $\gamma\beta = \sinh \theta$. Find θ , γ , and $\gamma\beta$ when $\beta = 3/5$.
87. Collinear dimensionless speeds combine according to

$$w = \frac{u + v}{1 + uv}.$$

For $u = 1/2$ and $v = 1/3$, find w . Show that if $u = \tanh \alpha$ and $v = \tanh \beta$, then $w = \tanh(\alpha + \beta)$.

88. The speed of an object approaching a terminal value is $v(t) = 60 \tanh(0.08t)$ ms^{-1} . Find $v(20)$ and the time when $v = 50$ ms^{-1} .

89. A sensor has response $S(c) = 5 \tanh(0.4c)$ volts for input $c \geq 0$.

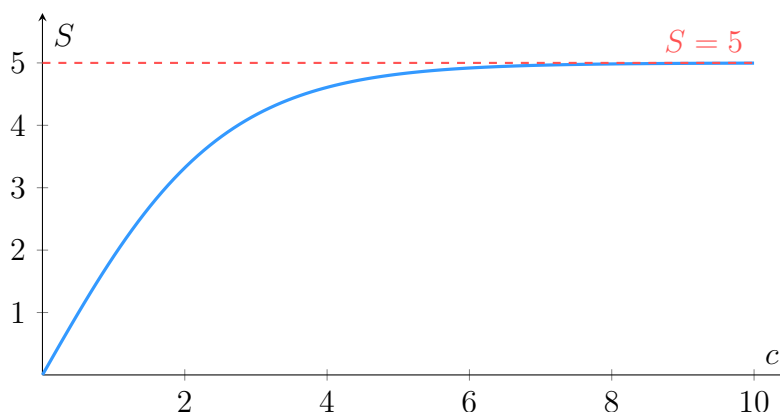


Figure 11.12: The saturating sensor response in Question 89.

Find $S(2)$, the input producing $S = 4$, and the limiting response as $c \rightarrow \infty$.

90. A neural-unit output is $y = \tanh(1.5x - 0.3)$. Find the input for which $y = 0$, the input for which $y = 0.8$, and the possible range of outputs.

91. A logistic curve can be written

$$P(t) = \frac{K}{2} \left[1 + \tanh\left(\frac{rt}{2}\right) \right].$$

Find $P(0)$ and show that $P(t) = 0.9K$ when $t = (\ln 9)/r$.

92. The temperature along a bar is modelled by $T(x) = 20 + 30 \operatorname{sech}(0.2x)$ degrees Celsius. Find $T(0)$, $T(5)$, and the two positions where $T = 25^\circ\text{C}$.

93. A pulse is modelled by $A(t) = 12 \operatorname{sech}(0.5(t - 4))$. Find the time and value of its peak, and find the two times at which its value is half the peak.

94. The tension in a catenary cable is $T(x) = H \cosh(x/a)$. If $H = 20$ kN and $a = 15$ m, find $T(10)$ and the two positions at which the tension is 30 kN.

95. Water-wave amplitude is modelled by $A(d) = 2 \operatorname{sech}(0.1d)$ metres, where d is depth in metres. Find $A(10)$ and the depth at which the amplitude is 1 m.

96. Voltage on a symmetric transmission line is modelled by

$$V(x) = 120 \frac{\cosh(0.1(20 - x))}{\cosh 2}, \quad 0 \leq x \leq 20.$$

Find $V(0)$, $V(10)$, and $V(20)$.

97. A heat-fin model is

$$T(x) = 25 + 75 \frac{\cosh(0.4(5-x))}{\cosh 2}, \quad 0 \leq x \leq 5.$$

Find the temperatures at the base, midpoint, and tip.

98. A path is parameterised by $x = 6 \cosh t$, $y = 4 \sinh t$. Verify that it lies on $x^2/36 - y^2/16 = 1$. Find the point when $t = \ln 2$, and recover t from the point $(15/2, 3)$.
99. A signal compressor uses $C(x) = \tanh x$. Calculate $C(0.5)$ and $C(2)$ to three decimal places. Compare the change in output with the change in input and explain the saturation effect.
100. A symmetric cable is modelled by $y = 20 \cosh(x/20) - 20$. The cable is attached at $x = \pm L$ where it is 10 m above its lowest point.
- (a) Find L exactly.
 - (b) Find the total span and give a decimal approximation.
 - (c) Find the cable height halfway between the centre and a tower.

Chapter 12

Differential Calculus

Chapter guide

Prerequisites. Functions and graphs, algebraic manipulation, trigonometric functions, exponentials, logarithms, and limits.

Learning outcomes. Readers should be able to interpret derivatives as limits and rates of change, apply differentiation rules, analyse stationary points and curve behaviour, and solve optimisation and related-rate problems.

Differential calculus is one of the two traditional divisions of calculus; the other is integral calculus. It provides a systematic way to describe how one quantity changes in relation to another.

The central focus of differential calculus is the derivative of a function, together with related ideas such as differentials and rates of change. The process of finding a derivative is called **differentiation**. The organisation of limits, derivatives, and applications follows standard introductory-calculus references (Strang and Herman 2016a; Neill 2018; Rodriguez 2020).

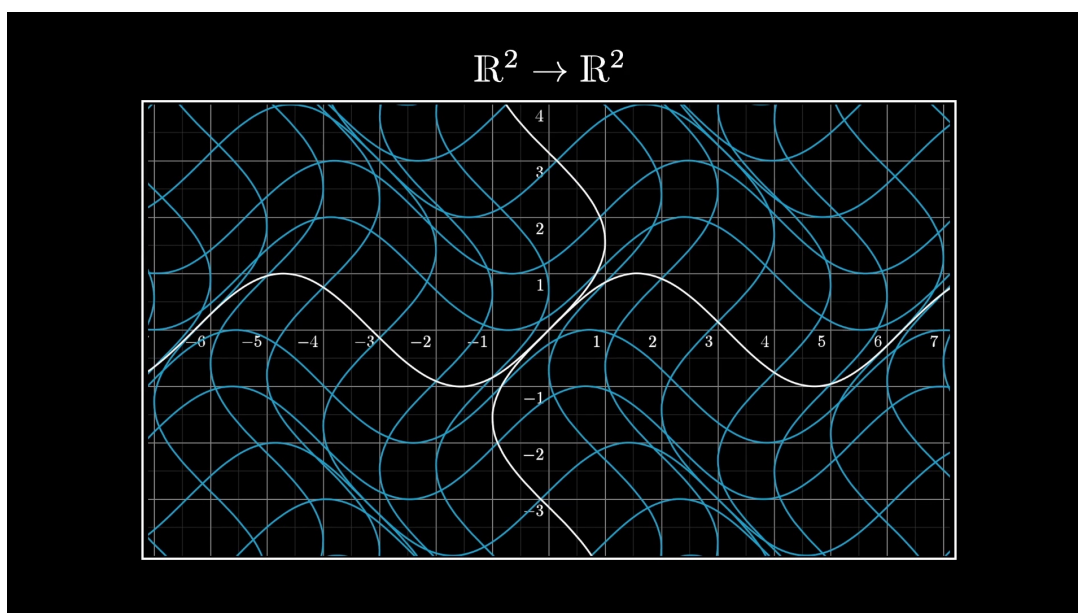


Figure 12.1: A transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$ deforms the coordinate grid. In higher dimensions, the derivative describes the local linear deformation near each point, extending the familiar idea of an instantaneous rate of change. Still by Grant Sanderson/3Blue1Brown from *The other way to visualize derivatives* (2018), licensed CC BY-NC-SA 4.0.

Differential calculus has applications in nearly all quantitative disciplines. For example, in physics, the derivative of the position of a moving body with respect to time is the velocity of the body, and the derivative of velocity with respect to time is acceleration. The reaction rate of a chemical reaction is a derivative, and in operations research, derivatives help determine efficient ways to transport materials and design factories.

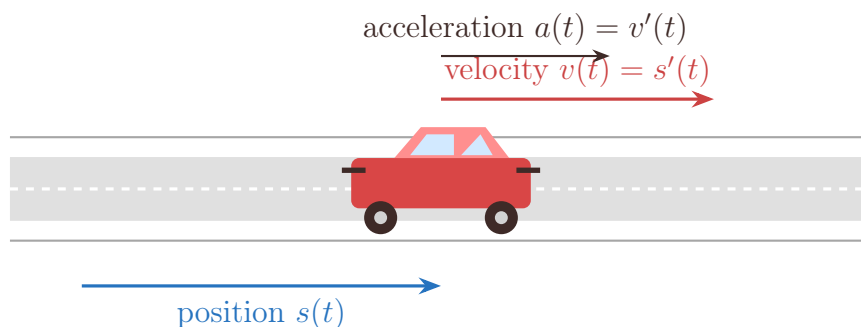


Figure 12.2: The motion of a racing car provides a natural application of differential calculus. If its position is $s(t)$, then its instantaneous velocity is $v(t) = s'(t)$ and its acceleration is $a(t) = v'(t) = s''(t)$. Original diagram created for this book.

This chapter begins with limits and the definition of the derivative, then develops techniques of differentiation and their applications.

12.1 Limits

A limit describes the value approached by a function as its input approaches a specified value. Limits provide the rigorous foundation for continuity, derivatives, and definite integrals. This chapter concentrates on limits of real-valued functions of one real variable.

12.1.1 Notation of limits

In formulae, a limit of a function is usually written as

$$\lim_{x \rightarrow c} f(x) = L$$

and is read as “the limit of $f(x)$ as x approaches c equals L ”. Informally, $f(x)$ can be made arbitrarily close to L by choosing x sufficiently close to c , without requiring $x = c$.

12.1.2 Geometric interpretation on a curve

To connect limits with rates of change, first consider the straight line in Figure 12.3.

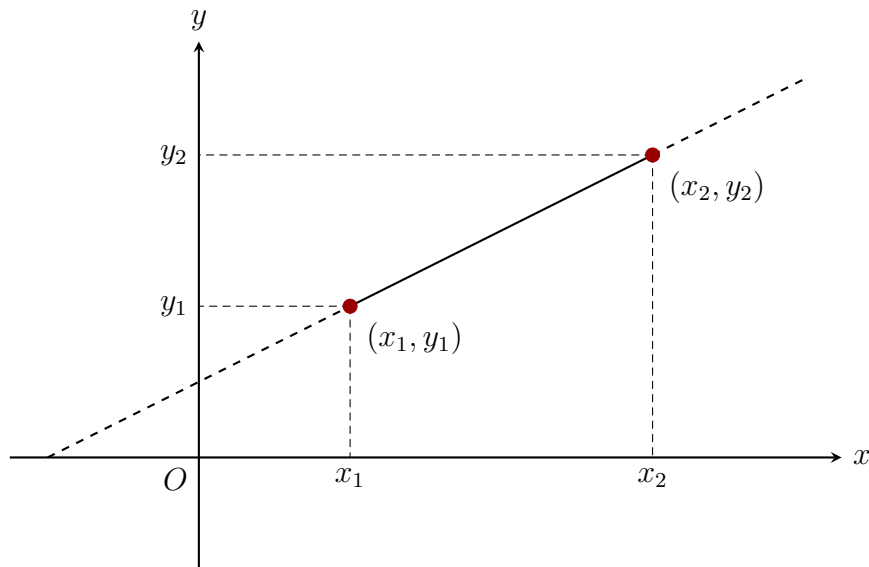


Figure 12.3: The gradient of a straight line determined by two points.

For this linear graph, the rate of change of y with respect to x is represented by the gradient m , where

$$m = \frac{\Delta y}{\Delta x}$$

or can be written as

$$m = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

The rate of change of a linear or constant function can usually be read directly from its formula.

For example, consider this function

$$y = 2x - 2$$

Its rate of change is 2. Similarly, every constant function has rate of change zero.

Now consider the non-linear function in Figure 12.4.

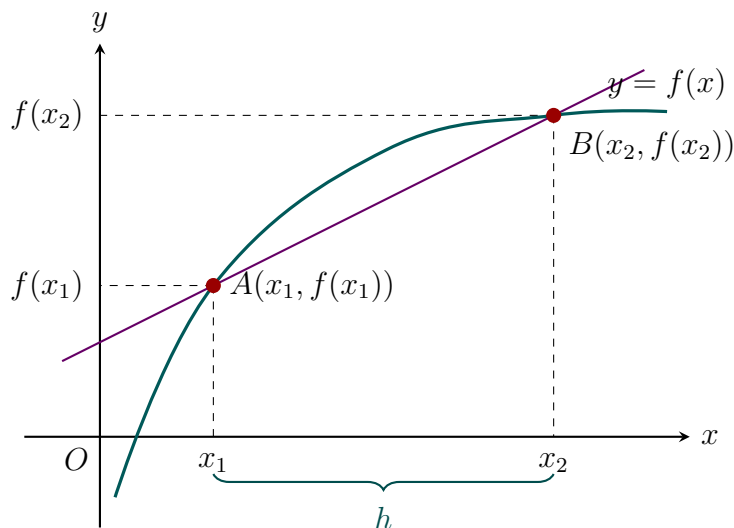


Figure 12.4: A secant line provides an estimate of the rate of change of a non-linear function over an interval.

Unlike a straight line, a non-linear graph does not have one constant gradient. The secant line through A and B gives the average rate of change over the interval from x_1 to x_2 . Allowing B to approach A will then lead to the gradient of the tangent at A .

From Figure 12.4, the gradient of AB is

$$m_{AB} = \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{f(x_1) - f(x_2)}{x_1 - x_2}$$

Bear in mind that

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{-1}{-1} \cdot \frac{f(x_1) - f(x_2)}{x_1 - x_2}$$

From Figure 12.4, $x_2 = x_1 + h$. Substituting this expression into the gradient formula gives

$$\begin{aligned} m_{AB} &= \frac{f(x_1 + h) - f(x_1)}{(x_1 + h) - x_1} \\ &= \frac{f(x_1 + h) - f(x_1)}{h} \end{aligned}$$

Hence,

$$\lim_{h \rightarrow 0} \frac{f(x_1 + h) - f(x_1)}{h}$$

As $h \rightarrow 0$, the point B approaches A , the secant line approaches the tangent line at A , and the secant gradient approaches the instantaneous rate of change of the function at x_1 .

12.2 First-Principles Differentiation

First-principles differentiation, also called *differentiation from the definition*, develops the derivative directly from the idea of a rate of change. It explains where the familiar differentiation rules come from and what a derivative actually measures.

Suppose $y = f(x)$. Between the points

$$A = (x, f(x)) \quad \text{and} \quad B = (x + h, f(x + h)),$$

the change in the independent variable is h , while the corresponding change in the function is

$$\Delta y = f(x + h) - f(x).$$

Therefore, the gradient of the secant line through A and B is

$$\boxed{\frac{\Delta y}{\Delta x} = \frac{f(x + h) - f(x)}{h}, \quad h \neq 0.} \quad (12.1)$$

This expression is called the *difference quotient*. It gives the average rate of change of f over the interval from x to $x + h$.

As h approaches zero, point B approaches point A . When the secant lines approach a unique limiting line, that limiting line is the tangent to the curve at A . Its gradient is the derivative.

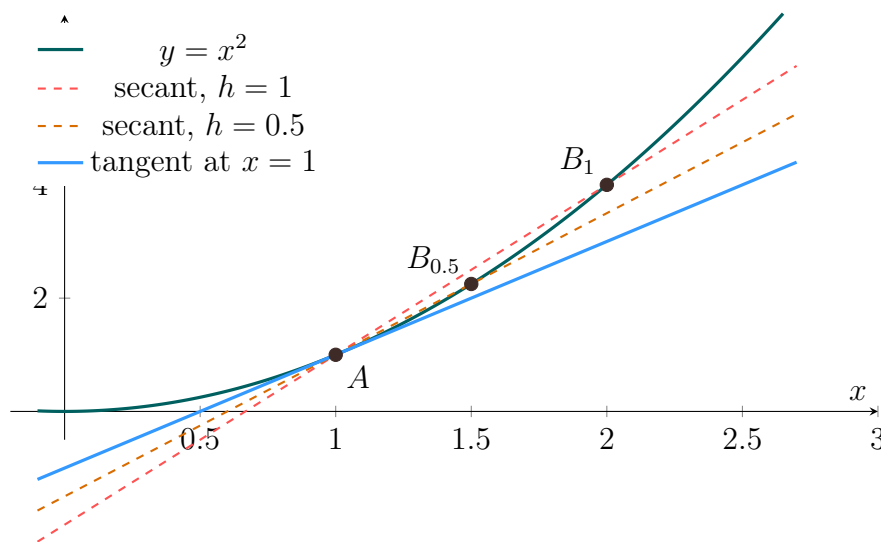


Figure 12.5: As h decreases, the secant line approaches the tangent line.

12.2.1 Derivative from first principles

Definition 12.1: Derivative.

The derivative of f at x is defined by

$$\boxed{f'(x) = \frac{dy}{dx} = \frac{d}{dx}f(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}}, \quad (12.2)$$

provided that this limit exists and is finite. At a particular point $x = a$,

$$\boxed{f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}}. \quad (12.3)$$

It is important that h is not set equal to zero at the beginning. The difference quotient contains division by h , so the expression must first be simplified for $h \neq 0$. Only then is the limit $h \rightarrow 0$ evaluated.

The derivative has several related interpretations:

- geometrically, $f'(a)$ is the gradient of the tangent to $y = f(x)$ at $x = a$;
- algebraically, it is the limiting value of the difference quotient;
- physically, it is an instantaneous rate of change;
- if $s(t)$ is position, then $s'(t)$ is instantaneous velocity.

12.2.2 A numerical view of the limiting process

Consider $f(x) = x^2$ at $x = 1$. For $h \neq 0$,

$$\begin{aligned} \frac{f(1+h) - f(1)}{h} &= \frac{(1+h)^2 - 1}{h} \\ &= \frac{2h + h^2}{h} \\ &= 2 + h. \end{aligned}$$

The following values show the difference quotient approaching 2 from both sides.

Table 12.1: Difference quotients for $f(x) = x^2$ at $x = 1$.

h	-0.5	-0.1	-0.01	0.01	0.1	0.5
$2 + h$	1.5	1.9	1.99	2.01	2.1	2.5

Thus,

$$f'(1) = \lim_{h \rightarrow 0} (2 + h) = 2.$$

12.2.3 Worked examples from first principles

Example 12.1: Derivative of $f(x) = x^2$

Using Equation (12.2),

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} \\
 &= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h} \\
 &= \lim_{h \rightarrow 0} \frac{h(2x + h)}{h} \\
 &= \lim_{h \rightarrow 0} (2x + h) \\
 &= 2x.
 \end{aligned}$$

Hence,

$$\boxed{\frac{d}{dx}(x^2) = 2x.}$$

Example 12.2: Derivative of a quadratic polynomial

Let

$$f(x) = 3x^2 - 5x + 4.$$

Then

$$\begin{aligned}
 f(x+h) &= 3(x+h)^2 - 5(x+h) + 4 \\
 &= 3x^2 + 6xh + 3h^2 - 5x - 5h + 4.
 \end{aligned}$$

Therefore,

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{6xh + 3h^2 - 5h}{h} \\
 &= \lim_{h \rightarrow 0} (6x + 3h - 5) \\
 &= 6x - 5.
 \end{aligned}$$

Thus,

$$\frac{d}{dx}(3x^2 - 5x + 4) = 6x - 5.$$

Example 12.3: Derivative of a reciprocal function

Let $f(x) = 1/x$, where $x \neq 0$. Then

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{\frac{1}{x+h} - \frac{1}{x}}{h} \\ &= \lim_{h \rightarrow 0} \frac{x - (x+h)}{h x(x+h)} \\ &= \lim_{h \rightarrow 0} \frac{-h}{h x(x+h)} \\ &= \lim_{h \rightarrow 0} \left(-\frac{1}{x(x+h)} \right) \\ &= -\frac{1}{x^2}. \end{aligned}$$

Therefore,

$$\frac{d}{dx} \left(\frac{1}{x} \right) = -\frac{1}{x^2}, \quad x \neq 0.$$

12.2.4 The equation of a tangent

Once $f'(a)$ is known, the tangent to $y = f(x)$ at the point $(a, f(a))$ is obtained from the point-gradient equation

$$y - f(a) = f'(a)(x - a). \quad (12.4)$$

For example, for $f(x) = x^2$ at $x = 1$,

$$f(1) = 1 \quad \text{and} \quad f'(1) = 2.$$

Hence, the tangent is

$$\begin{aligned} y - 1 &= 2(x - 1), \\ y &= 2x - 1. \end{aligned}$$

12.2.5 When a derivative does not exist

For a derivative to exist at $x = a$, the difference quotient must approach the same finite value from both sides. A derivative may fail to exist at a discontinuity, a corner, a cusp, or a vertical tangent.

Consider $f(x) = |x|$ at $x = 0$. Since $f(0) = 0$,

$$\frac{f(0+h) - f(0)}{h} = \frac{|h|}{h}.$$

For $h > 0$, this quotient is 1, whereas for $h < 0$, it is -1 . Therefore,

$$\lim_{h \rightarrow 0^+} \frac{|h|}{h} = 1 \quad \text{but} \quad \lim_{h \rightarrow 0^-} \frac{|h|}{h} = -1.$$

The two one-sided limits disagree, so $f'(0)$ does not exist.

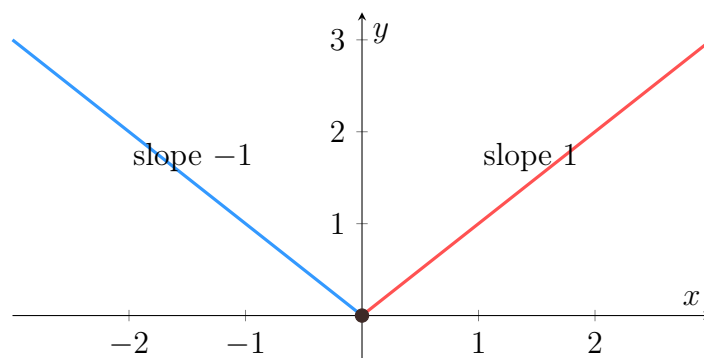


Figure 12.6: The corner of $y = |x|$ at the origin has unequal one-sided gradients.

Differentiability is stronger than continuity: if f is differentiable at a , then it is continuous at a . The converse is not always true, as $f(x) = |x|$ demonstrates.

12.3 Rules of differentiation

Different algebraic structures require different differentiation rules. The standard derivatives below are followed by the sum, difference, product, quotient, and chain rules.

12.3.1 Differentiation of standard functions

The following elementary derivatives will be used repeatedly.

1. **Constant rule.** If $f(x) = k$, where k is constant, then

$$\frac{dy}{dx} = 0$$

For example, if $f(x) = 2$, then $f'(x) = 0$.

2. **Power rule.** If $f(x) = x^n$, then, wherever the real-valued expression and derivative are defined,

$$\frac{dy}{dx} = nx^{n-1}$$

For example, if $f(x) = x^4$, then $f'(x) = 4x^3$.

3. **Constant-multiple rule.** If $f(x) = ax^n$, where a is constant, then

$$\frac{dy}{dx} = anx^{n-1}$$

For example, if $f(x) = 5x^5$, then $f'(x) = 25x^4$.

12.3.2 Sum and difference rule

If a and b are differentiable, then their sum and difference are differentiated term by term:

$$\frac{d}{dx}[f(x)] = \frac{d}{dx}[a(x) \pm b(x)] = \frac{d}{dx}[a(x)] \pm \frac{d}{dx}[b(x)]$$

This term-by-term rule is special to sums and differences; products, quotients, and compositions require their own rules.

Example 12.4

Differentiate $f(x) = 4x^3 + 5x^2$.

Using the above rule,

$$\begin{aligned} \frac{d}{dx}f(x) &= \frac{d}{dx}(4x^3 + 5x^2) \\ &= \frac{d}{dx}(4x^3) + \frac{d}{dx}(5x^2) \\ &= 12x^2 + 10x \\ &= 2x(6x + 5). \end{aligned}$$

12.3.3 Chain rule

For the composite function $y = f(g(x))$, set $u = g(x)$. If both component functions are differentiable at the relevant points, the chain rule gives

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

This indicates that if both f and g are differentiable at the relevant points, then $y = f[g(x)]$ is also differentiable at x .

Example 12.5

Differentiate $y = (x^2 + 5)^4$.

Set $u = x^2 + 5$. Then

$$y = u^4$$

Differentiating the outer function gives

$$\frac{dy}{du} = 4u^3$$

Differentiating the inner function gives

$$\frac{du}{dx} = 2x$$

Hence,

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{du} \times \frac{du}{dx} \\ &= 4u^3(2x) \\ &= 8xu^3 \end{aligned}$$

Substituting back for u gives

$$\frac{dy}{dx} = 8x(x^2 + 5)^3$$

The following subsection records the frequently used power form of the chain rule.

12.3.4 Power chain rule

The power rule states that, if $f(x)$ is differentiable at point x and $y = [f(x)]^n$, then the derivative of y is

$$\frac{dy}{dx} = n[f(x)]^{n-1} f'(x).$$

Example 12.6

Differentiate $y = 4(2x^3 - 5x^2)^7$.

Set $u = 2x^3 - 5x^2$. Then

$$y = 4u^7$$

Differentiating the outer function gives

$$\frac{dy}{du} = 28u^6$$

Differentiating the inner function gives

$$\frac{du}{dx} = 6x^2 - 10x$$

Therefore,

$$\frac{dy}{dx} = 28(2x^3 - 5x^2)^6 \cdot (6x^2 - 10x)$$

This derivative is correct, but it can be factorised further. First,

$$2x^3 - 5x^2 = x^2(2x - 5)$$

and

$$6x^2 - 10x = 2x(3x - 5)$$

Substitution gives

$$\begin{aligned} \frac{dy}{dx} &= 28(x^2(2x - 5))^6 \cdot 2x(3x - 5) \\ &= 28 \cdot x^{12}(2x - 5)^6 \cdot 2x(3x - 5) \\ &= 56x^{13}(2x - 5)^6(3x - 5) \end{aligned}$$

12.3.5 Product rule

The product rule stated that if $y = u(x)v(x)$, then the derivative of it can be represented as,

$$\frac{dy}{dx} = u(x)v'(x) + v(x)u'(x) \quad (12.5)$$

Example 12.7: Product rule with the chain rule

Differentiate

$$f(x) = x^2(x - 3)^5.$$

Let

$$u = x^2, \quad v = (x - 3)^5.$$

Then

$$u' = 2x$$

and, using the chain rule,

$$v' = 5(x - 3)^4.$$

Applying the product rule,

$$\begin{aligned} f'(x) &= x^2[5(x - 3)^4] + (x - 3)^5(2x) \\ &= 5x^2(x - 3)^4 + 2x(x - 3)^5. \end{aligned}$$

Both terms contain the common factor $x(x - 3)^4$, so

$$\begin{aligned} f'(x) &= x(x - 3)^4[5x + 2(x - 3)] \\ &= \boxed{x(x - 3)^4(7x - 6)}. \end{aligned}$$

The unfactorised answer

$$5x^2(x - 3)^4 + 2x(x - 3)^5$$

is also correct. Factoring is useful because it produces a shorter form and makes the stationary points easier to identify in later applications.

Example 12.8: A polynomial product

Differentiate

$$y = (2x + 1)(x^2 - 4x + 3).$$

Take

$$u = 2x + 1, \quad u' = 2, \quad v = x^2 - 4x + 3, \quad v' = 2x - 4.$$

Therefore,

$$\begin{aligned} \frac{dy}{dx} &= (2x + 1)(2x - 4) + (x^2 - 4x + 3)(2) \\ &= 4x^2 - 6x - 4 + 2x^2 - 8x + 6 \\ &= 6x^2 - 14x + 2. \end{aligned}$$

For a product of polynomials, one may alternatively expand the expression before differentiating. Both methods give the same derivative.

Example 12.9: A product containing a radical

Differentiate

$$y = x^3\sqrt{2x+1}.$$

First write the radical as a fractional power:

$$y = x^3(2x+1)^{1/2}.$$

Choose

$$u = x^3, \quad v = (2x+1)^{1/2}.$$

Hence,

$$u' = 3x^2, \quad v' = \frac{1}{2}(2x+1)^{-1/2}(2) = (2x+1)^{-1/2}.$$

The product rule gives

$$\begin{aligned} \frac{dy}{dx} &= x^3(2x+1)^{-1/2} + 3x^2(2x+1)^{1/2} \\ &= \frac{x^3 + 3x^2(2x+1)}{\sqrt{2x+1}} \\ &= \frac{x^2(7x+3)}{\sqrt{2x+1}}. \end{aligned}$$

When expansion is simpler

The product rule is always valid, but it is not always the shortest method. For example,

$$y = (x+2)(x-5)$$

may first be expanded:

$$\begin{aligned} y &= x^2 - 3x - 10, \\ \frac{dy}{dx} &= \boxed{2x - 3}. \end{aligned}$$

For two short polynomial factors, expansion is often convenient. The product rule is preferable when expansion would be lengthy or impossible, especially when powers, radicals, exponential functions, logarithms, or trigonometric functions are involved.

Products of three functions

The product rule can be extended. If

$$y = uvw,$$

then each factor is differentiated once:

$$\frac{dy}{dx} = u'vw + uv'w + uvw'. \quad (12.6)$$

For instance, if $y = x(x+1)(x-2)$, then

$$\begin{aligned} \frac{dy}{dx} &= (1)(x+1)(x-2) + x(1)(x-2) + x(x+1)(1) \\ &= 3x^2 - 2x - 2. \end{aligned}$$

12.3.6 Quotient rule

If

$$y = \frac{u(x)}{v(x)}, \quad v(x) \neq 0,$$

then

$$\boxed{\frac{dy}{dx} = \frac{v(x)u'(x) - u(x)v'(x)}{[v(x)]^2}}. \quad (12.7)$$

A useful verbal form is “bottom times derivative of top, minus top times derivative of bottom, all over bottom squared.” The order of the two terms in the numerator is important.

Example 12.10

Differentiate

$$f(x) = \frac{2x^2 + 1}{x^2 - 1}.$$

Let $u = 2x^2 + 1$ and $v = x^2 - 1$. Then $u' = 4x$ and $v' = 2x$, so

$$\begin{aligned} f'(x) &= \frac{(x^2 - 1)(4x) - (2x^2 + 1)(2x)}{(x^2 - 1)^2} \\ &= \boxed{-\frac{6x}{(x^2 - 1)^2}}. \end{aligned}$$

Example 12.11: simplifying first

For

$$y = \frac{x^3}{x^2} = x, \quad x \neq 0,$$

it is shorter to simplify before differentiating, giving $dy/dx = 1$ on the original domain $x \neq 0$. Algebraic simplification must not erase domain restrictions.

12.4 Second-order and higher-order derivatives

The derivative of a derivative is called a higher-order derivative. Common notations are

$$\begin{aligned} f'(x) &= \frac{dy}{dx}, \\ f''(x) &= \frac{d^2y}{dx^2}, \\ f'''(x) &= \frac{d^3y}{dx^3}, \\ f^{(n)}(x) &= \frac{d^ny}{dx^n}. \end{aligned}$$

The notation d^2y/dx^2 means “differentiate y twice with respect to x ”; it is not the square of dy/dx .

Example 12.12

If

$$y = x^4 - 3x^3 + 2x,$$

then

$$\begin{aligned} y' &= 4x^3 - 9x^2 + 2, \\ y'' &= 12x^2 - 18x, \\ y''' &= 24x - 18, \\ y^{(4)} &= 24, \\ y^{(5)} &= 0. \end{aligned}$$

Higher derivatives have important interpretations. If $s(t)$ is position, then $s'(t)$ is velocity and $s''(t)$ is acceleration. For a graph $y = f(x)$, the sign of $f''(x)$ describes its concavity, which will be used later in this chapter.

12.5 Implicit differentiation

An equation such as

$$x^2 + y^2 = 25$$

defines y implicitly because y is not isolated as a single function of x . To differentiate it, regard y as a differentiable function $y(x)$ and differentiate every term with respect to x . Each derivative involving y requires the chain rule.

For example,

$$\frac{d}{dx}(y^2) = 2y \frac{dy}{dx}.$$

Thus, from $x^2 + y^2 = 25$,

$$2x + 2y \frac{dy}{dx} = 0,$$

$$\frac{dy}{dx} = \boxed{-\frac{x}{y}}.$$

At $(3, 4)$, the tangent gradient is $-3/4$. The tangent and normal are

$$y - 4 = -\frac{3}{4}(x - 3),$$

$$y - 4 = \frac{4}{3}(x - 3),$$

respectively.

12.5.1 Products containing both x and y

When a term contains both variables, apply the product rule as well as the chain rule. For instance, differentiating

$$x^2y + xy^2 = 6$$

gives

$$2xy + x^2y' + y^2 + 2xyy' = 0,$$

$$y'(x^2 + 2xy) = -(2xy + y^2),$$

$$y' = \boxed{-\frac{2xy + y^2}{x^2 + 2xy}}.$$

12.5.2 Second derivative of an implicit function

Differentiate the first-derivative equation again, remembering that y and y' both depend on x . For the circle $x^2 + y^2 = 25$, begin with

$$2x + 2yy' = 0.$$

Differentiating once more,

$$\begin{aligned} 2 + 2[(y')^2 + yy''] &= 0, \\ y'' &= -\frac{1 + (y')^2}{y}. \end{aligned}$$

Since $y' = -x/y$ and $x^2 + y^2 = 25$,

$$y'' = -\frac{25}{y^3}.$$

12.6 Parametric differentiation

A curve may be given by

$$x = g(t), \quad y = h(t),$$

where t is a parameter. Provided that $dx/dt \neq 0$, the chain rule gives

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}. \quad (12.8)$$

This formula gives the gradient in terms of t . A horizontal tangent occurs when $dy/dt = 0$ and $dx/dt \neq 0$; a vertical tangent may occur when $dx/dt = 0$ and $dy/dt \neq 0$.

Example 12.13

Let

$$x = t^2 + t, \quad y = t^3.$$

Then

$$\frac{dx}{dt} = 2t + 1, \quad \frac{dy}{dt} = 3t^2,$$

so

$$\frac{dy}{dx} = \frac{3t^2}{2t + 1}.$$

At $t = 1$, the point is $(2, 1)$ and the gradient is 1, so the tangent is $y - 1 = x - 2$.

12.6.1 Second derivative for parametric equations

Differentiate dy/dx with respect to t and divide again by dx/dt :

$$\boxed{\frac{d^2y}{dx^2} = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{dx/dt}}. \quad (12.9)$$

For the preceding example,

$$\begin{aligned} \frac{d}{dt} \left(\frac{3t^2}{2t+1} \right) &= \frac{6t(t+1)}{(2t+1)^2}, \\ \frac{d^2y}{dx^2} &= \boxed{\frac{6t(t+1)}{(2t+1)^3}}. \end{aligned}$$

12.7 Differentiation of exponential and logarithmic functions

12.7.1 Exponential functions

For $a > 0$ and $a \neq 1$,

$$\frac{d}{dx} e^x = e^x, \quad (12.10)$$

$$\frac{d}{dx} e^{u(x)} = e^{u(x)} u'(x), \quad (12.11)$$

$$\frac{d}{dx} a^x = a^x \ln a, \quad (12.12)$$

$$\frac{d}{dx} a^{u(x)} = a^{u(x)} \ln a u'(x). \quad (12.13)$$

The special simplicity of e^x is one reason the number e is fundamental in calculus: its rate of change equals its current value.

Example 12.14

$$\begin{aligned} \frac{d}{dx} (3e^{-2x^2+1}) &= 3e^{-2x^2+1}(-4x) \\ &= \boxed{-12xe^{-2x^2+1}}, \\ \frac{d}{dx} (7^{3x-1}) &= \boxed{3(\ln 7)7^{3x-1}}. \end{aligned}$$

12.7.2 Logarithmic functions

For a differentiable positive function $u(x)$,

$$\frac{d}{dx} \ln x = \frac{1}{x}, \quad x > 0, \quad (12.14)$$

$$\frac{d}{dx} \ln |u(x)| = \frac{u'(x)}{u(x)}, \quad u(x) \neq 0, \quad (12.15)$$

$$\frac{d}{dx} \log_a x = \frac{1}{x \ln a}, \quad x > 0, a > 0, a \neq 1, \quad (12.16)$$

$$\frac{d}{dx} \log_a u(x) = \frac{u'(x)}{u(x) \ln a}. \quad (12.17)$$

The formula for $\ln x$ follows neatly from implicit differentiation. Let $y = \ln x$, so $e^y = x$. Then

$$e^y \frac{dy}{dx} = 1 \implies \frac{dy}{dx} = \frac{1}{e^y} = \frac{1}{x}.$$

Example 12.15

$$\begin{aligned} \frac{d}{dx} \ln(3x^2 - 1) &= \frac{6x}{3x^2 - 1}, \\ \frac{d}{dx} [5 \log_2(x + 4)] &= \frac{5}{(x + 4) \ln 2}. \end{aligned}$$

12.7.3 Logarithmic differentiation

Logarithmic differentiation is especially useful when a function contains many products, quotients, variable exponents, or radicals. The procedure is:

1. take natural logarithms of both sides;
2. expand using logarithm laws;
3. differentiate implicitly; and
4. multiply by y and substitute the original expression.

For example, let $y = x^x$ for $x > 0$. Then

$$\begin{aligned} \ln y &= x \ln x, \\ \frac{y'}{y} &= \ln x + 1, \\ y' &= x^x (\ln x + 1). \end{aligned}$$

More generally, if $y = [u(x)]^{v(x)}$ with $u(x) > 0$, then

$$\boxed{y' = u^v \left(v' \ln u + v \frac{u'}{u} \right)}. \quad (12.18)$$

As a second example, if

$$y = \frac{x^2 \sqrt{x+1}}{(x-2)^3},$$

then

$$\begin{aligned} \ln |y| &= 2 \ln |x| + \frac{1}{2} \ln |x+1| - 3 \ln |x-2|, \\ \frac{y'}{y} &= \frac{2}{x} + \frac{1}{2(x+1)} - \frac{3}{x-2}. \end{aligned}$$

Thus the derivative is the original function multiplied by the expression on the right.

12.8 Differentiation of trigonometric functions

The standard trigonometric derivative formulae assume that angles are measured in radians. Two fundamental limits are

$$\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1, \quad \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} = 0.$$

Using the addition formula for sine,

$$\begin{aligned} \frac{d}{dx} \sin x &= \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h} \\ &= \sin x \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} + \cos x \lim_{h \rightarrow 0} \frac{\sin h}{h} \\ &= \cos x. \end{aligned}$$

Similarly, $d(\cos x)/dx = -\sin x$. The complete standard list is

Function	Derivative
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\cot x$	$-\csc^2 x$
$\sec x$	$\sec x \tan x$
$\csc x$	$-\csc x \cot x$

Table 12.2: Derivatives of the six trigonometric functions.

The remaining four formulae follow from the quotient or reciprocal forms of the trigonometric functions. For example,

$$\begin{aligned}
 \frac{d}{dx} \tan x &= \frac{d}{dx} \left(\frac{\sin x}{\cos x} \right) \\
 &= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} \\
 &= \sec^2 x.
 \end{aligned}$$

12.8.1 Chain and power rules for trigonometric functions

If $u = u(x)$, attach the factor $u'(x)$:

$$\begin{aligned}
 \frac{d}{dx} \sin u &= \cos u \, u', \\
 \frac{d}{dx} \cos u &= -\sin u \, u', \\
 \frac{d}{dx} \tan u &= \sec^2 u \, u', \\
 \frac{d}{dx} \sec u &= \sec u \tan u \, u'.
 \end{aligned}$$

The same principle applies to powers. For example,

$$\begin{aligned}
 \frac{d}{dx} \sin^5(3x) &= 5 \sin^4(3x) \cos(3x)(3) \\
 &= \boxed{15 \sin^4(3x) \cos(3x)}, \\
 \frac{d}{dx} \cos^2(x^2) &= 2 \cos(x^2)[- \sin(x^2)](2x) \\
 &= \boxed{-4x \cos(x^2) \sin(x^2)}.
 \end{aligned}$$

12.9 Derivatives of inverse trigonometric functions

The notation $\sin^{-1} x$ means the inverse sine function, not $1/\sin x$. The latter is $\csc x$. Domain restrictions and principal-value ranges are essential because trigonometric functions are not one-to-one on their full domains.

To derive the inverse-sine formula, let

$$y = \sin^{-1} x, \quad -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}.$$

Then $\sin y = x$, so implicit differentiation gives

$$\cos y \frac{dy}{dx} = 1.$$

On the principal range, $\cos y \geq 0$, and $\cos y = \sqrt{1 - \sin^2 y} = \sqrt{1 - x^2}$. Therefore,

$$\boxed{\frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1 - x^2}}}.$$

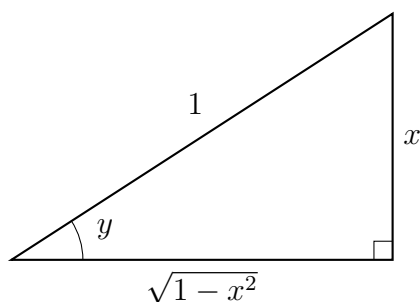


Figure 12.7: A right triangle for $\sin y = x$ on the principal range.

For a differentiable function $u = u(x)$, the chain-rule forms are

$$\frac{d}{dx} \sin^{-1} u = \frac{u'}{\sqrt{1 - u^2}}, \quad (12.19)$$

$$\frac{d}{dx} \cos^{-1} u = -\frac{u'}{\sqrt{1 - u^2}}, \quad (12.20)$$

$$\frac{d}{dx} \tan^{-1} u = \frac{u'}{1 + u^2}, \quad (12.21)$$

$$\frac{d}{dx} \cot^{-1} u = -\frac{u'}{1 + u^2}, \quad (12.22)$$

$$\frac{d}{dx} \sec^{-1} u = \frac{u'}{|u|\sqrt{u^2 - 1}}, \quad (12.23)$$

$$\frac{d}{dx} \csc^{-1} u = -\frac{u'}{|u|\sqrt{u^2 - 1}}. \quad (12.24)$$

For inverse sine and inverse cosine, the derivative is real when $|u| < 1$. For inverse secant and inverse cosecant, it is real when $|u| > 1$. The absolute value in the final two formulae is

necessary; it reflects the positive hypotenuse used when converting trigonometric quantities into algebraic expressions.

Example 12.16

$$\frac{d}{dx} \cos^{-1}(3x) = \boxed{-\frac{3}{\sqrt{1-9x^2}}}, \quad |x| < \frac{1}{3}.$$

Example 12.17: combined rules

If $f(x) = x^2 \cos^{-1}(e^{3x})$, then

$$f'(x) = 2x \cos^{-1}(e^{3x}) - x^2 \frac{3e^{3x}}{\sqrt{1-e^{6x}}}.$$

The real-valued domain must also satisfy $0 < e^{3x} \leq 1$, so $x \leq 0$; the displayed derivative is finite for $x < 0$.

12.10 Derivatives of hyperbolic functions

From the exponential definitions of the hyperbolic functions,

$$\begin{aligned} \frac{d}{dx} \sinh x &= \cosh x, & \frac{d}{dx} \cosh x &= \sinh x, \\ \frac{d}{dx} \tanh x &= \operatorname{sech}^2 x, & \frac{d}{dx} \coth x &= -\operatorname{csch}^2 x, \\ \frac{d}{dx} \operatorname{sech} x &= -\operatorname{sech} x \tanh x, & \frac{d}{dx} \operatorname{csch} x &= -\operatorname{csch} x \coth x. \end{aligned}$$

For a composite argument $u(x)$, multiply by $u'(x)$. For example,

$$\frac{d}{dx} \tanh(2x^2 - 1) = \boxed{4x \operatorname{sech}^2(2x^2 - 1)}.$$

12.11 Critical points and intervals of monotonicity

Definition 12.2: Critical number and critical point.

Let c belong to the domain of f . The number c is a *critical number* if

$$f'(c) = 0 \quad \text{or} \quad f'(c) \text{ does not exist.}$$

The corresponding point $(c, f(c))$ is a *critical point*. If $f'(c) = 0$, it is also called a *stationary point*.

A point at which the function itself is undefined is not a critical point, although that x -value may still divide the domain into intervals that must be studied separately.

12.11.1 Increasing and decreasing intervals

For a differentiable function on an interval,

$$\begin{aligned} f'(x) > 0 &\implies f \text{ is increasing,} \\ f'(x) < 0 &\implies f \text{ is decreasing,} \\ f'(x) = 0 \text{ throughout the interval} &\implies f \text{ is constant.} \end{aligned}$$

Critical numbers and excluded domain values divide the number line into test intervals. The sign of f' on each interval gives the direction of the curve.

Example 12.18

Let

$$f(x) = x^3 - 3x.$$

Then

$$f'(x) = 3x^2 - 3 = 3(x - 1)(x + 1),$$

so the critical numbers are $x = -1$ and $x = 1$. A sign chart gives

Interval	$(-\infty, -1)$	$(-1, 1)$	$(1, \infty)$
Sign of $f'(x)$	+	-	+
Behaviour of f	increasing	decreasing	increasing

Table 12.3: First-derivative sign chart for $f(x) = x^3 - 3x$.

12.12 Local and absolute extrema

A function has a local maximum at $x = c$ when $f(c)$ is at least as large as nearby values, and a local minimum when it is at most as large as nearby values. Every interior local extremum is a critical point, but a critical point need not be an extremum.

12.12.1 First derivative test

Suppose c is a critical number of a continuous function.

- If f' changes from positive to negative at c , then f has a local maximum at c .
- If f' changes from negative to positive at c , then f has a local minimum at c .
- If the sign of f' does not change, then c is not a local extremum.

For $f(x) = x^3 - 3x$, the preceding sign chart shows a local maximum at $(-1, 2)$ and a local minimum at $(1, -2)$.

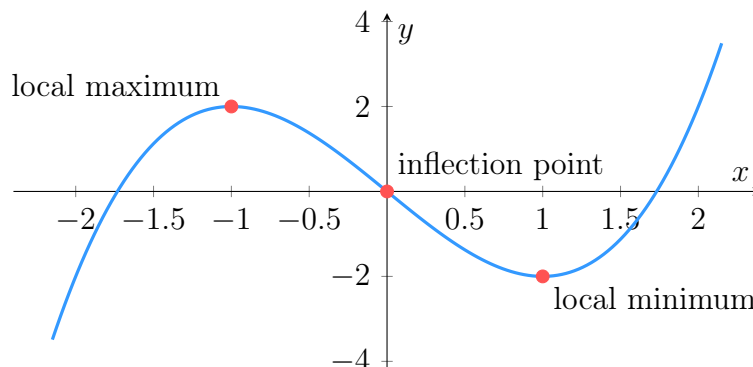


Figure 12.8: Local extrema and an inflection point of $y = x^3 - 3x$.

12.12.2 Second derivative test

If $f'(c) = 0$ and $f''(c)$ exists, then

$$f''(c) > 0 \implies \text{local minimum,}$$

$$f''(c) < 0 \implies \text{local maximum.}$$

If $f''(c) = 0$, the test is inconclusive. The point may be a maximum, a minimum, or neither, so the first derivative test or another argument is required.

For $f(x) = x^3 - 3x$, $f''(x) = 6x$. Therefore, $f''(-1) = -6 < 0$ confirms the local maximum and $f''(1) = 6 > 0$ confirms the local minimum.

12.12.3 Absolute extrema on a closed interval

The Extreme Value Theorem states that a continuous function on a closed, bounded interval $[a, b]$ attains both an absolute maximum and an absolute minimum. To find them:

1. find all critical numbers in (a, b) ;
2. evaluate f at each critical number and at both endpoints;
3. compare the resulting values.

For $f(x) = x^3 - 3x$ on $[-2, 3]$, evaluate

$$f(-2) = -2, \quad f(-1) = 2, \quad f(1) = -2, \quad f(3) = 18.$$

Thus the absolute maximum is 18 at $x = 3$, while the absolute minimum is -2 , attained at both $x = -2$ and $x = 1$.

12.13 Concavity and points of inflection

The second derivative measures how the first derivative changes.

$$\begin{aligned} f''(x) > 0 &\implies f \text{ is concave upward,} \\ f''(x) < 0 &\implies f \text{ is concave downward.} \end{aligned}$$

A point $(c, f(c))$ is a point of inflection when the graph is continuous there and its concavity changes across c . Possible candidates occur where $f''(c) = 0$ or $f''(c)$ does not exist, but the change in concavity must be verified.

For $f(x) = x^3 - 3x$, $f''(x) = 6x$. It is concave downward for $x < 0$ and concave upward for $x > 0$, so $(0, 0)$ is a point of inflection. Notice that $f'(0) = -3$; an inflection point need not have a horizontal tangent.

The function $f(x) = x^4$ provides an important warning. Although $f''(0) = 0$, its second derivative $12x^2$ is positive on both sides of the origin, so there is no inflection point.

12.14 Curve sketching with derivatives

Derivative information can determine the essential shape of a graph without plotting a large table of points. A systematic procedure is:

1. determine the domain and any symmetry;
2. find the x - and y -intercepts;
3. locate discontinuities and asymptotes;
4. find critical points and intervals of increase or decrease;
5. find intervals of concavity and points of inflection;
6. determine end behaviour; and
7. plot the important features and join them consistently.

12.14.1 Asymptotes

A vertical line $x = a$ is a vertical asymptote if at least one of the one-sided limits of $f(x)$ as $x \rightarrow a$ is infinite. A horizontal line $y = L$ is a horizontal asymptote if

$$\lim_{x \rightarrow \infty} f(x) = L \quad \text{or} \quad \lim_{x \rightarrow -\infty} f(x) = L.$$

An oblique asymptote has the form $y = mx + c$, where $m \neq 0$, and satisfies

$$\lim_{x \rightarrow \pm\infty} [f(x) - (mx + c)] = 0.$$

For rational functions, polynomial division is useful when the numerator's degree is exactly one more than the denominator's. A curve may cross a horizontal or oblique asymptote; an asymptote describes limiting behaviour, not an absolute prohibition against intersection.

For example,

$$f(x) = \frac{x^2 - 1}{x - 2} = x + 2 + \frac{3}{x - 2}.$$

Hence $x = 2$ is a vertical asymptote and $y = x + 2$ is an oblique asymptote.

12.14.2 Complete polynomial example

Consider

$$f(x) = x^4 - 8x^2.$$

The function is even, so its graph is symmetric about the y -axis. Its first derivative is

$$f'(x) = 4x^3 - 16x = 4x(x - 2)(x + 2).$$

The critical points are

$$(-2, -16), \quad (0, 0), \quad (2, -16).$$

The sign of f' shows that f decreases on $(-\infty, -2)$ and $(0, 2)$, and increases on $(-2, 0)$ and $(2, \infty)$. Thus $(-2, -16)$ and $(2, -16)$ are local minima, while $(0, 0)$ is a local maximum.

Next,

$$f''(x) = 12x^2 - 16 = 4(3x^2 - 4).$$

The graph is concave downward for $|x| < 2/\sqrt{3}$ and concave upward for $|x| > 2/\sqrt{3}$. The points of inflection are

$$\left(\pm \frac{2}{\sqrt{3}}, -\frac{80}{9} \right).$$

Its intercepts are $x = 0$ and $x = \pm 2\sqrt{2}$.

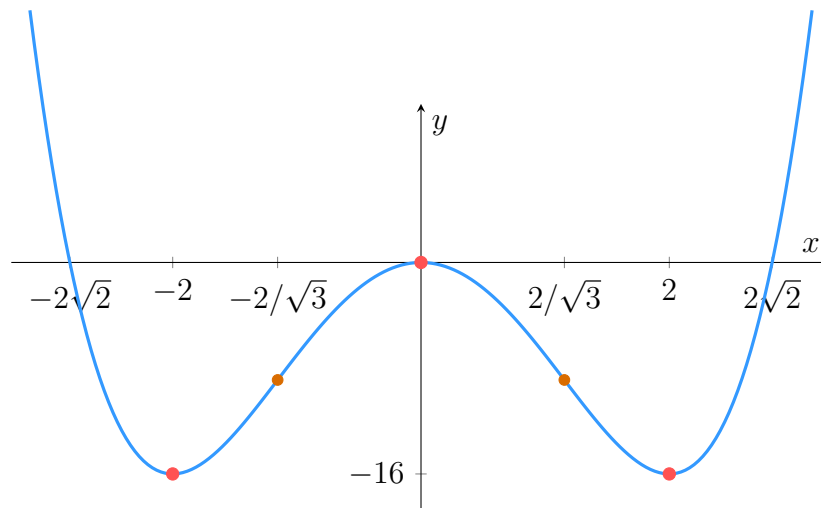


Figure 12.9: A derivative-based sketch of $f(x) = x^4 - 8x^2$.

12.15 Optimisation

Optimisation problems ask for the greatest or least possible value of a quantity subject to one or more constraints. A reliable procedure is:

1. identify the quantity to maximise or minimise;
2. introduce variables and write the objective function;
3. use the constraints to express the objective in one variable;
4. determine the physically meaningful domain;
5. find critical numbers and test them; and
6. interpret the result, including units and requested dimensions.

On a closed interval, endpoint values must also be tested. The second derivative test may classify an interior critical point, but comparison of all feasible candidates is the safest approach.

12.15.1 Maximum-area rectangle

A rectangle has perimeter 20 cm. Let its width be x and its length be y . The constraint is

$$2x + 2y = 20 \implies y = 10 - x,$$

where $0 < x < 10$. Its area is

$$A(x) = x(10 - x) = 10x - x^2.$$

Then

$$A'(x) = 10 - 2x = 0 \implies x = 5,$$

and $A''(x) = -2 < 0$. Hence the maximum-area rectangle is the 5 cm by 5 cm square, with area 25 cm^2 .

12.15.2 Open-box problem

Squares of side x are removed from the corners of an 8 cm by 5 cm card. Folding the sides produces an open box with dimensions

$$(8 - 2x) \times (5 - 2x) \times x.$$

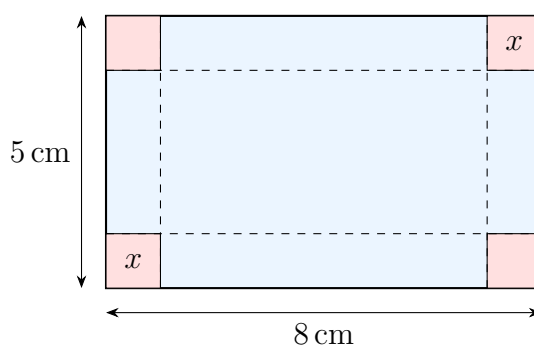


Figure 12.10: Corner squares removed to form an open box.

The volume is

$$V(x) = x(8 - 2x)(5 - 2x) = 4x^3 - 26x^2 + 40x,$$

with $0 < x < 5/2$. Now

$$\begin{aligned} V'(x) &= 12x^2 - 52x + 40 \\ &= 4(3x - 10)(x - 1). \end{aligned}$$

The candidates are $x = 1$ and $x = 10/3$, but only $x = 1$ lies in the physical domain. Since $V''(1) = 24 - 52 = -28 < 0$, the maximum volume is

$$\boxed{V(1) = 18 \text{ cm}^3},$$

and the box dimensions are 6 cm by 3 cm by 1 cm.

Other optimisation models include minimising material cost, finding the closest point on a curve, and maximising the volume of a solid subject to a fixed surface area.

12.16 Related rates

When several quantities change with time and are related by an equation, differentiate the equation with respect to time. The chain rule inserts a rate for every changing variable. A useful workflow is:

1. draw and label the situation;
2. record the known and required rates with their signs;
3. write an equation relating the variables;
4. eliminate extra variables if necessary;
5. differentiate with respect to time; and
6. substitute numerical values only after differentiating.

For example, if $x^2 - y^2 = 9$, then

$$2x \frac{dx}{dt} - 2y \frac{dy}{dt} = 0,$$

so

$$\boxed{\frac{dy}{dt} = \frac{x}{y} \frac{dx}{dt}}.$$

12.16.1 Expanding circle

The area of a circle increases at $6\pi \text{ cm}^2 \text{ s}^{-1}$. Since $A = \pi r^2$,

$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}.$$

At $r = 1.5 \text{ cm}$,

$$6\pi = 2\pi(1.5) \frac{dr}{dt} \quad \Longrightarrow \quad \boxed{\frac{dr}{dt} = 2 \text{ cm s}^{-1}}.$$

12.16.2 Filling a cylindrical tank

A cylindrical tank of fixed radius 6 cm is filled at $3 \text{ cm}^3 \text{ min}^{-1}$. From $V = \pi r^2 h$,

$$3 = \pi(6)^2 \frac{dh}{dt},$$

so

$$\boxed{\frac{dh}{dt} = \frac{1}{12\pi} \text{ cm min}^{-1}}.$$

12.16.3 Water draining from a cone

An inverted cone is 20 cm tall with opening radius 8 cm. Similar triangles give

$$\frac{r}{h} = \frac{8}{20} = \frac{2}{5}, \quad r = \frac{2}{5}h.$$

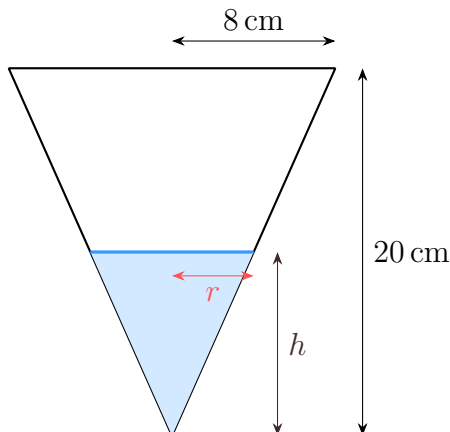


Figure 12.11: Similar triangles relate the water radius and depth.

Therefore,

$$V = \frac{1}{3}\pi r^2 h = \frac{4\pi}{75}h^3$$

and

$$\frac{dV}{dt} = \frac{4\pi}{25}h^2 \frac{dh}{dt}.$$

If water drains at $15 \text{ cm}^3 \text{ s}^{-1}$, then $dV/dt = -15$. When the depth is halfway down the cone, $h = 10$, so

$$\boxed{\frac{dh}{dt} = -\frac{15}{16\pi} \text{ cm s}^{-1}}.$$

The negative sign indicates that the water level is falling.

12.17 Further applications of differential calculus

The derivative is a general language for instantaneous change. Examples include

- velocity and acceleration from a position function;
- marginal cost, revenue, and profit in economics;
- growth and decay rates in population and radioactive models;
- sensitivity and error propagation in measurement;

- gradients used in numerical optimisation; and
- tangent-line approximation $f(x) \approx f(a) + f'(a)(x - a)$ near $x = a$.

For instance, since $\sqrt{101} = f(101)$ for $f(x) = \sqrt{x}$, linearising at $a = 100$ gives

$$\sqrt{101} \approx 10 + \frac{1}{20}(1) = 10.05.$$

Chapter summary and common pitfalls

Key ideas. The derivative is a local linear rate of change. Differentiation rules reduce complicated functions to standard derivatives, while signs of derivatives reveal monotonicity, curvature, and extrema.

Common pitfalls. Apply the chain rule to every composite layer, distinguish stationary from turning points, retain units in rate problems, and test endpoints as well as interior critical points in optimisation.

12.18 Practice and exercises

Foundations and core differentiation rules

- Let $f(x) = 2x^2 - 3x + 1$.
 - Simplify $[f(x+h) - f(x)]/h$ for $h \neq 0$.
 - Hence find $f'(x)$ from first principles.
 - Find the gradient and equation of the tangent at $x = 2$.
- Use first principles to differentiate each function on its natural domain.
 - $f(x) = x^2$;
 - $g(x) = 1/x$;
 - $h(x) = \sqrt{x}$.
- Investigate differentiability.
 - Explain why $f(x) = |x - 2|$ is not differentiable at $x = 2$.
 - Find a and b so that

$$g(x) = \begin{cases} ax + b, & x < 1, \\ x^2 + 1, & x \geq 1 \end{cases}$$
 is differentiable at $x = 1$.
- Differentiate and simplify.
 - $7x^6 - 4x^3 + 5x - 9$;
 - $3x^{5/2} - 4x^{-2} + 6$;
 - $5\sqrt{x} + 2/\sqrt{x}$;
 - $(2x - 1)(x^2 + 3)$, expanding first.

- Find the requested higher derivatives.

- For $y = x^4 - 3x^3 + 2x$, find y' , y'' , y''' , and $y^{(4)}$.

(b) For $y = x^5 - 5x^3$, find y'' and $y^{(5)}$.

(c) For $y = 1/x$, find y' , y'' , and y''' .

6. Use the product rule.

(a) $(x^2 + 1)(x - 3)$;

(b) $x^2 e^x$;

(c) $(2x - 1) \sin x$;

(d) $x^3 \sqrt{1 + x}$.

7. Use the quotient rule.

(a) $\frac{3x + 1}{x - 2}$;

(b) $\frac{x^2}{1 + x}$;

(c) $\frac{\sin x}{\cos x}$;

(d) $\frac{\ln x}{x}$.

8. Use the chain rule.

(a) $(3x^2 - 1)^5$;

(b) $\sqrt{4 - x^2}$;

(c) $(1 + 2x)^{-3}$;

(d) $(x^2 + x + 1)^{3/2}$.

9. Combine the product and chain rules.

(a) $x(1 + x^2)^4$;

(b) $e^x \sqrt{1 + x^2}$;

(c) $\sin x \cos 2x$.

10. Combine the quotient and chain rules.

(a) $\frac{(x^2 + 1)^3}{x - 1}$;

(b) $\frac{e^{2x}}{1 + x^2}$;

(c) $\frac{\ln(1 + x)}{\sqrt{x}}$.

11. Find the tangent and normal at the stated point.

- (a) $y = x^3 - 2x$ at $x = 1$;
 - (b) $y = \sqrt{x}$ at $x = 4$;
 - (c) $y = \ln x$ at $x = e$.
12. Determine the parameter.
- (a) The tangent to $y = x^2 + kx$ at $x = 1$ is parallel to $5x - y = 2$. Find k and the tangent equation.
 - (b) The tangent to $y = ax^2$ at $x = 2$ passes through $(0, -4)$. Find a .
13. Recover functions from derivative information.
- (a) Given $f'(x) = 3x^2 - 12x + 9$ and $f(0) = 5$, find f and its stationary points.
 - (b) Given $g''(x) = 6x - 4$, $g'(0) = 2$, and $g(0) = 1$, find g .
14. Use linear approximation to estimate
- (a) $\sqrt{99.6}$ near $x = 100$;
 - (b) $\sqrt[3]{8.12}$ near $x = 8$;
 - (c) $1/1.98$ near $x = 2$.
15. Each proposed derivative contains an error. Identify the error and give the correct derivative.
- (a) $\frac{d}{dx}(x^2 + 1)^5 = 5(x^2 + 1)^4$;
 - (b) $\frac{d}{dx}(x^2 \sin x) = 2x \cos x$;
 - (c) $\frac{d}{dx} \left(\frac{x+1}{x-1} \right) = \frac{2}{(x-1)^2}$.

Exponential, logarithmic, trigonometric, and hyperbolic functions

16. Differentiate the exponential functions.
- (a) e^{3x-1} ;
 - (b) $4e^{-x^2}$;
 - (c) $e^{\sqrt{x}}$;
 - (d) $\frac{e^x}{1 + e^x}$.
17. Differentiate functions with constant bases.
- (a) 2^{x^2} ;

(b) 5^{3x-1} ;

(c) 10^{-x} ;

(d) $3^{\sin x}$.

18. Differentiate the logarithmic functions and state any domain restrictions needed for the original function.

(a) $\ln(x^2 + 4)$;

(b) $\ln \sqrt{1+x}$;

(c) $\log_3(2x-1)$;

(d) $\ln |\sin x|$.

19. Use logarithmic differentiation.

(a) $y = x^x, x > 0$;

(b) $y = (x+1)^x, x > -1$;

(c) $y = x^{\sin x}, x > 0$.

20. Find y'/y first, then obtain y' .

(a) $y = \frac{(x^2+1)^3 \sqrt{x-1}}{x^2}, x > 1$;

(b) $y = \sqrt{\frac{x-1}{x+1}}, x > 1$.

(c) $y = \frac{(x^2+4)^2}{x^3(1-x)}, 0 < x < 1$.

21. Differentiate the basic trigonometric expressions.

(a) $\sin x$;

(b) $\cos x$;

(c) $\tan x$;

(d) $\cot x$;

(e) $\sec x$;

(f) $\csc x$;

(g) $\sin x + \cos x$.

22. Differentiate the composite trigonometric functions.

(a) $\sin(3x^2)$;

(b) $\cos^4(2x)$;

(c) $\tan \sqrt{x}$;

(d) $\sec(1 - 2x)$.

23. Differentiate and simplify.

(a) $x \sin x$;

(b) $e^x \cos x$;

(c) $\tan x/x$;

(d) $\sin x/(1 + \cos x)$.

24. Derive the stated rule by first rewriting the function as a quotient or reciprocal.

(a) $(\tan x)' = \sec^2 x$;

(b) $(\sec x)' = \sec x \tan x$;

(c) $(\cot x)' = -\csc^2 x$;

(d) $(\csc x)' = -\csc x \cot x$.

25. Differentiate the inverse trigonometric functions.

(a) $\sin^{-1}(2x)$;

(b) $\cos^{-1}(x/3)$;

(c) $\tan^{-1}(3x)$;

(d) $\tan^{-1} \sqrt{x}$;

(e) $\sec^{-1}(2x)$.

26. Differentiate expressions containing inverse trigonometric functions.

(a) $x \sin^{-1} x$;

(b) $e^x \tan^{-1} x$;

(c) $\sin^{-1}(x^2)$.

27. Differentiate the hyperbolic expressions.

(a) $\sinh x$;

(b) $\cosh x$;

(c) $\tanh x$;

(d) $\operatorname{sech} x$;

(e) $\operatorname{coth} x$;

(f) $\operatorname{cosech} x$;

(g) $\sinh x \cosh x$.

28. Differentiate the composite hyperbolic functions.

- (a) $\sinh(3x^2)$;
- (b) $\cosh^2 x$;
- (c) $e^x \tanh x$;
- (d) $\ln(\cosh x)$.

29. Find a formula for the n th derivative.

- (a) e^{2x} ;
- (b) $\sin 3x$;
- (c) xe^x ;
- (d) x^2e^x .

30. Find and classify the stationary points on the stated domains.

- (a) $f(x) = e^x(2 - x)$, $x \in \mathbb{R}$;
- (b) $g(x) = \ln x/x$, $x > 0$;
- (c) $h(x) = x^2e^{-x}$, $x \geq 0$;
- (d) $k(x) = x/(1 + x^2)$, $x \in \mathbb{R}$.

Implicit, inverse, and parametric differentiation

31. Find dy/dx by implicit differentiation.

- (a) $x^2 + y^2 = 25$;
- (b) $x^2/9 + y^2/4 = 1$;
- (c) $xy = 6$;
- (d) $x^2 - y^2 = 4$.

32. Differentiate the algebraic relations implicitly.

- (a) $x^2y + xy^2 = 6$;
- (b) $x^3 + y^3 = 3xy$;
- (c) $x^2 + xy + y^2 = 7$;
- (d) $x^2y^3 = 8$.

33. Differentiate the transcendental relations implicitly.

- (a) $\sin(x + y) = x$;
- (b) $e^{xy} = x + y$;

(c) $x \sin y + y \cos x = 2$;

(d) $\ln(x + y) = xy$.

34. Find the tangent and normal to each implicit curve.

(a) $x^2 + y^2 = 25$ at $(3, 4)$;

(b) $x^2 + xy + y^2 = 7$ at $(1, 2)$;

(c) $xy + \ln y = 2$ at $(2, 1)$;

(d) $x^2/9 + y^2/4 = 1$ at $(0, 2)$.

35. Find d^2y/dx^2 at the stated point.

(a) $x^2 + y^2 = 25$ at $(3, 4)$;

(b) $x^2 + xy + y^2 = 7$ at $(1, 2)$;

(c) $xy = 6$ at $(2, 3)$.

36. Use $(f^{-1})'(a) = 1/f'(f^{-1}(a))$.

(a) If $f(x) = 3x - 5$, find $(f^{-1})'(4)$.

(b) If $f(x) = x^3 + x$, find $(f^{-1})'(2)$.

(c) If $f(x) = \ln x$, find $(f^{-1})'(1)$.

(d) If $f(x) = x + \ln x$, find $(f^{-1})'(1)$.

37. For each parametrisation, find dy/dx .

(a) $x = t^2 + t, y = t^3$;

(b) $x = \cos t, y = \sin t$;

(c) $x = e^t, y = e^{-t}$;

(d) $x = t^2, y = t^3$;

(e) $x = \ln t, y = t^2, t > 0$.

38. Find d^2y/dx^2 .

(a) $x = t^2 + t, y = t^3$;

(b) $x = \cos t, y = \sin t$;

(c) $x = t - \sin t, y = 1 - \cos t$;

(d) $x = t, y = e^t$.

39. Find the tangent and normal to each parametric curve at the stated parameter value.

(a) $x = t^2 + 1, y = t^3 - 3t$ at $t = 1$;

- (b) $x = 2 \cos t$, $y = 3 \sin t$ at $t = \pi/4$;
(c) $x = e^t$, $y = te^t$ at $t = 0$.
40. Locate all horizontal and vertical tangents.
- (a) $x = 2e^t$, $y = e^t + 2e^{-t}$;
(b) $x = t^2 - 4t$, $y = t^3 - 3t$;
(c) $x = t^3 - 3t$, $y = t^2$.
41. Eliminate the parameter and verify that the parametric derivative agrees with the Cartesian derivative.
- (a) $x = t + 1$, $y = t^2$;
(b) $x = 2 \cos t$, $y = 2 \sin t$;
(c) $x = t^2$, $y = t^4$.
42. For $x = t^2$ and $y = t^3$, find the point or points at which the tangent has gradient
- (a) 1;
(b) 3.
43. Apply the chain rule through intermediate variables.
- (a) $x = t^2 + 1$ and $y = x^3$: find dy/dt .
(b) $y = e^{2t} + 1$ and $z = \ln y$: find dz/dt .
(c) $u = \sin x$ and $x = t^2$: find du/dt .
44. Differentiate with respect to time.
- (a) $u = x^2 + y^2$, where $x = \cos t$ and $y = \sin t$.
(b) $A = \pi r^2$, where $r = \sqrt{t + 1}$.
(c) $V = \frac{4}{3}\pi r^3$, where $r = t^2 + 1$.
45. Use differentials to estimate the stated change or percentage error.
- (a) The side of a cube is 10 ± 0.02 cm. Estimate the maximum percentage error in its volume.
(b) A sphere has radius 5 ± 0.01 cm. Estimate the maximum percentage error in its surface area.
(c) Estimate the change in $y = x^3$ when x changes from 2 to 2.03.
(d) A cylinder has measured radius 10 ± 0.02 cm and height 30 ± 0.05 cm. Estimate the maximum percentage error in its volume.

Extrema, concavity, and curve analysis

46. Find the critical numbers on the stated domains.

- (a) $x^3 - 3x$;
- (b) $x^4 - 4x^2$;
- (c) $x + 4/x$, $x > 0$;
- (d) $x^2 + 1/x$, $x > 0$.

47. Find the intervals on which each function increases and decreases.

- (a) $x^2 - 4x + 1$;
- (b) $x^3 - 3x$;
- (c) $\ln x - x$, $x > 0$.

48. Use the first derivative test to classify the critical points when

- (a) $f'(x) = (x + 2)(x - 1)$;
- (b) $g'(x) = x^2(x - 3)$;
- (c) $h'(x) = e^{-x}(2 - x)$.

49. Use the second derivative test where possible.

- (a) $f(x) = x^3 - 6x^2 + 9x$;
- (b) $g(x) = x^4 - 4x^2$;
- (c) $h(x) = \sin x$ on $[0, 2\pi]$;
- (d) $k(x) = x + 1/x$, $x > 0$.

50. Classify the stationary point at the origin for x^3 , x^4 , and x^5 . State which are stationary points of inflection. Generalise the classification for $y = x^n$, where $n > 1$ is an integer.

51. Find the absolute extrema on each closed interval.

- (a) $x^3 - 3x$ on $[-2, 3]$;
- (b) $x^2 - 4x + 5$ on $[0, 5]$;
- (c) $x + 4/x$ on $[1, 5]$;
- (d) e^x on $[-1, 1]$.

52. Find the intervals of concavity and all points of inflection.

- (a) $3x^2 - 4x^3$;
- (b) $x^4 - 4x^3$;

(c) $x^3 - 6x$.

53. Analyse concavity, paying attention to the domain.

(a) $\ln x$;

(b) $1/x$;

(c) e^{-x^2} .

54. Sketch each graph from its principal features.

(a) $y = x^2 - 4x + 3$;

(b) $y = x^3 - 3x$;

(c) $y = x^3 - 3x^2$.

55. Sketch $f(x) = x^4 - 8x^2$, showing its intercepts, stationary points, monotonicity, concavity, and inflection points. Hence state its range.

56. Find all vertical, horizontal, and oblique asymptotes.

(a) $\frac{x^2 - 1}{x - 2}$;

(b) $\frac{2x^2 + 1}{x^2 - 4}$;

(c) $\frac{x^2 + 1}{x}$;

(d) $\frac{x^3 + 1}{x^2 - 1}$, also identifying any removable discontinuity.

57. Analyse the stationary points and inflection points.

(a) xe^{-x} ;

(b) x^2e^{-x} for $x \geq 0$.

58. For $f(x) = \ln x/x$:

(a) find the limits as $x \rightarrow 0^+$ and $x \rightarrow \infty$;

(b) find the stationary point and intervals of monotonicity;

(c) find the intervals of concavity and point of inflection.

59. Find the absolute extrema.

(a) $\sin x + \cos x$ on $[0, 2\pi]$;

(b) $\sin 2x$ on $[0, \pi]$.

60. The parametric curve $x = 2e^t$, $y = e^t + 2e^{-t}$ has one stationary point. Find and classify it.
61. A differentiable function has $f'(x) > 0$ for $x < -1$, $f'(x) < 0$ for $-1 < x < 2$, and $f'(x) > 0$ for $x > 2$. Construct a variation table and classify the critical points.
62. Given $f'(x) = (x + 2)^2(x - 1)$, find the intervals of monotonicity and classify the stationary points $x = -2$ and $x = 1$. Also find the intervals of concavity and the inflection-point abscissae.
63. Find the unknown coefficients.
- (a) $f(x) = x^3 + ax^2 + bx$ is stationary at $x = 1$ and $x = 2$.
 - (b) $g(x) = x^2 + ax + b$ has minimum value -3 at $x = 2$.
64. Find every point at which the tangent is parallel to the stated line.
- (a) $y = x^3 - 2x$, parallel to $y = 4x + 1$;
 - (b) $y = \ln x$, parallel to $y = x/2$;
 - (c) $y = e^x$, parallel to the x -axis.
65. Find the cubic $f(x) = x^3 + ax^2 + bx + c$ that passes through $(0, 1)$ and is stationary at both $x = -1$ and $x = 1$. Classify both stationary points.

Optimisation and related rates

66. A rectangle has perimeter 20 cm.
- (a) Find the dimensions and greatest possible area.
 - (b) Generalise the result to a rectangle of perimeter P .
 - (c) If the area is fixed at A , find the rectangle with the smallest perimeter and state that perimeter.
67. A rectangular field beside a straight river requires fencing on only three sides.
- (a) If 120 m of fencing is available, find the dimensions and maximum area.
 - (b) Generalise the result for a fencing length of L metres.
68. Squares of side x are removed from the corners of an 8 cm by 5 cm card.
- (a) Find the cut size and resulting dimensions that maximise the volume of the open box.
 - (b) Find the volume when $x = 0.5$ cm and compare it with the maximum.

69. Optimise the material used for cylindrical containers.
- (a) A closed cylinder has volume $500\pi \text{ cm}^3$. Minimise its surface area.
 - (b) An open-topped cylinder has volume $108\pi \text{ cm}^3$. Minimise its surface area.
 - (c) State the minimum surface area obtained in each case.
70. Find the point or points on $y = x^2$ closest to
- (a) $(0, 3)$;
 - (b) $(0, 1)$.
71. A product has demand equation $p = 80 - 0.4x$, where x units are sold at RM p each.
- (a) State the economically meaningful interval for x .
 - (b) Find the sales level, price, and maximum revenue.
 - (c) Find and interpret the marginal revenue at the optimum.
72. A response to a dose $d > 0$ is modelled by $R(d) = d/(d^2 + 4)$.
- (a) Find the dose that maximises the response and state the maximum value.
 - (b) Find the limiting response as $d \rightarrow 0^+$ and as $d \rightarrow \infty$.
73. A rectangle is inscribed beneath a downward-opening parabola, above the x -axis, with its upper vertices on the parabola and its centre on the y -axis.
- (a) For $y = 12 - x^2$, find its maximum area.
 - (b) Generalise the result for $y = a - bx^2$, where $a, b > 0$.
74. A right circular cylinder is inscribed in a sphere of radius 5 cm.
- (a) Find the cylinder dimensions that maximise its volume.
 - (b) Find the ratio of this maximum cylinder volume to the sphere's volume.
75. A Norman window consists of a rectangle surmounted by a semicircle. Its outer perimeter is 20 m.
- (a) Find the dimensions that maximise the total area.
 - (b) Compare the optimal rectangular height h with the radius r , and state the maximum area.

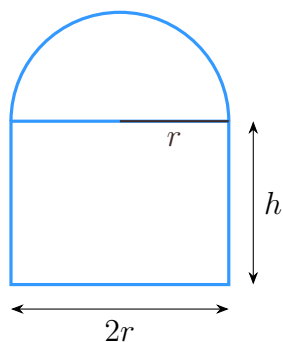


Figure 12.12: The Norman window in Question 75.

76. A particle has position $s(t) = t^3 - 6t^2 + 9t$ for $0 \leq t \leq 4$.
- Find its velocity and acceleration, times at rest, direction of motion, displacement, and total distance travelled.
 - Find its average velocity and greatest speed on the interval.
77. Solve the circle-rate problems.
- The area increases at $6\pi \text{ cm}^2 \text{ s}^{-1}$. Find dr/dt when $r = 1.5 \text{ cm}$.
 - The radius increases at 0.4 cm s^{-1} . Find the rates of change of circumference and area when $r = 5 \text{ cm}$.
78. A 10 m ladder leans against a wall. Its foot moves away from the wall at 2 m s^{-1} . When the foot is 6 m from the wall, find
- the rate at which the top descends;
 - the rate of change of the angle between the ladder and the ground.

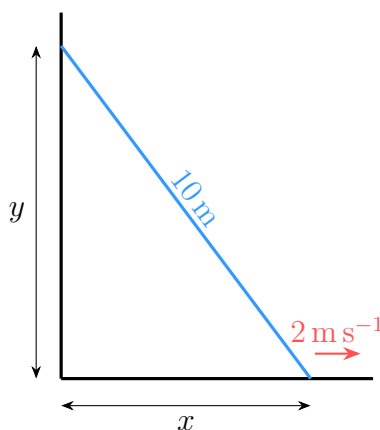


Figure 12.13: The sliding ladder in Question 78.

79. Water drains from an inverted cone of height 20 cm and opening radius 8 cm at $15 \text{ cm}^3 \text{ s}^{-1}$.

- (a) Find dh/dt when the water depth is 10 cm.
- (b) Find the corresponding water-surface radius and dr/dt .
80. A person 1.8 m tall walks away from a 6 m streetlight at 1.5 m s^{-1} .
- (a) Find the rates at which the shadow length and the shadow tip move.
- (b) Find the shadow length when the person is 10 m from the light.
- Applications and modelling**
81. A factory's total cost is $C(x) = 5000 + 25x + 0.02x^2$ ringgit.
- (a) Find the marginal cost, evaluate it at $x = 100$, and use it to estimate the cost of producing the 101st unit.
- (b) Find the average cost per unit at $x = 100$ and compare it with the marginal cost.
82. Demand for a product is $p = 120 - 0.5x$. Form the revenue function and find the output, price, and revenue at the revenue-maximising point.
83. For the demand in Question 82, suppose the cost is $C(x) = 2000 + 20x$.
- (a) Find the profit-maximising output and maximum profit.
- (b) Find the two break-even outputs.
84. A firm's average cost is $A(x) = 500/x + 20 + 0.1x$ for $x > 0$.
- (a) Find the production level that minimises average cost and state the minimum.
- (b) State the intervals on which the average cost decreases and increases.
85. A projectile has height $h(t) = 1 + 24t - 4.9t^2$ metres.
- (a) Find its velocity, greatest height, and time of greatest height.
- (b) Find the positive time at which it reaches the ground and its velocity at impact.
86. A car's stopping distance is $d(v) = 0.006v^2 + 0.25v$ metres, where v is in km h^{-1} .
- (a) Find $d(80)$ and $d'(80)$, then estimate the extra stopping distance caused by a 5 km h^{-1} increase in speed.
- (b) Calculate the exact increase and comment on the approximation.
87. A population is modelled by $P(t) = 5000e^{0.03t}$.
- (a) Find the population and its growth rate after 10 years.
- (b) Find its doubling time.

88. The concentration of a medicine is $C(t) = 20te^{-0.5t}$ mg L⁻¹ for $t \geq 0$.
- Determine when the concentration is increasing and decreasing.
 - Find when the concentration is greatest and calculate the maximum concentration.
89. A cooling object has temperature $T(t) = 22 + 68e^{-0.12t}$ degrees Celsius.
- Find the initial cooling rate and the limiting temperature.
 - Find its cooling rate after 10 minutes and the time at which it reaches 30°C.
90. A population obeys

$$P' = 0.30P \left(1 - \frac{P}{5000} \right), \quad P(0) = 600.$$

At what population is the growth rate greatest? Find that greatest rate and the time at which it occurs. Also find the first time at which the population reaches 4000.

91. The power delivered to a load resistance R is

$$P(R) = \frac{V^2 R}{(R + r)^2},$$

where V and r are positive constants. Show that power is maximised when $R = r$, then find the maximum for $V = 12$ V and $r = 4\Omega$. Also find the limiting power as $R \rightarrow 0^+$ and $R \rightarrow \infty$.

92. A camera is 20 m from a straight road. A car travels along the road at 15 m s^{-1} . If x is its distance along the road from the nearest point and θ is the camera angle, find $d\theta/dt$ and θ when $x = 30$ m.

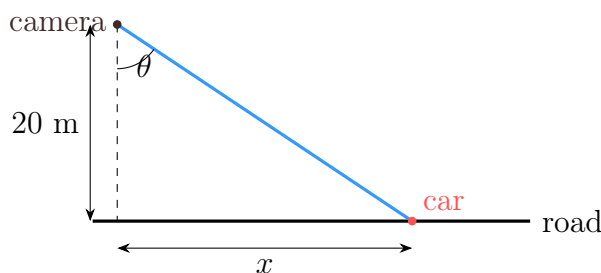


Figure 12.14: The tracking-camera geometry in Question 92.

93. Air enters a spherical balloon at $30 \text{ cm}^3 \text{ s}^{-1}$. Find the rates of change of its radius and surface area when the radius is
- 2 cm;
 - 4 cm.

94. A circular oil spill grows at $120 \text{ m}^2 \text{ min}^{-1}$ and has negligible initial area.
- Find the radial growth rate when the radius is 10 m and when the area is $400\pi \text{ m}^2$.
 - Find the time required to reach area $400\pi \text{ m}^2$.
95. A 10 m trough has triangular ends 4 m wide at the top and 2 m deep. Water enters at $0.5 \text{ m}^3 \text{ min}^{-1}$.
- Find the water-surface width and volume when the depth is 1 m.
 - Find the rate at which the water depth rises at that instant.

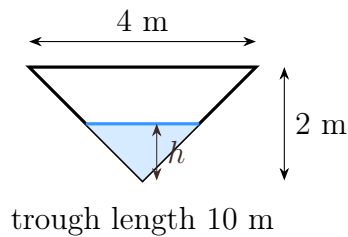


Figure 12.15: The triangular trough in Question 95.

96. A spherical component has measured radius $12 \pm 0.05 \text{ cm}$. Use differentials to estimate the maximum absolute and percentage errors in
- its volume;
 - its surface area.
97. A temperature sensor produces voltage $V = 0.04T + 0.5$.
- Find its sensitivity dV/dT and inverse calibration formula.
 - Find the temperature when $V = 1.7 \text{ V}$.
 - If the voltage uncertainty is $\pm 0.002 \text{ V}$, estimate the corresponding uncertainty in temperature.
98. A road profile is $h(x) = 0.002x^3 - 0.15x^2 + 2x$ metres.
- Find the gradient at $x = 10 \text{ m}$.
 - Locate and classify the stationary points, including their heights.
99. An open box is made by cutting squares of side x from a 30 cm by 20 cm sheet.
- Find the cut size and box dimensions that maximise volume.
 - Give the exact and approximate maximum volumes.

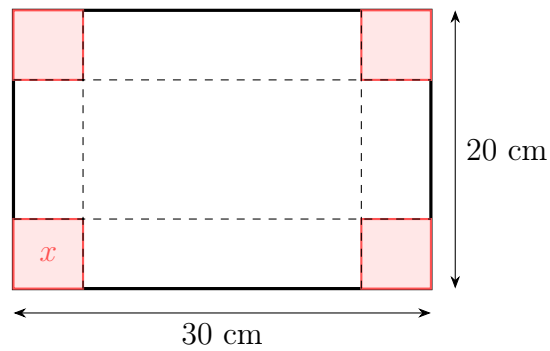


Figure 12.16: The sheet and corner cuts in Question 99.

100. A drone's distance from a transmitter is $d(t) = \sqrt{100 + t^2}$ metres, and the received signal is $S = 1000/d^2$.
- (a) Find $S(t)$ and its initial value.
 - (b) Find the rate of change of signal strength at $t = 10$ s.
 - (c) Show that the signal decreases for $t > 0$ and state its limiting value.

Chapter 13

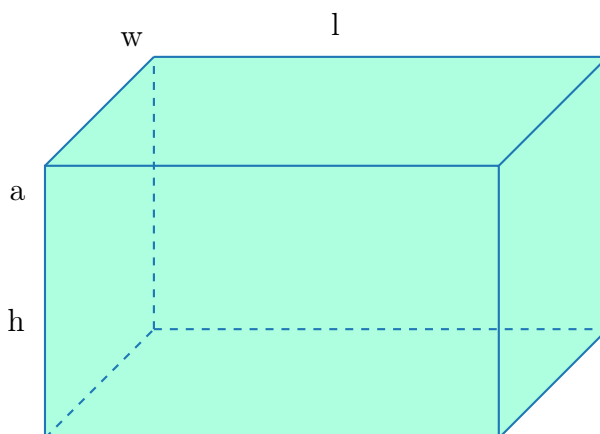
Integral Calculus

Chapter guide

Prerequisites. Differentiation, functions and graphs, algebraic identities, trigonometry, exponentials, and logarithms.

Learning outcomes. Readers should be able to find antiderivatives, apply standard integration techniques, evaluate definite integrals, and use integration to calculate accumulated quantities, areas, and volumes.

The volume of the cuboid below is found directly from its three perpendicular dimensions.



Its volume is

$$V = \text{length} \times \text{width} \times \text{height}.$$

An irregular solid, by contrast, may not have a single elementary volume formula.

Integration handles such problems by accumulating contributions from many small pieces. The same principle underlies calculations of area, displacement, mass, probability, and other accumulated quantities. This chapter follows standard introductory-calculus treatments of antiderivatives, definite integrals, and applications (Strang and Herman 2016a; Strang and Herman 2016b; Harvard University 2026).

13.1 Integration as antidifferentiation

In calculus, an antiderivative of a function f is a differentiable function F whose derivative is equal to the original function f .

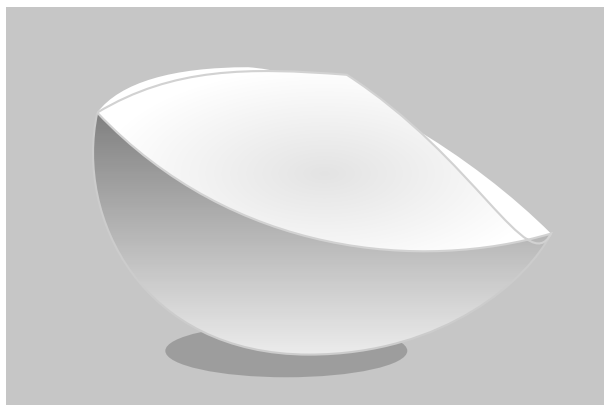


Figure 13.1: An irregular solid whose volume may be estimated by slicing.

Definition 13.1: Antiderivative.

Let f be a function defined on an interval. A differentiable function F is an *antiderivative* of f if

$$F'(x) = f(x).$$

Since the derivative of a constant is zero, every function of the form $F(x) + C$, where C is an arbitrary constant, is also an antiderivative of f . The family of all such antiderivatives is called the **indefinite integral** of f and is written as

$$\int f(x) dx = F(x) + C.$$

For example, if $f(x) = 10$, then $F(x) = 10x$ is one antiderivative because $F'(x) = 10$.

$$\begin{aligned} \int f(x) dx &= F(x) + C, \\ \int 10 dx &= 10x + C. \end{aligned}$$

To find an antiderivative, reverse an appropriate differentiation rule and include an arbitrary constant.

1. Identify a differentiation rule that could produce the integrand.
2. Reverse that rule. For example, $d(x^5)/dx = 5x^4$.
3. Include the constant of integration: $\int 5x^4 dx = x^5 + C$.

For instance, $x^2 - 3$ and $x^2 + 9$ both have derivative $2x$. They differ only by a constant, which explains the term C in an indefinite integral.

13.1.1 Sigma notation and limits of finite sums

Long additions occur frequently in algebra, probability, and calculus. Sigma notation provides a compact and precise way to describe such sums. The Greek capital letter sigma, Σ , is used because it corresponds to the letter “S” for sum.

The expression

$$\boxed{\sum_{k=m}^n a_k} \quad (13.1)$$

means that the term a_k is evaluated for every integer value of k from m to n , inclusive, and the resulting terms are added:

$$\sum_{k=m}^n a_k = a_m + a_{m+1} + a_{m+2} + \cdots + a_n.$$

Provided that $m \leq n$, the number of terms is

$$\boxed{n - m + 1.} \quad (13.2)$$

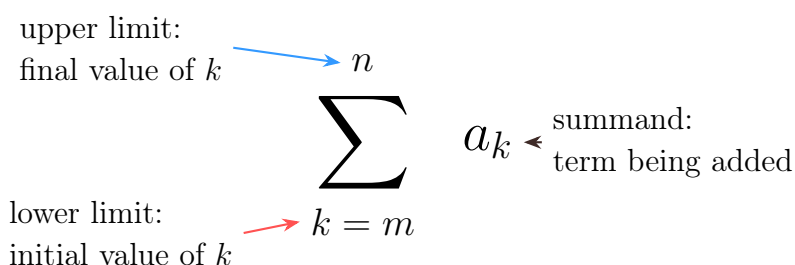


Figure 13.2: The parts of sigma notation. The letter k is called the index of summation.

Expanding and evaluating a finite sum

To expand a sum, substitute the consecutive integer values of the index into the summand. For example,

$$\begin{aligned} \sum_{k=1}^5 k &= 1 + 2 + 3 + 4 + 5 = 15, \\ \sum_{k=1}^3 (-1)^k k &= -1 + 2 - 3 = -2, \\ \sum_{k=1}^2 \frac{k}{k+1} &= \frac{1}{2} + \frac{2}{3} = \frac{7}{6}, \\ \sum_{k=4}^5 \frac{k^2}{k-1} &= \frac{16}{3} + \frac{25}{4} = \frac{139}{12}. \end{aligned}$$

The index is a *dummy variable*: changing its letter does not change the value of the sum. Thus,

$$\sum_{k=1}^n k^2 = \sum_{r=1}^n r^2 = \sum_{j=1}^n j^2.$$

Algebraic properties of finite sums

Let c be a constant and let (a_k) and (b_k) be finite sequences. Sigma notation obeys the usual distributive and associative laws:

$$\sum_{k=m}^n c = c(n - m + 1), \quad (13.3)$$

$$\sum_{k=m}^n c a_k = c \sum_{k=m}^n a_k, \quad (13.4)$$

$$\sum_{k=m}^n (a_k + b_k) = \sum_{k=m}^n a_k + \sum_{k=m}^n b_k, \quad (13.5)$$

$$\sum_{k=m}^n (a_k - b_k) = \sum_{k=m}^n a_k - \sum_{k=m}^n b_k. \quad (13.6)$$

A sum may also be split at any integer p satisfying $m \leq p < n$:

$$\boxed{\sum_{k=m}^n a_k = \sum_{k=m}^p a_k + \sum_{k=p+1}^n a_k.} \quad (13.7)$$

For example,

$$\begin{aligned} \sum_{k=1}^4 (3k - 2) &= 3 \sum_{k=1}^4 k - 2 \sum_{k=1}^4 1 \\ &= 3(10) - 2(4) \\ &= 22. \end{aligned}$$

Standard finite-sum formulae

The following identities are used repeatedly throughout calculus:

$$\sum_{k=1}^n 1 = n, \quad (13.8)$$

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}, \quad (13.9)$$

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}, \quad (13.10)$$

$$\sum_{k=1}^n k^3 = \left[\frac{n(n+1)}{2} \right]^2. \quad (13.11)$$

For a geometric sum with common ratio $r \neq 1$,

$$\boxed{\sum_{k=0}^n ar^k = a \frac{1 - r^{n+1}}{1 - r}.} \quad (13.12)$$

The formula for $1 + 2 + \cdots + n$ can be understood visually by placing two copies of a triangular arrangement together. The example below uses $n = 5$.

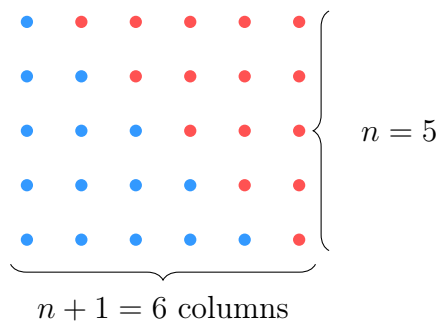


Figure 13.3: Twice the sum $1 + 2 + \cdots + n$ contains $n(n+1)$ dots, so $\sum_{k=1}^n k = n(n+1)/2$.

Changing the index

Sometimes a sum becomes easier to interpret after shifting its index. If $j = k - r$, then the limits and the summand must all be changed consistently. For example,

$$\begin{aligned} \sum_{k=3}^{n+2} (k-2)^2 &= \sum_{j=1}^n j^2 \\ &= \frac{n(n+1)(2n+1)}{6}. \end{aligned}$$

The first equality uses $j = k - 2$: when $k = 3$, $j = 1$, and when $k = n + 2$, $j = n$.

Telescoping sums

In a telescoping sum, most intermediate terms cancel. Consider

$$S_n = \sum_{k=1}^n \frac{1}{k(k+1)}.$$

Since

$$\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1},$$

we obtain

$$\begin{aligned}
S_n &= \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1}\right) \\
&= 1 - \frac{1}{n+1} \\
&= \frac{n}{n+1}.
\end{aligned}$$

Limits of finite sums

For each fixed positive integer n , the expression

$$S_n = \sum_{k=1}^n a_{k,n}$$

is a finite sum. However, as n increases, the number of terms and sometimes the terms themselves change. We may therefore study the sequence (S_n) through the limit

$$\lim_{n \rightarrow \infty} S_n.$$

Standard finite-sum formulae often convert such limits into ordinary rational-function limits.

Example 13.1

$$\begin{aligned}
\lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=1}^n k &= \lim_{n \rightarrow \infty} \frac{1}{n^2} \cdot \frac{n(n+1)}{2} \\
&= \lim_{n \rightarrow \infty} \frac{n+1}{2n} \\
&= \frac{1}{2}.
\end{aligned}$$

Example 13.2

$$\begin{aligned}
\lim_{n \rightarrow \infty} \frac{1}{n^3} \sum_{k=1}^n k^2 &= \lim_{n \rightarrow \infty} \frac{n(n+1)(2n+1)}{6n^3} \\
&= \lim_{n \rightarrow \infty} \frac{(n+1)(2n+1)}{6n^2} \\
&= \frac{1}{3}.
\end{aligned}$$

Example 13.3

Using the telescoping result above,

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{k(k+1)} = \lim_{n \rightarrow \infty} \frac{n}{n+1} = 1.$$

Connection with area and integration

An important type of finite sum has the form

$$\frac{1}{n} \sum_{k=1}^n f\left(\frac{k}{n}\right). \quad (13.13)$$

Here, the interval $[0, 1]$ is divided into n subintervals of width $1/n$. Each term is the area of a rectangle with width $1/n$ and height $f(k/n)$. As $n \rightarrow \infty$, the rectangles become narrower and their total area approaches the area beneath the graph. This is the central idea behind a Riemann sum.

For $f(x) = x^2$,

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \left(\frac{k}{n}\right)^2 &= \lim_{n \rightarrow \infty} \frac{1}{n^3} \sum_{k=1}^n k^2 \\ &= \frac{1}{3}. \end{aligned}$$

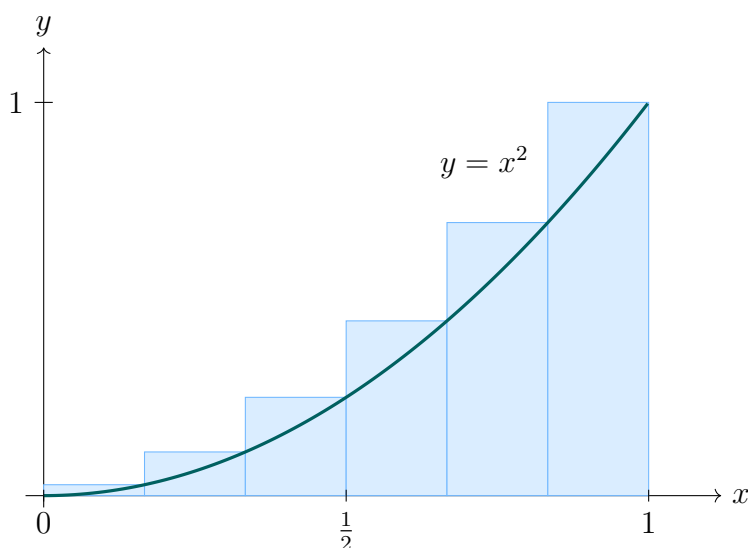


Figure 13.4: A right-endpoint finite sum with six rectangles approximating the area below $y = x^2$ on $[0, 1]$.

This geometric interpretation leads naturally to the formal study of Riemann sums in the next subsection.

13.1.2 Riemann summation

Riemann sums give a systematic method for approximating the net area between a curve and the x -axis. The main idea is to divide an interval into narrow subintervals, construct one rectangle over each subinterval, and add the signed areas of the rectangles.

Let f be continuous on $[a, b]$. Divide the interval into n equal subintervals:

$$a = x_0 < x_1 < x_2 < \cdots < x_n = b.$$

The common width of each subinterval is

$$\Delta x = \frac{b - a}{n}. \quad (13.14)$$

The partition points are therefore

$$x_k = a + k\Delta x = a + k \left(\frac{b - a}{n} \right), \quad k = 0, 1, \dots, n. \quad (13.15)$$

Choose a sample point x_k^* in the k -th subinterval $[x_{k-1}, x_k]$. A rectangle with width Δx and height $f(x_k^*)$ has signed area

$$f(x_k^*)\Delta x.$$

Adding all n rectangles gives the Riemann sum

$$S_n = \sum_{k=1}^n f(x_k^*)\Delta x. \quad (13.16)$$

If $f(x_k^*)$ is positive, the corresponding rectangle contributes positive area. If it is negative, the rectangle lies below the x -axis and contributes negative area.

Left, right, and midpoint sums

The type of Riemann sum depends on where the sample point is chosen within each subinterval.

Left endpoints

Choose x_{k-1} from the k -th subinterval:

$$L_n = \sum_{k=1}^n f(x_{k-1})\Delta x = \sum_{k=0}^{n-1} f(x_k)\Delta x. \quad (13.17)$$

Right endpoints

Choose x_k from the k -th subinterval:

$$R_n = \sum_{k=1}^n f(x_k)\Delta x. \quad (13.18)$$

Midpoints Choose the midpoint

$$x_k^* = a + \left(k - \frac{1}{2}\right) \Delta x :$$

$$M_n = \sum_{k=1}^n f\left(a + \left(k - \frac{1}{2}\right) \Delta x\right) \Delta x. \quad (13.19)$$

For a positive increasing function, the left sum is an underestimate and the right sum is an overestimate. The midpoint sum often gives a closer approximation because its sample points lie centrally within the subintervals.

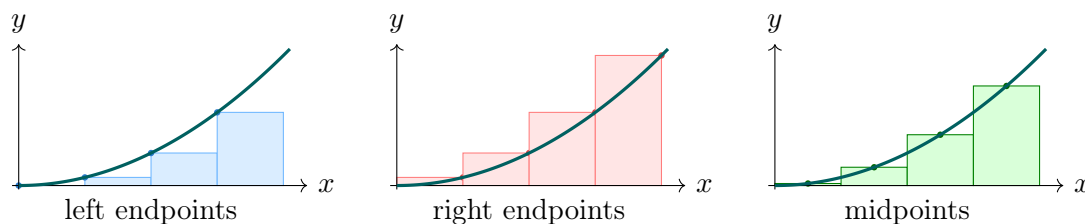


Figure 13.5: Left, right, and midpoint sums for $f(x) = x^2$ on $[0, 2]$ using four equal subintervals.

The definite integral as a limit

As n increases, $\Delta x = (b - a)/n$ approaches zero and the rectangles become narrower. If the Riemann sums approach a unique limit, independent of the choice of sample points, then f is integrable on $[a, b]$ and its definite integral is defined by

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x. \quad (13.20)$$

For right endpoints and equal subintervals, this becomes

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \frac{b-a}{n} \sum_{k=1}^n f\left(a + k \frac{b-a}{n}\right). \quad (13.21)$$

Here, f is the *integrand*, a and b are the lower and upper limits of integration, and x is the variable of integration. The definite integral represents net signed area: area above the x -axis is positive and area below it is negative.

Example 13.4: a finite left Riemann sum

Approximate

$$\int_0^2 x^2 dx$$

using a left Riemann sum with $n = 10$ equal subintervals.

The width is

$$\Delta x = \frac{2 - 0}{10} = \frac{1}{5},$$

and the left endpoints are

$$x_k = \frac{k}{5}, \quad k = 0, 1, \dots, 9.$$

Therefore,

$$\begin{aligned} L_{10} &= \sum_{k=0}^9 f(x_k) \Delta x \\ &= \sum_{k=0}^9 \left(\frac{k}{5}\right)^2 \left(\frac{1}{5}\right) \\ &= \frac{1}{125} \sum_{k=0}^9 k^2 \\ &= \frac{285}{125} \\ &= \boxed{2.28}. \end{aligned}$$

Because x^2 is increasing on $[0, 2]$, this left-endpoint sum underestimates the exact value of the integral.

Example 13.5: evaluating the limit exactly

For the same function, use a right Riemann sum with n subintervals. Now

$$\Delta x = \frac{2}{n}, \quad x_k = \frac{2k}{n}.$$

Hence,

$$\begin{aligned} R_n &= \sum_{k=1}^n \left(\frac{2k}{n}\right)^2 \frac{2}{n} \\ &= \frac{8}{n^3} \sum_{k=1}^n k^2 \\ &= \frac{8}{n^3} \left[\frac{n(n+1)(2n+1)}{6} \right] \\ &= \frac{4(n+1)(2n+1)}{3n^2}. \end{aligned}$$

Taking the limit,

$$\begin{aligned}
 \int_0^2 x^2 dx &= \lim_{n \rightarrow \infty} R_n \\
 &= \lim_{n \rightarrow \infty} \frac{4(n+1)(2n+1)}{3n^2} \\
 &= \boxed{\frac{8}{3}}.
 \end{aligned}$$

Thus, the approximation $L_{10} = 2.28$ approaches the exact value $8/3 \approx 2.667$ as the number of rectangles increases.

Recognising a Riemann sum

A limit written in sigma notation can often be converted into a definite integral by identifying Δx and the sample point x_k . For example,

$$\lim_{n \rightarrow \infty} \frac{3}{n} \sum_{k=1}^n \left(1 + \frac{3k}{n}\right)^2$$

has

$$\Delta x = \frac{3}{n}, \quad x_k = 1 + \frac{3k}{n}.$$

Since the interval begins at 1 and has total width 3, it ends at 4. Therefore,

$$\lim_{n \rightarrow \infty} \frac{3}{n} \sum_{k=1}^n \left(1 + \frac{3k}{n}\right)^2 = \int_1^4 x^2 dx.$$

Common errors

- Forgetting the factor Δx after the summation.
- Using $n - 1$ rectangles when the interval was divided into n subintervals.
- Mixing left-endpoint and right-endpoint indices within the same sum.
- Writing $x_k = k\Delta x$ when the interval begins at $a \neq 0$; the correct expression is $x_k = a + k\Delta x$.
- Treating a definite integral as ordinary unsigned area when part of the graph lies below the x -axis.

13.2 Indefinite integration

Indefinite integration is the reverse process of differentiation. If a function F satisfies

$$F'(x) = f(x),$$

then F is called an *antiderivative* of f . Since the derivative of any constant is zero, a function has an entire family of antiderivatives that differ only by a constant. We therefore write

$$\boxed{\int f(x) dx = F(x) + C,} \quad (13.22)$$

where C is the arbitrary constant of integration. Unlike a definite integral, an indefinite integral has no limits and does not represent a single numerical value. The following subsections introduce the standard antiderivatives used throughout integration.

13.2.1 Standard integrals

The standard rules of indefinite integration are obtained by reversing familiar differentiation rules. Every result should be checked by differentiating the proposed antiderivative and confirming that the original integrand is recovered.

Integrating a constant

If k is a constant, then

$$\boxed{\int k dx = kx + C.} \quad (13.23)$$

This follows because

$$\frac{d}{dx}(kx + C) = k.$$

In particular,

$$\int dx = \int 1 dx = x + C.$$

The power rule

For any real number $n \neq -1$,

$$\boxed{\int x^n dx = \frac{x^{n+1}}{n+1} + C.} \quad (13.24)$$

The formula is verified by differentiation:

$$\frac{d}{dx} \left(\frac{x^{n+1}}{n+1} + C \right) = \frac{n+1}{n+1} x^n = x^n.$$

The restriction $n \neq -1$ is essential because substituting $n = -1$ would require division by zero. This case has the logarithmic antiderivative

$$\boxed{\int \frac{1}{x} dx = \ln |x| + C, \quad x \neq 0.} \quad (13.25)$$

Constant multiples, sums, and differences

Constants may be taken outside an integral:

$$\boxed{\int k f(x) dx = k \int f(x) dx.} \quad (13.26)$$

Integration also distributes across sums and differences:

$$\boxed{\int [f(x) \pm g(x)] dx = \int f(x) dx \pm \int g(x) dx.} \quad (13.27)$$

Only one arbitrary constant is needed at the end, since several arbitrary constants can always be combined into a single constant C .

Frequently used algebraic forms

The basic rules give the following useful results. In the table, $a \neq 0$ and $n \neq -1$.

Integrand	Antiderivative
k	$kx + C$
x^n	$\frac{x^{n+1}}{n+1} + C$
$\frac{1}{x}$	$\ln x + C$
$(ax + b)^n$	$\frac{(ax + b)^{n+1}}{a(n+1)} + C$
$\frac{1}{ax + b}$	$\frac{1}{a} \ln ax + b + C$

Table 13.1: Standard algebraic antiderivatives.

The last two formulae include a factor of $1/a$ to reverse the derivative of the inner linear expression $ax + b$.

Rewriting before integrating

Radicals and reciprocal powers should usually be rewritten using indices before the power rule is applied:

$$\sqrt{x} = x^{1/2}, \quad \sqrt[3]{x^2} = x^{2/3}, \quad \frac{1}{x^m} = x^{-m}.$$

Algebraic fractions may also need to be simplified term by term. For instance,

$$\frac{x^2 + 1}{\sqrt{x}} = x^{3/2} + x^{-1/2}.$$

Example 13.6: integrating a radical

Evaluate

$$\int \sqrt{u} \, du.$$

Rewrite the radical and apply the power rule:

$$\begin{aligned} \int \sqrt{u} \, du &= \int u^{1/2} \, du \\ &= \frac{u^{3/2}}{3/2} + C \\ &= \boxed{\frac{2}{3}u^{3/2} + C}. \end{aligned}$$

Example 13.7: integrating a reciprocal power

Evaluate

$$\int \frac{12}{z^5} \, dz.$$

$$\begin{aligned} \int \frac{12}{z^5} \, dz &= 12 \int z^{-5} \, dz \\ &= 12 \left(\frac{z^{-4}}{-4} \right) + C \\ &= -3z^{-4} + C \\ &= \boxed{-\frac{3}{z^4} + C}. \end{aligned}$$

Differentiating $-3z^{-4}$ gives $12z^{-5}$, confirming the result.

Example 13.8: integrating a polynomial

Integrate each term separately:

$$\begin{aligned}\int (3x^2 - 4x + 5) dx &= 3 \int x^2 dx - 4 \int x dx + 5 \int 1 dx \\ &= x^3 - 2x^2 + 5x + C.\end{aligned}$$

Hence,

$$\boxed{\int (3x^2 - 4x + 5) dx = x^3 - 2x^2 + 5x + C.}$$

Example 13.9: simplifying an algebraic fraction

Evaluate

$$\int \frac{x^2 + 1}{\sqrt{x}} dx.$$

First rewrite the integrand:

$$\begin{aligned}\int \frac{x^2 + 1}{\sqrt{x}} dx &= \int (x^{3/2} + x^{-1/2}) dx \\ &= \frac{x^{5/2}}{5/2} + \frac{x^{1/2}}{1/2} + C \\ &= \boxed{\frac{2}{5}x^{5/2} + 2x^{1/2} + C}.\end{aligned}$$

Example 13.10: a linear expression raised to a power

Using the standard form for $(ax + b)^n$,

$$\begin{aligned}\int (3x - 2)^4 dx &= \frac{(3x - 2)^5}{3(5)} + C \\ &= \boxed{\frac{(3x - 2)^5}{15} + C}.\end{aligned}$$

The factor $1/3$ is required to compensate for the derivative of $3x - 2$.

13.2.2 Indefinite integrals of exponential functions

The derivative of an exponential function is proportional to the function itself. Reversing the corresponding differentiation rules gives

$$\boxed{\int e^x dx = e^x + C,} \quad \boxed{\int e^{kx} dx = \frac{1}{k}e^{kx} + C, \quad k \neq 0,} \quad (13.28)$$

and, for $a > 0$ with $a \neq 1$,

$$\boxed{\int a^x dx = \frac{a^x}{\ln a} + C,} \quad \boxed{\int a^{kx} dx = \frac{a^{kx}}{k \ln a} + C.} \quad (13.29)$$

The factor $1/k$ compensates for the derivative of the exponent. For example,

$$\int (3e^{4x} - 2^{5x}) dx = \frac{3}{4}e^{4x} - \frac{2^{5x}}{5 \ln 2} + C.$$

When the exponent is not linear, a substitution may be needed. For instance,

$$\begin{aligned} \int 2xe^{x^2} dx &= \int e^u du & (u = x^2, du = 2x dx) \\ &= \boxed{e^{x^2} + C}. \end{aligned}$$

13.2.3 Indefinite integrals of trigonometric functions

The following table collects the standard trigonometric antiderivatives. Absolute-value signs are required in logarithmic answers because the corresponding trigonometric expressions may be negative.

Integrand	Antiderivative	Integrand	Antiderivative
$\sin x$	$-\cos x + C$	$\cos x$	$\sin x + C$
$\sec^2 x$	$\tan x + C$	$\csc^2 x$	$-\cot x + C$
$\sec x \tan x$	$\sec x + C$	$\csc x \cot x$	$-\csc x + C$
$\tan x$	$\ln \sec x + C$	$\cot x$	$\ln \sin x + C$
$\sec x$	$\ln \sec x + \tan x + C$	$\csc x$	$-\ln \csc x + \cot x + C$

Table 13.2: Standard trigonometric antiderivatives.

For a linear argument $ax + b$, divide the result by the coefficient a . For example,

$$\int (3 \cos 2x - 4 \sin 5x + \sec^2 x) dx = \frac{3}{2} \sin 2x + \frac{4}{5} \cos 5x + \tan x + C.$$

The more complicated trigonometric integrals in later sections are first rewritten until one or more of these standard forms appears.

13.3 Integration by substitution method

Substitution reverses the chain rule. If $u = g(x)$, then $du = g'(x) dx$, and

$$\int f(g(x))g'(x) dx = \int f(u) du = F(g(x)) + C, \quad (13.30)$$

where $F'(u) = f(u)$. The aim is to replace the complete integral, including dx , by an expression involving only u and du .

13.3.1 Choosing a substitution

A useful first choice for u is an inner expression whose derivative also appears elsewhere in the integrand. Common signals include

$$[g(x)]^n g'(x), \quad \frac{g'(x)}{g(x)}, \quad e^{g(x)} g'(x), \quad \sin(g(x))g'(x).$$

The following process keeps the algebra organised.

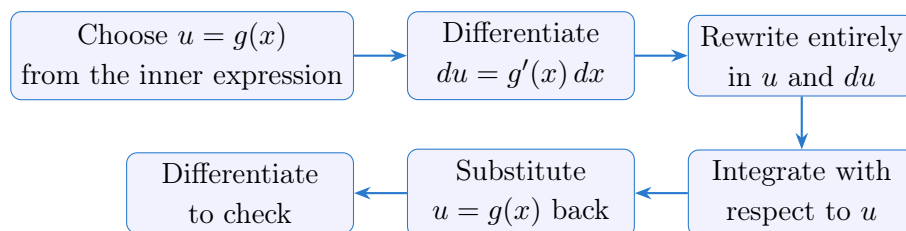


Figure 13.6: A reliable substitution workflow.

If only a constant multiple of $g'(x)$ is present, introduce the necessary constant and compensate for it outside the integral.

13.3.2 Algebraic and logarithmic forms

Example 13.11: a power of an inner function

Evaluate

$$\int 6x^2(2x^3 + 1)^4 dx.$$

Let $u = 2x^3 + 1$, so $du = 6x^2 dx$. Then

$$\begin{aligned}
 \int 6x^2(2x^3 + 1)^4 dx &= \int u^4 du \\
 &= \frac{u^5}{5} + C \\
 &= \boxed{\frac{(2x^3 + 1)^5}{5} + C}.
 \end{aligned}$$

Example 13.12: adjusting a constant

Evaluate

$$\int x^2(x^3 + 1)^2 dx.$$

With $u = x^3 + 1$, we have $du = 3x^2 dx$, so $x^2 dx = du/3$. Therefore,

$$\int x^2(x^3 + 1)^2 dx = \frac{1}{3} \int u^2 du = \boxed{\frac{(x^3 + 1)^3}{9} + C}.$$

Example 13.13: the logarithmic pattern

Whenever the numerator is the derivative of the denominator,

$$\boxed{\int \frac{g'(x)}{g(x)} dx = \ln |g(x)| + C.} \quad (13.31)$$

Thus,

$$\begin{aligned}
 \int \frac{x+1}{x^2+2x+3} dx &= \frac{1}{2} \int \frac{2x+2}{x^2+2x+3} dx \\
 &= \boxed{\frac{1}{2} \ln(x^2+2x+3) + C}.
 \end{aligned}$$

The absolute-value signs may be omitted here because $x^2 + 2x + 3 = (x+1)^2 + 2 > 0$ for every real x .

13.3.3 Trigonometric, exponential, and inverse-trigonometric forms

Substitution also reveals many standard integrals. For example,

$$\begin{aligned}\int x \cos(3x^2) dx &= \frac{1}{6} \int \cos u du && (u = 3x^2) \\ &= \boxed{\frac{1}{6} \sin(3x^2) + C},\end{aligned}$$

$$\begin{aligned}\int \frac{e^{3t}}{e^{3t} + 2} dt &= \frac{1}{3} \int \frac{1}{u} du && (u = e^{3t} + 2) \\ &= \boxed{\frac{1}{3} \ln(e^{3t} + 2) + C}.\end{aligned}$$

Two particularly useful inverse-trigonometric forms are

$$\boxed{\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C,} \quad a > 0, \quad (13.32)$$

$$\boxed{\int \frac{dx}{a^2 + x^2} = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C,} \quad a > 0. \quad (13.33)$$

For instance,

$$\begin{aligned}\int \frac{dx}{9 + x^2} &= \boxed{\frac{1}{3} \tan^{-1}\left(\frac{x}{3}\right) + C}, \\ \int \frac{dx}{\sqrt{16 - x^2}} &= \boxed{\sin^{-1}\left(\frac{x}{4}\right) + C}.\end{aligned}$$

13.3.4 Substitution in a definite integral

For a definite integral, either return to x before applying the original limits or change the limits to values of u . The second method is usually shorter:

$$\boxed{\int_a^b f(g(x))g'(x) dx = \int_{g(a)}^{g(b)} f(u) du.} \quad (13.34)$$

For example,

$$\begin{aligned}\int_0^1 2x(3 + x^2)^{1/2} dx &= \int_3^4 u^{1/2} du \\ &= \left[\frac{2}{3} u^{3/2} \right]_3^4 \\ &= \boxed{\frac{16}{3} - 2\sqrt{3}}.\end{aligned}$$

Once the limits have been converted to u -values, do not substitute x back into the

antiderivative.

13.4 Integration by parts method

Integration by parts reverses the product rule. Starting with

$$d(uv) = u dv + v du$$

and integrating both sides gives

$$\boxed{\int u dv = uv - \int v du.} \quad (13.35)$$

The method is most useful for a product in which one factor becomes simpler when differentiated and the other is easy to integrate. A common guide for choosing u is **LIPET**:

$$\begin{aligned} \text{Logarithmic} &\longrightarrow \text{Inverse trigonometric} \longrightarrow \text{Polynomial} \\ &\longrightarrow \text{Exponential} \longrightarrow \text{Trigonometric.} \end{aligned}$$

Choose u from the earliest suitable category and let the remaining factor, including dx , be dv . This is a guide rather than an inflexible rule: the new integral $\int v du$ must actually be simpler.

13.4.1 A single application

Evaluate $\int x \sin x dx$. Choose

$$u = x, \quad dv = \sin x dx, \quad du = dx, \quad v = -\cos x.$$

Then

$$\begin{aligned} \int x \sin x dx &= -x \cos x + \int \cos x dx \\ &= \boxed{-x \cos x + \sin x + C}. \end{aligned}$$

An apparently single factor may be treated as a product with 1. Thus,

$$\begin{aligned} \int \ln x dx &= \int (\ln x)(1) dx \\ &= x \ln x - \int x \left(\frac{1}{x}\right) dx \\ &= \boxed{x \ln x - x + C}. \end{aligned}$$

13.4.2 Repeated integration by parts

Polynomial factors eventually disappear under repeated differentiation. To evaluate $\int x^2 e^x dx$, apply the formula twice:

$$\begin{aligned}\int x^2 e^x dx &= x^2 e^x - 2 \int x e^x dx \\ &= x^2 e^x - 2 \left(x e^x - \int e^x dx \right) \\ &= \boxed{e^x(x^2 - 2x + 2) + C}.\end{aligned}$$

For products of a polynomial with an exponential or trigonometric function, the tabular arrangement below is a compact version of repeated integration by parts.

Sign	Differentiate	Integrate
+	x^2	e^x
−	$2x$	e^x
+	2	e^x
−	0	e^x

Multiplying diagonally with alternating signs gives $x^2 e^x - 2x e^x + 2e^x + C$.

13.4.3 A cyclic integral

Sometimes integration by parts returns the original integral. Let

$$I = \int e^x \sin x dx.$$

Applying integration by parts twice gives

$$\begin{aligned}I &= e^x \sin x - \int e^x \cos x dx \\ &= e^x \sin x - \left(e^x \cos x + \int e^x \sin x dx \right) \\ &= e^x(\sin x - \cos x) - I.\end{aligned}$$

Solving algebraically for I ,

$$\boxed{\int e^x \sin x dx = \frac{e^x}{2}(\sin x - \cos x) + C.}$$

Consistency in the choice of u and dv is essential in a cyclic calculation.

For definite integrals, the formula becomes

$$\boxed{\int_a^b u dv = [uv]_a^b - \int_a^b v du.} \quad (13.36)$$

13.5 Integration of trigonometric functions

Trigonometric integrals often require an identity before a standard antiderivative or substitution becomes visible.

13.5.1 Rewriting tools for integrating: trigonometric identities

The reciprocal and quotient identities are

$$\sec x = \frac{1}{\cos x}, \quad \csc x = \frac{1}{\sin x}, \quad \tan x = \frac{\sin x}{\cos x}, \quad \cot x = \frac{\cos x}{\sin x}.$$

13.5.2 Rewriting tools for integrating: Pythagorean identities

$$\boxed{\sin^2 x + \cos^2 x = 1, \quad 1 + \tan^2 x = \sec^2 x, \quad 1 + \cot^2 x = \csc^2 x.} \quad (13.37)$$

13.5.3 Rewriting tools for integrating: double angle identities

$$\sin 2x = 2 \sin x \cos x, \quad \cos 2x = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x.$$

13.5.4 Rewriting tools for integrating: power-reducing identities

Solving the double-angle identities for the squared functions gives

$$\boxed{\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}.} \quad (13.38)$$

These formulae are particularly useful when both powers of sine and cosine are even.

13.5.5 Integration of a certain form

Powers of sine and cosine

For an integral of the form

$$\int \sin^m x \cos^n x \, dx,$$

use the following guide.

Powers	Strategy
m odd	Save one factor $\sin x$, convert the remaining $\sin^2 x$ factors to $1 - \cos^2 x$, and use $u = \cos x$.
n odd	Save one factor $\cos x$, convert the remaining $\cos^2 x$ factors to $1 - \sin^2 x$, and use $u = \sin x$.
Both odd	Either of the preceding choices works.
Both even	Use the power-reducing identities, possibly more than once.

Table 13.3: Strategy for powers of sine and cosine.

For example,

$$\begin{aligned}
 \int \sin^3 x \, dx &= \int (1 - \cos^2 x) \sin x \, dx \\
 &= - \int (1 - u^2) \, du && (u = \cos x) \\
 &= \boxed{-\cos x + \frac{1}{3} \cos^3 x + C}.
 \end{aligned}$$

When both powers are even,

$$\begin{aligned}
 \int \sin^2 x \, dx &= \frac{1}{2} \int (1 - \cos 2x) \, dx \\
 &= \boxed{\frac{x}{2} - \frac{1}{4} \sin 2x + C}.
 \end{aligned}$$

Products with different angles

The product-to-sum identities are

$$\begin{aligned}
 \sin A \cos B &= \frac{1}{2} [\sin(A + B) + \sin(A - B)], \\
 \sin A \sin B &= \frac{1}{2} [\cos(A - B) - \cos(A + B)], \\
 \cos A \cos B &= \frac{1}{2} [\cos(A - B) + \cos(A + B)].
 \end{aligned}$$

Hence,

$$\begin{aligned}\int \sin 2x \cos 3x \, dx &= \frac{1}{2} \int (\sin 5x - \sin x) \, dx \\ &= \boxed{-\frac{1}{10} \cos 5x + \frac{1}{2} \cos x + C}.\end{aligned}$$

Powers of tangent and secant

For

$$\int \tan^m x \sec^n x \, dx,$$

the factors $\sec^2 x \, dx = d(\tan x)$ and $\sec x \tan x \, dx = d(\sec x)$ determine the main cases:

- If n is even, save a factor $\sec^2 x$ and convert the remaining secants using $\sec^2 x = 1 + \tan^2 x$.
- If m is odd, save a factor $\sec x \tan x$ and convert the remaining tangents using $\tan^2 x = \sec^2 x - 1$.
- If only an odd power of secant remains, use integration by parts or a reduction formula.

For example,

$$\begin{aligned}\int \tan^2 x \sec^4 x \, dx &= \int \tan^2 x (1 + \tan^2 x) \sec^2 x \, dx \\ &= \int (u^2 + u^4) \, du \\ &= \boxed{\frac{1}{3} \tan^3 x + \frac{1}{5} \tan^5 x + C}.\end{aligned}$$

The important odd-secant result is obtained by integration by parts:

$$\boxed{\int \sec^3 x \, dx = \frac{1}{2} \sec x \tan x + \frac{1}{2} \ln |\sec x + \tan x| + C.} \quad (13.39)$$

13.5.6 Guidelines in solving trigonometric integrations

Before committing to a method, use the following checklist:

1. Rewrite quotients using sine and cosine if this exposes a simple substitution.
2. Check for a derivative pair such as $\sec^2 x$ with $\tan x$ or $\sec x \tan x$ with $\sec x$.
3. Inspect the parity of the powers of sine, cosine, tangent, and secant.

4. Use product-to-sum identities when the angles differ.
5. Use integration by parts for isolated odd powers such as $\sec^3 x$ or for products involving x , exponentials, or logarithms.
6. Differentiate the final expression; trigonometric identities can make two correct answers look quite different.

13.6 Integration of hyperbolic functions

The hyperbolic functions introduced in Chapter 11 have differentiation rules that closely parallel those of the trigonometric functions. Reversing those rules gives a collection of useful standard antiderivatives. The notation $\sinh^{-1} x$ denotes the inverse hyperbolic sine, whereas $\operatorname{cosech} x = 1/\sinh x$ denotes a reciprocal.

13.6.1 Standard hyperbolic antiderivatives

The six direct derivative pairs are

Integrand	Antiderivative	Integrand	Antiderivative
$\sinh x$	$\cosh x + C$	$\cosh x$	$\sinh x + C$
$\operatorname{sech}^2 x$	$\tanh x + C$	$\operatorname{cosech}^2 x$	$-\coth x + C$
$\operatorname{sech} x \tanh x$	$-\operatorname{sech} x + C$	$\operatorname{cosech} x \coth x$	$-\operatorname{cosech} x + C$

Table 13.4: Direct hyperbolic antiderivatives.

Four further standard forms are

$$\int \tanh x \, dx = \ln(\cosh x) + C, \quad (13.40)$$

$$\int \coth x \, dx = \ln |\sinh x| + C, \quad (13.41)$$

$$\int \operatorname{sech} x \, dx = \tan^{-1}(\sinh x) + C, \quad (13.42)$$

$$\int \operatorname{cosech} x \, dx = \ln \left| \tanh \frac{x}{2} \right| + C. \quad (13.43)$$

For a linear argument $ax + b$, divide the antiderivative by a . For example,

$$\int (3 \sinh 2x - 4 \operatorname{sech}^2(3x)) \, dx = \boxed{\frac{3}{2} \cosh 2x - \frac{4}{3} \tanh 3x + C}.$$

13.6.2 Substitution patterns

The derivative pairs

$$d(\sinh x) = \cosh x \, dx, \quad d(\cosh x) = \sinh x \, dx, \quad d(\tanh x) = \operatorname{sech}^2 x \, dx$$

often reveal the appropriate substitution immediately. For instance,

$$\begin{aligned} \int \sinh^3 x \cosh x \, dx &= \int u^3 \, du && (u = \sinh x) \\ &= \boxed{\frac{1}{4} \sinh^4 x + C}, \end{aligned}$$

while

$$\begin{aligned} \int \tanh^3 x \operatorname{sech}^2 x \, dx &= \int u^3 \, du && (u = \tanh x) \\ &= \boxed{\frac{1}{4} \tanh^4 x + C}. \end{aligned}$$

The fundamental identity

$$\cosh^2 x - \sinh^2 x = 1$$

allows the remaining even power to be rewritten. Thus an odd power of $\cosh x$ is naturally paired with $u = \sinh x$, whereas an odd power of $\sinh x$ is naturally paired with $u = \cosh x$.

13.6.3 Even powers of hyperbolic sine and cosine

The power-reduction identities

$$\cosh^2 x = \frac{\cosh 2x + 1}{2}, \quad \sinh^2 x = \frac{\cosh 2x - 1}{2} \quad (13.44)$$

are used when both powers are even. They give

$$\int \cosh^2 x \, dx = \boxed{\frac{x}{2} + \frac{1}{4} \sinh 2x + C}, \quad (13.45)$$

$$\int \sinh^2 x \, dx = \boxed{-\frac{x}{2} + \frac{1}{4} \sinh 2x + C}. \quad (13.46)$$

For example,

$$\begin{aligned}\int \sinh^2(3x) dx &= \frac{1}{2} \int (\cosh 6x - 1) dx \\ &= \boxed{\frac{1}{12} \sinh 6x - \frac{x}{2} + C}.\end{aligned}$$

13.6.4 Antiderivatives involving inverse hyperbolic functions

Some algebraic integrals are most compactly expressed using inverse hyperbolic functions. For $a > 0$,

$$\int \frac{dx}{\sqrt{a^2 + x^2}} = \sinh^{-1}\left(\frac{x}{a}\right) + C, \quad (13.47)$$

$$\int \frac{dx}{a^2 - x^2} = \frac{1}{a} \tanh^{-1}\left(\frac{x}{a}\right) + C, \quad |x| < a, \quad (13.48)$$

$$\int \frac{dx}{\sqrt{x^2 - a^2}} = \cosh^{-1}\left(\frac{x}{a}\right) + C, \quad x > a. \quad (13.49)$$

These answers agree with logarithmic forms. In particular,

$$\sinh^{-1}\left(\frac{x}{a}\right) = \ln \left(\frac{x + \sqrt{x^2 + a^2}}{a} \right). \quad (13.50)$$

The constant term $-\ln a$ may be absorbed into the arbitrary constant of integration.

13.7 Integration by trigonometric substitution method

Trigonometric substitution is designed for radicals containing quadratic expressions. The substitution is chosen so that a Pythagorean identity removes the square root.

Expression	Substitution	Identity	Simplified radical
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$	$1 - \sin^2 \theta = \cos^2 \theta$	$a \cos \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta$	$1 + \tan^2 \theta = \sec^2 \theta$	$a \sec \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta$	$\sec^2 \theta - 1 = \tan^2 \theta$	$a \tan \theta$

Table 13.5: The three standard trigonometric substitutions, where $a > 0$.

After integrating in θ , use a right triangle to return to x .

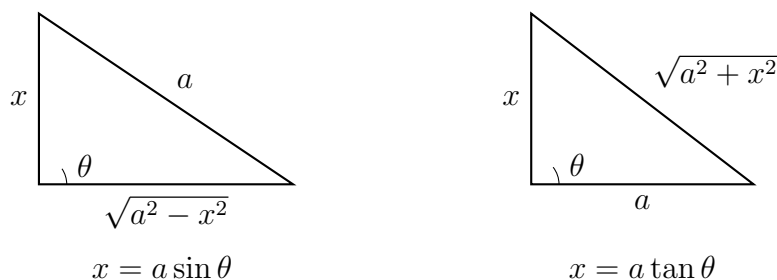


Figure 13.7: Reference triangles for two common substitutions.

Example 13.14: the form $\sqrt{a^2 - x^2}$

Evaluate $\int \sqrt{4 - x^2} dx$. Let $x = 2 \sin \theta$, so $dx = 2 \cos \theta d\theta$ and $\sqrt{4 - x^2} = 2 \cos \theta$. Therefore,

$$\begin{aligned}
 \int \sqrt{4 - x^2} dx &= 4 \int \cos^2 \theta d\theta \\
 &= 2\theta + \sin 2\theta + C \\
 &= 2\theta + 2 \sin \theta \cos \theta + C.
 \end{aligned}$$

Since $\theta = \sin^{-1}(x/2)$, $\sin \theta = x/2$, and $\cos \theta = \sqrt{4 - x^2}/2$,

$$\boxed{\int \sqrt{4 - x^2} dx = \frac{x}{2} \sqrt{4 - x^2} + 2 \sin^{-1}\left(\frac{x}{2}\right) + C.}$$

Example 13.15: the form $\sqrt{a^2 + x^2}$

For $a > 0$, consider

$$I = \int \sqrt{a^2 + x^2} \, dx.$$

Use $x = a \tan \theta$. On an interval where $-\pi/2 < \theta < \pi/2$, we have

$$dx = a \sec^2 \theta \, d\theta, \quad \sqrt{a^2 + x^2} = a \sec \theta.$$

Therefore,

$$\begin{aligned} I &= a^2 \int \sec^3 \theta \, d\theta \\ &= \frac{a^2}{2} (\sec \theta \tan \theta + \ln |\sec \theta + \tan \theta|) + C. \end{aligned}$$

Since

$$\tan \theta = \frac{x}{a}, \quad \sec \theta = \frac{\sqrt{a^2 + x^2}}{a},$$

back-substitution gives

$$\boxed{\int \sqrt{a^2 + x^2} \, dx = \frac{x}{2} \sqrt{a^2 + x^2} + \frac{a^2}{2} \ln |x + \sqrt{a^2 + x^2}| + C.} \quad (13.51)$$

Using the inverse-hyperbolic identity from the preceding section, the same answer may be written as

$$\boxed{\int \sqrt{a^2 + x^2} \, dx = \frac{x}{2} \sqrt{a^2 + x^2} + \frac{a^2}{2} \sinh^{-1} \left(\frac{x}{a} \right) + C.}$$

The two forms differ only by the constant $\frac{a^2}{2} \ln a$, which is absorbed into C .

Example 13.16: the form $\sqrt{x^2 - a^2}$

Evaluate

$$\int \frac{dx}{x\sqrt{x^2 - 9}}, \quad x > 3.$$

Let $x = 3 \sec \theta$. Then $dx = 3 \sec \theta \tan \theta \, d\theta$ and $\sqrt{x^2 - 9} = 3 \tan \theta$. Hence,

$$\begin{aligned}
\int \frac{dx}{x\sqrt{x^2-9}} &= \int \frac{3 \sec \theta \tan \theta}{(3 \sec \theta)(3 \tan \theta)} d\theta \\
&= \frac{1}{3} \int d\theta \\
&= \frac{\theta}{3} + C \\
&= \boxed{\frac{1}{3} \sec^{-1}\left(\frac{x}{3}\right) + C}.
\end{aligned}$$

Before using trigonometric substitution, always check for a simpler direct substitution. For example, $\int x\sqrt{x^2+4} dx$ is solved immediately by $u = x^2 + 4$.

13.8 Integration using partial fractions

Partial fractions decompose a rational function into simpler fractions whose antiderivatives are standard. First ensure that the rational function is *proper*: the degree of the numerator must be less than the degree of the denominator. If it is not, perform polynomial division before decomposing it.

13.8.1 Decomposition templates

After factoring the denominator over the real numbers, use the appropriate template.

$$\begin{aligned}
\frac{P(x)}{(x-a)(x-b)} &= \frac{A}{x-a} + \frac{B}{x-b}, \\
\frac{P(x)}{(x-a)^2} &= \frac{A}{x-a} + \frac{B}{(x-a)^2}, \\
\frac{P(x)}{(x-a)(x^2+px+q)} &= \frac{A}{x-a} + \frac{Bx+C}{x^2+px+q}.
\end{aligned}$$

Every power of a repeated factor must appear. An irreducible quadratic factor requires a linear numerator.

Example 13.17: distinct linear factors

Evaluate

$$\int \frac{5x+1}{x^2+x-2} dx.$$

Since $x^2+x-2 = (x-1)(x+2)$, write

$$\frac{5x+1}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}.$$

Multiplication by $(x - 1)(x + 2)$ gives

$$5x + 1 = A(x + 2) + B(x - 1).$$

Setting $x = 1$ gives $A = 2$, while setting $x = -2$ gives $B = 3$. Therefore,

$$\begin{aligned} \int \frac{5x + 1}{x^2 + x - 2} dx &= \int \left(\frac{2}{x - 1} + \frac{3}{x + 2} \right) dx \\ &= \boxed{2 \ln |x - 1| + 3 \ln |x + 2| + C}. \end{aligned}$$

Example 13.18: an irreducible quadratic factor

Evaluate

$$\int \frac{1}{x(x^2 + 1)} dx.$$

Use

$$\frac{1}{x(x^2 + 1)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 1}.$$

Equating coefficients gives $A = 1$, $B = -1$, and $C = 0$. Thus,

$$\begin{aligned} \int \frac{1}{x(x^2 + 1)} dx &= \int \left(\frac{1}{x} - \frac{x}{x^2 + 1} \right) dx \\ &= \boxed{\ln |x| - \frac{1}{2} \ln(x^2 + 1) + C}. \end{aligned}$$

Factor and cancel common factors before beginning a decomposition, but retain the original domain restrictions: cancellation does not restore a point at which the original expression was undefined.

13.9 Definite Integration

A definite integral has specified limits and represents a single number. Its variable of integration is a dummy variable, so

$$\int_a^b f(x) dx = \int_a^b f(t) dt.$$

Earlier sections defined this number as a limit of Riemann sums. The Fundamental Theorem of Calculus now provides an efficient way to evaluate it.

13.9.1 Fundamental Theorem of Calculus

The Fundamental Theorem of Calculus connects the two central ideas of calculus: differentiation, which measures instantaneous change, and integration, which measures accumulation. It shows that these processes are inverse operations under suitable conditions.

Let f be continuous on the closed interval $[a, b]$. The theorem has two closely related parts.

Part I: differentiating an accumulation function

Define the accumulation function

$$A(x) = \int_a^x f(t) dt, \quad a \leq x \leq b.$$

The letter t is a dummy variable of integration. The variable x appears as the upper limit and determines how much of the function has been accumulated. The first part of the theorem states that

$$A'(x) = \frac{d}{dx} \int_a^x f(t) dt = f(x). \quad (13.52)$$

Thus, the instantaneous rate at which accumulated area changes is equal to the current height of the function.

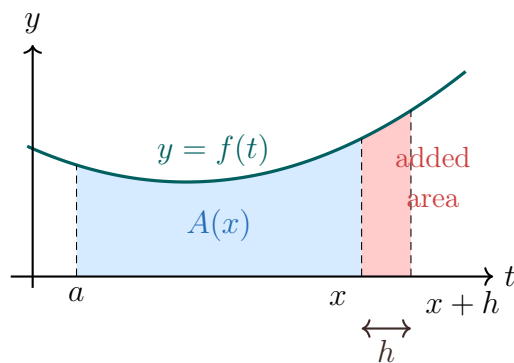


Figure 13.8: Increasing the upper limit from x to $x + h$ adds a narrow region whose area is approximately $f(x)h$.

To see why Part I is reasonable, consider the difference quotient

$$\begin{aligned} \frac{A(x+h) - A(x)}{h} &= \frac{1}{h} \left[\int_a^{x+h} f(t) dt - \int_a^x f(t) dt \right] \\ &= \frac{1}{h} \int_x^{x+h} f(t) dt. \end{aligned}$$

When h is small, $f(t)$ is close to $f(x)$ throughout the short interval $[x, x + h]$. Hence,

$$\int_x^{x+h} f(t) dt \approx f(x)h,$$

so the difference quotient approaches $f(x)$ as $h \rightarrow 0$.

Example 13.19: differentiating an accumulation function

If

$$G(x) = \int_2^x (t^3 - 4t) dt,$$

then Part I gives

$$G'(x) = x^3 - 4x.$$

There is no need to evaluate the integral first.

Variable limits and the chain rule

If the upper limit is a differentiable function $g(x)$, the chain rule gives

$$\frac{d}{dx} \int_a^{g(x)} f(t) dt = f(g(x))g'(x). \quad (13.53)$$

For example, if

$$H(x) = \int_0^{x^2} \sqrt{1+t^4} dt,$$

then

$$\begin{aligned} H'(x) &= \sqrt{1+(x^2)^4} (2x) \\ &= 2x\sqrt{1+x^8}. \end{aligned}$$

If the variable occurs in the lower limit, the sign is reversed:

$$\frac{d}{dx} \int_{g(x)}^b f(t) dt = -f(g(x))g'(x). \quad (13.54)$$

Combining both results,

$$\frac{d}{dx} \int_{u(x)}^{v(x)} f(t) dt = f(v(x))v'(x) - f(u(x))u'(x). \quad (13.55)$$

For instance,

$$\frac{d}{dx} \int_x^{x^2} e^{t^2} dt = 2xe^{x^4} - e^{x^2}.$$

Part II: evaluating a definite integral

Suppose F is any antiderivative of f on $[a, b]$, so that

$$F'(x) = f(x).$$

The second part of the Fundamental Theorem of Calculus states that

$$\boxed{\int_a^b f(x) dx = F(b) - F(a).} \quad (13.56)$$

This is often written using the evaluation notation

$$\boxed{\int_a^b f(x) dx = [F(x)]_a^b = F(b) - F(a).} \quad (13.57)$$

Part II replaces the limiting process of a Riemann sum with the much more efficient task of finding an antiderivative and evaluating it at the endpoints.

Why Part II follows from Part I

Let

$$A(x) = \int_a^x f(t) dt.$$

By Part I, $A'(x) = f(x)$. Since $F'(x) = f(x)$ as well, the functions A and F differ by a constant:

$$A(x) = F(x) + C.$$

Setting $x = a$ gives $A(a) = 0$, so $C = -F(a)$. Therefore,

$$A(x) = F(x) - F(a).$$

Finally, setting $x = b$ gives

$$\int_a^b f(t) dt = F(b) - F(a).$$

Example 13.20: confirming a Riemann-sum result

The previous subsection obtained

$$\int_0^2 x^2 dx = \frac{8}{3}$$

by taking the limit of a Riemann sum. Using Part II, an antiderivative of x^2 is $x^3/3$, so

$$\begin{aligned}\int_0^2 x^2 dx &= \left[\frac{x^3}{3} \right]_0^2 \\ &= \frac{2^3}{3} - \frac{0^3}{3} \\ &= \boxed{\frac{8}{3}}.\end{aligned}$$

The two methods agree, but the Fundamental Theorem makes the calculation much shorter.

Example 13.21: evaluating a trigonometric integral

Evaluate

$$\int_0^\pi \sin x \, dx.$$

Since an antiderivative of $\sin x$ is $-\cos x$,

$$\begin{aligned}\int_0^\pi \sin x \, dx &= [-\cos x]_0^\pi \\ &= -\cos \pi - (-\cos 0) \\ &= 1 + 1 \\ &= \boxed{2}.\end{aligned}$$

Basic properties of definite integrals

The evaluation formula immediately gives several useful properties:

$$\int_a^a f(x) \, dx = 0, \tag{13.58}$$

$$\int_b^a f(x) \, dx = -\int_a^b f(x) \, dx, \tag{13.59}$$

$$\int_a^b f(x) \, dx = \int_a^c f(x) \, dx + \int_c^b f(x) \, dx. \tag{13.60}$$

The definite integral is also linear. For constants r and s ,

$$\int_a^b [rf(x) + sg(x)] \, dx = r \int_a^b f(x) \, dx + s \int_a^b g(x) \, dx. \tag{13.61}$$

Net signed area versus total geometric area

The definite integral measures net signed area. A region below the x -axis contributes negatively. For example,

$$\begin{aligned}\int_{-1}^2 x \, dx &= \left[\frac{x^2}{2} \right]_{-1}^2 \\ &= 2 - \frac{1}{2} \\ &= \frac{3}{2}.\end{aligned}$$

However, the total geometric area between $y = x$ and the x -axis is

$$\frac{1}{2}(1)(1) + \frac{1}{2}(2)(2) = \frac{5}{2}.$$

To calculate total area when a function changes sign, split the interval at each x -intercept and make every contribution positive.

Why a definite integral has no arbitrary constant

An indefinite integral represents a family of antiderivatives and therefore includes $+C$. A definite integral is a single number. If $F(x) + C$ is used in the evaluation formula, the constant cancels:

$$[F(b) + C] - [F(a) + C] = F(b) - F(a).$$

Hence, $+C$ is never included in the final value of a definite integral.

13.9.2 Definite integration as a total change

If $r(x)$ is the rate of change of a quantity Q , so that $Q'(x) = r(x)$, then its accumulated change from $x = a$ to $x = b$ is

$$\boxed{Q(b) - Q(a) = \int_a^b r(x) \, dx.} \quad (13.62)$$

This interpretation is more general than area. The units of the integral are the units of the rate multiplied by the units of the independent variable. For example, integrating velocity in metres per second with respect to time in seconds gives displacement in metres.

Suppose a particle has velocity

$$v(t) = 3t^2 - 6t \quad \text{m s}^{-1}$$

for $0 \leq t \leq 3$. Its displacement is

$$\int_0^3 (3t^2 - 6t) dt = [t^3 - 3t^2]_0^3 = 0.$$

This does not mean that the particle remained at rest. Since $v(t) = 3t(t - 2)$ changes sign at $t = 2$, its total distance travelled is

$$\begin{aligned} \int_0^3 |v(t)| dt &= -\int_0^2 v(t) dt + \int_2^3 v(t) dt \\ &= 4 + 4 = \boxed{8 \text{ m}}. \end{aligned}$$

13.9.3 Calculating area under a graph

If $f(x) \geq 0$ on $[a, b]$, the area between $y = f(x)$ and the x -axis is

$$\boxed{A = \int_a^b f(x) dx.} \quad (13.63)$$

If the graph crosses the axis, split the interval at every intercept and make each contribution positive. Equivalently,

$$\boxed{A = \int_a^b |f(x)| dx.} \quad (13.64)$$

For example, $f(x) = x^2 - 4$ crosses the x -axis at $x = 2$ on the interval $[0, 4]$.

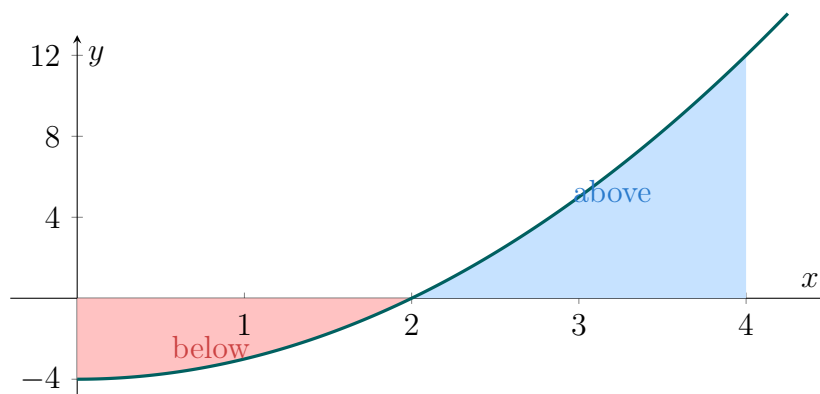


Figure 13.9: Total area treats both shaded regions as positive.

Therefore,

$$\begin{aligned} A &= -\int_0^2 (x^2 - 4) dx + \int_2^4 (x^2 - 4) dx \\ &= \frac{16}{3} + \frac{32}{3} = \boxed{16 \text{ square units}}. \end{aligned}$$

13.9.4 Calculating area between two curves

Suppose $f(x) \geq g(x)$ on $[a, b]$. A vertical strip has height $f(x) - g(x)$, so the enclosed area is

$$A = \int_a^b [\text{upper} - \text{lower}] dx = \int_a^b [f(x) - g(x)] dx. \quad (13.65)$$

For horizontal strips, if $x = R(y)$ is the right-hand boundary and $x = L(y)$ is the left-hand boundary, then

$$A = \int_c^d [\text{right} - \text{left}] dy = \int_c^d [R(y) - L(y)] dy. \quad (13.66)$$

Choose the direction that gives the simpler boundary expressions and the fewest separate integrals.

Example 13.22: a line and a parabola

Find the area enclosed by $y = 2 - x^2$ and $y = -x$. Their intersections satisfy

$$2 - x^2 = -x \implies (x - 2)(x + 1) = 0,$$

so $x = -1$ and $x = 2$. The parabola is above the line between these values.

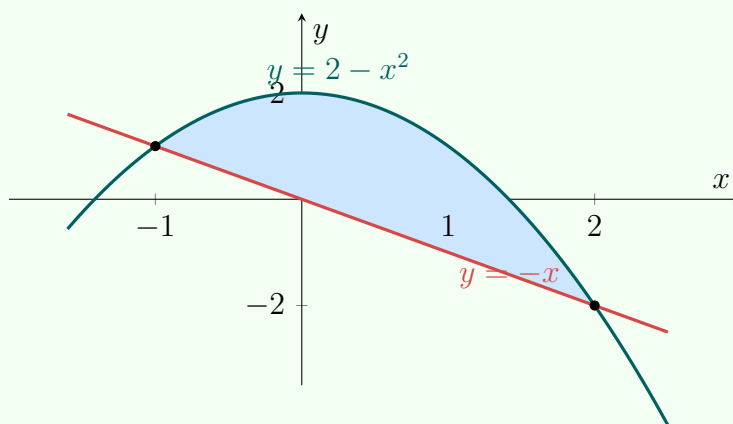


Figure 13.10: Vertical strips have height $(2 - x^2) - (-x)$.

Hence,

$$\begin{aligned} A &= \int_{-1}^2 (2 + x - x^2) dx \\ &= \left[2x + \frac{x^2}{2} - \frac{x^3}{3} \right]_{-1}^2 \\ &= \boxed{\frac{9}{2} \text{ square units}}. \end{aligned}$$

A dependable area procedure is: sketch the boundaries, find their intersections, decide between vertical and horizontal strips, write the strip length as a positive difference, and only then integrate.

13.9.5 Calculating area of a symmetrical curve

Symmetry can shorten a calculation. If f is even, so that $f(-x) = f(x)$, then

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx. \quad (13.67)$$

If f is odd, so that $f(-x) = -f(x)$, then

$$\int_{-a}^a f(x) dx = 0. \quad (13.68)$$

The second formula concerns signed integral, not total area. For example,

$$\int_{-1}^1 x^3 dx = 0,$$

but the total area between $y = x^3$ and the axis is

$$2 \int_0^1 x^3 dx = \left[\frac{1}{2} \right].$$

13.10 Application of integration

Integration calculates a volume by adding the volumes of thin cross-sections. If a solid has cross-sectional area $A(x)$ perpendicular to the x -axis, then

$$V = \int_a^b A(x) dx. \quad (13.69)$$

13.10.1 Volume of solids of revolution

If a plane region is rotated through one complete revolution about a fixed line in its plane, the resulting three-dimensional object is called a *solid of revolution*. The fixed line may be the x -axis, the y -axis, or any line parallel to one of these axes.

Choosing the direction of the slices

The disc and washer methods use slices perpendicular to the axis of revolution. The direction of the axis therefore determines the most natural variable of integration.

Axis	Slice	Thickness	Radius measured
Horizontal	Vertical	dx	Vertically
Vertical	Horizontal	dy	Horizontally

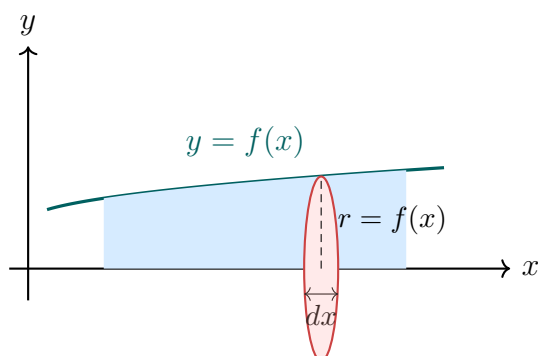
Table 13.6: Choosing slices for the disc or washer method.

Rotating the region under $y = f(x) \geq 0$ from $x = a$ to $x = b$ about the x -axis produces circular discs. Each disc has radius $f(x)$ and cross-sectional area $\pi[f(x)]^2$, so

$$V = \pi \int_a^b [f(x)]^2 dx. \quad (13.70)$$

Similarly, if $x = g(y) \geq 0$ is rotated about the y -axis from $y = c$ to $y = d$, then horizontal slices give

$$V = \pi \int_c^d [g(y)]^2 dy. \quad (13.71)$$

Figure 13.11: A thin vertical strip becomes a disc when rotated about the x -axis.

Deriving the disc formula from thin slices

Consider $y = 2x$ for $0 \leq x \leq 3$, rotated about the x -axis. A typical disc has radius $r = 2x$ and thickness Δx , so its approximate volume is

$$\Delta V \approx \pi r^2 \Delta x = \pi(2x)^2 \Delta x.$$

Adding n such discs and taking the limit gives

$$\begin{aligned}
 V &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \pi (2x_i^*)^2 \Delta x \\
 &= \pi \int_0^3 (2x)^2 dx \\
 &= 4\pi \left[\frac{x^3}{3} \right]_0^3 \\
 &= \boxed{36\pi \text{ cubic units}}.
 \end{aligned}$$

The limiting sum is exactly a definite integral: the area of each circular cross-section is accumulated along the length of the solid.

Worked examples about the coordinate axes

Example 13.23: a polynomial rotated about the x -axis

Rotate the region under $y = x^3$ for $0 \leq x \leq 2$ about the x -axis.

$$V = \pi \int_0^2 (x^3)^2 dx = \pi \left[\frac{x^7}{7} \right]_0^2 = \boxed{\frac{128\pi}{7}}.$$

Example 13.24: a radical function

The region bounded by $y = \sqrt{x}$, $x = 3$, and the x -axis is rotated about the x -axis. Thus,

$$V = \pi \int_0^3 (\sqrt{x})^2 dx = \pi \left[\frac{x^2}{2} \right]_0^3 = \boxed{\frac{9\pi}{2}}.$$

Example 13.25: horizontal discs about the y -axis

The region between $x = \sin y$ and the y -axis for $0 \leq y \leq \pi$ is rotated about the y -axis. The radius is a horizontal distance, $r(y) = \sin y$, so

$$\begin{aligned}
 V &= \pi \int_0^\pi \sin^2 y \, dy \\
 &= \pi \int_0^\pi \frac{1 - \cos 2y}{2} \, dy \\
 &= \pi \left[\frac{y}{2} - \frac{\sin 2y}{4} \right]_0^\pi \\
 &= \boxed{\frac{\pi^2}{2}}.
 \end{aligned}$$

Example 13.26: rewriting a curve for dy

The region between $y = x^2$, the y -axis, and $y = 4$ is rotated about the y -axis. Since $x = \sqrt{y}$ and $0 \leq y \leq 4$,

$$V = \pi \int_0^4 (\sqrt{y})^2 \, dy = \pi \left[\frac{y^2}{2} \right]_0^4 = \boxed{8\pi}.$$

Example 13.27: an exponential profile

The region under $y = 3e^{-x}$ for $0 \leq x \leq 1$ is revolved about the x -axis. Hence,

$$\begin{aligned}
 V &= \pi \int_0^1 (3e^{-x})^2 \, dx \\
 &= 9\pi \int_0^1 e^{-2x} \, dx \\
 &= 9\pi \left[-\frac{1}{2}e^{-2x} \right]_0^1 \\
 &= \boxed{\frac{9\pi}{2} (1 - e^{-2})}.
 \end{aligned}$$

13.10.2 Volume generated by area between two curves

If a region between two curves is rotated, its cross-section is generally a washer rather than a solid disc. If $R(x)$ is the outer radius and $r(x)$ is the inner radius, then

$$V = \pi \int_a^b (R(x)^2 - r(x)^2) \, dx. \quad (13.72)$$

For horizontal washers about a vertical axis,

$$V = \pi \int_c^d (R(y)^2 - r(y)^2) dy. \quad (13.73)$$

The radii are distances from the axis of rotation, not merely the equations of the curves. The cross-sectional area is the area of the outer circle minus the area of the inner circle.

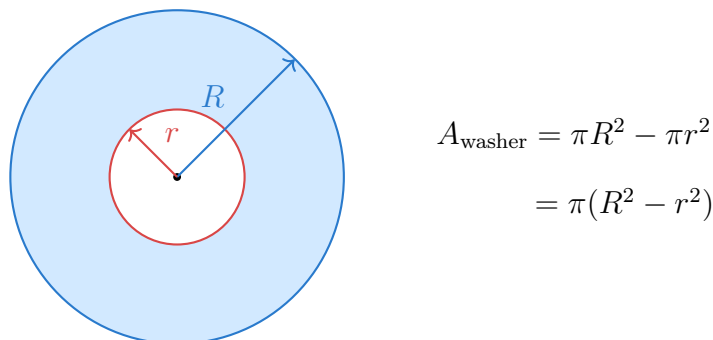


Figure 13.12: A washer has outer radius R and inner radius r .

As an example, rotate the region between $y = \sqrt{x}$ and $y = x/2$ for $0 \leq x \leq 4$ about the x -axis. Since $\sqrt{x} \geq x/2$ on this interval,

$$\begin{aligned} V &= \pi \int_0^4 \left[(\sqrt{x})^2 - \left(\frac{x}{2} \right)^2 \right] dx \\ &= \pi \int_0^4 \left(x - \frac{x^2}{4} \right) dx \\ &= \pi \left[\frac{x^2}{2} - \frac{x^3}{12} \right]_0^4 \\ &= \boxed{\frac{8\pi}{3} \text{ cubic units}}. \end{aligned}$$

Examples with bounds determined by intersections

Example 13.28: two power functions

The curves $y = x^2$ and $y = x^3$ meet at $x = 0$ and $x = 1$. On this interval, $x^2 \geq x^3$. Rotating the enclosed region about the x -axis gives

$$\begin{aligned} V &= \pi \int_0^1 [(x^2)^2 - (x^3)^2] dx \\ &= \pi \left[\frac{x^5}{5} - \frac{x^7}{7} \right]_0^1 \\ &= \boxed{\frac{2\pi}{35}}. \end{aligned}$$

Example 13.29: a reciprocal curve

The region bounded by $y = 4/x$, $y = 1/4$, and $x = 2$ is rotated about the x -axis. The right-hand limit is found from

$$\frac{4}{x} = \frac{1}{4} \implies x = 16.$$

Therefore,

$$\begin{aligned} V &= \pi \int_2^{16} \left[\left(\frac{4}{x} \right)^2 - \left(\frac{1}{4} \right)^2 \right] dx \\ &= \pi \left[-\frac{16}{x} - \frac{x}{16} \right]_2^{16} \\ &= \boxed{\frac{49\pi}{8}}. \end{aligned}$$

Example 13.30: a line and a parabola

For $y = 2x^2$ and $y = 3x - 1$, the intersections satisfy

$$2x^2 = 3x - 1 \implies x = \frac{1}{2}, 1.$$

The line is the outer radius on this interval. Hence,

$$\begin{aligned} V &= \pi \int_{1/2}^1 [(3x - 1)^2 - (2x^2)^2] dx \\ &= \boxed{\frac{\pi}{10}}. \end{aligned}$$

Example 13.31: washers about the y -axis

In the first quadrant, the curves $y = \sqrt[3]{x}$ and $y = x/4$ intersect at $(0, 0)$ and $(8, 2)$. To use horizontal washers, rewrite them as

$$x = y^3, \quad x = 4y.$$

For $0 \leq y \leq 2$, the outer radius is $R(y) = 4y$ and the inner radius is $r(y) = y^3$. Thus,

$$\begin{aligned} V &= \pi \int_0^2 [(4y)^2 - (y^3)^2] dy \\ &= \pi \left[\frac{16y^3}{3} - \frac{y^7}{7} \right]_0^2 \\ &= \frac{512\pi}{21}. \end{aligned}$$

13.10.3 Volume of solids of revolution about a line parallel to the x - or y -axis

For rotation about the horizontal line $y = k$, a vertical strip produces radii

$$R(x) = |\text{farther curve} - k|, \quad r(x) = |\text{nearer curve} - k|.$$

For rotation about the vertical line $x = h$, use horizontal strips and measure the radii from $x = h$. A labelled sketch is especially important because shifting the axis changes every radius.

The same plane region can generate different solids when the axis changes. Consider the cap bounded by $y = 2 - x^2$ and $y = 1$. Its intersections occur at $x = -1$ and $x = 1$.

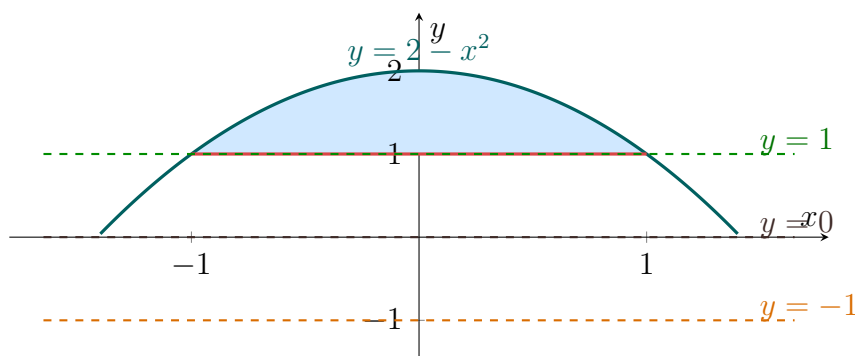


Figure 13.13: Changing the horizontal axis changes both radii and the volume.

Rotation about $y = 1$

The axis is already a boundary of the region, so the cross-sections are discs with radius

$$R(x) = (2 - x^2) - 1 = 1 - x^2.$$

Therefore,

$$V = \pi \int_{-1}^1 (1 - x^2)^2 dx = \boxed{\frac{16\pi}{15}}.$$

Rotation about the x -axis

The outer radius is $R(x) = 2 - x^2$ and the inner radius is $r(x) = 1$. Hence,

$$V = \pi \int_{-1}^1 [(2 - x^2)^2 - 1^2] dx = \boxed{\frac{56\pi}{15}}.$$

Rotation about $y = -1$

Both radii increase because the axis lies below the region:

$$R(x) = (2 - x^2) - (-1) = 3 - x^2, \quad r(x) = 1 - (-1) = 2.$$

Thus,

$$V = \pi \int_{-1}^1 [(3 - x^2)^2 - 2^2] dx = \boxed{\frac{32\pi}{5}}.$$

Axis	Outer radius	Inner radius	Volume
$y = 1$	$1 - x^2$	0	$\frac{16\pi}{15}$
$y = 0$	$2 - x^2$	1	$\frac{56\pi}{15}$
$y = -1$	$3 - x^2$	2	$\frac{32\pi}{5}$

Table 13.7: The same region rotated about three different horizontal axes.

The essential rule is always to measure each radius from the stated axis. Using the curve equations without accounting for the shifted axis gives an incorrect cross-sectional area.

13.11 Real-world application of integral calculus

The common feature of integration applications is accumulation from a rate or from thin pieces. Examples include

- displacement from velocity and velocity from acceleration;

- total fluid volume from a flow rate;
- mass from a varying linear, area, or volume density;
- work done by a variable force;
- area of an irregular surface and volume of a curved container; and
- probability from the area under a continuous probability-density function.

For example, if water enters a tank at the rate

$$r(t) = 12 + 3t^2 \quad \text{litres per minute,}$$

then the amount delivered during the first four minutes is

$$\begin{aligned} \int_0^4 (12 + 3t^2) dt &= [12t + t^3]_0^4 \\ &= \boxed{112 \text{ litres}}. \end{aligned}$$

The integral gives the accumulated amount directly; an initial amount in the tank would be added afterwards.

Chapter summary and common pitfalls

Key ideas. Indefinite integration produces families of antiderivatives, whereas a definite integral measures signed accumulation. Substitution, parts, identities, and partial fractions reverse different forms of differentiation.

Common pitfalls. Include the constant of integration in indefinite integrals, transform bounds when using substitution, distinguish signed area from geometric area, and attach the differential to the correct variable.

13.12 Practice and exercises

Give exact answers unless an approximation is requested. Include an arbitrary constant in every indefinite integral. Sketch or identify the relevant region before setting up an area or volume integral.

Finite sums, Riemann sums, and accumulation functions

1. Expand and evaluate.

(a) $\sum_{k=1}^4 (3k - 1);$

(b) $\sum_{k=0}^5 2^k;$

(c) $\sum_{k=1}^5 (-1)^{k+1} k.$

2. Write each expression in sigma notation.

(a) $2 + 5 + 8 + \cdots + 29;$

(b) $1^2 + 2^2 + \cdots + n^2;$

(c) $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{8 \cdot 9}.$

3. Evaluate using standard finite-sum formulae.

(a) $\sum_{k=1}^{20} k;$

(b) $\sum_{k=1}^{10} k^2;$

(c) $\sum_{k=1}^n (2k - 1).$

4. In $\sum_{k=2}^6 (k^2 - 1)$, set $j = k - 2$. Rewrite the sum using j and hence evaluate it.

5. Show that

$$\sum_{k=1}^n \left(\frac{1}{k} - \frac{1}{k+1} \right) = \frac{n}{n+1}.$$

Hence evaluate the sum when $n = 99$.

6. Evaluate the limit

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \left(\frac{k}{n} \right)^2.$$

7. Interpret the following as a definite integral and evaluate it:

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{2}{n} \sqrt{1 + \left(\frac{2k}{n} \right)^2}.$$

8. For $f(x) = x^2$ on $[0, 2]$ with four equal subintervals, calculate the left, right, and midpoint Riemann sums. Compare them with the exact integral.

9. For a positive increasing function, explain which of the left and right Riemann sums is an underestimate and which is an overestimate. State what changes when the function is decreasing.

10. If

$$F(x) = \int_1^{x^2} (t^3 + 1) dt,$$

find $F'(x)$.

11. If

$$G(x) = \int_{\sin x}^{2x} e^{t^2} dt,$$

find $G'(x)$.

12. Evaluate using the Fundamental Theorem of Calculus or symmetry.

(a) $\int_0^3 (2x + 1) dx;$

(b) $\int_{-\pi/2}^{\pi/2} \cos x dx;$

(c) $\int_{-3}^3 (x^5 + 2x^2) dx.$

Standard antiderivatives and substitution

13. Evaluate.

(a) $\int (6x^5 - 4x^{-3} + 7) dx;$

(b) $\int (3\sqrt{x} - 2x^{-2}) dx;$

(c) $\int \frac{x^4 + 2x}{x^2} dx.$

14. Rewrite the integrand before integrating.

(a) $\int \frac{(x+1)^2}{\sqrt{x}} dx;$

(b) $\int \frac{x^2 - 4}{x} dx;$

(c) $\int (1 + \sqrt{x})^2 dx.$

15. Evaluate the exponential integrals.

(a) $\int (3e^{2x} - 2 \cdot 5^x) dx;$

(b) $\int 4e^{-x/2} dx;$

(c) $\int 2^{3x+1} dx.$

16. Evaluate the trigonometric integrals.

(a) $\int (4 \sin 3x + 5 \cos 2x) dx;$

(b) $\int (2 \sec^2 x - 3 \csc^2 x) dx;$

(c) $\int (\sec x \tan x + \csc x \cot x) dx.$

17. Evaluate by substitution.

(a) $\int (3x - 1)^7 dx;$

(b) $\int \sqrt{5 - 2x} dx;$

(c) $\int \frac{dx}{4x + 3}.$

18. Use the logarithmic derivative pattern.

(a) $\int \frac{2x+5}{x^2+5x+7} dx;$

(b) $\int \frac{\cos x}{2+\sin x} dx;$

(c) $\int \frac{e^x}{1+e^x} dx.$

19. Evaluate.

(a) $\int x(x^2+4)^5 dx;$

(b) $\int x^3\sqrt{x^2+1} dx;$

(c) $\int \frac{x}{(x^2-9)^2} dx.$

20. Evaluate integrals involving roots or logarithms.

(a) $\int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx;$

(b) $\int \frac{dx}{\sqrt{x}(1+\sqrt{x})};$

(c) $\int \frac{\ln x}{x} dx.$

21. Evaluate the composite exponential integrals.

(a) $\int xe^{x^2} dx;$

(b) $\int \frac{e^{2x}}{1+e^{2x}} dx;$

(c) $\int \frac{2^x}{3+2^x} dx.$

22. Evaluate the composite trigonometric integrals.

(a) $\int \sin x \cos^4 x dx;$

(b) $\int \tan x \sec^2 x dx;$

(c) $\int \frac{\sin(\ln x)}{x} dx.$

23. Evaluate the inverse-trigonometric forms.

(a) $\int \frac{dx}{16+x^2};$

(b) $\int \frac{dx}{\sqrt{25-x^2}};$

(c) $\int \frac{dx}{1+9x^2}.$

24. Evaluate by changing the limits after substitution.

(a) $\int_0^1 2x(x^2+1)^3 dx;$

(b) $\int_0^{\pi/2} \sin x e^{\cos x} dx;$

(c) $\int_1^e \frac{(\ln x)^2}{x} dx.$

25. Evaluate, paying attention to orientation or symmetry.

(a) $\int_1^4 \frac{dx}{\sqrt{x}};$

(b) $\int_0^{\ln 2} \frac{e^x}{1+e^x} dx;$

(c) $\int_{-1}^1 \frac{x}{x^2+4} dx.$

26. Find the required function.

(a) $F'(x) = 3x^2 - 4x + 1, F(0) = 5;$

(b) $y' = 2e^{2x}, y(0) = 3;$

(c) $s'(t) = 6t - 4, s(2) = 7.$

27. The marginal cost of producing x components is $C'(x) = 0.04x + 12$ ringgit per component. If the fixed cost is RM 500, find $C(x)$ and the total cost of producing 100 components.

28. Let a , b , and k be constants.

(a) Evaluate $\int_0^1 (ax+b) dx.$

(b) Find k if $\int_0^2 kx dx = 12.$

(c) If $F(x) = \int_0^x (t+a) dt$ and $F(2) = 8$, find $a.$

29. Correct each proposed solution and explain the error.

(a) $\int x^{-1} dx = x^0/0 + C;$

(b) $\int (x^2 + 1)^3 dx = (x^2 + 1)^4 / (8x) + C;$

(c) after setting $u = x^2$ in $\int_0^1 2xe^{x^2} dx$, the limits remain “ $x = 0$ to $x = 1$ ”.

30. Evaluate the mixed substitution problems.

(a) $\int \frac{x+1}{x^2+2x+5} dx;$

(b) $\int \frac{dx}{\sqrt{2x+1}};$

(c) $\int_0^1 \frac{4x}{1+x^2} dx.$

Integration by parts and trigonometric integrals

31. Use integration by parts.

(a) $\int xe^x dx;$

(b) $\int x \cos 2x dx.$

32. Apply integration by parts repeatedly.

(a) $\int x^2 e^x dx;$

(b) $\int x^2 \sin x dx.$

33. Evaluate logarithmic integrals.

(a) $\int \ln x dx;$

(b) $\int x \ln x dx.$

34. Evaluate.

(a) $\int \tan^{-1} x dx;$

(b) $\int \sin^{-1} x dx.$

35. Solve the cyclic integrals.

(a) $\int e^x \sin x dx;$

(b) $\int e^{2x} \cos x dx.$

36. Evaluate the definite integrals by parts.

(a) $\int_0^1 x e^x dx;$

(b) $\int_0^\pi x \sin x dx.$

37. Derive a reduction step for $I_n = \int x^n \sin x dx$ and use repeated integration by parts to evaluate I_3 .

38. Evaluate the odd powers of sine.

(a) $\int \sin^3 x dx;$

(b) $\int \sin^5 x dx.$

39. Evaluate the odd powers of cosine.

(a) $\int \cos^3 x dx;$

(b) $\int \cos^5 x dx.$

40. Use power-reduction identities.

(a) $\int \sin^2 x dx;$

(b) $\int \cos^2 x dx.$

41. Evaluate.

(a) $\int \sin^2 x \cos^2 x dx;$

(b) $\int \sin^4 x dx.$

42. Use product-to-sum identities.

(a) $\int \sin 3x \sin 5x dx;$

(b) $\int \cos 4x \cos x dx.$

43. Evaluate integrals involving tangent and secant.

(a) $\int \tan^3 x \sec^2 x dx;$

(b) $\int \sec^4 x \, dx;$

(c) $\int \tan^2 x \, dx.$

44. Evaluate integrals involving cotangent and cosecant.

(a) $\int \cot^3 x \csc^2 x \, dx;$

(b) $\int \cot x \csc^2 x \, dx;$

(c) $\int \csc^2 x \, dx.$

45. Evaluate.

(a) $\int_0^{\pi/2} \sin^3 x \, dx;$

(b) $\int_0^{\pi/2} \sin^2 x \cos^2 x \, dx;$

(c) $\int_0^{\pi/4} \sec^2 x \, dx.$

Trigonometric substitution and partial fractions

46. Evaluate.

(a) $\int \frac{dx}{\sqrt{9-x^2}};$

(b) $\int \frac{dx}{x^2+25}.$

47. Evaluate $\int \sqrt{16-x^2} \, dx$, and hence find $\int_{-4}^4 \sqrt{16-x^2} \, dx$.

48. Evaluate.

(a) $\int \sqrt{x^2+9} \, dx;$

(b) $\int \frac{dx}{\sqrt{x^2+9}}.$

49. For $x > 3$, evaluate.

(a) $\int \frac{dx}{x^2\sqrt{x^2-9}};$

(b) $\int \frac{dx}{\sqrt{x^2-4}}.$

50. Decompose into partial fractions and integrate.

(a) $\int \frac{dx}{(x-1)(x+2)};$

(b) $\int \frac{5x+1}{x^2-x-2} dx.$

51. Evaluate.

(a) $\int \frac{dx}{x(x+1)};$

(b) $\int \frac{2x+3}{(x-1)^2} dx.$

52. Evaluate integrals with irreducible quadratic factors.

(a) $\int \frac{x}{x^2+4} dx;$

(b) $\int \frac{3x+5}{x^2+2x+5} dx.$

53. Divide first, then integrate.

(a) $\int \frac{x^2+1}{x+1} dx;$

(b) $\int \frac{2x^2+3x+1}{x+1} dx.$

54. Evaluate the definite integrals.

(a) $\int_0^1 \frac{dx}{(x+1)(x+2)};$

(b) $\int_0^1 \frac{3x+5}{x^2+3x+2} dx.$

55. Decompose

$$\frac{2x^2+1}{x(x^2+1)}$$

into partial fractions, state its excluded real value, and integrate it.

Definite integrals, total change, and areas

56. Evaluate.

(a) $\int_{-1}^2 (3x^2-2x) dx;$

(b) $\int_0^2 |x-1| dx;$

(c) $\int_{-2}^2 (x^3 + 2) dx.$

57. Given $\int_0^3 f(x) dx = 5$ and $\int_3^7 f(x) dx = -2$, find

(a) $\int_0^7 f(x) dx;$

(b) $\int_7^0 f(x) dx;$

(c) $\int_0^7 [2f(x) + 1] dx.$

58. Use symmetry to evaluate.

(a) $\int_{-a}^a x^7 \cos x dx;$

(b) $\int_{-2}^2 (x^4 + 3x^3) dx;$

(c) $\int_{-\pi}^{\pi} \cos x dx.$

59. Let $F(x) = \int_0^x (t^2 - 4t + 1) dt$. Find $F(x)$ and $F'(x)$, then determine the stationary values of F .

60. Find the average value of

(a) $f(x) = x^2$ on $[0, 3];$

(b) $f(x) = \sin x$ on $[0, \pi].$

Also find the positive $c \in [0, 3]$ for which c^2 equals the average value in part (a).

61. For each function, find both the net signed area and the total geometric area.

(a) $y = x^2 - 1$ on $[-2, 2];$

(b) $y = x(x - 2)$ on $[0, 3].$

62. Find the area under the graph and above the x -axis.

(a) $y = 4 - x^2$, $-2 \leq x \leq 2;$

(b) $y = \sqrt{x}$, $0 \leq x \leq 4.$

63. Find the total area between the graph and the x -axis.

(a) $y = x^2 - 4x$, $0 \leq x \leq 5;$

(b) $y = \sin x$, $0 \leq x \leq 2\pi$.

64. Find the area enclosed by

(a) $y = x + 2$ and $y = x^2$;

(b) $y = 2x$ and $y = x^2$.

65. Integrate with respect to y to find the area enclosed by

(a) $x = y^2$ and $x = 2y$;

(b) $x = 4 - y^2$ and $x = y^2$.

66. Find the area enclosed by

(a) $y = x^2$ and $y = 2 - x^2$;

(b) $y = |x|$ and $y = 2 - x^2$.

67. A tent-shaped cross-section is described by

$$f(x) = \begin{cases} x + 1, & -1 \leq x \leq 1, \\ 3 - x, & 1 < x \leq 3. \end{cases}$$

Find its total area above the x -axis.

68. For $k > 0$, find the area enclosed by $y = kx$ and $y = x^2$. Hence find k if the area is $9/2$ square units.

69. Differentiate.

(a) $\int_x^{x^2} \sqrt{1+t^3} dt$;

(b) $\int_0^{\ln x} \cos(t^2) dt$.

70. Solve for the stated positive or restricted constant.

(a) $\int_0^a x dx = 8$, $a > 0$;

(b) $\int_1^b \frac{dx}{x} = \ln 5$, $b > 1$;

(c) $\int_0^c \cos x dx = \frac{1}{2}$, $0 \leq c \leq \pi/2$.

71. Find the area bounded by $y = x^2$, the x -axis, and $x = 2$.

72. Find the total area between $y = x^3 - 4x$ and the x -axis on $[-3, 3]$.

73. Find the area enclosed by $y = x^2$ and $y = 4$.
74. A particle has velocity $v(t) = 3t^2 - 12t + 9$ for $0 \leq t \leq 4$. Find the times when it is at rest, its displacement, and its total distance travelled.
75. A particle has acceleration $a(t) = 6t - 4$ and initial velocity $v(0) = 2$. Find $v(t)$ and the displacement during the first three seconds. Determine whether the displacement equals the distance travelled.

Volumes of revolution

76. Find the volume formed by rotating about the x -axis the region under
- (a) $y = x^2$, $0 \leq x \leq 2$;
 - (b) $y = \sqrt{x}$, $0 \leq x \leq 4$.
77. The region bounded by $x = y^2$, the y -axis, $y = 0$, and $y = 2$ is rotated about the y -axis. Find the volume.
78. Find the volume formed by rotating the region between $y = 2x$ and $y = x^2$, $0 \leq x \leq 2$, about the x -axis.
79. The region between $y = x$ and $y = x^2$, $0 \leq x \leq 1$, is rotated about $y = 2$. Set up and evaluate the washer integral.
80. The region under $y = 4 - x^2$ and above the x -axis is rotated about $y = -1$. Find the volume.
81. The triangular region $0 \leq y \leq x \leq 2$ is rotated about the y -axis. Use washers perpendicular to the y -axis to find its volume.
82. The region under $y = \sqrt{x}$ for $0 \leq x \leq 4$ is rotated about the y -axis. Express x in terms of y and evaluate a washer integral.
83. Rotating the line $y = (r/h)x$, $0 \leq x \leq h$, about the x -axis forms a cone. Use integration to derive its volume.
84. Rotate the semicircle $y = \sqrt{R^2 - x^2}$ about the x -axis and derive the volume of a sphere of radius R .
85. A cylindrical block has radius 3 cm and height 5 cm. A coaxial cylindrical hole of radius 1 cm is drilled through it. Set up a washer integral and find the remaining volume.

Applications and modelling

86. Water enters a tank at $r(t) = 10 + 2t^2$ litres per minute. Find the amount delivered during the first three minutes and the final amount if the tank initially contains 40 litres.
87. A variable force $F(x) = 5 + 2x$ newtons moves an object from $x = 0$ to $x = 6$ metres. Find the work done.
88. A rod of length 8 m has linear density $\lambda(x) = 2 + 0.5x$ kilograms per metre, measured from one end. Find its total mass.
89. The marginal cost of producing x units is $C'(x) = 0.02x + 15$ ringgit per unit. The fixed cost is RM800. Find the cost of producing 200 units and the corresponding average cost.
90. A population grows at the instantaneous rate $P'(t) = 200e^{0.03t}$ people per year. Find the increase during the first ten years.
91. A reservoir's inflow rate varies as shown. The outflow rate is a constant 5 litres per minute for the entire 14 minutes.

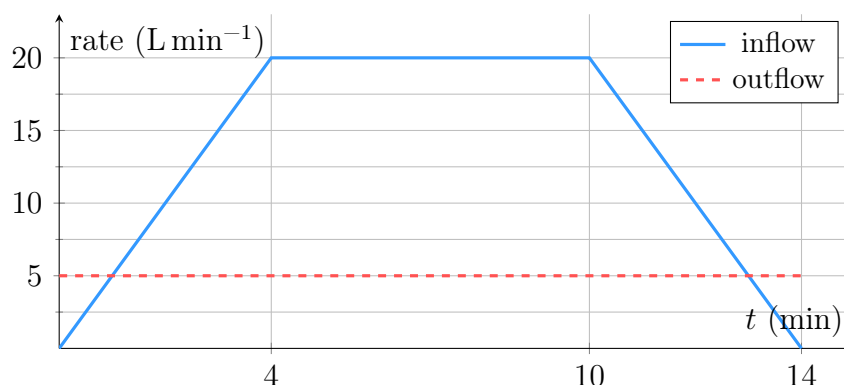


Figure 13.14: Reservoir flow rates for Question 91.

Find the total inflow and the net change in the reservoir's volume.

92. A medicine concentration is $C(t) = 40e^{-0.25t}$ milligrams per litre. Find the total exposure $\int_0^8 C(t) dt$ and the average concentration during the eight hours.
93. A solar array produces

$$P(t) = 1000 \sin\left(\frac{\pi t}{12}\right) \quad (0 \leq t \leq 12)$$

watts, where t is measured in hours. Find the energy produced in watt-hours and kilowatt-hours.

94. Rain falls at the rate $r(t) = 6 + 2t$ millimetres per hour for three hours. Find the total rainfall and the average rainfall rate.

95. A continuous random variable has density

$$f(x) = kx(1 - x), \quad 0 \leq x \leq 1.$$

Find k and $P(1/4 \leq X \leq 3/4)$.

96. A vessel has horizontal cross-sectional area

$$A(h) = \pi \left(1 + \frac{h}{4}\right)^2$$

square metres at height h . Find its volume from $h = 0$ to $h = 4$ m.

97. A parabolic arch is modelled by $y = 9 - x^2/4$, with distances in metres.

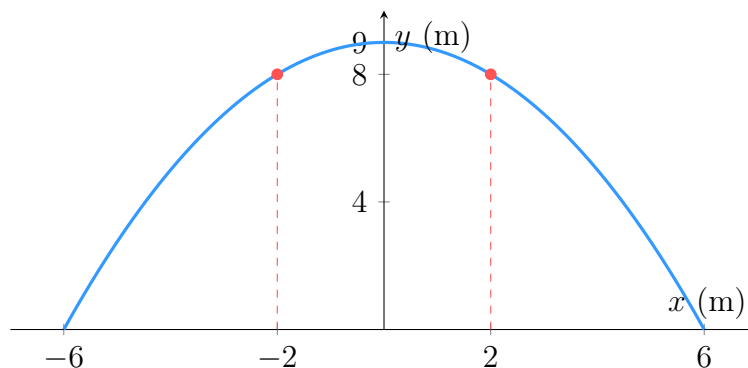


Figure 13.15: Parabolic arch and vehicle-edge positions for Question 97.

Find the cross-sectional area of the opening. A centred vehicle is 4 m wide; find its greatest permitted height if a vertical clearance of 0.5 m is required.

98. A sheet has variable width $w(x) = 2 + 0.1x$ metres for $0 \leq x \leq 10$ and uniform surface density 3 kg m^{-2} . Find its mass.
99. A spring obeys Hooke's law $F(x) = 250x$ newtons. Find the work required to stretch it from 0.05 m to 0.12 m beyond its natural length.
100. A workshop's marginal profit is $P'(x) = 60 - 0.4x$ ringgit per item, and $P(0) = -500$ ringgit.
- Find $P(x)$.
 - Find the production levels at which the workshop breaks even.
 - Find the production level giving maximum profit and state that profit.

Selected answers are provided in Appendix F.

Chapter 14

Differential Equations

Chapter guide

Prerequisites. Differentiation, integration, logarithms, exponentials, partial fractions, and algebraic rearrangement.

Learning outcomes. Readers should be able to classify common first-order differential equations, select and apply an appropriate solution method, impose initial conditions, verify solutions, and construct elementary models of changing systems.

Differential and integral calculus provide complementary ways of describing change and accumulation. Differential equations bring these ideas together by translating relationships observed in real systems into mathematical form.

Previously, differential and integral calculus were treated primarily as separate, dual operations on explicit functions. Differentiation provided the tools to evaluate instantaneous rates of change, while integration enabled the computation of net accumulation over an interval. However, physical systems rarely present themselves as explicit relations of the form $y = f(x)$. More frequently, physical principles, conservation laws, and empirical observations establish a direct relationship between a continuously varying quantity and the rate at which that quantity changes relative to independent variables such as time or spatial coordinates.

A differential equation relates an unknown function to one or more of its derivatives. Whereas an algebraic equation is solved for unknown numbers, a differential equation is solved for a function or a family of functions satisfying the stated relationship. Differential equations provide a key framework for mathematical modelling in engineering, physics, mathematical biology, and quantitative economics, particularly for deterministic continuous-time systems.

The presentation follows standard foundation and introductory-calculus references (Abu Bakar et al. 2025–2026; Strang and Herman 2016b). It covers the classification, formulation, and analytical solution of separable, homogeneous, linear, exact, Bernoulli, and affine-coefficient first-order equations, together with initial-value problems. The chapter concludes with applications to mixing, growth, and decay.

14.1 Introduction to differential equations

A **differential equation** is an equation involving an unknown function and one or more of its derivatives. In an application, the unknown function usually represents a quantity of interest, while its derivatives represent rates of change.

The **order** of a differential equation is the order of its highest derivative. When an equation is polynomial in its derivatives, its **degree** is the power of the highest-order derivative after the equation has been cleared of radicals and fractional powers involving derivatives. If the equation is not polynomial in its derivatives, its degree is not defined.

Differential equation	Order	Degree
$\frac{dy}{dx} + 3y = x$	1	1
$\frac{d^2y}{dx^2} + x^2 = 5\frac{dy}{dx}$	2	1
$\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^2 + y = 0$	2	3

A **solution** is a function that satisfies the equation on an interval. For example, differentiating

$$y = x^3 + C$$

gives $dy/dx = 3x^2$, so this one-parameter family is the **general solution** of that first-order equation. An additional condition can determine the arbitrary constant and produce a **particular solution**. For instance, $y(1) = 4$ gives $C = 3$ and hence $y = x^3 + 3$.

An equation together with a condition such as

$$y(x_0) = y_0$$

is called an **initial-value problem**. The condition identifies the member of the solution family whose graph passes through (x_0, y_0) .

A first-order differential equation assigns a gradient to each point in the plane. Drawing a short segment with that gradient at many points produces a *slope field*. Curves that remain tangent to these segments are solution curves. Figure 14.1 shows the slope field for $y' = -y$ together with several members of the family $y = Ce^{-x}$.

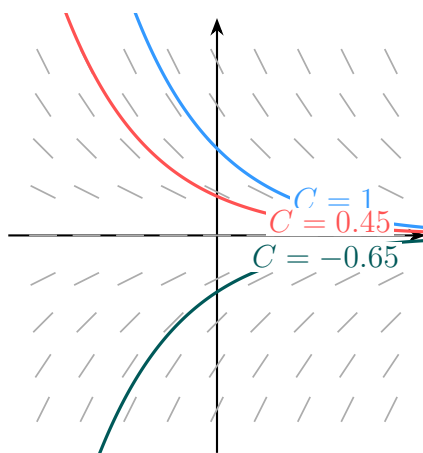


Figure 14.1: A slope field for $y' = -y$. Every solution $y = Ce^{-x}$ follows the local gradient segments.

Candidate solutions can be checked by direct substitution. If $y = \sin x$, then $y'' = -\sin x$, and therefore

$$y'' + y = -\sin x + \sin x = 0.$$

Thus $y = \sin x$ is a solution of $y'' + y = 0$. The methods in the remainder of this chapter concern first-order equations.

14.2 Separable differential equations

A first-order equation is **separable** if it can be written as

$$g(y) \frac{dy}{dx} = f(x), \quad (14.1)$$

or equivalently as $g(y) dy = f(x) dx$. Integration then gives

$$\int g(y) dy = \int f(x) dx + C. \quad (14.2)$$

The result is often an implicit relation between x and y ; it need not always be rearranged explicitly for y .

Example 14.1: a separable equation

Solve

$$y \frac{dy}{dx} = 3x^2.$$

Separating and integrating gives

$$\begin{aligned} y dy &= 3x^2 dx, \\ \frac{y^2}{2} &= x^3 + C, \\ y^2 &= 2x^3 + A. \end{aligned}$$

Thus the implicit general solution is

$$\boxed{y^2 = 2x^3 + A}.$$

If $y(5) = 1$, then $1 = 250 + A$, so $A = -249$ and the particular solution is

$$\boxed{y^2 = 2x^3 - 249}.$$

When separation requires division by an expression involving y , solutions that make that expression zero may be lost. Such constant or equilibrium solutions must be checked in the original equation before being discarded.

Example 14.2: separation with an initial condition

Solve

$$\frac{dy}{dx} = \frac{2y}{x^2 - 1}, \quad y(2) = 1.$$

For a nonzero solution,

$$\frac{dy}{y} = \frac{2 dx}{(x-1)(x+1)} = \left(\frac{1}{x-1} - \frac{1}{x+1} \right) dx.$$

Therefore

$$\ln |y| = \ln \left| \frac{x-1}{x+1} \right| + C,$$

and hence

$$y = A \frac{x-1}{x+1}.$$

The initial condition gives $A = 3$. Since the initial point lies in the interval $x > 1$,

$$\boxed{y = 3 \frac{x-1}{x+1}}, \quad x > 1.$$

The equilibrium solution $y = 0$, which was divided out, also satisfies the original differential equation but not the stated initial condition.

14.3 Homogeneous differential equations

A function $H(x, y)$ is **homogeneous of degree n** if

$$H(\lambda x, \lambda y) = \lambda^n H(x, y) \quad (14.3)$$

for every admissible scalar λ . Thus $x^2 - xy + y^2$ is homogeneous of degree 2, whereas $x^2 + y + 1$ is not homogeneous because its terms have different degrees.

A first-order equation

$$M(x, y) dx + N(x, y) dy = 0 \quad (14.4)$$

is called homogeneous when M and N are homogeneous functions of the same degree. It can usually be written as

$$\frac{dy}{dx} = F\left(\frac{y}{x}\right). \quad (14.5)$$

The substitution

$$y = vx, \quad \frac{dy}{dx} = v + x \frac{dv}{dx} \quad (14.6)$$

then reduces the equation to separable form. If the equation is more naturally expressed using x/y , one may instead use $x = vy$ and $dx/dy = v + y dv/dy$.

14.3.1 Procedure for the substitution $y = vx$

1. Verify that M and N have the same degree, or rewrite the equation as $dy/dx = F(y/x)$.
2. Put $y = vx$ and $dy/dx = v + x dv/dx$.
3. Separate v and x , then integrate.
4. Replace v by y/x and simplify.
5. Check any values that were divided out during the calculation.

Example 14.3: solving a homogeneous equation

Solve

$$(y^2 - x^2) dx + xy dy = 0.$$

After division by dx and substitution of $y = vx$,

$$\begin{aligned} v + x \frac{dv}{dx} &= \frac{1 - v^2}{v}, \\ x \frac{dv}{dx} &= \frac{1 - 2v^2}{v}. \end{aligned}$$

Hence

$$\frac{v}{1 - 2v^2} dv = \frac{dx}{x},$$

and integration gives

$$-\frac{1}{4} \ln |1 - 2v^2| = \ln |x| + C.$$

Substituting $v = y/x$ and absorbing constants gives

$$\boxed{x^4 - 2x^2y^2 = C.}$$

The straight-line solutions $y = \pm x/\sqrt{2}$ are included by taking $C = 0$.**Example 14.4: a homogeneous initial-value problem**

For

$$(x^2 + y^2) dx + xy dy = 0, \quad y(1) = -1,$$

the substitution $y = vx$ produces

$$\frac{v}{1 + 2v^2} dv = -\frac{dx}{x}.$$

Therefore

$$\frac{1}{4} \ln(1 + 2v^2) = -\ln |x| + C,$$

so $x^4 + 2x^2y^2 = C$. The initial point gives $C = 3$, and hence

$$x^4 + 2x^2y^2 = 3.$$

14.4 Linear first-order differential equations

A differential equation is **linear in y** when it can be written as

$$\frac{dy}{dx} + P(x)y = Q(x). \quad (14.7)$$

The dependent variable and its derivative occur only to the first power and are not multiplied together. The functions P and Q may still be non-linear functions of x .

The **integrating factor** is

$$\mu(x) = e^{\int P(x) dx}. \quad (14.8)$$

Since $\mu' = P\mu$, multiplication of Equation (14.7) by μ gives

$$\frac{d}{dx}(\mu y) = \mu Q.$$

Consequently,

$$y\mu = \int \mu Q dx + C. \quad (14.9)$$

Some equations are linear in x rather than in y . With x regarded as a function of y , the corresponding standard form is

$$\frac{dx}{dy} + P(y)x = Q(y), \quad \mu(y) = e^{\int P(y) dy}. \quad (14.10)$$

Example 14.5: a linear initial-value problem

Solve

$$x \frac{dy}{dx} + 2y = 3x, \quad y(1) = 5.$$

For $x > 0$, division by x gives

$$\frac{dy}{dx} + \frac{2}{x}y = 3, \quad \mu = e^{\int 2/x dx} = x^2.$$

Thus

$$\begin{aligned} \frac{d}{dx}(x^2y) &= 3x^2, \\ y &= x + \frac{C}{x^2}. \end{aligned}$$

The initial condition gives $C = 4$, so

$$y = x + \frac{4}{x^2}, \quad x > 0.$$

Example 14.6: a trigonometric integrating factor

For

$$y' + y \tan x = \sec x + 2x \cos x,$$

the integrating factor is $\mu = \sec x$ on any interval not crossing a zero of $\cos x$.

Therefore

$$\frac{d}{dx}(y \sec x) = \sec^2 x + 2x.$$

After integration,

$$y \sec x = \tan x + x^2 + C,$$

and hence

$$y = \sin x + x^2 \cos x + C \cos x.$$

Common checks for a linear equation

First divide by the coefficient of the derivative so that its coefficient is 1. Read P from this standard form. After multiplication by the integrating factor, verify that the complete left-hand side is the derivative of μy .

14.5 Non-homogeneous equations with linear coefficients

Here, *non-homogeneous* refers to coefficients that are affine linear expressions containing constants:

$$(a_1x + b_1y + c_1) dx + (a_2x + b_2y + c_2) dy = 0. \quad (14.11)$$

This differs from the use of the word for the linear equation $y' + P(x)y = Q(x)$ when $Q \neq 0$. The required method depends on

$$\Delta = a_1b_2 - a_2b_1. \quad (14.12)$$

14.5.1 Case 1: linearly independent coefficients

If $\Delta \neq 0$, the two coefficient lines have a unique intersection (h, k) . Translate that point to the new origin:

$$x = X + h, \quad y = Y + k, \quad dx = dX, \quad dy = dY. \quad (14.13)$$

Because (h, k) satisfies both line equations, the constant terms disappear and the

transformed equation is homogeneous in X and Y .

Example 14.7: translating to the intersection

Solve

$$(y - x - 2) dx + (4y + x - 3) dy = 0.$$

The coefficient lines intersect at $(-1, 1)$. Put

$$x = p - 1, \quad y = q + 1.$$

The equation becomes

$$(q - p) dp + (4q + p) dq = 0.$$

With $q = vp$ and $dq = v dp + p dv$,

$$(4v^2 + 2v - 1) \frac{dp}{p} + (4v + 1) dv = 0.$$

Integration gives

$$\ln |p| + \frac{1}{2} \ln |4v^2 + 2v - 1| = C,$$

and hence $4q^2 + 2pq - p^2 = C$. Returning to the original variables,

$$4(y - 1)^2 + 2(x + 1)(y - 1) - (x + 1)^2 = C,$$

or equivalently

$$4y^2 - 6y + 2xy - 4x - x^2 = C.$$

14.5.2 Case 2: linearly dependent coefficients

If $\Delta = 0$, the linear parts are proportional and the coefficient lines do not usually have a unique intersection. Use the repeated linear combination as a new variable, for example

$$z = a_1x + b_1y, \quad dz = a_1 dx + b_1 dy. \quad (14.14)$$

This normally reduces the equation to separable form in z and the remaining variable.

Example 14.8: using a repeated linear combination

Solve

$$(x - 2y + 3) dx + (2x - 4y - 1) dy = 0.$$

Put $z = x - 2y$, so $dx = dz + 2 dy$. Then

$$(z + 3)(dz + 2 dy) + (2z - 1) dy = 0,$$

$$(z + 3) dz + (4z + 5) dy = 0.$$

Thus

$$dy = - \left(\frac{1}{4} + \frac{7}{4(4z+5)} \right) dz.$$

Integration and multiplication by 16 give

$$16y + 4z + 7 \ln |4z + 5| = C.$$

Since $z = x - 2y$,

$$4x + 8y + 7 \ln |4x - 8y + 5| = C.$$

Test	Geometry	Reduction
$\Delta \neq 0$	Two lines meet at one point (h, k)	Translate with $x = X + h$, $y = Y + k$, then solve the homogeneous equation.
$\Delta = 0$	Parallel or coincident linear parts	Substitute the common combination $z = ax + by$, then separate variables.

14.6 Exact differential equations

The equation

$$M(x, y) dx + N(x, y) dy = 0 \quad (14.15)$$

is **exact** if there is a potential function $F(x, y)$ such that

$$dF = F_x dx + F_y dy = M dx + N dy.$$

Its solution is then $F(x, y) = C$. If M and N have continuous first partial derivatives on a rectangular region, exactness is equivalent to

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}. \quad (14.16)$$

To find F , integrate one coefficient while treating the other variable as constant:

$$F(x, y) = \int M(x, y) dx + g(y). \quad (14.17)$$

Differentiate this expression with respect to y and compare it with N to determine $g'(y)$.

Example 14.9: constructing a potential function

Solve

$$(3x^2y + 2x^2 - x) dx + (x^3 + 2y^3 - 2y) dy = 0.$$

Since $M_y = 3x^2 = N_x$, the equation is exact. Integrating M with respect to x gives

$$F = x^3y + \frac{2}{3}x^3 - \frac{1}{2}x^2 + g(y).$$

Comparison of F_y with N gives

$$g'(y) = 2y^3 - 2y, \quad g(y) = \frac{1}{2}y^4 - y^2.$$

Therefore

$$x^3y + \frac{2}{3}x^3 - \frac{1}{2}x^2 + \frac{1}{2}y^4 - y^2 = C.$$

Example 14.10: an exact initial-value problem

For

$$(y + 2xy^2 + 2x - 1)dx + (x + 2yx^2 - 3y + 2)dy = 0, \quad (x, y) = (1, 1),$$

we have $M_y = 1 + 4xy = N_x$. A potential function is

$$F = xy + x^2y^2 + x^2 - x - \frac{3}{2}y^2 + 2y.$$

At $(1, 1)$, $F = 5/2$, so

$$xy + x^2y^2 + x^2 - x - \frac{3}{2}y^2 + 2y = \frac{5}{2}.$$

14.7 Bernoulli differential equations

A **Bernoulli differential equation** has the form

$$\frac{dy}{dx} + P(x)y = Q(x)y^n, \quad n \neq 0, 1. \quad (14.18)$$

It is non-linear, but the substitution

$$u = y^{1-n} \quad (14.19)$$

converts it to a linear equation. Since

$$\frac{du}{dx} = (1-n)y^{-n} \frac{dy}{dx},$$

division of Equation (14.18) by y^n gives

$$\frac{du}{dx} + (1-n)P(x)u = (1-n)Q(x). \quad (14.20)$$

Solve this equation using an integrating factor, then replace u by y^{1-n} . If division by a power of y was used, check $y = 0$ separately because that solution may have been excluded. The same method works for an equation linearised in $x(y)$ by using $u = x^{1-n}$.

Example 14.11: linearising a Bernoulli equation

Solve

$$y(1 + xy^4) dx - dy = 0.$$

In derivative form,

$$\frac{dy}{dx} - y = xy^5.$$

Here $n = 5$. Put $u = y^{-4}$ to obtain

$$\frac{du}{dx} + 4u = -4x.$$

The integrating factor is e^{4x} , and therefore

$$\begin{aligned}\frac{d}{dx}(ue^{4x}) &= -4xe^{4x}, \\ ue^{4x} &= e^{4x} \left(\frac{1}{4} - x \right) + C.\end{aligned}$$

Hence

$$y^{-4} = \frac{1}{4} - x + Ce^{-4x},$$

so the non-zero solutions are

$$y = \left(\frac{1}{4} - x + Ce^{-4x} \right)^{-1/4}.$$

The function $y = 0$ is also a solution of the original equation.

14.8 Choosing a first-order method

Type	Recognition	Main step
Separable	$g(y)y' = f(x)$	Group the y -terms with dy and the x -terms with dx .
Homogeneous	$y' = F(y/x)$, or M, N have equal degree	Use $y = vx$ or $x = vy$.
Linear	$y' + P(x)y = Q(x)$	Multiply by $e^{\int P(x) dx}$.
Exact	$M_y = N_x$	Construct a potential F and set $F = C$.
Bernoulli	$y' + Py = Qy^n$, $n \neq 0, 1$	Use $u = y^{1-n}$, then solve the linear equation.
Affine	Both coefficients are linear in x, y with constants	Translate when $\Delta \neq 0$; use $z = ax + by$ when $\Delta = 0$.

An equation may fit more than one class. Simplify it before choosing a method; for example, the exact method is often shorter than a homogeneous substitution when both apply. Differentiate the final implicit relation to check it against the original equation.

14.9 Applications of differential equations in real-world problems

Differential equations become useful in applications only after the quantities in a physical situation have been translated into rates. A good model should identify the unknown quantity, specify its units, express how it changes, and include enough initial information to determine a particular solution. The following procedure is useful in many first-order models:

1. define the dependent variable and the independent variable;
2. express the net rate of change as

$$\text{rate of increase} - \text{rate of decrease};$$

3. form and solve the resulting differential equation;
4. apply the initial condition; and
5. interpret the answer within the physical interval for which the model is valid.

Two important applications at foundation level are mixing problems and exponential growth or decay.

14.9.1 Mixing problems

Mixing models describe the amount of a dissolved substance in a tank or vessel. The substance may be salt, sugar, a chemical pollutant, a medicine, or any other material carried by a liquid. The standard model assumes that the contents are mixed uniformly and instantaneously. Consequently, the concentration of the liquid leaving the vessel is equal to the concentration inside it at that instant.

Let

- $A(t)$ be the amount of dissolved substance at time t ;
- $V(t)$ be the volume of liquid in the vessel;
- q_{in} and q_{out} be the volume flow rates;
- c_{in} be the concentration of the incoming solution.

The concentration inside the vessel, and hence the concentration of the outgoing solution, is

$$c_{\text{out}}(t) = \frac{A(t)}{V(t)}.$$

The governing balance law is therefore

$$\boxed{\frac{dA}{dt} = c_{\text{in}}q_{\text{in}} - \frac{A(t)}{V(t)}q_{\text{out}}.} \quad (14.21)$$

The two terms on the right-hand side are the rate at which substance enters and the rate at which it leaves. If pure water enters, then $c_{\text{in}} = 0$ and the first term vanishes.

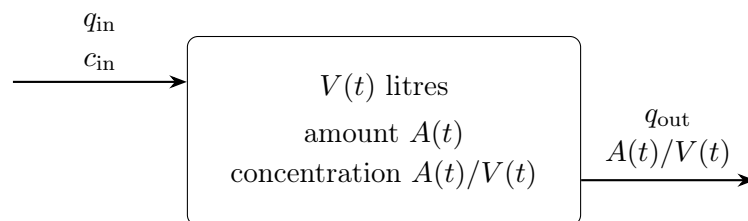


Figure 14.2: Rate balance for a uniformly mixed vessel.

If the flow rates are constant and the initial volume is V_0 , then

$$V(t) = V_0 + (q_{\text{in}} - q_{\text{out}})t. \quad (14.22)$$

The solution is physically meaningful only while the vessel contains a positive volume and has not exceeded its capacity. Thus, the time at which the vessel becomes empty or begins to overflow must be calculated together with the solution.

Constant-volume mixing

When $q_{\text{in}} = q_{\text{out}} = q$, the volume remains equal to V_0 . For a constant incoming concentration, equation (14.21) becomes

$$\frac{dA}{dt} + \frac{q}{V_0}A = c_{\text{in}}q.$$

Its solution, where $A(0) = A_0$, is

$$A(t) = c_{\text{in}}V_0 + (A_0 - c_{\text{in}}V_0)e^{-qt/V_0}. \quad (14.23)$$

The amount approaches $c_{\text{in}}V_0$ as t increases. Equivalently, the concentration inside the vessel approaches the incoming concentration.

Example 14.12: A constant-volume flavouring mixture

A food-processing vessel contains 150 L of liquid with 30 kg of a dissolved flavouring agent. A solution containing 5 kg L⁻¹ of the agent enters at 2 L min⁻¹, and the well-mixed liquid leaves at the same rate. Find the amount of flavouring agent after t minutes and after 180 seconds.

Solution. Let $A(t)$ be the amount of flavouring agent, measured in kilograms. Since the volume remains 150 L,

$$\begin{aligned} \text{rate in} &= (5)(2) = 10 \text{ kg min}^{-1}, \\ \text{rate out} &= \frac{A}{150}(2) = \frac{A}{75} \text{ kg min}^{-1}. \end{aligned}$$

Hence,

$$\frac{dA}{dt} + \frac{1}{75}A = 10, \quad A(0) = 30.$$

The integrating factor is $e^{t/75}$, so

$$\begin{aligned} \frac{d}{dt}(Ae^{t/75}) &= 10e^{t/75}, \\ Ae^{t/75} &= 750e^{t/75} + C, \\ A(t) &= 750 + Ce^{-t/75}. \end{aligned}$$

Using $A(0) = 30$ gives $C = -720$. Therefore,

$$A(t) = 750 - 720e^{-t/75}.$$

Since 180 seconds is 3 minutes,

$$A(3) = 750 - 720e^{-3/75} \approx \boxed{58.23 \text{ kg}}.$$

Increasing-volume mixing and overflow

If $q_{\text{in}} > q_{\text{out}}$, the volume increases. The denominator in the outgoing concentration must therefore be replaced by the changing volume from equation (14.22).

Example 14.13: A tank that is filling

A 120 L tank initially contains 40 L of liquid with 2 kg of dissolved fruit concentrate. Concentrated juice containing 1 kg L⁻¹ enters at 3 L min⁻¹, while the well-mixed solution leaves at 2 L min⁻¹. Find the amount $A(t)$ of concentrate, the amount after 10 minutes, and the time at which the tank begins to overflow.

Solution. The net volume flow rate is $3 - 2 = 1$ L min⁻¹, so

$$V(t) = 40 + t.$$

The amount satisfies

$$\frac{dA}{dt} = 3 - \frac{2A}{40 + t}, \quad A(0) = 2,$$

or

$$\frac{dA}{dt} + \frac{2}{40 + t}A = 3.$$

An integrating factor is

$$\exp\left(\int \frac{2}{40 + t} dt\right) = (40 + t)^2.$$

Therefore,

$$\begin{aligned} \frac{d}{dt} [A(40 + t)^2] &= 3(40 + t)^2, \\ A(40 + t)^2 &= (40 + t)^3 + C, \\ A(t) &= 40 + t + \frac{C}{(40 + t)^2}. \end{aligned}$$

The initial condition gives $C = -60800$, and hence

$$A(t) = 40 + t - \frac{60800}{(40 + t)^2}.$$

After 10 minutes,

$$A(10) = 50 - \frac{60800}{50^2} = \boxed{25.68 \text{ kg}}.$$

The tank begins to overflow when $V(t) = 120$:

$$40 + t = 120 \quad \implies \quad \boxed{t = 80 \text{ minutes}}.$$

Thus, the formula for $A(t)$ describes the stated system for $0 \leq t \leq 80$ unless the physical arrangement changes when overflow begins.

Decreasing-volume mixing and emptying

If $q_{\text{out}} > q_{\text{in}}$, the volume decreases. The model must stop when $V(t) = 0$, since the concentration $A(t)/V(t)$ is not defined once the vessel is empty.

Example 14.14: A basin that is draining

A basin initially contains 50 gal of liquid with 200 oz of dissolved pollutant. A stream containing 2 oz gal⁻¹ enters at 4 gal min⁻¹, while the well-mixed solution leaves at 7 gal min⁻¹. Find the amount of pollutant at time t , the amount after 0.25 hour, and the time at which the basin becomes empty.

Solution. The volume is

$$V(t) = 50 - 3t,$$

so the model is valid for $0 \leq t < 50/3$. The rate equation is

$$\frac{dA}{dt} = 8 - \frac{7A}{50 - 3t}, \quad A(0) = 200.$$

In standard linear form,

$$\frac{dA}{dt} + \frac{7}{50 - 3t}A = 8.$$

The integrating factor is

$$\exp\left(\int \frac{7}{50 - 3t} dt\right) = (50 - 3t)^{-7/3}.$$

Multiplying through and integrating gives

$$\begin{aligned} \frac{d}{dt} [A(50 - 3t)^{-7/3}] &= 8(50 - 3t)^{-7/3}, \\ A(50 - 3t)^{-7/3} &= 2(50 - 3t)^{-4/3} + C, \\ A(t) &= 2(50 - 3t) + C(50 - 3t)^{7/3}. \end{aligned}$$

Using $A(0) = 200$ gives

$$C = \frac{100}{50^{7/3}}.$$

Thus,

$$A(t) = 2(50 - 3t) + 100 \left(\frac{50 - 3t}{50} \right)^{7/3}.$$

Since 0.25 hour is 15 minutes,

$$A(15) = 10 + 100 \left(\frac{5}{50} \right)^{7/3} \approx \boxed{10.46 \text{ oz}}.$$

The basin becomes empty when

$$50 - 3t = 0 \quad \implies \quad t = \frac{50}{3} \text{ minutes} \approx 16.67 \text{ minutes}.$$

14.9.2 Exponential growth and decay

Suppose that the rate of change of a quantity is proportional to the amount present. If $P(t)$ denotes the amount at time t , then

$$\frac{dP}{dt} = kP, \tag{14.24}$$

where k is the constant of proportionality. Separating variables gives

$$\begin{aligned} \frac{1}{P} dP &= k dt, \\ \ln |P| &= kt + C. \end{aligned}$$

For a positive initial amount $P(0) = P_0$, the solution is

$$\boxed{P(t) = P_0 e^{kt}}. \tag{14.25}$$

The quantity grows when $k > 0$ and decays when $k < 0$. If a decay model is written as

$$\frac{dP}{dt} = -rP, \quad r > 0,$$

then $P(t) = P_0 e^{-rt}$. Its half-life $T_{1/2}$ is found from $P(T_{1/2}) = P_0/2$:

$$\boxed{T_{1/2} = \frac{\ln 2}{r}}. \tag{14.26}$$

Sometimes the decay constant is not supplied directly. If a proportion p of the initial amount has decayed after a time T , then the proportion remaining is $1 - p$, so

$$e^{kT} = 1 - p \quad \implies \quad \boxed{k = \frac{\ln(1 - p)}{T}}. \tag{14.27}$$

This calculation is different from simply writing $k = -p/T$ because the model changes continuously rather than in separate time steps.

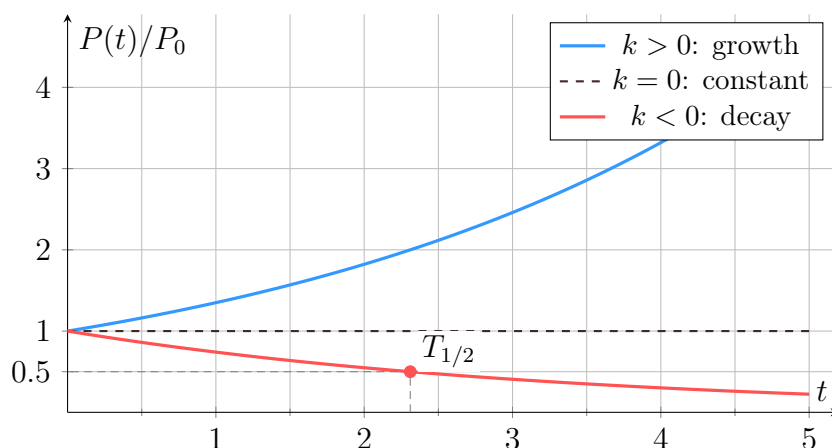


Figure 14.3: Normalised exponential growth and decay. For the decay curve, the half-life is the time at which $P(t)/P_0 = 1/2$.

Example 14.15: Decay when the rate constant is known

Hydrargyrum (mercury)-194 is modelled by

$$\frac{dP}{dt} = -0.645P,$$

where t is measured in hours. Find the general model satisfying $P(0) = P_0$. If $P_0 = 1350$ g, find the amount remaining after 45 minutes and the half-life.

Solution. Using equation (14.25),

$$P(t) = P_0 e^{-0.645t}.$$

Since 45 minutes is $3/4$ hour,

$$P\left(\frac{3}{4}\right) = 1350e^{-0.645(3/4)} \approx \boxed{832.23 \text{ g}}.$$

The half-life is

$$T_{1/2} = \frac{\ln 2}{0.645} \approx \boxed{1.075 \text{ hours}}.$$

Example 14.16: Determining the decay constant from data

An initial 630 mg sample of Uranium-238 loses 15.6% of its mass in 2.5 years. Assuming that its decay rate is proportional to the amount present, find a model for its mass, the mass remaining after 12 years, and its half-life.

Solution. After 2.5 years, the proportion remaining is

$$1 - 0.156 = 0.844.$$

Therefore,

$$630e^{2.5k} = 630(0.844),$$
$$k = \frac{\ln(0.844)}{2.5} \approx -0.067841.$$

The model is

$$P(t) = 630e^{-0.067841t},$$

where t is measured in years. Hence,

$$P(12) = 630e^{-0.067841(12)} \approx \boxed{279.12 \text{ mg}}.$$

Finally,

$$T_{1/2} = \frac{\ln 2}{0.067841} \approx \boxed{10.22 \text{ years}}.$$

Chapter summary and common pitfalls

Key ideas. A differential equation is classified before a solution method is chosen. A general solution contains arbitrary constants, while initial conditions select a particular solution and its physically relevant domain.

Common pitfalls. Do not divide by an expression that may be zero without checking the lost solutions, include constants after integration, use absolute values correctly in logarithms, and verify the final function in the original equation.

14.10 Practice and exercises

Foundations and method selection

1. State the order and degree of each differential equation. If the degree is not defined, explain why.

(a) $y'' - xy' + (1 - x)y = \sin y$;

(b) $5(y'')^2 - y = e^x$;

(c) $y'' + x(y')^5 + y = 0$;

(d) $\sqrt{y''} + y' = x$;

(e) $(y''')^2 + (y')^7 = 4$.

2. For $y' = 3x^2$, find the general solution. Then find the particular solutions satisfying

(a) $y(0) = 5$;

(b) $y(2) = 1$;

(c) $y(-1) = 0$.

Explain how the initial condition selects one member of the solution family.

3. Verify by substitution that

(a) $y = Ce^{-2x}$ satisfies $y' + 2y = 0$;

(b) $y = x^2 + C$ satisfies $y' = 2x$;

(c) $x^2 + y^2 = C$ satisfies $x + y y' = 0$ wherever $y \neq 0$;

(d) $y = (1 + Ce^{-x})^{-1}$ satisfies $y' = y(1 - y)$.

4. Classify each equation using every applicable description from: separable, homogeneous, linear, exact, Bernoulli, and affine linear coefficients.

(a) $y' = x(1 + y^2)$;

(b) $y' + \frac{2}{x}y = x^3$;

- (c) $(x + y) dx - x dy = 0$;
- (d) $(2xy + 1) dx + (x^2 + 2y) dy = 0$;
- (e) $y' - y = xy^4$;
- (f) $(x - y + 2) dx + (2x + y - 1) dy = 0$.

5. Rewrite each equation in the requested standard form.

- (a) $3xy' + 6y = x^2$ in the form $y' + P(x)y = Q(x)$;
- (b) $(1 + y^2) dx + (2xy - 3) dy = 0$ in the form $dx/dy + P(y)x = Q(y)$;
- (c) $y dx - (x + y) dy = 0$ as $dx/dy = F(x/y)$;
- (d) $(x^2 + y^2) dx - 2xy dy = 0$ as $dy/dx = F(y/x)$.

6. Match each recognising form to its principal substitution or test.

- (i) $y' = f(x)g(y)$;
- (ii) $y' = F(y/x)$;
- (iii) $y' + Py = Qy^n$;
- (iv) $M dx + N dy = 0$;
- (v) $(a_1x + b_1y + c_1)dx + (a_2x + b_2y + c_2)dy = 0$.

The available choices are separation, $y = vx$, $u = y^{1-n}$, $M_y = N_x$, and the determinant $\Delta = a_1b_2 - a_2b_1$. Briefly justify every match.

7. Find all constant solutions of

- (a) $y' = y(y - 3)$;
- (b) $y' = x(y + 2)(y - 1)$;
- (c) $y' = y - y^3$.

Explain why these solutions can be lost during separation.

8. State the largest interval containing the initial point on which the coefficients of the equation are continuous.

- (a) $y' + \frac{1}{x-2}y = x$, with $y(0) = 1$;
- (b) $y' + (\tan x)y = 1$, with $y(0) = 2$;
- (c) $y' = \frac{y}{x^2 - 9}$, with $y(4) = 1$.

9. For each equation, identify two methods that could be used and state which is shorter.

- (a) $(2xy - x^2) dx + (x^2 - y^2) dy = 0$;

(b) $(xy^2 - 2xy) dx + (x^2y - x^2) dy = 0$;

(c) $(x^4 + y^4) dx + 4xy^3 dy = 0$.

10. Choose an appropriate first-order method for each equation and give one sentence explaining the choice. Do not solve.

(a) $y' = \frac{x^2}{1 + y^2}$;

(b) $y' = \frac{x + y}{x}$;

(c) $y' + e^x y = x$;

(d) $(3x^2y + 1) dx + (x^3 + 2y) dy = 0$;

(e) $y' + xy = x^2y^3$;

(f) $(x + 2y - 3) dx + (2x - y + 4) dy = 0$.

Separable differential equations

11. Find the general solution.

(a) $y' = 4x^3$;

(b) $y y' = 2x$;

(c) $(1 + y^2)y' = 3x^2$;

(d) $y' = \frac{x}{y^2}$.

12. Solve by separation.

(a) $e^y y' = -\sin x$;

(b) $y' = e^{x-y}$;

(c) $y' \cos y = \cos x$;

(d) $y' = xe^{-y^2}$.

13. Solve, using partial fractions where necessary.

(a) $y' = \frac{2y}{x^2 - 1}$;

(b) $y' = \frac{x}{y(y + 1)}$;

(c) $y' = \frac{(y - 1)(y + 2)}{x}$.

14. Find the particular solution.

(a) $(x + 1)y' = x(y + 3)$, $y(0) = 1$;

- (b) $3y^2y' + 2x = 1$, $y(1) = 0$;
(c) $y' = x(1 + y^2)$, $y(0) = 1$.
15. For $y' = y(4 - y)$:
- (a) find the equilibrium solutions;
(b) solve for a non-equilibrium initial value $y(0) = y_0$;
(c) determine the limiting value when $0 < y_0 < 4$;
(d) describe what happens when $y_0 > 4$.
16. Obtain implicit solutions.
- (a) $(y^2 + 4)y' = x^2 + 1$;
(b) $y' = \frac{x + 1}{y + \sin y}$;
(c) $y' = \frac{x\sqrt{1 + x^2}}{1 + y^4}$.
17. The equation $y' = kx(y + 1)$ satisfies $y(0) = 2$ and $y(1) = 5$. Find k and the resulting particular solution.
18. Treat x as a function of y and solve.
- (a) $\frac{dx}{dy} = y(1 + x^2)$;
(b) $e^x \frac{dx}{dy} = 2y$;
(c) $(x + 2)\frac{dx}{dy} = 3y^2$.
19. A quantity satisfies $A' = -0.18A$ and $A(0) = 240$. Find the first time at which $A(t) < 50$.
20. Differentiate each proposed implicit solution and verify the corresponding equation.
- (a) $y^2 - x^4 = C$ for $y y' = 2x^3$;
(b) $e^y + \cos x = C$ for $e^y y' = \sin x$;
(c) $\ln |y - 1| - \ln |y + 2| = 3 \ln |x| + C$ for $y' = x^{-1}(y - 1)(y + 2)$.
21. Eliminate the arbitrary constant to form a differential equation for each family.
- (a) $y = x^2 + C$;
(b) $y = Ce^{3x}$;

- (c) $x^2 + y^2 = C$;
- (d) $y^2 = 2x^3 + C$.

22. Solve

$$y' = \frac{(y-2)(y+1)}{x}, \quad y(1) = 0.$$

State any equilibrium solutions and the interval containing $x = 1$ on which the particular solution is valid.

Homogeneous differential equations

23. For each equation, verify homogeneity, state the common degree, rewrite it as $y' = F(y/x)$, and solve.

- (a) $(y^2 - x^2) dx + xy dy = 0$;
- (b) $(x + 2y) dx + (2x + 3y) dy = 0$;
- (c) $(2xy - 2x^2) dx + (x^2 - 2y^2) dy = 0$.

24. Find the particular solution.

- (a) $(x^2 + y^2) dx + xy dy = 0$, $y(1) = -1$;
- (b) $(x + y) dx - x dy = 0$, $y(1) = 2$;
- (c) $(x^2 - xy + y^2) dx - xy dy = 0$, $y(1) = 1$.

25. Use $x = vy$ rather than $y = vx$.

- (a) $y dx - (x + y) dy = 0$;
- (b) $(x^2 + y^2) dx - xy dy = 0$;
- (c) $\frac{dx}{dy} = \frac{x + y}{y}$.

26. Find all straight-line solutions $y = mx$ of

- (a) $y' = \frac{x^2 + y^2}{2xy}$;
- (b) $y' = 1 + \frac{y^2}{x^2}$;
- (c) $y' = \frac{x - y}{x + y}$.

27. Show that $y = vx$ transforms

$$(x^2 + y^2) dy - xy dx = 0$$

into $v^3 dx + x(1 + v^2) dv = 0$. Hence solve the original equation.

28. Determine the value of a for which

$$(x^2 + axy + y^2) dx + (2x^2 - xy) dy = 0$$

has the straight-line solution $y = x$. Then solve the resulting homogeneous equation.

29. Decide which equation is homogeneous and solve it.

(a) $(2x^2 + xy) dy - (3xy + y^2) dx = 0$;

(b) $(x^2 + 2xy) dx - (x^2y + 3xy^2) dy = 0$.

30. Solve and leave the answers in implicit form.

(a) $y' = \frac{x + y}{x - y}$;

(b) $y' = \frac{2x - y}{x + 2y}$;

(c) $y' = \frac{y^2}{x^2 + xy}$.

31. Show that

$$(2xy - x^2) dx + (x^2 - y^2) dy = 0$$

is both homogeneous and exact. Solve it using both methods and compare the final relations.

32. Find all real values of k for which

$$(x^2 + kxy + y^2) dx + (2x^2 + kxy) dy = 0$$

is homogeneous. Explain why the value of k does not affect the homogeneity test.

33. Solve

$$(y - xe^{x/y}) dy + ye^{x/y} dx = 0, \quad y(0) = 1,$$

using a substitution based on x/y .

34. For each equation, decide whether $y = vx$ or $x = vy$ gives the simpler algebra, then solve.

(a) $y dx - (x + \sqrt{xy}) dy = 0$;

(b) $(x^2 - y^2) dy + 2xy dx = 0$;

(c) $y' = \frac{x^2 + xy}{y^2}$.

Linear first-order differential equations

35. Put each equation in standard linear form, find its integrating factor, and solve it.

(a) $x dy + (y + xy - e^x) dx = 0$;

(b) $dx = -(3 + 2x \tan y) dy$;

(c) $(1 - x)y' + y = x - x^3$;

(d) $(2xy - 2) dy + (1 + y^2) dx = 0$.

36. Solve the constant-coefficient equations.

(a) $3y' + 12y = 4$;

(b) $y' + 5y = 10e^{-x}$;

(c) $2y' - y = 6$.

37. Solve the variable-coefficient equations.

(a) $y' + \frac{2}{x}y = x^2$;

(b) $y' - \frac{3}{x}y = x^4$;

(c) $y' + 3x^2y = x^2$.

38. Solve on an interval where the trigonometric coefficients are continuous.

(a) $y' + y \tan x = \sec x$;

(b) $y' - y \cot x = \sin x$;

(c) $y' + 2y \tan x = \sin x \cos x$.

39. Regard x as a function of y .

(a) $\frac{dx}{dy} + \frac{x}{y} = y^2$;

(b) $\frac{dx}{dy} + 2x \tan y = -3$;

(c) $\frac{dx}{dy} + x \cot y = \sin y$.

40. Find the particular solution.

(a) $x dy - y dx = x^2 e^x dx$, $y(1) = 0$;

(b) $y' + \frac{2}{x}y = 3$, $y(1) = 5$;

(c) $\frac{dx}{dy} + \frac{4x}{1 - y} = 3$, $x(-1) = 0$.

41. Solve each initial-value problem and state the interval containing the initial point on which the solution is valid.

(a) $y' + \frac{y}{x-2} = x$, $y(0) = 1$;

(b) $y' + y \tan x = 1$, $y(0) = 2$;

(c) $y' - \frac{y}{x+1} = x^2$, $y(1) = 0$.

42. Verify directly that

$$y = x + \frac{C}{x^2}$$

is the general solution of $xy' + 2y = 3x$. Then determine the member passing through $(2, 3)$.

43. A linear equation has integrating factor $\mu(x) = e^{x^2}$ and satisfies

$$\frac{d}{dx}(e^{x^2}y) = 2xe^{x^2}.$$

Recover the original differential equation and find its general solution.

44. Determine k so that $y = x^2$ is a solution of

$$y' + \frac{k}{x}y = (k+2)x.$$

Is k uniquely determined? Explain.

45. For the homogeneous linear equation $y' + P(x)y = 0$:

(a) show that its general solution is $y = Ce^{-\int P(x) dx}$;

(b) show that the sum of any two solutions is again a solution;

(c) explain why this superposition property fails in general when $Q(x) \neq 0$.

46. Classify and solve.

(a) $y' = x - y$;

(b) $y' = \frac{y}{x} + x^2$;

(c) $(1+x^2)y' + 2xy = 1$.

Affine linear coefficients

47. For each equation, calculate Δ , classify the linear coefficients as independent or dependent, and identify either the translation point or the repeated linear combination.

(a) $(2y - x + 1) dx - (x - 2y - 3) dy = 0$;

(b) $(3y - 2x - 1) dx - (2x - 3y - 2) dy = 0$;

(c) $(x - y - 3) dx + (2y - x + 1) dy = 0$;

(d) $(x + y + 1) dx + (x - 2y + 4) dy = 0$;

(e) $(x - 2y + 3) dx + (2x - 4y - 3) dy = 0$.

48. Solve the equations with independent linear coefficients.

(a) $(y - x - 2) dx + (4y + x - 3) dy = 0$;

(b) $(x + 2y - 11) dx + (2x - 3y + 6) dy = 0$;

(c) $(2x + y - 1) dx + (x - 2y - 3) dy = 0$.

49. Solve the equations with dependent linear coefficients.

(a) $(x + y) dx + (x + y + 1) dy = 0$;

(b) $(x - 2y + 3) dx + (2x - 4y - 1) dy = 0$;

(c) $(x + y) dx + (3x + 3y - 4) dy = 0$.

50. Find the particular solution.

(a) $(x + y) dx + (x + y + 1) dy = 0$, $y(2) = 1$;

(b) $(x - 2y + 3) dx + (2x - 4y - 1) dy = 0$, $y(1) = 0$.

51. Translate the intersection to the origin and solve.

(a) $(2x + y - 4) dx + (x - y - 1) dy = 0$, through $(2, 0)$;

(b) $(x - 3y + 2) dx + (2x + y - 5) dy = 0$, through $(1, 1)$.

52. For

$$(x + ay + 1) dx + (2x + 4y - 3) dy = 0,$$

determine the values of a for which the linear parts are

(a) independent;

(b) dependent.

State the appropriate reduction in each case.

53. Show that the substitutions $x = p + 1$ and $y = q - 1$ transform

$$(2x + y - 1) dx + (x - 2y - 3) dy = 0$$

into a homogeneous equation. Solve and return to x and y .

54. The coefficient lines of an affine equation meet at $(3, 4)$. Explain geometrically why $x = X + 3$, $y = Y + 4$ removes both constant terms. Illustrate the translation by sketching the old and new origins.

55. Choose the most efficient substitution and justify it:

$$z = x + y, \quad z = x - 2y, \quad x = X + h, \quad y = Y + k.$$

- (a) $(x + y - 2) dx + (x + y + 2) dy = 0$;
- (b) $(x - 2y + 3) dx + (2x - 4y - 1) dy = 0$;
- (c) $(x - y - 3) dx + (2y - x + 1) dy = 0$.

56. Solve and differentiate the final implicit relation to check it.

$$(2y - x + 1) dx - (x - 2y - 3) dy = 0.$$

Exact differential equations

57. (a) State the exactness test for $M dx + N dy = 0$.

(b) Show that $(Ax + By) dx + (Cx + Dy) dy = 0$ is exact if and only if $B = C$.

(c) Show that $(Ax^2 + Bxy + Cy^2) dx + (Dx^2 + Exy + Fy^2) dy = 0$ is exact if and only if $B = 2D$ and $E = 2C$.

58. Test for exactness and solve.

- (a) $(2xy + x^3) dx + (x^2 + 2y^2) dy = 0$;
- (b) $(2x^3 + 3y) dx + (3x + y - 1) dy = 0$;
- (c) $(3x^2y + 2x^2 - x) dx + (x^3 + 2y^3 - 2y) dy = 0$.

59. Solve the exact equations involving exponential or trigonometric functions.

- (a) $(ye^x - 2x) dx + e^x dy = 0$;
- (b) $(xy \cos xy + \sin xy) dx + (x^2 \cos xy + e^y) dy = 0$;
- (c) $(2x \sin y - y \sin x) dx + (x^2 \cos y + \cos x) dy = 0$.

60. Find the particular solution.

- (a) $(y + 2xy^2 + 2x - 1) dx + (x + 2yx^2 - 3y + 2) dy = 0$, $(x, y) = (1, 1)$;
- (b) $2xe^{2y} dx + (2x^2e^{2y} - 2) dy = 0$, $y(2) = 0$;
- (c) $(xy \cos xy + \sin xy) dx + (x^2 \cos xy + e^y) dy = 0$, $y(0) = 1$.

61. Determine the parameter values that make the equation exact.

- (a) $(2xy + ay^2) dx + (x^2 + 4xy) dy = 0$;
(b) $(ye^x + kx) dx + (e^x + 2y) dy = 0$;
(c) $(3x^2y + by) dx + (x^3 + 2x + 4y^3) dy = 0$.
62. Describe all possible missing functions that preserve exactness, then write the solution in terms of an antiderivative of that function.
- (a) $(2xy + f(x)) dx + x^2 dy = 0$;
(b) $(e^x \sin y) dx + (e^x \cos y + g(y)) dy = 0$;
(c) $(3x^2 + y) dx + (x + h(y)) dy = 0$.
63. Show that each equation is both homogeneous and exact, then solve it by the shorter method.
- (a) $(xy^2 - 3y^3) dx + (x^2y - 9xy^2) dy = 0$;
(b) $(2xy - x^2) dx + (x^2 - y^2) dy = 0$.
64. Show that each equation is both separable and exact. Solve using both approaches.
- (a) $(xy^2 - 2xy) dx + (x^2y - x^2) dy = 0$;
(b) $2xe^{x^2} dx + \cos y dy = 0$.
65. Let
- $$F(x, y) = x^3y + \frac{2}{3}x^3 - \frac{1}{2}x^2 + \frac{1}{2}y^4 - y^2.$$
- (a) Calculate F_x and F_y .
(b) Write the corresponding exact differential equation.
(c) Find the change in F between $(0, 1)$ and $(1, 1)$.
66. Solve each equation using every applicable method and compare the amount of algebra required.
- (a) $(x^4 + y^4) dx + 4xy^3 dy = 0$;
(b) $(xy^2 - 3x^2y + x^3) dx + (x^2y - x^3 + 3y^3) dy = 0$;
(c) $(2xy^2 - 3x^2y + 2x^3) dx + (2x^2y - x^3 + 3y^3) dy = 0$.

Bernoulli differential equations

67. For each Bernoulli equation, identify $P(x)$, $Q(x)$, n , and the substitution $u = y^{1-n}$.
- (a) $y' + \frac{1}{x}y = \frac{1}{x}y^{-2}$;
(b) $y' - y = e^xy^2$;

$$(c) \quad y' + \frac{2}{x}y = \frac{5}{x^2}y^3;$$

$$(d) \quad y' - \frac{3}{x}y = -\frac{\cos x}{x^2}y^2.$$

68. Solve.

$$(a) \quad xy' + y = y^{-2};$$

$$(b) \quad y' - y = e^x y^2;$$

$$(c) \quad x^2 y' + 2xy = 5y^3;$$

$$(d) \quad y' + \frac{2}{x}y = x^2 y^4.$$

69. Solve and include any solution lost during division by a power of y .

$$(a) \quad y' - y = xy^5;$$

$$(b) \quad y' + \frac{1}{x}y = x^2 y^2;$$

$$(c) \quad y' - 2xy = -2xy^3;$$

$$(d) \quad 3(1 + x^2)y' = 2xy(y^3 - 1).$$

70. Find the particular solution.

$$(a) \quad x dy + y dx = x^3 y^6 dx, \quad y(1) = 1;$$

$$(b) \quad 2xyy' = y^2 - 2x^3, \quad y(2) = 1;$$

$$(c) \quad (y^4 - 2xy) dx + 3x^2 dy = 0, \quad y(2) = 1;$$

$$(d) \quad (x^2 + 2y^2) dx - y dy = 0, \quad y(0) = 1.$$

71. Treat x as the dependent variable and use a Bernoulli substitution.

$$(a) \quad \frac{dx}{dy} + \frac{x}{y} = yx^3;$$

$$(b) \quad \frac{dx}{dy} - 2x = e^y x^2.$$

72. For $y' + P(x)y = Q(x)y^n$, derive

$$u' + (1 - n)P(x)u = (1 - n)Q(x), \quad u = y^{1-n}.$$

State precisely why $y = 0$ must sometimes be checked separately.

73. Determine a so that $y = x^{-1}$ is a nonzero solution of

$$y' + \frac{2}{x}y = axy^3.$$

Then find the general nonzero solution.

74. Solve each equation using two distinct applicable methods.

(a) $x^2 y' - y^2 = xy$;

(b) $(x^2 + 6y^2) dx - 4xy dy = 0$;

(c) $(x^4 + y^4) dx + 4xy^3 dy = 0$.

75. The logistic equation is

$$P' = rP \left(1 - \frac{P}{K} \right), \quad r, K > 0.$$

Show that it is a Bernoulli equation with $n = 2$. Use $u = P^{-1}$ to derive its general nonzero solution, then impose $P(0) = P_0$.

Applications and modelling

76. A tank contains 1500 L of brine with 90 g of salt. Pure water enters at 30 L s^{-1} and the well-mixed solution leaves at the same rate.

(a) Form and solve the differential equation for the salt amount $A(t)$.

(b) Find the amount after 25 s.

(c) Find when the amount first reaches 35 g.

77. A 300 L tank initially contains 15 kg of dissolved chemical. Solution containing 0.10 kg L^{-1} enters at 16 L min^{-1} and the mixture leaves at the same rate.

(a) Find the amount $A(t)$.

(b) Find the concentration after 10 min.

(c) Determine the limiting concentration.

78. A vessel of capacity 400 L initially contains 120 L of water and 595 g of sugar. Syrup containing 5 g L^{-1} enters at 3 L s^{-1} , while mixture leaves at 2 L s^{-1} .

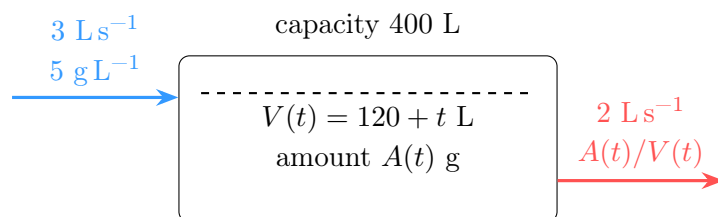


Figure 14.4: Variable-volume mixing model for Question 78.

(a) Form the differential equation for $A(t)$.

- (b) Find the amount after 2 min.
- (c) Determine when overflow begins and find the sugar amount then.
79. Use the initial tank in Question 78, but let the outflow rate be 6 L s^{-1} while the inflow remains 3 L s^{-1} .
- (a) Find $V(t)$ and the physical time interval.
- (b) Find $A(t)$ and the amount after 20 s.
- (c) Find the limiting concentration as the tank approaches empty.
80. A tank initially holds 200 L and 40 kg of salt. In separate trials, brine of concentration 0.15 kg L^{-1} enters at 4 L min^{-1} , while the outflow rate is
- (a) 4 L min^{-1} ;
- (b) 2 L min^{-1} ;
- (c) 7 L min^{-1} .
- For each trial, form the model, state the physical time interval, and find the salt amount after 20 min. Compare the three concentrations.
81. A lake contains $2.0 \times 10^8 \text{ L}$ of water and initially has a pollutant concentration of 8 mg L^{-1} . Clean water enters and mixed water leaves at $5.0 \times 10^6 \text{ L/day}$.
- (a) Model the pollutant amount.
- (b) Find the concentration after 30 days.
- (c) Find when the concentration falls below 1 mg L^{-1} .
82. A medicine is infused into the bloodstream at 12 mg h^{-1} . The body eliminates the medicine at a rate proportional to the amount present, with rate constant 0.18 h^{-1} . Initially no medicine is present.
- (a) Find the amount after t hours.
- (b) Determine the limiting amount.
- (c) Find when the amount reaches 90% of its limiting value.

83. A cup of coffee at 92°C is placed in a room at 22°C . Newton's law of cooling gives

$$T' = -k(T - 22).$$

If the coffee is 72°C after 10 min, find k , predict its temperature after 25 min, and find when it reaches 40°C .

84. A refrigerated sample at 4°C is moved into a laboratory at 24°C . After 6 min its temperature is 10°C .
- (a) Form a Newton heating model.
 - (b) Find the temperature after 15 min.
 - (c) Find when it first exceeds 20°C .
85. A bacterial culture grows at a rate proportional to its size. It contains 1200 cells initially and 2100 cells after 3 h. Find the model, the population after 8 h, and the doubling time.
86. A town has population 48 000 and initially grows continuously at 2.6% per year.
- (a) Form a continuous growth model.
 - (b) Predict the population after 12 years.
 - (c) Find when the population reaches 70 000.
87. An investment grows according to $A' = rA$. An initial RM 8,000 becomes RM 10,500 after five years.
- (a) Find the continuously compounded rate r .
 - (b) Predict the value after ten years.
 - (c) Find the doubling time.
88. Mercury-194 is modelled by $P' = -0.645P$, with time in hours. If $P(0) = 1350$ g, find the amount after 45 min, the half-life, and the time required for 95% of the sample to decay.
89. A 630 mg radioactive sample loses 15.6% of its mass in 2.5 years.
- (a) Determine its exponential decay model.
 - (b) Find the mass after 12 years.
 - (c) Find its half-life.
90. A wooden artefact contains 38% of the carbon-14 activity of living material. Use a carbon-14 half-life of 5730 years to estimate the age of the artefact. State the modelling assumption.
91. A drug dose of 250 mg has a half-life of 5.5 h.
- (a) Find the amount remaining after 12 h.
 - (b) Find when the amount first falls below 20 mg.

- (c) Determine the percentage eliminated after one day.

92. A fish population follows

$$P' = 0.30P \left(1 - \frac{P}{5000} \right), \quad P(0) = 600.$$

Solve the logistic equation, find $P(10)$, and determine when the population first reaches 4000.

93. A renewable resource satisfies $P' = 0.04P - 200$ with $P(0) = 8000$, where 200 units are harvested annually.

- (a) Solve the model.
- (b) Identify the equilibrium level.
- (c) Decide whether the resource grows or collapses from the stated initial value.
- (d) Repeat the qualitative conclusion for $P(0) = 4000$.

94. A falling object of mass m experiences gravity and a linear drag force of magnitude cv opposite its motion. Downward is positive.

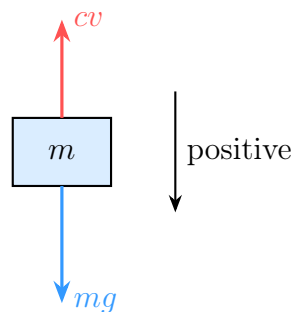


Figure 14.5: Forces on the falling object in Question 94.

- (a) Derive $mv' = mg - cv$.
- (b) Solve for $v(t)$ when the object is released from rest.
- (c) Find the terminal velocity.
- (d) Evaluate the terminal velocity for $m = 80$ kg, $c = 20$ kg s⁻¹, and $g = 9.81$ m s⁻².

95. In an RC circuit, charge $q(t)$ satisfies

$$R \frac{dq}{dt} + \frac{q}{C} = E, \quad q(0) = 0,$$

where E is constant.

- (a) Solve for $q(t)$.

- (b) Find the current $i(t) = q'(t)$.
- (c) Interpret the time constant RC .
- (d) Evaluate $q(0.5)$ for $R = 2 \text{ k}\Omega$, $C = 100 \mu\text{F}$, and $E = 12 \text{ V}$.

96. A reactant obeys the first-order law $C' = -kC$. Its concentration decreases from 0.80 mol L^{-1} to 0.50 mol L^{-1} in 18 min. Find k , the concentration after 45 min, and the time at which it reaches 0.10 mol L^{-1} .

97. A continuously stirred reactor of volume 600 L contains 18 kg of a chemical. Feed containing 0.04 kg L^{-1} enters at 5 L min^{-1} and mixture leaves at the same rate. The chemical also decomposes at rate $0.006A \text{ kg min}^{-1}$.

- (a) Form the linear equation for $A(t)$.
- (b) Solve it and find the long-term amount.
- (c) Find the amount after one hour.

98. Measurements of a decaying quantity are shown below.

$t \text{ (h)}$	0	2	4	6
$Q(t)$	100	81.9	67.0	54.9

Determine whether the data are consistent with $Q' = kQ$, estimate k , predict $Q(10)$, and estimate the half-life.

99. A 200 L tank initially contains 40 kg of salt. For the first 20 min, pure water enters at 5 L min^{-1} and mixture leaves at the same rate. After 20 min, the inlet is changed to brine containing 0.20 kg L^{-1} , with both flow rates unchanged.

- (a) Find the salt amount for $0 \leq t \leq 20$.
- (b) Use the amount at $t = 20$ as the new initial condition and find the amount for $t > 20$.
- (c) Determine the long-term concentration.

100. A water-treatment tank has constant volume 500 L. Initially the pollutant concentration is 60 mg L^{-1} . Water containing 10 mg L^{-1} enters at 2 L min^{-1} , mixed water leaves at 2 L min^{-1} , and a treatment process destroys pollutant at rate $0.004A \text{ mg min}^{-1}$.

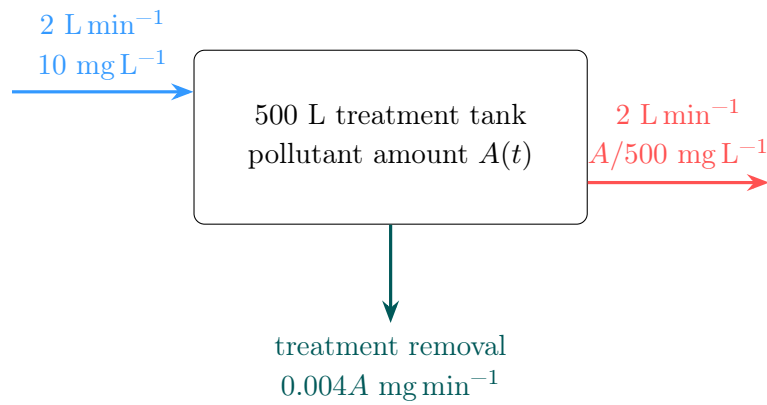


Figure 14.6: Combined flow and treatment removal for Question 100.

- (a) Derive the differential equation for $A(t)$.
- (b) Solve the initial-value problem.
- (c) Find the limiting concentration.
- (d) Determine when the concentration first falls below 8 mg L^{-1} .
- (e) Explain how the limiting value would change if the treatment process were switched off.

Chapter 15

Sequences and Series

Chapter guide

Prerequisites. Algebra, functions, limits, differentiation, factorials, and binomial coefficients.

Learning outcomes. Readers should be able to describe and analyse sequences, evaluate finite and infinite series, determine convergence in the cases studied, use sigma notation, and construct Taylor and Maclaurin approximations.

Sequences describe ordered lists, while series describe sums formed from the terms of those lists. Arithmetic and geometric progressions are familiar examples, but the same language also describes approximations, limiting processes, and functions represented by infinitely many polynomial terms.

This chapter develops limits of sequences, sigma notation, finite and infinite series, telescoping sums, binomial expansions, and power series. Taylor and Maclaurin series then connect these algebraic ideas to differentiation and numerical approximation. The progression from elementary sequences to power series follows standard pre-university and introductory-calculus references (Zaini bin Musa et al. 2020; Strang and Herman 2016b).

15.1 Sequences and their limits

A **sequence** is an ordered list of terms

$$a_1, a_2, a_3, \dots$$

The term a_k is called the k -th term. A sequence may be given explicitly, such as

$$a_k = 2k - 1,$$

or recursively, by specifying one or more initial terms and a rule relating later terms to earlier ones. For example,

$$a_1 = 3, \quad a_{k+1} = \frac{1}{2}a_k + 1.$$

The order matters: $(1, 2, 3)$ and $(3, 2, 1)$ contain the same numbers but are different sequences.

15.1.1 Convergent and divergent sequences

A sequence (a_k) **converges** to L if its terms can be made as close to L as desired by taking k sufficiently large. This is written

$$\lim_{k \rightarrow \infty} a_k = L. \quad (15.1)$$

If no finite limit exists, the sequence is **divergent**. Divergence may occur because the terms grow without bound, oscillate, or behave irregularly.

For example,

$$a_k = 1 - \frac{2}{k}$$

converges to 1, because $2/k \rightarrow 0$. In contrast,

$$b_k = \frac{k^2}{k+1} = \frac{k}{1+1/k}$$

grows without bound and is divergent. The sequence $(-1)^k$ is also divergent because it alternates between -1 and 1 .

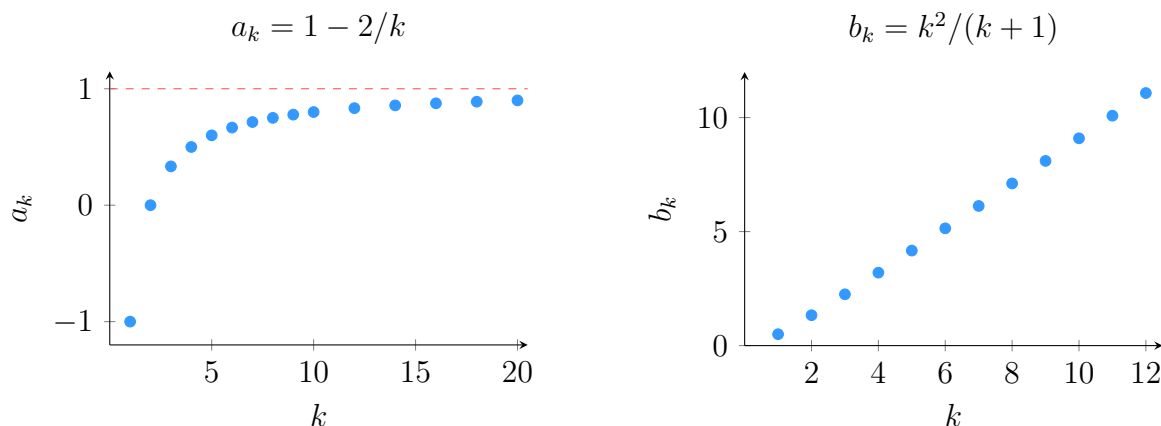


Figure 15.1: The left sequence approaches a horizontal limit; the right sequence grows without bound.

The usual limit laws apply whenever the relevant limits exist. If $a_k \rightarrow A$ and $b_k \rightarrow B$, then

$$\begin{aligned} a_k + b_k &\rightarrow A + B, \\ ca_k &\rightarrow cA, \\ a_k b_k &\rightarrow AB, \\ \frac{a_k}{b_k} &\rightarrow \frac{A}{B} \quad (B \neq 0). \end{aligned}$$

Example 15.1

Consider

$$a_k = 3, \quad b_k = \frac{1}{k}.$$

Since $a_k \rightarrow 3$ and $b_k \rightarrow 0$,

$$\lim_{k \rightarrow \infty} \frac{3k + 1}{k} = \lim_{k \rightarrow \infty} \left(3 + \frac{1}{k} \right) = 3.$$

For rational expressions in k , divide the numerator and denominator by the highest power of k . Thus

$$\lim_{k \rightarrow \infty} \frac{3k^2 - 1}{k^2 + k + 1} = \lim_{k \rightarrow \infty} \frac{3 - 1/k^2}{1 + 1/k + 1/k^2} = 3.$$

15.2 Series and sigma notation

Adding the terms of a sequence produces a **series**. A finite series is

$$a_1 + a_2 + \cdots + a_n,$$

while an infinite series is

$$a_1 + a_2 + a_3 + \cdots .$$

The Greek capital letter sigma denotes summation:

$$\sum_{k=m}^n a_k = a_m + a_{m+1} + \cdots + a_n. \quad (15.2)$$

The symbol k is a dummy index and may be replaced by another letter without changing the sum. The lower and upper limits state where the index begins and ends.

Examples.

$$\begin{aligned} 1 + 4 + 9 + \cdots + 100 &= \sum_{k=1}^{10} k^2, \\ 1 \cdot 2 + 2 \cdot 3 + \cdots + 19 \cdot 20 &= \sum_{k=1}^{19} k(k+1), \\ 1 + 3 + 5 + \cdots + (2n-1) &= \sum_{k=1}^n (2k-1). \end{aligned}$$

Useful summation properties include

$$\sum_{k=m}^n (a_k + b_k) = \sum_{k=m}^n a_k + \sum_{k=m}^n b_k, \quad (15.3)$$

$$\sum_{k=m}^n ca_k = c \sum_{k=m}^n a_k, \quad (15.4)$$

$$\sum_{k=m}^n c = (n - m + 1)c. \quad (15.5)$$

The number of terms from $k = m$ to $k = n$ is $n - m + 1$. This is especially important when a sum begins at 0 rather than 1.

15.3 Finite series

The following standard formulae are used repeatedly:

$$\sum_{r=1}^n r = \frac{n(n+1)}{2}, \quad (15.6)$$

$$\sum_{r=1}^n r^2 = \frac{n(n+1)(2n+1)}{6}, \quad (15.7)$$

$$\sum_{r=1}^n r^3 = \left[\frac{n(n+1)}{2} \right]^2. \quad (15.8)$$

Polynomial summands are expanded and separated before these results are applied.

Example 15.2

Evaluate

$$\sum_{r=1}^{10} (r+1)(7-r).$$

Since $(r+1)(7-r) = -r^2 + 6r + 7$,

$$\begin{aligned} \sum_{r=1}^{10} (r+1)(7-r) &= -\sum_{r=1}^{10} r^2 + 6\sum_{r=1}^{10} r + 7\sum_{r=1}^{10} 1 \\ &= -385 + 6(55) + 70 \\ &= 15. \end{aligned}$$

The negative value of some later terms is part of the stated algebraic sum.

Example 15.3

Find the sum of the first 20 odd squares:

$$\begin{aligned}
 1^2 + 3^2 + 5^2 + \cdots + 39^2 &= \sum_{r=1}^{20} (2r-1)^2 \\
 &= \sum_{r=1}^{20} (4r^2 - 4r + 1) \\
 &= 4 \sum_{r=1}^{20} r^2 - 4 \sum_{r=1}^{20} r + 20 \\
 &= 10\,660.
 \end{aligned}$$

More generally,

$$\sum_{r=1}^n (2r-1)^2 = \frac{n(4n^2-1)}{3}.$$

Example 15.4

Since $r(r+1) = r^2 + r$,

$$\begin{aligned}
 \sum_{r=1}^n r(r+1) &= \frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} \\
 &= \frac{n(n+1)(n+2)}{3}.
 \end{aligned}$$

15.3.1 Changing limits and indices

Changing the lower limit changes both the first term and the number of terms. For example,

$$\begin{aligned}
 \sum_{r=1}^{n+1} (2r+1) &= 3 + 5 + \cdots + (2n+3), \\
 \sum_{r=0}^{n+1} (2r+1) &= 1 + 3 + \cdots + (2n+3).
 \end{aligned}$$

The second sum has one additional term. A shifted sum can be converted to a standard one by adding or subtracting boundary terms. For instance,

$$\sum_{r=n}^{2n} (2r+1) = \sum_{r=1}^{2n} (2r+1) - \sum_{r=1}^{n-1} (2r+1).$$

15.4 Telescoping series and the method of differences

A finite sum **telescopes** when most intermediate terms cancel. If

$$a_k = f(k) - f(k-1),$$

then

$$\sum_{k=1}^n a_k = [f(1) - f(0)] + [f(2) - f(1)] + \cdots + [f(n) - f(n-1)] \quad (15.9)$$

$$= f(n) - f(0). \quad (15.10)$$

Example 15.5

Consider

$$\sum_{k=1}^n \left(\frac{1}{2k-1} - \frac{1}{2k+1} \right).$$

Writing out the first and last terms shows the cancellation:

$$\left(1 - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{5} \right) + \left(\frac{1}{5} - \frac{1}{7} \right) + \cdots + \left(\frac{1}{2n-1} - \frac{1}{2n+1} \right).$$

Hence

$$\sum_{k=1}^n \left(\frac{1}{2k-1} - \frac{1}{2k+1} \right) = 1 - \frac{1}{2n+1} = \frac{2n}{2n+1}.$$

Example 15.6

Partial fractions often reveal a telescoping structure:

$$\frac{1}{(k+1)(k+2)} = \frac{1}{k+1} - \frac{1}{k+2}.$$

Therefore,

$$\begin{aligned} \sum_{k=1}^n \frac{1}{(k+1)(k+2)} &= \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \cdots + \left(\frac{1}{n+1} - \frac{1}{n+2} \right) \\ &= \frac{1}{2} - \frac{1}{n+2} \\ &= \frac{n}{2(n+2)}. \end{aligned}$$

Sometimes cancellation occurs in two interlaced chains. For example,

$$\sum_{k=1}^n \left(\frac{1}{k} - \frac{1}{k+2} \right) = 1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2}.$$

15.5 Infinite series and partial sums

The expression

$$\sum_{k=1}^{\infty} a_k$$

is defined through its sequence of partial sums

$$S_n = \sum_{k=1}^n a_k.$$

The infinite series converges to S when

$$\lim_{n \rightarrow \infty} S_n = S.$$

If $\sum a_k$ converges, then necessarily $a_k \rightarrow 0$. The converse is not true: the harmonic series $\sum_{k=1}^{\infty} 1/k$ diverges even though $1/k \rightarrow 0$.

15.5.1 Geometric series

A geometric series has a constant ratio r between successive terms:

$$a + ar + ar^2 + \cdots.$$

For $r \neq 1$, its first n terms sum to

$$S_n = a \frac{1 - r^n}{1 - r}. \quad (15.11)$$

If $|r| < 1$, then $r^n \rightarrow 0$ and

$$\sum_{k=0}^{\infty} ar^k = \frac{a}{1 - r}. \quad (15.12)$$

If $|r| \geq 1$, the infinite geometric series diverges, except for the trivial case $a = 0$.

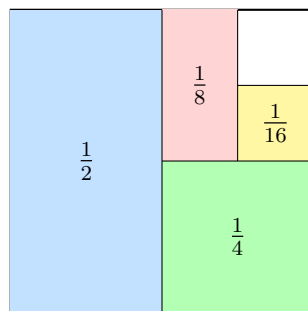


Figure 15.2: Successively halving the remaining area illustrates $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = 1$.

Example 15.7

The series

$$6 + 18 + 54 + \cdots$$

has ratio 3, so its terms do not approach zero and the infinite series diverges. By contrast,

$$6 + 2 + \frac{2}{3} + \frac{2}{9} + \cdots$$

has ratio $1/3$ and sum $6/(1 - 1/3) = 9$.

15.6 Binomial expansions

A **binomial** is an expression with two terms, such as $a + b$. Its powers can be expanded systematically using factorials and combinations.

15.6.1 Factorials and binomial coefficients

For a non-negative integer n ,

$$n! = n(n-1)(n-2) \cdots 2 \cdot 1, \quad 0! = 1.$$

Factorials satisfy the recursive identity

$$(n+1)! = (n+1)n!.$$

For example,

$$\frac{8!}{3!} = 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4, \quad \frac{n!}{(n-3)!} = n(n-1)(n-2).$$

The binomial coefficient is

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}, \quad 0 \leq r \leq n. \quad (15.13)$$

In particular,

$$\binom{n}{0} = 1, \quad \binom{n}{1} = n, \quad \binom{n}{2} = \frac{n(n-1)}{2}.$$

Lemma 15.1 (Pascal's identity). *For integers $n \geq 1$ and $1 \leq r \leq n$,*

$$\binom{n}{r-1} + \binom{n}{r} = \binom{n+1}{r}. \quad (15.14)$$

Proof. Writing both terms over a common denominator gives

$$\begin{aligned} \binom{n}{r-1} + \binom{n}{r} &= \frac{n!}{(r-1)!(n-r+1)!} + \frac{n!}{r!(n-r)!} \\ &= \frac{n!r + n!(n-r+1)}{r!(n-r+1)!} \\ &= \frac{(n+1)!}{r!(n+1-r)!} \\ &= \binom{n+1}{r}. \end{aligned}$$

□

This explains why every interior entry of Pascal's triangle is the sum of the two entries immediately above it.

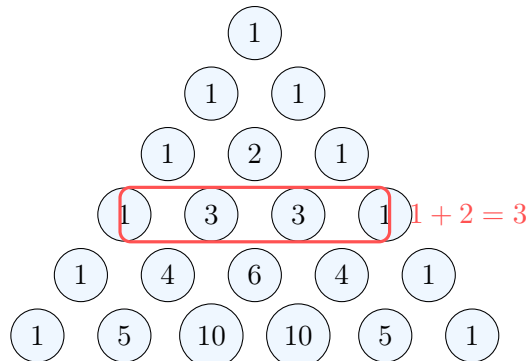


Figure 15.3: The rows of Pascal's triangle give the coefficients of $(a+b)^n$ for $n = 0, 1, 2, \dots$

15.6.2 The binomial theorem for positive integer powers

Theorem 15.2 (Binomial theorem). *For every positive integer n ,*

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k. \quad (15.15)$$

Proof. We use mathematical induction on n . For $n = 1$,

$$\sum_{k=0}^1 \binom{1}{k} a^{1-k} b^k = a + b = (a+b)^1,$$

so the base case holds. Assume the formula holds for some positive integer m . Multiplying the induction hypothesis by $a + b$ and collecting equal powers gives

$$\begin{aligned}(a + b)^{m+1} &= \sum_{k=0}^m \binom{m}{k} a^{m+1-k} b^k + \sum_{k=0}^m \binom{m}{k} a^{m-k} b^{k+1} \\ &= a^{m+1} + \sum_{k=1}^m \left(\binom{m}{k} + \binom{m}{k-1} \right) a^{m+1-k} b^k + b^{m+1} \\ &= \sum_{k=0}^{m+1} \binom{m+1}{k} a^{m+1-k} b^k,\end{aligned}$$

where the last equality uses Pascal's identity together with $\binom{m+1}{0} = \binom{m+1}{m+1} = 1$. Thus the formula holds for $m + 1$, and hence for every positive integer n . $\square$

The general term is

$$T_{k+1} = \binom{n}{k} a^{n-k} b^k, \quad k = 0, 1, \dots, n. \quad (15.16)$$

The expansion has $n + 1$ terms. The power of a decreases from n to 0, the power of b increases from 0 to n , and the total degree of every term is n . The coefficients are symmetric because $\binom{n}{k} = \binom{n}{n-k}$.

Example 15.8

$$\begin{aligned}(x + 2y)^3 &= x^3 + 3x^2(2y) + 3x(2y)^2 + (2y)^3 \\ &= x^3 + 6x^2y + 12xy^2 + 8y^3.\end{aligned}$$

Similarly,

$$(2x - 3y)^4 = 16x^4 - 96x^3y + 216x^2y^2 - 216xy^3 + 81y^4.$$

Example 15.9: a numerical approximation

Writing $2.02 = 2 + 0.02$ gives

$$\begin{aligned}(2.02)^6 &= 2^6 + \binom{6}{1} 2^5(0.02) + \binom{6}{2} 2^4(0.02)^2 + \dots \\ &= 67.937289638464 \\ &\approx 67.9373.\end{aligned}$$

Because the expansion terminates, this value could be calculated exactly; the small powers of 0.02 merely make later terms negligible for the stated accuracy.

15.6.3 Finding a particular term or coefficient

It is often unnecessary to expand the entire expression. To find a specified term, substitute the appropriate value of k into the general term.

Example 15.10

The fifth term of

$$(x^3 + \sqrt{y})^7$$

corresponds to $k = 4$:

$$T_5 = \binom{7}{4} (x^3)^3 (\sqrt{y})^4 = 35x^9y^2.$$

To find the coefficient of x^2 in

$$\left(\frac{3}{x} - 5x^3\right)^6,$$

write

$$T_{k+1} = \binom{6}{k} 3^{6-k} (-5)^k x^{-6+4k}.$$

The condition $-6 + 4k = 2$ gives $k = 2$. Hence the coefficient is

$$\binom{6}{2} 3^4 (-5)^2 = 30\,375.$$

If the binomial itself contains several powers of x , first treat the whole second expression as one quantity. For example, with $u = 2x - x^2$,

$$\begin{aligned} (1 + 2x - x^2)^5 &= (1 + u)^5 \\ &= 1 + 5u + 10u^2 + 10u^3 + \dots \\ &= 1 + 10x + 35x^2 + 40x^3 + \dots \end{aligned}$$

15.7 The generalised binomial series

For any real exponent α ,

$$(1 + x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha - 1)}{2!} x^2 + \frac{\alpha(\alpha - 1)(\alpha - 2)}{3!} x^3 + \dots \quad (15.17)$$

If α is a non-negative integer, the expansion terminates and is valid for every real x . Otherwise it is an infinite series and converges for

$$|x| < 1.$$

When the expression is $(a + bx)^\alpha$, first factor out a :

$$(a + bx)^\alpha = a^\alpha \left(1 + \frac{b}{a}x\right)^\alpha.$$

The validity condition becomes $|bx/a| < 1$.

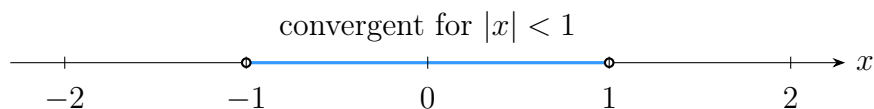


Figure 15.4: The generalised binomial series is centred at 0 and has radius of convergence 1.

Example 15.11

Up to the term in x^3 ,

$$\begin{aligned}\sqrt{1+2x} &= (1+2x)^{1/2} \\ &= 1 + x - \frac{1}{2}x^2 + \frac{1}{2}x^3 + \cdots, \quad |x| < \frac{1}{2},\end{aligned}$$

and

$$(1+x)^{-4} = 1 - 4x + 10x^2 - 20x^3 + \cdots, \quad |x| < 1.$$

Example 15.12

To approximate $\sqrt[3]{8.064}$, write

$$\sqrt[3]{8.064} = 2(1.008)^{1/3}.$$

Using $u = 0.008$,

$$\begin{aligned}2(1+u)^{1/3} &\approx 2\left(1 + \frac{u}{3} - \frac{u^2}{9} + \frac{5u^3}{81}\right) \\ &\approx 2.0053.\end{aligned}$$

15.7.1 Expansions obtained from partial fractions

Rational functions can often be decomposed into geometric or binomial series. For example,

$$f(x) = \frac{17-x}{(2-x)(3+x)} = \frac{3}{2-x} + \frac{4}{3+x}.$$

Now

$$\begin{aligned}\frac{3}{2-x} &= \frac{3}{2} \left(1 - \frac{x}{2}\right)^{-1}, & |x| < 2, \\ \frac{4}{3+x} &= \frac{4}{3} \left(1 + \frac{x}{3}\right)^{-1}, & |x| < 3.\end{aligned}$$

Therefore the common interval is $|x| < 2$, and

$$f(x) = \frac{17}{6} + \frac{11}{36}x + \frac{113}{216}x^2 + \frac{179}{1296}x^3 + \cdots.$$

15.8 Power series

A **power series** centred at a is an infinite series of the form

$$\sum_{n=0}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots. \quad (15.18)$$

It behaves like an infinite polynomial. For a given series there is usually a radius R such that the series converges when $|x-a| < R$ and diverges when $|x-a| > R$. The endpoints $x = a \pm R$ must be checked separately.

For example,

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots, \quad |x| < 1,$$

is the infinite geometric series with first term 1 and ratio x . Substituting or rescaling x produces many related expansions:

$$\begin{aligned}\frac{1}{1+x} &= 1 - x + x^2 - x^3 + \cdots, & |x| < 1, \\ \frac{1}{2-x} &= \frac{1}{2} + \frac{x}{4} + \frac{x^2}{8} + \frac{x^3}{16} + \cdots, & |x| < 2.\end{aligned}$$

Inside its interval of convergence, a power series may be differentiated and integrated term by term:

$$\frac{d}{dx} \sum_{n=0}^{\infty} c_n(x-a)^n = \sum_{n=1}^{\infty} n c_n(x-a)^{n-1}, \quad (15.19)$$

$$\int \sum_{n=0}^{\infty} c_n(x-a)^n dx = C + \sum_{n=0}^{\infty} \frac{c_n}{n+1} (x-a)^{n+1}. \quad (15.20)$$

15.9 Taylor series

Suppose a function has derivatives of every required order near $x = a$. Its Taylor series about a is

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n. \quad (15.21)$$

The first few terms are

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \cdots.$$

The centre a is the anchor point of the approximation. The Taylor polynomial agrees with the function and its first several derivatives at that point, so it is normally most accurate near a .

15.9.1 Taylor expansion of the sine function

For $f(x) = \sin x$ about $a = \pi/4$, write $h = x - \pi/4$. Since the derivatives cycle through $\sin x$, $\cos x$, $-\sin x$, and $-\cos x$,

$$\sin x = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}h - \frac{\sqrt{2}}{4}h^2 - \frac{\sqrt{2}}{12}h^3 + \cdots.$$

To approximate $\sin 48^\circ$, use $h = 3^\circ = \pi/60$:

$$\sin 48^\circ \approx 0.7431446,$$

which agrees with the calculator value 0.7431448 to six decimal places.

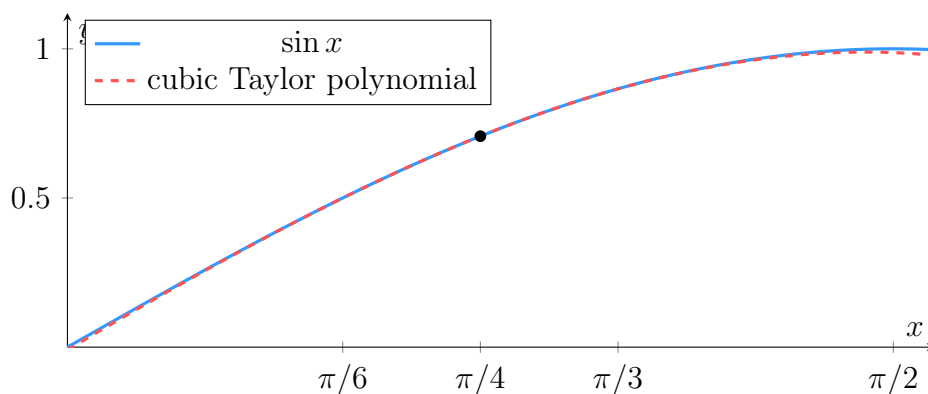


Figure 15.5: A Taylor polynomial follows the function closely near its centre $x = \pi/4$.

15.9.2 Approximating a square root

For $g(x) = \sqrt{x}$ about $a = 4$,

$$g'(x) = \frac{1}{2\sqrt{x}}, \quad g''(x) = -\frac{1}{4x^{3/2}}, \quad g'''(x) = \frac{3}{8x^{5/2}}.$$

Thus

$$\sqrt{x} = 2 + \frac{1}{4}(x-4) - \frac{1}{64}(x-4)^2 + \frac{1}{512}(x-4)^3 + \cdots.$$

Setting $x = 5$ gives $\sqrt{5} \approx 2.2324$, or 2.23 to two decimal places.

15.10 Maclaurin series

A Maclaurin series is a Taylor series centred at 0:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n. \quad (15.22)$$

Several standard expansions should be recognised:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots, \quad -\infty < x < \infty, \quad (15.23)$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots, \quad -\infty < x < \infty, \quad (15.24)$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots, \quad -\infty < x < \infty, \quad (15.25)$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots, \quad -1 < x \leq 1. \quad (15.26)$$

The logarithmic series can be obtained by integrating the geometric series for $1/(1+x)$ term by term.

15.10.1 Substitution and multiplication of series

Known series can be transformed algebraically. For example,

$$e^{-3x^2} = 1 - 3x^2 + \frac{9x^4}{2} - \frac{27x^6}{6} + \cdots,$$

so

$$x^4 e^{-3x^2} = x^4 - 3x^6 + \frac{9}{2}x^8 + \cdots.$$

Similarly,

$$\begin{aligned} \sin^2 x &= x^2 - \frac{1}{3}x^4 + \frac{2}{45}x^6 + \cdots, \\ \cos^2 x &= 1 - x^2 + \frac{1}{3}x^4 - \frac{2}{45}x^6 + \cdots. \end{aligned}$$

Adding the series gives $\sin^2 x + \cos^2 x = 1$, as expected.

Example 15.13: Logarithmic calculation

Since

$$3 - 2x - x^2 = 3 \left(1 - \frac{2}{3}x - \frac{1}{3}x^2 \right),$$

we may write

$$\begin{aligned} \ln(3 - 2x - x^2) &= \ln 3 + \ln \left(1 - \frac{2}{3}x - \frac{1}{3}x^2 \right) \\ &= \ln 3 - \frac{2}{3}x - \frac{5}{9}x^2 - \frac{26}{81}x^3 + \cdots. \end{aligned}$$

Only terms capable of contributing to the required power need to be retained during multiplication or composition.

15.10.2 Integration of a series

Series are particularly useful when an antiderivative is not elementary. For example,

$$\frac{\sin x}{x} = 1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \frac{x^6}{7!} + \cdots.$$

Integrating term by term gives

$$\int \frac{\sin x}{x} dx = C + x - \frac{x^3}{3 \cdot 3!} + \frac{x^5}{5 \cdot 5!} - \frac{x^7}{7 \cdot 7!} + \cdots.$$

Likewise,

$$\frac{e^x}{x} = \frac{1}{x} + 1 + \frac{x}{2!} + \frac{x^2}{3!} + \cdots,$$

so, on an interval not crossing $x = 0$,

$$\int \frac{e^x}{x} dx = C + \ln|x| + x + \frac{x^2}{4} + \frac{x^3}{18} + \cdots.$$

15.10.3 Accuracy of a Taylor approximation

The degree- n Taylor polynomial is

$$T_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k.$$

If $|f^{(n+1)}(t)| \leq M$ between a and x , the Lagrange remainder satisfies

$$|f(x) - T_n(x)| \leq \frac{M|x-a|^{n+1}}{(n+1)!}. \quad (15.27)$$

Accuracy generally improves when more terms are retained and when x is closer to the centre. The interval of convergence must still be respected; an algebraically correct power series need not represent the function outside that interval.

Chapter summary and common pitfalls

Key ideas. A sequence concerns individual terms, whereas a series concerns their sum. Convergence determines whether an infinite process has a finite limit, and power series approximate functions within an interval of validity.

Common pitfalls. Distinguish the n th term from a partial sum, check convergence before using an infinite-sum formula, and state the centre, order, and range of validity of a series approximation.

15.11 Practice and exercises**Sequences and limits**

1. Write the first five terms of each sequence.

- (a) $a_n = 2n - 1$;
- (b) $a_n = (-1)^n/n$;
- (c) $a_1 = 2$ and $a_{n+1} = 2a_n + 1$.

2. Find a formula for the n th term of each sequence.

- (a) 4, 7, 10, 13, ...;
- (b) $1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \dots$;
- (c) 2, 6, 18, 54, ...

3. Determine whether each sequence converges. If it does, find its limit.

- (a) $1 + 1/n$;
- (b) $n/(n + 1)$;
- (c) $n + (-1)^n$;
- (d) $(-1)^n$.

4. Evaluate each limit.

- (a) $\lim_{n \rightarrow \infty} \frac{3n^2 - 1}{n^2 + n + 1}$;
- (b) $\lim_{n \rightarrow \infty} \frac{2n^3 + n}{5n^3 - 4}$;
- (c) $\lim_{n \rightarrow \infty} \frac{n + 1}{n^2 + 1}$.

5. Evaluate each limit, rationalising or using a standard limit where appropriate.

- (a) $\lim_{n \rightarrow \infty} (\sqrt{n^2 + 3n} - n)$;

- (b) $\lim_{n \rightarrow \infty} (1 + 1/n)^n$;
(c) $\lim_{n \rightarrow \infty} 3^n/n^2$.
6. Discuss the effect of oscillation in each sequence.
- (a) $\sin n/n$;
(b) $\cos(n\pi)$;
(c) $(-1)^n/n$.
7. An arithmetic sequence satisfies $a_1 = 5$ and $a_{n+1} = a_n + 3$. Find a_n , a_{20} , and the sum of its first 20 terms.
8. A geometric sequence has first term 81 and common ratio $1/3$. Find a_n , a_6 , and the sum of its first six terms.
9. Determine the unknown parameters.
- (a) An arithmetic sequence has $a_3 = 11$ and $a_8 = 31$. Find a_1 , the common difference, and a_n .
(b) A geometric sequence has positive common ratio, $g_2 = 6$, and $g_5 = 162$. Find its first term and common ratio.
10. For $a_n = (-1)^n + 1/n$, find the limits of the even and odd subsequences. Hence decide whether (a_n) converges.
11. Evaluate the following factorial-ratio limits.
- (a) $\lim_{n \rightarrow \infty} \frac{n!}{(n+1)!}$;
(b) $\lim_{n \rightarrow \infty} \frac{(n+2)!}{n!}$;
(c) $\lim_{n \rightarrow \infty} \frac{n^2}{n!}$.
12. State whether each sequence is increasing, decreasing, or neither; state whether it is bounded; and find its limit if one exists.
- (a) $n/(n+1)$;
(b) $2 + (-1)^n/n$;
(c) $1/n$.
13. A sequence is defined by $a_1 = 0$ and $a_{n+1} = \frac{1}{2}a_n + 3$. Find an explicit formula for a_n , calculate a_5 , and determine $\lim_{n \rightarrow \infty} a_n$.

14. Let $a_n = n^2$ and $b_n = a_{n+1} - a_n$. Find b_n and hence evaluate $\sum_{k=1}^n b_k$.
15. Find the smallest positive integer n satisfying each condition.

(a) $|(4 + 1/n) - 4| < 0.01$;

(b) $\left| \frac{2n+1}{n+4} - 2 \right| < 0.01$.

Sigma notation and finite sums

16. Express each sum using sigma notation.

(a) $2 + 5 + 8 + \cdots + 29$;

(b) $1^2 + 2^2 + \cdots + n^2$;

(c) $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{8 \cdot 9}$.

17. Expand and evaluate.

(a) $\sum_{k=1}^4 (2k + 3)$;

(b) $\sum_{k=0}^5 3 \cdot 2^k$;

(c) $\sum_{k=1}^5 (-1)^k k$.

18. Using the standard formulae for $\sum k$ and $\sum k^2$, show that

$$\sum_{k=1}^n (3k^2 - 2k + 1) = \frac{n(2n^2 + n + 1)}{2}.$$

19. Evaluate

$$\sum_{k=1}^{50} k, \quad \sum_{k=1}^{50} k^2, \quad \sum_{k=1}^{50} k^3.$$

20. Derive closed forms for

(a) $\sum_{k=1}^n k(k+1)$;

(b) $\sum_{k=1}^n (2k-1)^2$.

21. Re-index $\sum_{r=3}^7 (r-2)^2$ so that the new index begins at 1, and then evaluate the sum.

22. Simplify

$$\sum_{k=1}^{2n} k - 2 \sum_{k=1}^n k.$$

23. Use partial fractions to evaluate

$$\sum_{k=1}^n \left(\frac{1}{k} - \frac{1}{k+1} \right).$$

24. Show that

$$\sum_{k=1}^n \frac{1}{(2k-1)(2k+1)} = \frac{n}{2n+1}.$$

25. Evaluate

$$\sum_{r=n+1}^{2n} (2r-1).$$

26. Prove using sigma notation that the sum of the first n positive even integers is $n(n+1)$ and the sum of the first n positive odd integers is n^2 .

27. Simplify

$$\sum_{r=0}^n (n+1+2r).$$

28. Evaluate the finite geometric sums.

(a) $\sum_{k=0}^8 3 \cdot 2^k;$

(b) $\sum_{k=1}^n 5(1/2)^{k-1}.$

29. Find a formula for $\sum_{k=0}^n (-1/2)^k$, and use it to evaluate the sum when $n = 5$.

30. Find the number of terms in each series.

(a) $2 + 5 + 8 + \cdots$ has sum 155;

(b) $3 + 6 + 12 + \cdots$ has sum 3069.

Geometric and telescoping series

31. State whether each series converges. Find the sum when it does.

(a) $12 + 6 + 3 + \cdots;$

(b) $5 - 10 + 20 - \cdots;$

(c) $3 + 1 + \frac{1}{3} + \cdots$.

32. Find the interval of convergence and sum of each geometric power series.

(a) $\sum_{k=0}^{\infty} (x/3)^k$;

(b) $\sum_{k=0}^{\infty} (2x)^k$.

33. Convert each recurring decimal to a rational number.

(a) $0.\overline{27}$;

(b) $0.14555\dots$

34. Find S_n and S_{∞} for each geometric series.

(a) $a = 4$, $r = 1/4$;

(b) $a = 6$, $r = -1/2$.

35. Apply the n th-term test to each series. State clearly when the test is inconclusive.

(a) $\sum_{n=1}^{\infty} \frac{n+1}{n}$;

(b) $\sum_{n=1}^{\infty} (-1)^n$;

(c) $\sum_{n=1}^{\infty} \frac{1}{n}$.

36. Evaluate $\sum_{k=1}^{\infty} 1/[k(k+1)]$ by telescoping.

37. Find the finite and infinite sums of

$$\sum_{k=1}^n \frac{1}{(3k-2)(3k+1)}.$$

38. Determine the unknown geometric data.

(a) $1 + q + q^2 + \cdots$ has sum 5.

(b) An infinite geometric series has second term 3 and sum 12. Find its first term and common ratio.

39. Evaluate

$$\sum_{k=0}^{\infty} [3(1/2)^k + 2(-1/3)^k].$$

40. An infinite geometric series has sum 12, and the sum of its first two terms is 9. Find all possible real first terms and common ratios.
41. Insert three positive geometric means between 2 and 162.
42. Solve $1 + 2 + 4 + \cdots + 2^{n-1} = 1023$ for n .

43. If $\sum_{n=1}^{\infty} a_n = S$, express

$$\sum_{n=1}^{\infty} \left(2a_n + \frac{3}{4^n} \right)$$

in terms of S .

44. Find a closed form for $\sum_{k=1}^n 1/[k(k+2)]$ and hence evaluate the corresponding infinite series.
45. A convergent geometric series has sum 10, while its tail from the third term onwards has sum 2.5. Find all possible real first terms and common ratios.

Factorials and the binomial theorem

46. Simplify.

- (a) $(n+2)!/n!$;
(b) $1/n! - 1/(n+1)!$;
(c) $(2n)!/(2n-2)!$.

47. Solve for the positive integer n .

- (a) $(n+1)! = 30(n-1)!$;
(b) $(n+1)! + n! = 72(n-1)!$.

48. Construct row $n = 6$ of Pascal's triangle and explain its symmetry.

49. Expand fully.

- (a) $(x+2)^5$;
(b) $(p-2q)^5$.

50. Expand $(2x-3)^4$ in descending powers of x .

51. Write the general term in $(a+b)^n$. Hence find the fourth term in the expansion of $(x+2)^7$.

52. Find the indicated coefficient.

- (a) the coefficient of x^5 in $(1 + 2x)^8$;
 - (b) the coefficient of x^4 in $(3 - x)^7$.
53. Find the constant term in each expansion.
- (a) $(x^2 + 1/x)^6$;
 - (b) $(x - 2/x^2)^6$.
54. Find the middle term or terms of
- (a) $(a + b)^{10}$;
 - (b) $(x + 1)^9$.
55. Determine the unknown parameter.
- (a) The coefficient of x^2 in $(1 + kx)^5$ is 90.
 - (b) In $(2 + x)^n$, the coefficient of x is three times the constant term.
56. Use the binomial theorem to prove
- (a) $\sum_{r=0}^n \binom{n}{r} = 2^n$;
 - (b) $\sum_{r=0}^n (-1)^r \binom{n}{r} = 0$ for $n \geq 1$;
 - (c) $\sum_{r=0}^n r \binom{n}{r} = n2^{n-1}$.
57. Use a binomial expansion to approximate
- (a) $(1.02)^5$;
 - (b) $(0.99)^6$.
58. Find
- (a) the coefficient of x^{12} in $(x^2 - 1/x)^9$;
 - (b) the constant term in $(x - 1/x)^8$.
59. Find the sum of the coefficients and the alternating sum of the coefficients of $(2x - 1)^7$.
60. In six independent trials, each with success probability p , use the binomial theorem to write the probability of exactly two successes.

Generalised binomial and power series

61. Write the first four terms of
- (a) $(1 + x)^{-1}$;
 - (b) $(1 + x)^{1/2}$.

62. Expand $(1-2x)^{-1/2}$ up to and including the term in x^3 . State the interval of convergence.
63. Expand $\sqrt{4+x}$ up to and including the term in x^3 , and state the interval of convergence.
64. Expand $(8-x)^{1/3}$ up to and including the term in x^3 , and state the interval of convergence.
65. Find the first four terms and interval of convergence of
- (a) $1/(2-x)$;
 - (b) $1/(3+2x)$.

66. Express

$$\frac{x+7}{x^2-x-6}$$

in partial fractions. Hence obtain its Maclaurin series up to the term in x^3 and state the interval of convergence.

67. Find the Maclaurin series of $e^x/(1-x)$ up to the term in x^3 .
68. Write the first three non-zero terms of
- (a) $\sin(x^2)$;
 - (b) e^{-2x} ;
 - (c) $\ln(1+x^2)$.

69. Differentiate the geometric series to show that

$$\frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} nx^{n-1}, \quad |x| < 1.$$

Hence evaluate $\sum_{n=1}^{\infty} n/2^n$.

70. Integrate the geometric series to obtain a series for $-\ln(1-x)$. Hence evaluate $\sum_{n=1}^{\infty} 1/(n2^n)$.
71. Find the interval of convergence of
- (a) $\sum_{n=0}^{\infty} ((x-2)/3)^n$;
 - (b) $\sum_{n=0}^{\infty} ((x+1)/2)^n$.

72. Use a generalised binomial expansion to estimate, to five decimal places,

- (a) $\sqrt{0.98}$;
 - (b) $(1.03)^{2/3}$.
73. Use the first four terms of a suitable series to approximate
- (a) $1/1.05$;
 - (b) $\sqrt{4.1}$.
74. Find
- (a) the coefficient of x^4 in $(1 - 2x)^{-3}$;
 - (b) the coefficient of x^3 in $(1 + x)^{1/2}$.
75. Estimate $\ln(1.2)$ using terms through x^4 in the series for $\ln(1 + x)$, and give a bound for the magnitude of the next term.

Taylor and Maclaurin series

76. State the degree-3 Taylor polynomial of a function about $x = a$. Hence write the degree-3 Taylor polynomial of e^x about $x = 1$.
77. Find the Taylor expansion of $\cos x$ about $x = \pi/3$ up to the term in $(x - \pi/3)^3$.
78. Write the Taylor expansion of $\ln x$ about $x = 1$ through the fourth power of $x - 1$. Hence estimate $\ln(1.1)$.
79. Write the first four non-zero terms of the Maclaurin series for e^x , $\sin x$, $\cos x$, and $\ln(1 + x)$.
80. Find the first four non-zero terms of
- (a) xe^{-3x^2} ;
 - (b) $x \sin(3x)$.
81. Find the coefficient of x^3 in $(1 - x + x^2)e^{-2x}$. Also verify from their Maclaurin series that $\cos(2x) = 1 - 2\sin^2 x$ through terms in x^6 .
82. Obtain the first three non-constant terms of the following antiderivatives by termwise integration.
- (a) $\int \frac{\cos x - 1}{x^2} dx$;
 - (b) $\int \frac{\sin x}{x} dx$.
83. Use power series to estimate

- (a) $\int_0^{0.1} e^{-x^2} dx;$
 (b) $\int_0^{0.5} \cos(x^2) dx;$
 (c) $\int_0^{1/3} \frac{dx}{1+x^6}.$

84. Use the Lagrange remainder to bound the error when $\sin x$ on $|x| \leq 0.5$ is approximated by $x - x^3/6$.
 85. Determine whether the degree-4 Maclaurin polynomial for e^x approximates $e^{0.1}$ with absolute error less than 10^{-6} .

Applications and modelling

86. A graduate's monthly salary is RM 2,400 in the first year and rises by RM 150 each year. Find the monthly salary in the tenth year and the sum of the ten annual monthly-salary rates.
 87. A stadium has 20 seats in its first row and four more seats in each successive row. There are 15 rows. Find the number of seats in the last row and the total seating capacity.

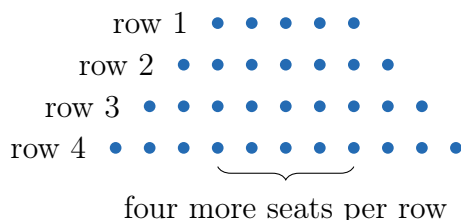


Figure 15.6: The stadium rows form an arithmetic sequence; the dots are schematic rather than the actual seat counts.

88. A ball is dropped from a height of 10 m and rebounds to 70% of its preceding height. Find its height after the fifth bounce and the total vertical distance travelled before it comes to rest.
 89. A patient receives 100 mg of a medicine at the end of each hour. Immediately before the next dose, 80% of the medicine remains. If A_n is the amount immediately after the n th dose, find A_n and its limiting value.
 90. RM 500 is deposited at the end of every month into an account earning 1% per month. Find the balance immediately after the twelfth deposit, assuming interest is credited before each new deposit.

91. A population satisfies $P_0 = 1000$ and $P_{n+1} = 1.05P_n + 200$. Find an explicit formula for P_n and estimate P_{10} .
92. A square sheet has one quarter of its current area removed at each stage, leaving three congruent scaled copies of the previous pattern. Find the fraction of the original area remaining after six stages and the total fraction eventually removed.

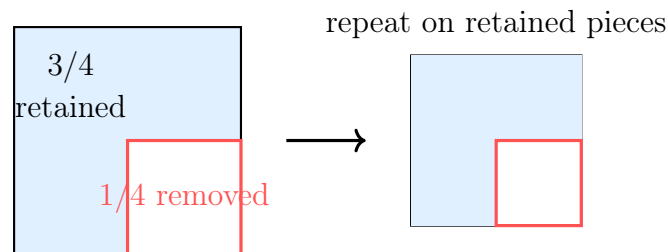


Figure 15.7: Each stage retains a fraction $3/4$ of the area present at the preceding stage.

93. A cooling object follows $T(t) = 20 + 80e^{-0.1t}$ degrees Celsius. Use the first four terms of the Maclaurin series for $e^{-0.1}$ to estimate $T(1)$.
94. A cube is heated through 50°C . Its linear expansion factor is $1 + \alpha\Delta T$, where $\alpha = 12 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$. Use a binomial expansion to estimate the percentage increase in volume, neglecting terms containing α^2 and higher powers.
95. RM 10,000 is invested for five years at 4% compound interest. Estimate its value using terms through the quadratic term of $(1.04)^5$, and compare it with the calculator value.
96. A genetic trait appears independently with probability 0.2 in each of five offspring. Find the probability that
- exactly three offspring have the trait;
 - at least one offspring has the trait.
97. Components are defective independently with probability 0.01. For a batch of ten, find the probabilities of
- no defective components;
 - exactly one defective component.
98. A small-angle sensor uses $\sin x$ with x in radians. Approximate $\sin(0.2)$ using terms through x^5 and bound the error using the next non-zero term.
99. At height h above a spherical body of radius R , gravitational strength relative to its surface value is $(1 + h/R)^{-2}$. For $R = 6400 \text{ km}$ and $h = 400 \text{ km}$, use terms through $(h/R)^2$ to estimate the ratio and compare it with the exact value.

100. A new production process makes 1000 units in its first week, and weekly output is 90% of the preceding week's output. Find the total output during the first 12 weeks, the limiting total output, and the first week in which output is below 300 units.

Part IV

Vectors, Discrete Mathematics, and Computing

Chapter 16

Vectors

Chapter guide

Prerequisites. Coordinate geometry, trigonometry, algebra, simultaneous equations, and elementary differentiation and integration.

Learning outcomes. Readers should be able to perform vector operations in two and three dimensions, use dot and cross products, describe lines and planes, calculate distances and relative motion, and analyse vector-valued functions.

Describing a displacement requires two pieces of information: how far to travel and in which direction. A vector records both magnitude and direction, whereas an ordinary scalar records magnitude alone.

For example, “walk 5 metres north” describes a vector, whereas “walk 5 metres” specifies only a scalar distance. Scalars are sufficient for quantities such as mass and temperature, but directional quantities require vectors.

This chapter introduces Euclidean vectors, their different forms and operations, geometric applications, and some elementary ideas from vector calculus. The geometric and calculus-based treatment draws on standard multivariable calculus and Euclidean-geometry references (Auroux 2010; Marsden and Tromba 2026).

16.1 Euclidean vectors

The word *vector* has several mathematical meanings. This chapter begins with Euclidean vectors, the directed quantities used in plane and spatial geometry.

Definition 16.1: Euclidean vector.

A Euclidean vector, or simply a vector, is a geometric object possessing both magnitude and direction. It is commonly represented by a directed line segment from an initial point A to a terminal point B , denoted by $\overrightarrow{AB}$.

16.1.1 Physical quantities

It is useful first to distinguish scalar physical quantities from vector quantities.

Definition 16.2: Physical quantity.

A physical quantity is a measurable property of a material or system. It is expressed by a numerical magnitude together with an appropriate unit of measurement.

For the purposes of this chapter, physical quantities are classified as follows.

1. Scalar quantities

- They are specified by magnitude alone.
- Examples include mass, temperature, and time.

2. Vector quantities

- They are specified by both magnitude and direction.
- Examples include displacement, velocity, and force.

16.1.2 Vector notation

Geometrically, a vector is represented by a directed line segment, as shown below.

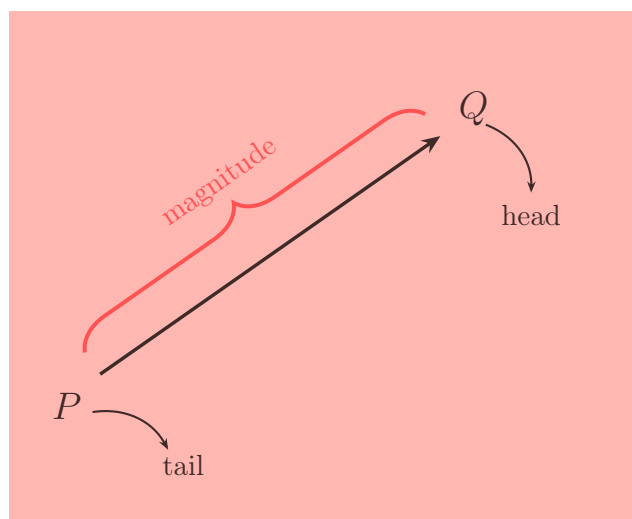


Figure 16.1: A directed line segment representing a vector.

Common vector notations include the following:

- a directed segment, such as $\overrightarrow{PQ}$;
- a bold lowercase letter, such as $\mathbf{a}$;
- a lowercase letter with an arrow, such as $\vec{a}$; and
- an underlined lowercase letter, such as $\underline{a}$, in texts that adopt this convention.

16.2 Vectors in two-dimensional space

Two-dimensional vectors establish the coordinate methods that will later be extended to three dimensions.

Consider the points $A(1, 2)$ and $B(3, 5)$. To move from A to B , the horizontal coordinate increases by

$$3 - 1 = 2,$$

while the vertical coordinate increases by

$$5 - 2 = 3.$$

Therefore, the vector from A to B is found by subtracting the coordinates of the initial point from those of the terminal point:

$$\begin{aligned}\overrightarrow{AB} &= (3 - 1)\mathbf{i} + (5 - 2)\mathbf{j} \\ &= 2\mathbf{i} + 3\mathbf{j}.\end{aligned}\tag{16.1}$$

Equivalently, the same vector may be written in column or ordered-pair form as

$$\overrightarrow{AB} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} = (2, 3).\tag{16.2}$$

In general, if $A(x_1, y_1)$ and $B(x_2, y_2)$, then

$$\boxed{\overrightarrow{AB} = (x_2 - x_1)\mathbf{i} + (y_2 - y_1)\mathbf{j} = \begin{bmatrix} x_2 - x_1 \\ y_2 - y_1 \end{bmatrix}}.\tag{16.3}$$

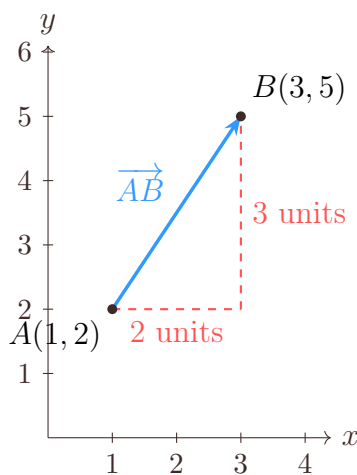


Figure 16.2: The vector from $A(1, 2)$ to $B(3, 5)$.

16.2.1 Position vectors

Let O denote the origin and let $P(a, b)$ be a point in the Cartesian plane. The **position vector** of P is the displacement vector from the origin to P . It is denoted by

$$\overrightarrow{OP} = \mathbf{p} = a\mathbf{i} + b\mathbf{j} = \begin{bmatrix} a \\ b \end{bmatrix} = (a, b).\tag{16.4}$$

Thus, the components of a position vector are precisely the coordinates of its endpoint.

By the Pythagorean theorem, the magnitude of $\overrightarrow{OP}$ is

$$\left| \overrightarrow{OP} \right| = |\mathbf{p}| = \sqrt{a^2 + b^2}. \quad (16.5)$$

If $\overrightarrow{OP}$ makes an angle α with the positive x -axis, then

$$\cos \alpha = \frac{a}{\sqrt{a^2 + b^2}}, \quad \sin \alpha = \frac{b}{\sqrt{a^2 + b^2}}. \quad (16.6)$$

When $a \neq 0$,

$$\tan \alpha = \frac{b}{a}. \quad (16.7)$$

The quadrant containing P must be considered when determining α from the tangent ratio.

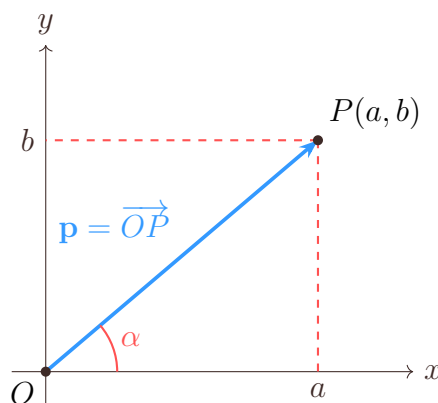


Figure 16.3: The position vector of the point $P(a, b)$.

16.2.2 Free vectors

A **free vector** is a vector determined only by its magnitude and direction. Unlike a position vector, it is not attached to a fixed initial point and may be translated parallel to itself without changing the vector.

For example, suppose

$$\overrightarrow{P_1P_2} = \overrightarrow{Q_1Q_2} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

Although the directed line segments begin at different points, both represent the same free vector because they have identical components, magnitude, and direction. In general, if

$$\overrightarrow{AB} = \overrightarrow{CD},$$

then the two directed line segments are different representations of one free vector.

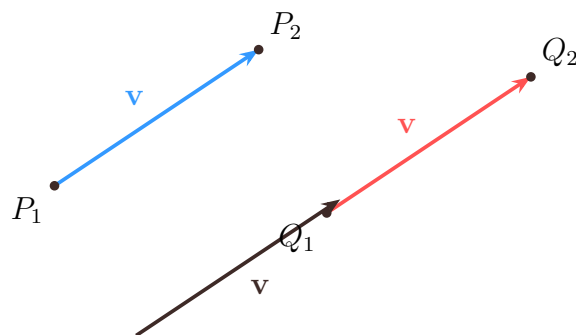


Figure 16.4: A free vector may be represented at different positions without changing its magnitude or direction.

16.2.3 Parallel vectors

Two nonzero vectors are said to be *parallel* if one is a nonzero scalar multiple of the other. Thus, for nonzero vectors $\mathbf{a}$ and $\mathbf{b}$,

$$\mathbf{a} \parallel \mathbf{b} \iff \mathbf{a} = k\mathbf{b} \text{ for some } k \in \mathbb{R}, k \neq 0. \quad (16.8)$$

If $k > 0$, the vectors point in the same direction. If $k < 0$, they point in opposite directions, but they are still parallel.

Example 16.1: Identifying parallel vectors

Let $\mathbf{x}$ be a nonzero vector and suppose that

$$\overrightarrow{AB} = 2\mathbf{x}, \quad \overrightarrow{PQ} = 4\mathbf{x}, \quad \overrightarrow{MN} = -3\mathbf{x}.$$

Since

$$\overrightarrow{PQ} = 2\overrightarrow{AB},$$

$\overrightarrow{AB}$ and $\overrightarrow{PQ}$ are parallel and point in the same direction. Moreover,

$$\overrightarrow{MN} = -\frac{3}{4}\overrightarrow{PQ},$$

so $\overrightarrow{PQ}$ and $\overrightarrow{MN}$ are also parallel, although they point in opposite directions.

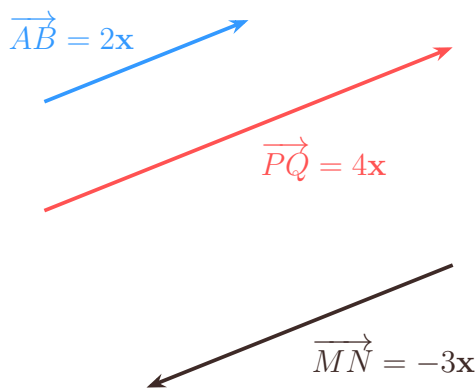


Figure 16.5: Parallel vectors may point in the same or in opposite directions.

Parallel vectors can also be used to establish collinearity. For example, if

$$\overrightarrow{AB} = k\overrightarrow{BC}, \quad k \neq 0,$$

then $\overrightarrow{AB}$ and $\overrightarrow{BC}$ are parallel. Because the corresponding lines share the point B , they must be the same straight line. Hence, the points A , B , and C are collinear.

16.2.4 Equal vectors

Two vectors are equal if and only if they have the same magnitude and the same direction. Thus,

$$\boxed{\mathbf{a} = \mathbf{b} \iff \|\mathbf{a}\| = \|\mathbf{b}\| \text{ and } \mathbf{a}, \mathbf{b} \text{ point in the same direction.}} \quad (16.9)$$

The initial points of the vectors do not need to coincide. Since vectors are free vectors, translating a vector without rotating or resizing it produces an equal vector.

In component form, the vectors

$$\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} \quad \text{and} \quad \mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j}$$

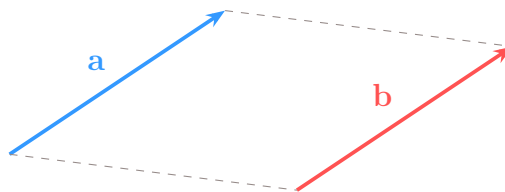
are equal precisely when their corresponding components are equal:

$$\boxed{\mathbf{a} = \mathbf{b} \iff a_1 = b_1 \text{ and } a_2 = b_2.} \quad (16.10)$$

For example, if

$$\overrightarrow{AB} = 3\mathbf{i} + 2\mathbf{j} \quad \text{and} \quad \overrightarrow{PQ} = 3\mathbf{i} + 2\mathbf{j},$$

then $\overrightarrow{AB} = \overrightarrow{PQ}$, even when $A \neq P$ and $B \neq Q$.



$$\|\mathbf{a}\| = \|\mathbf{b}\| \text{ and both have the same direction}$$

Figure 16.6: Equal vectors may have different initial points.

16.2.5 Unit vectors

A *unit vector* is a vector whose magnitude is one. Unit vectors are useful when only a direction is required, because they describe direction without introducing any additional scaling.

Let $\mathbf{r}$ be a nonzero vector. Dividing $\mathbf{r}$ by its magnitude produces a unit vector in the direction of $\mathbf{r}$:

$$\hat{\mathbf{r}} = \frac{\mathbf{r}}{\|\mathbf{r}\|}, \quad \mathbf{r} \neq \mathbf{0}. \quad (16.11)$$

Indeed,

$$\|\hat{\mathbf{r}}\| = \left\| \frac{\mathbf{r}}{\|\mathbf{r}\|} \right\| = \frac{\|\mathbf{r}\|}{\|\mathbf{r}\|} = 1.$$

The standard unit vectors in two-dimensional Cartesian space are

$$\mathbf{i} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \mathbf{j} = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

which point along the positive x - and y -axes, respectively.

Example 16.2: Finding a unit vector

Let

$$\mathbf{r} = 3\mathbf{i} + 4\mathbf{j}.$$

Its magnitude is

$$\|\mathbf{r}\| = \sqrt{3^2 + 4^2} = 5.$$

Therefore, the unit vector in the direction of $\mathbf{r}$ is

$$\hat{\mathbf{r}} = \frac{3\mathbf{i} + 4\mathbf{j}}{5} = \frac{3}{5}\mathbf{i} + \frac{4}{5}\mathbf{j}.$$

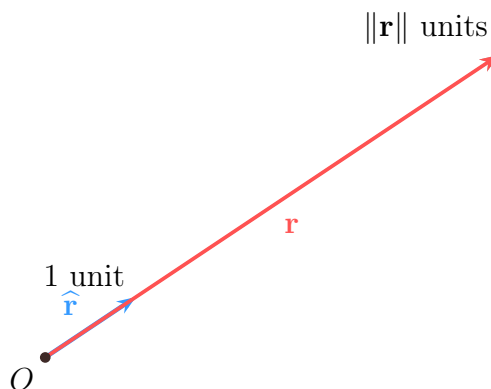


Figure 16.7: The unit vector $\hat{\mathbf{r}}$ has the same direction as $\mathbf{r}$ but magnitude one.

16.2.6 Negative vectors

For any vector $\mathbf{a}$, its *negative vector*, denoted by $-\mathbf{a}$, has the same magnitude as $\mathbf{a}$ but points in the opposite direction. Therefore,

$$\boxed{\|-\mathbf{a}\| = \|\mathbf{a}\|, \quad -\mathbf{a} \text{ points opposite to } \mathbf{a}.} \quad (16.12)$$

If

$$\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j},$$

then every component changes sign:

$$\boxed{-\mathbf{a} = -a_1\mathbf{i} - a_2\mathbf{j}.} \quad (16.13)$$

For instance, if $\mathbf{a}$ represents a displacement of 5 units eastward, then $-\mathbf{a}$ represents a displacement of 5 units westward. The two vectors are parallel, but they have opposite directions.

Negative vectors satisfy

$$-(-\mathbf{a}) = \mathbf{a} \quad \text{and} \quad \mathbf{a} + (-\mathbf{a}) = \mathbf{0}.$$

Example 16.3

If

$$\mathbf{a} = 4\mathbf{i} - 3\mathbf{j},$$

then

$$-\mathbf{a} = -4\mathbf{i} + 3\mathbf{j}.$$

Both vectors have magnitude 5, but their directions differ by 180° .

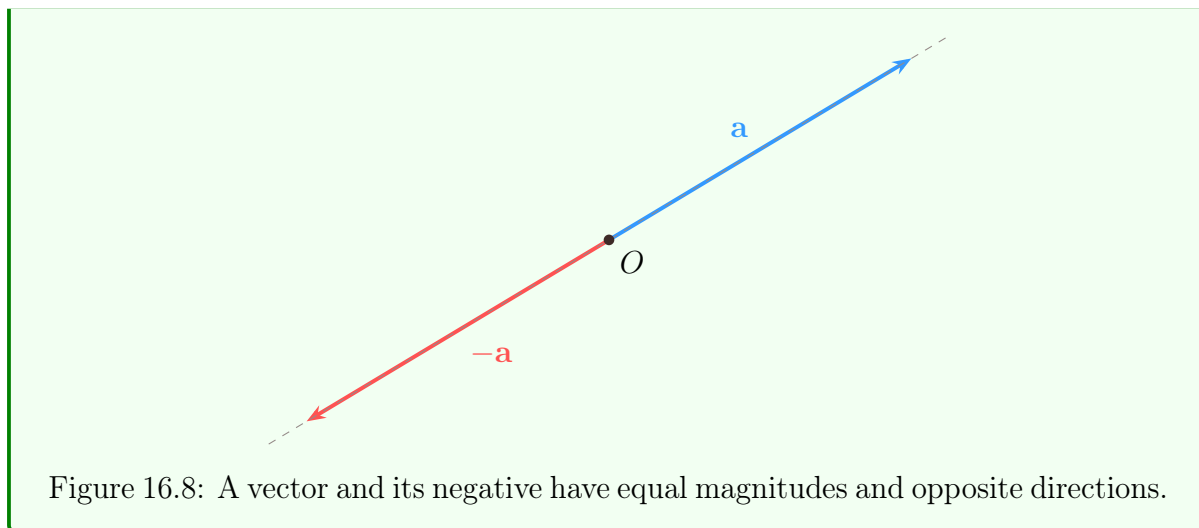


Figure 16.8: A vector and its negative have equal magnitudes and opposite directions.

16.2.7 Zero vectors

The *zero vector*, also called the *null vector*, is the vector whose components are all zero. It is denoted by $\mathbf{0}$ and, in two dimensions, is written as

$$\mathbf{0} = 0\mathbf{i} + 0\mathbf{j} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \quad (16.14)$$

Its magnitude is zero:

$$\|\mathbf{0}\| = 0.$$

Unlike a nonzero vector, the zero vector has no defined direction. Geometrically, its initial and terminal points coincide. Thus,

$$\overrightarrow{AB} = \mathbf{0} \iff A = B.$$

The zero vector is the additive identity of a vector space. Adding it to any vector leaves that vector unchanged:

$$\mathbf{a} + \mathbf{0} = \mathbf{0} + \mathbf{a} = \mathbf{a}. \quad (16.15)$$

For example,

$$\overrightarrow{KS} + \mathbf{0} = \overrightarrow{KS}.$$

A vector added to its negative gives the zero vector:

$$\mathbf{a} + (-\mathbf{a}) = \mathbf{0}. \quad (16.16)$$

Scalar multiplication also satisfies

$$0\mathbf{a} = \mathbf{0} \quad \text{and} \quad k\mathbf{0} = \mathbf{0}$$

for every vector $\mathbf{a}$ and every scalar k .

16.3 Arithmetic properties of vectors

Vector arithmetic combines vectors or changes their magnitudes while preserving information about direction. The basic operations are addition, subtraction, and multiplication by a scalar.

16.3.1 Vector addition

Let

$$\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} \quad \text{and} \quad \mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j}.$$

Their sum is obtained by adding corresponding components:

$$\boxed{\mathbf{a} + \mathbf{b} = (a_1 + b_1)\mathbf{i} + (a_2 + b_2)\mathbf{j}.} \quad (16.17)$$

The vector $\mathbf{a} + \mathbf{b}$ is called the *resultant vector*. It can be constructed geometrically using either the head-to-tail rule or the parallelogram law.

The head-to-tail rule

Translate $\mathbf{b}$ without changing its magnitude or direction until its tail is placed at the head of $\mathbf{a}$. The vector drawn from the tail of $\mathbf{a}$ to the head of $\mathbf{b}$ is the resultant $\mathbf{a} + \mathbf{b}$.

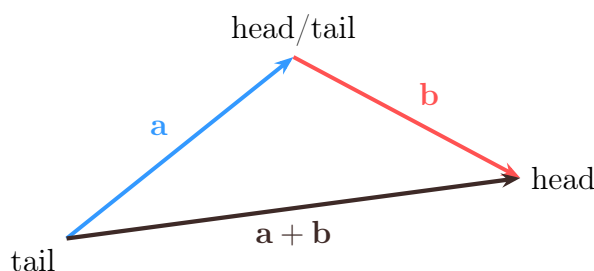


Figure 16.9: Vector addition by the head-to-tail rule.

Reversing the order produces the same resultant: $\mathbf{b} + \mathbf{a} = \mathbf{a} + \mathbf{b}$.

The parallelogram law

Place the tails of $\mathbf{a}$ and $\mathbf{b}$ at the same point and use the vectors as adjacent sides of a parallelogram. The diagonal drawn from their common tail is the resultant.

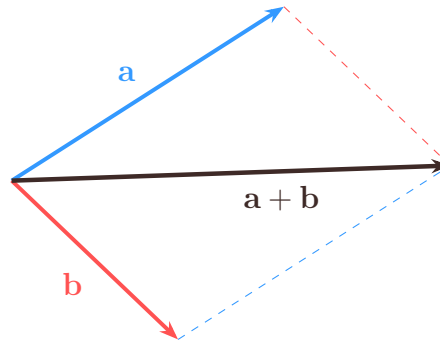


Figure 16.10: Vector addition by the parallelogram law.

Vector addition is both commutative and associative:

$$\begin{aligned}\mathbf{a} + \mathbf{b} &= \mathbf{b} + \mathbf{a}, \\ (\mathbf{a} + \mathbf{b}) + \mathbf{c} &= \mathbf{a} + (\mathbf{b} + \mathbf{c}).\end{aligned}$$

16.3.2 Vector subtraction

Subtracting $\mathbf{b}$ from $\mathbf{a}$ means adding the negative of $\mathbf{b}$:

$$\boxed{\mathbf{a} - \mathbf{b} = \mathbf{a} + (-\mathbf{b}).} \quad (16.18)$$

In component form,

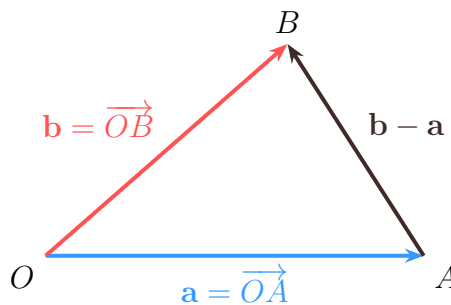
$$\boxed{\mathbf{a} - \mathbf{b} = (a_1 - b_1)\mathbf{i} + (a_2 - b_2)\mathbf{j}.} \quad (16.19)$$

If $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$, the triangle rule gives

$$\overrightarrow{OA} + \overrightarrow{AB} = \overrightarrow{OB}.$$

Therefore,

$$\boxed{\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \mathbf{b} - \mathbf{a}.} \quad (16.20)$$

Figure 16.11: The displacement from A to B is $\mathbf{b} - \mathbf{a}$.

16.3.3 Multiplication by a scalar

Let k be a scalar and $\mathbf{a}$ a vector. Multiplying $\mathbf{a}$ by k changes its magnitude by the factor $|k|$:

$$\|k\mathbf{a}\| = |k| \|\mathbf{a}\|. \quad (16.21)$$

If $k > 0$, then $k\mathbf{a}$ points in the same direction as $\mathbf{a}$. If $k < 0$, it points in the opposite direction. If $k = 0$, the result is the zero vector.

In component form,

$$k\mathbf{a} = (ka_1)\mathbf{i} + (ka_2)\mathbf{j}. \quad (16.22)$$

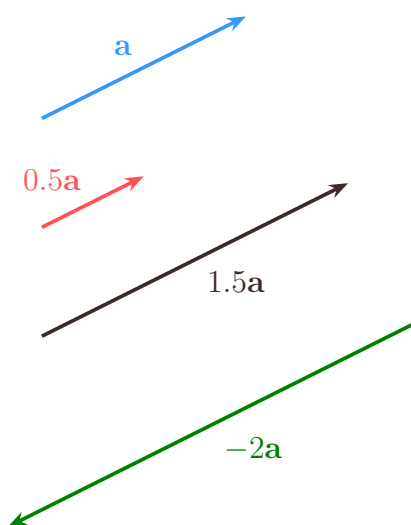


Figure 16.12: The effect of positive and negative scalar multiplication.

Scalar multiplication is distributive and associative:

$$\begin{aligned} k(\mathbf{a} + \mathbf{b}) &= k\mathbf{a} + k\mathbf{b}, \\ (k + m)\mathbf{a} &= k\mathbf{a} + m\mathbf{a}, \\ k(m\mathbf{a}) &= (km)\mathbf{a}. \end{aligned}$$

16.4 Resultant vectors

A *resultant vector* is the single vector obtained by adding two or more vectors. If the vectors are $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$, then

$$\mathbf{R} = \mathbf{v}_1 + \mathbf{v}_2 + \dots + \mathbf{v}_n = \sum_{r=1}^n \mathbf{v}_r. \quad (16.23)$$

Resultants are used to combine vector quantities such as displacement, velocity, acceleration, force, momentum, and electric field. Although a scale drawing can provide a

useful illustration, resolving every vector into components is generally the most accurate method.

16.4.1 Position-vector applications

If the position vectors of points A and B are

$$\overrightarrow{OA} = \mathbf{a} \quad \text{and} \quad \overrightarrow{OB} = \mathbf{b},$$

then the vector from A to B is

$$\boxed{\overrightarrow{AB} = \mathbf{b} - \mathbf{a}.} \quad (16.24)$$

Collinearity

Three distinct points P , Q , and R are collinear if the two vectors sharing the common point Q are nonzero scalar multiples:

$$\boxed{\overrightarrow{PQ} = k\overrightarrow{QR}, \quad k \neq 0.} \quad (16.25)$$

For example, suppose the position vectors of A , B , and C are $6\mathbf{a}$, $3\mathbf{b}$, and $4\mathbf{a} + \mathbf{b}$, respectively. Then

$$\begin{aligned} \overrightarrow{AC} &= (4\mathbf{a} + \mathbf{b}) - 6\mathbf{a} = -2\mathbf{a} + \mathbf{b}, \\ \overrightarrow{CB} &= 3\mathbf{b} - (4\mathbf{a} + \mathbf{b}) = -4\mathbf{a} + 2\mathbf{b} = 2\overrightarrow{AC}. \end{aligned}$$

Equivalently,

$$\overrightarrow{AC} = \frac{1}{2}\overrightarrow{CB}.$$

Hence, A , C , and B are collinear, and $AC : CB = 1 : 2$. The following diagram illustrates the position vectors and the common straight line. The origin O does not need to lie on the line containing A , C , and B .

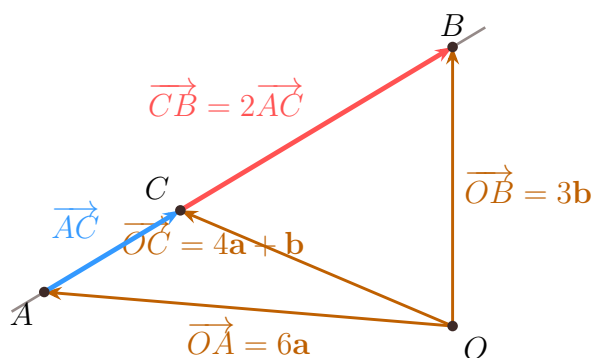


Figure 16.13: Since $\overrightarrow{CB} = 2\overrightarrow{AC}$, the points A , C , and B are collinear.

Internal division

Let C divide the line segment AB internally in the ratio

$$AC : CB = m : n,$$

where the position vectors of A and B are $\mathbf{a}$ and $\mathbf{b}$. Since

$$n\overrightarrow{AC} = m\overrightarrow{CB},$$

the position vector of C is

$$\boxed{\overrightarrow{OC} = \frac{n\mathbf{a} + m\mathbf{b}}{m + n}.} \quad (16.26)$$

For example, if $M = (3, 5)$ and $N = (1, -4)$ and K divides MN internally with $MK : KN = 2 : 5$, then

$$\overrightarrow{OK} = \frac{5(3, 5) + 2(1, -4)}{7} = \left(\frac{17}{7}, \frac{17}{7}\right).$$

The midpoint is the special case $m = n = 1$:

$$\boxed{\overrightarrow{OC} = \frac{\mathbf{a} + \mathbf{b}}{2}.} \quad (16.27)$$

External division

If C divides the line through A and B externally in the ratio $AC : CB = m : n$, then

$$\boxed{\overrightarrow{OC} = \frac{m\mathbf{b} - n\mathbf{a}}{m - n}, \quad m \neq n.} \quad (16.28)$$

For example, if $A = (2, 3)$ and $B = (7, 2)$, the point dividing AB externally in the ratio $1 : 3$ has position vector

$$\overrightarrow{OC} = \frac{(7, 2) - 3(2, 3)}{1 - 3} = \left(-\frac{1}{2}, \frac{7}{2}\right).$$

16.4.2 Describing vector directions

The direction of a two-dimensional vector may be stated using:

1. an angle measured anticlockwise from the positive x -axis;
2. a compass direction, such as north-east or S 30° E;
3. a three-figure bearing measured clockwise from north.

If a vector $\mathbf{v}$ has magnitude V and makes an angle θ anticlockwise from the positive x -axis, then

$$\mathbf{v} = V \cos \theta \mathbf{i} + V \sin \theta \mathbf{j}. \quad (16.29)$$

If its direction is given by a bearing β , then

$$\mathbf{v} = V \sin \beta \mathbf{i} + V \cos \beta \mathbf{j}. \quad (16.30)$$

The signs of the components depend on the direction. For example, a velocity of magnitude 5 m s^{-1} directed S 75° E is

$$\mathbf{v} = 5 \sin 75^\circ \mathbf{i} - 5 \cos 75^\circ \mathbf{j}.$$

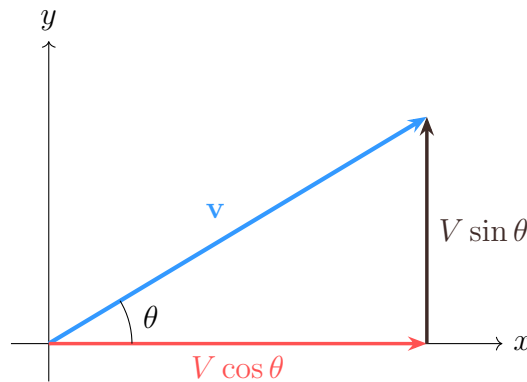


Figure 16.14: Resolving a vector into horizontal and vertical components.

16.4.3 Calculating a resultant by components

Suppose

$$\mathbf{v}_r = x_r \mathbf{i} + y_r \mathbf{j} \quad (r = 1, 2, \dots, n).$$

The components of the resultant are

$$R_x = \sum_{r=1}^n x_r, \quad R_y = \sum_{r=1}^n y_r, \quad (16.31)$$

so that

$$\mathbf{R} = R_x \mathbf{i} + R_y \mathbf{j}.$$

Its magnitude and direction are

$$\|\mathbf{R}\| = \sqrt{R_x^2 + R_y^2}, \quad \theta = \text{atan2}(R_y, R_x). \quad (16.32)$$

The function atan2 uses the signs of both components to place the angle in the correct quadrant. When using $\tan^{-1}(R_y/R_x)$, the quadrant must be checked separately.

16.4.4 Resultant of two vectors

If two vectors have magnitudes A and B with included angle ϕ , the parallelogram law and cosine rule give

$$R = \sqrt{A^2 + B^2 + 2AB \cos \phi}. \quad (16.33)$$

If α is the angle between the resultant and the vector of magnitude A , then

$$\tan \alpha = \frac{B \sin \phi}{A + B \cos \phi}. \quad (16.34)$$

16.4.5 Worked examples

Example 16.4: Successive displacements

A person walks 2 m east, 3 m north, and 6 m west. The resultant displacement is

$$\mathbf{R} = 2\mathbf{i} + 3\mathbf{j} - 6\mathbf{i} = -4\mathbf{i} + 3\mathbf{j} \text{ m.}$$

Therefore,

$$\|\mathbf{R}\| = 5 \text{ m,}$$

and the direction is N 53.13°W, or bearing 306.87°.

Example 16.5: Forces specified by bearings

Two forces have magnitudes 5 N and 8 N and bearings 090° and 060°. Their resultant is

$$\begin{aligned} \mathbf{R} &= 5\mathbf{i} + 8 \sin 60^\circ \mathbf{i} + 8 \cos 60^\circ \mathbf{j} \\ &= (5 + 4\sqrt{3})\mathbf{i} + 4\mathbf{j} \text{ N.} \end{aligned}$$

Thus,

$$\|\mathbf{R}\| = \sqrt{(5 + 4\sqrt{3})^2 + 4^2} \approx 12.58 \text{ N,}$$

with bearing

$$\beta = \tan^{-1} \left(\frac{5 + 4\sqrt{3}}{4} \right) \approx 71.46^\circ.$$

Hence, the resultant is approximately 12.58 N on a bearing of 071.46°.

Example 16.6: A raindrop in moving air

A raindrop falls vertically at 3 m s^{-1} while the air moves eastward at 4 m s^{-1} . Its resultant velocity is

$$\mathbf{v} = 4\mathbf{i} - 3\mathbf{j} \text{ m s}^{-1}.$$

It therefore moves at 5 m s^{-1} , directed 36.87° below the positive x -axis, or S 53.13° E.

16.5 Relative vectors

In the physical world, motion does not occur in isolation. Whenever the position or velocity of an object is stated, the observer or coordinate system relative to which it is measured must also be specified. This coordinate system is called a *reference frame*.

For example, suppose a passenger is sitting in a train travelling east at 15 m s^{-1} . The passenger is stationary relative to the train but moves east at 15 m s^{-1} relative to the Earth. Both statements are correct because they use different reference frames.

16.5.1 Reference frames and relative position

Relative position indicates the position of one point with respect to another. If the position vectors of A and B relative to a common origin O are

$$\mathbf{r}_{A/O} = \overrightarrow{OA} \quad \text{and} \quad \mathbf{r}_{B/O} = \overrightarrow{OB},$$

then the position of B relative to A is

$$\boxed{\mathbf{r}_{B/A} = \mathbf{r}_{B/O} - \mathbf{r}_{A/O} = \overrightarrow{AB}.} \quad (16.35)$$

Reversing the order reverses the vector:

$$\boxed{\mathbf{r}_{A/B} = -\mathbf{r}_{B/A}.} \quad (16.36)$$

For three points A , B , and C , the head-to-tail rule gives the relative-position identity

$$\boxed{\mathbf{r}_{C/A} = \mathbf{r}_{B/A} + \mathbf{r}_{C/B}.} \quad (16.37)$$

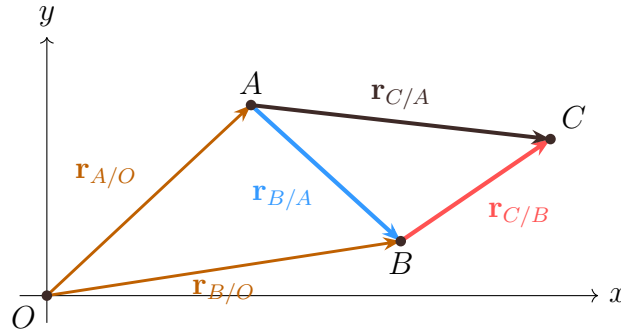


Figure 16.15: Relative positions between three points measured from a common reference frame.

Example 16.7

Suppose

$$\mathbf{r}_{A/O} = 3\mathbf{i} + 2\mathbf{j} \quad \text{and} \quad \mathbf{r}_{B/O} = 5\mathbf{i} + \mathbf{j}.$$

Then

$$\mathbf{r}_{B/A} = (5\mathbf{i} + \mathbf{j}) - (3\mathbf{i} + 2\mathbf{j}) = 2\mathbf{i} - \mathbf{j}.$$

The distance between A and B is therefore

$$\|\mathbf{r}_{B/A}\| = \sqrt{2^2 + (-1)^2} = \sqrt{5}.$$

16.5.2 Relative velocity

If two objects move in the same reference frame, the apparent velocity of one object as observed from the other is obtained by subtraction. If $\mathbf{v}_A$ and $\mathbf{v}_B$ are their velocities relative to the ground, then

$$\boxed{\mathbf{v}_{A/B} = \mathbf{v}_A - \mathbf{v}_B, \quad \mathbf{v}_{B/A} = \mathbf{v}_B - \mathbf{v}_A.} \quad (16.38)$$

Consequently, $\mathbf{v}_{A/B} = -\mathbf{v}_{B/A}$. The first subscript names the object being observed, whereas the second names the observer or reference body. The same idea applies in one, two, or three dimensions.

In one-dimensional motion, signs must be assigned before using Equation (16.38). For example, take east as positive. If A travels east at 50 km h^{-1} and B travels east at 60 km h^{-1} , then

$$\mathbf{v}_{B/A} = (60 - 50)\mathbf{i} = 10\mathbf{i} \text{ km h}^{-1}.$$

An observer on A therefore sees B moving east at only 10 km h^{-1} . If instead A travels west at 50 km h^{-1} , then $\mathbf{v}_A = -50\mathbf{i}$ and $\mathbf{v}_{B/A} = 110\mathbf{i} \text{ km h}^{-1}$.

For two-dimensional motion, either construct a velocity triangle or resolve each velocity into perpendicular components. With east represented by $\mathbf{i}$ and north by $\mathbf{j}$, a speed v on

bearing β has the component form

$$\mathbf{v} = v \sin \beta \mathbf{i} + v \cos \beta \mathbf{j}. \quad (16.39)$$

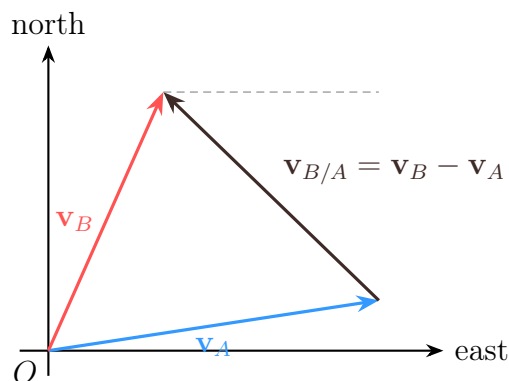


Figure 16.16: The relative-velocity vector completes the velocity triangle.

Example 16.8: apparent motion

A car travels east at 40 km h^{-1} , while a train travels north at 75 km h^{-1} . The train's velocity relative to the car is

$$\begin{aligned} \mathbf{v}_{T/C} &= 75\mathbf{j} - 40\mathbf{i}, \\ \|\mathbf{v}_{T/C}\| &= \sqrt{40^2 + 75^2} = 85 \text{ km h}^{-1}. \end{aligned}$$

Its apparent direction is $\tan^{-1}(40/75) \approx 28.1^\circ$ west of north. In compass notation, the direction is approximately N 28.1° W.

16.5.3 Relative position during motion

Suppose the initial position vectors are $\mathbf{r}_A(0)$ and $\mathbf{r}_B(0)$ and both objects move with constant velocities. Then

$$\mathbf{r}_A(t) = \mathbf{r}_A(0) + \mathbf{v}_A t, \quad (16.40)$$

$$\mathbf{r}_B(t) = \mathbf{r}_B(0) + \mathbf{v}_B t. \quad (16.41)$$

Subtracting gives the fundamental relative-motion equation

$$\mathbf{r}_{B/A}(t) = \mathbf{r}_{B/A}(0) + \mathbf{v}_{B/A} t. \quad (16.42)$$

Thus a two-body problem may be reduced to a one-body problem: hold A fixed and allow B to move along a straight relative path with velocity $\mathbf{v}_{B/A}$.

Closest approach

Let $\mathbf{r}_0 = \mathbf{r}_{B/A}(0)$ and $\mathbf{u} = \mathbf{v}_{B/A} \neq \mathbf{0}$. The squared separation is

$$d(t)^2 = \|\mathbf{r}_0 + \mathbf{u}t\|^2.$$

At the closest point, the relative-position vector is perpendicular to the relative-velocity vector. Equivalently,

$$\boxed{t_{\min} = -\frac{\mathbf{r}_0 \cdot \mathbf{u}}{\|\mathbf{u}\|^2}, \quad d_{\min} = \|\mathbf{r}_0 + \mathbf{u}t_{\min}\|.} \quad (16.43)$$

If only future motion is considered and $t_{\min} < 0$, the bodies are already moving apart, so their minimum future separation occurs at $t = 0$.

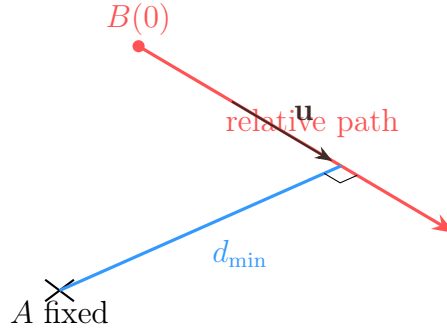


Figure 16.17: At closest approach, the separation is perpendicular to the relative path.

Collision and interception

The bodies collide precisely when their relative separation becomes zero:

$$\boxed{\mathbf{r}_0 + \mathbf{u}t = \mathbf{0} \quad \text{for some } t \geq 0.} \quad (16.44)$$

Therefore, $-\mathbf{r}_0$ must be a non-negative scalar multiple of $\mathbf{u}$. Geometrically, the relative path must point directly from the initial relative position towards the reference body. In an interception problem, choose the pursuer's velocity so that this condition is satisfied; the interception time must agree in every component equation.

16.6 Vectors in three-dimensional space

Three mutually perpendicular axes form the Cartesian coordinate system in space. The coordinate planes are

$$xy\text{-plane} : z = 0, \quad yz\text{-plane} : x = 0, \quad xz\text{-plane} : y = 0.$$

A point $P(a, b, c)$ has position vector

$$\boxed{\overrightarrow{OP} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k} = \langle a, b, c \rangle.} \quad (16.45)$$

The component operations introduced in two dimensions extend directly:

$$\langle a, b, c \rangle + \langle d, e, f \rangle = \langle a + d, b + e, c + f \rangle,$$

$$\lambda \langle a, b, c \rangle = \langle \lambda a, \lambda b, \lambda c \rangle,$$

$$\|\langle a, b, c \rangle\| = \sqrt{a^2 + b^2 + c^2}.$$

For $\mathbf{v} \neq \mathbf{0}$, the corresponding unit vector is

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|}. \quad (16.46)$$

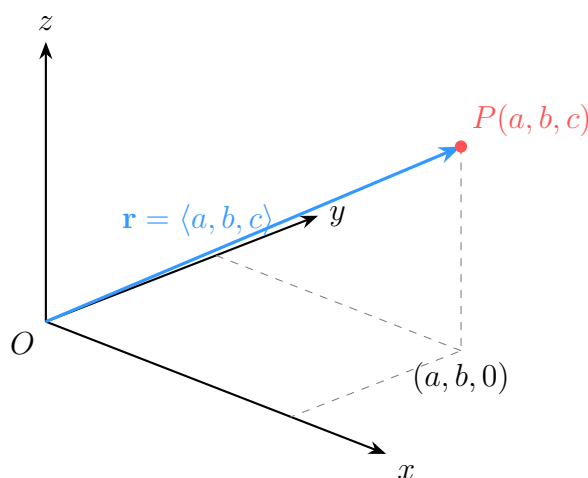


Figure 16.18: A position vector and its projection onto the xy -plane.

If $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$, then

$$\overrightarrow{PQ} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle, \quad (16.47)$$

$$d(P, Q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}, \quad (16.48)$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right). \quad (16.49)$$

16.6.1 Direction angles and direction cosines

Let a non-zero vector $\mathbf{v} = \langle a, b, c \rangle$ make angles α , β , and γ with the positive x -, y -, and z -axes, respectively. Then

$$\cos \alpha = \frac{a}{\|\mathbf{v}\|}, \quad \cos \beta = \frac{b}{\|\mathbf{v}\|}, \quad \cos \gamma = \frac{c}{\|\mathbf{v}\|}. \quad (16.50)$$

The numbers $\cos \alpha$, $\cos \beta$, and $\cos \gamma$ are the direction cosines. Since $a^2 + b^2 + c^2 = \|\mathbf{v}\|^2$,

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1. \quad (16.51)$$

The component ratio $a : b : c$ is called a direction ratio. Multiplying all three entries by the same non-zero scalar gives the same direction. For example, a vector with direction ratio $3 : 6 : 2$ and magnitude 14 has unit direction $\langle 3, 6, 2 \rangle / 7$, so the vector is $\langle 6, 12, 4 \rangle$.

16.7 Multiplication of two vectors

There are two principal products of vectors. The dot product produces a scalar and measures directional agreement; the cross product is defined in three dimensions and produces a vector perpendicular to the original pair.

16.7.1 Dot product

For $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$, the dot product is

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3. \quad (16.52)$$

If θ is the angle between two non-zero vectors, where $0 \leq \theta \leq \pi$, then the geometric form is

$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta. \quad (16.53)$$

Consequently,

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}. \quad (16.54)$$

The sign of the dot product reveals the angle type: it is positive for an acute angle, zero for a right angle, and negative for an obtuse angle. In particular,

$$\mathbf{a} \perp \mathbf{b} \iff \mathbf{a} \cdot \mathbf{b} = 0$$

for non-zero vectors. The main algebraic properties are

$$\begin{aligned} \mathbf{a} \cdot \mathbf{b} &= \mathbf{b} \cdot \mathbf{a}, \\ \mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) &= \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}, \\ (\lambda \mathbf{a}) \cdot \mathbf{b} &= \lambda(\mathbf{a} \cdot \mathbf{b}), \\ \mathbf{a} \cdot \mathbf{a} &= \|\mathbf{a}\|^2. \end{aligned}$$

Example 16.9

Let $\mathbf{a} = \langle 2, 1, 1 \rangle$ and $\mathbf{b} = \langle 1, 2, 1 \rangle$. Then

$$\mathbf{a} \cdot \mathbf{b} = 5, \quad \|\mathbf{a}\| = \|\mathbf{b}\| = \sqrt{6},$$

so $\cos \theta = 5/6$ and $\theta \approx 33.6^\circ$.

16.7.2 Scalar and vector projections

The scalar projection of $\mathbf{b}$ onto a non-zero vector $\mathbf{a}$ is the signed length of the component of $\mathbf{b}$ in the direction of $\mathbf{a}$:

$$\text{comp}_{\mathbf{a}} \mathbf{b} = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|} = \mathbf{b} \cdot \hat{\mathbf{a}}. \quad (16.55)$$

The corresponding vector projection is

$$\text{proj}_{\mathbf{a}} \mathbf{b} = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|^2} \mathbf{a}. \quad (16.56)$$

The rejection, or component perpendicular to $\mathbf{a}$, is

$$\mathbf{b}_{\perp} = \mathbf{b} - \text{proj}_{\mathbf{a}} \mathbf{b}.$$

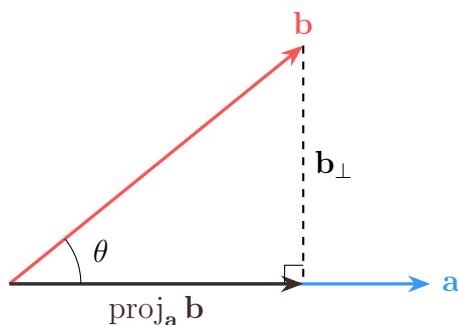


Figure 16.19: Resolution of $\mathbf{b}$ into components parallel and perpendicular to $\mathbf{a}$.

Example 16.10: force on an incline

A cart of mass 400 kg rests on a slope inclined at 30° . Ignoring other forces, the component of its weight down the slope is

$$mg \sin 30^\circ = 400(9.81) \left(\frac{1}{2} \right) = 1962 \text{ N}.$$

An equal force directed up the slope is required to prevent sliding.

16.7.3 Cross product

For $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$,

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \langle a_2 b_3 - a_3 b_2, a_3 b_1 - a_1 b_3, a_1 b_2 - a_2 b_1 \rangle. \quad (16.57)$$

Its direction is determined by the right-hand rule, and its magnitude is

$$\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\| \|\mathbf{b}\| \sin \theta. \quad (16.58)$$

The cross product is perpendicular to both original vectors:

$$(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{a} = 0, \quad (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{b} = 0.$$

Unlike the dot product, it is not commutative:

$$\begin{aligned} \mathbf{a} \times \mathbf{b} &= -(\mathbf{b} \times \mathbf{a}), \\ \mathbf{a} \times (\mathbf{b} + \mathbf{c}) &= \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}, \\ \mathbf{a} \times \mathbf{a} &= \mathbf{0}. \end{aligned}$$

For non-zero vectors,

$$\mathbf{a} \parallel \mathbf{b} \iff \mathbf{a} \times \mathbf{b} = \mathbf{0}.$$

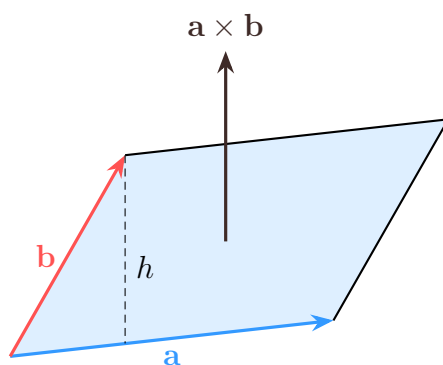


Figure 16.20: The magnitude of $\mathbf{a} \times \mathbf{b}$ equals the area of the parallelogram.

Hence,

$$\text{area of a parallelogram} = \|\mathbf{a} \times \mathbf{b}\|, \quad (16.59)$$

$$\text{area of a triangle} = \frac{1}{2} \|\mathbf{a} \times \mathbf{b}\|. \quad (16.60)$$

The scalar triple product gives the signed volume of a parallelepiped:

$$V = |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|. \quad (16.61)$$

Three vectors are coplanar if and only if their scalar triple product is zero.

Example 16.11

For $\mathbf{a} = \langle 1, -1, 0 \rangle$ and $\mathbf{b} = \langle 0, 1, 2 \rangle$,

$$\mathbf{a} \times \mathbf{b} = \langle -2, -2, 1 \rangle, \quad \|\mathbf{a} \times \mathbf{b}\| = 3.$$

The parallelogram area is 3 square units and the corresponding triangle area is $3/2$

square units.

16.8 Lines in three-dimensional space

A line is determined by one point on the line and a non-zero direction vector. If $P_0(x_0, y_0, z_0)$ lies on the line and $\mathbf{v} = \langle a, b, c \rangle$ is parallel to it, then every point on the line has position vector

$$\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}, \quad t \in \mathbb{R}. \quad (16.62)$$

This is the vector equation of the line. Equating components gives the parametric equations

$$x = x_0 + at, \quad y = y_0 + bt, \quad z = z_0 + ct. \quad (16.63)$$

If a , b , and c are all non-zero, elimination of t gives the symmetric or Cartesian form

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}. \quad (16.64)$$

If a direction component is zero, the associated coordinate is constant and must be written separately. For example, the direction $\langle 2, 0, -3 \rangle$ gives $y = y_0$ rather than a fraction with zero denominator.

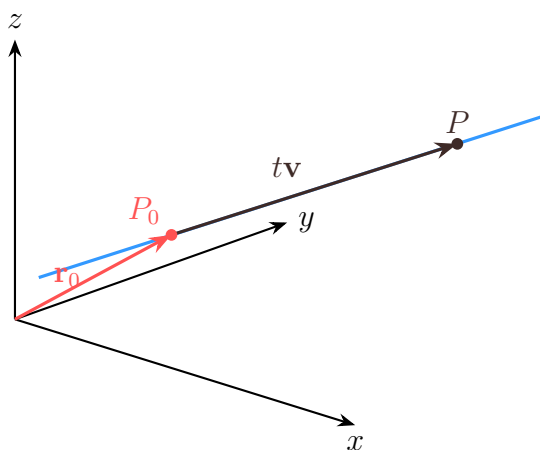


Figure 16.21: Every point on a line is obtained from a fixed position vector plus a scalar multiple of its direction vector.

If two points P and Q are given, a suitable direction vector is $\overrightarrow{PQ} = \mathbf{r}_Q - \mathbf{r}_P$. Either point may be used as the fixed point, and reversing the direction vector describes the same line.

Example 16.12

The line through $P(5, -2, 4)$ parallel to $\langle 3, 12, -4 \rangle$ is

$$\mathbf{r} = \langle 5, -2, 4 \rangle + t\langle 3, 12, -4 \rangle.$$

Thus $x = 5 + 3t$, $y = -2 + 12t$, and $z = 4 - 4t$. On the xy -plane, $z = 0$, so $t = 1$ and the line meets that plane at $(8, 10, 0)$.

16.8.1 Intersection and relative position of two lines

Let

$$\ell_1 : \mathbf{r} = \mathbf{a} + s\mathbf{u}, \quad \ell_2 : \mathbf{r} = \mathbf{b} + t\mathbf{v}.$$

The lines are

- parallel when $\mathbf{u} \times \mathbf{v} = \mathbf{0}$;
- identical when they are parallel and $\mathbf{b} - \mathbf{a}$ is also parallel to their direction;
- intersecting when $\mathbf{a} + s\mathbf{u} = \mathbf{b} + t\mathbf{v}$ has a consistent solution; or
- skew when they are neither parallel nor intersecting.

When solving for an intersection, the same values of s and t must satisfy all three component equations. Agreement in only two components is not sufficient.

The acute angle θ between two lines is the acute angle between their direction vectors:

$$\cos \theta = \frac{|\mathbf{u} \cdot \mathbf{v}|}{\|\mathbf{u}\| \|\mathbf{v}\|}, \quad 0 \leq \theta \leq \frac{\pi}{2}. \quad (16.65)$$

The absolute value selects the acute angle. Without it, the supplementary angle $\pi - \theta$ may be obtained instead.

16.9 Planes in three-dimensional space

A plane is determined by a point P_0 and a non-zero normal vector $\mathbf{n}$. If P is any point on the plane, then $\overrightarrow{P_0P} = \mathbf{r} - \mathbf{r}_0$ lies within the plane and is perpendicular to $\mathbf{n}$. Therefore,

$$\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) = 0. \quad (16.66)$$

If $\mathbf{n} = \langle a, b, c \rangle$ and $P_0(x_0, y_0, z_0)$, this becomes

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0, \quad (16.67)$$

or, after expansion,

$$ax + by + cz = d. \quad (16.68)$$

The coefficients a , b , and c are the components of a normal vector. Multiplying the entire equation by a non-zero constant does not change the plane.

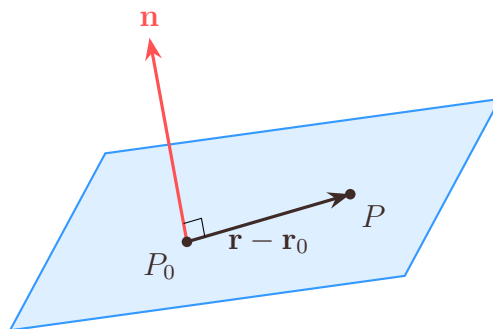


Figure 16.22: A normal vector is perpendicular to every direction lying in the plane.

16.9.1 Constructing a plane

If three non-collinear points A , B , and C are known, two directions in the plane are $\overrightarrow{AB}$ and $\overrightarrow{AC}$. Hence a normal vector is

$$\boxed{\mathbf{n} = \overrightarrow{AB} \times \overrightarrow{AC}.} \quad (16.69)$$

Substitute this normal and any one of the three points into Equation (16.67). The cross product is non-zero exactly when the three points are non-collinear.

A plane may also be represented parametrically by two non-parallel direction vectors $\mathbf{u}$ and $\mathbf{v}$ lying in it:

$$\boxed{\mathbf{r} = \mathbf{r}_0 + s\mathbf{u} + t\mathbf{v}, \quad s, t \in \mathbb{R}.} \quad (16.70)$$

16.9.2 Lines and planes

Let a line have direction $\mathbf{v}$ and a plane have normal $\mathbf{n}$. Then

$$\begin{aligned} \text{line parallel to plane} &\iff \mathbf{v} \cdot \mathbf{n} = 0, \\ \text{line perpendicular to plane} &\iff \mathbf{v} = \lambda \mathbf{n} \text{ for some } \lambda \neq 0. \end{aligned}$$

To find a line–plane intersection, substitute the line’s parametric coordinates into the plane equation and solve for the parameter. One solution gives one point of intersection; no solution means the line is parallel and separate; an identity means the entire line lies in the plane.

If ϕ is the acute angle between a line and a plane, then it is complementary to the angle between $\mathbf{v}$ and $\mathbf{n}$. Thus,

$$\boxed{\sin \phi = \frac{|\mathbf{v} \cdot \mathbf{n}|}{\|\mathbf{v}\| \|\mathbf{n}\|}.} \quad (16.71)$$

16.9.3 Relations and intersections of planes

For planes with normal vectors $\mathbf{n}_1$ and $\mathbf{n}_2$:

$$\begin{aligned} \text{parallel} &\iff \mathbf{n}_1 \times \mathbf{n}_2 = \mathbf{0}, \\ \text{perpendicular} &\iff \mathbf{n}_1 \cdot \mathbf{n}_2 = 0. \end{aligned}$$

Two parallel planes are either distinct or coincident. Two non-parallel planes intersect in a line whose direction vector is

$$\boxed{\mathbf{v} = \mathbf{n}_1 \times \mathbf{n}_2.} \quad (16.72)$$

One point on the intersection line is found by solving the two plane equations after assigning a convenient value to one coordinate. The vector equation of the intersection line then follows from that point and $\mathbf{v}$.

The acute angle θ between two planes is the acute angle between their normal vectors:

$$\boxed{\cos \theta = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{\|\mathbf{n}_1\| \|\mathbf{n}_2\|}.} \quad (16.73)$$

16.10 Perpendicular distances in space

Projection and cross-product formulae provide compact expressions for the shortest distances between points, lines, and planes.

16.10.1 Distance from a point to a line

Let P be a point and let the line pass through Q with direction $\mathbf{v}$. The area of the parallelogram formed by $\overrightarrow{QP}$ and $\mathbf{v}$ is both $\|\overrightarrow{QP} \times \mathbf{v}\|$ and $\|\mathbf{v}\|d$. Therefore,

$$\boxed{d(P, \ell) = \frac{\|\overrightarrow{QP} \times \mathbf{v}\|}{\|\mathbf{v}\|}.} \quad (16.74)$$

The same formula gives the distance between two parallel lines: choose one point from each line and use their common direction vector.

16.10.2 Distance between skew lines

For $\ell_1 : \mathbf{r} = \mathbf{a} + s\mathbf{u}$ and $\ell_2 : \mathbf{r} = \mathbf{b} + t\mathbf{v}$, where the directions are not parallel, the vector $\mathbf{u} \times \mathbf{v}$ is perpendicular to both lines. Projecting $\mathbf{b} - \mathbf{a}$ onto this common normal gives

$$\boxed{d(\ell_1, \ell_2) = \frac{|(\mathbf{b} - \mathbf{a}) \cdot (\mathbf{u} \times \mathbf{v})|}{\|\mathbf{u} \times \mathbf{v}\|}.} \quad (16.75)$$

16.10.3 Distance from a point to a plane

For the plane $ax + by + cz + d = 0$ and point $P(x_0, y_0, z_0)$,

$$d(P, \pi) = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}. \quad (16.76)$$

This is the absolute scalar projection of a point-to-plane vector onto the normal direction.

If parallel planes are written with the same left-hand coefficients,

$$ax + by + cz = d_1, \quad ax + by + cz = d_2,$$

then

$$d(\pi_1, \pi_2) = \frac{|d_2 - d_1|}{\sqrt{a^2 + b^2 + c^2}}. \quad (16.77)$$

The coefficients must first be scaled so that the two normal vectors are identical. A line has a positive distance from a plane only when it is parallel to that plane; otherwise it intersects the plane and the minimum distance is zero.

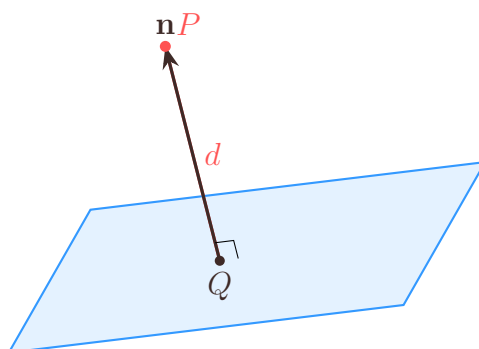


Figure 16.23: The shortest point-to-plane distance is measured in the normal direction.

16.11 Vector-valued functions and space curves

A vector-valued function assigns a vector to each value of a scalar parameter. In two and three dimensions it has the form

$$\begin{aligned} \mathbf{r}(t) &= f(t)\mathbf{i} + g(t)\mathbf{j}, \\ \mathbf{r}(t) &= f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}, \end{aligned}$$

respectively. Although f , g , and h are real-valued functions, $\mathbf{r}(t)$ is a vector. Its terminal point traces a plane curve or a space curve as t varies. The parameter often represents time, but it may be any real variable.

The domain of $\mathbf{r}$ is the intersection of the domains of all its components:

$$\boxed{D_{\mathbf{r}} = D_f \cap D_g \cap D_h.} \quad (16.78)$$

For example,

$$\mathbf{r}(t) = \left\langle \sqrt{t-1}, \ln(4-t), \frac{1}{t+1} \right\rangle$$

requires $t \geq 1$, $t < 4$, and $t \neq -1$. Hence its domain is $[1, 4)$.

16.11.1 Tracing and identifying a curve

For $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$, the equations $x = x(t)$, $y = y(t)$, and $z = z(t)$ are parametric equations. A Cartesian description may sometimes be obtained by eliminating t . The order in which points are reached gives the orientation of the curve, information that can be lost after eliminating the parameter.

For instance,

$$\mathbf{r}(t) = \langle 2 \cos t, -3 \sin t \rangle, \quad 0 \leq t \leq 2\pi,$$

traces

$$\frac{x^2}{4} + \frac{y^2}{9} = 1.$$

Since $\mathbf{r}(0) = (2, 0)$ and the initial motion is downward, the ellipse is traced clockwise. Similarly,

$$\mathbf{r}(t) = \langle 3 \cos t, 2 \sin t, 3 \rangle$$

traces an ellipse in the horizontal plane $z = 3$.

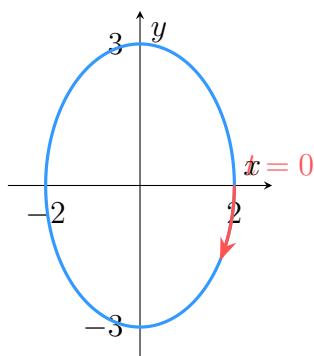


Figure 16.24: The parameter determines both the ellipse and its clockwise orientation.

16.11.2 Limits and continuity

Limits are evaluated componentwise:

$$\boxed{\lim_{t \rightarrow a} \mathbf{r}(t) = \left\langle \lim_{t \rightarrow a} f(t), \lim_{t \rightarrow a} g(t), \lim_{t \rightarrow a} h(t) \right\rangle,} \quad (16.79)$$

provided that every component limit exists and is finite. The function is continuous at $t = a$ when

$$\lim_{t \rightarrow a} \mathbf{r}(t) = \mathbf{r}(a),$$

which is equivalent to continuity of every component at a .

16.12 Differential calculus of vectors

The derivative of a vector-valued function is defined by

$$\mathbf{r}'(t) = \lim_{h \rightarrow 0} \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h}. \quad (16.80)$$

When the component functions are differentiable,

$$\mathbf{r}'(t) = f'(t)\mathbf{i} + g'(t)\mathbf{j} + h'(t)\mathbf{k}. \quad (16.81)$$

The derivative is tangent to the curve and points in its direction of increasing parameter. The principal differentiation rules are

$$\begin{aligned} \frac{d}{dt}(\mathbf{r} + \mathbf{s}) &= \mathbf{r}' + \mathbf{s}', \\ \frac{d}{dt}(f\mathbf{r}) &= f'\mathbf{r} + f\mathbf{r}', \\ \frac{d}{dt}(\mathbf{r} \cdot \mathbf{s}) &= \mathbf{r}' \cdot \mathbf{s} + \mathbf{r} \cdot \mathbf{s}', \\ \frac{d}{dt}(\mathbf{r} \times \mathbf{s}) &= \mathbf{r}' \times \mathbf{s} + \mathbf{r} \times \mathbf{s}', \\ \frac{d}{dt}\mathbf{r}(g(t)) &= g'(t)\mathbf{r}'(g(t)). \end{aligned}$$

If $\mathbf{r}'(t_0) \neq \mathbf{0}$, the tangent line at $P = \mathbf{r}(t_0)$ is

$$\mathbf{R}(s) = \mathbf{r}(t_0) + s\mathbf{r}'(t_0), \quad s \in \mathbb{R}. \quad (16.82)$$

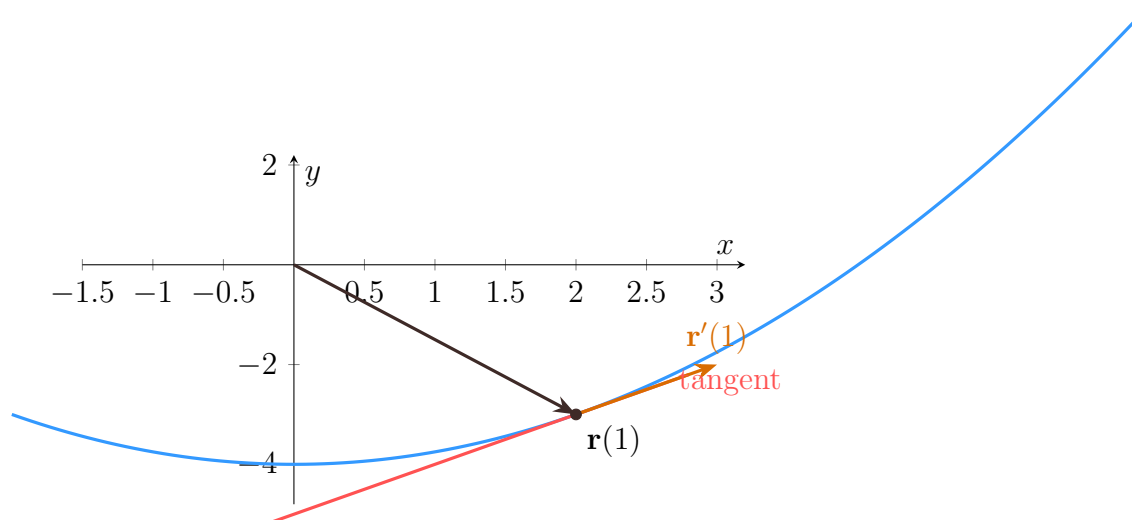


Figure 16.25: At $t = 1$, the derivative vector is tangent to the curve.

16.12.1 Velocity, speed, and acceleration

If $\mathbf{r}(t)$ is the position of a particle, then

$$\boxed{\mathbf{v}(t) = \mathbf{r}'(t), \quad v(t) = \|\mathbf{v}(t)\|, \quad \mathbf{a}(t) = \mathbf{v}'(t) = \mathbf{r}''(t).} \quad (16.83)$$

Velocity and acceleration are vectors, whereas speed is a non-negative scalar. The velocity is tangent to the path. Acceleration describes the change in velocity and need not be tangent to the path.

Example 16.13

For

$$\mathbf{r}(t) = 3t\mathbf{i} + 2t^2\mathbf{j} - \sin t \mathbf{k},$$

we have

$$\begin{aligned} \mathbf{v}(t) &= 3\mathbf{i} + 4t\mathbf{j} - \cos t \mathbf{k}, \\ v(t) &= \sqrt{9 + 16t^2 + \cos^2 t}, \\ \mathbf{a}(t) &= 4\mathbf{j} + \sin t \mathbf{k}. \end{aligned}$$

16.13 Integral calculus of vectors

Integration is also performed componentwise. If $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$, then

$$\boxed{\int \mathbf{r}(t) dt = \left(\int f(t) dt \right) \mathbf{i} + \left(\int g(t) dt \right) \mathbf{j} + \left(\int h(t) dt \right) \mathbf{k} + \mathbf{C},} \quad (16.84)$$

where $\mathbf{C} = \langle C_1, C_2, C_3 \rangle$ is an arbitrary constant vector. For a definite integral,

$$\int_a^b \mathbf{r}(t) dt = \left\langle \int_a^b f(t) dt, \int_a^b g(t) dt, \int_a^b h(t) dt \right\rangle. \quad (16.85)$$

In motion problems, successive integration gives

$$\begin{aligned} \mathbf{v}(t) &= \int \mathbf{a}(t) dt, \\ \mathbf{r}(t) &= \int \mathbf{v}(t) dt. \end{aligned}$$

Initial conditions determine both constant vectors.

Example 16.14: recovering motion from acceleration

Suppose

$$\mathbf{a}(t) = \langle 4t, 6t, 1 \rangle, \quad \mathbf{v}(0) = \langle 1, -1, 1 \rangle, \quad \mathbf{r}(0) = \langle 1, 0, 0 \rangle.$$

Integrating and applying the initial conditions gives

$$\begin{aligned} \mathbf{v}(t) &= \langle 2t^2 + 1, 3t^2 - 1, t + 1 \rangle, \\ \mathbf{r}(t) &= \left\langle \frac{2}{3}t^3 + t + 1, t^3 - t, \frac{1}{2}t^2 + t \right\rangle. \end{aligned}$$

16.14 Circular motion in vector form

Consider uniform circular motion of radius R about the origin with angular speed ω . If the initial angular position is θ_0 , then

$$\mathbf{r}(t) = R \cos(\theta_0 + \omega t) \mathbf{i} + R \sin(\theta_0 + \omega t) \mathbf{j}. \quad (16.86)$$

Positive ω gives anticlockwise motion and negative ω gives clockwise motion. Differentiation yields

$$\mathbf{v}(t) = -R\omega \sin(\theta_0 + \omega t) \mathbf{i} + R\omega \cos(\theta_0 + \omega t) \mathbf{j}, \quad (16.87)$$

$$\mathbf{a}(t) = -R\omega^2 \cos(\theta_0 + \omega t) \mathbf{i} - R\omega^2 \sin(\theta_0 + \omega t) \mathbf{j}. \quad (16.88)$$

Therefore,

$$\|\mathbf{v}\| = R|\omega|, \quad \mathbf{a} = -\omega^2 \mathbf{r}, \quad \|\mathbf{a}\| = R\omega^2 = \frac{v^2}{R}. \quad (16.89)$$

The velocity is tangent to the circle and the acceleration points towards the centre; hence they are perpendicular at every instant.

For a circle centred at (h, k) ,

$$\mathbf{r}(t) = [h + R \cos(\theta_0 + \omega t)]\mathbf{i} + [k + R \sin(\theta_0 + \omega t)]\mathbf{j}. \quad (16.90)$$

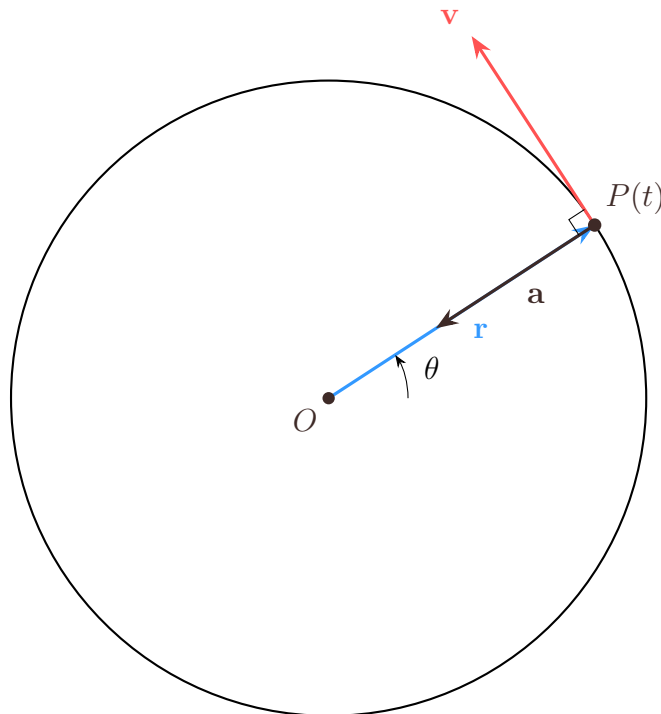


Figure 16.26: In uniform circular motion, velocity is tangential and acceleration is centripetal.

16.15 Linear and projectile motion in vector form

For motion under a constant acceleration vector $\mathbf{a}$, integration gives the vector forms of the familiar constant-acceleration equations:

$$\mathbf{v}(t) = \mathbf{v}_0 + \mathbf{a}t, \quad \mathbf{r}(t) = \mathbf{r}_0 + \mathbf{v}_0t + \frac{1}{2}\mathbf{a}t^2. \quad (16.91)$$

These equations describe motion along a line when $\mathbf{v}_0$ and $\mathbf{a}$ are parallel. Otherwise, the path is generally curved.

16.15.1 Projectile motion

In the ideal projectile model, air resistance and the rotation of the Earth are neglected and gravitational acceleration is constant. Take the positive y -direction upwards, so

$$\mathbf{a} = -g\mathbf{j}, \quad g \approx 9.81 \text{ m s}^{-2}.$$

If a projectile is launched from (x_0, y_0) with speed u at angle θ above the horizontal, its initial velocity is

$$\mathbf{v}_0 = u \cos \theta \mathbf{i} + u \sin \theta \mathbf{j}.$$

Therefore,

$$\mathbf{v}(t) = u \cos \theta \mathbf{i} + (u \sin \theta - gt) \mathbf{j}, \quad (16.92)$$

$$\mathbf{r}(t) = (x_0 + u \cos \theta t) \mathbf{i} + \left(y_0 + u \sin \theta t - \frac{1}{2}gt^2 \right) \mathbf{j}. \quad (16.93)$$

The horizontal and vertical motions share the same time parameter but may be analysed independently. Eliminating t from the component equations gives

$$y - y_0 = (x - x_0) \tan \theta - \frac{g(x - x_0)^2}{2u^2 \cos^2 \theta}, \quad (16.94)$$

which is a parabola.

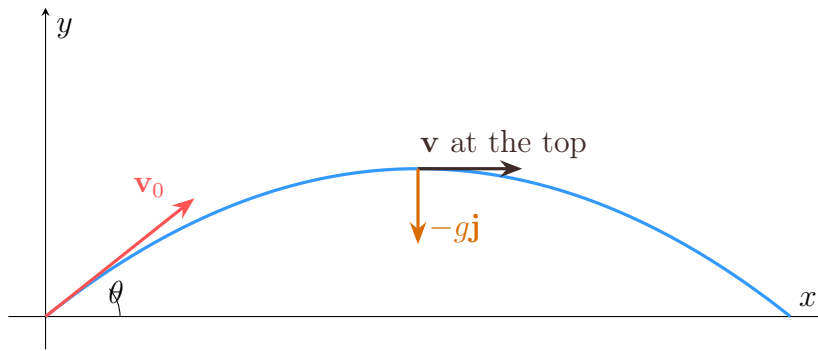


Figure 16.27: The horizontal velocity remains constant while gravity changes the vertical velocity.

At maximum height, the vertical velocity is zero, giving

$$t_H = \frac{u \sin \theta}{g}, \quad H - y_0 = \frac{u^2 \sin^2 \theta}{2g}. \quad (16.95)$$

If the projectile lands at the same vertical level from which it was launched, then

$$T = \frac{2u \sin \theta}{g}, \quad \mathcal{R} = \frac{u^2 \sin 2\theta}{g}. \quad (16.96)$$

For unequal launch and landing heights, solve the vertical position equation for its non-negative time and substitute that value into the horizontal position.

Example 16.15: horizontal launch from a cliff

A projectile is launched horizontally from a 100 m cliff and lands 90 m from its base. Taking the launch point as the origin,

$$-100 = -\frac{1}{2}(9.81)t^2 \quad \Rightarrow \quad t \approx 4.515 \text{ s.}$$

Since $90 = u_x t$, its initial horizontal speed is

$$u_x = \frac{90}{4.515} \approx 19.9 \text{ m s}^{-1}.$$

Chapter summary and common pitfalls

Key ideas. Vectors encode magnitude and direction. Dot products measure alignment, cross products produce perpendicular directions, and vector equations describe lines, planes, relative motion, and curves.

Common pitfalls. Preserve vector direction when subtracting, distinguish scalar and vector projections, use consistent units in relative motion, and choose the correct point and direction or normal vector in three-dimensional geometry.

16.16 Practice and exercises

1. State whether each quantity is scalar or vector:

- (a) momentum;
- (b) density;
- (c) electric charge;
- (d) temperature.

2. Find the magnitude and a unit vector in the direction of each vector:

- (a) $5\mathbf{i} - 3\mathbf{j}$;
- (b) $2\mathbf{i} + \mathbf{j}$;
- (c) $7\mathbf{i} - 4\mathbf{j}$;
- (d) $12\mathbf{i} + 3\mathbf{j}$.

3. Find the unit vector having the same direction as $\mathbf{i} + 2\mathbf{j}$.

4. If

$$\overrightarrow{AB} = \begin{pmatrix} 2 \\ n \end{pmatrix} \quad \text{and} \quad \|\overrightarrow{AB}\| = \sqrt{29},$$

find the possible values of n .

5. Simplify each vector expression:

- (a) $\overrightarrow{DB} + \overrightarrow{CA} + \overrightarrow{AD}$;
- (b) $\overrightarrow{CA} + \overrightarrow{AD} + \overrightarrow{DB}$;
- (c) $\overrightarrow{AD} - \overrightarrow{DC} + \overrightarrow{CE} - \overrightarrow{FE}$;
- (d) $\overrightarrow{BA} - \overrightarrow{EA} + \overrightarrow{FE}$.

6. The vertices A, B, C, D occur consecutively around a rectangle. Determine which statements are true:

- (a) $\overrightarrow{BC} = \overrightarrow{DA}$;
- (b) $BD = AC$;
- (c) $\overrightarrow{AB} + \overrightarrow{CD} = \mathbf{0}$;
- (d) $\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{CA}$;
- (e) $\overrightarrow{AC} + \overrightarrow{CD} = \overrightarrow{AD}$;
- (f) $\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AD} + \overrightarrow{DC}$.

7. Let $ABCDE$ be a pentagon satisfying

$$\overrightarrow{AB} = \mathbf{a}, \quad \overrightarrow{BC} = \mathbf{b}, \quad \overrightarrow{CD} = \mathbf{c}, \quad \overrightarrow{AE} = \mathbf{e}.$$

Let $WXYZ$ be a square satisfying

$$\overrightarrow{YX} = \mathbf{x}, \quad \overrightarrow{YZ} = \mathbf{y}.$$

Express the following in terms of the given vectors:

- (a) $\overrightarrow{AC}$;
- (b) $\overrightarrow{CE}$;
- (c) $\overrightarrow{WZ}$;
- (d) $\overrightarrow{XZ}$.

8. The consecutive directed sides of an oriented quadrilateral $PQRS$ are

$$\overrightarrow{PQ} = \mathbf{a}, \quad \overrightarrow{QR} = \mathbf{b}, \quad \overrightarrow{RS} = \mathbf{c}, \quad \overrightarrow{SP} = \mathbf{d}.$$

Show that

$$\mathbf{a} + \mathbf{b} + \mathbf{c} + \mathbf{d} = \mathbf{0}.$$

9. $OABC$ is a rectangle with

$$\overrightarrow{OA} = \mathbf{a}, \quad \overrightarrow{OB} = \mathbf{b}, \quad \overrightarrow{OC} = \mathbf{c}.$$

Points P, Q, R, S, T are the midpoints of OA, AB, BC, OC, CA , respectively.

- (a) Express $\overrightarrow{OP}$, $\overrightarrow{OQ}$, $\overrightarrow{OR}$, and $\overrightarrow{OS}$ in terms of $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$.
- (b) Show that $\overrightarrow{PQ}$ is parallel to $\overrightarrow{SR}$.

10. Points O, A, B, C satisfy

$$\overrightarrow{OA} = 10\mathbf{a}, \quad \overrightarrow{OB} = 5\mathbf{b}, \quad \overrightarrow{OC} = 4\mathbf{a} + 3\mathbf{b}.$$

Show that A, B, C are collinear.

11. Points A, B, C have position vectors $\mathbf{a}$, $\mathbf{b}$, and $3\mathbf{a}$, respectively, and O is the origin. Point M lies on AB such that

$$\overrightarrow{AM} = \frac{1}{3}\overrightarrow{AB},$$

while N lies on OM produced such that

$$\overrightarrow{ON} = \frac{9}{5}\overrightarrow{OM}.$$

- (a) Find the position vector of N in terms of $\mathbf{a}$ and $\mathbf{b}$.
(b) Find $\overrightarrow{BN}$ and $\overrightarrow{NC}$, and hence show that B, N, C are collinear.
12. Find the position vector of P when P divides $\overrightarrow{AB}$ in the stated ratio. Use both a vector method and the section formula.
- (a) $A(3, 5)$ and $B(-2, 3)$, in the ratio $3 : 2$ externally;
(b) $A(1, 3)$ and $B(5, 8)$, in the ratio $-5 : 3$.
13. Determine each resultant velocity using both vector components and a geometrical construction:
- (a) 20 m s^{-1} north and 30 m s^{-1} east;
(b) 3 m s^{-1} in the direction $S70^\circ E$ and 4 m s^{-1} in the direction $S10^\circ W$;
(c) 15 m s^{-1} and 8 m s^{-1} , both in the direction $N29.3^\circ W$.
14. The resultant of two coplanar forces is 15 N at 30° from the positive x -axis. Determine the other force when one force is
- (a) 20 N at 0° from the x -axis;
(b) 12 N at 60° from the x -axis.
15. Sketch each situation and determine the resultant acceleration:
- (a) 10 km s^{-2} south and 30 km s^{-2} west;
(b) 45 km s^{-2} west and 70 km s^{-2} south;
(c) 15 km s^{-2} in the direction $S60^\circ W$ and 35 km s^{-2} in the direction $N30^\circ E$.
16. Find the magnitude and direction of the resultant of forces of 12 N and 15 N acting on bearings 060° and 180° , respectively.
17. Sally and Lina pull a toy with forces of equal magnitude. Their directions make an angle of 40° . If the resultant force is 50 N , determine the magnitude of each force.

18. A passenger on a bus moving with velocity

$$(10\mathbf{i} - 20\mathbf{j}) \text{ m s}^{-1}$$

observes a car moving with velocity

$$(16\mathbf{i} + 8\mathbf{j}) \text{ m s}^{-1}.$$

Find the apparent velocity of the car relative to the passenger.

19. One ship sails east at 24 km h^{-1} , while another sails north at 32 km h^{-1} . Find the velocity of the second ship relative to the first.
20. A car travels east at 50 km h^{-1} , while a train travels south at 120 km h^{-1} . Find the velocity and direction of the train relative to the car.
21. Two bodies A and B have velocities $\mathbf{v}_A$ and $\mathbf{v}_B$. Explain the difference between ${}_A\mathbf{v}_B$ and ${}_B\mathbf{v}_A$, where the subscript denotes the chosen reference body.
22. State whether each statement is true or false:
- (a) Two bodies moving in the same direction with the same speed have zero relative velocity.
 - (b) In one dimension, when two bodies move in opposite directions, the magnitude of their relative velocity equals the sum of their individual speeds.
 - (c) In a two-body motion problem, the reference frame is often taken to be at rest relative to one of the bodies.
 - (d) $\mathbf{v}_C = \mathbf{v}_D - {}_C\mathbf{v}_D$.
 - (e) A collision requires the relative initial position vector to be parallel to the relative velocity vector, with the motion directed towards zero separation.
23. A girl rides a horse along a straight path at 5 km h^{-1} . A second rider moves at 3 km h^{-1} along a perpendicular straight path. Find the velocity of the second rider relative to the first.
24. Four particles A, B, C, D satisfy the following relative positions:
- A is 2 m east of B ;
 - A is 4 m north-east of C ;
 - D is 1 m in the direction $S60^\circ E$ from C .

Draw a single diagram showing all four particles and their relative positions.

25. At the initial instant, a lorry is 20 km due north of a car. The car travels east at 75 km h^{-1} , while the lorry travels at 50 km h^{-1} in the direction $S60^\circ E$. Find
- the time at which the vehicles are closest;
 - their minimum separation.
26. Ship A has velocity $(3\mathbf{i} - 4\mathbf{j}) \text{ ms}^{-1}$, and ship B has velocity $(-2\mathbf{i} + 8\mathbf{j}) \text{ ms}^{-1}$. At a certain instant, B is 26 km due east of A .
- Find the velocity of A relative to B in terms of $\mathbf{i}$ and $\mathbf{j}$.
 - Find its magnitude and direction, giving the direction to the nearest degree.
 - Draw the current positions and the direction in which A travels relative to B .
 - Find the shortest distance between the ships, to the nearest kilometre.
 - Find the separation of the ships after 3 minutes.
27. A yacht in distress is 8 km in the direction $S40^\circ W$ from a harbour and drifts in the direction $S10^\circ E$ at 4 km h^{-1} . In what direction should a lifeboat travel to intercept it if the lifeboat's speed is 30 km h^{-1} ?
28. A disabled ship drifts east at 3 knots. At 11.00 a.m., a rescue boat leaves a port 40 nautical miles south-west of the ship. The rescue boat travels at 10 knots. Find the bearing on which it should sail to overtake the ship and the time at which the interception occurs.
29. For each pair of points, find the distance $d(A, B)$, the midpoint of AB , and the vector $\overrightarrow{AB}$:
- $A(-4, 0, 1)$ and $B(3, -2, 1)$;
 - $A(0, 0, 0)$ and $B(-8, -1, 4)$.
30. Let
- $$\mathbf{p} = 4\mathbf{i} - 4\mathbf{j} + 3\mathbf{k}, \quad \mathbf{q} = 2a\mathbf{i} - \mathbf{j} + b\mathbf{k}, \quad \mathbf{r} = 3\mathbf{i} - 9\mathbf{j} + (8 - b)\mathbf{k},$$
- where a, b are constants.
- Find a and b if $2\mathbf{p} + \mathbf{q} = \mathbf{r}$.
 - Find the unit vector in the direction of $\mathbf{p}$.
 - Find a vector of magnitude $\sqrt{41}/4$ in the direction of $\mathbf{p}$.
31. Find a unit vector in the direction of each vector:
- $\langle \frac{1}{2}, -\frac{1}{2}, \frac{1}{4} \rangle$;

(b) $\langle 1, 6, -3 \rangle$.

32. Given

$$\mathbf{a} = \mathbf{i} + \mathbf{j} + \mathbf{k}, \quad \mathbf{b} = 2\mathbf{i} - 2\mathbf{j}, \quad \mathbf{c} = -\mathbf{i} - 2\mathbf{j} + 5\mathbf{k},$$

(a) find $2\mathbf{a} + \mathbf{b}$ and show that it is parallel to $2\mathbf{i} + \mathbf{k}$;

(b) show that $\|\mathbf{a} + \mathbf{b} + \mathbf{c}\| = 7$.

33. Find a vector of magnitude 5 having

(a) direction angles 70.53° , 131.81° , and 48.16° ;

(b) approximate direction cosines 0.21, -0.45 , and 0.87 .

34. The direction cosines of the position vector of P are $-1/3$, $2/3$, and $2/3$. Determine the coordinates of P if

$$\|\overrightarrow{OP}\| = 6,$$

where O is the origin.

35. Find $\mathbf{v} \cdot \mathbf{w}$ and state whether $\mathbf{v}$ and $\mathbf{w}$ are orthogonal:

(a) $\mathbf{v} = -\mathbf{j}$, $\mathbf{w} = \mathbf{k}$;

(b) $\mathbf{v} = \mathbf{i} - 3\mathbf{j}$, $\mathbf{w} = 5\mathbf{i} + 2\mathbf{j}$;

(c) $\mathbf{v} = \langle 3, -2, 4 \rangle$, $\mathbf{w} = \langle 0, -3, 7 \rangle$;

(d) if $\mathbf{u} = 3\mathbf{i} - 2\mathbf{j} - \mathbf{k}$ and $\mathbf{v} = -\mathbf{i} + \mathbf{j} + \mathbf{k}$, evaluate

$$(\|\mathbf{u}\|\|\mathbf{v}\|) \cdot (\|\mathbf{v}\|\|\mathbf{u}\|).$$

36. Find the dot product of vectors $\mathbf{g}$ and $\mathbf{h}$ if their magnitudes are 1 and 2, respectively, and the angle between them is 135° .

37. Let

$$\mathbf{a} = 3\mathbf{i} + \mathbf{j} - \mathbf{k}, \quad \mathbf{b} = \mathbf{i} + 2\mathbf{j} - 2\mathbf{k}.$$

Find

(a) $\mathbf{a} \cdot \mathbf{b}$;

(b) the angle between $\mathbf{a}$ and $\mathbf{b}$;

(c) a scalar k such that $\mathbf{a}$ is orthogonal to $\mathbf{a} - k\mathbf{b}$.

38. Justify whether each statement is true or false:

(a) If the dot product of two non-zero vectors is zero, the vectors are perpendicular.

- (b) The dot product of two vectors is always a vector.
- (c) The dot product of a vector with itself is the square of its magnitude.
- (d) Reversing the order of vectors in a dot product changes the result.
- (e) A positive dot product implies that the vectors are parallel and point in the same direction.
- (f) The expression $\mathbf{a} \cdot \mathbf{b} + 2$ is a scalar.
- (g) The distance between points A and B in three-dimensional space is

$$\|\overrightarrow{OB} - \overrightarrow{OA}\| = \|\overrightarrow{OA} - \overrightarrow{OB}\|.$$

39. Given

$$\mathbf{a} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k}, \quad \mathbf{b} = -2\mathbf{i} + \mathbf{j} - \mathbf{k},$$

find

- (a) $\mathbf{a} \cdot \mathbf{a}$;
- (b) $\mathbf{a} \cdot \mathbf{b}$;
- (c) $\mathbf{a} \cdot (\mathbf{a} + \mathbf{b})$;
- (d) the angle between $\mathbf{a} + \mathbf{b}$ and $\mathbf{a} - \mathbf{b}$.

40. Given

$$\mathbf{a} = -2\mathbf{i} + 3\mathbf{j} + \mathbf{k}, \quad \mathbf{b} = 7\mathbf{i} + 4\mathbf{j} + 5\mathbf{k}, \quad \mathbf{c} = \mathbf{i} - 5\mathbf{j} + 2\mathbf{k},$$

evaluate

- (a) $\mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$;
- (b) $(2\mathbf{a} + \mathbf{b}) \cdot 3\mathbf{c}$;
- (c) $\text{proj}_{\mathbf{c}} \mathbf{b}$;
- (d) $\text{proj}_{\mathbf{b}}(\mathbf{a} + \mathbf{c})$.

41. Given

$$\mathbf{g} = 2\mathbf{i} + 9\mathbf{j} + 5\mathbf{k}, \quad \mathbf{h} = -3\mathbf{i} + 4\mathbf{j} - 6\mathbf{k},$$

show that $\mathbf{g}$ and $\mathbf{h}$ are perpendicular.

42. Let

$$\mathbf{a} = \mathbf{i} + \mathbf{j}, \quad \mathbf{b} = 2\mathbf{i} + \mathbf{k}, \quad \mathbf{c} = 2\mathbf{j} - \mathbf{k}.$$

Find

- (a) m if $\mathbf{a} + m\mathbf{b}$ is parallel to $\mathbf{c}$;
- (b) n if $\mathbf{a} + n\mathbf{b}$ is perpendicular to $\mathbf{c}$.

43. Find the component of $\mathbf{p} = \mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$ in the direction of $\mathbf{q}$, where
- (a) $\mathbf{q} = -4\mathbf{i} + 3\mathbf{j}$;
 - (b) $\mathbf{q} = 6\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$.
44. Treating A and B as position vectors, find the scalar and vector projections of B along A for each pair:
- (a) $A(4, -2, 3)$ and $B(1, 3, 5)$;
 - (b) $A(9, 1, -4)$ and $B(0, 6, -5)$;
 - (c) $A(11, 2, 1)$ and $B(7, -6, -5)$.
45. A car is stationary on a road inclined at 30° to the horizontal. Its weight has magnitude 10.8 N.
- (a) Sketch the situation.
 - (b) Find the scalar component of the weight along the road.
 - (c) Find the vector projection of the weight onto the road.
46. A force of magnitude $22g$ makes an angle of 12° with an inclined plane and pulls a box up the plane. Find the magnitude of the component of the force acting parallel to the plane.
47. Let $\mathbf{a}$ and $\mathbf{b}$ be non-zero unequal vectors with angle θ between them. Show that
- $$\|\text{proj}_{\mathbf{a}} \mathbf{b} + \text{proj}_{\mathbf{b}} \mathbf{a}\|^2 = \|\text{proj}_{\mathbf{b}} \mathbf{a}\|^2 + \|\text{proj}_{\mathbf{a}} \mathbf{b}\|^2 + 2(\text{proj}_{\mathbf{a}} \mathbf{b}) \cdot (\text{proj}_{\mathbf{b}} \mathbf{a}).$$
48. Determine whether each statement is true or false for vectors $\mathbf{A}$ and $\mathbf{B}$:
- (a) $\mathbf{A} \times \mathbf{B}$ is a vector.
 - (b) $\mathbf{A} \times \mathbf{B}$ is parallel to $\mathbf{A}$.
 - (c) $\mathbf{A} \times \mathbf{B}$ is perpendicular to $\mathbf{B}$.
 - (d) $\mathbf{A} \times \mathbf{B} = \mathbf{B} \times \mathbf{A}$.
 - (e) $\|\mathbf{A} \times \mathbf{B}\| = \|\mathbf{A}\|\|\mathbf{B}\|$ indicates that $\mathbf{A}$ and $\mathbf{B}$ are orthogonal.
49. Find the scalars s, t if
- $$(\mathbf{i} + s\mathbf{j} + t\mathbf{k}) \times (3\mathbf{i} + 2\mathbf{j} + \mathbf{k}) = \mathbf{0}.$$

50. Find a vector perpendicular to both $\mathbf{m}$ and $\mathbf{n}$:

- (a) $\mathbf{m} = 2\mathbf{i} + 4\mathbf{j} + 3\mathbf{k}$, $\mathbf{n} = 3\mathbf{i} + 3\mathbf{j}$;
 (b) $\mathbf{m} = 3\mathbf{i} - 5\mathbf{j} + 8\mathbf{k}$, $\mathbf{n} = 3\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$.

51. Find a unit vector perpendicular to both

$$3\mathbf{i} - 2\mathbf{j} + \mathbf{k} \quad \text{and} \quad 7\mathbf{i} - 3\mathbf{j} + 4\mathbf{k}.$$

52. Points A, B, C have position vectors

$$2\mathbf{i} + \mathbf{j} + \mathbf{k}, \quad 3\mathbf{i} - \mathbf{j} + \mathbf{k}, \quad \mathbf{i} - \mathbf{j} + \mathbf{k},$$

respectively. Find

- (a) the lengths AB and AC ;
 (b) $\angle BAC$, correct to the nearest degree;
 (c) the area of triangle ABC .

53. Find the area of each triangle:

- (a) triangle FEG , where

$$\overrightarrow{OE} = \langle 2, -8, 7 \rangle, \quad \overrightarrow{OF} = \langle 4, 5, 1 \rangle, \quad \overrightarrow{OG} = \langle -2, 1, 7 \rangle;$$

- (b) triangle OEF , where

$$\overrightarrow{OE} = \langle -1, 9, 3 \rangle, \quad \overrightarrow{OF} = \langle 6, 2, -7 \rangle.$$

54. Find the volume of the parallelepiped with adjacent sides $\overrightarrow{AB}$, $\overrightarrow{AC}$, and $\overrightarrow{AD}$:

- (a) $A(0, 0, 0)$, $B(1, -1, 2)$, $C(0, 3, -1)$, $D(3, -4, 1)$;
 (b) $A(2, 1, -1)$, $B(3, 0, 2)$, $C(4, -2, 1)$, $D(5, -3, 0)$.

55. The position vectors of A, B, C, D are

$$\begin{aligned} \mathbf{r}_A &= \mathbf{i} + 2\mathbf{j} - \mathbf{k}, & \mathbf{r}_B &= 2\mathbf{i} + 3\mathbf{j} + \mathbf{k}, \\ \mathbf{r}_C &= 3\mathbf{i} + \mathbf{j} + 3\mathbf{k}, & \mathbf{r}_D &= 2\mathbf{i} + \mathbf{k}. \end{aligned}$$

Determine $\mathbf{r}_B - \mathbf{r}_A$, $\mathbf{r}_C - \mathbf{r}_B$, $\mathbf{r}_D - \mathbf{r}_C$, and $\mathbf{r}_D - \mathbf{r}_A$. Hence, find the area of parallelogram $ABCD$.

56. Find the vector, parametric, and Cartesian equations of each line:

- (a) the line through the point with position vector $2\mathbf{i} + \mathbf{j} - \mathbf{k}$, parallel to $-5\mathbf{i} - 2\mathbf{j} - \mathbf{k}$;

(b) the line through $(-1, 4, 2)$, parallel to

$$\frac{x-2}{4} = \frac{y}{5} = \frac{z+2}{3};$$

(c) the line through points with position vectors $\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ and $-5\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$.

57. Find the parametric equations of the line through $P_1(5, -2, 4)$ and $P_2(2, 6, 1)$. Determine, where possible, the points at which it intersects the three coordinate planes.

58. Show that

$$\frac{x-4}{3} = \frac{y-6}{4} = \frac{z+9}{-12} \quad \text{and} \quad \frac{x-1}{-6} = \frac{y-2}{-8} = \frac{z-3}{24}$$

represent the same straight line.

59. Find a unit vector parallel to a line perpendicular to both

$$\begin{aligned} L_1 : \quad & \frac{x-2}{4} = \frac{4-y}{5} = \frac{z-1}{5}, \\ L_2 : \quad & x = 3 + 2t, \quad y = 6 - 6t, \quad z = 1 - 2t, \quad t \in \mathbb{R}. \end{aligned}$$

60. Determine whether each pair of lines intersects. If it does, find the point of intersection.

(a)

$$\begin{aligned} l_1 : \quad & x = 1 + 2t, \quad y = 1 - 4t, \quad z = 5 - t, \quad t \in \mathbb{R}, \\ l_2 : \quad & x = 4 - v, \quad y = -1 + 6v, \quad z = 4 + v, \quad v \in \mathbb{R}; \end{aligned}$$

(b)

$$\begin{aligned} l_1 : \quad & x = 1 - 6t, \quad y = 3 + 2t, \quad z = 1 - 2t, \quad t \in \mathbb{R}, \\ l_2 : \quad & x = 2 + 2v, \quad y = 6 + v, \quad z = 2 + v, \quad v \in \mathbb{R}; \end{aligned}$$

(c)

$$\begin{aligned} l_1 : \quad & x - 2 = \frac{y-3}{2} = \frac{z-1}{2}, \\ l_2 : \quad & \frac{x-5}{-2} = y + 1 = \frac{z+13}{6}. \end{aligned}$$

61. Find the angles between the lines

$$\begin{aligned} l_1 : \quad & x = 7 - 2t, \quad y = 4 + 3t, \quad z = 5t, \quad t \in \mathbb{R}, \\ l_2 : \quad & x = -1 + 4v, \quad y = 3 - 4v, \quad z = 1 + v, \quad v \in \mathbb{R}. \end{aligned}$$

62. Find vector and Cartesian equations of the plane satisfying each condition:

- (a) it passes through $P(-2, 5, -8)$ and has normal vector
- (i) $\mathbf{i}$;
 - (ii) $\mathbf{j}$;
 - (iii) $\mathbf{k}$;
- (b) it passes through $P(3, -2, 5)$ and is perpendicular to $\langle 4, 2, -7 \rangle$;
- (c) it is parallel to $x - 2y + 2z = 1$ and passes through $A(1, -1, 2)$;
- (d) it passes through $A(2, 5, -1)$, $B(2, -3, 1)$, and $C(1, 2, 2)$.

63. Find a vector equation of the plane through $A(4, 6, 3)$ that contains the vectors

$$\mathbf{u} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}, \quad \mathbf{v} = 2\mathbf{j} + \mathbf{k}.$$

Determine whether $P(2, -1, -4)$ lies on the plane.

64. Find parametric equations for the line of intersection of

$$x - 2y + 2z = 1 \quad \text{and} \quad 3x - y - z = 2.$$

65. Determine whether each pair of planes is parallel (distinct or coincident), perpendicular, or neither:

- (a) $-8x - 6y + 2z = 1$ and $z = 4x + 3y$;
- (b) $x + 4y - 3z = 1$ and $-3x + 6y + 7z = 0$;
- (c) $2x + 2y - z = 4$ and $6x - 3y + 2z = 5$;
- (d) $\sqrt{2}x - \sqrt{8}y + 3\sqrt{2}z = 6$ and $2\sqrt{3}y - \sqrt{3}x - \sqrt{27}z + 3\sqrt{6} = 0$.

66. Justify each answer:

- (a) Can two planes intersect at exactly one point?
- (b) How can vector operations determine whether three vectors are coplanar?
- (c) What information is sufficient to determine the equation of a plane?
- (d) What is the condition for a line to be perpendicular to a plane?

67. Find the shortest distance from P to the given line:

- (a) $P(1, -2, 2)$ and

$$\frac{x}{1} = \frac{2y}{1} = \frac{z}{-1};$$

- (b) $P(a, 0, -a)$ and

$$\frac{x+a}{2} = \frac{y-a}{2} = \frac{z-a}{2}, \quad a \neq 0;$$

(c) $P(2, 2, 0)$ and

$$x = -t, \quad y = t, \quad z = -1 + t, \quad t \in \mathbb{R};$$

(d) $P(0, 4, 1)$ and

$$x = 2 + t, \quad y = 2 + t, \quad z = t, \quad t \in \mathbb{R}.$$

68. Find the exact perpendicular distance between each pair of lines:

(a)

$$\frac{x+1}{3} = \frac{y-2}{-2} = \frac{z-1}{1} \quad \text{and} \quad \frac{x-2}{5} = \frac{y+1}{1} = \frac{z}{3};$$

(b)

$$\frac{x+1}{1} = \frac{y-3}{2} = \frac{z+2}{3}$$

and the line through $(1, 3, -2)$ and $(0, 1, -1)$.

69. Let l_1 be the line through A, B and l_2 the line through C, D . Find the shortest distance between l_1 and l_2 , correct to two decimal places:

(a) $A(1, -2, 3), B(2, 0, 5), C(4, 1, -1), D(-2, 3, 4)$;

(b) $A(1, 3, 0), B(0, 4, 5), C(-2, -1, 2), D(5, 1, 0)$.

70. Find the shortest distance from $Q(1, 1, 5)$ to the line

$$x = 1 + t, \quad y = 3 - t, \quad z = 2t, \quad t \in \mathbb{R}.$$

71. Point A has coordinates $(1, 2, 1)$. The line l and plane π are

$$l: \frac{x-2}{3} = \frac{y+1}{-1} = \frac{z+1}{2}, \quad \pi: 2x + y - z = -2.$$

(a) Find the position vector of the point where l meets π .

(b) Find the perpendicular distance from A to l .

(c) Find the acute angle between l and π .

72. Find the shortest distance from each point to its corresponding plane:

(a) $(2, 1, 1)$ and $x + 2y + 2z = 11$;

(b) $(0, -1, 0)$ and $4y + 3z = -10$.

73. Find the shortest distance from the origin to the plane through

$$P(1, -1, 4), \quad Q(2, 0, 1), \quad R(0, 2, 3).$$

74. Find the perpendicular distance from $(1, 2, 1)$ to each plane:

(a) the plane through $(0, 0, 0)$, $(1, 2, 4)$, and $(-2, -1, 1)$;

(b) the plane through $(-3, 5, 1)$ that contains the line

$$\mathbf{r} = (2\mathbf{i} + \mathbf{j} + 3\mathbf{k}) + t(3\mathbf{i} + \mathbf{j} + 5\mathbf{k}), \quad t \in \mathbb{R}.$$

75. Find the shortest distance between the planes

$$x - 2y + 2z = 1 \quad \text{and} \quad 2x - 4y + 4z = 3.$$

76. Find the domain of each vector-valued function:

(a) $\mathbf{r}(t) = \sqrt{t-1}\mathbf{i} + \sqrt{2-t}\mathbf{j}$;

(b) $\mathbf{r}(t) = \tan t\mathbf{i} + (t^2 + 8t)\mathbf{j}$;

(c) $\mathbf{r}(t) = t^2\mathbf{i} + \tan t\mathbf{j} + 3\mathbf{k}$;

(d) $\mathbf{r}(t) = e^t\mathbf{i} + \frac{1}{t}\mathbf{j} + (t+1)^{-3}\mathbf{k}$;

(e) $\mathbf{r}(t) = \frac{1}{t+1}\mathbf{i} + \ln t\mathbf{j} + \sqrt{2-t}\mathbf{k}$;

(f) $\mathbf{r}(t) = \cos t\mathbf{i} + \sin t\mathbf{j} + 5\mathbf{k}$.

77. Sketch the two position vectors listed for each vector-valued function:

(a) $\mathbf{r}(t) = 3t\mathbf{i} + (1 - 9t^2)\mathbf{j}$; $\mathbf{r}(0)$ and $\mathbf{r}(1)$;

(b) $\mathbf{r}(t) = (t^3 - 1)\mathbf{i} + (t^2 + 2)\mathbf{j}$; $\mathbf{r}(1)$ and $\mathbf{r}(2)$;

(c) $\mathbf{r}(t) = (3 + t)\mathbf{i} + (2 - t)\mathbf{j} + (1 + 2t)\mathbf{k}$; $\mathbf{r}(-1)$ and $\mathbf{r}(0)$.

78. Eliminate the parameter to obtain a Cartesian equation, and interpret each curve:

(a) $x = 1 + t$, $y = 1 - t$;

(b) $x = 9 \cos t$, $y = 9 \sin t$;

(c) $x = 2 \cos t$, $y = 2 + 2 \sin t$.

79. Sketch each curve in $\mathbb{R}^3$, treating a missing component as zero:

(a) $\mathbf{F}(t) = 2t\mathbf{i} + t^2\mathbf{j}$, $t \geq 0$;

(b) $\mathbf{F}(t) = \sin t\mathbf{i} - \cos t\mathbf{j}$, $0 \leq t \leq 2\pi$.

80. Find the Cartesian equation and sketch each curve:

- (a) $\mathbf{r}(t) = \langle \sin t, 4 \cos t \rangle$, $0 \leq t \leq 2\pi$;
 (b) $\mathbf{r}(t) = \langle 3 \cos t, 2 \sin t \rangle$, $0 \leq t \leq 2\pi$;
 (c) $\mathbf{r}(t) = \langle 2 \cos t, 2 \sin t \rangle$, $0 \leq t \leq \pi$.
81. Give a vector-valued parametrisation for each curve with the stated orientation:
- (a) the line segment from $A(-2, 1)$ to $B(4, 5)$, travelling from A to B for $0 \leq t \leq 1$;
 (b) the circle $x^2 + y^2 = 25$, travelled clockwise from $(5, 0)$;
 (c) the ellipse $x^2/9 + y^2/4 = 1$, travelled anticlockwise from $(3, 0)$.
82. Sketch and describe the curve determined by each vector-valued function, indicating its orientation:
- (a) $\mathbf{F}(t) = \cos t \mathbf{i} + \sin t \mathbf{j} + 2\mathbf{k}$, $0 \leq t \leq 2\pi$;
 (b) $\mathbf{F}(t) = 6 \sin t \mathbf{i} + 4\mathbf{j} + 25 \cos t \mathbf{k}$, $-2\pi \leq t \leq 2\pi$;
 (c) $\mathbf{F}(t) = (t^2 + 1)\mathbf{i} + t\mathbf{j} + 3\mathbf{k}$, $t \in \mathbb{R}$;
 (d) $\mathbf{F}(t) = t\mathbf{i} - 3 \sin t \mathbf{j} + 3 \cos t \mathbf{k}$, $t \geq 0$;
 (e) $\mathbf{F}(t) = -3 \sin t \mathbf{i} + 3 \cos t \mathbf{j} - t\mathbf{k}$, $t \geq 0$;
 (f) $\mathbf{F}(t) = \frac{1}{2} \sin t \mathbf{i} - 3t\mathbf{j} + \cos t \mathbf{k}$, $t \geq 0$;
 (g) $\mathbf{F}(t) = e^t \cos t \mathbf{i} + e^t \sin t \mathbf{j}$, $0 \leq t \leq \pi$.
83. Explain how the paths

$$\mathbf{r}_1(t) = \langle \cos t, \sin t \rangle \quad \text{and} \quad \mathbf{r}_2(t) = \langle \sin t, \cos t \rangle$$

differ as they travel around the unit circle.

84. Suppose

$$\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}.$$

Explain why $\lim_{t \rightarrow a} \mathbf{r}(t)$ is a vector rather than a scalar.

85. Evaluate each limit:

(a)

$$\lim_{t \rightarrow 2} \left(t\mathbf{i} + \frac{t^2 - 4}{t^2 - 2}\mathbf{j} + \frac{1}{t}\mathbf{k} \right);$$

(b)

$$\lim_{t \rightarrow 0} \left(t^2\mathbf{i} + 3t\mathbf{j} + \frac{1 - \cos t}{t}\mathbf{k} \right);$$

(c)

$$\lim_{t \rightarrow \pi} (t\mathbf{i} + \cos t \mathbf{j} + \sin t \mathbf{k});$$

(d)

$$\lim_{t \rightarrow 1} \left(\sqrt{t} \mathbf{i} + \frac{\ln t}{t^2 - 1} \mathbf{j} + \frac{1}{t - 1} \mathbf{k} \right).$$

86. State the condition required for a vector-valued function $\mathbf{r}(t)$ to be continuous at $t = a$.

87. Check the continuity of each function at $t = 0$ and $t = 2$, where applicable:

(a) $\mathbf{r}(t) = |t|\mathbf{i} + \frac{1}{t-2}\mathbf{j} + \sqrt{t+2}\mathbf{k};$

(b) $\mathbf{r}(t) = \frac{t}{\sin t}\mathbf{i} + t^2\mathbf{j} + \ln(8-3t)\mathbf{k}.$

Vector calculus and motion

88. Let

$$\mathbf{r}(t) = \ln t \mathbf{i} + e^{-3t} \mathbf{j} + t^2 \mathbf{k}.$$

Find its domain, $\mathbf{r}'(t)$, and $\mathbf{r}''(t)$.

89. Given

$$\mathbf{r}(t) = \langle t^2, t^3, t \rangle, \quad \mathbf{s}(t) = \langle e^{3t}, e^{2t}, e^t \rangle,$$

find

(a) $\frac{d}{dt}(\mathbf{r} \cdot \mathbf{s});$

(b) $\frac{d}{dt}(\mathbf{r} \times \mathbf{s}).$

90. The curve

$$\mathbf{r}(t) = \sin t \mathbf{i} + 2 \cos t \mathbf{j}$$

is an ellipse. Find the vector equation of its tangent line at $t = \pi/4$.

91. For

$$\mathbf{r}(t) = 2t\mathbf{i} + (t^2 - 4)\mathbf{j}, \quad -2 \leq t \leq 3,$$

eliminate the parameter, state the orientation, and draw $\mathbf{r}(1)$ and $\mathbf{r}'(1)$ on the same sketch.

92. A particle has position

$$\mathbf{r}(t) = 3t\mathbf{i} + 2t^2\mathbf{j} - \sin t \mathbf{k}.$$

Find its velocity, speed, and acceleration at time t .

93. Evaluate

$$(a) \int (\sin t \mathbf{i} + 6\mathbf{j} + 4t\mathbf{k}) dt;$$

$$(b) \int_0^2 \left(12t^2\mathbf{i} + 4e^{2t}\mathbf{j} + \frac{1}{t+1}\mathbf{k} \right) dt.$$

94. A particle satisfies

$$\mathbf{a}(t) = \langle 4t, 6t, 1 \rangle, \quad \mathbf{v}(0) = \langle 1, -1, 1 \rangle, \quad \mathbf{r}(0) = \langle 1, 0, 0 \rangle.$$

Find $\mathbf{v}(t)$ and $\mathbf{r}(t)$.

95. A particle moves anticlockwise around a circle of radius 20 m centred at the origin with angular speed $\pi/6 \text{ rad s}^{-1}$, starting from $(20, 0)$.

(a) Find its position, velocity, and acceleration vectors.

(b) Find its position after 3 seconds.

(c) Find its speed and acceleration magnitude.

96. A rider travels clockwise on a circular platform of radius 8 m centred at $(3, -2)$. Initially, the rider's angular position is $\pi/3$, and the angular speed has magnitude 0.4 rad s^{-1} . Write a position-vector function for the rider.

97. Show directly from Equation (16.86) that, in uniform circular motion,

$$\mathbf{r}(t) \cdot \mathbf{v}(t) = 0 \quad \text{and} \quad \mathbf{a}(t) = -\omega^2 \mathbf{r}(t).$$

98. A ball is projected from ground level at 40 m s^{-1} and 40° above the horizontal. Using $g = 9.81 \text{ m s}^{-2}$, find

(a) its position vector at maximum height;

(b) its total time of flight;

(c) its horizontal range.

99. A projectile is launched horizontally from a cliff 100 m high and lands 90 m from the foot of the cliff. Find its initial speed and its velocity immediately before impact.

100. A stone is launched from the origin with initial velocity

$$\mathbf{v}_0 = 21\mathbf{i} + 13\mathbf{j} \text{ m s}^{-1}.$$

Determine whether it passes through the point $(21, 8.095)$ when $g = 9.81 \text{ m s}^{-2}$.

Brief answers and guidance are provided in Appendix F.

Chapter 17

Introduction to Graph Theory

Bonus chapter.

Chapter guide

Prerequisites. Sets, elementary counting, logical reasoning, and basic algebraic notation.

Learning outcomes. Readers should be able to describe graphs using standard terminology, analyse paths and connectivity, recognise important graph families, and apply algorithms for colouring, shortest paths, and spanning trees.

Graph theory studies networks of objects and the connections between them. Despite the word *graph*, these structures are different from the graphs of functions studied earlier. A graph-theoretic graph consists of points, called vertices, joined by lines or curves, called edges. The exact position of a vertex and the shape or length of an edge usually do not matter; only the pattern of connections is important.

Graphs are used to model road systems, social networks, electrical circuits, communication links, chemical structures, scheduling problems, and many other situations in which relationships are more important than physical distance.

The terminology and algorithms used here follow a standard introductory discrete-mathematics treatment (Levin 2024).

Figure 17.1 demonstrates the central modelling idea. The physical route map and the abstract graph have different shapes, but they represent the same connections between stations.

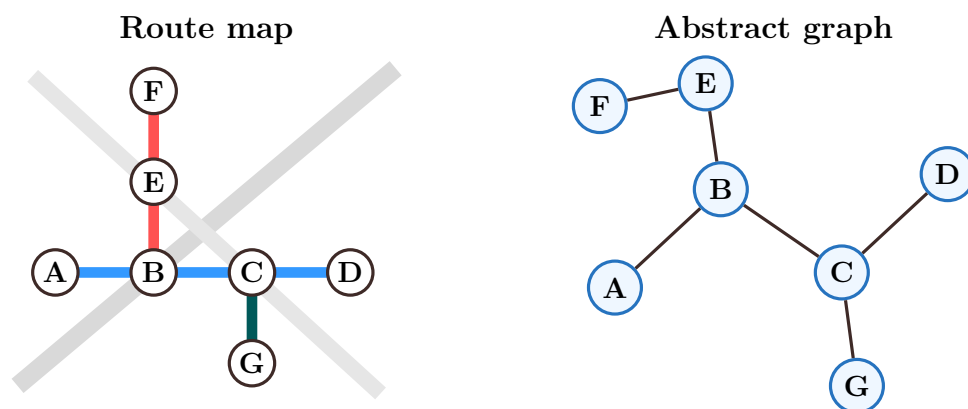


Figure 17.1: A transport map and its abstract graph. Distances and line shapes change, but the vertex–edge relationships are preserved.

17.1 Graphs, vertices, and edges

Definition 17.1: Graph.

A graph G is an ordered pair

$$G = (V, E), \quad (17.1)$$

where V is a non-empty set of *vertices* and E is a set of *edges* joining pairs of vertices.

For example, let

$$\begin{aligned} V &= \{A, B, C, D, E\}, \\ E &= \{AB, AC, BC, BD, CD, DE\}. \end{aligned}$$

The graph has five vertices and six edges.

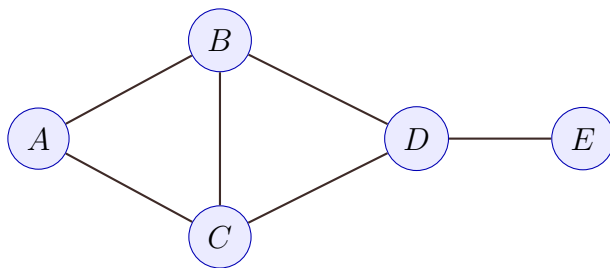


Figure 17.2: A graph with vertex set $V = \{A, B, C, D, E\}$.

The *order* of a finite graph is the number of vertices, $|V|$, and its *size* is the number of edges, $|E|$. Thus, the graph above has order 5 and size 6.

Two vertices are *adjacent* if an edge joins them. An edge is *incident* with each of its endpoints. In Figure 17.2, A is adjacent to B and C , while the edge DE is incident with D and E .

17.1.1 Simple graphs and related structures

A *simple graph* has no loop and no repeated edge between the same pair of vertices. A *loop* joins a vertex to itself, while *parallel edges* join the same pair of vertices more than once. A graph that permits parallel edges is often called a *multigraph*.

Unless stated otherwise, the graphs in this chapter are finite, simple, and undirected.

17.2 Degree of a vertex

Definition 17.2: Degree.

The degree of a vertex v , written $\deg(v)$, is the number of edges incident with v . A loop, when loops are permitted, contributes two to the degree.

In Figure 17.2,

$$\begin{aligned} \deg(A) &= 2, & \deg(B) &= 3, & \deg(C) &= 3, \\ \deg(D) &= 3, & \deg(E) &= 1. \end{aligned}$$

A vertex of degree 0 is *isolated*, and a vertex of degree 1 is a *pendant vertex* or *leaf*.

17.2.1 The handshaking lemma

Every edge has two endpoints, so it contributes 2 to the total degree. Consequently,

$$\sum_{v \in V} \deg(v) = 2|E|. \quad (17.2)$$

This result is called the *handshaking lemma*. It also shows that the number of vertices of odd degree must be even.

Example 17.1: Finding the number of edges

A graph has vertex degrees

$$2, 3, 3, 4, 4, 4.$$

The sum of the degrees is

$$2 + 3 + 3 + 4 + 4 + 4 = 20.$$

Therefore,

$$2|E| = 20 \implies |E| = 10.$$

17.3 Directed and weighted graphs

In an undirected graph, an edge AB may be followed in either direction. In a *directed graph*, or *digraph*, each edge has a specified direction and is represented by an arrow. A directed edge from A to B is written (A, B) .

For a vertex v in a digraph:

- the *indegree*, $\deg^-(v)$, counts edges entering v ;
- the *outdegree*, $\deg^+(v)$, counts edges leaving v .

Since every directed edge leaves one vertex and enters another,

$$\sum_{v \in V} \deg^-(v) = \sum_{v \in V} \deg^+(v) = |E|. \quad (17.3)$$

A *weighted graph* assigns a numerical value to each edge. Depending on the application, a weight may represent distance, travel time, cost, capacity, or another measurable quantity.

17.4 Walks, trails, paths, and cycles

Definition 17.3: Walk, trail, and path.

A *walk* is a sequence of adjacent vertices. A *trail* is a walk that does not repeat an edge. A *path* is a walk that does not repeat a vertex.

For the graph in Figure 17.2:

- A, B, C, B, D is a walk, but not a trail because BC is repeated;
- A, B, C, D, B is a trail, but not a path because B is repeated;
- A, C, D, E is a path.

A walk is *closed* if it begins and ends at the same vertex. A closed trail is called a *circuit*. A *cycle* is a closed path in which no vertex is repeated except the first and last vertex. For instance,

$$A, B, C, A$$

is a cycle of length 3 in Figure 17.2.

The length of an unweighted walk is the number of edges it uses. In a weighted graph, the weight or length of a path is the sum of the weights of its edges.

17.5 Connected graphs and components

Definition 17.4: Connected graph.

An undirected graph is connected if there is a path between every pair of vertices. A maximal connected part of a graph is called a connected component.

Removing a particular vertex or edge may disconnect a connected graph. Such a vertex is called a *cut vertex*, and such an edge is called a *bridge*. Bridges are especially important in transport and communication networks because their failure separates the network.

The graph in Figure 17.2 is connected, and DE is a bridge. Removing DE isolates E from the remaining vertices.

17.6 Some important families of graphs

Several standard graphs occur frequently.

- The *path graph* P_n consists of n vertices arranged in a single path.
- The *cycle graph* C_n consists of a cycle containing n vertices.
- The *complete graph* K_n joins every pair of distinct vertices.
- The *complete bipartite graph* $K_{m,n}$ separates its vertices into sets of sizes m and n , with every vertex in one set joined to every vertex in the other.

- The *star graph* $K_{1,n}$ has one central vertex joined to n leaves.

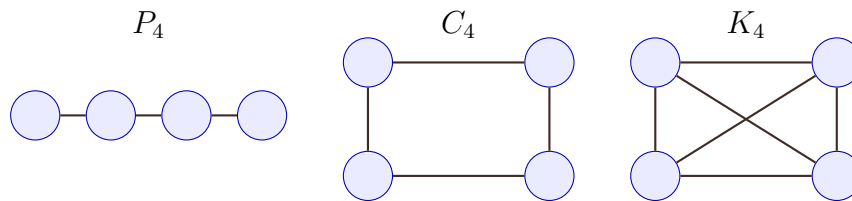


Figure 17.3: The path graph P_4 , cycle graph C_4 , and complete graph K_4 .

Every vertex in K_n has degree $n - 1$. By the handshaking lemma,

$$|E(K_n)| = \frac{n(n-1)}{2}. \quad (17.4)$$

Similarly,

$$|E(K_{m,n})| = mn. \quad (17.5)$$

17.7 Bipartite graphs

Definition 17.5: Bipartite graph.

A graph is bipartite if its vertex set can be divided into two disjoint sets X and Y so that every edge joins a vertex in X to a vertex in Y . There are no edges between two vertices within the same set.

Bipartite graphs model relationships between two different types of object, such as students and courses, workers and jobs, or machines and tasks.

A graph is bipartite if and only if it contains no cycle of odd length. Thus, every tree is bipartite, while a triangle is not.

17.8 Adjacency matrices

A finite graph may be represented by a matrix. Suppose the vertices are listed as $v_1, v_2, \dots, v_n$.

Definition 17.6: Adjacency matrix.

The adjacency matrix of a simple graph is the $n \times n$ matrix $A = (a_{ij})$ defined by

$$a_{ij} = \begin{cases} 1, & \text{if } v_i \text{ is adjacent to } v_j, \\ 0, & \text{otherwise.} \end{cases} \quad (17.6)$$

Using the vertex order A, B, C, D, E , the adjacency matrix of Figure 17.2 is

$$\begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

For a simple undirected graph:

- the adjacency matrix is symmetric;
- every diagonal entry is zero; and
- the sum of a row is the degree of the corresponding vertex.

More generally, the entry in row i , column j of A^k counts the number of walks of length k from v_i to v_j .

17.9 Trees and spanning trees

Definition 17.7: Tree.

A tree is a connected graph containing no cycle. A forest is a graph whose connected components are trees.

If a tree has n vertices, then it has exactly

$$n - 1 \tag{17.7}$$

edges. The following statements are equivalent for a finite graph T :

1. T is a tree;
2. every pair of vertices is joined by exactly one path;
3. T is connected and has $|V| - 1$ edges;
4. T has no cycle and has $|V| - 1$ edges.

A *spanning tree* of a connected graph contains all the vertices of the original graph and enough edges to form a tree. Removing edges from a connected network until no cycle remains produces a spanning tree.

Example 17.2: Recognising a tree

A connected graph has 12 vertices and 11 edges. It is a tree, because a connected graph with $n - 1$ edges contains no cycle. If one additional edge is inserted, exactly one cycle is created.

17.10 Euler trails and circuits

An Euler problem asks whether every edge can be used exactly once.

Definition 17.8: Euler trail and Euler circuit.

An Euler trail is a trail containing every edge of a graph exactly once. An Euler circuit is an Euler trail that begins and ends at the same vertex.

For a connected graph after isolated vertices are ignored:

- an Euler circuit exists if and only if every vertex has even degree;
- an open Euler trail exists if and only if exactly two vertices have odd degree.

If there are exactly two odd vertices, an Euler trail must start at one and finish at the other.

Example 17.3: Testing for an Euler trail

Suppose a connected graph has degree sequence

$$2, 2, 3, 3, 4, 4.$$

Exactly two vertices have odd degree, so the graph has an Euler trail but no Euler circuit. The trail must begin at one degree-3 vertex and end at the other.

17.11 Hamilton paths and cycles

A Hamilton problem asks whether every vertex can be visited exactly once.

Definition 17.9: Hamilton path and Hamilton cycle.

A Hamilton path visits every vertex exactly once. A Hamilton cycle is a cycle that visits every vertex exactly once before returning to the starting vertex.

Euler questions concern edges, whereas Hamilton questions concern vertices. Unlike the Euler criteria, there is no elementary test involving only vertex degrees that completely decides whether an arbitrary graph has a Hamilton cycle.

The complete graph K_n has a Hamilton cycle for every $n \geq 3$. A path graph P_n has a Hamilton path, but it has no Hamilton cycle when $n > 2$.

17.12 Planar graphs

Definition 17.10: Planar graph.

A graph is planar if it can be drawn in the plane without any edges crossing except at their endpoints. A particular crossing-free drawing is called a plane graph.

In a connected plane graph, the edges divide the plane into regions called faces. The unbounded region outside the drawing also counts as a face. Euler's formula states that

$$|V| - |E| + |F| = 2. \quad (17.8)$$

Example 17.4: Counting faces

A connected plane graph has 8 vertices and 12 edges. Therefore,

$$\begin{aligned} 8 - 12 + |F| &= 2, \\ |F| &= 6. \end{aligned}$$

Thus, the drawing has six faces, including the exterior face.

For a simple connected planar graph with at least three vertices,

$$|E| \leq 3|V| - 6. \quad (17.9)$$

Since K_5 has 5 vertices and 10 edges, but

$$10 > 3(5) - 6 = 9,$$

K_5 is not planar. For a simple bipartite planar graph with at least three vertices, the stronger bound

$$|E| \leq 2|V| - 4$$

applies. This shows that $K_{3,3}$ is also non-planar.

17.13 Vertex colouring**Definition 17.11: Proper vertex colouring and chromatic number.**

A proper vertex colouring assigns colours to vertices so that adjacent vertices receive different colours. The minimum number of colours required is the chromatic number, written $\chi(G)$.

Some standard chromatic numbers are

$$\begin{aligned} \chi(K_n) &= n, \\ \chi(C_n) &= \begin{cases} 2, & n \text{ even,} \\ 3, & n \text{ odd,} \end{cases} \\ \chi(T) &= 2 \quad \text{for every tree } T \text{ with at least one edge.} \end{aligned}$$

A graph is bipartite precisely when it can be properly coloured using at most two colours.

Colouring has applications in timetable construction, frequency assignment, register allocation in computing, and any problem where conflicting objects must receive different resources.

17.14 Shortest paths in weighted graphs

For a weighted graph with non-negative edge weights, Dijkstra's algorithm finds the shortest paths from a chosen starting vertex.

1. Assign distance 0 to the starting vertex and ∞ to every other vertex.
2. Select the unvisited vertex with the smallest tentative distance.
3. Use its outgoing edges to update the tentative distances of its neighbours.
4. Mark the selected vertex as visited and repeat until the required destination is visited.

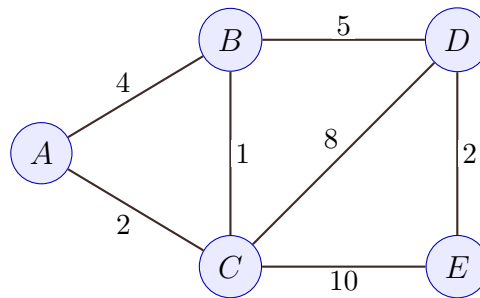


Figure 17.4: A weighted graph for shortest-path and spanning-tree problems.

Example 17.5: A shortest path

For Figure 17.4, the shortest route from A to E is

$$A \longrightarrow C \longrightarrow B \longrightarrow D \longrightarrow E,$$

with total weight

$$2 + 1 + 5 + 2 = 10.$$

17.15 Minimum spanning trees

In a connected weighted graph, a *minimum spanning tree* is a spanning tree having the smallest possible total edge weight. Kruskal's algorithm constructs one as follows:

1. Arrange the edges in increasing order of weight.
2. Add the lightest available edge that does not create a cycle.
3. Continue until $|V| - 1$ edges have been selected.

For the graph in Figure 17.4, Kruskal's algorithm selects edges of weights

$$1, 2, 2, 5,$$

for a total weight of 10. The chosen edges may be BC , AC , DE , and BD .

Chapter summary and common pitfalls

Key ideas. A graph records adjacency rather than geometric shape. Degree, paths, cycles, connectivity, trees, planarity, colouring, and weighted algorithms describe different structural properties of a network.

Common pitfalls. Distinguish a path from a trail and a circuit from a cycle, do not infer adjacency from the drawing alone, and verify algorithmic choices against the graph's weights and constraints.

17.16 Practice and exercises

Unless otherwise stated, graphs are finite, simple, and undirected. Give reasons for classifications and algorithmic choices. When more than one shortest path, colouring, or spanning tree is possible, one valid answer is sufficient unless all possibilities are requested.

Graph language, degree, and standard families

1. Let

$$V = \{A, B, C, D, E, F\}, \quad E = \{AB, AC, BC, BD, CD, DE\}.$$

Find the order, size, degree of every vertex, adjacent vertices of D , pendant vertices, and isolated vertices.

2. A graph has 7 vertices and 11 edges. Six vertex degrees are 1, 2, 2, 3, 4, 4. Find the remaining degree and check that it is possible for a simple graph on seven vertices.

3. Decide whether each sequence is graphical. Give either a construction or a reason it is impossible.

(a) 3, 3, 2, 2, 2;

(b) 4, 4, 1, 1, 0;

(c) 2, 2, 2, 2;

(d) 3, 3, 3, 1.

4. Prove that a simple graph of order n has at most $n(n-1)/2$ edges. Hence find the largest possible size of a simple graph with 12 vertices.

5. For each of P_7 , C_7 , K_7 , and $K_{1,6}$, state its order, size, and degree sequence.

6. Let G be a simple graph of order 8 and size 11.

(a) Find the size of its complement $\overline{G}$.

(b) If $\deg_G(v) = 3$, find $\deg_{\overline{G}}(v)$.

(c) Describe $\overline{K_5}$ and $\overline{C_5}$.

7. An r -regular graph has n vertices.

- (a) Show that it has $nr/2$ edges and that nr must be even.
- (b) How many edges has a 3-regular graph on 10 vertices?
- (c) Explain why no 3-regular graph on 9 vertices exists.

8. A digraph has arcs

$$(A, B), (A, C), (C, A), (B, C), (C, D), (D, B).$$

Find the indegree and outdegree of every vertex and verify both directed degree-sum identities.

- 9. A multigraph has one loop at A , two parallel edges joining A to B , and one edge BC . Find the degree of each vertex and verify the handshaking lemma, remembering how a loop contributes.
- 10. In Figure 17.2, find the subgraph induced by $\{B, C, D, E\}$. State its vertex set, edge set, order, and size.
- 11. Explain why equal degree sequences do not guarantee isomorphism. Use C_6 and the disjoint union of two triangles as an example.
- 12. Classify K_4 using as many of the following terms as apply: simple, directed, connected, regular, complete, bipartite, planar, and weighted.
- 13. Explain the distinctions among a subgraph, a spanning subgraph, and an induced subgraph. Give a small example of each using K_4 .
- 14. Show that the complement of $K_{m,n}$ is the disjoint union $K_m \cup K_n$.
- 15. Prove from the handshaking lemma that every finite undirected graph has an even number of odd-degree vertices.

Walks, connectivity, and bipartite graphs

- 16. In Figure 17.2, classify each sequence as a walk, trail, path, circuit, or cycle, listing every applicable term.
 - (a) A, B, C, A ;
 - (b) A, B, D, C, B ;
 - (c) E, D, B, C, A ;
 - (d) A, B, C, B, D .

17. For Figure 17.2, find the distances from A to every vertex. Hence find the eccentricity of A , the diameter, radius, and all central vertices.
18. In Figure 17.2, determine the components remaining after
- (a) deleting vertex D ;
 - (b) deleting edge DE ;
 - (c) deleting vertex B .
19. Find every bridge and cut vertex in Figure 17.2. Justify each choice.
20. Find the length of a shortest path from A to E in Figure 17.2, give two shortest paths, and count all simple A – E paths.
21. A graph has vertex set $\{1, 2, 3, 4, 5, 6, 7\}$ and edge set

$$\{12, 23, 31, 45, 56\}.$$

Find all connected components and state whether the graph is connected.

22. Explain why every connected graph of order n has at least $n - 1$ edges. When does equality occur?
23. A graph has bipartition $X = \{1, 2, 3\}$ and $Y = \{a, b\}$, with edges $1a, 1b, 2a, 3b$. Find every degree, its size, and a proper two-colouring.
24. Determine which graphs are bipartite and justify the answers using odd cycles: C_{10} , C_{11} , $K_{2,5}$, and a triangle with a pendant edge.
25. Prove that if a simple graph G is disconnected, then its complement $\overline{G}$ is connected.

Adjacency matrices and counting walks

26. Write the adjacency matrix of

$$V = \{P, Q, R, S\}, \quad E = \{PQ, PR, QR, QS, RS\},$$

using the order P, Q, R, S .

27. The adjacency matrix of a graph, in the order A, B, C, D, E , is

$$\begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

Find every degree, the size, and all pendant or isolated vertices.

28. For the path P_3 with vertex order 1, 2, 3, write A and calculate A^2 . Interpret the entries $(A^2)_{11}$, $(A^2)_{13}$, and $(A^2)_{22}$.
29. Explain why the number of triangles in a simple graph is $\text{tr}(A^3)/6$. Use the formula to find the number of triangles in K_4 .
30. Write the adjacency matrix of the digraph in Question 8, taking rows as initial vertices and columns as terminal vertices. Explain how row and column sums encode the directed degrees.
31. Write a weighted adjacency matrix for a triangle whose edge weights are $w(AB) = 4$, $w(AC) = 7$, and $w(BC) = 2$. Use 0 on the diagonal and ∞ when no edge is available.
32. If A is the adjacency matrix of a simple graph of order n , show that the adjacency matrix of its complement is $J - I - A$. Apply this to find the edge set of $\overline{P_4}$.
33. A graph has adjacency matrix

$$\begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}.$$

Identify the graph and determine whether it is regular, connected, and bipartite.

34. Use the matrix in Question 27 to find the number of walks of length 2 from A to D , the number of walks of length 2 from A to E , and the number of walks of length 3 from A to E .
35. For a simple undirected graph with adjacency matrix A , prove that A is symmetric with zero diagonal and that $A\mathbf{1}$ is the vector of vertex degrees.

Trees, forests, and spanning trees

36. A tree has 42 vertices. Find its number of edges and the sum of its vertex degrees.
37. A forest has 18 vertices and 4 connected components. Find its number of edges.
38. A tree has three vertices of degree 3, four vertices of degree 2, and all remaining vertices of degree 1. Find the number of leaves, the order, and the size of the tree.
39. Construct a tree with degree sequence 4, 2, 2, 1, 1, 1, 1 by giving an edge set on vertices $A, \dots, G$.
40. Decide which edge set forms a spanning tree of Figure 17.2.

- (a) $\{AB, AC, BD, DE\}$;
 - (b) $\{AB, AC, BC, DE\}$.
41. Prove that removing any edge from a tree disconnects it and adding any new edge to a tree creates exactly one cycle.
42. A connected graph has 15 vertices and 20 edges. How many edges must be removed to obtain a spanning tree?
43. Prove that every tree is bipartite by colouring vertices according to the parity of their distance from a fixed vertex.
44. Prove that every tree with at least two vertices has at least two leaves. Hint: consider the endpoints of a longest path.
45. In K_4 with vertices 1, 2, 3, 4, find two edge-disjoint spanning trees.

Euler and Hamilton problems

46. For each connected graph degree sequence, decide whether an Euler circuit, only an open Euler trail, or neither is possible.
- (a) 2, 2, 2, 2;
 - (b) 1, 2, 2, 3;
 - (c) 1, 3, 3, 3, 4.
47. Does Figure 17.2 have an Euler trail? If edge BE is added, identify the possible endpoints of an open Euler trail.
48. A connected street network has vertex degrees 2, 2, 2, 4, 4, 4. Can an inspector start at a depot, traverse each street exactly once, and return to the depot? Explain.
49. A connected graph has exactly two odd vertices u and v . Explain why every Euler trail must start at one of them and finish at the other.
50. A connected graph has four odd vertices A, B, C, D . What is the minimum number of added edges needed to make every degree even? Give two possible pairings.
51. Give an example of
- (a) a graph with an Euler circuit but no Hamilton cycle;
 - (b) a graph with a Hamilton cycle but no Euler circuit.
52. For $n > 2$, state which of the following exist in P_n : an Euler trail, Euler circuit, Hamilton path, and Hamilton cycle.

- 53. For $n \geq 3$, state which Euler and Hamilton structures exist in C_n .
- 54. For $n \geq 3$, determine when K_n has an Euler circuit and when it has a Hamilton cycle.
- 55. Prove that a connected graph with an Euler circuit cannot contain a bridge.

Planarity and colouring

- 56. A connected plane graph has 10 vertices and 15 edges. Find the number of faces, including the exterior face.
- 57. A connected plane graph has 18 vertices and 11 faces. Find its number of edges.
- 58. Every face of a connected simple plane graph is triangular and the graph has 12 vertices. Find its numbers of edges and faces.
- 59. A connected simple bipartite plane graph has 10 vertices and every face boundary has length 4. Find its numbers of edges and faces.
- 60. Use the planar edge bound to prove that K_5 is non-planar.
- 61. Use the bipartite planar edge bound to prove that $K_{3,3}$ is non-planar.
- 62. A simple graph has 7 vertices and 15 edges. Explain why $|E| \leq 3|V| - 6$ does not decide whether the graph is planar.
- 63. Find the chromatic number of C_8 , C_9 , K_6 , $K_{1,7}$, and a non-trivial tree.
- 64. Show that a graph with at least one edge is bipartite if and only if its chromatic number is 2.
- 65. Give a proper colouring of Figure 17.2 and prove that its chromatic number is 3.

Shortest paths and minimum spanning trees

- 66. Apply Dijkstra's algorithm to Figure 17.4, starting at A . Give the final distance and one predecessor for every vertex.
- 67. Using Figure 17.4, find a shortest path from A to D , from A to E , and from B to E .
- 68. Apply Kruskal's algorithm to Figure 17.4. List the selected edges and the total weight of a minimum spanning tree.
- 69. Suppose the weight of CE in Figure 17.4 is changed from 10 to 3. Find the new minimum spanning tree and a shortest A – E path.

70. Form a shortest-path tree rooted at A for Figure 17.4. Is this tree also a minimum spanning tree? Explain why the two concepts are not identical in general.
71. A square has all four boundary edges of weight 1 and both diagonals of weight 2. Find the minimum spanning-tree weight and the number of distinct minimum spanning trees.
72. Explain why distinct edge weights guarantee a unique minimum spanning tree. Show by example that repeated weights do not necessarily imply multiple minimum spanning trees.
73. Explain why Dijkstra's algorithm requires non-negative edge weights. What can go wrong after a vertex has been permanently labelled?
74. A directed weighted network has arcs

$$S \rightarrow A : 4, \quad S \rightarrow B : 1, \quad B \rightarrow A : 2, \quad A \rightarrow T : 1, \quad B \rightarrow T : 7.$$

Find the shortest directed path from S to T .

75. Choose the most appropriate graph-theoretic method for each task: finding a cheapest route between two cities, connecting all sites with minimum total cable, traversing every street once, and assigning channels to conflicting transmitters.

Applications and modelling

76. At a conference, 24 people each shake hands with exactly seven others. How many handshakes occur? Explain why the degree condition is consistent with the handshaking lemma.
77. A social network contains 12 users and 27 mutual friendships. Find its average degree. Could exactly five users have odd degree?
78. Ten teams play a round-robin competition in which every pair meets once. Model the schedule as a graph and find the total number of matches and the degree of each team vertex.
79. An airline records each direct one-way route as a directed edge. If its network has 38 routes, find the total indegree and total outdegree. Interpret a vertex with indegree 0 and outdegree 5.
80. Course prerequisites are represented by arcs $A \rightarrow B$, $B \rightarrow C$, $C \rightarrow A$, and $C \rightarrow D$. Explain why the first three courses cannot be completed in an order satisfying all prerequisites.

81. A university has 120 students and 8 clubs. Join a student to each club they attend. Identify a natural bipartition. If the graph has 310 edges, what does that number represent?
82. A road inspector's connected street network has degree sequence 2, 2, 2, 4, 4, 4. Determine whether every street can be inspected once in a closed route and state where the route may begin.
83. A postal route network has exactly two odd intersections, the post office P and depot D . If every street must be used exactly once, where must the route start and end?
84. In Figure 17.2, interpret vertices as communication stations. Which single link is a bridge, and what failure occurs if it is removed?
85. A fibre network connects 30 buildings and is known to be a tree. How many links does it use? What structural change occurs when one new link is installed between previously non-adjacent buildings?
86. A connected planar utility diagram has 14 junctions and 20 pipes. If the drawing has no crossings, how many regions does it form?
87. Seven regions surround a circular lake so that each region touches only its two neighbours. Model the adjacency graph and find the minimum number of map colours required.
88. Four examinations conflict pairwise, and a fifth examination conflicts only with the first. Model the conflicts and find the minimum number of time slots.
89. Three transmitters interfere pairwise, and a fourth interferes only with the third. Find a minimum channel assignment and state the number of channels required.
90. Five program variables have interference graph C_5 . What is the minimum number of registers needed if adjacent vertices cannot share a register?
91. A web-link graph has adjacency matrix A . Explain what $(A^2)_{ij} = 6$ means when edges are directed, and why this can help rank possible two-click routes.
92. A contact network uses unweighted edges. Explain how the graph distance between two people measures the smallest possible number of transmissions linking them.
93. An airline uses one hub joined directly to eight regional airports, with no regional-to-regional links. Identify the graph, its size, the hub degree, and every bridge.
94. A methane molecule is idealised as one carbon atom joined to four hydrogen atoms. Identify the graph and give its degree sequence and chromatic number.

95. A delivery driver uses the weighted road network in Figure 17.4. Find the least travel cost from A to E and give a route attaining it.
96. The total weight of all links in Figure 17.4 is 32. A company retains only a minimum spanning tree. Find the retained weight and the saving.
97. A warehouse has nine locations arranged in a 3×3 grid, with edges between horizontally or vertically adjacent locations. Give a Hamilton path beginning at the upper-left location and ending at the lower-right location.
98. A maintenance robot must traverse every corridor in a connected building graph. Explain which vertex-degree information decides whether it can finish where it started, finish elsewhere without repeating an edge, or must repeat at least one corridor.
99. The weighted campus network in Figure 17.5 connects a gate G , laboratory L , library B , cafeteria C , hall H , and residence R .
- Find a shortest route from G to R and its weight.
 - Find a minimum spanning tree and its total weight.
 - If all nine links were built, compare the total weight with the minimum spanning-tree weight.

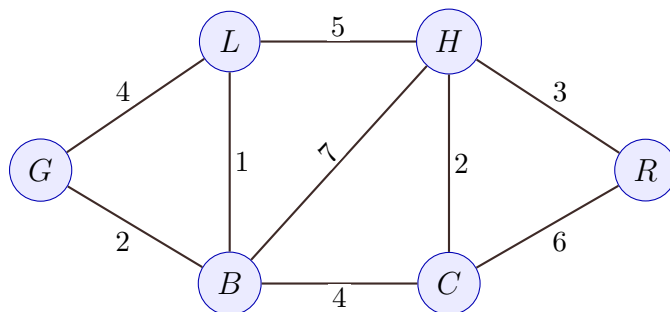


Figure 17.5: A weighted campus network for Questions 99 and 100.

100. Use the unweighted structure of Figure 17.5.
- Find every vertex degree and decide whether an Euler trail or Euler circuit exists.
 - Assuming the displayed drawing is a plane graph, find the number of faces.
 - Identify whether any existing edge is a bridge.
 - Show that adding the new link LC makes an Euler circuit possible.

Brief answers are provided in Appendix F.

Chapter 18

Introduction to Logical Reasoning

Chapter guide

Prerequisites. Set notation and familiarity with ordinary mathematical statements.

Learning outcomes. Readers should be able to translate statements into symbolic logic, construct truth tables, negate quantified statements, test arguments, identify common fallacies, and select an elementary proof method, including mathematical induction.

Mathematics is concerned not only with obtaining answers, but also with explaining why those answers must be correct. Logical reasoning provides the language and structure needed to distinguish a valid argument from an unsupported claim. It is therefore fundamental to proof, problem solving, computer programming, scientific investigation, and careful decision-making.

Every mathematical argument is built from statements, assumptions, and rules of inference. In ordinary language, these components are sometimes hidden by ambiguity or incomplete information. Symbolic logic makes their structure explicit, allowing an argument to be examined independently of its subject matter.

This chapter introduces propositions and truth values, logical connectives, truth tables, conditional statements, quantifiers, valid argument forms, and elementary methods of proof. The objective is not to replace ordinary reasoning with symbols, but to use symbols as a precise tool for organising and checking thought. The terminology and proof methods follow introductory texts in logic and discrete mathematics (Hammack 2018; Levin 2024).

18.1 Statements and truth values

Definition 18.1: Statement or proposition.

A *statement*, also called a *proposition*, is a declarative sentence that has exactly one truth value: it is either true or false.

For example, each of the following is a statement:

- “7 is a prime number.” This statement is true.
- “Every square number is odd.” This statement is false.
- “The graph of $y = 2x + 1$ is a straight line.” This statement is true.

Questions, commands, and exclamations are not statements because they do not assert something that can be assigned a truth value. Thus, “What is x ?”, “Factorise this polynomial”, and “What a beautiful graph!” are not propositions.

An expression such as

$$x + 3 = 7$$

is called an *open sentence*. Its truth depends on the value assigned to x . Once a value is specified, or once the variable is quantified, it may become a statement. For example, “There exists a real number x such that $x + 3 = 7$ ” is true.

Propositions are commonly represented by letters such as p , q , and r . The symbols T and F denote the truth values true and false, respectively.

18.2 Negation and compound statements

Definition 18.2: Negation.

The *negation* of a proposition p is the proposition “not p ”, written $\neg p$. It has the opposite truth value from p .

If p is the proposition “12 is divisible by 4”, then $\neg p$ is “12 is not divisible by 4”. Since p is true, $\neg p$ is false.

Simple propositions may be joined to form compound propositions. The most frequently used logical connectives are listed below.

Symbol	Name	Read as	When it is true
$\neg p$	Negation	not p	when p is false
$p \wedge q$	Conjunction	p and q	when both p and q are true
$p \vee q$	Disjunction	p or q	when at least one of p and q is true
$p \rightarrow q$	Implication	if p , then q	except when p is true and q is false
$p \leftrightarrow q$	Biconditional	p if and only if q	when p and q have the same truth value

Table 18.1: The principal logical connectives.

Unless explicitly stated otherwise, the logical word “or” is inclusive. The proposition $p \vee q$ is therefore true when p is true, when q is true, or when both are true. An exclusive or, meaning exactly one but not both, may be expressed as

$$(p \vee q) \wedge \neg(p \wedge q).$$

18.2.1 Truth tables

A truth table lists the truth value of a compound proposition for every possible combination of truth values of its component propositions. With n independent proposition letters, there are 2^n possible rows.

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \rightarrow q$	$p \leftrightarrow q$
T	T	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

Table 18.2: Truth tables for the main logical connectives.

The implication $p \rightarrow q$ often requires the most care. It makes a promise: whenever p is true, q must also be true. The promise is broken only when p occurs but q does not. If p is false, the implication is regarded as true because no counterexample to the promise has occurred.

18.2.2 Order of logical operations

Parentheses should be used whenever an expression might be ambiguous. When parentheses are omitted, the conventional order of precedence is

$$\neg, \quad \wedge, \quad \vee, \quad \rightarrow, \quad \leftrightarrow.$$

Thus $\neg p \vee q$ means $(\neg p) \vee q$, not $\neg(p \vee q)$.

18.3 Conditional statements

In the implication

$$p \rightarrow q,$$

p is called the *hypothesis* or *antecedent*, while q is called the *conclusion* or *consequent*. The statement may also be expressed in several equivalent ways:

- p implies q ;
- p is sufficient for q ;
- q is necessary for p ;
- q whenever p ;
- p only if q .

For example, “If an integer is divisible by 6, then it is divisible by 3” states that divisibility by 6 is sufficient for divisibility by 3. Divisibility by 3 is necessary, but not sufficient, for divisibility by 6.

18.3.1 Converse, inverse, and contrapositive

Every implication $p \rightarrow q$ has three related conditional statements.

Name	Form	Description
Original	$p \rightarrow q$	If p , then q .
Converse	$q \rightarrow p$	If q , then p .
Inverse	$\neg p \rightarrow \neg q$	If not p , then not q .
Contrapositive	$\neg q \rightarrow \neg p$	If not q , then not p .

The original implication is logically equivalent to its contrapositive. The converse is logically equivalent to the inverse. In general, the original statement and its converse are not equivalent.

Let

p : an integer is divisible by 4,
 q : the integer is even.

The implication $p \rightarrow q$ is true, as is its contrapositive: if the integer is not even, then it is not divisible by 4. The converse is false because an even integer, such as 6, need not be divisible by 4.

18.4 Logical equivalence

Definition 18.3: Logical equivalence.

Two compound propositions P and Q are *logically equivalent*, written

$$P \equiv Q,$$

if they have the same truth value for every possible assignment of truth values to their variables.

Logical equivalences allow complicated propositions to be rewritten without changing their meaning. Important laws include the following.

Law	First form	Second form
Identity	$p \wedge \text{T} \equiv p$	$p \vee \text{F} \equiv p$
Domination	$p \vee \text{T} \equiv \text{T}$	$p \wedge \text{F} \equiv \text{F}$
Idempotent	$p \vee p \equiv p$	$p \wedge p \equiv p$
Double negation	$\neg(\neg p) \equiv p$	
Complement	$p \vee \neg p \equiv \text{T}$	$p \wedge \neg p \equiv \text{F}$
Commutative	$p \vee q \equiv q \vee p$	$p \wedge q \equiv q \wedge p$
Associative	$(p \vee q) \vee r \equiv p \vee (q \vee r)$	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
Distributive	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
De Morgan	$\neg(p \wedge q) \equiv \neg p \vee \neg q$	$\neg(p \vee q) \equiv \neg p \wedge \neg q$
Absorption	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$
Implication	$p \rightarrow q \equiv \neg p \vee q$	
Biconditional	$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$	

Table 18.3: Selected laws of logical equivalence.

For example, De Morgan's law may be verified by a truth table.

p	q	$\neg(p \wedge q)$	$\neg p \vee \neg q$	Equivalent?
T	T	F	F	yes
T	F	T	T	yes
F	T	T	T	yes
F	F	T	T	yes

The two central columns are identical, so $\neg(p \wedge q) \equiv \neg p \vee \neg q$.

18.5 Tautologies, contradictions, and contingencies

Definition 18.4: Tautology.

A *tautology* is a compound proposition that is true for every possible assignment of truth values.

The proposition $p \vee \neg p$ is a tautology. It expresses the law of the excluded middle: either p is true or it is not.

Definition 18.5: Contradiction.

A *contradiction* is a compound proposition that is false for every possible assignment of truth values.

The proposition $p \wedge \neg p$ is a contradiction because a statement cannot be both true and false in the same interpretation.

Definition 18.6: Contingency.

A *contingency* is a proposition that is true for some assignments of truth values and false for others.

For example, $p \wedge q$ is a contingency. Its truth depends on the values of p and q .

18.6 Predicates and quantifiers

Definition 18.7: Predicate.

A *predicate* is an open sentence containing one or more variables. It becomes a proposition when values are assigned to the variables or when the variables are quantified.

If $P(x)$ denotes “ $x^2 \geq 0$ ”, then the truth of $P(x)$ depends on the domain of x . Over the real numbers, $P(x)$ is true for every x . Over the complex numbers, the comparison symbol $\geq$ is not generally defined in the same way. The domain is therefore an essential part of a quantified statement.

The two principal quantifiers are:

- the *universal quantifier* $\forall$, read “for every” or “for all”;
- the *existential quantifier* $\exists$, read “there exists”.

For example,

$$\forall x \in \mathbb{R}, \quad x^2 \geq 0$$

is true, whereas

$$\exists x \in \mathbb{R}, \quad x^2 = -1$$

is false.

18.6.1 Negating quantified statements

The negation of “every object has a property” is “at least one object does not have the property”. Similarly, the negation of “there exists an object with a property” is “no object has the property”. Symbolically,

$$\neg(\forall x P(x)) \equiv \exists x \neg P(x), \tag{18.1}$$

$$\neg(\exists x P(x)) \equiv \forall x \neg P(x). \tag{18.2}$$

For example, the negation of

$$\forall x \in \mathbb{R}, \quad x^2 + x + 1 > 0$$

is

$$\exists x \in \mathbb{R} \text{ such that } x^2 + x + 1 \leq 0.$$

Notice that the inequality is also negated: “not greater than” becomes “less than or equal to”.

18.6.2 The order of quantifiers

When several quantifiers occur, their order matters. Over the real numbers,

$$\forall x \exists y \text{ such that } y > x$$

is true because, after any x is chosen, one may choose $y = x + 1$. However,

$$\exists y \forall x \text{ such that } y > x$$

is false because there is no real number greater than every real number.

18.7 Arguments and validity

Definition 18.8: Argument.

An *argument* consists of one or more propositions called *premises*, followed by a proposition called the *conclusion*.

Definition 18.9: Valid argument.

An argument is *valid* if it is impossible for all its premises to be true while its conclusion is false.

Validity concerns the structure of an argument rather than the actual truth of its premises. A valid argument with false premises may have a false conclusion. To guarantee a true conclusion, an argument must be valid and all of its premises must be true; such an argument is called *sound*.

Several standard valid forms occur repeatedly in mathematics.

Rule	Premises	Conclusion
Modus ponens	$p \rightarrow q, p$	Therefore q .
Modus tollens	$p \rightarrow q, \neg q$	Therefore $\neg p$.
Hypothetical syllogism	$p \rightarrow q, q \rightarrow r$	Therefore $p \rightarrow r$.
Disjunctive syllogism	$p \vee q, \neg p$	Therefore q .
Conjunction	p, q	Therefore $p \wedge q$.
Simplification	$p \wedge q$	Therefore p .

Table 18.4: Common valid rules of inference.

Example 18.1: Identifying an argument form

Suppose the following premises are known:

- (i) If an integer is divisible by 6, then it is divisible by 3.
- (ii) The integer n is not divisible by 3.

Let p mean “ n is divisible by 6” and let q mean “ n is divisible by 3”. The premises have the forms $p \rightarrow q$ and $\neg q$. By modus tollens, $\neg p$ follows. Therefore n is not divisible by 6.

An argument with premises $P_1, P_2, \dots, P_n$ and conclusion C is valid precisely when

$$(P_1 \wedge P_2 \wedge \dots \wedge P_n) \rightarrow C$$

is a tautology. A truth table may therefore be used to test validity.

18.7.1 Common invalid argument forms

Two arguments may resemble valid inference rules while remaining invalid.

Affirming the consequent.

$$p \rightarrow q, \quad q, \quad \therefore p.$$

For example, “If a number is divisible by 4, it is even. The number is even. Therefore it is divisible by 4.” The conclusion does not follow; 6 is a counterexample.

Denying the antecedent.

$$p \rightarrow q, \quad \neg p, \quad \therefore \neg q.$$

For example, “If a shape is a square, it has four sides. The shape is not a square. Therefore it does not have four sides.” A non-square rectangle shows that the reasoning is invalid.

18.8 Methods of mathematical proof

A proof is a finite argument that establishes a conclusion from definitions, accepted assumptions, and previously established results. The method chosen depends on the logical form of the statement.

A proof-writing standard

Before beginning a calculation, state the claim with its domain and quantifiers clear. Introduce arbitrary objects and record every hypothesis that will be used. Each subsequent step should follow from a definition, an assumption, an algebraic operation that preserves equivalence, or a previously proved result. Finally, state how the last step establishes the original claim. Examples may suggest a theorem, but finitely many examples cannot prove a universal statement.

Figure 18.1 gives a practical first guide. It does not replace mathematical judgement: a statement may admit several valid proofs, and the shortest method is not always apparent at the beginning.

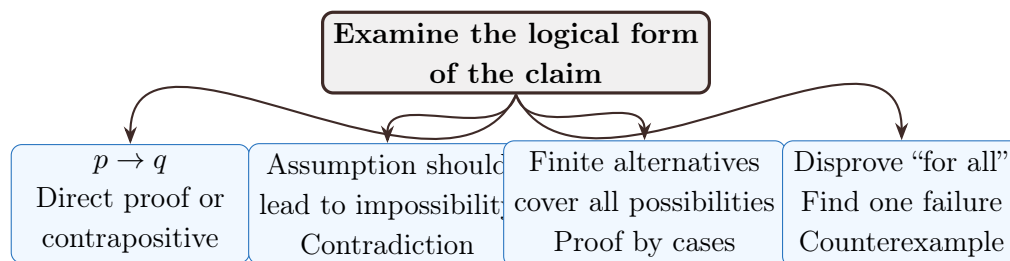


Figure 18.1: A guide to selecting an elementary proof method.

18.8.1 Direct proof

To prove $p \rightarrow q$ directly, assume p and use definitions and known results to derive q .

Example 18.2: Sum of two odd integers

Prove that the sum of two odd integers is even.

Proof. Let the two odd integers be

$$a = 2m + 1, \quad b = 2n + 1,$$

where $m, n \in \mathbb{Z}$. Then

$$\begin{aligned} a + b &= (2m + 1) + (2n + 1) \\ &= 2(m + n + 1). \end{aligned}$$

Since $m + n + 1$ is an integer, $a + b$ is divisible by 2 and is therefore even. $\square$

18.8.2 Proof by contrapositive

Since $p \rightarrow q$ is equivalent to $\neg q \rightarrow \neg p$, one may prove an implication by proving its contrapositive.

Example 18.3: Parity of a square

Prove that if n^2 is even, then n is even.

Proof. We prove the contrapositive: if n is odd, then n^2 is odd. Write $n = 2k + 1$ for some $k \in \mathbb{Z}$. Then

$$\begin{aligned} n^2 &= (2k + 1)^2 \\ &= 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1, \end{aligned}$$

which is odd. Therefore the contrapositive is true, and hence the original implication is true. $\square$

18.8.3 Proof by contradiction

To prove a proposition p by contradiction, assume $\neg p$ and show that this assumption leads to an impossibility.

Example 18.4: No smallest positive rational number

Proof. Suppose, for contradiction, that there is a smallest positive rational number r . Since $r > 0$, the number $r/2$ is also a positive rational number. However,

$$0 < \frac{r}{2} < r,$$

contradicting the assumption that r is the smallest positive rational number. Therefore no smallest positive rational number exists. $\square$

18.8.4 Proof by cases

If a hypothesis naturally divides into a finite number of possibilities, each case may be proved separately. The cases must be exhaustive, and the same stated conclusion must be reached in every case.

Example 18.5: An absolute-value inequality

Prove that $|x| \geq x$ for every real number x .

Proof. Let $x \in \mathbb{R}$. Exactly one of the following cases holds.

- If $x \geq 0$, then $|x| = x$, so $|x| \geq x$.
- If $x < 0$, then $|x| = -x > x$, so again $|x| \geq x$.

Since the cases cover all real numbers, the statement is proved. $\square$

18.8.5 Disproving a universal statement

A universal statement $\forall x P(x)$ is disproved by finding a single counterexample: one value of x for which $P(x)$ is false. For instance, the claim “Every prime number is odd” is false because 2 is prime and even.

18.8.6 Proof of an equivalence

To prove a biconditional $p \leftrightarrow q$, prove both implications:

$$p \rightarrow q \quad \text{and} \quad q \rightarrow p.$$

Proving only one direction is insufficient.

Example 18.6: An integer and its square

Prove that an integer n is even if and only if n^2 is even.

Proof. First suppose that n is even. Then $n = 2k$ for some $k \in \mathbb{Z}$, so $n^2 = 2(2k^2)$ is even. Conversely, suppose that n^2 is even. If n were odd, then $n = 2k + 1$ for some $k \in \mathbb{Z}$, and

$$n^2 = 2(2k^2 + 2k) + 1$$

would be odd, a contradiction. Hence n is even. Both implications have been proved, so the biconditional follows. $\square$

18.8.7 Proof by mathematical induction

Mathematical induction proves a statement $P(n)$ for every integer $n \geq n_0$. It requires two logically separate steps:

1. verify the *base case* $P(n_0)$;
2. for an arbitrary $k \geq n_0$, assume $P(k)$ and use that assumption to prove $P(k + 1)$.

The assumption $P(k)$ in the second step is the *induction hypothesis*. It is temporary and may be used only to establish the next case.

Theorem 18.1. *For every positive integer n ,*

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Proof. For $n = 1$, the left-hand side is 1 and the right-hand side is $1(2)/2 = 1$, so the base case holds. Now let $k \geq 1$ and assume

$$1 + 2 + \cdots + k = \frac{k(k+1)}{2}.$$

Adding $k + 1$ to both sides gives

$$\begin{aligned} 1 + 2 + \cdots + k + (k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{(k+1)(k+2)}{2}. \end{aligned}$$

This is precisely the required formula with $n = k + 1$. Therefore the formula holds for every positive integer n by mathematical induction. $\square$

18.9 Logic and sets

Set operations have direct logical interpretations. Let A and B be subsets of a universal set U . For any $x \in U$,

$$\begin{aligned} x \in A \cap B &\iff (x \in A) \wedge (x \in B), \\ x \in A \cup B &\iff (x \in A) \vee (x \in B), \\ x \in A^c &\iff \neg(x \in A), \\ x \in A \setminus B &\iff (x \in A) \wedge \neg(x \in B). \end{aligned}$$

Similarly,

$$A \subseteq B \iff \forall x \in U, (x \in A) \rightarrow (x \in B),$$

and

$$A = B \iff \forall x \in U, (x \in A) \leftrightarrow (x \in B).$$

The logical De Morgan laws correspond exactly to the set identities

$$\begin{aligned} (A \cap B)^c &= A^c \cup B^c, \\ (A \cup B)^c &= A^c \cap B^c. \end{aligned}$$

This correspondence explains why reasoning about truth conditions and reasoning about membership in sets often follow the same algebraic pattern.

Chapter summary and common pitfalls

Key ideas. Logical form separates the validity of an argument from its subject matter. Equivalences rewrite propositions, quantifiers specify scope, and proof methods are selected from the structure of the claim. Induction establishes an infinite sequence of cases through a base case and an implication from each case to the next.

Common pitfalls. An implication is not equivalent to its converse, negating a quantified statement changes both the quantifier and predicate, and examples can establish existence but cannot prove a universal claim.

18.10 Practice and exercises

1. Determine which of the following are propositions. For each proposition, state its truth value.
 - (a) 17 is a prime number.
 - (b) Solve the equation $x + 4 = 9$.
 - (c) $x^2 = 9$.
 - (d) Every rectangle is a parallelogram.
 - (e) Is 0 an even integer?
2. Write the negation of each statement without using the phrase “it is not the case that”.
 - (a) $x > 5$.
 - (b) Every student completed the exercise.
 - (c) At least one real number satisfies $x^2 = 2$.
 - (d) n is even and positive.
3. Let p be true, q be false, and r be true. Determine the truth value of each proposition.
 - (a) $p \wedge q$
 - (b) $\neg q \vee r$
 - (c) $(p \rightarrow q) \vee r$
 - (d) $p \leftrightarrow (q \vee r)$
4. Construct a truth table for each proposition.
 - (a) $(p \wedge q) \rightarrow p$
 - (b) $(p \rightarrow q) \wedge (q \rightarrow p)$

$$(c) \neg(p \vee q) \leftrightarrow (\neg p \wedge \neg q)$$

5. Let p mean “the integer n is divisible by 8” and let q mean “ n is even”. Write in words:

$$(a) p \rightarrow q;$$

$$(b) q \rightarrow p;$$

$$(c) \neg q \rightarrow \neg p.$$

Identify the converse and contrapositive of $p \rightarrow q$.

6. Write the converse, inverse, and contrapositive of the statement: “If a function is differentiable at $x = a$, then it is continuous at $x = a$.”

7. Use logical equivalences to simplify each expression.

$$(a) \neg(p \rightarrow q)$$

$$(b) p \vee (p \wedge q)$$

$$(c) \neg(\neg p \vee q)$$

$$(d) (p \rightarrow q) \wedge p$$

8. Classify each proposition as a tautology, contradiction, or contingency.

$$(a) p \vee \neg p$$

$$(b) (p \rightarrow q) \wedge p \wedge \neg q$$

$$(c) p \leftrightarrow q$$

9. Translate each statement into symbols. State the domain clearly.

(a) Every integer has a square that is non-negative.

(b) There is a real number whose square is 5.

(c) No integer is strictly between 2 and 3.

10. Negate each quantified statement and simplify the result.

$$(a) \forall x \in \mathbb{R}, x^2 + 1 > 0;$$

$$(b) \exists n \in \mathbb{Z}, n^2 = 3;$$

$$(c) \forall x \in A, P(x) \rightarrow Q(x).$$

11. Determine whether each argument is valid. Name the inference rule or invalid form involved.

- (a) If n is divisible by 10, then n is divisible by 5. The integer n is divisible by 10. Therefore n is divisible by 5.
- (b) If a graph is a tree, then it is connected. The graph is not a tree. Therefore it is not connected.
- (c) Either $x < 0$ or $x \geq 0$. It is not true that $x < 0$. Therefore $x \geq 0$.
12. Test the validity of the following argument with a truth table:
- $$p \rightarrow q, \quad q \rightarrow r, \quad \neg r, \quad \therefore \neg p.$$
13. Prove directly that the product of two odd integers is odd.
14. Prove by contrapositive that if $3n + 2$ is even, then n is even, where $n \in \mathbb{Z}$.
15. Prove by contradiction that there is no greatest negative real number.
16. Give a counterexample to each statement.
- (a) If ab is even, then both a and b are even.
- (b) Every continuous function is differentiable.
- (c) For all real x , $x^2 > x$.
17. Prove using element-wise logical reasoning that
- $$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C).$$
18. Explain why proving $p \rightarrow q$ does not by itself prove $p \leftrightarrow q$. Give a mathematical example in which $p \rightarrow q$ is true but $q \rightarrow p$ is false.

Additional grouped practice

The questions below group related symbolic and proof skills under a single number so that each exercise examines a complete line of reasoning.

19. Classify each sentence as a proposition, an open sentence, or neither. If it is a proposition, state its truth value:
- (a) $2 + 5 = 8$;
- (b) $x - 3 > 0$;
- (c) Close the window;
- (d) There exists an integer n with $n^2 = 16$;

- (e) What is the domain of f ?
20. Write clear negations of:
- (a) $a \leq b$;
 - (b) the graph is connected or planar;
 - (c) every prime number is odd;
 - (d) exactly one of p and q is true.
21. Let p mean “Ali studies”, q mean “Ali passes”, and r mean “Ali celebrates”. Translate between words and symbols:
- (a) $p \wedge (q \rightarrow r)$;
 - (b) $(p \wedge q) \rightarrow r$;
 - (c) Ali passes only if he studies;
 - (d) Ali neither studies nor celebrates.
22. If $p = \text{F}$, $q = \text{T}$, and $r = \text{F}$, determine the truth value of
- (a) $\neg p \wedge q$;
 - (b) $p \vee (q \wedge r)$;
 - (c) $(p \rightarrow q) \wedge (q \rightarrow r)$;
 - (d) $p \leftrightarrow (r \vee \neg q)$.
23. Construct truth tables for
- (a) $\neg p \vee q$;
 - (b) $\neg(p \wedge q)$;
 - (c) $(p \vee q) \wedge \neg q$;
 - (d) $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$.
24. Construct an eight-row truth table for $(p \vee q) \rightarrow r$ and identify all rows on which it is false.
25. Insert parentheses according to the standard order of precedence, then evaluate when $p = \text{T}$, $q = \text{F}$, and $r = \text{T}$:
- (a) $\neg p \vee q \wedge r$;
 - (b) $p \rightarrow q \vee r$;
 - (c) $p \wedge q \leftrightarrow r$.

26. Rewrite “If n is divisible by 12, then n is divisible by 3” using the phrases
- (a) sufficient condition;
 - (b) necessary condition;
 - (c) only if;
 - (d) whenever.
27. Write the converse, inverse, and contrapositive of:
- (a) if n is a multiple of 6, then n is even;
 - (b) if $x > 2$, then $x^2 > 4$.
28. Decide whether each condition is necessary, sufficient, both, or neither for an integer n to be divisible by 6:
- (a) n is divisible by 12;
 - (b) n is divisible by 3;
 - (c) n is divisible by both 2 and 3;
 - (d) n is even.
29. Use standard equivalence laws to simplify:
- (a) $p \wedge (p \vee q)$;
 - (b) $p \vee (\neg p \wedge q)$;
 - (c) $(p \wedge q) \vee (p \wedge \neg q)$;
 - (d) $\neg\neg(p \vee q)$.
30. Apply De Morgan’s laws to negate:
- (a) $p \wedge q \wedge r$;
 - (b) $(p \vee q) \wedge r$;
 - (c) $x > 0$ and $x \leq 5$;
 - (d) n is even or divisible by 3.
31. Rewrite without implication or biconditional symbols:
- (a) $p \rightarrow q$;
 - (b) $p \leftrightarrow q$;
 - (c) $\neg(p \rightarrow q)$;
 - (d) $(p \rightarrow q) \wedge (q \rightarrow r)$.

32. Prove algebraically or by a truth table that

(a) $p \vee (p \wedge q) \equiv p$;

(b) $p \rightarrow q \equiv \neg q \rightarrow \neg p$;

(c) $\neg(p \leftrightarrow q)$ is exclusive or.

33. Classify as tautology, contradiction, or contingency:

(a) $(p \wedge q) \rightarrow p$;

(b) $p \wedge \neg p$;

(c) $(p \rightarrow q) \vee (q \rightarrow p)$;

(d) $(p \vee q) \wedge \neg p$.

34. Determine whether each collection is satisfiable. If it is, give one assignment:

(a) $p \vee q, \neg p$;

(b) $p \rightarrow q, p, \neg q$;

(c) $p \leftrightarrow q, q \leftrightarrow r, \neg p$.

35. Find the truth set in the stated domain:

(a) $P(x) : x^2 = 9, x \in \mathbb{R}$;

(b) $Q(n) : n^2 < n, n \in \mathbb{Z}$;

(c) $R(x) : 1/(x - 2) > 0, x \in \mathbb{R} \setminus \{2\}$.

36. Translate into symbols, specifying each domain:

(a) every real number has a non-negative square;

(b) some integer is its own additive inverse;

(c) no prime number is negative;

(d) exactly one real solution satisfies $x^2 = 0$.

37. Negate and simplify:

(a) $\forall x \in \mathbb{R}, x^2 \geq x$;

(b) $\exists n \in \mathbb{Z}, n^2 + n = 1$;

(c) $\forall x \in A, P(x) \vee Q(x)$;

(d) $\exists x \in A, P(x) \wedge \neg Q(x)$.

38. Express in symbols and negate:

(a) every student solved at least one problem;

- (b) some student solved every problem;
 - (c) no student solved every problem.
39. Over $\mathbb{R}$, determine the truth values of
- (a) $\forall x \exists y, x + y = 0$;
 - (b) $\exists y \forall x, x + y = 0$;
 - (c) $\forall x \exists y, xy = 1$;
 - (d) $\exists y \forall x, xy = 0$.
40. Give a counterexample when possible:
- (a) $\forall x \in \mathbb{R}, x^2 \geq x$;
 - (b) $\forall n \in \mathbb{Z}$, if n^2 is even then n is odd;
 - (c) $\forall a, b \in \mathbb{R}, |a + b| = |a| + |b|$.
41. Identify each valid inference rule or invalid form:
- (a) $p \rightarrow q, p \therefore q$;
 - (b) $p \rightarrow q, \neg q \therefore \neg p$;
 - (c) $p \rightarrow q, q \therefore p$;
 - (d) $p \vee q, \neg p \therefore q$.
42. Test by truth table whether $p \rightarrow q, p \vee r, \neg q \therefore r$ is valid.
43. Derive the conclusion and name the rules used:
- $$p \rightarrow q, \quad q \rightarrow r, \quad p, \quad r \rightarrow s.$$
44. Explain the error in each argument:
- (a) If it rains, the road is wet. The road is wet, so it rained.
 - (b) If n is divisible by 4, it is even. The number is not divisible by 4, so it is odd.
 - (c) Either p or q . We know p , so q is false.
45. Prove directly:
- (a) the sum of two even integers is even;
 - (b) the sum of an odd and an even integer is odd;
 - (c) the square of an odd integer is odd.
46. Prove directly that if $a \mid b$ and $a \mid c$, then $a \mid (mb + nc)$ for all integers m, n .

47. Prove by contrapositive:

- (a) if n^2 is even, then n is even;
- (b) if ab is odd, then both a and b are odd.

48. Prove by contradiction that $\sqrt{2}$ is irrational.

49. Prove by cases that $n(n+1)$ is even for every integer n .

50. Prove both directions of: an integer n is odd if and only if n^2 is odd.

51. Disprove each universal statement with a counterexample:

- (a) every prime number is odd;
- (b) $(a+b)^2 = a^2 + b^2$ for all real a, b ;
- (c) every function with a derivative at a has derivative 0.

52. Translate the set statements into logic:

- (a) $A \subseteq B$;
- (b) $x \in A \cap B$;
- (c) $x \in A \setminus B$;
- (d) $x \notin A \cup B$.

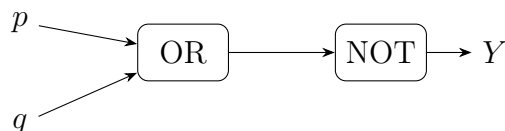
53. Prove using membership logic:

- (a) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$;
- (b) $(A \cap B)^c = A^c \cup B^c$.

54. Let $U = \{1, 2, 3, 4, 5\}$, $A = \{1, 3, 5\}$, and $B = \{2, 3, 4\}$. Determine the truth values of

- (a) $3 \in A \cap B$;
- (b) $1 \in A \setminus B$;
- (c) $5 \in A^c \cup B$;
- (d) $A \subseteq B$.

55. For the circuit below, write the output proposition and construct its truth table.



56. Construct truth tables for the Boolean gates NAND and NOR, and express each using $\wedge$, $\vee$, and $\neg$.
57. A program executes a command when $(p \wedge \neg q) \vee r$ is true. List every input combination that executes the command.
58. Three statements are made:

$$p \vee q, \quad p \rightarrow r, \quad q \rightarrow \neg r.$$

Find all truth assignments satisfying all three.

59. Determine whether each set of premises is consistent:
- (a) $p \rightarrow q, p, \neg q$;
 - (b) $p \vee q, \neg p, \neg q$;
 - (c) $p \rightarrow q, q \rightarrow r, \neg r, \neg p$.
60. From $p \vee q, p \rightarrow r, q \rightarrow r$, derive r and name the argument form.
61. A student writes: “Assume n^2 is even. Then $n^2 = 2k$, so $n = \sqrt{2k}$, which is even.” Explain why this proof is invalid and state a valid proof strategy.
62. For each claim, choose a suitable method from direct proof, proof by contrapositive, proof by contradiction, proof by cases, or a counterexample:
- (a) if n^2 is odd, then n is odd;
 - (b) $\sqrt{5}$ is irrational;
 - (c) $|x| \geq 0$ for all real x ;
 - (d) not every even integer is divisible by 4.
63. Prove the existential statement: there are consecutive positive integers whose squares sum to 25.
64. Prove that the equation $2x + 3 = 11$ has a unique real solution.
65. Prove that n is divisible by 6 if and only if n is divisible by both 2 and 3.
66. Prove by cases that the product of two consecutive integers is divisible by 2.
67. Prove that the sum of a rational number and an irrational number is irrational.
68. Prove that if $0 < a < b$, then $a^2 < b^2$.

69. Prove that $A \subseteq B$ and $B \subseteq C$ imply $A \subseteq C$.
70. If A has n elements, explain logically why its power set has 2^n elements.
71. Express exclusive or in two equivalent ways and verify the equivalence:

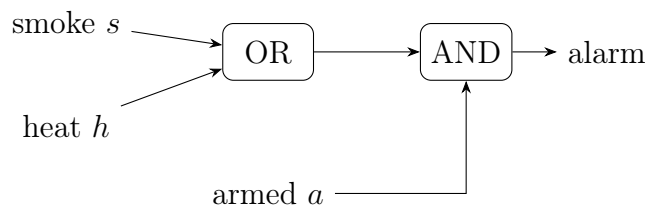
$$(p \vee q) \wedge \neg(p \wedge q), \quad (p \wedge \neg q) \vee (\neg p \wedge q).$$

72. Show that NAND alone can express negation and conjunction.
73. Determine whether $(p \rightarrow q) \wedge (p \rightarrow r)$ and $p \rightarrow (q \wedge r)$ are logically equivalent.
74. Write a conjunctive normal form and a disjunctive normal form for $p \leftrightarrow q$.
75. Test the validity of the argument

$$p \vee q, \quad p \rightarrow r, \quad q \rightarrow s, \quad \neg r, \quad \therefore s.$$

Application-based reasoning

76. A laboratory door opens when a valid card is detected and either a PIN is correct or a supervisor override is active. Let these conditions be c, p , and s . Write and simplify the access rule, then state when access is denied.
77. An alarm sounds according to the diagram.



Write the alarm rule and list the conditions under which it sounds.

78. A database query selects records satisfying $(M \vee P) \wedge \neg A$, where M means Mathematics, P means Physics, and A means absent. Translate the query into words and rewrite it in disjunctive form.
79. A web search for documents containing “calculus” and either “physics” or “engineering”, but not “statistics”, is required. Write a Boolean search expression.
80. A diagnostic rule says: “If a patient has the disease, then the test is positive.” Explain why a positive result alone does not prove the patient has the disease, naming the logical fallacy.

81. A forecast states: “If a storm arrives, the match is cancelled.” The match is cancelled. What, if anything, follows about the storm?
82. A regulation says that being at least 18 is necessary for holding a licence. Express this as an implication and explain why being 18 is not necessarily sufficient.
83. Scholarship eligibility requires a grade of at least 80 and either financial need or a research award. Define propositions and write the eligibility and ineligibility rules.
84. Three examinations A, B, C must satisfy: A is before B , and B is before C . Prove logically that A must be before C and name the inference pattern.
85. A network is operational if the server is working and at least one of two links is active. Write the condition and its negation using De Morgan’s laws.
86. A program accepts an integer n when $n > 0$ and (n is even or n is divisible by 3). Determine whether $n = -6, 2, 9, 25$ are accepted.
87. Write a spreadsheet-style logical formula for awarding “Pass” exactly when a score is at least 50 and attendance is at least 80%. State the negation of the pass condition.
88. Translate and negate: “Every respondent selected at least one of options A, B, and C.”
89. A scientific claim is “If the model is correct, then the observed value lies in the predicted interval.” An observation lies outside the interval. What conclusion follows, assuming the implication?
90. In economics, positive profit is sufficient for revenue to exceed cost. Express this statement and its contrapositive.
91. Use the definition of a tree to assess: “If a graph has $n - 1$ edges, then it is a tree.” State a missing necessary assumption or give a counterexample.
92. A theorem states: differentiability at a implies continuity at a . Classify the following reasoning: “The function is not continuous at a , therefore it is not differentiable at a .”
93. Events A and B satisfy “at least one occurs, but not both.” Express the event using set operations and logical connectives.
94. In a survey of 100 people, 62 use service A , 48 use service B , and 25 use both. Use inclusion–exclusion to decide whether the statement “Everyone uses at least one service” is true.

95. A proof claims that all positive real numbers are equal by dividing both sides of an equation by $a - b$ after assuming $a = b$. Identify the invalid step and explain it logically.
96. An automated rule base contains $p \rightarrow q$, $q \rightarrow r$, and $\neg r$. Determine what it can validly infer about p and q .
97. Design a two-input digital circuit whose output is true exactly when the inputs differ. Give its Boolean expression and truth table.
98. A password is valid when it has at least eight characters, contains a digit, and contains either an uppercase letter or a symbol. Write the validity rule and its negation.
99. A decision system follows this tree: applicants with qualification q proceed; among them, those with experience e or training t are accepted. Express the final acceptance condition and list its minimal satisfying combinations.
100. A project is approved if at least two of the three committee members vote yes. Write a proposition using p, q, r , construct its truth table, and state whether the rule is monotone: can changing a no vote to yes ever change approval to rejection?

Brief answers and guidance are provided in Appendix F.

Chapter 19

Computing with R

Bonus chapter.

Chapter guide

Prerequisites. Basic computer use, arithmetic, functions, probability, statistics, and matrices.

Learning outcomes. Readers should be able to write reproducible R scripts, work with vectors and data frames, import and summarise data, create standard graphics, evaluate distributions and tests, and perform elementary matrix calculations.

R is a programming language and computing environment designed for data analysis, probability, statistics, and graphics. It allows a calculation to be recorded as code, checked, repeated with new data, and shared with another reader. This makes R useful not only as a calculator but also as a tool for reproducible mathematical work (R Core Team 2026).

R and RStudio are not the same product. R is the programming language that performs the computations while RStudio is an integrated development environment (IDE) that provides an editor, console, plot window, file browser, and workspace viewer around R, like Visual Studio Code. Other editors may be used, and R can be run without RStudio. In every case, R itself must be installed before an editor can execute R code.

This chapter uses base R wherever possible. Package-specific commands are identified explicitly, since they work only after the corresponding package has been installed and loaded. Statistical terminology follows the conventions used elsewhere in the book (Moore et al. 2021).

19.1 A reproducible R workflow

Typing a command directly into the console is convenient for a quick check, but a substantial analysis should be written in a script. A script is a plain text record of the commands used, so errors can be corrected without repeating the entire analysis manually.

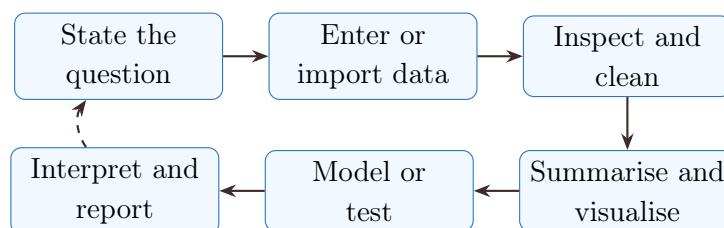


Figure 19.1: A statistical analysis is an iterative, reproducible workflow.

A good script normally contains comments describing the purpose of each stage, uses meaningful object names, keeps the original data unchanged, and ends with an interpretation rather than only a numerical output.

19.2 Objects, assignment, and arithmetic

An R command is evaluated when it is sent to the console. In printed R output, the prompt symbol `>` indicates that R is ready for a command; it is not part of the command itself. The notation `[1]` at the beginning of an output line indicates that the first displayed value is the first element of the result.

The preferred assignment operator is `<-`. An equals sign also works in many assignment contexts, but using `<-` makes the direction of assignment clear.

```
n <- 15
radius <- 4.2
area <- pi * radius^2
area

# Everything after # is a comment.
```

Object names are case-sensitive: `height`, `Height`, and `HEIGHT` are three different names. A name should begin with a letter and may contain letters, digits, underscores, and periods. Names such as `sample_mean` are preferable to vague names such as `x1`.

Useful workspace commands include

Command	Purpose
<code>ls()</code>	list objects in the current workspace
<code>rm(x)</code>	remove the object <code>x</code>
<code>rm(list = ls())</code>	remove every workspace object
<code>class(x)</code>	report the class of <code>x</code>
<code>str(x)</code>	display the internal structure of <code>x</code>
<code>help(mean)</code> or <code>?mean</code>	open documentation for a function
<code>args(mean)</code>	show the principal arguments of a function

R uses the ordinary arithmetic operators `+`, `-`, `*`, `/`, and `^`. Parentheses should be used whenever the intended order might be unclear.

```
(20 - 4) / 2
sqrt(100)
exp(2)
log(10)      # natural logarithm
log10(1000)  # base-10 logarithm
log2(64)
sin(pi / 2)
```

Functions are called by writing their name followed by parentheses. Named arguments, such as `mean = 10`, make a command easier to read and do not depend on the argument order.

19.3 Vectors and indexing

A vector stores values of one basic type. The function `c()` combines values, a colon creates a sequence of consecutive integers, `seq()` creates a more general sequence, and `rep()` repeats values.

```
a <- c(1, 7, 32, 16)
b <- 1:10
c <- 20:15
d <- seq(from = 1, to = 5, by = 0.5)
e <- seq(from = 0, to = 10, length.out = 5)
f <- rep(0, times = 5)
g <- rep(1:3, times = 4)
```

The basic vector types include numeric, character, and logical values. If different types are combined, R converts them to a common type. For example, `c(2, "three")` is a character vector rather than a mixed numeric and character container.

19.3.1 Vectorised calculations

Most arithmetic and mathematical functions act element by element.

```
c(1, 2, 3) + c(4, 5, 6)
sqrt(c(100, 225, 400))
c(2, 4, 6) > 3
```

If vector lengths differ, R may recycle the shorter vector. For example, when a length-four vector is added to `c(10,20)`, R reuses the values as `c(10,20,10,20)`. Recycling is convenient when intentional, but it can hide an error. R warns when the longer length is not a multiple of the shorter length.

19.3.2 Selecting elements

Square brackets select part of a vector. Positive indices keep positions, negative indices omit positions, and logical conditions keep values for which the condition is true.

```
x <- c(4, 9, 1, 7, 12, 3)
x[3]           # third element
x[2:4]         # positions 2 through 4
x[c(1, 6)]     # first and sixth elements
x[-c(2, 5)]    # omit positions 2 and 5
x[x > 5]       # values greater than 5
which(x > 5)   # positions of values greater than 5
```

```
length(x[x > 5])
```

R distinguishes a missing value, written `NA`, from the numerical values zero and `NaN`. A missing value usually propagates through a calculation unless it is removed explicitly.

```
y <- c(4, NA, 8, 10)
is.na(y)
mean(y)           # returns NA
mean(y, na.rm = TRUE) # ignores the missing value
```

19.4 Descriptive statistics

For a numeric vector `x`, base R supplies the most common descriptive statistics directly.

Command	Quantity	Important detail
<code>sum(x)</code>	sum	use <code>na.rm = TRUE</code> if needed
<code>mean(x)</code>	arithmetic mean	affected by extreme values
<code>median(x)</code>	median	resistant to extreme values
<code>min(x), max(x)</code>	extremes	return one value each
<code>range(x)</code>	minimum and maximum	returns a length-two vector
<code>var(x)</code>	sample variance	denominator is $n - 1$
<code>sd(x)</code>	sample standard deviation	square root of <code>var(x)</code>
<code>quantile(x)</code>	sample quantiles	default includes quartiles
<code>summary(x)</code>	compact summary	depends on the object's class

The four panels in Figure 19.2 illustrate the kinds of questions answered by common base-R graphics. A histogram displays a distribution, a boxplot summarises centre and spread, a scatter plot displays association, and a Q–Q plot helps assess a distributional assumption.

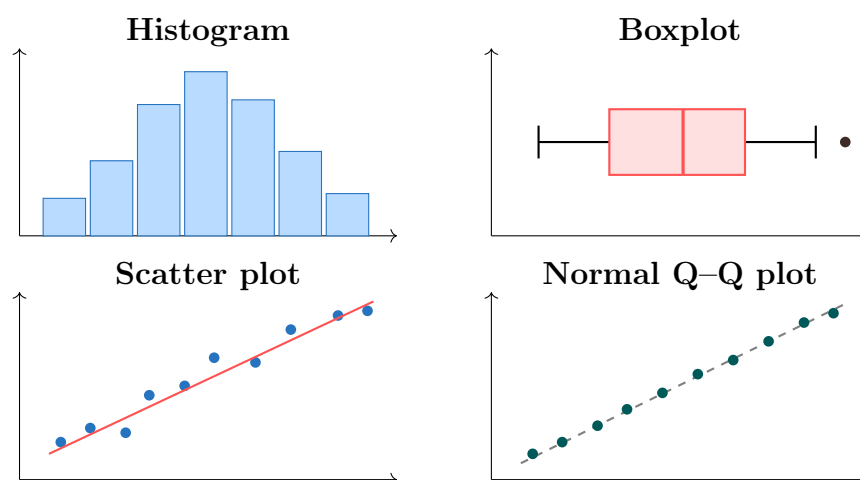


Figure 19.2: Four common graphical outputs produced in a statistical analysis with R.

```
marks <- c(72, 81, 68, 90, 76, 85)
mean(marks)
median(marks)
var(marks)
sd(marks)
sort(marks)
summary(marks)
```

The fact that `var()` uses $n - 1$ is important when comparing R output with a formula or calculator that may instead compute a population variance.

19.5 Data frames and data input

A data frame is a rectangular table in which columns may have different types. Each row usually represents an observational unit, while each column represents a variable.

```
students <- data.frame(
  name = c("Aina", "Ben", "Chen", "Devi"),
  score = c(78, 84, 69, 91),
  group = c("A", "B", "A", "B")
)

students$score
students[2, ]
students[, c("name", "score")]
students[students$score >= 80, ]
```

The commands `head()`, `tail()`, `dim()`, `names()`, `class()`, `str()`, and `summary()` should be used immediately after a data set is created or imported. They catch problems such as an unexpected column name, a numeric column imported as text, or a wrong number of observations.

19.5.1 Importing a comma-separated file

A comma-separated values file can be imported with `read.csv()`. Relative paths make a project easier to move between computers.

```
dahlia <- read.csv("data/dahlia.csv")
head(dahlia)
dim(dahlia)
names(dahlia)
str(dahlia)
summary(dahlia)
```

The interactive command `file.choose()` may be useful during an initial exploration, but a saved script is more reproducible when it contains a stable relative path. Data

can be exported with `write.csv(dahlia, "outputs/dahlia-clean.csv", row.names = FALSE)`.

Character columns representing categories can be converted to factors:

```
dahlia$species <- factor(dahlia$species)
levels(dahlia$species)
table(dahlia$species)
```

19.5.2 Built-in data sets

The command `data()` lists many data sets supplied with R. `data("Nile")` loads the annual Nile-flow time series, and `help(Nile)` describes it. The data set `iris` contains four flower measurements and a species label.

```
data("Nile")
mean(Nile)
sd(Nile)
summary(Nile)

data("iris")
head(iris)
str(iris)
table(iris$Species)
```

19.6 Base R graphics

A graph should answer a question rather than merely decorate an analysis. Titles and axis labels should identify the variables and, where applicable, their units.

Command	Typical purpose
<code>plot(x)</code>	index or time-series plot
<code>plot(x,y)</code>	scatter plot of two quantitative variables
<code>hist(x)</code>	distribution of one quantitative variable
<code>boxplot(x)</code>	centre, spread, and potential outliers
<code>boxplot(y ~ g)</code>	compare a quantitative variable by group
<code>barplot(table(g))</code>	frequencies of categories
<code>qqnorm(x)</code>	normal quantile–quantile plot

```
hist(iris$Sepal.Length,
     main = "Distribution of sepal length",
     xlab = "Sepal length (cm)",
     col = "lightblue", border = "white")

plot(iris$Sepal.Length, iris$Sepal.Width,
```

```

xlab = "Sepal length (cm)",
ylab = "Sepal width (cm)",
pch = 19, col = "navy")

boxplot(Sepal.Length ~ Species, data = iris,
        xlab = "Species", ylab = "Sepal length (cm)")

```

The formula `Sepal.Length ~ Species` is read as “sepal length modelled or grouped by species”. When several plots must be compared, `par(mfrow = c(1, 2))` divides the graphics device into one row and two columns; `par(mfrow = c(1, 1))` restores a single panel.

19.7 Probability distributions in R

R uses a consistent four-function naming convention for probability distributions. The first letter specifies the task and the remainder specifies the distribution.

Prefix	Meaning	Example for a binomial variable
d	probability mass or density	<code>dbinom(x,n,p)</code>
p	cumulative probability $P(X \leq x)$	<code>pbinom(x,n,p)</code>
q	quantile corresponding to a cumulative probability	<code>qbinom(q,n,p)</code>
r	generate random observations	<code>rbinom(m,n,p)</code>

Common suffixes include `binom`, `pois`, `norm`, `unif`, and `exp`. Thus the same pattern applies across the binomial, Poisson, normal, uniform, and exponential distributions.

19.7.1 Discrete distributions

For $X \sim B(5, 0.1)$,

```

dbinom(3, size = 5, prob = 0.1)      # P(X = 3)
pbinom(2, size = 5, prob = 0.1)      # P(X <= 2)
1 - pbinom(1, size = 5, prob = 0.1)  # P(X >= 2)
pbinom(1, 5, 0.1) - pbinom(0, 5, 0.1) # P(X = 1)

```

For $Y \sim \text{Poisson}(3)$, use `dpois()`, `ppois()`, `qpois()`, and `rpois()` in the same way. The complement is often simpler than adding many upper-tail probabilities:

```

1 - ppois(5, lambda = 3) # P(Y > 5)
ppois(5, lambda = 3, lower.tail = FALSE)

```

19.7.2 Continuous distributions

For a continuous variable, `dnorm()` gives a density height rather than the probability of one exact value. Probabilities are obtained from cumulative functions. If $X \sim N(70, 8^2)$, then

```
pnorm(82, mean = 70, sd = 8, lower.tail = FALSE)
pnorm(78, 70, 8) - pnorm(62, 70, 8)
qnorm(0.90, mean = 70, sd = 8)
```

Uniform and exponential calculations use `punif()`, `qunif()`, `pexp()`, and `qexp()`. In R, the exponential parameter is the rate λ , so the mean is $1/\lambda$.

Random simulation should be reproducible. Calling `set.seed()` before a random-number function causes the same pseudorandom sequence to be generated when the script is rerun.

```
set.seed(13)
sample <- rnorm(n = 100, mean = 50, sd = 6)
mean(sample)
hist(sample)
```

19.8 Hypothesis testing in R

R performs arithmetic correctly, but it does not choose the hypotheses or interpret the scientific question. A complete test should include:

1. the null and alternative hypotheses;
2. the significance level and assumptions;
3. the test statistic and its reference distribution;
4. the p-value or critical-value decision;
5. a conclusion stated in the context of the problem.

A small p-value is evidence against H_0 . Failing to reject H_0 does not prove that H_0 is true; it means that the observed sample does not provide sufficient evidence against it at the chosen significance level.

19.8.1 A one-sample z-test

If the population standard deviation σ is known and the normal or large-sample conditions are appropriate, the statistic is

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}.$$

Base R can calculate the statistic and p-value directly. For a sample of 40 GPAs with $\bar{x} = 3.05$, known $\sigma = 0.5$, and null mean 3.2:


```
xbar <- 3.05
mu0 <- 3.2
sigma <- 0.5
n <- 40

z <- (xbar - mu0) / (sigma / sqrt(n))
p_value <- 2 * pnorm(-abs(z))
c(z = z, p_value = p_value)
```

This gives $z \approx -1.897$ and $p \approx 0.0578$. At the 5% level there is insufficient evidence that the mean differs from 3.2. The package-specific command `BSDA::z.test()` offers a convenient alternative when the BSDA package is available, but the manual base-R calculation makes the formula and tail choice explicit.

19.8.2 A one-sample t-test

When σ is unknown, replace it by the sample standard deviation and use a t distribution with $n - 1$ degrees of freedom. The base-R function is `t.test()`.

```
plants <- c(14, 14, 16, 13, 12, 17,
           15, 14, 15, 13, 15, 14)

result <- t.test(plants, mu = 15,
                 alternative = "two.sided",
                 conf.level = 0.95)

result
result$statistic
result$p.value
result$conf.int
```

The sample mean is approximately 14.333, the test statistic is $t \approx -1.685$ with 11 degrees of freedom, and $p \approx 0.120$. Hence the sample does not provide sufficient evidence at the 5% level that the population mean differs from 15. The corresponding 95% confidence interval is approximately (13.462, 15.204).

The alternative argument controls the direction:

$$\begin{aligned} \text{alternative} = \text{"two.sided"} &\iff H_1 : \mu \neq \mu_0, \\ \text{alternative} = \text{"less"} &\iff H_1 : \mu < \mu_0, \\ \text{alternative} = \text{"greater"} &\iff H_1 : \mu > \mu_0. \end{aligned}$$

The direction must be selected from the research question before examining the sample result.

19.8.3 Checking normality

For a small-sample t procedure, the population distribution should be reasonably normal and should not contain severe outliers. The Shapiro–Wilk test and graphical checks provide complementary information.

```
shapiro.test(plants)

par(mfrow = c(1, 2))
hist(plants, main = "Plant heights", xlab = "Height")
qqnorm(plants)
qqline(plants, col = "red", lwd = 2)
par(mfrow = c(1, 1))
```

For the Shapiro–Wilk test, H_0 states that the population is normal. A p-value below the chosen level provides evidence against normality. A larger p-value does not prove normality, especially with a small sample, so the histogram, quantile–quantile plot, sample size, and data-collection process must also be considered. In base R, `shapiro.test()` accepts sample sizes from 3 to 5000.

19.9 Matrices in R

R stores a matrix as a vector with row and column dimensions. Unless `byrow = TRUE` is specified, `matrix()` fills entries down the columns.

```
A <- matrix(1:12, nrow = 3, ncol = 4)
B <- matrix(1:12, nrow = 3, ncol = 4, byrow = TRUE)
A
B
```

Vectors may be combined by rows or columns.

```
a <- c(1, 2, 3)
b <- c(10, 20, 30)
c <- c(100, 200, 300)

Rmat <- rbind(a, b, c)
Cmat <- cbind(a, b, c)
```

For `M[i, j]`, the index before the comma selects rows and the index after the comma selects columns. Leaving an index empty selects every row or column.

```
A[2, 3]
A[1:2, ]
A[, c(1, 3)]
A[-1, , drop = FALSE]
rownames(A) <- c("r1", "r2", "r3")
colnames(A) <- c("c1", "c2", "c3", "c4")
```

The argument `drop = FALSE` preserves a matrix structure when a subset would otherwise simplify to a vector.

19.9.1 Matrix operations

R distinguishes elementwise multiplication from matrix multiplication:

Command	Operation
<code>A + B</code> , <code>A - B</code>	elementwise addition or subtraction
<code>k * A</code>	scalar multiplication
<code>A * B</code>	elementwise multiplication
<code>A %*% B</code>	matrix product
<code>t(A)</code>	transpose
<code>diag(A)</code>	diagonal entries
<code>det(A)</code>	determinant of a square matrix
<code>solve(A)</code>	inverse of a nonsingular square matrix
<code>solve(A,b)</code>	solution of $Ax = b$

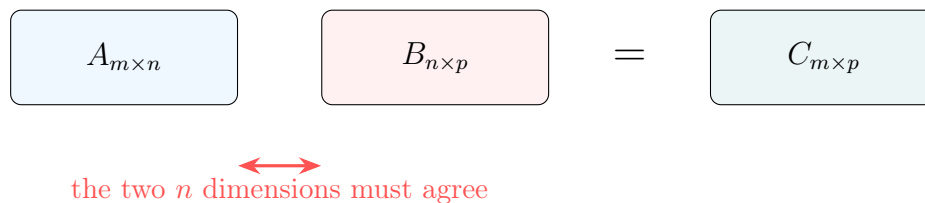


Figure 19.3: The inner dimensions must agree in a matrix product.

Summary functions may be applied across rows or columns. In `apply(M, margin, function)`, margin 1 means rows and margin 2 means columns.

```
sum(A)
apply(A, 1, sum)
apply(A, 2, mean)
rowSums(A)
colMeans(A)
```

For a square matrix, `solve(A)` should not be interpreted as dividing by A . It computes an inverse numerically when the matrix is nonsingular. Solving a system with `solve(A,b)` is generally preferable to forming `solve(A) %*% b` explicitly.

Example 19.1: solving a linear system

Consider

$$\begin{aligned}x + 3y + 5z &= 3, \\2x + 4y + 6z &= 7, \\3x + 2y + 4z &= 1.\end{aligned}$$

```
coef <- matrix(c(1, 3, 5,
                 2, 4, 6,
                 3, 2, 4),
              nrow = 3, byrow = TRUE)
rhs <- c(3, 7, 1)
solve(coef, rhs)
```

The solution is $(x, y, z) = (2/3, 43/6, -23/6)$.

19.10 Two complete data examples

19.10.1 Flower measurements

The following base-R script performs the principal tasks from import through description and visualisation using the built-in `iris` data. It can be adapted to an imported flower data set by replacing `iris` with the imported data-frame name and adjusting the column names.

```
data("iris")

# Inspect the data
head(iris)
dim(iris)
names(iris)
str(iris)
summary(iris)

# Summarise and compare variables
sd(iris$Sepal.Length)
table(iris$Species)
aggregate(Sepal.Length ~ Species, data = iris, mean)

# Visualise
hist(iris$Sepal.Length,
     xlab = "Sepal length (cm)",
     main = "Iris sepal lengths")
plot(iris$Sepal.Length, iris$Sepal.Width,
     xlab = "Sepal length (cm)",
     ylab = "Sepal width (cm)")
boxplot(Sepal.Length ~ Species, data = iris,
```

```
xlab = "Species", ylab = "Sepal length (cm)")
```

19.10.2 A time series

The Nile data illustrate the difference between a distribution plot and a time plot. A histogram shows how flow values are distributed; a line plot preserves their chronological order.

```
data("Nile")
summary(Nile)

par(mfrow = c(1, 2))
plot(Nile, ylab = "Annual flow", main = "Nile time series")
hist(Nile, xlab = "Annual flow", main = "Flow distribution")
par(mfrow = c(1, 1))
```

These plots answer different questions and should not be treated as interchangeable. The time plot can reveal changes, cycles, or unusual years; the histogram summarises shape, centre, and spread without retaining time order.

19.11 Good practice and common errors

1. Keep code in a script and use comments to state the purpose of each step.
2. Use relative file paths and keep raw data separate from generated output.
3. Inspect imported data before calculating statistics.
4. Record a random seed whenever simulation is part of the analysis.
5. Distinguish `=` from `==`: the first is commonly used for assignment or function arguments, while the second tests equality.
6. Use `&` and `|` for elementwise logical conditions, and use parentheses to make compound conditions explicit.
7. Check missing values rather than allowing `NA` to propagate unnoticed.
8. For probabilities, identify whether the event is exact, cumulative, an interval, or a complement before choosing a command.
9. For hypothesis tests, choose the alternative hypothesis before examining the p-value and report a contextual conclusion.
10. For matrices, check dimensions and distinguish `*` from `%*%`.

Chapter summary and common pitfalls

Key ideas. A reproducible analysis records data preparation, calculation, graphics, and interpretation in a script. R operates on typed objects, often applies functions elementwise, and uses specialised functions for probability and inference.

Common pitfalls. Distinguish assignment from comparison, inspect missing values and data types, set working paths explicitly, read warning messages, and interpret statistical output rather than reporting it without context.

19.12 Practice and exercises

Unless an exact value is requested, give numerical results to at least four significant figures. Write executable base-R code and briefly interpret every statistical result.

R basics and vectors

1. State whether each statement is true or false, correcting each false statement.
 - (a) RStudio is the language that performs R calculations.
 - (b) R object names are case-sensitive.
 - (c) The symbol `#` begins a comment.
 - (d) The prompt `>` must be typed at the start of each command.
2. Write R commands to evaluate
 - (a) $(18 - 6)/3 + 5^2$;
 - (b) $\sqrt{225} + \log_{10}(1000)$;
 - (c) e^3 ;
 - (d) $\cos \pi + \sin(\pi/2)$.
3. Create objects named `radius` and `height` containing 3.5 and 8. Use them to calculate the volume $\pi r^2 h$ of a cylinder. Add a suitable comment and then remove `height` from the workspace.
4. Construct each vector without typing every element individually.
 - (a) 5, 10, 15, ..., 160;
 - (b) 87, 86, 85, ..., 56;
 - (c) five repetitions of -2 ;
 - (d) 1, 2, 3, 1, 2, 3, 1, 2, 3.
5. Let `a <- c(2,5,8,11)` and `b <- c(1,2,3,4)`. Find `a+b`, `a*b`, `a^2`, and `sqrt(a+b)`.

6. For `x <- c(4,9,1,7,12,3)`, predict the output of `x[2:4]`, `x[c(1,6)]`, `x[x>5]`, and `x[-c(2,5)]`.
7. For the vector in Question 6, write commands to find the positions of values greater than 5, the number of such values, and their mean.
8. Let `y <- c(4,NA,8,10)`. Explain the output of `mean(y)`, then find the mean of the observed values and the position of every missing value.
9. For $x = (3, 5, 5, 7, 10)$, use R to find the sum, mean, median, range, sample variance, and sample standard deviation.
10. Predict the result of `c(1,2,3,4)+c(10,20)`. Explain vector recycling and give an example that produces a recycling warning.

Data frames, input, and graphics

11. Create a data frame containing the names Aina, Ben, and Chen; scores 78, 84, 69; and groups A, B, and A. Extract the score column, the second row, and the rows with scores at least 75.
12. Load `iris`. Write commands to display its first six rows, dimensions, column names, structure, summary, and species frequencies.
13. Write code to import `data/dahlia.csv` as `dahlia`, inspect it with five appropriate commands, and export it to `outputs/dahlia-copy.csv` without row numbers.
14. Suppose `results` has columns `name`, `score`, and `attendance`. Write base-R code to select students having `score >= 50` and `attendance >= 80`, retaining only their names and scores.
15. State which graph is most appropriate for each purpose and give the corresponding base-R command: the distribution of one quantitative variable, the relationship between two quantitative variables, a quantitative variable compared across categories, and a chronological time series.
16. Using `Nile`, write code to calculate its mean, standard deviation, and summary, then display a time plot and histogram side by side. Explain what information is lost in the histogram.

Probability calculations

17. Explain the roles of the prefixes `d`, `p`, `q`, and `r` in R distribution functions.
18. If $X \sim B(8, 0.2)$, use R to find $P(X = 2)$, $P(X \leq 2)$, and $P(X \geq 1)$.

19. If $X \sim B(5, 0.1)$, find the probabilities of exactly three, more than one, fewer than two, at least two, and at most two successes.
20. If $Y \sim \text{Poisson}(3)$, find $P(Y = 0)$, $P(Y \leq 2)$, and $P(Y \geq 1)$.
21. If $X \sim N(70, 8^2)$, find $P(X > 82)$, $P(62 < X < 78)$, and the 90th percentile.
22. If $U \sim U(2, 10)$, find $P(4 < U < 7)$ and the mean. Write R code to generate 200 observations from this distribution.
23. If T is exponential with rate 0.25, find $P(T > 6)$ and the median. Give the appropriate R commands.
24. Explain why `set.seed(13)` is useful before a simulation. Write code to generate 1000 binomial observations with $n = 20$ and $p = 0.3$, then compare their sample mean with the theoretical mean.

Hypothesis testing

25. List the principal stages of a hypothesis test and explain why a large p-value does not prove the null hypothesis.
26. A manager claims that the mean hourly wage is below RM9.78. A sample of 40 employees has mean RM9.60, and the known population variance is 2.20. Perform the one-sided z-test at $\alpha = 0.10$ using base R and state the conclusion.
27. A sample of 40 GPAs has mean 3.05. Test $H_0 : \mu = 3.2$ against $H_1 : \mu \neq 3.2$ when the population standard deviation is 0.5. Give the z statistic, p-value, and conclusion at the 5% level.
28. For the plant heights

14, 14, 16, 13, 12, 17, 15, 14, 15, 13, 15, 14,

use `t.test()` to test $H_0 : \mu = 15$ against a two-sided alternative. Report the sample mean, test statistic, degrees of freedom, p-value, 95% confidence interval, and conclusion.

29. Write the `t.test()` syntax for two-sided, lower-tailed, and upper-tailed tests of a vector `x` against `mu0`.
30. The heights

165, 172, 168, 174, 169, 171, 175, 167, 170, 173, 166, 168, 172, 169, 171

are tested against a population mean of 170 cm. Perform a two-sided one-sample t-test and interpret the result.

31. Give R commands for a Shapiro–Wilk test, histogram, and normal quantile–quantile plot with reference line. State the null hypothesis of the Shapiro–Wilk test and explain why graphical checks remain necessary.

Matrices and integrated applications

32. Let `A<-matrix(1:12,nrow=3,ncol=4)`. Write out A , find `A[2,3]`, extract its second row, and extract columns 1 and 4.
33. Compare `matrix(1:6,nrow=2)` with `matrix(1:6,nrow=2,byrow=TRUE)`.
34. Let `a <- c(1,2,3)` and `b <- c(10,20,30)`. Find `rbind(a,b)` and `cbind(a,b)` and state their dimensions.

35. Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix}.$$

Construct them in R and compare `A*B` with `A%*%B`.

36. For the matrix A in Question 35, find A^T , $\det A$, and A^{-1} using R.

37. Use `solve()` to solve

$$x + 3y + 5z = 3, \quad 2x + 4y + 6z = 7, \quad 3x + 2y + 4z = 1.$$

38. Let `C <- matrix(1:9,nrow=3)`. Find its row sums and column sums using both `apply()` and the specialised functions `rowSums()` and `colSums()`.
39. Using `iris`, produce a compact analysis containing inspection commands, the standard deviation of sepal length, species counts, mean sepal length by species, a histogram, a scatter plot, and a grouped boxplot.
40. A matrix stores two test scores for three students:

$$S = \begin{pmatrix} 72 & 85 \\ 91 & 68 \\ 88 & 79 \end{pmatrix}.$$

Construct S , label its rows and columns, calculate each student's mean and each test's mean, and identify the student with the greatest average.

Data handling and exploratory programming

41. Predict the values and classes produced by each expression. Explain any type coercion.

(a) `c(2,5.5)`;

- (b) `c(2, "5");`
- (c) `c(TRUE, 4);`
- (d) `as.numeric(c("3", "7"))`.

42. Let `x <- c(-2, 0, 3, 7)`. Predict the outputs of `x > 0 & x < 5` and `x == 0 | x == 7`. Explain the difference between `=`, `==`, `&`, and `|`.

43. Create the named vector

```
marks <- c(Aina=72, Ben=81, Chen=76, Devi=90).
```

Select Ben's mark by name, select all marks exceeding 75, and return the names of the students whose marks exceed 75.

44. For `z <- c(8, 3, 8, 1, 5)`, predict `sort(z)`, `unique(z)`, and `order(z)`. Explain what the last result represents.

45. Let `x <- c(12, NA, 15, 18, NA, 10)`. Write code to count the missing values, extract the observed values, calculate their mean, and replace each missing value by that mean without altering the original vector.

46. Compare the mean and median of 10, 11, 11, 12, 12 before and after adding the value 100. Which measure is more resistant to the extreme observation?

47. For $x = (2, 4, 5, 7, 9, 12, 15)$, use R's default quantile convention to find the first quartile, median, third quartile, and interquartile range.

48. Create a data frame with products A, B, C; prices 12, 8, 15; and quantities 5, 9, 4. Add a revenue column, calculate total revenue, and select products generating at least RM 65 revenue.

49. Convert `c("A", "B", "A", "C", "B", "A")` to a factor with levels A, B, C. Produce frequency and relative-frequency tables and identify the modal level.

50. Scores 70, 80, 75, 90, 85, 65 belong respectively to groups A, A, B, B, B, A. Create a data frame and use `aggregate()` to find the mean score of each group.

51. A data frame has names A–E, scores 45, 72, 88, 61, 90, and attendance percentages 95, 78, 85, 92, 76. Write code to select

- (a) students with score at least 50 and attendance at least 80;
- (b) students with score above 80 or attendance above 90.

52. Write base-R code for a 2×2 graphics layout containing a histogram of `x`, boxplot of `x`, scatter plot of `x` against `y`, and bar chart of `table(g)`. Include informative labels and restore the one-panel layout afterwards.
53. Explain how the choice of histogram breaks can change the visual impression of a distribution. Write two commands that draw histograms of `x` using 5 and 20 breaks.
54. For `x <- c(10,11,12,12,13,14,30)`, use the 1.5 IQR rule to identify any boxplot outlier. Give an R command that returns the flagged value directly.
55. For yearly values `y <- c(10,12,11,15,18)`, write code for a labelled line plot with visible points. State the overall pattern shown.

Probability, simulation, and statistical inference

56. If $X \sim B(20, 0.3)$, use `qbinom()` to find the smallest integer k satisfying $P(X \leq k) \geq 0.90$. Verify the answer using two cumulative probabilities.
57. If $X \sim B(12, 0.25)$, find $P(X = 4)$, $P(X \leq 3)$, and $P(X \geq 1)$. State the mean and standard deviation.
58. If $Y \sim \text{Poisson}(4)$, find $P(3 \leq Y \leq 7)$, $P(Y > 6)$, and the 95th percentile.
59. If $X \sim N(100, 15^2)$, find the central 95% interval, $P(|X - 100| < 15)$, and the R commands used.
60. If $U \sim U(5, 20)$, find its first quartile, third quartile, and 90th percentile. Generate 500 observations reproducibly.
61. If T is exponential with rate 0.2, find $P(T > 10)$ and the 90th percentile. Use the memoryless property to find $P(T > 15 \mid T > 5)$.
62. Simulate 10,000 rolls of a fair die and estimate the probability of an even result. Compare the estimate with the theoretical value.
63. Use independent uniform observations on $[-1, 1]^2$ to estimate π from the proportion satisfying $x^2 + y^2 \leq 1$. Set a seed and explain why the estimator is multiplied by 4.
64. A sample of size 64 has mean 50, and the known population standard deviation is 8. Construct a manual 95% z-confidence interval for the population mean using `qnorm()`.
65. For a calculated z statistic of 1.75, find the lower-tailed, upper-tailed, and two-sided p-values. Give the corresponding R commands.

66. For `x <- c(21,23,20,24,22,25,19,23)`, use `t.test()` to construct a 95% confidence interval for the population mean. Report the sample mean and standard deviation.
67. Explain the decisions at significance levels 0.05 and 0.01 if a test returns (a) $p = 0.03$ and (b) $p = 0.20$. Distinguish statistical significance from practical importance.
68. A Shapiro–Wilk test gives $p = 0.03$ and the Q–Q plot shows marked curvature. What conclusion is appropriate at the 5% level? How would your interpretation differ if $p = 0.40$ but the plot showed one severe outlier?

Matrix computing and diagnostic reasoning

69. If A is 3×2 and B is 2×4 , state the dimensions of $A \% * \% B$ and explain why $B \% * \% A$ is undefined. When is $A*B$ permitted?
70. Construct the 3×3 identity matrix with `diag()` and verify in R that multiplying a compatible matrix by it leaves the matrix unchanged.
71. Let

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} 5 & 1 \\ 7 & 0 \end{pmatrix}.$$

Use one call to `solve(A,B)` to solve both systems represented by the columns of B .

72. For $M = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$, calculate the determinant and predict what `solve(M)` does. Explain the mathematical reason.
73. Create the matrix $M = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$ by rows, label its rows `r1–r3` and columns `c1–c3`, then extract rows `r1`, `r3` and column `c2` while preserving matrix form.
74. The transformation matrix $T = \begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix}$ acts on the point $(3, 4)^T$. Calculate the image using `%*%` and interpret the horizontal and vertical scale factors.
75. Correct each faulty fragment and explain the error.
- (a) `Mean(x);`
 - (b) `x = = 3;`
 - (c) `mean(x,na.rm=TURE);`
 - (d) `A*B`, when a matrix product was intended.

Applications and mini-projects

76. A monthly budget has categories rent, food, transport, utilities, and leisure with costs RM 1,200, RM 450, RM 300, RM 250, and RM 180. Create a named vector, find the total, calculate each percentage share, and identify the largest category.
77. Room temperatures are 21.4, 22.1, NA, 23.0, 22.7, 21.9, NA. Count the missing readings, calculate the observed mean, and create a cleaned copy in which missing values are replaced by that mean. Explain why the raw vector should be retained.
78. Create a student data frame with names Aina, Ben, Chen, Devi, Ethan; scores 78, 49, 88, 65, 91; and attendance 85, 92, 76, 81, 95. A student passes only when score is at least 50 and attendance at least 80. List the students who pass and find the pass rate.
79. Shaft diameters in millimetres are 11.98, 12.01, 12.04, 11.97, 12.00, 12.02, 12.06, 11.99. The tolerance interval is [11.97, 12.03]. Find the mean, standard deviation, passing measurements, and passing proportion.
80. Clinic waiting times in minutes are 6, 8, 5, 7, 12, 9, 4, 15, 8, 6. Calculate descriptive statistics, draw a histogram and boxplot side by side, and identify any value flagged by the 1.5 IQR rule.
81. Regional sales are 120, 135, 128, 142, 150, 138 for regions North, North, South, South, East, East. Use a data frame and `aggregate()` to calculate regional means and totals, then display the totals in a bar chart.
82. Annual rainfall values are 120, 135, 128, 142, 150, 138. Plot them in chronological order and as a histogram. Identify the maximum and explain which plot preserves the ordering of years.
83. In a poll, each of 20 independent respondents supports a proposal with probability 0.58. Use R to find the probability of at least 12 supporters and the expected number of supporters.
84. Manufacturing defects occur at a mean rate of 2.5 per batch. Find the probability of between two and five defects inclusive and the probability of no defect.
85. Delivery times are normally distributed with mean 35 minutes and standard deviation 4 minutes. Find the probability of delivery within 40 minutes, the probability of exceeding 42 minutes, and the 95th percentile.
86. Service time is exponential with rate 0.15 per minute. Find the probability that service exceeds 8 minutes and the median service time. Simulate and plot 1000 service times reproducibly.

87. Ten independent components each work with probability 0.9. Find the probability that at least nine work. Simulate 5000 systems and compare the simulated reliability with the exact result.
88. A wage survey of 50 workers has mean RM 412 with known population standard deviation RM 35. Test $H_0 : \mu = 400$ against $H_1 : \mu > 400$ at the 5% level using a manual z-test.
89. Battery lives in hours are 9.8, 10.1, 9.9, 10.3, 10.0, 9.7, 10.2, 10.1, 9.9, 10.4. Use a two-sided one-sample t-test to assess a claimed mean of 10 hours. Report the confidence interval and contextual conclusion.
90. Plant lengths are 12.1, 12.4, 12.0, 12.3, 12.2, 12.5, 15.8 cm. Write a normality-checking workflow and explain how the value 15.8 should influence the choice and interpretation of a small-sample t procedure.
91. Products A and B require respectively (2, 5) and (3, 4) units of labour and material. If 100 units of A and 80 units of B are planned, use matrix multiplication to find total labour and material requirements.
92. Use the rotation matrix $R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ to rotate the points (1, 0), (2, 1), and (-1, 3) through 90° anticlockwise. Store the points as columns of one matrix and perform a single matrix product.

93. Let

$$A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

be a network adjacency matrix. Calculate A^2 in R and interpret its diagonal and off-diagonal entries as two-step walks.

94. Four students have three test scores:

$$\begin{pmatrix} 72 & 80 & 76 \\ 88 & 91 & 85 \\ 65 & 70 & 68 \\ 90 & 87 & 93 \end{pmatrix}.$$

Label the matrix, calculate student and test means, and identify the student with the greatest mean.

95. Write a reproducible CSV workflow that imports `data/survey.csv`, inspects names and types, reports missing-value counts for every column, and exports a cleaned object to `outputs/survey-clean.csv` without row names.

96. Using Figure 19.1, write a seven-point checklist for a reproducible analysis report, including the research question, data source, cleaning decisions, seed, numerical summaries, graphics, and contextual conclusion.
97. Using `iris`, calculate mean sepal length by species, identify the species with the greatest mean, and create a grouped boxplot and labelled scatter plot of sepal length against sepal width.
98. Using `Nile`, compare the mean of the first 50 observations with the mean of the last 50. Draw the full time plot with horizontal lines for both means and explain why this comparison cannot be recovered from a histogram alone.
99. Temperatures and daily energy use are

(22, 18), (24, 20), (26, 23), (28, 27), (30, 31), (32, 36), (34, 42).

Create the vectors, reproduce the scatter plot in Figure 19.4, label both axes, calculate the mean energy use, and identify the temperature associated with maximum use. Describe the relationship without claiming causation.

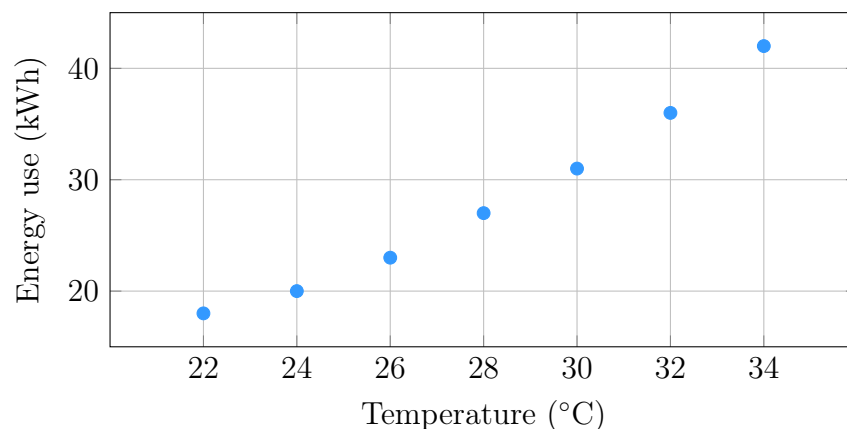


Figure 19.4: Temperature and daily energy use for Question 99.

100. Package weights, in grams, are

48.2, 50.1, NA, 49.7, 51.0, 48.9, 50.4,
49.5, 50.2, 48.7, 51.3, NA.

Build a complete base-R analysis that

- (a) preserves the raw vector and extracts observed values;
- (b) reports the missing count, mean, and standard deviation;
- (c) draws a histogram and Q-Q plot;
- (d) tests $H_0 : \mu = 50$ against a two-sided alternative at 5%;

- (e) reports the confidence interval and a contextual conclusion.

Brief answers and representative R code are provided in Appendix F.

Appendix A

Notation Glossary

This glossary collects notation used repeatedly throughout the book. It is intended as a quick reference rather than a substitute for the definitions in the relevant chapters. Some symbols are context-dependent; the surrounding text should always be checked when a symbol admits more than one meaning. Appendix B gives a more detailed account of set notation.

Notation and unit conventions

Variables and scalar quantities are printed in italic type, while named functions and unit symbols are upright. Vectors are written in bold lower-case type, such as **a**, and matrices are represented by upper-case letters, such as *A*. Directed segments retain the arrow notation $\overrightarrow{AB}$.

For complex numbers, this book uses lower-case letters such as *z* and *w*, with $\text{Re}(z)$, $\text{Im}(z)$, and $\bar{z}$ for the real part, imaginary part, and conjugate. Physical quantities place a space between the numerical value and the upright unit symbol, for example 5 m, 12 m s⁻¹, and 20 °C. Plane angles use 30° without an intervening space. Malaysian currency is written as RM 5,000. Decimal points are used, and digits in numbers of four or more digits are grouped in threes with a comma. SI names and symbols follow the SI Brochure (Bureau International des Poids et Mesures 2026).

General notation

Symbol	Name or reading	Meaning or use
$=$	Equals	The two expressions represent the same value or object.
$\neq$	Not equal to	The two expressions do not represent the same value or object.
$\approx$	Approximately equal to	Equality after rounding or numerical approximation.
$\propto$	Proportional to	One quantity is a constant multiple of another.
$ x $	Absolute value or modulus	Distance of the real number x from zero.
$n!$	Factorial	$n! = n(n-1) \cdots 2 \cdot 1$, with $0! = 1$.
$\binom{n}{r}$	Binomial coefficient	Number of ways to choose r objects from n without regard to order.
$\sum$	Summation	Adds a sequence of terms over a stated index range.
$\prod$	Product notation	Multiplies a sequence of factors over a stated index range.
Δ	Delta or change	A finite change; it also denotes a discriminant when explicitly defined.

Sets and logic

Symbol	Name or reading	Meaning or use
$\mathbb{N}, \mathbb{W}$	Natural and whole numbers	The convention in this book is $\mathbb{N} = \{1, 2, 3, \dots\}$ and $\mathbb{W} = \{0, 1, 2, \dots\}$.
$\mathbb{Z}, \mathbb{Q}$	Integers and rational numbers	Standard number sets.
$\mathbb{R}, \mathbb{C}$	Real and complex numbers	The principal scalar number systems used in this book.
$x \in A$	Is an element of	The object x belongs to the set A .
$A \subseteq B$	Is a subset of	Every element of A is also an element of B .
$A \cup B, A \cap B$	Union and intersection	Elements in either set; elements common to both sets.
$A \setminus B$	Set difference	Elements in A that are not in B .
A^c	Complement	Elements in the universal set that are not in A .
$\emptyset$	Empty set	The set containing no elements.
$\forall, \exists, \exists!$	For all; there exists; there exists uniquely	Quantifiers used in mathematical statements.
$P \Rightarrow Q$	Implies	Whenever P is true, Q must also be true.
$P \Leftrightarrow Q$	If and only if	Each statement implies the other.
$\neg P, P \wedge Q, P \vee Q$	Not; and; or	Common operations of propositional logic.

Functions and algebra

Symbol	Name or reading	Meaning or use
$f: X \rightarrow Y$	Function from X to Y	X is the domain and Y is the codomain.
$f(x)$	Value of f at x	The output assigned to the input x .
$D(f), R(f)$	Domain and range	The permitted input set and the set of attained outputs.
$f \circ g$	Composition	$(f \circ g)(x) = f(g(x))$; the function g acts first.
f^{-1}	Inverse function	Reverses a one-to-one function; it does not mean $1/f$.
$f^{\circ n}$	n th iterate	The function f composed with itself n times.
$\deg p$	Degree of a polynomial	Highest exponent with a non-zero coefficient.
$\Delta = b^2 - 4ac$	Quadratic discriminant	Determines the nature of the roots of $ax^2 + bx + c = 0$.
$\log_b x, \ln x$	Logarithm; natural logarithm	Logarithm to base b ; logarithm to base e .
e	Euler's number	The base of the natural logarithm, approximately 2.71828.

Trigonometry and complex numbers

Symbol	Name or reading	Meaning or use
$\sin \theta, \cos \theta, \tan \theta$	Trigonometric functions	Sine, cosine, and tangent of the angle θ .
$\sin^{-1} x, \cos^{-1} x, \tan^{-1} x$	Inverse trigonometric functions	Principal angles whose trigonometric values equal x .
$\sinh x, \cosh x, \tanh x$	Hyperbolic functions	Hyperbolic sine, cosine, and tangent.
i	Imaginary unit	The complex number satisfying $i^2 = -1$.
$\bar{z}$	Complex conjugate	If $z = a + bi$, then $\bar{z} = a - bi$.
$ z $	Modulus of z	Distance of z from the origin in the Argand plane.
$\arg z, \operatorname{Arg} z$	Argument; principal argument	An angle of z ; its chosen principal value.
$\operatorname{cis} \theta$	Cosine-sine form	$\operatorname{cis} \theta = \cos \theta + i \sin \theta$.

Calculus and differential equations

Symbol	Name or reading	Meaning or use
$\lim_{x \rightarrow a} f(x)$	Limit	Value approached by $f(x)$ as x approaches a .
$f'(x), \frac{dy}{dx}$	First derivative	Instantaneous rate of change of a function.
$f''(x), \frac{d^2y}{dx^2}$	Second derivative	Rate of change of the first derivative.
$\frac{d^ny}{dx^n}$	n th derivative	Higher-order derivative with respect to x .
$\int f(x) dx$	Indefinite integral	Family of antiderivatives of f .
$\int_a^b f(x) dx$	Definite integral	Signed accumulation of f from a to b .
C	Constant of integration	Arbitrary constant appended to an indefinite integral.
y', y''	Derivatives in differential equations	First and second derivatives of the dependent variable.
$\mu(x)$	Integrating factor	Multiplier used to solve a linear first-order differential equation.

Sequences and series

Symbol	Name or reading	Meaning or use
u_n, a_n	n th term	A term of a sequence indexed by n .
S_n	Partial sum	Sum of the first n terms of a series.
$\sum_{n=1}^{\infty} a_n$	Infinite series	Limit of its sequence of partial sums, when that limit exists.
$T_n(x)$	Taylor polynomial	Degree- n polynomial approximation to a function.
$R_n(x)$	Remainder or error term	Difference between a function and its degree- n approximation.

Probability and statistics

Symbol	Name or reading	Meaning or use
$P(A)$	Probability of A	Number between 0 and 1 measuring the likelihood of event A .
$X \sim \mathcal{D}$	Is distributed as	The random variable X follows distribution $\mathcal{D}$.
$E(X), \text{Var}(X)$	Expectation and variance	Mean and spread of a random variable.
μ, σ, σ^2	Population parameters	Population mean, standard deviation, and variance.
$\bar{x}, s, s^2$	Sample statistics	Sample mean, standard deviation, and variance.
H_0, H_1	Null and alternative hypotheses	Competing statements in a hypothesis test.
α	Significance level	Probability threshold used when testing H_0 .
p -value	Observed significance	Probability, assuming H_0 , of a result at least as extreme as the observation.
Z, T	Standardised test statistics	Statistics compared with the standard normal or Student t distribution.

Matrices and vectors

Symbol	Name or reading	Meaning or use
A^T	Transpose of A	Matrix obtained by interchanging rows and columns.
$\det A$, $ A $	Determinant	Scalar associated with a square matrix.
A^{-1}	Matrix inverse	Matrix satisfying $AA^{-1} = A^{-1}A = I$, when it exists.
I , O	Identity and zero matrices	Multiplicative identity and additive identity for matrices.
$\mathbf{a}$	Vector	Quantity possessing magnitude and direction.
$\ \mathbf{a}\ $	Magnitude or norm	Length of the vector $\mathbf{a}$.
$\hat{\mathbf{a}}$	Unit vector	Vector of magnitude 1 in the direction of $\mathbf{a}$.
$\mathbf{a} \cdot \mathbf{b}$	Dot product	Scalar product used for angles and projections.
$\mathbf{a} \times \mathbf{b}$	Cross product	Vector perpendicular to both vectors in three dimensions.
$\mathbf{i}, \mathbf{j}, \mathbf{k}$	Cartesian unit vectors	Unit vectors along the positive coordinate axes.
$\mathbf{r}(t)$	Vector-valued function	Position or curve expressed as a function of a parameter.

Common units

Symbol	Name or reading	Meaning or use
m, cm, km	Length	Metre, centimetre, and kilometre.
s, min, h	Time	Second, minute, and hour.
kg, g, mg	Mass	Kilogram, gram, and milligram.
m s^{-1}	Speed or velocity	Metres per second.
m s^{-2}	Acceleration	Metres per second squared.
km h^{-1}	Speed	Kilometres per hour.
mg L^{-1}	Concentration	Milligrams per litre.
N, J, W	Derived SI units	Newton, joule, and watt.
$^\circ$, $^\circ\text{C}$	Angle and temperature	Degree and degree Celsius.

Graph theory and computing

Symbol	Name or reading	Meaning or use
$G = (V, E)$	Graph	A vertex set V together with an edge set E .
$\deg(v)$	Degree of a vertex	Number of edges incident with v , with loops counted appropriately.
K_n, C_n	Complete graph and cycle graph	Standard graph families on n vertices.
$K_{m,n}$	Complete bipartite graph	Every vertex in one part is joined to every vertex in the other part.
$\leftarrow$	Assignment in R	Stores the value on the right in the object on the left.
NA	Missing value in R	Marker for unavailable or missing data.

When a symbol is reused in different chapters, its local definition takes priority. For example, C may denote a constant of integration, a set, a matrix, or a labelled point depending on the context.

Appendix B

List of Set Notations

This appendix summarises the set notation most commonly encountered in A-level mathematics, particularly in functions, inequalities, probability, and Venn-diagram problems. Unless stated otherwise, the convention

$$\mathbb{N} = \{1, 2, 3, \dots\} \quad \text{and} \quad \mathbb{W} = \{0, 1, 2, 3, \dots\}$$

is used. Some textbooks include 0 in $\mathbb{N}$, so the convention should always be checked when consulting another source.

B.1 Basic notation

Symbol	Meaning	Example or remark
$\{ \}$	Set braces	$A = \{1, 2, 3\}$ lists the elements of A .
$x \in A$	x is an element of A	If $A = \{1, 2, 3\}$, then $2 \in A$.
$x \notin A$	x is not an element of A	For the same set, $5 \notin A$.
$\emptyset$	Empty set	The unique set containing no elements; $n(\emptyset) = 0$.
U	Universal set	The set of all objects being considered in a particular problem.
$A = B$	Equal sets	A and B contain exactly the same elements.
$A \neq B$	Unequal sets	At least one element belongs to one set but not the other.
$n(A)$ or $ A $	Cardinality of A	The number of elements in a finite set; for example, $n(\{2, 4, 6\}) = 3$.
$\{x \in S \mid P(x)\}$	Set-builder notation	The set of all x in S for which the condition $P(x)$ is true.
$\mid$ or $:$	Such that	$\{x \in \mathbb{R} \mid x > 0\}$ means all positive real numbers.
$\dots$	Continuation of a pattern	$\{2, 4, 6, \dots\}$ denotes the positive even integers.

Table B.1: Basic symbols used to describe sets.

B.2 Relationships between sets

Symbol	Meaning	Definition or example
$A \subseteq B$	A is a subset of B	Every element of A is also an element of B . Equality is allowed.
$A \subsetneq B$	A is a proper subset of B	$A \subseteq B$ and $A \neq B$.
$A \not\subseteq B$	A is not a subset of B	At least one element of A does not belong to B .
$A \supseteq B$	A is a superset of B	Equivalent to $B \subseteq A$.
$A \cap B = \emptyset$	A and B are disjoint	The sets have no elements in common.

Table B.2: Symbols describing relationships between sets.

Membership and inclusion must not be confused. If $A = \{1, 2, 3\}$, then

$$1 \in A \quad \text{but} \quad \{1\} \subseteq A.$$

The first statement compares an element with a set, whereas the second compares one set with another.

B.3 Operations on sets

Symbol	Operation	Meaning
$A \cup B$	Union	Elements belonging to A , to B , or to both.
$A \cap B$	Intersection	Elements belonging to both A and B .
$A \setminus B$	Set difference	Elements belonging to A but not to B . The order matters.
A^c or A'	Complement	Elements of the universal set U that do not belong to A .

Table B.3: Common operations on sets.

For a universal set U ,

$$A^c = U \setminus A.$$

The A-level inclusion–exclusion formula for two finite sets is

$$n(A \cup B) = n(A) + n(B) - n(A \cap B).$$

B.4 Standard number sets

Symbol	Name	Description
$\mathbb{N}$	Natural numbers	$\{1, 2, 3, \dots\}$ under the convention used in this book.
$\mathbb{W}$	Whole numbers	$\{0, 1, 2, 3, \dots\}$. This symbol is common in school mathematics but is not universal.
$\mathbb{Z}$	Integers	$\{\dots, -2, -1, 0, 1, 2, \dots\}$.
$\mathbb{Z}^+$	Positive integers	$\{1, 2, 3, \dots\}$.
$\mathbb{Z}^-$	Negative integers	$\{-1, -2, -3, \dots\}$.
$\mathbb{Q}$	Rational numbers	Numbers expressible as a/b , where $a, b \in \mathbb{Z}$ and $b \neq 0$.
$\mathbb{R} \setminus \mathbb{Q}$	Irrational numbers	Real numbers that cannot be expressed as a ratio of two integers.
$\mathbb{R}$	Real numbers	All numbers represented on the real number line.
$\mathbb{C}$	Complex numbers	Numbers of the form $a+bi$, where $a, b \in \mathbb{R}$ and $i^2 = -1$.

Table B.4: Standard number-set symbols.

These number sets satisfy

$$\mathbb{N} \subsetneq \mathbb{W} \subsetneq \mathbb{Z} \subsetneq \mathbb{Q} \subsetneq \mathbb{R} \subsetneq \mathbb{C}.$$

B.5 Interval notation

Let $a, b \in \mathbb{R}$ with $a < b$.

Interval	Set-builder form	Description
(a, b)	$\{x \in \mathbb{R} \mid a < x < b\}$	Open interval; neither endpoint is included.
$[a, b]$	$\{x \in \mathbb{R} \mid a \leq x \leq b\}$	Closed interval; both endpoints are included.
$[a, b)$	$\{x \in \mathbb{R} \mid a \leq x < b\}$	a is included and b is excluded.
$(a, b]$	$\{x \in \mathbb{R} \mid a < x \leq b\}$	a is excluded and b is included.
$(-\infty, a)$	$\{x \in \mathbb{R} \mid x < a\}$	All real numbers less than a .
$(-\infty, a]$	$\{x \in \mathbb{R} \mid x \leq a\}$	All real numbers less than or equal to a .
(a, ∞)	$\{x \in \mathbb{R} \mid x > a\}$	All real numbers greater than a .
$[a, \infty)$	$\{x \in \mathbb{R} \mid x \geq a\}$	All real numbers greater than or equal to a .
$(-\infty, \infty)$	$\mathbb{R}$	The complete real number line.

Table B.5: Intervals of real numbers.

Infinity is not a real endpoint, so $-\infty$ and ∞ always use parentheses rather than square brackets.

B.6 Logical symbols used with sets

Symbol	Meaning	Example
$\forall$	For every or for all	$\forall x \in A$ means “for every x in A .”
$\exists$	There exists	$\exists x \in A$ means that at least one such element exists.
$\Rightarrow$	Implies	$x \in A \cap B \Rightarrow x \in A$.
$\Leftrightarrow$	If and only if	$A = B \Leftrightarrow A \subseteq B$ and $B \subseteq A$.

Table B.6: Logical notation commonly appearing in set statements.

B.7 Useful equivalences and identities

For any element x and sets A and B ,

$$\begin{aligned}
x \in A \cup B &\Leftrightarrow x \in A \text{ or } x \in B, \\
x \in A \cap B &\Leftrightarrow x \in A \text{ and } x \in B, \\
x \in A \setminus B &\Leftrightarrow x \in A \text{ and } x \notin B, \\
x \in A^c &\Leftrightarrow x \in U \text{ and } x \notin A.
\end{aligned}$$

The elementary identity and complement laws are

$$\begin{aligned}
A \cup \emptyset &= A, & A \cap U &= A, \\
A \cup U &= U, & A \cap \emptyset &= \emptyset, \\
A \cup A^c &= U, & A \cap A^c &= \emptyset, \\
(A^c)^c &= A.
\end{aligned}$$

De Morgan's laws are

$$(A \cup B)^c = A^c \cap B^c, \quad (A \cap B)^c = A^c \cup B^c.$$

Appendix C

Trigonometric and Hyperbolic Functions and Identities

This appendix collects the trigonometric and hyperbolic formulae most commonly required in A-level and foundation mathematics. All identities are understood to hold wherever the expressions involved are defined. Unless degrees are shown explicitly, angles are measured in radians.

C.1 Angle measure

$$180^\circ = \pi \text{ radians}, \quad \theta_{\text{rad}} = \frac{\pi}{180} \theta_{\text{deg}}, \quad \theta_{\text{deg}} = \frac{180}{\pi} \theta_{\text{rad}}.$$

C.2 Trigonometric functions

For a right-angled triangle with respect to an acute angle θ ,

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}, \quad \cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}, \quad \tan \theta = \frac{\text{opposite}}{\text{adjacent}}.$$

On the unit circle, the point corresponding to an angle θ has coordinates $(\cos \theta, \sin \theta)$. This definition extends sine and cosine to all real angles.

C.2.1 The six trigonometric functions

Function	Quotient or reciprocal form	Right-triangle form
$\sin \theta$	$\sin \theta$	Opposite/Hypotenuse
$\cos \theta$	$\cos \theta$	Adjacent/Hypotenuse
$\tan \theta$	$\frac{\sin \theta}{\cos \theta}$	Opposite/Adjacent
$\csc \theta$ or $\operatorname{cosec} \theta$	$\frac{1}{\sin \theta}$	Hypotenuse/Opposite
$\sec \theta$	$\frac{1}{\cos \theta}$	Hypotenuse/Adjacent
$\cot \theta$	$\frac{\cos \theta}{\sin \theta}$	Adjacent/Opposite

Table C.1: Definitions of the six trigonometric functions.

Let $k \in \mathbb{Z}$. The real domains, ranges, periods, and symmetries are summarised below.

Function	Domain	Range	Period	Parity
$\sin x$	$\mathbb{R}$	$[-1, 1]$	2π	Odd
$\cos x$	$\mathbb{R}$	$[-1, 1]$	2π	Even
$\tan x$	$x \neq \frac{\pi}{2} + k\pi$	$\mathbb{R}$	π	Odd
$\csc x$	$x \neq k\pi$	$(-\infty, -1] \cup [1, \infty)$	2π	Odd
$\sec x$	$x \neq \frac{\pi}{2} + k\pi$	$(-\infty, -1] \cup [1, \infty)$	2π	Even
$\cot x$	$x \neq k\pi$	$\mathbb{R}$	π	Odd

Table C.2: Properties of the trigonometric functions.

C.2.2 Exact values at standard angles

		0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
		0°	30°	45°	60°	90°
$\sin \theta$	0		$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1		$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan \theta$	0		$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	Undefined
$\csc \theta$	Undefined		2	$\sqrt{2}$	$\frac{2\sqrt{3}}{3}$	1
$\sec \theta$	1		$\frac{2\sqrt{3}}{3}$	$\sqrt{2}$	2	Undefined
$\cot \theta$	Undefined		$\sqrt{3}$	1	$\frac{\sqrt{3}}{3}$	0

Table C.3: Exact trigonometric values in the first quadrant.

C.3 Fundamental trigonometric identities

C.3.1 Reciprocal and quotient identities

$$\begin{aligned} \csc x &= \frac{1}{\sin x}, & \sec x &= \frac{1}{\cos x}, & \cot x &= \frac{1}{\tan x}, \\ \tan x &= \frac{\sin x}{\cos x}, & \cot x &= \frac{\cos x}{\sin x}. \end{aligned}$$

C.3.2 Pythagorean identities

$$\sin^2 x + \cos^2 x = 1, \quad 1 + \tan^2 x = \sec^2 x, \quad 1 + \cot^2 x = \csc^2 x.$$

C.3.3 Symmetry and periodicity

$$\begin{aligned} \sin(-x) &= -\sin x, & \cos(-x) &= \cos x, & \tan(-x) &= -\tan x, \\ \csc(-x) &= -\csc x, & \sec(-x) &= \sec x, & \cot(-x) &= -\cot x, \\ \sin(x + 2\pi) &= \sin x, & \cos(x + 2\pi) &= \cos x, & \tan(x + \pi) &= \tan x. \end{aligned}$$

C.3.4 Complementary and related-angle identities

$$\begin{aligned} \sin\left(\frac{\pi}{2} - x\right) &= \cos x, & \cos\left(\frac{\pi}{2} - x\right) &= \sin x, \\ \tan\left(\frac{\pi}{2} - x\right) &= \cot x, & \cot\left(\frac{\pi}{2} - x\right) &= \tan x, \\ \sec\left(\frac{\pi}{2} - x\right) &= \csc x, & \csc\left(\frac{\pi}{2} - x\right) &= \sec x. \end{aligned}$$

$$\begin{aligned} \sin(\pi - x) &= \sin x, & \cos(\pi - x) &= -\cos x, & \tan(\pi - x) &= -\tan x, \\ \sin(\pi + x) &= -\sin x, & \cos(\pi + x) &= -\cos x, & \tan(\pi + x) &= \tan x, \\ \sin(2\pi - x) &= -\sin x, & \cos(2\pi - x) &= \cos x, & \tan(2\pi - x) &= -\tan x. \end{aligned}$$

C.4 Compound-angle and multiple-angle identities**C.4.1 Addition and subtraction formulae**

$$\begin{aligned} \sin(A \pm B) &= \sin A \cos B \pm \cos A \sin B, \\ \cos(A \pm B) &= \cos A \cos B \mp \sin A \sin B, \\ \tan(A \pm B) &= \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}. \end{aligned}$$

The paired signs must be read consistently. For example,

$$\cos(A + B) = \cos A \cos B - \sin A \sin B, \quad \cos(A - B) = \cos A \cos B + \sin A \sin B.$$

C.4.2 Double-angle identities

$$\begin{aligned}\sin 2x &= 2 \sin x \cos x, \\ \cos 2x &= \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x, \\ \tan 2x &= \frac{2 \tan x}{1 - \tan^2 x}.\end{aligned}$$

C.4.3 Half-angle and power-reduction identities

$$\begin{aligned}\sin^2 x &= \frac{1 - \cos 2x}{2}, & \cos^2 x &= \frac{1 + \cos 2x}{2}, \\ \sin \frac{x}{2} &= \pm \sqrt{\frac{1 - \cos x}{2}}, & \cos \frac{x}{2} &= \pm \sqrt{\frac{1 + \cos x}{2}}, \\ \tan \frac{x}{2} &= \frac{\sin x}{1 + \cos x} = \frac{1 - \cos x}{\sin x}.\end{aligned}$$

The sign in a half-angle square-root formula is determined by the quadrant containing $x/2$.

C.4.4 Triple-angle identities

$$\begin{aligned}\sin 3x &= 3 \sin x - 4 \sin^3 x, \\ \cos 3x &= 4 \cos^3 x - 3 \cos x, \\ \tan 3x &= \frac{3 \tan x - \tan^3 x}{1 - 3 \tan^2 x}.\end{aligned}$$

C.5 Product–sum transformations

C.5.1 Products expressed as sums

$$\begin{aligned}2 \sin A \cos B &= \sin(A + B) + \sin(A - B), \\ 2 \cos A \sin B &= \sin(A + B) - \sin(A - B), \\ 2 \cos A \cos B &= \cos(A + B) + \cos(A - B), \\ 2 \sin A \sin B &= \cos(A - B) - \cos(A + B).\end{aligned}$$

C.5.2 Sums expressed as products

$$\begin{aligned}\sin A + \sin B &= 2 \sin \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right), \\ \sin A - \sin B &= 2 \cos \left(\frac{A+B}{2} \right) \sin \left(\frac{A-B}{2} \right), \\ \cos A + \cos B &= 2 \cos \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right), \\ \cos A - \cos B &= -2 \sin \left(\frac{A+B}{2} \right) \sin \left(\frac{A-B}{2} \right).\end{aligned}$$

C.6 The auxiliary-angle form

Let $R = \sqrt{a^2 + b^2}$. Then

$$\begin{aligned}a \sin x + b \cos x &= R \sin(x + \alpha), & R \cos \alpha &= a, & R \sin \alpha &= b, \\ a \cos x + b \sin x &= R \cos(x - \alpha), & R \cos \alpha &= a, & R \sin \alpha &= b.\end{aligned}$$

Consequently,

$$-R \leq a \sin x + b \cos x \leq R.$$

C.7 Inverse trigonometric functions

The restricted ranges below make each trigonometric function one-to-one and therefore define its principal inverse.

Inverse function	Equivalent statement	Domain	Principal range
$y = \sin^{-1} x$	$x = \sin y$	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$y = \cos^{-1} x$	$x = \cos y$	$[-1, 1]$	$[0, \pi]$
$y = \tan^{-1} x$	$x = \tan y$	$\mathbb{R}$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Table C.4: Principal inverse trigonometric functions.

Useful inverse-function identities include

$$\begin{aligned}
\sin(\sin^{-1} x) &= x & (-1 \leq x \leq 1), \\
\cos(\cos^{-1} x) &= x & (-1 \leq x \leq 1), \\
\tan(\tan^{-1} x) &= x & (x \in \mathbb{R}), \\
\sin^{-1} x + \cos^{-1} x &= \frac{\pi}{2} & (-1 \leq x \leq 1).
\end{aligned}$$

C.8 Exponential forms of trigonometric functions

Euler's formula gives

$$e^{ix} = \cos x + i \sin x.$$

Hence,

$$\cos x = \frac{e^{ix} + e^{-ix}}{2}, \quad \sin x = \frac{e^{ix} - e^{-ix}}{2i}.$$

C.9 Hyperbolic functions

C.9.1 Exponential definitions

Function	Exponential form	Quotient or reciprocal form
$\sinh x$	$\frac{e^x - e^{-x}}{2}$	$\sinh x$
$\cosh x$	$\frac{e^x + e^{-x}}{2}$	$\cosh x$
$\tanh x$	$\frac{e^x - e^{-x}}{e^x + e^{-x}}$	$\frac{\sinh x}{\cosh x}$
$\operatorname{cosech} x$ or $\operatorname{csch} x$	$\frac{e^x - e^{-x}}{2}$	$\frac{1}{\sinh x}$
$\operatorname{sech} x$	$\frac{e^x + e^{-x}}{e^x - e^{-x}}$	$\frac{\cosh x}{\sinh x}$
$\coth x$	$\frac{e^x + e^{-x}}{e^x - e^{-x}}$	$\frac{\cosh x}{\sinh x}$

Table C.5: Definitions of the six hyperbolic functions.

The exponential definitions immediately give

$$e^x = \cosh x + \sinh x, \quad e^{-x} = \cosh x - \sinh x.$$

C.9.2 Domains, ranges, and symmetry

Function	Domain	Range	Parity
$\sinh x$	$\mathbb{R}$	$\mathbb{R}$	Odd
$\cosh x$	$\mathbb{R}$	$[1, \infty)$	Even
$\tanh x$	$\mathbb{R}$	$(-1, 1)$	Odd
$\operatorname{cosech} x$	$\mathbb{R} \setminus \{0\}$	$\mathbb{R} \setminus \{0\}$	Odd
$\operatorname{sech} x$	$\mathbb{R}$	$(0, 1]$	Even
$\operatorname{coth} x$	$\mathbb{R} \setminus \{0\}$	$(-\infty, -1) \cup (1, \infty)$	Odd

Table C.6: Properties of the real hyperbolic functions.

Unlike trigonometric functions, the real hyperbolic functions are not periodic.

C.10 Fundamental hyperbolic identities

C.10.1 Reciprocal, quotient, and quadratic identities

$$\begin{aligned} \operatorname{sech} x &= \frac{1}{\cosh x}, & \operatorname{cosech} x &= \frac{1}{\sinh x}, & \operatorname{coth} x &= \frac{1}{\tanh x}, \\ \tanh x &= \frac{\sinh x}{\cosh x}, & \operatorname{coth} x &= \frac{\cosh x}{\sinh x}. \end{aligned}$$

$$\boxed{\cosh^2 x - \sinh^2 x = 1, \quad 1 - \tanh^2 x = \operatorname{sech}^2 x, \quad \operatorname{coth}^2 x - 1 = \operatorname{cosech}^2 x.}$$

C.10.2 Symmetry identities

$$\begin{aligned} \sinh(-x) &= -\sinh x, & \cosh(-x) &= \cosh x, & \tanh(-x) &= -\tanh x, \\ \operatorname{cosech}(-x) &= -\operatorname{cosech} x, & \operatorname{sech}(-x) &= \operatorname{sech} x, & \operatorname{coth}(-x) &= -\operatorname{coth} x. \end{aligned}$$

C.11 Hyperbolic compound-angle identities

$$\begin{aligned} \sinh(A \pm B) &= \sinh A \cosh B \pm \cosh A \sinh B, \\ \cosh(A \pm B) &= \cosh A \cosh B \pm \sinh A \sinh B, \\ \tanh(A \pm B) &= \frac{\tanh A \pm \tanh B}{1 \pm \tanh A \tanh B}. \end{aligned}$$

Unlike the circular cosine formula, the sign in the hyperbolic cosine formula is the same as the sign on the left.

C.11.1 Double-angle and power identities

$$\begin{aligned}\sinh 2x &= 2 \sinh x \cosh x, \\ \cosh 2x &= \cosh^2 x + \sinh^2 x = 2 \cosh^2 x - 1 = 1 + 2 \sinh^2 x, \\ \tanh 2x &= \frac{2 \tanh x}{1 + \tanh^2 x}, \\ \cosh^2 x &= \frac{\cosh 2x + 1}{2}, & \sinh^2 x &= \frac{\cosh 2x - 1}{2}.\end{aligned}$$

C.11.2 Triple-angle identities

$$\begin{aligned}\sinh 3x &= 3 \sinh x + 4 \sinh^3 x, \\ \cosh 3x &= 4 \cosh^3 x - 3 \cosh x, \\ \tanh 3x &= \frac{3 \tanh x + \tanh^3 x}{1 + 3 \tanh^2 x}.\end{aligned}$$

C.12 Hyperbolic product–sum transformations

C.12.1 Products expressed as sums

$$\begin{aligned}2 \sinh A \cosh B &= \sinh(A + B) + \sinh(A - B), \\ 2 \cosh A \sinh B &= \sinh(A + B) - \sinh(A - B), \\ 2 \cosh A \cosh B &= \cosh(A + B) + \cosh(A - B), \\ 2 \sinh A \sinh B &= \cosh(A + B) - \cosh(A - B).\end{aligned}$$

C.12.2 Sums expressed as products

$$\begin{aligned}\sinh A + \sinh B &= 2 \sinh \left(\frac{A + B}{2} \right) \cosh \left(\frac{A - B}{2} \right), \\ \sinh A - \sinh B &= 2 \cosh \left(\frac{A + B}{2} \right) \sinh \left(\frac{A - B}{2} \right), \\ \cosh A + \cosh B &= 2 \cosh \left(\frac{A + B}{2} \right) \cosh \left(\frac{A - B}{2} \right), \\ \cosh A - \cosh B &= 2 \sinh \left(\frac{A + B}{2} \right) \sinh \left(\frac{A - B}{2} \right).\end{aligned}$$

C.13 Inverse hyperbolic functions

The three principal inverse hyperbolic functions most commonly required at A-level are listed below.

Function	Domain	Range	Logarithmic form
$\sinh^{-1} x$	$\mathbb{R}$	$\mathbb{R}$	$\ln \left(x + \sqrt{x^2 + 1} \right)$
$\cosh^{-1} x$	$[1, \infty)$	$[0, \infty)$	$\ln \left(x + \sqrt{x^2 - 1} \right)$
$\tanh^{-1} x$	$(-1, 1)$	$\mathbb{R}$	$\frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$

Table C.7: Principal inverse hyperbolic functions.

The inverses of the reciprocal hyperbolic functions may be written as

$$\begin{aligned} \operatorname{sech}^{-1} x &= \cosh^{-1} \left(\frac{1}{x} \right) = \ln \left(\frac{1 + \sqrt{1 - x^2}}{x} \right), & 0 < x \leq 1, \\ \operatorname{cosech}^{-1} x &= \sinh^{-1} \left(\frac{1}{x} \right) = \ln \left(\frac{1}{x} + \sqrt{1 + \frac{1}{x^2}} \right), & x \neq 0, \\ \operatorname{coth}^{-1} x &= \tanh^{-1} \left(\frac{1}{x} \right) = \frac{1}{2} \ln \left(\frac{x+1}{x-1} \right), & |x| > 1. \end{aligned}$$

Here, the superscript -1 denotes a functional inverse. It must not be confused with a reciprocal; for example,

$$\sinh^{-1} x \neq \frac{1}{\sinh x} \quad \text{and} \quad \frac{1}{\sinh x} = \operatorname{cosech} x.$$

C.14 Circular and hyperbolic comparison

The sign changes in the following pairs are especially important:

Trigonometric identity	Hyperbolic identity
$\cos^2 x + \sin^2 x = 1$	$\cosh^2 x - \sinh^2 x = 1$
$\cos(A+B) = \cos A \cos B - \sin A \sin B$	$\cosh(A+B) = \cosh A \cosh B + \sinh A \sinh B$
$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$	$\tanh(A+B) = \frac{\tanh A + \tanh B}{1 + \tanh A \tanh B}$
$\cos 2x = 1 - 2 \sin^2 x$	$\cosh 2x = 1 + 2 \sinh^2 x$

Table C.8: Key differences between circular and hyperbolic identities.

Appendix D

Core Formula Reference

This appendix collects frequently used formulae from calculus, probability, statistics, and vector geometry. It is intended as a quick reference rather than a substitute for the definitions, conditions, and derivations in the corresponding chapters.

D.1 Differentiation

If $u = u(x)$ and $v = v(x)$ are differentiable and c is constant, then

$$\begin{aligned}\frac{d}{dx}(cu) &= cu', & \frac{d}{dx}(u \pm v) &= u' \pm v', \\ \frac{d}{dx}(uv) &= u'v + uv', & \frac{d}{dx}\left(\frac{u}{v}\right) &= \frac{u'v - uv'}{v^2}, \\ \frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx}.\end{aligned}$$

For a parametrised curve $x = x(t)$, $y = y(t)$ with $dx/dt \neq 0$,

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}, \quad \frac{d^2y}{dx^2} = \frac{d}{dt}(dy/dx) \bigg/ \frac{dx}{dt}.$$

D.2 Integration

The principal techniques are

$$\begin{aligned}\text{substitution: } & \int f(g(x))g'(x) dx = \int f(u) du, \\ \text{integration by parts: } & \int u dv = uv - \int v du.\end{aligned}$$

The Fundamental Theorem of Calculus gives

$$\int_a^b f(x) dx = F(b) - F(a), \quad F'(x) = f(x).$$

For geometric area, split the interval wherever the graph crosses the axis and integrate the absolute vertical separation of the relevant curves.

Function	Derivative
x^n	nx^{n-1}
e^x	e^x
a^x	$a^x \ln a$
$\ln x$	$1/x$
$\log_a x$	$1/(x \ln a)$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\cot x$	$-\csc^2 x$
$\sec x$	$\sec x \tan x$
$\csc x$	$-\csc x \cot x$
$\arcsin x$	$1/\sqrt{1-x^2}$
$\arccos x$	$-1/\sqrt{1-x^2}$
$\arctan x$	$1/(1+x^2)$
$\sinh x$	$\cosh x$
$\cosh x$	$\sinh x$
$\tanh x$	$\operatorname{sech}^2 x$

Table D.1: Frequently used derivatives. Apply the chain rule when the argument is a function of x .

Integrand	Antiderivative
$x^n, n \neq -1$	$x^{n+1}/(n+1) + C$
$1/x$	$\ln x + C$
e^x	$e^x + C$
a^x	$a^x/\ln a + C$
$\sin x$	$-\cos x + C$
$\cos x$	$\sin x + C$
$\sec^2 x$	$\tan x + C$
$\csc^2 x$	$-\cot x + C$
$\sec x \tan x$	$\sec x + C$
$\csc x \cot x$	$-\csc x + C$
$1/(1+x^2)$	$\arctan x + C$
$1/\sqrt{1-x^2}$	$\arcsin x + C$
$\sinh x$	$\cosh x + C$
$\cosh x$	$\sinh x + C$

Table D.2: Frequently used indefinite integrals.

D.3 Probability and statistics

For events A and B ,

$$\begin{aligned} P(A^c) &= 1 - P(A), \\ P(A \cup B) &= P(A) + P(B) - P(A \cap B), \\ P(A \mid B) &= \frac{P(A \cap B)}{P(B)}, \\ P(A \cap B) &= P(A)P(B) \quad \text{when } A \text{ and } B \text{ are independent.} \end{aligned}$$

For a discrete random variable X ,

$$E(X) = \sum_x xP(X = x), \quad \text{Var}(X) = E(X^2) - [E(X)]^2.$$

Model	Probability rule	$E(X)$	$\text{Var}(X)$
Binomial $B(n, p)$	$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$	np	$np(1 - p)$
Poisson $\text{Po}(\lambda)$	$P(X = x) = e^{-\lambda} \lambda^x / x!$	λ	λ
Exponential $\text{Exp}(\lambda)$	$P(X > x) = e^{-\lambda x}, x \geq 0$	$1/\lambda$	$1/\lambda^2$

Table D.3: Selected probability distributions.

If $X \sim N(\mu, \sigma^2)$, standardise using

$$Z = \frac{X - \mu}{\sigma}.$$

For a sample mean based on independent observations,

$$E(\bar{X}) = \mu, \quad \text{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}} \quad \text{or} \quad \frac{s}{\sqrt{n}} \quad \text{when } \sigma \text{ is estimated.}$$

A two-sided confidence interval for a population mean has the general form

$$\bar{x} \pm (\text{critical value}) \times \text{SE}(\bar{X}).$$

The choice between a normal and a t critical value depends on the assumptions and whether the population standard deviation is known.

D.4 Vectors and three-dimensional geometry

For $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$,

$$\|\mathbf{a}\| = \sqrt{a_1^2 + a_2^2 + a_3^2}, \quad \hat{\mathbf{a}} = \frac{\mathbf{a}}{\|\mathbf{a}\|}.$$

$$\begin{aligned}
\mathbf{a} \cdot \mathbf{b} &= \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta, \\
\text{comp}_{\mathbf{a}} \mathbf{b} &= \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|}, \\
\text{proj}_{\mathbf{a}} \mathbf{b} &= \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|^2} \mathbf{a}, \\
\|\mathbf{a} \times \mathbf{b}\| &= \|\mathbf{a}\| \|\mathbf{b}\| \sin \theta.
\end{aligned}$$

Consequently,

$$\begin{aligned}
\text{area of a parallelogram} &= \|\mathbf{a} \times \mathbf{b}\|, \\
\text{area of a triangle} &= \frac{1}{2} \|\mathbf{a} \times \mathbf{b}\|, \\
\text{volume of a parallelepiped} &= |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|.
\end{aligned}$$

A line through the point with position vector $\mathbf{a}$ and direction $\mathbf{d}$ is

$$\mathbf{r} = \mathbf{a} + t\mathbf{d}.$$

A plane through the point with position vector $\mathbf{a}$ and normal $\mathbf{n}$ is

$$\mathbf{n} \cdot (\mathbf{r} - \mathbf{a}) = 0.$$

The distance from a point with position vector $\mathbf{p}$ to the line $\mathbf{r} = \mathbf{a} + t\mathbf{d}$ is

$$\frac{\|(\mathbf{p} - \mathbf{a}) \times \mathbf{d}\|}{\|\mathbf{d}\|}.$$

The distance from (x_0, y_0, z_0) to the plane $Ax + By + Cz + D = 0$ is

$$\frac{|Ax_0 + By_0 + Cz_0 + D|}{\sqrt{A^2 + B^2 + C^2}}.$$

Appendix E

Statistical Tables

Table 1: Cumulative Binomial Probabilities

If p is the probability of success in a single trial, n is the number of trials, and r is the greatest permitted number of successes, the table gives

$$P(X \leq r) = \sum_{x=0}^r \binom{n}{x} p^x (1-p)^{n-x}.$$

An empty entry represents a probability below 0.00005; a displayed value of 1.0000 represents a probability above 0.99995. $n = 2$

r	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0	1.0000	.9801	.9604	.9409	.9216	.9025	.8836	.8649	.8464	.8281
1	1.0000	.9999	.9996	.9991	.9984	.9975	.9964	.9951	.9936	.9919
2	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
$n = 5$										
r	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0	1.0000	.9510	.9039	.8587	.8154	.7738	.7339	.6957	.6591	.6240
1	1.0000	.9990	.9962	.9915	.9852	.9774	.9681	.9575	.9456	.9326
2	1.0000	1.0000	.9999	.9997	.9994	.9988	.9980	.9969	.9955	.9937
3	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9999	.9998	.9997
4	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
$n = 10$										
r	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0	1.0000	.9044	.8171	.7374	.6648	.5987	.5386	.4840	.4344	.3894
1	1.0000	.9957	.9838	.9655	.9418	.9139	.8824	.8483	.8121	.7746
2	1.0000	.9999	.9991	.9972	.9938	.9885	.9812	.9717	.9599	.9460
3	1.0000	1.0000	1.0000	.9999	.9996	.9990	.9980	.9964	.9942	.9912
4	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9998	.9997	.9994	.9990
5	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999
6	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
$n = 20$										
r	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0	1.0000	.8179	.6676	.5438	.4420	.3585	.2901	.2342	.1887	.1516
1	1.0000	.9831	.9401	.8802	.8103	.7358	.6605	.5869	.5169	.4516
2	1.0000	.9990	.9929	.9790	.9561	.9245	.8850	.8390	.7879	.7334
3	1.0000	1.0000	.9994	.9973	.9926	.9841	.9710	.9529	.9294	.9007
4	1.0000	1.0000	1.0000	.9997	.9990	.9974	.9944	.9893	.9817	.9710
5	1.0000	1.0000	1.0000	1.0000	.9999	.9997	.9991	.9981	.9962	.9932
6	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9997	.9997	.9994	.9987
7	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9998
8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
$n = 50$										
r	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0	1.0000	.6050	.3642	.2181	.1299	.0769	.0453	.0266	.0155	.0090
1	1.0000	.9106	.7358	.5553	.4005	.2794	.1900	.1265	.0827	.0532
2	1.0000	.9862	.9216	.8108	.6767	.5405	.4162	.3108	.2260	.1605
3	1.0000	.9984	.9822	.9372	.8609	.7604	.6473	.5327	.4253	.3303
4	1.0000	.9999	.9968	.9832	.9510	.8964	.8206	.7290	.6290	.5277
5	1.0000	1.0000	.9995	.9963	.9856	.9622	.9224	.8650	.7919	.7072
6	1.0000	1.0000	.9999	.9993	.9964	.9882	.9711	.9417	.8981	.8404
7	1.0000	1.0000	1.0000	.9999	.9992	.9968	.9906	.9780	.9562	.9232
8	1.0000	1.0000	1.0000	1.0000	.9999	.9992	.9973	.9927	.9833	.9672
9	1.0000	1.0000	1.0000	1.0000	1.0000	.9998	.9993	.9978	.9944	.9875
10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9998	.9994	.9983	.9957
11	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9995	.9987
12	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9996
13	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999
14	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Table 1: Cumulative Binomial Probabilities—continued

$n = 100$									
r	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0	.3660	.1326	.0476	.0169	.0059	.0021	.0007	.0002	.0001
1	.7358	.4033	.1946	.0872	.0371	.0152	.0060	.0023	.0009
2	.9206	.6767	.4198	.2321	.1183	.0566	.0258	.0113	.0048
3	.9816	.8590	.6472	.4295	.2578	.1430	.0744	.0367	.0173
4	.9966	.9492	.8179	.6289	.4360	.2768	.1632	.0903	.0474
5	.9995	.9845	.9192	.7884	.6160	.4407	.2914	.1799	.1045
6	.9999	.9959	.9688	.8936	.7660	.6064	.4443	.3032	.1940
7	1.0000	.9991	.9894	.9525	.8720	.7483	.5988	.4471	.3128
8	1.0000	.9998	.9968	.9810	.9369	.8537	.7340	.5926	.4494
9	1.0000	1.0000	.9991	.9932	.9718	.9225	.8380	.7220	.5875
10	1.0000	1.0000	.9998	.9978	.9885	.9624	.9092	.8243	.7118
11	1.0000	1.0000	1.0000	.9993	.9957	.9832	.9531	.8972	.8124
12	1.0000	1.0000	1.0000	.9998	.9985	.9931	.9776	.9441	.8862
13	1.0000	1.0000	1.0000	1.0000	.9995	.9974	.9901	.9718	.9355
14	1.0000	1.0000	1.0000	1.0000	.9999	.9991	.9959	.9867	.9659
15	1.0000	1.0000	1.0000	1.0000	1.0000	.9997	.9984	.9942	.9831
16	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9994	.9976	.9922
17	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9998	.9991	.9966
18	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9997	.9986
19	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9995
20	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9998
21	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999
22	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

$n = 2$									
r	0.1	0.15	0.2	0.25	0.3	0.35	0.4	0.45	0.5
0	.8100	.7225	.6400	.5625	.4900	.4225	.3600	.3025	.2500
1	.9900	.9775	.9600	.9375	.9100	.8775	.8400	.7975	.7500
2	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

$n = 5$									
r	0.1	0.15	0.2	0.25	0.3	0.35	0.4	0.45	0.5
0	.5905	.4437	.3277	.2373	.1681	.1160	.0778	.0503	.0312
1	.9185	.8352	.7373	.6328	.5282	.4284	.3370	.2562	.1875
2	.9914	.9734	.9421	.8965	.8369	.7648	.6826	.5931	.5000
3	.9995	.9978	.9933	.9844	.9692	.9460	.9130	.8688	.8125
4	1.0000	.9999	.9997	.9990	.9976	.9947	.9898	.9815	.9688
5	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

$n = 10$									
r	0.1	0.15	0.2	0.25	0.3	0.35	0.4	0.45	0.5
0	.3487	.1969	.1074	.0563	.0282	.0135	.0060	.0025	.0010
1	.7361	.5443	.3758	.2440	.1493	.0860	.0464	.0233	.0107
2	.9298	.8202	.6778	.5256	.3828	.2616	.1673	.0996	.0547
3	.9872	.9500	.8791	.7759	.6496	.5138	.3823	.2660	.1719
4	.9984	.9901	.9672	.9219	.8497	.7515	.6331	.5044	.3770
5	.9999	.9986	.9936	.9803	.9527	.9051	.8338	.7384	.6230
6	1.0000	.9999	.9991	.9965	.9894	.9740	.9452	.8980	.8281
7	1.0000	1.0000	.9999	.9996	.9984	.9952	.9877	.9726	.9453
8	1.0000	1.0000	1.0000	1.0000	.9999	.9995	.9983	.9955	.9893
9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9997	.9990
10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

$n = 20$									
r	0.1	0.15	0.2	0.25	0.3	0.35	0.4	0.45	0.5
0	.1216	.0388	.0115	.0032	.0008	.0002			
1	.3917	.1756	.0692	.0243	.0076	.0021	.0005	.0001	
2	.6769	.4049	.2061	.0913	.0355	.0121	.0036	.0009	.0002
3	.8670	.6477	.4114	.2252	.1071	.0444	.0160	.0049	.0013
4	.9568	.8298	.6296	.4148	.2375	.1182	.0510	.0189	.0059
5	.9887	.9327	.8042	.6172	.4164	.2454	.1256	.0553	.0207
6	.9976	.9781	.9133	.7858	.6080	.4166	.2500	.1299	.0577
7	.9996	.9941	.9679	.8982	.7723	.6010	.4159	.2520	.1316
8	.9999	.9987	.9900	.9591	.8867	.7624	.5956	.4143	.2517
9	1.0000	.9998	.9974	.9861	.9520	.8782	.7553	.5914	.4119
10	1.0000	1.0000	.9994	.9961	.9829	.9468	.8725	.7507	.5881
11	1.0000	1.0000	.9999	.9991	.9949	.9804	.9435	.8692	.7483
12	1.0000	1.0000	1.0000	.9998	.9987	.9940	.9790	.9420	.8684
13	1.0000	1.0000	1.0000	1.0000	.9997	.9985	.9935	.9786	.9423
14	1.0000	1.0000	1.0000	1.0000	1.0000	.9997	.9984	.9936	.9793
15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9997	.9985	.9941
16	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9997	.9987
17	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9998
18	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Table 1: Cumulative Binomial Probabilities—continued

$n = 50$									
r	0.1	0.15	0.2	0.25	0.3	0.35	0.4	0.45	0.5
0	.0052	.0003							
1	.0338	.0029	.0002						
2	.1117	.0142	.0013	.0001					
3	.2503	.0460	.0057	.0005					
4	.4312	.1121	.0185	.0021	.0002				
5	.6161	.2194	.0480	.0070	.0007	.0001			
6	.7702	.3613	.1034	.0194	.0025	.0002			
7	.8779	.5188	.1904	.0453	.0073	.0008	.0001		
8	.9421	.6681	.3073	.0916	.0183	.0025	.0002		
9	.9755	.7911	.4437	.1637	.0402	.0067	.0008	.0001	
10	.9906	.8801	.5836	.2622	.0789	.0160	.0022	.0002	
11	.9968	.9372	.7107	.3816	.1390	.0342	.0057	.0006	
12	.9990	.9699	.8139	.5110	.2229	.0661	.0133	.0018	.0002
13	.9997	.9868	.8894	.6370	.3279	.1163	.0280	.0045	.0005
14	.9999	.9947	.9393	.7481	.4468	.1878	.0540	.0104	.0013
15	1.0000	.9981	.9692	.8369	.5692	.2801	.0955	.0220	.0033
16	1.0000	.9993	.9856	.9017	.6839	.3889	.1561	.0427	.0077
17	1.0000	.9998	.9937	.9449	.7822	.5060	.2369	.0765	.0164
18	1.0000	.9999	.9975	.9713	.8594	.6216	.3356	.1273	.0325
19	1.0000	1.0000	.9991	.9861	.9152	.7264	.4465	.1974	.0595
20	1.0000	1.0000	.9997	.9937	.9522	.8139	.5610	.2862	.1013
21	1.0000	1.0000	.9999	.9974	.9749	.8813	.6701	.3900	.1611
22	1.0000	1.0000	1.0000	.9990	.9877	.9290	.7660	.5019	.2399
23	1.0000	1.0000	1.0000	.9996	.9944	.9604	.8438	.6134	.3359
24	1.0000	1.0000	1.0000	.9999	.9976	.9793	.9022	.7160	.4439
25	1.0000	1.0000	1.0000	1.0000	.9991	.9900	.9427	.8034	.5561
26	1.0000	1.0000	1.0000	1.0000	.9997	.9955	.9686	.8721	.6641
27	1.0000	1.0000	1.0000	1.0000	.9999	.9981	.9840	.9220	.7601
28	1.0000	1.0000	1.0000	1.0000	1.0000	.9993	.9924	.9556	.8389
29	1.0000	1.0000	1.0000	1.0000	1.0000	.9997	.9966	.9765	.8987
30	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9986	.9884	.9405
31	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9995	.9947	.9675
32	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9998	.9978	.9836
33	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9991	.9923
34	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9997	.9967
35	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9987
36	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9995
37	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9998
38	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Table 1 gives binomial probabilities only for a limited range of values of n and p since, in practice, either the more compact Poisson table or the Normal table can usually provide an adequate approximation.

As a reasonable working rule:

- (i) use the Poisson approximation if $p < 0.1$, putting $m = np$;
- (ii) use the Normal approximation if $0.1 \leq p \leq 0.9$ and $np > 5$, putting $\mu = np$ and $\sigma = \sqrt{np(1-p)}$;
- (iii) use the Poisson approximation if $p > 0.9$, putting $m = n(1-p)$ and working in terms of failures.

For $p > 0.5$, work in terms of failures, whose probability is $q = 1 - p$.

Example E.1

If 40 or more of 50 seeds are to germinate when the germination rate is 70%, this is equivalent to 10 or fewer failures. Thus the required probability is $1 - P(11 \text{ or more failures}) = 1 - 0.9211 = 0.0789$.

Table 1: Cumulative Binomial Probabilities—continued

$n = 100$									
r	0.1	0.15	0.2	0.25	0.3	0.35	0.4	0.45	0.5
0									
1	.0003								
2	.0019								
3	.0078	.0001							
4	.0237	.0004							
5	.0576	.0016							
6	.1172	.0047	.0001						
7	.2061	.0122	.0003						
8	.3209	.0275	.0009						
9	.4513	.0551	.0023						
10	.5832	.0994	.0057	.0001					
11	.7030	.1635	.0126	.0004					
12	.8018	.2473	.0253	.0010					
13	.8761	.3474	.0469	.0025	.0001				
14	.9274	.4572	.0804	.0054	.0002				
15	.9601	.5683	.1285	.0111	.0004				
16	.9794	.6725	.1923	.0211	.0010				
17	.9900	.7633	.2712	.0376	.0022	.0001			
18	.9954	.8372	.3621	.0630	.0045	.0001			
19	.9980	.8935	.4602	.0995	.0089	.0003			
20	.9992	.9337	.5595	.1488	.0165	.0008			
21	.9997	.9607	.6540	.2114	.0288	.0017			
22	.9999	.9779	.7389	.2864	.0479	.0034	.0001		
23	1.0000	.9881	.8109	.3711	.0755	.0066	.0003		
24	1.0000	.9939	.8686	.4617	.1136	.0121	.0006		
25	1.0000	.9970	.9125	.5535	.1631	.0211	.0012		
26	1.0000	.9986	.9442	.6417	.2244	.0351	.0024	.0001	
27	1.0000	.9994	.9658	.7224	.2964	.0558	.0046	.0002	
28	1.0000	.9997	.9800	.7925	.3768	.0848	.0084	.0004	
29	1.0000	.9999	.9888	.8505	.4623	.1236	.0148	.0008	
30	1.0000	1.0000	.9939	.8962	.5491	.1730	.0248	.0015	
31	1.0000	1.0000	.9969	.9307	.6331	.2331	.0398	.0030	.0001
32	1.0000	1.0000	.9984	.9554	.7107	.3029	.0615	.0055	.0002
33	1.0000	1.0000	.9993	.9724	.7793	.3803	.0913	.0098	.0004
34	1.0000	1.0000	.9997	.9836	.8371	.4624	.1303	.0166	.0009
35	1.0000	1.0000	.9999	.9906	.8839	.5458	.1795	.0272	.0018
36	1.0000	1.0000	.9999	.9948	.9201	.6269	.2386	.0429	.0033
37	1.0000	1.0000	1.0000	.9973	.9470	.7024	.3068	.0651	.0060
38	1.0000	1.0000	1.0000	.9986	.9660	.7699	.3822	.0951	.0105
39	1.0000	1.0000	1.0000	.9993	.9790	.8276	.4621	.1343	.0176
40	1.0000	1.0000	1.0000	.9997	.9875	.8750	.5433	.1831	.0284
41	1.0000	1.0000	1.0000	.9999	.9928	.9123	.6225	.2415	.0443
42	1.0000	1.0000	1.0000	.9999	.9960	.9406	.6967	.3087	.0666
43	1.0000	1.0000	1.0000	1.0000	.9979	.9611	.7635	.3828	.0967
44	1.0000	1.0000	1.0000	1.0000	.9989	.9754	.8211	.4613	.1356
45	1.0000	1.0000	1.0000	1.0000	.9995	.9850	.8689	.5413	.1841
46	1.0000	1.0000	1.0000	1.0000	.9997	.9912	.9070	.6196	.2421
47	1.0000	1.0000	1.0000	1.0000	.9999	.9950	.9362	.6931	.3086
48	1.0000	1.0000	1.0000	1.0000	.9999	.9973	.9577	.7596	.3822
49	1.0000	1.0000	1.0000	1.0000	1.0000	.9985	.9729	.8173	.4602
50	1.0000	1.0000	1.0000	1.0000	1.0000	.9993	.9832	.8654	.5398
51	1.0000	1.0000	1.0000	1.0000	1.0000	.9996	.9900	.9040	.6178
52	1.0000	1.0000	1.0000	1.0000	1.0000	.9998	.9942	.9338	.6914
53	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9968	.9559	.7579
54	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9983	.9716	.8159
55	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9991	.9824	.8644
56	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9996	.9894	.9033
57	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9998	.9939	.9334
58	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9966	.9557
59	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9982	.9716
60	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9991	.9824
61	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9995	.9895
62	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9998	.9940
63	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9967
64	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9982
65	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9991
66	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9996
67	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9998
68	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999
69	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Table 2: Cumulative Poisson Probabilities

The table gives the probability that r or fewer random events occur in an interval when the average number of events is m :

$$P(X \leq r) = \sum_{x=0}^r e^{-m} \frac{m^x}{x!}.$$

An empty entry represents a probability below 0.00005; a displayed value of 1.0000 represents a probability above 0.99995.

r	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
0	.9048	.8187	.7408	.6703	.6065	.5488	.4966	.4493	.4066	.3679
1	.9953	.9825	.9631	.9384	.9098	.8781	.8442	.8088	.7725	.7358
2	.9998	.9989	.9964	.9921	.9856	.9769	.9659	.9526	.9371	.9197
3	1.0000	.9999	.9997	.9992	.9982	.9966	.9942	.9909	.9865	.9810
4	1.0000	1.0000	1.0000	.9999	.9998	.9996	.9992	.9986	.9977	.9963
5	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9998	.9997	.9994
6	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999
7	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

r	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0	.3329	.3012	.2725	.2466	.2231	.2019	.1827	.1653	.1496	.1353
1	.6990	.6626	.6268	.5918	.5578	.5249	.4932	.4628	.4337	.4060
2	.9004	.8795	.8571	.8335	.8088	.7834	.7572	.7306	.7037	.6767
3	.9743	.9662	.9569	.9463	.9344	.9212	.9068	.8913	.8747	.8571
4	.9946	.9923	.9893	.9857	.9814	.9763	.9704	.9636	.9559	.9473
5	.9990	.9985	.9978	.9968	.9955	.9940	.9920	.9896	.9868	.9834
6	.9999	.9997	.9996	.9994	.9991	.9987	.9981	.9974	.9966	.9955
7	1.0000	1.0000	.9999	.9999	.9998	.9997	.9996	.9994	.9992	.9989
8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9999	.9998	.9998
9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

r	2.1	2.2	2.3	2.4	2.5	2.6	2.7	2.8	2.9	3
0	.1225	.1108	.1003	.0907	.0821	.0743	.0672	.0608	.0550	.0498
1	.3796	.3546	.3309	.3084	.2873	.2674	.2487	.2311	.2146	.1991
2	.6496	.6227	.5960	.5697	.5438	.5184	.4936	.4695	.4460	.4232
3	.8386	.8194	.7993	.7787	.7576	.7360	.7141	.6919	.6696	.6472
4	.9379	.9275	.9162	.9041	.8912	.8774	.8629	.8477	.8318	.8153
5	.9796	.9751	.9700	.9643	.9580	.9510	.9433	.9349	.9258	.9161
6	.9941	.9925	.9906	.9884	.9858	.9828	.9794	.9756	.9713	.9665
7	.9985	.9980	.9974	.9967	.9958	.9947	.9934	.9919	.9901	.9881
8	.9997	.9995	.9994	.9991	.9989	.9985	.9981	.9976	.9969	.9962
9	.9999	.9999	.9999	.9998	.9997	.9996	.9995	.9993	.9991	.9989
10	1.0000	1.0000	1.0000	1.0000	.9999	.9999	.9999	.9998	.9998	.9997
11	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9999
12	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Table 2: Cumulative Poisson Probabilities—continued

<i>r</i>	3.1	3.2	3.3	3.4	3.5	3.6	3.7	3.8	3.9	4
0	.0450	.0408	.0369	.0334	.0302	.0273	.0247	.0224	.0202	.0183
1	.1847	.1712	.1586	.1468	.1359	.1257	.1162	.1074	.0992	.0916
2	.4012	.3799	.3594	.3397	.3208	.3027	.2854	.2689	.2531	.2381
3	.6248	.6025	.5803	.5584	.5366	.5152	.4942	.4735	.4532	.4335
4	.7982	.7806	.7626	.7442	.7254	.7064	.6872	.6678	.6484	.6288
5	.9057	.8946	.8829	.8705	.8576	.8441	.8301	.8156	.8006	.7851
6	.9612	.9554	.9490	.9421	.9347	.9267	.9182	.9091	.8995	.8893
7	.9858	.9832	.9802	.9769	.9733	.9692	.9648	.9599	.9546	.9489
8	.9953	.9943	.9931	.9917	.9901	.9883	.9863	.9840	.9815	.9786
9	.9986	.9982	.9978	.9973	.9967	.9960	.9952	.9942	.9931	.9919
10	.9996	.9995	.9994	.9992	.9990	.9987	.9984	.9981	.9977	.9972
11	.9999	.9999	.9998	.9998	.9997	.9996	.9995	.9994	.9993	.9991
12	1.0000	1.0000	1.0000	.9999	.9999	.9999	.9999	.9998	.9998	.9997
13	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9999
14	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

<i>r</i>	4.1	4.2	4.3	4.4	4.5	4.6	4.7	4.8	4.9	5
0	.0166	.0150	.0136	.0123	.0111	.0101	.0091	.0082	.0074	.0067
1	.0845	.0780	.0719	.0663	.0611	.0563	.0518	.0477	.0439	.0404
2	.2238	.2102	.1974	.1851	.1736	.1626	.1523	.1425	.1333	.1247
3	.4142	.3954	.3772	.3594	.3423	.3257	.3097	.2942	.2793	.2650
4	.6093	.5898	.5704	.5512	.5321	.5132	.4946	.4763	.4582	.4405
5	.7693	.7531	.7367	.7199	.7029	.6858	.6684	.6510	.6335	.6160
6	.8786	.8675	.8558	.8436	.8311	.8180	.8046	.7908	.7767	.7622
7	.9427	.9361	.9290	.9214	.9134	.9049	.8960	.8867	.8769	.8666
8	.9755	.9721	.9683	.9642	.9597	.9549	.9497	.9442	.9382	.9319
9	.9905	.9889	.9871	.9851	.9829	.9805	.9778	.9749	.9717	.9682
10	.9966	.9959	.9952	.9943	.9933	.9922	.9910	.9896	.9880	.9863
11	.9989	.9986	.9983	.9980	.9976	.9971	.9966	.9960	.9953	.9945
12	.9997	.9996	.9995	.9993	.9992	.9990	.9988	.9986	.9983	.9980
13	.9999	.9999	.9998	.9998	.9997	.9997	.9996	.9995	.9994	.9993
14	1.0000	1.0000	1.0000	.9999	.9999	.9999	.9999	.9999	.9998	.9998
15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9999
16	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

<i>r</i>	5.2	5.4	5.6	5.8	6	6.2	6.4	6.6	6.8	7
0	.0055	.0045	.0037	.0030	.0025	.0020	.0017	.0014	.0011	.0009
1	.0342	.0289	.0244	.0206	.0174	.0146	.0123	.0103	.0087	.0073
2	.1088	.0948	.0824	.0715	.0620	.0536	.0463	.0400	.0344	.0296
3	.2381	.2133	.1906	.1700	.1512	.1342	.1189	.1052	.0928	.0818
4	.4061	.3733	.3422	.3127	.2851	.2592	.2351	.2127	.1920	.1730
5	.5809	.5461	.5119	.4783	.4457	.4141	.3837	.3547	.3270	.3007
6	.7324	.7017	.6703	.6384	.6063	.5742	.5423	.5108	.4799	.4497
7	.8449	.8217	.7970	.7710	.7440	.7160	.6873	.6581	.6285	.5987
8	.9181	.9027	.8857	.8672	.8472	.8259	.8033	.7796	.7548	.7291
9	.9603	.9512	.9409	.9292	.9161	.9016	.8858	.8686	.8502	.8305
10	.9823	.9775	.9718	.9651	.9574	.9486	.9386	.9274	.9151	.9015
11	.9927	.9904	.9875	.9841	.9799	.9750	.9693	.9627	.9552	.9467
12	.9972	.9962	.9949	.9932	.9912	.9887	.9857	.9821	.9779	.9730
13	.9990	.9986	.9980	.9973	.9964	.9952	.9937	.9920	.9898	.9872
14	.9997	.9995	.9993	.9990	.9986	.9981	.9974	.9966	.9956	.9943
15	.9999	.9998	.9998	.9996	.9995	.9993	.9990	.9986	.9982	.9976
16	1.0000	.9999	.9999	.9999	.9998	.9997	.9996	.9995	.9993	.9990
17	1.0000	1.0000	1.0000	1.0000	.9999	.9999	.9999	.9998	.9997	.9996
18	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9999	.9999
19	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Table 2: Cumulative Poisson Probabilities—continued

<i>r</i>	7.2	7.4	7.6	7.8	8	8.2	8.4	8.6	8.8	9
0	.0007	.0006	.0005	.0004	.0003	.0003	.0002	.0002	.0002	.0001
1	.0061	.0051	.0043	.0036	.0030	.0025	.0021	.0018	.0015	.0012
2	.0255	.0219	.0188	.0161	.0138	.0118	.0100	.0086	.0073	.0062
3	.0719	.0632	.0554	.0485	.0424	.0370	.0323	.0281	.0244	.0212
4	.1555	.1395	.1249	.1117	.0996	.0887	.0789	.0701	.0621	.0550
5	.2759	.2526	.2307	.2103	.1912	.1736	.1573	.1422	.1284	.1157
6	.4204	.3920	.3646	.3384	.3134	.2896	.2670	.2457	.2256	.2068
7	.5689	.5393	.5100	.4812	.4530	.4254	.3987	.3728	.3478	.3239
8	.7027	.6757	.6482	.6204	.5925	.5647	.5369	.5094	.4823	.4557
9	.8096	.7877	.7649	.7411	.7166	.6915	.6659	.6400	.6137	.5874
10	.8867	.8707	.8535	.8352	.8159	.7955	.7743	.7522	.7294	.7060
11	.9371	.9265	.9148	.9020	.8881	.8731	.8571	.8400	.8220	.8030
12	.9673	.9609	.9536	.9454	.9362	.9261	.9150	.9029	.8898	.8758
13	.9841	.9805	.9762	.9714	.9658	.9595	.9524	.9445	.9358	.9261
14	.9927	.9908	.9886	.9859	.9827	.9791	.9749	.9701	.9647	.9585
15	.9969	.9959	.9948	.9934	.9918	.9898	.9875	.9848	.9816	.9780
16	.9987	.9983	.9978	.9971	.9963	.9953	.9941	.9926	.9909	.9889
17	.9995	.9993	.9991	.9988	.9984	.9979	.9973	.9966	.9957	.9947
18	.9998	.9997	.9996	.9995	.9993	.9991	.9989	.9985	.9981	.9976
19	.9999	.9999	.9999	.9998	.9997	.9997	.9995	.9994	.9992	.9989
20	1.0000	1.0000	1.0000	.9999	.9999	.9999	.9998	.9998	.9997	.9996
21	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9999	.9999	.9998
22	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999
23	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

<i>r</i>	9.2	9.4	9.6	9.8	10	11	12	13	14	15
0	.0001	.0001	.0001	.0001						
1	.0010	.0009	.0007	.0006	.0005	.0002	.0001			
2	.0053	.0045	.0038	.0033	.0028	.0012	.0005	.0002	.0001	
3	.0184	.0160	.0138	.0120	.0103	.0049	.0023	.0011	.0005	.0002
4	.0486	.0429	.0378	.0333	.0293	.0151	.0076	.0037	.0018	.0009
5	.1041	.0935	.0838	.0750	.0671	.0375	.0203	.0107	.0055	.0028
6	.1892	.1727	.1574	.1433	.1301	.0786	.0458	.0259	.0142	.0076
7	.3010	.2792	.2584	.2388	.2202	.1432	.0895	.0540	.0316	.0180
8	.4296	.4042	.3796	.3558	.3328	.2320	.1550	.0998	.0621	.0374
9	.5611	.5349	.5089	.4832	.4579	.3405	.2424	.1658	.1094	.0699
10	.6820	.6576	.6329	.6080	.5830	.4599	.3472	.2517	.1757	.1185
11	.7832	.7626	.7412	.7193	.6968	.5793	.4616	.3532	.2600	.1848
12	.8607	.8448	.8279	.8101	.7916	.6887	.5760	.4631	.3585	.2676
13	.9156	.9042	.8919	.8786	.8645	.7813	.6815	.5730	.4644	.3632
14	.9517	.9441	.9357	.9265	.9165	.8540	.7720	.6751	.5704	.4657
15	.9738	.9691	.9638	.9579	.9513	.9074	.8444	.7636	.6694	.5681
16	.9865	.9838	.9806	.9770	.9730	.9441	.8987	.8355	.7559	.6641
17	.9934	.9919	.9902	.9881	.9857	.9678	.9370	.8905	.8272	.7489
18	.9969	.9962	.9952	.9941	.9928	.9823	.9626	.9302	.8826	.8195
19	.9986	.9983	.9978	.9972	.9965	.9907	.9787	.9573	.9235	.8752
20	.9994	.9992	.9990	.9987	.9984	.9953	.9884	.9750	.9521	.9170
21	.9998	.9997	.9996	.9995	.9993	.9977	.9939	.9859	.9712	.9469
22	.9999	.9999	.9998	.9998	.9997	.9990	.9970	.9924	.9833	.9673
23	1.0000	1.0000	.9999	.9999	.9999	.9995	.9985	.9960	.9907	.9805
24	1.0000	1.0000	1.0000	1.0000	1.0000	.9998	.9993	.9980	.9950	.9888
25	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9997	.9990	.9974	.9938
26	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9995	.9987	.9967
27	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9998	.9994	.9983
28	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9997	.9991
29	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9996
30	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9998
31	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999
32	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Table 2: Cumulative Poisson Probabilities—continued

r	16	17	18	19	20	21	22	23	24	25
3	.0001									
4	.0004	.0002	.0001							
5	.0014	.0007	.0003	.0002	.0001					
6	.0040	.0021	.0010	.0005	.0003	.0001	.0001			
7	.0100	.0054	.0029	.0015	.0008	.0004	.0002	.0001		
8	.0220	.0126	.0071	.0039	.0021	.0011	.0006	.0003	.0002	.0001
9	.0433	.0261	.0154	.0089	.0050	.0028	.0015	.0008	.0004	.0002
10	.0774	.0491	.0304	.0183	.0108	.0063	.0035	.0020	.0011	.0006
11	.1270	.0847	.0549	.0347	.0214	.0129	.0076	.0044	.0025	.0014
12	.1931	.1350	.0917	.0606	.0390	.0245	.0151	.0091	.0054	.0031
13	.2745	.2009	.1426	.0984	.0661	.0434	.0278	.0174	.0107	.0065
14	.3675	.2808	.2081	.1497	.1049	.0716	.0477	.0311	.0198	.0124
15	.4667	.3715	.2867	.2148	.1565	.1111	.0769	.0520	.0344	.0223
16	.5660	.4677	.3751	.2920	.2211	.1629	.1170	.0821	.0563	.0377
17	.6593	.5640	.4686	.3784	.2970	.2270	.1690	.1228	.0871	.0605
18	.7423	.6550	.5622	.4695	.3814	.3017	.2325	.1748	.1283	.0920
19	.8122	.7363	.6509	.5606	.4703	.3843	.3060	.2377	.1803	.1336
20	.8682	.8055	.7307	.6472	.5591	.4710	.3869	.3101	.2426	.1855
21	.9108	.8615	.7991	.7255	.6437	.5577	.4716	.3894	.3139	.2473
22	.9418	.9047	.8551	.7931	.7206	.6405	.5564	.4723	.3917	.3175
23	.9633	.9367	.8989	.8490	.7875	.7160	.6374	.5551	.4728	.3939
24	.9777	.9594	.9317	.8933	.8432	.7822	.7117	.6346	.5540	.4734
25	.9869	.9748	.9554	.9269	.8878	.8377	.7771	.7077	.6319	.5529
26	.9925	.9848	.9718	.9514	.9221	.8826	.8324	.7723	.7038	.6294
27	.9959	.9912	.9827	.9687	.9475	.9175	.8775	.8274	.7677	.7002
28	.9978	.9950	.9897	.9805	.9657	.9436	.9129	.8726	.8225	.7634
29	.9989	.9973	.9941	.9882	.9782	.9626	.9398	.9085	.8679	.8179
30	.9994	.9986	.9967	.9930	.9865	.9758	.9595	.9360	.9042	.8633
31	.9997	.9993	.9982	.9960	.9919	.9848	.9735	.9564	.9322	.8999
32	.9999	.9996	.9990	.9978	.9953	.9907	.9831	.9711	.9533	.9285
33	.9999	.9998	.9995	.9988	.9973	.9945	.9895	.9813	.9686	.9502
34	1.0000	.9999	.9998	.9994	.9985	.9968	.9936	.9882	.9794	.9662
35	1.0000	1.0000	.9999	.9997	.9992	.9982	.9962	.9927	.9868	.9775
36	1.0000	1.0000	.9999	.9998	.9996	.9990	.9978	.9956	.9918	.9854
37	1.0000	1.0000	1.0000	.9999	.9998	.9995	.9988	.9974	.9950	.9908
38	1.0000	1.0000	1.0000	1.0000	.9999	.9997	.9993	.9985	.9970	.9943
39	1.0000	1.0000	1.0000	1.0000	.9999	.9999	.9996	.9992	.9983	.9966
40	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9998	.9996	.9990	.9980
41	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9998	.9995	.9988
42	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9997	.9993
43	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9998	.9996
44	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9998
45	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999
46	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999
47	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Table 2: Cumulative Poisson Probabilities—continued

r	26	27	28	29	30	32	34	36	38	40
9	.0001	.0001								
10	.0003	.0002	.0001							
11	.0008	.0004	.0002	.0001	.0001					
12	.0018	.0010	.0006	.0003	.0002					
13	.0038	.0022	.0013	.0007	.0004	.0001				
14	.0076	.0046	.0027	.0016	.0009	.0003	.0001			
15	.0142	.0088	.0054	.0033	.0019	.0007	.0002	.0001		
16	.0248	.0160	.0101	.0063	.0039	.0014	.0005	.0002		
17	.0411	.0274	.0179	.0115	.0073	.0028	.0010	.0003	.0001	
18	.0646	.0445	.0300	.0199	.0129	.0052	.0020	.0007	.0002	.0001
19	.0968	.0687	.0478	.0326	.0219	.0093	.0037	.0014	.0005	.0002
20	.1387	.1015	.0727	.0511	.0353	.0159	.0068	.0027	.0010	.0004
21	.1905	.1436	.1060	.0767	.0544	.0260	.0116	.0049	.0019	.0007
22	.2517	.1952	.1483	.1104	.0806	.0406	.0191	.0085	.0035	.0014
23	.3209	.2559	.1998	.1529	.1146	.0610	.0302	.0141	.0062	.0026
24	.3959	.3242	.2599	.2042	.1572	.0881	.0460	.0224	.0103	.0045
25	.4739	.3979	.3272	.2637	.2084	.1228	.0674	.0345	.0166	.0076
26	.5519	.4744	.3997	.3301	.2673	.1656	.0953	.0513	.0259	.0123
27	.6270	.5509	.4749	.4014	.3329	.2162	.1306	.0736	.0389	.0193
28	.6967	.6247	.5500	.4753	.4031	.2741	.1733	.1023	.0565	.0294
29	.7593	.6935	.6226	.5492	.4757	.3380	.2235	.1379	.0796	.0432
30	.8134	.7553	.6903	.6206	.5484	.4061	.2804	.1806	.1089	.0617
31	.8589	.8092	.7515	.6874	.6186	.4765	.3427	.2303	.1448	.0855
32	.8958	.8546	.8051	.7479	.6845	.5468	.4089	.2861	.1875	.1153
33	.9249	.8918	.8505	.8011	.7444	.6150	.4772	.3470	.2365	.1514
34	.9472	.9213	.8879	.8465	.7973	.6792	.5454	.4115	.2914	.1939
35	.9637	.9441	.9178	.8841	.8426	.7379	.6117	.4778	.3510	.2424
36	.9756	.9612	.9411	.9144	.8804	.7901	.6744	.5442	.4138	.2963
37	.9840	.9737	.9587	.9381	.9110	.8352	.7319	.6087	.4784	.3547
38	.9897	.9825	.9717	.9562	.9352	.8732	.7834	.6699	.5430	.4160
39	.9936	.9887	.9810	.9697	.9537	.9044	.8283	.7263	.6059	.4790
40	.9961	.9928	.9875	.9795	.9677	.9293	.8664	.7771	.6657	.5419
41	.9976	.9955	.9920	.9864	.9779	.9488	.8981	.8217	.7211	.6033
42	.9986	.9973	.9950	.9911	.9852	.9636	.9237	.8599	.7712	.6618
43	.9992	.9984	.9969	.9944	.9903	.9747	.9439	.8919	.8155	.7162
44	.9996	.9991	.9981	.9965	.9937	.9827	.9596	.9181	.8538	.7657
45	.9998	.9995	.9989	.9978	.9960	.9884	.9714	.9391	.8861	.8097
46	.9999	.9997	.9994	.9987	.9975	.9924	.9801	.9555	.9128	.8479
47	.9999	.9998	.9996	.9992	.9985	.9951	.9864	.9680	.9343	.8804
48	1.0000	.9999	.9998	.9996	.9991	.9969	.9909	.9775	.9514	.9075
49	1.0000	1.0000	.9999	.9998	.9995	.9981	.9940	.9844	.9647	.9297
50	1.0000	1.0000	.9999	.9999	.9997	.9988	.9961	.9894	.9747	.9474
51	1.0000	1.0000	1.0000	.9999	.9998	.9993	.9976	.9929	.9822	.9613
52	1.0000	1.0000	1.0000	1.0000	.9999	.9996	.9985	.9953	.9877	.9719
53	1.0000	1.0000	1.0000	1.0000	.9999	.9998	.9991	.9970	.9916	.9800
54	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9994	.9981	.9944	.9860
55	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9997	.9988	.9963	.9903
56	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9998	.9993	.9976	.9934
57	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9995	.9985	.9956
58	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9997	.9990	.9971
59	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9998	.9994	.9981
60	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9996	.9988
61	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9998	.9992
62	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9995
63	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9997
64	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9998
65	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999
66	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999
67	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

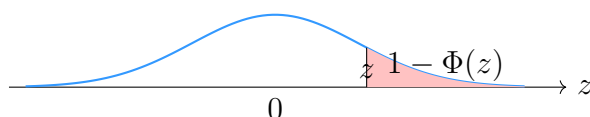
For values of m greater than 30, the table of areas under the Normal curve may be used to obtain approximate Poisson probabilities, putting $\mu = m$ and $\sigma = \sqrt{m}$.

Table 3: Areas in the Upper Tail of the Normal Distribution

The function tabulated is $1 - \Phi(z)$, where $\Phi(z)$ is the cumulative distribution function of a standard Normal variable:

$$1 - \Phi(z) = \frac{1}{\sqrt{2\pi}} \int_z^{\infty} e^{-t^2/2} dt.$$

Thus $1 - \Phi(z)$ is the probability that a standardised Normal variable is greater than $z = (x - \mu)/\sigma$.



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641
0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
2.3	.0107	.0104	.0102	.0099	.0096	.0093	.0091	.0088	.0086	.0084
2.4	.0082	.0079	.0077	.0075	.0073	.0071	.0069	.0067	.0065	.0063
2.5	.0062	.0060	.0058	.0057	.0055	.0053	.0052	.0050	.0049	.0048
2.6	.0046	.0045	.0044	.0042	.0041	.0040	.0039	.0037	.0036	.0035
2.7	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0028	.0027	.0026
2.8	.0025	.0024	.0024	.0023	.0022	.0021	.0021	.0020	.0019	.0019
2.9	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014	.0013
3.0	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010	.0010
3.1	.0009	.0009	.0009	.0008	.0008	.0008	.0007	.0007	.0007	.0007
3.2	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005	.0005	.0005
3.3	.0004	.0004	.0004	.0004	.0004	.0004	.0003	.0003	.0003	.0003
3.4	.0003	.0003	.0003	.0003	.0002	.0002	.0002	.0002	.0002	.0002
3.5	.0002	.0002	.0002	.0002	.0002	.0001	.0001	.0001	.0001	.0001
3.6	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001
3.7	.0001	.0001	.0001	.0000	.0000	.0000	.0000	.0000	.0000	.0000
3.8	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000
3.9	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000
4.0	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000

$$1 - \Phi(5.0) = 0.0000002867.$$

$$1 - \Phi(5.5) = 0.0000000190.$$

$$1 - \Phi(6.0) = 0.0000000010.$$

Appendix F

Answers to Practice Exercises

This appendix brings together the answer keys supplied for the practice sections throughout all chapters. Equivalent mathematical forms are also acceptable unless a particular form was requested.

Chapter 1: Number System

These answers correspond to Section 1.9.

1. Yes; for example, $0 = 0/1$.
2. (a) Yes. If $r + s$ were rational, then $s = (r + s) - r$ would be rational, a contradiction. (b) Not necessarily: $0 \cdot s = 0$. If $r \neq 0$ is rational, then rs is irrational.
3. (a) True; (b) false; (c) false; (d) true; (e) false; (f) false; (g) false; (h) false; (i) false, for example $(\sqrt{2})^2 = 2$; (j) false because $\sqrt{18}/\sqrt{2} = 3$.
4. (a) 15, rational; (b) $4\sqrt{5}$, irrational; (c) $2\sqrt{2}/5$, irrational; (d) $3/5$, rational.
5. (a) $a = \pm\sqrt{7}$, irrational; (b) $b = \pm 4$, rational; (c) $c = \pm 0.3$, rational; (d) $d = \pm\sqrt{13}/3$, irrational.
6. Examples: (a) $-11/4, -5/2, -9/4$; (b) $0.202, 0.205, 0.208$; (c) $2/5, 5/12, 9/20$.
7. Examples: (a) $3/2, \sqrt{2}$; (b) $1/10, \sqrt{2}/10$; (c) $1/2, \sqrt{2}/3$; (d) $2, (\sqrt{3} + \sqrt{5})/2$.
8. (a) $9/20$; (b) $4/11$; (c) $23/18$; (d) $2/625$. If $x = 0.\overline{18}$, then $100x - x = 18$, so $x = 18/99 = 2/11$.
9. $-7/3 \approx -2.333$, $-9/4 = -2.25$, 5 , 5.4 . Also $2 < \sqrt{6} < 3$, $3 < \sqrt{11} < 4$, and $4 < \sqrt{19} < 5$.
10. (a) $(-3, 0)$; (b) $(-1, e]$; (c) $\mathbb{R}$.
11. (a) $(3, 4)$; (b) $(-\infty, 5)$; (c) $[0, 1)$; (d) $(-2, 3]$; (e) $(-5, 1)$; (f) $(0, 4]$; (g) $[2, 7]$; (h) $[-2, \infty)$.
12. (a) distributive; (b) commutative addition; (c) associative multiplication; (d) multiplicative inverse; (e) commutative addition; (f) associative addition; (g) additive inverse; (h) additive identity.
13. Line 3 is wrong: subtracting $(500 - 3)$ gives $1273 - 500 + 3 = 776$.
14. (a) 10; (b) $1/25$; (c) -125 .
15. (a) $x^5/(2y^2)$; (b) $3y^{13}/2$; (c) $y^{-1/2} = 1/\sqrt{y}$.
16. (a) $\sqrt{15}/3$; (b) $39\sqrt{2}$; (c) $\sqrt{3}\sqrt[3]{4}/4$; (d) $\sqrt{5}$; (e) $2\sqrt{2}$; (f) $7 - 2\sqrt{10}$; (g) $29\sqrt{6}/6$.
17. (a) $7/2$; (b) 4; (c) $\log_3 256$; (d) 2; (e) -3 ; (f) $5/2$.
18. (a) $x = 6$; (b) $x = -1$; (c) $x = 9$; (d) $x = 0, 2$; (e) $x = 5/4$; (f) $x = 3$; (g) $x = \pm 1$.
19. (a) $y = 25$; (b) $x = 5 + e$; (c) $x = 9/22$; (d) no real solution satisfying the domains; (e) $x = 5$; (f) $x = 3$; (g) $x = 1, 8$; (h) $x = 4$.
20. (a) two distinct irrational real roots; (b) one repeated rational root; (c) no real roots.
21. The intersection equation is $3x^2 - 2mx + 3 = 0$. Thus the line intersects the curve when $m^2 \geq 9$, that is, $m \leq -3$ or $m \geq 3$. If “cut” is intended to mean two distinct intersection points, then $m < -3$ or $m > 3$.
22. (a) $x = \pm 1/\sqrt{2}$; (b) $t = 8/5, 5$; (c) $x = -17/2$.
23. $t = \ln(1/2)/\ln(0.93) \approx 9.55$ years.
24. (a) 3^{7x} ; (b) $-24/125$; (c) $65 \cdot 4^{n-4}$; (d) a^2b^5 ; (e) p^6q^5 ; (f) $128y^2/x^4$; (g) $16rs^2$; (h) u^6/v^4 ; (i) $x^3y^5/27$; (j) $3d^5/c^7$; (k) $3/\sqrt[4]{b}$; (l) $z^{7/6}$.
25. (a) $8^{2p-1} = 2^{6p-3} = 64^p/8$. (b) $25^{q+1} = 5^{2q+2} = 25(5^{2q})$. (c) the expression is $3^k(9 + 3 + 1) = 13 \cdot 3^k$.
26. (a) $x = -7$; (b) $x = 1$; (c) $x = -1$.

27. (a) $h_0 = 2.5$ m, $h_7 \approx 0.623$ m; first below 0.5 m at $n = 9$. (b) $N(5) = 58,320$ cells, reached at $t = 5$ h. (c) $P(18) \approx 2.663$ million. (d) RM 25,342.18 approximately.
28. (a) $PR = 6$ cm. (b) area $= 13/2$ cm² and hypotenuse $= \sqrt{38}$ cm. (c) $y = 29 + 12\sqrt{5}$. (d) (i) $x = -1$; (ii) $x = 13/4$; (iii) $x = 3/4$.
29. (a) $\mathbb{N}$; (b) $\mathbb{W}$; (c) $\mathbb{Z}$; (d) $\mathbb{Q}$; (e) $\mathbb{N}$; (f), (g) irrational real.
30. (a) $-2.3 < -9/4 < -\sqrt{5} < -2$; (b) $2/3 < 0.67 < \sqrt{0.45} < 7/10$.
31. (a) $(-4, 3] = \{x \in \mathbb{R} : -4 < x \leq 3\}$. (b) $x \geq 2$, or $\{x \in \mathbb{R} : x \geq 2\}$. (c) $(-4, 6)$, or $-4 < x < 6$. (d) $x < -2$ or $x \geq 4$.
32. $A \cap B = (1, 4)$; $A \cup B = [-3, 7]$; $A \cap C = [-3, 0]$; $(A \cup B) \cap C = [-3, 0]$; $A \setminus B = [-3, 1]$.
33. Mathematics only 19; Physics only 12; at least one 43; neither 7.
34. (a) $\{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 3\}, \{2, 4\}, \{3, 4\}$. (b) 16 subsets and 15 proper subsets. (c) $\emptyset \subseteq S$ is true; $\emptyset \in S$ is false.
35. $(A \cup B)^c = \{6\} = A^c \cap B^c$ and $(A \cap B)^c = \{1, 3, 4, 5, 6\} = A^c \cup B^c$.
36. (a) commutative addition; (b) associative multiplication; (c) distributive; (d) additive identity; (e) multiplicative inverse.
37. (a) $x = -2, 8$; (b) $-4 < x < 3$; (c) $x \leq -6$ or $x \geq -2$.
38. (a) 7.25×10^{-5} ; (b) 8.43×10^6 ; (c) 1.28×10^3 ; (d) 4×10^{-10} .
39. (a) $7/9$; (b) $1/6$; (c) $27/11$; (d) $37/300$.
40. If $\sqrt{3} = a/b$ in lowest terms, then $a^2 = 3b^2$, forcing both a and b to be divisible by 3, a contradiction. If $5 + 2\sqrt{3}$ were rational, then $\sqrt{3}$ would be rational.
41. (a) 8; (b) $1/9$; (c) 27; (d) 4.
42. (a) x^5y ; (b) b^5/a^5 ; (c) $2p^7/q^5$.
43. (a) $x\sqrt{x}$; (b) $a^3\sqrt{a^2}$; (c) 16; (d) $1/\sqrt[4]{y^3}$.
44. (a) $6\sqrt{2}$; (b) $11\sqrt{2}$; (c) $6\sqrt{5}$; (d) $3\sqrt[3]{2}$.
45. (a) $10 + 2\sqrt{21}$; (b) 18; (c) $15 - 6\sqrt{6}$.
46. (a) $5\sqrt{3}/3$; (b) $3 - \sqrt{5}$; (c) $(\sqrt{7} + \sqrt{2})/5$; (d) $7 + 4\sqrt{3}$.
47. (a) $x = 5$; (b) $x = 3$; (c) $x = 1$.
48. (a) $x = 3$; (b) $x = 4$; (c) $x = 0, 2$.
49. (a) $x > 4$; (b) $x \geq -3/2$; (c) $x < -1$.
50. (a) 10; (b) 8; (c) a^2 .
51. (a) -3 ; (b) -3 ; (c) 5; (d) $1/2$.
52. (a) $3\log_a x + \log_a y - \frac{1}{2}\log_a z$; (b) $2\ln|x-1| - 4\ln|x+2|$.
53. (a) $\ln(x^2y/z^3)$; (b) $\log_a(\sqrt{p}q^2)$.
54. $\log_7 20 = \ln 20 / \ln 7 \approx 1.5395$.
55. (a) $x = 9$; (b) $x = 3$; (c) $x = 5/2$.
56. (a) $x \geq 16$; (b) $1 < x < 2$; (c) $-2 < x < 0$.
57. (a) $x = \ln 10 / \ln 3$; (b) $x = \frac{1}{2} \ln(17/5)$; (c) $x = \ln 14 / \ln(7/2)$.
58. Using change of base, $(\ln b / \ln a)(\ln c / \ln b) = \ln c / \ln a = \log_a c$.
59. (a) $3/2$; (b) $-2/3$; (c) $1/2$.
60. $t = \ln(A/P) / \ln(1+r)$; for doubling at 8%, $t = \ln 2 / \ln 1.08 \approx 9.01$ years.
61. (a) $(-1, 3]$; (b) $(-3, 3)$; (c) $(-\infty, -4] \cup [1, \infty)$.
62. (a) $\{2n : n \in \mathbb{N}\}$; (b) $\{x \in \mathbb{Q} : -1 < x < 1\}$; (c) $\{x \in \mathbb{R} : |x-2| \geq 4\}$.
63. (a) $[-8/3, 4]$; (b) $(-6, -3) \cup (1, 4)$; (c) $x = -1, 3$.
64. (a) $k = \pm 6$; (b) $|k| > 6$; (c) $|k| < 6$.
65. (a) $(-\infty, 7/3]$; (b) $(-5, \infty)$; (c) $(-2, 2)$; (d) $[1, \infty) \setminus \{4\}$.
66. (a) never; (b) sometimes; (c) sometimes; (d) always.
67. From outermost to innermost: $\mathbb{R} \supset \mathbb{Q} \supset \mathbb{Z} \supset \mathbb{W} \supset \mathbb{N}$. Place $\sqrt{2}$ in $\mathbb{R} \setminus \mathbb{Q}$, $3/5$ in $\mathbb{Q} \setminus \mathbb{Z}$, -4 in $\mathbb{Z} \setminus \mathbb{W}$, 0 in $\mathbb{W} \setminus \mathbb{N}$, and 9 in $\mathbb{N}$.
68. (a) HCF 42, LCM 252; (b) HCF 6, LCM 360.
69. (a) divisible by both 3 and 9; (b) divisible by 4, 5, and 11; (c) $a = 0$ or 9.
70. To 3 s.f.: 38.7, with $38.65 \leq x < 38.75$. To 2 d.p.: 38.75, with $38.745 \leq x < 38.755$.
71. $12.35 \leq L < 12.45$ cm; maximum percentage error is about 0.40%.
72. (a) 9; (b) 6; (c) 3.
73. (a) $2\sqrt{13}$; compare squares $45 < 52$. (b) $\log_2 7$; it is $> 5/2$, whereas $\log_3 10 < 5/2$. (c) $3^{1/4}$; their twelfth powers are 16 and 27.
74. The domain requires $x > 1$. Then $x^2 - 1 = 8$, so $x = 3$ and $\sqrt{x^2 - 1} = 2\sqrt{2}$.

75. If $\sqrt{2} + \sqrt{3}$ were rational, squaring would make $5 + 2\sqrt{6}$ rational, contradicting the irrationality of $\sqrt{6}$. Also $1/(\sqrt{3} - \sqrt{2}) = \sqrt{3} + \sqrt{2}$.
76. $-22 \leq T \leq -18$, or $[-22, -18]$, with both number-line endpoints closed.
77. Sugar = 350 g; the sugar fraction is $2/7$.
78. $7.6(50,000) = 380,000$ cm = 3.8 km.
79. $2^{36} = 64 \cdot 2^{30}$ bytes, so the capacity is 64 GiB.
80. The ratio is $10^{6.8-4.3} = 10^{2.5} = 100\sqrt{10} \approx 316.2$.
81. $\text{pH} = -\log_{10}(2.5 \times 10^{-4}) \approx 3.602$; at pH 8.2, $[H^+] = 10^{-8.2} \approx 6.31 \times 10^{-9}$ mol L $^{-1}$.
82. $I/I_0 = 10^{35/10} = 10^{3.5} \approx 3162.3$.
83. After 12 years, about 59,459 residents; the population first exceeds 60,000 in year 13.
84. Value after five years $\approx$ RM 42,338.43; it falls below half during the fifth year.
85. Value after eight years $\approx$ RM 17,461.10; doubling time $\approx$ 14.78 years.
86. Price $\approx$ RM 307.31; total increase $\approx$ 22.93%.
87. $M(t) = 160(1/2)^{t/6}$; $M(15) = 20\sqrt{2} \approx 28.28$ mg.
88. Height after six bounces = $3(0.7)^6 \approx 0.353$ m; first below 0.2 m at bounce 8.
89. After six hours there are 128,000 cells; 100,000 is exceeded after about 5.73 h, approximately 5 h 44 min.
90. Football only 28, both 17, badminton only 21, neither 14; probability of neither = $14/80 = 7/40$.
91. Class size = $28 + 22 - 15 = 35$; exactly one subject = 20, so the probability is $20/35 = 4/7$.
92. $11.97 \leq d \leq 12.03$ mm; 11.96 fails, 11.98 passes, and 12.031 fails.
93. Side = $\sqrt{180} = 6\sqrt{5}$ cm; diagonal = $6\sqrt{10}$ cm.
94. Area = $(8 + \sqrt{3})(8 - \sqrt{3}) = 61$ cm 2 ; diagonal = $\sqrt{134}$ cm.
95. $R = 12(\sqrt{5} + 1)/4 = 3(\sqrt{5} + 1)$ ohms.
96. Distance = $(3.00 \times 10^8)(2.5 \times 10^{-3}) = 7.5 \times 10^5$ m = 750,000 m.
97. $8(1.496 \times 10^{11}) = 1.1968 \times 10^{12}$ m = 1.1968×10^9 km.
98. $n > \ln(1.2)/\ln(1.006) \approx 30.48$; the first whole month is month 31.
99. $M(10) \approx 32.58$ g; $M = 20$ at $t \approx 15.85$ years.
100. $A(t) = 5000(1.05)^t$, $B(t) = 6500(1.03)^t$. They are equal at $t \approx 13.64$ years. Plan B is larger before then; Plan A is larger afterwards.

Chapter 2: Functions

These answers correspond to Section 2.16.

- (i) True; (ii) false: the shift is right 2 and up 3; (iii) true; (iv) false: $x \neq 0$; (v) true.
- (i) real; (ii) 3, 1; (iii) 0; (iv) $x = 0$; (v) $x - 2$.
- (a) $D_f = [-5, 10]$ and $R_f = (-4, 6]$. For (b): (i) $D = [-5, 10]$, $R = [-6, 4]$; (ii) $D = (-10, 5]$, $R = [-6, 4]$; (iii) $D = (-7, 8]$, $R = (-4, 6]$; (iv) $D = (-10, 5]$, $R = [-8, 2]$; (v) $D = [-3, 12]$, $R = [-3, 7]$; (vi) $D = (-9, 6]$, $R = [-8, 2]$. The sketches follow the indicated reflections and translations, with endpoint inclusion preserved.
- Five examples are: reflect in the x -axis then shift up 2; reflect in the y -axis then shift up 2; shift right 3, reflect in the x -axis, then shift up 1; shift left 3, reflect in the y -axis, then shift up 1; shift up 5, reflect in the x -axis, then shift up 7. All give $y = -x/3 + 2$; the representation is not unique.
- (a) $-3(x-2) - 3 = -3x + 3$; (b) $-\frac{1}{2}(x+1)^2 + 2$; (c) $3(x-2)^2 - 1$; (d) $\frac{1}{2}(x+3)^3 + 2$; (e) $2(x-1)^3 - 3$; (f) $\frac{2}{x-2} - 1$; (g) $-\frac{3}{x+2} - 2$.
- (a) $g = \frac{1}{2}(x-2)^2 - 3$: right 2, down 3. (b) $g = -(x-2)^2 + 2$: reflect in the x -axis, right 2, up 2. (c) left 3, down 2. (d) $g = -f(x-3) + 2$: reflect in the x -axis, right 3, up 2.
- (a) Intercepts $(1/3, 0)$, $(0, 1)$; $D = R = \mathbb{R}$. (b) vertex $(1, 4)$, roots $-1, 3$, y -intercept 3, $R = (-\infty, 4]$. (c) vertex $(1, 1)$, no real roots, y -intercept 3, $R = [1, \infty)$. (d) intercepts $(1 - \sqrt[3]{2}, 0)$, $(0, 1)$; $D = R = \mathbb{R}$. (e) asymptotes $x = 3, y = 2$; intercepts $(2, 0)$, $(0, 4/3)$. (f) asymptotes $x = -2, y = 3$; intercepts $(-5/3, 0)$, $(0, 5/2)$. (g) $A = -1/2$; asymptotes $x = -1/2, y = -1/2$ and intercepts $(1, 0)$, $(0, 1)$.
- (i) True; (ii) false; (iii) true; (iv) false: $x \geq 3$; (v) true; (vi) false: left 2; (vii) true; (viii) false: $x > -4$.
- (i) > 7 ; (ii) $(0, \infty)$; (iii) up 4; (iv) $(-\infty, 9]$, $[0, \infty)$; (v) down 5; (vi) the x -axis, then left 2; (vii) $y = x$.
- (i) $D_{\ln} = (0, \infty)$. (ii) $e^x > 0$ and approaches, but never reaches, the x -axis. (iii) the vertex is $(2, 0)$. (iv) $x \geq 5$. (v) their domains and ranges are interchanged.
- $f: D = \mathbb{R}, R = \{2\sqrt{2}\}$; $g: D = R = \mathbb{R}$; $h: D = R = [0, \infty)$. Their common point is $(2, 2\sqrt{2})$.
- (a) $\ln(x+3) - 4$; (b) $-\ln(2-x) + 3$; (c) $e^{x+3} + 4$; (d) $-e^{-2(x-1)} - 2$; (e) $-\sqrt{-x-2} + 3$; (f) $\sqrt{5}\sqrt{3-x} - 1$; (g) $2|x-4| - 2$; (h) $-3|x+2| + 4$.
- (a) reflect in the y -axis, right 1, reflect in the x -axis, up 2; (b) left 2, reflect in the x -axis, down 3; (c) reflect in the y -axis, right 1, down 3; (d) right $3/2$, up 3.
- (a) $D = (-\infty, 3)$, $R = \mathbb{R}$, vertical asymptote $x = 3$, $x = 3 - e^{-2}$, $y(0) = -2 - \ln 3$. (b) $D = (-\infty, -2)$, $R = \mathbb{R}$, asymptote $x = -2$, $x = -2 - e^{-4}$; no y -intercept. (c) $D = \mathbb{R}$, $R = (-\infty, 2)$, asymptote $y = 2$, $x = 2 - \ln 2$, $y(0) = 2 - e^2$. (d) $D = (-\infty, 3]$, $R = (-\infty, -2]$, no x -intercept, $y(0) = -\sqrt{3} - 2$. (e) vertex $(1/2, 2)$, roots 0, 1, $R = (-\infty, 2]$. (f) $D = (-\infty, -5/2)$, asymptote $x = -5/2$, $x = -(e^3 + 5)/2$; no y -intercept. (g) $D = \mathbb{R}$, $R = (-\infty, 2)$, asymptote $y = 2$, $x = (2 - \ln 2)/3$, $y(0) = 2 - e^2$. (h) endpoint $(-2, -2)$, $D = [-2, \infty)$, $R = [-2, \infty)$, intercept $(0, 0)$. (i) vertex $(-3, -5)$, $D = \mathbb{R}$, $R = (-\infty, -5]$, no x -intercept, $y(0) = -20$.
- (a) $R = \{-9/5, 9/5\}$. (b) $R = \mathbb{R}$. (c) $R = \mathbb{R}$. (d) $R = (-\infty, -7) \cup [1, 2) \cup (3, \infty)$. (e) $R = \{-5\} \cup [-3, \infty)$. (f) $R = (-2, -1] \cup \{2\} \cup (3, \infty)$.
- $f + g = -\sqrt{2-2x} - 4$ for $x < -1$, $4x - 1$ for $-1 \leq x < 2$, and $(x+2)^2$ for $x \geq 2$. Its range is $(-\infty, -6) \cup [-5, 7) \cup [16, \infty)$.
- $q - p = 4 - 2x$ for $x \leq -3$, $-x^2$ for $-3 < x \leq 1$, and 2 for $x > 1$. Its range is $(-9, 0] \cup \{2\} \cup [10, \infty)$.
- (a) Replace f and g in each operation; the domain of the sum, difference, and product is $[-3/2, \infty)$, while the quotient also excludes $x = \pm\sqrt{2}$. (b) the common domain is $(-5, \infty)$; the quotient also excludes $x = 1, 5$.
- (a) $[3/4, \infty)$; (b) $\mathbb{R} \setminus \{-1, 5\}$; (c) $(-2/3, \infty)$; (d) $[-10/3, \infty) \setminus \{-3/2, 2, 4\}$.
- (a) $f \circ g = 2x^2 - 4x - 3$, $g \circ f = 4x^2 - 16x + 15$; values 27, -1. (b) $f \circ g = 2/[1 - 2\sqrt{1-2x}]$.

- $g \circ f = \sqrt{1 - 4/(1 - 2x)}$; values $2/(1 - 2\sqrt{7})$ and $\sqrt{7/3}$. (c) $f \circ g = 2$, $g \circ f = 3$; values 2, 3. (d) $f \circ g = \ln(6 - e^{x+2}) - 2$, $g \circ f = -x$; values $\ln(6 - e^{-1}) - 2$ and -2 .
21. (a), (d) are functions; (b), (c) are not functions of x .
22. (a) $\mathbb{R} \setminus \{-3, 3\}$; (b) $(-\infty, 5/2]$; (c) $(-\infty, -2) \cup (2, \infty)$; (d) $[-1, \infty) \setminus \{2\}$.
23. (a) $[2, \infty)$; (b) $(-\infty, 5]$; (c) $[-3, \infty)$; (d) $(-\infty, 4)$.
24. $f(-2) = 15$, $f(a) = 2a^2 - 3a + 1$, $f(a+1) = 2a^2 + a$, and $f(a+1) - f(a) = 4a - 1$.
25. (a) right 4, up 3; (b) left 2, reflect in the x -axis; (c) reflect in the y -axis, down 5; (d) reflect in the y -axis, right 3, reflect in the x -axis, up 1.
26. (a) $(x-3)^2 - 4$; (b) $-(x+2)^2$; (c) $3x^2 + 1$.
27. (a) gradient $-2/3$; intercepts $(6, 0)$, $(0, 4)$. (b) gradient -4 ; intercepts $(7/4, 0)$, $(0, 7)$. (c) vertical line, x -intercept $(5, 0)$, no y -intercept. (d) gradient 0, y -intercept $(0, -3)$, no x -intercept.
28. (a) $y = 3x - 7$; (b) $y = -x + 3$; (c) $y = 2x + 2$; (d) $y = 2x$.
29. (a) vertex $(2, -5)$, axis $x = 2$, y -intercept -1 . (b) $(-1, 4)$, $x = -1$, y -intercept 1. (c) $(3, -2)$, $x = 3$, y -intercept 7. (d) $(4, 1)$, $x = 4$, y -intercept 33. (e) $(-3, 6)$, $x = -3$, y -intercept $3/2$. (f) $(-2, -11)$, $x = -2$, y -intercept -3 .
30. (a) $(x+4)^2 - 13$; (b) $2(x-3)^2 - 13$; (c) $-(x-2)^2 + 5$; (d) $3(x+3)^2 - 34$; (e) $-2(x+2)^2 + 13$; (f) $\frac{1}{2}(x-3)^2 - \frac{1}{2}$.
31. (a) $x = 2, 3$: two roots. (b) $x = -1/2$: one repeated root. (c) no real roots. (d) $x = 1/2, 3$: two roots. (e) $x = 2/3$: one repeated root. (f) $x = 1 \pm \sqrt{2}$: two roots.
32. $2(x+2)(x-5) = 2x^2 - 6x - 20$; y -intercept -20 ; vertex $(3/2, -49/2)$.
33. Intercepts $(-2, 0)$, $(0, 0)$, $(2, 0)$; the function is odd; left end down and right end up.
34. (a) -2 has multiplicity 2 and touches; 1 has multiplicity 1 and crosses. (b) 3 has multiplicity 3 and crosses with flattening; -1 has multiplicity 2 and touches.
35. (a) asymptotes $x = 3$, $y = 2$; intercepts $(-1/2, 0)$, $(0, -1/3)$. (b) asymptotes $x = -2$, $y = -1$; intercepts $(2, 0)$, $(0, 1)$. (c) no vertical or horizontal asymptotes; $y = x + 1$ except for a hole at $(1, 2)$; intercepts $(-1, 0)$, $(0, 1)$. (d) asymptotes $x = -1$, $y = 3$; intercepts $(2, 0)$, $(0, -6)$. (e) asymptotes $x = 4$, $y = 2$; intercepts $(1, 0)$, $(0, 1/2)$. (f) vertical asymptotes $x = -2, 2$, horizontal asymptote $y = 0$; the only intercept is $(0, 0)$.
36. (a) $x = 3/2$; (b) $x = 11/3$; (c) $x = 2/3$; (d) $x = 1$; (e) $x = -4$; (f) $x = 1 \pm \sqrt{5}$.
37. Values $1/4, 1/2, 1, 2, 4$; intercept $(0, 1)$; asymptote $y = 0$; domain $\mathbb{R}$, range $(0, \infty)$.
38. (a) $2 \ln |x| + \frac{1}{2} \ln(x+1) - \ln |x-3|$; (b) $\ln(x^2/\sqrt{x+4})$; (c) $2^5 = 32$; (d) $2 + 4 \log_3 |x| - \frac{1}{2} \log_3 y$; (e) $\log_5(a^3b/c^2)$; (f) $\log_{10}(0.001) = -3$.
39. (a) $x = 4$; (b) $x = 11$; (c) $x = 9$; (d) $x = 5$; (e) $x = 3$; (f) $x = 2$.
40. (a) $x = 5, -2$; (b) $-5 < x < 3$; (c) $x \leq -2$ or $x \geq 8$; (d) $x = 2, -10/3$; (e) $-2 \leq x \leq 3$; (f) $x < 1$ or $x > 7$.
41. Reflect in the x -axis, stretch vertically by 2, right 1, up 3; domain $\mathbb{R}$, range $(-\infty, 3)$, asymptote $y = 3$, y -intercept $3 - 2/e$.
42. (a) $x = 3$; (b) $x = -3/2$; (c) $x = \ln 7/(2 \ln 5)$; (d) $x = 5$; (e) $x = 1/2$; (f) $x = (1 + \ln 6)/2$.
43. (a) $3 \ln x + \frac{1}{2} \ln y - 2 \ln |z|$; (b) $2 \log_a |u+1| - \frac{1}{2} \log_a v$; (c) $3 + 2 \log_2 |x| - \log_2 y$; (d) $\frac{2}{3} \ln |p| + \frac{1}{3} \ln q - 4 \ln |r|$; (e) $\log_b m - 2 \log_b |n| - \frac{1}{2} \log_b k$. These forms use the domains of the original expressions: in particular, $x, y > 0$ and $z \neq 0$ in (a); $u \neq -1$ and $v > 0$ in (b); $x \neq 0$ and $y > 0$ in (c); $p, r \neq 0$ and $q > 0$ in (d); and $m, k > 0$, $n \neq 0$, $b > 0$, $b \neq 1$ in (e).
44. (a) $\ln(x^3y/z^2)$; (b) $\log_a(\sqrt{p}/q)$; (c) $\ln(u^2\sqrt{v})$; (d) $\log_3(rs^2/t)$; (e) $\log_b(x^4/\sqrt[3]{y})$.
45. (a) $x = 6$; (b) $x = 4$; (c) $x = 1$; (d) $x = 8$; (e) $x = 3$; (f) $x = 5$.
46. (a) $(-\infty, 7/2)$; (b) $(-\infty, -4) \cup (1, \infty)$; (c) $(-2, -1) \cup (-1, \infty)$; (d) $(-\infty, 5)$; (e) $(-\infty, -2) \cup (2, \infty)$; (f) $(1/2, 1) \cup (1, \infty)$.
47. Reflect $\sqrt{x}$ in the y -axis, shift right 2, stretch vertically by 3, reflect in the x -axis, and shift up 4; endpoint $(2, 4)$, domain $(-\infty, 2]$, range $(-\infty, 4]$.
48. (a) $x = 3$; (b) $x = 1, 2$; (c) $x = 7$.

49. (a) $4 - x$ for $x < 4$, $x - 4$ for $x \geq 4$. (b) $5 - 2x$ for $x \leq 3$, $2x - 7$ for $x > 3$. (c) $-2x - 1$ for $x < -2$, 3 for $-2 \leq x < 1$, $2x + 1$ for $x \geq 1$.
50. Vertex $(-3, 5)$, axis $x = -3$, roots $-11/2, -1/2$, y -intercept -1 , domain $\mathbb{R}$, range $(-\infty, 5]$.
51. (a) $(-1, 1), (2, 4)$; (b) $(2, 4)$; (c) $(-1, 1)$.
52. For every $x \in \mathbb{R}$, $\ln(e^x) = x$; for every $x > 0$, $e^{\ln x} = x$. Hence the two functions are inverses, and their graphs are reflections in $y = x$.
53. e^x : $D = \mathbb{R}$, $R = (0, \infty)$, intercept $(0, 1)$, asymptote $y = 0$. $\ln x$: $D = (0, \infty)$, $R = \mathbb{R}$, intercept $(1, 0)$, asymptote $x = 0$. $\sqrt{x}$: $D = R = [0, \infty)$, intercept $(0, 0)$. $|x|$: $D = \mathbb{R}$, $R = [0, \infty)$, intercept $(0, 0)$. $1/x$: $D = R = \mathbb{R} \setminus \{0\}$, asymptotes $x = 0, y = 0$, no intercepts.
54. (a) $a = 3, b = 2$; (b) $h = 3, k = -2$; (c) $c = 2, d = 3$; (d) $a = 4, k = 3$; (e) $h = -2, k = 1$.
55. (a) $x = \frac{1}{2} \ln 7$; (b) $x = \pm 3$; (c) $x = 0, \ln 3$; (d) $x = 0, \ln 4$; (e) $x = 3$; (f) $x = 1/4, 4$.
56. $-1, 0, 9/4, 3, -1$.
57. Domain $\mathbb{R}$; range $(-\infty, 4)$; intercepts $(-3, 0), (0, 0), (5, 0)$; jump discontinuities at $x = 0$ and $x = 2$.
58. $b = 3$ and $a = 3$.
59. $P(t) = 4$ for $0 < t \leq 1$, 6.50 for $1 < t \leq 2$, 9 for $2 < t \leq 3$, and 11.50 for $3 < t \leq 4$.
60. $f + g = \sqrt{x-1} + x + 2$, $f - g = \sqrt{x-1} - x - 2$, $fg = (x+2)\sqrt{x-1}$, and $f/g = \sqrt{x-1}/(x+2)$; all have domain $[1, \infty)$.
61. (a) $x + 3, x \neq 3$; (b) $(x+3)/(x+2), x \neq \pm 2$; (c) $1/(\sqrt{x}+2), x \geq 0, x \neq 4$.
62. $f \circ g = 3x^2 + 5, g \circ f = 9x^2 - 6x + 3$; values $17, 6$.
63. (a) $(-\infty, 5/2]$; (b) $(0, e^2) \cup (e^2, \infty)$; (c) $(-\infty, -3) \cup (3, \infty)$.
64. $f \circ g \circ h = 2x^2 + 1$ and $h \circ g \circ f = 4(x+1)^2$; values $9, 36$.
65. $f^{\circ 2} = 4x - 9, f^{\circ 3} = 8x - 21, f^{\circ n}(x) = 2^n(x-3) + 3$.
66. $g(x) = 3x - 4$.
67. (a) $f^{-1}(x) = (x+4)/5$, domain $\mathbb{R}$; (b) $g^{-1}(x) = 3 - 2x$, domain $\mathbb{R}$.
68. $f^{-1}(x) = 2 + \sqrt{x-1}$. f : domain $[2, \infty)$, range $[1, \infty)$; f^{-1} : domain $[1, \infty)$, range $[2, \infty)$.
69. $f^{-1}(x) = (x+3)/(x-2)$; $D_f = \mathbb{R} \setminus \{1\}$, $R_f = \mathbb{R} \setminus \{2\}$, with these sets interchanged for the inverse.
70. Both $f(g(x))$ and $g(f(x))$ simplify to x .
71. (a) yes; (b) no; (c) yes; (d) yes.
72. For $f \circ g = 15, x = 1/2, -7/2$; for $g \circ f = 5, x = \pm\sqrt{2}$.
73. $a = 3, b = 8$ and $(g \circ f)(x) = 6x + 15$.
74. The rule is $y = x + 2$. It is one-to-one; reversing all arrows gives $x = y - 2$, and the inverse relation is also a function.
75. $f(g(x)) = \sqrt{x^2} = x$ because $x \geq 0$; $g(f(x)) = x$. The restrictions make $\sqrt{x^2} = x$ and ensure that the two functions are one-to-one inverses.
76. $F(d) = 4 + 1.8d$; $F(12) = \text{RM } 25.60$; at most 14 whole kilometres.
77. $C(g) = 25$ for $0 \leq g \leq 10$ and $C(g) = 25 + 3(g-10)$ for $g > 10$; costs RM 25 and RM 43.
78. $F(25) = 77$; $F^{-1}(x) = \frac{5}{9}(x-32)$; $95^\circ\text{F} = 35^\circ\text{C}$.
79. $P(x) = 12x - 1200$; break-even at $x = 100$; profit for $x > 100$.
80. Initial height 1 m; maximum 21 m at $t = 2$ s.
81. $A(x) = x(80 - 2x)$; maximum 800 m^2 when the dimensions are 20 m by 40 m.
82. Ground contacts $x = \pm 6$, so the span is 12 m; $h(3) = 9$ m.
83. $M(4) = 25.3125 \text{ mg L}^{-1}$; it first falls below 20 mg L^{-1} at the fifth whole hour.
84. $P(10) \approx 7401$; it exceeds 7000 after about 8.58 years (during the ninth year).
85. $V(t) = 8000(1.05)^t$; $V(6) \approx \text{RM } 10,720.77$; doubling time $\ln 2 / \ln 1.05 \approx 14.21$ years.
86. $\text{pH} \approx 4.495$; for $\text{pH } 6, [H^+] = 10^{-6}$.
87. The increase is $10 \log_{10} 100 = 20 \text{ dB}$.
88. $T(10) \approx 42.48^\circ\text{C}$; $T = 30^\circ\text{C}$ at about 17.83 min.
89. $I(3) = 100$; $d = 5$; physically $d > 0$ because distance from the source cannot be zero or negative.
90. $A(50) = 24$ and $A(200) = 15$. The horizontal asymptote $y = 12$ is the limiting unit cost; $x = 0$ is a non-physical vertical asymptote.
91. $R(20) = 50$; $c = 80$; the limiting response is 100.

92. $d(80) = 58.4$ m; the positive solution of $d(v) = 60$ is $v \approx 81.31 \text{ km h}^{-1}$.
93. Restricting $r \geq 0$ makes the area function one-to-one and selects the positive square root; the radius is 10 m.
94. $49.92 \leq x \leq 50.08$; 49.91 fails and 50.06 passes.
95. $x = -5, 19$; distance at most 3 gives $4 \leq x \leq 10$.
96. $C(x) = 18$ for $0 \leq x \leq 100$, and $C(x) = 18 + 0.28(x - 100)$ for $x > 100$; $C(250) = \text{RM } 60$.
97. $C(0.8) = 6$, $C(4) = 13.50$, and $C(8) = 20.50$. The sketch has the endpoint conventions shown by the inequalities.
98. $(D \circ V)(T) = 10T$; the display reads 300 at 30°C .
99. $S^{-1}(y) = (y - 2)/(5 - y)$, for $2 \leq y < 5$; a reading of 4.4 gives $c = 4$.
100. Break-even at $d = 500/3 \approx 166.67$ km. Plan A is cheaper below this distance and Plan B above it.

Chapter 3: Polynomials

These answers correspond to Section 3.11.

- (a) $4x^5 - 3x + 1$ is a polynomial of degree 5 with leading coefficient 4; $2x^{-1} + x$ is not a polynomial; $\sqrt{2}x^3 - x$ is a polynomial of degree 3 with leading coefficient $\sqrt{2}$; $3^x + 1$ is not a polynomial. (b) Degree 6, leading coefficient -3 , constant term -7 ; $p(x) \rightarrow -\infty$ as $x \rightarrow \pm\infty$.
- (a) $f + 2g = 2x^3 + 2x^2 + 5x$;
 $f - g = 2x^3 - x^2 - 4x + 6$;
 $fg = 2x^5 + 6x^4 - 5x^3 + x^2 + 14x - 8$. (b)
 $p(-2) = 23$.
- (a) $x^3 - 2x^2 - 5x + 6$; (b)
 $(x - 2)^2(x + 3) = x^3 - x^2 - 8x + 12$; (c)
 $x^2(x - 1)(x + 1) = x^4 - x^2$.
- (a) Quotient $x^2 + 3x + 10$, remainder $19x - 7$;
(b) quotient $x + 3$, remainder 0. (c) Quotient
 $x^2 + 3x + 10$, remainder $19x - 7$.
- (a) Quotient $4x^2 + 2x + 3$, remainder 6; (b)
quotient $\frac{5}{3}x^2 + \frac{1}{9}x - \frac{28}{27}$, remainder $\frac{55}{27}$.
- (a) 2; (b) 11; (c) $125/27$.
- (a) $k = -3/2$; (b) $a = 60$; (c) $m = -3$.
- (a) The value at $x = 1/2$ is zero and the
quotient is $x^2 + 4x + 3$. (b) $(x - 1)(x - 5)(x + 3)$.
- (a) The remainder is $R(x) = 3x - 1$. (b)
 $R(0) = -1$.
- (a) $x = 1/2, -1, -3$; (b) $x = -1, 1, 4$; (c)
 $x = 1, 5, -3$.
- (a) $x = \pm 1, \pm 2$; (b) $(x - 2)(2x - 3)(x + 2)$, with
zeros $2, 3/2, -2$.
- The roots are -2 (multiplicity 1, crossed), 1
(multiplicity 2, touched), and 3 (multiplicity 3,
crossed with flattening). Since the degree is
even with positive leading coefficient, both ends
rise. The y -intercept is -54 .
- (a) $\frac{3}{5(x+3)} + \frac{2}{5(x-2)}$; (b) $\frac{2}{x-1} + \frac{3}{x+2}$; (c)
 $\frac{11}{5(x-2)} - \frac{1}{5(x+3)}$.
- (a) $\frac{3}{x-1} + \frac{8}{(x-1)^2}$; (b) $\frac{4}{x} + \frac{-2x+3}{x^2+1}$; (c)
 $2x + 1 + \frac{5}{x} - \frac{2}{x+1}$.
- (a) $\frac{1}{3(x-1)} - \frac{1}{x+1} + \frac{5}{3(x+2)}$. (b) Combining
the right-hand side over $x(x+1)(x^2+1)$
produces the numerator $4x^3 + 2x^2 + 3x + 1$. (c)
$$\frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{(x-1)^3}$$
$$+ \frac{Dx+E}{x^2+4} + \frac{Fx+G}{(x^2+4)^2}.$$
- (a) Polynomial, degree 6, leading coefficient 5;
(b) not a polynomial; (c) polynomial, degree 2,
leading coefficient $\sqrt{3}$; (d) not a polynomial; (e)
polynomial, degree 0, leading coefficient -8 ; (f)
zero polynomial, degree undefined.
- (a) 8, $-4, 5$; (b) 5, $-2, 7$; (c) 6, 1, 0; (d) 5, 8, 4,
where each triple is degree, leading coefficient,
and constant term.
- $7x^5 - 2x^3 - x + 5$; coefficients 7, 0, $-2, 0, -1, 5$.
- $p(0) = -6, p(1) = -2, p(-1) = -6, p(2) = 24$.
- $p(x) = 4x^2 - x + 1$.
- (a) $2x^3 - 2x^2 - 5x + 5$; (b) $x^3 + 6x^2 + x - 8$; (c)
 $-x^3 + 2x^2 + 3x - 4$.
- (a) $x^2 - 2x - 8$; (b) $3x^3 - 2x^2 - 3x + 2$; (c)
 $x^4 + 2x^2 + 9$.
- $f(g(x)) = x^2 + 6x + 8; g(f(x)) = x^2 + 2$.
- $f(x) = 3x^2 + 2x + 1; g(x) = x^2 - 4x + 5$.
- $\deg(pq) = 6, \deg(p+q) = 4, \deg(p^2q) = 10$;
 $\deg(p - q^2) \leq 4$, with equality unless the
degree-4 terms cancel.
- (a) even; (b) odd; (c) neither; (d) both even and
odd.
- Root -2 has multiplicity 3 and is crossed with
flattening; root 1 has multiplicity 2 and is
touched; the y -intercept is 24.
- (a) $x(x+3)(x-2) = x^3 + x^2 - 6x$; (b)
 $-2(x-1)^2(x+4)$; (c) $x^2 - 5$.
- (a) left end down, right end up; (b) both ends
down; (c) left end up, right end down; (d) both
ends up.
- $a = 2, b = 1$.
- (a) Quotient $2x^2 - 3x - 2$, remainder $8x - 5$; (b)
quotient $x^2 + 1$, remainder 0; (c) quotient
 $3x - 1$, remainder $6x + 3$.

32. (a) Quotient $2x^4 + 4x^3 + 5x^2 + 10x + 21$, remainder 38; (b) quotient $x^3 - x^2 + 5x - 5$, remainder 6; (c) quotient $x^4 + 2x^3 + 4x^2 + 8x + 16$, remainder 0.
33. (a) Quotient $2x^3 + x^2$, remainder 3; (b) quotient $x^2 + x - \frac{4}{3}$, remainder $\frac{29}{3}$.
34. $2x^3 + x^2 + 14$.
35. $a = 3$, $b = -1$, remainder -2 .
36. (a) 31; (b) 1; (c) 13; (d) 65.
37. (a) $-\frac{31}{8}$; (b) $\frac{596}{81}$.
38. $a = -6$, $b = 11$.
39. $-3x + 1$.
40. $2x^2 + 1$.
41. $x + 2$.
42. (a) $k = -5$; (b) $k = 6$; (c) $k = 3$.
43. $k = -1$.
44. $12 - 3x$.
45. 280093.
46. (a) $(x + 2)(x - 3)(x + 3)$; (b) $(2x - 5)(x - 2)(x + 2)$; (c) $(x - 4)(x - 1)(x + 1)$.
47. (a) $(x - 2)(x + 1)(x + 3)$; (b) $(x - 1)(x + 1)(x - 2)(x - 3)$; (c) $(3x + 1)(x - 2)(x + 2)$.
48. (a) $\pm\frac{1}{2}, \pm 1, \pm\frac{3}{2}, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12$; (b) $\pm\frac{1}{3}, \pm\frac{2}{3}, \pm\frac{4}{3}, \pm 1, \pm 2, \pm 4$.
49. (a) $(x - 1)(x - 2)(x + 3)$; (b) $(2x + 1)(x - 2)(x + 2)$; (c) $(x - 5)(x - 1)(x + 2)$.
50. (a) $(x - 1)(x + 1)(x - 3)(x + 3)$; (b) $(2x^2 + 1)(x - \sqrt{3})(x + \sqrt{3})$; (c) $(x - 1)(x + 1)(x^2 + 1)(x - 2)(x + 2)(x^2 + 4)$.
51. (a) $x^2(x - 1)^2$; (b) $(x - 1)^2(x + 2)^2$; (c) $x^2(x - 1)^3$.
52. (a) $x = 2, -1, -3$; (b) $x = \pm 1, \pm 3$; (c) $x = -\frac{1}{2}, \pm 2$.
53. (a) $(x + 2)(x - 1)^2 = x^3 - 3x + 2$; (b) $4x(x + 1)(x - 3)$; (c) $(x^2 - 4)^2$.
54. (a) -3 has multiplicity 2 and is touched; 1 has multiplicity 3 and is crossed with flattening. (b) -1 has multiplicity 1 and is crossed; 2 has multiplicity 4 and is touched. The leading coefficient is negative and the degree is 5.
55. Positive on $(-\infty, -2) \cup (3, \infty)$ and negative on $(-2, 1) \cup (1, 3)$; it is zero at $-2, 1, 3$.
56. (a) $\frac{13}{3(x-2)} + \frac{8}{3(x+1)}$; (b) $\frac{3}{5(x+2)} + \frac{17}{5(x-3)}$; (c) $-\frac{1}{7(x-4)} + \frac{15}{7(x+3)}$.
57. (a) $\frac{2}{x+2} - \frac{3}{(x+2)^2}$; (b) $\frac{1}{x-1} + \frac{6}{(x-1)^2} + \frac{12}{(x-1)^3}$.
58. (a) $\frac{15}{2(x-1)} - \frac{9x-1}{2(x^2+1)}$; (b) $\frac{37}{25(x-1)} + \frac{14}{5(x-1)^2} + \frac{13x+18}{25(x^2+4)}$.
59. (a) $x + 2 + \frac{15}{2(x-1)} - \frac{3}{2(x+1)}$; (b) $3x^2 - 2x + 2 + \frac{1}{x} - \frac{1}{x+1}$.
60. (a) $\frac{A}{x-2} + \frac{B}{(x-2)^2} + \frac{Cx+D}{x^2+9}$; (b) $\frac{7-x}{(x-1)(x+2)}$; (c) multiplication by $n(n+1)$ gives the identity immediately.
61. $P(x) = -x^2 + 40x - 300 = -(x-10)(x-30)$; break-even levels 10 and 30; positive profit for $10 < x < 30$.
62. $h(2) = 45$ m; the projectile reaches the ground at $t = 5$ s.
63. $x(x+3) = 70$, so $x = 7$; dimensions 7 m $\times$ 10 m.
64. $V(x) = x(12-2x)(8-2x) = 4x^3 - 40x^2 + 96x$; $V(1) = 60$ cm³. Its zeros are 0, 4, 6, but the physical interval is $0 \leq x \leq 4$, so the physical endpoints are 0 and 4.
65. Frame area $(x+4)(x+3) - x(x-1) = 8x + 12$; at $x = 10$, the area is 92 cm².
66. Path area $4x^2 + 32x$. Solving $4x^2 + 32x = 80$ gives $x = 2$ m.
67. $n(n+1)(n+2) = 120$ and $n^3 + 3n^2 + 2n - 120 = (n-4)(n^2 + 7n + 30)$; the integers are 4, 5, 6.
68. $t(t+1)(t+2) = 120$, so $t = 4$ h.
69. Ground contacts $x = 2, 8$; span 6 units; $h(5) = 9$ units.
70. $p(x) = x^2 + 2x + 1 = (x+1)^2$.
71. $p(n) = n^2 + 4n + 3$; hence $p(4) = 35$.
72. $(x+4)^2 - x^2 = 8x + 16 = 96$, so $x = 10$ cm.
73. $x^2(x+3) = 54$; the positive solution is $x = 3$ cm.

74. $P(x) = (x-1)(x-2)(x-3)$; break-even levels 1, 2, 3; $P(x) > 0$ on $(1, 2) \cup (3, \infty)$ for $x \geq 0$.
75. $V(x) = (x+4)(x+2)(x-1)$; the other edges are $x+2$ and $x-1$, with $x > 1$; $V(3) = 70$ cubic units.
76. Zeros $-3, 1, 4$. The roots -3 and 4 have multiplicity 1 and are crossed; 1 has multiplicity 2 and is touched.
77. $C(t) = \frac{3}{t+2} + \frac{4}{t+5}$; $C(0) = \frac{23}{10}$ and $C(5) = \frac{29}{35}$.
78. $V(x) = (x+1)(x+2)(x+4)$; the other dimensions are $x+1$ and $x+4$; $V(2) = 72$ cubic units.
79. $D(10) = 4231$, which reproduces the stored decimal digits.
80. $T(3) = 0$ and $T(x) = (x-3)(x+1)(2x-5)$; the non-negative zeros are $x = \frac{5}{2}$ and $x = 3$.
81. $s(t) = t(t-3)^2$; the particle is at the origin at $t = 0$ and $t = 3$, with $t = 3$ a double root.
82. Area $= \pi[(x+2)^2 - x^2] = 4\pi(x+1)$; at $x = 3$ it is $16\pi \text{ m}^2$.
83. The remainder is $S(2) = 10$, equal to the sensor output at the calibration setting.
84. $L_N = 1 - \frac{1}{N+1} = \frac{N}{N+1}$; $L_9 = \frac{9}{10}$.
85. The fraction is improper. Division and cancellation give $A(x) = 2 - \frac{1}{x+1}$; $A(1) = \frac{3}{2}$, and $A(x) \rightarrow 2$ as $x \rightarrow \infty$.
86. (a) $R(p) = p(800 - 20p) = -20p^2 + 800p$. (b) $R(p) = 6000$ gives $(p-10)(p-30) = 0$, so $p = 10$ or 30 . (c) $0 \leq p \leq 40$; the zeros correspond to a free ticket and to a price at which predicted attendance is zero.
87. (a) $q(v) = 16 - (v-60)^2/200 = -v^2/200 + 3v/5 - 2$. (b) $v = 60 \pm 10\sqrt{2}$, approximately 45.86 and 74.14 km h^{-1} . (c) Maximum score 16 at 60 km h^{-1} .
88. $x(x+1)(x+4) = 160$, so $x^3 + 5x^2 + 4x - 160 = (x-4)(x^2 + 9x + 40) = 0$. The physical solution is $x = 4$, giving dimensions $4 \text{ m} \times 5 \text{ m} \times 8 \text{ m}$.
89. Solving the three coefficient equations gives $V(h) = h^3 + h^2 + 4h$; hence $V(4) = 96$.
90. $C(t) = \frac{3}{t+1} + \frac{2}{t+4}$; $C(0) = 7/2$, $C(2) = 4/3$, and $C(t) \rightarrow 0$.
91. The first differences are 3, 11, 25, 45, the second differences are 8, 14, 20, and the constant third difference is 6. Thus $p(n) = n^3 + n^2 + n + 2$, and $p(5) = 157$.
92. Since $y(-x) = -x - 0.002x^3 = -y(x)$, the correction is odd. At $x = 10$, $y = 12$ and $y - x = 2$; at $x = -10$, $y = -12$ and $y - x = -2$.
93. The baseline is reached at $t = 0, 3, 8$. The index is above the baseline on $(0, 3) \cup (8, \infty)$ and below it on $(3, 8)$. Also $P(5) = 440$.
94. $H(f) = \frac{f+1}{f+3} = 1 - \frac{2}{f+3}$, with the original restrictions $f \neq -2, -3$. Here $H(1) = 1/2$, the removable discontinuity is at $f = -2$, and $H(f) \rightarrow 1$.
95. The length is $30 - x$, so $x(30 - x) = 176$ and $(x-8)(x-22) = 0$. Since the width does not exceed the length, the dimensions are $8 \text{ m} \times 22 \text{ m}$.
96. $h(t) = -t^3 + 6t^2 + 16t$. Its zeros are $-2, 0, 8$; the physical flight interval is $0 \leq t \leq 8$, and $h(4) = 96$.
97. $T(n) = (n-1)(n-2)(n+3)$. The non-negative zeros are 1 and 2. For $n \geq 0$, $T(n) > 0$ on $[0, 1) \cup (2, \infty)$ and $T(n) < 0$ on $(1, 2)$.
98. Modulo $x^2 - 1$, $M(x) \equiv x + 2 - 3x + 5 = 7 - 2x$.
99. Division by $x + 2$ gives $2x^2 + 5x + 3$, so $C(x) = (x+2)(x+1)(2x+3)$. Also $C(2) = 84$.
100. (a) $A(x) = x(12 - 2x) = 12x - 2x^2$, with $0 < x < 6$. (b) $A(x) = -2(x-3)^2 + 18$, so the maximum area is 18 cm^2 at $x = 3 \text{ cm}$. (c) $A(x) = 16$ gives $x = 2 \text{ cm}$ or $x = 4 \text{ cm}$.

Chapter 4: Inequalities

These answers correspond to Section 4.9.

1. (a) $(-\infty, 3]$; (b) $(-2, 5)$; (c) $x \geq 1$; (d) $x < -4$ or $x \geq 2$; (e) $(-\infty, 6) \cup (6, \infty)$; (f) $[-7, 0)$.
2. (a) $(1, 4]$; (b) $[3, 4)$; (c) $(-3, -1)$; (d) $[-5, \infty)$; (e) $\mathbb{R}$; (f) $[2, 11)$.
3. (a) $(-\infty, -4]$; (b) $[-1/3, \infty)$; (c) $[12, \infty)$; (d) $(-14/9, \infty)$.
4. (a) $(-4/3, 2]$; (b) $[41, 59)$; (c) $(-1, 2)$.
5. (a) $[-5, -2]$; (b) $(-1, 0)$.
6. (a) $x > -4$; the direction reverses. (b) False: $ac > bc$. (c) False: $1/a > 1/b$. (d) False, for example $-3 < -2$ but $9 > 4$.
7. (a) $(-3, 4]$; (b) $[4, \infty)$; (c) $(-\infty, -2) \cup [2, \infty)$.
8. (a) $5 \leq 3x - 1 < 14$; (b) $-6 < -2x + 4 \leq 0$; (c) $4 \leq x^2 < 25$; (d) $1/5 < 1/x \leq 1/2$.
9. (a) $-2, -1, 0, 1, 2, 3, 4$; seven integers. (b) $1, 2, 3, 4$; four integers. (c) $-4, -3, \dots, 3, 4$; nine integers.
10. (a) $-2 \leq x < 4$, $[-2, 4)$. (b) $x \leq 1$ or $x > 3$, $(-\infty, 1] \cup (3, \infty)$. (c) $-1 < x \leq 5$, $(-1, 5]$.
11. (a) $\mathbb{R}$; (b) $\emptyset$; (c) $\mathbb{R}$.
12. (a) Four: $1, 2, 3, 4$. (b) Seven: $-6, -5, -4, 3, 4, 5, 6$. (c) Seven: $-3, -2, -1, 0, 1, 2, 3$.
13. (a) $(-\infty, 17]$; (b) $(-\infty, 4/11)$; (c) $(-\infty, 12/7]$.
14. (a) $(-\infty, 23/4]$; (b) $(8, \infty)$; (c) $\emptyset$.
15. Adding c preserves the difference $b - a$; multiplying it by a positive number preserves its sign, whereas multiplication by a negative number reverses the sign. Suitable numerical examples are $2 < 5 \Rightarrow 5 < 8$, $2(3) < 5(3)$, and $2(-1) > 5(-1)$.
16. (a) $(-\infty, -2) \cup (2, \infty)$; (b) $[0, 8]$; (c) $(-\infty, -1) \cup (7, \infty)$; (d) $(-\infty, 1) \cup (3, \infty)$.
17. $(-4, 2)$.
18. $(-\infty, -2) \cup (5, \infty)$.
19. (a) $-6 < k < 6$; (b) $k \geq 4$; (c) $-4 < k < 4$.
20. (a) $\mathbb{R}$; (b) $\emptyset$; (c) $\mathbb{R}$.
21. (a) $\{3\}$; (b) $\mathbb{R} \setminus \{-1/2\}$; (c) $\{-4\}$.
22. (a) $[2 - 2\sqrt{3}, 2 + 2\sqrt{3}]$; (b) $\emptyset$; (c) $[-2, 1]$.
23. (a) $(-3, 1/2) \cup (2, \infty)$; (b) $[-1, 1] \cup [5, \infty)$; (c) $(-\sqrt{3}, 1) \cup (\sqrt{3}, 3)$.
24. $(-\infty, -4) \cup (-3, 2)$.
25. $[-2, 1] \cup [3, \infty)$.
26. $(-\infty, -5] \cup [-1, 5]$.
27. (a) $(-\infty, -1)$; (b) $[-3, \infty)$; (c) $[-2, 4]$.
28. (a) $(-\infty, -3) \cup (-1, 1) \cup (3, \infty)$; (b) $\{-2, 2\}$; (c) $(-2, 1)$.
29. (a) $(-\infty, -2] \cup [1, 4]$; (b) $(-\infty, -3) \cup (-3, 2)$; (c) $(-\infty, -5] \cup [1/2, \infty)$.
30. (a) $[-2, -1] \cup [1, 2]$; (b) $(-\infty, -3) \cup (-2, 2) \cup (3, \infty)$; (c) $(-\infty, -1] \cup [1, \infty)$.
31. (a) $k = 7$; (b) $\{a, b\} = \{-2, 6\}$.
32. $[(3 - \sqrt{29})/2, (3 + \sqrt{29})/2]$.
33. (a) $k \leq 9$; (b) $k < -4$ or $k > 4$; (c) no real value of k .
34. (a) $[-1, 1] \cup [4, \infty)$; (b) $(-\infty, -3) \cup (-1, 2)$.
35. $p(x) = (x + 2)(x - 1)^2(x - 4)$, and $p(x) \leq 0$ on $[-2, 4]$.
36. (a) $[-7, -1) \cup [4, \infty)$; (b) $(-1, 2) \cup (5, \infty)$; (c) $[-5/2, 4)$; (d) $(0, \infty)$.
37. (a) $[-5, 0] \cup (3, \infty)$; (b) $(-3, 3]$; (c) $(-\infty, -12] \cup (-2, 3)$.
38. $[0, 4) \cup (4, \infty)$.
39. $[-1/\sqrt{2}, 0) \cup [1/\sqrt{2}, \infty)$.
40. $[-3, 2] \cup (4, \infty)$.
41. $(-\infty, -1/2) \cup (2/3, \infty)$.
42. $(-\infty, -8) \cup (-5, \infty)$.
43. $[-3, 2) \cup [5, \infty)$.
44. (a) $(-1, 1) \cup (1, \infty)$; (b) $(-\infty, -2) \cup (-2, 2]$; (c) $[-3, 3) \cup (3, \infty)$.
45. (a) $(-\infty, -1) \cup (-1, 2]$; (b) $(1, \infty)$; (c) $(-2, \infty)$.
46. (a) $(-3, -1) \cup (1/2, 4)$; (b) $(-\infty, -5] \cup \{1\} \cup (2, \infty)$; (c) $(-2, 2)$.
47. (a) $k = 5$; (b) $k = 3$.
48. (a) $(0, 3)$; (b) $[-2, 0)$; (c) $(-\infty, -2) \cup (1, \infty)$.

49. (a) $(-\infty, -2) \cup [1, 4)$; (b) $(0, 3) \cup [5, \infty)$; (c) $(-1, 2]$.
50. $(-2, 1) \cup [4, \infty)$.
51. (a) True. (b) False; for example, $1 + (-1) = 0$. (c) True. (d) The standard interval form is used for $b > 0$; if $b \leq 0$, neither inequality has a solution. (e) True.
52. (a) $y = -11, -5$; (b) no solution; (c) $y = -4/5, 18/5$.
53. (a) $y = -1, 2$; (b) $x = 1, 2$; (c) $x = 5/4, 5$.
54. (a) $x = -1/2, 4$; (b) $x = 2/5$; (c) $x = -5/3, 1$.
55. (a) $x = 2, 3$; (b) $x = \pm 3$; (c) $x = -1, 3$.
56. $x = -1/3, 3$.
57. $x = -2, 5/7$.
58. (a) $(-\sqrt{5}, -1) \cup (1, \sqrt{5})$; (b) $(-\infty, -4] \cup [6, \infty)$; (c) $(-\infty, 0]$.
59. (a) $(-\infty, -14) \cup (6, \infty)$; (b) $[1, 9]$.
60. $(-2, 0)$.
61. (a) $(-\infty, 8/5] \cup [10, \infty)$; (b) $(1, 2)$; (c) $(-\infty, 7/4] \cup [5/2, \infty)$.
62. $[0, 4]$.
63. $(-\infty, 0] \cup \{2\} \cup [4, \infty)$.
64. $(-\infty, -1) \cup (-1, 11/4] \cup [13/2, \infty)$.
65. $(-8, -2/3)$.
66. $(5/7, 7/3)$.
67. (a) $[-2, 3]$; (b) $(-\infty, -1) \cup (5, \infty)$.
68. (a) $[-4, -2] \cup [2, 4]$; (b) $(-\infty, -5/2) \cup (-1/2, 3/2) \cup (7/2, \infty)$; (c) $x = -5, 1$.
69. (a) $(a - r, a + r)$; (b) $a = 3, r = 5$; (c) $((5 - k)/2, (5 + k)/2)$.
70. (a) $x = -1, 3$; (b) $x = \pm 1, \pm 3$; (c) $x = -2, 0$.
71. (a) $(-4/3, 6)$; (b) $[-1/4, 3/2]$; (c) $(-\infty, -6] \cup [2/3, \infty)$.
72. (a) $(1/2, \infty)$; (b) $[5/2, 3) \cup (3, 7/2]$; (c) $(11/5, 4) \cup (4, 13)$.
73. (a) $4 - 2x$ for $x < 2$, and $2x - 4$ for $x \geq 2$. (b) -3 for $x < -1$, $2x - 1$ for $-1 \leq x < 2$, and 3 for $x \geq 2$; the equation has solution $x = 1$. (c) $4 - x^2$ for $-2 < x < 2$, and $x^2 - 4$ for $x \leq -2$ or $x \geq 2$.
74. (a) $[-1, 1)$; (b) $[-4, -1) \cup (2, 4]$; (c) $(-3, -1) \cup [3, 7]$.
75. Squaring gives $(|a| + |b|)^2 - (a + b)^2 = 2(|ab| - ab) \geq 0$. Applying the triangle inequality to $a = (a - b) + b$ and then exchanging a, b yields $||a| - |b|| \leq |a - b|$.
76. $|h - 100| \leq 2$; $98 \leq h \leq 102$ cm.
77. $|d - 12.00| \leq 0.03$; $11.97 \leq d \leq 12.03$ mm.
78. Safe: $|T + 18| \leq 2$, or $-20 \leq T \leq -16$. Alarm: $|T + 18| > 2$.
79. $68n + 75 \leq 800$, so $n \leq 725/68$; at most 10 passengers.
80. $1200 + 45n \leq 3000$, so at most 40 attendees.
81. $25 + 0.08m \leq 40 + 0.03m$, giving $0 \leq m \leq 300$ minutes.
82. $P(x) > 0$ for $10 < x < 50$.
83. $3 - \sqrt{3} \leq t \leq 3 + \sqrt{3}$ seconds.
84. $x(x + 4) \geq 96$ with $x > 0$, so $x \geq 8$ m.
85. $0 \leq v \leq 30$.
86. At $x = \pm 2$, the arch height is $12 - 4/9 = 104/9 \approx 11.556$ m. With clearance, the vehicle height is at most $199/18 \approx 11.056$ m.
87. $5 \leq 40/(t + 2) \leq 10$ gives $2 \leq t \leq 6$.
88. $500/x + 8 \leq 13$ gives $x \geq 100$ components.
89. $1 \leq 12/(R + 2) \leq 3$ gives $2 \leq R \leq 10$ ohms.
90. $1/4 \leq (4 + x)/(20 + x) \leq 2/5$ gives $4/3 \leq x \leq 20/3$ litres.
91. $|x - 12| \leq 7$, so $5 \leq x \leq 19$ km.
92. $[49.6, 50.4] \cap [49.9, 50.5] = [49.9, 50.4]$ mm.
93. $|x - 2| < |x - 10|$ gives $x < 6$.
94. $3 - 2\sqrt{2} \leq t \leq 3 + 2\sqrt{2}$.
95. Accepted: -2.5 and 0.4 ; flagged: -3.1 and 2.6 .
96. $|s - 80| \leq 8$ and $s \geq 75$ give $75 \leq s \leq 88$.
97. In the physical domain $0 < x < 4$,
- $$x(12 - 2x)(8 - 2x) \geq 60$$
- gives $1 \leq x \leq (9 - \sqrt{21})/2$ cm.
98. $900 + 50p \leq 1200$; $p = 0, 1, 2, 3, 4, 5, 6$.

99. $|24 \times 10^{-6}(T - 20)| \leq 0.001$, so

$$-\frac{65}{3} \leq T \leq \frac{185}{3}$$

degrees Celsius, approximately

$$-21.667 \leq T \leq 61.667.$$

100. Non-loss values are the integers $10 \leq n \leq 30$.

The maximum occurs at $n = 20$, giving profit $P(20) = 100$.

Chapter 5: Matrices

These answers correspond to Section 5.8.

1. (a) scalar 1×1 matrix; (b) zero matrix; (c) identity matrix; (d) lower-triangular matrix; (e) row matrix; (f) diagonal matrix; (g) symmetric matrix; (h) square matrix; (i) skew-symmetric matrix. (j) Scalar, diagonal, symmetric, and square; (k) skew-symmetric and square; (l) upper triangular and square; (m) column matrix.

2. (a) $\begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}$; (b) $\begin{pmatrix} a & 0 & 0 \\ b & c & 0 \\ d & e & f \end{pmatrix}$; (c) $\begin{pmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{pmatrix}$; (d) $\begin{pmatrix} a & b & 0 \\ c & d & e \\ 0 & f & g \end{pmatrix}$.

3. (a) $\begin{pmatrix} 3 & 2 \\ 3 & 2 \end{pmatrix}$; (b) undefined; (c) $\begin{pmatrix} 6 & -18 \\ 6 & 4 \\ 10 & -12 \end{pmatrix}$; (d)

$\begin{pmatrix} 6 & 18 \\ -6 & 12 \\ 18 & 24 \end{pmatrix}$; (e) $M \begin{pmatrix} 3 & 2 \\ 3 & 2 \end{pmatrix}$; (f), (g) undefined;

(h) $\begin{pmatrix} -1 & -1 \\ 14 & -4 \end{pmatrix}$; (i) A . (j) $2E - F = \begin{pmatrix} -2 & 6 \\ -2 & 1 \end{pmatrix}$

and $3F + E = \begin{pmatrix} 13 & -4 \\ -1 & 18 \end{pmatrix}$. (k) PQ is 2×4 , QR is 3×2 , and RP is 4×3 ; PR , QP , and RQ are undefined.

4. (a) False: the identity property is being used, not associativity. (b) True. (c) True, interpreted as $(m+k)A$. (d) False: multiplication is not generally commutative and DC may be undefined. (e) False; either factor may be non-zero when the product is zero. (f) True by distributivity. (g) True by associativity. (h) True. (i) False; take $A = \mathbf{0}$ and any $B \neq C$.

5. The valid order is

$$\begin{pmatrix} 1 & -3 & 2 \\ 4 & 1 & 5 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} \begin{pmatrix} 5 & 2 \end{pmatrix},$$

giving $\begin{pmatrix} 40 & 16 \\ 165 & 66 \end{pmatrix}$.

6. $p = -3$ and $q = 2$.

7. Both sides equal

$$\begin{pmatrix} au + bx & cu + dx \\ av + by & cv + dy \\ aw + bz & cw + dz \end{pmatrix}.$$

8. (a) 22; (b) -1 .

9. (a) True, since $\det A^T = \det A$; (b) false, because one row interchange changes the sign; (c) true; (d) true, since the first and third columns coincide; (e) true, since a triangular determinant is the product of its diagonal entries. (f) $\det B = 6(-1)(-2) = 12$; the row replacement does not change the determinant.

$$A^{-1} = \begin{pmatrix} 5/22 & 0 & 1/22 \\ -3/110 & 1/5 & -1/22 \\ -19/55 & 1/5 & 1/11 \end{pmatrix},$$

$$B^{-1} = \frac{1}{7} \begin{pmatrix} -3 & -2 & 4 \\ 5 & 1 & -2 \\ 4 & 5 & -3 \end{pmatrix}.$$

Both methods give the same matrices.

11. (a) $(x, y) = (23/14, 1/14)$; (b) $(x, y, z) = (-25/3, -34/3, 7/3)$; (c) $(x, y) = (3, 1)$; (d) $(x, y, z) = (1, 2, 3)$.
12. (a) $(x, y) = (-1, 2)$; (b) $(x, y, z) = (2, -2, 2)$; (c) $(x, y) = (2, 1)$; (d) $(x, y, z) = (1, 2, 3)$.
13. (a) $(x, y) = (1, 2)$; (b) $(x, y, z) = (3/11, 2/11, -1/11)$; (c) $\Delta = 13$, $\Delta_x = 18$, $\Delta_y = 1$, so $(x, y) = (18/13, 1/13)$; (d) $\Delta = 7$, $\Delta_x = 7$, $\Delta_y = 14$, $\Delta_z = 21$, so $(x, y, z) = (1, 2, 3)$.
14. The system is $x + y + 2z = 36$, $3x + 2y + z = 43$, $3x + y + 4z = 74$, or

$$\begin{pmatrix} 1 & 1 & 2 \\ 3 & 2 & 1 \\ 3 & 1 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 36 \\ 43 \\ 74 \end{pmatrix}.$$

Marker pen RM 7, examination pad RM 5, file RM 12.

15. The system is $4x + 5y + 2z = 8500$, $3x + 2y + z = 4550$, $x + y + z = 2150$. Thus $(x, y, z) = (750, 900, 500)$ ringgit.

16. Order 2×3 ; $a_{12} = -1$, $a_{23} = 5$; second column $(-1, 3)^T$; first row $(2, -1, 4)$.
17. $(x, y, z) = (4, 3, 4)$.
18. $\begin{pmatrix} 1 & 0 & -1 & -2 \\ 3 & 2 & 1 & 0 \\ 5 & 4 & 3 & 2 \end{pmatrix}$.
19. $X = \begin{pmatrix} 2 & 1 \\ 2/3 & 1 \end{pmatrix}$.
20. $AB = \begin{pmatrix} 10 & 5 \\ 2 & -2 \end{pmatrix}$.
21. $AB = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ and $BA = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$; matrix multiplication is not commutative.
22. $A^2 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$, $A^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$, and $A^5 = \begin{pmatrix} 1 & 5 \\ 0 & 1 \end{pmatrix}$.
23. $\mathbf{0}$.
24. $A^T = \begin{pmatrix} 1 & 3 \\ -2 & 4 \\ 0 & 5 \end{pmatrix}$ and $(A+B)^T = \begin{pmatrix} 1 & 2 \\ -1 & 7 \\ 2 & 9 \end{pmatrix}$, which equals $A^T + B^T$.
25. $S = \begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}$ and $K = \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix}$.
26. (a) $b = -3$, with a, c arbitrary; (b) $a = c = 0$ and $b = 3$.
27. The determinant is 22; $k = 2$.
28. $\det A = -12$, $\det(A^2) = 144$, and $\det(3A) = -324$.
29. $k = 0, \pm\sqrt{5}$.
30. (a) -10 ; (b) $-1/2$; (c) -16 ; (d) 125.
31. $M_{12} = -5$, $C_{12} = 5$, $C_{23} = -5$, and $C_{31} = -17$.
32. $\text{adj } A = \begin{pmatrix} 1 & -2 & 2 \\ 2 & 1 & -1 \\ -2 & 4 & 1 \end{pmatrix}$ and $A^{-1} = \frac{1}{5} \text{adj } A$.
33. $A^{-1} = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -2 & 4 \end{pmatrix}$; $(x, y) = (2, 1)$.
34. Both products are I_2 .
35. Invertible when $k \neq \pm\sqrt{6}$; then $A^{-1} = \frac{1}{k^2 - 6} \begin{pmatrix} k & -2 \\ -3 & k \end{pmatrix}$.
36. $X = \begin{pmatrix} -6 & -1 \\ 5 & 1 \end{pmatrix}$.
37. $X = \begin{pmatrix} 3 & -1 \\ 2 & -1 \end{pmatrix}$.
38. (a) False: $(AB)^{-1} = B^{-1}A^{-1}$; (b) true; (c) false: $(kA)^{-1} = k^{-1}A^{-1}$; (d) true.
39. RREF $\left(\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & 1 \end{array} \right)$; $(x, y) = (3, 1)$.
40. RREF $\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}$; rank 2.
41. The ranks are 1, 3, and 0, respectively.
42. (a) Unique: $(x, y) = (2, -1)$; (b) no solution; (c) infinitely many: $(x, y) = (3 - 2t, t)$.
43. (a) Inconsistent, so no solution; (b) dependent, with $(x, y) = (2 - t, t)$.
44. $k = 4$ gives infinitely many solutions; $k \neq 4$ gives no solution. No value gives a unique solution.
45. The inverse and Cramer methods require a square coefficient matrix with non-zero determinant. Gauss-Jordan elimination can expose inconsistency, free variables, and solution sets even when no inverse exists.
46. $\begin{pmatrix} 2 & -1 & 3 \\ 1 & 4 & -1 \\ 3 & 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \\ 5 \end{pmatrix}$.
47. $x + 2y - z = 4$, $3y + 5z = -2$, and $2x - y = 7$.
48. $(a, b, c, d) = (2, 3, 3, 4)$.
49. $D^n = \text{diag}(2^n, (-1)^n, 3^n)$ and $D^4 = \text{diag}(16, 1, 81)$.
50. $P^2 = P$ (idempotent), $N^2 = \mathbf{0}$ (nilpotent), and $R^2 = I$ (involutory).
51. -36 ; expansion along the first row is shortest.
52. $\frac{1}{4} \begin{pmatrix} 3 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 3 \end{pmatrix}$.
53. $\begin{pmatrix} 3 & -2 & 1 \\ -2 & 2 & -1 \\ 1 & -1 & 1 \end{pmatrix}$.
54. $EA = \begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix}$ swaps the rows; $AE = \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix}$ swaps the columns.
55. $(ABC)^T = ((AB)C)^T = C^T(AB)^T = C^TB^TA^T$.
56. $3(-1)(-2) = 6$.

57. $A^5 = 31A - 30I = \begin{pmatrix} -30 & 31 \\ -62 & 63 \end{pmatrix}.$

58. $a = 2, b = 2.$

59. Pivots: x, y ; free variable: $z = t$; rank 2;
 $(x, y, z) = (-1 - 2t, 4 + 3t, t).$

60. $R_2 \leftarrow R_2 - 3R_1$, followed by $R_1 \leftarrow R_1 - 2R_2$,
gives I_2 .

61. $(x, y, z) = t(3, -1, 1)$; a basis vector is $(3, -1, 1).$ 74.

62. $(x, y, z) = (2 - t/2, 1 - t/2, t).$

63. $AB = \begin{pmatrix} 4 & 4 \\ 10 & 8 \end{pmatrix}, \det A = -2,$
 $A^{-1} = \begin{pmatrix} -2 & 1 \\ 3/2 & -1/2 \end{pmatrix},$ and $(x, y) = (1, 2).$

64. $Sw = (81.0, 76.3, 85.2)^T$. The third student has
the highest weighted mark, 85.2.

65. $Ec = (2755, 2316)^T$ ringgit, so the combined
spending is RM 5,071.

66. With holding row vector $h = (5000, 3000, 2000),$
 $hR = (360, 30)$. The scenario profits are RM 360
and RM 30.

67.

$$RC = \begin{pmatrix} 7 & 8 \\ 6 & 5 \end{pmatrix},$$

a 2×2 matrix of raw-material units per
product. Hence

$$(RC)p = \begin{pmatrix} 134 \\ 100 \end{pmatrix}.$$

68.

$$T \begin{pmatrix} 120 \\ 80 \\ 40 \end{pmatrix} = \begin{pmatrix} 128 \\ 76 \\ 44 \end{pmatrix}.$$

The corrected red, green, and blue intensities
are 128, 76, and 44.

69.

$$\begin{pmatrix} 1 & 1 & 1 \\ 2 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1200 \\ 0 \\ 100 \end{pmatrix},$$

giving $(x, y, z) = (260, 520, 420)$ N.

70. The system is

$$\begin{aligned} x + y + z &= 50, \\ 0.02x + 0.05y + 0.08z &= 2.2, \\ y &= 2z. \end{aligned}$$

Thus the required volumes are 20 L, 20 L, and
10 L.

71. If $v = (s, r)^T$, then $Pv = v$ gives $0.3s = 0.2r$.
Together with $s + r = 1000$, this gives
 $v = (400, 600)^T$.

72. The system is $x + y + z = 29, x - y = 4$, and
 $y = 2z$. Hence $(x, y, z) = (14, 10, 5)$ nurses.

73. Solving the three equations gives
 $(x, y, z) = (500, 300, 200)$ subscriptions.

$$\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}^{-1} = \frac{1}{5} \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix}.$$

Applying it to $(13, 14)^T$ gives $(x, y) = (5, 3).$

75. Since T is triangular, $\det T = 2(3)(1) = 6$.
Volume is scaled by $|\det T|$, so the transformed
volume is 120 m^3 .

76. Revenues $(89, 80)^T$ ringgit; combined revenue
RM 169.

77. Closing stock $(125, 65, 95)^T$; the second material
is lowest.

78. Route costs $(960, 900, 1020)^T$ ringgit; total
RM 2,880.

79. $(2, 1, 3)N = (11, 12)$, so the totals are 11 protein
units and 12 fibre units.

80. 150 adult tickets and 200 student tickets.

81. 18 L of the 20% solution and 12 L of the 50%
solution.

82. $(I_1, I_2) = (4, 3).$

83. 30 tables and 20 chairs.

84. Sandwich RM 4, drink RM 3, fruit cup RM 2.

85. Conservation gives $30 + x = 10 + y$ and
 $15 + y = 35 + x$; together with $y = 2x$, this gives
 $x = 20$ and $y = 40$ vehicles per hour.

86. $Kv = (11, 8)^T, K^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix},$ and
 $K^{-1}(11, 8)^T = (3, 5)^T.$

87. After one transition: $(80, 20)^T$; after two:
 $(70, 30)^T.$

88. $v_1 = (200, 180)^T$ and $v_2 = (360, 244)^T.$

89. $(I - A)x = d$ gives $x = (100, 80)^T.$

90. Adjusted image $\begin{pmatrix} 70 & 120 \\ 170 & 220 \end{pmatrix}$; transpose
 $\begin{pmatrix} 70 & 170 \\ 120 & 220 \end{pmatrix}.$

91. Point vector $(20, 19, 15)^T$; the ranking is team 1,
team 2, team 3.

92. Degree vector $(2, 3, 3, 2)^T$; $(A^2)_{14} = 2$ two-step walks. $2b = 2c + d$. The least positive solution is $(a, b, c, d) = (1, 5, 3, 4)$.
93. Image vertices $(0, 0)$, $(2, 0)$, $(1, 1)$. The area changes from $1/2$ to 1 , a factor of $|\det T| = 2$.
94. $R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, $F = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$; $Rv = (-1, 3)^T$ and $FRv = (-1, -3)^T$.
95. Area $= \frac{1}{2}|4(5) - 1(2)| = 9$ square units.
96. The balance equations are $3a = c$, $8a = 2d$, and
97. $(a, b, c) = (1, 2, 1)$, so $p(x) = x^2 + 2x + 1$.
98. $(a, b, c) = (1, 0, 1)$, so $y = x^2 + 1$.
99. $(x, y, z) = (40, 30, 30)$, which gives totals 100, 200, and 130 in the three equations.
100. $(x, y, z) = (30, 40, 20)$. The total cost is $(5000, 4000, 6000)(30, 40, 20)^T = \text{RM } 430,000$.

Chapter 6: Trigonometry

These answers correspond to Section 6.8.

1. (a) $5\pi/4$; (b) $-5\pi/6$; (c) $9\pi/2$.

2. (a) 210° ; (b) -225° ; (c) 660° .

3. (a) Coterminal angle 330° , reference angle 30° ;
 (b) coterminal angle $5\pi/4$, reference angle $\pi/4$;
 (c) coterminal angle 160° , reference angle 20° .
 (d) The quadrantal angles are $5\pi/2$, 270° ,
 -270° , 3π , and $-\pi$. (e) For example, 480° ,
 840° , -240° , and -600° .

4. (a) $1/2$; (b) $-\sqrt{2}/2$; (c) $-1/\sqrt{3}$; (d) -2 ; (e) 2 ;
 (f) -1 ; (g) $\sqrt{2}/2$; (h) $-\sqrt{3}$; (i) $1/2$; (j) $-\sqrt{2}/2$;
 (k) $1/2$; (l) $\sqrt{3}$.

5. (a) With $r = 13$,

$$\sin \theta = \frac{12}{13}, \quad \cos \theta = -\frac{5}{13}, \quad \tan \theta = -\frac{12}{5},$$

and $\csc \theta = 13/12$, $\sec \theta = -13/5$,
 $\cot \theta = -5/12$. (b)

$$\sin \theta = \frac{3}{\sqrt{13}}, \quad \cos \theta = \frac{2}{\sqrt{13}}, \quad \tan \theta = \frac{3}{2},$$

with $\csc \theta = \sqrt{13}/3$, $\sec \theta = \sqrt{13}/2$, and
 $\cot \theta = 2/3$. Also $\cot(\pi/2 - \theta) = 3/2$ and
 $\csc(\theta - \pi/2) = -\sqrt{13}/2$.

6. (a) $\cos \theta = 4/5$, $\tan \theta = -3/4$, $\csc \theta = -5/3$,
 $\sec \theta = 5/4$, and $\cot \theta = -4/3$. (b) θ lies in
 quadrant III, and

$$\sin \theta = -\frac{3}{\sqrt{13}}, \quad \cos \theta = -\frac{2}{\sqrt{13}}, \quad \tan \theta = \frac{3}{2},$$

with $\csc \theta = -\sqrt{13}/3$, $\sec \theta = -\sqrt{13}/2$, and
 $\cot \theta = 2/3$.

7. (a) $\cos x$; (b) $\cos x$; (c) $-\cot x$; (d) $-\csc x$.

Reflection in the x -axis preserves the unit-circle
 horizontal coordinate and reverses the vertical
 coordinate; exchanging the reference-triangle
 legs gives the cofunction identity. Cosine and
 secant are even; sine, cosecant, tangent, and
 cotangent are odd.

8. Both equal $(\sqrt{6} - \sqrt{2})/4$.

9. $h = 10 \tan 61.7^\circ + 1.5 \approx 20.072$ m.

10. $h = 8.8 \tan 49^\circ \approx 10.123$ m.

11. Amplitude 3; period π ; shift $\pi/2$ right; midline
 $y = 1$; range $[-2, 4]$.

12. Domain $\mathbb{R} \setminus \{\pi/2 + k\pi : k \in \mathbb{Z}\}$; range $\mathbb{R}$; period
 π ; odd; zeros $x = k\pi$; vertical asymptotes
 $x = \pi/2 + k\pi$.

13. Replace $\csc x$ by $1/\sin x$ and $\cot x$ by
 $\cos x/\sin x$; the left side becomes
 $(1 - \cos^2 x)/\sin x = \sin x$.

14. The numerator is
 $\sin x/\cos x + \cos x/\sin x = 1/(\sin x \cos x)$, equal
 to the denominator $\sec x \csc x$.

15. Combining over the common denominator
 $\sin x(1 + \cos x)$ gives

$$\frac{\sin^2 x + (1 + \cos x)^2}{\sin x(1 + \cos x)} = \frac{2(1 + \cos x)}{\sin x(1 + \cos x)} = 2 \csc x.$$

16. $\sin 75^\circ = \cos 15^\circ = (\sqrt{6} + \sqrt{2})/4$;
 $\tan 105^\circ = -(2 + \sqrt{3})$.

17. $\sin A = 12/13$ and $\cos B = -24/25$, so
 $\sin(A + B) = -323/325$.

18. $\sin 2A = -24/25$, $\cos 2A = -7/25$, and
 $\tan 2A = 24/7$.

19.

$$\begin{aligned} \cos 3x &= \cos(2x + x) \\ &= (2 \cos^2 x - 1) \cos x - 2 \sin^2 x \cos x \\ &= 4 \cos^3 x - 3 \cos x. \end{aligned}$$

20. (a) $\cos 112.5^\circ = -\frac{1}{2}\sqrt{2 - \sqrt{2}}$; (b)
 $\tan 67.5^\circ = 1 + \sqrt{2}$; (c) $\sin 22.5^\circ = \frac{1}{2}\sqrt{2 - \sqrt{2}}$;
 (d) $\cos 22.5^\circ = \frac{1}{2}\sqrt{2 + \sqrt{2}}$; (e)
 $\tan 22.5^\circ = \sqrt{2} - 1$.

21. (a) $\frac{1}{2}(\sin 5x + \sin x)$; (b) $\frac{1}{2}(\cos 3x - \cos 7x)$; (c)
 $\cos 9x + \cos 3x$; (d) $\frac{1}{2}(\sin 5x - \sin x)$; (e)
 $\frac{1}{2}(\sin 9x + \sin 3x)$; (f) $\frac{1}{2}(\cos 3x - \cos 5x)$.

22. (a) $2 \sin 4x \sin x$; (b) $2 \cos(9x/2) \cos(3x/2)$; (c)
 $2 \sin 3x \cos x$; (d) $-2 \sin 4x \sin x$; (e)
 $2 \cos(9x/2) \cos(3x/2)$; (f) $2 \cos 5x \sin 2x$.

23. The numerator is $2 \sin 3x \cos x$ and the
 denominator is $-2 \sin 3x \sin x$; their quotient is
 $-\cot x$ wherever defined.

24. $x = \pi/3, 2\pi/3$.

25. (a) $x = 3\pi/8, 5\pi/8, 11\pi/8, 13\pi/8$. (b) $x = \pi/3, 2\pi/3, 4\pi/3, 5\pi/3$.
26. $x = \pi/18, 7\pi/18, 13\pi/18$.
27. $x = 5\pi/12 + k\pi$ or $x = 7\pi/12 + k\pi, k \in \mathbb{Z}$.
28. $x = \pi/2, 3\pi/2$.
29. (a) $x = 90^\circ$ or $x = \sin^{-1}(1/4) \approx 14.478^\circ$ or $x = 180^\circ - \sin^{-1}(1/4) \approx 165.522^\circ$. (b) $x = -2\pi/3, -\pi/3, \pi/3, 2\pi/3, 4\pi/3, 5\pi/3, 7\pi/3, 8\pi/3$.
30. (a) $x = k\pi \pm \pi/6, k \in \mathbb{Z}$. (b) $x = -3\pi/4, -\pi/4, \pi/4, 3\pi/4$.
31. $3 \cos x + 4 \sin x = 5 \cos(x - \alpha)$, where $\alpha = \tan^{-1}(4/3)$; maximum 5, minimum -5.
32. $\cos x - 3 \sin x = \sqrt{10} \cos(x + \tan^{-1} 3)$. The solutions are $x = 0$ and $x = 2\pi - 2 \tan^{-1} 3$.
33. $\sin^{-1} \quad [-1, 1] \quad [-\pi/2, \pi/2]$
 $\cos^{-1} \quad [-1, 1] \quad [0, \pi]$
 $\tan^{-1} \quad \mathbb{R} \quad (-\pi/2, \pi/2)$
 $\sec^{-1} \quad |x| \geq 1 \quad [0, \pi] \setminus \{\pi/2\}$
 $\csc^{-1} \quad |x| \geq 1 \quad [-\pi/2, \pi/2] \setminus \{0\}$
34. (a) $\pi/4$; (b) $2\pi/3$; (c) $-\pi/6$; (d) $2\pi/3$; (e) $\pi/6$.
35. (a) $3/5$; (b) $-4/7$; (c) 5.
36. (a) $\pi/6$; (b) $\pi/4$; (c) $-\pi/4$.
37. Let $\theta = \sin^{-1} x$. Then $\theta \in [-\pi/2, \pi/2]$ and $\cos^{-1} x = \pi/2 - \theta$ because $\cos(\pi/2 - \theta) = \sin \theta = x$. Hence the sum is $\pi/2$.
38. $\cos(\sin^{-1} x) = \sqrt{1 - x^2}$ for $-1 \leq x \leq 1$, and $\tan(\sin^{-1} x) = x/\sqrt{1 - x^2}$ for $-1 < x < 1$.
39. $\sin(\cos^{-1} x) = \sqrt{1 - x^2}$ and $\cos(\tan^{-1} x) = 1/\sqrt{1 + x^2}$.
40. $-3/5$.
41. The tangent of the sum is

$$\frac{1/7 + 1/13}{1 - (1/7)(1/13)} = \frac{2}{9}.$$

Both angles and their sum are acute, so the principal inverse tangent gives the stated identity.

42. The two inverse values must both equal $\pi/4$; hence $x = \sqrt{2}/2$.
43. $x = 1/\sqrt{3}$ and $x = -1/2$, respectively.

44. The inverse sine returns only a principal value, so angles outside its principal range may be replaced by a different angle having the same

sine. The equality is guaranteed precisely for $-\pi/2 \leq \theta \leq \pi/2$.

45. (a) Expansion gives $2 \sin^2 x + 2 \cos^2 x = 2$. (b) Cross-multiplication gives $(1 - \sin x)(1 + \sin x) = \cos^2 x$. (c) The left side is $(1/\cos x - \cos x)/(\sin x/\cos x) = \sin x$.
46. The sine and shifted-cosine graphs coincide because $\cos(\pi/2 - x) = \sin x$. The tangent graph has zeros $k\pi$ and vertical asymptotes $\pi/2 + k\pi$; the sine and cosine graphs have no vertical asymptotes.

47.

$$f(x) = 1 - \cos^2 x + \cos x = \frac{5}{4} - \left(\cos x - \frac{1}{2}\right)^2.$$

Its period is 2π and its range is $[-1, 5/4]$. Its zeros satisfy

$$\cos x = \frac{1 - \sqrt{5}}{2},$$

so $x = 2k\pi \pm \cos^{-1}((1 - \sqrt{5})/2), k \in \mathbb{Z}$.

48. α is in quadrant IV and β is in quadrant II. Thus

$$\cos(\alpha - \beta) = \frac{12}{13} \left(-\frac{12}{13}\right) + \left(-\frac{5}{13}\right) \frac{5}{13} = -1,$$

so $\sec(\alpha - \beta) = -1$.

49. Multiplying by $\cos \theta$ gives

$$7 \sin \theta + 2 \cos^2 \theta = 5,$$

hence $2 \sin^2 \theta - 7 \sin \theta + 3 = 0$. The valid root is $\sin \theta = 1/2$, giving $\theta = \pi/6, 5\pi/6$.

50. Either $\cos \theta = 0$ or $\sin \theta = 1/3$. Thus

$$\theta = 90^\circ, 270^\circ, 19.471^\circ, 160.529^\circ.$$

51. Since $(2 \tan \theta - 1)(\tan \theta + 3) = 0$,

$$\theta = 26.565^\circ, 108.435^\circ, 206.565^\circ, 288.435^\circ.$$

52. $\sin 6x + \sin 2x = 2 \sin 4x \cos 2x$. Hence

$$\begin{aligned} \sin 6x + \sin 2x - \sin 4x &= \sin 4x(2 \cos 2x - 1), \\ &= 4 \cos 2x \cos 3x \sin x. \end{aligned}$$

53.

$$\begin{aligned} 5 \sin x - 12 \cos x &= 13 \sin(x - \alpha), \\ \alpha &= \tan^{-1}(12/5). \end{aligned}$$

The maximum 13 occurs at $x = \alpha + \pi/2 + 2k\pi$, and the minimum -13 at $x = \alpha + 3\pi/2 + 2k\pi$. The acute solution is

$$x = \alpha + \sin^{-1}(2/13) \approx 76.230^\circ.$$

54. $3 \cos x + 4 \sin x = 5 \cos(x - \alpha)$, where $\alpha = \tan^{-1}(4/3)$. The maximum 5 occurs at $x = \alpha + 2k\pi$ and the minimum -5 at $x = \alpha + \pi + 2k\pi$. There is no acute solution of the stated equation because the expression is greater than 2 throughout $0 < x < \pi/2$.

55. $8 \cos x - 15 \sin x = 17 \cos(x + \alpha)$, where $\alpha = \tan^{-1}(15/8)$. Its maximum and minimum are 17 and -17 , attained at $x = -\alpha + 2k\pi$ and $x = \pi - \alpha + 2k\pi$, respectively. The acute solution is

$$x = \cos^{-1}(2/17) - \alpha \approx 21.316^\circ.$$

56. $D_f = \mathbb{R}$, $D_g = [-1/3, 1/3]$, and $D_h = [-1/3, 1/3]$.
57. $\sin \theta = 4/5$, $\cos \theta = 3/5$, $\cot \theta = 3/4$, $\sec \theta = 5/3$, and $\csc \theta = 5/4$.
58. $\cos \theta = 5/13$, $\sin \theta = 12/13$, $\tan \theta = 12/5$, $\cot \theta = 5/12$, and $\csc \theta = 13/12$.
59. (a) $12/13$; (b) $(4 - 3\sqrt{3})/10$; (c) -9 ; (d) $(\sqrt{6} + \sqrt{2})/4$.
60. (a) $1/2$; (b) $-24/25$; (c) $(6 + 4\sqrt{5})/15$; (d) $\sqrt{3} + 3\sqrt{2}/4$.
61. (a) $\sqrt{2x-1}/x$, valid for $x \geq 1/2$. (b) $\sqrt{2/(1-x)}$, valid for $x < 1$.
62. The first formula follows because sine is odd and its principal range is symmetric about zero. For the second, if $\theta = \cos^{-1} x$, then $\pi - \theta$ lies in $[0, \pi]$ and has cosine $-x$. Finally, the complementary-angle identity applied to $1/x$ gives $\sec^{-1} x + \csc^{-1} x = \pi/2$ on $|x| \geq 1$ with the stated principal ranges.
63. (a) $\pi - \tan^{-1} 3$; (b) $\tan^{-1}(1/3)$. In part (a), the sum lies in quadrant II even though the tangent formula produces the negative value -3 ; quadrant information is therefore needed.
64. (a) $-\pi/6$; (b) $2\pi/3$; (c) $-\pi/6$; (d) $2\pi/3$.
65. (a) $4/5$; (b) $-12/5$; (c) $3/5$; (d) $13/12$.
66. The joint is $(4 \cos 30^\circ, 4 \sin 30^\circ) = (2\sqrt{3}, 2)$, and the end effector is

$$(2\sqrt{3} + 3 \cos 90^\circ, 2 + 3 \sin 90^\circ) = (2\sqrt{3}, 5).$$

Its distance from the origin is $\sqrt{37} \approx 6.083$ units and its direction is $\tan^{-1}(5/(2\sqrt{3})) \approx 55.285^\circ$.

67. The total displacement is approximately 18.894 km east and 5.785 km north. Its magnitude is 19.760 km and its direction is 17.024° north of east.
68. The ground speed is $\sqrt{180^2 + 40^2} \approx 184.391$ km h $^{-1}$. The direction is $\tan^{-1}(40/180) \approx 12.529^\circ$ east of north.
69. The midline is 12 cm, amplitude 5 cm, frequency 2 Hz, and period 0.5 s. The first maximum occurs at $t = 1/6$ s. During the first period, $s = 12$ at $t = 1/24$ s and $t = 7/24$ s.
70. The range is $[2.2, 3.4]$ L and the period is 4 s. The first time at 3.1 L is $t = 1/3$ s. During the first cycle, $1/3 \leq t \leq 5/3$.
71. $\cos \theta \geq 1.8/2.4 = 0.75$, so $0^\circ \leq \theta \leq \cos^{-1}(0.75) \approx 41.410^\circ$.
72. $\angle B = 180^\circ - 52^\circ - 73^\circ = 55^\circ$. By the sine rule, $BC \approx 148.323$ m and $AC \approx 154.184$ m.
73. (a) $\theta_2 = \sin^{-1}(\sin 45^\circ/1.5) \approx 28.126^\circ$. (b) The critical angle is $\sin^{-1}(1/1.5) \approx 41.810^\circ$.
74. The maximum torque is $0.35(120) = 42$ N m. The condition $\sin \theta \geq 30/42$ gives $45.585^\circ \leq \theta \leq 134.415^\circ$.
75. $\theta = \tan^{-1}(20^2/(80 \cdot 9.8)) \approx 27.031^\circ$. For $\theta = 30^\circ$, $v = \sqrt{80(9.8) \tan 30^\circ} \approx 21.275$ m s $^{-1}$.
76. The height of A is $35 \tan 34^\circ \approx 23.608$ m. The height of B is
- $$35(\tan 34^\circ + \tan 24^\circ) \approx 39.191 \text{ m.}$$
77. If d is the nearer horizontal distance and h is the height, then $h = d \tan 74.2^\circ = (d + 35) \tan 51.8^\circ$. Thus $d \approx 19.653$ ft and $h \approx 69.451$ ft.
78. Taking the pole bases as $x = 0$ and $x = 25$, the wires have equations $y = 10 - (2/5)x$ and $y = (3/5)x$. Their intersection has $x = 10$ and height $y = 6$ m.
79. The elevation angles at altitudes $a + b$ and b are $\cot^{-1}(x/(a + b))$ and $\cot^{-1}(x/b)$, respectively, so their difference gives the formula. A physical domain is $x > 0$, $a > 0$, and $b > 0$.
80. By the cosine rule, $\cos \theta = (2^2 + 3^2 - 4^2)/(2 \cdot 2 \cdot 3) = -1/4$. Hence $\theta \approx 104.478^\circ$, or 104° to the nearest degree.
81. The distance is $60/\tan 12^\circ \approx 282.278$ m.

82. The minimum horizontal run is
 $0.9/\tan 5^\circ \approx 10.287$ m.

83.

$$h(t) = 20 - 18 \cos\left(\frac{\pi t}{20}\right).$$

Thus $h(10) = 20$ m. Setting $h = 29$ gives
 $\cos(\pi t/20) = -1/2$, so $t = 40/3$ s or $80/3$ s.

84.

$$d(t) = 3.3 + 1.9 \cos\left(\frac{\pi t}{6}\right).$$

Hence $d(3) = 3.3$ m. The first two times when
 $d = 2$ are approximately $t = 4.439$ h and
 $t = 7.561$ h after high tide.

85. The amplitude is 1.5 h, the midline is 12 h, the
period is 365 days, and the phase shift is 172
days to the right. The model predicts a longest
day of 13.5 h on day 172 and a shortest day of
10.5 h near day 354.5.

86. Amplitude 0.004; angular frequency 440π
 rad s^{-1} ; frequency 220 Hz; period $1/220$ s;
phase shift $1/1320$ s to the left.

87. Maximum magnitude 240 V; frequency 50 Hz;
first positive time $t = 1/600$ s.

88. At 30° , $R = 200\sqrt{3}/9.8 \approx 35.348$ m. The
greatest range occurs at $\theta = 45^\circ$ and is
 $400/9.8 \approx 40.816$ m.

89. The cosine rule gives

$$d = \sqrt{8^2 + 11^2 - 2(8)(11)\cos 65^\circ} \approx 10.518 \text{ km.}$$

90. Pitch angle $\tan^{-1}(3.2/5.5) \approx 30.192^\circ$; rafter
length $\sqrt{3.2^2 + 5.5^2} \approx 6.363$ m.

91. The angle of elevation is
 $\tan^{-1}(120/160) \approx 36.870^\circ$, and the line-of-sight
distance is 200 m.

92.

$$x(t) = 5 \cos(\pi t/6), \quad y(t) = 5 \sin(\pi t/6).$$

At $t = 4$, the position is $(-5/2, 5\sqrt{3}/2)$.

93. The maximum is 31° at $t = 2$ (February), and
the minimum is 23° at $t = 8$ (August). In May,
 $T(5) = 27^\circ$.

94. $\sin 8t \cos 3t = \frac{1}{2}(\sin 11t + \sin 5t)$; the angular
frequencies are 11 and 5 rad s^{-1} .

95.

$$\cos(102\pi t) + \cos(98\pi t) = 2 \cos(100\pi t) \cos(2\pi t).$$

The carrier frequency is 50 Hz and the beat
frequency is 2 Hz.

96. $6 \cos t + 8 \sin t = 10 \cos(t - \alpha)$, where
 $\alpha = \tan^{-1}(4/3)$. The amplitude is 10 and the
phase shift is α to the right.

97. The angle is

$$\tan^{-1}(17/20) - \tan^{-1}(5/20) \approx 26.328^\circ.$$

98. The angle is $\cos^{-1}(6/10) \approx 53.130^\circ$, and the
height reached is 8 m.

99. The width is

$$2000 \cot 27^\circ - 2000 \cot 39^\circ \approx 1455.427 \text{ m.}$$

100. The maximum is 80 kW at $t = 12$, and the
minimum is 20 kW at $t = 0$ and $t = 24$. Solving
 $\sin(\pi(t - 6)/12) \geq 1/2$ gives $8 \leq t \leq 16$. A
limitation is that real output is zero at night
and is affected by cloud, shading, season, and
equipment losses, so it need not follow a smooth
sinusoid.

Chapter 7: Linearisation

These answers correspond to Section 7.8. Graphical estimates may vary slightly.

- (a) Linear, $y = 4x + 7$. (b) Non-linear; the first differences are 3, 5, 7, 9. (c) Linear, $y = 2x + 1$; compare $\Delta y/\Delta x$ because the x -spacing is unequal.
- Every pair gives gradient 3, so $y = 3x + 1$.
- A suitable visual fit is close to $y = 2x + 2$; the stated least-squares line is $y = 1.983x + 2.043$.
- The temperature falls by 1.6°C per minute and the estimated initial temperature is 84°C . $T(25) = 44^\circ\text{C}$; this is extrapolation.
- The gradient is 3 and the intercept is 1, so output = 3(pressure) + 1. Gradient units are mV kPa^{-1} .
- The residuals are 0.2, -0.2 , 0.3, -0.3 . They are small and occur on both sides of zero without an obvious pattern.
- Model A. Its residuals are small and unpatterned; model B has a curved negative-positive-negative pattern.
- $x = 6$ and $x = 18$ are interpolations; $x = 2$ and $x = 25$ are extrapolations.
- It estimates capacity at age zero. The value may reflect calibration, rounding, or production variation, so the fitted intercept remains informative even if 102% is not a literal nominal capacity.
- Force the origin only when theory requires zero response at zero input and the measuring system has no offset. Mass against volume for a homogeneous material is a possible example; Celsius against Fahrenheit is not.
- The point (4, 12) is the possible outlier. The line predicts 9, so its residual is about 3. Check transcription, equipment, procedure, and whether a different physical condition occurred.
- Association may be strong but non-linear. A straight line would leave systematic residuals, so the variables should be transformed or a different model used.
- (a) Horizontal x^2 , vertical y . (b) Horizontal x , vertical $\ln y$. (c) Horizontal $1/x$, vertical $1/y$. (d) Horizontal $\log_{10} x$, vertical $\log_{10} y$.
- Exact data satisfying a linear equation lie on one line and can determine it algebraically. Measured data generally scatter, so a line of best fit is used and uncertainty is acknowledged.
- Parabola: plot against x^2 . Exponential decay: plot $\ln y$ against x . Reciprocal curve: plot against $1/x$. Power law: plot $\log y$ against $\log x$.
- (a) $Y = y$, $X = x^3$: $Y = aX + b$. (b) $Y = y$, $X = \sqrt{x}$: $Y = aX + b$. (c) $Y = y$, $X = 1/x$: $Y = aX + b$.
- (a) $xy = ax^2 + b$: $Y = xy$, $X = x^2$. (b) $x^2y = ax^2 + b$: $Y = x^2y$, $X = x^2$. (c) $xy = ax + b$: $Y = xy$, $X = x$.
- (a) $y/x = ax + b$: $Y = y/x$, $X = x$. (b) $y/x^2 = ax + b$: $Y = y/x^2$, $X = x$. (c) Dividing by x gives $y = ax + b$.
- (a) $1/y = x/a + b/a$. (b) $1/y = ax + b$. (c) $1/y = x^2/a + b/a$.
- (a) $\log y = n \log x + \log a$. (b) $\log y = x \log b + \log A$. (c) $\ln y = kx + \ln A$.
- Plot $1/y$ against x ; gradient a , intercept b .
- $x/y = ax + b$, so plot x/y against x . Gradient a and intercept b .
- $1/y = x^2/a + b/a$. Gradient $1/a$ and intercept b/a .
- $1/y = (b/a)(1/x) + 1/a$. Gradient b/a and intercept $1/a$.
- Real logarithms require $x > 0$ and $y > 0$.
- (a) $\log_{10} y = x \log_{10} b + \log_{10} A$; hence $A = 10^c$, $b = 10^m$. (b) $\ln y = x \ln b + \ln A$; hence $A = e^c$, $b = e^m$.
- $\ln y = -kx + \ln A$. Gradient $-k$, intercept $\ln A$; the negative gradient indicates decay.
- $\ln(y - c) = n \ln x + \ln a$, requiring $x > 0$ and $y - c > 0$.
- (a) $Y = \sqrt{y}$, $X = x$. (b) $Y = 1/y$, $X = 1/x$. (c) $Y = x/y$, $X = x$.
- Plot y against $1/x$ to obtain $Y = bX + a$, or use $xy = ax + b$ and plot xy against x .
- $xy = mx^2 + c$, so $y = mx + c/x$.

32. (a) $y > 0$. (b) $x > 0$ and $y > 0$. (c) $x \neq 0$. (d) $y \neq 0$.
33. Changing log base multiplies both log coordinates by the same constant, leaving the gradient n unchanged. The intercept is $\log_{10} a$ or $\ln a$, so its numerical value changes.
34. (a) Gradient a , intercept b . (b) Gradient $1/a$, intercept b/a . (c) Gradient n , intercept $\ln a$. (d) Gradient k , intercept $\ln A$.
35. Respectively: y against x^4 ; y against $1/x$; $1/y$ against x ; $\log y$ against $\log x$; and $\log y$ against x .
36. $a = 3$, $b = 1$.
37. $a = 2.5$, $b = -4$, and $y(4) = 9$.
38. Since $1/y = x/a + b/a$, $a = 8$ and $b = 4$.
39. $n = 2.4$ and $a = 10^{-0.6990} \approx 0.200$.
40. $A = 10^{0.4771} \approx 3$ and $b = 10^{0.1761} \approx 1.5$.
41. The gradient is $k = (1.0 - 2.6)/(5 - 1) = -0.4$ and the intercept is 3. Thus $A = e^3 \approx 20.1$; the model is decay.
42. $a = 3$, $b = -2$.
43. $a = 5$, $b = 2$.
44. $a = 5$, $b = 2$.
45. $a = 0.3$, $b = 0.6$.
46. Plotting y against x^2 gives $y = 3x^2 + 2$; hence $a = 3$, $b = 2$.
47. The transformed line is $xy = 2x^2 - 3$; hence $a = 2$, $b = -3$.
48. $y = 3x^2$.
49. $y = 5(2^x)$. The common-log gradient is $\log_{10} 2 \approx 0.3010$ and the intercept is $\log_{10} 5 \approx 0.6990$.
50. $y = 80(1/2)^x = 80e^{-(\ln 2)x}$.
51. (a) $y = 2x^3 - 1$. (b) $y = 4/(x + 2)$. (c) $y = 3x^{3/2}$.
52. Gradient 2, intercept 2; hence $y = 2x^2 + 2$.
53. The gradient has units h^{-1} . Strictly, one takes the logarithm of a numerical ratio such as $M/(1 \text{ unit})$, so the log argument is dimensionless.
54. The estimates are $15.9/8 = 1.9875$ and $8.2/4 = 2.05$. Widely separated points reduce the relative effect of coordinate-reading errors.
55. $C = 80e^{-0.25t}$ and $C(6) \approx 17.9$.
56. Least squares gives $y \approx 2.007x^2 + 2.941$. Thus $a \approx 2.01$, $b \approx 2.94$, suggesting $y \approx 2x^2 + 3$.
57. $\ln C \approx -0.2995t + 3.9136$, so $C_0 \approx 50.1$, $k \approx 0.300$, and $C(5) \approx 11.2$.
58. Plot $\ln(T - 22)$ against t . A fit gives $A \approx 49.6$, $k \approx 0.218 \text{ min}^{-1}$, and $T(15) \approx 23.9^\circ\text{C}$.
59. Plot y against x^2 and $\ln y$ against $\ln x$. The first is exactly $y = 2x^2 + 3$; the log-log points are not exactly collinear because of the additive constant. The first model is better.
60. Model A has small residuals scattered around zero. Model B has a curved arch, showing that part of the trend remains unexplained.
61. With all points, $y \approx 2.186x + 1.619$. Omitting (3, 13) gives $y \approx 2.017x + 1.058$. The point has strong influence, but it should be omitted only after a contextual or measurement-based justification.
62. Predictions are 0.8, 2.3, 3.8, 5.3, so the residuals are 0.2, -0.2, 0.4, -0.3. The largest absolute residual is 0.4 at (2, 4.2).
63. Since $10^{0.602} \approx 4.00$, the model is $y \approx 4.00x^{1.30}$ and $y(5) \approx 32.4$.
64. The relationship may change beyond the observed range, and small uncertainty in a transformed gradient can produce a large back-transformed error at $x = 50$.
65. Measure several positive lengths and repeated periods, form $\ln L$ and $\ln T$, then fit $\ln T = n \ln L + \ln a$. The gradient gives n and $a = e^{\text{intercept}}$. Timing reaction, length measurement, or damping is a possible uncertainty.
66. $T = 2L^{1/2}$, so $n = 1/2$ and $T(2) = 2\sqrt{2} \approx 2.83$ s.
67. $F = 3x + 2$. The gradient is 3 N cm^{-1} and the intercept is a 2 N preload or zero offset. The original data are already linear.
68. $a = 0.6$, $n = 1.8$, so $v = 0.6r^{1.8}$ and $v(2) \approx 2.09$.
69. $a = 10^{-1.3010} \approx 0.05$, $n = 3$, so $M = 0.05L^3$ and $M(20) = 400$.
70. $N_0 = 500$, $k = 0.35$, and the doubling time is $\ln 2/0.35 \approx 1.98$ time units.
71. The half-life is $\ln 2/0.18 \approx 3.85$ years and $M(8) = 60e^{-1.44} \approx 14.2$.

72. $\ln C = \ln 60 - 0.12t$ and $C(8) = 60e^{-0.96} \approx 23.0$ mg L⁻¹.
73. $\ln(T - 22) = \ln 58 - 0.15t$ and $T(12) \approx 31.6^\circ\text{C}$.
74. $V_0 = 12$ V, $RC = 1/0.25 = 4$ s, and $V(6) = 12e^{-1.5} \approx 2.68$ V.
75. $I/I_0 = e^{-0.35x} = 0.1$, so $x = \ln 10/0.35 \approx 6.58$ cm.
76. An inverse-square law produces log-log gradient -2 . Here $I = 180d^{-2}$, so $I(5) = 7.2$.
77. $g = 4\pi^2/4.024 \approx 9.81$ ms⁻².
78. Plot T against $r^{3/2}$. The gradient is $a = 0.5$, so $T(4) = 0.5(4^{3/2}) = 4$.
79. Plot R against $1/d^2$; the gradient is a . $R = 0.060/(0.5)^2 = 0.24$ Ω .
80. $Q = 1.2r^4$ and $Q(1.5) = 6.075 \approx 6.08$.
81. $1/v = -1/u + 1/f$. The gradient is -1 , the intercept is $1/f$, and therefore f is the reciprocal of the intercept.
82. $P = 240/V + 5$, so $P(8) = 35$. The constant $b = 5$ represents a pressure offset or background reading.
83. $\ln k = \ln A - (E/R)(1/T)$. Thus $E = 5200(8.314) \approx 4.32 \times 10^4$ J mol⁻¹, or 43.2 kJ mol⁻¹.
84. $r = 0.08(3)^{1.5} \approx 0.416$. The exponent means a 1% increase in C produces approximately a 1.5% increase in rate.
85. $1/V_{\max} = 0.25$, so $V_{\max} = 4$. Then $K_m/V_{\max} = 0.75$, giving $K_m = 3$.
86. The annual growth is 8% and $P(6) = 2000(1.08)^6 \approx 3174$.
87. $\log_{10} V = \log_{10} 12000 + t \log_{10} 0.82$. $V(6) \approx \text{RM } 3,648.08$.
88. $\ln A = \ln P + t \ln(1 + r)$, so the intercept gives $P = e^c$ and $r = e^m - 1$. Here $A(10) \approx \text{RM } 8,954.24$.
89. $\ln P = \ln 240 + t \ln 1.045$ and $P(8) \approx 341.30$.
90. $t = 0.8(40)^{0.75} \approx 12.7$ h. A positive exponent means the time increases with n ; this differs from a usual decreasing learning curve and should be checked against context.
91. $C = 4.5m^{0.6}$, so $C(15) \approx 22.8$. A 1% increase in m corresponds approximately to a 0.6% increase in cost.
92. $F = 2.0 \times 10^{-5}(30)^2 = 0.0180$ in the force units. Doubling speed multiplies drag by $2^2 = 4$.
93. Plot d against v^2 . Then $d(25) = 0.04(25)^2 + 3 = 28$ m. The intercept 3 m represents a fixed baseline distance or measurement offset in this simplified model.
94. Resistance changes by 0.38 units per degree and is 100 units at 0°C . Outside the calibrated range the sensor response may become non-linear or the material may change behaviour.
95. Plot I against $1/d^2$; $I(6) = 720/36 = 20$.
96. $F = 1361r^{-2}$, so $F(1.52) \approx 589$ W m⁻². The expected log-log gradient is -2 .
97. $D = 0.012(10^6)^{0.28} \approx 0.574$. Since the exponent is 0.28, a 1% increase in E produces only about a 0.28% increase in D .
98. $P = 0.42v^{3.1}$ and $P(8) \approx 265$. The fitted exponent 3.1 is close to, but slightly greater than, the ideal cubic exponent 3.
99. Use exactly the same observations in both transformed plots. Fit each line, inspect residuals against the corresponding transformed horizontal variable, and compare pattern and relative scatter rather than relying only on visual straightness or unlike scales.
100. The fitted line is $\ln M \approx -0.1450t + 4.7794$. Hence $M_0 \approx 119$ mg, $k \approx 0.145$ d⁻¹, the half-life is about 4.78 d, and $M(15) \approx 13.5$ mg. Day 15 lies beyond the measured range, and environmental conditions or the decay mechanism may change.

Chapter 8: Geometry and Shapes

These answers correspond to Section 8.7.

- (a) Centre $(3, -2)$, radius 5; (b) $(0, 4)$, 3; (c) $(-1, 5)$, $2\sqrt{3}$; (d) $(0, 0)$, $3\sqrt{2}$.
- (a) $(x - 3)^2 + (y + 4)^2 = 36$; (b) $(x + 2)^2 + (y - 5)^2 = 16$; (c) $(x - 2)^2 + (y + 3)^2 = 25$; (d) $(x + 1)^2 + (y - 2)^2 = -5$, so there is no real circle.
- (a) $(x + 2)^2 + (y - 3)^2 = 16$; (b) $x^2 + y^2 = 25$; (c) $(x - 1)^2 + (y + 1)^2 = 16$; (d) $(x - 2)^2 + (y - 3)^2 = 16$.
- (a) $(x - 2)^2 + (y + 1)^2 = 13$; (b) $(x - 1)^2 + (y - 1)^2 = 16$; (c) $(x - 1)^2 + (y - 2)^2 = 25$.
- (a) $x^2 + y^2 = 1$; (b) $x^2 + y^2 - 4x - 6y = 0$; (c) $(x - 4)^2 + (y - 5/2)^2 = 25/4$.
- $(2, 4)$ on; $(5, 3)$ on; $(0, 0)$ inside; $(8, -1)$ outside; $(-2, -4)$ on.
- (a) $(\pm 4, 0)$, $(0, \pm 4)$; (b) $(-1, 0)$, $(5, 0)$, $(0, -1 \pm \sqrt{6})$; (c) $(0, 0)$, $(6, 0)$, $(0, 4)$.
- (a) $(4, 3)$, $(-4, 3)$; (b) $(3, 2)$, $(-1, -2)$; (c) $(2, 2)$, $(2, -4)$.
- (a) $k = \pm 5$; (b) $c = \pm 4$; (c) $k = -3 \pm 3\sqrt{5}$.
- (a) $x + 3y = 10$; (b) $x - 2y + 4 = 0$; (c) $3x + 4y = 23$.
- (a) $y = 3x$; (b) $y = -2x + 7$; (c) $y - 5 = \frac{4}{3}(x - 1)$.
- Circle $(x - 2)^2 + (y + 1)^2 = 13$; tangent $y - 2 = \frac{2}{3}x$; normal $y - 2 = -\frac{3}{2}x$.
- (a) $\sqrt{41}$; (b) $\sqrt{17}$; (c) $4\sqrt{3}$.
- Tangents $y = \pm \frac{5}{12}(x - 13)$; contact points $(25/13, \mp 60/13)$ respectively; each length 12.
- (a) Radical axis $x = 4$; common chord endpoints $(4, \pm 3)$. (b) Radical axis $2x + y - 4 = 0$.
- The distances from $(2, -1)$ are $18/5$, 5, and 6. Hence the lines are respectively secant, tangent, and external.
- (a) $(x - 3)^2 + (y - 3)^2 = 9$; (b) $(x - 4)^2 + (y - 8)^2 = 16$; (c) $(x - 2)^2 + (y - 25/8)^2 = (25/8)^2$.
- (a) $(x - 2)^2 + (y + 1)^2 = 25$, a circle; (b) $x = 1$, the perpendicular bisector; (c) $(x - 2)^2 + y^2 = 4$, an Apollonius circle.
- The radius vector (x_1, y_1) is normal to the tangent. Thus $(x_1, y_1) \cdot (x - x_1, y - y_1) = 0$, giving $xx_1 + yy_1 = x_1^2 + y_1^2 = r^2$.
- Centre $(10, 6)$, radius 6, diameter 12; northern tangent $y = 12$; path area $\pi(8^2 - 6^2) = 28\pi \text{ m}^2$.
- (a) Vertex $(0, 0)$, focus $(0, 1)$, directrix $y = -1$, axis $x = 0$, upwards. (b) $(3, -1)$, $(3, -5/4)$, $y = -3/4$, axis $x = 3$, downwards. (c) $(-1, 2)$, $(1, 2)$, $x = -3$, axis $y = 2$, right. (d) $(-1, -1)$, $(-3/4, -1)$, $x = -5/4$, axis $y = -1$, right.
- (a) $(x + 4)^2 = 4(y + 2)$, vertex $(-4, -2)$, $a = 1$; (b) $(x - 3)^2 = 8(y - 2)$, $(3, 2)$, $a = 2$; (c) $(y - 2)^2 = -8(x + 1)$, $(-1, 2)$, $a = -2$; (d) $(x + 3)^2 = -4(y - 1)$, $(-3, 1)$, $a = -1$.
- (a) $x^2 = 12y$; (b) $(y + 1)^2 = 12(x - 2)$; (c) $(x + 2)^2 = -12(y - 4)$; (d) $(y - 2)^2 = 8(x - 1)$.
- (a) $(y - 2)^2 = -4(x + 1)$, focus $(-2, 2)$; (b) $(x - 2)^2 = -\frac{4}{3}(y - 2)$, focus $(2, 5/3)$; (c) $x^2 = 4(y + 2)$, focus $(0, -1)$.
- (a) Focus $(1, 1)$; endpoints $(-5, 1)$, $(7, 1)$; length 12. (b) Focus $(-2, -3)$; endpoints $(-2, 5)$, $(-2, -11)$; length 16. (c) Focus $(5, 0)$; endpoints $(5, \pm 10)$; length 20.
- (a) $(2 \pm 2\sqrt{3}, 4 \pm 2\sqrt{3})$ with matching signs; (b) $(2, 1 \pm 2\sqrt{6})$; (c) $(-3, 3)$, $(5, 3)$.
- A point on the parabola is at focal distance $y - k + a \geq a$. Equality occurs only at the vertex (h, k) . Thus $r = a$, with contact at the vertex.
- The latus endpoints are $(3 + a, 2 \pm 2a)$, so its circle has centre $(3 + a, 2)$ and radius $2|a|$. Its distance from the directrix $x = 3 - a$ is also $2|a|$; the contact point is $(3 - a, 2)$.
- (a) $x^2 = 8y$; (b) $y^2 = 8(x - 1)$; (c) $(y - 1)^2 = -12(x - 1)$.
- (a) $p = 3$; (b) $p = 4$; (c) $p = 4$.
- $x^2 = 8y$: upwards; $y^2 = -12x$: left; $(x - 2)^2 = -4(y + 1)$: downwards; $(y + 3)^2 = 16(x - 1)$: right.

32. Two intersections for $k > 0$, one at the vertex for $k = 0$, and none for $k < 0$.
33. Translate right 2 and down 1. The focus moves from $(0, 1)$ to $(2, 0)$ and the directrix from $y = -1$ to $y = -2$.
34. $(\pm 2, 1)$. Substitution also gives $y = -5$, which is rejected because $x^2 = 4y$ requires $y \geq 0$.
35. Endpoints $(\pm 2\sqrt{ak}, k)$, so the length is $4\sqrt{ak}$. For $a = 1$ and length 8, $k = 4$.
36. Since $4|a| = 12$, $a = \pm 3$. The equations are $(x - 3)^2 = 12(y + 2)$ and $(x - 3)^2 = -12(y + 2)$, with foci $(3, 1)$ and $(3, -5)$.
37. (a) Centre $(0, 0)$, horizontal; vertices $(\pm 5, 0)$, co-vertices $(0, \pm 3)$, foci $(\pm 4, 0)$. (b) $(2, -1)$, vertical; vertices $(2, 4), (2, -6)$, co-vertices $(-2, -1), (6, -1)$, foci $(2, 2), (2, -4)$. (c) $(-3, 2)$, horizontal; vertices $(-9, 2), (3, 2)$, co-vertices $(-3, 2 \pm 2\sqrt{5})$, foci $(-7, 2), (1, 2)$. (d) $(1, 4)$, vertical; vertices $(1, 0), (1, 8)$, co-vertices $(-2, 4), (4, 4)$, foci $(1, 4 \pm \sqrt{7})$.
38. (a) $(x - 3)^2/4 + y^2/9 = 1$; foci $(3, \pm\sqrt{5})$. (b) $(x - 3)^2/9 + (y + 2)^2/16 = 1$; foci $(3, -2 \pm \sqrt{7})$. (c) $(x + 2)^2/9 + (y - 3)^2/4 = 1$; foci $(-2 \pm \sqrt{5}, 3)$.
39. (a) $(x - 2)^2/25 + (y + 2)^2/21 = 1$; (b) $(x - 1)^2/10 + (y - 2)^2 = 1$; (c) $(x - 1)^2/9 + (y - 2)^2 = 1$; (d) $(x - 4)^2/5 + (y - 6)^2/9 = 1$.
40. (a) Centre $(2, 1)$, $a = 4$, $c = 3$, $b^2 = 7$: $(x - 2)^2/16 + (y - 1)^2/7 = 1$. (b) $x^2/9 + y^2/25 = 1$.
41. (a) $(x + 1)^2/25 + (y - 2)^2/9 = 1$; (b) $(x - 3)^2/16 + (y - 1)^2/36 = 1$.
42. (a) $x^2/4 + y^2/9 = 1$, foci $(0, \pm\sqrt{5})$; (b) $x^2/9 + y^2/16 = 1$, foci $(0, \pm\sqrt{7})$.
43. (a) At each focus $(\pm 4, 0)$ the latus endpoints are $(\pm 4, \pm 9/5)$; length $18/5$. (b) At each focus $(2, 2)$ and $(2, -4)$ the endpoints have $x = 2 \pm 16/5$; length $32/5$.
44. $(4, 0)$ and $(0, 3)$ are on; $(2, 2)$ and $(0, 0)$ are inside; $(5, 0)$ is outside.
45. (a) $(\pm 5, 0)$; (b) $(3, \pm 12/5)$; (c) $(\pm 5/\sqrt{2}, \pm 3/\sqrt{2})$ with matching signs.
46. At a vertex the two focal distances are $a - c$ and $a + c$, whose sum is $2a$. Applying Pythagoras to a co-vertex and a focus gives $a^2 = b^2 + c^2$, hence $c^2 = a^2 - b^2$.
47. (a) Axes 12, 8; area 24π . (b) Axes 10, 6; area 15π . (c) Axes 6, 4; area 6π .
48. At $(0, 3)$ the distances are 5 and 5. At $(3, 12/5)$ they are $13/5$ and $37/5$. Both sums equal $10 = 2a$.
49. (a) $k = 25/4$; (b) $k = 25/4$.
50. The rectangle areas corresponding to Question 47 are (a) 96; (b) 60; (c) 24 square units.
51. Combining the equations gives $y = 0$, then $x = \pm 5$. Thus the only contacts are $(\pm 5, 0)$, and both curves have vertical tangents there.
52. $(x - 2)^2/36 + (y + 1)^2/16 = 1$; vertices $(-4, -1), (8, -1)$; co-vertices $(2, -5), (2, 3)$; foci $(2 \pm 2\sqrt{5}, -1)$.
53. (a) Centre $(0, 0)$, horizontal; vertices $(\pm 4, 0)$, foci $(\pm\sqrt{41}, 0)$, asymptotes $y = \pm 5x/4$. (b) $(-2, 1)$, horizontal; vertices $(-5, 1), (1, 1)$, foci $(-2 \pm \sqrt{34}, 1)$, asymptotes $y - 1 = \pm \frac{5}{3}(x + 2)$. (c) $(-2, 0)$, vertical; vertices $(-2, \pm 6)$, foci $(-2, \pm 3\sqrt{5})$, asymptotes $y = \pm 2(x + 2)$. (d) $(1, 3)$, vertical; vertices $(1, -1), (1, 7)$, foci $(1, -3), (1, 9)$, asymptotes $y - 3 = \pm \frac{2}{\sqrt{5}}(x - 1)$.
54. (a) $y^2/4 - (x - 4)^2/25 = 1$, centre $(4, 0)$, vertical. (b) $(y + 1)^2/4 - (x + 2)^2/(1/4) = 1$, centre $(-2, -1)$, vertical. (c) $(x + 1)^2/4 - (y - 2)^2/9 = 1$, centre $(-1, 2)$, horizontal.
55. (a) $(x - 1)^2/9 - (y - 2)^2/16 = 1$; (b) $(y - 1)^2/16 - (x + 2)^2/9 = 1$; (c) $x^2/9 - y^2/36 = 1$.
56. (a) $x^2/16 - y^2/9 = 1$; (b) $(y + 1)^2/9 - (x - 2)^2/(9/4) = 1$.
57. (a) Foci $(\pm 5, 0)$; endpoints $(\pm 5, \pm 9/4)$; length $9/2$. (b) Foci $(0, \pm 6)$; endpoints $(\pm 11/5, \pm 6)$; length $22/5$.
58. Standard form is $(y + 1)^2/4 - (x + 2)^2/(1/4) = 1$. Thus $2b = 1$ and the asymptotes are $y + 1 = \pm 4(x + 2)$.
59. (a) $(\pm 3, 0)$; (b) $(5, \pm 8/3)$; (c) no real intersections.
60. $c/a = 5/3$, so $b^2 = c^2 - a^2 = 16a^2/9$ and $b = 4a/3$. Taking $a = 3, b = 4$ gives $(y - 2)^2/9 - (x - 1)^2/16 = 1$.
61. (a) $b^2 = 9$; (b) $b^2 = 16$.

62. At $(3, 0)$ the focal distances are 2 and 8. At $(5, 16/3)$ they are $16/3$ and $34/3$. Both absolute differences equal $6 = 2a$.
63. (a) $a = 5$; (b) $b = 8/3$.
64. The auxiliary rectangle has corners $(-3, -4), (-3, 2), (7, -4), (7, 2)$. Its diagonals give $y + 1 = \pm \frac{3}{5}(x - 2)$.
65. Rearranging gives $y/x = \pm \frac{b}{a} \sqrt{1 - a^2/x^2}$. As $|x| \rightarrow \infty$, this tends to $\pm b/a$, giving the stated asymptotes.
66. Here $c = 5$ and $2a = 6$, so $a = 3$, $b = 4$, and $x^2/9 - y^2/16 = 1$.
67. $x^2/16 - y^2/16 = 1$; foci $(\pm 4\sqrt{2}, 0)$; asymptotes $y = \pm x$.
68. (a) horizontal hyperbola; (b) vertical hyperbola; (c) the intersecting lines $y = \pm 2x/3$; (d) no real curve.
69. (a) $y = -x^2 + 5x - 1$, a downward parabola; (b) $y = 2(x - 1)^2 - 3$; (c) $(x + 1)^2 = 4(y - 4)$.
70. (a) $x^2 + y^2 = 9$; (b) $(x - 2)^2 + (y + 1)^2 = 16$; (c) $(x + 3)^2 + (y - 2)^2 = 25$.
71. (a) $x^2/9 + y^2/16 = 1$; (b) $(x - 2)^2 + (y + 5)^2/16 = 1$; (c) $(x + 1)^2/25 + (y - 3)^2/4 = 1$.
72. (a) $(y + 4)^2/9 - (x - 1)^2/4 = 1$; (b) $x^2/16 - y^2/9 = 1$; (c) $(y - 1)^2/25 - (x + 2)^2/4 = 1$. In every case the parameter excludes $\pi/2 + n\pi$.
73. (a) $y = x^2 - 2$ with $|x| \geq 2$; (b) $y = x^2 + 2$, with every real x attained.
74. (a) $(5, -1), (2, 2), (-1, -1), (2, -4)$. (b) $(2\sqrt{3}, 1), (0, 2), (-4, 0)$.
75. (a) $t = \tan^{-1}(4/3) + 2n\pi$; (b) $t = \pi/4 + 2n\pi$; $s = 2$.
76. (a) Full unit circle, anticlockwise. (b) Full unit circle, clockwise. (c) Upper half of the ellipse from $(3, 0)$ to $(-3, 0)$, anticlockwise.
77. (a) $t = \pi/4, 5\pi/4$ modulo 2π ; points $(2\sqrt{2}, 2\sqrt{2})$ and $(-2\sqrt{2}, -2\sqrt{2})$. (b) $t = 0, 2$; points $(0, 0), (2, 4)$. (c) $t = \pi/2, 3\pi/2$; points $(0, \pm 3)$.
78. One choice is (a) $x = 2 + 5 \cos t$, $y = -1 + 5 \sin t$; (b) $x = -1 + 4 \cos t$, $y = 3 + 2 \sin t$; (c) $x = 2 + 4t$, $y = -1 + 2t^2$; (d) $x = 3 \sec t$, $y = 4 \tan t$.
79. The unit circle is traced anticlockwise twice.
80. Positions: $(50, 0), (0, 30), (-50, 0), (0, -30)$. The first return is at $t = 60$ s.
81. $x = 18 \cos(\pi t/20)$, $y = 20 + 18 \sin(\pi t/20)$. Heights: 38, 20, and 2 m.
82. Area $= \pi(50^2 - 45^2) = 475\pi$ m²; midline lap length $= 2\pi(47.5) = 95\pi$ m.
83. At depths 2 and 4 m the surface levels are $y = -3$ and $y = -1$. The widths are respectively 8 m and $4\sqrt{6}$ m.
84. Each route has length $\sqrt{6^2 - 2^2} = 4\sqrt{2}$ km. The contact points are $(14/3, 3 \pm 4\sqrt{2}/3)$.
85. Entry and exit points are $(2 \pm 5\sqrt{2}, -1 \pm 5\sqrt{2})$ with matching signs. The road length inside is 20 units.
86. With $x^2 = 4ay$, the rim point $(2.4, 0.6)$ gives $a = 2.4$. The focus is $(0, 2.4)$ m, so the receiver is 2.4 m above the vertex.
87. Ground intersections are $x = \pm 12$, so the width is 24 m. At the vehicle edges $x = \pm 4$, the arch height is 8 m; allowing the clearance gives maximum vehicle height 7.5 m.
88. $x^2 = 12y$ has $a = 3$, so the focus is $(0, 3)$. At $y = 4$, $x = \pm 4\sqrt{3}$, giving width $8\sqrt{3}$ m.
89. $x^2/3600 + y^2/1600 = 1$; foci $(\pm 20\sqrt{5}, 0)$ m; area 2400π m².
90. $a - c = 40$, $a + c = 160$, so $a = 100$, $c = 60$, $b = 80$. One equation is $x^2/10000 + y^2/6400 = 1$, with the star at $(60, 0)$; distances are in millions of kilometres.
91. $a = 15$, $b = 9$, $c = 12$. The foci are $(\pm 12, 0)$ m and are 24 m apart.
92. $a = 13$, $b = 5$, $c = 12$. The foci are $(\pm 12, 0)$ and the constant total path length is $2a = 26$ units.
93. Minimum width = 12 m. At $y = 80$, $x = \pm 6\sqrt{2}$, so the width is $12\sqrt{2}$ m. The asymptotes are $y - 50 = \pm 5x$.
94. $c = 6$, $a = 4$, $b^2 = 20$, so $x^2/16 - y^2/20 = 1$. The right branch contains points closer to $(6, 0)$.
95. Vertices $(0, \pm 4)$; foci $(0, \pm 5)$; asymptotes $y = \pm 4x/3$. The requested distance is 1 unit.
96. $x = 100 + 30 \cos(\pi t/6)$, $y = 50 + 30 \sin(\pi t/6)$. Positions after 3, 6, 9 minutes are $(100, 80), (70, 50), (100, 20)$.
97. Period 90 min; foci $(\pm 2000\sqrt{6}, 0)$ km. Positions are $(7000, 0), (0, 5000), (-7000, 0)$.

98. $t = x/6$ gives $y = 2x - x^2/12$. The maximum is more clearance.
 (12, 12) and the stream meets the ground at (0, 0) and (24, 0). 100. (a) The site is an ellipse with axes 20 and 16;
 the fountain is a circle of diameter 8. (b) The
 99. At $x = \pm 3$, the parabolic clearance is road intersections are $(\pm 5\sqrt{7}/2, 6)$. (c) No:
 $55/8 = 6.875$ m and the semicircular clearance every fountain point has $|y| \leq 4$. (d)
 is $\sqrt{55} \approx 7.416$ m. The semicircular design gives $x = 10 \cos t, y = 8 \sin t$.

Chapter 9: Probability and Statistics

These answers correspond to Section 9.15.

1. $6(9)(5) = 270$ meals.
2. (a) $10^4 = 10,000$; (b) ${}_{10}P_4 = 5040$; (c) $9(9)(8)(7) = 4536$.
3. (a) $11!/(2!2!) = 9,979,200$; (b) $13!/(2!2!2!) = 778,377,600$.
4. Treat the jazz discs as one block:
 $9!3! = 2,177,280$.
5. (a) $5! = 120$; (b) $9!/2 = 181,440$.
6. There are $\binom{10}{5} = 252$ committees. (a) $\binom{6}{3}\binom{4}{2}/252 = 10/21$. (b) $[\binom{6}{3}\binom{4}{2} + \binom{6}{4}\binom{4}{1} + \binom{6}{5}]/252 = 31/42$.
7. $(125 + 35 - 17)/250 = 143/250 = 0.572$.
8. (a) $3/36 = 1/12$; (b) $6/36 = 1/6$; (c) $9/36 = 1/4$.
9. The intersection contains $20 + 21 - 30 = 11$ students. (a) $11/30$; (b) $9/30 = 3/10$; (c) $11/21$.
10. (a) $(5/12)(7/11) = 35/132$; (b) $[\binom{5}{2} + \binom{7}{2}]/\binom{12}{2} = 31/66$; (c) $1 - \binom{7}{2}/\binom{12}{2} = 15/22$.
11. (a) $P(L) = (1/4)(2/3) + (3/4)(1/3) = 5/12$; (b) $P(B | L) = (1/4)/(5/12) = 3/5$.
12. (a) $0.7(0.8) + 0.3(0.3) = 0.65$; (b) $0.56/0.65 = 56/65 \approx 0.8615$.
13. $k = 0.15$, $E(X) = 1.56$, and $\text{Var}(X) = 1.4064$. Thus $E(Y) = -3.88$ and $\text{Var}(Y) = 5.6256$.
14. $E(G) = 6(1/6) - 1(5/6) = 1/6$ ringgit. The positive expectation makes the game favourable in the long run.
15. The density integrates to 1.
 $P(2 \leq X \leq 3.5) = 35/96 \approx 0.3646$,
 $P(X < 3) = 1/2$, and
 $P(X > 5) = 2/27 \approx 0.0741$.
16. $k = 3$ and
$$F(t) = \begin{cases} 0, & t < 1, \\ 1 - t^{-3}, & t \geq 1. \end{cases}$$

Also $E(T) = 3/2$ and
 $P(2.5 \leq T \leq 3.5) \approx 0.0407$.
17. (a) 0.2182; (b) 0.4114; (c) 0.9885; (d) mean 4, standard deviation $\sqrt{3.2} \approx 1.7889$.
18. $X \sim B(500, 0.04)$; mean 20 and standard deviation $\sqrt{19.2} \approx 4.3818$.
19. The one-minute mean is 3. (a) 0.9161; (b) 0.5768; (c) 0.5380.
20. (a) For mean 4, $P(X \leq 1) = 0.0916$. (b) For mean 8, $P(X = 3) = 0.0286$. (c) For mean 12, $P(X = 0) = e^{-12} \approx 0.00000614$.
21. With $\lambda = 3$, the approximation gives $P(X \geq 4) = 0.3528$.
22. (a) 0.0075; (b) 0.0094; (c) 0.8186; (d) $a \approx 1.03$.
23. The standard deviation is 10. (a) 0.9772; (b) 0.4332; (c) 0.6827.
24. Using $z_{0.30} \approx -0.5244$ and $z_{0.85} \approx 1.0364$ gives $\mu \approx 51.72$ and $\sigma \approx 12.81$.
25. With mean 24 and standard deviation $\sqrt{16.8}$, (a) 0.0050; (b) 0.9898; (c) 0.0331.
26. (a) $5/22 \approx 0.2273$; (b) $3/11 \approx 0.2727$; (c) $1/2$. Mean 55 seconds and standard deviation $110/\sqrt{12} \approx 31.7543$ seconds.
27. (a) 0.4866; (b) 0.1599; (c) 0.2903.
28. (a) Population: all paint-thickness points on that car; sample: the 30 selected points. (b) Population: all classrooms in the school; sample: the seven selected classrooms. (c) Population: all seniors; sample: the 100 surveyed seniors.
29. Simple random sampling is broadly representative but needs a full frame; systematic sampling is convenient but can follow periodic patterns; stratified sampling represents subgroups but needs correct strata; cluster sampling is economical but clusters may differ; convenience sampling is quick but biased; snowball sampling reaches hidden groups but depends on participants' networks.
30. $E(\bar{X}) = 72$, $\text{SE}(\bar{X}) = 18/9 = 2$, and $P(69 < \bar{X} < 75) = P(-1.5 < Z < 1.5) \approx 0.8664$.
31. Since the population standard deviation is unknown, use a Student t interval with 199 degrees of freedom. With $t_{0.995, 199} \approx 2.6008$, the 99% interval is $101.0 \pm 2.6008(2.78/\sqrt{200})$, approximately (100.489, 101.511) g.
32. With $t_{0.95, 11} \approx 1.796$, the interval is $2.71 \pm 1.796(0.51/\sqrt{12})$, approximately (2.446, 2.974).

33. $8439 \pm 1.645(100/5)$, or approximately (8406.10, 8471.90) lb.
34. Since the population standard deviation is unknown, $t = (407 - 400)/(38/\sqrt{172}) \approx 2.416$ with 171 degrees of freedom. The one-tailed p-value is approximately 0.0084. Reject H_0 ; the data support a mean exceeding RM 400.
35. $t = 1.667$ with 24 degrees of freedom and two-tailed $p \approx 0.1086$. Do not reject H_0 at 1%.
36. $t = \sqrt{10} \approx 3.162$ with 9 degrees of freedom and $p \approx 0.0058$. Reject H_0 ; the data support production above 250 units per hour.
37. $z \approx -6.642$, with two-tailed $p \approx 3.1 \times 10^{-11}$. The critical-value and p-value methods both reject H_0 . The 95% interval is approximately $(98.082, 98.318)^\circ\text{F}$, which excludes 98.6°F .
38. (a) $11!/(4!4!2!) = 34,650$; (b) $10!/(3!3!2!) = 50,400$.
39. (a) ${}_{12}P_3 = 1320$; (b) $\binom{12}{4} = 495$; (c) $\binom{5}{2}\binom{7}{2} + \binom{5}{3}\binom{7}{1} + \binom{5}{4} = 285$.
40. (a) $3!4!3!2! = 1728$; (b) $6!4! = 17,280$; (c) $9! - 2(8!) = 282,240$.
41. (a) $7! = 5040$; (b) $2(6!) = 1440$; (c) $5040 - 1440 = 3600$.
42. (a) $\binom{13}{2}\binom{39}{3} = 712,842$; (b) $\binom{52}{5} - \binom{48}{5} = 886,656$; (c) $13\binom{4}{3}(12)\binom{4}{2} = 3744$.
43. There are ten ordered outcomes with sum at least 9. (a) $3/10$; (b) $7/10$; (c) $1/5$.
44. The denominator is $\binom{15}{3} = 455$. (a) $120/455 = 24/91$; (b) $\binom{11}{3}/455 = 33/91$; (c) $[\binom{6}{2}\binom{9}{1} + \binom{6}{3}]/455 = 31/91$.
45. $P(A \cup B) = 0.70$, $P(A | B) = 0.625$, and $P(B | A) = 5/11 \approx 0.4545$. They are not independent because $0.25 \neq 0.55(0.40)$.
46. (a) $12/120 = 0.10$; (b) $18/60 = 0.30$; (c) $42/90 = 7/15$. The variables are not independent because $P(\text{late} | \text{car}) = 0.30 \neq 0.25$.
47. $P(D) = 0.60(0.02) + 0.30(0.04) + 0.10(0.07) = 0.031$. Given a defect, the probabilities for A, B, C are respectively $12/31$, $12/31$, and $7/31$.
48. (a) $52 + 47 + 40 - 20 - 18 - 15 + 8 = 94$; (b) $(20 - 8) + (18 - 8) + (15 - 8) = 29$; (c) 6.
49. The triangular area is 18 and the square area is 100, so $P(R) = 0.18$. Within $X \leq 4$, the relevant areas are 16 and 40, giving conditional probability 0.40.
50. (a) $26(36^7) = 2,037,468,266,496$; (b) $\binom{7}{2}10^226^6 = 648,723,129,600$; (c) $26(36^7 - 26^7) = 1,828,641,201,920$.
51.
$$F(x) = \begin{cases} 0, & x < -1, \\ 0.10, & -1 \leq x < 0, \\ 0.40, & 0 \leq x < 1, \\ 0.80, & 1 \leq x < 2, \\ 1, & x \geq 2. \end{cases}$$
- $P(-1 < X \leq 1) = 0.70$; a median is 1; the mode is 1.
52. $P(X = 0, 1, 2, 3) = (1, 3, 3, 1)/8$; $E(X) = 3/2$ and $\text{Var}(X) = 3/4$.
53. (a) $12(0.10) + 4(0.25) = \text{RM } 2.20$; (b) RM 2.20; (c) RM 0.80 per game.
54. $k = 3/2$ and $F(x) = \frac{3}{2}x - \frac{1}{2}x^3$ for $0 \leq x \leq 1$. The requested probability is 0.6480 and $E(X) = 3/8$.
55. $k = 1/2$ and $F(x) = x^2/4$ for $0 \leq x \leq 2$. The median is $\sqrt{2}$, $E(X) = 4/3$, and $\text{Var}(X) = 2/9$.
56. Since $np = 12$ and $np(1 - p) = 8.4$, $p = 0.30$ and $n = 40$.
57. (a) 0.2311; (b) 0.2528; (c) 0.7086; (d) 0.9862.
58. $1 - 0.92^n \geq 0.95$ gives $n \geq \ln(0.05)/\ln(0.92) \approx 35.93$; hence $n = 36$.
59. The means are 1.2, 1.2, and 7.2, respectively. (a) 0.3012; (b) 0.2169; (c) 0.8445.
60. $P(X = 2 | X \leq 3) = 9/26 \approx 0.3462$ and $P(X \geq 4 | X \geq 2) \approx 0.4405$.
61. With $\lambda = 3$, (a) 0.0498; (b) 0.4232; (c) 0.1847.
62. (a) 1.960; (b) 1.645; (c) 2.326.
63. The corresponding standard scores are -1 and 2 , giving $\mu = 50$ and $\sigma = 10$.
64. (a) 0.9545; (b) about 46 packets; (c) $500 - 1.645(4) \approx 493.42$ g.
65. Mean 55, standard deviation $\sqrt{24.75}$, with limits 49.5 and 60.5. The approximation is 0.7311.
66. (a) systematic; (b) stratified; (c) cluster; (d) voluntary-response or convenience sampling.
67. (a) It excludes non-users; sample randomly from the whole student body. (b) It excludes people

- without suitable access; offer offline response modes. (c) It underrepresents students who miss early lectures; sample across times and programmes.
68. $E(\bar{X}) = 120$, $SE = 5$, and $P(115 < \bar{X} < 128) = P(-1 < Z < 1.6) \approx 0.7865$.
69. (a) Standard errors 4 and 2. (b) The probabilities are approximately 0.5467 and 0.8664.
70. The standard error is $12/8 = 1.5$, so $P(\bar{X} > 53) = P(Z > 2) \approx 0.0228$.
71. $42.5 \pm 1.96(8/8)$, giving (40.54, 44.46).
72. $\bar{x} = 12.05$, $s \approx 0.2449$, and $t_{0.975,7} \approx 2.365$. The interval is approximately (11.845, 12.255).
73. $n \geq [1.96(15)/3]^2 = 96.04$, so the minimum is 97.
74. The intervals are approximately (71.710, 78.290), (71.080, 78.920), and (69.848, 80.152). Greater confidence produces a wider interval.
75. (a) $H_0 : \mu = 50$, $H_1 : \mu < 50$; (b) $H_0 : \mu = 120$, $H_1 : \mu \neq 120$; (c) $H_0 : \mu = 8$, $H_1 : \mu > 8$.
76. Type I: conclude that the process underfills when its mean is 500 mL. Type II: fail to detect a true mean below 500 mL. Power is the probability of correctly detecting a specified underfilling mean.
77. $z = 2.40$ and $p \approx 0.0082$. Reject H_0 at 1%; there is evidence that $\mu > 100$.
78. $t \approx 1.571$ with 15 degrees of freedom and $p \approx 0.1369$. Do not reject H_0 .
79. Reject at 5% because $0.032 < 0.05$; do not reject at 1% because $0.032 > 0.01$. The smaller significance level demands stronger evidence.
80. $z = -1.60$, $p \approx 0.1096$, and the 95% interval is (18.22, 20.18). All three approaches give the decision not to reject H_0 .
81. (a) 0.0512; (b) 0.6723; (c) 0.9421.
82. (a) $\binom{8}{2}\binom{6}{2}\binom{4}{2} = 2520$; (b) $2(2520) = 5040$.
83. (a) $e^{-1.8} \approx 0.1653$; (b) with mean 3.6, $P(X \geq 3) \approx 0.6973$; (c) with mean 12.6, $P(X = 5) \approx 0.0089$.
84. The ten-minute mean is 2. (a) 0.6767; (b) 0.1429; (c) 0.3158.
85. (a) $1 - e^{-20/30} \approx 0.4866$; (b) $e^{-2} \approx 0.1353$; (c) by memorylessness, $e^{-1} \approx 0.3679$.
86. The parallel stage works with probability $1 - (0.10)(0.15) = 0.985$. (a) The system reliability is $0.985(0.95) = 0.9358$. (b) The failure probability is $1 - 0.93575 = 0.0643$. (c) After the upgrade, the reliability is $0.985(0.98) = 0.9653$.
87. (a) $\lambda = \ln 2/18 \approx 0.03851$ per month; (b) $e^{-24\lambda} \approx 0.3969$; (c) $-\ln(0.9)/\lambda \approx 2.736$ months.
88. (a) $1/3$; (b) $6/15 = 0.4$; mean 7.5 minutes and standard deviation $15/\sqrt{12} \approx 4.330$ minutes.
89. (a) 0.0455; (b) about 9 bottles; (c) $1 + 2.326(0.015) \approx 1.0349$ L.
90. (a) $68 + 1.282(12) \approx 83.38$; (b) 0.0098; (c) approximately (44.48, 91.52).
91. $P(+) = 0.02(0.95) + 0.98(0.10) = 0.117$. Thus $P(D | +) \approx 0.1624$ and $P(D' | -) \approx 0.9989$.
92. (a) Expected claim $= 1000(0.10) + 5000(0.05) = \text{RM } 350$; (b) RM 470; (c) RM 130 expected profit.
93. (a) $X \sim B(105, 0.92)$; (b) $E(X) = 96.6$; (c) $P(X > 100) \approx 0.0706$.
94. (a) (6.812, 7.988) mg L^{-1} . (b) $z = -2.00$, $p \approx 0.0228$; reject H_0 . (c) The sample supports a true mean below 8 mg L^{-1} .
95. (a) With $t_{0.975,11} \approx 2.201$, the interval is (490.56, 513.44) MPa. (b) $t \approx 0.385$, $p \approx 0.7076$; do not reject H_0 .
96. $t = -1.75$ with 8 degrees of freedom and one-tailed $p \approx 0.0591$. Do not reject H_0 at 5%. The observations should be independent and the population approximately normal.
97. (a) binomial; (b) Poisson; (c) exponential; (d) normal; (e) uniform. The choices follow from fixed independent trials, event counts over time, memorylessness, accumulated small effects, and equal likelihood over an interval, respectively.
98. Select 50 students at random from each year group, using separate complete lists and random numbers. Library-only sampling overrepresents students who use the library and may overestimate study time.
99. (a) $0.50(0.04) + 0.30(0.07) + 0.20(0.12) = 0.065$; (b) $0.024/0.065 \approx 0.3692$; (c) $0.935^{10} \approx 0.5106$.
100. (a) Using $t_{0.975,15} \approx 2.131$, the interval is (46.57, 50.83). (b) $t = -1.30$, $p \approx 0.1066$; do not reject H_0 . (c) A Type II error would be failing to conclude that the true mean is below 50 when it actually is below 50.

Chapter 10: Complex Numbers

These answers correspond to Section 10.12.

1. $\mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$, with the irrational numbers in $\mathbb{R} \setminus \mathbb{Q}$. Thus $\mathbb{C}$ contains every displayed number system, while non-real complex numbers lie outside $\mathbb{R}$.
2. (a) $x = (1 \pm 3i)/2$; (b) $x = 1 \pm 2i$.
3. (a) $5 + 5i$: real part 5, imaginary part 5. (b) $3 + 2i$: real part 3, imaginary part 2. (c) $-i$: real part 0, imaginary part -1 .
4. $i^{4n} = 1$. Divide the exponent by 4: remainders 0, 1, 2, 3 give 1, i , -1 , $-i$, respectively. The same remainder rule works for negative exponents after reduction modulo 4.
5. The factors are conjugates, so their product is $(-1)^2 + (\sqrt{2})^2 = 3$. Hence $x = 3$ and $y = 0$.
6. A point on the negative imaginary axis has principal argument $-\pi/2$.
7. (a) $|z| = 6$, $\text{Arg } z = -\pi/3$, $z = 6 \text{cis}(-\pi/3) = 6e^{-\pi i/3}$. (b) $|z| = \sqrt{3}$, $\text{Arg } z = \pi/2$, $z = \sqrt{3} \text{cis}(\pi/2) = \sqrt{3}e^{\pi i/2}$. (c) $|z| = 4$, $\text{Arg } z = 5\pi/6$, $z = 4 \text{cis}(5\pi/6) = 4e^{5\pi i/6}$.
8. (a) $3e^{-\pi i/4}$; (b) $6e^{-5\pi i/6}$; (c) $8e^{2\pi i/3}$.
9. (a) $\frac{3}{2} - \frac{3\sqrt{3}}{2}i$; (b) $-\sqrt{2} - \sqrt{2}i$; (c) $-2\sqrt{3} + 2i$; (d) $-3i$.
10. (a) Product $2\sqrt{2}e^{11\pi i/12}$; quotient $\sqrt{2}e^{-7\pi i/12}$. (b) Product $2\sqrt{2}e^{11\pi i/12}$; quotient $2^{-1/2}e^{-5\pi i/12}$.
11. (a) $2\sqrt{2}$; (b) $-\pi/4$.
12. (a) $2e^{\pi i/2} = 2i$. (b) $32e^{-2\pi i/3} = -16 - 16\sqrt{3}i$.
13. (a) $5\pi/12$. (b) Modulus $243/8$ and principal argument $7\pi/12$.
14. (a) $\cos 2x + i \sin 2x$; (b) $\cos 5x + i \sin 5x$; (c) $\cos 2x - i \sin 2x$.
15. $z_1^9 = -4,194,304 + 4,194,304i$ and $z_2^8 = -32,768 + 32,768\sqrt{3}i$.
16. $z_1 + z_2 = -(\sqrt{3} + 2\sqrt{2}) - (1 + 2\sqrt{2})i$, so

$$|z_1 + z_2| = 2\sqrt{5 + \sqrt{6} + \sqrt{2}}.$$

Comparing the squared modulus with $2^2 + 4^2 + 2(2)(4)\cos(\pi/12)$ gives $\cos(\pi/12) = (\sqrt{6} + \sqrt{2})/4$ and hence the stated value of $\cos(11\pi/12)$.

17. Euler's formula and its conjugate give the first two identities. Replacing x by nx gives (a) and (b). Expansion of $(\cos x + i \sin x)^4$ gives

$$\cos 4x = 8 \cos^4 x - 8 \cos^2 x + 1, \quad \sin 4x = 4 \sin x \cos x (\cos^2 x - \sin^2 x).$$

18. Equating real and imaginary parts of $(\cos \theta + i \sin \theta)^5$ gives

$$\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta,$$

$$\sin 5\theta = 16 \sin^5 \theta - 20 \sin^3 \theta + 5 \sin \theta.$$

19. $2 + 2\sqrt{3}i = 4 \operatorname{cis}(\pi/3)$, so its square roots are $\sqrt{3} + i$ and $-\sqrt{3} - i$.
20. (a) With $\rho = (3\sqrt{2})^{1/3}$, the roots are $\rho \operatorname{cis}(\pi/4 + 2k\pi/3)$, $k = 0, 1, 2$. (b) $2 \operatorname{cis}(-\pi/9 + 2k\pi/3)$.
(c) $3 \operatorname{cis}(-\pi/4 + 2k\pi/3)$.
21. The roots are $e^{k\pi i/5}$, $k = 0, \dots, 9$. Two lie on the real axis, none on the imaginary axis, and two lie strictly in each quadrant.
22. The required roots are $2 \operatorname{cis}(3\pi/10)$ and $2 \operatorname{cis}(7\pi/10)$.
23. (a) $-2 \pm 3i$; (b) $1 \pm 2i$; (c) $-2 \pm 3i$; (d) $\pm 3i$.
24. (a) $x^2 - 4x + 29 = 0$; (b) $x^2 + 2x + 4 = 0$; (c) $2x^2 - 6x + 5 = 0$.
25. (a) $x^3 - 4x^2 + 14x - 20$. (b) $x^4 - 4x^3 + 6x^2 - 4x + 5$.
26. (a) Every real k . (b) $k = \pm 6$. (c) $3 - 2\sqrt{2} < k < 3 + 2\sqrt{2}$.
27. (a) $z = 1, i$; (b) the repeated root $z = -i$; (c) $z = 1, 1 + i$.
28. (a) Sum 3, product $13/2$. (b) The other root is $-2 - 5i$ and the equation is $z^2 + 4z + 29 = 0$.
(c) $-2 \pm 3i$.
29. Over the reals,

$$z^4 + 6z^2 + 25 = (z^2 - 2z + 5)(z^2 + 2z + 5).$$

Its complex roots are $1 \pm 2i$ and $-1 \pm 2i$.
30. The roots are $2 \pm 3i$. They are reflections in the real axis, have common real part 2, are 6 units apart, and both have modulus $\sqrt{13}$.
31. Comparing with $z^2 + 4z + 13 = 0$ gives $a = 3$ and $b = 15$.
32. $r = -3 \pm 4i$. The decay rate is 3 and the angular frequency is 4.
33. (a) $z = 3 + i$; (b) $z = 2 + 2i$; (c) $z = 2 - 2i$.
34. (a) $4 + 3i$; (b) $12 - 5i$; (c) $-3 + 4i$.
35. $z + w = 1 + i$ and $zw = 10 + 11i$. Thus both sides of the first identity equal $1 - i$, and both sides of the second equal $10 - 11i$. In general, conjugation preserves addition and multiplication because it changes only the sign of every imaginary part.
36. The points lie respectively in quadrants I, II, III, and IV.
37. (a) $\sqrt{2} \operatorname{cis}(\pi/4)$. (b) $2 \operatorname{cis}(5\pi/6)$. (c) $2\sqrt{2} \operatorname{cis}(-3\pi/4)$. (d) $4 \operatorname{cis}(-\pi/2)$. Replace **cis** by the corresponding exponential in each case.
38. (a) $-1 + \sqrt{3}i$; (b) $\frac{3\sqrt{3}}{2} - \frac{3}{2}i$; (c) $-2\sqrt{2} + 2\sqrt{2}i$; (d) $-5i$.
39. (a) $2 \operatorname{cis}(-2\pi/3)$; (b) $3 \operatorname{cis}(-3\pi/4)$; (c) $2e^{\pi i/3}$.
40. The principal arguments are $3\pi/4$, $-\pi/2$, and $-2\pi/3$. Add $2k\pi$, $k \in \mathbb{Z}$, to obtain all arguments.

41. $\bar{z} = 5e^{-2\pi i/3}$, $1/z = \frac{1}{5}e^{-2\pi i/3}$, and $-z = 5e^{-\pi i/3}$.
42. The product is $6 \operatorname{cis}(-\pi/12)$, or
- $$\frac{3}{2}(\sqrt{6} + \sqrt{2}) - \frac{3}{2}(\sqrt{6} - \sqrt{2})i.$$
43. $3 \operatorname{cis}(\pi/2) = 3i$.
44. $\frac{3}{2} \operatorname{cis}(-3\pi/4) = -\frac{3\sqrt{2}}{4}(1 + i)$.
45. $AB = 5$ and the midpoint is $-1 + \frac{7}{2}i$.
46. $P = (A + 2B)/3 = 5 - i$.
47. Area $= \frac{1}{2}|3(4) - 1(1)| = 11/2$.
48. $u\bar{v} = (1 + 2i)(4 + 2i) = 10i$, whose real part is zero; the vectors are perpendicular.
49. $2e^{k\pi i/3}$ for $k = 0, 1, \dots, 5$.
50. u^n is real when $n \equiv 0 \pmod{6}$ and purely imaginary when $n \equiv 3 \pmod{6}$. The least positive exponent in the final part is 5.
51. Add and subtract the equations $e^{ix} = \cos x + i \sin x$ and $e^{-ix} = \cos x - i \sin x$.
52. $\cos x = (e^{ix} + e^{-ix})/2$ and $\sin x = (e^{ix} - e^{-ix})/(2i)$. At $x = \pi/3$, the values are $1/2$ and $\sqrt{3}/2$.
53. (a) 0; (b) $e^{\pi i/6} = \frac{\sqrt{3}}{2} + \frac{1}{2}i$; (c) -1 .
54. For $k \in \mathbb{Z}$: (a) $x = 2k\pi$; (b) $x = (2k + 1)\pi$; (c) $x = \pi/2 + 2k\pi$; (d) $x = -\pi/2 + 2k\pi$.
55. (a) $\frac{e^2}{2}(1 + \sqrt{3}i)$; (b) $-e^{-1}$; (c) $-\frac{3\sqrt{3}}{2} + \frac{3}{2}i$.
56. $\cos(2i) = \cosh 2 = (e^2 + e^{-2})/2$ and $\sin(-i) = -i \sinh 1$.
57. Substitute iy into $\cos z = (e^{iz} + e^{-iz})/2$ and $\sin z = (e^{iz} - e^{-iz})/(2i)$.
58. $\operatorname{Log}(1 + i) = \ln \sqrt{2} + i\pi/4$; $\operatorname{Log}(-4) = \ln 4 + i\pi$; $\operatorname{Log}(-i) = -i\pi/2$.
59. (a) $z = 2k\pi i$; (b) $z = (2k + 1)\pi i$; (c) $z = \ln 2 + i(\pi/2 + 2k\pi)$, where $k \in \mathbb{Z}$.
60. $\operatorname{Log}((-1)(-1)) = \operatorname{Log}(1) = 0$, whereas $\operatorname{Log}(-1) + \operatorname{Log}(-1) = 2\pi i$.
61. $\cos 3x = 4 \cos^3 x - 3 \cos x$, $\sin 3x = 3 \sin x - 4 \sin^3 x$, $\cos 4x = 8 \cos^4 x - 8 \cos^2 x + 1$, and $\sin 4x = 4 \sin x \cos x (\cos^2 x - \sin^2 x)$.
62. $\cos 15^\circ = \sin 75^\circ = (\sqrt{6} + \sqrt{2})/4$ and $\sin 15^\circ = \cos 75^\circ = (\sqrt{6} - \sqrt{2})/4$.
63. (a) Centre $2 - i$, radius 3. (b) $|z + 2i| = 3$: centre $-2i$, radius 3. (c) Centre $-1 + 3i$, radius $\sqrt{5}$.
64. (a) Closed disk, centre $1 + i$, radius 2. (b) Exterior of the circle centred at -2 , radius 3, boundary omitted. (c) Annulus centred at i , with inner boundary omitted and outer boundary included.

65. The equation is $y = x - 4$. It is the perpendicular bisector of the segment joining $1 + i$ and $5 - 3i$.
66. Squaring gives $(x - 4)^2 + y^2 = 4$, so the centre is 4 and the radius is 2.
67. A ray beginning at $1 - i$ and directed at angle $\pi/3$; the initial point is omitted because the argument there is undefined.
68. A closed angular sector with vertex -2 , opening to the right between directions $-\pi/4$ and $\pi/4$. The vertex is excluded.
69. The part of the vertical strip $-1 \leq x \leq 3$ lying above $y = 2$; the vertical edges are included and the horizontal edge is omitted.
70. The rectangle $[-2, 3] \times (1, 4)$: vertical edges included and horizontal edges excluded.
71. The boundary is $x = -1$ and the required half-plane is $x > -1$.
72. $z = 4 + 3i$ and $z = -4 + 3i$.
73. $z = 5$ and $z = 5 + 2i$.
74. $z = 3 \pm 4i$.
75. The images are -2 , $-1 + i$, $-3 + i$, and $-2 + 2i$. The map enlarges by $\sqrt{2}$, rotates by $\pi/4$, then translates 2 units left.
76. The image is $|w - (3 + 2i)| = 4$.
77. The closed right half of the disk centred at $1 + i$ with radius 2, including the diameter on $x = 1$.
78. (a) $(x - 1)^2 + y^2 = 4$, a circle. (b) $y = x + 2$, a line. (c) $x + y = 2$, a line.
79. The three phasors sum to $120(1 + e^{2\pi i/3} + e^{4\pi i/3}) = 0$, expressing the balance of the three phases.
80. $Z = 8 + 6i = 10 \operatorname{cis}(\tan^{-1}(3/4)) \Omega$ and $I = 100/Z = 8 - 6i = 10 \operatorname{cis}(-\tan^{-1}(3/4))$ amperes.
81. The resultant is $3 + 4i$, with amplitude 5 and phase $\tan^{-1}(4/3)$.
82. $H(\pi/3) = 1/2 + \sqrt{3}i/2 = e^{\pi i/3}$, so the gain is 1 and the phase is $\pi/3$.
83. The scale factor is 2 and the rotation is $\pi/6$. The image is $(1 + 2\sqrt{3}) + (2 - \sqrt{3})i$.
84. The images are $2 - i$, $2 - i + \sqrt{2}(1 + i)$, and $2 - i + \sqrt{2}(-1 + i)$. The map rotates by $\pi/4$ and then translates by $2 - i$.
85. The resultant is $6 + 8i = 10 \operatorname{cis}(\tan^{-1}(4/3))$ km.
86. $z_1 = 1$, $z_2 = 1 + i$, $z_3 = i$, and $z_4 = 0$. The robot traces the vertices of a unit square and returns to its starting point.
87. The vertices are $2 + i$, $-1 + 2i$, $-2 - i$, and $1 - 2i$.
88. With $\zeta = e^{2\pi i/5}$, the phasors are $A, A\zeta, A\zeta^2, A\zeta^3, A\zeta^4$ and their sum is 0.

89. $I = 130/(12 + 5i) = 120/13 - (50/13)i = 10 \operatorname{cis}(-\tan^{-1}(5/12))$ amperes. The power factor is $12/13$.
90. Resonance requires $\omega^2 = 1/(LC) = 1000$, so $\omega = 10\sqrt{10} \operatorname{rad s}^{-1}$.
91. The roots are $-2 \pm 3i$. The decay rate is 2 and the angular frequency is 3.
92. $B = (1 + 2i)e^{i\pi/3} = \frac{1}{2} - \sqrt{3} + i(1 + \frac{\sqrt{3}}{2})$. Both A and B have modulus $\sqrt{5}$.
93. Successively, the images are $3 + 2i$, $-3 + 2i$, and $3 - 2i$.
94. $20 \operatorname{cis}(150^\circ) = -10\sqrt{3} + 10i \operatorname{km}$.
95. The sum is $4\sqrt{3}$, so the amplitude is $4\sqrt{3}$ and the phase is 0. The waves interfere constructively.
96. $w = (3 - 4i)/25$. Thus $|w| = 1/5 = 1/|z|$ and $\operatorname{Arg} w = -\operatorname{Arg} z = -\tan^{-1}(4/3)$.
97. $z_1 = i$, $z_2 = -1 + i$, $z_3 = -i$, and $z_4 = -1 - i$. The observed values are bounded and have entered a two-cycle.
98. The tips are 1 and i , and their distance is $|1 - i| = \sqrt{2}$.
99. Equidistance gives $\operatorname{Re}(z) = 0$. Intersecting with $|z + 3| = 5$ gives $z = \pm 4i$.
100. The equal-distance locus is the imaginary axis. Its intersections with $|z + 4| = 5$ are $z = \pm 3i$. Their directions from A are $\pm \tan^{-1}(3/4)$, which lie between $-\pi/4$ and $\pi/4$; hence both points satisfy all three conditions.

Chapter 11: Hyperbolic Functions

These answers correspond to Section 11.7.

1. $\sinh x = (e^x - e^{-x})/2$, $\cosh x = (e^x + e^{-x})/2$,
 $\tanh x = (e^x - e^{-x})/(e^x + e^{-x})$,
 $\operatorname{sech} x = 2/(e^x + e^{-x})$, $\operatorname{cosech} x = 2/(e^x - e^{-x})$,
and $\coth x = (e^x + e^{-x})/(e^x - e^{-x})$.
2. 0, 1, 0, 1; $\operatorname{cosech} 0$ and $\coth 0$ are undefined.
3. $\sinh x = 4/3$, $\cosh x = 5/3$, $\tanh x = 4/5$,
 $\operatorname{sech} x = 3/5$, $\operatorname{cosech} x = 3/4$, and $\coth x = 5/4$.
4. (a) $13/12$; (b) $5/13$; (c) $12/13$; (d) $12/5$; (e) $13/5$.
5. $\sinh x = -12/5$, $\tanh x = -12/13$,
 $\operatorname{sech} x = 5/13$, $\operatorname{cosech} x = -5/12$, and
 $\coth x = -13/12$.
6. $\sinh x = -3/4$, $\cosh x = 5/4$, $\operatorname{sech} x = 4/5$,
 $\operatorname{cosech} x = -4/3$, and $\coth x = -5/3$.
7. (a) $3/4, 5/4$; (b) $4/5, 3/5$; (c) $8/15, 17/15$.
8. Adding and subtracting the definitions gives the
two stated formulae. Hence the requested
expressions are $2 \cosh 2x$ and $2 \sinh 2x$.
9. (a) 1; (b) $\sinh^2 x$; (c) $\operatorname{sech}^2 x$; (d) $\operatorname{cosech}^2 x$.
10. Substitute $-x$ into the exponential definitions.
The results are respectively odd, even, odd, and
even.
11. (a) Domain and range $\mathbb{R}$, odd, intercept $(0, 0)$,
no horizontal asymptote. (b) Domain $\mathbb{R}$, range
 $[1, \infty)$, even, y -intercept $(0, 1)$, no horizontal
asymptote. (c) Domain $\mathbb{R}$, range $(-1, 1)$, odd,
intercept $(0, 0)$, asymptotes $y = \pm 1$. (d)
Domain $\mathbb{R}$, range $(0, 1]$, even, y -intercept $(0, 1)$,
asymptote $y = 0$.
12. (a) $x = 0$; (b) $x = \ln 2$; (c)
 $x = \ln(2/3) = -\ln(3/2)$.
13. (a) $x = 0$; (b) $x = \pm \ln 2$; (c) no real solution.
14. (a) $x = \frac{1}{2} \ln 3$; (b) $x = -\ln 2$; (c) no real
solution.
15. (a) $x = \pm \ln 2$; (b) $x = \ln(3/2)$; (c) $x = -\ln 2$.
The cosecant and cotangent equations exclude
 $x = 0$.
16. Expanding the two squares from the
exponential definitions leaves $4/4 = 1$.
17. Divide the fundamental identity by $\cosh^2 x$ and
by $\sinh^2 x$, respectively; the second division
requires $x \neq 0$.
18. (a) $\sinh x \cosh y + \cosh x \sinh y$; (b)
 $\sinh x \cosh y - \cosh x \sinh y$; (c)
 $\cosh x \cosh y + \sinh x \sinh y$; (d)
 $\cosh x \cosh y - \sinh x \sinh y$; (e)
 $(\tanh x + \tanh y)/(1 + \tanh x \tanh y)$.
19. (a) $4/3$; (b) $5/3$; (c) $7/24$; (d) $25/24$; (e) $4/5$.
20. $\tanh(a + b) = 5/7$, $\tanh(a - b) = 1/5$, and
 $\tanh 2a = 4/5$.
21. $\sinh 2x = 15/8$, $\cosh 2x = 17/8$, and
 $\tanh 2x = 15/17$.
22. $\sinh x = \sqrt{3}$, $\cosh x = 2$, and $\tanh x = \sqrt{3}/2$.
23. (a) $(\cosh 2x + 1)/2$; (b) $(\cosh 2x - 1)/2$; (c)
 $\cosh 2x + 4$; (d) $\cosh 2x + 6$.
24. Expand the left sides with the addition
formulae. The unwanted terms cancel, leaving
the stated right sides.
25. (a) $\operatorname{sech} x$; (b) $\tanh x$ where defined; (c) e^{2x} ; (d)
1.
26. Let $s = \sinh x$. Then $12s^2 + 7s - 12 = 0$, so
 $\sinh x = 3/4$ or $-4/3$.
27. (a) $x = \sinh^{-1} 1$ or $x = \sinh^{-1}(-1/2)$; (b)
 $x = \pm \cosh^{-1} 2$; (c) $x = \tanh^{-1}(1/3) = \frac{1}{2} \ln 2$.
The algebraic roots $\cosh x = 1/2$ and
 $\tanh x = -1$ are inadmissible.
28. (a) $x = \ln(4 + \sqrt{17})$; (b) $x = \pm \ln(2 + \sqrt{3})$; (c)
 $x = \frac{1}{2} \ln 7$.
29. (a) $x = \pm \frac{1}{2} \cosh^{-1} 5$; (b) $x = \frac{1}{2} \sinh^{-1} 3$; (c)
 $x = \frac{1}{2} \tanh^{-1}(1/2) = \frac{1}{4} \ln 3$.
30. (a) $x = -\ln 2$; (b) $x = -2, 4$; (c) $x = 2$.
31. Substitute $2x + x$ into the addition formulae
and use $\cosh^2 x = 1 + \sinh^2 x$ or
 $\sinh^2 x = \cosh^2 x - 1$.
32. $\sinh 3x = 665/432$, $\cosh 3x = 793/432$, and
 $\cosh 5x = 60073/15552$.
33. (a) $8 \cosh^4 x - 8 \cosh^2 x + 1$; (b)
 $4 \sinh x \cosh x (\cosh^2 x + \sinh^2 x)$; (c)
 $(3 \tanh x + \tanh^3 x)/(1 + 3 \tanh^2 x)$.
34. Add the formulae for $\cosh((n + 1)x)$ and
 $\cosh((n - 1)x)$. The recurrence gives
 $C_4 = 8 \cosh^4 x - 8 \cosh^2 x + 1$.

35. (a) AM–GM gives $(e^x + e^{-x})/2 \geq 1$. (b) $\cosh^2 x - \sinh^2 x = 1$ and $\cosh x > 0$. (c) Divide part (b) by $\cosh x$.
36. An increasing odd curve through $(0, 0)$, with domain and range $\mathbb{R}$ and no asymptotes.
37. An even U-shaped curve with minimum $(0, 1)$, domain $\mathbb{R}$, and range $[1, \infty)$.
38. An increasing odd curve through $(0, 0)$ with asymptotes $y = -1$ and $y = 1$.
39. An even bell-shaped curve with maximum $(0, 1)$ and asymptote $y = 0$.
40. $\operatorname{cosech} x$ has vertical asymptote $x = 0$ and horizontal asymptote $y = 0$; $\coth x$ has vertical asymptote $x = 0$ and horizontal asymptotes $y = \pm 1$. Both exclude $x = 0$.
41. (a) Right 1, vertical stretch 2, up 3. (b) Horizontal compression by factor $1/2$, reflection in the x -axis, up 1. (c) Left 2, down 1. (d) Horizontal stretch by factor 2 and vertical stretch by 3.
42. (a) x -intercept and centre $(2, 0)$; y -intercept $(0, -\sinh 2)$; range $\mathbb{R}$; no asymptotes. (b) turning point $(0, 2)$, range $(-\infty, 2]$, y -intercept 2, and roots $x = \pm \cosh^{-1} 2$. (c) centre and y -intercept $(0, 1)$; x -intercept $(-\frac{1}{4} \ln 2, 0)$; range $(-2, 4)$; asymptotes $y = -2, 4$.
43. (a) $(0, 0)$; (b) $(\pm \cosh^{-1} 2, 2)$; (c) $(\tanh^{-1}(1/2), 1/2)$.
44. (a) none; (b) two; (c) none; (d) one.
45. (a) The fundamental identity gives $X^2 - Y^2 = 1$. (b) $P(0) = (1, 0)$ and $P(\ln 2) = (5/4, 3/4)$. (c) $\cosh$ is even and $\sinh$ is odd.
- 46.

x	-2	-1	0	1	2
$\sinh x$	-3.627	-1.175	0	1.175	3.627
$\cosh x$	3.762	1.543	1	1.543	3.762
$\tanh x$	-0.964	-0.762	0	0.762	0.964

The first and third rows are odd, the second is even, and $\tanh x$ remains bounded while $|\sinh x|$ and $\cosh x$ grow exponentially.

47. As $x \rightarrow \infty$: $\sinh x, \cosh x \rightarrow \infty$, $\tanh x \rightarrow 1$, and $\operatorname{sech} x \rightarrow 0$. As $x \rightarrow -\infty$: $\sinh x \rightarrow -\infty$, $\cosh x \rightarrow \infty$, $\tanh x \rightarrow -1$, and $\operatorname{sech} x \rightarrow 0$.
48. (a) $\operatorname{sech} x$; (b) $\coth x$; (c) $\sinh x$; (d) $\cosh x$.
49. The graphs are reflections in $y = x$. The graph of $y = \sinh^{-1} x$ is therefore the reflection of $y = \sinh x$ in that line.
50. Both functions are even and fail the horizontal-line test on $\mathbb{R}$. Their principal inverses use the restrictions $x \geq 0$.
51. $\sinh^{-1} x$ is the inverse hyperbolic sine, whereas $1/\sinh x = \operatorname{cosech} x$ is a reciprocal.
52. Domains/ranges: $\sinh^{-1} : \mathbb{R} \rightarrow \mathbb{R}$; $\cosh^{-1} : [1, \infty) \rightarrow [0, \infty)$; $\tanh^{-1} : (-1, 1) \rightarrow \mathbb{R}$; $\operatorname{cosech}^{-1} : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R} \setminus \{0\}$; $\operatorname{sech}^{-1} : (0, 1] \rightarrow [0, \infty)$; $\coth^{-1} : (-\infty, -1) \cup (1, \infty) \rightarrow \mathbb{R} \setminus \{0\}$.
53. All four values are 0.
54. (a) $\ln 2$; (b) $\ln(3/2)$; (c) $-\ln 2$.
55. (a) $\ln 2$; (b) $\ln(3/2)$; (c) $\ln(2 + \sqrt{3})$.
56. (a) $\frac{1}{2} \ln 3$; (b) $-\ln 2$; (c) $\ln(3/2)$.
57. (a) $\ln 2$; (b) $-\ln(3/2)$; (c) $\ln(1 + \sqrt{2})$.
58. (a) $\ln 2$; (b) $\ln(3/2)$; (c) $\ln(2 + \sqrt{3})$.
59. (a) $\frac{1}{2} \ln 3$; (b) $-\ln 2$; (c) $\ln(3/2)$.
60. Put $y = \sinh^{-1} x$, write $x = (e^y - e^{-y})/2$, solve the resulting quadratic in e^y , and take the positive root.
61. With $y \geq 0$, solving $x = (e^y + e^{-y})/2$ gives $e^y = x + \sqrt{x^2 - 1}$, hence $\cosh^{-1} x = \ln(x + \sqrt{x^2 - 1})$ for $x \geq 1$.
62. Solving $x = (e^{2y} - 1)/(e^{2y} + 1)$ gives $\tanh^{-1} x = \frac{1}{2} \ln((1+x)/(1-x))$, with $-1 < x < 1$.
63. $\operatorname{cosech}^{-1} x = \sinh^{-1}(1/x)$ for $x \neq 0$; $\operatorname{sech}^{-1} x = \cosh^{-1}(1/x)$ for $0 < x \leq 1$; $\coth^{-1} x = \tanh^{-1}(1/x)$ for $|x| > 1$. Substitution into the preceding logarithmic forms gives the required formulae.
64. Each composition equals x , on the respective domains $\mathbb{R}$, $[1, \infty)$, $(-1, 1)$, and $(0, 1]$.
65. (a) x ; (b) $|x|$; (c) x ; (d) $|x|$.
66. $\cosh u = \sqrt{1 + x^2}$, $\tanh u = x/\sqrt{1 + x^2}$, and $\operatorname{sech} u = 1/\sqrt{1 + x^2}$.
67. $\sinh u = \sqrt{x^2 - 1}$, $\tanh u = \sqrt{x^2 - 1}/x$, and $\operatorname{cosech} u = 1/\sqrt{x^2 - 1}$ for $x > 1$.
68. $\cosh u = 1/\sqrt{1 - x^2}$, $\sinh u = x/\sqrt{1 - x^2}$, and $\operatorname{sech} u = \sqrt{1 - x^2}$.

69. (a) $x = 4/3$; (b) $x = 5/4$; (c) $x = 2/3$.
70. (a) $x = 1$; (b) $x = 1/5$; (c) $x = 6/5$.
71. Apply the logarithmic forms or the fact that $\sinh$ and $\tanh$ are odd and one-to-one.
72. Both formulae equal $|x|$ because the principal inverses return nonnegative values while $\cosh$ and sech are even.
73. Let $u = \sinh^{-1} x$. Then $\tanh u = x/\sqrt{1+x^2}$, so applying $\tanh^{-1}$ proves the result.
74. Let $u = \cosh^{-1} x$. Then $\sinh u = \sqrt{x^2 - 1}$ and $u \geq 0$, giving the identity.
75. Apply $\tanh$ to the sum and use the addition formula. For $u, v \in (-1, 1)$, the resulting fraction also lies in $(-1, 1)$, so the principal-value identity holds.
76. Possible forms are (a) $\sinh^{-1}(3/4)$; (b) $\cosh^{-1} 2$; (c) $\tanh^{-1}(1/2)$; (d) $\cosh^{-1}(3/2)$.
77. (a) $x \geq 2$; (b) $x \in (-\sqrt{3}, -1) \cup (1, \sqrt{3})$; (c) $x \in [0, 1)$; (d) $x \in (-\infty, -1/2) \cup (1/2, \infty)$.
78. (a) $2x\sqrt{1+x^2}$; (b) $1+2x^2$; (c) $2x/(1+x^2)$ for $|x| < 1$; (d) $4x^3 - 3x$ for $x \geq 1$.
79. Since $\sinh^{-1}(3/4) = \ln 2$, the solution is $x = 0$.
80. (a) $x = 5/12$; (b) no real solution because $\cosh^{-1}(3/4)$ is undefined; (c)
$$x = \frac{2/5 - 3/\sqrt{34}}{1 - 6/(5\sqrt{34})}.$$
81. At $x = 0$, $y = 0$; at $x = 12$, $y = 12(\cosh 1 - 1)$. The required positions are $x = \pm 12 \cosh^{-1}(3/2)$.
82. Centre height 8 m; tower height $8 \cosh(5/4)$ m; sag $8[\cosh(5/4) - 1]$ m.
83. $S = 20 \sinh(3/5) \approx 12.733$ m. If $S = 15$, then $L = 20 \sinh^{-1}(3/4) = 20 \ln 2$ m.
84. Central height 15 m; span $10 \cosh^{-1} 4$ m; $h(4) = 20 - 5 \cosh(4/5) \approx 13.313$ m.
85. (a) $(5/4, 3/4)$; (b) $t = \ln 2$; (c) both coordinates increase for $t > 0$, so the point moves along the right branch away from $(1, 0)$.
86. $\theta = \tanh^{-1}(3/5) = \ln 2$, $\gamma = 5/4$, and $\gamma\beta = 3/4$.
87. $w = 5/7$. The second result is exactly the addition formula for $\tanh(\alpha + \beta)$.
88. $v(20) = 60 \tanh(1.6) \approx 55.300 \text{ ms}^{-1}$; $t = 12.5 \tanh^{-1}(5/6) \approx 14.987 \text{ s}$.
89. $S(2) = 5 \tanh(0.8) \approx 3.320 \text{ V}$; $c = 2.5 \tanh^{-1}(4/5) = 2.5 \ln 3 \approx 2.747$; the limiting response is 5 V.
90. $x = 1/5$ when $y = 0$; $x = 1/5 + (2/3) \ln 3$ when $y = 0.8$; the range is $(-1, 1)$.
91. $P(0) = K/2$. At $t = (\ln 9)/r$, the argument is $\ln 3$ and $\tanh(\ln 3) = 4/5$, giving $P = 0.9K$.
92. $T(0) = 50^\circ\text{C}$; $T(5) = 20 + 30 \operatorname{sech} 1 \approx 39.442^\circ\text{C}$; $x = \pm 5 \cosh^{-1} 6$.
93. Peak 12 at $t = 4$; half-peak times $t = 4 \pm 2 \cosh^{-1} 2$.
94. $T(10) = 20 \cosh(2/3) \approx 24.612 \text{ kN}$; $x = \pm 15 \cosh^{-1}(3/2) \text{ m}$.
95. $A(10) = 2 \operatorname{sech} 1 \approx 1.296 \text{ m}$; $d = 10 \cosh^{-1} 2 \approx 13.170 \text{ m}$.
96. $V(0) = 120 \text{ V}$, $V(10) = 120 \cosh 1 / \cosh 2 \approx 49.219 \text{ V}$, and $V(20) = 120 / \cosh 2 \approx 31.896 \text{ V}$.
97. Base 100°C ; midpoint $25 + 75 \cosh 1 / \cosh 2 \approx 55.762^\circ\text{C}$; tip $25 + 75 / \cosh 2 \approx 44.935^\circ\text{C}$.
98. Substitution gives $\cosh^2 t - \sinh^2 t = 1$. At $t = \ln 2$, the point is $(15/2, 3)$; conversely that point gives $t = \ln 2$.
99. $C(0.5) \approx 0.462$ and $C(2) \approx 0.964$. Although the input rises by 1.5, the output rises by only about 0.502 because $\tanh x$ approaches its upper asymptote 1.
100. (a) $L = 20 \cosh^{-1}(3/2) \text{ m}$. (b) Span $= 40 \cosh^{-1}(3/2) \approx 38.497 \text{ m}$. (c) The halfway height is $10(\sqrt{5} - 2) \approx 2.361 \text{ m}$.

Chapter 12: Differential Calculus

These answers correspond to Section 12.18.

- (a) $4x + 2h - 3$; (b) $f'(x) = 4x - 3$; (c) gradient 5, tangent $y - 3 = 5(x - 2)$.
- (a) $2x$; (b) $-1/x^2$; (c) $1/(2\sqrt{x})$, $x > 0$.
- (a) The one-sided derivatives are -1 and 1 . (b) Continuity gives $a + b = 2$ and matching derivatives give $a = 2$; hence $a = 2$, $b = 0$.
- (a) $42x^5 - 12x^2 + 5$; (b) $\frac{15}{2}x^{3/2} + 8x^{-3}$; (c) $\frac{5}{2\sqrt{x}} - x^{-3/2}$; (d) $6x^2 - 2x + 6$.
- (a) $y' = 4x^3 - 9x^2 + 2$, $y'' = 12x^2 - 18x$, $y''' = 24x - 18$, $y^{(4)} = 24$; (b) $y'' = 20x^3 - 30x$, $y^{(5)} = 120$; (c) $-x^{-2}$, $2x^{-3}$, $-6x^{-4}$.
- (a) $3x^2 - 6x + 1$; (b) $e^x(x^2 + 2x)$; (c) $2\sin x + (2x - 1)\cos x$; (d) $x^2(7x + 6)/(2\sqrt{1+x})$.
- (a) $-7/(x - 2)^2$; (b) $x(x + 2)/(1 + x)^2$; (c) $\sec^2 x$; (d) $(1 - \ln x)/x^2$.
- (a) $30x(3x^2 - 1)^4$; (b) $-x/\sqrt{4 - x^2}$; (c) $-6(1 + 2x)^{-4}$; (d) $\frac{3}{2}(2x + 1)\sqrt{x^2 + x + 1}$.
- (a) $(1 + x^2)^3(1 + 9x^2)$; (b) $e^x(x^2 + x + 1)/\sqrt{1 + x^2}$; (c) $\cos x \cos 2x - 2\sin x \sin 2x$.
- (a) $\frac{(x^2 + 1)^2(5x^2 - 6x - 1)}{(x - 1)^2}$; (b) $\frac{2e^{2x}(x^2 - x + 1)}{(1 + x^2)^2}$; (c) $\frac{1}{2x^{3/2}} \left(\frac{2x}{1 + x} - \ln(1 + x) \right)$.
- (a) Tangent $y = x - 2$; normal $y = -x$. (b) Tangent $y - 2 = (x - 4)/4$; normal $y - 2 = -4(x - 4)$. (c) Tangent $y - 1 = (x - e)/e$; normal $y - 1 = -e(x - e)$.
- (a) $k = 3$ and the tangent is $y = 5x - 1$; (b) $a = 1$.
- (a) $f = x^3 - 6x^2 + 9x + 5$; local maximum $(1, 9)$ and local minimum $(3, 5)$. (b) $g = x^3 - 2x^2 + 2x + 1$.
- (a) 9.98; (b) 2.01; (c) 0.505.
- (a) Missing inner derivative: $10x(x^2 + 1)^4$. (b) The product rule gives $2x \sin x + x^2 \cos x$. (c) The numerator sign is reversed: $-2/(x - 1)^2$.
- (a) $3e^{3x-1}$; (b) $-8xe^{-x^2}$; (c) $e^{\sqrt{x}}/(2\sqrt{x})$; (d) $e^x/(1 + e^x)^2$.
- (a) $2x(\ln 2)2^{x^2}$; (b) $3(\ln 5)5^{3x-1}$; (c) $-(\ln 10)10^{-x}$; (d) $(\ln 3)3^{\sin x} \cos x$.
- (a) $2x/(x^2 + 4)$, all real x ; (b) $1/[2(1 + x)]$, $x > -1$; (c) $2/[(2x - 1)\ln 3]$, $x > 1/2$; (d) $\cot x$, where $\sin x \neq 0$.
- (a) $x^x(\ln x + 1)$; (b) $(x + 1)^x[\ln(x + 1) + x/(x + 1)]$; (c) $x^{\sin x}[\cos x \ln x + \sin x/x]$.
- (a) $\frac{y'}{y} = \frac{6x}{x^2 + 1} + \frac{1}{2(x - 1)} - \frac{2}{x}$; multiply by y for y' . (b) $y'/y = 1/(x^2 - 1)$, so $y' = y/(x^2 - 1)$. (c) $y'/y = 4x/(x^2 + 4) - 3/x + 1/(1 - x)$; multiply by y .
- (a) $\cos x$; (b) $-\sin x$; (c) $\sec^2 x$; (d) $-\csc^2 x$; (e) $\sec x \tan x$; (f) $-\csc x \cot x$; (g) $\cos x - \sin x$.
- (a) $6x \cos(3x^2)$; (b) $-8 \cos^3(2x) \sin 2x$; (c) $\sec^2 \sqrt{x}/(2\sqrt{x})$; (d) $-2 \sec(1 - 2x) \tan(1 - 2x)$.
- (a) $\sin x + x \cos x$; (b) $e^x(\cos x - \sin x)$; (c) $[x \sec^2 x - \tan x]/x^2$; (d) $1/(1 + \cos x)$.
- Quotient or reciprocal differentiation gives (a) $\sec^2 x$; (b) $\sec x \tan x$; (c) $-\csc^2 x$; (d) $-\csc x \cot x$.
- (a) $2/\sqrt{1 - 4x^2}$; (b) $-1/\sqrt{9 - x^2}$; (c) $3/(1 + 9x^2)$; (d) $1/[2\sqrt{x}(1 + x)]$; (e) $1/(|x|\sqrt{4x^2 - 1})$, for $|x| > 1/2$.
- (a) $\sin^{-1} x + x/\sqrt{1 - x^2}$; (b) $e^x[\tan^{-1} x + 1/(1 + x^2)]$; (c) $2x/\sqrt{1 - x^4}$.
- (a) $\cosh x$; (b) $\sinh x$; (c) $\operatorname{sech}^2 x$; (d) $-\operatorname{sech} x \tanh x$; (e) $-\operatorname{cosech}^2 x$; (f) $-\operatorname{cosech} x \coth x$; (g) $\cosh^2 x + \sinh^2 x = \cosh(2x)$.
- (a) $6x \cosh(3x^2)$; (b) $2 \cosh x \sinh x$; (c) $e^x(\tanh x + \operatorname{sech}^2 x)$; (d) $\tanh x$.
- (a) $2^n e^{2x}$; (b) $3^n \sin(3x + n\pi/2)$; (c) $e^x(x + n)$; (d) $e^x[x^2 + 2nx + n(n - 1)]$.
- (a) Local maximum $(1, e)$. (b) Maximum $(e, 1/e)$. (c) Endpoint minimum $(0, 0)$ and maximum $(2, 4e^{-2})$. (d) Minimum $(-1, -1/2)$ and maximum $(1, 1/2)$; increasing on $(-1, 1)$ and decreasing outside it.
- (a) $-x/y$; (b) $-4x/(9y)$; (c) $-y/x$; (d) x/y .

32. (a) $-(2xy + y^2)/(x^2 + 2xy)$; (b) $(y - x^2)/(y^2 - x)$; (c) $-(2x + y)/(x + 2y)$; (d) $-2y/(3x)$.
33. (a) $\sec(x + y) - 1$; (b) $[1 - ye^{xy}]/[xe^{xy} - 1]$; (c) $[y \sin x - \sin y]/[x \cos y + \cos x]$; (d) $[y(x + y) - 1]/[1 - x(x + y)]$.
34. (a) Tangent $y - 4 = -\frac{3}{4}(x - 3)$; normal $y - 4 = \frac{4}{3}(x - 3)$. (b) Tangent $y - 2 = -\frac{4}{5}(x - 1)$; normal $y - 2 = \frac{5}{4}(x - 1)$. (c) Tangent $y - 1 = -\frac{1}{3}(x - 2)$; normal $y - 1 = 3(x - 2)$. (d) Tangent $y = 2$; normal $x = 0$.
35. (a) $y'' = -25/y^3$, hence $-25/64$. (b) $y'' = -42/125$ at $(1, 2)$. (c) $y'' = 12/x^3$, hence $y'' = 3/2$ at $(2, 3)$.
36. (a) $1/3$; (b) $1/4$; (c) e ; (d) $1/2$.
37. (a) $3t^2/(2t + 1)$; (b) $-\cot t$; (c) $-e^{-2t}$; (d) $3t/2$; (e) $2t^2$.
38. (a) $6t(t + 1)/(2t + 1)^3$; (b) $-\csc^3 t$; (c) $-1/[4 \sin^4(t/2)]$; (d) e^t .
39. (a) At $(2, -2)$: tangent $y = -2$, normal $x = 2$. (b) At $(\sqrt{2}, 3\sqrt{2}/2)$: tangent gradient $-3/2$ and normal gradient $2/3$. (c) At $(1, 0)$: tangent $y = x - 1$, normal $y = 1 - x$.
40. (a) Horizontal tangent at $(2\sqrt{2}, 2\sqrt{2})$; no vertical tangent. (b) Horizontal tangents at $(-3, -2)$ and $(5, 2)$; vertical tangent at $(-4, 2)$. (c) Horizontal tangent at $(0, 0)$; vertical tangents at $(-2, 1)$ and $(2, 1)$.
41. (a) $y = (x - 1)^2$ and $dy/dx = 2(x - 1) = 2t$. (b) $x^2 + y^2 = 4$ and $dy/dx = -x/y = -\cot t$. (c) $y = x^2$ for $x \geq 0$ and $dy/dx = 2t^2 = 2x$; the derivative at the origin is 0 by the Cartesian form.
42. (a) $t = 2/3$, giving $(x, y) = (4/9, 8/27)$; (b) $t = 2$, giving $(x, y) = (4, 8)$.
43. (a) $6t(t^2 + 1)^2$; (b) $2e^{2t}/(e^{2t} + 1)$; (c) $2t \cos(t^2)$.
44. (a) $du/dt = 0$; (b) $dA/dt = \pi$; (c) $dV/dt = 8\pi t(t^2 + 1)^2$.
45. (a) Approximately 0.6%; (b) approximately 0.4%; (c) $dy \approx 0.36$; (d) $|dV|/V \approx 2(0.02/10) + 0.05/30$, so the maximum percentage error is approximately 0.567%.
46. (a) $x = \pm 1$; (b) $x = 0, \pm\sqrt{2}$; (c) $x = 2$; (d) $x = 2^{-1/3} = 1/\sqrt[3]{2}$.
47. (a) Decreasing for $x < 2$, increasing for $x > 2$. (b) Increasing on $(-\infty, -1)$ and $(1, \infty)$, decreasing on $(-1, 1)$. (c) Increasing on $(0, 1)$ and decreasing on $(1, \infty)$.
48. (a) Local maximum at $x = -2$ and local minimum at $x = 1$. (b) $x = 0$ is not an extremum; $x = 3$ is a local minimum. (c) Local maximum at $x = 2$.
49. (a) Local maximum $(1, 4)$ and local minimum $(3, 0)$. (b) Local maximum $(0, 0)$ and local minima $(\pm\sqrt{2}, -4)$. (c) Maximum at $x = \pi/2$ and minimum at $x = 3\pi/2$. (d) Minimum $(1, 2)$; there is no maximum on $x > 0$.
50. x^3 and x^5 have stationary inflection points at the origin; x^4 has a local minimum there and no inflection. In general, even n gives a local minimum, while odd $n > 1$ gives a stationary point of inflection.
51. (a) Maximum 18 at $x = 3$; minimum -2 at $x = -2, 1$. (b) Minimum 1 at $x = 2$; maximum 10 at $x = 5$. (c) Minimum 4 at $x = 2$; maximum $29/5$ at $x = 5$. (d) Minimum e^{-1} at $x = -1$ and maximum e at $x = 1$.
52. (a) Up on $(-\infty, 1/4)$, down on $(1/4, \infty)$; inflection $(1/4, 1/8)$. (b) Up on $(-\infty, 0) \cup (2, \infty)$, down on $(0, 2)$; inflections $(0, 0)$ and $(2, -16)$. (c) Down for $x < 0$, up for $x > 0$; inflection $(0, 0)$.
53. (a) $\ln x$ is concave down on $(0, \infty)$. (b) $1/x$ is concave down on $(-\infty, 0)$ and up on $(0, \infty)$; $x = 0$ is not an inflection point. (c) e^{-x^2} is down for $|x| < 1/\sqrt{2}$ and up otherwise, with inflections $(\pm 1/\sqrt{2}, e^{-1/2})$.
54. (a) Roots 1, 3, vertex $(2, -1)$, upward-opening parabola. (b) Roots $0, \pm\sqrt{3}$; local maximum $(-1, 2)$ and local minimum $(1, -2)$; inflection $(0, 0)$. (c) Roots 0 (double) and 3; local maximum $(0, 0)$, local minimum $(2, -4)$, and inflection point $(1, -2)$. It increases on $(-\infty, 0) \cup (2, \infty)$ and decreases on $(0, 2)$.
55. Intercepts $0, \pm 2\sqrt{2}$; minima $(\pm 2, -16)$; maximum $(0, 0)$; down for $|x| < 2/\sqrt{3}$ and up otherwise; inflections $(\pm 2/\sqrt{3}, -80/9)$. The range is $[-16, \infty)$.
56. (a) Vertical $x = 2$, oblique $y = x + 2$. (b) Vertical $x = \pm 2$, horizontal $y = 2$. (c) Vertical $x = 0$, oblique $y = x$. (d) Vertical $x = 1$, oblique $y = x$, and a removable discontinuity at $(-1, -3/2)$.

57. (a) Maximum $(1, e^{-1})$; inflection $(2, 2e^{-2})$. (b) Endpoint minimum $(0, 0)$, maximum $(2, 4e^{-2})$; inflection abscissae $2 \pm \sqrt{2}$.
58. (a) The limits are $-\infty$ and 0 , respectively. (b) Increasing on $(0, e)$ and decreasing on (e, ∞) ; maximum $(e, 1/e)$. (c) Concave down on $(0, e^{3/2})$ and up afterwards; inflection $(e^{3/2}, 3/(2e^{3/2}))$.
59. (a) Maximum $\sqrt{2}$ at $x = \pi/4$ and minimum $-\sqrt{2}$ at $x = 5\pi/4$. (b) Maximum 1 at $x = \pi/4$ and minimum -1 at $x = 3\pi/4$.
60. The stationary point is $(2\sqrt{2}, 2\sqrt{2})$ and is a local minimum.
61. Increasing, then decreasing, then increasing; local maximum at $x = -1$ and local minimum at $x = 2$.
62. Decreasing on $(-\infty, 1)$ and increasing on $(1, \infty)$. The point $x = -2$ is stationary but not an extremum; $x = 1$ is a local minimum. Since $f''(x) = 3x(x+2)$, the graph is concave up on $(-\infty, -2) \cup (0, \infty)$ and down on $(-2, 0)$, with inflection abscissae -2 and 0 .
63. (a) $a = -9/2$, $b = 6$; (b) $a = -4$, $b = 1$.
64. (a) $(\pm\sqrt{2}, 0)$; (b) $(2, \ln 2)$; (c) none.
65. $f(x) = x^3 - 3x + 1$; local maximum $(-1, 3)$ and local minimum $(1, -1)$.
66. (a) 5 cm by 5 cm; area 25 cm^2 . (b) A square of side $P/4$; area $P^2/16$. (c) A square of side $\sqrt{A}$; the minimum perimeter is $4\sqrt{A}$.
67. (a) Width 30 m, river-parallel length 60 m; maximum area 1800 m^2 . (b) Width $L/4$, river-parallel length $L/2$, maximum area $L^2/8$.
68. (a) $x = 1$ cm; box 6 cm by 3 cm by 1 cm; volume 18 cm^3 . (b) At $x = 0.5$ cm the volume is 14 cm^3 , which is 4 cm^3 below the maximum.
69. (a) $r = \sqrt[3]{250}$ cm and $h = 2r$. (b) $r = h = \sqrt[3]{108}$ cm. (c) The minimum areas are $6\pi(250)^{2/3} \text{ cm}^2$ for the closed cylinder and $3\pi(108)^{2/3} \text{ cm}^2$ for the open one.
70. (a) $(\pm\sqrt{5/2}, 5/2)$; minimum distance $\sqrt{11}/2$. (b) $(\pm 1/\sqrt{2}, 1/2)$; minimum distance $\sqrt{3}/2$.
71. (a) $0 \leq x \leq 200$. (b) $x = 100$ units, price RM 40, maximum revenue RM 4,000. (c) $R'(100) = 0$; revenue has zero instantaneous change at the optimum.
72. (a) $d = 2$; maximum response $1/4$. (b) Both limiting responses are 0 .
73. (a) Half-width $x = 2$ m, width 4 m, height 8 m; maximum area 32 m^2 . (b) The optimal half-width is $\sqrt{a/(3b)}$, the height is $2a/3$, and the maximum area is $4a^{3/2}/(3\sqrt{3b})$.
74. Radius $5\sqrt{2/3}$ cm and height $10/\sqrt{3}$ cm; maximum volume $500\pi/(3\sqrt{3}) \text{ cm}^3$. The cylinder-to-sphere volume ratio is $1/\sqrt{3}$.
75. (a) $r = h = 20/(4 + \pi)$ m, so the width is $40/(4 + \pi)$ m. (b) The optimal rectangular height equals the semicircle radius, and the maximum area is $200/(4 + \pi) \text{ m}^2$.
76. (a) $v = 3(t-1)(t-3)$, $a = 6t - 12$; rest at $t = 1, 3$; forward on $(0, 1)$ and $(3, 4)$, backward on $(1, 3)$; displacement 4 and total distance 12 . (b) Average velocity $= 1 \text{ unit s}^{-1}$; greatest speed $= 9 \text{ units s}^{-1}$, attained at $t = 0$ and $t = 4$.
77. (a) $dr/dt = 2 \text{ cm s}^{-1}$. (b) $dC/dt = 0.8\pi \text{ cm s}^{-1}$ and $dA/dt = 4\pi \text{ cm}^2 \text{ s}^{-1}$.
78. (a) $dy/dt = -3/2 \text{ m s}^{-1}$; the top descends at 1.5 m s^{-1} . (b) If θ is the angle with the ground, then $d\theta/dt = -1/4 \text{ rad s}^{-1}$.
79. (a) $dh/dt = -15/(16\pi) \text{ cm s}^{-1}$. (b) $r = 4$ cm and $dr/dt = -3/(8\pi) \text{ cm s}^{-1}$.
80. (a) Shadow length rate $9/14 \text{ m s}^{-1}$; shadow-tip rate $15/7 \text{ m s}^{-1}$. (b) The shadow length is $30/7$ m.
81. (a) $C'(x) = 25 + 0.04x$; $C'(100) = 29$. The marginal estimate is RM 29; the exact additional cost is RM 29.02. (b) $C(100)/100 = \text{RM } 77$ per unit, which is RM 48 greater than the marginal cost at that output.
82. $R = 120x - 0.5x^2$; maximum at $x = 120$, price RM 60, revenue RM 7,200.
83. (a) $P = 100x - 0.5x^2 - 2000$; maximum at $x = 100$, with profit RM 3,000. (b) Break-even outputs $x = 100 \pm 10\sqrt{60}$, approximately 22.54 and 177.46 units.
84. (a) $x = 50\sqrt{2} \approx 70.71$ units; minimum average cost $20 + 10\sqrt{2} \approx \text{RM } 34.14$. (b) The average cost decreases on $(0, 50\sqrt{2})$ and increases on $(50\sqrt{2}, \infty)$.
85. (a) $v = 24 - 9.8t$; maximum at $t = 120/49 \approx 2.449$ s with height $1489/49 \approx 30.388$ m. (b) Ground time

- $(120 + \sqrt{14890})/49 \approx 4.939$ s; impact velocity $-\sqrt{595.6} \approx -24.405$ m s⁻¹.
86. (a) $d(80) = 58.4$ m and $d'(80) = 1.21$ m per km h⁻¹. The estimated extra distance is 6.05 m. (b) The exact increase is $d(85) - d(80) = 6.20$ m, so the linear estimate is low by 0.15 m.
87. (a) $P(10) = 5000e^{0.3} \approx 6749.29$ and $P'(10) = 150e^{0.3} \approx 202.49$ people per year. (b) Doubling time $\ln 2/0.03 \approx 23.10$ years.
88. (a) Increasing for $0 < t < 2$ and decreasing for $t > 2$. (b) Maximum at $t = 2$ h; $C_{\max} = 40/e \approx 14.72$ mg L⁻¹.
89. (a) $T'(0) = -8.16$ °C min⁻¹ and $T \rightarrow 22$ °C. (b) $T'(10) = -8.16e^{-1.2} \approx -2.458$ °C min⁻¹; the temperature reaches 30 °C at $t = \ln(8.5)/0.12 \approx 17.83$ min.
90. Greatest growth at $P = 2500$, at rate 375 units per time. Since $P = 5000/[1 + (22/3)e^{-0.3t}]$, this occurs at $t = (10/3)\ln(22/3) \approx 6.64$. The population reaches 4000 at $t = (10/3)\ln(88/3) \approx 11.26$.
91. $P'(R) = V^2(r - R)/(R + r)^3$, so the maximum occurs at $R = r$. The maximum is $V^2/(4r) = 9$ W. The power tends to 0 as $R \rightarrow 0^+$ and as $R \rightarrow \infty$.
92. $\tan \theta = x/20$, so $d\theta/dt = 300/(400 + x^2)$. At $x = 30$, $d\theta/dt = 3/13$ rad s⁻¹ and $\theta = \tan^{-1}(3/2) \approx 0.983$ rad.
93. (a) At $r = 2$ cm, $dr/dt = 15/(8\pi)$ cm s⁻¹ and $dS/dt = 30$ cm² s⁻¹. (b) At $r = 4$ cm, $dr/dt = 15/(32\pi)$ cm s⁻¹ and $dS/dt = 15$ cm² s⁻¹.
94. (a) At $r = 10$, $dr/dt = 6/\pi$ m min⁻¹; at area 400π , $r = 20$ and $dr/dt = 3/\pi$ m min⁻¹. (b) The required time is $400\pi/120 = 10\pi/3 \approx 10.47$ min.
95. Similar triangles give surface width $2h$ and $V = 10h^2$. (a) At $h = 1$, the width is 2 m and the volume is 10 m³. (b) $dh/dt = 1/40 = 0.025$ m min⁻¹.
96. (a) Maximum absolute volume error $|dV| \approx 4\pi(12)^2(0.05) = 28.8\pi \approx 90.48$ cm³; percentage error $\approx 1.25\%$. (b) Maximum absolute surface-area error $|dS| \approx 8\pi(12)(0.05) = 4.8\pi \approx 15.08$ cm²; percentage error $\approx 0.833\%$.
97. (a) Sensitivity $dV/dT = 0.04$ V °C⁻¹ and $T = 25(V - 0.5)$. (b) $T = 30$ °C. (c) Temperature uncertainty $|dT| \approx 0.002/0.04 = 0.05$ °C.
98. (a) $h'(10) = -0.4$. (b) The stationary abscissae are $x = 25 \mp 5\sqrt{105}/3$, approximately 7.922 and 42.078 m. Their heights are approximately 7.425 m and -32.425 m; the first is a local maximum and the second a local minimum.
99. (a) $x = (25 - 5\sqrt{7})/3 \approx 3.924$ cm. The box dimensions are approximately 22.153 cm by 12.153 cm by 3.924 cm. (b) Maximum volume $(10000 + 7000\sqrt{7})/27 \approx 1056.31$ cm³.
100. (a) $S(t) = 1000/(100 + t^2)$ and $S(0) = 10$. (b) $S'(t) = -2000t/(100 + t^2)^2$, so $S'(10) = -0.5$ signal units per second. (c) $S'(t) < 0$ for $t > 0$ and $S(t) \rightarrow 0$.

Chapter 13: Integral Calculus

These answers correspond to Section 13.12.

1. (a) 26; (b) 63; (c) 3.
2. (a) $\sum_{k=1}^{10} (3k-1)$; (b) $\sum_{k=1}^n k^2$; (c) $\sum_{k=1}^8 \frac{1}{k(k+1)}$.
3. (a) 210; (b) 385; (c) n^2 .
4. $\sum_{j=0}^4 [(j+2)^2 - 1] = 85$.
5. The intermediate terms cancel, leaving $1 - 1/(n+1) = n/(n+1)$; for $n = 99$, the value is $99/100$.
6. $1/3$.
7. $\int_0^2 \sqrt{1+x^2} dx = \sqrt{5} + \frac{1}{2} \ln(2 + \sqrt{5})$.
8. Left = $7/4$, right = $15/4$, midpoint = $21/8$; the exact value is $8/3$.
9. Increasing: left underestimates and right overestimates. Decreasing: the roles are reversed.
10. $F'(x) = 2x(x^6 + 1)$.
11. $G'(x) = 2e^{4x^2} - e^{\sin^2 x} \cos x$.
12. (a) 12; (b) 2; (c) 36.
13. (a) $x^6 + 2x^{-2} + 7x + C$; (b) $2x^{3/2} + 2/x + C$; (c) $x^3/3 + 2 \ln|x| + C$.
14. (a) $\frac{2}{5}x^{5/2} + \frac{4}{3}x^{3/2} + 2\sqrt{x} + C$; (b) $x^2/2 - 4 \ln|x| + C$; (c) $x + \frac{4}{3}x^{3/2} + x^2/2 + C$.
15. (a) $\frac{3}{2}e^{2x} - \frac{2 \cdot 5^x}{\ln 5} + C$; (b) $-8e^{-x/2} + C$; (c) $2^{3x+1}/(3 \ln 2) + C$.
16. (a) $-\frac{4}{3} \cos 3x + \frac{5}{2} \sin 2x + C$; (b) $2 \tan x + 3 \cot x + C$; (c) $\sec x - \csc x + C$.
17. (a) $(3x-1)^8/24 + C$; (b) $-(5-2x)^{3/2}/3 + C$; (c) $\frac{1}{4} \ln|4x+3| + C$.
18. (a) $\ln(x^2 + 5x + 7) + C$; (b) $\ln|2 + \sin x| + C$; (c) $\ln(1 + e^x) + C$.
19. (a) $(x^2 + 4)^6/12 + C$; (b) $(x^2 + 1)^{5/2}/5 - (x^2 + 1)^{3/2}/3 + C$; (c) $-1/[2(x^2 - 9)] + C$.
20. (a) $2e^{\sqrt{x}} + C$; (b) $2 \ln(1 + \sqrt{x}) + C$; (c) $(\ln x)^2/2 + C$.
21. (a) $e^{x^2}/2 + C$; (b) $\frac{1}{2} \ln(1 + e^{2x}) + C$; (c) $\ln(3 + 2^x)/\ln 2 + C$.
22. (a) $-\cos^5 x/5 + C$; (b) $\tan^2 x/2 + C$; (c) $-\cos(\ln x) + C$.
23. (a) $\frac{1}{4} \tan^{-1}(x/4) + C$; (b) $\sin^{-1}(x/5) + C$; (c) $\frac{1}{3} \tan^{-1}(3x) + C$.
24. (a) $15/4$; (b) $e - 1$; (c) $1/3$.
25. (a) 2; (b) $\ln(3/2)$; (c) 0.
26. (a) $F = x^3 - 2x^2 + x + 5$; (b) $y = e^{2x} + 2$; (c) $s = 3t^2 - 4t + 3$.
27. $C(x) = 0.02x^2 + 12x + 500$; $C(100) = \text{RM } 1,900$.
28. (a) $a/2 + b$; (b) $k = 6$; (c) $a = 3$.
29. (a) $\int x^{-1} dx = \ln|x| + C$; the power rule excludes exponent -1 . (b) Expand first: $x^7/7 + 3x^5/5 + x^3 + x + C$. (c) The new limits are $u = 0$ and $u = 1$, giving $e - 1$.
30. (a) $\frac{1}{2} \ln(x^2 + 2x + 5) + C$; (b) $\sqrt{2x+1} + C$; (c) $2 \ln 2$.
31. (a) $e^x(x-1) + C$; (b) $\frac{x}{2} \sin 2x + \frac{1}{4} \cos 2x + C$.
32. (a) $e^x(x^2 - 2x + 2) + C$; (b) $-x^2 \cos x + 2x \sin x + 2 \cos x + C$.
33. (a) $x \ln x - x + C$; (b) $\frac{x^2}{2} \ln x - \frac{x^2}{4} + C$.
34. (a) $x \tan^{-1} x - \frac{1}{2} \ln(1 + x^2) + C$; (b) $x \sin^{-1} x + \sqrt{1 - x^2} + C$.
35. (a) $\frac{1}{2} e^x (\sin x - \cos x) + C$; (b) $\frac{1}{5} e^{2x} (2 \cos x + \sin x) + C$.

36. (a) 1; (b) π .
37. $I_n = -x^n \cos x + nx^{n-1} \sin x - n(n-1)I_{n-2}$;
 $I_3 = -x^3 \cos x + 3x^2 \sin x + 6x \cos x - 6 \sin x + C$.
38. (a) $-\cos x + \cos^3 x/3 + C$; (b)
 $-\cos x + \frac{2}{3} \cos^3 x - \frac{1}{5} \cos^5 x + C$.
39. (a) $\sin x - \sin^3 x/3 + C$; (b)
 $\sin x - \frac{2}{3} \sin^3 x + \frac{1}{5} \sin^5 x + C$.
40. (a) $x/2 - \sin 2x/4 + C$; (b)
 $x/2 + \sin 2x/4 + C$.
41. (a) $x/8 - \sin 4x/32 + C$; (b)
 $3x/8 - \sin 2x/4 + \sin 4x/32 + C$.
42. (a) $\sin 2x/4 - \sin 8x/16 + C$; (b)
 $\sin 3x/6 + \sin 5x/10 + C$.
43. (a) $\tan^4 x/4 + C$; (b) $\tan x + \tan^3 x/3 + C$;
(c) $\tan x - x + C$.
44. (a) $-\cot^4 x/4 + C$; (b) $-\cot^2 x/2 + C$; (c)
 $-\cot x + C$.
45. (a) $2/3$; (b) $\pi/16$; (c) 1.
46. (a) $\sin^{-1}(x/3) + C$; (b) $\frac{1}{5} \tan^{-1}(x/5) + C$.
47. $\frac{x}{2} \sqrt{16 - x^2} + 8 \sin^{-1}(x/4) + C$; the definite
integral is 8π .
48. (a) $\frac{x}{2} \sqrt{x^2 + 9} + \frac{9}{2} \ln|x + \sqrt{x^2 + 9}| + C$; (b)
 $\ln|x + \sqrt{x^2 + 9}| + C$.
49. (a) $\sqrt{x^2 - 9}/(9x) + C$; (b)
 $\ln(x + \sqrt{x^2 - 4}) + C$ for $x > 3$.
50. (a) $\frac{1}{3} \ln \left| \frac{x-1}{x+2} \right| + C$; (b)
 $\frac{11}{3} \ln|x - 2| + \frac{4}{3} \ln|x + 1| + C$.
51. (a) $\ln|x| - \ln|x + 1| + C$; (b)
 $2 \ln|x - 1| - 5/(x - 1) + C$.
52. (a) $\frac{1}{2} \ln(x^2 + 4) + C$; (b)
 $\frac{3}{2} \ln(x^2 + 2x + 5) + \tan^{-1}((x + 1)/2) + C$.
53. (a) $x^2/2 - x + 2 \ln|x + 1| + C$; (b)
 $x^2 + x + C$.
54. (a) $\ln(4/3)$; (b) $\ln 6$.
55. $\frac{1}{x} + \frac{x}{x^2 + 1}$; excluded value $x = 0$; integral
 $\ln|x| + \frac{1}{2} \ln(x^2 + 1) + C$.
56. (a) 6; (b) 1; (c) 8.
57. (a) 3; (b) -3; (c) 13.
58. (a) 0; (b) $64/5$; (c) 0.
59. $F = x^3/3 - 2x^2 + x$ and $F' = x^2 - 4x + 1$.
The stationary pairs are
 $(2 - \sqrt{3}, -10/3 + 2\sqrt{3})$ and
 $(2 + \sqrt{3}, -10/3 - 2\sqrt{3})$.
60. (a) 3; (b) $2/\pi$; $c = \sqrt{3}$.
61. (a) net = $4/3$, total = 4; (b) net = 0, total
= $8/3$.
62. (a) $32/3$; (b) $16/3$.
63. (a) 13; (b) 4.
64. (a) $9/2$; (b) $4/3$.
65. (a) $4/3$; (b) $16\sqrt{2}/3$.
66. (a) $8/3$; (b) $7/3$.
67. 4 square units.
68. Area = $k^3/6$; hence $k = 3$.
69. (a) $2x\sqrt{1 + x^6} - \sqrt{1 + x^3}$; (b)
 $\cos((\ln x)^2)/x$.
70. (a) $a = 4$; (b) $b = 5$; (c) $c = \pi/6$.
71. $8/3$.
72. $41/2$.
73. $32/3$.
74. At rest at $t = 1, 3$; displacement = 4; total
distance = 12.
75. $v(t) = 3t^2 - 4t + 2$; displacement = 15.
Since $v(t) > 0$ throughout, the distance is
also 15.
76. (a) $32\pi/5$; (b) 8π .
77. $32\pi/5$.
78. $64\pi/15$.

79. $V = \pi \int_0^1 [(2-x^2)^2 - (2-x)^2] dx = 8\pi/15$.
80. $V = \pi \int_{-2}^2 [(5-x^2)^2 - 1] dx = 832\pi/15$.
81. $V = \pi \int_0^2 (4-y^2) dy = 16\pi/3$.
82. $x = y^2$; $V = \pi \int_0^2 (16-y^4) dy = 128\pi/5$.
83. $V = \pi \int_0^h (r^2 x^2/h^2) dx = \pi r^2 h/3$.
84. $V = \pi \int_{-R}^R (R^2 - x^2) dx = 4\pi R^3/3$.
85. $V = \pi \int_0^5 (3^2 - 1^2) dx = 40\pi \text{ cm}^3$.
86. Delivered = 48 L; final amount = 88 L.
87. $W = \int_0^6 (5 + 2x) dx = 66 \text{ J}$.
88. $m = \int_0^8 (2 + 0.5x) dx = 32 \text{ kg}$.
89. $C(x) = 0.01x^2 + 15x + 800$;
 $C(200) = \text{RM } 4,200$; average cost = RM 21.
90. Increase = $\frac{20000}{3}(e^{0.3} - 1) \approx 2332.39$
 people.
91. Total inflow = 200 L; total outflow = 70 L;
 net change = +130 L.
92. Exposure = $160(1 - e^{-2}) \text{ mg h L}^{-1}$; average
 = $20(1 - e^{-2}) \text{ mg L}^{-1}$.
93. Energy = $24000/\pi \approx 7639.44 \text{ Wh} \approx 7.639$
 kW h.
94. Total rainfall = 27 mm; average rate = 9
 mm h^{-1} .
95. $k = 6$; $P(1/4 \leq X \leq 3/4) = 11/16$.
96. $V = \int_0^4 A(h) dh = 28\pi/3 \text{ m}^3$.
97. The roots are $x = \pm 6$ and the opening area
 is 72 m^2 . At $x = \pm 2$, the arch height is 8 m,
 so the permitted vehicle height is 7.5 m.
98. $m = 3 \int_0^{10} (2 + 0.1x) dx = 75 \text{ kg}$.
99. $W = \int_{0.05}^{0.12} 250x dx = 119/80 \approx 1.4875 \text{ J}$.
100. (a) $P(x) = 60x - 0.2x^2 - 500$; (b)
 $x = 150 \pm 100\sqrt{2}$ (approximately 8.58 and
 291.42); (c) maximum at $x = 150$, with
 profit RM 4,000.

Chapter 14: Differential Equations

These answers correspond to Section 14.10.

- (a) order 2, degree 1; (b) order 2, degree 2; (c) order 2, degree 1; (d) order 2, degree 1 after isolating and squaring the radical; (e) order 3, degree 2.
- The general solution is $y = x^3 + C$. (a) $y = x^3 + 5$; (b) $y = x^3 - 7$; (c) $y = x^3 + 1$. Each initial condition fixes the single value of C .
- Direct differentiation gives respectively $y' = -2Ce^{-2x}$, $y' = 2x$, $2x + 2yy' = 0$, and $y' = Ce^{-x}/(1 + Ce^{-x})^2 = y(1 - y)$.
- (a) separable; (b) linear; (c) homogeneous, linear, and affine linear coefficients; (d) exact; (e) Bernoulli with $n = 4$; (f) affine linear coefficients with independent linear parts.
- (a) $y' + (2/x)y = x/3$; (b) $dx/dy + [2y/(1 + y^2)]x = 3/(1 + y^2)$; (c) $dx/dy = x/y + 1$; (d) $dy/dx = [1 + (y/x)^2]/[2(y/x)]$.
- (i) separation; (ii) $y = vx$; (iii) $u = y^{1-n}$; (iv) test $M_y = N_x$; (v) calculate $\Delta = a_1b_2 - a_2b_1$.
- (a) $y = 0, 3$; (b) $y = -2, 1$; (c) $y = -1, 0, 1$. They may be lost when a corresponding factor is divided out during separation.
- (a) $(-\infty, 2)$; (b) $(-\pi/2, \pi/2)$; (c) $(3, \infty)$.
- (a) homogeneous and exact; exact is shorter. (b) separable and exact; exact is slightly shorter. (c) homogeneous, exact, and Bernoulli; exact is shortest.
- (a) separable; (b) homogeneous, or linear after expansion; (c) linear; (d) exact; (e) Bernoulli; (f) translate the intersection of the affine coefficient lines.
- (a) $y = x^4 + C$; (b) $y^2 = 2x^2 + C$; (c) $y + y^3/3 = x^3 + C$; (d) $2y^3 = 3x^2 + C$.
- (a) $e^y = \cos x + C$; (b) $e^y = e^x + C$; (c) $\sin y = \sin x + C$; (d) $\int_0^y e^{u^2} du = x^2/2 + C$.
- (a) $y = C(x - 1)/(x + 1)$ on intervals excluding $x = \pm 1$; (b) $y^3/3 + y^2/2 = x^2/2 + C$; (c) $\ln|(y - 1)/(y + 2)| = 3 \ln|x| + C$, together with $y = 1, -2$.
- (a) $y = 4e^x/(x + 1) - 3$, $x > -1$; (b) $y^3 = x - x^2$; (c) $y = \tan(x^2/2 + \pi/4)$ on the interval containing 0 before the first pole.
- Equilibria are $y = 0, 4$. For $y_0 \neq 0, 4$, $y = 4/[1 + ((4 - y_0)/y_0)e^{-4x}]$. If $0 < y_0 < 4$, then $y \rightarrow 4$ from below; if $y_0 > 4$, it decreases towards 4 for increasing x .
- (a) $y^3/3 + 4y = x^3/3 + x + C$; (b) $y^2/2 - \cos y = x^2/2 + x + C$; (c) $y + y^5/5 = (1 + x^2)^{3/2}/3 + C$.
- $k = 2 \ln 2$ and $y = 3e^{(\ln 2)x^2} - 1 = 3 \cdot 2^{x^2} - 1$.
- (a) $\tan^{-1} x = y^2/2 + C$; (b) $e^x = y^2 + C$; (c) $(x + 2)^2 = 2y^3 + C$.
- $A(t) = 240e^{-0.18t}$; the threshold occurs at $t = \ln(240/50)/0.18 \approx 8.715$, and $A < 50$ thereafter.
- Differentiation gives (a) $yy' = 2x^3$; (b) $e^y y' = \sin x$; (c) $y' = (y - 1)(y + 2)/x$.
- (a) $y' = 2x$; (b) $y' = 3y$; (c) $x + yy' = 0$; (d) $yy' = 3x^2$.
- $y = 2(1 - x^3)/(1 + 2x^3)$, $x \in (0, \infty)$. The equilibria are $y = 2, -1$.
- The coefficient functions in (a) and (c) are homogeneous of degree 2; those in (b) are homogeneous of degree 1. (a) $x^4 - 2x^2y^2 = C$; (b) $(x + y)(x + 3y) = C$; (c) $3x^2y - 2x^3 - 2y^3 = C$.
- (a) $x^4 + 2x^2y^2 = 3$; (b) $y = x(\ln x + 2)$, $x > 0$; (c) the initial point lies on the straight-line solution $y = x$.
- (a) $x = y(\ln|y| + C)$; (b) $\ln|x| - y^2/(2x^2) = C$; (c) $x = y(\ln|y| + C)$.
- (a) $m = \pm 1$; (b) no real m ; (c) $m = -1 \pm \sqrt{2}$.
- $\ln|y| - x^2/(2y^2) = C$.
- $a = -3$. Non-straight solutions satisfy $\ln|y/x - 1| - y/x = \ln|x| + C$; also $y = x$.
- Equation (a) is homogeneous and gives $2 \ln|y/x| + y/x = \ln|x| + C$, together with $y = 0$. Equation (b) is not homogeneous.
- (a) $\tan^{-1}(y/x) - \frac{1}{2} \ln(x^2 + y^2) = C$; (b) $x^2 - xy - y^2 = C$; (c) $\ln|y| + y/x = C$.

31. Both methods give $3x^2y - x^3 - y^3 = C$.
32. Every real k works because all terms have total degree 2.
33. $e^{x/y} + \ln y = 1$ on the branch containing $(0, 1)$.
34. (a) Using $x = vy$, $2\sqrt{x/y} = \ln|y| + C$. (b) Using $y = vx$, $y^3 - 3x^2y = C$. (c) Using $y = vx$, $\int_{y/x}^{y/x} v^2/(1+v-v^3) dv = \ln|x| + C$, together with $y = mx$ where $m^3 - m - 1 = 0$.
35. (a) $\mu = xe^x$, $y = e^x/(2x) + C/(xe^x)$. (b) $\mu = \sec^2 y$, $x = -3 \sin y \cos y + C \cos^2 y$. (c) $\mu = (1-x)^{-1}$, $y = (1-x)[C - x^2/2 - 2x - 2 \ln|1-x|]$. (d) $\mu = 1 + y^2$, $x = (2y + C)/(1 + y^2)$.
36. (a) $y = 1/3 + Ce^{-4x}$; (b) $y = (5/2)e^{-x} + Ce^{-5x}$; (c) $y = Ce^{x/2} - 6$.
37. (a) $y = x^3/5 + C/x^2$; (b) $y = x^5/2 + Cx^3$; (c) $y = 1/3 + Ce^{-x^3}$.
38. (a) $y = \sin x + C \cos x$; (b) $y = (x + C) \sin x$; (c) $y = \cos^2 x [C - \ln|\cos x|]$.
39. (a) $x = y^3/4 + C/y$; (b) $x = -3 \sin y \cos y + C \cos^2 y$; (c) $x = [y/2 - \sin 2y/4 + C]/\sin y$.
40. (a) $y = x(e^x - e)$; (b) $y = x + 4/x^2$; (c) $x = 1 - y - (1 - y)^4/8$.
41. (a) $y = [x^3/3 - x^2 - 2]/(x - 2)$ on $(-\infty, 2)$. (b) $y = \cos x [2 + \ln|\sec x + \tan x|]$ on $(-\pi/2, \pi/2)$. (c) $y = (x + 1)[x^2/2 - x + \ln(x + 1) + 1/2 - \ln 2]$, $x > -1$.
42. Direct substitution verifies the family. Through $(2, 3)$, $C = 4$, so $y = x + 4/x^2$.
43. The equation is $y' + 2xy = 2x$ and $y = 1 + Ce^{-x^2}$.
44. Every real k works: substitution gives $(k + 2)x$ on both sides.
45. (a) $y = Ce^{-\int P(x) dx}$. (b) $L[y_1 + y_2] = 0$. (c) If $L[y_1] = L[y_2] = Q$, then $L[y_1 + y_2] = 2Q$, not generally Q .
46. (a) $y = x - 1 + Ce^{-x}$; (b) $y = x^3/2 + Cx$; (c) $y = (x + C)/(1 + x^2)$.
47. (a) $\Delta = 0$, $z = 2y - x$; (b) $\Delta = 0$, $z = 3y - 2x$; (c) $\Delta = 1$, intersection $(5, 2)$; (d) $\Delta = -3$, intersection $(-2, 1)$; (e) $\Delta = 0$, $z = x - 2y$.
48. (a) $4(y - 1)^2 + 2(x + 1)(y - 1) - (x + 1)^2 = C$. (b), with $X = x - 3$, $Y = y - 4$, $X^2 + 4XY - 3Y^2 = C$. (c), with $X = x - 1$, $Y = y + 1$, $X^2 + XY - Y^2 = C$.
49. (a) $y + (x + y)^2/2 = C$. (b), $z = x - 2y$: $y + z/4 + (7/16) \ln|4z + 5| = C$. (c), $z = x + y$: $y + z/2 + \ln|z - 2| = C$.
50. (a) $y + (x + y)^2/2 = 11/2$. (b) $y + (x - 2y)/4 + (7/16) \ln|4x - 8y + 5| = 1/4 + (7/8) \ln 3$.
51. (a), $X = x - 5/3$, $Y = y - 2/3$: $X^2 + XY - Y^2/2 = -1/3$. (b), $X = x - 13/7$, $Y = y - 9/7$: $\frac{1}{2} \ln(X^2 - XY + Y^2) + \frac{5}{\sqrt{3}} \tan^{-1} \frac{2Y - X}{\sqrt{3}X} = C$, with C fixed by $(X, Y) = (-6/7, -2/7)$.
52. $\Delta = 4 - 2a$. Use a translation for $a \neq 2$ and $z = x + 2y$ for $a = 2$.
53. $(2p + q) dp + (p - 2q) dq = 0$, hence $(x - 1)^2 + (x - 1)(y + 1) - (y + 1)^2 = C$.
54. The translation sends $(3, 4)$ to $(0, 0)$; because both affine expressions vanish at $(3, 4)$, both constant terms disappear.
55. (a) $z = x + y$; (b) $z = x - 2y$; (c) translate the intersection $(5, 2)$.
56. With $z = 2y - x$, $y - z/3 + (2/9) \ln|3z + 5| = C$.
57. (a) $M_y = N_x$. (b) $B = C$. (c) $B = 2D$ and $E = 2C$.
58. (a) $x^2y + x^4/4 + (2/3)y^3 = C$; (b) $x^4/2 + 3xy + y^2/2 - y = C$; (c) $x^3y + (2/3)x^3 - x^2/2 + y^4/2 - y^2 = C$.
59. (a) $ye^x - x^2 = C$; (b) $x \sin(xy) + e^y = C$; (c) $x^2 \sin y + y \cos x = C$.
60. (a) $xy + x^2y^2 + x^2 - x - (3/2)y^2 + 2y = 5/2$; (b) $x^2e^{2y} - 2y = 4$; (c) $x \sin(xy) + e^y = e$.
61. (a) $a = 2$; (b) every real k ; (c) $b = 2$.
62. (a) $x^2y + \int f(x) dx = C$; (b) $e^x \sin y + \int g(y) dy = C$; (c) $x^3 + xy + \int h(y) dy = C$. In each case the named function may be any differentiable function.
63. (a) $x^2y^2 - 6xy^3 = C$; (b) $3x^2y - x^3 - y^3 = C$.
64. (a) $x^2(y^2/2 - y) = C$; (b) $e^{x^2} + \sin y = C$.
65. (a) $F_x = 3x^2y + 2x^2 - x$, $F_y = x^3 + 2y^3 - 2y$. (b) $F_x dx + F_y dy = 0$. (c) $7/6$.

66. (a) $x^5 + 5xy^4 = C$; (b) $x^4 - 4x^3y + 2x^2y^2 + 3y^4 = C$; (c) $2x^4 - 4x^3y + 4x^2y^2 + 3y^4 = C$.
67. (a) $P = 1/x$, $Q = 1/x$, $n = -2$, $u = y^3$. (b) $P = -1$, $Q = e^x$, $n = 2$, $u = y^{-1}$. (c) $P = 2/x$, $Q = 5/x^2$, $n = 3$, $u = y^{-2}$. (d) $P = -3/x$, $Q = -\cos x/x^2$, $n = 2$, $u = y^{-1}$.
68. (a) $y^3 = 1 + C/x^3$; (b) $y^{-1} = Ce^{-x} - e^x/2$; (c) $y^{-2} = 2/x + Cx^4$; (d) $y^{-3} = x^3 + Cx^6$.
69. Add $y = 0$ in every part. (a) $y^{-4} = Ce^{-4x} - x + 1/4$; (b) $y^{-1} = Cx - x^3/2$; (c) $y^{-2} = 1 + Ce^{-2x^2}$; (d) $y^{-3} = 1 + C(1 + x^2)$.
70. (a) $y = [2/(5x^3 - 3x^5)]^{1/5}$. (b) $y = \sqrt{(9/2)x - x^3}$ on the branch through $(2, 1)$. (c) $y = [x^2/(x + 2)]^{1/3}$. (d) $y = \sqrt{(17/16)e^{4x} - x^2/2 - x/4 - 1/16}$.
71. (a) $x^{-2} = y^2[C - 2\ln|y|]$. (b) $x^{-1} = Ce^{-2y} - e^y/3$.
72. Multiplication by $(1 - n)y^{-n}$ gives $u' + (1 - n)Pu = (1 - n)Q$. Division by a power of y can remove $y = 0$, so it must be checked in the original equation.
73. $a = 1$ and $y^{-2} = x^2 + Cx^4$.
74. (a) $y = x/(C - \ln|x|)$, together with $y = 0$; (b) $y^2 + x^2/2 = Cx^3$; (c) $x^5 + 5xy^4 = C$.
75. $P(t) = K/[1 + Ce^{-rt}] = K/[1 + ((K - P_0)/P_0)e^{-rt}]$, together with $P = 0$.
76. $A' = -A/50$, $A = 90e^{-t/50}$ g. $A(25) \approx 54.588$ g, and $A = 35$ at $t = 50 \ln(18/7) \approx 47.223$ s.
77. $A' = 1.6 - (4/75)A$ and $A = 30 - 15e^{-4t/75}$ kg. At 10 min the concentration is ≈ 0.07067 kg L⁻¹; the limiting concentration is 0.10 kg L⁻¹.
78. With $V = 120 + t$, $A' = 15 - 2A/(120 + t)$ and $A = 5(120 + t) - 72000/(120 + t)^2$. $A(120) = 1198.75$ g. Overflow begins at $t = 280$ s, when $A = 1999.55$ g.
79. $V = 120 - 3t$, $0 \leq t < 40$ s, and $A = 5V - V^2/2880$. Thus $A(20) = 298.75$ g and $A/V \rightarrow 5$ g L⁻¹.
80. (a) $A = 30 + 10e^{-t/50}$, $A(20) \approx 36.703$ kg, concentration ≈ 0.18352 kg L⁻¹. (b) $A = 0.3(100 + t) + 1000/(100 + t)$, $A(20) = 44.333$ kg, concentration ≈ 0.18472 kg L⁻¹. (c) $V = 200 - 3t$, $0 \leq t < 200/3$, $A = 0.15V + 10(V/200)^{7/3}$, $A(20) \approx 25.351$ kg, concentration ≈ 0.18108 kg L⁻¹.
81. $A = 1.6 \times 10^9 e^{-0.025t}$ mg. Concentration after 30 days is ≈ 3.779 mg L⁻¹; it falls below 1 mg L⁻¹ after $40 \ln 8 \approx 83.178$ days.
82. $A = (200/3)(1 - e^{-0.18t})$ mg. The limit is 200/3 mg; 90% is reached after $\ln 10/0.18 \approx 12.792$ h.
83. $k = \ln(7/5)/10 \approx 0.03365$ min⁻¹, $T = 22 + 70e^{-kt}$. Then $T(25) \approx 52.18^\circ\text{C}$ and $T = 40^\circ\text{C}$ at $t \approx 40.36$ min.
84. $T = 24 - 20e^{-kt}$, $k = \ln(10/7)/6 \approx 0.05945$ min⁻¹. $T(15) \approx 15.80^\circ\text{C}$ and $T = 20^\circ\text{C}$ at $t \approx 27.07$ min.
85. $P = 1200e^{kt}$, $k = \ln(7/4)/3$. $P(8) \approx 5337$ and doubling time ≈ 3.716 h.
86. $P = 48000e^{0.026t}$. $P(12) \approx 65575$; $P = 70000$ at $t \approx 14.51$ years.
87. $r = \ln(21/16)/5 \approx 0.05439$ year⁻¹. $A(10) = \text{RM } 13,781.25$; doubling time ≈ 12.745 years.
88. $P = 1350e^{-0.645t}$. After 45 min, $P \approx 832.23$ g; half-life ≈ 1.075 h; 95% decays after ≈ 4.645 h.
89. $P = 630(0.844)^{t/2.5}$. $P(12) \approx 279.12$ mg; half-life ≈ 10.217 years.
90. Age = $5730 \ln(1/0.38)/\ln 2 \approx 7999$ years, assuming a constant decay rate and original activity equal to living material.
91. $A = 250 \cdot 2^{-t/5.5}$ mg. $A(12) \approx 55.10$ mg; it falls below 20 mg after ≈ 20.04 h; about 95.14% is eliminated after one day.
92. $P = 5000/[1 + (22/3)e^{-0.30t}]$. $P(10) \approx 3663$; $P = 4000$ at $t \approx 11.262$.
93. For $P(0) = 8000$, $P = 5000 + 3000e^{0.04t}$ and it grows. The equilibrium is 5000. For $P(0) = 4000$, $P = 5000 - 1000e^{0.04t}$ and the model reaches zero at $25 \ln 5$.
94. $mv' = mg - cv$ and $v = (mg/c)(1 - e^{-ct/m})$. Terminal velocity is mg/c , equal to 39.24 m s⁻¹ for the stated data.
95. $q = CE(1 - e^{-t/(RC)})$ and $i = (E/R)e^{-t/(RC)}$. The time constant RC gives 63.2% of the final charge. Here $q(0.5) \approx 1.1015 \times 10^{-3}$ C.
96. $k = \ln(1.6)/18 \approx 0.02611$ min⁻¹. $C(45) \approx 0.2471$ mol L⁻¹; $C = 0.10$ at $t \approx 79.64$ min.

97. $A' + (43/3000)A = 0.20$ and $A = 600/43 + (18 - 600/43)e^{-43t/3000}$. The limit is $600/43 \approx 13.953$ kg; $A(60) \approx 15.666$ kg.
98. The ratios are approximately 0.819 every two hours, so $k \approx -0.09984 \text{ h}^{-1}$. Thus $Q(10) \approx 36.85$ and the half-life is ≈ 6.94 h.
99. For $0 \leq t \leq 20$, $A = 40e^{-t/40}$ kg. For $t > 20$, $A = 40 + 40(e^{-1/2} - 1)e^{-(t-20)/40}$. The limiting concentration is 0.20 kg L^{-1} .
100. $A' + 0.008A = 20$, $A(0) = 30000$, so $A = 2500 + 27500e^{-0.008t}$ mg. The limiting concentration is 5 mg L^{-1} ; it falls below 8 mg L^{-1} after $125 \ln(55/3) \approx 363.59$ min. Without treatment the limit is 10 mg L^{-1} .

Chapter 15: Sequences and Series

These answers correspond to Section 15.11.

1. (a) 1, 3, 5, 7, 9; (b) $-1, 1/2, -1/3, 1/4, -1/5$; (c) 2, 5, 11, 23, 47.
2. (a) $a_n = 3n + 1$; (b) $a_n = (-1)^{n+1}/n$; (c) $a_n = 2 \cdot 3^{n-1}$.
3. (a) Converges to 1; (b) converges to 1; (c) diverges and is unbounded; (d) diverges by oscillation.
4. (a) 3; (b) $2/5$; (c) 0.
5. (a) $3/2$; (b) e ; (c) $+\infty$.
6. (a) Converges to 0 by the squeeze theorem; (b) diverges between -1 and 1 ; (c) converges to 0 because the oscillation has shrinking magnitude.
7. $a_n = 3n + 2$, $a_{20} = 62$, and $S_{20} = 670$.
8. $a_n = 81(1/3)^{n-1}$, $a_6 = 1/3$, and $S_6 = 364/3$.
9. (a) $d = 4$, $a_1 = 3$, $a_n = 4n - 1$; (b) $r = 3$, $g_1 = 2$.
10. $a_{2m} \rightarrow 1$ and $a_{2m-1} \rightarrow -1$. Since the subsequences have different limits, (a_n) diverges.
11. (a) 0; (b) $+\infty$; (c) 0.
12. (a) Increasing, bounded, limit 1; (b) neither increasing nor decreasing, bounded, limit 2; (c) decreasing, bounded, limit 0.
13. $a_n = 6 - 6/2^{n-1}$, $a_5 = 45/8$, and the limit is 6.
14. $b_n = 2n + 1$ and $\sum_{k=1}^n b_k = a_{n+1} - a_1 = n^2 + 2n$.
15. (a) $n = 101$; (b) $n = 697$.
16. (a) $\sum_{k=1}^{10} (3k - 1)$; (b) $\sum_{k=1}^n k^2$; (c) $\sum_{k=1}^8 1/[k(k+1)]$.
17. (a) 32; (b) 189; (c) -3 .
18. Substitution of the standard sums gives $3n(n+1)(2n+1)/6 - 2n(n+1)/2 + n = n(2n^2 + n + 1)/2$.
19. 1275, 42925, and 1625625, respectively.
20. (a) $n(n+1)(n+2)/3$; (b) $n(4n^2 - 1)/3$.
21. With $k = r - 2$, the sum is $\sum_{k=1}^5 k^2 = 55$.
22. n^2 .
23. $n/(n+1)$.
24. Use $1/[(2k-1)(2k+1)] = \frac{1}{2}[1/(2k-1) - 1/(2k+1)]$; the sum is $n/(2n+1)$.
25. $3n^2$.
26. $\sum_{k=1}^n 2k = 2[n(n+1)/2] = n(n+1)$ and $\sum_{k=1}^n (2k-1) = 2[n(n+1)/2] - n = n^2$.
27. $(n+1)(2n+1)$.
28. (a) 1533; (b) $10(1 - 2^{-n})$.
29. $S_n = \frac{2}{3}[1 - (-1/2)^{n+1}]$; $S_5 = 21/32$.
30. (a) 10 terms; (b) 10 terms.
31. (a) Converges to 24; (b) diverges because $|r| = 2$; (c) converges to $9/2$.
32. (a) $|x| < 3$, with sum $3/(3-x)$; (b) $|x| < 1/2$, with sum $1/(1-2x)$.
33. (a) $3/11$; (b) $131/900$.
34. (a) $S_n = \frac{16}{3}(1 - 4^{-n})$, $S_\infty = 16/3$; (b) $S_n = 4[1 - (-1/2)^n]$, $S_\infty = 4$.
35. (a) Diverges because the terms tend to 1; (b) diverges because the terms do not approach zero; (c) the term test is inconclusive because $1/n \rightarrow 0$; the harmonic series nevertheless diverges.
36. 1.

37. The finite sum is $\frac{1}{3}(1 - 1/(3n + 1))$ and the infinite sum is $1/3$.
38. (a) $q = 4/5$; (b) first term 6, common ratio $1/2$.
39. $15/2$.
40. $(a, r) = (6, 1/2)$ or $(18, -1/2)$.
41. 6, 18, 54.
42. $n = 10$.
43. $2S + 1$.
44. $S_n = \frac{1}{2} \left(\frac{3}{2} - \frac{1}{n+1} - \frac{1}{n+2} \right)$, and $S_\infty = 3/4$.
45. $(a, r) = (5, 1/2)$ or $(15, -1/2)$.
46. (a) $(n+2)(n+1)$; (b) $n/(n+1)!$; (c) $(2n)(2n-1)$.
47. (a) $n = 5$; (b) $n = 8$.
48. 1, 6, 15, 20, 15, 6, 1. The row is symmetric because $\binom{6}{r} = \binom{6}{6-r}$.
49. (a) $x^5 + 10x^4 + 40x^3 + 80x^2 + 80x + 32$; (b) $p^5 - 10p^4q + 40p^3q^2 - 80p^2q^3 + 80pq^4 - 32q^5$.
50. $16x^4 - 96x^3 + 216x^2 - 216x + 81$.
51. $T_{r+1} = \binom{n}{r} a^{n-r} b^r$; the fourth term is $280x^4$.
52. (a) 1792; (b) 945.
53. (a) 15; (b) 60.
54. (a) $252a^5b^5$; (b) $126x^5$ and $126x^4$.
55. (a) $k = \pm 3$; (b) $n = 6$.
56. Substitute $x = 1$ in $(1+x)^n$ for (a), $x = -1$ for (b), and differentiate before setting $x = 1$ for (c). The results are 2^n , 0, and $n2^{n-1}$.
- 57.
- (a) $1.1040808032 \approx 1.104081$,
(b) $0.9414801494 \approx 0.941480$.
58. (a) 36; (b) 70.
59. Sum of coefficients = 1; alternating sum = -2187 .
60. $\binom{6}{2}p^2(1-p)^4 = 15p^2(1-p)^4$.
61. (a) $1 - x + x^2 - x^3 + \cdots$; (b) $1 + x/2 - x^2/8 + x^3/16 + \cdots$.
62. $1 + x + \frac{3}{2}x^2 + \frac{5}{2}x^3 + \cdots$, for $|x| < 1/2$.
63. $2 + x/4 - x^2/64 + x^3/512 + \cdots$, for $|x| < 4$.
64. $2 - x/12 - x^2/288 - 5x^3/20736 + \cdots$, for $|x| < 8$.
65. (a) $1/2 + x/4 + x^2/8 + x^3/16 + \cdots$, $|x| < 2$;
(b) $1/3 - 2x/9 + 4x^2/27 - 8x^3/81 + \cdots$, $|x| < 3/2$.
66. $\frac{x+7}{x^2-x-6} = \frac{2}{x-3} - \frac{1}{x+2}$, and $-\frac{7}{6} + \frac{x}{36} - \frac{43x^2}{216} + \frac{49x^3}{1296} + \cdots$, $|x| < 2$.
67. $1 + 2x + \frac{5}{2}x^2 + \frac{8}{3}x^3 + \cdots$.
68. (a) $x^2 - x^6/6 + x^{10}/120 + \cdots$; (b) $1 - 2x + 2x^2 + \cdots$; (c) $x^2 - x^4/2 + x^6/3 + \cdots$.
69. Termwise differentiation gives the stated identity; setting $x = 1/2$ gives $\sum n/2^n = 2$.
70. $-\ln(1-x) = \sum_{n=1}^{\infty} x^n/n$ for $|x| < 1$; setting $x = 1/2$ gives $\ln 2$.
71. (a) $-1 < x < 5$; (b) $-3 < x < 1$. Both endpoints diverge in each case.
72. (a) 0.98995; (b) 1.01990.
73. (a) 0.952375; (b) 2.024846.
74. (a) 240; (b) $1/16$.
75. $\ln(1.2) \approx 0.1822667$. The next term has magnitude $0.2^5/5 = 0.000064$.
76. $T_3(x) = \sum_{k=0}^3 f^{(k)}(a)(x-a)^k/k!$; for e^x about 1 it is $e[1 + (x-1) + (x-1)^2/2 + (x-1)^3/6]$.

77. With $h = x - \pi/3$,

$$\cos x = \frac{1}{2} - \frac{\sqrt{3}}{2}h - \frac{1}{4}h^2 + \frac{\sqrt{3}}{12}h^3 + \dots$$
78. With $h = x - 1$,

$$\ln x = h - h^2/2 + h^3/3 - h^4/4 + \dots;$$

$$\ln(1.1) \approx 0.0953083.$$
- 79.
- $$e^x = 1 + x + x^2/2 + x^3/6 + \dots,$$
- $$\sin x = x - x^3/6 + x^5/120 - x^7/5040 + \dots,$$
- $$\cos x = 1 - x^2/2 + x^4/24 - x^6/720 + \dots,$$
- $$\ln(1+x) = x - x^2/2 + x^3/3 - x^4/4 + \dots.$$
80. (a) $x - 3x^3 + \frac{9}{2}x^5 - \frac{9}{2}x^7 + \dots$; (b)
 $3x^2 - \frac{9}{2}x^4 + \frac{81}{40}x^6 - \frac{243}{560}x^8 + \dots$
81. The coefficient is $-16/3$. Both sides of the identity expand to
 $1 - 2x^2 + \frac{2}{3}x^4 - \frac{4}{45}x^6 + \dots$
82. (a) $C - x/2 + x^3/72 - x^5/3600 + \dots$; (b)
 $C + x - x^3/18 + x^5/600 + \dots$
83. (a) 0.099668 approximately; (b) 0.496884 approximately; (c) 0.333268 approximately.
84. Regard $x - x^3/6$ as the degree-4 polynomial, whose x^4 coefficient is zero. Then
 $|R_4(x)| \leq 0.5^5/5! = 1/3840 \approx 0.0002604$.
85. Yes. The remainder satisfies

$$\frac{e^{0.1}(0.1)^5}{5!} \approx 9.21 \times 10^{-8} < 10^{-6}.$$
86. Tenth-year monthly salary RM 3,750; sum of the ten annual rates RM 30,750.
87. Last row 76 seats; total capacity 720 seats.
88. The height after the fifth bounce is
 $10(0.7)^5 = 1.6807$ m. The total distance is

$$10 + 20 \sum_{n=1}^{\infty} (0.7)^n = \frac{170}{3} \approx 56.67$$
 m.
89. $A_n = 500(1 - 0.8^n)$ mg and $A_n \rightarrow 500$ mg.
90. $500 \frac{1.01^{12} - 1}{0.01} \approx \text{RM } 6,341.25$.
91. $P_n = 5000(1.05)^n - 4000$ and
 $P_{10} \approx 4144.47$.
92. Remaining fraction $(3/4)^6 = 729/4096$; the total fraction eventually removed is 1.
93. $e^{-0.1} \approx 1 - 0.1 + 0.1^2/2 - 0.1^3/6 = 0.904833$; hence $T(1) \approx 92.39^\circ\text{C}$.
94. Relative volume increase $\approx 3\alpha\Delta T = 0.0018$, or 0.18%.
95. Quadratic estimate:
 $10000[1 + 5(0.04) + 10(0.04)^2] = \text{RM } 12,160$.
 Calculator value: RM 12,166.53 approximately.
96. (a) $\binom{5}{3}(0.2)^3(0.8)^2 = 0.0512$; (b)
 $1 - (0.8)^5 = 0.67232$.
97. (a) $(0.99)^{10} \approx 0.90438$; (b)
 $10(0.01)(0.99)^9 \approx 0.09135$.
98. $\sin(0.2) \approx 0.2 - 0.2^3/6 + 0.2^5/120 = 0.1986693$. The next-term magnitude is
 $0.2^7/7! \approx 2.54 \times 10^{-9}$.
99. Here $h/R = 1/16$. The quadratic estimate is $1 - 2(1/16) + 3(1/16)^2 = 0.886719$; the exact value is
 $(17/16)^{-2} = 256/289 \approx 0.885813$.
100. First 12 weeks: $10000(1 - 0.9^{12}) \approx 7175.70$ units; limiting total 10000 units; first below 300 units in week 13.

Chapter 16: Vectors

These answers correspond to Section 16.16.

- (a) Vector; (b) scalar; (c) scalar; (d) scalar.
- (a) Magnitude $\sqrt{34}$, unit vector $\frac{5\mathbf{i} - 3\mathbf{j}}{\sqrt{34}}$; (b) $\sqrt{5}$, $\frac{2\mathbf{i} + \mathbf{j}}{\sqrt{5}}$; (c) $\sqrt{65}$, $\frac{7\mathbf{i} - 4\mathbf{j}}{\sqrt{65}}$; (d) $3\sqrt{17}$, $\frac{4\mathbf{i} + \mathbf{j}}{\sqrt{17}}$.
- $\frac{\mathbf{i} + 2\mathbf{j}}{\sqrt{5}}$.
- $4 + n^2 = 29$, so $n = \pm 5$.
- (a) $\overrightarrow{CB}$; (b) $\overrightarrow{CB}$; (c) $2\mathbf{r}_D + \mathbf{r}_F - \mathbf{r}_A - 2\mathbf{r}_C$; (d) $2\mathbf{r}_E - \mathbf{r}_B - \mathbf{r}_F$, where $\mathbf{r}_X$ denotes the position vector of X .
- (a) False; (b) true; (c) true; (d) false; (e) true; (f) true.
- (a) $\mathbf{a} + \mathbf{b}$; (b) $\mathbf{e} - \mathbf{a} - \mathbf{b}$; (c) $-\mathbf{x}$; (d) $\mathbf{y} - \mathbf{x}$.
- The directed sides form a closed route, so $\overrightarrow{PQ} + \overrightarrow{QR} + \overrightarrow{RS} + \overrightarrow{SP} = \overrightarrow{PP} = \mathbf{0}$.
- (a) $\overrightarrow{OP} = \frac{1}{2}\mathbf{a}$, $\overrightarrow{OQ} = \frac{1}{2}(\mathbf{a} + \mathbf{b})$, $\overrightarrow{OR} = \frac{1}{2}(\mathbf{b} + \mathbf{c})$, and $\overrightarrow{OS} = \frac{1}{2}\mathbf{c}$. (b) $\overrightarrow{PQ} = \overrightarrow{SR} = \frac{1}{2}\mathbf{b}$.
- $\overrightarrow{AC} = 3(\mathbf{b} - 2\mathbf{a})$ and $\overrightarrow{AB} = 5(\mathbf{b} - 2\mathbf{a})$; hence $\overrightarrow{AC} = \frac{3}{5}\overrightarrow{AB}$.
- (a) $\overrightarrow{ON} = \frac{6}{5}\mathbf{a} + \frac{3}{5}\mathbf{b}$. (b) $\overrightarrow{BN} = \frac{2}{5}(3\mathbf{a} - \mathbf{b})$ and $\overrightarrow{NC} = \frac{3}{5}(3\mathbf{a} - \mathbf{b})$, so the vectors are parallel and B, N, C are collinear.
- (a) $P = (-12, -1)$. (b) Using the directed ratio $-5 : 3$ gives $P = \frac{-5B + 3A}{-2} = \left(11, \frac{31}{2}\right)$.
- (a) $30\mathbf{i} + 20\mathbf{j}$; magnitude $10\sqrt{13}$ and direction $N56.3^\circ E$. (b) Approximately $2.124\mathbf{i} - 4.965\mathbf{j}$; magnitude 5.401 and direction $S23.2^\circ E$. (c) Magnitude 23 m s^{-1} in the direction $N29.3^\circ W$.
- The second force is (a) $-7.010\mathbf{i} + 7.500\mathbf{j}$ N, with magnitude 10.266 N and direction 133.1° from the positive x -axis; (b) $6.990\mathbf{i} - 2.892\mathbf{j}$ N, with magnitude 7.565 N and direction 337.5° .
- (a) $-30\mathbf{i} - 10\mathbf{j}$; magnitude $10\sqrt{10}$ and direction $S71.6^\circ W$. (b) $-45\mathbf{i} - 70\mathbf{j}$; magnitude $5\sqrt{277}$ and direction $S32.7^\circ W$. (c) Approximately $4.510\mathbf{i} + 22.811\mathbf{j}$; magnitude 23.252 and direction $N11.2^\circ E$.
- The resultant is approximately $10.392\mathbf{i} - 9\mathbf{j}$ N, with magnitude 13.748 N and bearing 130.9° .
- If each force has magnitude F , then $50 = 2F \cos 20^\circ$; hence $F \approx 26.60$ N.
- $\mathbf{v}_{\text{car/bus}} = 6\mathbf{i} + 28\mathbf{j} \text{ m s}^{-1}$.
- $-24\mathbf{i} + 32\mathbf{j} \text{ km h}^{-1}$; magnitude 40 km h^{-1} in the direction $N36.9^\circ W$.
- $-50\mathbf{i} - 120\mathbf{j} \text{ km h}^{-1}$; magnitude 130 km h^{-1} in the direction $S22.6^\circ W$.
- $\mathbf{A}\mathbf{v}_B = \mathbf{v}_B - \mathbf{v}_A$, whereas $\mathbf{B}\mathbf{v}_A = \mathbf{v}_A - \mathbf{v}_B = -\mathbf{A}\mathbf{v}_B$.
- (a) True; (b) true; (c) true; (d) true; (e) true, provided the relative motion reaches zero separation at a positive time.
- Taking the first rider's direction as east and the second rider's direction as north gives $-5\mathbf{i} + 3\mathbf{j} \text{ km h}^{-1}$. Its magnitude is $\sqrt{34} \text{ km h}^{-1}$ and its direction is $N59.0^\circ W$.
- One suitable coordinate diagram takes $C = (0, 0)$, giving $A = (2\sqrt{2}, 2\sqrt{2})$, $B = (2\sqrt{2} - 2, 2\sqrt{2})$, and $D = (\sqrt{3}/2, -1/2)$, with east as the positive x -direction.
- The relative position and velocity of the lorry from the car are $\mathbf{r}_0 = 20\mathbf{j}$ and

- $\mathbf{v} = (-31.699\mathbf{i} - 25\mathbf{j}) \text{ km h}^{-1}$. Thus (a) $t = -\mathbf{r}_0 \cdot \mathbf{v} / \|\mathbf{v}\|^2 \approx 0.3068 \text{ h}$, or 18.4 min; (b) the minimum separation is approximately 15.70 km.
26. (a) $5\mathbf{i} - 12\mathbf{j} \text{ ms}^{-1}$. (b) Magnitude 13 ms^{-1} , direction $S22.6^\circ E$. (c) The relative path begins 26 km west of B and proceeds south-east. (d) The shortest distance is 24 km. (e) After 3 min the separation is approximately 25.19 km.
27. The required course is approximately $S34.1^\circ W$ (bearing 214.1°). Interception occurs after about 0.2933 h, or 17.6 min.
28. The rescue boat should sail on bearing 057.2° . The interception time is approximately 5.228 h after 11.00 a.m., or about 4.14 p.m.
29. (i) $d = \sqrt{53}$, midpoint $(-1/2, -1, 1)$, $\overrightarrow{AB} = \langle 7, -2, 0 \rangle$; (ii) $d = 9$, midpoint $(-4, -1/2, 2)$, $\overrightarrow{AB} = \langle -8, -1, 4 \rangle$.
30. (a) $a = -5/2$, $b = 1$. (b) $\frac{4\mathbf{i} - 4\mathbf{j} + 3\mathbf{k}}{\sqrt{41}}$. (c) $\mathbf{i} - \mathbf{j} + \frac{3}{4}\mathbf{k}$.
31. (a) $\langle 2/3, -2/3, 1/3 \rangle$; (b) $\langle 1, 6, -3 \rangle / \sqrt{46}$.
32. (a) $2\mathbf{a} + \mathbf{b} = 4\mathbf{i} + 2\mathbf{k} = 2(2\mathbf{i} + \mathbf{k})$. (b) $\mathbf{a} + \mathbf{b} + \mathbf{c} = 2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$, whose magnitude is 7.
33. (a) Approximately $\frac{5}{3}\mathbf{i} - \frac{10}{3}\mathbf{j} + \frac{10}{3}\mathbf{k}$. (b) Using the stated rounded direction cosines gives approximately $1.05\mathbf{i} - 2.25\mathbf{j} + 4.35\mathbf{k}$.
34. $P = (-2, 4, 4)$.
35. (a) 0, orthogonal; (b) -1 , not orthogonal; (c) 34, not orthogonal; (d) $-6\sqrt{42}$.
36. $\mathbf{g} \cdot \mathbf{h} = 2 \cos 135^\circ = -\sqrt{2}$.
37. (a) 7; (b) $\cos^{-1}\left(\frac{7}{3\sqrt{11}}\right) \approx 45.2^\circ$; (c) $k = 11/7$.
38. (a) True; (b) false; (c) true; (d) false; (e) false; (f) true; (g) true.
39. (a) 14; (b) -9 ; (c) 5; (d) $\cos^{-1}\left(\frac{8}{\sqrt{76}}\right) \approx 23.4^\circ$.
40. (a) -12 ; (b) -99 ; (c) $\langle -1/10, 1/2, -1/5 \rangle$; (d) $\mathbf{0}$.
41. $\mathbf{g} \cdot \mathbf{h} = -6 + 36 - 30 = 0$, so the vectors are perpendicular.
42. (a) $m = -1/2$; (b) $n = 2$.
43. The scalar components are (a) 1; (b) -1 .
44. (a) Scalar $13/\sqrt{29}$; vector $\frac{13}{29}\langle 4, -2, 3 \rangle$. (b) Scalar $26/\sqrt{98}$; vector $\frac{13}{49}\langle 9, 1, -4 \rangle$. (c) Scalar $60/\sqrt{126}$; vector $\frac{10}{21}\langle 11, 2, 1 \rangle$.
45. Taking the unit vector up the road as $\mathbf{u} = \langle \cos 30^\circ, \sin 30^\circ \rangle$, the scalar component of the weight is -5.4 N . The vector projection is $-5.4\mathbf{u} = \langle -2.7\sqrt{3}, -2.7 \rangle \text{ N}$; it acts down the road.
46. The component parallel to the plane is $22g \cos 12^\circ \approx 211.1 \text{ N}$.
47. Apply $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + 2\mathbf{u} \cdot \mathbf{v}$ with $\mathbf{u} = \text{proj}_{\mathbf{a}} \mathbf{b}$ and $\mathbf{v} = \text{proj}_{\mathbf{b}} \mathbf{a}$.
48. (a) True; (b) false; (c) true; (d) false; (e) true for non-zero vectors.
49. The vectors must be parallel. Since the first component fixes the scale factor as $1/3$, $s = 2/3$ and $t = 1/3$.
50. Suitable answers are (a) $-3\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$; (b) $-7\mathbf{i} + 15\mathbf{j} + 12\mathbf{k}$.
51. Either of the two unit vectors $\pm \frac{-\mathbf{i} - \mathbf{j} + \mathbf{k}}{\sqrt{3}}$.
52. (a) $AB = AC = \sqrt{5}$; (b) $\angle BAC = \cos^{-1}(3/5) \approx 53^\circ$; (c) area = 2 square units.
53. (a) $\sqrt{2098}$ square units; (b) $\sqrt{8018}/2$ square units.
54. (a) 16 cubic units; (b) 4 cubic units.

55. The four differences are $\langle 1, 1, 2 \rangle$, $\langle 1, -2, 2 \rangle$, $\langle -1, -1, -2 \rangle$, and $\langle 1, -2, 2 \rangle$. The area is $3\sqrt{5}$ square units.
56. (a) $\mathbf{r} = \langle 2, 1, -1 \rangle + t\langle -5, -2, -1 \rangle$;
 $x = 2 - 5t$, $y = 1 - 2t$, $z = -1 - t$;
 $\frac{x-2}{-5} = \frac{y-1}{-2} = \frac{z+1}{-1}$. (b)
 $\mathbf{r} = \langle -1, 4, 2 \rangle + t\langle 4, 5, 3 \rangle$ and
 $\frac{x+1}{4} = \frac{y-4}{5} = \frac{z-2}{3}$. (c)
 $\mathbf{r} = \langle 1, -1, 2 \rangle + t\langle -2, 1, 0 \rangle$; $x = 1 - 2t$,
 $y = -1 + t$, $z = 2$; $\frac{x-1}{-2} = y + 1$, $z = 2$.
57. $\mathbf{r} = \langle 5, -2, 4 \rangle + t\langle -3, 8, -3 \rangle$. The coordinate-plane intersections are $(0, 34/3, -1)$, $(17/4, 0, 13/4)$, and $(1, 26/3, 0)$.
58. The second direction vector is -2 times the first, and $\langle 1, 2, 3 \rangle - \langle 4, 6, -9 \rangle = -\langle 3, 4, -12 \rangle$; hence the equations describe the same line.
59. A suitable unit vector is $\pm \frac{20\mathbf{i} + 9\mathbf{j} - 7\mathbf{k}}{\sqrt{530}}$.
60. (a) They intersect at $(5, -7, 3)$; (b) at $(-2, 4, 0)$; (c) at $(1, 1, -1)$.
61. The acute angle is $\cos^{-1}\left(\frac{15}{\sqrt{38}\sqrt{33}}\right) \approx 64.9^\circ$; the supplementary angle is 115.1° .
62. (a)(i) $x = -2$ and $\mathbf{r} = \langle -2, 5, -8 \rangle + s\mathbf{j} + t\mathbf{k}$; (ii) $y = 5$ and $\mathbf{r} = \langle -2, 5, -8 \rangle + s\mathbf{i} + t\mathbf{k}$; (iii) $z = -8$ and $\mathbf{r} = \langle -2, 5, -8 \rangle + s\mathbf{i} + t\mathbf{j}$. (b) $4x + 2y - 7z + 27 = 0$ and, for example, $\mathbf{r} = \langle 3, -2, 5 \rangle + s\langle 1, -2, 0 \rangle + t\langle 7, 0, 4 \rangle$. (c) $x - 2y + 2z = 7$ and $\mathbf{r} = \langle 1, -1, 2 \rangle + s\langle 2, 1, 0 \rangle + t\langle -2, 0, 1 \rangle$. (d) $9x + y + 4z = 19$ and $\mathbf{r} = \langle 2, 5, -1 \rangle + s\langle 0, -8, 2 \rangle + t\langle -1, -3, 3 \rangle$.
63. $\mathbf{r} = \langle 4, 6, 3 \rangle + s\langle 2, -1, 3 \rangle + t\langle 0, 2, 1 \rangle$. Its Cartesian equation is $7x + 2y - 4z = 28$. The point P lies on the plane ($s = -1$, $t = -4$).
64. One parametrisation is $x = 3 + 4t$, $y = 4 + 7t$, $z = 3 + 5t$, $t \in \mathbb{R}$.
65. (a) Parallel and distinct; (b) perpendicular; (c) neither; (d) coincident.
66. (a) No: two planes are parallel, coincident, or meet in a line. (b) Their scalar triple product is zero precisely when they are coplanar. (c) Examples include a point and a normal vector, three non-collinear points, or a point with two non-parallel direction vectors. (d) The line's direction vector must be parallel to the plane's normal.
67. (a) $\sqrt{65}/3$; (b) $|a|\sqrt{78}/3$; (c) $\sqrt{78}/3$; (d) $\sqrt{78}/3$.
68. (a) $11\sqrt{26}/39$; (b) $4\sqrt{5}/5$.
69. (a) 3.90; (b) 3.14.
70. $\sqrt{5}$.
71. (a) The intersection point is $(-4, 1, -5)$. (b) The distance is $4\sqrt{42}/7$. (c) The acute angle is approximately 19.1° .
72. (a) $5/3$; (b) $6/5$.
73. The plane is $2x + y + z = 5$; the distance is $5\sqrt{6}/6$.
74. (a) $3\sqrt{14}/14$; (b) $31\sqrt{14}/126$.
75. $1/6$.
76. (a) $[1, 2]$; (b) $\mathbb{R} \setminus \{\pi/2 + k\pi : k \in \mathbb{Z}\}$; (c) the same set; (d) $\mathbb{R} \setminus \{-1, 0\}$; (e) $(0, 2]$; (f) $\mathbb{R}$.
77. (a) $\mathbf{r}(0) = \langle 0, 1 \rangle$, $\mathbf{r}(1) = \langle 3, -8 \rangle$; (b) $\mathbf{r}(1) = \langle 0, 3 \rangle$, $\mathbf{r}(2) = \langle 7, 6 \rangle$; (c) $\mathbf{r}(-1) = \langle 2, 3, -1 \rangle$, $\mathbf{r}(0) = \langle 3, 2, 1 \rangle$.
78. (a) $x + y = 2$, a straight line; (b) $x^2 + y^2 = 81$, a circle of radius 9; (c) $x^2 + (y - 2)^2 = 4$, a circle centred at $(0, 2)$.
79. (a) $z = 0$, $y = x^2/4$ with $x \geq 0$; (b) $z = 0$, $x^2 + y^2 = 1$, beginning at $(0, -1, 0)$ and travelling anticlockwise when viewed from the positive z -axis.
80. (a) $x^2 + y^2/16 = 1$, clockwise from $(0, 4)$; (b) $x^2/9 + y^2/4 = 1$, anticlockwise from $(3, 0)$;

- (c) $x^2 + y^2 = 4$, the upper semicircle from $(2, 0)$ to $(-2, 0)$.
81. (a) $\mathbf{r}(t) = \langle -2 + 6t, 1 + 4t \rangle$, $0 \leq t \leq 1$; (b) $\mathbf{r}(t) = \langle 5 \cos t, -5 \sin t \rangle$; (c) $\mathbf{r}(t) = \langle 3 \cos t, 2 \sin t \rangle$, with $0 \leq t \leq 2\pi$ in (b) and (c).
82. (a) A unit circle in the plane $z = 2$, anticlockwise from $(1, 0, 2)$. (b) An ellipse $x^2/36 + z^2/625 = 1$ in the plane $y = 4$, traversed twice over the stated interval. (c) The parabola $x = y^2 + 1$ in the plane $z = 3$, with y increasing. (d) A circular helix of radius 3 about the x -axis, moving in the positive x -direction. (e) A circular helix of radius 3 about the z -axis, moving downward. (f) An elliptical helix about the y -axis, moving in the negative y -direction. (g) A planar logarithmic spiral $r = e^t$, $z = 0$, moving anticlockwise outward from radius 1 to radius e^π .
83. Both trace the unit circle. $\mathbf{r}_1$ begins at $(1, 0)$ and moves anticlockwise; $\mathbf{r}_2$ begins at $(0, 1)$ and moves clockwise.
84. The limit is taken componentwise:
 $\lim_{t \rightarrow a} \mathbf{r}(t) = \langle \lim f(t), \lim g(t), \lim h(t) \rangle$,
 which has vector components.
85. (a) $2\mathbf{i} + \frac{1}{2}\mathbf{k}$; (b) $\mathbf{0}$; (c) $\pi\mathbf{i} - \mathbf{j}$; (d) no finite vector limit, because the $\mathbf{k}$ component is unbounded.
86. Every component must be continuous at a , equivalently $\lim_{t \rightarrow a} \mathbf{r}(t) = \mathbf{r}(a)$.
87. (a) Continuous at $t = 0$ but undefined, hence not continuous, at $t = 2$. (b) Undefined at $t = 0$ (although the first component has a removable limit) and continuous at $t = 2$.
88. Domain $(0, \infty)$; $\mathbf{r}'(t) = \langle 1/t, -3e^{-3t}, 2t \rangle$ and $\mathbf{r}''(t) = \langle -1/t^2, 9e^{-3t}, 2 \rangle$.
89. (a) $(2t + 3t^2)e^{3t} + (3t^2 + 2t^3)e^{2t} + (1 + t)e^t$.
 (b) Write $\frac{d}{dt}(\mathbf{r} \times \mathbf{s}) = A(t)\mathbf{i} + B(t)\mathbf{j} + C(t)\mathbf{k}$,
- where
- $$A(t) = (t^3 + 3t^2)e^t - (1 + 2t)e^{2t},$$
- $$B(t) = (1 + 3t)e^{3t} - (t^2 + 2t)e^t,$$
- $$C(t) = (2t + 2t^2)e^{2t} - (3t^3 + 3t^2)e^{3t}.$$
90. $\mathbf{R}(\lambda) = \left\langle \frac{\sqrt{2}}{2}, \sqrt{2} \right\rangle + \lambda \langle 1, -2 \rangle$.
91. $y = x^2/4 - 4$, with $-4 \leq x \leq 6$ and motion from left to right as t increases. Also $\mathbf{r}(1) = \langle 2, -3 \rangle$ and $\mathbf{r}'(1) = \langle 2, 2 \rangle$.
92. $\mathbf{v}(t) = \langle 3, 4t, -\cos t \rangle$,
 $\|\mathbf{v}(t)\| = \sqrt{9 + 16t^2 + \cos^2 t}$, and
 $\mathbf{a}(t) = \langle 0, 4, \sin t \rangle$.
93. (a) $-\cos t \mathbf{i} + 6t \mathbf{j} + 2t^2 \mathbf{k} + \mathbf{C}$; (b) $32\mathbf{i} + 2(e^4 - 1)\mathbf{j} + \ln 3 \mathbf{k}$.
94. $\mathbf{v}(t) = \langle 2t^2 + 1, 3t^2 - 1, t + 1 \rangle$ and
 $\mathbf{r}(t) = \langle \frac{2}{3}t^3 + t + 1, t^3 - t, \frac{1}{2}t^2 + t \rangle$.
95. (a) With $\omega = \pi/6$, $\mathbf{r} = \langle 20 \cos \omega t, 20 \sin \omega t \rangle$,
 $\mathbf{v} = \langle -20\omega \sin \omega t, 20\omega \cos \omega t \rangle$, and
 $\mathbf{a} = -\omega^2 \mathbf{r}$. (b) $\mathbf{r}(3) = \langle 0, 20 \rangle$ m. (c) Speed $10\pi/3 \text{ m s}^{-1}$ and acceleration magnitude $5\pi^2/9 \text{ m s}^{-2}$.
96. Let $\theta(t) = \pi/3 - 0.4t$. Then
 $\mathbf{r}(t) = \langle 3 + 8 \cos \theta(t), -2 + 8 \sin \theta(t) \rangle$.
97. Differentiating $\mathbf{r} = R\langle \cos \omega t, \sin \omega t \rangle$ gives
 $\mathbf{v} = R\omega \langle -\sin \omega t, \cos \omega t \rangle$ and $\mathbf{a} = -\omega^2 \mathbf{r}$.
 Their dot product is zero.
98. (a) At maximum height the position is approximately $80.31\mathbf{i} + 33.69\mathbf{j}$ m. (b) The flight time is approximately 5.242 s. (c) The range is approximately 160.62 m.
99. The flight time is $\sqrt{200/9.81} \approx 4.515$ s.
 The initial speed is approximately 19.93 m s^{-1} , and the impact velocity is approximately $19.93\mathbf{i} - 44.29\mathbf{j} \text{ m s}^{-1}$.
100. Yes. The horizontal coordinate equals 21 when $t = 1$ s, and then
 $y = 13 - \frac{1}{2}(9.81) = 8.095$ m.

Chapter 17: Introduction to Graph Theory

These answers correspond to Section 17.16.

- Order 6, size 6; $\deg(A), \dots, \deg(F) = (2, 3, 3, 3, 1, 0)$. The neighbours of D are B, C, E ; E is pendant and F is isolated.
- The degree sum must be 22, so the missing degree is 6. It is possible: join one vertex to all six others, then use residual edges BC, BD, BE, CD, CF for residual degrees 3, 3, 2, 1, 1, 0.
- (a) Graphical, for example with edges AB, AC, AD, BC, BE, DE ; (b) not graphical, since a degree-4 vertex cannot coexist with an isolated vertex on five vertices; (c) graphical, realised by C_4 ; (d) not graphical: three degree-3 vertices force the fourth vertex to have degree 3, not 1.
- There are only $\binom{n}{2} = n(n-1)/2$ unordered pairs of distinct vertices, and a simple graph has at most one edge per pair. For $n = 12$, the maximum is 66.
- P_7 : order 7, size 6, degrees 2, 2, 2, 2, 1, 1. C_7 : order 7, size 7, all degrees 2. K_7 : order 7, size 21, all degrees 6. $K_{1,6}$: order 7, size 6, degrees 6, 1, 1, 1, 1, 1, 1.
- (a) $\binom{8}{2} - 11 = 17$; (b) $7 - 3 = 4$; (c) $\overline{K_5}$ is five isolated vertices, while $\overline{C_5} \cong C_5$.
- (a) The degree sum is $nr = 2|E|$, so $|E| = nr/2$ and nr is even; (b) 15 edges; (c) $9 \cdot 3 = 27$ is odd, contradicting the handshaking lemma.

8.

v	A	B	C	D
$\deg^-(v)$	1	2	2	1
$\deg^+(v)$	2	1	2	1

Both columns of totals sum to $6 = |E|$.

- $(\deg A, \deg B, \deg C) = (4, 3, 1)$. Their sum is 8, twice the four edges; the loop contributes 2 to $\deg A$.
- The induced subgraph has vertices $\{B, C, D, E\}$ and edges $\{BC, BD, CD, DE\}$, so its order and size are both 4.
- Both graphs have six vertices of degree 2, but C_6 is connected whereas $C_3 \cup C_3$ has two components. Isomorphisms preserve connectedness.
- K_4 is simple, undirected, connected, 3-regular, complete, and planar. It is neither bipartite nor weighted as stated.
- A subgraph deletes any vertices or edges; a spanning subgraph keeps all four vertices; an induced subgraph contains every K_4 edge between its chosen vertices. Examples are one edge, a spanning path, and the triangle induced by any three vertices, respectively.
- Edges within each original bipartition class are absent from $K_{m,n}$ and therefore present in its complement; all cross-edges are removed. Hence $\overline{K_{m,n}} = K_m \cup K_n$.
- The total degree is even. The sum of the even degrees is even, so the sum of the odd degrees is even; a sum of odd integers is even only when their number is even.

16. (a) A closed walk, trail, circuit, and 3-cycle; (b) a walk and trail, but not a path because B repeats; (c) a walk, trail, and path; (d) only a walk, since edge BC is repeated.
17. From A , the distances to A, B, C, D, E are $0, 1, 1, 2, 3$. Thus $e(A) = 3$, the diameter is 3, the radius is 2, and the centres are B, C, D .
18. (a) Components $\{A, B, C\}$ and $\{E\}$; (b) components $\{A, B, C, D\}$ and $\{E\}$; (c) one component $\{A, C, D, E\}$.
19. The only bridge is DE , and the only cut vertex is D . Deleting either separates E from the rest of the graph.
20. The shortest length is 3, for example $A - B - D - E$ or $A - C - D - E$. There are four simple paths: these two and $A - B - C - D - E$, $A - C - B - D - E$.
21. The components are $\{1, 2, 3\}$, $\{4, 5, 6\}$, and $\{7\}$, so the graph is disconnected.
22. A connected graph contains a spanning tree, which uses $n - 1$ edges. Equality occurs exactly when the connected graph is itself a tree.
23. Degrees are $2, 1, 1, 2, 2$ for $1, 2, 3, a, b$, respectively; the size is 4. Colour all vertices of X one colour and all vertices of Y a second colour.
24. C_{10} and $K_{2,5}$ are bipartite. C_{11} and the triangle with a pendant edge are not, because each contains an odd cycle.
25. If u, v lie in different components of G , then uv is an edge of $\overline{G}$. If they lie in the same component, choose w in another component; then $u - w - v$ is a path in $\overline{G}$. Thus $\overline{G}$ is connected.
26.
$$\begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}.$$
27. The degrees are $(2, 3, 3, 3, 1)$, whose sum 12 gives size 6. Vertex E is pendant; there is no isolated vertex.

28.

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad A^2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}.$$

The three requested entries count, respectively, one closed two-step walk at 1, one two-step walk from 1 to 3, and two closed two-step walks at 2.

29. Every triangle produces six directed closed walks of length 3: three starting vertices and two directions. In K_4 , $\text{tr}(A^3) = 24$, so there are $24/6 = 4$ triangles.

30.
$$\begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$
 Row sums are outdegrees and column sums are indegrees.

31. $\begin{pmatrix} 0 & 4 & 7 \\ 4 & 0 & 2 \\ 7 & 2 & 0 \end{pmatrix}$.
32. $J - I$ has 1 in every off-diagonal position; subtracting A exchanges adjacency and non-adjacency. For P_4 with edges 12, 23, 34, the complement has edges 13, 14, 24.
33. The graph is C_4 . It is connected, 2-regular, and bipartite, with opposite vertices in the same bipartition class.
34. The requested numbers are 2, 0, and 2. The two length-3 walks to E are $A - B - D - E$ and $A - C - D - E$.
35. Undirected adjacency gives $a_{ij} = a_{ji}$; simplicity gives $a_{ii} = 0$. The i th entry of $A\mathbf{1}$ is the sum of row i , hence $\deg(v_i)$.
36. A tree on 42 vertices has 41 edges and degree sum 82.
37. A forest with n vertices and c components has $n - c$ edges, so the answer is $18 - 4 = 14$.
38. If there are ℓ leaves, then $\ell + 3(3) + 4(2) = 2[(\ell + 7) - 1]$, giving $\ell = 5$. The order is 12 and the size is 11.
39. One construction is $\{AB, AC, AD, AE, BF, CG\}$. Its degrees are 4, 2, 2, 1, 1, 1, 1.
40. (a) is a spanning tree: it is connected on all five vertices and has four edges. (b) is not: ABC is a cycle and DE is a separate component.
41. The unique path between the endpoints of a tree edge uses that edge, so deletion disconnects them. A new edge joins two vertices already connected by a unique path, producing exactly that path plus the new edge as one cycle.
42. A spanning tree uses 14 edges, so $20 - 14 = 6$ edges must be removed.
43. Fix a root and place vertices at even distance in one class and at odd distance in the other. Adjacent vertices have distances of opposite parity, so this is a bipartition.
44. If an endpoint of a longest path had another neighbour outside the path, the path could be extended; if it had another neighbour on the path, a cycle would result. Hence both endpoints are leaves.
45. For example, $T_1 = \{12, 23, 34\}$ and $T_2 = \{13, 14, 24\}$ are edge-disjoint spanning trees.
46. (a) Euler circuit; (b) open Euler trail only, with the degree-1 and degree-3 vertices as endpoints; (c) neither, because there are four odd vertices.
47. The original graph has four odd vertices B, C, D, E , so it has no Euler trail. Adding BE makes only C, D odd, so an open Euler trail can start at one and finish at the other.
48. Yes. Every degree is even, so an Euler circuit exists and may start at the depot.
49. Every internal visit to a vertex uses one entering and one leaving edge. Only the start and finish can have unmatched incident edges, so they must be the two odd vertices u, v .

50. Two added edges suffice, pairing the four odd vertices. Examples are AB with CD , or AC with BD .
51. (a) Two triangles sharing exactly one vertex: all degrees are even, but the shared cut vertex prevents a Hamilton cycle; (b) K_4 , which has a Hamilton cycle but four odd degrees and hence no Euler circuit.
52. P_n has an Euler trail and a Hamilton path, but no Euler circuit and no Hamilton cycle for $n > 2$.
53. C_n has an Euler circuit and a Hamilton cycle, and therefore also an Euler trail and Hamilton path.
54. K_n has an Euler circuit exactly when n is odd, since every degree is $n - 1$. It has a Hamilton cycle for every $n \geq 3$.
55. Traversing a bridge moves from one side of the cut to the other. Returning to the starting side would require traversing that bridge a second time, contradicting an Euler circuit.
56. $F = 2 - V + E = 2 - 10 + 15 = 7$.
57. $E = V + F - 2 = 18 + 11 - 2 = 27$.
58. Since $3F = 2E$ and $V - E + F = 2$, one obtains $E = 3V - 6 = 30$ and $F = 20$.
59. Since $4F = 2E$, Euler's formula gives $E = 2V - 4 = 16$ and $F = 8$.
60. K_5 has 10 edges, but a simple planar graph on five vertices can have at most $3(5) - 6 = 9$; hence K_5 is non-planar.
61. $K_{3,3}$ is bipartite with 6 vertices and 9 edges, but a bipartite planar graph can have at most $2(6) - 4 = 8$ edges.
62. Here $15 = 3(7) - 6$, so the necessary inequality is satisfied. Satisfaction is not sufficient for planarity, so no conclusion follows.
63. The chromatic numbers are, respectively, 2, 3, 6, 2, 2.
64. A bipartition gives a proper two-colouring. Conversely, the two colour classes of a proper two-colouring contain no internal edge, so they form a bipartition. At least one edge rules out chromatic number 1.
65. For example, colour A, D blue, B, E red, and C green. The triangle ABC requires at least three colours, so $\chi = 3$.
66. Final distances from A are $d(A) = 0$, $d(C) = 2$, $d(B) = 3$, $d(D) = 8$, $d(E) = 10$. One predecessor list is $C \leftarrow A$, $B \leftarrow C$, $D \leftarrow B$, $E \leftarrow D$.
67. $A - C - B - D$, weight 8; $A - C - B - D - E$, weight 10; $B - D - E$, weight 7.
68. Kruskal selects $BC(1)$, $AC(2)$, $DE(2)$, and $BD(5)$, for total weight 10.
69. The new minimum spanning tree is $\{BC, AC, DE, CE\}$, of weight $1 + 2 + 2 + 3 = 8$. The shortest $A-E$ path is $A - C - E$, of weight 5.

70. One shortest-path tree uses AC, CB, BD, DE , and in this example it is also an MST of weight 10. In general, a shortest-path tree minimises root-to-vertex distances, while an MST minimises total tree weight.
71. Any three boundary edges form an MST of weight 3. There are four choices, according to which boundary edge is omitted.
72. With distinct weights, the usual exchange argument prevents two different minimum choices at the first differing edge. The converse is false: a triangle with $AB = AC = 1$ and $BC = 2$ has the unique MST $\{AB, AC\}$ despite a repeated weight.
73. A negative edge discovered later may reduce the distance to a vertex that Dijkstra has already permanently labelled. The algorithm does not revisit such a vertex, so its final label can be wrong.
74. The shortest path is $S - B - A - T$, with weight $1 + 2 + 1 = 4$.
75. Dijkstra's algorithm; Kruskal's algorithm; Euler-trail analysis; proper vertex colouring.
76. The number of handshakes is $24(7)/2 = 84$. The degree sum 168 is even, as required.
77. The average degree is $2(27)/12 = 4.5$. Exactly five odd-degree users are impossible because the number of odd-degree vertices is even.
78. The schedule is K_{10} . It has $\binom{10}{2} = 45$ matches and every team vertex has degree 9.
79. Both the total indegree and total outdegree are 38. The stated vertex is a source airport with five departing direct routes and no arriving direct route.
80. $A \rightarrow B \rightarrow C \rightarrow A$ is a directed cycle. Each course would have to precede itself indirectly, so no prerequisite-respecting order exists.
81. The bipartition is students and clubs. The 310 edges represent 310 student–club memberships.
82. All degrees are even, so a closed Euler route exists and may begin at any intersection, including the depot.
83. The route must start at P and end at D , or start at D and end at P .
84. Link DE is the bridge. Its failure isolates station E from all other stations.
85. A tree on 30 vertices has 29 links. Adding one new link creates exactly one cycle, providing a redundant route around that cycle.
86. $F = 2 - 14 + 20 = 8$ regions.
87. The adjacency graph is C_7 , whose odd cycle requires three colours.
88. The first four examinations form K_4 , so four slots are necessary and sufficient; the fifth can share any slot except the first exam's.
89. Three channels are necessary for the triangle. For example, assign channels 1, 2, 3 to the three pairwise-interfering transmitters and channel 1 to the fourth.

90. Since C_5 is an odd cycle, three registers are necessary and sufficient.
91. It means that there are six directed walks of length 2 from page i to page j , hence six possible two-click routes, counting distinct intermediate choices and repeated-walk possibilities.
92. Graph distance is the minimum number of contact edges in any path between the people, so it is the smallest number of transmissions that could carry information or infection from one to the other.
93. The graph is $K_{1,8}$, with size 8 and hub degree 8. Every one of its eight edges is a bridge.
94. The graph is $K_{1,4}$, with degree sequence 4, 1, 1, 1, 1 and chromatic number 2.
95. The least cost is 10, attained by $A - C - B - D - E$.
96. The MST retains weight 10, so the saving is $32 - 10 = 22$.
97. Number rows and columns from the upper left. One path is $(1, 1), (1, 2), (1, 3), (2, 3), (2, 2), (2, 1), (3, 1), (3, 2), (3, 3)$.
98. All even degrees permit a closed Euler circuit. Exactly two odd degrees permit only an open Euler trail between those vertices. More than two odd vertices means that some corridor must be repeated if all are to be covered in one route.
99. (a) Two shortest routes are $G - B - L - H - R$ and $G - B - C - H - R$, each of weight 11; (b) an MST is LB, GB, CH, HR, BC , of weight $1 + 2 + 2 + 3 + 4 = 12$; (c) all links weigh 34, so the MST saves 22.
100. (a) Degrees are $(\deg G, \deg L, \deg B, \deg C, \deg H, \deg R) = (2, 3, 4, 3, 4, 2)$. There is an open Euler trail from L to C or conversely, but no Euler circuit; (b) $F = 2 - 6 + 9 = 5$; (c) there is no bridge, since every edge lies on a cycle; (d) adding LC makes both odd degrees even, so the connected graph then has an Euler circuit.

Chapter 18: Introduction to Logical Reasoning

These answers correspond to Section 18.10.

1. (a) Proposition, true; (b) not a proposition; (c) open sentence; (d) proposition, true; (e) not a proposition.
2. (a) $x \leq 5$; (b) at least one student did not complete the exercise; (c) no real number satisfies $x^2 = 2$; (d) n is not even or n is not positive.
3. (a) false; (b) true; (c) true; (d) true.
4. Using row order $(p, q) = (T, T), (T, F), (F, T), (F, F)$: (a) T, T, T, T ; (b) T, F, F, T ; (c) T, T, T, T .
5. (a) If n is divisible by 8, then n is even. (b) If n is even, then n is divisible by 8. (c) If n is not even, then n is not divisible by 8. Part (b) is the converse and part (c) is the contrapositive.
6. Converse: if the function is continuous at a , then it is differentiable at a . Inverse: if it is not differentiable at a , then it is not continuous at a . Contrapositive: if it is not continuous at a , then it is not differentiable at a .
7. (a) $p \wedge \neg q$; (b) p ; (c) $p \wedge \neg q$; (d) $p \wedge q$.
8. (a) tautology; (b) contradiction; (c) contingency.
9. (a) $\forall n \in \mathbb{Z}, n^2 \geq 0$; (b) $\exists x \in \mathbb{R}, x^2 = 5$; (c) $\neg \exists n \in \mathbb{Z}, 2 < n < 3$.
10. (a) $\exists x \in \mathbb{R}, x^2 + 1 \leq 0$; (b) $\forall n \in \mathbb{Z}, n^2 \neq 3$; (c) $\exists x \in A, P(x) \wedge \neg Q(x)$.
11. (a) Valid, modus ponens; (b) invalid, denying the antecedent; (c) valid, disjunctive syllogism.
12. Valid. Hypothetical syllogism gives $p \rightarrow r$ from the first two premises, and modus tollens with $\neg r$ gives $\neg p$.
13. Let the odd integers be $2m + 1$ and $2n + 1$. Their product is

$$(2m + 1)(2n + 1) = 2(2mn + m + n) + 1,$$

which is odd.

14. Prove the contrapositive. If $n = 2k + 1$ is odd, then $3n + 2 = 6k + 5 = 2(3k + 2) + 1$, which is odd. Therefore, if $3n + 2$ is even, then n is even.
15. If a greatest negative real number r existed, then $r/2$ would also be negative and would satisfy $r/2 > r$, a contradiction.
16. Possible counterexamples are (a) $a = 2, b = 3$; (b) $f(x) = |x|$ at $x = 0$; (c) $x = 0$.
17. For any x ,

$$\begin{aligned}x \in A \setminus (B \cup C) &\iff x \in A \wedge x \notin B \cup C \\&\iff x \in A \wedge x \notin B \wedge x \notin C \\&\iff x \in (A \setminus B) \cap (A \setminus C).\end{aligned}$$

18. A biconditional requires both $p \rightarrow q$ and $q \rightarrow p$. Divisibility by 4 implies evenness, but evenness does not imply divisibility by 4; for example, 6 is even but is not divisible by 4.
19. (a) proposition, false; (b) open sentence; (c) neither; (d) proposition, true; (e) neither.
20. (a) $a > b$; (b) the graph is neither connected nor planar; (c) at least one prime is not odd; (d) p and q have the same truth value.
21. (a) Ali studies, and if he passes then he celebrates. (b) If Ali studies and passes, then he celebrates. (c) $q \rightarrow p$. (d) $\neg p \wedge \neg r$.
22. (a) true; (b) false; (c) false; (d) true.
23. In row order $(T, T), (T, F), (F, T), (F, F)$: (a) T, F, T, T ; (b) F, T, T, T ; (c) F, T, F, F ; (d) T, T, T, T .
24. It is false precisely for $(p, q, r) = (T, T, F), (T, F, F), (F, T, F)$.
25. (a) $(\neg p) \vee (q \wedge r)$, false; (b) $p \rightarrow (q \vee r)$, true; (c) $(p \wedge q) \leftrightarrow r$, false.
26. Divisibility by 12 is sufficient for divisibility by 3; divisibility by 3 is necessary for divisibility by 12; n is divisible by 12 only if it is divisible by 3; n is divisible by 3 whenever it is divisible by 12.
27. (a) Converse: $\text{even} \rightarrow \text{multiple of 6}$; inverse: $\text{not a multiple of 6} \rightarrow \text{not even}$; contrapositive: $\text{not even} \rightarrow \text{not a multiple of 6}$. (b) Replace the two clauses similarly: converse $x^2 > 4 \rightarrow x > 2$, inverse $x \leq 2 \rightarrow x^2 \leq 4$, contrapositive $x^2 \leq 4 \rightarrow x \leq 2$.
28. (a) sufficient but not necessary; (b) necessary but not sufficient; (c) both; (d) necessary but not sufficient.
29. (a) p ; (b) $p \vee q$; (c) p ; (d) $p \vee q$.
30. (a) $\neg p \vee \neg q \vee \neg r$; (b) $(\neg p \wedge \neg q) \vee \neg r$; (c) $x \leq 0$ or $x > 5$; (d) n is odd and not divisible by 3.
31. (a) $\neg p \vee q$; (b) $(p \wedge q) \vee (\neg p \wedge \neg q)$; (c) $p \wedge \neg q$; (d) $(\neg p \vee q) \wedge (\neg q \vee r)$.
32. (a) absorption gives p ; (b) both columns match because an implication is equivalent to its contrapositive; (c) it is true exactly when p and q differ.
33. (a) tautology; (b) contradiction; (c) tautology; (d) contingency.
34. (a) satisfiable, for example $p = F, q = T$; (b) unsatisfiable; (c) satisfiable with $p = q = r = F$.
35. (a) $\{-3, 3\}$; (b) $\emptyset$; (c) $(2, \infty)$.
36. (a) $\forall x \in \mathbb{R}, x^2 \geq 0$. (b) $\exists n \in \mathbb{Z}, n = -n$. (c) $\forall p \in \mathbb{P}, p \geq 0$. (d) $\exists x \in \mathbb{R} [x^2 = 0 \wedge \forall y \in \mathbb{R} (y^2 = 0 \rightarrow y = x)]$.
37. (a) $\exists x \in \mathbb{R}, x^2 < x$; (b) $\forall n \in \mathbb{Z}, n^2 + n \neq 1$; (c) $\exists x \in A, \neg P(x) \wedge \neg Q(x)$; (d) $\forall x \in A, \neg P(x) \vee Q(x)$.
38. Let $L(s, p)$ mean that student s solved problem p . (a) $\forall s \exists p L(s, p)$; negation $\exists s \forall p \neg L(s, p)$. (b) $\exists s \forall p L(s, p)$; negation $\forall s \exists p \neg L(s, p)$. (c) $\neg \exists s \forall p L(s, p)$; its negation is the formula in (b).
39. (a) true, choose $y = -x$; (b) false; (c) false because $x = 0$; (d) true, choose $y = 0$.

40. (a) $x = 1/2$; (b) $n = 2$; (c) $a = 1, b = -1$.
41. (a) modus ponens; (b) modus tollens; (c) affirming the consequent, invalid; (d) disjunctive syllogism.
42. Valid. If q is false, $p \rightarrow q$ gives $\neg p$; then $p \vee r$ gives r .
43. Modus ponens gives q , then r , then s ; the conclusion is s .
44. (a) affirming the consequent; (b) denying the antecedent; (c) inclusive or does not imply that q is false when p is true.
45. Write even integers as $2a, 2b$ and odd integers as $2a + 1$. Then (a) $2a + 2b = 2(a + b)$; (b) $(2a + 1) + 2b = 2(a + b) + 1$; (c) $(2a + 1)^2 = 2(2a^2 + 2a) + 1$.
46. If $b = ak$ and $c = a\ell$, then $mb + nc = a(mk + n\ell)$, so $a \mid (mb + nc)$.
47. (a) If n were odd, n^2 would be odd; hence even n^2 implies even n . (b) If either factor is even, the product is even; hence an odd product has two odd factors.
48. Assuming $\sqrt{2} = a/b$ in lowest terms forces both a and b to be even, contradicting lowest terms.
49. If n is even, $n(n + 1)$ has an even factor; if n is odd, $n + 1$ is even. Thus the product is even.
50. Odd $n = 2k + 1$ gives odd n^2 ; conversely, the contrapositive says even n gives even n^2 .
51. (a) 2 is prime and even; (b) $a = b = 1$; (c) $f(x) = x$ has derivative 1.
52. (a) $\forall x(x \in A \rightarrow x \in B)$; (b) $x \in A \wedge x \in B$; (c) $x \in A \wedge x \notin B$; (d) $x \notin A \wedge x \notin B$.
53. (a) Membership gives $x \in A \wedge (x \in B \vee x \in C)$, which distributes to the right side. (b) Negating $x \in A \wedge x \in B$ gives $x \notin A \vee x \notin B$.
54. (a) true; (b) true; (c) false; (d) false.
55. $Y = \neg(p \vee q)$. In row order TT, TF, FT, FF , the output is F, F, F, T .
56. NAND is $\neg(p \wedge q)$ with outputs F, T, T, T ; NOR is $\neg(p \vee q)$ with outputs F, F, F, T .
57. The executing inputs (p, q, r) are TTT, TFT, FTT, FFT , and $TF F$.
58. The satisfying assignments are (T, F, T) and (F, T, F) .
59. (a), (b) inconsistent; (c) consistent, for example $p = q = r = F$.
60. This is proof by cases, also called constructive dilemma: either premise of the disjunction implies r .
61. $\sqrt{2k}$ need not be an even integer. Use the contrapositive: the square of an odd integer is odd.
62. Suitable choices are (a) contrapositive; (b) contradiction; (c) cases; (d) counterexample.
63. The consecutive integers 3 and 4 satisfy $3^2 + 4^2 = 25$.
64. Solving gives $x = 4$. If x and y both solve it, then $2x + 3 = 2y + 3$, hence $x = y$; the solution is unique.

65. A multiple of 6 is a multiple of 2 and 3. Conversely, divisibility by the coprime numbers 2 and 3 implies divisibility by their product 6.
66. One of $n, n + 1$ is even in either parity case, so their product is divisible by 2.
67. If rational r plus irrational s were rational, subtracting r would make s rational, a contradiction.
68. $b^2 - a^2 = (b - a)(b + a) > 0$, so $a^2 < b^2$.
69. If $x \in A$, then $x \in B$, and hence $x \in C$; therefore $A \subseteq C$.
70. Each element independently has two choices—included or omitted. The multiplication principle gives 2^n subsets.
71. Both expressions are true only on rows TF and FT , so they are equivalent forms of exclusive or.
72. $\neg p = p \uparrow p$, and $p \wedge q = (p \uparrow q) \uparrow (p \uparrow q)$.
73. Yes. Both simplify to $\neg p \vee (q \wedge r)$.
74. CNF: $(\neg p \vee q) \wedge (\neg q \vee p)$. DNF: $(p \wedge q) \vee (\neg p \wedge \neg q)$.
75. Valid. From $p \rightarrow r$ and $\neg r$, infer $\neg p$; from $p \vee q$, infer q ; then $q \rightarrow s$ gives s .
76. Access is $c \wedge (p \vee s)$. It is denied when $\neg c \vee (\neg p \wedge \neg s)$.
77. The alarm is $a \wedge (s \vee h)$; it sounds when armed and at least one of smoke or heat is detected.
78. Records are present, study Mathematics or Physics, and are not absent. DNF: $(M \wedge \neg A) \vee (P \wedge \neg A)$.
79. $C \wedge (P \vee E) \wedge \neg S$, using the initial letters of the four search terms.
80. This affirms the consequent. Other causes can produce a positive result, so disease does not follow from positivity alone.
81. Nothing definite follows about the storm; concluding that it arrived would affirm the consequent.
82. If L means holds a licence and A means at least 18, then $L \rightarrow A$. Age alone may not meet tests, fees, or other requirements.
83. With g, n, r denoting grade, need, and award, eligibility is $g \wedge (n \vee r)$; ineligibility is $\neg g \vee (\neg n \wedge \neg r)$.
84. From $A < B$ and $B < C$, infer $A < C$ by transitivity or hypothetical syllogism.
85. Operational: $s \wedge (\ell_1 \vee \ell_2)$. Not operational: $\neg s \vee (\neg \ell_1 \wedge \neg \ell_2)$.
86. -6 rejected; 2 accepted; 9 accepted; 25 rejected.
87. For example, `IF(AND(score>=50,attendance>=80%),"Pass","Fail")`. Failure means score < 50 or attendance < 80%.
88. $\forall x[A(x) \vee B(x) \vee C(x)]$. Its negation is $\exists x[\neg A(x) \wedge \neg B(x) \wedge \neg C(x)]$.

89. By modus tollens, the model is not correct, assuming the stated implication and observation.
90. $P \rightarrow (R > C)$; contrapositive: $R \leq C \rightarrow \neg P$.
91. The statement is false without connectedness. A disconnected graph may contain a cycle and an isolated vertex while still having $n - 1$ edges.
92. It is valid modus tollens, equivalently the contrapositive of the theorem.
93. $(A \cup B) \setminus (A \cap B)$, corresponding to $(p \vee q) \wedge \neg(p \wedge q)$.
94. $|A \cup B| = 62 + 48 - 25 = 85$, so the claim is false; 15 use neither.
95. Since $a = b$, the divisor $a - b$ is zero. Division by zero is the invalid step.
96. From $q \rightarrow r$ and $\neg r$, infer $\neg q$; then from $p \rightarrow q$, infer $\neg p$.
97. XOR: $(p \wedge \neg q) \vee (\neg p \wedge q)$, with outputs F, T, T, F in row order TT, TF, FT, FF .
98. Validity is $\ell \wedge d \wedge (u \vee s)$. Its negation is $\neg \ell \vee \neg d \vee (\neg u \wedge \neg s)$.
99. Acceptance is $q \wedge (e \vee t)$. The minimal satisfying combinations are q, e and q, t .
100. Approval is

$$(p \wedge q) \vee (p \wedge r) \vee (q \wedge r).$$

The input–output rows may be written compactly as

$$\begin{aligned} TTT, TTF, TFT, TFF &\mapsto T, T, T, F, \\ FTT, FTF, FFT, FFF &\mapsto T, F, F, F. \end{aligned}$$

The rule is monotone: changing no to yes cannot turn approval into rejection.

Chapter 19: Computing with R

These answers correspond to Section 19.12.

1. (a) False: R is the language and computing environment; RStudio is an interface for working with R. (b) True. (c) True. (d) False: the prompt is displayed by the console and is not part of the command.
2. (a) $(18-6)/3+5^2$ gives 29. (b) `sqrt(225)+log10(1000)` gives 18. (c) `exp(3)` gives approximately 20.0855. (d) `cos(pi)+sin(pi/2)` gives 0, apart from possible floating-point rounding.
3. For example, `radius <- 3.5; height <- 8; volume <- pi*radius^2*height`. Thus the volume is $98\pi \approx 307.876$. Add a comment beginning with #, then use `rm(height)`.
4. (a) `seq(5,160,by=5)`; (b) `87:56`; (c) `rep(-2,5)`; (d) `rep(1:3,3)`.
5. `a+b` is (3, 7, 11, 15); `a*b` is (2, 10, 24, 44); `a^2` is (4, 25, 64, 121); and `sqrt(a+b)` is approximately (1.7321, 2.6458, 3.3166, 3.8730).
6. The outputs are respectively (9, 1, 7), (4, 3), (9, 7, 12), and (4, 1, 7, 3).
7. `which(x>5)` gives (2, 4, 5); `length(x[x>5])` gives 3; and `mean(x[x>5])` gives $28/3 \approx 9.3333$.
8. `mean(y)` returns NA because a missing value is present. `mean(y,na.rm=TRUE)` gives $22/3 \approx 7.3333$, and `which(is.na(y))` gives position 2.
9. Sum 30; mean 6; median 5; range (3, 10); sample variance 7; sample standard deviation $\sqrt{7} \approx 2.6458$.
10. The result is (11, 22, 13, 24) because (10, 20) is recycled to (10, 20, 10, 20). For example, `1:4 + 1:3` gives a warning because 4 is not a multiple of 3.
11. One solution is

```
students <- data.frame(
  name = c("Aina", "Ben", "Chen"),
  score = c(78, 84, 69),
  group = c("A", "B", "A")
)
students$score
students[2, ]
students[students$score >= 75, ]
```

12. Use the following commands:

```
data("iris")
head(iris); dim(iris); names(iris)
str(iris); summary(iris)
table(iris$Species)
```

13. Use `dahlia <- read.csv("data/dahlia.csv")` and, for example, `head`, `dim`, `names`, `str`, and `summary`. Export with `write.csv(dahlia, "outputs/dahlia-copy.csv", row.names=FALSE)`.

14. `results[results$score>=50 & results$attendance>=80, c("name","score")]`.
15. Distribution: `hist(x)`. Two quantitative variables: `plot(x,y)`. Quantitative response by category: `boxplot(y~g)`. Chronological series: `plot(x)` or `plot(time,x,type="l")`.
16. Use `mean(Nile)`, `sd(Nile)`, and `summary(Nile)`. Set `par(mfrow=c(1,2))`, call `plot(Nile)` and `hist(Nile)`, then restore `par(mfrow=c(1,1))`. The histogram discards chronological order.
17. `d` gives a mass or density, `p` gives a cumulative probability, `q` gives a quantile, and `r` generates random observations.
18. Using `dbinom(2,8,.2)`, `pbinom(2,8,.2)`, and `1-dbinom(0,8,.2)`, the answers are approximately 0.2936, 0.7969, and 0.8322.
19. Exactly three: 0.0081; more than one: 0.08146; fewer than two: 0.91854; at least two: 0.08146; at most two: 0.99144. Appropriate commands use `dbinom()` and `pbinom()` with complements where needed.
20. `dpois(0,3)` gives 0.04979; `ppois(2,3)` gives 0.42319; and `1-dpois(0,3)` gives 0.95021.
21. The answers are approximately 0.06681, 0.68269, and 80.2524. Use `pnorm(82,70,8,lower.tail=FALSE)`, `pnorm(78,70,8)-pnorm(62,70,8)`, and `qnorm(.9,70,8)`.
22. $P(4 < U < 7) = 3/8 = 0.375$ and $E(U) = 6$. Generate observations with

`runif(200,min=2,max=10).`

23. $P(T > 6) = e^{-1.5} \approx 0.22313$ and the median is $\ln 2/0.25 \approx 2.77259$. Use `pexp(6,rate=.25,lower.tail=FALSE)` and `qexp(.5,rate=.25)`.
24. The seed makes the pseudorandom output reproducible. For example, `set.seed(13); sim <- rbinom(1000,20,.3); mean(sim)`. The sample mean should be close to the theoretical mean $np = 6$.
25. State the hypotheses, choose the significance level and method, calculate the statistic and p-value, make the decision, and interpret it in context. A large p-value means insufficient evidence against H_0 ; it does not establish that H_0 is true.
26. With `z <- (9.60-9.78)/(sqrt(2.20)/sqrt(40))`, $z \approx -0.7675$. The lower-tail p-value `pnorm(z)` is approximately 0.2214. Since $p > 0.10$, do not reject H_0 ; the sample does not sufficiently support the manager's claim.
27. $z \approx -1.8974$ and `2*pnorm(-abs(z))` gives $p \approx 0.05778$. Do not reject H_0 at the 5% level; there is insufficient evidence that the mean differs from 3.2.
28. `t.test(plants,mu=15)` gives $\bar{x} \approx 14.3333$, $t \approx -1.6848$, $df = 11$, $p \approx 0.1201$, and a 95% confidence interval of approximately (13.4624, 15.2042). Do not reject H_0 at the 5% level.
29. Use

```
t.test(x, mu = mu0, alternative = "two.sided")
t.test(x, mu = mu0, alternative = "less")
t.test(x, mu = mu0, alternative = "greater")
```

30. The sample mean is exactly 170, so $t = 0$, $df = 14$, and $p = 1$. The 95% confidence interval is approximately (168.3787, 171.6213). There is no evidence that the population mean differs from 170 cm.

31. Use `shapiro.test(x)`, `hist(x)`, `qqnorm(x)`, and `qqline(x)`. The Shapiro–Wilk null hypothesis is that the population is normal. Graphs remain necessary because the test may have little power in a small sample and may flag negligible deviations in a large sample.

32.

$$A = \begin{pmatrix} 1 & 4 & 7 & 10 \\ 2 & 5 & 8 & 11 \\ 3 & 6 & 9 & 12 \end{pmatrix}.$$

`A[2,3]` is 8; row 2 is (2, 5, 8, 11); columns 1 and 4 form $\begin{pmatrix} 1 & 10 \\ 2 & 11 \\ 3 & 12 \end{pmatrix}$.

33. The default column-filled matrix is $\begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix}$; with `byrow=TRUE` it is $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$.

34. `rbind(a,b)` is $\begin{pmatrix} 1 & 2 & 3 \\ 10 & 20 & 30 \end{pmatrix}$, of dimension 2×3 . `cbind(a,b)` is $\begin{pmatrix} 1 & 10 \\ 2 & 20 \\ 3 & 30 \end{pmatrix}$, of dimension 3×2 .

35. Elementwise multiplication gives

$$A * B = \begin{pmatrix} 2 & 0 \\ 3 & 8 \end{pmatrix},$$

whereas matrix multiplication gives

$$A \%* \% B = \begin{pmatrix} 4 & 4 \\ 10 & 8 \end{pmatrix}.$$

36. $A^T = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$, $\det A = -2$, and $A^{-1} = \begin{pmatrix} -2 & 1 \\ 3/2 & -1/2 \end{pmatrix}$. The R commands are `t(A)`, `det(A)`, and `solve(A)`.

37. With the coefficient matrix entered by rows, `solve(coef,rhs)` gives

$$(x, y, z) = \left(\frac{2}{3}, \frac{43}{6}, -\frac{23}{6} \right).$$

38. $C = \begin{pmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{pmatrix}$. The row sums are (12, 15, 18) and the column sums are (6, 15, 24). Use `apply(C,1,sum)`, `apply(C,2,sum)`, `rowSums(C)`, and `colSums(C)`.

39. A suitable script uses `head`, `dim`, `names`, `str`, and `summary`; then `sd(iris$Sepal.Length)`,

`table(iris$Species)`, and `aggregate(Sepal.Length~Species,data=iris,mean)`. The requested plots use `hist`, `plot`, and the grouped `boxplot` shown in the chapter.

40. Label with, for example,

```
rownames(S) <- c("Student1", "Student2", "Student3")
colnames(S) <- c("Test1", "Test2")
```

The student means are

(78.5, 79.5, 83.5),

and the test means are approximately (83.6667, 77.3333). Student 3 has the greatest average. The relevant commands are

```
rowMeans(S)
colMeans(S)
which.max(rowMeans(S))
```

41. (a) Numeric vector (2, 5.5); (b) character vector `c("2", "5")` because all elements are coerced to character; (c) numeric vector (1, 4) because `TRUE` is coerced to 1; (d) numeric vector (3, 7).
42. The outputs are `c(FALSE, FALSE, TRUE, FALSE)` and `c(FALSE, TRUE, FALSE, TRUE)`. The symbol `=` is used for assignment in some contexts or for named arguments, `==` tests equality, `&` is elementwise conjunction, and `|` is elementwise disjunction.
43. `marks["Ben"]` gives 81. The command `marks[marks>75]` gives Ben 81, Chen 76, and Devi 90. Finally, `names(marks)[marks>75]` returns their names.
44. `sort(z)` is (1, 3, 5, 8, 8); `unique(z)` is (8, 3, 1, 5); and `order(z)` is (4, 2, 5, 1, 3), the original positions that arrange the values in ascending order.

45

```
missing_n <- sum(is.na(x))      # 2
observed <- x[!is.na(x)]
observed_mean <- mean(observed) # 13.75
cleaned <- x
cleaned[is.na(cleaned)] <- observed_mean
```

46. Initially the mean is 11.2 and the median is 11. After adding 100, the mean is 26 and the median is 11.5. The median is much more resistant to the extreme value.
47. With R's default convention, $Q_1 = 4.5$, median = 7, $Q_3 = 10.5$, and IQR = 6. Use `quantile(x)` and `IQR(x)`.

48

```
products <- data.frame(
  product = c("A", "B", "C"),
  price = c(12, 8, 15),
  quantity = c(5, 9, 4)
)
products$revenue <- products$price * products$quantity
sum(products$revenue)
products[products$revenue >= 65, ]
```


Revenues are (60, 72, 60), total revenue is RM 192, and only product B generates at least RM 65.

49. First define

```
group <- factor(c("A", "B", "A", "C", "B", "A"),
               levels = c("A", "B", "C"))
table(group)
prop.table(table(group))
```

The frequency table gives A: 3, B: 2, C: 1; the relative frequencies are (0.5, 0.3333, 0.1667). Thus A is the modal level.

50. `aggregate(score~group, data=d, mean)` gives mean scores 71.6667 for A and 83.3333 for B.

51. With the conditions applied to the data frame, (a) students C and D are selected; (b) students A, C, D, and E are selected. Representative conditions are `score>=50 & attendance>=80` and `score>80 | attendance>90`.

52. For example,

```
par(mfrow = c(2, 2))
hist(x, main = "Distribution of x", xlab = "x")
boxplot(x, main = "Boxplot of x", ylab = "x")
plot(x, y, xlab = "x", ylab = "y", pch = 19)
barplot(table(g), xlab = "Group", ylab = "Frequency")
par(mfrow = c(1, 1))
```

53. Wider or differently positioned bins can conceal or exaggerate clusters, gaps, and skewness. Compare `hist(x, breaks=5)` with `hist(x, breaks=20)`.

54. $Q_1 = 11.5$, $Q_3 = 13.5$, and $IQR = 2$, giving fences 8.5 and 16.5. Thus 30 is flagged. The direct command is `boxplot.stats(x)$out`.

55. Use, for example, `plot(y, type="o", pch=19, xlab="Year", ylab="Value")`. The values show an overall increase, with a small decline from the second to the third year.

56. The required integer is $k = 9$: `qbinom(.90, 20, .3)` gives 9, `pbinom(8, 20, .3)` is approximately 0.8867, and `pbinom(9, 20, .3)` is approximately 0.9520.

57. The probabilities are approximately 0.19358, 0.64878, and 0.96832, using `dbinom(4, 12, .25)`, `pbinom(3, 12, .25)`, and `pbinom(0, 12, .25, lower.tail=FALSE)`. The mean is 3 and the standard deviation is 1.5.

58.

$$P(3 \leq Y \leq 7) \approx 0.71076, \quad P(Y > 6) \approx 0.11067.$$

Use `ppois(7, 4) - ppois(2, 4)` and `ppois(6, 4, lower.tail=FALSE)`. The 95th percentile is `qpois(.95, 4) = 8`.

59. The central 95% interval is approximately (70.6005, 129.3995). The command is `qnorm(c(.025, .975), 100, 15)`. Also, `pnorm(115, 100, 15) - pnorm(85, 100, 15)` gives 0.68269.

60. The first quartile, third quartile, and 90th percentile are 8.75, 16.25, and 18.5. Generate observations with `set.seed(13); u <- runif(500,5,20)`.
61. $P(T > 10) = e^{-2} \approx 0.13534$ and the 90th percentile is $-\ln(0.1)/0.2 \approx 11.5129$. By memorylessness, $P(T > 15 \mid T > 5) = P(T > 10) \approx 0.13534$.
62. One solution is `set.seed(13); d <- sample(1:6,10000,replace=TRUE); mean(d%%2==0)`. The estimate should be close to the theoretical value $1/2$.
63. For example,

```
set.seed(13)
x <- runif(100000, -1, 1)
y <- runif(100000, -1, 1)
pi_hat <- 4 * mean(x^2 + y^2 <= 1)
```

The square has area 4 and the unit circle has area π , so the in-circle proportion estimates $\pi/4$.

64. The standard error is $8/\sqrt{64} = 1$. With `zstar <- qnorm(.975)`, the interval is 50 ± 1.959964 , or approximately (48.0400, 51.9600).
65. The lower-, upper-, and two-sided p-values are approximately 0.95994, 0.04006, and 0.08012. Use `pnorm(1.75)`, `pnorm(1.75,lower.tail=FALSE)`, and `2*pnorm(-abs(1.75))`.
66. The sample mean is 22.125 and the sample standard deviation is approximately 2.0310. The command `t.test(x,conf.level=.95)` gives a confidence interval of approximately (20.4270, 23.8230).
67. For $p = 0.03$, reject at 5% but not at 1%. For $p = 0.20$, reject at neither level. Statistical significance concerns evidence against H_0 ; practical importance concerns whether the effect is large enough to matter in context.
68. With $p = 0.03$, reject the normality null at 5%; the curved Q–Q plot supports concern about non-normality. With $p = 0.40$, normality is not rejected, but the severe plotted outlier must still be investigated because a non-significant test does not guarantee a suitable model.
69. $A \% * \% B$ is 3×4 . The product $B \% * \% A$ is undefined because the inner dimensions 4 and 3 disagree. For two matrix operands, elementwise $A*B$ requires the same dimensions; it is therefore invalid for these A and B .
70. Use `I <- diag(3)`. For any compatible 3-row matrix M , $I \% * \% M$ equals M ; for any compatible 3-column matrix, $M \% * \% I$ equals M .
- 71.

$$\text{solve}(A, B) = \begin{pmatrix} 1.6 & 0.6 \\ 1.8 & -0.2 \end{pmatrix}.$$

Each column is the solution for the corresponding right-hand side in B .

72. $\det(M) = 0$. The rows and columns are linearly dependent, so no inverse exists and `solve(M)` reports that the system is exactly singular.
73. Representative code is

```
M <- matrix(1:9, nrow = 3, byrow = TRUE)
rownames(M) <- c("r1", "r2", "r3")
colnames(M) <- c("c1", "c2", "c3")
M[c("r1", "r3"), "c2", drop = FALSE]
```

The extracted matrix is $\begin{pmatrix} 2 \\ 8 \end{pmatrix}$.

74.

$$T \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 6 \\ 2 \end{pmatrix}.$$

The horizontal coordinate is doubled and the vertical coordinate is halved.

75. Correct forms are `mean(x)`, `x==3`, `mean(x,na.rm=TRUE)`, and `A%*%B`. The errors are case sensitivity, separation of the equality operator, a misspelt logical constant, and confusing elementwise with matrix multiplication.
76. The total is RM 2,380. Percentage shares are approximately 50.42%, 18.91%, 12.61%, 10.50%, and 7.56%. Rent is largest. Suitable commands are `sum(budget)`, `100*budget/sum(budget)`, and `names(which.max(budget))`.
77. There are two missing readings and the observed mean is 22.22°C. Use a copied vector before replacement. Retaining the raw vector preserves an auditable record and prevents an imputation decision from being mistaken for an observed measurement.
78. Aina, Devi, and Ethan pass. The pass rate is $3/5 = 0.60$. In R, create `pass <- score>=50 & attendance>=80`, then use `students[pass,]` and `mean(pass)`.
79. The mean is 12.00875 mm and sample standard deviation is about 0.03044 mm. The passing measurements are 11.98, 12.01, 11.97, 12.00, 12.02, 11.99, so the passing proportion is $6/8 = 0.75$.
80. Mean = 8, median = 7.5, sample standard deviation = 3.3333, and range = (4, 15). The upper boxplot fence is 12.875, so 15 is flagged. Use `par(mfrow=c(1,2))`, `hist(wait)`, and `boxplot(wait)`.
81. Regional means are North 127.5, South 135, and East 144. Totals are North 255, South 270, and East 288. Use `aggregate(sales~region,d,mean)`, `aggregate(sales~region,d,sum)`, and a bar plot of the totals.
82. The maximum is 150 in year 5. A line or index plot preserves time order; a histogram retains only the distribution of the six values.
83. `pbinom(11,20,.58,lower.tail=FALSE)` gives approximately 0.52290. The expected number is $20(0.58) = 11.6$.
84. `ppois(5,2.5)-ppois(1,2.5)` gives approximately 0.67068, while `dpois(0,2.5)` gives approximately 0.082085.
85. The three results are approximately 0.89435, 0.04006, and 41.5794 minutes, using `pnorm(40, 35,4)`, `pnorm(42,35,4,lower.tail=FALSE)`, and `qnorm(.95,35,4)`.

86. $P(T > 8) = e^{-1.2} \approx 0.30119$, and the median is $\ln 2 / 0.15 \approx 4.6210$ minutes. For simulation use `set.seed(13); service <- rexp(1000, rate=.15); hist(service)`.
87. The exact reliability is `pbinom(8, 10, .9, lower.tail=FALSE)  $\approx 0.73610$` . Simulate with `set.seed(13); mean(rbinom(5000, 10, .9) >= 9)`; the result should be close to the exact probability.
88. $z = (412 - 400) / (35 / \sqrt{50}) \approx 2.4244$ and the upper-tail p-value is approximately 0.007668. Reject H_0 at 5%; the data support a mean wage above RM 400.
89. The sample mean is 10.04 hours, $t \approx 0.5695$, $df = 9$, and $p \approx 0.5830$. The 95% confidence interval is approximately (9.8811, 10.1989). Do not reject H_0 ; the sample does not show that the mean battery life differs from 10 hours.
90. Use `shapiro.test(plants)`, `hist(plants)`, `qqnorm(plants)`, and `qqline(plants)`. The value 15.8 is a severe possible outlier relative to the cluster near 12; verify it, investigate its source, and treat a small-sample t conclusion cautiously rather than deleting it automatically.
91. With $R = \begin{pmatrix} 2 & 3 \\ 5 & 4 \end{pmatrix}$ and $\mathbf{x} = (100, 80)^T$,

$$R\mathbf{x} = \begin{pmatrix} 440 \\ 820 \end{pmatrix}.$$

Thus 440 labour units and 820 material units are required.

92. Store the points as $P = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 3 \end{pmatrix}$. Then

$$RP = \begin{pmatrix} 0 & -1 & -3 \\ 1 & 2 & -1 \end{pmatrix},$$

giving rotated points (0, 1), (-1, 2), and (-3, -1).

- 93.

$$A^2 = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

Each diagonal 2 counts the two ways to leave a vertex and return in two steps; each off-diagonal 1 counts the one two-step walk through the third vertex.

94. Student means are 76, 88, approximately 67.6667, and 90. Test means are 78.75, 82, and 80.5. Student 4 has the greatest average.
95. One documented choice is

```
survey_raw <- read.csv("data/survey.csv")
head(survey_raw); names(survey_raw); str(survey_raw)
colSums(is.na(survey_raw))
survey_clean <- survey_raw[complete.cases(survey_raw), ]
write.csv(survey_clean, "outputs/survey-clean.csv",
          row.names = FALSE)
```

Removing incomplete rows is only one cleaning rule and should be justified rather than applied automatically.

96. A suitable checklist records: the research question; source and sampling context; raw-data location; inspection and cleaning decisions; random seed when simulation is used; numerical summaries and labelled graphics; method, assumptions, and a contextual conclusion.
97. `aggregate(Sepal.Length~Species,iris,mean)` gives approximately 5.006, 5.936, and 6.588 for setosa, versicolor, and virginica. Virginica has the greatest mean. Use the grouped `boxplot()` and labelled `plot()` commands from the chapter.
98. Use `m1<-mean(head(Nile,50))` and `m2<-mean(tail(Nile,50))`, followed by `plot(Nile)` and horizontal lines such as `abline(h=c(m1,m2),lty=2,col=c("blue","red"))`. A histogram discards which observations belong to the early and late periods, so it cannot display this temporal comparison.
99. Representative code is

```
temperature <- seq(22, 34, by = 2)
energy <- c(18, 20, 23, 27, 31, 36, 42)
plot(temperature, energy, pch = 19,
     xlab = "Temperature (degrees C)",
     ylab = "Energy use (kWh)")
mean(energy)
temperature[which.max(energy)]
```

The mean energy use is $197/7 \approx 28.1429$ kW h. The maximum, 42 kW h, occurs at 34°C. The scatter plot shows a strong positive association over these observations, but the plot alone does not establish that temperature causes the change.

100. A complete compact script is

```
weights_raw <- c(48.2, 50.1, NA, 49.7, 51.0, 48.9,
                50.4, 49.5, 50.2, 48.7, 51.3, NA)
observed <- weights_raw[!is.na(weights_raw)]
sum(is.na(weights_raw))
mean(observed)
sd(observed)

par(mfrow = c(1, 2))
hist(observed, xlab = "Weight (g)")
qqnorm(observed); qqline(observed)
par(mfrow = c(1, 1))

result <- t.test(observed, mu = 50)
result
```

There are two missing values. For the ten observed weights, $\bar{x} = 49.8$ g and $s \approx 0.9989$ g. `t.test(observed,mu=50)` gives $t \approx -0.6332$, $df = 9$, $p \approx 0.5424$, and a 95% confidence interval of approximately (49.0854, 50.5146) g. The histogram and Q-Q plot should accompany the assumption check. At 5%, do not reject H_0 ; the sample provides insufficient evidence that the mean package weight differs from 50 g.

References

- Abu Bakar, A. S., F. Hussain, N. Hashim and H. S. Mohamad Sukri (2025–2026). *FAC1004 Advanced Mathematics II*. PASUM lecture notes. Centre for Foundation Studies in Science, Universiti Malaya.
- Auroux, D. (2010). *18.02SC Multivariable Calculus*. MIT OpenCourseWare. URL: <https://ocw.mit.edu/courses/18-02sc-multivariable-calculus-fall-2010/> (visited on 27/08/2026).
- Bureau International des Poids et Mesures (2026). *The International System of Units (SI Brochure)*. Bureau International des Poids et Mesures. DOI: 10.59161/AUEZ1291. URL: <https://www.bipm.org/en/publications/si-brochure> (visited on 27/08/2026).
- Centre for Foundation Studies in Science, Universiti Malaya (2025–2026). *FAC1001 Advanced Mathematics I: Tutorials 1–11*. Tutorial sheets. Universiti Malaya.
- Chai, M., S. K. Ting and Y. J. Aw (2024). *Mathematics for Matriculation Science: Semester 1*. Revised edition. SAP Publications (M) Sdn. Bhd.
- Dawkins, P. (16th Nov. 2022). *Circles*. Paul's Online Math Notes, Lamar University. URL: <https://tutorial.math.lamar.edu/classes/alg/circles.aspx> (visited on 27/08/2026).
- Encyclopaedia Britannica (2026). *Function*. Encyclopaedia Britannica. URL: <https://www.britannica.com/science/function-mathematics> (visited on 22/08/2026).
- Hammack, R. (2018). *Book of Proof*. 3rd ed. Richard Hammack. URL: <https://richardhammack.github.io/BookOfProof/> (visited on 27/08/2026).
- Harvard University (2026). *Calculus Applied!* Harvard Online and edX. URL: <https://harvardonline.harvard.edu/course/calculus-applied> (visited on 27/08/2026).
- Hashim, N. and H. S. Mohamad Sukri (2025–2026). *FAC1001 Advanced Mathematics I*. PASUM lecture notes. Centre for Foundation Studies in Science, Universiti Malaya.
- Jalaluddin, J. F., N. Hashim and H. S. Mohamad Sukri (2025–2026). *FAD1014 Mathematics II*. PASUM lecture notes. Centre for Foundation Studies in Science, Universiti Malaya.
- Jalaluddin, J. F., M. S. Mazlin, N. Hashim and H. S. Mohamad Sukri (2025–2026). *FAD1013 Mathematics I*. PASUM lecture notes. Centre for Foundation Studies in Science, Universiti Malaya.
- Levin, O. (2024). *Discrete Mathematics: An Open Introduction*. 4th ed. Open Math Books. URL: <https://discrete.openmathbooks.org/> (visited on 27/08/2026).
- London School of Economics and Political Science (2026). *Maths Essentials*. edX. URL: <https://www.edx.org/learn/math/the-london-school-of-economics-and-political-science-maths-essentials> (visited on 27/08/2026).
- Marsden, J. E. and A. J. Tromba (2026). *The Geometry of Euclidean Space*. Course notes. California Institute of Technology. URL: https://www.cds.caltech.edu/~marsden/wiki/uploads/math1c-08/textbooks/Key_Points.pdf (visited on 27/08/2026).
- Ministry of Education Malaysia (2018). *Dokumen Standard Kurikulum dan Pentaksiran: KSSM Additional Mathematics Forms 4 and 5*. Curriculum Development Division. URL:

- <https://bpk.moe.gov.my/kurikulum/kssm/kssm-tingkatan-4-dan-5/440-dskp-kssm-additional-mathematics-form-4-and-5/file> (visited on 27/08/2026).
- Moore, D. S., G. P. McCabe and B. A. Craig (2021). *Introduction to the Practice of Statistics*. 10th ed. Macmillan Learning.
- Murdoch, J. and J. A. Barnes (1998). *Statistical Tables*. 4th ed. Palgrave.
- Neill, H. (2018). *Calculus: A Complete Introduction*. John Murray Press.
- O'Connor, J. J. and E. F. Robertson (May 2000). *Lejeune Dirichlet (1805–1859)*. MacTutor History of Mathematics Archive, University of St Andrews. URL: <https://mathshistory.st-andrews.ac.uk/Biographies/Dirichlet/> (visited on 27/08/2026).
- R Core Team (2026). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. Vienna, Austria. URL: <https://www.r-project.org/> (visited on 27/08/2026).
- Rodriguez, C. (2020). *18.100A: Real Analysis*. Lecture notes prepared by P. Dote. MIT OpenCourseWare. URL: https://ocw.mit.edu/courses/18-100a-real-analysis-fall-2020/mit18_100af20_lec_full.pdf (visited on 27/08/2026).
- Schwarz, K. (4th Oct. 2021). *Functions, Part I*. Stanford University. URL: <https://web.stanford.edu/class/archive/cs/cs103/cs103.1222/lectures/06/> (visited on 27/08/2026).
- Strang, G. and E. Herman (2016a). *Calculus Volume 1*. OpenStax. URL: <https://openstax.org/details/books/calculus-volume-1> (visited on 27/08/2026).
- (2016b). *Calculus Volume 2*. OpenStax. URL: <https://openstax.org/details/books/calculus-volume-2> (visited on 27/08/2026).
- Wong Mee Kiong, Zaini bin Musa, Azizah binti Kamar, Saripah binti Ahmad, Nurbaiti binti Ahmad Zaki and Zefry Hanif bin Burham@Borhan (2019). *Additional Mathematics Form 4*. Penerbitan Pelangi Sdn. Bhd.
- Yusof, S. N., S. F. Hassan and D. Selanaithi (2025–2026). *FAD1013 Mathematics I*. PASUM lecture notes. Centre for Foundation Studies in Science, Universiti Malaya.
- Zaini bin Musa, Wong Mee Kiong, Azizah binti Kamar, Zakry bin Ismail, Nurbaiti binti Ahmad Zaki, Zefry Hanif bin Burham@Borhan and Saripah binti Ahmad (2020). *Additional Mathematics Form 5*. Abadi Ilmu Sdn. Bhd.

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Dirichlet portrait

Figure 2.1. Unknown creator, *Johann Peter Gustav Lejeune Dirichlet*. Source: Wikimedia Commons, `File:Dirichlet.jpg`. The file is identified as public domain and carries the Creative Commons Public Domain Mark 1.0. The PNG used in this book is a format conversion of the Commons image.

Transformation still

Figure 12.1. Grant Sanderson, 3Blue1Brown, *The other way to visualize derivatives* (2018), image `intro_2.jpg`. Source: `3blue1brown.com/lessons/derivatives-and-transforms/`. Licensed under the Creative Commons Attribution–NonCommercial–ShareAlike 4.0 International licence (CC BY-NC-SA 4.0): <https://creativecommons.org/licenses/by-nc-sa/4.0/>.

Complex Gamma visualisation

Figure 10.1. Original computational graphic generated for this book from the complex Gamma function using SciPy and Matplotlib. The reproducible source is `scripts/generate-original-figures.py`.

Racing-car diagram

Figure 12.2. Original TikZ diagram created for this book.

Publication Notes

Supplementary project materials

All user-supplied illustrated notes, snippets/screenshots of lecture slides, and diagrams were used as guides for selected explanations, exercises, and diagrams concerning sets, functions, matrices, inequalities, sequences and series, probability, statistical inference, differential equations, calculus, conic sections, parametric equations, vectors, reference frames, relative position, and statistical computing with R. Because these private project files have no identifiable author, publication date, publisher, or public retrieval location, they are described here rather than included as formal APA reference-list entries.

Editorial note

Unless otherwise stated, the surrounding exposition, derivations, notation, worked calculations, exercises, and TikZ diagrams were independently written and typeset for this book. Reference materials were reorganised, corrected where necessary, and adapted to maintain consistent notation throughout the document.

This book was developed and typeset in Prism with assistance from OpenAI language models. AI-assisted work included editorial restructuring, copy-editing, LaTeX drafting, and the implementation of selected diagrams. The author remained responsible for selecting the scope, reviewing the mathematics, adapting source material, and approving the final presentation.

About the Author

Asfa Izzat was born in Ipoh, Perak. Although mathematics was not initially his strongest subject at school, this did not prevent him from developing a lasting interest in it. His experience as a student shaped the central aim of this book: to present foundation mathematics as a connected body of ideas that can be learned progressively through clear explanations, worked examples, and carefully arranged practice. He began developing the book while studying at the Centre for Foundation Studies in Science, Universiti Malaya.

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